The Fermat tower in the metabelian approximation

Grothendieck, *La Longue Marche*, §51 and its refinements

LM7_FERMAT_TOWER — 13 November 2026

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INCOMPLETE: the exact residual questions are retained in the paper and ledger.

We give proofs and reproducible calculations for the integral schemes, the Fermat modules and splitting, the cuspidal motive, the arithmetic Kummer images, and the covering comparisons. We preserve the arithmetic plus-Fermat model of the manuscript. Full profinite realization questions and the remaining cubic section calculation are explicitly unresolved; the finite and rational calculations below do not replace those questions.

Notation fixed on this page. C4 means cote 140-4. A.1 gives the expanded dictionary, including the changing early and final versions.

Our notation	Meaning and manuscript symbol	C4 scan; folio
Π_0, F, M	$F \xrightarrow{\sim} \Pi_0$, $x \mapsto \lambda_0$, $y \mapsto \lambda_1$; $M = \Pi_0^{\mathrm{ab}}$	34–38; 664–666
$\mathfrak{S}_{\mathrm{d}}, \mathcal{S}$	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0'$ and its indicated finite schematic stage; his $\mathfrak{S}$ and underlined $\mathcal{G}^+$	34–38; 664–666
$\gamma = \mathbb{O}_{12}^*$	Units of his ring scheme O_{12}, with multiplier μ	51–58; 673–676v
$N, \Gamma, \mathcal{M}$	His corrected $N_{\rho,\sigma}$ and final $\Gamma_{\rho,\sigma}, M_{\rho,\sigma}$	62, 85–88; 678v, 690–691v
X_n, U_n	$X_n : x_0^n + x_1^n + x_\infty^n = 0$; U_n deletes its coordinate cusps	97–99; 696–697
Λ, c	$\widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$; a specified generator of $\widehat{F}'/\widehat{F}''$	comparison with 95–99; 695–697
$G_{\mathbb{Q}}, \chi, \kappa_a$	Arithmetic $\mathbb{I}\Gamma_{\mathbb{Q}}$; cyclotomic multiplier; Kummer cocycle with chosen roots	156–170; 725v–732v

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Scope, choices, and verification

We use the supplied second-pass transcriptions and inspect the available scans at the mathematical decision points recorded in the readings ledger. A transcription is evidence for a reading, and a proved identity is evidence for the mathematical meaning; the latter does not settle unread handwriting. The manuscript passage for each numbered statement is also recorded in `LM7_FERMAT_TOWER_results.csv`. All named groups in an argument carry the discrete, profinite, pro-ℓ, or algebraic interpretation stated there. Prime-to-characteristic and fppf assertions are specified separately.

We label a proved calculation *Proved*, a false proposed assertion together with its proved replacement *Refuted*, and a reduction to an opened theorem *Known*. An *Open* label records an uncompleted step of this return, with its proved cases and attempted approaches. Such a label does not assert that the literature has no answer. The report has a row for each of the three items, three leads, lettered tasks, and follow-ups in the request. The bibliography records which original texts were opened; a bibliographic identification alone is labelled unverified.

The programs use integer arithmetic, exact rational linear algebra, finite permutation groups, and finite-field arithmetic. Their saved standard output and machine-readable witnesses accompany the source. Finite verification is used with the displayed general proofs, and a Lie-algebra computation is transferred to a group only where the requisite comparison is proved. The source and complete reproduction instructions are in `code/`.

A. Integral structure, level twelve, and the congruence actions

We use constant finite group schemes, including in residue characteristic dividing their order. Thus C_2 below means the disjoint union of two copies of $\operatorname{Spec}\mathbb{Z}$; the morphism $C_2 \to \mu_2$, $i \mapsto (-1)^i$, will always be written explicitly when it is used. All scheme exact sequences in this section are exact for the fppf topology. Our finite-stage notation distinguishes a group from its schematic envelope by specifying their comparison morphism.

A.1. Dictionary and the finite-stage comparison maps [Proved]

Fix the following matrices, over $\mathbb{Z}$:

$$\rho = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}, \qquad \sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \tau = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \omega = -I.$$

Put

$$\lambda_0 = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \quad \lambda_1 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix}, \quad \Pi_0 = \langle \lambda_0, \lambda_1 \rangle, \quad M = \Pi_0^{\mathrm{ab}} = \mathbb{Z}e_0 \oplus \mathbb{Z}e_1.$$

The following dictionary fixes the successive versions in §51.

Our symbol	Definition and manuscript symbol	C4 scan; folio
$\mathfrak{S}_{\mathrm{d}}$	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0'$, the determinant-one part of the discrete $\mathfrak{S}$; $\Pi_0' = [\Pi_0, \Pi_0]$	33–34; 664–664v
$\mathcal{S}$	Pushout of $1 \to M \to \mathfrak{S}_{\mathrm{d}} \to H \to 1$ along $M \to \underline{M} = \mathbb{G}_a^2$; his $\underline{\mathcal{G}}^+$	34–38; 664–666
H	$\mathrm{SL}_2(\mathbb{Z})/\Pi_0$, of order 12; the full determinant-sign quotient has order 24	33, 66; 664, 680v
$\mathbb{O}_N$	Additive pushout $\mathbb{Z}_{\mathrm{const}} \amalg_{N\mathbb{Z}} \underline{N\mathbb{Z}}$, with its ring law in A.5	51–54; 673–674v
γ	$\mathbb{O}_{12}^*$; $\mu : \gamma \to \mathbb{G}_m$ is evaluation; $\varepsilon_2, \varepsilon_3 : \gamma \to C_2$ are the two residue characters	55–58; 675–676v
M_0	$\{c_0e_0 + c_1e_1 : c_0 + c_1 \equiv 0 \pmod 3\}$, a lattice with its own vector group $\underline{M}_0$	60–62; 677v–678v
N	The corrected $\underline{N}_{\rho,\sigma}$, with the loop equation in $\underline{M}_0$	62; 678v
Z, G	$\underline{Z}_{\rho,\sigma}$ and $\underline{G}_{\rho,\sigma}$, namely $\mathbb{G}_a \rtimes \gamma$ and $\mathcal{S} \rtimes \gamma$	75–76; 685–685v
$\Gamma_{\mathrm{e}}, \mathcal{M}_{\mathrm{e}}$	The versions of $\Gamma_{\rho,\sigma}, \underline{M}_{\rho,\sigma}$ in (123)–(127)	70–72; 682v–683v
$N_\rho, N_\rho^?$	The smooth closure of the ρ-normalizer, and its final bisection (212); A.13 records the smaller literal version (171)	78–82, 94; 686v–688v, 694v
$N_\sigma, N_\sigma^\sharp$	σ-normalizer and its intersection with $G^\sharp = \underline{M} \rtimes \gamma$	83–84; 689–689v
$\Gamma, \mathcal{M}$	The final fibre products (183)–(184), retaining the central C_2	85–88; 690–691v
$\mathcal{M}(0)$, $\mathcal{M}(1/2)$, $\mathcal{M}(-\bar{j})$	The three graph subgroups in A.15	88–94; 691v–694v

Proof. The assertions about the discrete groups are proved in A.2. A section $h \mapsto s_h$ of $\mathfrak{S}_{\mathrm{d}} \to H$ gives the finite-stage scheme $\coprod_{h \in H} \mathbb{A}^2_{\mathbb{Z}}$, with multiplication

$$(m, h)(n, k) = (m + a_h n + c(h, k), hk), \qquad c(h, k) = s_h s_k s_{hk}^{-1} \in M.$$

Changing the section to $d_h s_h$ gives the explicit isomorphism $(m, h) \mapsto (m - d_h, h)$. Thus this construction has a specified, section-independent meaning. An integral cocycle is obtained from

$$\rho \longmapsto (R, 0), \quad \sigma \longmapsto (S, (1/3, -1/3)), \qquad R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Choose words s_h lifting the twelve elements of H and let t_h be their affine translations. Then $c(h, k) = t_h + a_h t_k - t_{hk} \in \mathbb{Z}^2$, since the indicated word lies in M. The affine representation kills ω; the retained component h records it. The pair consisting of the component and affine image is faithful: on its possible kernel in M, the affine image is translation by the lattice vector itself.

The envelope $E_{\mathfrak{S}_{\mathrm{d}}}$ of §50 is the inverse limit of the analogous stages obtained from nM. The transition from any of these stages to the stage above is induced by the inclusion $nM \hookrightarrow M$

and is a morphism of affine group schemes. It gives the comparison $E_{\mathfrak{S}_{\mathrm{d}}} \to \mathcal{S}$; the formulas below concern this specified finite stage. On profinite points the continuous map $\widehat{\mathfrak{S}_{\mathrm{d}}} \to \mathcal{S}(\widehat{\mathbb{Z}})$ uses one common H-component and the coordinate maps $\widehat{M} \to M \otimes \mathbb{Z}_p$ at every prime. This describes its image and the meaning of the diagonal points used later in the manuscript. Likewise $\widehat{\mathbb{Z}}^* \to \gamma(\widehat{\mathbb{Z}})$ sends μ to its single residue class r mod 12 and the unique $k \in \widehat{\mathbb{Z}}$ with $\mu = r + 12k$. The proof of A.5 shows directly that this is a continuous homomorphism. These comparison maps retain the integral and profinite data rather than identifying all points of the schemes with one profinite group. □

A.2. The subgroup, its free basis, and the action of the quotient [Proved]

The subgroup Π_0 is normal and free on the displayed λ_0, λ_1. Its indices in $\mathrm{SL}_2(\mathbb{Z})$ and $\mathrm{GL}_2(\mathbb{Z})$ are 12 and 24. Moreover

$$H \simeq \mathrm{SL}_2(\mathbb{Z}/4)/V_4 \simeq \mathrm{Dic}_3, \qquad V_4 = \langle \lambda_0 \bmod 4, \lambda_1 \bmod 4 \rangle.$$

Conjugation on M sends ρ, σ, τ respectively to $R, S, -I$; in particular $\tau' = \omega\sigma\tau$ acts as $-S$. For every divisor N of 24, the action of $\mathrm{GL}_2(\mathbb{Z})/\Pi_0$ on M/NM has the following orders.

N	1	2	3	4	6	8	12	24
$\lvert\,\mathrm{im}\,\rvert$	1	6	12	12	12	12	12	12
$\lvert\,\ker\,\rvert$	24	4	2	2	2	2	2	2

This proves the index and lattice-action portion of Lead 2 (§51.1–4; C4 scans 34–38, 64–67; fols. 664–666, 679v–681).

Proof. Matrix multiplication gives

$$\lambda_0 = \omega\sigma\rho\sigma\rho, \quad \lambda_1 = \omega\rho\sigma\rho\sigma,$$

$$\rho\lambda_0\rho^{-1} = \lambda_1, \quad \rho\lambda_1\rho^{-1} = (\lambda_1\lambda_0)^{-1}, \quad \sigma\lambda_0\sigma^{-1} = \lambda_1, \quad \sigma\lambda_1\sigma^{-1} = \lambda_0.$$

Conjugation by τ inverts each λ_i. These identities prove normality and the stated action after abelianization. We use the amalgam presentation $\mathrm{SL}_2(\mathbb{Z}) = C_4 *_{C_2} C_6$, with factors generated by σ, ρ and $\sigma^2 = \rho^3 = \omega$; it is recorded in [KLG13, §4, equation (1)]. Adding $\lambda_0 = 1$ forces $\sigma\rho\sigma^{-1} = \rho^{-1}$; $\lambda_1 = 1$ is its conjugate. The resulting group has the twelve normal forms $\rho^i\sigma^j$, $0 \le i < 6$, $0 \le j < 2$. Modulo four, V_4 has four elements, is normal, and the images of ρ, σ in $\mathrm{SL}_2(\mathbb{Z}/4)/V_4$ retain orders six and four. Since $|\mathrm{SL}_2(\mathbb{Z}/4)| = 48$, this realizes all twelve normal forms and proves the index claim and the stated quotient. Surjectivity of the reduction map follows also from elementary row operations over $\mathbb{Z}/4$.

The Bass–Serre tree of the amalgam has trivial Π_0-stabilizers: each factor, and therefore every conjugate of a factor, injects in H. Its quotient by Π_0 has $12/4 + 12/6 = 5$ vertices and $12/2 = 6$ edges. It is connected. Choosing a maximal tree shows directly that its fundamental group is free of rank $6 - 5 + 1 = 2$, and lifting paths identifies this group with Π_0. Since λ_0, λ_1 generate it, they are a free basis. For completeness, a surjective endomorphism of a finitely generated free group is injective: free groups are residually finite by completing the finite labelled path of a nonempty reduced word to permutations; on the finite set of subgroups of any fixed finite index, inverse image under a surjection is an injective permutation. A nontrivial element in its kernel would then lie in every finite-index subgroup, a contradiction. This applies to the map from a chosen free basis to the displayed generating pair.

The matrices R, S satisfy $R^3 = S^2 = (RS)^2 = I$ and give the six distinct matrices of S_3 modulo every $N \ge 2$. The additional $-I$ is already I modulo two. For $N \ge 3$, $-I$ is outside those six matrices: direct inspection of I, R, R^2, S, RS, R^2S proves this over $\mathbb{Z}/N$. Thus the

image has order twelve for each displayed $N \geq 3$. The central element ω is the order-two kernel integrally, and the table follows. The program `a_finite.py` constructs the twenty-four cosets modulo four and propagates the matrix action by a coupled breadth-first traversal; it also constructs the above quotient graph and checks its connectivity and Euler characteristic. This is a finite certificate for the presentation argument, not a finite test used in place of freeness. □

A.3. The precise source of level twelve and the group of order thirty-two [Proved]

Let Π_{00} be the inverse image in Π_0 of M_0, and put $\delta = \lambda_1\lambda_0^{-1}$. Then

$$\Pi_{00} = \langle\langle \delta \rangle\rangle_{\mathrm{SL}_2(\mathbb{Z})} = \Pi_0 \cap \mathrm{SL}_2(\mathbb{Z})', \qquad [\mathrm{SL}_2(\mathbb{Z}) : \Pi_{00}] = 36.$$

Its congruence level is exactly 12, and the reduction map gives

$$K_{32} := \Pi_{00}/\Gamma(12) = V_4 \times Q_8 \ \subset\ \mathrm{SL}_2(\mathbb{Z}/4) \times \mathrm{SL}_2(\mathbb{F}_3) = \mathrm{SL}_2(\mathbb{Z}/12).$$

Here Q_8 is the quaternion subgroup of $\mathrm{SL}_2(\mathbb{F}_3)$. The subgroup K_{32} is normal in $\mathrm{GL}_2(\mathbb{Z}/12)$. This is the group of order 32 asked for at C4 scan 64, folio 679v; its exact congruence origin is the additional index-three lattice $M_0 \subset M$, together with the level-four realization of Π_0.

Proof. With $e_i = [\lambda_i]$ we have

$$M_0 = \mathbb{Z}(e_1 - e_0) + \mathbb{Z}(3e_1) = (1 - R)M + (1 - S)M.$$

The normal closure J of δ is contained in Π_{00}. In Π_0/J, the relation $\lambda_0 = \lambda_1$ and its ρ-conjugate give $\lambda_1 = (\lambda_1\lambda_0)^{-1}$, so the quotient is cyclic of order at most three. The map $M \to \mathbb{Z}/3$, $(c_0, c_1) \mapsto c_0 + c_1$, shows that its order is exactly three. Hence $J = \Pi_{00}$. Abelianizing the amalgam gives $\mathrm{SL}_2(\mathbb{Z})^{\mathrm{ab}} \simeq \mathbb{Z}/12$, with $[\rho] = 2$ and $[\sigma] = 3$ after choosing a generator. Each $[\lambda_i] = 4$. The restriction to Π_0 therefore has kernel precisely Π_{00}, proving the intersection formula. In particular the two occurrences of 12 in Lead 2 are related by the exact sequence $0 \to C_3 \to C_{12} \to H^{\mathrm{ab}} = C_4 \to 0$.

Chinese remainders give the displayed product decomposition. The matrix of δ is

$$\delta = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \qquad \delta^2 \equiv 5I \pmod{12}.$$

Modulo four its normal closure is V_4: conjugation cyclically permutes the three nonzero elements. Modulo three its normal closure is the eight-element group consisting of $\pm I$ and the six trace-zero determinant-one matrices; each of those six squares to $-I$. It is closed, as one checks on two anticommuting generators, and is the quaternion group. To see that the normal closure in the product is the whole product, take commutators of δ with $(g, 1)$: these generate $V_4 \times 1$. We can then isolate $(1, \delta \bmod 3)$, whose normal closure is $1 \times Q_8$. Consequently the image of Π_{00} is $V_4 \times Q_8$. Its full inverse image has index $48 \cdot 24/32 = 36$, equal to the already proved index of Π_{00}; the two subgroups coincide. This proves $\Gamma(12) \subset \Pi_{00}$.

For minimality, the upper transvection $T^m = \left(\begin{smallmatrix} 1 & m \\ 0 & 1 \end{smallmatrix}\right)$ belongs to Π_{00} only when $4 \mid m$, because the only such element of V_4 is I, and only when $3 \mid m$, because Q_8 has no nonidentity element of order three. Every $\Gamma(m)$ contains T^m, so $\Gamma(m) \subset \Pi_{00}$ implies $12 \mid m$. Finally V_4 is normalized by $\mathrm{GL}_2(\mathbb{Z}/4)$, and the trace description of Q_8 is invariant under every invertible conjugator. Their product is normal in the full level-twelve group. All thirty-two matrices, their orders, their conjugates, and the normal-closure assertions are exhaustively checked and printed by `a_finite.py`. □

A.4. All congruence arrows and the scope of Lead 2 [Proved]

The exact finite quotients obtained from A.3 have orders

group	order	quotient by K_{32}
$\mathrm{SL}_2(\mathbb{Z}/12)$	1152	36
$\mathrm{im}(\mathrm{GL}_2(\mathbb{Z}) \to \mathrm{GL}_2(\mathbb{Z}/12))$	2304	72
$\mathrm{GL}_2(\mathbb{Z}/12)$	4608	144

The middle image is $\det^{-1}\{1,-1\}$. Thus reduction induces isomorphisms

$$\mathrm{SL}_2(\mathbb{Z})/\Pi_{00} \simeq \mathrm{SL}_2(\mathbb{Z}/12)/K_{32}, \qquad \mathrm{GL}_2(\mathbb{Z})/\Pi_{00} \simeq \overset{-1}{\det}\{1,-1\}/K_{32}.$$

There is also a surjective homomorphism $\mathfrak{S}_{\mathrm{d}} \to \mathrm{SL}_2(\mathbb{Z}/12)/K_{32}$ with kernel M_0. Reduction to $\mathrm{SL}_2(\mathbb{Z}/12)$ itself does not factor through $\mathfrak{S}_{\mathrm{d}}$. These are the actual morphisms relating the action on M to the order-32 subgroup. The action images in A.2 have order at most 12; none is the order-32 group. They determine the lattice M_0 by $M_0 = (1-R)M + (1-S)M$, whose inverse image is used in A.3. This proves the strongest precise form of Lead 2 and the relation requested in item 1 (C4 scans 64–67; fols. 679v–681).

Proof. The determinant map on $\mathrm{GL}_2(\mathbb{Z}/12)$ is onto its four units, with kernel of order 1152. Integral invertible matrices have determinant ± 1, and the elementary generators of $\mathrm{SL}_2(\mathbb{Z}/12)$ lift to integral matrices. Adjoining τ therefore gives exactly the middle group. Taking inverse images of K_{32} proves each isomorphism and each kernel claim. For the last assertion, direct multiplication gives

$$[\lambda_0, \lambda_1] \equiv \begin{pmatrix} 1 & 4 \\ 4 & 5 \end{pmatrix} \not\equiv I \pmod{12}.$$

This element is killed in $\mathfrak{S}_{\mathrm{d}}$ but survives under reduction, proving the asserted failure of factorization by an explicit element. It dies after quotient by K_{32}, as the exact kernel calculation requires. This also verifies the qualification in the margin of scan 67. The scan 66 notation for the Sylow-two group is transcribed as D_4; the six order-four elements and unique involution prove that its abstract group is Q_8, independently of the notation intended by that handwriting. □

A.5. The ring scheme and its exact unit sequence [Proved]

For a positive integer N, the ring scheme $\mathbb{O}_N$ is $\coprod_{r \in \mathbb{Z}/N} \mathbb{A}^1_{\mathbb{Z}}$. A section is locally written $n + N[x]$, with the equivalence

$$(n, x) \sim (n + Nk, x - k) \quad (k \in \mathbb{Z}).$$

Addition is the additive pushout law, and multiplication is

$$(n + N[x])(m + N[y]) = nm + N[ny + mx + Nxy]. \tag{A.5.1}$$

Its unit scheme and unit exact sequence are

$$\mathbb{O}_N^* = \coprod_{r \in (\mathbb{Z}/N)^*} \operatorname{Spec} \mathbb{Z}[t, (r + Nt)^{-1}],$$

$$1 \longrightarrow D(1 + Nt) \longrightarrow \mathbb{O}_N^* \longrightarrow (\mathbb{Z}/N)^*_{\mathrm{const}} \longrightarrow 1. \tag{A.5.2}$$

The right arrow is smooth and surjective. The corresponding additive sequence is $0 \to \mathbb{G}_a \to \mathbb{O}_N \to (\mathbb{Z}/N)_{\mathrm{const}} \to 0$. These prove (64)–(66), (74), and (78)–(79), with the stated choice of constant finite schemes (C4 scans 51–55; fols. 673–675).

More generally, for a unital associative ring A and a two-sided ideal J that is finite free as an abelian group, the additive pushout $A[J]$ has a ring structure and exact sequences

$$0 \to \underline{J} \to A[J] \to (A/J)_{\mathrm{const}} \to 0, \quad 1 \to (1+\underline{J})^* \to A[J]^* \to ((A/J)^*)_{\mathrm{const}} \to 1. \tag{A.5.3}$$

The second surjection is smooth. This supplies the general version of (69)–(73), including a proof of its local surjectivity.

Proof. The pushout is the quotient of $\mathbb{Z}_{\mathrm{const}} \times \mathbb{G}_a$ by the constant subgroup $k \mapsto (Nk, -k)$. Choosing one integer representative of every residue identifies its components as claimed. Formula (A.5.1) is unchanged under either replacement $(n, x) \mapsto (n + Nk, x - k)$; direct expansion proves associativity and distributivity. Equivalently it is the multiplication inherited from the ring with square-zero-free ideal coordinate $N\mathbb{Z} \otimes R$, whose internal product is $N[x]\,N[y] = N[Nxy]$.

A unit must have unit residue and unit evaluation $n + Nx$. Conversely choose integers r, r' with $rr' = 1 + Nh$ and set

$$y = -\frac{h + r'x}{r + Nx}.$$

Then $(r + N[x])(r' + N[y]) = 1$, proving both sufficiency and the displayed localized coordinate ring. In residue class one the coordinate is $1 + N[t]$, with multiplication $t \star t' = t + t' + Ntt'$; this is the kernel in (A.5.2). Every geometric fibre of every component is nonempty: when the characteristic divides N, its representative is nonzero; otherwise at most one value of t is excluded. The arrow is an open subscheme of an affine-line bundle over the constant base, so it is smooth and surjective, and therefore fppf locally admits sections. This proves exactness without asserting surjectivity on points over every ring.

For the general construction, take $(A_{\mathrm{const}} \times \underline{J})/J_{\mathrm{const}}$, where $j \mapsto (j, -j)$, with multiplication $(a, x)(b, y) = (ab, ay + xb + xy)$. Each component is an affine space of rank $\operatorname{rank} J$. Fix a component $b \in (A/J)^*$ and a lift $a \in A$. On this component multiplication on J on the left and on the right is a pair of square matrices. The unit locus is the open set where both determinants are invertible. Indeed, a chosen lift a' of b^{-1} leaves both inverse errors in J; invertibility of left and right multiplication on J corrects the two errors, producing a left and a right inverse, which coincide.

To prove that this open set has a point in every geometric fibre, work over an algebraically closed field k. In the finite-dimensional algebra of pairs of endomorphisms of J_k, take the subalgebra generated by the left-right multiplication pairs from A and J_k. The image of J_k is an ideal, and the image of a is a unit modulo this ideal. In any finite-dimensional algebra, units lift modulo every ideal: reduce by the nilpotent Jacobson radical, decompose the semisimple quotient into matrix blocks, and choose arbitrary invertible entries in the blocks killed by the ideal; units then lift across the nilpotent radical by the finite geometric series. Lifting the adjustment back to J_k gives $a + j$ with both multiplication operators invertible. The preceding correction supplies its inverse. Thus every component-unit fibre is nonempty. It is open in affine space, so the map in (A.5.3) is smooth and surjective. The kernel is exactly the units of the residue-one component. This also proves the pair description (73): in the finite free unitization of J, a right inverse is a two-sided inverse, by the determinant criterion for multiplication on a finite free module. □

A.6. Three literal ring formulas and their exact replacements [Refuted]

The evaluation criterion in (67b), the multiplication printed in (68), and the full-kernel assertion (77) are false for the pushout (66). Their correct replacements are A.5 and

$$\ker(\mathbb{O}_N^* \xrightarrow{\mu} \mathbb{G}_m) = \coprod_{r \in (\mathbb{Z}/N)^*} \operatorname{Spec} \mathbb{Z}[t]/(r + Nt - 1). \tag{A.6.1}$$

The residue-one component alone is ${}_N\mathbb{G}_a$. After inverting N, evaluation and residue give an isomorphism of ring schemes $\mathbb{O}_N \simeq \mathbb{G}_a \times (\mathbb{Z}/N)_{\mathrm{const}}$, and of unit schemes $\mathbb{O}_N^* \simeq \mathbb{G}_m \times (\mathbb{Z}/N)_{\mathrm{const}}^*$. The putative global section of the unit-residue map using the four integers $\pm 1, \pm 5$ in (83) is also false over $\mathbb{Z}$. These are mathematical corrections to C4 scans 51–56, fols. 673–675v; they do not settle an unattached handwritten reading.

Proof. Over $\mathbb{F}_2$, the sections $0{+}2[0]$ and $0{+}2[1]$ are distinct points of the same affine component of $\mathbb{O}_2$, while their evaluations coincide. Thus evaluation does not detect equality. For (68), the product $12[1]\,12[1]$ evaluates to 144 over $\mathbb{Z}$; the printed formula evaluates to 12. Formula (A.5.1) inserts exactly the missing factor N.

Evaluation on the r-component is $r+Nt$, giving (A.6.1). For $N = 12$, the section $5{+}12[-1/3]$ over $\mathbb{Q}$ has evaluation one and residue five. It is a nonidentity element of the full kernel, although ${}_{12}\mathbb{G}_{a,\mathbb{Q}} = 0$. After N is inverted, $(\mu, r) \mapsto r + N[(\mu - r)/N]$ is the inverse of the displayed comparison, proving the stronger structural assertion. Finally a $\mathbb{Z}$-point of $\mathbb{O}_{12}^*$ evaluates to ± 1, and hence has residue 1 or 11. The other two residue components have no integral point. Thus there is no global section of the four-component residue map over $\mathbb{Z}$; the numbers ± 5 can be used as affine coordinate representatives only. All these counterexamples are checked by `a_schemes.py`. □

A.7. Integral coordinates on the multiplier scheme [Proved]

Write $a = \nu_2, b = \nu_3 \in \{0,1\}_{\mathrm{const}}$, so that $\varepsilon_2 = (-1)^a$ and $\varepsilon_3 = (-1)^b$. The four charts of γ have coordinates

$$r_0 = 1 - 6a + 4b, \qquad \mu = r_0 + 12k \in R^*.$$

The regular functions used in the manuscript are

$$\nu = -3a + 2b + 6k, \quad \nu' = -a + b + 3k, \quad \nu'' = -a + b + 2k, \tag{A.7.1}$$

and satisfy, integrally,

$$\mu = 1 + 2\nu = 1 - 2a + 4\nu' = 1 - 2b + 6\nu'', \qquad 2\nu' - a = 3\nu'' - b. \tag{A.7.2}$$

For the alternative representative $\mu_0 = (-5)^a 5^b$ used in (84)–(92), with $\mu = \mu_0 + 12\nu_0$, the required formulas instead are

$$\nu' = -a + b - 6ab + 3\nu_0, \qquad \nu'' = -a + b - 4ab + 2\nu_0. \tag{A.7.3}$$

The transcribed coefficients $-3ab, -2ab$ are false. The substitutions (A.7.3) correct the corresponding expansions in (138)–(141) as well. The map $\gamma \to \operatorname{Spec}\mathbb{Z}$ is smooth, flat, of relative dimension one, with four geometric components at every prime.

Proof. The representatives $1, -5, 5, -1$ are congruent to $(-1)^a$ mod 4 and $(-1)^b$ mod 3. Each equation in (A.7.2) follows by expansion. Conversely the simultaneous data in (63) give $k = \nu' - \nu''$, and substitution recovers (A.7.1) without division by two or three. Thus these are integral isomorphisms of schemes. Expanding $(-5)^a 5^b$ on the four constant components gives $1 - 6a + 4b - 24ab$, whence (A.7.3). The discrepancy is detected over $\mathbb{F}_7$ by $a = b = 1, \nu_0 = 0$: $\mu = -25 \equiv 3$ is a unit, whereas either of the transcribed formulas reconstructs $\mu = 1$. The exact polynomial checks are in `a_schemes.py`. Smoothness and the component claim follow from the four open affine-line charts of A.5. In characteristics two and three, each chart is the entire affine line since its evaluation is a constant nonzero element. These calculations concern C4 scans 51, 57–58, 74–75; fols. 673, 676–676v, 684v–685. □

A.8. The first requested exact sequence, over every base ring [Proved]

Set $\delta = e_1 - e_0$ and $e = 3e_1$, a basis (δ, e) of M_0. For $\alpha = a_0e_0 + a_1e_1$ and $\beta = b_0e_0 + b_1e_1$, put

$$F(\alpha, \beta) = (1 - R)\alpha - (1 - S)\beta = (-a_0 - a_1 + b_0 - b_1)\delta + a_1e. \tag{A.8.1}$$

Define N by $F(\alpha, \beta) = \nu''e$ *in* $\underline{M}_0$, with multiplier in γ. Then

$$(t, s, \underline{\mu}) \longmapsto \Big(\underline{\mu}, \alpha = te_0 + \nu''\delta, \beta = te_0 + s(e_0 + e_1)\Big) \tag{A.8.2}$$

is an isomorphism of schemes $\mathbb{G}_a^2 \times \gamma \simeq N$, and its group law is

$$(t, s, \underline{\mu})(t', s', \underline{\mu}') = (t + \mu t',\ s + \mu s',\ \underline{\mu\mu'}). \tag{A.8.3}$$

Consequently the sequence requested at the beginning of §51.4 is split exact over $\mathbb{Z}$:

$$1 \longrightarrow N^{0+} = \mathbb{G}_a^2 \longrightarrow N \longrightarrow \mathbb{O}_{12}^* \longrightarrow 1. \tag{A.8.4}$$

The middle group is smooth and flat of relative dimension three, including at two and three. This proves (104)–(105) after the author's correction in scans 60–62, fols. 677v–678v.

Proof. Multiplication by the displayed R, S gives (A.8.1). The two coordinates of its required equality say exactly $a_1 = \nu''$ and $a_0 + a_1 = b_0 - b_1$. Therefore the inverse to (A.8.2) is $t = a_0 + a_1$, $s = b_1$, with no division by an integer. In particular $F : M^2 \to M_0$ is a split surjection of free lattices.

We verify the group law and its automorphism meaning. For $\underline{\mu} \in \gamma(R)$, define an automorphism $u_{\underline{\mu}}$ of $\mathcal{S}_R$ by

$$u_{\underline{\mu}}(\rho) = \lambda_1^{\nu}\rho, \qquad u_{\underline{\mu}}(\sigma) = \omega^a\sigma, \qquad u_{\underline{\mu}}(m) = \mu m \quad (m \in \underline{M}). \tag{A.8.5}$$

Exponents on the lattice denote additive vector-group coordinates. These formulas define the morphism on every component of the finite-stage scheme. Indeed $(1 + R + R^2)e_1 = 0$ gives $(\lambda_1^{\nu}\rho)^3 = \omega$, and $(\omega^a\sigma)^2 = \omega$. Substituting in $\lambda_0 = \omega\sigma\rho\sigma\rho$ and $\lambda_1 = \omega\rho\sigma\rho\sigma$ gives multiplication by $1 + 2\nu = \mu$ on both lattice generators. The commutator subgroup of the lattice is therefore killed. These checks construct a compatible map on the twelve charts of $\mathcal{S}$, using the scalar map on their affine coordinates; they do not infer uniqueness merely from the values of two sections in a nonreduced fibre. The identity

$$\nu(\underline{\mu\mu'}) = \nu(\underline{\mu}) + \mu\nu(\underline{\mu}')$$

holds as an integral polynomial on every pair of components, by (A.7.1). The sign a adds modulo two. Thus $u_{\underline{\mu}}u_{\underline{\mu}'} = u_{\underline{\mu\mu'}}$; the formula for the inverse follows, so these are automorphisms.

The pair of elements

$$\Big(\mathrm{Int}(te_0)u_{\underline{\mu}}, \quad (-s(e_0 + e_1))u_{\underline{\mu}}\Big) \tag{A.8.6}$$

records the effective automorphism and the σ-normalizer conjugator of (A.8.2). To check this directly, $(1 - R)\delta = 3e_1$ and $\nu = 3\nu'' - b$ give the first generator formula after the prescribed τ^b correction; the second follows from $te_0 - \beta = -s(e_0 + e_1)$ and $(1 - S)(e_0 + e_1) = 0$. Each factor in (A.8.6) multiplies with scalar action μ on its one-dimensional translation coordinate. Their fibre product over γ therefore has exactly (A.8.3), which proves the transported group law on the manuscript's data. Its section is $t = s = 0$. Smoothness follows from (A.8.2) and A.7. □

A.9. The premature exact sequences and their flat replacement [Refuted]

The surjection onto $\mathbb{G}_m \times C_2^2$ in (54), and the surjection $M^2 \to M$ in (59), are false as written. The latter has cokernel $\mathbb{Z}/3$, and the correct integral lattice sequence is

$$0 \to P \to M^2 \xrightarrow{F} M \to \mathbb{Z}/3 \to 0, \quad P = \mathbb{Z}(e_0, e_0) \oplus \mathbb{Z}(0, e_0 + e_1). \tag{A.9.1}$$

The raw scheme obtained by imposing the loop equation only in $\underline{M}$ is

$$\widetilde{Z} \simeq \operatorname{Spec}_\gamma \mathcal{O}_\gamma[t, s, z]/(3z). \tag{A.9.2}$$

Its projection to γ is split surjective, with kernel $\mathbb{G}_a^2 \times {}_3\mathbb{G}_a$ as a scheme. The closed subscheme $z = 0$ is exactly N of A.8 and is the schematic closure of $\widetilde{Z}_{\mathbb{Z}[1/3]}$. The fine kernel sequence (56), after the sign in (55) is corrected to $1 - R$, is exact with raw kernel $\mathbb{G}_a^2 \times {}_3\mathbb{G}_a$ and quotient ${}_4\mathbb{G}_a$. These are the full corrections to (54)–(59), (94)–(103) (C4 scans 48–49, 59–62; fols. 671v–672, 677–678v).

Proof. The sum of the two coordinates of $F(\alpha, \beta)$ is $3a_1$. Its image is therefore M_0, and (A.8.1) proves surjectivity onto that lattice and the stated kernel. Tensoring this map with a ring does not preserve the monomorphism $\underline{M_0} \to \underline{M}$: in its basis (δ, e) the determinant is -3. Imposing the equation in $\underline{M}$ instead yields $3(a_1 - \nu'') = 0$; the other equation determines b_0 from a_0, a_1, b_1. Take $z = a_1 - \nu''$, $t = a_0 + a_1$, $s = b_1$ to obtain (A.9.2). Its ideal of functions vanishing after inverting three is (z), which proves the closure assertion. In particular the characteristic three raw fibre has dimension four, while the corrected fibre has dimension three. The kernel version has dimensions three and two. Finite point enumeration over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$ verifies these counts in `a_schemes.py`.

For (54), over the algebraic closure of $\mathbb{F}_2$, equation (63) forces $\mu = 1$ on every component, so any other nonzero element of that field has no lift, even fppf locally. Equivalently, the already specified map to the naive character base factors through γ; the correct exact target is the one in A.8. For (56), put $a = b = 0$ and $w = \nu'$. Its equations are $4w = 0$ and $F(\alpha, \beta) = 2we_1$. A section of the projection to ${}_4\mathbb{G}_a$ is $\alpha = 2we_0 - 2we_1$, $\beta = 0$: $F(\alpha, 0) = -6we_1 = 2we_1$ since $4w = 0$. The kernel is $F = 0$, already computed. The literal plus sign in $(1 + R)\alpha$ in (55) is inconsistent with (57), for example at $\alpha = \beta = e_0, w = 0$ over $\mathbb{Z}$. We have stated which replacement makes the sequence exact.

Scan 61 (100)–(101) contains another sign discrepancy visible in the attached image: the zero equation (96) requires $F(\alpha, \beta) - \nu'' e = 0$, or equivalently $(1 - R)(\alpha - \nu''\delta) - (1 - S)\beta = 0$. The plus sign in the shifted α gives $6e_1$ on $\alpha = \delta, \beta = 0, \nu'' = 1$ over $\mathbb{Z}$. Thus this replacement is forced by (94)–(96); it is not an unannounced choice between readings. □

A.10. The integral comparison with the fine scheme of LM1 [Proved]

The fine scheme of scan 41, with component bits a, b, coordinate $w = \nu'$, and loop equation $3a_1 = 2w - a + b$, is isomorphic over $\mathbb{Z}$ to the corrected N of A.8. The comparison and its inverse are

$$k = w - a_1, \quad \nu'' = a_1, \quad t = a_0 + a_1, \quad s = b_1, \qquad w = 3k - a + b, \quad a_1 = 2k - a + b, \tag{A.10.1}$$

with α, β recovered by (A.8.2). In particular it is an isomorphism on fibres at two and three and on points over every ring. The alternate LM1 coordinate using $\nu = 2w + a$ is carried to the present coordinate using $\nu = 2w - a$ by $w_{\text{here}} = w_{\text{LM1}} + a$. This settles the finite-stage comparison left in LM1 E.20 for this specific parameter scheme (C4 scans 40–41, 62, 71–72; fols. 667v–668, 678v, 683–683v).

Proof. The equation $2w-3a_1 = a-b$ implies $w = 3(w-a_1)-a+b$ and $a_1 = 2(w-a_1)-a+b$. Thus (A.10.1) is integral in both directions. Substituting gives $\mu = 1-2a+4w = 1-6a+4b+12k$, the same inverted element in both coordinate rings. The remaining loop coordinate equation is exactly $a_0 + a_1 = b_0 - b_1$, whose solutions are (A.8.2). This proves the claimed scheme isomorphism, including all nilpotent test rings. Both groups use the automorphism and conjugator multiplication (A.8.6), so the comparison also respects their group structures. The final coordinate translation follows by equating $2w_{\mathrm{here}} - a = 2w_{\mathrm{LM1}} + a$ as integral functions; substitution, not cancellation in a characteristic-two ring, proves the equality. □

A.11. Effective automorphisms and the early quotient sequences [Proved]

The effective group, its geometric extension, and the early quotient groups are

$$Z = \mathbb{G}_a \rtimes_\mu \gamma, \quad G = \mathcal{S} \rtimes_u \gamma, \quad \Gamma_{\mathrm{e}} = \mathbb{G}_a \rtimes_\mu \gamma,$$
$$\mathcal{M}_{\mathrm{e}} \simeq (\mathbb{G}_a \times \mathcal{S}) \rtimes \gamma, \qquad \mathcal{M}_{\mathrm{e}}^{\sharp} \simeq \mathbb{G}_a^3 \rtimes_\mu \gamma. \tag{A.11.1}$$

The γ-action on $\mathcal{S}$ is (A.8.5). In N, the subgroup L_0 is the t-axis and the quotient is the s-coordinate. We obtain the exact sequences

$$\begin{aligned} &1 \to L_0 \to N \to \Gamma_{\mathrm{e}} \to 1, && 1 \to \mathbb{G}_a \to \Gamma_{\mathrm{e}} \to \gamma \to 1, \\ &1 \to \mathcal{S} \to \mathcal{M}_{\mathrm{e}} \to \Gamma_{\mathrm{e}} \to 1, && 1 \to \underline{M} \to \mathcal{M}_{\mathrm{e}}^{\sharp} \to \Gamma_{\mathrm{e}} \to 1, \\ &1 \to L_0 \to Z \to \gamma \to 1, && 1 \to \mathcal{S} \to G \to \gamma \to 1. \end{aligned} \tag{A.11.2}$$

All are split and exact on points over every ring. The lattice kernel sequences (127)–(132) are $0 \to \mathbb{Z} \to P \to \mathbb{Z} \to 0$ and their vector-group base changes; the maps are $(t,s) \mapsto s$ and $t \mapsto (t,0)$. These prove (123)–(127), (131)–(132), (142), (145)–(150) (C4 scans 70–76; fols. 682v–685v).

Proof. Under (A.8.6), the first factor is $\mathrm{Int}(te_0)u_{\underline{\mu}}$ and the second has coordinate $-s$. The kernel of the effective projection is the s-axis; its image is the displayed Z. Inner conjugation by te_0 embeds $L_0 \simeq \mathbb{G}_a$ into Z, and (A.8.5) normalizes it by the scalar μ. Thus the quotients are represented by the displayed affine schemes, with the stated sections.

For the crossed quotient defining $\mathcal{M}_{\mathrm{e}}$, take $\mathcal{S} \rtimes N$ with the action of N through Z and quotient by $t \mapsto (-te_0, (t,0,1))$. This is a normal subgroup because its action on $\mathcal{S}$ equals inner conjugation by te_0. Every coset has exactly one representative with $t = 0$; its remaining coordinates are $(x, s, \underline{\mu})$ and multiply as in (A.11.1). The same calculation with $\underline{M}$ in place of $\mathcal{S}$ gives the sharp subgroup. All kernel and section assertions now follow by coordinate projection, so they are exact on arbitrary ring points and hence as fppf sheaves. The kernel lattice P was computed in A.9; its quotient by the diagonal e_0-line has the free generator $(0, e_0 + e_1)$. This proves every lattice and vector-group sequence asserted here.

The ambiguous exponent in scan 76 (151) is fixed by the verified formula $\lambda_1^{\nu}\rho$ in scan 93, and independently by $u(\lambda_i) = \mu\lambda_i$ in A.8. The candidate exponent $6\nu''$ fails this test, since the required relation is $\nu = 3\nu'' - b$. We retain this correction explicitly in the readings ledger. □

A.12. The two normalizers and the central extension of the multiplier scheme [Proved]

Let $d = \nu''$ and set, in $G = \mathcal{S} \rtimes \gamma$,

$$v_{\underline{\mu}} = \sigma^b \delta^{-d} u_{\underline{\mu}}.$$

For the raw normalizer of (152), before the closure in (159), the literal sequence (155) is

$$1 \to C_6 \times {}_3\mathbb{G}_a \to N_\rho^{\mathrm{raw}} \to \gamma \to 1, \qquad N_\rho^{\mathrm{raw}} \simeq {}_3\mathbb{G}_a \times N_\rho. \tag{A.12.0}$$

The smooth closure of the ρ-normalizer is represented by $\{\rho^k v_{\underline{\mu}} : k \in \mathbb{Z}/6\}$; hence it has the exact sequence

$$1 \to C_6 = L_\rho \to N_\rho \to \gamma \to 1. \tag{A.12.1}$$

The final subgroup specified in (212), scans 93–94, is

$$N_\rho^? = \{\omega^i v_{\underline{\mu}} : i \in \mathbb{Z}/2\}, \qquad 1 \to C_2 = \mu_\rho \to N_\rho^? \to \gamma \to 1. \tag{A.12.2}$$

Its multiplication on $C_2 \times \gamma$ is

$$(i, \underline{\mu})(i', \underline{\mu}') = (i + i' + c(\underline{\mu}, \underline{\mu}'), \underline{\mu\mu}'), \qquad c = (a+b)b' \pmod 2. \tag{A.12.3}$$

Thus (A.12.2) is the pullback along $(\varepsilon_2, \varepsilon_3) : \gamma \to C_2^2$ of the order-eight dihedral central extension $1 \to C_2 \to D_8 \to C_2^2 \to 1$. Moreover

$$N_\sigma^\sharp = \mathbb{G}_a \rtimes_\mu \gamma, \qquad N_\sigma = (C_4 \times \mathbb{G}_a) \rtimes \gamma, \tag{A.12.4}$$

where γ acts by ε_2 on C_4 and by μ on $\mathbb{G}_a$. These assertions prove (155), (159), and (173)–(181), and compute the extension asked for after (196) (C4 scans 78–84, 87, 93–94; fols. 686v–689v, 691, 694–694v).

Proof. The identities $(1-R)\delta = 3e_1$ and $\nu - 3d = -b$ give

$$\delta^{-d} u_{\underline{\mu}}(\rho) \delta^d = \lambda_1^{-b} \rho.$$

For $b = 1$, direct multiplication in the modular group gives $\sigma\lambda_1^{-1}\rho\sigma^{-1} = \rho^{-1}$. For $b = 0$ the displayed expression is ρ. Thus $v_{\underline{\mu}}$ has the required normalizing action $\rho \mapsto \rho^{\varepsilon_3}$. In the finite quotient H, the centralizer of ρ is its cyclic subgroup of order six. Within each of these six components, the remaining translation condition is $(1-R)m = 0$. Its kernel has equations equivalent to $3z = 0$, because $\det(1-R) = 3$. Explicitly $m = z\delta$, with $3z = 0$. The cyclic component C_6 fixes this vector. Conjugation by $v_{\underline{\mu}}$ acts on it as $\mu\varepsilon_3$, and $\mu\varepsilon_3 = 1 + 6\varepsilon_3 d$; hence it also fixes $z\delta$. This proves the central direct product (A.12.0) and the original sequence (155), including its extra fibre in characteristic three. The closure of its characteristic-not-three part is the zero translation. Translation by $v_{\underline{\mu}}$ therefore gives exactly the smooth normalizer stated above, including all its components, with kernel L_ρ.

The identity

$$d(\underline{\mu\mu}') = \varepsilon_3(\underline{\mu}')d(\underline{\mu}) + \mu d(\underline{\mu}') \tag{A.12.5}$$

is an integral polynomial identity on the sixteen pairs of charts; it follows by multiplying $\mu = \varepsilon_3 + 6d$ and using (A.7.1). In particular its proof does not cancel six in a test ring. Moving $\sigma^{b'}$ past δ^{-d} uses $S\delta = -\delta$, and $u_{\underline{\mu}}(\sigma) = \omega^a\sigma$. The two central factors are $\omega^{ab'}$ and $\omega^{bb'}$. Equation (A.12.5) then gives

$$v_{\underline{\mu}} v_{\underline{\mu}'} = \omega^{(a+b)b'} v_{\underline{\mu\mu}'}.$$

Consequently the multiplication in N_ρ is $(k, \underline{\mu})(k', \underline{\mu}') = (k + \varepsilon_3 k' + 3c, \underline{\mu\mu}')$. Restricting k to $0, 3$ proves (A.12.2)–(A.12.3). The bilinear function $((a,b),(a',b')) \mapsto (a+b)b'$ is a two-cocycle. The resulting eight-element group has five involutions and two elements of order four, and is generated by two involutions with noncommuting product of order four. This proves it is D_8; the full cocycle and element-order checks are in `a_finite.py`. Exactness of (A.12.2) holds even on ring points, because $v_{\underline{\mu}}$ is a global section as a scheme, although it is not a group section.

For σ, the centralizer in H is C_4, generated by σ. Its translation kernel is $\ker(1-S) = \mathbb{Z}(e_0 + e_1)$, also after every base change. These translations commute with σ, so the geometric centralizer is $C_4 \times \mathbb{G}_a$. Equation (A.8.5) supplies a group section over γ and its action on

both factors, which proves (A.12.4) and all the stated exact sequences. The sharp intersection retains just the translation component. Similarly $N_\rho \cap G^\sharp$ is the section over $\gamma' = \ker \varepsilon_3$, since the component $\rho^k\sigma^b$ is the identity precisely when $k = b = 0$. Put $\gamma'' = \ker \varepsilon_2$, $\gamma^0 = \gamma' \cap \gamma''$, and $\gamma''' = \{a = b\}$. The neutral partial section of (166)–(167) is $v(\gamma^0)$. The partial section of (168)–(169) is exactly

$$\underline{\mu} \longmapsto \omega^b v_{\underline{\mu}} \quad (\underline{\mu} \in \gamma'''),$$

because its cocycle vanishes on $a = b$, and at $\underline{\mu} = -1$ it is τ'. Conjugation by τ' sends $v_{\underline{\mu}}$ on γ' to $\omega^a v_{\underline{\mu}}$, as (A.12.3) shows. It preserves the partial section precisely on γ^0. This proves the prime and double-prime intersections and the normalization assertions of (160)–(170) with their exact domains. □

A.13. The literal definition (171) has only half the stated image [Refuted]

The group defined in (171) from $Z_\rho^{0\sharp}$, ω, and τ' maps onto

$$\gamma''' = \{\varepsilon_2 = \varepsilon_3\} \subset \gamma,$$

and is the inverse image of this subgroup in the final $N_\rho^?$ of (212). Thus (172) with target all of γ is false for the literal definition (171). Its exact replacement is $1 \to C_2 \to N^?_{\rho,\mathrm{literal}} \to \gamma''' \to 1$; the final fibres (212) yield the larger exact sequence (A.12.2). The surviving last line of (194), with N_ρ and no question mark, requires the replacement $1 \to C_3 \to N_\rho/\mu_\rho \to \gamma \to 1$; the isomorphism to γ holds for $N_\rho^?/\mu_\rho$. The final Γ in (184) also satisfies $\Gamma/C_2 \simeq \mathbb{G}_a \rtimes \gamma$; the unquotiented version of this isomorphism in the third row of (195) is false. These corrections concern C4 scans 82, 85–87, 94; fols. 688v, 690–691, 694v.

Proof. The identity component $Z_\rho^{0\sharp}$ maps isomorphically to $\gamma^0 = \{a = b = 0\}$ by A.12. The central ω has trivial character, and τ' has multiplier -1, hence $(a, b) = (1, 1)$. The literal generated subgroup therefore has image in $a = b$. Conversely multiplication by -1 carries the whole 00 chart to the whole 11 chart, so the image is exactly γ'''. At $\underline{\mu} = -1$, $d = 0$ and $\tau' = \omega v_{-1}$; hence both lifts over each of these charts occur. This proves the asserted inclusion and its exact image and kernel. Over $\mathbb{F}_2$, the γ-point with $a = 1, b = 0, k = 0$ has evaluation one and has no lift to this literal subgroup. This finite counterexample is recorded in `a_schemes.py`.

Quotienting (A.12.1) by its central order-two subgroup leaves kernel $C_6/C_2 = C_3$, which proves the correction to (194). The final fibre product defining Γ has kernel $C_2 \times \mathbb{G}_a$ over γ, by A.12. Its quotient by the indicated central C_2 is $N_\sigma^\sharp$, giving the claimed isomorphism. On $\mathbb{F}_2$ points the two groups have 32 and 16 elements, respectively; the explicit projection has kernel of order two. Thus the missing quotient in (195) is detected by an actual map and a finite counterexample. □

A.14. The final fibre products and every surviving extension sequence [Proved]

Using the final (212), define

$$\Gamma = N_\rho^? \times_\gamma N_\sigma^\sharp, \quad \mathcal{M} = \Gamma \times_\gamma G, \quad \mathcal{M}^\sharp = \Gamma \times_\gamma G^\sharp. \tag{A.14.1}$$

The coordinates on Γ are $(i, s, \underline{\mu})$ and its law is

$$(i, s, \underline{\mu})(i', s', \underline{\mu}') = (i + i' + (a + b)b',\ s + \mu s',\ \underline{\mu\mu}'). \tag{A.14.2}$$

There are central quotient morphisms, with the indicated kernels,

$$\Gamma \twoheadrightarrow \Gamma_{\mathrm{e}}, \qquad \mathcal{M} \twoheadrightarrow \mathcal{M}_{\mathrm{e}}, \qquad \mathcal{M}^\sharp \twoheadrightarrow \mathcal{M}^\sharp_{\mathrm{e}}, \qquad \ker = C_2 = \mu_\rho. \tag{A.14.3}$$

In addition to A.8, A.11, and A.12, the final exact sequences are

$$\begin{gathered} 1 \to \mathcal{S} \to \mathcal{M} \to \Gamma \to 1, \qquad 1 \to \underline{M} \to \mathcal{M}^\sharp \to \Gamma \to 1, \\ 1 \to C_2 \times \mathbb{G}_a \times \mathcal{S} \to \mathcal{M} \to \gamma \to 1, \\ 1 \to C_2 \times \mathbb{G}_a \times \underline{M} \to \mathcal{M}^\sharp \to \gamma \to 1, \qquad 1 \to C_2 \times \mathbb{G}_a \to \Gamma \to \gamma \to 1. \end{gathered} \tag{A.14.4}$$

For $X = \Gamma, \mathcal{M}, \mathcal{M}^\sharp$, the auxiliary group $S_0X = Z \times_\gamma X$ has split exact sequence $1 \to L_0 \to S_0X \to X \to 1$. These prove (182)–(193) and (196), with the corrected quotient versions of (194)–(195). The central extensions are precisely the pullbacks of the explicit D_8 extension in A.12.

Proof. Every map to γ in (A.14.1) is a homomorphism, and the multiplication of the fibre product is componentwise. Insert (A.12.3)–(A.12.4) to obtain (A.14.2). Forgetting i is a homomorphism, since its cocycle is central, and its kernel is the constant C_2. The resulting three groups have exactly the coordinates and actions in (A.11.1), proving the actual comparisons (A.14.3) between the early and final revisions. The projection $\mathcal{M} \to \Gamma$ has the group section $(a, b) \mapsto (a, b, u_{\underline{\mu}})$; its kernel is $\mathcal{S}$. Replacing $\mathcal{S}$ by $\underline{M}$ gives the sharp row. Kernels over γ are the products of the three displayed geometric kernels, proving (A.14.4). These projections are onto on ring points: use $v_{\underline{\mu}}$, $u_{\underline{\mu}}$, and zero translation as a scheme section. Finally $Z \to \gamma$ has the group section (A.8.5), so the auxiliary projections are split exact. This proves exactness and representability directly for the entire final diagram.

The central factor μ_ρ here is $(\omega, 1, 1)$ in (a, b, U)-coordinates. The central subgroup μ_G is $(1, 1, \omega)$ and has a separate inclusion. The quotient map in (A.14.3) removes the first of these, exactly as its kernel states; no central factors have been combined without a map. □

A.15. The three distinguished subgroups: all requested points and fibres [Proved]

The final subgroups of scans 88–94 are the graph subgroups

$$\mathcal{M}(0) = \{U = u_{\underline{\mu}}\}, \qquad \mathcal{M}(1/2) = \{U = b_\sigma\}, \qquad \mathcal{M}(-\bar{j}) = \{U = a_\rho\} \tag{A.15.1}$$

inside $\mathcal{M} = \{(a_\rho, b_\sigma, U)\}$. Projection on (a_ρ, b_σ) identifies each with Γ as a group scheme over $\mathbb{Z}$. They are smooth and flat of relative dimension two, with eight geometric components at every prime.

For any connected ring R, their common point set is exactly

$$\{(a, b, k, i, s) : a, b, i \in \{0, 1\},\ k, s \in R,\ 1 - 6a + 4b + 12k \in R^*\}\,. \tag{A.15.2}$$

For a general ring the bits are locally constant functions on its spectrum. The group law is (A.14.2), with multiplier product

$$\begin{aligned} a'' &= a + a' \pmod 2, \qquad b'' = b + b' \pmod 2, \\ k'' &= \frac{r_0 r_0' - r_0''}{12} + r_0 k' + r_0' k + 12kk'. \end{aligned} \tag{A.15.3}$$

The numerator is an integer divisible by twelve on every component. In particular the requested rings have the following complete parametrizations, with $s \in R$ and $i \in C_2$ always free.

R	Allowed multipliers and component bits; $k = (\mu - r_0)/12$
$\mathbb{Z}$	$\mu = 1 : (a, b, k) = (0, 0, 0)$; $\mu = -1 : (a, b, k) = (1, 1, 0)$.
$\mathbb{Z}[1/2]$	$\mu = \pm 2^m$ $(m \in \mathbb{Z})$; a arbitrary, $(-1)^b = \mu \bmod 3$.
$\mathbb{Z}[1/3]$	$\mu = \pm 3^m$ $(m \in \mathbb{Z})$; b arbitrary, $(-1)^a = \mu \bmod 4$.
$\mathbb{Z}[1/6]$	$\mu = \pm 2^m 3^n$ $(m, n \in \mathbb{Z})$; a, b arbitrary.
$\mathbb{F}_2$	a, b, k arbitrary; $\mu = 1$; $\lvert\mathcal{M}(p)(\mathbb{F}_2)\rvert = 32$.
$\mathbb{F}_3$	a, b, k arbitrary; $\mu = (-1)^b$; $\lvert\mathcal{M}(p)(\mathbb{F}_3)\rvert = 72$.

Here p denotes any of the three displayed points, not a prime. As abstract groups, the integral points are $C_2 \times (\mathbb{Z} \rtimes \{\pm 1\})$; the two finite groups are $D_8 \times C_2^2$ over $\mathbb{F}_2$ and $C_3 \times (C_3 \rtimes D_8)$ over $\mathbb{F}_3$, where D_8 acts on the second C_3 through ε_3.

Proof. The maps $\Gamma \to G$ given respectively by $(a_\rho, b_\sigma) \mapsto u_{\underline{\mu}}, b_\sigma, a_\rho$ are homomorphisms. Their graphs have the stated inverse projection, proving all three group-scheme isomorphisms, not just equality of dimensions or of point counts. Formula (A.15.2) follows from $\Gamma \simeq C_2 \times \mathbb{G}_a \times \gamma$ as a scheme. Equations (A.14.2) and multiplication of $r_0 + 12k$ give (A.15.3); its carry term is integral because residue classes multiply according to the two sign bits.

For $\mathbb{Z}$, invertibility forces $\mu = \pm 1$ and the congruences in (A.7.2) force exactly the displayed bits. The cocycle then vanishes since $a = b$. For $\mathbb{Z}[1/2]$, the unit group is $\{\pm 2^m\}$, and divisibility by twelve in that ring is equivalent to the displayed congruence modulo three; modulo four imposes no condition because two is inverted. The argument with the primes reversed proves the next row, and inverting six proves the fourth. Uniqueness of k in these torsion-free rings follows directly. In the two finite fields, twelve vanishes and all representatives r_0 are units, giving $4 \cdot 2 \cdot 2 \cdot 2 = 32$ and $4 \cdot 3 \cdot 2 \cdot 3 = 72$ points. These were also counted directly from the original lattice equations by `a_finite.py`.

There is an explicit ring isomorphism $\mathbb{O}_N(\mathbb{Z}/q) \to \mathbb{Z}/Nq$, $r + N[t] \mapsto r + Nt$. Therefore $\gamma(\mathbb{F}_2) = (\mathbb{Z}/24)^* \simeq C_2^3$ and $\gamma(\mathbb{F}_3) = (\mathbb{Z}/36)^* \simeq C_3 \times C_2^2$. The residue maps split as maps of these finite abelian groups, with kernels C_2 and C_3. Pulling back the D_8 extension yields the claimed finite-group descriptions, including its action on s. Finally each of the eight charts is an open subset of affine two-space, and over characteristics two and three it is the whole affine two-space. This proves smoothness, flatness, and the component count on the actual fibres.

Under (A.14.3), the first two graphs contain μ_ρ, while the third intersects it trivially. Indeed $U = a_\rho$ forces $U = \omega$ when $a_\rho = \omega$. Its image in $\mathcal{M}/\mu_\rho$ is a bisection over Γ_{e}, with kernel μ_G. This gives explicitly the change of descriptions in scans 91–94 rather than identifying the central quotients with all three unquotiented graphs. □

A.16. An actual comparison with the groups of §48 [Proved]

Let H_{48} be the group of pairs of quadratic-normalizer matrices

$$A = c\rho + dI, \quad B = \begin{pmatrix} u & t \\ -et & eu \end{pmatrix}, \quad Q_3 = c^2 + cd + d^2, \quad Q_2 = t^2 + u^2,$$

with Q_2, Q_3 units and $e^2 = (e')^2 = 1$, satisfying $\det B = eQ_2 = e'Q_3$. Its character map is

$$H_{48} \to \gamma_{48} = \mathbb{G}_m \times \mu_2^2, \quad (A, B) \mapsto (\det B, e, e').$$

There is a specified comparison of character schemes

$$\iota : \gamma \longrightarrow \gamma_{48}, \qquad \underline{\mu} \longmapsto (\mu, (-1)^a, (-1)^b), \tag{A.16.1}$$

which is an isomorphism over $\mathbb{Z}[1/6]$. Define the actual base change $H_{48}^\dagger = H_{48} \times_{\gamma_{48}} \gamma$. Projection followed by the zero-translation section is a morphism of affine group schemes

$$H_{48}^\dagger \longrightarrow \Gamma/\mu_\rho, \quad (A, B, \underline{\mu}) \longmapsto (0, \underline{\mu}). \tag{A.16.2}$$

Its kernel is $ST_\rho \times ST_\sigma$, the product of the two norm-one quadratic groups, and its image is exactly the displayed copy of γ. For the full groups the analogous map $\mathcal{M}_{48} \times_{\gamma_{48}} \gamma \to \mathcal{M}/\mu_\rho$ has kernel $ST_\rho \times ST_\sigma \times \mathrm{SL}_2$ and image the canonical γ section.

The §48 identifications $\mathcal{M}_{48}(0) \simeq H_{48}$ and $\mathcal{M}_{48}(1/2) \simeq H_{48}$ hold by their graph maps. The graph $U = A$ is defined there over $e' = 1$, and is identified with $H_{48}|_{e'=1}$. Its comparison is the

restriction of (A.16.2) over $b = 0$. Thus all three §51 groups are related to the earlier groups by these explicit base changes, character maps, and central quotients. The resulting schemes are not isomorphic: their generic relative dimensions are respectively two and three, and their bad fibres have the distinct descriptions proved below.

Proof. Solving $A\rho = \rho A$ and $B\sigma = e\sigma B$ gives the two displayed matrix forms, and their determinants give exactly the stated equations. The character map is fppf surjective: given its three coordinates, adjoining square roots of the required unit norms and taking scalar elements in the quadratic algebras supplies lifts. The extensions adjoining square roots are finite free and faithfully flat, including in characteristic two. Its kernel is precisely the two norm-one groups. Formula (A.16.1) is a homomorphism by the ring and sign formulas in A.5–A.7. Over $\mathbb{Z}[1/6]$, its inverse recovers the constant signs and sets $k = (\mu - r_0)/12$, proving the asserted isomorphism.

The base-changed character map retains its exact kernel, so its composition with the group section of Γ/μ_ρ has the kernel and image claimed in (A.16.2). For the full matrix triple, the additional determinant-prescribed U coordinate has kernel SL_2. The section $U = \mathrm{diag}(\det B, 1)$ gives the graph at zero, and $U = B$ gives the graph at one-half. For $U = A$, the determinant condition reads $e' = 1$, proving exactly the specified restriction. This derives the comparison from the equations used in LM1 B.2–B.5 instead of importing their conclusion.

Here is also a direct fibre computation. Eliminating the common determinant gives

$$H_{48} = \operatorname{Spec} \mathbb{Z}[c, d, t, u, e, e', Q_2^{-1}]/(Q_3 - ee'Q_2, e^2 - 1, (e')^2 - 1).$$

Relative to the sign scheme μ_2^2 its hypersurface is smooth: the gradient in c, d, t, u is $(2c + d, c + 2d, -2ee't, -2ee'u)$ and cannot vanish on $Q_2 \neq 0$. For characteristic different from two, use the last two entries; in characteristic two the first two become d, c and cannot both vanish because Q_3 is a unit. The geometric fibres are nonempty, so this is a smooth surjective map of relative dimension three. Thus the generic dimension is three. At three, $Q_3 = (c - d)^2$; setting $v = c - d \neq 0$ shows four smooth geometric components $\mathbb{G}_m^2 \times \mathbb{G}_a$. At two put $x = e - 1, y = e' - 1$. The sign algebra is $\mathbb{F}_2[x, y]/(x^2, y^2)$; after reduction, $Q_2 = (t + u)^2$ and the remaining equation is the norm-one equation $a^2 + ab + b^2 = 1$. Hence the reduced fibre is

$$\mathbb{G}_a \times \mathbb{G}_m \times R^1_{\mathbb{F}_4/\mathbb{F}_2}\mathbb{G}_m.$$

It is geometrically connected, and smoothness over the sign algebra gives generic length four before reduction. Restricting $e' = 1$ replaces that length by two. The full $\mathcal{M}_{48}$ is, as a scheme, the product of this scheme with SL_2, by $U = \mathrm{diag}(\det B, 1)h$. These fibre calculations and the explicit morphisms above prove the asserted comparison with A.15, whose characteristic-two fibre has eight reduced affine-plane components. They also prove flatness of the §48 models and their failure of smoothness exactly at two. No nonexistence of further mathematical comparisons is asserted. □

A.17. Linear congruence descent from the actual braid automorphism [Proved]

Use the standard tangential and braid normalization

$$B_3 = \langle \beta_1, \beta_2 : \beta_1\beta_2\beta_1 = \beta_2\beta_1\beta_2 \rangle, \quad x = \beta_1^2, \quad y = \beta_2^2.$$

For $(\mu, f) \in \widehat{\mathrm{GT}_0}$, where $f \in \widehat{\langle x, y \rangle}'$, let

$$\Phi(\beta_1) = \beta_1^\mu, \qquad \Phi(\beta_2) = f^{-1}\beta_2^\mu f. \tag{A.17.1}$$

For every $n \geq 2$, define the continuous surjection

$$q_n : \widehat{B}_3 \longrightarrow \mathrm{SL}_2(\mathbb{Z}/n), \quad \beta_1 \mapsto U = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \beta_2 \mapsto W = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

There is a unique matrix

$$V_n(\Phi) = \begin{pmatrix} \mu & t_n(\Phi) \\ 0 & 1 \end{pmatrix} \quad \text{such that} \quad q_n\Phi = \mathrm{Int}(V_n(\Phi))q_n. \tag{A.17.2}$$

The maps $\Phi \mapsto V_n(\Phi)$ are continuous homomorphisms, compatible under reduction. In particular all linear congruence kernels are invariant. They give a specified action on the full $\mathrm{GL}_2(\mathbb{Z}/n)$ by the same conjugation formula. This supplies a direct proof of the descent needed in LM2 C.d1 and its requested independent check at $n = 12$.

Proof. The group-theoretic input is that (A.17.1) is an automorphism of $\widehat{B}_3$ for an actual $\widehat{\mathrm{GT}}_0$ pair. Lochak–Schneps prove the braid identity in Lemma 5 of [LS94]; Harbater–Schneps, §1.3, Proposition 4, proof of (i)⇒(iii), explicitly applies that lemma to the full profinite group, [HS00]. Both proofs were opened and checked. The input uses the two cycle relations and the automorphism requirement; no pentagon is used here. The following reduction from that theorem to (A.17.2) is elementary and also proves the requisite generation condition cannot be omitted.

Write $A = q_n\Phi(\beta_1) = U^\mu$ and $B = q_n\Phi(\beta_2) = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Conjugacy to W^μ gives $a + d = 2$, $ad - bc = 1$. The lower-left entry of $ABA = BAB$ gives

$$c(1 + \mu c) = 0. \tag{A.17.3}$$

The pair A, B generates all of $\mathrm{SL}_2(\mathbb{Z}/n)$, because q_n is onto and Φ is an automorphism. For each prime $p \mid n$, the element c mod p cannot vanish: otherwise both matrices would be upper triangular, and could not generate $\mathrm{SL}_2(\mathbb{F}_p)$. Thus c is a unit modulo n. Equation (A.17.3) yields $c = -\mu^{-1}$; the determinant and trace then give

$$d = 2 - a, \quad b = \mu(a-1)^2, \quad t_n = \mu(1-a).$$

Direct multiplication proves $V_nUV_n^{-1} = A$, $V_nWV_n^{-1} = B$. Agreement on the topological generators proves (A.17.2) on the full profinite group. For surjectivity of q_n, elementary row operations over each local ring $\mathbb{Z}/p^r$ reduce a determinant-one matrix to diagonal form, and diagonal determinant-one matrices are products of elementary matrices. Chinese remainders combine these operations. Powers of U, W give all the elementary matrices.

A matrix commuting with U, W is scalar, by comparing its four entries. A scalar relating two matrices with lower-right entry one is the identity. This proves uniqueness. The product $V_n(\Phi)V_n(\Psi)$ has the same normalization and implements the composite automorphism, which proves the homomorphism assertion. The formula for a in terms of $q_n(f)$ proves continuity; uniqueness proves compatibility in n. Taking the same inner conjugation on $\mathrm{GL}_2(\mathbb{Z}/n)$ is now a well-typed extension of the action on its determinant-one subgroup.

For comparison with the modular group, the central braid $z = (\beta_1\beta_2)^3$ maps to ω, and $\widehat{B}_3/\overline{\langle z^2 \rangle} \simeq \widehat{\mathrm{SL}_2(\mathbb{Z})}$. The cited proof also gives $\Phi(z) = z^\mu$, so this quotient is preserved. The subgroup $\widehat{\Pi}_0$ is the inverse image of V_4 modulo four, preserved by the matrices in A.18. Alternatively its generators are $\lambda_0 = \omega U^{-2}$, $\lambda_1 = \omega W^{-2}$; their images are respectively λ_0^μ and a conjugate of λ_1^μ by an element of $\widehat{\Pi}_0'$. To verify the last assertion, evaluate the commutator word f in $U^2 = \omega\lambda_0^{-1}$ and $W^2 = \omega\lambda_1^{-1}$: its central factors cancel. This constructs the comparison to the action on $M/12M$ used below.

The program `a_finite.py` independently enumerates every signed matrix-coboundary solution used in LM2 C.d1 at level twelve: there are twelve for each of the four multipliers and no failed intertwiner. The program `a_braid.py` additionally enumerates the eight possible values of f in the finite commutator image, for every multiplier. All thirty-two pairs satisfy the braid relation; eight pairs fail generation and are explicitly rejected. The remaining twenty-four pairs give the twelve distinct matrices in A.18. This records both the computation and the exact reason the discarded matrix solutions cannot be automorphisms of the profinite source. □

A.18. Exact images at level twelve, with the genuine normalization stated [Proved]

For the normalization (A.17.1)–(A.17.2), the images of $G_{\mathbb{Q}}$, $\widehat{\mathrm{GT}}$, and $\widehat{\mathrm{GT}}_0$ in $\mathrm{GL}_2(\mathbb{Z}/12)$ are all

$$\mathcal{V}_{12} = \left\{ \begin{pmatrix} \mu & t \\ 0 & 1 \end{pmatrix} : \mu \in \{1,5,7,11\}, \quad t \in \{0,4,8\} \right\}, \qquad |\mathcal{V}_{12}| = 12. \tag{A.18.1}$$

Their specified actions in $\mathrm{Aut}(\mathrm{GL}_2(\mathbb{Z}/12))$ therefore have image $\mathrm{Ad}(\mathcal{V}_{12})$ of order 12. Their images in $\mathrm{Out}(\mathrm{GL}_2(\mathbb{Z}/12))$ are trivial. The canonical outer images on $\mathrm{SL}_2(\mathbb{Z}/12)$ are all $(\mathbb{Z}/12)^*/((\mathbb{Z}/12)^*)^2 = C_2^2$, of order four. On $M/12M$ the genuine image of each group is the scalar subgroup $\{\mu I : \mu \in (\mathbb{Z}/12)^*\}$, also of order four.

For $G_{\mathbb{Q}}$, choose the tangential base point at $\lambda = 0$ with tangent $+1$ and the real interval path to the tangent at one; this is the Belyi normalization in (A.17.1). The kernel of its normalized level-twelve matrix representation is exactly

$$G_{\mathbb{Q}(\zeta_{12}, \sqrt[3]{2})}. \tag{A.18.2}$$

The extended group generated by the geometric modular group and any of the three arithmetic groups acts on the same linear congruence quotient with image $\mathrm{Ad}(\mathrm{GL}_2(\mathbb{Z}/12))$ of order 1152. These statements give the precise actions requested after LM2; the order 32 remains the normal subgroup K_{32} of A.3.

Proof. Modulo four, $q_4(x) = U^2$ and $q_4(y) = W^2$ commute, so $q_4(f) = I$. Hence $q_4\Phi(\beta_2) = W^\mu$, whose upper-left entry is one. The formula in A.17 gives $t \equiv 0 \pmod 4$. Thus the whole $\widehat{\mathrm{GT}}_0$ image lies in $\mathcal{V}_{12}$. This set is a group under matrix multiplication and has twelve elements. To prove equality for the arithmetic subgroup, we now compute a compatible finite geometric cover, including its base-point comparison.

Consider the Legendre curve $E_\lambda : y^2 = x(x-1)(x-\lambda)$ over $\mathbb{P}^1 - \{0,1,\infty\}$, and its finite etale cover of cyclic order-three subgroups. Its four points are the four x-coordinates of nonzero three-torsion modulo sign. With the oriented vanishing cycles above the intervals $[0,\lambda]$ and $[\lambda,1]$, a positive loop around either colliding pair of branch points is a full twist. On the double cover it acts by the square of the transvection in the corresponding vanishing cycle. Taking their intersection to be one gives the matrices U^2 and W^2. This constructs the geometric monodromy comparison from x, y in (A.17.1) to the projectivization of $E[3]$. Changing the simultaneous homology basis conjugates both sides; the four-point permutation representation and its coefficient field are unchanged. The central signs used to define λ_i disappear under this specified projectivization.

We also specify the arithmetic fibre functor. At the tangent $\lambda = 0, +1$, the fibre is obtained by adjoining formal roots $\lambda^{1/m}$ and letting $G_{\mathbb{Q}}$ act on their coefficients, fixing the symbols $\lambda^{1/m}$. The real interval path gives the usual second generator. In every abelian Kummer covering generated by $\lambda^{1/m}$ and $(1-\lambda)^{1/m}$, the leading coefficients at the two tangents are one. Thus the path cocycle has zero abelianization; the induced based Galois action has precisely $x \mapsto x^\chi$,

$y \mapsto f^{-1}y^{\chi}f$ with $f \in \widehat{F}_2'$. The arithmetic compatibility and the inclusions $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}} \subset \widehat{\mathrm{GT}}_0$ are the tangential actions described in [HS00, §0.3 and §1.2–1.3]. Consequently the arithmetic conjugation of this four-point Legendre monodromy is the projective reduction of V_{12}. There is no extra projective conjugator ambiguity here: a matrix centralizing the projective images of U^2, W^2 modulo three centralizes their actual matrices, since their trace two is nonzero; comparing entries then makes it scalar.

The three-division polynomial, obtained from the tangent-and-chord addition formula on the displayed cubic, is

$$\psi_3(x) = 3x^4 - 4(1+\lambda)x^3 + 6\lambda x^2 - \lambda^2. \tag{A.18.3}$$

At zero it is $x^3(3x-4)$. The root with constant term $4/3$ lies in $\mathbb{Q}[[\lambda]]$, by the formal implicit-function theorem. For the other roots put $\lambda = s^3$, $x = s^2 z$. Dividing the polynomial by s^6 gives

$$3s^2z^4 - 4(1+s^3)z^3 + 6sz^2 - 1. \tag{A.18.4}$$

Each root c of $c^3 = -1/4$ is simple at $s = 0$, and uniquely lifts to $z \in \mathbb{Q}(c)[[s]]$. Hence the exact field of all Puiseux coefficients is $L = \mathbb{Q}(\zeta_3, \sqrt[3]{2})$: its containment follows by the unique recursion for the coefficients, and the reverse containment follows from the three leading coefficients, because $(-2c)^3 = 2$. The polynomial $X^3 - 2$ is irreducible by Eisenstein at two, and its discriminant -108 is not a square. Therefore $[L : \mathbb{Q}] = 6$ and the arithmetic action on the four lines has image S_3, fixing the one rational branch.

Modulo four, V_{12} is $\mathrm{diag}(\chi, 1)$, so its remaining field is $\mathbb{Q}(i)$. The field L has unique quadratic subfield $\mathbb{Q}(\sqrt{-3})$, by the subgroup lattice of S_3, and therefore $L \cap \mathbb{Q}(i) = \mathbb{Q}$. The pair consisting of V mod 4 and [V mod 3] determines every $V \in \mathcal{V}_{12}$: a scalar with lower-right entry one is the identity, and Chinese remainders recover both entries. Thus the arithmetic kernel fixes exactly the degree-twelve field $L(i) = \mathbb{Q}(\zeta_{12}, \sqrt[3]{2})$, proving (A.18.1)–(A.18.2). The two containing GT groups have the same image by the already proved upper bound. The formal polynomial calculations (A.18.3)– (A.18.4), the leading field, and all twelve matrices are checked in `a_braid.py`.

The centralizer of $\mathrm{SL}_2(\mathbb{Z}/12)$ in $\mathrm{GL}_2(\mathbb{Z}/12)$ consists of its four scalar units. The normalization $V_{22} = 1$ makes their intersection with $\mathcal{V}_{12}$ trivial. This proves the genuine automorphism image count and the trivial outer image on the full GL group. A conjugation by V is inner on SL exactly when $V \in \mathrm{SL}_2(\mathbb{Z}/12) \cdot ((\mathbb{Z}/12)^* I)$; taking determinants gives its unit square class. All four units modulo twelve square to one, proving the outer SL count. The scalar action on $M/12M$ follows from the generator comparison at the end of A.17, and cyclotomic surjectivity realizes all four scalars. Finally the geometric image is SL and the determinants of $\mathcal{V}_{12}$ are all units, so $\langle \mathrm{SL}_2(\mathbb{Z}/12), \mathcal{V}_{12} \rangle = \mathrm{GL}_2(\mathbb{Z}/12)$. Quotienting its order 4608 by the scalar centralizer of order four gives 1152.

This specifies which choices give a genuine action and which quotients remove them. The congruence action is obtained from $\widehat{B}_3$ and its modular quotient; A.4 proves that the reduction to the unquotiented GL or SL group cannot instead be defined by first passing through $\mathfrak{S}_{\mathrm{d}}$. No equality between the normalized upper-right entry and the Kummer cocycle based at $1/2$ is used in this proof. □

A.19. The normalized cyclotomic arithmetic image in the minimum approximation [Proved]

In the tangential normalization of A.17–A.18, the arithmetic homomorphism into the final §51 minimum approximation has image in $\mathcal{M}(0)$: its effective coordinate is $U = u_{\underline{\mu}}$. Its restriction to $G_{\mathbb{Q}^{\mathrm{cyc}}}$ has image in the abelian group

$$\ker(\mathcal{M}(0)(\widehat{\mathbb{Z}})^{\Delta} \longrightarrow \widehat{\mathbb{Z}}^*) = C_2 \times \widehat{\mathbb{Z}}. \tag{A.19.1}$$

After the central quotient by μ_ρ, this kernel is $\widehat{\mathbb{Z}}$ under addition. In particular the closed commutator subgroup of $G_{\mathbb{Q}^{\mathrm{cyc}}}$ maps to the identity in this normalized minimum approximation. Here the superscript Δ requires the single global component choice described in A.1; it is essential to (A.19.1).

Proof. The congruence-normalized braid action has conjugator $f \in \widehat{F}_2'$. In the modular quotient its image is in $\widehat{\Pi}_0'$ by the evaluation calculation in A.17, so it vanishes in the minimum approximation. Thus both elementary generators U, W are sent to their μ-th powers. The automorphism (A.8.5) has exactly these images on diagonal profinite points: from $U = (\sigma\rho)^{-1}$, $W = (\rho\sigma)^{-1}$ and $\lambda_0 = \omega U^{-2}$, $\lambda_1 = \omega W^{-2}$, the possible central factor is $\omega^{\nu+a} = 1$, since $\nu = (\mu-1)/2 \equiv a \pmod 2$. This proves $U = u_{\underline{\mu}}$ for the effective coordinate, which is precisely the graph defining $\mathcal{M}(0)$. The retained normalizer conjugators give its other two coordinates as in (183), so its projection to those coordinates is the isomorphism of A.15.

For a diagonal cyclotomic point with $\mu = 1$, its residue is one and the unique quotient $(\mu-1)/12$ is zero in $\widehat{\mathbb{Z}}$: hence $a = b = k = 0$. Formula (A.14.2) on this fibre is $(i, s)(i', s') = (i + i', s + s')$. This proves (A.19.1), its central quotient, and the commutator assertion. It also explains why replacing diagonal profinite points by arbitrary $\gamma(\mathbb{F}_2)$-points would give the wrong kernel. □

B. The Fermat tower and its homology

We use the arithmetic model actually written in (220):

$$X_n^+ :\ x_0^n + x_1^n + x_\infty^n = 0, \qquad D_n = \{x_0 x_1 x_\infty = 0\}, \qquad U_n^+ = X_n^+ \setminus D_n.$$

The notation $X_n : x^n + y^n = z^n$, used in our geometric chain calculations, is related to this model explicitly in B.1. Throughout this part, $F = \langle x, y\rangle$, $c = [x, y] = xyx^{-1}y^{-1}$, and $F'' = [F', F']$. In a profinite group, a commutator subgroup means its closed version. Put

$$A_n = (\mathbb{Z}/n)^2, \quad R_n = \mathbb{Z}[A_n] = \mathbb{Z}[X^{\pm1}, Y^{\pm1}]/(X^n - 1, Y^n - 1), \quad K_n = \ker(F \longrightarrow A_n).$$

Capital letters X, Y in a group ring denote the images of x, y. We write $M = \mathbb{Z}^3/\mathbb{Z}(1, 1, 1)$, with basis the classes e_0, e_1 and $e_\infty = -e_0 - e_1$.

B.1. Arithmetic models and the precise problem [Proved]

The morphism

$$q_n : U_n^+ \longrightarrow U_1^+, \qquad [x_0 : x_1 : x_\infty] \longmapsto [x_0^n : x_1^n : x_\infty^n]$$

is defined over $\mathbb{Q}$, is equivariant for coordinate permutations, and has deck group μ_n^3/μ_n. The coordinate

$$z = -x_0/x_\infty, \qquad \psi(z) = [-z : z - 1 : 1]$$

identifies U_1^+ with $\mathbb{P}^1 - \{0, 1, \infty\}$. For $\alpha_n^n = -1$, the geometric comparison

$$\iota_n : X_n^+ \longrightarrow X_n, \qquad [x_0 : x_1 : x_\infty] \longmapsto [x_0 : x_1 : \alpha_n x_\infty]$$

is an isomorphism over $\mathbb{Q}(\alpha_n)$. At odd levels one can take $\alpha_n = -1$ over $\mathbb{Q}$; at level four the displayed comparison uses $\alpha_4 = \zeta_8$. In particular $X_4^+(\mathbb{R}) = \varnothing$, whereas $[0 : 1 : 1] \in X_4(\mathbb{Q})$.

The *position du problème* in §51.11 is to identify the affine representation

$$\widetilde{\mathcal{M}} \longrightarrow G_{\rho,\sigma}/\mu_G \longrightarrow \widehat{\mathbb{Z}}^* \times S_3$$

with the action on a fibre of the universal abelian Fermat covering, and separately to interpret its specialization at $z = 1/2$. Its translation group is $\widehat{M}(1)$. *Manuscript: §51.11, folios 695–697, scans 95–99, (218)–(230); §51.14, folio 723, scan 151, (315)–(317).*

Proof. The asserted equations follow on taking n-th powers. On the chart $x_\infty \neq 0$, division by α_n gives $(x_0/\alpha_n x_\infty)^n + (x_1/\alpha_n x_\infty)^n = 1$; the induced coordinate on the base is z. Compatible choices, for example $\alpha_n = \exp(\pi i/n)$ after fixing an embedding in $\mathbb{C}$, commute with the power maps. Thus all later geometric comparisons have an actual map behind them. The even-level real assertion follows because a sum of three real fourth powers vanishes only when all three coordinates vanish. Equations (228)–(230) explicitly identify the deck lattice and its cyclotomic action; (219) singles out the remaining rank-one affine specialization. These are the data just stated. □

B.2. The universal abelian covering [Proved]

Over $\mathbb{C}$, q_n corresponds to the subgroup K_n of $\pi_1(U_1) = F$. Its monodromy is $x \mapsto (1, 0)$, $y \mapsto (0, 1)$. Consequently the pro-étale tower $\varprojlim_n U_n^+$ is the universal abelian geometric covering, with deck module $\widehat{M}(1)$. Its deck and coordinate-permutation actions form $(\mu_n^3/\mu_n) \rtimes S_3$ at finite level. *Manuscript: §51.11, folios 696–697, scans 97–99, (220)–(230); §51.13, folio 715, scan 135.*

Proof. In the standard geometric chart, $u^n = z$ and $v^n = 1 - z$. Analytic continuation around 0 multiplies u by ζ_n and leaves v unchanged; continuation around 1 does the reverse. These two elements generate A_n, so the covering is connected and its subgroup is exactly K_n. Every finite quotient of $\widehat{\mathbb{Z}}^2$ factors through some A_n; this proves the universal assertion, rather than only the existence of a collection of abelian coverings. A change of compatible geometric roots identifies the deck group with $\widehat{\mathbb{Z}}^2$ by a change of basis; intrinsically it is $(\widehat{\mathbb{Z}}^3/\widehat{\mathbb{Z}}) \otimes \widehat{\mathbb{Z}}(1)$. Permutations act on the three coordinates, commute with power maps, and conjugate the diagonal multipliers by permuting them. This proves the stated semidirect product and the compatibility of the action with the inverse system. □

B.3. The withdrawn freeness and its exact replacement [Refuted]

For $n > 1$, the action of μ_n^3 on X_n^+ is not free: its diagonal μ_n fixes the whole curve. The quotient group scheme $A_n^{\mathrm{geom}} = \mu_n^3/\mu_n$ acts freely on U_n^+, and $U_n^+ \to U_1^+$ is an A_n^{geom}-torsor. At a coordinate cusp its stabilizer in A_n^{geom} is a copy of μ_n. Thus the quotient action on the compact curve is not free. These are assertions about actions on schemes; the corresponding integral homology modules are computed in B.5–B.7. *Manuscript: §51.11, folio* 696′*, scan 98, deletion before (224) and the inserted U_n in the paragraph following (225).*

Proof. A multiplier (a_0, a_1, a_∞) fixes a point with all coordinates invertible precisely when the three multipliers agree projectively. Its class in the quotient is then the identity. At a point $[0 : b : c]$, with $bc \neq 0$, the equality required is $a_1 = a_\infty$; a_0 is unrestricted. Modulo the diagonal this is exactly μ_n. This calculation also describes the stabilizer functors, and hence applies to group schemes, including their nonreduced fibres. On a torus chart the power map is given by adjoining independent n-th roots of two units. Its coordinate algebra is locally free of rank n^2, and the standard root-of-unit torsor identity proves faithful flatness and the torsor assertion. Over $\mathbb{Z}[1/n]$ it is finite étale. At residue characteristics dividing n it is still a torsor for the fppf topology; this does not assert étaleness or smoothness of those Fermat fibres. The finite checks in `code/b_freeness.py` enumerate the full and quotient stabilizers over finite fields for $1 \leq n \leq 8$. They give full-group stabilizer order n on the open part, quotient stabilizer order one there, and quotient stabilizer order n at a cusp. □

B.4. The discrete Alexander module [Proved]

Conjugation induces a free rank-one $R = \mathbb{Z}[X^{\pm 1}, Y^{\pm 1}]$-module structure

$$F'/F'' = R\,c.$$

With the convention for c fixed above, the injective Fox-chain map is

$$R\,c \longrightarrow R^2, \qquad fc \longmapsto ((1-Y)f, (X-1)f),$$

and its image is $\ker\big((X-1,\ Y-1) : R^2 \to R\big)$.

Proof. The graph with one vertex and oriented edges x, y has fundamental group F. Its universal abelian covering has vertex module R, edge module R^2, and the displayed boundary. Its first homology is the abelianization of its fundamental group F'. The chain of $xyx^{-1}y^{-1}$ is $(1-Y, X-1)$. For a cycle (a, b), reduction of $(X-1)a + (Y-1)b = 0$ modulo $X-1$ gives $(Y-1)\bar{b} = 0$ in the domain $\mathbb{Z}[Y^{\pm 1}]$. Thus $b = (X-1)f$. Substitution and cancellation of the nonzero divisor $X-1$ give $a = (1-Y)f$. The same cancellation proves uniqueness. This supplies both the generator and the proof of freeness. □

B.5. All-cusp-deleted integral homology [Proved]

There are canonical R_n-linear descriptions

$$H_1(U_n, \mathbb{Z}) = \ker\left(R_n^2 \xrightarrow{(X-1,\ Y-1)} R_n\right), \tag{B.1}$$

$$0 \longrightarrow R_n/\mathbb{Z}\mathcal{N}_n \xrightarrow{f \mapsto fc} H_1(U_n, \mathbb{Z}) \xrightarrow{q_{n*}} n\mathbb{Z}^2 \longrightarrow 0, \qquad \mathcal{N}_n = \sum_{a,b=0}^{n-1} X^a Y^b. \tag{B.2}$$

The deck action on the right-hand term is trivial. The group $H_1(U_n, \mathbb{Z})$ has rank $n^2 + 1$, and over $\mathbb{C}$ the trivial deck character occurs twice and every other character occurs once.

Proof. The finite covering of the two-circle graph has n^2 vertices and $2n^2$ edges, with boundary as in (B.1). A spanning tree proves that its cycle group is free of rank $2n^2 - n^2 + 1$. The inclusion $K_n \subset F$ induces $K_n^{\mathrm{ab}} \to F^{\mathrm{ab}}$. Its image is $n\mathbb{Z}^2$; its kernel is the image of F', because $F' \subset K_n$ and $K_n/F' = n\mathbb{Z}^2$. B.4 identifies that image with the orbit module generated by c. The kernel of $R_n \to R_n^2,\ f \mapsto ((1-Y)f, (X-1)f)$ consists of elements fixed by both translations. Equality of all coefficients says precisely that it is $\mathbb{Z}\mathcal{N}_n$. This proves (B.2), including its exactness over $\mathbb{Z}$. On a deck character (ζ_n^a, ζ_n^b), the boundary in (B.1) has rank one unless $a = b = 0$; in that case it has rank zero. This proves the character formula directly. □

B.6. The compact integral homology module [Proved]

Define

$$S_X = \sum_{i=0}^{n-1} X^i, \quad S_Y = \sum_{j=0}^{n-1} Y^j, \quad P_n = \sum_{0 \le j \le i < n} X^i Y^j, \quad I_n = (S_X, S_Y, P_n) \subset R_n.$$

The compact homology has the exact integral presentation

$$H_1(X_n, \mathbb{Z}) \simeq R_n/I_n, \qquad 1 \longmapsto c. \tag{B.3}$$

Equivalently, (B.1) is divided by the $3n$ cusp cycles

$$Y^j(S_X,0),\quad X^i(0,S_Y),\quad X^i(S_{XY},XS_{XY})\quad (0\le i,j<n),\qquad S_{XY}=\sum_{i=0}^{n-1}(XY)^i. \tag{B.4}$$

The cusp subgroup is primitive of rank $3n-1$. The rank of (B.3) is $(n-1)(n-2)$, and its complex characters are exactly those (a,b) with $a,b,a+b\neq 0 \pmod n$, each once.

Proof. Filling a puncture kills its oriented peripheral loop in first homology. Lifting the peripheral words $x^n,y^n,(xy)^n$ gives precisely (B.4). The homology sequence of the pair consisting of the compact surface and its punctured surface identifies the subgroup of peripheral classes with $\mathbb{Z}[D_n]/\mathbb{Z}\sum_{P\in D_n}P$. The quotient is the integral homology of an oriented compact surface, so is torsion-free. This proves primitivity and rank $3n-1$.

We give an integral derivation of the ring presentation. First fill the $2n$ cusps over $0,1$. The resulting affine curve $u^n+v^n=1$ retracts onto the inverse image of the segment $[0,1]$. To see this, take a closed regular neighbourhood of that segment in the z-plane. Outside it the finite map is unramified and its ends are collars. Inside it the local charts at the only branch points are $w\mapsto w^n$; their lifted discs and the intervening strips are a regular neighbourhood of the lifted segment. Shrinking these collars, discs and strips gives the stated retraction. That graph has vertices $A_n/\langle X\rangle$ and $A_n/\langle Y\rangle$, and one edge for each element of A_n; it is the complete bipartite graph on two sets of n vertices. Its cycles are the integral matrices whose row and column sums vanish. The map

$$R_n\longrightarrow\{\text{such matrices}\},\qquad f\longmapsto(1-X)(1-Y)f$$

is surjective: subtracting the last row and column expresses every such matrix as an integral sum of elementary rectangles. Its kernel is (S_X,S_Y), as one sees by successively taking differences of rows and columns. The elementary rectangle is the commutator cycle, up to the fixed choice of its orientation.

Now fill the n remaining cusps. In the original two-edge chain module the identities

$$(1-Y)P_n=S_X-YS_{XY},\qquad (X-1)P_n=S_Y-S_{XY}$$

give

$$(S_{XY},XS_{XY})=-Y^{-1}P_n\,c+Y^{-1}(S_X,0)+Y^{-1}(0,S_Y).$$

All infinity cusp cycles are translates of this one. Thus their classes in the bipartite graph kill exactly the ideal generated by P_n, proving (B.3). For a character with both $a,b\neq 0$, the last two displayed identities make P_n zero precisely when $a+b\neq 0$; when $a+b=0$, $(X-1)P_n=-n$ is nonzero. The character statement follows. Its cardinality is $(n-1)(n-2)$, agreeing with the topological rank. □

B.7. Finite integral computations and module freeness [Proved]

The exact calculations for $1\le n\le 8$ give

n	1	2	3	4	5	6	7	8
$g(X_n)$	0	0	1	3	6	10	15	21
$\operatorname{rk} H_1(U_n)$	2	5	10	17	26	37	50	65
$\operatorname{rk} H_1(X_n)$	0	0	2	6	12	20	30	42
$\operatorname{rk}\langle\text{cusps}\rangle$	2	5	8	11	14	17	20	23

For every $n>1$, $H_1(U_n,\mathbb{Z})$ is not a free R_n-module. For every $n\ge 3$, $H_1(X_n,\mathbb{Z})$ is cyclic but not free as an R_n-module. At $n=1,2$ the compact module is zero. The exact comparison maps, deck actions and S_3-actions are supplied as integer matrices in `code/b_fermat_modules.json`.

Proof. The program builds the lifted graph, chooses the tree obtained from the paths $x^i y^j$, and uses each remaining edge as a fundamental cycle. It checks the Fox boundary, computes the deck transformations and the two based substitutions in B.9, and expresses every cusp cycle in this integral basis. It checks that the nonzero Smith entries of the cusp matrix and of the compact annihilator presentation are all one. It also checks that the map to F^{ab} has Smith entries n, n. These are full integer presentations, rather than character counts. All output rows displayed here were generated and checked by that program.

A free R_n-module has integral rank divisible by n^2. The affine rank n^2+1 fails this test for $n > 1$. The positive compact rank $(n-1)(n-2)$ is strictly below n^2. These rank conclusions apply through the explicit presentations (B.1) and (B.3); they do not replace the scheme-action assertion of B.3. □

B.8. The full profinite rank-one theorem [Proved]

Let $\widehat{F}$ be the free profinite group on x, y, and put

$$\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]] = \varprojlim_{m,n} (\mathbb{Z}/m)[A_n].$$

There are continuous Λ-linear isomorphisms

$$\widehat{F}'/\widehat{F}'' \xrightarrow{\sim} \varprojlim_n H_1(U_n, \widehat{\mathbb{Z}}) \xrightarrow{\sim} \varprojlim_n H_1(X_n, \widehat{\mathbb{Z}}) \simeq \Lambda\, c. \tag{B.5}$$

The transition maps on homology are pushforwards of the covering maps. The same limits can be formed diagonally with coefficients $\mathbb{Z}/n$. This is the full profinite assertion, including the prime-to-ℓ factors of the completed group ring.

Proof. We first prove the group-to-tower comparison. Let $\widehat{K_n} = \ker(\widehat{F} \to A_n)$ and $H = \bigcap_n \widehat{K_n} = \widehat{F}'$. The map $H \to \varprojlim_n \widehat{K_n}^{\mathrm{ab}}$ is surjective: compatible abelianized cosets are nonempty compact subsets of $\widehat{K_n}$, nested along a cofinal divisibility chain; their intersection supplies a lift in H. Its kernel contains H'. To prove the reverse inclusion, take a continuous homomorphism $f : H \to C$ to a finite abelian group. Choose an open normal subgroup V of $\widehat{F}$ with $V \cap H \subset \ker f$. Since H is normal, $[V, H] \subset V \cap H$, and f extends to HV by $f(hv) = f(h)$. Compactness gives $\widehat{K_n} \subset HV$ for a sufficiently divisible n. It follows that every element in $\bigcap_n \widehat{K_n'}$ is annihilated by f. Finite abelian quotients separate H/H', proving the first isomorphism. The covering graph identifies $\widehat{K_n}^{\mathrm{ab}}$ with $H_1(U_n, \widehat{\mathbb{Z}})$: the finite-index topology on $K_n \subset F$ is its full profinite topology, because the intersection of the finitely many F-conjugates of a finite-index subgroup of K_n is finite-index normal in F. The extra quotient $n\widehat{\mathbb{Z}}^2$ in (B.2) has transition maps given by inclusion and has inverse limit $\bigcap_n n\widehat{\mathbb{Z}}^2 = 0$. Thus its rank-two contribution at every finite level does not survive in (B.5).

For the second isomorphism, work first modulo m. The kernel of $H_1(U_n, \mathbb{Z}/m) \to H_1(X_n, \mathbb{Z}/m)$ is the peripheral module of B.6. Under $U_{dn} \to U_n$, a small loop around a cusp maps to d times a small loop. Thus the image of that kernel is zero when $m \mid d$. The inverse system is pro-zero. All groups are finite, so taking the inverse limit preserves the short exact sequences. This gives the second isomorphism modulo m, and passage to the inverse limit over m gives the assertion.

Finally use (B.3) modulo m. It stays exact because its integral quotient is free. Under $R_{dn} \to R_n$,

$$S_{X,dn} \mapsto dS_{X,n}, \quad S_{Y,dn} \mapsto dS_{Y,n}, \quad P_{dn} \mapsto dP_n + \binom{d}{2}\mathcal{N}_n.$$

The last formula counts pairs $0 \le j \le i < dn$ with each prescribed pair of residues modulo n. A value of d divisible by $2m$ kills every image generator of I_{dn} modulo m. Hence $\varprojlim_n I_n(\mathbb{Z}/m) = 0$,

and the finite inverse-limit exact sequence identifies $\varprojlim_n H_1(X_n, \mathbb{Z}/m)$ with $\varprojlim_n (\mathbb{Z}/m)[A_n]$, by the compatible generator c. Passing to m proves freeness. This argument agrees with [Ots16, Theorem 6.1], whose full profinite statement was opened and checked. It does not use a power-series-domain argument. The diagonal $(m, n) = (r, r)$ is cofinal in the divisibility product, which proves the coefficient formulation in the statement. □

B.9. The exact permutation comparison [Proved]

On Λc the selected S_3-action is semilinear:

$$r(f(X,Y)c) = f(Y, X^{-1}Y^{-1})c, \qquad s(f(X,Y)c) = -f(Y,X)c, \tag{B.6}$$

where $r^3 = s^2 = (rs)^2 = 1$. It agrees with the coordinate-permutation action on the Fermat tower after a compatible deck translation. Changing that translation conjugates the action by the corresponding deck operator. The same comparison is represented integrally at every finite level by the matrices of B.7.

Proof. Conjugation by the modular generators acts on F by $\rho : x \mapsto y,\ y \mapsto x^{-1}y^{-1}$ and $\sigma : x \mapsto y,\ y \mapsto x$. Replace ρ by $x\rho$; its based substitution is

$$r : x \mapsto xyx^{-1}, \quad y \mapsto y^{-1}x^{-1}.$$

Direct free reduction gives $r(c) = c$ and $s(c) = c^{-1}$. Their actions on F^{ab} are

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The defects in r^3 and $(rs)^2$ are inner automorphisms by elements of F'. Such automorphisms act trivially on F'/F'' and on K_n^{ab}, since $F' \subset K_n$. Formula (B.6) therefore gives a genuine S_3-action. At the level of deck extensions this is the complement constructed in B.11. The coordinate permutations provide another complement of the same A_n-extension. B.10 proves that there is a unique deck translation conjugating the two. Uniqueness makes these translations compatible as n varies. Passing to homology and then to (B.5) gives the claimed comparison, with its ambiguity stated. □

B.10. The two cohomology groups governing uniqueness [Proved]

For every abelian group I, with trivial S_3-action on I, put $M_I = I^3/I(1,1,1)$. Then

$$H^0(S_3, M_I) = H^1(S_3, M_I) = 0. \tag{B.7}$$

The same assertions hold for continuous cohomology with a profinite I. For any profinite group B acting on I and commuting with the permutation action,

$$H^0(B \times S_3, M_I) = H^1(B \times S_3, M_I) = 0.$$

Manuscript: §51.12–13, folios 702, 715–716, scans 109, 136–138, (295)–(297); §51.14, folio 721′, scan 148, (309).

Proof. Use coordinates in which r, s act by R, S above. An invariant (u, v) satisfies $u = v$, $u = -v$, and $u = 2v$. Thus $2v = 3v = 0$, so $u = v = 0$, over every I. A crossed homomorphism is specified by $f(r) = a = (a_0, a_1)$, $f(s) = b$. The relation $s^2 = 1$ gives $b = (t, -t)$. The relation $r^3 = 1$ imposes no condition, because $1 + R + R^2 = 0$. The relation $(rs)^2 = 1$ gives

$$a_0 = 2a_1 + 3t.$$

The element $m = (-a_1 - 2t, -a_1 - t)$ satisfies $(R-1)m = a$ and $(S-1)m = b$. Hence the cocycle is a coboundary, and the conjugating element is unique by the invariant calculation. Every formula uses only integral addition, so it also proves the continuous profinite assertion. For $B \times S_3$, first subtract the unique coboundary that kills the restriction to S_3. The cocycle identity for commuting $b \in B, s \in S_3$ then says that $f(b)$ is S_3-invariant. It is zero. This proves the last assertion without a spectral-sequence convergence assumption. □

B.11. An explicit integral splitting [Proved]

The extension $\mathfrak{S}^+/\mu_{\mathfrak{S}}$ of S_3 by M splits. In the affine realization

$$\rho = (R, 0), \qquad \sigma = (S, t), \qquad t = (1/3, -1/3),$$

a complement is generated by

$$\rho' = \lambda_0 \rho = (R, (1, 0)), \qquad \sigma' = \sigma.$$

It fixes $p = (2/3, 1/3)$. In particular its integral extension class, and the profinite class after completion, are zero. *Manuscript: §51.12, folios 700–702, scans 105–109; §51.13, folio* 715′*, scan 136.*

Proof. For affine multiplication $(A, u)(B, v) = (AB, u + Av)$, $1 + R + R^2 = 0$ gives $(\rho')^3 = 1$, $(1 + S)t = 0$ gives $(\sigma')^2 = 1$, and

$$RS = \begin{pmatrix} -1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad (1, 0) + Rt = (4/3, 2/3), \qquad (1 + RS)(4/3, 2/3) = 0$$

gives $(\rho'\sigma')^2 = 1$. Their linear parts generate all of S_3, so this is a section. Its translations lie in the original affine cosets, since the only changed generator was multiplied by $\lambda_0 \in M$. Solving $(1 - R)p = (1, 0)$ and $(1 - S)p = t$ gives the stated fixed point.

For complete cocycle data choose $h = r^i s^j$ and put $t_h = R^i(jt)$. The original integral cocycle is

$$c(h, k) = t_h + A_h t_k - t_{hk}.$$

The section just found has translation $p - A_h p$. Thus the integral cochain $b_h = p - A_h p - t_h$ satisfies $c + \delta b = 0$. This proves vanishing as a cohomology class, not merely satisfaction of a few relations. All formulas commute with reduction modulo N and with profinite completion. □

B.12. Splitting and uniqueness for the full quotient [Proved]

The continuous extension

$$1 \longrightarrow \widehat{M} \longrightarrow G_{\rho,\sigma}/\mu_G \longrightarrow \widehat{\mathbb{Z}}^* \times S_3 \longrightarrow 1$$

has zero class in continuous H^2. Its set of continuous splittings is a torsor under $\widehat{M}$, by conjugation. For the canonical scalar lift u_μ of the manuscript, put $\nu = (\mu - 1)/2 \in \widehat{\mathbb{Z}}$. A section commuting with the complement of B.11 is

$$s(\mu) = (-\nu, -\nu)\, u_\mu.$$

*Manuscript: §51.12, folios 704–*704′*, scans 113–114, (264)–(266); §51.13, folios 715–*716′*, scans 135–138.*

Proof. The manuscript's scalar automorphism sends ρ to $\lambda_1^\nu \rho$ and fixes σ modulo the central involution, and it acts by multiplication by μ on M. In the affine realization it is conjugation by

$$(\mu I, w_\mu), \qquad w_\mu = (-\nu/3, \nu/3).$$

Indeed $(1-R)w_\mu = \nu(0,1)$ and $\mu t + (1-S)w_\mu = t$. The product law follows from $\nu(\mu\mu') = \nu(\mu) + \mu\nu(\mu')$. After multiplying by the integral translation $(-\nu, -\nu)$, its affine translation becomes $(1-\mu)p$. These are precisely the scalar homotheties about p; they commute with the finite complement fixing p, and multiply according to $\mu\mu'$. This gives a continuous section of the full product. The thirds in this verification are computed in $\widehat{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{Q}$; the correcting cochain is integral at every prime, including 3. The half is integral because every profinite unit is odd at 2.

Two sections differ by a continuous 1-cocycle with coefficients $\widehat{M}$, acted on by (μ, h) as μA_h. B.10 makes that cocycle a coboundary and makes the conjugating element unique. Thus the asserted torsor and the vanishing of the actual H^2-class both hold. □

B.13. The finite 2- and 3-primary checks [Proved]

For $N \mid 24$, the actual extension class of B.12 in coefficients M/NM is zero, with trivial 2- and 3-primary parts. The independent normalized-cochain computation gives

$$\left|H^0(S_3, M/NM)\right| = \left|H^1(S_3, M/NM)\right| = 1, \qquad \left|H^2(S_3, M/NM)\right| = \gcd(N, 3).$$

In particular the zero answer for the actual class at $N = 3$ is a genuine cocycle computation in a nonzero obstruction group.

Proof. The integral cochain of B.11 is included in `code/b_splitting.json`. For the full product we use the reference affine lift

$$t_{\mu,h} = w_\mu + \mu t_h, \qquad b_{\mu,h} = (-\nu, -\nu) + \mu b_h.$$

The program verifies $c + \delta b = 0$ modulo N for every pair of elements of $(\mathbb{Z}/6N)^* \times S_3$. The modulus $6N$ retains the halves and thirds in these reference formulas. For $N = 1, 2, 3, 4, 6, 8, 12, 24$ the numbers of checked pairs are, respectively,

$$144,\ 576,\ 1296,\ 2304,\ 5184,\ 9216,\ 20736,\ 82944.$$

The normalized differentials $C^0 \to C^1 \to C^2 \to C^3$ are integer matrices. Their nonzero Smith diagonals are $1^2,\ 1^8,\ 1^{41}, 3$. For an integer matrix with nonzero Smith entries d_i, its kernel modulo N has cardinality $N^{b-r} \prod_i \gcd(N, d_i)$, where b is its column count and r its rank. Applying this formula to those three matrices gives the displayed cohomology orders. The code checks consecutive differentials compose to zero and verifies the full-product cocycle by exact rational arithmetic before reduction. □

B.14. Identification and rigidity of the arithmetic tower [Proved]

Let

$$E = \pi_1^{\mathrm{et}}([U_1^+/S_3], \bar{b}), \qquad E_{\mathrm{ab}} = E/\overline{[\pi_1(U_{1,\overline{\mathbb{Q}}}), \pi_1(U_{1,\overline{\mathbb{Q}}})]}.$$

Its exact extension

$$1 \to \widehat{M}(1) \to E_{\mathrm{ab}} \to \Gamma_{\mathbb{Q}} \times S_3 \to 1$$

splits, with one conjugacy class of splittings and a unique $\widehat{M}(1)$-conjugator between any two. The affine representation on a Fermat fibre has target $\widehat{M} \rtimes (\widehat{\mathbb{Z}}^* \times S_3)$. The representation

supplied by the quotient $G_{\rho,\sigma}/\mu_G$, with the same geometric translation map and cyclotomic character, agrees with it after a unique translation of the fibre. This proves the identification posed in B.1. *Manuscript: §51.13, folios 715–718′, scans 135–142, (294)–(298).*

Proof. The coordinate-permutation action is defined over $\mathbb{Q}$. Consequently the arithmetic quotient of the orbifold group by the geometric fundamental group is $\Gamma_{\mathbb{Q}} \times S_3$. The fibres of $U_n^+ \to U_1^+$ form compatible torsors under $M/nM(1)$. Choose one point in their inverse-limit fibre. Monodromy then acts by affine transformations with linear action $\chi(\gamma)A_h$, and its restriction to the geometric group is the abelianization map of B.2. Thus it factors through E_{ab}. The map from E_{ab} to the pullback of this affine group over $\Gamma_{\mathbb{Q}} \times S_3$ is an isomorphism: it is the identity on its kernel $\widehat{M}(1)$ and on its quotient, so injectivity and surjectivity follow by lifting a quotient element and correcting in the kernel. The pullback of a semidirect product is split.

B.10 with $B = \Gamma_{\mathbb{Q}}$, acting through χ, proves uniqueness of the conjugacy class and of its conjugator. The algebraically constructed representation and the Fermat representation have the same restriction to $\widehat{M}(1)$ and the same quotient action. Their translation difference is therefore a cocycle on $\Gamma_{\mathbb{Q}} \times S_3$. It is the unique coboundary given by B.10, which proves the claimed comparison. This identifies the resulting unbased torsor canonically; a based homomorphism still requires the fibre point chosen above. The group E in this argument is the arithmetic fundamental group, with quotient the actual Galois group. No equality of that group with a group of all loop-preserving outer automorphisms is used.

The source changes its tilde notation here. At scan 136 it describes the geometric-abelian quotient, while at scans 139–140, in (298), it uses $\widetilde{\mathcal{M}} = \mathcal{M}/\mu_{\mathcal{M}}$ for the orbifold group. Our notation retains the successive maps $\mathcal{M} \to E \to E_{\mathrm{ab}}$. Thus the argument uses the displayed kernels, rather than identifying the two quotients because they share a symbol in the manuscript. □

Scope of the two ranks. The map in B.14 first uses the quotient $\widehat{F} \to \widehat{F}^{\mathrm{ab}} = \widehat{M}(1)$. The Λ-module in B.8 is obtained by taking the homology *of the kernels* of its finite quotients. This is the exact passage from the universal abelian covering to the commutator module seen by the metabelian fundamental group. It is this map that connects the rank-two construction in §51 to the universal homology module.

C. The relevant 1-motives and the cuspidal extension

We keep the arithmetic plus model X_n^+ and its reduced divisor D_n of $3n$ coordinate cusps. All groups and motives in this part are over characteristic zero. The geometric comparison of B.1 transports the computations on $x^n + y^n = z^n$ to this model; its field and its effect on the arithmetic descent are retained. Write $J_n = \mathrm{Pic}^0(X_n^+)$ and $\mathcal{J}_n = \mathrm{Pic}^0(X_n^+, D_n)$, the degree-zero generalized Jacobian with reduced modulus D_n.

C.1. The actual Fermat 1-motive and its realizations [Known]

Put

$$L_n = \mathrm{Div}^0_{D_n}(X^+_{n,\overline{\mathbb{Q}}}), \qquad \mathcal{P}_n = [L_n \xrightarrow{D \mapsto \mathcal{O}(D)} J_n].$$

This is a Deligne 1-motive, with its natural descent over $\mathbb{Q}$. Its Cartier dual, using the principal polarization of J_n, is

$$\mathcal{A}_n = [0 \longrightarrow \mathcal{J}_n].$$

There is an exact sequence of semiabelian varieties

$$1 \longrightarrow T_{D_n} := (\mathrm{Res}_{D_n/\mathbb{Q}}\,\mathbb{G}_m)/\mathbb{G}_m \longrightarrow \mathcal{J}_n \longrightarrow J_n \longrightarrow 1. \tag{C.1}$$

The character lattice of T_{D_n} is L_n, and its cocharacter lattice is $\mathbb{Z}[D_n]/\mathbb{Z}\sum_{P\in D_n} P$. The Betti and Tate realizations of $\mathcal{A}_n$ are

$$T_{\mathbb{Z}}(\mathcal{A}_n) = H_1(U_n^+(\mathbb{C}),\mathbb{Z}), \qquad T_\ell(\mathcal{A}_n) = H_1^{\mathrm{et}}(U^+_{n,\overline{\mathbb{Q}}},\mathbb{Z}_\ell). \tag{C.2}$$

Its de Rham realization is the dual of $H^1_{\mathrm{dR}}(U_n^+/\mathbb{Q})$. Its weights are -2, with rank $3n-1$, and -1, with rank $(n-1)(n-2)$; its total rank is n^2+1. *Source: Deligne, [Del74, 10.1.3–10.1.8, 10.2.10–10.2.11, 10.3.4–10.3.8].*

Proof. We spell out the reduction to the cited construction. A point of the generalized Picard group is a line bundle on X_n^+, together with a trivialization along D_n, modulo scalar automorphisms of the line bundle. The fibre over a fixed degree-zero line bundle is thus the product of the nonzero fibres at the cusps modulo simultaneous scaling. This proves (C.1) and both of its lattice descriptions.

Deligne's definition 10.3.4 assigns to a smooth open curve $X - D$ the complex $[\ker(\mathbb{Z}[D]\to\mathbb{Z}) \to \mathrm{Pic}^0(X)]$. For $X = X_n^+$, this is precisely $\mathcal{P}_n$. Construction 10.3.6 identifies its m-torsion realization with $H^1(U_n^+,\mu_m)$: a representative is a pair $(\mathcal{L},D)$ with $\mathcal{L}^{\otimes m}\simeq\mathcal{O}(D)$, modulo pairs $(\mathcal{O}(E),mE)$. This is the same complex defining the finite realization of $[L_n\to J_n]$. Thus the comparison includes the cusp lattice and has no missing toric summand. Construction 10.2.11 dualizes this complex to the extension by $\mathrm{Hom}(L_n,\mathbb{G}_m)$ whose class evaluated at D is $\mathcal{O}(D)$; the line-bundle description above identifies this extension with $\mathcal{J}_n$. Dualizing the finite realizations with values in μ_m gives $H_1(U_n^+,\mathbb{Z}/m)$, and their inverse limits give (C.2).

The Betti construction 10.1.3 and the comparison 10.3.8 give the integral homology realization; 10.1.7–10.1.8 and 10.3.9 give the de Rham comparison. Concretely, its dual Hodge filtration has

$$F^1H^1_{\mathrm{dR}}(U_n^+) = H^0(X_n^+,\Omega^1_{X_n^+}(\log D_n)), \quad \dim F^1 = g_n + 3n - 1.$$

The torus in (C.1) contributes weight -2 and the abelian variety weight -1. Their ranks are $3n-1$ and $2g_n$, respectively. B.6 gives $2g_n=(n-1)(n-2)$. Every construction used here is functorial for the deck and permutation automorphisms, which proves compatibility of all the realizations with those actions. □

C.2. The rank-two identification in the lead [Refuted]

The assertion that the rank-two module $\widehat{M}(1)$ of §51, or its reduction modulo a nontrivial integer m, is the full Tate realization of the generalized Jacobian of X_n^+ with its cusps is false for $n>1$. Its exact replacement is

$$\widehat{M}(1) = T_\bullet(T), \qquad T = (\mathbb{G}_m^3)/\mathbb{G}_m = \mathcal{J}_1, \tag{C.3}$$

and the covering pushforward has the exact realization

$$0 \longrightarrow (R_n/\mathbb{Z}\mathcal{N}_n)\otimes\widehat{\mathbb{Z}} \longrightarrow T_\bullet(\mathcal{J}_n) \xrightarrow{q_{n*}} n\widehat{M}(1) \longrightarrow 0. \tag{C.4}$$

The Galois structure on the first term is the one transported by its inclusion into the geometric homology in (C.2). At level $n=2$, the ranks in (C.4) are $3,5,2$. *Manuscript: §51.11, folio 697, scan 99, (228); §51.13–14, folios 719–720, scans 143–146, (299)–(305).*

Proof. At $n = 1$ the compact curve is a projective line, so its ordinary Jacobian vanishes. Formula (C.1) then gives (C.3), with cocharacter lattice M. Abel–Jacobi sends a peripheral loop to the corresponding cocharacter; hence its Tate realization is the geometric abelianization used in B.2 and B.14. For arbitrary n, functoriality of the generalized Jacobian gives the norm map induced by q_n. On the Betti realization it is the map $K_n^{\mathrm{ab}} \to F^{\mathrm{ab}}$. B.5 computes its kernel and its image exactly. Tensoring (B.2) with the flat ring $\widehat{\mathbb{Z}}$ and using (C.2) proves (C.4). This is an explicit comparison morphism; at level two its matrix and its Smith entries $2, 2$ are supplied in B.7.

The geometric module in the middle has rank $n^2 + 1$; the proposed module has rank two. Modulo $m > 1$, their orders are m^{n^2+1} and m^2, respectively. Thus even an abstract isomorphism of the proposed finite realizations fails, with $n = m = 2$ giving the finite orders 32 and 4. The finite order comparison is checked in `code/b_freeness.py`. The norm comparison (C.4) gives the actual relationship that replaces that proposed identification. □

C.3. Complex multiplication with its actual ring map [Proved]

For $n \geq 3$, over a field containing μ_n, the deck action gives a homomorphism

$$R_n \longrightarrow \mathrm{End}(J_n)$$

whose kernel is I_n of B.6. The induced commutative semisimple algebra

$$\mathcal{E}_n = \mathbb{Q}[A_n]/(I_n \otimes \mathbb{Q}) \ \subset \ \mathrm{End}^0(J_n)$$

has dimension $(n-1)(n-2) = 2 \dim J_n$. Its factors are cyclotomic fields indexed by the Galois orbits of characters (a, b) with $a, b, a + b \neq 0 \pmod{n}$. This is the precise complex-multiplication algebra supplied by the Fermat automorphisms. The action on $\mathcal{J}_n$ restricts to the permutation action on the torus lattice in (C.1).

Proof. Automorphisms of the curve act on divisor classes and therefore on the Jacobian. An endomorphism of a complex abelian variety is zero as soon as it is zero on integral first homology: after lifting to the complex universal cover it is a complex linear map, and the integral lattice spans that cover over $\mathbb{R}$. B.6 identifies this homology with R_n/I_n, so the kernel is exactly I_n. The rational group algebra of a finite abelian group is a product of number fields, and its character decomposition selects exactly the characters in B.6. Each has multiplicity one. The dimension assertion follows by counting them. Complex conjugation pairs (a, b) with $(-a, -b)$; none of the selected characters is real: a real character would have a, b of order at most two and would force one of $a, b, a + b$ to vanish. Thus the selected cyclotomic factors are CM fields. This proves the full degree required for complex multiplication. The line-bundle-with-trivialization description of $\mathcal{J}_n$ shows directly that deck and coordinate automorphisms permute the cuspidal fibres, giving the torus action claimed. The notation $\mathbb{Z}[\mu_n]^2$ in Lead 3 is therefore replaced by this explicit group-ring homomorphism and its kernel, with no interpretation of that ambiguous expression left implicit. □

C.4. Exact order of cuspidal differences [Proved]

For every $n \geq 3$ and every two distinct geometric coordinate cusps P, Q of X_n^+, the class $[P - Q] \in J_n$ has exact order n. Consequently the subgroup generated by all cuspidal degree-zero divisor classes has exponent exactly n. At $n = 1, 2$ that subgroup is zero.

Proof. Let $\ell_Q = 0$ be the tangent line at a coordinate cusp Q. For example, at $Q = [0 : q : 1]$, with $q^n = -1$, its equation is $y - qz = 0$. Substitution into $x^n + y^n + z^n = 0$ gives $x^n = 0$; the

intersection divisor is exactly nQ. The same holds at every coordinate cusp. Therefore

$$\operatorname{div}(\ell_P/\ell_Q) = n(P-Q), \tag{C.5}$$

and the order divides n.

We prove exactness, including the even levels. At $Q=[0:q:1]$, the functions z/ℓ_Q and x/ℓ_Q have their only poles at Q, of orders n and $n-1$, respectively. Thus the pole semigroup $H(Q)$ contains $\langle n-1, n\rangle$. The latter numerical semigroup has exactly $(n-1)(n-2)/2$ positive gaps: the least element in residue class $r \pmod{n-1}$ is rn, for $0 \le r < n-1$, and counting the smaller positive members of those residue classes gives this number. Riemann–Roch gives exactly g_n gaps for $H(Q)$: for $a \ge 2g_n$, $\ell(aQ) = a+1-g_n$, and every successive dimension increase is zero or one. B.6 gives $g_n = (n-1)(n-2)/2$. Since the contained semigroup already has all g_n gaps, it equals $H(Q)$.

A proper positive divisor d of n satisfies $d \le n/2 < n-1$, hence is a gap. A principal divisor $d(P-Q)$ would supply a nonconstant function with its only pole at Q, of order d, a contradiction. The exact order is therefore n. The genus-zero cases have zero Jacobian. This gives a proof for every n, without a parity restriction. □

C.5. The extension class and its exact isogeny exponent [Proved]

The class of (C.1) is the homomorphism

$$L_n \longrightarrow J_n^\vee, \qquad D \longmapsto [D], \tag{C.6}$$

under the principal polarization and the extension–character correspondence $\operatorname{Ext}^1(J_n, T_{D_n}) \simeq \operatorname{Hom}(L_n, J_n^\vee)$. For every $n \ge 3$, this class has exact order n. Its pullback by $[n]: J_n \to J_n$ is split, and the resulting isogeny

$$T_{D_n} \times J_n \longrightarrow \mathcal{J}_n$$

has kernel isomorphic to $J_n[n]$ and degree n^{2g_n}. The splitting of that pullback descends to $\mathbb{Q}$. There is no single positive integer that kills these extension classes at every Fermat level.

Proof. Push (C.1) out by the torus character associated with $D = \sum a_P P$. The fibre of the resulting $\mathbb{G}_m$-extension over a line bundle $\mathcal{L}$ is the nonzero part of $\bigotimes_P \mathcal{L}(P)^{\otimes a_P}$. This is the restriction of the Poincare bundle corresponding to $[D] \in J_n^\vee$; it proves (C.6). Conversely, these character pushouts determine a torus extension, because a split torus is a product of copies of $\mathbb{G}_m$. Descent retains exactly the Galois-equivariant homomorphisms in (C.6); morphisms from an abelian variety to a torus vanish, so there is no ambiguity in descending a splitting. This is also the Cartier-duality description in [Del74, 10.2.11–10.2.12].

C.4 says exactly that (C.6) is killed by n, and that no proper divisor of n kills it. Pullback by the multiplication map $[m]$ multiplies the associated point of $J_n^\vee$ by m, as is immediate from the Poincare bundle. Thus $[n]^*$(C.1) is the trivial torus extension. Projection from that pullback to $\mathcal{J}_n$ has kernel $J_n[n]$, proving the degree and the isogeny statement. A splitting over $\overline{\mathbb{Q}}$ is unique because $\operatorname{Hom}(J_n, T_{D_n}) = 0$, so it is Galois-invariant and descends. Finally C.4 gives exponents n for unbounded n, proving the last assertion.

This statement describes the order in the algebraic extension group. Its realizations give a splitting of the weight filtration in the isogeny category, and an integral comparison whose finite defect is controlled by n. It does not replace that specified extension group by an unspecified Galois cohomology group. □

C.6. Rohrlich's theorem and the requested computations [Known]

For $n \geq 3$, the geometric cuspidal subgroup is

$$C_n \simeq \begin{cases} (\mathbb{Z}/n)^{3n-7}, & n \text{ odd}, \\ (\mathbb{Z}/n)^{3n-7} \times \mathbb{Z}/2, & n \text{ even}. \end{cases}$$

This is [Roh77, Theorem 1, p. 101], *Points at infinity on the Fermat curves*. We opened the original theorem and its definitions on p. 95. The odd case is also reproduced with an independent proof in [Vél79, Theorem 1, p. 28-02]. For the requested levels the exact orders of the extension class are

n	3	4	5	7
g_n	1	3	6	15
$\operatorname{rk} T_{D_n}$	8	11	14	20
$\operatorname{rk} T_\bullet(\mathcal{J}_n)$	10	17	26	50
ord[(C.1)]	3	4	5	7

At $n = 3, 5, 7$, the computed nontrivial Smith factors of the checked cuspidal presentation are respectively $3^2,\ 5^8,\ 7^{14}$.

Proof. Rohrlich defines $\mathcal{D}^\infty$ to be the degree-zero divisors supported on the $3n$ coordinate cusps of $x^n + y^n = z^n$, and $\mathcal{F}^\infty$ to be the principal divisors in this lattice. The geometric isomorphism ι_n of B.1 identifies their quotient with the image of $L_n \to J_n$. By (C.6), that is precisely the image of the homomorphism classifying the generalized-Jacobian extension. The exponent of the displayed group is n, also in the even case because $2 \mid n$. Therefore multiplication by n kills the extension class, and no smaller positive integer does. C.5 then constructs the splitting isogeny, over $\mathbb{Q}$ itself. This is the reduction from the original theorem to the asserted weight-extension statement.

For independent integral computations at the requested odd levels, choose a cusp P_*, and identify the degree-zero cusp lattice with the free lattice on $P - P_*$, for $P \neq P_*$. Velu's proof determines its principal-divisor sublattice modulo n, with six relation classes. In the standard ordering a_i, b_i, c_i, these can be written

$$\sum_i (a_i - P_*), \quad \sum_i (b_i - P_*), \quad \sum_i (c_i - P_*),$$
$$\sum_i i(a_i - b_i), \quad \sum_i i(a_i - c_i), \quad \sum_i i^2(a_i + b_i + c_i - 3P_*).$$

The same displayed presentation appears in [BM22, Theorem 29], which was also opened. Together with n times each basis vector it is the relation matrix used by `code/c_cuspidal_exponent.py`. Its exact integer Smith form gives the three listed odd-level groups, hence their exponents. Formula (C.6) identifies the image of the extension-class homomorphism with that same cuspidal group, so the exponent is again the order of the extension class.

For $n = 4$ the program does not apply an odd-level presentation. It checks instead the independent exact certificate of C.4: the tangent divisor is $4(P - Q)$, the pole semigroup $\langle 3, 4\rangle$ has gaps $1, 2, 5$, and the proper divisors $1, 2$ of 4 are gaps. Therefore the class has exact order four. The same certificate is checked for all four requested levels and, as an additional arithmetic verification, for every $3 \leq n \leq 32$. The proof of C.4 supplies its validity for all levels. □

C.7. The toric motive actually constructing the tower [Proved]

Let $T = \mathbb{G}_m^3/\mathbb{G}_m$, and let $\psi : U_1^+ \hookrightarrow T$ be the map of B.1. There is a Cartesian square over $\mathbb{Q}$

$$\begin{array}{ccc} U_n^+ & \longrightarrow & T \\ {\scriptstyle q_n}\downarrow & & \downarrow{\scriptstyle [n]} \\ U_1^+ & \xrightarrow{\psi} & T. \end{array} \tag{C.7}$$

For a point $e \in U_1^+(k)$, the 1-motive

$$\mathcal{K}_e = [\mathbb{Z} \xrightarrow{1\mapsto\psi(e)} T]$$

has finite realizations

$$0 \to T[n] \to T_{\mathbb{Z}/n}(\mathcal{K}_e) \to \mathbb{Z}/n \to 0, \tag{C.8}$$

whose fibre over 1 is $q_n^{-1}(e)$. The compatible fibres give exactly the Fermat torsor and its affine Galois action of B.14. Thus the rank-two module of §51 is the weight--2 part of this toric Kummer construction. *Manuscript: §51.13–14, folios 719–724, scans 143–153, (299)–(323), especially (317) and (323).*

Proof. A point $[a_0 : a_1 : a_\infty]$ of T lies over $[-e : e-1 : 1]$ under $[n]$ precisely when its n-th powers have that projective ratio. Their sum is zero, which is the equation of U_n^+. This proves the Cartesian square scheme-theoretically by its equations on a torus chart. Deligne's construction 10.1.5 represents $T_{\mathbb{Z}/n}([\mathbb{Z} \to T])$ by pairs (a, t), with $t^n = \psi(e)^a$, modulo $(a, t) \sim (a + nb, t\psi(e)^b)$. Projection to a mod n gives (C.8), and the fibre at $a = 1$ is the torsor of n-th roots of $\psi(e)$. By (C.7) it is the stated Fermat fibre. All formulas commute with automorphisms of k and with coordinate permutations, proving the claimed action and its compatibility with the representation in B.14.

This construction has weights $0, -2$, with quotient $\mathbb{Z}$ and toric Tate kernel of rank two. The generalized-Jacobian construction in C.1 is connected to it by the norm morphism (C.4); that map records the full additional kernel at level n. □

C.8. Lead 3 and the reading of the manuscript [Refuted]

Lead 3's proposed identification of §51.13–14 with a splitting of the generalized Jacobian of every Fermat curve is false. The source's explicit construction is (C.7) and the Kummer fibre (C.8). Its further elliptic-curve construction starts with an elliptic curve E/S and the three nonzero two-torsion points $D = E[2] \setminus \{0\}$, and has exact sequence

$$1 \to \mathbb{G}_m \otimes (p_*\mathbb{Z}_D/\mathbb{Z}_S) \to \mathcal{J}_{E,D} \to E \to 1. \tag{C.9}$$

The Tate kernel of (C.9) has rank two, as written in (303). It is identified with the torus of the three-point quotient $E/\{\pm 1\}$ punctured at the images of D. The weight-extension result for the Fermat Jacobians is C.5, with exact and unbounded exponents n; the rank-two group extension of B.12 has zero class and unique splitting conjugacy class. *Manuscript: §51.13, folios 719–720, scans 143–145, (299)–(304); §51.14, folios 720′–724, scans 146–153, (305)–(323).*

Proof. The divisor used in scan 144 is explicitly the degree-three subscheme of nonzero two-torsion points on the elliptic curve, and (302) has quotient that elliptic curve. Applying the line-bundle construction of C.1 to (E, D) proves (C.9). The three-point quotient map $E \to E/\{\pm 1\}$ identifies the two degree-three etale divisors. Hence it identifies their cocharacter lattices $p_*\mathbb{Z}_D/\mathbb{Z}_S$ and their associated tori. On the generalized Jacobians this torus isomorphism

is the pullback of a trivial line bundle with prescribed nonzero values at the three marked points; the three values are carried to the corresponding points of D, modulo common scaling. This is the explicit comparison explaining the repeated rank-two Tate module. The degree-three divisor is etale over bases on which 2 is invertible, precisely the qualification made at scan 152.

The other torus comparison written at scan 147 is also explicit:

$$[x_0 : x_1 : x_\infty] \longmapsto (x_1/x_\infty, x_\infty/x_0, x_0/x_1) \in \ker(\mathbb{G}_m^3 \xrightarrow{\text{product}} \mathbb{G}_m).$$

It is an isomorphism: its first two coordinates recover x_1/x_∞ and x_0/x_∞. For an even permutation it commutes with coordinate permutation; for an odd permutation its value is the inverse of the permuted value. This proves the manuscript's marginal qualification that it is not equivariant for the two untwisted full S_3-actions. The S_3-module used in B.10 is always the quotient lattice, so no product-one torus action is substituted in the vanishing proof.

At scans 151 and 153 the manuscript supplies ψ and the fibre of the universal multiplication tower at the marked section e. They are exactly the equations proved in C.7. For the full Fermat generalized Jacobian, the torus has rank $3n-1$ and the comparison with this rank-two torus is the noninjective norm morphism (C.4). Thus the proposed equality fails by that explicit morphism and its nonzero kernel already at $n = 2$. The exponents in C.5 further show that a fixed bounded torsion correction cannot describe all levels.

The supplied table of contents retains the unread initial letters in *[??]tivisation*; the body heading at folio 708, scan 121, reads *Retour sur la tour de Fermat*. The mathematics settles the interpretation just described. It does not settle the missing letters of the table-of-contents word. □

D. The arithmetic action, its Kummer coordinates, and the two CM factors

We use $G_k = \mathrm{Gal}(\overline{k}/k)$, $K = \mathbb{Q}^{\mathrm{cyc}} = \mathbb{Q}^{\mathrm{ab}}$, and the cyclotomic character $\chi : G_{\mathbb{Q}} \to \widehat{\mathbb{Z}}^\times$. Compatible roots of unity identify $T\mathbb{G}_m$ with $\widehat{\mathbb{Z}}(1)$. A coordinate in this Tate module depends on that choice. The group in scans 156–170 is the arithmetic $G_{\mathbb{Q}}$; the larger loop-automorphism groups enter only in the explicitly designated comparisons below. The relevant scans 157,166–170 have been inspected as images, independently of their transcription. A positive-weight calculation always refers to the kernel of the multiplier and to its lower-central filtration.

The homomorphism of scan 157 and the rational-prime questions

D.1. The actual affine Kummer homomorphism [Proved]

Choose compatible roots $a^{1/n}$ of $a = -1/2$. The homomorphism of §51.15, (326), (329)–(335), scans 156–160, is, after a choice of origin in its torsor,

$$\rho_a : G_{\mathbb{Q}} \longrightarrow \widehat{\mathbb{Z}}^\times \ltimes \widehat{\mathbb{Z}}(1), \qquad g \longmapsto (\chi(g), \kappa_a(g)), \qquad g(a^{1/n})/a^{1/n} = \zeta_n^{\kappa_a(g) \bmod n}.$$

Its image is the closed subgroup of index two

$$\{(u, v) : (-1)^v = \chi_{-8}(u)\}, \tag{D.1}$$

where χ_{-8} is the quadratic character of $\mathbb{Q}(\sqrt{-2})$, with values $+1$ at $1, 3 \pmod 8$ and -1 at $5, 7 \pmod 8$. Changing the root system translates the origin and changes κ_a by $(\chi - 1)b$. On G_K its image is exactly $2\widehat{\mathbb{Z}}(1)$.

Proof. The cocycle identity is obtained by applying g to $h(a^{1/n})$: $\kappa_a(gh) = \kappa_a(g) + \chi(g)\kappa_a(h)$. This is precisely the multiplication in the displayed affine group. In the homogeneous Fermat model of the manuscript, the point $1/2$ has coordinates $(-1/2, -1/2, 1)$; its fibre is the diagonal pair of root torsors of $-1/2$. Dividing the two coordinates removes the diagonal ambiguity and gives the rank-one torsor of (329)–(335). Thus this construction supplies the stated morphism, rather than merely a group of the same abstract form. The unread signs in (332), scan 159, do not change this argument: the homogeneous equation and the specified point determine the two nth powers to be $-1/2$.

The square root $\sqrt{-2}/2$ lies in K, and hence the parity of $\kappa_a(g)$ is exactly the quadratic action on $\sqrt{-2}$. On G_K the roots of -1 are fixed, so $\kappa_a = -\kappa_2$. D.2 below proves that this has image $2\widehat{\mathbb{Z}}(1)$. Finally χ is onto. Every fibre over u is a coset of that translation image; the parity constraint therefore describes the entire fibre, proving (D.1). □

D.2. Exact joint images and independence for distinct rational primes [Proved]

For distinct positive rational primes $p_1, \dots, p_r$, the continuous homomorphism

$$(\kappa_{p_1}, \dots, \kappa_{p_r}) : G_K \longrightarrow \widehat{\mathbb{Z}}(1)^r$$

has image $2\widehat{\mathbb{Z}}(1)^r$, of index 2^r. Consequently every odd-primary projection is onto, the 2-primary image is $2\mathbb{Z}_2^r$, and the characters are linearly independent over $\widehat{\mathbb{Z}}$. On $G_{\mathbb{Q}}$ the full affine image consists exactly of the pairs $(u, (v_i))$ satisfying $(-1)^{v_i} = \chi_{\mathbb{Q}(\sqrt{p_i})}(u)$ for every i. These assertions answer the individual and simultaneous questions of scans 168–169; integral surjectivity as written there is false.

Proof. Let $a = \prod p_i^{e_i} > 0$ and $n \geq 1$. We first compute the exact relation

$$a \in K^{\times n} \quad \Longleftrightarrow \quad n \mid 2e_i \quad \text{for every } i. \tag{D.2}$$

Choose the positive real root $\alpha = a^{1/n}$. Any other root differs by an element of K, so membership in K is independent of this choice. A finite subfield $\mathbb{Q}(\alpha)$ of the abelian extension $K/\mathbb{Q}$ is Galois and totally real. Every conjugate of α is a real root of $X^n - a$, of which there are at most two. Thus $\alpha^2 \in \mathbb{Q}$ and valuation at p_i gives $n \mid 2e_i$. Conversely this divisibility makes α a product of rational numbers and square roots of rational primes. Such square roots are cyclotomic: $\sqrt{2} = \zeta_8 + \zeta_8^{-1}$, and the quadratic Gauss sum for an odd prime has square $(-1)^{(p-1)/2}p$. Multiplication by i removes the latter sign. This proves (D.2).

For completeness, the finite Kummer pairing used here needs no rank estimate. Let B be the subgroup generated by the classes of the p_i in $K^\times/K^{\times n}$ and let $L = K(p_i^{1/n} : 1 \leq i \leq r)$. The pairing $\mathrm{Gal}(L/K) \times B \to \mu_n$, $(g, a) \mapsto g(a^{1/n})/a^{1/n}$, is nondegenerate on both sides: a Galois element fixing every root is trivial, and a root fixed by every element belongs to K. Both groups have exponent dividing n; injections into their finite character groups therefore have equal orders and are isomorphisms. By (D.2) the annihilator in $(\mathbb{Z}/n)^r$ of the relation subgroup is exactly $2(\mathbb{Z}/n)^r$. These are all finite images. Compactness gives the inverse-limit image stated above. A linear relation vanishes on $2\widehat{\mathbb{Z}}^r$ only when all coefficients vanish, since multiplication by two on $\widehat{\mathbb{Z}}$ is injective. Parity and surjectivity of χ give the last assertion just as in D.1.

The program `d_kummer_soule.py` enumerates these annihilators for every $n \leq 24$ and records the indices for $r = 1, 2, 3, 4$. This checks, in particular, the finite obstruction to the proposed surjectivity. □

D.3. General openness and two corrections to the manuscript [Proved]

Let k be a field finitely generated over $\mathbb{Q}$ and $A \subset k^\times$ a finitely generated subgroup of torsion-free rank r. The Kummer homomorphism on $G_{k(\mu_\infty)}$ has open image in $\mathrm{Hom}(A, \widehat{\mathbb{Z}}(1)) \simeq \widehat{\mathbb{Z}}(1)^r$.

This proves the openness requested under (355), scans 166–167, also for normal connected schemes essentially of finite type over $\mathbb{Q}$ and their groups of units. The stronger surjectivity proposed in scan 167 is false even for the full unit group of a normal arithmetic scheme of dimension two. The criterion for cyclotomic surjectivity proposed in scan 163 is also false.

Proof. Torsion in A contributes no homomorphism to the torsion-free Tate module, so choose a free complement. Write $H = \mathrm{Gal}(k(\mu_\infty)/k)$, an open subgroup of $\widehat{\mathbb{Z}}^\times$: the algebraic closure of $\mathbb{Q}$ inside k is a number field. Choose an integer m such that H contains all units congruent to one modulo m. There is $u \in H$ with $u - 1 = dv$, where $d = \mathrm{lcm}(2, m)$ and $v \in \widehat{\mathbb{Z}}^\times$: take $u_p = 1 + p^{v_p(m)}$ at $p \mid m$, $u_p = 2$ at odd $p \nmid m$, and $u_2 = 3$ when $2 \nmid m$.

Suppose $\alpha^n = a \in A$ with $\alpha \in k(\mu_\infty)$. Put $\tau(\alpha)/\alpha = \zeta_n^{c_\tau}$. Since H is abelian, comparison of its two cocycle products gives $(u-1)c_\tau = (\chi(\tau) - 1)c_\sigma$ for an element σ with character u. Setting $b = v^{-1}c_\sigma$ modulo n shows that $\alpha^d \zeta_n^{-b}$ is fixed by H. Thus

$$a^d \in k^{\times n}. \tag{D.3}$$

Choose a normal finitely generated $\mathbb{Z}$-algebra R with fraction field k containing A as units. Its unit group is finitely generated. One proof is to take a normal projective compactification over $\operatorname{Spec}\mathbb{Z}$: a unit has divisor supported on the finitely many codimension-one boundary components, while the kernel of this divisor map is the unit group of the ring of global functions of the compactification. That ring is finite over $\mathbb{Z}$; its units are finitely generated by the number-field unit theorem. Normalization is finite here, so this compactification argument applies to R.

If $b^n = a^d$ in k, both b and b^{-1} are integral over R, hence belong to R. In the free lattice $U = R^\times/\mu(R)$, the saturation of A modulo A is finite; let e annihilate it. Equation (D.3) then implies $de\, a \in nA$ in additive lattice notation. The kernel of $A/nA \to k(\mu_\infty)^\times/k(\mu_\infty)^{\times n}$ has order at most $(de)^r$. The perfect finite pairing in D.2 therefore bounds all finite indices of the Kummer image by this same number. A closed subgroup of $\widehat{\mathbb{Z}}^r$ with uniformly bounded finite indices has finite index: any collection of more distinct cosets is separated in one finite quotient. This proves openness.

For a normal connected scheme, the Kummer covers of units are finite etale and the generic-point map is onto their monodromy groups; the preceding field computation consequently gives the same image. For the counterexample take $S_1 = \operatorname{Spec}\mathbb{Z}[1/2][t]$ and $A = \mathcal{O}(S_1)^\times = \{\pm 2^j\}$. Its generic fibre has the Kummer image $2\widehat{\mathbb{Z}}$ of D.2. Moreover A is saturated in $\mathbb{Q}[t]^\times$ modulo its torsion, so the proposed saturated variant fails too. Finally $k = \mathbb{Q}(\sqrt{5})$ has only the roots of unity ± 1, but its cyclotomic image has index two because $k \subset \mathbb{Q}(\mu_5)$. □

D.4. Exact local inertia images [Proved]

Embed $G_{\mathbb{Q}_\ell}$ into $G_\mathbb{Q}$ and put $I_\ell^+ = I_\ell \cap \ker\chi$. For a positive rational prime p, the image of $\kappa_{p,\ell}$ on I_ℓ^+ is $\ell^{e_{p,\ell}}\mathbb{Z}_\ell(1)$, where

$$e_{p,\ell} = \begin{cases} 0 & \ell > 2, p = \ell, \\ v_\ell(p^{\ell-1} - 1) - 1 & \ell > 2, p \neq \ell, \\ 1 & \ell = p = 2, \\ v_2(p^2 - 1) - 2 & \ell = 2, p \text{ odd}. \end{cases} \tag{D.4}$$

Every image in scan 170 is therefore open, and the formula determines exactly which are onto. The program `d_local_cm.py` records every pair $p, \ell \le 50$; for example the image for $(p, \ell) = (3, 11)$ has index 11, and for $(19, 7)$ index 49.

Proof. The fixed field of I_ℓ^+ is $L = \mathbb{Q}_\ell^{\mathrm{ur}}(\mu_{\ell^\infty})$, abelian over $\mathbb{Q}_\ell$. For odd ℓ, choose a cyclotomic multiplier with $u - 1 \in \mathbb{Z}_\ell^\times$. The cocycle argument in D.3, now with $d = 1$, proves that an ℓ^kth root of $a \in \mathbb{Q}_\ell^\times$ belongs to L precisely when $a \in \mathbb{Q}_\ell^{\times \ell^k}$. The decomposition $\mathbb{Q}_\ell^\times = \ell^{\mathbb{Z}} \times \mu_{\ell-1} \times (1 + \ell\mathbb{Z}_\ell)$ and the ℓ-adic logarithm give the first two cases.

For $\ell = 2$ use multiplier -1. The same argument proves that a having a 2^kth root in L implies $a^2 \in \mathbb{Q}_2^{\times 2^k}$, equivalently $a = \pm b^{2^{k-1}}$ for $b \in \mathbb{Q}_2^\times$. Conversely every quadratic extension of $\mathbb{Q}_2$ lies in L: the square classes are generated by $-1, 2, 5$, the first two have cyclotomic square roots and $\mathbb{Q}_2(\sqrt{5})$ is unramified. Thus $\pm b^{2^{k-1}}$ indeed has a 2^kth root in L. For $a = 2$ the largest such k is one. For odd p, write $p = \epsilon u$, $\epsilon \in \{\pm 1\}$, $u \in 1 + 4\mathbb{Z}_2$; logarithms give the largest k as $v_2(u-1) - 1 = v_2(p^2 - 1) - 2$. Finite Kummer duality converts these root heights into the displayed closed images. □

The universal Fermat cocycle and its actual comparison maps

D.5. Full profinite universal measure, and the precise correction to Lead 1 [Known]

Put $\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$. The compatible projective Fermat homology of Otsubo's Theorem 6.1 is a free cyclic Λ-module. For its specified generator c, write $g(c) = F_g c$; then $F_{gh} = g(F_h)F_g$. On G_K this is a homomorphism into $\Lambda^\times$. The comparison with Anderson's relative-path measure B_g is the identity

$$(1 - s^r)(1 - t^m)F_g = (1 - s^{\chi(g)r})(1 - t^{\chi(g)m})B_g. \tag{D.5}$$

Here r, m are integers and s, t are the two deck generators. This is Otsubo's (7.2). In particular $F_g = B_g$ on G_K. Its ℓ-power specialization is related to Ihara's power series by the explicit derived-module basis comparison below.

The morphisms relating this invariant to D.1 are the exact quotient

$$1 \longrightarrow \widehat{\pi}'/\widehat{\pi}'' \longrightarrow \widehat{\pi}/\widehat{\pi}'' \longrightarrow \widehat{\pi}^{\mathrm{ab}} \longrightarrow 1 \tag{D.6}$$

and the root torsors obtained by evaluating the abelian covering at $1/2$. For rational primes the additional exact comparison is

$$p = \prod_{a=1}^{p-1}(1 - \zeta_p^a), \qquad \kappa_p\big|_{G_K} = \sum_{a=1}^{p-1} \kappa_{1-\zeta_p^a}\big|_{G_K}. \tag{D.7}$$

Lead1's proposed Kummer comparison is made exact by (D.5)–(D.7) and the coefficient extraction in D.23 below. In particular the rank-one affine coordinate is obtained from the full universal measure by a specified continuous map. This comparison retains the factor two in D.2 and the prime-to-ℓ deck components that a pure pro-ℓ series discards.

Proof. We use [Ots16, Theorem 6.1, Corollary 6.2, equation(7.2)], whose full text was opened. The non-zero-divisor assertion in Corollary 6.2 permits cancellation in (D.5); when $\chi = 1$ the two measures coincide. The maps to ℓ-power deck quotients and $\mathbb{Z}_\ell$ coefficients are continuous quotient maps of completed group rings; hence this comparison is profinite before any pro-ℓ specialization. Part B constructs the derived-module comparison in (D.6); evaluation of its abelian quotient at the displayed point gives D.1 by the explicit root equations. Finally (D.7) is $\Phi_p(1) = p$ and functoriality of roots under multiplication. Applying valuations at the distinct primes to its rational norm products, followed by the exact pairing of D.2, proves independence through these cyclotomic elements. The possible quadratic relations, including the factor two, are retained; an assertion of unrestricted integral independence of all circular units would not give the required exact image.

Otsubo's Remark 8.2 identifies the original results precisely as Ihara1986, Theorem 7 for prime-power level, and Anderson1989, Theorem 6 in general. Anderson's complete original has now been retrieved from the official GDZ scan and checked. Its §§1.3–1.4 use the full ring $\widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}(1)^2]]$; Proposition 5.5.1, p.88, constructs the all-degree torsion comparison from relative interval homology and states $e_N(B_\sigma\,\sigma(v)) = \sigma(e_N(v))$. Theorem 6, pp.118–124, then interpolates Gauss sums using a specified Frobenius lift and the hyperadelic gamma function; it is not by itself the torsion comparison.

The original comparison to Ihara is [And89, Proposition 13.3.2 and §§13.4–13.6]. For odd ℓ it constructs a Galois-equivariant map $\theta : \pi'_\ell/\pi''_\ell \simeq \varprojlim_n T_\ell J_{\ell^n}$ and a basis-change unit $Q(u,v)$, with exact formula

$$F_\sigma^{\mathrm{Ih}}(u,v) = S_\sigma(u)S_\sigma(v)B_{\sigma,\ell}(u,v)\frac{Q((1+u)^{\chi_\ell(\sigma)}-1,(1+v)^{\chi_\ell(\sigma)}-1)}{Q(u,v)},$$

$$S_\sigma(T) = \frac{(1+T)^{\chi_\ell(\sigma)}-1}{T}.$$

The map θ sends a deck transformation to its translation on the compatible division points of the divisor $(\delta_n) - (\infty_n)$, where $\delta_n = (t_n, -t_n(1-t^{-1})_n)$ in §13.5. That section records the plus-model twist for $\ell = 2$; RemarkII in §13.3 proves $F_\sigma^{\mathrm{Ih}} = B_{\sigma,\ell}$ on the cyclotomic kernel for every ℓ. These maps specify both the basis coboundary and the arithmetic twist.

The original Ihara1986 and Coleman1989 proofs remain to be obtained; the separate coefficient formula used below was checked in Ihara–Kaneko–Yukinari's primary paper and now also in Anderson's original. The retrieved original [Iha00, Supplement] corrects Ihara1986, Theorem 7 by replacing the Jacobi indices (a,b,c) with $(-a,-b,-c)$, and corrects the signs in Theorem 10. It also instructs replacing Anderson1989, p.125 line11's group-ring identification by its inverse, and records the inverse deck convention on p.85. These corrections explain why a convention must be attached to the original theorem numbers; D.6 uses the explicitly checked Otsubo convention. □

D.6. Frobenius specialization with nonsplit primes and the plus model [Known]

Let $p \nmid n$, $f = \mathrm{ord}_n(p)$, $q = p^f$, and let $\theta : \mathbb{F}_q^\times \to \mu_n$ be fixed. For $a, b, a+b \neq 0$ modulo n, put

$$j_n^{a,b} = -\sum_{t\in\mathbb{F}_q} \theta(t)^a\theta(1-t)^b,$$

with nontrivial characters extended by zero. Otsubo's Theorem 8.1 identifies this number with the character specialization of the product of F_{Frob_p} along the orbit $(a,b),(pa,pb),\dots,(p^{f-1}a,p^{f-1}b)$. For $x^n + y^n = z^n$ the sum of these $j_n^{a,b}$ is the trace of geometric Frobenius on H^1 over $\mathbb{F}_q$. For the manuscript's $x_0^n + x_1^n + x_\infty^n = 0$ it is instead the sum of

$$j_{n,+}^{a,b} = \theta(-1)^{a+b} j_n^{a,b}. \tag{D.8}$$

Proof. The cited theorem [Ots16, Theorem 8.1] is formulated for an arbitrary residue degree; its product, rather than a single value at p on a non-fixed character, is essential. The comparison with H^1 uses the ordinary arithmetic Frobenius action on the Tate module and its dual geometric action on cohomology; these have the same characteristic polynomial in this convention. The change of models is defined over $\mathbb{Q}(\epsilon_n)$ by $(x_0 : x_1 : x_\infty) \mapsto (x_0 : x_1 : \epsilon_n x_\infty)$, $\epsilon_n^n = -1$. Its Frobenius discrepancy is $\epsilon_n^{q-1} = (-1)^{(q-1)/n}$. Projectively this multiplies the first two coordinates by the same element. The (a,b) character is therefore multiplied by $\theta(-1)^{a+b}$, which proves (D.8). This retains the arithmetic descent of scan 171, including even n. □

D.7. Complete finite-field computation in the requested range [Proved]

The files `d_jacobi.py`, `d_jacobi_field.cpp`, `d_jacobi.out` and `d_jacobi.json` compute every nondegenerate character value for every prime $p \le 50$ and $3 \le n \le 12$ with $p \nmid n$, over the exact field $\mathbb{F}_{p^{\mathrm{ord}_n(p)}}$. Every value satisfies $j\bar{j} = q$. Character sums and independent point counts agree for both models. Levels 1 and 2 have zero projective H^1, and the output separately retains the bad-reduction pairs $p \mid n$.

Proof. For non-semiprimitive pairs the code finds a monic polynomial for which multiplication by its residue class has order $q-1$. This certifies that all nonzero elements of the quotient ring form a cyclic group, hence that the quotient is the required finite field. It enumerates the cyclotomic-number array $C_{ij} = \#\{t \neq 0,1 : \theta(t) = \zeta_n^i, \theta(1-t) = \zeta_n^j\}$. Discrete Fourier summation gives each j as an integral polynomial modulo Φ_n. All norm identities are checked in this exact quotient ring. Independently, the standard curve has $n^2 C_{00} + 2n + n\,[\theta(-1) = 1]$ points, and the plus curve has $n^2 C_{ee} + 3n\,[e = 0]$, where $\theta(-1) = \zeta_n^e$.

For the larger semiprimitive fields, $p^h \equiv -1 \pmod{n}$, $q = p^{2h}$, the code uses and checks the exact identity $j = -p^h$. Indeed the characters are trivial on $\mathbb{F}_{p^h}^\times$. In the quadratic extension, summing the additive character over trace fibres gives the Gauss sum $G(\theta^a) = p^h \theta^a(\tau)$ for a nonzero trace-zero element τ; the quotient $G(\theta^a)G(\theta^b)/G(\theta^{a+b})$ is p^h. This proves the shortcut. The smaller semiprimitive cases are also enumerated directly. The largest direct field is $\mathbb{F}_{47^5}$; it was actually enumerated. For an explicit arithmetic-twist check, $(p,n) = (43,6)$ gives standard trace 32 and plus-model trace -64, with respective point counts 12 and 108. These computations verify the finite-field side of D.6 and its character conventions; they do not claim a numerical oracle evaluating an arbitrary abstract Galois element independently of the theorem. □

Circular Kummer characters, regularity, and the first irregular prime

D.8. The checked Ihara–Anderson–Coleman coefficient formula [Known]

For odd ℓ, let $U = \log(1+u)$, $V = \log(1+v)$, $W = -U - V$. On $G_{\mathbb{Q}(\mu_{\ell^\infty})}$ the universal series has the expansion

$$F_g(u,v) = \exp\left(\sum_{m \ge 3 \text{ odd}} \frac{\kappa_m(g)}{(1-\ell^{m-1})m!} (U^m + V^m + W^m) \right), \tag{D.9}$$

where κ_m is the compatible Kummer character of $\epsilon_n^{(m)} = \prod_{a \in (\mathbb{Z}/\ell^n)^\times} (\zeta_{\ell^n}^a - 1)^{\langle a^{m-1} \rangle_n}$. The integral factor $1 - \ell^{m-1}$ is a unit. Vandiver at ℓ is equivalent to surjectivity of all these κ_m for $m = 3, 5, \ldots, \ell - 2$.

Proof. This is the reduction of [IKY87, Theorems A_2 and B, equations(9)–(11)] to the root convention of D.1. Their character is defined by the same quotient of a chosen ℓ^nth root and its Galois translate, so no unnamed rescaling is involved. Their Theorem B identifies $\beta_m = (1-\ell^{m-1})^{-1}\kappa_m$, and Theorem A_2 gives the exponential with $\beta_m/m!$. The paragraph immediately following Theorem B proves the stated Vandiver equivalence. Both pages were inspected visually. Reducing modulo ℓ permits replacing $\zeta^a - 1$ by $1 - \zeta^a$: the resulting sign is an ℓth power. These are the exact characters computed next.

Anderson's original §11.6, Proposition 11.6.1 and formula(13.7.3), pp.117–118 and 130, give an independent normalization check. His β_m sums i^{m-1} times the Kummer cocycle of $1 - \zeta_{\ell^n}^{-i}$ over every $0 < i < \ell^n$. On the cyclotomic kernel this is the cocycle of $\zeta_{\ell^n}^i - 1$, since their

ratio is $\zeta_{\ell^n}^{-i}$ and all of its ℓ-power roots are fixed. Splitting the sum into $\ell \nmid i$ and $i = \ell j$ gives $\beta_m = \kappa_m + \ell^{m-1}\beta_m$. Thus his exponential coefficient is exactly the one in (D.9). □

D.9. An exact obstruction module, not an irregular-prime guess [Known]

Let $\mathcal{F}_\ell \subset \mathbb{Z}_\ell[[u,v]]^\times$ be the group defined by the five conditions of Ichimura–Kaneko1989, §1.2: congruence $F \equiv 1 \pmod{uvw}$, inversion $F\overline{F} = 1$, symmetry in u, v, w, symmetry of $F * F$ in four variables with product of 1+variables equal to one, and

$$\prod_{\zeta^\ell=1} F(\zeta(1+u)-1, \zeta(1+v)-1) = F((1+u)^\ell - 1, (1+v)^\ell - 1).$$

Their Theorem 1 identifies equality $\operatorname{Im} F = \mathcal{F}_\ell$ with Vandiver at ℓ. Their Theorem 2 computes the characteristic power series of the defect $(\mathcal{F}_\ell / \operatorname{Im} F)(-1)$ as that of $\operatorname{Hom}_{\mathbb{Z}_\ell}(A_\infty^+, \mathbb{Q}_\ell/\mathbb{Z}_\ell)$ over the cyclotomic Iwasawa algebra. Thus the relevant obstruction is the plus class-group module, not merely divisibility of a Bernoulli numerator. At every regular odd prime this defect is zero. D.10 proves the same equality at every odd prime at most50, including 37.

Proof. We use [IK89, Theorems1–2]; the defining conditions and both theorem statements were checked on the original pages 314–315. Equation(D.9) identifies their F and Kummer characters with ours. The plus part A_∞^+ is the direct limit of the ℓ-Sylow parts of the ideal class groups of $\mathbb{Q}(\mu_{\ell^n})$, with the authors' transition maps. The stated characteristic-series equality records exactly the size invariant proved there; it is not promoted to an isomorphism of modules. Regularity implies that the full class number at the first level is prime to ℓ, hence its plus part is, and so Vandiver holds. For the bounded range, the certificates below establish the equivalent surjectivities directly. □

D.10. Soulé characters modulo every prime at most50 [Proved]

For every odd prime $\ell \le 50$ and each odd $3 \le m \le \ell - 2$, the program `d_kummer_soule.py` exhibits an explicit nonzero value of $\kappa_m \bmod \ell$. Consequently all these individual characters are onto $\mathbb{Z}_\ell$, their distinct Teichmuller-weight components are independent, and Vandiver holds throughout this range. At $\ell = 37$ all seventeen characters are certified by the single prime $q = 5477$. The Bernoulli computation finds the irregular pair $(37, 32)$ and no other irregular pair below 50; it causes no missing plus-part image here.

Proof. For a prime $q \equiv 1 \pmod{\ell^2}$, a primitive root $g \bmod q$, and $r = g^{(q-1)/\ell}$, the code evaluates $c_m(r) = \prod_{a=1}^{\ell-1}(1-r^a)^{a^{m-1} \bmod \ell}$ and finds the unique $k \bmod \ell$ such that $c_m(r)^{(q-1)/\ell} = r^k$. A nonzero k proves that the corresponding cyclotomic element is not an ℓth power. All choices, exponents and residues are retained in the JSON, so each witness is checked by modular exponentiation alone.

This is also a certificate on the infinite cyclotomic field, not only at its first layer. In $\mathbb{Q}(\zeta_\ell)^\times/\ell$ the class of c_m has Teichmuller character $b \mapsto b^{1-m}$. The only order-ℓ subextension of the cyclotomic $\mathbb{Z}_\ell$-extension has Kummer class ζ_ℓ, of character $b \mapsto b$. These characters differ in the indicated range. Thus the nonzero class cannot disappear in that extension. The character takes a unit value in $\mathbb{Z}_\ell$, proving surjectivity; projection by the distinct idempotents of $\mathbb{F}_\ell^\times$ proves joint independence of these weight components. D.8 now proves Vandiver without inferring it from the irregularity test.

The witness primes, grouped by ℓ, are

ℓ	5	7	11	13	17	19	23
q	101	197, 491	727	677, 2029	3469	10831	12697

ℓ	29	31	37	41	43	47
q	10093, 13457	15377	5477	16811	22189	8837.

For $\ell = 3$ the specified weight list is empty; for $\ell = 2$ there is likewise no weight in the requested range. No omitted character is hidden by these conventions. □

Metabelian leading terms and the lifting issue

D.11. Exact special-derivation images through weight 10 [Proved]

Let L be the free Lie algebra over $\mathbb{Q}$ on x, y, with the three peripheral elements $x, y, -x-y$. Modulo inner derivations normalize a positive-degree special derivation by $D(x) = 0$, $D(y) = [f, y]$. For weight $n \geq 2$, it is special exactly when some $g \in L_n$ satisfies

$$[f, y] = [g, x + y]. \tag{D.10}$$

The metabelian quotient is $L/L'' = \mathbb{Q}x \oplus \mathbb{Q}y \oplus \mathbb{Q}[X, Y]c$, where $c = [x, y]$, $X = \operatorname{ad} x$, $Y = \operatorname{ad} y$; the last summand is abelian. The following exact rational ranks were computed, with source witnesses, by `d_metabelian.py`:

n	1	2	3	4	5	6	7	8	9	10
special outer source	0	0	1	0	3	0	6	4	13	12
its metabelian image	0	0	1	0	3	0	5	0	7	0
S_3-invariant source	0	0	1	0	1	0	2	1	4	2
its metabelian image	0	0	1	0	1	0	1	0	2	0

(D.11)

Thus all three weight-five directions survive. The extra invariant direction in weight seven dies. In weight nine two invariant directions survive. These are images of derivations of the free Lie algebra; special derivations defined only on L/L'' have dimension $n - 2$ for every $n \geq 3$, and are a larger calculation.

Proof. Subtracting the inner derivation that conjugates x gives the normalization. For $n \geq 2$, the centralizer of x has no component of degree n, so it is unique. The condition at the third peripheral element gives (D.10), with a harmless simultaneous sign determined by the convention for g. The free metabelian description follows by Jacobi: the commuting operators X, Y generate L'/L'' from c, and its monomials are independent, for instance in the usual upper-triangular matrix realization over $\mathbb{Q}[X, Y]$.

The program uses the Lyndon basis in each degree and expands its brackets in the free associative algebra. It forms the exact matrix of $(f, g) \mapsto [f, y] - [g, x + y]$ and computes its rational nullspace. The invariant subspace is separately defined by the checked LM2 equations

$$f(x, y) + f(y, x) = 0, \qquad [x, f(x, y)] - [x + y, f(-x - y, y)] = 0. \tag{D.12}$$

These follow by conjugating the normalized outer derivation by the transposition and the three-cycle of the three peripheral elements and then renormalizing $D(x) = 0$. Thus the matrix computes the actual infinitesimal centralizer, rather than the two linear cycle equations. Projection of each Lyndon bracket to $\mathbb{Q}[X, Y]c$ gives the image ranks. Every pivot-image vector retains an exact preimage in the JSON.

In the metabelian algebra (D.10) reads $Yp = (X + Y)q$, hence $(X + Y) \mid p$. This gives the stated $n - 2$ comparison dimension. At weight five the full image consists of $(X + Y)$ times all degree-two polynomials; the invariant image is spanned by $X^3 + 2X^2Y + 2XY^2 + Y^3$. At weight seven its invariant image is spanned by $X^5 + 3X^4Y + 5X^3Y^2 + 5X^2Y^3 + 3XY^4 + Y^5$. At weight nine one extra independent vector is $X^2Y^2(X + Y)^3$; the output gives the other vector and both preimages. The zero even-weight images, including weights eight and ten with nonzero sources, are computed by this projection, not assumed from parity. □

D.12. The reviewed pro-ℓ lifting assertion and its precise scope [Proved]

For a free pro-ℓ group on finitely many peripheral generators, every integral special derivation is the leading term of a special automorphism. For type $(0,3)$ and $\ell \geq 5$, every S_3-invariant integral leading term lifts to an element of the centralizer of the geometric S_3 in the pro-ℓ outer special-automorphism group. Thus the image ranks in (D.11) are ranks of actual pro-ℓ special-automorphism images, and of their centralizer for $\ell \geq 5$.

Proof. Write Γ_m for the closed lower-central series. Its associated graded is the free Lie algebra over $\mathbb{Z}_\ell$. Start with conjugators representing the given degree-n Lie elements. The product-of-peripheral-generators equation holds modulo Γ_{n+2} because its leading error is the special-derivation equation. At stage m the error lies in Γ_m/Γ_{m+1}; changing the conjugators by elements of Γ_{m-1} changes it by $\sum_i [a_i, x_i]$. This map is onto L_m: iterated Jacobi rewrites every Lie monomial of degree m as a sum $[x_i, b_i]$. In the two-generator normalization fixing x, the two available operators are $[\ , y]$ and $[\ , x+y]$; their sum still spans $[x, L_{m-1}] + [y, L_{m-1}] = L_m$. We can therefore cancel each error without altering the prescribed leading term.

The corrections converge in the free pro-ℓ group. Indeed every finite ℓ-group quotient is nilpotent, so Γ_m eventually lies in every open normal subgroup. The resulting endomorphism is the identity on the Frattini quotient and hence is onto by the pro-ℓ basis theorem; a finitely generated profinite group is Hopfian, so it is an automorphism. This independently proves the reviewed assertion.

For the invariant assertion, the group of IA automorphisms in this pro-ℓ setting is pro-ℓ, as are the successive kernels. A finite group of order prime to ℓ acts on it. For a quotient by a stable closed normal subgroup, fixed elements lift to fixed elements: for finite ℓ-groups this follows by induction on a central series and averaging the additive obstruction, and the profinite assertion follows by compactness. Apply this to the order-six group S_3 and successive filtration quotients. This proves the claimed lift. No such convergence argument applies to the full free profinite group: a finite nonnilpotent quotient need not kill any Γ_m. □

D.13. The remaining full profinite image comparison in D(e) [Open]

The maps from actual full profinite loop-preserving outer automorphisms, and from $\widehat{\mathrm{GT}}_0$, to their pro-ℓ leading terms land in the second and fourth rows of (D.11), respectively. D.12 proves the reverse inclusions for the pro-ℓ groups, not for those full profinite source groups. The full profinite realization of the extra weight-five metabelian directions, and of the extra invariant weight-nine direction, remains unproved here. Consequently D(e), as asked for the three full profinite groups, remains undecided in this return.

D.39 extends a new representative through the entire all-prime mixed Fermat homology tower, jointly with the retained degree-eight quotient. Its common quotient stabilizes at exponent eight. The exact two-adic system has cokernel killed by sixteen, and the half-average gives every odd-prime component. The resulting multiplier-one automorphism retains the normalized G6 and T3 actions, replaces the prior J3 and J12 actions by the specified new components, and retains its checked J48 seed at every later level. Both compact CM matrices and the fixed-fibre pair survive. The exact source kernel is the intersection of the closed L-commutator subgroup with D9K; the nonzero degree-sixteen word remains in it. The projection to the earlier finite joint group is strict, detected by x^{144}. D.45 proves that this recorded action has no full profinite lift. D.46–D.47 subsequently construct a replacement on their exact relative sources; its full profinite realization remains unresolved.

D.40 extends that action through the actual image in every finite perfect quotient of the marked source. Their exact common quotient with the retained tower is C_6, and the normalized conjugators pair with the identity on all perfect coordinates. A further actual S_7 quotient gives

the fibre product over the common sign C_2. The exact remaining full-source kernel maps onto A_{10} under the computed S_{10} quotient. These maps retain both compact-cover and fixed-fibre CM coordinates; D.45 excludes a full profinite lift of this recorded action.

D.41 then extends the action through all actual symmetric quotients simultaneously. Their even image is the full product of the alternating factors indexed by distinct marked kernels, and the common quotient with D.40 is exactly the sign C_2. Both projection kernels are computed. The remaining full-source kernel maps onto $\mathrm{PSL}_2(\mathbb{F}_7)$ under the specified $\mathrm{PGL}_2(\mathbb{F}_7)$ map and kills the canonical perfect mod-seven coordinate. This retains the actual CM data; D.45 excludes a full profinite lift of the recorded action.

D.42 extends this actual source action through the full relative pronilpotent kernel of the specified projective-linear quotient. Its maximal common marked quotient has base order 54 and kernel $\mathbb{Z}_3^{10}$. The nonzero common conjugator is retained by an actual coprime centralizer lift, and every projection kernel is computed. The remaining source kernel maps onto $\mathrm{PSL}_2(\mathbb{F}_{11})$ under the specified $\mathrm{PGL}_2(\mathbb{F}_{11})$ map. D.45 supplies a finite sphere quotient that excludes every full profinite lift of this recorded action.

D.43 extends the retained action simultaneously through all relative pronilpotent completions whose marked finite base has nonabelian simple even subgroup. The exact common quotient with the retained tower is $C_{54} \times_{C_6} D_{\mathscr{J}}$, with its actual source-kernel formula. One compatible centralizer lift supplies the conjugator over the whole family. The exact remaining source kernel maps onto $A_7 \times A_7$ under an explicit quotient whose involution exchanges the two simple factors. D.45 excludes every full profinite lift of this recorded action.

D.44 extends the same quotient action to the complete family of anabelian even bases with their relative pronilpotent kernels. The common quotient and both source kernels are exact, and an actual $A_5^4 \rtimes S_4$ marking detects residual kernel A_5^4. D.45 then excludes a genuine full profinite lift of the entire recorded normalized D.39 action and every D.40–D.44 extension retaining it. Its sphere projection through B/D_9F has a nonzero degree-nine obstruction after every lower centralizer and multiplier freedom is included. The elliptic projection has an explicit corrected lift to $B/D_{10}B'$. D.46–D.47 subsequently supply a replacement realizing the prescribed CM pair on their exact relative sources. The full source image questions in D.13 and D.19 remain to be determined.

D.46 constructs a replacement on the entire relative pronilpotent completion of the actual rank-97 subgroup N. Its full relative pro-two component is an actual arithmetic commutator action with identity sphere projection and compact elliptic scalar η_*; every odd relative-prime component is the identity. The two actions join through explicit coprime centralizer lifts. The resulting G_2, G_3 actions determine the prescribed pair $T_4 = I, W_3 = [i]$, and the reconstructed parameter has an actual $\widehat{F}'$ preimage. Thus the replacement solves the entire rank-four relative pro-two lifting step. Its exact remaining full-source kernel has image A_7 under the displayed S_7 quotient. Its full profinite realization remains in the lifting problem.

D.47 extends this replacement coherently over every Fermat level. For $M = \widehat{N} \cap \widehat{F}'$, it proves both inclusions identifying the closed relative pro-prime source with the inverse system over the actual L_m. The common marked quotient is $\mathcal{H}_\infty$, and the simultaneous kernel is $\prod_p M^{(p)}$. The source kernel lies in $M' \subset \widehat{F}''$, so the action descends to the full metabelian quotient, with every profinite deck coordinate retained. Its coefficient-two action is the actual arithmetic h action; every odd coefficient action is identity. The actual marked cubic-four and quartic-three cover maps give $(-I, I)$, outside the proved Galois joint graph. The Heisenberg quotient and full Alexander-module argument also proves this in the outer action space. All comparison kernels to D.46 are computed. The remaining full-source kernel still has exact A_7 image in the displayed S_7 quotient. The full GT0 and full loop-preserving $\widehat{F}$ lifting questions remain unresolved.

D.48 extends the entire D.47 action over the central double cover $\mathcal{E} = \widehat{B}/[\widehat{N}, \widehat{B}]$, jointly with every Fermat level. The new subgroup M^+ has index two in M; its relative pro-two source is unchanged. The comparison kernel is the product of the explicitly identified odd-prime kernels. The actual h action fixes the enlarged base, and the coprime centralizer corrections retain the complete D.47 action and parameter. On the new finite mixed quotient $\mathcal{J}_3$, the exact old source-kernel image is C_3^{96}; the new source kernel has trivial image there. Its full common quotient with D.47 is $\mathcal{E} \times_Q G_3$, and the identity action on $\mathcal{J}_3$ agrees with the retained action on that whole quotient. The remaining full-source kernel has exact A_7 image; full profinite realization remains unresolved.

D.49 extends this same action over the maximal marked quotient fixed pointwise by the actual arithmetic h, and over every relative-prime source above that base. Its exact comparison with D.48 retains both kernel layers, including an actual A_5 quotient of the base residual. The full congruence quotient $\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$, with one global sign, factors through the identity-three component; the new action is therefore pointwise identity on this entire quotient. Its remaining source kernel has exact image $[V_{3,2}, A]$ in the specified mixed homology quotient over G_3. It contains an actual cyclic moving subspace of dimension $64 \cdot 3^{97}$; the dimension of the full arithmetic span A is retained as unevaluated. The earlier central iteration stabilizes by the exact integral equality $[L, \widehat{B}] = L$. The remaining mixed source and full profinite realization questions stay active.

D.50 extends this same retained action through the entire two-adic homology tower over the actual marked G_3. The exact residual image is the integral moving summand U^-, and every common quotient is computed from the original source kernels. A regular-edge Fitting projection and its explicit crossed homomorphism repair the initial peripheral endomorphism to an automorphism; the cyclic norm correction fixes the original cusp. Compatible conjugators and invertible reduction modulo two give all higher lifts and their continuous inverse limit. The full parameter, CM pair, metabelian and congruence actions are retained. The remaining source kernel has a nonzero image containing $W_1 \wedge V$ in the next actual class-two exponent-four quotient. Extending across that remaining kernel and attaining the full profinite source remain active steps.

D.51 retains this entire chosen homology action and both full conjugators and extends them through the moving part of the next actual class-two exponent-four quotient. The exact old source image is $Z = W \wedge V$; its invariant and moving dimensions are respectively $m^2(q-1)/2$ and $w\rho/2$. The actual common quotient is $\mathcal{T}_2/Z$. Characteristic descent, exterior preservation of Z^A, and Frattini generation prove the new action on the joint source. Its projection adds the whole moving central summand and retains every previous parameter, CM, metabelian and congruence comparison. The remaining exact image is $Z^A \neq 0$. In the separate order-1080 verification, the complete corrected cusp word has 4,159,444 letters. That example's fixed homology matrix has intersection-pairing defect of rank 72, excluding its extension to the unquotiented class-two peripheral source. The arithmetic invariant obstruction and the full profinite realization remain to be evaluated.

D.52 solves the entire invariant boundary equation in the separate marked order-1080 source. Its 120-dimensional space consists of allowable conjugator pairs. A retained weight-60 parameter solves all thirteen actual quadratic equations; the complete 6,352,056-letter corrected word, both torsion relations and generation are verified. The new homology action has deviation rank 96 over $\mathbb{F}_2$ and preserves the mod-two ribbon form. It replaces the excluded finite representative while retaining the actual invariant quotient. It also has a complete two-adic homology continuation with the entire new class-two action retained, exact common quotient and both projection kernels. Its next exponent-eight residual is twice the commutator module, witnessed by a nonzero 272-letter commutator square. The arithmetic G_3 correction and the full profinite realization remain to be evaluated through their own actual source maps.

D.53 computes the full actual moving peripheral-pair space, its exact homology image and kernel, and a positive arithmetic dimension bound. Its integral lifts give actual doubled automorphisms of the complete D.50 source. D.54 lifts every such difference to the full joint class-two source, through an explicit compatible square module. The fixed arithmetic cusp defect is preserved literally, and the actual affine quadratic map descends through every same-action ambiguity. Its arithmetic zero locus and full profinite realization remain to be evaluated. In the separate order-1080 source, the full finite homology zero locus has exactly864 elements; all262,144 parameters, the original equations, generation and the complete long cusp words have been checked.

D.55 gives the exact full-group norm from the arithmetic cusp obstruction to the compact orientation-bivector defect. Its local image lies in the norm kernel, and its residual quotient has explicit Tate-norm and compact-projection maps. Every point of the reduced boundary zero set has invertible homology, so generation follows there. The actual ordinary relative-H_2 orientation comparison is an isomorphism. The arithmetic zero set and residual norm kernel remain unevaluated, with all maps retained for the next attack.

D.56 evaluates an actual irreducible C_3^4 quotient of the arithmetic normal span and the unique parameter $(2, 1, 2, 1)$. The marked quotient has order3888 and actual action $\mathrm{Int}(t^6)$. Its exact A_4 comparison kernel gives surjective integral homology and class-two boundary maps. The smaller arithmetic correction space has dimension40, with invariant central target dimension2560. Its actual quadratic zero set and full profinite realization remain unevaluated; every source and comparison kernel is retained.

D.57 now solves the complete ten-coordinate boundary equation on the actual order-3888 arithmetic quotient, for both integral rank-nine branches and all sixteen framed cases. The literal genus-three cover and degree-eight elliptic map, exhaustive integral conjugator search, separate plus-central alignment and augmentation correction give the asserted automorphism of $\widehat{B}/(R_h \cap M_* \cap L_{*,2})$. Its entire W_h action and prescribed normalized parameter are retained. The exact remaining quotient kernels carry the full G_3 invariant and full profinite realization questions onward.

D.58 retains the entire D.57 action and its conjugators while restoring the full integral arithmetic homology. The actual projective repair stays in $\mathcal{L} = \ker(U^- \to U_*^-)$. It then extends through the remaining moving class-two module, with exact residual kernel $\mathcal{K}_0 = \ker(Z \to Z_*)^A$. All 2560 solved invariant coordinates are retained; the remaining dimension is $m^2(|A| - 1)/2 - 2560$. The next actual invariant boundary and the full profinite realization remain to be evaluated across this specified source map.

Established comparisons and attempted lifting. Taking the maximal pro-ℓ quotient is characteristic, so it defines the displayed continuous homomorphisms of outer automorphism groups. Taking a leading term is injective on each filtration quotient by its definition; the peripheral and centralizer equations give (D.10)–(D.12). D.12 realizes every admissible leading vector after the change of source to the pro-ℓ group. A proposed full profinite lift would have to solve the same conjugacy and generation equations in every finite quotient, including nonnilpotent quotients. The lower-central correction sequence controls none of the residual errors in such a quotient. We also checked the weaker route of solving just the metabelian Lie equations: its larger dimension $n - 2$ proves that this route discards further lifting conditions already at weight four. These two failed transfers are exact gaps, not counterexamples to a full profinite lifting theorem.

For the Galois source, (D.9) gives at most one rational direction in each odd weight and zero in even weights in this metabelian image. D.10 supplies nonzero directions through weight ten at $\ell = 37$; E.3 supplies the rational general-prime comparison. The weight-five pro-ℓ image therefore distinguishes the whole special group from its S_3 centralizer and from

Galois. D.20–D.21 below add the full profinite beta/gamma compatibility and prove equality of the rational metabelian graded images of Galois and full $\widehat{\mathrm{GT}}$ through weight ten. The pentagon used there is an additional equation on $\widehat{\mathrm{GT}}_0$. The exact missing lift test for the latter is D.22; the pro-ℓ construction alone supplies only the subfamily of its tests with nilpotent quotients. D.24 gives an explicit test of the general automorphism-lifting route and its peripheral limitation. D.25 constructs nonarithmetical finite GT shadows with generation verified. D.26 lifts all its selected actions through the complete abelian-kernel tower. D.27 proves that the twenty-three nonidentity selected actions are nevertheless nongenuine, and constructs an alternate action in the same extra CM pair that survives both characteristic class-two tests and extends over the full relative pro-three kernel. D.28–D.29 further extend the alternate homology class through a compatible mixed-prime quotient, the entire class-two pro-two tower, and every odd relative pro-prime kernel. D.30 excludes the two recorded later central representatives using a complete lower-centralizer calculation, then constructs a further central replacement. D.31 replaces that central representative again by an explicitly compared action on $\mathcal{S}_2^{(7)}$, whose kernel over Q has class three and third subgroup C_2^{1860}; it retains the same homology and CM data and combines with every odd relative kernel and the mixed quotient. The canonical map to D.30's target has kernel C_2^{1620}, and the induced action differs from the earlier one already on $H^{(1)}$. D.32 then proves that this recorded degree-seven action has no peripheral lift even to the full relative pro-two source: its next central layer, all lower centralizer freedoms, and the multiplier are exhausted by an explicit separating functional. D.33 constructs a new degree-eight replacement on the exact quotient $\mathcal{S}_2^{(8)}$, with class-four kernel, third subgroup C_2^{9105} and fourth subgroup C_2^{1620}. Its canonical projection to D.31 has kernel C_2^{7245}, and its induced action changes the recorded image of s first in degree five. A restricted stable correction extends that new action over the entire class-two tower while retaining degree eight, all odd relative kernels and the mixed quotient. D.34 identifies the full relative pro-two source with the exact outer deck centralizer of the marked genus-one three-cusp cover, including the full multiplier pair and the cubic arithmetic twist. D.35 constructs actual odd-deck boundary lifts and proves that the whole compact CM torus is already realized by actual Galois action centralizing all deck maps. Its exact reduction map retains the finite-action fibre over the recorded D.33 action. D.45 proves that particular fibre empty by an actual source-inner transport; the second CM factor remains to be realized with a replacement action. D.36 supplies the second marked elliptic cover and the exact shared cyclic-twelve homology maps. Its arithmetic joint image has order six, with the cubic and tangent twists retained; it also computes the gauge-invariant discrepancy between the D.33 cover sign and its fixed-fibre T_4 coordinate. D.37 then constructs an actual generating shadow on the shared congruence/Fermat-twelve exponent-twelve homology quotient. Its exact pre-gauge G_6 conjugators agree with the retained D.33 components, while its actual normalized compact-cover matrices are $-I$ on cubic four-torsion and I on quartic three-torsion, both evaluated through one cyclic-twelve cover. The two-cover arithmetic graph fails on this finite shadow. D.37 also computes both universal relative-prime quotients and the exact mixed kernel $(\mathbb{Z}/4)^{768} \oplus (\mathbb{Z}/3)^{864}$. D.38 constructs a new action on the exact fibre product of this Fermat quotient with D.33's degree-eight quotient. It proves both kernel maps and the common quotient, retains cubic Tate sixteen and quartic Tate three, and changes the degree-eight action by exactly the displayed cusp-inner gauge. D.45 proves that this recorded action has no full profinite lift. D.46–D.47 subsequently construct a replacement on their exact relative sources; its full profinite realization remains unresolved. D.47 proves that its full metabelian outer action is outside Galois through the exact cover maps. Its realization by the requested full profinite source groups, and genuine CM-pair attainability in those groups, remain undecided. □

The integral signs, CM coordinates, and the genuine congruence action

D.14. What the first integral sign congruences say [Proved]

The norm equations give $\epsilon_2 = \chi_4(\mu)$ and $\epsilon_3 = \chi_{-3}(\mu)$, already at moduli 4 and 3. On the Galois image $\mu = \chi$. Thus these signs add no independent condition to the full universal datum (χ, F_g): the exact comparison is evaluation of χ modulo 4 and 3. The unadorned series restricted to G_K has $\chi = 1$, and consequently both signs there are one.

Proof. For a unit Gaussian norm $a^2 + b^2$, exactly one of a, b is odd; its value is one modulo 4. For the Eisenstein norm $c^2 + cd + d^2$, reduction modulo 3 is $(c - d)^2$, hence is one for a unit. The signed norm equalities in LM2 C.d2 therefore force the stated signs. Conversely the local norm images are $1 + 4\mathbb{Z}_2$ and $1 + 3\mathbb{Z}_3$: the norm on sufficiently deep principal units is the trace under logarithm, and inspection at moduli 8 and 9 supplies the first step. These finite images are also enumerated inside `d_local_cm.py`. Evaluating the already specified multiplier proves the asserted factorization; it does not equate an unmarked beta series with its missing multiplier. □

D.15. The two normalizer coordinates are the projective CM representations [Known]

Use the rigidified modular stack $\mathcal{M}_{1,1}/\{\pm 1\} \simeq [U_{0,3}/S_3]$ and the positive tangential point $\lambda = 0$ with tangent1 and the standard branch-cut paths on $0 < \lambda < 1$. Choose the same etale paths and homology basis in the modular congruence quotient and in LM2 C.d4. Then its coordinates $T = A^{-1}V$ and $W = \beta^{-1}V$, each modulo its single global sign, are respectively the projective adelic Tate representations of

$$E_3 : y^2 = x^3 + 1, \qquad E_4 : y^2 = x^3 - x. \tag{D.13}$$

Their determinants are χ; their actions on the CM orders are complex conjugation according to χ_{-3} and χ_4. Changing the paths conjugates the indicated representations by their transported bases. Quadratic twists of the displayed curves make exactly the same representations modulo the global sign.

Proof. The quotient of an elliptic curve by $[-1]$ is a four-pointed projective line, with the identity among the four branch points. Ordering the other three points gives the Legendre parameter and is a torsor under S_3. Conversely the double cover of this four-pointed line gives the elliptic curve; its two possible covering automorphisms are exactly ± 1. These two constructions are inverse on families and their isomorphisms after rigidification, proving the displayed stack equivalence over $\mathbb{Q}$.

The projective Tate local system consequently has geometric monodromy $\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$. At the elliptic points its inertia generators are the matrices R, S of LM2, satisfying $R^3 = S^2 = -I$. Transport the Galois action at either elliptic point to the cusp by a fixed path. The resulting path cocycle is a conjugator A, respectively β, with $u_0(R) = AR^{\chi_{-3}}A^{-1}$ and $u_0(S) = \epsilon_2 \beta S \beta^{-1}$. These are precisely the conjugators in C.d4: their prescribed reductions modulo 2 remove the finite inertia ambiguity. More explicitly, the centralizer of either torsion generator in the free profinite product $C_2 \amalg C_3$ is its cyclic factor, whose intersection with the level-two free subgroup is trivial. Hence two such normalized path conjugators coincide. Transport of the Tate action now gives exactly $A^{-1}V$ and $\beta^{-1}V$, proving the formulas.

At $\lambda = 1/2$, the change $X = 4x - 2$, $Y = 8y$ gives $Y^2 = X^3 - 4X$, the quadratic twist by 2 of E_4. At $\lambda^2 - \lambda + 1 = 0$, shifting x by $(1 + \lambda)/3$ gives $y^2 = x^3 + (2\lambda - 1)/9$; this constant is the cube of $-(2\lambda - 1)/3$, so it is a quadratic twist of E_3 over $\mathbb{Q}(\sqrt{-3})$. The descent

to $\mathbb{Q}$ is fixed by its action on the three nonzero two-torsion points: for $j = 0$ the splitting field $\mathbb{Q}(\sqrt{-3})$ forces the cubic constant to be a rational cube up to a quadratic twist, and for $j = 1728$ rational two-torsion forces the coefficient to be a negative square. Thus the descended projective representations are those in (D.13). This is a typed arithmetic comparison, not an inference from the shapes of the two centralizer rings. The CM reciprocity used next is [LR22, Corollary 3.4], or equivalently [Mil07, Theorem 3.10], both opened. □

D.16. Exact joint CM image: one additional quadratic relation [Proved]

Let $K_3 = \mathbb{Q}(\sqrt{-3})$, $K_4 = \mathbb{Q}(i)$ and $L = K_3K_4$. In the product of the two normalizer groups of D.15 with common determinant and the modulo-two conditions of LM2 C.d4, the joint Galois image has index two. Its exact additional condition is equality of the two restrictions to

$$M = \mathbb{Q}(i, \sqrt[4]{3}); \qquad K_3^{\mathrm{ab}} \cap K_4^{\mathrm{ab}} = \mathbb{Q}^{\mathrm{ab}} M. \tag{D.14}$$

Thus the two-tori lead has an open image, but the Galois image does not fill the proposed full pair of equal norms. The missing coset is a single C_2; the existence of an element of $\widehat{\mathrm{GT}}_0$ realizing that coset is a separate unresolved realization question.

Proof. Both quadratic rings have class number one, as follows already from their Euclidean norms. CM reciprocity identifies the Weber-function torsion fields with the ray class fields; taking all levels in [LR22, Corollary 3.4] gives K_j^{ab}. The field defined by the Tate representation modulo ± 1 over K_j contains these Weber values (x^3 for E_3, x^2 for E_4), and is contained in K_j^{ab} since the Tate action over K_j is linear over its commutative CM order. It is therefore exactly K_j^{ab}. An element outside G_{K_j} cannot act as $\pm I$, because it conjugates a nonreal CM endomorphism to its distinct complex conjugate. Hence the field of the projective representation over $\mathbb{Q}$ is also exactly K_j^{ab}.

More precisely the image fills the specified individual factor. Reciprocity identifies $\mathrm{Gal}(K_j^{\mathrm{ab}}/K_j)$ with $\widehat{\mathcal{O}}_{K_j}^{\times}/\mathcal{O}_{K_j}^{\times}$. Reduction modulo 2 maps $\mu_6/\{\pm 1\}$ isomorphically to $(\mathcal{O}_{K_3}/2)^{\times}$, and $\mu_4/\{\pm 1\}$ isomorphically to $(\mathcal{O}_{K_4}/2)^{\times}$. Thus removing that finite root of unity gives an isomorphism with the units congruent to one modulo 2, modulo the single global sign. The curves in (D.13) have their full two-torsion over the stated fields, so their projective representations give exactly this isomorphism. The determinant is restriction to the cyclotomic subfield by the Weil pairing. Outside G_L, the two conjugation signs are already forced by that determinant and are all realized by $G_{\mathbb{Q}}$.

It remains to compute the intersection; this step determines the joint image, not merely its openness. Put $E = K_3^{\mathrm{ab}} \cap K_4^{\mathrm{ab}}$. It contains $\mathbb{Q}^{\mathrm{ab}}$. In $G = \mathrm{Gal}(E/\mathbb{Q})$ the two distinct index-two subgroups fixing K_3, K_4 are abelian. Their intersection is central, and together they generate G. Choosing representatives x, y of their two nontrivial cosets, the closed commutator group is generated by the central element $[x, y]$. Since x^2 and y^2 are central, this element has order at most two. Therefore $[E : \mathbb{Q}^{\mathrm{ab}}] \leq 2$.

The splitting field M of $X^4 - 3$ has dihedral group of order eight: its automorphisms are (a, b), $0 \leq a < 4$, $0 \leq b < 2$, acting by $\sqrt[4]{3} \mapsto i^a \sqrt[4]{3}$, $i \mapsto (-1)^b i$, with product $(a, b)(c, d) = (a + (-1)^b c, b + d)$. The subgroups fixing K_4 and K_3 are respectively $b = 0$ and $a + b = 0 \pmod 2$, both abelian. Thus $M \subset E$, and M is nonabelian over $\mathbb{Q}$. This realizes the bound and proves (D.14). Galois theory now identifies the joint image with the fibre product over this exact intersection. The common-determinant target is the fibre product over $\mathbb{Q}^{\mathrm{ab}}$, so its index is two. The program checks the dihedral group and this finite index explicitly. □

D.17. The local indices at two and three [Proved]

For $\mathcal{O}_L = \mathbb{Z}[i, z]$, $z^2 - z + 1 = 0$, the paired local norm map to the equal-norm units of

$\mathcal{O}_{K_3} \otimes \mathbb{Z}_p$ and $\mathcal{O}_{K_4} \otimes \mathbb{Z}_p$ has index two at each $p = 2, 3$, and is onto at every other prime. Its two local obstruction classes are identified by the global root-of-unity pairs, leaving the single global relation of D.16. Exact finite enumeration gives

p^k	2	4	8	16	32	3	9	27
equal-norm pairs	6	48	384	3072	24576	24	648	17496
norm image	6	24	192	1536	12288	12	324	8748.

(D.15)

Proof. At either bad prime one quadratic subfield is unramified and the other ramified. Given a compatible pair (a, b), first lift its coordinate in the unramified subfield through the ramified quadratic norm. It is a norm because the quadratic Hilbert-symbol obstruction equals the Hilbert symbol of its rational norm, and that rational norm is already a norm from the ramified quadratic subfield. Its valuation is zero, so a lift is a unit. The remaining discrepancy has norm one in the ramified subfield. Hilbert90 writes it as $c/\bar{c}$. It can be corrected by a unit from the biquadratic field exactly when the valuation of c in that ramified subfield is even: norms from its unramified quadratic extension have even valuation, and multiplying c by a rational scalar changes its valuation by an even integer. This gives precisely one C_2 obstruction. The unit norm in an unramified extension is onto, so no further obstruction remains.

The pairs $(-1, 1)$ and $(1, i)$ represent the nonzero class at each prime. The identity $N_{L/K_3}(z + i\bar{z}) = -1$, $N_{L/K_4}(z + i\bar{z}) = -i$ shows that their sum is zero. At other primes the integral norm torus has connected smooth kernel; norm surjectivity on residue fields and on successive principal-unit quotients proves surjectivity. The code enumerates the rings directly: for $a+bz+i(c+dz)$ its two norms are $(a+bz)^2+(c+dz)^2$ and $(a+ci)^2+(a+ci)(b+di)+(b+di)^2$. It tests invertibility by the rational norm and records the entire image size. The signs above are also tested at every displayed level. The global roots of unity contribute the same nonzero class at both bad primes, hence remove only the diagonal C_2. This agrees with, and does not replace, the field-intersection proof of D.16. □

D.18. The genuine cusp congruence action retains the full rank-one Kummer coordinate [Known]

In the positive tangential normalization $\lambda = 0$, tangent1, take $q^{1/n}$ with positive real leading coefficient $256^{-1/n}$ and $\zeta_n = \exp(2\pi i/n)$. Use the resulting Tate basis, matched to the Legendre cycles over $[0, \lambda]$ and $[\lambda, 1]$, with positive local monodromy $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. The genuine normalized congruence representation is

$$V_g = \begin{pmatrix} \chi(g) & -8\kappa_2(g) \\ 0 & 1 \end{pmatrix}. \tag{D.16}$$

Compatible roots may be chosen so that $\kappa_{-1/2} = (\chi - 1)/2 - \kappa_2$. Consequently the continuous inverse comparison on this image is

$$(\chi, V_{12}) \longmapsto \left(\chi, \frac{\chi - 1}{2} + \frac{V_{12}}{8}\right). \tag{D.17}$$

Multiplication by eight is a homeomorphism from $\widehat{\mathbb{Z}}$ onto $8\widehat{\mathbb{Z}}$, so no part of this Kummer coordinate is lost in the genuine normalized congruence action. In the basis of the manuscript with local monodromy $\left(\begin{smallmatrix} 1 & -2 \\ 0 & 1 \end{smallmatrix}\right)$, conjugation by $\mathrm{diag}(-1, 1)$ reverses the off-diagonal sign. On *outer* congruence actions this translation is forgotten: the comparison retains only χ modulo squares.

Proof. The Legendre invariant is $j(\lambda) = 256(1-\lambda+\lambda^2)^3/(\lambda^2(1-\lambda)^2)$. Inverting $j(q) = q^{-1} + 744 + \cdots$ gives $q = \lambda^2/256\,(1 + \lambda\mathbb{Q}[[\lambda]])$. Tate uniformization gives the exact torsion sequence with basis $\zeta_n, q^{1/n}$ and matrix $\left(\begin{smallmatrix}\chi & \kappa_q \\ 0 & 1\end{smallmatrix}\right)$; see [Jos, §11, Theorem 11.17 and the preceding uniformization]. For the formal field used here, the comparison follows by base change of the universal Tate equation and its torsion representatives over $\mathbb{Z}((q))$: the convergent uniformization identities are identities of Laurent series with integral coefficients, and remain identities after $q = q(\lambda)$. Thus the torsion representatives are $\zeta_n^a q^{b/n}$ over $\overline{\mathbb{Q}((\lambda))}$. The tangential section fixes compatible $\lambda^{1/n}$ and the unique roots with constant coefficient one of the last power-series unit. Its only remaining Kummer action on $q^{1/n}$ is therefore that on $256^{-1/n}$, namely $-8\kappa_2$. The Legendre curve is a quadratic twist of the split Tate curve at this cusp; that twist multiplies the entire matrix by a single sign and disappears in the projective congruence local system. Normalizing its lower-right entry to one yields (D.16). To fix the residual shear, let $K(m) = \int_0^{\pi/2}(1 - m\sin^2\theta)^{-1/2}d\theta$. The branch-cut periods have ratio $\tau = iK(1-\lambda)/K(\lambda)$, purely imaginary and positive for $0 < \lambda < 1$. Hence $q = \exp(2\pi i\tau)$ and its chosen roots are positive real. A shear would replace $q^{1/n}$ by $\zeta_n^b q^{1/n}$ and change the off-diagonal entry by $b(1-\chi)$; positivity for every n forces $b = 0$. This proves that the Tate basis is the specified global Legendre branch-cut basis, rather than merely a basis with the same local monodromy.

Choose $(-1)^{1/n} = \zeta_{2n}$. Its Kummer cocycle is $(\chi - 1)/2$, which proves (D.17) by multiplication of root systems. The stated change of basis explicitly handles the orientation convention. Finally conjugation by V_g on the special linear congruence quotient has outer class determined by its determinant modulo squares, as established in Part A; its unipotent upper-triangular factor is inner. Thus the often proposed claim that the rank-one translation is visible only metabelianly is false for genuine normalized actions, while the precise outer quotient does discard it. □

D.19. The residual GT and full profinite realization questions [Open]

D.1–D.4 settle item3, including general openness, the quadratic failure of surjectivity, joint independence, and the local inertia images. D.15–D.18 settle the stated Galois comparison of the two CM coordinates and the rank-one congruence coordinate. The unresolved parts of D(e) and D-after-LM2(1) concern the full profinite GT beta/gamma presentation and actual realization outside Galois: whether the extra leading metabelian directions of D.11 lift from $\widehat{\mathrm{GT}}_0$ or the full profinite loop-preserving group, and whether $\widehat{\mathrm{GT}}_0$ supplies the missing index-two CM coset of D.16. These assertions remain on the coverage ledger; none follows from solving only matrix or Lie equations.

D.39 extends a new representative through the entire all-prime mixed Fermat homology tower, jointly with the retained degree-eight quotient. Its common quotient stabilizes at exponent eight. The exact two-adic system has cokernel killed by sixteen, and the half-average gives every odd-prime component. The resulting multiplier-one automorphism retains the normalized G6 and T3 actions, replaces the prior J3 and J12 actions by the specified new components, and retains its checked J48 seed at every later level. Both compact CM matrices and the fixed-fibre pair survive. The exact source kernel is the intersection of the closed L-commutator subgroup with D9K; the nonzero degree-sixteen word remains in it. The projection to the earlier finite joint group is strict, detected by x^{144}. D.45 proves that this recorded action has no full profinite lift. D.46–D.47 subsequently construct a replacement on their exact relative sources; its full profinite realization remains unresolved.

D.40 extends that action through the actual image in every finite perfect quotient of the marked source. Their exact common quotient with the retained tower is C_6, and the normalized conjugators pair with the identity on all perfect coordinates. A further actual S_7 quotient gives

the fibre product over the common sign C_2. The exact remaining full-source kernel maps onto A_{10} under the computed S_{10} quotient. These maps retain both compact-cover and fixed-fibre CM coordinates; D.45 excludes a full profinite lift of this recorded action.

D.41 then extends the action through all actual symmetric quotients simultaneously. Their even image is the full product of the alternating factors indexed by distinct marked kernels, and the common quotient with D.40 is exactly the sign C_2. Both projection kernels are computed. The remaining full-source kernel maps onto $\mathrm{PSL}_2(\mathbb{F}_7)$ under the specified $\mathrm{PGL}_2(\mathbb{F}_7)$ map and kills the canonical perfect mod-seven coordinate. This retains the actual CM data; D.45 excludes a full profinite lift of the recorded action.

D.42 extends this actual source action through the full relative pronilpotent kernel of the specified projective-linear quotient. Its maximal common marked quotient has base order 54 and kernel $\mathbb{Z}_3^{10}$. The nonzero common conjugator is retained by an actual coprime centralizer lift, and every projection kernel is computed. The remaining source kernel maps onto $\mathrm{PSL}_2(\mathbb{F}_{11})$ under the specified $\mathrm{PGL}_2(\mathbb{F}_{11})$ map. D.45 supplies a finite sphere quotient that excludes every full profinite lift of this recorded action.

D.43 extends the retained action simultaneously through all relative pronilpotent completions whose marked finite base has nonabelian simple even subgroup. The exact common quotient with the retained tower is $C_{54} \times_{C_6} D_{\mathscr{J}}$, with its actual source-kernel formula. One compatible centralizer lift supplies the conjugator over the whole family. The exact remaining source kernel maps onto $A_7 \times A_7$ under an explicit quotient whose involution exchanges the two simple factors. D.45 excludes every full profinite lift of this recorded action.

D.44 extends the same quotient action to the complete family of anabelian even bases with their relative pronilpotent kernels. The common quotient and both source kernels are exact, and an actual $A_5^4 \rtimes S_4$ marking detects residual kernel A_5^4. D.45 then excludes a genuine full profinite lift of the entire recorded normalized D.39 action and every D.40–D.44 extension retaining it. Its sphere projection through B/D_9F has a nonzero degree-nine obstruction after every lower centralizer and multiplier freedom is included. The elliptic projection has an explicit corrected lift to $B/D_{10}B'$. D.46–D.47 subsequently supply a replacement realizing the prescribed CM pair on their exact relative sources. The full source image questions in D.13 and D.19 remain to be determined.

D.46 constructs a replacement on the entire relative pronilpotent completion of the actual rank-97 subgroup N. Its full relative pro-two component is an actual arithmetic commutator action with identity sphere projection and compact elliptic scalar η_*; every odd relative-prime component is the identity. The two actions join through explicit coprime centralizer lifts. The resulting G_2, G_3 actions determine the prescribed pair $T_4 = I, W_3 = [i]$, and the reconstructed parameter has an actual $\widehat{F}'$ preimage. Thus the replacement solves the entire rank-four relative pro-two lifting step. Its exact remaining full-source kernel has image A_7 under the displayed S_7 quotient. Its full profinite realization remains in the lifting problem.

D.47 extends this replacement coherently over every Fermat level. For $M = \widehat{N} \cap \widehat{F}'$, it proves both inclusions identifying the closed relative pro-prime source with the inverse system over the actual L_m. The common marked quotient is $\mathcal{H}_\infty$, and the simultaneous kernel is $\prod_p M^{(p)}$. The source kernel lies in $M' \subset \widehat{F}''$, so the action descends to the full metabelian quotient, with every profinite deck coordinate retained. Its coefficient-two action is the actual arithmetic h action; every odd coefficient action is identity. The actual marked cubic-four and quartic-three cover maps give $(-I, I)$, outside the proved Galois joint graph. The Heisenberg quotient and full Alexander-module argument also proves this in the outer action space. All comparison kernels to D.46 are computed. The remaining full-source kernel still has exact A_7 image in the displayed S_7 quotient. The full GT0 and full loop-preserving $\widehat{F}$ lifting questions remain unresolved.

D.48 extends the entire D.47 action over the central double cover $\mathcal{E} = \widehat{B}/[\widehat{N}, \widehat{B}]$, jointly with every Fermat level. The new subgroup M^+ has index two in M; its relative pro-two source is unchanged. The comparison kernel is the product of the explicitly identified odd-prime kernels. The actual h action fixes the enlarged base, and the coprime centralizer corrections retain the complete D.47 action and parameter. On the new finite mixed quotient $\mathcal{J}_3$, the exact old source-kernel image is C_3^{96}; the new source kernel has trivial image there. Its full common quotient with D.47 is $\mathcal{E} \times_Q G_3$, and the identity action on $\mathcal{J}_3$ agrees with the retained action on that whole quotient. The remaining full-source kernel has exact A_7 image; full profinite realization remains unresolved.

D.49 extends this same action over the maximal marked quotient fixed pointwise by the actual arithmetic h, and over every relative-prime source above that base. Its exact comparison with D.48 retains both kernel layers, including an actual A_5 quotient of the base residual. The full congruence quotient $\mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}$, with one global sign, factors through the identity-three component; the new action is therefore pointwise identity on this entire quotient. Its remaining source kernel has exact image $[V_{3,2}, A]$ in the specified mixed homology quotient over G_3. It contains an actual cyclic moving subspace of dimension $64 \cdot 3^{97}$; the dimension of the full arithmetic span A is retained as unevaluated. The earlier central iteration stabilizes by the exact integral equality $[L, \widehat{B}] = L$. The remaining mixed source and full profinite realization questions stay active.

D.50 extends this same retained action through the entire two-adic homology tower over the actual marked G_3. The exact residual image is the integral moving summand U^-, and every common quotient is computed from the original source kernels. A regular-edge Fitting projection and its explicit crossed homomorphism repair the initial peripheral endomorphism to an automorphism; the cyclic norm correction fixes the original cusp. Compatible conjugators and invertible reduction modulo two give all higher lifts and their continuous inverse limit. The full parameter, CM pair, metabelian and congruence actions are retained. The remaining source kernel has a nonzero image containing $W_1 \wedge V$ in the next actual class-two exponent-four quotient. Extending across that remaining kernel and attaining the full profinite source remain active steps.

D.51 retains this entire chosen homology action and both full conjugators and extends them through the moving part of the next actual class-two exponent-four quotient. The exact old source image is $Z = W \wedge V$; its invariant and moving dimensions are respectively $m^2(q-1)/2$ and $w\rho/2$. The actual common quotient is $\mathcal{T}_2/Z$. Characteristic descent, exterior preservation of Z^A, and Frattini generation prove the new action on the joint source. Its projection adds the whole moving central summand and retains every previous parameter, CM, metabelian and congruence comparison. The remaining exact image is $Z^A \neq 0$. In the separate order-1080 verification, the complete corrected cusp word has 4,159,444 letters. That example's fixed homology matrix has intersection-pairing defect of rank 72, excluding its extension to the unquotiented class-two peripheral source. The arithmetic invariant obstruction and the full profinite realization remain to be evaluated.

D.52 solves the entire invariant boundary equation in the separate marked order-1080 source. Its 120-dimensional space consists of allowable conjugator pairs. A retained weight-60 parameter solves all thirteen actual quadratic equations; the complete 6,352,056-letter corrected word, both torsion relations and generation are verified. The new homology action has deviation rank 96 over $\mathbb{F}_2$ and preserves the mod-two ribbon form. It replaces the excluded finite representative while retaining the actual invariant quotient. It also has a complete two-adic homology continuation with the entire new class-two action retained, exact common quotient and both projection kernels. Its next exponent-eight residual is twice the commutator module, witnessed by a nonzero 272-letter commutator square. The arithmetic G_3 correction and the full profinite realization remain to be evaluated through their own actual source maps.

D.53 computes the full actual moving peripheral-pair space, its exact homology image and kernel, and a positive arithmetic dimension bound. Its integral lifts give actual doubled automorphisms of the complete D.50 source. D.54 lifts every such difference to the full joint class-two source, through an explicit compatible square module. The fixed arithmetic cusp defect is preserved literally, and the actual affine quadratic map descends through every same-action ambiguity. Its arithmetic zero locus and full profinite realization remain to be evaluated. In the separate order-1080 source, the full finite homology zero locus has exactly864 elements; all262,144 parameters, the original equations, generation and the complete long cusp words have been checked.

D.55 gives the exact full-group norm from the arithmetic cusp obstruction to the compact orientation-bivector defect. Its local image lies in the norm kernel, and its residual quotient has explicit Tate-norm and compact-projection maps. Every point of the reduced boundary zero set has invertible homology, so generation follows there. The actual ordinary relative-H_2 orientation comparison is an isomorphism. The arithmetic zero set and residual norm kernel remain unevaluated, with all maps retained for the next attack.

D.56 evaluates an actual irreducible C_3^4 quotient of the arithmetic normal span and the unique parameter $(2, 1, 2, 1)$. The marked quotient has order3888 and actual action $\mathrm{Int}(t^6)$. Its exact A_4 comparison kernel gives surjective integral homology and class-two boundary maps. The smaller arithmetic correction space has dimension40, with invariant central target dimension2560. Its actual quadratic zero set and full profinite realization remain unevaluated; every source and comparison kernel is retained.

D.57 now solves the complete ten-coordinate boundary equation on the actual order-3888 arithmetic quotient, for both integral rank-nine branches and all sixteen framed cases. The literal genus-three cover and degree-eight elliptic map, exhaustive integral conjugator search, separate plus-central alignment and augmentation correction give the asserted automorphism of $\widehat{B}/(R_h \cap M_* \cap L_{*,2})$. Its entire W_h action and prescribed normalized parameter are retained. The exact remaining quotient kernels carry the full G_3 invariant and full profinite realization questions onward.

D.58 retains the entire D.57 action and its conjugators while restoring the full integral arithmetic homology. The actual projective repair stays in $\mathcal{L} = \ker(U^- \to U_*^-)$. It then extends through the remaining moving class-two module, with exact residual kernel $\mathcal{K}_0 = \ker(Z \to Z_*)^A$. All 2560 solved invariant coordinates are retained; the remaining dimension is $m^2(|A| - 1)/2 - 2560$. The next actual invariant boundary and the full profinite realization remain to be evaluated across this specified source map.

Concrete residual instances. The first extra metabelian special vector occurs at weight five, where the free special source has dimension three and its invariant part dimension one; exact preimages are in `d_metabelian.json`. The first extra invariant metabelian vector occurs at weight nine. D.12 supplies their pro-ℓ realizations and identifies the nonnilpotent-quotient obstruction to transferring that construction. For the CM problem, D.17 supplies an explicit finite matrix coset, represented locally by $(-1, 1)$ at one bad prime and the identity at the other, whose two obstruction classes disagree. D.16 proves that this coset has no Galois lift. Its possible lift to an actual full profinite $\widehat{\mathrm{GT}}_0$ element must solve the conjugator and generation equations before congruence reduction; the matrix reconstruction of LM2 C.d4 does not do so. The source search reached the original Ihara Part II through the RIMS1224 PostScript file, and the exact modern profinite lift criterion through Dolgushev–Guynee. Part I's RIMS1177 catalogue entry has no download; the PDF, PostScript and DVI locations returned 404. D.20–D.22 record the additional results actually obtained. D.25–D.33 carry out finite and infinite quotient tests for the missing CM coset: the twenty-three initially selected nonidentity shadows and two later central representatives are excluded, while successive explicitly distinguished replacements

supply actual compatible actions on the stated quotient systems. D.31 constructs an actual action on $\mathcal{S}_2^{(7)}$, combined with all odd relative pro-prime kernels and D.28's mixed quotient through its exact common quotient. D.32 proves that its recorded T_2 action is obstructed at the next degree, including every same-action conjugator and multiplier choice. D.33 then changes the lower action explicitly and supplies a compatible degree-eight replacement with a class-four kernel, including the whole class-two tower and the same mixed-prime quotient. The remaining relative-source kernel is exactly $\gamma_3 N^{(2)} \cap D_9(K^{(2)})$, and an actual nonzero triple commutator is exhibited in it. The entire CM pair has not been excluded. D.34–D.35 give exact structural maps for the relative problem: the marked elliptic cover, the full outer deck centralizer, actual odd-deck boundary lifts, and a genuine arithmetic surjection onto the single compact CM torus. The inverse image of the recorded D.33 action under (D.35.5a) retains the full finite coordinates modulo the specified cusp-inner subgroup. D.45 proves this particular fibre empty, using the full torsion and quotient-preservation maps and an actual cusp-inner transport. D.36 gives the actual quartic-cover and common cyclic-twelve maps, the arithmetic joint degree-six fibre, and the retained gauge-invariant sign discrepancy. The opened cover-to-fixed-fibre source equations retain their stated Galois scope. D.37 supplies a further exact mixed Fermat-twelve shadow, retaining D.33 on G_6 and evaluating both actual cover matrices as $(-I, I)$. Its universal relative-prime quotient maps and mixed kernel are explicit. D.38 then proves compatibility for a new Fermat action with the retained degree-eight component, through their exact maximal common quotient. The simultaneous finite group, both projection kernels and a nonzero word in the remaining full-source kernel are computed. The unresolved steps are the complete lift tests for the two indicated metabelian vectors and for the missing CM coset. □

The original profinite beta/gamma relations and their graded consequence

D.20. Ihara's exact cyclic-cover compatibility and gamma multiplication [Known]

Write a full profinite GT element as (λ, f) acting by $x \mapsto x^\lambda$, $y \mapsto f^{-1}y^\lambda f$. For $n \geq 1$ let $H_n = \ker(\widehat{F}_2 \to \mathbb{Z}/n)$, where $x \mapsto 1$, $y \mapsto 0$, and define the continuous epimorphism

$$\phi_n : H_n \longrightarrow \widehat{F}_2, \qquad x^n \mapsto x, \quad y \mapsto y, \quad x^{-j}yx^j \mapsto 1 \quad (0 < j < n).$$

For every $g \in G_{\mathbb{Q}}$ one has

$$\phi_n(f_g) = y^{-\kappa_n(g)} f_g, \qquad \phi_n \circ g|_{H_n} = g \circ \phi_n. \tag{D.18}$$

Here the roots of the positive integer n use the positive real branches of D.2. These are [Iha00, Theorem 1 and Proposition 1].

For every $\sigma = (\lambda, f) \in \widehat{\mathrm{GT}}$, put $c_n(f) = \exp_y(\phi_n(f)) \in \widehat{\mathbb{Z}}$. The continuous cover maps above satisfy

$$\phi_n(f) = y^{c_n(f)} f, \qquad \phi_n \sigma|_{H_n} = \sigma \phi_n, \qquad c_{mn}(f) = c_m(f) + c_n(f). \tag{D20.1}$$

The full profinite assertion follows from [Enr07, Theorem8.1 and Remark8.6]; the exact comparison of its generators with ours is proved below. On the Galois subgroup $c_n(f_g) = -\kappa_n(g)$.

The full profinite beta function of $\sigma \in \widehat{\mathrm{GT}}$ has Ihara's factorization

$$B_\sigma(s,t) = \frac{\Gamma_\sigma(s)\Gamma_\sigma(t)}{\Gamma_\sigma(s+t)}, \qquad s, t \in \mathbb{Q}/\mathbb{Z}. \tag{D.19}$$

The coefficient ring and completed group-ring interpretation are those of the hyperadelic gamma function; equality precedes every prime-power specialization. For Galois elements Anderson's multiplication formula, in Ihara's corrected convention, is

$$\frac{\prod_{nc=0}\Gamma_g(s+c)/\Gamma_g(c)}{\Gamma_g(ns)} = 1 \otimes \exp(-2\pi i\, ns\kappa_n(g)). \tag{D.20}$$

Ihara Part II, Introduction and §2, states (D.19) and (D.20), attributes (D.20) to Anderson's Corollary 8.6.3, and gives the following exact relations among its intermediate subgroups: $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}_K \subset \widehat{\mathrm{GT}}_{A'}$ and $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}_A \subset \widehat{\mathrm{GT}}_{A'}$. Here K means all the commuting squares in (D.18), A means (D.20), and A' means the two distribution identities in the proof below. Combining these proved inclusions with (D20.1) gives the stronger equality

$$\widehat{\mathrm{GT}}_K = \widehat{\mathrm{GT}}_{A'} = \widehat{\mathrm{GT}}. \tag{D20.2}$$

Proof. Ihara's earlier original notes [Iha97, Sections2–4, pp.46–52] give the profinite beta definition, the associativity statement yielding gamma factorization, and a determinant proof sketch for multiplier one. We read all ten pages. The general normalization here retains the opened Part II and Anderson comparisons.

The original RIMS1224 version was downloaded, decompressed, converted from PostScript, and read. Its proof of Theorem 1 identifies H_n with the fundamental group of $\mathbb{P}^1 - \{0, \mu_n, \infty\}$, and ϕ_n with the map induced by filling the punctures different from 1, after the change $t \mapsto t^n$. Coefficientwise Galois action on Puiseux series commutes with this change, proving the second equality of (D.18). Applying the two sides to y shows that $\phi_n(f_g)f_g^{-1}$ centralizes $y^{\chi(g)}$, hence is a power of y. Its exponent is the logarithmic Kummer quasi-cocycle $\Psi_n^{(0)}(g)$, characterized in the original Theorem 1 by $(n^{-r})^{g-1} = \exp(2\pi i r\Psi_n^{(0)}(g))$ for $r \in \mathbb{Q}$. Thus $\Psi_n^{(0)} = -\kappa_n$, proving the first equality with its sign.

We now give the full-GT reduction on these same profinite groups. Enriquez uses the Schreier generators $x^n, x^j y x^{-j}$ and a filling map δ_n killing the generators with $0 < j < n$. The identity

$$x^{-j}yx^j = x^{-n}(x^{n-j}yx^{-(n-j)})x^n$$

shows that its continuous extension is exactly ϕ_n. The cited theorem therefore gives the first equality in (D20.1), with precisely our c_n. Its proof uses the actual cyclotomic configuration covers, the commuting strand-doubling maps of Lemmas8.2–8.4, and the pentagon. Lemma8.5 removes the central factors; erasing strand zero gives $\delta_n(f) = y^{c_n(f)}f$.

Here is the resulting intertwining map explicitly. Since λ is a unit modulo n, σ preserves H_n. On x^n, both maps $H_n \to \widehat{F}_2$ give x^λ; on y, both give $f^{-1}y^\lambda f$. For $z_j = x^{-j}yx^j$, write $j\lambda = a + nb$ with $1 \le a < n$ and $b \in \widehat{\mathbb{Z}}$. Then

$$\sigma(z_j) = k^{-1}x^{-nb}z_a^\lambda x^{nb}k, \qquad k = x^{-j\lambda}fx^{j\lambda} \in H_n.$$

Applying ϕ_n gives 1, also equal to $\sigma\phi_n(z_j)$. These free profinite generators prove the second equality.

Finally $\phi_n(H_{mn}) \subset H_m$ and

$$\phi_m\phi_n|_{H_{mn}} = \phi_{mn} : H_{mn} \longrightarrow \widehat{F}_2.$$

Indeed x^{mn} maps to x, while each $x^{-j}yx^j$ with $0 < j < mn$ is killed at the first step when $n \nmid j$, and at the second when $j = nk$, $0 < k < m$. Applying this identity to f, using $f \in \widehat{F}_2' \subset H_{mn}$, gives $y^{c_{mn}(f)}f = y^{c_n(f)+c_m(f)}f$. The embedded procyclic subgroup generated by y has exponent

group $\widehat{\mathbb{Z}}$, proving additivity. Ihara's Proposition2 now applies to every full-GT element, and the inclusions already displayed give (D20.2).

For completeness the identities defining A' are

$$\Psi^{(0)}_{mn} = \Psi^{(0)}_m + \Psi^{(0)}_n, \qquad \Psi^{(in)}_{mn} = \Psi^{(i)}_m \quad (i \neq 0 \bmod m).$$

The functions $\Psi^{(i)}_n$ are defined by the action of f on $(1-\zeta_n^{-i}t^{1/n})^r$; Proposition 2 proves these identities directly from (D.18). Proposition 3 proves that they are exactly the logarithmic derivative of (D.20), using Part I's Theorem 2, whose formula is reproduced there. The original Introduction explicitly says that the proof of (D.19) uses the five-cycle relation. These statements specify which equation is used and which group it constrains.

Anderson's original Corollary 8.6.3, p.103, was also visually checked. With $\mathcal{W} = \prod_p W(\overline{\mathbb{F}}_p)$ from his §7.1, it states, in $(\overline{\mathbb{Q}} \otimes \mathcal{W})^\times$,

$$\prod_{nb=a} \Gamma_g(b) = ((n^{\langle -a\rangle})^{g-1} \otimes 1)\Gamma_g(a) \prod_{nc=0} \Gamma_g(c).$$

The roots of n are positive real. Put $a = ns$ and divide by the two displayed products; their Kummer factor is $\exp(-2\pi i\, ns\kappa_n(g)) \otimes 1$. Ihara's coefficient convention places $\mathcal{W}$ first; the tensor flip $z \otimes w \mapsto w \otimes z$ gives exactly (D.20), including its sign.

There is a precise obstruction to reversing the logarithmic derivative in the hyperadelic ring. For complementary nonzero idempotents $e, e' \in \widehat{\mathbb{Z}}$, $ee' = 0$, the nonconstant element $ex^{e'}$ has derivative zero. It is nonconstant already after projection to a coefficient prime supported on e and a finite deck quotient supported on e'. Thus the kernel of the derivation contains more than constants. Ihara's final Remark explicitly records that $\widehat{\mathrm{GT}}_K \subset \widehat{\mathrm{GT}}_A$ was not proved there. Formula (D20.1) establishes the full-GT cover and distribution identities; the gamma multiplication formula (D.20) retains its proved Galois scope. □

D.21. Galois and full GT have the same rational metabelian graded image through weight ten [Proved]

For every prime ℓ, the rational graded images of the actual groups $G_{\mathbb{Q}}$ and $\widehat{\mathrm{GT}}$ in the metabelian outer action, in weights 1 through 10 and after removing the cyclotomic multiplier, both have dimensions

$$0, 0, 1, 0, 1, 0, 1, 0, 1, 0. \tag{D.21}$$

In weight $n \geq 3$ odd the common line, expressed in the $p(X, Y)[x, y]$ coordinate of D.11, is

$$\mathbb{Q}_\ell \, \frac{(X+Y)^n - X^n - Y^n}{XY}. \tag{D.22}$$

The inclusion of this line in the $\widehat{\mathrm{GT}}_0$ upper-bound space is the explicit polynomial inclusion in D.11. The latter upper bound has an additional line in weight nine.

Proof. We give the graded comparison explicitly. Map the free profinite group to its pro-ℓ quotient, take logarithms in each rational nilpotent quotient, and then take the first nonzero homogeneous term of a filtered GT element. Expanding the two-cycle, three-cycle and five-cycle equations gives their infinitesimal equations for this term. The infinitesimal pure-braid generators are the initial classes of the logarithms of the pure-braid generators. Their relations are $[t_{ij}, t_{kl}] = 0$ for four distinct indices and $[t_{ij}, t_{ik} + t_{jk}] = 0$; the successive split fibrations forgetting a strand identify their free Lie kernels, so these relations give the associated graded pure-braid Lie algebra used in the expansion. Thus the displayed map sends the leading term into $\mathfrak{grt}_1(\mathbb{Q}_\ell)$.

Furusho's [Fur11, Corollary 0.3, Theorem 0.5, Proposition 5.1 and Lemma 5.2] gives the gamma constraint on its metabelian image. Here is the comparison with our coordinate, including the factor that matters. For a Lie element $h = h_x x + h_y y$, put $M(h) = (h_y y)^{\mathrm{ab}}$. Direct expansion gives

$$M([x,y]) = XY, \quad M([x,h]) = XM(h), \quad M([y,h]) = YM(h) \quad (h \in L').$$

It follows that $M(p(X,Y)[x,y]) = XYp(X,Y)$ and that M kills L''. Furusho's gamma constraint is $M(h) = \gamma(X) + \gamma(Y) - \gamma(X+Y)$. In homogeneous degree n it is a scalar multiple of $X^n + Y^n - (X+Y)^n$. Since multiplication by XY is injective, this bounds the image by the line (D.22). Equivalently the actual module action has leading multiplier $-XYp$, because $D([x,y]) = [x,[p[x,y],y]] = -XYp[x,y]$. This is the leading term of the full profinite beta multiplier in (D.19), rather than an unlabelled identification of the two series.

The program `d_gamma.py` checks $M(h) = XY\bar{h}$ on all 224 Lyndon basis words of degrees 2 through 10, and solves for each gamma polynomial in the exact invariant image basis of `d_metabelian.json`. In weight nine its coordinates in that basis are $(15/2, 9/2)$. The first basis vector $X^2Y^2(X+Y)^3$ is independent of the gamma line, verified by an exact rank-two calculation. Thus the comparison with the extra vector is computed in the same coordinate system as D.11.

The special equation forces $X+Y$ to divide p, as proved in D.11. At $Y = -X$ the numerator of (D.22) is $-(1+(-1)^n)X^n$, so every positive even-degree scalar must vanish in characteristic zero. Weight one vanishes directly. For each odd weight 3,5,7,9, E.3 supplies a nonzero Galois class for every ℓ, using [HM03, Theorem 8.4, §8.4, Lemma 8.5 and Remark 8.6]; at $\ell = 37$ D.10 also supplies direct finite-field certificates. Since $G_{\mathbb{Q}} \subset \widehat{\mathrm{GT}}$, these lower bounds equal the upper bounds just proved. This proves (D.21) for actual profinite source groups with rational graded target. It does not replace either source by its pro-unipotent automorphism group. □

D.22. The exact finite-quotient criterion for the remaining GT0 lifts [Known]

Let $N \triangleleft B_3$ have finite index and lie in PB_3, and let $\mathrm{GT}(N)$ be the finite set of pairs satisfying the two hexagon equations, the commutator condition on f, the unit condition on λ, and surjectivity onto B_3/N. Write $R_{K,N}$ for reduction from a finer such quotient. Then the image of the actual full profinite group is exactly

$$\mathrm{Im}\big(\widehat{\mathrm{GT}}_0 \longrightarrow \mathrm{GT}(N)\big) = \bigcap_{K \triangleleft B_3,\ [B_3:K]<\infty,\ K \subset N} R_{K,N}\big(\mathrm{GT}(K)\big). \tag{D.23}$$

This is [DG24, Theorem 5.2, Corollary 5.4 and Proposition 5.5]. It includes all finite quotients of the braid group, including the nonnilpotent ones. Consequently the finite Lie and CM solutions in D.11 and D.17 specify candidates and concrete finite tests, but their realization in $\widehat{\mathrm{GT}}_0$ requires the intersection in (D.23).

Proof. One inclusion is reduction of a genuine element. For the other, the fibres over a fixed member of the right side form an inverse system of nonempty finite sets indexed by all finer quotients. Any finite collection is dominated by their intersection, so the system has the finite intersection property. Compactness gives a compatible family. Theorem 5.2 identifies these compatible pairs with the profinite group of hexagon-preserving invertible automorphisms, and Proposition 5.5 identifies that group with Harbater–Schneps' $\widehat{\mathrm{GT}}_0$. Surjectivity in each finite quotient supplies the invertibility condition; it cannot be omitted. This proves the equality, including the exact hypothesis that the pro-ℓ lift of D.12 alone does not supply. □

D.23. The exact coefficient map from the profinite universal measure to every rational Kummer character [Proved]

Use the positive interval paths and the Ihara convention of D.20. Write $c = [x, y]$, $\Lambda = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$, and let s, t act on $\widehat{F}_2'/\widehat{F}_2'' = \Lambda c$ by conjugation by x, y. For $g \in G_{\mathbb{Q}}$ put $f_g \bmod \widehat{F}_2'' = u_g c$ and $A_\lambda(s) = (s^\lambda - 1)/(s - 1)$, where $\lambda = \chi(g)$. Then the universal measures of D.5 satisfy the exact identities

$$F_g = A_\lambda(s)A_\lambda(t)(1 + (s-1)(t-1)u_g), \qquad B_g = 1 + (s-1)(t-1)u_g. \tag{D.24}$$

For a positive integer n, reduce u_g by $t = 1$, $s^n = 1$, and write $u_{g,n} = \sum_{j=0}^{n-1} a_j(g)s^j \in \widehat{\mathbb{Z}}[\mathbb{Z}/n]$. The coefficient map is

$$\kappa_n(g) = a_0(g) - a_{n-1}(g). \tag{D.25}$$

Consequently the homomorphism of scan 157 is recovered from (χ, F) by

$$g \longmapsto \left(\chi(g), \frac{\chi(g) - 1}{2} - (a_0(g) - a_1(g))\right), \tag{D.26}$$

where the two coefficients in (D.26) use $n = 2$. For $p \neq \ell$, the extraction map in (D.25) on the $\mathbb{Z}_\ell$ coefficient component does not factor through the quotient with only pro-ℓ deck groups; the proof gives an explicit element of its kernel on which extraction is 1. For every odd ℓ, this failure already occurs on actual Galois elements: a pro-ℓ tame-inertia subgroup at 2 has trivial full cyclotomic character and trivial normalized pro-ℓ action on $\pi_1(\mathbb{P}^1 - \{0, 1, \infty\})$, but its $\kappa_{2,\ell}$ image is all of $\mathbb{Z}_\ell$.

Proof. All divisions in (D.24) are well-defined continuous operations on their stated images. The product $(s-1)(t-1)$ is a non-zero-divisor by Otsubo's Corollary 6.2, hence its individual factors are also non-zero-divisors. The finite geometric sums defining A_m for integers m extend by continuity to $\lambda \in \widehat{\mathbb{Z}}$. For a unit λ, $A_\lambda(s)A_{\lambda^{-1}}(s^\lambda) = 1$, by multiplication by $s - 1$ and cancellation, so A_λ is a unit. The inverse on a closed divisible image is continuous because the completed group ring is compact and Hausdorff.

In additive notation for the derived module, $f_g^{-1}y^\lambda f_g = (t^\lambda - 1)u_g c \cdot y^\lambda$. The commutator identities give

$$[x^\lambda, y^\lambda] = A_\lambda(s)A_\lambda(t)c, \qquad [x^\lambda, (t^\lambda - 1)u_g c] = (s^\lambda - 1)(t^\lambda - 1)u_g c.$$

This proves the first identity of (D.24); Otsubo's comparison (D.5) proves the second. To fix the path convention, the lift of $[x, y]$ along the positive interval has homology chain $(s-1)(1-t)\delta = -(1-s)(1-t)\delta$. Thus it is the negative of Otsubo's Pochhammer generator. The factor is the constant -1, so their Galois multipliers agree for every χ, not only on the cyclotomic kernel. The leading derivation convention in D.11 uses $D(y) = [f, y]$, so its f is the negative of the leading term of this Ihara conjugator; this accounts for the sign $-XYp$ in D.21.

Consider the Schreier generators $e_j = x^{-j}yx^j$ of H_n, besides x^n. In H_n^{ab}, $c = e_{n-1} - e_0$. Conjugation by x^a sends this to $e_{-a-1} - e_{-a}$, with indices modulo n; conjugation by y acts trivially on this abelianization. Therefore the coefficient of y in $\phi_n(u_g c)$ is $a_{n-1} - a_0$. By Ihara's exact equality (D.18), that coefficient is $-\kappa_n(g)$, proving (D.25). Formula (D.26) now follows from the chosen roots $(-1/2)^{1/m} = \zeta_{2m}/2^{1/m}$ and $\kappa_{-1} = (\chi - 1)/2$. The formulas give a continuous map from the actual metabelian action to the affine image of D.1, with no assertion that arbitrary coefficients satisfy Galois relations.

Here is the promised prime-to-ℓ kernel element. In $\mathbb{Z}_\ell[[\widehat{\mathbb{Z}}^2]]$, take the normalized Haar idempotent e_p of the $\mathbb{Z}_p$ factor of the first deck group. Its images at finite p^r-levels are $p^{-r}\sum_{j=0}^{p^r-1} s^j$; these are compatible since p is a unit in $\mathbb{Z}_\ell$. Projection to the pro-ℓ deck quotient

sends e_p to 1. At $t = 1, s^p = 1$ its image is $p^{-1}\sum_{j=0}^{p-1} s^j$. Hence $1 - e_p$ lies in the kernel of the pro-ℓ deck projection but its coefficient extraction is 1. This proves the exact failure of that factorization on the ambient module.

For the asserted arithmetic witness, take an odd ℓ and a pro-ℓ Sylow subgroup J of inertia at 2. Its tame quotient is $\mathbb{Z}_\ell(1)$. Every root of unity of odd order lies in the maximal unramified extension of $\mathbb{Q}_2$, and the action on the 2-power roots has pro-2 image. Therefore $\chi(J) = 1$. The extensions obtained by adjoining $2^{1/\ell^r}$ over the maximal unramified field are totally tamely ramified of degree ℓ^r; thus $\kappa_{2,\ell}(J) = \mathbb{Z}_\ell$. The pro-$\ell$ outer action of the thrice-punctured line is unramified at 2, as in E.3 and [HM03, §8.4]. It follows that the normalized Ihara action of any $g \in J$ is inner. An inner automorphism fixing x has conjugator in the centralizer $\langle x\rangle$. Comparing its action on y with $f_g^{-1} y f_g$, where $f_g \in F'_\ell$, shows that f_g differs from a power of x by a power of y. Abelianization forces both powers to vanish, so this normalized action is the identity. Hence its pure pro-ℓ universal measure is 1, while the full profinite coefficient extraction ranges over $\mathbb{Z}_\ell$. This proves the asserted failure on actual Galois elements, even when the whole cyclotomic character is retained.

The program `d_kummer_measure.py` independently computes abelian Fox derivatives and Schreier coefficients on 160 explicit free words at all 24 levels $1 \le n \le 24$, giving3840 exact checks of (D.25)'s coefficient identity. It also checks210 commutator-action identities for positive and negative integer exponents, which extend continuously as above. Its output does not substitute computation for the Galois theorem (D.18); that theorem supplies the arithmetic meaning of the checked coefficients. □

An additional full profinite lifting test

D.24. What unrestricted profinite lifting supplies, with an exact peripheral failure [Known]

Put $P = \widehat{F}_2/\widehat{F}_2''$. The canonical homomorphism

$$\mathrm{Aut}(\widehat{F}_2) \longrightarrow \mathrm{Aut}(P)$$

is surjective. Its kernel contains an automorphism which does not preserve the conjugacy class of the peripheral subgroup generator x, although it acts identically on the entire metabelian quotient. Thus the unrestricted lifting theorem supplies lifts in the displayed source; it supplies no peripheral conclusion for the chosen lift. The existence of peripheral lifts for the additional directions in D.13 remains a separate unsolved step of that calculation.

Proof. We use the profinite Gaschütz lemma in Bleher–Chinburg–Lubotzky, Lemma 2.3 and Corollary 2.4 [BCL22]. Applied to a quotient of a two-generated free profinite group, it lifts any generating pair of the quotient to a generating pair of the free profinite group. For an automorphism of the quotient, lift the images of x, y in this way. The resulting endomorphism of $\widehat{F}_2$ is surjective and hence invertible, by Hopficity of finitely generated profinite groups. This proves the asserted surjectivity and explains the imported theorem in our precise setting.

Here is a lift on which the peripheral condition fails. Let

$$q : \widehat{F}_2 \longrightarrow A_5, \qquad q(x) = (12345), \quad q(y) = (123),$$

and let $t = (12) \in S_5$. The map

$$(p, q) : \widehat{F}_2 \longrightarrow P \times A_5$$

is onto. Both projections are onto, and Goursat's argument identifies any proper subdirect product with a nontrivial common quotient of P and A_5. The latter group is simple and

perfect, whereas every quotient of P has trivial second derived subgroup. There is no such common quotient, proving the assertion for this actual map.

The continuous automorphism $\mathrm{id}_P \times \mathrm{Ad}(t)$ of $P \times A_5$ lifts by the same lemma to an automorphism α of $\widehat{F}_2$. It lies in the kernel of the displayed map on automorphism groups. The two elements $q(x)$ and $tq(x)t^{-1}$ belong to the two different conjugacy classes of five-cycles in A_5. Consequently $\alpha(x)$ cannot be conjugate to x in $\widehat{F}_2$. A loop-preserving lift here would have multiplier one, because its action on $P^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ is the identity. It would therefore have to send x to a conjugate of x, which this lift does not do.

The program `d_lifting.py` verifies the order sixty quotient, generation by the specified pair, simplicity by the normal closures of all five conjugacy classes, perfectness, and the two disjoint classes of twelve five-cycles. It also checks all 3600 multiplication identities for $\mathrm{Ad}(t)$. This example diagnoses the missing condition in the attempted unrestricted-lifting route. It makes no claim that a different lift of a given metabelian automorphism cannot preserve all three loops. □

The CM field and explicit nonarithmetical finite shadows

D.25. A finite CM entanglement field and nonarithmetical GT0 shadows [Proved]

Put $r_0 = (3/4)^{1/4}$ and $M' = \mathbb{Q}(i, r_0)$. There are exact equalities

$$\mathbb{Q}(E_3[4]) = M' = \mathbb{Q}(x(E_4[3])), \qquad M'\mathbb{Q}^{\mathrm{ab}} = M\mathbb{Q}^{\mathrm{ab}}, \tag{D.25.1}$$

where $E_3 : y^2 = x^3 + 1$, $E_4 : y^2 = x^3 - x$, and M is D.16's field. The projective $E_3[12]$ representation recovers the full $E_3[4]$ representation: choose its unique global sign fixing the rational point $(0, 1)$ of order three. Thus the extra relation of D.16 is already detected by T mod 12 and W mod 3, with this sign rule.

Here is a finite refinement of the matrix problem, including generation and the two hexagons. Write

$$B = \langle s, r : s^2 = r^3 = 1 \rangle, \quad Q = \mathrm{SL}_2(\mathbb{Z}/12)/\{\pm I\}, \quad N = \ker(B \to Q), \quad G_j = B/[N, N]N^j.$$

Use $s \mapsto \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and $r \mapsto \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ projectively. Then $|Q| = 576$, N is free of rank 97, and $|G_j| = 576j^{97}$ for $j = 2, 3, 6$. Among the 48 pairs of matrix conjugators with multiplier one and $V_{12} = I$,24 satisfy the CM relation and 24 violate it. Exact computation in G_2 and G_3 gives

condition	CM-compatible	extra CM coset
matrix conjugators	24	24
cusp relation in G_2	12	12
cusp relation in G_3	24	24
automorphisms of G_6	12	12.

(D.25.2)

Each surviving row has an explicit certificate whose induced matrix on N^{ab}/p has rank 97 for $p = 2, 3$. The twelve extra pairs yield twelve distinct nonarithmetical finite $\widehat{\mathrm{GT}}_0$ shadows for the pullback of $[N, N]N^6$ to B_3. One representative has

$$A_{12} = I, \qquad \beta_{12} = \begin{pmatrix} 9 & 8 \\ 4 & 9 \end{pmatrix}, \qquad V_{12} = I. \tag{D.25.3}$$

Here "nonarithmetical" means that no element of $G_{\mathbb{Q}}$ has that finite shadow. D.27 proves that these twelve *selected* shadows have no genuine full profinite lift. It also constructs a different shadow with the same CM pair (D.25.3) that survives both class-two refinements; genuine realization of that CM pair remains undecided.

Proof. For a nonzero two-torsion point of E_3 with x-coordinate $e \in \{-1, -\zeta_3, -\zeta_3^2\}$, the coordinates of its halves satisfy $x = e(1 \pm \sqrt{3})$. Their y-coordinates have squares $-9 \pm 6\sqrt{3}$. For the plus sign choose y with

$$y = \sqrt{3}(\sqrt{3} - 1)r_0.$$

The other square root is obtained using the product $3\sqrt{-3}$, up to sign. These formulas prove $\mathbb{Q}(E_3[4]) = \mathbb{Q}(i, r_0)$ in both directions. For E_4 the three-division polynomial is $3x^4 - 6x^2 - 1$. Choose two roots a, b with $a^2 = 1 + 2/\sqrt{3}$, $b^2 = 1 - 2/\sqrt{3}$ and $ab = i/\sqrt{3}$. Then $w = a + ib$ satisfies $w^4 = 4/3$; conversely

$$a = (w + 2/w)/2, \qquad b = (w - 2/w)/(2i).$$

The splitting field is therefore the same M'. Since $r_0 = 3^{1/4}/\sqrt{2}$, adjoining the abelian field $\mathbb{Q}(\sqrt{2})$ proves the second equality in (D.25.1).

The single global sign in this comparison is essential. For $P \in E_3[4]$ and $Q_0 = (0, 1) \in E_3[3]$, the addition law gives

$$y(P) = \frac{x(P)^2}{4}\big(x(P - Q_0) - x(P + Q_0)\big).$$

The points on the right belong to $E_3[12]$, proving directly that its projective coordinate field contains the full four-torsion field. Equivalently every Galois matrix on $E_3[3]$ fixes Q_0, so exactly one of the two simultaneous signs on $E_3[12]$ has that property. The separate projective $E_3[4]$ field is only $\mathbb{Q}(\zeta_{12})$; the displayed addition formula is the precise map supplying the missing sign.

We describe the finite calculation so that its certificates can be verified without enumerating $576 \cdot 6^{97}$ elements. Let $F = \ker(B \to S_3)$ and $H = q(F) \subset Q$; then $|H| = 96$. The graph $N\backslash\mathcal{T}$ of the Bass–Serre tree of $C_2 * C_3$ has edges Q and vertices $Q/\langle s\rangle \amalg Q/\langle r\rangle$. There are 576 edges and 480 vertices. A spanning tree gives97 integral fundamental cycles, a basis of N^{ab}. Write e_q for the oriented edge from its s-vertex to its r-vertex. The path cocycle into $\mathbb{Z}[Q]$ is

$$P(s) = 0, \quad P(r) = e_1 - e_r, \quad P(gh) = P(g) + q(g)P(h). \tag{D.25.4}$$

Its restriction identifies N^{ab} with the cycle group. Reduction modulo j realizes G_j faithfully as the image of B in $(\mathbb{Z}/j)[Q] \rtimes Q$.

For $A \in C_H(r)$ and $\beta \in C_H(s)$ choose the recorded paths P_A, P_β. There are respectively six and eight such choices. A lift of the cusp-preserving relation

$$\beta s \beta^{-1} A r A^{-1} = sr$$

exists modulo p exactly when

$$(1-s)P_\beta + s\big((1-r)P_A + (A-1)P(r)\big) \in (1-s)C_p + s(1-r)C_p, \quad C_p = H_1(N\backslash\mathcal{T}, \mathbb{F}_p). \tag{D.25.5}$$

This is an explicit linear system. The right side has rank 96 at $p = 2$ and 97 at $p = 3$. Its solution counts are those in (D.25.2). For every surviving pair the code gives corrections $a, b \in C_p$ and checks generation as follows. The induced equivariant map on edges sends

$$e_q \longmapsto q\big(P_A + a - P_\beta - b + e_A\big).$$

Its restriction to the 97 fundamental cycles has rank 97 in each characteristic. The homomorphism already induces the identity on Q, so it is an automorphism of G_p. Chinese remainders combine the two recorded corrections into single conjugators in G_6, giving an automorphism there. The code also multiplies the resulting group elements in the semidirect model and checks

both hexagon identities directly. All spanning-tree choices, matrix pairs, cycle coefficients and ranks are reproducible from `d_cm_shadow.py`; its JSON retains the certificates.

The CM test in these coordinates is particularly small. Use the unique representative of an element of H whose modulo-four image lies in the four-group generated by U^2, W^2. For every one of the six A choices, $A_4 = I$. The rational order-three line is $\ker(R-I)$ modulo 3, spanned by $(1,-1)$. If A_3^{-1} acts on this line by -1, the global-sign normalization changes $T_4 = I$ to $-I$, and the restriction to M' sends r_0 to $-r_0$; otherwise it fixes r_0. On the other factor β_3^{-1} is in $\{\pm I, \pm S\}$; its class is nontrivial exactly for $\pm S$, again sending r_0 to $-r_0$. Comparing these two bits gives the 24/24 split. In (D.25.3) the bits are zero and one, respectively.

The conjugator bits are invariants of the finite action, not arbitrary choices in a matrix centralizer. The same linear computation of (D.25.4) gives

$$q\big(C_{G_2}(s) \cap \operatorname{im} F\big) = 1, \qquad q\big(C_{G_3}(r) \cap \operatorname{im} F\big) = 1. \tag{D.25.6}$$

For example the first test asks whether $(1-s)P_\beta$ lies in $(1-s)C_2$; only $q(\beta) = 1$ survives. The second uses $(1-r)P_A + (A-1)P(r)$ and $(1-r)C_3$, again with only the identity surviving. Thus G_6 uniquely determines the reductions of both normalized conjugators in Q.

Finally we give the exact passage to finite GT shadows. Put $u = (sr)^{-1}$, $v = sus$, $x = u^2$, $y = v^2$; these are the images of the standard braid generators and their squares under $B_3 \to B_3/Z(B_3) = B$. Let α be one of the finite automorphisms just constructed. It fixes u and satisfies $\alpha(s) = \beta s \beta^{-1} = sf$, where $f = s\beta s\beta^{-1}$. Choose a lift of β to the discrete free group $F = \langle x, y\rangle$ and write its abelianization as $a_0 x + b_0 y$. Since s swaps x, y, $f^{\mathrm{ab}} = (b_0 - a_0)(x - y)$. Set $k = b_0 - a_0$, $\beta' = x^k \beta$, and $f' = s\beta' s(\beta')^{-1}$. Then $f' \in [F, F]$. The gauge $\alpha' = \mathrm{Int}(x^k)\alpha$ still fixes u and satisfies

$$\alpha'(s) = sf', \qquad \alpha'(r) = (f')^{-1} r, \qquad \alpha'(v) = (f')^{-1} v f'. \tag{D.25.7}$$

The last two equalities, with $r = uv$ and $s = uvu$, are exactly the two $m = 0$ hexagons (2.1)–(2.2) of [BDHP26, Definitions2.1 and 2.3]. The pullback $\widetilde{N}$ of $[N,N]N^6$ to B_3 is normal, finite index, and contained in PB_3, since $G_6 \to S_3$ is the standard permutation quotient. Here $\widetilde{N}_{\mathrm{ord}} = 36$: the cusp sr has orders 24,36,72 in G_2, G_3, G_6, respectively, its square has order 36 in G_6, and the braid center is killed. Thus $m = 0$ gives the required unit multiplier one, f' has the required commutator image, and α' being an automorphism gives surjectivity. This proves that $(0, f' \bmod \widetilde{N})$ is a finite GT shadow.

The gauge multiplies A, β, V on the left by the same x^k, so it fixes $T = A^{-1}V$ and $W = \beta^{-1}V$. Equation(D.25.6) makes their reductions recoverable from the finite shadow. The twelve different extra pairs have different such coordinates, hence give distinct shadows. No Galois element can give one of them: its two restrictions to the single field M' in (D.25.1) must agree. This proves nonarithmeticality without assuming that an arbitrary finite shadow lifts to $\widehat{\mathrm{GT}}_0$. The exact all-finer-quotient test is D.22. The later calculation D.27 rejects these specific selected shadows as genuine full profinite actions while retaining an alternate homology correction for the same extra CM pair. □

Isolated shadows and a complete tower of homology refinements

D.26. The CM witnesses survive every abelian-kernel refinement [Proved]

Keep B, Q, N, F and G_j from D.25, and let $\widetilde{N}_j \triangleleft B_3$ be the inverse image of $[N,N]N^j$. Every $\widetilde{N}_j$ is an isolated object of the gentle GT-shadow groupoid: every shadow with that target induces an automorphism of G_j.

Put $C = N^{\mathrm{ab}} \simeq \mathbb{Z}^{97}$, and let s, r act on C by conjugation. For $t = sr$ define $c = [t^{12}] \in C$. The integral maps

$$L : C \oplus C \longrightarrow C, \quad L(b,a) = (1-s)b + s(1-r)a, \qquad \widetilde{L}(b,a,h) = L(b,a) - hc \tag{D.26.1}$$

have Smith invariant factors $(1^{96}, 2)$ and (1^{97}), respectively. In particular $\operatorname{coker} L = C_2$ and $\widetilde{L}$ is onto over $\mathbb{Z}$.

There is an exact sequence and a specified quotient map

$$1 \longrightarrow \widehat{\mathbb{Z}}^{97} \longrightarrow \mathcal{G} := \varprojlim_{j} G_j \longrightarrow Q \longrightarrow 1, \qquad \widehat{B} \twoheadrightarrow \mathcal{G} = \widehat{B}/\overline{[\widehat{N}, \widehat{N}]}. \tag{D.26.2}$$

Each of D.25's twenty-four certified automorphisms of G_6 lifts to a continuous automorphism of $\mathcal{G}$ inducing the identity on Q and sending t to t^{λ}, for some $\lambda \in \widehat{\mathbb{Z}}^{\times}$ with $\lambda \equiv 1 \pmod{72}$. A single cusp gauge supplies a compatible family of GT shadows at all $\widetilde{N}_j$. For each of the twelve extra CM pairs, its member at level six can be chosen to be one of D.25's nonarithmetical shadows, with $m = (\lambda - 1)/2 \equiv 0 \pmod{36}$. Thus all twelve extra CM pairs survive the entire family of abelian-kernel refinements of N.

This family does not exhaust the finite quotients in D.22: there is an explicit finite quotient of B below G_6 whose kernel does not contain $[N, N]$. D.27 proves that all twenty-three nonidentity selected G_6 actions nevertheless have no genuine full profinite lift, and gives an alternate extra-CM homology action surviving both class-two tests.

Proof. First we verify isolation directly from finite shadow equations. Let (m, f) be a shadow at $\widetilde{N}_j$, put $\lambda = 2m + 1$, and choose a representative $f \in [F, F]$. The hexagons give a surjection $\phi : B_3 \to G_j$ and send the braid center to its λth power, which is trivial in G_j. Hence ϕ factors through B. The cusp t has order $12j$ in G_j: its image in Q has order twelve, and the cycle c is primitive in C. The last assertion also follows from the integral cycle coordinates in the calculation below, one of which is one and all the others are zero. Thus $(\widetilde{N}_j)_{\mathrm{ord}} = 6j$ and the shadow's multiplier is a unit modulo twelve.

Evaluate the same word f in the linear matrices U^2, W^2 of A.17 modulo twelve. Its reduction modulo four is I, since U^2, W^2 generate an abelian four-group modulo four. Consequently the linear matrices $A = U^{\lambda}$ and $D = f^{-1} W^{\lambda} f$ satisfy the braid identity with positive sign. Indeed the projective identity first gives $ADA = \pm DAD$, and modulo four the choice $\lambda \equiv \pm 1$ gives the positive identity for U^{λ}, W^{λ}; an invertible matrix cannot equal its negative modulo four. The pair A, D generates all of $\mathrm{SL}_2(\mathbb{Z}/12)$: its generated subgroup surjects onto Q, and a proper such subgroup would be a complement to $\{\pm I\}$. That complement cannot exist, because both lifts $\pm S$ of the order-two element $s \in Q$ have square $-I$.

Write $D = \left(\begin{smallmatrix} a & b \\ c_0 & d \end{smallmatrix}\right)$. The trace, determinant, braid identity and generation now give

$$a + d = 2, \quad ad - bc_0 = 1, \quad c_0(1 + \lambda c_0) = 0, \quad c_0 \in (\mathbb{Z}/12)^{\times}.$$

Thus $c_0 = -\lambda^{-1}$ and the matrix $V = \left(\begin{smallmatrix} \lambda & \lambda(1-a) \\ 0 & 1 \end{smallmatrix}\right)$ conjugates U, W to A, D, respectively. This repeats the elementary matrix argument of A.17 using only the finite shadow hypotheses. It proves $q\phi = \mathrm{Int}(V)q$. In particular $\phi(N) \subset C/jC$. Since this subgroup is abelian of exponent j, ϕ kills $[N, N]N^j$. It therefore induces a surjective endomorphism of the finite group G_j, hence an automorphism. Its kernel is precisely $\widetilde{N}_j$. This is isolation in [DG24, Definition3.13].

For the integral assertion use the same oriented graph and spanning tree as in D.25. The ninety-seven fundamental cycles form a matrix with the identity matrix in the non-tree-edge coordinates, so projection to those coordinates is an exact integral inverse on C. The columns of L are $(1-s)C$ and $s(1-r)C$ in this basis. The program `d_cm_refine.py` computes their integral Smith form as $(1^{96}, 2)$. Appending the column $-P(t^{12})$ gives Smith form (1^{97}); an

independent Hermite-normal-form calculation of this augmented matrix returns I_{97}. The program retains the complete integer coordinates and reproduces the spanning tree. Thus these are integer-image computations, not deductions from ranks modulo two and three.

We construct the lift, retaining the multiplier's full residue. Fix one of the twenty-four certified matrix pairs A_0, β_0 and choose the fixed integral paths P_A, P_β of D.25. Let $E \in C$ be its integral cusp error, the left side of (D.25.5). With cycle corrections (b, a) the exact cusp equation for multiplier $\lambda = 1 + 12h$ is

$$E + L(b, a) - hc = 0. \tag{D.26.3}$$

To see the exponent, $P(t^{1+12h}) = P(t) + hc$, since $q(t)c = c$. This proves (D.26.3) in every finite quotient, and by continuity for $h \in \mathbb{Z}_p$.

At $p = 2, 3$ begin with D.25's correction modulo p and $h = 0 \pmod p$. Suppose coefficients have been constructed modulo p^k; their error in (D.26.3) is divisible by p^k. Surjectivity of $\widetilde{L}$ supplies a correction $(\delta b, \delta a, \delta h)$ modulo p equal to the negative error divided by p^k. Add p^k times that correction. This constructs compatible solutions over $\mathbb{Z}_p$. Their induced matrix on $C \otimes \mathbb{Z}_p$ reduces to D.25's invertible matrix modulo p, hence is invertible. The associated map on the quotient Q is the identity, so it is an automorphism of the whole extension. The initial values $h = 0 \pmod 2$ and $h = 0 \pmod 3$ are preserved. Accordingly the combined multiplier satisfies $\lambda \equiv 1 \pmod{72}$, not just modulo twelve. As a check of the power and sign conventions, the program carries every one of the twenty-four solutions through exponents $2, 4, 8, 16$ and $3, 9, 27, 81$, and directly verifies both general-m hexagons in their faithful group models: there are 384 successful identity checks.

For $p > 3$ take the identity automorphism on $\mathcal{G}_p = \varprojlim_k G_{p^k}$, with multiplier one, but choose conjugators with the required images A_0, β_0 in Q. Such choices exist. For example, an arbitrary lift a_0 of $A_0 \in C_Q(r)$ has $a_0 r a_0^{-1} r^{-1} = z \in C \otimes \mathbb{Z}_p$ and $(1 + r + r^2)z = 0$. Multiplication by three is invertible, and the identity $(1 - r)(2 + r) = 3 - (1 + r + r^2)$ shows that $z = (1 - r)((2 + r)z/3)$. Adjusting a_0 by the negative of this cycle gives a lift centralizing r. The same argument for s uses $(1 - s)z = 2z$ when $(1 + s)z = 0$ and division by two. These are compatible p-adic choices, and both belong to the preimage of $H = q(F)$.

The extensions $\mathcal{G}_p$ have the same quotient Q, and their fibre product over Q is precisely $\mathcal{G}$. This follows either from the path model or from Chinese remainders on C/jC. The constructed automorphisms and conjugators therefore assemble to $\alpha \in \mathrm{Aut}(\mathcal{G})$ and $A, \beta \in \overline{F}$, with $\alpha(s) = \beta s \beta^{-1}$, $\alpha(r) = ArA^{-1}$ and $\alpha(u) = u^\lambda$. Here $\overline{F}$ denotes the image of $\widehat{F}$ in $\mathcal{G}$. Its abelianization is $\overline{F}^{\mathrm{ab}} = \widehat{\mathbb{Z}}x \oplus \widehat{\mathbb{Z}}y$, because the kernel $\overline{[\widehat{N}, \widehat{N}]}$ is contained in $\widehat{F}'$.

Write $\beta^{\mathrm{ab}} = ax + by$, set $k = b - a$, and conjugate α by x^k. With $\beta' = x^k\beta$ and $f' = s\beta' s(\beta')^{-1}$, the same abelianization calculation as in D.25 gives $f' \in \overline{F}'$. The resulting automorphism satisfies

$$\alpha'(u) = u^\lambda, \qquad \alpha'(v) = (f')^{-1} v^\lambda f', \qquad \alpha'(s) = sf' = (f')^{-1} s. \tag{D.26.4}$$

Since $r = uv$, $s = uvu = vuv$, and $m = (\lambda - 1)/2$, its two hexagons are exactly

$$u^\lambda (f')^{-1} v^\lambda f' = (f')^{-1} r x^{-m}, \qquad (f')^{-1} v^\lambda f' u^\lambda = vuy^{-m} f'. \tag{D.26.5}$$

Reduction to every G_j gives a GT shadow: the center is killed, the unit condition holds, f' has commutator image, and the induced map is an automorphism. The single global pair makes these reductions compatible.

For precision concerning D.25's finite gauge, choose its discrete representative of β to agree with the constructed profinite β both in G_6 and in $\overline{F}^{\mathrm{ab}}/36$. Such a representative exists because F is dense in $\overline{F}$ and this is a finite joint quotient. Then its integer $b - a$ agrees with k

modulo 36, the order of x in G_6. Thus its D.25 gauge is exactly the reduction of (D.26.4), with $m = 0 \pmod{36}$. The gauge preserves $T = A^{-1}V$ and $W = \beta^{-1}V$. Consequently all twelve extra CM pairs remain nonarithmetical at level six in the compatible tower. This specifies the finite gauges that extend; it does not make an assertion about an independently chosen gauge.

Finally we locate the remaining finite tests by an actual quotient. Choose a free basis of the rank 97 group N and map its first two generators to $(12), (23) \in S_3$ and the others to the identity. Let J be the normal core in B of the intersection of this map's kernel with $[N, N]N^6$. Then $J \triangleleft B$ has finite index, $J \subset [N, N]N^6$, but $[N, N] \not\subset J$, since the commutator of the first two basis elements has nontrivial image in S_3. The quotient map $B \to B/J$ therefore cannot factor through any G_j: every such factorization would kill $[N, N]$. This also proves the kernel in (D.26.2) is nontrivial and that the homology tower is not cofinal in the system of D.22.

The source search supplies no additional lifting assertion for these quotients. In particular isolation gives reduction *homomorphisms*, as stated in [DG24, Remark 3.16]; the all-finer-quotient criterion is its Theorem 5.2 and Corollary 5.4. The abelian sufficient criterion in [DLL24, TheoremB.2] requires the pure quotient to be abelian. Here its image already surjects onto $H \simeq C_2^2 \times \mathrm{SL}_2(\mathbb{F}_3)$: the isomorphism sends the unique lift with modulo-four image in V_4 to its modulo-four and modulo-three components. The latter group's second derived subgroup has order two. Similarly the dihedral quotients of [BDHP26, Remark 3.2] embed their F images into products of three dihedral groups and are metabelian. No such pure quotient can surject onto this H, since surjections preserve the derived-series images. These explicit maps explain why those sufficient families do not supply the remaining lift. □

Nonabelian refinement tests at two and three

D.27. Class-two obstructions, an alternate CM witness, and relative pro-prime lifts [Proved]

Keep B, Q, N from D.25, write $\gamma_3 N = [N, [N, N]]$, and put

$$K_3 = \gamma_3 N\, N^3, \qquad K_2 = \gamma_3 N\, [N, N]^2 N^4, \qquad H_p = B/K_p. \tag{D.27.1}$$

These are characteristic refinements of the kernel N, with

$$|H_3| = 576 \cdot 3^{4753}, \qquad |H_2| = 576 \cdot 2^{4850}. \tag{D.27.2}$$

The central layers over N^{ab}/p have dimensions 4656 at three and 4753 at two.

At three, exact averaging over the normal subgroup $O_2(Q) \simeq C_2^2 \times Q_8$ reduces the central correction system to a 135×270 matrix of rank 127. Among the twenty-four *chosen certificates* of D.25, four admit class-two lifts with the recorded conjugators: two CM-compatible pairs and the two extra pairs

$$(A, \beta) = \left(I, \begin{pmatrix} 9 & 8 \\ 4 & 9 \end{pmatrix}\right), \quad \left(I, \begin{pmatrix} 9 & 4 \\ 8 & 9 \end{pmatrix}\right) \pmod{12}. \tag{D.27.3}$$

Both extra lifts have zero central correction. Their fibre products with G_2 give nonarithmetical finite shadows of order $576 \cdot 2^{97} \cdot 3^{4753}$.

At two the exact central correction matrix has rank 4737. Allowing the multiplier $\lambda = 1 + 72\eta$, $\eta \in \mathbb{F}_2$, raises the rank to 4738. The fixed-conjugator test has two survivors: the identity pair and the extra pair

$$A = 5I, \qquad \beta = I \pmod{12}. \tag{D.27.4}$$

Both use $\lambda = 1$. Its fibre product with G_3 gives a nonarithmetical finite shadow of order $576 \cdot 2^{4850} \cdot 3^{97}$.

There are actual infinite nonabelian extensions of these three witnesses. Let $\mathcal{P}_p$ be the relative pro-p completion of B over Q, defined by retaining the full pro-p completion $N^{(p)}$ of N as kernel. There is an exact sequence

$$1 \longrightarrow N^{(p)} \longrightarrow \mathcal{P}_p \longrightarrow Q \longrightarrow 1, \qquad N^{(p)} \text{ free pro-}p \text{ of rank97.} \tag{D.27.5}$$

The two pairs (D.27.3) extend to automorphisms of $G_2 \times_Q \mathcal{P}_3$, and (D.27.4) extends to an automorphism of $\mathcal{P}_2 \times_Q G_3$. After one cusp gauge, each gives compatible finite GT shadows on every finite characteristic refinement retaining the indicated G_6 and CM data. These are automorphisms of the specified profinite quotients. The complete lower-conjugator freedom preserving each selected G_6 action has dimension 52 at two and 34 at three. Adjoining its correction columns raises the ranks to 4748 and 130, respectively, and leaves exactly the same survivors. Their intersection consists only of the identity action. Consequently *all twenty-three nonidentity selected G_6 actions in D.25 have no genuine full profinite lift.* This includes all twelve selected extra-CM shadows, despite their compatible lifts through D.26's entire homology tower.

There is, however, a different homology correction for the first pair in (D.27.3) that lifts to both H_2 and H_3. Its five exact mod-two obstruction forms vanish, the induced kernel map has rank 97, and the corrected cusp word of length 470844 evaluates to the identity in the full Magnus model. It has multiplier $\lambda = 1$ and gives an automorphism of

$$H_2 \times_Q H_3, \qquad |H_2 \times_Q H_3| = 576 \cdot 2^{4850} \cdot 3^{4753}. \tag{D.27.9}$$

The same witness extends to $H_2 \times_Q \mathcal{P}_3$, with the identity action on $\mathcal{P}_3$ and prescribed elliptic conjugators. A single cusp gauge gives compatible nonarithmetical shadows on every finite characteristic refinement retaining its G_6 and CM data. D.30 proves that this particular recorded central representative has no genuine full lift, jointly retaining its G_6 data, and constructs a further explicit central replacement. Genuine realization of the CM pair remains undecided.

Proof. The characteristic quotients in (D.27.1) have explicit free-group models. At three, use the class-two group

$$(v, z)(v', z') = (v + v', z + z' + \tfrac{1}{2} v \wedge v'), \quad v \in \mathbb{F}_3^{97},\ z \in \bigwedge^2 \mathbb{F}_3^{97}. \tag{D.27.6}$$

It has exponent three and is freely generated, in the class-two exponent-three variety, by $(e_i, 0)$. Collection of commutators gives the asserted kernel $\gamma_3 N\, N^3$, dimension $97 + \binom{97}{2} = 4753$. At two use the truncated noncommutative Magnus expansion $a_i \mapsto 1 + X_i$ over $\mathbb{F}_2$, modulo degree three. The multiplication is

$$(v, w)(v', w') = (v + v', w + w' + v(v')^{\mathsf{T}}). \tag{D.27.7}$$

The kernel over $v \in \mathbb{F}_2^{97}$ is the space of symmetric 97×97 matrices: the diagonal entries are a_i^2, and the off-diagonal basis entries are $[a_i, a_j]$. They are independent central elements of order two, proving the kernel description in (D.27.1) and the order $2^{97+97+\binom{97}{2}} = 2^{4850}$. These normal forms also prove that the models are faithful.

Here is an exact evaluation of the required words in these models. In the graph of D.25 retain its spanning tree and orient every edge from its s-vertex to its r-vertex. During evaluation of a word in B, keep its current image $q \in Q$. Multiplication by s changes the state to qs and contributes no edge. Multiplication by r appends $e_q - e_{qr}$; multiplication by r^{-1} appends $e_q - e_{qr^2}$. Delete tree edges and read the remaining oriented edges as free generators. For

a closed N-word this is exactly the tree-collapse isomorphism $N \simeq \pi_1(N\backslash\mathcal{T}) \simeq F_{97}$. The program constructs the 97 basis words in s, r from the fundamental cycles and verifies that they evaluate to the standard generators. Thus evaluating a cusp defect uses actual group words, including all degree-two terms.

Let $M(q)$ be the exact matrix of q on the cycle group modulo p. The central action is $z \mapsto M(q)zM(q)^{\mathsf{T}}$, on alternating matrices at three and symmetric matrices at two. Left-multiplying the two elliptic conjugators by central elements changes the cusp defect by

$$(1-s)z_\beta + s(1-r)z_A. \tag{D.27.8}$$

At three the uncorrected defect is computed from the word $\beta s\beta^{-1}ArA^{-1}(sr)^{-1}$ in (D.27.6). Its degree-one coordinate is zero by D.25. At two the same word is evaluated in (D.27.7), with a symmetric degree-two coordinate.

The subgroup $O_2(Q)$ has order 32. Indeed the unique lift of $H = q(F)$ whose modulo-four component lies in V_4 identifies H with $V_4 \times \mathrm{SL}_2(\mathbb{F}_3)$; $V_4 \times Q_8$ is normal in Q. Averaging over it at three is an exact idempotent. Its image on the central layer has dimension 135. This can be computed both from the explicit projector and from the integral graph character

$$\chi_C(q) = 576\,[q=1] - \#(Q/\langle s\rangle)^q - \#(Q/\langle r\rangle)^q + 1, \quad \chi_{\wedge^2 C}(q) = \tfrac{1}{2}(\chi_C(q)^2 - \chi_C(q^2)).$$

The averaged integral lattice is a direct summand over $\mathbb{Z}_{(3)}$, so its rank remains135 modulo three. The program selects an explicit basis of that image and computes the rank 127 matrix in (D.27.8).

Averaging loses no obstruction. If P is the averaging operator, then $z - Pz = \frac{1}{32}\sum_{g\in O_2(Q)}(1-g)z$. Writing each g as a word in s, r telescopes each $(1-g)z$ into $(1-s)a + (1-r)b$. The identity

$$(1-r)b = s(1-r)b + (1-s)(1-r)b$$

converts this to (D.27.8). The program performs that telescoping construction, reconstructs the central correction in all 4656 coordinates, and checks the full equation. The resulting four successful certificates are retained in `d_cm_class2.json`.

At two the program uses exact integer bit vectors for all 4753 coordinates. The multiplier must retain $m = 0 \pmod{36}$ at G_6. The two possibilities in the new quotient are $\lambda = 1 + 72\eta$ modulo 144. Since $t^{12} \in N$ and N/K_2 has exponent four, $t^{72} = (t^{12})^2$ in H_2; its central coordinate is cc^{T}, where $c = [t^{12}] \bmod 2$. Appending this single column gives the rank 4738 augmented system. For every successful solution the code checks the complete4753-coordinate equation. It stores the exact central coefficients and multiplier bit in `d_cm_class2_two.json`.

We now account for *all* conjugator choices retaining a selected G_6 action. Equation (D.25.6) forces their images $q(A), q(\beta) \in Q$: the difference of two conjugators inducing the same finite action centralizes the corresponding elliptic generator, and the indicated centralizers have trivial image in Q. The remaining differences therefore lie in the kernel N. At two, right-multiplication $\beta \mapsto \beta c$ can preserve the G_2 action precisely when the degree-one class of c lies in $(N^{\mathrm{ab}}/2)^s$, a space of dimension 52. For each basis class the program evaluates the actual word $[c, s]$ in the Magnus model and adds its central defect, transported by $q(\beta)$, to (D.27.8). Changes of lift of c are already included among its central columns. The analogous right freedom of A contributes no further obstruction: every order-three centralizer element lifts through the two-kernel by the coprime conjugacy argument proved below. At three the roles are reversed. The space $(N^{\mathrm{ab}}/3)^r$ has dimension 34; its actual defects $[c, r]$, transported by $sq(A)$, are appended to the averaged system. The order-two centralizer lifts through the three-kernel. The resulting exact ranks are 4748 and 130. These calculations exhaust the lower-conjugator freedom, and their saved certificates show the survivor sets asserted in the statement.

Here is why their disjointness away from the identity proves the stated genuine nonliftability. A genuine GT_0 action on the completed braid quotient induces an automorphism of $\widehat{B} = C_2 \amalg C_3$ with its specified cusp power. The torsion theorem of Herfort–Ribes, [HR85, Theorem A, p. 155; Theorem A′, pp. 157–159], makes the images of s, r conjugates of s, r respectively: their orders distinguish the two factors, and the given action on $B^{\mathrm{ab}} = C_6$ distinguishes r from r^{-1}. The induced identity on S_3 lets the conjugators be chosen in $\widehat{F}$: multiply a conjugator by an appropriate power of its elliptic generator to remove its image in the corresponding S_3 centralizer. Undo the known D.25 cusp gauge by the inner automorphism $\mathrm{Int}(x^{-k})$. Since $x = u^2$ centralizes the cusp u, the prescribed cusp power is retained. The resulting automorphism induces the selected action on G_6 and the identity on Q, hence preserves $\widehat{N}$ and its characteristic subgroups defining H_2, H_3. It must therefore give elliptic-conjugacy lifts to *both* of these quotients, with exactly the conjugator freedom just exhausted. The multiplier condition $m = 0 \pmod{36}$ is included: at two it is $\lambda = 1$ or73 modulo 144, while at three its change acts trivially on the exponent-three kernel. Only the identity selected action survives both systems. This argument proves nonexistence of a genuine full lift of each of the other twenty-three selected actions. It does not rely on finite-quotient torsion conjugacy, which need not follow from the full profinite theorem.

Every successful correction is an automorphism, with a separate generation proof. The maps already induce the identity on Q, preserve N, and retain the invertible map on N^{ab}/p certified in D.25. The Frattini subgroup of each finite p-kernel is its subgroup of powers and commutators, so that map is onto the whole p-kernel. Hence it is onto the extension and therefore an automorphism. The two prime kernels have coprime orders; the map from B to their fibre product over Q is onto by the same property on N. This proves the stated combined finite quotient orders and automorphisms. The cusp gauge of D.25 supplies a commutator representative and both hexagons, so the finite actions give actual shadows.

The alternate certificate is found and verified separately in `d_cm_quadratic_search.py`. The mod-two degree-one cusp equation has a98-dimensional affine solution space in 194 conjugator coordinates. The orthogonal complement of the rank 4748 correction image has dimension five. Applying those five linear functionals to the actual degree-two word gives five quadratic polynomials on the affine solution space. The program computes them directly from the graph paths, including the inverse-generator square terms. In the fixed spanning-tree basis its successful homology masks are

$$b = \texttt{8ca9620ec63b26ee68fbb210}_{16}, \qquad a = \texttt{60da0c00a2c8c0f44be10df0}_{16}. \tag{D.27.10}$$

They occur on the 48th tested affine vector; all five obstructions vanish and the independently computed kernel rank is 97. The exact central corrections and right s-centralizer coefficients are saved in `d_cm_quadratic_search.json`. Substituting them as explicit products of the 97 basis words, their squares and commutators gives a470844-letter cusp word. Direct evaluation returns $q = 1$, $v = 0$, and zero in every one of the 4753 central coordinates. Thus existence does not depend on random search success rates or solely on the five projected equations. Retain the original mod-three certificate, whose central correction is zero. The coprime kernel product and the Frattini generation argument already proved give (D.27.9).

We next prove the infinite relative lifts, rather than inferring them from these finite calculations. Define $\mathcal{P}_p$ as the inverse limit of B/L over the characteristic finite p-power-index subgroups L of N. Such subgroups are cofinal among the finite p-quotients of the finitely generated free group N: take the intersection of all normal subgroups of p-power index bounded by a given integer. This proves (D.27.5) and gives the canonical maps $\widehat{B} \twoheadrightarrow \mathcal{P}_p \twoheadrightarrow G_p$.

For the two pairs (D.27.3), the recorded mod-three conjugator A is exactly the identity. The cusp equation then says $\beta s \beta^{-1} = s$ in G_3, so the recorded automorphism of G_3 is the identity. Its specific centralizing β lifts to an element of $C_{\mathcal{P}_3}(s)$. To prove this, lift it arbitrarily; its

conjugate of s and s are two complements of order two in an extension with pro-three kernel. They are conjugate by that kernel. For a finite p-kernel and a cyclic group of order prime to p, this conjugacy follows by induction on the kernel: pass to an invariant central elementary abelian subgroup and average the additive cocycle there. Compactness gives the result for a pro-p kernel. Thus the identity automorphism of $\mathcal{P}_3$ has the required elliptic conjugators and reduces to the specified mod-three action. Its fibre product with D.25's mod-two automorphism proves the first assertion following (D.27.5). Its fibre product with the alternate mod-two class-two automorphism instead proves the asserted action on $H_2 \times_Q \mathcal{P}_3$.

For (D.27.4), the recorded mod-two β is exactly the identity; the cusp equation gives $ArA^{-1} = r$ in G_2. The same coprime conjugacy proof, now with an order-three complement and a pro-two kernel, lifts its prescribed A to $C_{\mathcal{P}_2}(r)$. Combining the identity action on $\mathcal{P}_2$ with the recorded mod-three action proves the other assertion.

Finally choose a lift of each combined β to $\widehat{F}$, write its abelianization as $ax + by$, and conjugate the quotient automorphism by x^{b-a}. The resulting $f = s\beta' s(\beta')^{-1}$ is the image of an element of $\widehat{F}'$ and satisfies both hexagons on every finite quotient. This single choice supplies compatible shadows; a discrete representative agreeing in the finite quotient and in $\widehat{F}^{\mathrm{ab}}/36$ gives the corresponding D.25 gauge. Their CM coordinates are unchanged. Their nonarithmeticality therefore follows from the same common-field restriction in D.25.

The strongest new inverse system retains the entire free pro-three kernel and the explicit class-two two-kernel. Its quotient maps are $\widehat{B} \twoheadrightarrow H_2 \times_Q \mathcal{P}_3 \twoheadrightarrow H_2 \times_Q H_3$. Higher two-primary and mixed-prime refinements remain in the all-finite-quotient intersection of D.22. The alternate certificate changes the initial homology action in the same CM matrix pair, explaining exactly why the rejection of the twenty-three selected actions does not reject that matrix pair. The surviving alternate action is recorded separately and is not substituted silently for any original D.25 certificate. □

A compatible mixed-prime refinement

D.28. An extra CM action through a mixed two–three kernel [Proved]

Keep $B = C_2 * C_3$, $q : B \twoheadrightarrow Q$, N, G_3, H_2 and $\mathcal{P}_3$ from D.25–D.27. Let $V = \mathbb{F}_4$, viewed additively as $\mathbb{F}_2^2$. The Q-action on V factors through

$$Q \longrightarrow \mathrm{SL}_2(\mathbb{F}_3)/Q_8 \simeq C_3; \qquad s|_V = I_2, \qquad r|_V = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \tag{D.28.1}$$

In the ordered rank 97 graph basis of D.25, define $\varphi : N^{\mathrm{ab}}/2 \twoheadrightarrow V$ by the two binary row masks

$$\texttt{1860e0d6cb6b6efc923133e5a}_{16}, \qquad \texttt{1231c145bdbd87ed2d1f1a136}_{16}, \tag{D.28.2}$$

where bit i is the coefficient of basis vector i, starting at $i = 0$. This is a Q-equivariant epimorphism. Its kernel is independent of multiplying φ by a nonzero scalar in $\mathbb{F}_4$. Put

$$L = \ker(N \longrightarrow V), \quad H = B/L, \quad J = B/([L, L]L^3). \tag{D.28.3}$$

These are specified discrete finite quotients, with

$$|H| = 2304, \quad \operatorname{rank} L = 385, \qquad 1 \longrightarrow \mathbb{F}_3^{385} \longrightarrow J \longrightarrow H \longrightarrow 1. \tag{D.28.4}$$

The extra matrix pair

$$A = I, \qquad \beta = \begin{pmatrix} 9 & 8 \\ 4 & 9 \end{pmatrix} \pmod{12} \tag{D.28.5}$$

admits a cusp-preserving automorphism of J with multiplier $\lambda = 1$. It can be chosen to agree with D.27's alternate action on the exact common quotient

$$C = H \times_Q G_3. \tag{D.28.6}$$

Consequently there is one compatible automorphism, with specified elliptic conjugators and multiplier one, of

$$\mathcal{R} = J \times_C (H_2 \times_Q \mathcal{P}_3). \tag{D.28.7}$$

The canonical continuous map $\widehat{B} \twoheadrightarrow \mathcal{R}$ is onto. One global cusp gauge gives compatible nonarithmetical GT_0 shadows on every finite characteristic refinement of its G_6 quotient within $\mathcal{R}$.

This refinement retains a mixed-prime action in the kernel:

$$\ker(J \to Q) \simeq \mathbb{F}_3^{385} \rtimes C_2^2, \qquad \dim_{\mathbb{F}_3}(\mathbb{F}_3^{385})^{C_2^2} = 97. \tag{D.28.8}$$

The semidirect-product identification is noncanonical; the kernel, its normal three-subgroup and their action are canonical in (D.28.3). In particular J is not a quotient of any product over Q of relative pro-prime completions whose kernel is a direct product of pro-p groups. The maximal quotient of J with nilpotent kernel over Q is exactly C. The full profinite realization of (D.28.5) remains the all-finer-quotient question of D.22.

Proof. Write $C_N = H_1(N\backslash\mathcal{T}, \mathbb{F}_2)$, where $\mathcal{T}$ is the Bass–Serre tree used in D.25. A group of order three acts irreducibly on the two-dimensional space in (D.28.1), and $\mathrm{End}_Q(V) = \mathbb{F}_4$. The exact equations for a matrix $Z : C_N \to V$ are

$$ZM(s) = Z, \qquad ZM(r) = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} Z. \tag{D.28.9}$$

The program `a_mixed_module_explore.py` solves these equations over $\mathbb{F}_2$. Their solution space has dimension two, and the first basis matrix is (D.28.2). Both displayed equations are checked on all 97 columns. The matrix has rank two, so it is onto. Every nonzero solution is obtained from it by a nonzero scalar in its endomorphism field: this gives the stated independence of its kernel. It also proves that L is preserved by any automorphism of $\widehat{B}$ inducing the identity on Q, through the induced Q-equivariant action on $N^{\mathrm{ab}}/2$. Normality of L in B follows directly from (D.28.9).

For clarity, the finite extension H is computed with its extension class, not replaced by a split extension. Let $P(q)$ be the fixed path chain to the representative of $q \in Q$ in D.25. For $q, q' \in Q$ the chain $P(q) + qP(q') - P(qq')$ is a cycle. Set

$$c(q, q') = \varphi(P(q) + qP(q') - P(qq')) \in V.$$

Then H consists of the 2304 pairs (q, v) with law

$$(q, v)(q', v') = (qq', v + qv' + c(q, q')). \tag{D.28.10}$$

The cocycle identity follows by cancelling the four path expressions in the associativity equation. This is the actual quotient B/L: the path construction sends its generators to $(s, 0), (r, 0)$ and restricts to φ on N. Its kernel is therefore exactly L. The code also checks that these generators reach all 2304 pairs. The H-cover of the graph has 2304 edges and $2304/2 + 2304/3 = 1920$ vertices. Its free fundamental group has rank $2304 - 1920 + 1 = 385$, proving (D.28.4) after abelianization modulo three.

Here are the complete finite lifting equations. Put $C_L = H_1(L\backslash\mathcal{T}, \mathbb{F}_3)$ and use the oriented graph's spanning-tree basis. Let $a, b \in H$ be candidate reductions of A, β, and put

$$s' = bsb^{-1}, \quad r' = ara^{-1}, \quad t = sr.$$

The finite checks first require $s'r' = t$ and that the assignment $s \mapsto s'$, $r \mapsto r'$ is an automorphism α_H of H. They evaluate its map on every one of the 2304 elements and verify both generator transitions. If $a_1, b_1 \in C_L$ are the left correction coordinates of the two conjugators, their cusp equation is

$$(1 - s')b_1 + s'(1 - r')a_1 = -d, \tag{D.28.11}$$

where the actual path-chain defect is

$$d = (1 - s')P(b) + s'\big((1 - r')P(a) + aP(r)\big) - P(t). \tag{D.28.12}$$

All actions in these formulas are the left permutation action on the edges of the H-cover. Thus both sides are elements of the specified385-dimensional cycle space; membership is checked by reconstruction from their chord coordinates.

The compatibility conditions determine which finite solution we use. In the coordinates (D.28.10), the corrected conjugators of D.27 reduce to

$$a = (1, (1, 0)), \qquad b = (q(\beta), (1, 1)). \tag{D.28.13}$$

To obtain the second coordinate one must include D.27's right s-centralizer correction. The program evaluates (D.28.2) on its full corrected homology mask, and obtains the bit values1 and 3 in (D.28.13). The desired reductions in G_3 are $A_3 = 1$ and

$$P(\beta_3) = P(q(\beta)) + 2c_{39}, \tag{D.28.14}$$

where c_{39} is the basis cycle numbered39 in D.25. These are exactly D.27's retained mod-three conjugators.

The inclusion $L \hookrightarrow N$ induces the map $C_L \to N^{\mathrm{ab}}/3$ obtained by summing each edge chain over the four edges above each edge of the Q-cover. It has rank 97. Applying this explicit map to the two corrected path chains gives194 further linear conditions imposing (D.28.14). Together with (D.28.11) the resulting system has rank 482 in 770 variables, and hence an affine solution space of dimension 288. No arithmetic or profinite lifting assumption enters this finite system.

The file `a_mixed_prime_compatible.json` contains its two complete385-coordinate solution vectors. The 22nd tested vector in the deterministic enumeration has invertible kernel action. The test of invertibility uses the actual equivariant graph map: the base edge is sent to the path

$$e_a + P(a) + a_1 - P(b) - b_1, \tag{D.28.15}$$

and edge h is sent to its translate by $\alpha_H(h)$. The induced385 by 385 matrix has rank 385 over $\mathbb{F}_3$. The program additionally reconstructs all 385 free basis words of L, checks that each gives its claimed fundamental cycle, and evaluates each basis image from the substituted s, r words. All 385 columns agree with the matrix from (D.28.15). Finally the corrected54658-letter cusp word

$$\beta s\beta^{-1} ArA^{-1}(sr)^{-1}$$

is evaluated directly: its H-coordinate and all 2304 edge-chain coordinates are zero. These checks certify existence independently of the random search method used to locate the recorded vectors.

The resulting homomorphism $B \to J$ preserves L because its reduction is α_H. It kills $[L,L]L^3$, and so induces an endomorphism of J. Its action on the kernel $L/[L,L]L^3$ is the invertible matrix just verified, and its action on H is an automorphism. It is therefore surjective on J, hence an automorphism of the finite group. The conjugators lie in the image of $F = \ker(B \to S_3)$, since their images in Q do; every kernel correction lies in $L \subset F$. The multiplier is exactly one.

We prove the common-quotient assertion, which is needed to combine the witnesses. Write $M = L^{\mathrm{ab}}/3$ and $V = N/L$. The kernel of $J \to Q$ is an extension $1 \to M \to \overline{N} \to V \to 1$. Since $|V| = 4$ is invertible in $\mathbb{F}_3$, its extension class vanishes. Explicitly, for a normalized additive two-cocycle $b(v,w)$, put $h(v) = \frac{1}{4}\sum_t b(v,t)$. Summing its cocycle identity over $t \in V$ gives $b(v,w) = vh(w) + h(v) - h(vw)$, so changing the section by $-h$ splits the extension. This proves the semidirect product in (D.28.8), with its stated ambiguity.

The finite covering map $L\backslash\mathcal{T} \to N\backslash\mathcal{T}$ supplies a transfer. On homology its composite with projection is multiplication by four; averaging is an exact projection onto invariants because four is invertible. More directly, the cellular chain complex of the quotient graph is the coinvariant complex of the covering graph. The coinvariant functor is exact by averaging, so taking its first homology gives

$$M_V \xrightarrow{\sim} N^{\mathrm{ab}}/3. \tag{D.28.16}$$

The explicit edge-sum map above is this isomorphism on coinvariants. Consequently $\dim M^V = \dim M_V = 97$, whereas $\dim M = 385$; the nontrivial summand has dimension 288. These dimensions and nontriviality are independently checked using the two matrices of V on C_L.

In a nilpotent finite quotient, the images of the two- and three-Sylow subgroups commute. Thus every quotient of $\overline{N}$ which is nilpotent kills $[V,M]$. Conversely $\overline{N}/[V,M] \simeq V \times M_V$ is nilpotent. The maps $J \to H$ and $J \to G_3$ are defined because $[L,L]L^3 \subset L$ and $[L,L]L^3 \subset [N,N]N^3$. Their joint map is onto $H \times_Q G_3$, by coprimality of its two kernel orders. Equation (D.28.16) identifies its kernel with $[V,M]$, proving (D.28.6) as the maximal quotient of J with nilpotent kernel over Q.

The kernel over Q of $H_2 \times_Q \mathcal{P}_3$ is the direct product of a finite two-group and a pro-three group. Every finite quotient of that kernel is nilpotent. Its maximal common quotient with J is therefore at most C. Both factors map onto C, so it is exactly C. Apply the elementary subdirect-product description to the closed image of $\widehat{B}$ in their product: its associated common quotient is the quotient by the product of their kernels. The equality just proved identifies that quotient with C, and gives the surjection in (D.28.7). The matrices and conjugators (D.28.13)–(D.28.14) show that the two automorphisms agree on C. They consequently define an automorphism of the indicated fibre product.

Choose one lift of its combined β to $\widehat{F}$. The cusp gauge of D.25, applied once to this lift, gives a commutator representative and both hexagons throughout the compatible inverse system. The CM pair is unchanged under that gauge. Its two actions on the common CM field in D.25 disagree, so none of these finite shadows retaining that pair is induced by a Galois element. The quotient (D.28.7) and its surjection from $\widehat{B}$ are the precise source and target of the construction; no lift of this quotient automorphism to $\widehat{B}$ is inferred. □

An entire class-two two-primary tower and every odd pro-prime kernel

D.29. A compatible infinite nonabelian lift of the alternate CM pair [Proved]

For $k \geq 1$ define the characteristic quotients

$$H^{(k)} = B/(\gamma_3 N\,[N,N]^{2^k} N^{2^{k+1}}), \qquad |H^{(k)}| = 576 \cdot 2^{4753k+97}. \tag{D.29.1}$$

Thus $H^{(1)} = H_2$ of D.27. The alternate CM pair $A = I$, $\beta = \left(\begin{smallmatrix} 9 & 8 \\ 4 & 9 \end{smallmatrix}\right)$ has actual cusp-preserving automorphisms on $H^{(2)}$ and $H^{(3)}$. Their corrected cusp words have lengths 1816936 and 4448112, respectively, and are checked in the complete truncated Magnus models over $\mathbb{Z}/4$ and $\mathbb{Z}/8$. Their multipliers are 1 and 289 modulo 288 and 576. They retain D.27's homology data, including its right s-centralizer correction, while changing the central coordinates as explicitly recorded.

There is a compatible infinite lift to

$$\mathcal{P}_2^{[2]} := \varprojlim_k H^{(k)}, \qquad 1 \longrightarrow N^{(2)}/\overline{\gamma_3 N^{(2)}} \longrightarrow \mathcal{P}_2^{[2]} \longrightarrow Q \longrightarrow 1. \tag{D.29.2}$$

It induces the particular $H^{(1)}$ action obtained by reducing the recorded $\mathbb{Z}/4$ witness. This is a specified replacement of D.27's central representative within the same homology class; it is not asserted to extend every previously recorded central representative.

Let $\mathcal{P}_{\mathrm{odd}}$ be the fibre product over Q of the relative completions $\mathcal{P}_p$ for every odd prime p. The new action extends to an automorphism of

$$\mathcal{R} = \mathcal{P}_2^{[2]} \times_Q \mathcal{P}_{\mathrm{odd}}, \qquad \ker(\mathcal{R} \to Q) = \left(N^{(2)}/\overline{\gamma_3 N^{(2)}}\right) \times \prod_{p \text{ odd}} N^{(p)}. \tag{D.29.3}$$

Its action on every odd relative completion is the identity, with the prescribed conjugators at three. Its multiplier has $\lambda_2 \in 1 + 16\mathbb{Z}_2$ and $\lambda_p = 1$ for every odd p. A single profinite cusp gauge supplies compatible nonarithmetical shadows on every finite characteristic refinement retaining its G_6 and CM data.

The mixed-prime witness J of D.28 agrees with this action on the exact common quotient $C = H \times_Q G_3$ constructed there. Hence it further extends to

$$J \times_C \mathcal{R}. \tag{D.29.4}$$

The realization of this CM pair by an automorphism of the full $\widehat{B}$, and hence by genuine profinite GT_0, remains undecided. No replacement of the all-finite-quotient test of D.22 by (D.29.2)–(D.29.4) is made.

Proof. The truncated Magnus model over $\mathbb{Z}/2^k$ has multiplication

$$(v, w)(v', w') = (v + v', w + w' + v(v')^{\mathsf{T}}). \tag{D.29.5}$$

The image of a free generator is $(e_i, 0)$, its inverse is $(-e_i, e_i e_i^{\mathsf{T}})$, and its ath power is $(ae_i, \binom{a}{2} e_i e_i^{\mathsf{T}})$. The pair of diagonal coordinates a mod 2^k and $\binom{a}{2}$ mod 2^k determines a mod 2^{k+1}: replacing a by $a + 2^k$ changes the second coordinate by 2^{k-1} modulo 2^k. The independent off-diagonal commutators have order 2^k. Collection in a free class-two group therefore proves faithfulness, the kernel in (D.29.1), and its order $2^{(k+1)97 + k\binom{97}{2}}$ over Q. The graph-word evaluation of D.27 applies with integer coefficients modulo 2^k, so it computes actual words in these quotients.

First consider the lift from $\mathbb{Z}/2$ to $\mathbb{Z}/4$. Inside the latter model, let C_1 have basis a_i^2 and $[a_i, a_j]$. It is an abelian group $(\mathbb{Z}/4)^{4753}$: the commutator of two squares is a fourth power of a basic commutator, which vanishes. Its elements are identified by the upper-triangular Magnus coefficients, with $v_i = 2w_{ii}$ and $w_{ji} = -w_{ij}$ for $i \neq j$. Its subgroup $2C_1$ is the new kernel over $H^{(1)}$.

Fix the full D.27 conjugators, including their recorded central and right-centralizer words, and write $g_s = \beta s \beta^{-1}$, $g_r = ArA^{-1}$. Conjugation gives integral linear automorphisms S, R of C_1. Changing the conjugators on the left by $b, a \in C_1$ changes the cusp error by

$$(1 - S)b + S(1 - R)a - hc_1, \tag{D.29.6}$$

where c_1 is the C_1 coordinate of t^{72} and the multiplier is changed by $72h$. This formula is exact because C_1 is abelian; it includes the actual g_s, g_r actions rather than replacing them by their Q images. Modulo two its matrix L_0 has rank 4738, as in D.27, and its kernel has dimension $9507 - 4738 = 4769$.

The code forms all fifteen exact dual functionals on $\operatorname{coker} L_0$. For a relation $z \in \ker L_0$ it computes the new obstruction

$$z \longmapsto \tfrac{1}{2} Lz \pmod{\operatorname{im} L_0}, \qquad \ker L_0 \longrightarrow \operatorname{coker} L_0. \tag{D.29.7}$$

Choosing a different zero-one lift of z changes this class by an L_0 image, so the map is well-defined. The exact rank is 15. Its saved relation certificates, followed by an ordinary L_0 correction, solve the full $\mathbb{Z}/4$ equation. Generation is unchanged on $N^{\mathrm{ab}}/2$, where the rank remains97; the finite two-kernel Frattini argument gives an automorphism. Direct evaluation of the resulting 1816936-letter cusp word has zero in every degree-one and degree-two coordinate. These are the actual results of `d_cm_exponent4.py`, with all coefficients saved in its JSON output.

We prove the infinite extension by exact stabilization. For $k \geq 1$ put

$$C_k = \ker(H^{(k+1)} \to H^{(k-1)}), \quad H^{(0)} = G_2, \quad C_k = \langle a_i^{2^k}, [a_i, a_j]^{2^{k-1}} \rangle \simeq (\mathbb{Z}/4)^{4753}. \tag{D.29.8}$$

The commutator of two displayed powers has exponent 2^{2k} and vanishes modulo 2^{k+1}, so C_k is abelian. Its subgroup $2C_k$ is exactly $\ker(H^{(k+1)} \to H^{(k)})$. Its conjugation action factors through G_2: commutators of an element of $N^2[N, N]$ with its generators vanish in this quotient. Thus C_k is here a G_2-module; the reduced module $C_k/2C_k$ is the Q-module used in D.27.

Here are the formulas proving stabilization, including the normalization of its diagonal coordinates. Suppose the image of a_i under conjugation by a fixed word is (v_i, W_i) in the Magnus model. Its 2^kth power has degree-two coordinate $2^k W_i + \binom{2^k}{2} v_i v_i^{\mathsf{T}}$. After division by 2^{k-1} its coordinate is

$$2W_i + (2^k - 1) v_i v_i^{\mathsf{T}} \pmod 4. \tag{D.29.9}$$

The diagonal coordinate must be multiplied by the inverse of $2^k - 1$ modulo four to express it in the basis $a_j^{2^k}$. The image of a basic commutator is $v_i v_j^{\mathsf{T}} - v_j v_i^{\mathsf{T}}$. For $k \geq 2$ the factor $2^k - 1$ is always3 modulo four. The vectors v_i mod 4 depend only on the fixed image in Q, and W_i mod 2 depends only on the fixed conjugator image in G_2. Consequently these matrices are identical for every $k \geq 2$.

The multiplier column stabilizes as well. Changing λ by $72 \cdot 2^{k-1} h$ changes the cusp power by $(t^{12})^{3 \cdot 2^k h}$. If t^{12} has Magnus coordinates (v, W), its normalized coordinate for $k \geq 2$ is

$$2W + v v^{\mathsf{T}} \pmod 4, \tag{D.29.10}$$

again with diagonal multiplier3. This follows by the power formula in (D.29.5). In particular the multiplier retains $\lambda \equiv 1 \pmod{72}$ at every finite stage.

The program `d_cm_pro2_class2.py` computes this one constant matrix modulo four and its map (D.29.7). The latter again has rank 15, with explicit basis certificates. Hence its image contains twice the entire target module. To verify this last implication, let $2e$ be any target. Surjectivity of (D.29.7) supplies $z \in \ker L_0$ with $Lz/2 - e \in \operatorname{im} L_0$; an additional correction $2z'$ then makes $L(z + 2z') = 2e$ modulo four. The program also verifies an actual next lift: a 4448112-letter cusp word vanishes in the complete $\mathbb{Z}/8$ Magnus model. This yields the stated multiplier289 and kernel order 2^{14356}.

We now construct a compatible infinite family, rather than infer one from a few finite examples. Start with the recorded $H^{(2)}$ action. Lift its two conjugators and multiplier arbitrarily one level higher; its cusp error lies in $2C_2$. The surjectivity just proved corrects

that error by elements of C_2 and the specified multiplier increment. These corrections fix the action on $H^{(1)}$. More generally, at stage k the same constant matrix corrects the error in $2C_k$, preserves the action on $H^{(k-1)}$, and produces an automorphism of $H^{(k+1)}$. The homology parities never change, so the stabilization hypotheses remain satisfied at every step. Although the chosen action at level k can change once, every level $k-1$ has already stabilized. Taking these stabilized reductions gives compatible actions, conjugators and multiplier on the entire inverse system. The multiplier increments tend to zero two-adically. Every finite action is an automorphism by the same rank 97 Frattini argument, so their inverse limit is an automorphism.

The characteristic quotients in (D.29.1) are cofinal among the finite two-group quotients of the free class-two completion of N. This proves the exact kernel and quotient map in (D.29.2). The construction fixes the reduction of the recorded $H^{(2)}$ witness to $H^{(1)}$; it does not claim that every central coordinate of the $H^{(2)}$ witness remains fixed. Both finite representatives are retained in the output files, so the comparison is explicit.

For every odd p, the prescribed image $q(\beta)$ centralizes s in Q. It lifts to $C_{\mathcal{P}_p}(s)$ by coprime order-two conjugacy, using the proof in D.27. At three retain the particular G_3 conjugator there; at the other odd primes lift its Q image. Take $A = 1$ and the identity action on each $\mathcal{P}_p$. These actions have the same elliptic reductions and cusp multiplier component1. Finite kernels of distinct prime-power order have no common nontrivial quotient, so the map from $\widehat{B}$ to their fibre product with (D.29.2) is onto. Passing to all finite subproducts proves the exact quotient map and kernel in (D.29.3). The two-primary multiplier is 1 modulo 16; its odd components are 1. This supplies a single $\lambda \in \widehat{\mathbb{Z}}^{\times}$ agreeing on all common finite cusp quotients.

Choose one lift of the combined β to $\widehat{F}$. The cusp gauge of D.26 then makes its associated f a commutator and supplies both hexagons at every finite level. It fixes the CM coordinates. They still disagree on the common field of D.25, so every member retaining the G_6 and CM data is nonarithmetical. D.28's additional mixed-prime quotient was verified with these same homology parities, including the right-centralizer correction, and the same G_3 conjugators. Its exact common quotient with (D.29.3) is the C computed there. Thus the two actions agree on C and define (D.29.4). This last step uses the specified quotient maps and compatibility, not only the equality of their CM matrix pairs. □

A rank-four nonabelian obstruction and its compatible replacement

D.30. The first new rank-four layer and an actual class-three kernel [Proved]

The quotient map

$$Q \longrightarrow Q_0 := Q/O_2(Q) \simeq S_3 \times C_3 \tag{D.30.1}$$

has kernel $O_2(Q) \simeq C_2^2 \times Q_8$ of order 32. Put $K = \ker(B \to Q_0)$. Then K is free of rank 4, $N \triangleleft K$ has index 32, and the relative pro-two completion $\mathcal{P}_2$ has free pro-two kernel $K^{(2)}$ over Q_0. Its index 32 subgroup is the rank 97 group $N^{(2)}$ used previously.

Write $D_j(K)$ for the mod-two Zassenhaus filtration and set

$$T_0 = B/D_6(K), \qquad T_1 = B/D_7(K). \tag{D.30.2}$$

Their orders are $576 \cdot 2^{303}$ and $576 \cdot 2^{993}$, respectively. The new central layer $Z = D_6(K)/D_7(K)$ has dimension 690. The image

$$W = \gamma_3 N\, D_7(K)/D_7(K) \subset Z \tag{D.30.3}$$

has dimension 240. Central cusp corrections have rank 227 on W and 648 on Z.

Both the specific alternate H_2 representative recorded in D.27 and the specific changed $H^{(1)}$ representative retained in D.29 have no genuine full profinite lift. This conclusion uses their retained G_6 action jointly with their T_0 action. It follows after exhausting the entire lower N-centralizer freedom: the image of N in T_0 has central Frattini subgroup of dimension 225 and Frattini quotient of dimension 78; its full s-centralizer is generated by 152 explicitly computed elements. Their defects raise the Z correction rank to 654, and each recorded cusp target remains outside that image. Separating linear functionals are saved with the exact rank certificates. This excludes these representatives, not every action with their CM pair or their G_6 data.

There is a further explicit replacement with the same homology masks, right-centralizer data, G_3 conjugators and extra CM pair. Its simultaneous correction matrix has rank 28: fifteen conditions retain the $\mathbb{Z}/4$ class-two lift and thirteen remove the new W obstruction. The resulting 2944468-letter cusp word vanishes both in the full rank 97 $\mathbb{Z}/4$ model and in the full rank-four Magnus model through degree six. Its multiplier is 1. Thus it gives compatible automorphisms on $H^{(2)}$ and T_1, with exact common quotient

$$C_* = B/(\gamma_3 N\, D_7(K)), \qquad |C_*| = 576 \cdot 2^{753}, \quad |H^{(2)} \times_{C_*} T_1| = 576 \cdot 2^{9843}. \tag{D.30.4}$$

The replacement extends through the entire stable tower of D.29 while retaining its T_1 action and conjugators. Its two-primary target is the precise quotient

$$\mathcal{S}_2 = \mathcal{P}_2/(\overline{\gamma_3 N^{(2)}} \cap D_7(K^{(2)})) \simeq \mathcal{P}_2^{[2]} \times_{C_*} T_1. \tag{D.30.5}$$

Its kernel over Q is nilpotent of class exactly three: its third lower-central subgroup is $W \simeq C_2^{240}$, and its fourth is trivial. The same action extends over every odd relative pro-prime kernel and the compatible mixed-prime quotient J of D.28. One global cusp gauge gives nonarithmetical finite shadows on the characteristic refinements retaining the G_6 and CM data. The genuine full profinite realization of this replacement and of the extra CM pair remains undecided.

Proof. The map (D.30.1) is induced by reduction modulo two and by reduction modulo three followed by $\mathrm{SL}_2(\mathbb{F}_3) \to \mathrm{SL}_2(\mathbb{F}_3)/Q_8 = C_3$. Both maps kill the global sign, and Chinese remainders make their joint image $S_3 \times C_3$. Its kernel has order 32 and is exactly the subgroup used in D.27. The images of s and r retain orders two and three. Thus K acts freely on the Bass–Serre tree. Its quotient graph has 18 edges and 15 vertices, so its fundamental group has rank 4. The same graph argument gives $[K : N] = 32$ and rank 97 for N.

The induced pro-two topology on N is its full pro-two topology. Indeed the intersection of the finitely many K-conjugates of a normal subgroup of two-power index in N still has two-power index in N, and then in K. Conversely restriction of a finite two-group quotient of K gives one of N. Taking the finite normal cores under B preserves this property. This proves the relative-completion comparison and its specified kernel maps, without identifying completions from ranks alone.

The Zassenhaus quotient is evaluated by the Magnus map over $\mathbb{F}_2$, retaining all noncommutative monomials of degree less than its index. The exact Hall-power basis is [Efr23, Proposition 4.4 and its proof]; its Lyndon standard factorization is Lemma 2.2 there. For four free generators the dimensions in degrees one through six are

$$4, 10, 20, 70, 204, 690. \tag{D.30.6}$$

For degree six the program uses the 670 ordinary Lyndon brackets and the squares of the 20 degree-three Lyndon brackets. It checks their independence in the 4096-dimensional associative tensor space. Summing (D.30.6) gives308 through degree five and 998 through degree six. Dividing the corresponding kernel orders by $[K : N] = 32$ proves the orders in (D.30.2).

The map $K/D_2(K) \to O_2(Q)^{\mathrm{ab}}$ is an isomorphism of four-dimensional spaces, so $N \subset D_2(K)$. The group $O_2(Q)$ has exponent four and central commutator subgroup of order two, hence $D_3(K) \subset N$. Its degree-two relation space $R_2 \subset D_2(K)/D_3(K)$ has dimension nine. All these maps are also checked from the actual 97 words of the original N basis in the new rank-four graph. Their degree-one terms vanish and their degree-two terms span exactly R_2. The filtration laws give

$$\gamma_3 N \subset D_6(K), \qquad [N,N]^2 N^4 \subset D_8(K). \tag{D.30.7}$$

These laws follow directly from multiplication of the Magnus series: commutator degrees add and squaring doubles the least degree. The degree-six image of $\gamma_3 N$ is therefore $[R_2,[R_2,R_2]]$. The code forms all 729 ordered triple brackets and finds rank 240. It retains a basis made of actual triple commutators of N words. This proves (D.30.3) and provides a quotient map $H^{(1)} \to T_0$.

The initial selected cusp word is evaluated in the rank-four model. Its terms below degree six vanish, and its degree-six term lies in W, as (D.30.7) requires. For central corrections $b, a \in Z$ the cusp-error map is

$$(1-s)b + s(1-r)a. \tag{D.30.8}$$

Here conjugation on Z factors through Q_0; the kernel K acts trivially. Tensor substitution of the exact four-dimensional graph matrices gives ranks 227 on W and 648 on Z. Both selected targets fail these fixed-conjugator tests.

We next exhaust the remaining lower conjugators, so the negative conclusion has a precise source. Let P be the image of N in T_0. It is class two of exponent four by (D.30.7), and has order 2^{303}. Its Frattini subgroup $A_0 = P^2[P,P]$ is central elementary abelian. The program evaluates all 97 generator squares and 4656 pairwise commutators in the rank-four Magnus model. Their exact span has dimension 225. Collection modulo that span gives the 78-dimensional elementary abelian quotient $V_0 = P/A_0$; the dimensions sum to 303, and the constructed words generate all of P.

Write s also for its conjugation action. The connecting map

$$V_0^s \longrightarrow A_0/(1-s)A_0, \qquad \bar{c} \longmapsto [c,s] \bmod (1-s)A_0 \tag{D.30.9}$$

is well-defined. It is additive because A_0 is central in P, and changing a lift changes the defect by $(1-s)A_0$. Exact computation gives

$$\dim A_0^s = 117, \quad \dim V_0^s = 42, \quad \operatorname{rank}(D.30.9) = 7. \tag{D.30.10}$$

Choose a basis of its 35-dimensional kernel and correct each lift by an element of A_0 to centralize s. Together with a basis of A_0^s, these are 152 actual generators of $C_P(s)$: their images generate the quotient and their central kernel is precisely A_0^s. The code computes every corrected generator with full degree-six terms, not just its image in P.

Its further defect in Z is additive modulo $(1-s)Z$, since $N \subset K$ acts trivially on Z. Thus adjoining these 152 columns accounts for the entire lower s-centralizer freedom. The other elliptic centralizer lifts through the two-kernel by coprime order-three conjugacy. Moreover $q_0(A) = q_0(\beta) = 1$ for the specified CM pair, so no omitted conjugation changes the new defect columns. The full augmented matrix has rank 654. Each of the two selected target vectors is outside its image; the saved separating functional annihilates every column and takes value one on that target.

There is no remaining multiplier column. The actual rank-four word $(sr)^6$ has degree-one vector $(1,1,0,1)$. Its fourth power has a nonzero degree-four term, while its eighth power has no term through degree six. Thus $t = sr$ has order 48 in both T_0 and T_1. Every multiplier change preserving the selected T_0 action is already trivial on T_1.

For a genuine full lift, the original Herfort–Ribes torsion theorem used in D.27 supplies elliptic conjugators in $\widehat{F}$. The retained G_6 action forces their Q coordinates by (D.25.6). This use of G_6 is essential: T_0 alone does not retain all its homology data. Differences between the chosen and new conjugators therefore lie in N, and the preceding finite centralizer computation exhausts their reductions. Undoing the specified cusp gauge retains the cusp power. The two nonzero separating values prove the stated absence of a genuine lift for these selected representatives. No assertion that arbitrary torsion in T_1 is conjugate into its elliptic factors is used.

We now construct the replacement. It changes central coordinates explicitly, retaining the homology data. Use $C_1 \simeq (\mathbb{Z}/4)^{4753}$ from D.29, with basis the squares and commutators of the 97 N generators. Their images in $K/D_7(K)$ have degree at least four, so commute there as well. Hence their cusp effects are linear in both target groups. The degree-six action on these images is computed from the actual elliptic conjugates, including their degree-one and degree-two path terms; replacing those conjugates by their Q_0 matrices would lose terms. For the A-column one may use the action of t in place of the product of the two elliptic conjugates: their discrepancy has degree six, whose conjugation is trivial on elements starting in degree four.

The 4769-dimensional relation space of the mod-two central correction matrix has two exact obstruction maps. The first is D.29's fifteen-dimensional obstruction to lifting that correction modulo four. The second is the thirteen-dimensional quotient of W by (D.30.8). The joint map has rank 28. The code solves for first obstruction zero and second obstruction equal to the selected cusp defect. An upper central correction then preserves the full $\mathbb{Z}/4$ cusp relation. Its image starts in $D_8(K)$, so it does not affect the degree-six test. Finally the retained triple-commutator basis of W removes that degree-six error. Those corrections vanish identically in every class-two quotient of N. This gives one pair of actual words satisfying both equations simultaneously.

The result is verified directly by `d_cm_rank4_joint.py`. Its 2944468-letter B word collapses to 562814 letters in the rank-four free basis. Every degree-one and degree-two coordinate vanishes in the rank 97 $\mathbb{Z}/4$ model, and every coordinate in degrees one through six vanishes in the rank-four mod-two model. The multiplier is 1. The map on $N^{\mathrm{ab}}/2$ retains rank 97. The map on $K/D_2(K)$ is the identity, because its identification with $O_2(Q)^{\mathrm{ab}}$ is preserved and the action on Q is the identity. The finite two-kernel Frattini argument therefore proves automorphisms of both quotients.

The kernel defining $H^{(2)}$ is $\gamma_3 N[N,N]^4 N^8$. The last two factors lie in $D_{16}(K)$, hence in $D_7(K)$. Therefore its product with $D_7(K)$ is exactly $\gamma_3 N D_7(K)$. This proves the common-quotient formula (D.30.4). The kernel of $T_1 \to C_*$ is exactly W of order 2^{240}, giving all three stated finite orders. The common conjugator words prove compatibility on C_*, and the normal-subgroup product formula proves surjectivity onto the fibre product.

Restart D.29's stable iteration from this new seed. Every later correction in C_k, $k \geq 2$, lies in $D_8(K)$: its generators are powers N^{2^k} and $[N,N]^{2^{k-1}}$. The corresponding multiplier change acts trivially on T_1. Thus the entire iteration retains this finite action and its exact conjugators. The inverse-limit normal-subgroup formula gives (D.30.5). Its third lower-central subgroup over Q is the image of $\gamma_3 N$, namely W. Since $[N, \gamma_3 N] \subset D_8(K)$, its fourth lower-central subgroup is trivial. The nonzero dimension 240 proves that the class is exactly three.

The odd-prime centralizer construction of D.29 is unchanged. The mixed quotient J still agrees on $H \times_Q G_3$: all changes lie in $N^2[N,N]$ or $\gamma_3 N$ on the two-primary side, and the three-primary conjugators are retained as separately chosen components of the fibre product. The displayed discrete two-primary correction word need not itself have the original G_3 abelianization. The maximal common quotient of J with the new primewise system is the

same one, because its finite kernel over Q has maximal nilpotent quotient computed in D.28. The combined conjugators therefore lift to a single pair in the resulting fibre product. The one global cusp gauge proves the stated compatible finite shadow family and its nonarithmeticality on the refinements retaining the CM data. Further finite quotients remain in D.22's genuine-lift test. □

A compatible degree-seven replacement

D.31. A class-three two-primary kernel with third subgroup of rank 1860 [Proved]

Retain $K, N, \mathcal{P}_2$ and the Zassenhaus filtration from D.30. Put

$$T_2 = B/D_8(K), \qquad U = \gamma_3 N\, D_8(K)/D_8(K), \qquad C_{**} = B/(\gamma_3 N\, D_8(K)). \tag{D.31.1}$$

The group U is elementary abelian of dimension 1860. Its degree-six quotient has dimension 240 and its degree-seven subgroup dimension 1620. Conjugation makes U an actual Q-module. Its cusp-correction map has rank 1810 and cokernel dimension 50.

The fixed homology and CM data of D.30 admit another explicit central replacement. Its fifty degree-seven conditions, together with the fifteen class-two $\mathbb{Z}/4$ conditions, form a map of rank 65 on the 4769-dimensional central relation space. The resulting 4,903,248-letter cusp word vanishes in both complete models: the rank-97 class-two Magnus model over $\mathbb{Z}/4$, and the rank-four mod-two Magnus model through degree seven. Its multiplier is 1. This gives compatible automorphisms of $H^{(2)}$ and T_2, with

$$|T_2| = 576 \cdot 2^{3333}, \quad |C_{**}| = 576 \cdot 2^{1473}, \quad |H^{(2)} \times_{C_{**}} T_2| = 576 \cdot 2^{11463}. \tag{D.31.2}$$

The same stable correction construction extends this seed to the precise quotient

$$\mathcal{S}_2^{(7)} = \mathcal{P}_2/(\overline{\gamma_3 N^{(2)}} \cap D_8(K^{(2)})) \simeq \mathcal{P}_2^{[2]} \times_{C_{**}} T_2. \tag{D.31.3}$$

Its kernel over Q has class exactly three, with third lower-central subgroup C_2^{1860} and trivial fourth subgroup. Its multiplier has two-adic component in $1+16\mathbb{Z}_2$. The odd-prime components can all be 1, with the original odd-primary conjugators retained separately. The action combines with every odd relative pro-prime completion and D.28's mixed quotient J.

There is a canonical surjection

$$\pi : \mathcal{S}_2^{(7)} \longrightarrow \mathcal{S}_2 \tag{D.31.4}$$

to D.30's target, whose kernel is C_2^{1620}. The recorded new automorphism induces an automorphism α_6' there, with $\pi\alpha_7 = \alpha_6'\pi$. It is an explicitly different central representative from the recorded D.30 automorphism α_6: their reductions already differ on the image of s in $H^{(1)}$. The nonzero central difference is recorded in the reduction certificate. Thus this step replaces the earlier action within its fixed homology class; it does not assert that the earlier central coordinates were all retained.

One global cusp gauge gives compatible nonarithmetical finite shadows on characteristic refinements retaining the G_6 and CM data. D.32 below proves that the recorded T_2 action has no peripheral lift to the full relative pro-two source, hence no genuine full profinite lift. The attainability of the same extra CM coset by a different action remains undecided.

Proof. Modulo $D_4(K)$ every element of N is a product of the nine chosen degree-two relation lifts and elements of $D_3(K)/D_4(K)$, whose twenty Lyndon brackets are retained. A triple commutator modulo $D_8(K)$ uses only degree patterns $(2,2,2)$ and permutations of $(3,2,2)$. All non-Lie correction terms begin in degree eight. Products of these degree-six and degree-seven

elements are additive, and their squares vanish. Conjugation by a degree-one element of K adds a degree-seven term $[L_1, W]$. Jacobi expresses it using the same generators, because $[L_1, R_2] \subset L_3$. Thus the normal closure contributes no omitted degree-seven generators.

The program retains the 240 actual triple-commutator words chosen in D.30, including their degree-seven terms. It adjoins the brackets

$$[r_i, [r_j, z_k]], \qquad 1 \le i, j \le 9, \quad 1 \le k \le 20, \tag{D.31.5}$$

where the r_i are the degree-two relation lifts and the z_k are the degree-three Lyndon lifts. Their leading degree-seven terms have rank 1620. The earlier 240 basis elements have independent degree-six terms, so the combined rank is 1860. This proves the exact description of U, with its displayed filtration. Since $[N, U] \subset [D_2(K), D_6(K)] \subset D_8(K)$, its conjugation action factors through $Q = B/N$.

The action retains the extension between degrees six and seven. The 240 inhomogeneous basis words are conjugated by the actual B words s and sr and expanded through degree seven. Only the pure degree-seven basis uses the leading tensor action. The actual elliptic images may use these same module actions: $q(\beta)$ centralizes s in Q, so $\beta s \beta^{-1} s^{-1} \in N$, whose conjugation on U is trivial; likewise the A conjugate of r differs from r by an element of N. This proves the precise action entering

$$(1 - s)b + s(1 - r)a \quad (b, a \in U). \tag{D.31.6}$$

Its exact rank is 1810. The code obtains all fifty annihilating functionals from this actual module.

For simultaneous correction, use C_1 from D.29. Its generators have degree at least four in K, so its image is abelian modulo $D_8(K)$. Its effect in that quotient depends only on its coefficients modulo two. In the N class-two $\mathbb{Z}/4$ model it is the earlier abelian module of exponent four. The degree-seven cusp effects use the actual elliptic conjugate path through degree three and the images of the 97 N generators through degree five. Here the degree-two action of N does matter on C_1; the calculation therefore does not replace it by the leading Q_0 tensor matrices.

On the 4769-dimensional relation space, impose the fifteen conditions for lifting the correction modulo four and the fifty functionals from (D.31.6). The joint rank is 65. Its solved target retains zero in the first fifteen coordinates and cancels the new cusp obstruction in the last fifty. Functionals on U are extended linearly for evaluating individual columns; this introduces no assumption, because the class-two conditions put the final combined error in U. An upper twice-central correction then restores the complete $\mathbb{Z}/4$ equation and lies in $D_8(K)$, so it changes no term of the other equation. Finally the actual triple-commutator words representing U remove its error. They vanish in every class-two quotient of N.

The resulting 4,903,248-letter B word is evaluated directly in both group models. All rank-97 degree-one and degree-two coefficients modulo four vanish, and all rank-four coefficients in degrees one through seven modulo two vanish. Its multiplier is 1. The retained homology map on $N^{\mathrm{ab}}/2$ has rank 97; the action on $K/D_2(K)$ is the identity through $O_2(Q)^{\mathrm{ab}}$. The finite two-kernel Frattini argument proves automorphisms of both finite quotients, in addition to their cusp equations.

The next restricted Witt dimension is $\dim D_7(K)/D_8(K) = (4^7 - 4)/7 = 2340$, using the same Hall-power basis in [Efr23, Proposition 4.4]. The sum through degree seven is 3338. Hence $|T_2| = 18 \cdot 2^{3338} = 576 \cdot 2^{3333}$. The kernel of $T_2 \to C_{**}$ is exactly U, of order 2^{1860}. As in D.30,

$$(\gamma_3 N [N, N]^4 N^8) D_8(K) = \gamma_3 N D_8(K), \tag{D.31.7}$$

because the last two factors lie in $D_{16}(K)$. This proves the exact common quotient, the remaining orders in (D.31.2), and the canonical surjection from B onto the finite fibre product. The same two-primary conjugator words give agreement there.

Restart the stable iteration from the new seed. Its G_2 action has not changed, so the stabilized correction matrices and their fifteen-dimensional surjectivity certificate are unchanged. Every later C_k correction, $k \geq 2$, lies in $D_8(K)$, so the entire T_2 action and its conjugators stay fixed. The cusp t still has order 48 in T_2: the degree-one term of t^6 is nonzero, its fourth power has a nonzero degree-four term, and its eighth power starts in degree eight. The later multiplier increments are divisible by 144 and therefore act trivially on T_2. They converge to a two-adic multiplier in $1 + 16\mathbb{Z}_2$. The inverse-limit normal-subgroup formula gives (D.31.3). Its third lower-central subgroup over Q is U, and its fourth is trivial by $[N, \gamma_3 N] \subset D_8(K)$. The nonzero third subgroup proves that the class is exactly three.

The inclusion $D_8(K) \subset D_7(K)$ gives (D.31.4). Its kernel is the degree-seven part of the image of $\gamma_3 N$, of dimension 1620. The new automorphism preserves this quotient: it preserves the preimages of N and K over Q, their lower-central subgroups, and the characteristic Zassenhaus filtration. The comparison of recorded actions is checked separately in `d_cm_degree7_reduction.py`. Let z be the lower beta central-change vector and M_s the rank-97 homology matrix. Its associated symmetric matrix Z has nonzero difference

$$Z + M_s Z M_s^{\mathsf{T}}. \tag{D.31.8}$$

The twice-central and triple-commutator corrections vanish in $H^{(1)}$, so (D.31.8) is precisely the change of the s image there. This proves $\alpha_6' \neq \alpha_6$ by an explicit quotient map, while the original G_6 homology data remain fixed.

For odd primes choose the original conjugators as separate components, using the coprime centralizer construction. The discrete two-primary word is not asserted to preserve its odd-primary abelianization. The new two-primary kernel is a pro-two group, so the maximal common quotient with J is still the $H \times_Q G_3$ proved in D.28 after adjoining the odd components. The homology and separately chosen three-primary data agree there. The exact fibre product consequently carries one combined action and compatible conjugators. The global cusp gauge supplies the claimed finite shadow family. The quotients (D.31.1)–(D.31.4) specify the retained information. D.32 applies the next finite quotient to the recorded action and proves a complete obstruction there; the quotient constructions proved here remain valid. □

The next obstruction for the recorded degree-seven action

D.32. An exhaustive degree-eight obstruction for the recorded representative [Proved]

Let α_7 be the specific automorphism of $T_2 = B/D_8(K)$ recorded in D.31, and put

$$T_3 = B/D_9(K), \qquad Z_8 = D_8(K)/D_9(K) \simeq C_2^{8230}. \tag{D.32.1}$$

The natural map $T_3 \to T_2$ has kernel Z_8 and $|T_3| = 576 \cdot 2^{11563}$. The recorded α_7 has no lift to T_3 that sends s, r to conjugates of s, r and sends the cusp $t = sr$ to a power compatible with its recorded lower image.

This is a complete test for that lower action. The fixed-conjugator correction map on Z_8 has rank 7753. The multiplier contributes one more independent column. The full lower centralizer is

$$C_{N/D_8(K)}(s) \simeq C_2^{1666}; \tag{D.32.2}$$

its degree-four through degree-seven dimensions are $37, 102, 348, 1179$. Adding all its defects gives rank 7777. An explicit linear functional annihilates that entire image and evaluates to one on the recorded cusp error.

Moreover, α_7 has no automorphism lift to the full relative pro-two group $\mathcal{P}_2$ that maps t to a power of itself. Consequently the specific compatible quotient action recorded in D.31 has no genuine full profinite lift. Its quotient automorphisms remain valid. The attainability of the same CM pair by a different action is still undecided.

Proof. The exact error and the next layer. The seed program reconstructs the recorded D.31 conjugators, including every upper central and triple-commutator correction in their stated order. It evaluates the full $4,903,248$-letter cusp word in the rank-four Magnus algebra through degree eight. Degrees one through seven vanish. The degree-eight coefficient has 32,296 nonzero monomials. No change of the lower representative enters this evaluation.

The restricted Hall basis in [Efr23, Proposition 4.4] consists in this degree of 8160 ordinary degree-eight Lyndon brackets, 60 squares of degree-four brackets, six fourth powers of degree-two brackets, and four eighth powers of the generators. Its dimension is 8230. The dimension sum through degree eight is 11568; multiplying by $|Q_0| = 18$ proves the order in the statement. Conjugation on this central K-layer factors through Q_0. Since the two recorded conjugators lie in K, their corrections have precisely the map

$$(b, a) \longmapsto (1 - s)b + s(1 - r)a, \qquad b, a \in Z_8. \tag{D.32.3}$$

Its rank is 7753.

The cusp has order 48 in T_2 and order 96 in T_3. Indeed the nonzero degree-one term of t^6 has eighth power in degree eight and sixteenth power in degree sixteen. Direct evaluation of t^{48} gives 6561 nonzero degree-eight monomials. Every multiplier change retaining the lower action therefore contributes either zero or this one column. The change by 144 is compatible also with the recorded finite-shadow multiplier and has the same column, since $144 \equiv 48 \pmod{96}$. Adding it raises the rank to 7754; the target remains outside. This includes the multiplier freedom that first becomes visible in this transition.

The retained homology quotient. In $K/\Phi(N)$ the quotient by $N/\Phi(N)$ is $O_2(Q) = V_4 \times Q_8$, of class two and exponent four. The image of $\gamma_3 K$ in the elementary abelian module $N/\Phi(N)$ is generated by the conjugates of the 64 words $[a_i, [a_j, a_k]]$, with $1 \le i, j, k \le 4$. Successive multiplication by the four augmentation operators gives the exact dimensions

$$87,\ 67,\ 43,\ 19,\ 4,\ 0 \quad \text{for } \gamma_j K, \quad 3 \le j \le 8. \tag{D.32.4}$$

The other factors of $D_8(K)$ are fourth powers from $\gamma_2 K$, squares from $\gamma_4 K$, and eighth powers from K. Their half-powers already lie in N, so they vanish in $N/\Phi(N)$. Thus $D_8(K) \subset \Phi(N)$ and the natural map from B to G_2 factors through T_2.

Exhausting the lower centralizer. The code uses the exact involution on the chosen free basis of K:

$$a_1 \mapsto a_1^{-1}, \quad a_2 \mapsto a_2^{-1}, \quad a_3 \mapsto a_4^{-1}, \quad a_4 \mapsto a_3^{-1}. \tag{D.32.5}$$

Its Magnus substitution retains every coefficient through degree eight. It is checked against the actual Bass–Serre words.

For $n \ge 3$, the kernel of $N/D_{n+1}(K) \to N/D_n(K)$ is the full restricted layer $Z_n = D_n(K)/D_{n+1}(K)$. It is central in the former group. The connecting map

$$C_{N/D_n(K)}(s) \longrightarrow Z_n/(1 - s)Z_n \tag{D.32.6}$$

which sends an element to its commutation defect is therefore a genuine group homomorphism. The program maintains a full polycyclic sequence for the source, including all its central

coordinates. It eliminates the connecting columns in descending sequence order. Each remaining generator has its own earlier leading coordinate and uses only later pivot generators for its correction. The resulting products generate the complete kernel: their filtered leading coordinates are independent, so their ordered products give at least the required number of elements, while the kernel order is exactly the source order divided by the computed image order. Lifting these products and adjoining $\ker(1 - s : Z_n \to Z_n)$ gives the entire next centralizer.

The initial centralizer in $N/D_3(K)$ has dimension six. The successive exact counts are

n	3	4	5	6	7
$\dim Z_n$	20	70	204	690	2340
$\dim Z_n^s$	11	40	105	355	1179
rank(connecting map)	1	6	3	13	7
$\log_2 \lvert C_{N/D_{n+1}(K)}(s)\rvert$	16	50	152	494	1666

(D.32.7)

Every lifted generator is directly checked fixed at its asserted level. The final generators all have degree at least four. Their products commute and their squares vanish modulo $D_8(K)$, proving (D.32.2), including its exponent and the displayed degree distribution.

Changing the right s-conjugator by these elements adds their degree-eight defects. The preceding conjugator belongs to K, which acts trivially on Z_8, so no further conjugation factor is suppressed. This exhausts the finite conjugator choices as well. An s-conjugator for any proposed lift can be normalized into F by multiplying it by s, since its image in S_3 centralizes the transposition. Its lower difference from the recorded conjugator then belongs to F and centralizes s in T_2. Projecting this difference to G_2 and applying D.25's projected-centralizer calculation puts it in $N/D_8(K)$. Thus (D.32.2) includes every such lower difference, not only ones initially chosen inside N. The defect map into the cokernel of (D.32.3) is a group homomorphism. All 1666 generators are included. The order-three centralizer lifts across a finite two-kernel, by conjugacy of the two order-three complements, and contributes no additional classes. The final rank is 7777. The certificate contains a functional on the 65536-dimensional degree-eight tensor space, verifies its vanishing on every row of the image basis and every lower-centralizer defect, and verifies its value one on the target. This proves the negative finite conjugator test. The parametrization also accounts for generation. Its tuples define homomorphisms from B that induce the recorded lower action and the identity on Q_0. The relative pro-two universal property extends them to $\mathcal{P}_2$, and the characteristic filtration gives the induced maps of T_3. Their maps on $K/D_2(K)$ are the recorded isomorphism, so the finite Frattini theorem makes them surjective, hence automorphisms. The proved emptiness occurs before any further generation condition is needed.

Promotion to the actual relative pro-two source. Let S_0 be the preimage of $\langle s\rangle$ in $B \to Q_0$. Its index is nine. The Bass–Serre quotient has nine edges, six vertices over the order-two factor, and three over the order-three factor. Three of the first six vertices have stabilizer C_2, and all other stabilizers are trivial. Its first Betti number is one. Thus

$$S_0 \simeq C_2 * C_2 * C_2 * \mathbb{Z}, \qquad S \simeq C_2 \amalg_2 C_2 \amalg_2 C_2 \amalg_2 \mathbb{Z}_2, \tag{D.32.8}$$

where S is its closure in $\mathcal{P}_2$. The second map is exactly the pro-two completion: K has index two in S_0, and the normal core in B of any normal subgroup of K with finite two-power index still has two-power index in K. Consequently the relative topology induces the full pro-two topology on S_0.

The original [HR85, §2, Theorem A′ and Theorem B′] applies with the class of finite two-groups. It proves that the involutions of S are conjugate into the three vertex factors, whose generators are themselves discrete B-conjugates of s. As S is a Sylow pro-two subgroup,

every involution of $\mathcal{P}_2$ is therefore conjugate to s. The same graph also gives $C_{\mathcal{P}_2}(s) = \langle s \rangle$: the normalizer quotient of S is C_3, acting cyclically on its three distinct involution classes; an element centralizing s projects into $C_{Q_0}(s)$ and hence normalizes the full preimage S of $\langle s \rangle$. Its action must fix the class of s, so its image in this C_3 is trivial. It therefore lies in S, where Theorem B′ computes its centralizer.

Now an automorphism lifting α_7 must send s to a conjugate. Its image of r is conjugate to r with the prescribed lower conjugator, because the kernel of $\mathcal{P}_2 \to T_2$ is pro-two and the order-three complements are conjugate. Normalize the s-conjugator into the preimage F of the identity in S_3 by multiplying it by s when necessary. The corresponding conjugator of D.31 is already in F. Their lower difference centralizes s. Because T_2 retains G_2, D.25's projected-centralizer calculation forces this difference into $N/D_8(K)$. The complete test (D.32.2)–(D.32.7) therefore applies to every such lift, giving a contradiction. The finite maps and the full source have now been related explicitly. This excludes the recorded representative, while leaving changes to its lower action within the same CM pair available for further testing. □

A degree-eight replacement compatible with the infinite class-two tower

D.33. A class-four two-primary kernel with third subgroup of rank 9105 [Proved]

This addresses the genuine realization question in Part D(e), manuscript §51.16, folio 734 verso, scan 174, and the two-CM-coordinate follow-up to §51.15, folios 725–727, scans 156–160. Use $K, N, Q, \mathcal{P}_2$ and $H^{(k)}$ from D.29–D.32, and set

$$T_3 = B/D_9(K), \qquad U_8 = \gamma_3 N\, D_9(K)/D_9(K), \qquad C_{***} = B/(\gamma_3 N\, D_9(K)). \tag{D.33.1}$$

The elementary abelian group U_8 has dimension 9105, with successive degree-six, degree-seven and degree-eight dimensions $240, 1620, 7245$. The image of $\gamma_4 N$ in T_3 has dimension 1620, and the image of $\gamma_5 N$ is trivial. There is an explicit new peripheral automorphism of $H^{(2)} \times_{C_{***}} T_3$, with multiplier 1, retaining the homology and extra CM pair of D.27. The exact orders are

$$|T_3| = 576 \cdot 2^{11563}, \qquad |C_{***}| = 576 \cdot 2^{2458}, \qquad |H^{(2)} \times_{C_{***}} T_3| = 576 \cdot 2^{18708}. \tag{D.33.2}$$

Its complete $12,921,196$-letter cusp word is the identity in both full finite models. The independently collapsed words have $2,128,528$ letters in the rank-four K basis and $836,592$ letters in the rank-97 N basis.

The T_3 component of this action extends to the precise quotient

$$\mathcal{S}_2^{(8)} = \mathcal{P}_2/(\overline{\gamma_3 N^{(2)}} \cap D_9(K^{(2)})) \simeq \mathcal{P}_2^{[2]} \times_{C_{***}} T_3. \tag{D.33.3}$$

The first lookahead may replace the initial $H^{(2)}$ central coordinates; the entire T_3 action is retained. The kernel over Q has class exactly four, with

$$\gamma_3 = C_2^{9105}, \qquad \gamma_4 = C_2^{1620}, \qquad \gamma_5 = 1. \tag{D.33.4}$$

The compatible multiplier has two-adic component in $1 + 32\mathbb{Z}_2$; all odd-prime components can be 1. An actual first continuation to the rank-97 class-two model over $\mathbb{Z}/8$ has multiplier 1 and an $18,077,708$-letter cusp word. It retains the entire T_3 action. The resulting action combines with all odd relative pro-prime completions and D.28's mixed quotient J along their specified common quotient $H \times_Q G_3$.

There is a canonical surjection

$$\mathcal{S}_2^{(8)} \longrightarrow \mathcal{S}_2^{(7)} \quad \text{with kernel } C_2^{7245}. \tag{D.33.5}$$

The induced action on its target replaces the recorded D.31 action: the images of s first differ in degree five of the K Magnus expansion. Thus D.32's obstruction to that recorded action is retained. The new action supplies compatible nonarithmetical shadows on characteristic refinements retaining the G_6 and CM data. Attainability of the same CM pair by a genuine full profinite peripheral automorphism remains undecided. D.45 later proves that this particular recorded degree-eight action has no full relative pro-two peripheral lift.

Proof. We first determine the common quotient using actual group generators. The images of $D_j(K)$ in $N/\Phi(N)$, for $3 \le j \le 8$, have dimensions $88, 68, 43, 19, 4, 0$. Choose independent images starting with the deepest layer, using the retained Hall-power words, and then adjoin the nine degree-two relation words. This gives the following basis of the 97-dimensional Frattini quotient:

least K degree	2	3	4	5	6	7
number of generators	9	20	25	24	15	4.

(D.33.6)

All generators are explicit words in B. They are a basis of $N^{(2)}$: their images generate its Frattini quotient, so the endomorphism of the free pro-two group of rank 97 sending its original basis to these words is surjective, hence an automorphism. The change of basis and its homology matrix are retained in the certificate. This argument makes no assertion that the weighted basis is an abstract discrete free basis.

Collection in these actual generators bounds the image of $\gamma_3 N$ modulo $D_9(K)$. Commutators of length at least five have degree at least ten. The contributing basic commutators of length three and four have weighted counts

degree and pattern	number
$6 : (2,2,2)$	$(9^3 - 9)/3 = 240$
$7 : (2,2,3)$	$9^2 \cdot 20 = 1620$
$8 : (2,2,4)$	$9^2 \cdot 25 = 2025$
$8 : (2,3,3)$	$9 \cdot 20^2 = 3600$
$8 : (2,2,2,2)$	$(9^4 - 9^2)/4 = 1620.$

(D.33.7)

Products are additive on this subgroup and squares vanish, since its least degree is six. These counts therefore give the upper bound $240 + 1620 + 7245 = 9105$. The actual words from D.31 provide the first 1860 independent vectors, retaining their degree-eight terms. In degree eight the words $[r_i, [r_j, z]]$, with z ranging over all 70 restricted degree-four Hall-power words of K, have rank 3645. Adjoining $[z_i, [r_j, z_k]]$, with z_i, z_k ranging over the 20 degree-three Lyndon words, raises the rank to 7245. Their leading terms are realized by precisely these commutator words. Thus the lower bound equals the collection bound. Fourfold commutators in the nine r_i have rank 1620. This proves (D.33.4) for the image in T_3, including the exact fourth subgroup rather than only its ambient Lie span.

The N kernel of T_3 has order 2^{11563} by D.32. Quotienting by U_8 gives 2^{2458}. There is also an explicit class-two description of this common quotient. In the weighted basis (D.33.6), retain each generator and each central square or commutator until its doubled weighted degree reaches nine. Of the 4753 square/commutator coordinates, 45 have weight four, 2271 have weights five through eight, and 2437 have weight at least nine. Consequently the normal-form order is

$$2^{97+2316+45} = 2^{2458}. \tag{D.33.8}$$

The map from this presentation onto the actual quotient has the same order on both sides, so it is an isomorphism. In particular the exact kernel of $H^{(2)} \to C_{***}$ is $(\mathbb{Z}/4)^{2437} \times (\mathbb{Z}/2)^{2271}$: take the coordinates of weight at least nine, and the squares of those of weights five through

eight. They are actual words in $D_9(K)$. Furthermore

$$(\gamma_3 N[N,N]^4 N^8) D_9(K) = \gamma_3 N D_9(K), \tag{D.33.9}$$

since the two exponent factors lie in $D_{16}(K)$. This is the exact common-quotient identity. It proves the fibre-product assertion, surjectivity of B onto that product, and all orders in (D.33.2).

We next specify a correcting group whose law is genuinely abelian in both targets. Keep the nine original N generators with independent degree-two terms. Replace each of the other 88 generators by itself times the inverse of the product of those nine that cancels its degree-two term. These triangular Nielsen operations give an actual discrete free basis. Its 4708 squares and commutators involving at least one of the latter 88 generators have degree at least five. The remaining 45 have degree four; their squares have degree eight. Adjoin the 9105 actual generators of U_8. In the joint quotient these words generate

$$\mathcal{A} \simeq (\mathbb{Z}/4)^{4708} \times (\mathbb{Z}/2)^{9150}. \tag{D.33.10}$$

They commute in T_3 because each has degree at least five, and commute in $H^{(2)}$ because their images lie in its abelian subgroup C_1. The 4708 high words and the 45 twice-low words have independent images in C_1, of order 4 and 2 respectively. Their kernel in the joint group is exactly U_8, proving (D.33.10). Equivalently this is the full image of $D_5(K) \cap \Phi(N)$. Indeed $D_5(K)/D_9(K)$ has dimension $204 + 690 + 2340 + 8230 = 11464$, and its homology image has dimension 43. Its kernel over that image, together with the 7145 binary dimensions of the kernel in (D.33.8), has order $2^{11421+7145} = 2^{18566}$, equal to (D.33.10).

Let $S = \beta s \beta^{-1}$ and let T denote conjugation by t. For left corrections $b\beta, aA$ in $\mathcal{A}$, the exact change of the cusp error is

$$(1 - S)b + (S - T)a. \tag{D.33.11}$$

To verify the formula, multiply $bSb^{-1}aRa^{-1}t^{-1}$, where $R = ArA^{-1}$, and move its four correction factors to the left. The original error SRt^{-1} lies in $\mathcal{A}$, so its conjugation on this abelian group is trivial. This gives (D.33.11). Changing the multiplier by 144 adds the actual t^{-144} coordinate. Its degree-eight term equals that of t^{48}, and it is retained in every calculation. No quadratic terms have been removed: they vanish because the chosen corrections commute in each of the two full target groups.

Conjugation on U_8 now retains the degree-eight effect of N. In particular $[N, U_8]$ is the fourth subgroup of dimension 1620. The program uses the actual degree-two inner factor in $\beta s \beta^{-1} = ks$, the complete involution on the four K generators, and the actual t-conjugate generator words. The resulting map (D.33.11) on U_8^2 has rank 8981 and cokernel dimension 124. All 124 annihilating functionals are obtained by exact elimination.

For the high C_1 words, retain their complete order-four coordinates in $H^{(2)}$ and their full degree-five through degree-eight expansions in T_3. The parity map on the 9416 lower variables has rank 4687. The upper variables are twice every original C_1 coordinate in both conjugators and the multiplier-144 variable, hence 9507 variables. In the combined space consisting of the 4753 upper class-two coordinates and the 124 U_8 obstruction coordinates, their image has rank 4738. Its cokernel has dimension $15 + 124 = 139$.

The exact Bockstein calculation on the lower parity relations has rank 139: the combined lower matrix has rank $4687 + 139 = 4826$. The functionals on U_8 are extended linearly to the ambient tensor space. This is legitimate here because satisfying the full class-two equation puts the final error in U_8, by (D.33.9). Each class-two part of an upper annihilator evaluates evenly on a high lower column: twice that high word has zero T_3 image. Division by two in this even

image is the explicit Bockstein used by the program. Thus these are simultaneous equations for the actual order-four group, rather than just equations for its first binary quotient.

Solving these equations, then the upper equation, then the U_8 equation gives the recorded multiplier-one words. The two independent complete-word evaluations in the statement verify every coefficient, not merely the annihilating functionals. The induced homology map on $N/\Phi(N)$ is unchanged and has rank 97. The quotient action on Q is unchanged. Characteristicity of $D_9(K)$ and of the defining class-two subgroup of N makes the resulting maps endomorphisms of both stated quotients. The finite two-kernel Frattini argument makes them automorphisms.

To continue while retaining T_3, use D.29's stable two-level module $C_k \simeq (\mathbb{Z}/4)^{4753}$. At $k = 2$, restrict the lower corrections to the squares of the 4708 high C_1 words. Their degree is at least ten. Exclude the lower multiplier-144 change. The restricted parity rank is 4687 on 9416 variables; its relation space has dimension 4729. Its Bockstein onto the full 15-dimensional upper obstruction space has rank 15. This exact computation uses the unchanged G_2 data, on which the stable module action depends. All upper C_3 words have degree at least sixteen, as does t^{288}.

An actual first continuation gives the stated $18,077,708$-letter cusp identity over $\mathbb{Z}/8$. The complete new conjugator corrections and t^{288} are also expanded through degree eight; each is the identity in T_3. For subsequent stages $k \geq 3$, all lower and upper corrections already lie in $D_{16}(K)$. D.29's one-level-lookahead construction therefore applies while fixing T_3 at every step. The initial $H^{(2)}$ central coordinates can change once in this procedure; after lookahead each lower level stabilizes. The displayed degree-eight action remains fixed throughout. The multiplier increments are divisible by 288, so their two-adic limit lies in $1 + 32\mathbb{Z}_2$.

Taking the inverse limit gives (D.33.3), with the specified convergent conjugators. Its third and fourth subgroups are their computed images in T_3; the fifth vanishes in both factors. This proves (D.33.4). The inclusion $D_9(K) \subset D_8(K)$ gives (D.33.5), whose kernel is the 7245-dimensional pure degree-eight part of U_8. Direct expansion of $bS(b)^{-1}$ for the recorded correction b gives a first nonzero term in degree five. Later stable corrections vanish there, proving the stated comparison with D.31 by this explicit quotient map.

Choose the original odd-primary conjugators as separate components. The new two-primary words need not preserve their own odd-primary abelianization. The unchanged G_2 and separately retained G_3 coordinates agree with D.28 on $H \times_Q G_3$, the exact maximal common quotient already proved there. The corresponding fibre product thus carries one action. The global cusp gauge of D.26 and D.29 produces compatible finite shadows, with their nonarithmetical CM value retained whenever the quotient retains those coordinates.

Finally the remaining source kernel is explicit and nonzero. Choose three distinct weight-three generators from (D.33.6). Their nonzero basic triple commutator belongs to $\gamma_3 N^{(2)} \cap D_9(K^{(2)})$: nonvanishing follows from its nonzero ordinary free Lie term in that free pro-two basis. Hence (D.33.3) retains a nontrivial relative pro-two kernel. D.45 and its source-transport supplement exclude a lift of this recorded action across the full kernel; another action with the prescribed CM pair remains in the realization problem. □

The exact elliptic cover and the remaining relative automorphism group

D.34. A marked elliptic-cover and centralizer description of the relative pro-two source [Proved]

This addresses Part D(e) and the CM follow-up to LM2, manuscript §51.16, folio 734 verso, scan 174, and §51.15, folios 725–727, scans 156–160. Retain $B = \langle s, r \mid s^2 = r^3 = 1 \rangle$, $t = sr$,

Q_0, K and $\mathcal{P}_2$ from D.30. The index-18 subgroup K is the fundamental group of

$$\mathcal{E}_0^\circ : \quad w^3 = z(z-1), \qquad z \notin \{0, 1, \infty\}. \tag{D.34.1}$$

Its smooth compactification has genus one. Its three removed points are denoted P_0, P_1, P_∞. The regular Q_0 dessin has permutation cycles $2^9, 3^6, 6^3$, and $Q_0 \simeq S_3 \times C_3$ acts by the following explicit deck maps. For $\zeta = \exp(2\pi i/3)$ set

$$\delta(z,w) = (z, \zeta w), \quad s(z,w) = (1-z, w), \quad r_0(z,w) = \left(\frac{1}{1-z}, \frac{-w}{1-z}\right), \qquad r = \delta r_0. \tag{D.34.2}$$

Here $s^2 = r_0^3 = (sr_0)^2 = 1$, δ is central, $t^2 = \delta^2$, and the standard positive generators $x = t^{-2}$ and $y = sxs$ of $\Gamma(2)$ both act as δ.

In the fixed rank-four basis a_1, a_2, a_3, a_4 of D.30, the three cusp generators, corresponding respectively to $z = 0, 1, \infty$, are

$$c_0 = a_1 a_4^{-1} a_2 = t^6, \qquad c_1 = a_1^{-1} a_3 a_2^{-1}, \qquad c_2 = a_3^{-1} a_4. \tag{D.34.3}$$

The negative choice of orientation implicit in $t^6 = x^{-3}$ is the same at all three cusps. Their full Q_0 conjugacy maps are specified in the proof and in the 54-entry word certificate.

Put $K_2 = K^{(2)}$ and let $\Lambda \simeq Q_0 \subset \mathrm{Out}(K_2)$ be the outer conjugation action. Restriction is an isomorphism of profinite groups

$$\mathrm{Aut}_{Q_0}(\mathcal{P}_2) \xrightarrow{\sim} \{a \in \mathrm{Aut}(K_2) : \bar{a} \in C_{\mathrm{Out}(K_2)}(\Lambda)\}. \tag{D.34.4}$$

The subscript on the left means that the induced map on the displayed quotient Q_0 is the identity. Under this isomorphism, the exact normalized boundary condition is

$$a(c_0) = c_0^{\lambda_2} \quad \Longleftrightarrow \quad \alpha(t) = t^{(\lambda_2, 1)}, \qquad \lambda_2 \in \mathbb{Z}_2^\times, \tag{D.34.5}$$

where $\overline{\langle t \rangle} \simeq \mathbb{Z}_2 \times C_3$, and the two entries are its two-adic and order-three exponents. Both sides then send every c_i to a K_2-conjugate of $c_i^{\lambda_2}$.

Filling the three cusps gives the exact map

$$K_2 \longrightarrow T_2\mathcal{E}_0, \qquad 0 \longrightarrow (\mathbb{Z}_2^2)_{\mathrm{cusp}} \longrightarrow H_1(K_2, \mathbb{Z}_2) \longrightarrow T_2\mathcal{E}_0 \longrightarrow 0. \tag{D.34.6}$$

The sequence has a canonical Q_0-equivariant splitting over $\mathbb{Z}_2$, using the central order-three idempotent. On its rank-two elliptic factor, t^2 has matrix

$$Z = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}, \qquad Z^2 + Z + I = 0. \tag{D.34.7}$$

Every automorphism in (D.34.5) therefore acts by $\lambda_2 I$ on the cusp factor and by a unit $a + bZ \in \mathbb{Z}_2[\zeta]^\times$ on the elliptic factor, with

$$a^2 - ab + b^2 = \lambda_2. \tag{D.34.8}$$

The retained T_3 component of D.33 has the following exact reduction on the latter factor:

$$M_{16} = \begin{pmatrix} 3 & 8 \\ 8 & 11 \end{pmatrix} = 11I + 8Z \pmod{16}, \qquad \det M_{16} = 1 \pmod{16}. \tag{D.34.9}$$

All the stable corrections of D.33 retain this matrix.

For the arithmetic comparison, let $c = \sqrt[3]{4} > 0$, $E_3 : Y^2 = X^3 + 1$, and $P = (0, 1) \in E_3[3]$. The exact isomorphism with origin P_0 on $\mathcal{E}_0$ is

$$\phi = T_P \circ \phi_0, \qquad \phi_0(z, w) = (cw, 2z - 1), \qquad \phi : \mathcal{E}_0 \longrightarrow E_3. \tag{D.34.10}$$

It is defined over $\mathbb{Q}(c)$. If $\Phi = T_2\phi$ and $\kappa_{4,3}(g)$ is determined by $g(c)/c = \zeta^{\kappa_{4,3}(g)}$, then the actual Galois representations satisfy

$$\Phi\rho_{\mathcal{E}_0}(g)\Phi^{-1} = [\zeta]^{-\kappa_{4,3}(g)}\rho_{E_3}(g), \qquad \kappa_{4,3} = 2\kappa_{2,3}. \tag{D.34.11}$$

Thus this comparison retains the cubic Kummer twist. Equation (D.34.11) concerns actual Galois action; (D.34.4)–(D.34.9) give the specified maps for the larger relative automorphism group.

Proof. The program computes the regular right-multiplication permutations of s, r, t on the 18 explicitly listed elements of Q_0. Their respective cycle lengths are nine twos, six threes and three sixes. The compact dessin therefore has Euler characteristic $9 + 6 - 18 + 3 = 0$, and genus one. Removing its three faces leaves the graph of Euler characteristic -3 whose four generators were retained in D.30.

There is also an algebraic identification retaining the actual monodromy character. The homomorphism $Q_0 \to S_3 \times C_3$ sends a word to its action on the three $\Gamma(2)$ cusps and to its exponent sum in r modulo three. All 18 pairs are distinct, and multiplication is checked on all 18^2 pairs. Thus K is the kernel on $\Gamma(2)$ of the character sending $x = t^{-2}$ and $y = sxs$ to $1 \in \mathbb{Z}/3$. For the map (D.34.1) to the z line, a positive loop about either zero or one multiplies w by ζ. The third positive loop does the same, since $z(z - 1)$ has pole order two at infinity. This is exactly the indicated character, proving the marked cover comparison. The full induced pro-two topology is the one proved in D.30.

The maps s, r_0 in (D.34.2) preserve $w^3 = z(z - 1)$ and satisfy the displayed S_3 relations by direct substitution. They commute with δ. Their 18 products are distinct, since the six base maps are distinct and each has three different scalar lifts. The invariant map to the orbifold line is

$$j_0(z) = \frac{4(z^2 - z + 1)^3}{27z^2(z - 1)^2}. \tag{D.34.12}$$

It is invariant under both base generators. Consequently these maps realize the exact Q_0 cover of the orbifold, with the stated s, r and cyclic character.

For completeness every cusp map is retained at the word level. Choose the recorded short B representatives g_i of the three left cosets of $\langle t \rangle$ in Q_0, with $g_0 = 1$. Then $c_i = g_i t^6 g_i^{-1}$ gives (D.34.3). For a representative h of $q \in Q_0$, write $qg_i = g_j t^a$ in Q_0, $0 \le a < 6$, and put

$$k_{q,i} = hg_i t^{-a} g_j^{-1} \in K. \quad \text{Then} \quad hc_i h^{-1} = k_{q,i} c_j k_{q,i}^{-1}. \tag{D.34.13}$$

The certificate records all 54 resulting K words. Each equality is independently reduced in $C_2 * C_3$. It also records all section cocycles $h_q h_{q'} h_{qq'}^{-1} \in K$ and verifies their conjugation identity on all four K generators, giving 1296 exact checks. This specifies the outer Q_0 action, including its extension data.

The 18 matrices of that action on $H_1(K, \mathbb{F}_2)$ are distinct. Thus an element of $\mathcal{P}_2$ centralizing K_2 has trivial image in Q_0, and then lies in the center of the nonabelian free pro-two group K_2, which is trivial. Conjugation consequently embeds $\mathcal{P}_2$ into $\mathrm{Aut}(K_2)$. Its image is exactly the inverse image of Λ under $\pi : \mathrm{Aut}(K_2) \to \mathrm{Out}(K_2)$: it contains every inner automorphism and one representative of every member of Λ.

Restriction now gives (D.34.4) in both directions. An automorphism of $\mathcal{P}_2$ over Q_0 restricts to an automorphism whose outer class centralizes Λ. Conversely, conjugation by any such

$a \in \mathrm{Aut}(K_2)$ preserves $\pi^{-1}(\Lambda)$ and induces the identity on Λ. Transporting this conjugation through the preceding embedding gives the required automorphism of $\mathcal{P}_2$. The restriction is exactly a, because $a\,\mathrm{Int}(k)a^{-1} = \mathrm{Int}(a(k))$. Faithfulness makes these operations inverse. Compactness makes the displayed continuous bijections isomorphisms of profinite groups.

We verify the boundary condition without assuming a root uniqueness statement for the whole group. The stabilizer in Q_0 of $[c_0] = (1,1,0,1)$ in $H_1(K,\mathbb{F}_2)$ is precisely $\langle t\rangle$, of order six. The element c_0 is primitive in K_2: (D.34.3) permits replacing a_4 by c_0, with inverse $a_4 = a_2 c_0^{-1} a_1$. The free-product centralizer theorem [HR85, Theorem B′] therefore gives $C_{K_2}(c_0) = \overline{\langle c_0\rangle}$. For any element centralizing c_0, its Q_0 image is a power of t; remove that power and use this last equality. We obtain

$$C_{\mathcal{P}_2}(c_0) = \overline{\langle t\rangle} \simeq \mathbb{Z}_2 \times C_3. \tag{D.34.14}$$

The procyclic description follows because its kernel over its order-six image is $\langle t^6\rangle \simeq \mathbb{Z}_2$. If $a(c_0) = c_0^{\lambda_2}$, then $\alpha(t)$ centralizes c_0, since it centralizes $c_0^{\lambda_2}$ and this element generates the same closed subgroup. Inside (D.34.14), its sixth power fixes its two-adic exponent as λ_2, while the unchanged Q_0 image fixes its order-three exponent as 1. This proves (D.34.5), including the converse. Equation (D.34.13), and the fact that $\alpha(h)h^{-1} \in K_2$, then prove the assertion at every cusp.

Filling the punctures is an actual morphism of curves. On topological fundamental groups its kernel is the normal closure of the c_i. Its compact genus-one quotient is $\mathbb{Z}^2$, so pro-two completion gives (D.34.6). The identification with $T_2\mathcal{E}_0$ uses the cofinal system of covers $[2^n]$ of this pointed elliptic curve. In the displayed K basis the integral filling matrix and a section are

$$P = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{pmatrix}, \qquad J = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}. \tag{D.34.15}$$

The cusp lattice is generated by $(1,1,0,-1)$ and $(-1,-1,1,0)$. These, followed by the two columns of J, form an integral basis. Thus P is the precise quotient map, rather than a rational identification of lattices.

Let $z_0 = t^2$. The matrix $PM_{z_0}J$ is (D.34.7). On the rank-four lattice the idempotent $(1+z_0+z_0^2)/3$ has image exactly the cusp lattice; its complementary image maps isomorphically by P to the elliptic quotient. This proves the canonical splitting. The latter matrix has irreducible polynomial X^2+X+1 modulo two. A vector and its Z translate are therefore a basis over $\mathbb{Z}_2$, so its centralizer ring is exactly $\mathbb{Z}_2[Z]$. Its determinant is $a^2 - ab + b^2$.

The determinant equals the common cusp multiplier. One can see this directly from the fundamental class of the punctured torus relative to its boundary circles. Its boundary is the sum of the three oriented boundary classes, so a common power λ_2 acts by λ_2 on that relative fundamental class. Naturality of the intersection pairing on the compact rank-two quotient then gives $M^{\mathsf{T}} \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) M = \lambda_2 \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. For a two-by-two matrix this is exactly (D.34.8). The same relative pair and its orientation are covered by [Wil20, Theorem 3.3 and §1.3]; here the integral filling map has been given explicitly.

For (D.34.9), the program reconstructs the exact D.33 conjugators. It calculates their action on all four K generators in abelianization modulo sixteen, obtaining

$$M_K = \begin{pmatrix} 7 & 12 & 2 & 2 \\ 12 & 15 & 10 & 10 \\ 14 & 6 & 5 & 4 \\ 14 & 6 & 4 & 5 \end{pmatrix} \pmod{16}. \tag{D.34.16}$$

This matrix commutes with every Q_0 matrix modulo sixteen and fixes both cusp vectors. Multiplication PM_KJ gives (D.34.9). The quotient T_3 retains $H_1(K,\mathbb{Z}/16)$ because the abelian image of $D_9(K)$ is $16\mathbb{Z}_2^4$. D.33 retains its entire T_3 action, so it retains this Tate component as well.

Finally (D.34.10) follows by substituting $(2z-1)^2 = 4z(z-1)+1 = (cw)^3+1$; its inverse before translation is $z=(Y+1)/2$, $w=X/c$. The point P has order three: its horizontal tangent gives $2P=-P$. Under ϕ_0, δ is $[\zeta]$, s is $[-1]$, and r_0 is translation by $-P$. After translating the origin by P, the deck maps become

$$s = T_{-P}[-1], \qquad r = T_{-P}[\zeta], \qquad t = [-\zeta]. \tag{D.34.17}$$

These are the claimed translations of order three combined with μ_6. Substitution in (D.34.12) gives the Belyi map $(X^3+4)^3/(27X^6)$, invariant under these same deck transformations. For $g \in G_{\mathbb{Q}}$, the exact equality of morphisms $g(\phi) = [\zeta]^{\kappa_{4,3}(g)}\phi$ holds: P is rational and is fixed by $[\zeta]$. Functoriality on Tate modules proves (D.34.11). The equality $\kappa_{4,3} = 2\kappa_{2,3}$ follows from $c = (\sqrt[3]{2})^2$. This proves the arithmetic comparison with its specified field of definition and twisting character. □

An actual odd-deck lift and the exact remaining involution condition

D.35. Odd-deck boundary lifts and genuine realization of the compact CM torus [Proved]

This continues Part D(e), the CM follow-up to LM2, manuscript §51.16, folio 734 verso, scan 174, and §51.15, folios 725–727, scans 156–160. Retain the marked curve, K_2, Λ, the canonical cusp/elliptic splitting and the matrix Z of D.34. Let $\Lambda_o \simeq C_3^2$ be the unique order-nine subgroup of $\Lambda \simeq S_3 \times C_3$. Its preimage in $\mathcal{P}_2$ has an order-nine complement Δ_9 to K_2.

For every $\eta \in \mathcal{O}_{3,2}^{\times}$, where $\mathcal{O}_{3,2} = \mathbb{Z}_2[Z] \simeq \mathbb{Z}_2[\zeta_3]$, there is an actual continuous automorphism of the free pro-two group K_2 which centralizes the conjugation action of Δ_9 pointwise, preserves each of the three peripheral conjugacy classes with the same power

$$\lambda_2 = N_{\mathcal{O}_{3,2}/\mathbb{Z}_2}(\eta), \tag{D.35.1}$$

and acts on the canonical homology splitting by $\lambda_2 I_{\mathrm{cusp}} \oplus \eta_{\mathrm{ell}}$. Here the displayed CM identification sends ζ_3^2 to Z, because $t^2 = \delta^2$ in D.34; all evaluations below use that map. In particular the norm-one unit

$$\eta_* = \frac{-3-8\zeta_3^2}{7} = \frac{1-2\zeta_3^2}{1-2\zeta_3}, \qquad \eta_*(Z) = \frac{-3I-8Z}{7} \tag{D.35.2}$$

has an actual such lift with $\lambda_2 = 1$ and with the compact Tate-module reduction M_{16} of D.33–D.34. This assertion retains that Tate component; equality with the complete D.33 finite action is not asserted.

Here is the exact residual group. Let $\mathcal{O}_\partial$ be the subgroup of $\mathrm{Out}(K_2)$ preserving each cusp conjugacy class with a common unit power. Set

$$C_o = C_{\mathcal{O}_\partial}(\Lambda_o), \qquad \iota(a) = \bar{s}a\bar{s}^{-1} \quad (a \in C_o). \tag{D.35.3}$$

Then ι is an involution and

$$C_{\mathcal{O}_\partial}(\Lambda) = C_o^{\iota}, \qquad C_o \longrightarrow \mathcal{O}_{3,2}^{\times} \text{ is onto.} \tag{D.35.4}$$

The involution induces the identity on this torus target. In fact the fixed subgroup also maps onto the entire torus:

$$C_o^{\iota} \longrightarrow \mathcal{O}_{3,2}^{\times} \text{ is onto.} \tag{D.35.4a}$$

This stronger assertion is realized by the actual Galois action on $\mathcal{E}_0^\circ$ over $\mathbb{Q}(\zeta_3)$, including the norm-one unit (D.35.2). The exact relation to the full relative peripheral source is

$$1 \longrightarrow \langle \mathrm{Int}(c_0) \rangle \longrightarrow \mathrm{Aut}_{Q_0,t}(\mathcal{P}_2) \longrightarrow C_o^\iota \longrightarrow 1, \qquad \langle \mathrm{Int}(c_0) \rangle \simeq \mathbb{Z}_2. \tag{D.35.5}$$

The middle group acts identically on Q_0 and sends t to $t^{(\lambda_2,1)}$, with the full two-component exponent of D.34. The remaining condition retains the prescribed finite action, not only its compact Tate image. More precisely put $R = T_3 = \mathcal{P}_2/D_9(K_2)$ and let $\alpha_{33,R}$ be the action retained by D.33. Reduction defines an actual homomorphism

$$C_o^\iota \longrightarrow \mathrm{Aut}_{Q_0,t}(R)/\langle \mathrm{Int}(c_0) \rangle. \tag{D.35.5a}$$

The fibre over $[\alpha_{33,R}]$ has precisely the outer classes of normalized relative lifts of that recorded finite action. D.45 and its source-transport supplement prove that this particular fibre is empty. Realization of the full missing CM pair by another finite action remains undecided. Equations (D.35.3)–(D.35.5a) retain both the order-two deck symmetry and the specified finite coordinates.

Proof. We first give the marked surface comparison and the actual boundary automorphisms used in the lift. In the original free basis of D.34 put

$$a = a_1, \quad b = a_2^{-1}, \quad c = a_1 a_4^{-1} a_3 a_1^{-1}, \quad d = a_1 a_3^{-1} a_1 a_2 a_1^{-1}. \tag{D.35.6}$$

The inverse change of basis is

$$a_1 = a, \quad a_2 = b^{-1}, \quad a_3 = a b^{-1} a^{-1} d^{-1} a, \quad a_4 = a b^{-1} a^{-1} d^{-1} c^{-1} a. \tag{D.35.7}$$

Direct free reduction verifies both compositions. It also gives

$$c_0 = cd[a,b], \qquad c_1 = b^{-1} a^{-1} d^{-1} a b, \qquad c_2 = a^{-1} c^{-1} a. \tag{D.35.8}$$

Thus the first boundary is conjugate to $R = [a,b]cd$, and the other two are inverse conjugates of d, c respectively. These exact words identify the original three peripheral classes with the standard surface pair; inversion of the latter two generators does not change their common power under any automorphism.

Let $\mathcal{A}_\partial$ be the closed subgroup of $\mathrm{Aut}(K_2)$ preserving the three peripheral classes individually with a common power, and let $\mathcal{A}_\partial^{\mathrm{perm}}$ allow their permutation. Their homology maps are continuous. In a basis with the cusp lattice first and a chosen rank-two handle section last, the image of the pure subgroup is exactly

$$\left\{ \begin{pmatrix} \lambda I_2 & B \\ 0 & M \end{pmatrix} : \lambda \in \mathbb{Z}_2^\times, \quad B \in M_2(\mathbb{Z}_2), \quad M \in \mathrm{GL}_2(\mathbb{Z}_2), \quad \det M = \lambda \right\}. \tag{D.35.9}$$

Necessity follows from the peripheral homology and the relative fundamental-class calculation in D.34. We prove surjectivity using actual automorphisms, rather than lifting a matrix formally.

In the basis (a,b,c,d) define

	a	b	c	d
U	a	ba	c	d
L	ab	b	c	d
H	a	b	d	$d^{-1}cd$
θ	ca	b	bcb^{-1}	cdc^{-1}.

(D.35.10)

Each row is an automorphism of the discrete free group and hence of its free pro-two completion. For U, L the inverse replaces the extra handle letter by its inverse; for H the inverse sends (c, d) to (cdc^{-1}, c). The inverse of θ is

$$a \mapsto b^{-1}c^{-1}ba, \quad b \mapsto b, \quad c \mapsto b^{-1}cb, \quad d \mapsto b^{-1}c^{-1}bdb^{-1}cb. \tag{D.35.11}$$

Free reduction verifies both inverse compositions in all four cases. The first three rows fix R, while $\theta(R) = cRc^{-1}$. The row H interchanges the c, d peripheral classes; U, L, θ are pure. Thus all the stated homology operations come from the specified peripheral automorphism groups.

The handle actions of U, L generate a dense subgroup of $\mathrm{SL}_2(\mathbb{Z}_2)$. Indeed their powers supply both elementary matrices with every two-adic coefficient in the closure, and elementary row reduction over $\mathbb{Z}_2$ generates $\mathrm{SL}_2(\mathbb{Z}_2)$. Compactness of the source makes the homology image closed. Conjugating θ by H supplies the other cusp row, and conjugating these two shears by the handle rotation $UL^{-1}U$ supplies the other handle column. The four resulting integral shear vectors have determinant ± 1 in $M_2(\mathbb{Z})$. Their powers and products therefore give every $B \in M_2(\mathbb{Z}_2)$. All these shears are pure, including those obtained by conjugating with H.

Finally every $\lambda \in \mathbb{Z}_2^{\times}$ occurs as the cyclotomic character of an actual element of $G_{\mathbb{Q}}$. Use its action on the geometric pro-two fundamental group of $\mathcal{E}_0^{\circ}/\mathbb{Q}$, based at the rational tangent with local parameter w at P_0. All three removed points are rational, so this is a pure peripheral automorphism with common power λ. This assertion follows directly by acting on finite étale covers and their tangential fibres; the action on a punctured formal parameter is the action on its compatible 2^n-th roots. Compose such an actual automorphism with the determinant-one handle operations and the four shears already constructed. This adjusts its handle matrix and off-diagonal block to any prescribed pair in (D.35.9), proving the claimed homology surjectivity.

We record why the needed correction kernel is pro-two. For a finitely generated free pro-two group, the kernel of its automorphism action on the Frattini quotient is pro-two. One proof uses the characteristic lower exponent-two central series: after the action on its first quotient is fixed, each successive automorphism kernel embeds into an additive space of homomorphisms to the next elementary abelian central layer. The finite quotients are consequently two-groups, and the inverse limit is pro-two. The kernel of integral homology on $\mathcal{A}_{\partial}^{\mathrm{perm}}$ is a closed subgroup of this kernel. It fixes all peripheral labels, since their three distinct nonzero mod-two homology vectors are fixed, and its common multiplier is one.

The coprime complement facts needed here admit the same elementary inverse-limit proof. In a finite extension with a normal two-group and odd quotient, successive elementary abelian chief factors have vanishing positive cohomology for the odd quotient, by averaging with its invertible order. Induction supplies a complement and shows that any two complements are conjugate by the normal two-group. Finite solutions form nonempty compact sets; their inverse limit supplies a complement and conjugacy for a normal pro-two kernel and a finite odd quotient. Apply this first to the preimage of Λ_o in $\mathcal{P}_2$ to obtain Δ_9. Conjugation embeds this complement into $\mathcal{A}_{\partial}^{\mathrm{perm}}$. Its homology action is faithful, because the complete Q_0 action is already faithful modulo two in D.34.

For a prescribed unit η, put $\lambda = N(\eta)$ and form the integral homology map $T = \lambda I_{\mathrm{cusp}} \oplus \eta_{\mathrm{ell}}$ using D.34's canonical splitting. The central order-three deck map is the identity on the cusp lattice and has the indicated CM action on the elliptic lattice. The translation subgroup of order three permutes the cusp classes and is the identity on the elliptic quotient. It follows that T centralizes the homology action of Δ_9. Equation (D.35.9) supplies an actual $a \in \mathcal{A}_{\partial}$ with homology T.

Write h for integral homology. In the preimage $h^{-1}(h(\Delta_9)) \subset \mathcal{A}_{\partial}^{\mathrm{perm}}$, the two groups Δ_9 and $a\Delta_9 a^{-1}$ are complements to the pro-two kernel of h. The preceding conjugacy proof gives

$l \in \ker h$ such that $la\Delta_9(la)^{-1} = \Delta_9$. Because T centralizes $h(\Delta_9)$ and h is injective on that finite subgroup, la centralizes it pointwise. The correction l is pure with multiplier one, so la has precisely the required homology and boundary powers. This proves the actual lifting assertion.

In (D.35.2), multiplying $1 - 2\zeta_3^2$ by its conjugate gives norm seven, and expansion gives $(1 - 2\zeta_3^2)^2/7 = (-3 - 8\zeta_3^2)/7$. Thus $N(\eta_*) = 1$. Substitution of the matrix Z gives

$$\eta_*(Z) = \begin{pmatrix} 5/7 & -8/7 \\ 8/7 & -3/7 \end{pmatrix} \equiv \begin{pmatrix} 3 & 8 \\ 8 & 11 \end{pmatrix} \pmod{16}, \tag{D.35.12}$$

as asserted. This exact calculation only compares the compact Tate component, so it does not identify the other D.33 coordinates.

It remains to verify every map in (D.35.3)–(D.35.5). The subgroup Λ_o is normal in Λ, and together with $\bar{s}$ generates Λ. The pure peripheral subgroup is normalized by $\bar{s}$, which permutes the three cusp labels. Hence conjugation by $\bar{s}$ defines the displayed involution, and its fixed subgroup is exactly the centralizer of all of Λ. The odd-deck construction proves the asserted surjection from C_o to the CM unit group. On the compact Tate lattice s acts by $-I$, so conjugation by it fixes the torus target. On the cusp factor all maps in question are scalar, so their multipliers are also unchanged.

For the right arrow of (D.35.5), restrict to K_2 as in D.34 and take the outer class. Its image lies in C_o^ι. Conversely choose a representative a of any outer class there. The boundary condition gives $a(c_0) = kc_0^{\lambda_2}k^{-1}$ for some $k \in K_2$. Replacing a by $\mathrm{Int}(k^{-1})a$ makes this equality exact, without changing its outer class. D.34's two-way extension map then gives an automorphism over Q_0 with $t \mapsto t^{(\lambda_2,1)}$. Finally an element of the kernel is $\mathrm{Int}(k)$ on K_2. Its homology forces $\lambda_2 = 1$, and its exact boundary condition forces $k \in C_{K_2}(c_0) = \langle c_0 \rangle$. The free group's trivial center identifies these inner automorphisms with $\mathbb{Z}_2$. This proves exactness, including the actual full multiplier pair.

We now prove the stronger torus surjectivity (D.35.4a) arithmetically. Write $k = \mathbb{Q}(\zeta_3)$, $\mathcal{O} = \mathbb{Z}_2[\zeta_3]$ and $U = 1 + 2\mathcal{O}$. On $E_3 : Y^2 = X^3 + 1$, all three nonzero two-torsion points are k-rational, with $X = -1, -\zeta_3, -\zeta_3^2$. The CM automorphism $[\zeta_3]$ is also defined over k and permutes those three points cyclically. Its action modulo two has irreducible polynomial $X^2 + X + 1$. Consequently T_2E_3 is free of rank one over $\mathcal{O}$, and its G_k image is a closed subgroup $H \subset U$.

There is a small exact Frobenius computation proving $H = U$. Direct counting gives

p	$\#E_3(\mathbb{F}_p)$	a_p	two CM Frobenius scalars
7	12	-4	$-1 + 2\zeta_3,\ -3 - 2\zeta_3$
13	12	2	$3 + 4\zeta_3,\ -1 - 4\zeta_3.$

(D.35.13)

Both primes split in k. The characteristic polynomial of arithmetic Frobenius on T_2E_3 is $X^2 - a_pX + p$; its two roots in $\mathcal{O}$ are the displayed scalars. This trace formula follows from the degree of Frobenius and of $1 - \mathrm{Frob}$ together with the determinant-degree equality [Mil21, II, Proposition 6.4; IV, Proposition 9.1 and the proof of Theorem 9.4; V, Proposition 8.1]. Conjugating an actual Frobenius by complex conjugation replaces its CM scalar by its conjugate and interchanges the two primes over p. Thus H contains both listed roots for each row, independently of the choice of the first prime above p.

Modulo eight, these four elements generate all sixteen elements of $U/(1 + 8\mathcal{O})$. Their multiplication is checked directly using $(a + b\zeta_3)(c + d\zeta_3) = (ac - bd) + (ad + bc - bd)\zeta_3$. Moreover $1 + 8\mathcal{O} \subset U^2$: for $v \in 8\mathcal{O}$, the integral convergent series $\exp(\frac{1}{2}\log(1 + v)) \in 1 + 4\mathcal{O}$ is a square root of $1 + v$. Hence $HU^2 = U$. For completeness this implies $H = U$ without an openness assumption: a proper closed subgroup of an abelian pro-two group has a nontrivial

finite two-group quotient, and that quotient has a quotient of order two, which kills all squares. Such a quotient contradicts $HU^2 = U$.

The extension $k(\sqrt[3]{4})/k$ has degree three: a real cubic root of four cannot lie in the quadratic field k. It is Galois since $\zeta_3 \in k$. Its intersection with the pro-two division field $k(E_3[2^\infty])$ is k. Therefore the joint image of the actual characters $(\rho_{E_3}, \kappa_{4,3})$ is $U \times C_3$. The exact twist (D.34.11) now gives

$$\rho_{\mathcal{E}_0}(G_k) = \mu_3\, U = \mathcal{O}^\times. \tag{D.35.14}$$

Every deck map in (D.34.2) is defined over k. Functoriality of the geometric fundamental group consequently makes the actual outer G_k action commute with all of Λ, while all three rational cusps are preserved with common multiplier χ_2. This supplies (D.35.4a). Its precise normalized lift is the one already constructed in (D.35.5): inner conjugation by K_2 makes the c_0 image the exact power, and the restriction/extension isomorphism of D.34 gives the automorphism over Q_0 with full exponent $(\chi_2, 1)$. Inner conjugation by K_2 fixes the compact Tate action. Thus no compact component changes in making this boundary normalization.

The involution itself also has a concrete quotient geometry. In the model (D.34.1), s is $z \mapsto 1 - z$ and fixes w. Its quotient is the w line, since $z^2 - z - w^3 = 0$. The three interior fixed points satisfy $z = 1/2$ and $1 + 4w^3 = 0$; the fourth fixed point is P_∞. The two removed points P_0, P_1 have common image $w = 0$, and the removed P_∞ has image $w = \infty$. The quotient orbifold is therefore the sphere with these two punctures and three order-two cone points. Its fundamental group has the presentation

$$\langle q_1, q_2, q_3, h_0, h_\infty \mid q_1^2 = q_2^2 = q_3^2 = 1,\ q_1 q_2 q_3 h_0 h_\infty = 1 \rangle \simeq C_2 * C_2 * C_2 * \mathbb{Z}. \tag{D.35.15}$$

The covering character sends each q_i and h_∞ to the nontrivial element of C_2, and h_0 to the identity. Its kernel is the marked K, with the original cusp classes given by lifting these puncture loops. Pro-two completion gives the exact preimage of $\langle s \rangle$ in $\mathcal{P}_2$ already constructed in D.32: $C_2 \amalg_2 C_2 \amalg_2 C_2 \amalg_2 \mathbb{Z}_2$. This identifies the geometric involution and the previously used Bass–Serre subgroup by their covering map, rather than just by their abstract presentations.

Finally $D_9(K_2)$ is characteristic, so the group in (D.35.5) acts on R. Its kernel $\langle \mathrm{Int}(c_0) \rangle$ maps onto the displayed procyclic inner subgroup of the finite target. Quotienting gives exactly (D.35.5a). If a representative over R differs from $\alpha_{33,R}$ by a power of this inner automorphism, that power lifts to $\mathbb{Z}_2$ and is removed before reduction. This proves the stated description of the actual fibre. The single-torus arithmetic image (D.35.14) does not prescribe these additional finite coordinates.

The accompanying program checks the two inverse basis changes, all three boundary words, each displayed automorphism and inverse, the four independent cusp shears, the generated group $\mathrm{SL}_2(\mathbb{Z}/16)$ of order 3072, the exact norm-one lift (D.35.12), and every point count and unit-group generation check in (D.35.13). The general two-adic and odd-deck arguments above prove existence beyond these finite checks. □

The second marked elliptic cover and the exact joint arithmetic fibre

D.36. A quartic cover, a shared degree-twelve cover, and the retained CM discrepancy [Proved]

This addresses Part D(e) and the two-torus follow-up to LM2, manuscript §51.15, folios 725–727, scans 156–160, and §51.16, folio 734 verso, scan 174. Write $U = \mathbb{P}^1 \setminus \{0, 1, \infty\}$, $F = \pi_1(U) = \langle x, y \rangle$, and let K_n be the kernel of $F \to \mathbb{Z}/n$ sending both x, y to one. Its marked cover is

$$C_n^\circ : \quad v^n = z(z-1), \qquad z \in U. \tag{D.36.1}$$

Here $C_3 = \mathcal{E}_0$ of D.34. The smooth compactification of C_4° is an elliptic curve with four removed points. The following rational maps are inverse isomorphisms of compact curves:

$$\begin{aligned} \psi_0 : C_4 \longrightarrow E_4, \qquad E_4 : Y^2 = X^3 - X, \\ (z,v) \longmapsto \left(\frac{z}{v^2}, \frac{v}{z-1}\right), \qquad (X,Y) \longmapsto \left(\frac{X^2}{X^2-1}, \frac{Y}{X^2-1}\right). \end{aligned} \tag{D.36.2}$$

The four omitted points map exactly to $E_4[2]$. Translation by $(0,0)$ after ψ_0 sends the point over $z = 0$ to the origin and is defined over $\mathbb{Q}$. The deck map $v \mapsto iv$ becomes $[i] : (X,Y) \mapsto (-X, iY)$; the swap $z \mapsto 1 - z$ becomes $T_{(0,0)}[-1]$.

Restriction to $\widehat{K_n}$ and filling all cusp subgroups give actual maps from automorphisms of $\widehat{F}$ preserving each of the three ordered cusp conjugacy classes with one common multiplier to the homology of C_n. For $n = 3, 4$ their compact targets are precisely $T\mathcal{E}_0$ and TE_4 with the marked isomorphisms above. With the positive z tangent and the lift $v = \zeta_8 z^{1/4}(1-z)^{1/4}$, the restricted Galois action on the quartic compact Tate module is

$$\rho_4^{\mathrm{res}}(g) = [i]^{\kappa_{-1,4}(g)} \rho_{E_4}(g), \qquad \kappa_{-1,4}(g) = \frac{\chi_8(g) - 1}{2} \pmod 4. \tag{D.36.3}$$

The cubic lift with leading coefficient -1 is rational and has no corresponding tangent correction. Over $L = \mathbb{Q}(\zeta_{12})$, the factor in (D.36.3) is a global sign and disappears on projective three-torsion.

The common cyclic cover and its two maps are

$$C_{12} \longrightarrow C_3, \ (z,v) \mapsto (z, v^4), \qquad C_{12} \longrightarrow C_4, \ (z,v) \mapsto (z, v^3). \tag{D.36.4}$$

It has genus five and four punctures. On compact integral homology the joint map is onto and has kernel $\mathbb{Z}^6$:

$$0 \longrightarrow \mathbb{Z}^6 \longrightarrow H_1(C_{12}, \mathbb{Z}) \longrightarrow H_1(C_3, \mathbb{Z}) \oplus H_1(C_4, \mathbb{Z}) \longrightarrow 0. \tag{D.36.5}$$

All four nonzero Smith invariants are one. The S_3 normal core of the cyclic-twelve character has deck group $(\mathbb{Z}/4)^2 \times \mathbb{Z}/3$, of order 48; the full Fermat-twelve abelian cover has deck group $(\mathbb{Z}/12)^2$, of order 144, and maps to this core with kernel of order three.

There is an exact arithmetic joint image over L. Put $\mathcal{O} = \mathbb{Z}_2[\zeta_3]$. For a norm-one-modulo-four unit $\eta \in (\mathcal{O}/4)^\times$, let $\tau(\eta) \in \mu_3$ be the unique element with the same reduction modulo two, and set

$$\epsilon(\eta) = \tau(\eta)^{-1}\eta \in \{I, -I\}. \tag{D.36.6}$$

Then the actual image of $(\rho_{\mathcal{E}_0,4}, \mathbb{P}\rho_{E_4,3})(G_L)$ is exactly the six-element graph

$$\left\{ \left(\eta, [i]^{(1-\epsilon(\eta))/2}\right) : N(\eta) = 1 \pmod 4 \right\}. \tag{D.36.7}$$

The exponent treats ϵ as the scalar sign ± 1. The joint fixed field is $L(\sqrt[3]{4}, r_0)$, of degree six over L, where $r_0 = (3/4)^{1/4}$; the common subfield of the two torsion fields is $L(r_0)$, of degree two. The normalized unit (D.36.6) is precisely the actual $E_3[4]$ action after D.34's cubic twist is removed.

For the family retained in D.33 these three specified coordinates are

$$\epsilon_{\mathrm{cover}} = -1, \qquad T_4^{\mathrm{LM2}} = I, \qquad W_3^{\mathrm{LM2}} = [i]. \tag{D.36.8}$$

The first sign is unchanged by every Ihara gauge $\mathrm{Int}(x^k)$: this multiplies its cover Tate action by ζ_3^k, while the exact simultaneous change $(A, \beta, V) \mapsto (x^k A, x^k \beta, x^k V)$ fixes $T = A^{-1}V$ and

$W = \beta^{-1}V$. The actual norm-one Galois element supplied by D.35 can realize the entire D.33 compact Tate-sixteen matrix and the displayed W_3; its actual T_4 is $-I$. Thus the full prescribed finite action is still an additional requirement. Equality of these cover and fixed-fibre signs for every genuine full profinite $\widehat{\mathrm{GT}}_0$ element has not been proved here.

Proof. Substitution of (D.36.2) gives $Y^2 - X^3 + X = 0$, using $v^4 = z(z-1)$, and substitution of its inverse gives the original equation. Both rational maps are inverse on dense open sets and therefore extend to inverse maps of the smooth projective curves. At $z = 0$, v is a local parameter, $z = -v^4 + O(v^8)$, and the image is $(0, 0)$. At $z = 1$ the coordinates have pole orders two and three, giving the origin. The two points over infinity have $z/v^2 \to \pm 1$ and $v/(z-1) \to 0$, giving $(1, 0), (-1, 0)$. These are exactly the four two-torsion points. The point $(0, 0)$ has order two, so the stated translation gives the pointed isomorphism over $\mathbb{Q}$.

The two deck calculations are exact substitutions. Multiplying v by i sends (X, Y) to $(-X, iY)$. Replacing z by $1 - z$ sends it to $(-1/X, -Y/X^2)$. The addition law on E_4 gives $T_{(0,0)}(X, Y) = (-1/X, Y/X^2)$, proving the swap formula. Since $[i]$ fixes $(0, 0)$, these formulas also hold after the pointed translation.

We describe the source and all integral quotient maps explicitly. Choose coset representatives $1, x, \ldots, x^{n-1}$ for K_n. Its Schreier basis consists of

$$a = x^n, \quad b_j = x^j y x^{-(j+1 \bmod n)}, \qquad 0 \le j < n. \tag{D.36.9}$$

The peripheral words are x^n, y^n, and $x^j((xy)^{-1})^{n/d}x^{-j}$ for $0 \le j < d$, where $d = \gcd(n, 2)$. Their $d + 2$ homology vectors sum to zero and span a primitive rank-$d + 1$ lattice. Thus the compact quotient has rank $n - d$ and genus $(n - d)/2$. This proves all stated genera, puncture counts and free ranks. It also proves the exact profinite filling map by killing the normal closures of these actual cusp words. An automorphism preserving the three ordered cusp classes acts on F^{ab} by the common scalar unit, so it preserves $\widehat{K}_n$ and the family of lifted cusp subgroups. Restriction and filling are consequently defined for that entire pure source, without a Galois realization assumption. The swap also preserves the cyclic character $(1, 1)$; a three-cycle does not preserve K_4, as it sends that character to $(1, -2)$. The later normal-core construction is the precise enlargement on which the complete S_3 action is defined.

The quartic arithmetic tangent factor follows within the arithmetic fundamental group. Let s_z be its section at the positive z tangent. In the displayed Puiseux lift, $s_z(g)$ multiplies the leading coefficient by $g(\zeta_8)/\zeta_8 = i^{\kappa_{-1,4}(g)}$. The positive loop x multiplies v by i. Therefore the section at the rational v tangent on the cover is $x^{-\kappa_{-1,4}(g)}s_z(g)$ modulo $\widehat{K}_4$. Changing the chosen lift of this exponent only changes the section by a member of $\widehat{K}_4$, whose inner action is trivial on compact homology. Conjugation and the deck formula $x \mapsto [i]$ give (D.36.3), with the displayed sign of the exponent. For the cubic cover the lift is $v = -z^{1/3}(1 - z)^{1/3}$, whose leading coefficient is rational. For $g \in G_L$, $\chi_8(g)$ is 1 or 5 modulo eight, so the quartic factor is I or $-I$. Its projective action is therefore the actual projective E_4 action.

Inclusion of the actual Schreier words (D.36.9) for $n = 12$ in those for $n = 3, 4$ gives the homology maps in (D.36.4). The certificate records their punctured matrices and primitive cusp quotients. In its compact bases the two matrices are

$$\begin{aligned} M_3 &= \begin{pmatrix} 1 & 0 & -1 & 1 & 0 & -1 & 1 & 0 & -1 & 1 \\ 0 & 1 & -1 & 0 & 1 & -1 & 0 & 1 & -1 & 0 \end{pmatrix}, \\ M_4 &= \begin{pmatrix} 1 & 0 & -1 & 0 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 & 0 & 1 & 0 & -1 & 0 & 1 \end{pmatrix}. \end{aligned} \tag{D.36.10}$$

The common cover uses compatible lifts v^4, v^3; converting its cubic leading coefficient ζ_6 to D.34's coefficient -1 is exactly the deck map δ. This is defined over L and does not alter the

scalar sign (D.36.6). The stack of the two matrices has rank four and Smith diagonal $(1,1,1,1)$, proving (D.36.5). Its deck-equivariance is checked against the full order-twelve matrix and the elliptic matrices

$$D_3 = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \qquad D_4 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \tag{D.36.11}$$

The S_3 orbit of the cyclic character is generated, up to units, by $(1,1),(1,-2),(-2,1)$. Modulo four their common kernel is zero. Modulo twelve it is exactly $\{(0,0),(4,8),(8,4)\}$. This proves the normal-core orders and the actual quotient from the Fermat-twelve deck group. It does not replace the smaller normal core by the full abelian cover.

For the arithmetic graph, D.35 proves $\rho_{\mathcal{E}_0}(G_{\mathbb{Q}(\zeta_3)}) = \mathcal{O}^\times$. The subgroup fixing $L = \mathbb{Q}(\zeta_3, i)$ is cut out by $\chi_4 = 1$, equivalently by the norm modulo four. Hence its image on $\mathcal{E}_0[4]$ is exactly the six-element norm kernel in $(\mathcal{O}/4)^\times$. Reduction modulo two identifies its unique order-three factor. D.34's formula $\rho_{\mathcal{E}_0} = \zeta_3^{-\kappa_{4,3}}\rho_{E_3}$ and the trivial $G_{\mathbb{Q}(\zeta_3)}$ action on $E_3[2]$ show that this factor is exactly $\tau = \zeta_3^{-\kappa_{4,3}}$. The residual principal unit has norm one modulo four and is therefore exactly I or $-I$. This proves (D.36.6) as an equality with the actual $E_3[4]$ representation, including its cubic twist.

D.25 gives the common field $M' = \mathbb{Q}(i, r_0)$ for full $E_3[4]$ and projective $E_4[3]$. Over L it is quadratic, and its nontrivial automorphism sends r_0 to $-r_0$. On $E_3[4]$ this is $-I$. To identify its other action without a matrix-sign assumption, put $w_0 = 1/r_0$ and use the exact three-division roots from D.25:

$$a = \frac{w_0 + 2/w_0}{2}, \qquad b = \frac{w_0 - 2/w_0}{2i}. \tag{D.36.12}$$

The involution sends w_0 to $-w_0$ and consequently sends every x-coordinate $\pm a, \pm b$ to its negative. This is precisely the projective action of $[i]$ on $E_4[3]$. For a nonzero point $P \in E_4[3]$, the basis $(P, [i]P)$ realizes it by the matrix D_4 mod 3; independence follows from the irreducibility of X^2+1 over $\mathbb{F}_3$. This proves the graph (D.36.7). The cubic extension $L(\sqrt[3]{4})/L$ is independent of the quadratic extension $L(r_0)/L$, and the product map to the two distinct factors μ_3 and $\{\pm I\}$ is injective. Its kernel field is therefore exactly $L(\sqrt[3]{4}, r_0)$.

D.34 computes the D.33 compact matrix modulo sixteen; modulo four it is $-I$, so its order-three factor is one and its normalized sign is -1. D.25–D.33 retain $A_{12} = V_{12} = I$ and $\beta_{12} = \left(\begin{smallmatrix} 9 & 8 \\ 4 & 9 \end{smallmatrix}\right)$ as the fixed CM coordinates, giving $T_4 = I$ and $W_3 = [D_4]$. The gauge comparison is exact: $x = t^{-2}$ acts on the cubic compact homology as $\delta = [\zeta_3]$, while its simultaneous left multiplication on A, β, V cancels in T, W. Removing the unique order-three factor in (D.36.6) cancels that same ζ_3^k and leaves the cover sign unchanged. This proves every part of (D.36.8), including its stability under the specified gauge. The norm-one unit of D.35 has this full Tate-sixteen matrix; an actual Galois element realizing it fixes L and consequently has $W_3 = [i]$ and $T_4 = -I$ by (D.36.7).

We record the source checks that delimit the remaining transfer. Nakamura, Tsunogai and Yasuda [NTY10, Proposition 4.4] computes a beta-function cover action for full $\widehat{\mathrm{GT}}$. Their fixed-fibre comparison, Theorem 4.16, and braid power-conjugacy Proposition 4.13 are stated for $G_{\mathbb{Q}}$; the introduction, p.432, expressly leaves the broader implication undecided. The later full mapping-class-group actions in [LNS26, Theorem 1.1] come with an explicit qualification on p.2 concerning compatibility with the gluing rules. These opened results do not supply the missing equivariant comparison for the specified $\widehat{\mathrm{GT}}_0$ source. The shared cover and the exact finite coordinates above provide a concrete further test; no nonempty full-source fibre follows from the arithmetic graph. □

A shared Fermat-twelve quotient and both actual cover matrices

D.37. A generating mixed Fermat shadow with exact elliptic-cover evaluations [Proved]

Retain B, F, Q, N, x, y, t from D.25, and put

$$\begin{aligned} F_{12} = \ker(F \longrightarrow (\mathbb{Z}/12)^2), \qquad L = N \cap F_{12}, \\ H = B/L, \qquad J_m = B/[L,L]L^m. \end{aligned} \tag{D.37.1}$$

The character sends x, y to the indicated ordered basis. Then

$$\begin{aligned} |H| = 6912, \qquad \operatorname{rank} L = 1153, \\ \ker(H \to Q) \simeq C_2^2 \times C_3, \qquad |J_m| = 6912m^{1153}. \end{aligned} \tag{D.37.2}$$

There is an explicitly certified automorphism of J_{12} with multiplier one and the extra CM pair

$$A_{12} = I, \qquad \beta_{12} = \begin{pmatrix} 9 & 8 \\ 4 & 9 \end{pmatrix}, \qquad V_{12} = I. \tag{D.37.3}$$

Before the common Ihara gauge, its two conjugators project to exactly the retained G_6 conjugators of D.33: the actual pro-two component, including its right-s-centralizer correction, and the separately specified pro-three component. The constructed J_{12} action also retains its recorded J_6 action. Agreement with D.33's larger degree-eight quotient is not asserted.

The cusp has order 288 in J_{12}. Its pullback to B_3 therefore has $N_{\mathrm{ord}} = 144$. The recorded gauge makes f a commutator word and gives both $m = 0$ hexagons and generation, so (D.37.3) occurs in an actual nonarithmetical finite GT0 shadow.

Restriction to the three marked cyclic covers of D.36 and filling their peripheral classes give actual maps from this finite source. In the bases recorded there, the normalized shadow has

$$\rho_{C_3,4} = -I, \qquad \rho_{C_4,3} = I. \tag{D.37.4}$$

Both matrices are reductions of its one compact action on $C_{12} : v^{12} = z(z-1)$, through the two explicit integral maps of D.36. Its cubic Teichmüller factor is one. Consequently its normalized cubic sign is -1 and its quartic projective i-exponent is zero; these two actual cover coordinates violate the arithmetic graph of D.36. Its retained fixed-fibre coordinates remain $T_4 = I$ and $W_3 = [i]$.

The maximal common quotients with the full relative pro-two and pro-three sources over Q are also explicit. Let H_4, H_3 be the congruence quotient joined with the affine Fermat characters modulo $4, 3$, and let $L_j = \ker(B \to H_j)$. Then $|H_4| = 2304$, $|H_3| = 1728$, and

$$\begin{aligned} J_{12}^{\langle 2 \rangle} = B/[L_4, L_4]L_4^4, \quad \operatorname{rank} L_4 = 385, \\ J_{12}^{\langle 3 \rangle} = B/[L_3, L_3]L_3^3, \quad \operatorname{rank} L_3 = 289. \end{aligned} \tag{D.37.5}$$

Here the superscript means the universal quotient whose kernel over Q is a p-group. Their maximal common quotient is Q, and the canonical map has the exact kernel

$$\begin{aligned} \ker\left(J_{12} \longrightarrow J_{12}^{\langle 2 \rangle} \times_Q J_{12}^{\langle 3 \rangle}\right) \\ \simeq (\mathbb{Z}/4)^{768} \oplus (\mathbb{Z}/3)^{864}. \end{aligned} \tag{D.37.6}$$

These maps compute the mixed part of this finite test. They do not decide its lift to the full profinite group or the attainability of the entire CM pair.

The manuscript passage addressed is Part D(e), section 51.16, folio 734 verso, scan 174, together with D-after-LM2(1), section 51.15, folios 725–727, scans 156–160. The comparison keeps the two exact cover maps and the fixed-fibre maps of D.36.

Proof. We first give the quotient maps, including the affine marking. The following matrices and translations define a homomorphism from B to $\mathbb{Z}^2 \rtimes S_3$:

$$s \longmapsto \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, 0 \right), \qquad r \longmapsto \left(\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, e_1 \right). \tag{D.37.7}$$

Both defining torsion relations hold. The linear part is the specified S_3 quotient. Direct multiplication gives t^2 the translation $-e_1$, so $x = t^{-2}$ and $y = sxs$ give e_1, e_2. The restriction to F is therefore its ordered abelianization, with kernel F'. Its image contains every integral translation and both linear generators; thus (D.37.7) identifies B/F' with the displayed affine group by actual inverse generators. Reducing its translations modulo 12 and joining it with Q gives precisely B/L.

An element of the joint image is recorded as (q, a, b), where $q \in Q$ and $(a, b) \in (\mathbb{Z}/12)^2$; multiplication uses the linear matrix determined by q mod 2. Exact enumeration from the two displayed generators gives 6912 elements, with

$$\ker(H \to Q) = \{(a, b) : a, b \equiv 0 \pmod 2,\ a + b \equiv 0 \pmod 3\}. \tag{D.37.8}$$

This is $C_2^2 \times C_3$. Reduction of the same recorded states modulo 4 and modulo 3 gives orders 2304 and 1728; their fibre product over Q has order 6912, so the reduction map identifies that fibre product with H. The cusp t has order 24. The quotient maps to the full Fermat-twelve affine group and to Q have kernels of orders 8 and 12, respectively. The kernel in (D.37.8) is abelian of exponent six, so $[N, N]N^6 \subset L$. Conversely, $[L, L]L^{12} \subset [N, N]N^6$. These inclusions give the actual comparison chain

$$J_{12} \longrightarrow G_6 \longrightarrow H \longrightarrow Q.$$

The kernel L is torsion-free since it lies in F. The graph $L\backslash\mathcal{T}$ has 6912 edges and $6912/2 + 6912/3 = 5760$ vertices. Its spanning tree gives 1153 fundamental cycles and free generators. In particular, all ranks in (D.37.2) and (D.37.5) follow from these actual covering graphs. The full Fermat subgroup F_{12} has rank 145, and $[F_{12} : L] = 8$, which also checks the same rank by the free-group covering formula.

Write e_h for the oriented edge indexed by $h \in H$. The path cocycle is

$$P(s) = 0, \qquad P(r) = e_1 - e_r, \qquad P(ab) = P(a) + aP(b). \tag{D.37.9}$$

Its restriction to L is the integral cycle group. The recorded cycle matrix C has an identity submatrix on its 1153 non-tree edges. Thus reduction modulo any m faithfully represents J_m as the image in $(\mathbb{Z}/m)[H] \rtimes H$: its kernel on L is exactly $[L, L]L^m$, including for the nonfield coefficient rings $\mathbb{Z}/4$ and $\mathbb{Z}/12$.

There are six possible Q conjugators for r and eight for s in D.25's normalized matrix problem. Each has twelve lifts to H. Of the resulting $48 \cdot 144$ ordered choices, 576 satisfy

$$\beta s \beta^{-1} A r A^{-1} = t \tag{D.37.10}$$

already in H, and each induces an automorphism there. The code verifies this last assertion on all 6912 states and both generating transitions. It records the induced crossed homomorphism on the six linear S_3 elements; the resulting affine map is a permutation.

Put $s' = \beta s \beta^{-1}$ and $r' = ArA^{-1}$ in H. For corrections b, a in the cycle lattice, the complete cusp defect is the affine linear expression

$$\begin{aligned} \mathcal{E}(a, b) &= E + (1 - s')b + s'(1 - r')a, \\ E &= P_\beta - s'P_\beta \\ &\quad + s'\big(P_A - r'P_A + AP(r)\big) - P(t). \end{aligned} \tag{D.37.11}$$

Every term is evaluated in the exact graph module. Since s', r' generate H, the correction image has the same coinvariants as the original s, r image. Both its mod-two and mod-three ranks are 1152, with one-dimensional cokernel. Exhausting all 576 base choices gives

linear cusp condition	base choices	matrix pairs	extra pairs
mod2	288	24	12
mod3	192	48	24
both, with the same base choice	96	24	12.

(D.37.12)

These are linear solvability counts. Generation below is verified for the specified final witness. The cocycle of t^{72} has zero image in each of the two cokernels, so allowing the compatible lower-level multiplier changes $\lambda = 1 + 72j$ does not alter this table. The witness uses $\lambda = 1$ throughout.

We retain more than the two matrix coordinates. Under $H \to Q$, the graph-chain map sends e_h to $e_{q(h)}$. On cycle coordinates this gives an explicit integral 97×1153 matrix S. Its mod-two rank is 95. The target mod-two conjugators are the D.27 homology masks used throughout D.33, with the recorded right-centralizer term on β. Every later D.33 correction lies in $\Phi(N^{(2)})$, so this is its exact G_2 component. For the separate odd component, the target is $A = 1$ and the D.25 path to β plus twice its cycle number 39; this is D.33's specified G_3 centralizer representative. The proof uses that specified odd component, rather than reducing D.33's discrete pro-two word modulo three.

Appending the two exact conjugator projections to (D.37.11) gives ranks 1247 and 1248 at two and three, and affine dimensions 1059 and 1058. The selected base conjugators have affine coordinates

$$A = (I, (6, 0)), \qquad \beta = \left(\begin{pmatrix} 3 & 4 \\ 8 & 3 \end{pmatrix}, (6, 0) \right), \tag{D.37.13}$$

where the matrix is the canonical global-sign representative of (D.37.3). The generated mod-two witness is the first recorded choice; at three the successive kernel ranks are $1099, 1045, 1135, 1063, 1081, 1135, 1153$. The seventh choice is therefore the recorded generating one.

Here generation is an exact group assertion. If α_H is the verified base automorphism, the graph map sends

$$e_h \longmapsto \alpha_H(h)(P_A + a - P_\beta - b + e_A). \tag{D.37.14}$$

Its restrictions to all 1153 fundamental cycles have rank 1153 at both primes. The induced endomorphism of J_6 is already an automorphism of H and is invertible on its kernel, so it is an automorphism. Every cycle is also converted to a literal B word and checked integrally. Independent substitution of the two generator images into all those words reproduces the entire kernel matrices.

The exponent-four step retains this very J_6 action. Let F_0 be the integral correction matrix in (D.37.11), and let z be the chosen mod-two correction. The upper equation is

$$F_0 w = -\frac{E + F_0 z}{2} \pmod 2, \qquad z_4 = z + 2w. \tag{D.37.15}$$

The numerator is even in every cycle coordinate. The recorded witness has zero upper obstruction on its first test. This equation has rank 1152 and affine dimension 1154. The resulting z_4 is combined with the same mod-three solution by $z_{12} = 9z_4 + 4z_3 \pmod{12}$. Invertibility modulo two makes the kernel matrix invertible modulo four; together with the verified mod-three rank it is invertible over $\mathbb{Z}/12$.

The final cusp word has 673392 letters in s, r, r^{-1}. It evaluates to zero in all 6912 edge coordinates modulo 12, and all 1153 basis-image checks are repeated on the final pair. The

class of t^{24} has a nonzero reduction at both primes; its mod-twelve order is 12. This proves the asserted cusp order 288.

For clarity, the finite GT normalization is also checked on the same pair. Its integral F abelianization is $\beta^{\mathrm{ab}} = 882x + 180y$. Set $k = -702$ and $\beta' = x^k\beta$. Then

$$f' = s\beta' s(\beta')^{-1} \in F', \qquad \alpha' = \mathrm{Int}(x^k)\alpha. \tag{D.37.16}$$

The exact abelianization of f' is $(0, 0)$. The gauge fixes $u = t^{-1}$ and gives $\alpha'(s) = sf'$, $\alpha'(r) = (f')^{-1}r$ and $\alpha'(v) = (f')^{-1}vf'$, as in D.25. Both corresponding $m = 0$ hexagons are multiplied and checked in the faithful J_{12} model. The pullback of $[L, L]L^{12}$ to B_3 is normal, finite index, and contained in PB_3. The braid center is killed, and $x = t^{-2}$ has order144. Thus the commutator, unit, two-hexagon and surjectivity conditions of [BDHP26, Definitions 2.1 and 2.3] all hold for this actual finite shadow. The same left gauge on A, β, V leaves $T = A^{-1}V$ and $W = \beta^{-1}V$ unchanged. Its projection to G_6 retains D.25's conjugator-rigidity test, proving that (D.37.3) is an invariant of this shadow and has no Galois realization.

We next prove (D.37.4) through the actual cover maps. For $n = 3, 4, 12$ put $K_n = \ker(F \to \mathbb{Z}/n)$, with both loops mapping to one. There are inclusions $L \subset F_{12} \subset K_{12} \subset K_n$. The geometric Fermat marking also has an explicit map from the manuscript's plus model. Over $\mathbb{Q}(\zeta_{24})$, on $x_0^{12} + x_1^{12} + x_\infty^{12} = 0$ it is

$$z = -\frac{x_0^{12}}{x_\infty^{12}}, \qquad v = \zeta_{24}\frac{x_0x_1}{x_\infty^2}, \qquad v^{12} = z(z-1).$$

With $x_\infty = 1$, choose the positive-interval lifts $x_0 = \zeta_{24}z^{1/12}$ and $x_1 = \zeta_{24}^{-1}(1-z)^{1/12}$; this gives exactly D.36's cyclic-twelve tangent $v \sim \zeta_{24}z^{1/12}$. The maps to C_3, C_4 are $v \mapsto v^4, v^3$. Thus the even-degree twist and the field of this map are retained. The finite multiplier here is one modulo288, so every Galois element inducing this shadow would fix that field. For $m \mid 12$,

$$[L, L]L^{12} \subset [K_n, K_n]K_n^m. \tag{D.37.17}$$

Consequently restriction of the finite automorphism and abelianization give the actual action on $H_1(K_n, \mathbb{Z}/m)$. The ordered peripheral classes are preserved downstairs; their lifted conjugacy classes are permuted with the same powers. Filling their span therefore gives the compact action. The code verifies this span preservation directly as well.

There is no ambiguous free-basis conversion in this calculation. Collapsing the tree of the six-edge S_3 graph gives chord generators b_1, b_2 with $x = b_1$, $y = b_2^{-1}$, checked in both directions. The actual normalized f' becomes a 99308-letter word in $x^{\pm1}, y^{\pm1}$. For the Schreier basis

$$a = x^n, \qquad b_j = x^j y x^{-((j+1) \bmod n)} \quad (0 \le j < n),$$

let D_n denote conjugation by x on punctured homology, and let v_n be the exact homology class of f'. Since $\alpha'(x) = x$ and $\alpha'(y) = (f')^{-1}yf'$, its homology matrix has columns

$$e_a, \qquad e_{b_j} + D_n^j(D_n - I)v_n. \tag{D.37.18}$$

The program computes this matrix and independently substitutes x and $(f')^{-1}yf'$ into every Schreier word. The integral filling matrices P_n, J_n from D.36 give $P_nH_nJ_n$. These are exactly $-I$ modulo four at $n = 3$ and I modulo three at $n = 4$. The two equations

$$M_nH_{12}^{\mathrm{c}} = H_n^{\mathrm{c}}M_n \pmod{m_n}, \qquad (n, m_n) = (3, 4), (4, 3), \tag{D.37.19}$$

are checked using the same compact cyclic-twelve action and the actual matrices M_n of D.36. Thus these are evaluations of one source action under the specified maps.

The order-three lift of the cubic mod-two matrix is I, so its normalized principal-unit sign is -1. The quartic matrix is I, giving projective exponent zero. For an actual Galois element with this finite multiplier, D.36's tangent and cubic-twist comparisons impose its six-element arithmetic graph, where these two signs agree. Their computed disagreement gives a second direct certificate of nonarithmeticality. It supplies no unproved cover-to-fixed-fibre equality for GT0: both maps have instead been evaluated on the same finite shadow, giving the four coordinates stated above.

Finally, the universal quotient statements are exact group calculations. The two subgroup sequences are

$$1 \to L \to L_4 \to C_3 \to 1, \qquad 1 \to L \to L_3 \to C_2^2 \to 1.$$

Their homology five-term sequences give

$$(L^{\mathrm{ab}}/4)_{C_3} \simeq L_4^{\mathrm{ab}}/4, \qquad (L^{\mathrm{ab}}/3)_{C_2^2} \simeq L_3^{\mathrm{ab}}/3, \tag{D.37.20}$$

because the complementary group order is a unit in the coefficient ring. Equivalently, averaging its finite action gives the same explicit projection on each module. In a relative pro-two quotient of J_{12}, the three-primary module and the normal closure of that C_3 must be killed. The remaining module is exactly the first coinvariant module in (D.37.20); its extension over H_4 is $B/[L_4, L_4]L_4^4$. This extension has a two-group kernel over Q, so it is itself a quotient of the full relative pro-two source. This proves both factorizations and universality. The same argument with C_2^2 proves the pro-three assertion.

Any common quotient of those two relative quotients has a kernel over Q that is both a two-group and a three-group, hence is trivial. They therefore have maximal common quotient Q. Goursat's argument applied to their common surjection from J_{12} gives the full fibre product in (D.37.6). On the coefficient kernel, averaging over C_3 and over C_2^2 splits off modules of ranks 385 and 289. The complementary ranks are $1153 - 385 = 768$ and $1153 - 289 = 864$. Since a direct summand of a finite free module over $\mathbb{Z}/4$ is free, these complementary modules are exactly those in (D.37.6). This proves the stated kernel, not only its order. The existence of a lift of the recorded action through the remaining full source is still the exact all-finite-quotient question of D.22. □

The common quotient with the degree-eight action

D.38. A simultaneous mixed Fermat and degree-eight shadow [Proved]

Retain L, L_4, J_{12} from D.37 and the marked free rank-four group K from D.34. Put

$$E_2 = B/[L_4, L_4]L_4^4, \qquad T_3 = B/D_9(K). \tag{D.38.1}$$

The maximal common marked quotient of J_{12} and T_3, with their displayed maps from B, is

$$C = B/([L_4, L_4]L_4^4 D_9(K)). \tag{D.38.2}$$

The image of $D_9(K)$ in E_2 lies in $L_4^{\mathrm{ab}}/4$ and has exact group type $C_4^{51} \oplus C_2^{265}$. Consequently

$$\ker(C \to H_4) \simeq C_4^{69} \oplus C_2^{265}, \qquad |C| = 2304\, 2^{403} = 18\, 2^{410}. \tag{D.38.3}$$

There is an actual generating finite GT0 shadow on

$$\mathcal{J} = B/([L, L]L^{12} \cap D_9(K)) \simeq J_{12} \times_C T_3. \tag{D.38.4}$$

Its multiplier is one. Before the common Ihara gauge, its T_3 component is exactly the recorded D.33 action, and its two conjugators have exactly the prescribed common images in C. Its new

J_{12} component retains D.37's normalized G_6 and J_3 actions. It replaces the recorded D.37 J_{12} action: both normalized generator images differ already modulo two.

The common gauge exponent is $k = -774$. Since $x^k = c_0^{258}$, the normalized T_3 component is $\mathrm{Int}(c_0^{258})\alpha_{33}$. It therefore has exactly the same image in D.35's quotient by $\langle \mathrm{Int}(c_0) \rangle$. The compact evaluations, in D.36's bases, are

$$\rho_{C_3,16} = \begin{pmatrix} 3 & 8 \\ 8 & 11 \end{pmatrix}, \qquad \rho_{C_4,3} = I. \tag{D.38.5}$$

The cubic reduction modulo four is $-I$, so the same arithmetic-graph violation and the same extra fixed-fibre pair $T_4 = I,\ W_3 = [i]$ persist.

The orders and projection kernels are

$$\begin{gathered} |\mathcal{J}| = 2^{13472}3^{1156}, \\ |\ker(\mathcal{J} \to J_{12})| = 2^{11158}, \qquad |\ker(\mathcal{J} \to T_3)| = 2^{1903}3^{1154}. \end{gathered} \tag{D.38.6}$$

Its cusp still has order 288 and its finite-shadow N_{ord} is 144. For the two exact source maps

$$\widehat{B} \twoheadrightarrow \mathcal{J}, \qquad \widehat{F} \twoheadrightarrow \ker(\mathcal{J} \to S_3),$$

the kernel is the closure of the intersection in (D.38.4), taken in the respective source. It contains the explicit nonzero word

$$[a_1^{12}, a_2^{12}]^2, \tag{D.38.7}$$

where a_1, a_2 are the first two marked generators of K. This word has a nonzero degree-sixteen mod-two Magnus term. Thus the simultaneous finite action leaves an explicit nontrivial deck kernel for the full lifting test. D.45 and its source-transport supplement exclude a full profinite lift of this particular action, whose degree-eight component differs from D.33 by the stated source cusp-inner map.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160. The comparison retains D.33's degree-eight component and the actual two cover maps of D.37; it does not assert compatibility with D.33's entire infinite class-two system.

Proof. We compute the common quotient before solving any conjugator equation. Let $V = L_4^{\mathrm{ab}}/4 \simeq (\mathbb{Z}/4)^{385}$, the normal coefficient subgroup of E_2, and let G be the image of K there. The quotient $A_0 = K/L_4$ has order 128. It embeds in

$$O_2(Q) \times (\mathbb{Z}/4)^2, \tag{D.38.8}$$

because the affine linear part of every K element is the identity. The two factors have class at most two and exponent four. Thus A_0 has those properties, and its commutator group has exponent two. The program also verifies these statements on every pair of its 128 elements.

It follows that $\gamma_3 G \subset V$, every element of G has order dividing 16, and every element of $\gamma_2 G$ has order dividing 8. The latter assertion uses $(\gamma_2 G)^2 \subset V$. Write $U = \gamma_3 G$, and let I be the augmentation ideal for the action of the four actual marked K generators on V. Since V is abelian, for every $j \geq 3$,

$$\gamma_j G = I^{j-3}U. \tag{D.38.9}$$

The Zassenhaus product $D_n(G) = \prod_{i2^r \geq n} (\gamma_i G)^{2^r}$ therefore gives the exact equality

$$W := \mathrm{im}(D_9 K \to E_2) = \gamma_9 G + 2\gamma_5 G = I^6 U + 2I^2 U. \tag{D.38.10}$$

Indeed G^{16}, $(\gamma_2 G)^8$ and $(\gamma_3 G)^4$ vanish, and the remaining terms are contained in the two displayed summands. The equality is functorial under the surjection $K \twoheadrightarrow G$, since every lower-central commutator and every power in the Zassenhaus product has a preimage of the same form.

All groups in (D.38.10) are computed in one integral graph module. The H_4 graph has 2304 edges and 1920 vertices; its 385 non-tree edges give the identity minor of its cycle matrix. The four marked K words are evaluated in the faithful model $(\mathbb{Z}/4)[H_4] \rtimes H_4$. Their twenty-four basic triple commutators lie in V. Taking their module span and repeatedly adjoining their four translates gives their full normal closure U. This is $\gamma_3 G$: making the basic generator commutators central is precisely the quotient of class at most two.

The successive exact group types are

j	number of C_4 factors in $\gamma_j G$	number of C_2 factors
3	375	1
4	355	11
5	316	38
6	252	85
7	178	134
8	106	146
9	51	127

(D.38.11)

Adjoining $2\gamma_5 G$ to the last row gives $W \simeq C_4^{51} \oplus C_2^{265}$. The calculation performs elimination over $\mathbb{Z}/4$: odd pivots give independent order-four generators, and the remaining even rows reduce to exact binary elimination. It also checks membership of every generator translate, including the full s, r action. This verifies the normal closure and every subsequent module product, rather than only their rational ranks.

Quotienting the free coefficient module V by W therefore leaves $C_4^{385-51-265} \oplus C_2^{265}$, proving (D.38.3). The universal relative pro-two quotient of J_{12} over Q is E_2, by D.37. On the other hand, $D_9 K \subset N$, and $T_3 \to Q$ has a two-group kernel. Every common quotient of J_{12} and T_3 over Q must consequently factor through E_2. The largest common quotient of E_2 and $B/D_9 K$ is exactly E_2/W, by the definition of the product of the two normal kernels. This proves (D.38.2), in both directions, and the normal-subgroup Chinese remainder isomorphism proves (D.38.4).

We now impose the exact common images of the recorded D.33 conjugators. Both long words are reconstructed from its retained textual coefficients, without rerunning or changing any earlier solver. Their SHA256 hashes agree with the original reconstruction certificate, and their full cusp word has the recorded length 12921196. Evaluation in E_2 gives paths P_A^* and P_β^*. Their cusp defect lies in W, providing a direct check that the recorded T_3 action gives its stated common-quotient action.

The projection $H \to H_4$ sends each graph edge to its projected edge. On the cycle bases it gives an exact matrix

$$S_4 : \mathbb{Z}^{1153} \longrightarrow \mathbb{Z}^{385}. \tag{D.38.12}$$

Let a, b be the two unknown L cycle corrections, and let Z be the 385×316 matrix of the recorded generators of W. In addition to D.37's complete cusp equation modulo four, impose

$$\begin{aligned} S_4 a - Z u_A &= (P_A^* - \pi P_A)_{\text{cycles}}, \\ S_4 b - Z u_\beta &= (P_\beta^* - \pi P_\beta)_{\text{cycles}}. \end{aligned} \tag{D.38.13}$$

These equations state equality of the actual conjugators in C, including all central coordinates. The prior exact G_2 conjugator constraints are included as twice their binary equations. There are 2117 equations and 2938 unknowns over $\mathbb{Z}/4$.

Unit-pivot elimination has rank 1588. The residual even system, divided by two, has binary rank 263. The full solution space is an explicitly parametrized torsor under

$$C_4^{1087} \oplus C_2^{263}. \tag{D.38.14}$$

The first recorded solution has a full rank 1153 map on $L^{\mathrm{ab}}/2$. It is combined with exactly the same mod-three correction as D.37. Thus it gives an automorphism of J_{12}: its mod-four kernel matrix is invertible because its mod-two reduction is, and its unchanged mod-three matrix was already verified to be invertible.

The resulting 669040-letter cusp word is evaluated in all 6912 coordinates modulo12. Every one of the 1153 free-kernel basis images is independently computed by generator substitution and agrees with the graph map. The program also checks (D.38.13) against the stored W coefficients and against the separate exact membership reduction.

The two component automorphisms agree on C because their conjugators do. More explicitly, their compositions from B to C agree on s, r; one kills the J_{12} kernel and the other kills D_9K, so this common homomorphism factors through their product kernel. It is onto because either component is an automorphism. The product of the two component automorphisms therefore restricts to an automorphism of the fibre product in (D.38.4). This is generation for the actual joint finite group, not a rank assertion for a linear approximation.

Both component conjugators lie over the identity permutation, so their compatible pairs lift to actual words of F under the surjection $B \to \mathcal{J}$. For the recorded J_{12} representative, $\beta^{\mathrm{ab}} = 666x - 108y$. The gauge exponent is $-108 - 666 = -774$. Every other F lift with the same J_{12} image changes this abelianization by an element of $144\mathbb{Z}^2$: the kernel is $[L, L]L^{12}$ and $L \subset F_{12}$. Write the abelianization of a compatible lift as $(666 + 144u, -108 + 144v)$. With the displayed gauge $k = -774$, the word $f = s(x^k\beta)s(x^k\beta)^{-1}$ has abelianization

$$f^{\mathrm{ab}} = (144(v - u), 144(u - v)) \equiv (0, 0) \pmod{144}.$$

Both x and $y = sxs$ have order 144 in $\mathcal{J}$: x has orders 144 and 48 in the two components. Putting $\epsilon = v - u$, the word

$$f_0 = x^{-144\epsilon} y^{144\epsilon} f \in F'$$

has the same joint finite image as f. Thus the joint f coset contains an actual commutator word, including the effect of every compatible T_3 lift.

The two $m = 0$ hexagons follow as in D.25 from the torsion and cusp equations, and are also checked in the new faithful J_{12} component. They hold in T_3 by the same identities for its recorded action and the common inner gauge. Together with the joint automorphism this verifies the finite GT0-shadow conditions of [BDHP26, Definitions 2.1 and 2.3]. The preimage of the joint kernel under $B_3 \to B$ is normal, has finite index, and lies in PB_3, since the joint quotient retains $B \to S_3$. The braid centre maps to one. The cusp has order $\mathrm{lcm}(288, 96) = 288$, and both pure loops have order 144; hence the pulled-back kernel has $N_{\mathrm{ord}} = 144$ and the multiplier is one.

The gauge is particularly explicit on the degree-eight component: $x^{-774} = t^{1548} = (t^6)^{258} = c_0^{258}$. It therefore changes the recorded T_3 automorphism by exactly the inner automorphism stated in the theorem. It fixes its class under D.35's specified cusp-inner quotient. On the other component, the normalized new and old images of each of s, r have nonzero differences on 5188 edge coordinates modulo12, and already differ modulo two. Their projections to G_6 agree, and their reductions modulo three on the full H module agree. These checks prove the asserted replacement and both retained normalized actions.

The compact-cover evaluation is repeated using the new normalized 97388-letter F word. All Schreier-generator substitutions for C_3, C_4, C_{12} and both compact projection identities pass. They again give $-I$ on cubic four-torsion and I on quartic three-torsion.

For the finer cubic component, D_9K maps to $16K^{\mathrm{ab}}$, so T_3 retains the full $K^{\mathrm{ab}}/16$ action. The exact compact change of basis from D.34 to D.36 is

$$B_{\mathrm{c}} = \begin{pmatrix} -1 & 1 \\ -1 & 0 \end{pmatrix}. \tag{D.38.15}$$

It is obtained by rewriting the four marked K words in the cyclic-three Schreier basis and applying the two integral filling maps. Its determinant is one, it intertwines the marked deck matrices, and $B_{\mathrm{c}} M_{16} B_{\mathrm{c}}^{-1} = M_{16}$ for $M_{16} = \left(\begin{smallmatrix} 3 & 8 \\ 8 & 11 \end{smallmatrix}\right)$. The gauge is a K-inner cusp conjugation, so it acts trivially on this homology. This proves (D.38.5) through the exact comparison maps. D.36's arithmetic graph therefore still excludes a Galois realization; the unchanged G_6 CM pair gives the independent exclusion of D.25.

For the orders, the restricted free-Lie dimensions of rank four in degrees one through eight are

$$4,\ 10,\ 20,\ 70,\ 204,\ 690,\ 2340,\ 8230.$$

Their sum is 11568, giving $|T_3| = 18\,2^{11568}$. Equations (D.37.2), (D.38.3) and the fibre-product formula give every order in (D.38.6).

Both kernels in (D.38.4) are normal and have finite index in B, and their intersection is contained in F. Consequently their completions give exactly the two source maps in the statement; the open embedding $\widehat{F} \hookrightarrow \widehat{B}$ identifies their kernels. Finally, each marked a_i^{12} lies in L. Indeed the K image in H embeds in $O_2(Q) \times (\mathbb{Z}/12)^2$ and has exponent dividing 12; the two specific words are also checked directly. Thus (D.38.7) lies in $[L, L]$. In the free mod-two Magnus algebra, the first term of $a_i^{12} - 1$ is X_i^4. The first term of the commutator is consequently $X_1^4 X_2^4 + X_2^4 X_1^4$, and squaring gives four distinct degree-sixteen monomials. It is nonzero and belongs to $D_{16}K \subset D_9K$. This proves both its inclusion in the stated remaining kernel and its nontriviality in the full free pro-two, hence full profinite, source. The D.45 obstruction, transported through the exact cusp-inner gauge above, excludes a full profinite lift of this recorded finite shadow. The inverse-system test of D.22 remains required for a replacement realizing its CM pair. □

An action on the entire mixed Fermat homology tower

D.39. An integral all-prime homology lift retaining the degree-eight and CM data [Proved]

Use the marked groups B, F, K, L, L_4, H, H_4 of D.34–D.38, and write closures inside $\widehat{B}$. Set

$$\begin{aligned} \mathcal{H} &= \widehat{B}/\overline{[L,L]} = \varprojlim_m B/[L,L]L^m, \\ \mathcal{T} &= \widehat{B}/(\overline{[L,L]} \cap \overline{D_9K}), \qquad T_3 = B/D_9K. \end{aligned} \tag{D.39.1}$$

The maximal common marked quotient of $\mathcal{H}$ and T_3 is

$$C_8 = B/([L_4, L_4]D_9K), \qquad \mathcal{T} \simeq \mathcal{H} \times_{C_8} T_3. \tag{D.39.2}$$

Its coefficient kernel and order are

$$\ker(C_8 \to H_4) \simeq C_8^7 \oplus C_4^{62} \oplus C_2^{265}, \qquad |C_8| = 2304\,2^{410} = 18\,2^{417}. \tag{D.39.3}$$

Here C_n in an abelian group type denotes a cyclic group of order n. The canonical map from this common quotient to D.38's common quotient has kernel C_2^7.

There is an actual continuous automorphism of $\mathcal{T}$, with multiplier exactly one, satisfying the peripheral and two hexagon equations in this quotient. It gives one compatible generating

family of finite GT0 shadows throughout this homology tower. Its normalized G_6 and T_3 actions are exactly those of D.38. Its normalized action on every odd-primary homology component $\varprojlim_n B/[L,L]L^{p^n}$ is the identity. In particular, it replaces D.38's J_3 and J_{12} actions, while retaining the fixed-fibre pair $T_4 = I,\ W_3 = [i]$.

The construction retains an explicitly checked seed on

$$\mathcal{J}_{48} = J_{48} \times_{C_8} T_3, \qquad J_{48} = B/[L,L]L^{48}, \tag{D.39.4}$$

with

$$\begin{gathered} |\mathcal{J}_{48}| = 2^{15771}3^{1156}, \qquad |\ker(\mathcal{J}_{48} \to J_{48})| = 2^{11151}, \\ |\ker(\mathcal{J}_{48} \to T_3)| = 2^{4202}3^{1154}. \end{gathered} \tag{D.39.5}$$

Its cusp has order 1152 and its finite-shadow N_{ord} is 576. The recorded seed has $\beta^{\mathrm{ab}} = (1482, 1044)$ and gauge $k = -438$. Its normalized degree-eight action is

$$\mathrm{Int}(c_0^{146})\alpha_{33} = \mathrm{Int}(c_0^{258})\alpha_{33}, \tag{D.39.6}$$

the second expression being D.38's action. The equality follows from the exact order 16 of $c_0 = t^6$ in T_3. Through the marked compact-cover maps of D.36 and D.38, its actual matrices are

$$\rho_{C_3,16} = \begin{pmatrix} 3 & 8 \\ 8 & 11 \end{pmatrix}, \qquad \rho_{C_4,3} = I. \tag{D.39.7}$$

Thus every finite refinement in this family retaining $\mathcal{J}_{48}$ has the same nonarithmetical CM data.

The exact remaining kernel of each source map

$$\widehat{B} \twoheadrightarrow \mathcal{T}, \qquad \widehat{F} \twoheadrightarrow \ker(\mathcal{T} \to S_3)$$

is $\overline{[L,L]} \cap \overline{D_9K}$, with the open embedding of $\widehat{F}$ understood. It contains the nonzero word $[a_1^{12}, a_2^{12}]^2$ of D.38. The canonical projection from $\mathcal{T}$ to D.38's underlying finite joint group is strict: x^{144} is killed there and survives in $\mathcal{T}$. The present action induces a replacement of its J_{12} component, as specified above. D.45 proves that this recorded action has no lift to an actual full profinite GT0 element. The prescribed CM pair remains a target for a different action.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160. It establishes compatibility across every prime and every finite homology level, together with the retained degree-eight action.

Proof. We first specify the integral source and its prime components. The finite-index subgroup L is free of rank 1153. Its induced profinite topology is its full topology: for a finite-index subgroup of L, its normal core in B is again finite-index. Thus its closure is $\widehat{L}$, and

$$\overline{L}/\overline{[L,L]} \simeq \widehat{\mathbb{Z}}^{1153}. \tag{D.39.8}$$

The graph of D.37 has 6912 edges and 5760 vertices. Its integer cycle matrix has the identity on its 1153 chosen non-tree edges. Consequently its path model identifies $B/[L,L]$ with the subgroup of $\mathbb{Z}[H] \rtimes H$ whose elements have the specified path boundary. Reducing the coefficients modulo m, or completing them in $\mathbb{Z}_p$ or $\widehat{\mathbb{Z}}$, gives the corresponding groups without losing any cycle coordinate.

Put $\mathcal{H}_p = \varprojlim_n J_{p^n}$. There is a canonical isomorphism

$$\mathcal{H} \longrightarrow \prod_{p,H} \mathcal{H}_p, \tag{D.39.9}$$

where the product consists of tuples with the same image in H. To prove surjectivity, choose one word lifting their common base element and subtract its path in every component; the remaining cycle vectors form an element of $\prod_p \mathbb{Z}_p^{1153} = \widehat{\mathbb{Z}}^{1153}$. The same cycle-coordinate description proves injectivity. This also proves the inverse-limit assertion in (D.39.1).

The stabilized common quotient. Work first in $E_8 = B/[L_4, L_4]L_4^8$, with coefficient subgroup $V_8 = (\mathbb{Z}/8)^{385}$, and let G be the image of K. The quotient $A_0 = K/L_4$ has order 128, class two and exponent four; its derived group has order two. These facts are checked on all its elements using the faithful H_4 graph model, and also follow from D.38's embedding in $O_2(Q) \times (\mathbb{Z}/4)^2$ together with the displayed generators. Hence $\gamma_3 G$ lies in the abelian subgroup V_8. With I the augmentation ideal of the actual four K-generator actions and $U = \gamma_3 G$, one has $\gamma_j G = I^{j-3}U$ for $j \geq 3$.

The full Zassenhaus product in this exponent-eight module is

$$W_8 := \mathrm{im}(D_9 K \to E_8) = G^{16} + (\gamma_2 G)^8 + 4\gamma_3 G + 2\gamma_5 G + \gamma_9 G. \tag{D.39.10}$$

Each displayed summand lies in V_8, so the sum is an abelian-module sum. To exhaust the product, the terms with $i = 1, 2$ begin at powers $16, 8$, respectively; those with $i = 3, 4$ are contained in $4\gamma_3 G$; those with $5 \leq i \leq 8$ are contained in $2\gamma_5 G$; and all terms with $i \geq 9$ lie in $\gamma_9 G$. Further powers are already in these summands. Surjectivity of $K \to G$ lifts every commutator and power appearing in this description.

The two power summands in (D.39.10) are computed on their entire groups. Choose actual lifts g_h of all 128 elements h of A_0, and denote their coefficient actions by M_h. Every element of G is vg_h, with $v \in V_8$, and

$$(vg_h)^{16} = (1 + M_h + \cdots + M_h^{15})v + g_h^{16}.$$

For h of order one or two the norm vanishes modulo eight. For h of order four it is $4(1 + M_h + M_h^2 + M_h^3)$. The program includes each actual g_h^{16} and every column of this norm on the full coefficient module. Their span is C_2^{327}.

For the derived subgroup, select an actual basic commutator c projecting to the nontrivial element of A_0'. Its intersection with V_8 is generated by the following explicit vectors: c^2; the conjugates of the six basic K commutators by all g_h, subtracting c when their base image is nontrivial; and $(1 - M_c)V_8$. These vectors belong to $\gamma_2 G$. Conversely their span, together with the coset represented by c, is a normal subgroup containing every basic commutator. Normality under the four generators and M_c is checked, and conjugation by V_8 is included by $(1 - M_c)V_8$. This proves equality with $\gamma_2 G$, including all conjugator freedom. Its coefficient intersection has type C_8^{381}. The eighth powers in the nontrivial coset are exactly

$$(vc)^8 = 4(1 + M_c)v + c^8, \qquad v \in \gamma_2 G \cap V_8;$$

the eighth powers in the identity coset vanish. Their span is C_2^{189}.

The remaining summands are obtained by exact module elimination over $\mathbb{Z}/8$. The successive group types are as follows; the entries give the numbers of cyclic factors of the indicated orders.

j	C_8	C_4	C_2
3	375	1	0
4	355	11	10
5	316	38	18
6	252	85	18
7	178	134	28
8	106	146	84
9	51	127	134

(D.39.11)

The normal closure of the twenty-four basic triple commutators gives the first row. Repeated multiplication by I gives the subsequent rows. Every translate by a marked K generator is checked. Summing all five terms in (D.39.10) gives

$$W_8 \simeq C_8^{51} \oplus C_4^{265} \oplus C_2^{62}. \tag{D.39.12}$$

Its full s, r stability is separately verified. Reduction modulo four gives exactly D.38's recorded D_9 image. Quotienting V_8 by (D.39.12) proves (D.39.3), and the kernel of reduction to the earlier common quotient is the quotient of $4V_8$ by its intersection with W_8. It has order 2^7 and exponent two.

These finite computations determine the common quotient at every higher precision. Indeed $L_4 \subset N \subset D_2K$, whence

$$L_4^8 \subset D_{16}K \subset D_9K. \tag{D.39.13}$$

In $\mathcal{E}_2 = \varprojlim_n B/[L_4, L_4]L_4^{2^n}$, the image of D_9K therefore contains $8\mathbb{Z}_2^{385}$ and is exactly the inverse image of W_8 modulo eight. D.37's relative-prime argument also applies integrally: the maximal relative pro-two quotient of $\mathcal{H}$ over Q is $\mathcal{E}_2$. Explicitly, the complementary group $L_4/L \simeq C_3$ has vanishing positive-degree homology over $\mathbb{Z}_2$, and its coinvariants on $L^{\mathrm{ab}} \otimes \mathbb{Z}_2$ are $L_4^{\mathrm{ab}} \otimes \mathbb{Z}_2$. Killing its normal closure and the odd coefficient factors gives this map; every quotient over Q with pro-two kernel kills them and factors through the same map. Since $T_3 \to Q$ has two-group kernel, every common marked quotient factors through $\mathcal{E}_2$. Dividing it by the just-computed image of D_9K proves (D.39.2) in both directions. The quotient by the intersection of the two closed normal kernels then gives its asserted fibre product.

The exact integral two-adic correction. We retain the actual conjugators of D.33 in T_3. They are reconstructed from their recorded coefficients; both SHA256 values and their 12 921 196-letter cusp reconstruction agree with the original certificate. Their images in E_8 determine their exact common images in C_8. Their cusp defect belongs to W_8, which is checked independently of the next solver.

Let A_H, β_H be D.37's chosen conjugator images in H, let P_A, P_β be their canonical integral paths, and put $s_* = \beta_H s \beta_H^{-1}$ and $r_* = A_H r A_H^{-1}$. Thus $s_* r_* = t$. Left corrections by cycle vectors $a, b \in \mathbb{Z}_2^{1153}$ give the complete cusp equation

$$(1 - s_*)b + (s_* - t)a = -E, \tag{D.39.14}$$

where E is the integral defect of the two canonical paths. All terms are written in the identity-minor cycle basis. This equation is exact because the coefficient subgroup is abelian; it includes the full chosen paths and their base action.

Let S_4 and S_Q be the integer cycle maps induced by $H \to H_4$ and $H \to Q$. Let $\mathcal{W} \subset \mathbb{Z}_2^{385}$ be the inverse image of W_8. Its basis is the 378 recorded rows of (D.39.12), together with $8e_j$ on the seven nonpivot columns. After the recorded column permutation, its basis matrix is triangular with diagonal

$$1^{51}, \quad 2^{265}, \quad 4^{62}, \quad 8^7. \tag{D.39.15}$$

It has index 2^{410} and contains $8\mathbb{Z}_2^{385}$. Write Z for its column matrix. In addition to (D.39.14), impose

$$\begin{aligned} S_4 b - Z u_\beta &= r_\beta, \quad & S_4 a - Z u_A &= r_A, \\ S_Q b - 2v_\beta &= r_{\beta,2}, \quad & S_Q a - 2v_A &= r_{A,2}. \end{aligned} \tag{D.39.16}$$

The first row prescribes the actual D.33 conjugators in C_8. The second row prescribes exactly the retained G_2 conjugators. Each right-hand side is the difference between the target path and the canonical path under the indicated map, in its recorded integer residue representative. The

variables u_A, u_β have 385 coordinates and v_A, v_β have 97 coordinates. Thus (D.39.14)–(D.39.16) form a signed integer matrix M of size 2117×3270.

The exact elimination modulo 256 gives full row rank, with invariant factors

$$1^{1588}, \quad 2^{457}, \quad 4^{64}, \quad 8^{7}, \quad 16. \tag{D.39.17}$$

These determine the integral $\mathbb{Z}_2$ invariant factors. To see why the finite calculation suffices, lift its elementary row and column operations to $\mathbb{Z}_2$. The transformed matrix is $[D\ 0] + 256R$, with D the diagonal in (D.39.17). Factoring D on the left leaves a square leading block congruent to the identity modulo 16, hence invertible. The remaining columns are also divisible by the corresponding diagonal entries. Thus the exact image has the displayed invariant factors. In particular

$$16\mathbb{Z}_2^{2117} \subset M\mathbb{Z}_2^{3270}, \qquad \operatorname{rank}_{\mathbb{Z}_2} \ker M = 1153. \tag{D.39.18}$$

The recorded solution z modulo 256 has a full-rank 1153×1153 kernel map modulo two. It lifts to an exact solution retaining z modulo 16: its integral residual is divisible by 256, so dividing that residual by 16 places it in the image in (D.39.18). This proves existence without assuming that an arbitrary mod-two solution lifts. The certificate also constructs the lift recursively. From a solution modulo 2^{k+4}, solve

$$M\delta = \frac{b - Mz}{2^k} \pmod{256}, \qquad z_{\text{new}} = z + 2^k\delta \pmod{2^{k+8}}, \tag{D.39.19}$$

where b denotes the whole signed right-hand side. The new right-hand side is divisible by 16, so (D.39.18) makes this finite equation solvable. Beginning with $k = 4$ and increasing k by four gives compatible coefficients with a two-adic limit. Two exact steps, retaining moduli 16 and 256 and solving at moduli 4096 and 65536, are checked using the actual signed integer matrix. The multiplier remains exactly one.

The resulting homomorphism of $\mathcal{H}_2$ induces the prescribed base automorphism and a matrix in $\mathrm{GL}_{1153}(\mathbb{Z}_2)$ on its coefficient kernel. The mod-two invertibility just verified proves this integral invertibility. It is therefore an actual automorphism. Equations (D.39.16) retain exactly the common conjugators needed to pair it with T_3.

One formula for every odd prime. The selected base action on H is precisely $\mathrm{Int}(x^6)$, checked on all 6912 elements; moreover $A_H = x^6$. Put $h = x^{-6}\beta_H$. It centralizes s in H. Choose its canonical word c and path P_c. The vector $\delta = P_c - sP_c$ is an integral cycle and satisfies $s\delta = -\delta$. At every odd prime, replace the path of c by

$$P_{c,p} = P_c - \tfrac{1}{2}\delta = \tfrac{1}{2}(P_c + sP_c). \tag{D.39.20}$$

This is an element of $\mathcal{H}_p$ with the same base image, and it centralizes s exactly. Indeed its base commutes with s, its path is s-fixed, and $P_s = 0$. Set $A_p = x^6$ and $\beta_p = x^6 c_p$. Both generator images are now their conjugates by x^6, so the resulting automorphism is exactly $\mathrm{Int}(x^6)$. It fixes t, and its coefficient map is an invertible permutation action. Formula (D.39.20) is one formula over $\mathbb{Z}[1/2]$; it simultaneously supplies all odd-prime components at every precision.

The two-adic and odd-prime conjugators have the same images A_H, β_H in H. Isomorphism (D.39.9) therefore assembles them into actual conjugators and an automorphism of $\mathcal{H}$. Equations (D.39.16) ensure their equality with the D.33 conjugators in C_8. The two component automorphisms agree there on s, r, hence on the whole common quotient. Their product consequently gives an automorphism of $\mathcal{T}$. This proves generation on the entire joint group and every displayed finite reduction.

A single compatible gauge and the finite seed. The image $\mathcal{F} = \ker(\mathcal{T} \to S_3)$ has abelianization $\widehat{\mathbb{Z}}^2$ in the marked x, y basis, because the source kernel in (D.39.1) is contained in $\overline{[L,L]} \subset \widehat{F}'$. For the assembled conjugator write $\beta^{\mathrm{ab}} = (a, b)$ and set $k = b - a \in \widehat{\mathbb{Z}}$. Then $\beta' = x^k\beta$ and

$$f = s\beta' s(\beta')^{-1} \in \mathcal{F}' \tag{D.39.21}$$

have an actual lift of f in $\widehat{F}'$. This follows directly by abelianizing: s interchanges x and y, so $f^{\mathrm{ab}} = (b - a - k, a + k - b) = 0$. The normalized automorphism is $\mathrm{Int}(x^k)\alpha$. It still fixes t; its torsion relations give the two $m = 0$ hexagons as in D.25. Thus one common profinite gauge works at all levels.

There is also an exact description of its odd components. In the p-adic abelianization, (D.39.20) gives equal coordinates for c_p, hence $k_p = -6$ for every odd p. The recorded finite two-adic seed gives $k_2 + 6 \equiv 0 \pmod 4$. In $\mathcal{H}_p$, the closed subgroup generated by x is an extension of its order-twelve base image by a pro-p group. Therefore $x^{k+6} = 1$: its p-component exponent is zero, and the prime-to-p part of the order divides the prime-to-p part of 12, on which $k + 6$ also vanishes. It follows that the normalized odd component is $\mathrm{Int}(x^{k+6}) = 1$ on the whole group.

For a literal finite check, combine the retained two-adic coefficients modulo 16 with (D.39.20) modulo three by the Chinese remainder map to $\mathbb{Z}/48$. The resulting 2 676 134-letter cusp word vanishes in all 6912 coordinates. Substitution on every one of the 1153 kernel basis words agrees with the computed graph matrix. Its mod-two rank is 1153, and its mod-three matrix is exactly the invertible inner action by x^6, with its inverse checked. Both $m = 0$ hexagons are checked separately. The integral lift above retains these mod-sixteen coefficients, and the odd formula retains their mod-three coefficients. Thus this entire J_{48} representative remains unchanged in the limit.

Its literal β word has abelianization $(1482, 1044)$ and gauge -438. Every lift with its J_{48} image changes these coordinates by elements of $576\widehat{\mathbb{Z}}^2$, since the kernel is $\overline{[L,L]L^{48}}$ and $L \subset F_{12}$. Consequently the global k reduces to -438 modulo 576. The loop x has order 576 in J_{48}, as checked by its 576th, 288th and 192nd powers. It has order 48 in T_3. Thus the same finite gauge acts on the joint seed. In particular the finite coset of f contains a discrete commutator word: the residual abelianization of a compatible F lift is a multiple of $(576, -576)$, removable by powers of x^{576} and y^{576} in the joint kernel.

The difference from D.38's gauge is $-438 - (-774) = 336 = 7 \cdot 48$. The normalized T_3 actions therefore agree exactly, proving (D.39.6). Projection of the two new normalized generator images to G_6 also agrees exactly with D.38; this is checked in all 97 coefficient coordinates. On the whole J_3 module each generator instead has the identity action. Comparing with D.38 gives 4544 nonzero mod-three edge-coordinate differences for each of s, r. Both J_{12} images are replaced as well. These are comparisons under the actual quotient maps, including the common gauge.

The normalized 390 284-letter F word is evaluated on the cyclic-three, cyclic-four and cyclic-twelve covers of D.36. All Schreier substitutions and both compact projection maps are checked. They give $-I$ on cubic four-torsion and I on quartic three-torsion. For the cubic Tate-sixteen refinement, the retained T_3 map and the exact basis matrix $\left(\begin{smallmatrix} -1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ of D.38 give precisely (D.39.7). The preserved G_6 map gives the separately marked fixed-fibre pair. D.36's arithmetic graph excludes the pair in (D.39.7), and D.25 excludes the fixed-fibre pair from the Galois image.

For every m divisible by 48, the common quotient of J_m and T_3 is still C_8, by (D.39.13) and the relative-prime factorization. The automorphism preserves every subgroup $m\widehat{\mathbb{Z}}^{1153}$ of the coefficient kernel. Hence its reductions to $J_m \times_{C_8} T_3$ are compatible automorphisms. Their pulled-back kernels under $B_3 \to B$ are normal, finite-index, and contained in PB_3; the

braid centre is killed. The common commutator and torsion/cusp equations give the finite GT0-shadow conditions of [BDHP26, Definitions 2.1 and 2.3]. The seed cusp has order 1152, since $x = t^{-2}$ has order 576 and t has even order. Its two pure loops therefore have order 576 and the pulled-back kernel has $N_{\mathrm{ord}} = 576$.

Finally $|J_{48}| = 6912\,48^{1153}$ and $|T_3| = 18\,2^{111568}$. Dividing their product by (D.39.3) proves every order in (D.39.5). The same normal-kernel argument proves the exact source maps in the statement. The word $[a_1^{12}, a_2^{12}]^2$ lies in $[L, L]$ and has the nonzero degree-sixteen Magnus term computed in D.38, so it also belongs to D_9K and survives in the full free pro-two source. The canonical map to D.38 has kernel

$$\frac{\overline{[L,L]L^{12}} \cap \overline{D_9K}}{\overline{[L,L]} \cap \overline{D_9K}}. \tag{D.39.22}$$

It contains the nonzero class of x^{144}: that element is killed by both finite components of D.38, but has abelianization $(144, 0)$ in the present $\mathcal{F}$. Its survival is also checked in the new J_{48} seed. These exact kernels specify the source of the lifting test; D.45 proves that its fibre over this recorded action is empty. □

Simultaneous perfect quotients and a further mixed extension

D.40. An extension through every finite perfect source quotient [Proved]

Let $\mathscr{P}$ be the set of finite-index normal subgroups M of the marked group $B = C_2 * C_3$ for which B/M is perfect. Define the actual marked quotient

$$\mathcal{R} = \overline{\operatorname{im}\left(B \longrightarrow \prod_{M \in \mathscr{P}} B/M\right)}, \qquad K_{\mathcal{R}} = \bigcap_{M \in \mathscr{P}} \overline{M} \ \subset \widehat{B}. \tag{D.40.1}$$

Then $\mathcal{R} = \widehat{B}/K_{\mathcal{R}}$ and its maximal prosolvable quotient is exactly

$$\mathcal{R}/\mathcal{R}' \simeq B^{\mathrm{ab}} \simeq C_6. \tag{D.40.2}$$

Here $\mathcal{R}'$ denotes the closed subgroup generated by commutators. Its finite images are perfect. The isomorphism in (D.40.2) uses the original marked images of s and r, of orders two and three.

Write $R_T = \overline{[L,L]} \cap \overline{D_9K}$ for D.39's source kernel and retain its normalized automorphism α_{39} of $\mathcal{T} = \widehat{B}/R_T$. There are canonical marked isomorphisms

$$\begin{aligned} \mathcal{U} &= \widehat{B}/(R_T \cap K_{\mathcal{R}}) \simeq \mathcal{T} \times_{C_6} \mathcal{R}, \\ \mathcal{V} &= \widehat{B}/(R_T \cap K_{\mathcal{R}} \cap \ker \sigma_7) \simeq \mathcal{U} \times_{C_2} S_7, \end{aligned} \tag{D.40.3}$$

where σ_7 is the actual surjection

$$s \longmapsto (1\,2)(3\,4)(5\,6), \qquad r \longmapsto (1\,4\,6)(2\,7\,5), \tag{D.40.4}$$

and the last fibre product uses the common sign map. On these groups the maps

$$(\alpha_{39}, 1_{\mathcal{R}}), \qquad (\alpha_{39}, 1_{\mathcal{R}}, 1_{S_7}) \tag{D.40.5}$$

are actual continuous peripheral automorphisms with multiplier one and the two hexagon equations. Their simultaneous loop conjugators, common gauge and commutator are constructed below. Thus the D.39 action extends coherently through every finite perfect quotient of the full marked source and through (D.40.4).

In particular it acts as the identity on

$$\mathcal{P}_{\geq 5} = \prod_{p \geq 5} \mathrm{PSL}_2(\mathbb{Z}_p) \tag{D.40.6}$$

under the actual modular maps $s \mapsto \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$, $r \mapsto \left[\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right]$. The resulting projection $\mathcal{V} \to \mathcal{T} \times \mathcal{P}_{\geq 5}$ is onto. Its source kernel in $\widehat{B}$ before passing to $\mathcal{V}$ is exactly $R_T \cap \ker \rho_{\geq 5}$. Some additional exact projection kernels are

$$\begin{aligned} \ker(\mathcal{U} \to \mathcal{T}) &= \mathcal{R}', \\ \ker(\mathcal{V} \to \mathcal{U}) &= A_7, \\ \ker(\mathcal{V} \to \mathcal{T}) &= \mathcal{R}' \times A_7. \end{aligned} \tag{D.40.7}$$

The remaining full-source kernel $R_T \cap K_{\mathcal{R}} \cap \ker \sigma_7$ is nontrivial. Its restriction under the actual surjection

$$\begin{aligned} \sigma_{10}(s) &= (1\,2)(3\,4)(5\,6)(7\,8)(9\,10), \\ \sigma_{10}(r) &= (1\,3\,9)(4\,10\,8)(5\,6\,7) \end{aligned} \tag{D.40.8}$$

maps onto A_{10}. This is a computed remaining source kernel, not a nonlifting assertion for the complete CM pair. D.45 excludes a full profinite lift of this recorded action; a replacement realizing the same CM pair remains open. The original $\widehat{F}$ source maps onto $\ker(\mathcal{V} \to S_3)$ with the same displayed kernel. The compact-cover and fixed-fibre CM data remain exactly those of D.39.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160. The finite-subdirect argument, the two explicit central covers, and all source maps are proved here.

Proof. The finite-subdirect argument. Let H be the image of B in a finite product $G_1 \times \cdots \times G_j$ of its perfect quotients. Every coordinate map is onto. Write H_∞ for the stable term of the finite derived series. It is perfect and H/H_∞ is the maximal solvable quotient. For each coordinate kernel N_i, the image of H_∞ is G_i: every term of the derived series has that image, and the series stabilizes. Hence $H = H_\infty N_i$, so every N_i maps onto H/H_∞. On the other hand

$$[N_1, N_2, \ldots, N_j] \subseteq \bigcap_i N_i = 1. \tag{D.40.9}$$

For $j = 1$ this just says that the single coordinate kernel is trivial. Taking images in H/H_∞ shows that this finite solvable quotient is nilpotent.

A finite nilpotent quotient of B is generated by an element of order dividing two and an element of order dividing three. They lie in different Sylow factors and commute. The quotient is therefore cyclic of order dividing six. In particular $H' = H_\infty$ is perfect. Every finite quotient of $\mathcal{R}$ factors through one such finite coordinate image: an open kernel in the closed product image contains the intersection of finitely many coordinate kernels. It follows that every finite solvable quotient of $\mathcal{R}$ is a quotient of C_6, and that every finite image of $\mathcal{R}'$ is perfect.

Determining both factors of the common cyclic quotient. The upper bound six is attained.

The following two marked matrix pairs give actual surjections from B:

G	s	r
$\mathrm{SL}_3(\mathbb{F}_7)$	$\begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 4 & 5 & 1 \\ 1 & 2 & 3 \\ 2 & 0 & 1 \end{pmatrix}$
$\mathrm{SL}_4(\mathbb{F}_3)$	$\begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	$\begin{pmatrix} 0 & 2 & 2 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 2 \\ 0 & 1 & 2 & 1 \end{pmatrix}$

(D.40.10)

Both pairs have determinant one and satisfy $s^2 = r^3 = 1$. Their generation is verified by actual words and orbit lower bounds. In their faithful actions on nonzero vectors, the recorded stabilizer-chain orbit lengths are

G	orbit lengths	$\lvert G\rvert$
$\mathrm{SL}_3(\mathbb{F}_7)$	342, 336, 49	5 630 688
$\mathrm{SL}_4(\mathbb{F}_3)$	80, 78, 72, 27	12 130 560.

(D.40.11)

Each generator used in these orbit calculations is stored as a word in the displayed s, r. A separate standard-library verifier evaluates all those words as matrices, checks their permutations, and computes the orbits of the generators fixing all earlier base points. Each orbit is a lower bound for the orbit of the actual stabilizer in $\langle s, r\rangle$. The products in (D.40.11) equal the upper bounds

$$|\,\mathrm{SL}_n(\mathbb{F}_q)| = q^{n(n-1)/2} \prod_{a=2}^{n} (q^a - 1),$$

obtained by counting ordered bases and dividing by the $q - 1$ determinant values. Thus the displayed words generate the full groups; no bare group-order output is needed for this inference.

For all matrix commutator identities in this proof use $[a, b] = aba^{-1}b^{-1}$. These two groups are perfect by elementary matrix identities. Row elimination, using elementary transvections and

$$\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} = \left(E_{12}(a)E_{21}(-a^{-1})E_{12}(a)\right)\left(E_{12}(-1)E_{21}(1)E_{12}(-1)\right),$$

shows that elementary matrices generate each special linear group. For three distinct indices, $[E_{ij}(a), E_{jk}(b)] = E_{ik}(ab)$. Every elementary generator in dimensions three and four is therefore a commutator. The verifier also checks these identities over both fields.

In the first group, $z = 2I$ is central of order three; replace the displayed r by zr. In the second, $z = -I$ is central of order two; replace s by zs. The torsion relations still hold. Each new map ϕ_1 is a central-character twist of the original epimorphism ϕ_0. On B' the two maps agree, and $\phi_0(B') = G$ because G is perfect. Thus the twisted maps are also onto. Their generation is additionally verified by the same word-and-orbit procedure.

For each pair the joint image is $G \times C_d$, $d = 3$ or 2, under $(g, \epsilon) \mapsto (g, z^\epsilon g)$. Indeed $B \to G \times C_d$ is onto: a common quotient of the perfect group G and the cyclic group is trivial, so the elementary subdirect-product argument gives the full product. Equivalently, the ratio $\phi_1(b)\phi_0(b)^{-1}$ gives the specified central character. On s, r the two characters are exactly

$$\chi_2(s) = 1, \quad \chi_2(r) = 0, \qquad \chi_3(s) = 0, \quad \chi_3(r) = 1. \tag{D.40.12}$$

All four perfect quotients are coordinates of $\mathcal{R}$. Their central ratios therefore give a surjection $\mathcal{R} \to C_2 \times C_3 = C_6$. Together with the upper bound already proved, this establishes (D.40.2), with its exact marked map.

The common quotient and simultaneous conjugators. Every finite quotient of $\mathcal{T}$ is solvable. The group H is an extension of Q by $C_2^2 \times C_3$, and Q is solvable: reduction modulo four has two-group kernel over S_3, while $\mathrm{SL}_2(\mathbb{F}_3)$ has normal Q_8 with cyclic quotient. D.39's homology group is therefore an extension of a solvable finite group by an abelian profinite group. Also T_3 is an extension of the solvable Q_0 by a finite two-group. The displayed embedding of $\mathcal{T}$ in their product proves the assertion. Consequently every common marked quotient of $\mathcal{T}$ and $\mathcal{R}$ factors through the just-computed C_6. Conversely R_T is contained in $\widehat{F}' \subset \widehat{B}'$, so $\mathcal{T}$ retains precisely the original map $B \to B^{\mathrm{ab}} = C_6$. The maximal common marked quotient is therefore exactly C_6.

For completeness, the subdirect-product argument used here has explicit kernels. For a closed subgroup $J \subset X \times Y$ surjecting onto both factors, set $K_X = \{x : (x,1) \in J\}$ and $K_Y = \{y : (1,y) \in J\}$. These are closed normal subgroups and the pairs in J induce a canonical isomorphism $X/K_X \simeq Y/K_Y$. Its inverse identifies J with that fibre product. Applying this to the image of $\widehat{B}$ proves the first isomorphism of (D.40.3), including the quotient by the intersection of the two source kernels.

The automorphism α_{39} fixes B^{ab}: its images of s, r are their conjugates. It hence pairs with $1_{\mathcal{R}}$ on the common C_6. There are also compatible actual conjugators, not only compatible automorphisms. Let A, β be the raw D.39 conjugators and let $k = \beta_y^{\mathrm{ab}} - \beta_x^{\mathrm{ab}} \in \widehat{\mathbb{Z}}$ be its actual global gauge. Their retained H coordinates are $(6,0)$, and $k \equiv 0 \pmod 6$. Since $x = t^{-2}$ and $y = sxs$ have images $(0,1)$ in $C_2 \times C_3 = B^{\mathrm{ab}}$, an element of F with abelianization (a,b) maps to $(0, a+b \bmod 3)$. Thus

$$A,\ \beta,\ x^k A,\ x^k \beta \quad \text{all have image zero in } B^{\mathrm{ab}}. \tag{D.40.13}$$

The finite seed checks these residues, and D.39's integral and odd-prime limits retain them.

Before gauge, use the paired conjugators

$$(A, x_{\mathcal{R}}^{-k}), \qquad (\beta, x_{\mathcal{R}}^{-k}). \tag{D.40.14}$$

Their two components agree in C_6, by (D.40.13). On $\mathcal{R}$ they give the inner automorphism $\mathrm{Int}(x_{\mathcal{R}}^{-k})$, which fixes the cusp. The common gauge by x^k changes them to $(x^k A, 1)$ and $(x^k \beta, 1)$, and changes the automorphism to $(\alpha_{39}, 1)$. The resulting f is $(f_{39}, 1)$.

Both conjugators lie in the image of $\widehat{F}$, as seen under the retained map to S_3. The full source kernel is contained in $\widehat{F}'$, so the marked abelianization remains $\widehat{\mathbb{Z}}^2$. D.39's gauge calculation then gives an actual lift of this common f in $\widehat{F}'$. The torsion and exact cusp equations give the same two $m = 0$ hexagons. The component automorphisms and their inverses preserve the fibre product, proving generation in every finite coordinate image and compatibility in the limit.

Every good-prime congruence level. Use the exact two matrices in (D.40.6). Their product is $\left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, so x is $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. These are the same modular markings as D.25. They generate every projective special linear quotient modulo an odd integer: their elementary upper and lower unipotents generate the local matrix groups by row elimination, and the Chinese remainder map supplies the independent local coefficients.

For $p \geq 5$ and every $e \geq 1$, put $D = \mathrm{diag}(2, 2^{-1})$ over $\mathbb{Z}/p^e$. The exact commutator identities

$$[D, E_{12}(a/3)] = E_{12}(a), \qquad [D, E_{21}(-4a/3)] = E_{21}(a) \tag{D.40.15}$$

show that $\mathrm{SL}_2(\mathbb{Z}/p^e)$, and its projective quotient, are perfect. Their canonical kernels therefore belong to $\mathscr{P}$. Projection to these coordinates and passage to the inverse limit give the actual surjection $\mathcal{R} \to \mathcal{P}_{\geq 5}$. Surjectivity onto the whole product follows from the same Chinese remainder generation at every finite collection of primes and exponents.

Every finite quotient of $\mathcal{P}_{\geq 5}$ is perfect, while every finite quotient of $\mathcal{T}$ is solvable. Their only common quotient is trivial, since a finite perfect solvable group is trivial and profinite groups are separated by finite quotients. The same subdirect-product argument proves that their actual joint source image is $\mathcal{T} \times \mathcal{P}_{\geq 5}$, with source kernel $R_T \cap \ker \rho_{\geq 5}$. Formula (D.40.5) acts as the identity on the congruence component. The proof uses the actual global k: its odd components are -6. It does not substitute the integer finite-seed gauge -438 at unrelated prime levels.

The S_7 extension and the next explicit kernel. The certificate enumerates all 5040 elements generated by (D.40.4), and all 3 628 800 elements generated by (D.40.8). The torsion relations are checked independently. In the first quotient t has cycle type $(4, 3)$ and $x = t^{-2}$ has order six. In the second t has cycle type $(4, 5, 1)$ and x has order ten.

We give the elementary normal-subgroup facts needed for the kernel argument. For $n \geq 5$, every nontrivial normal subgroup of A_n contains a three-cycle. A permutation with a cycle of length at least four gives a three-cycle by taking its commutator with a three-cycle on three consecutive entries. For a permutation having a three-cycle and other entries, the commutator with $(a\, b\, d)$, where a, b are successive entries and d is outside that cycle, gives either a five-cycle or a double transposition. A permutation consisting of transpositions gives a double transposition by the same calculation on two of its pairs. A double transposition, commuted with a three-cycle using one entry of each pair and a fifth entry, gives a five-cycle, reducing to the first case. A sole three-cycle already has the required form.

All three-cycles are conjugate in A_n: correct an odd conjugating permutation by a transposition of two unused entries. They generate A_n, since each pair of transpositions is a product of three-cycles. Thus A_n is simple and perfect. An element centralizing A_n in S_n fixes every three-cycle and hence every three-element support; intersecting such supports fixes every point. Its centralizer is trivial. It follows that the normal subgroups of S_n are exactly $1, A_n, S_n$: a normal subgroup intersecting A_n trivially centralizes it.

There is no continuous surjection $\mathcal{U} \to S_7$. Such a map would factor through a finite marked image $J \subset T_0 \times G_1 \times \cdots \times G_j$, where T_0 is a finite solvable quotient of $\mathcal{T}$ and each G_i is perfect. Let $N_0, N_1, \ldots, N_j$ be the coordinate kernels. The image of N_0 in S_7 contains A_7, because its quotient is solvable. For $i \geq 1$, J' maps onto G_i, so $J = J'N_i$. The normal image of N_i therefore contains an odd permutation and must be all of S_7. But the iterated commutator of these coordinate kernels lies in their trivial intersection, whereas its image contains A_7. This is a contradiction.

The largest common marked quotient of $\mathcal{U}$ and the actual S_7 quotient is consequently C_2: the only alternatives among quotients of S_7 are S_7 and the trivial group, and the common sign quotient is retained. This proves the second isomorphism in (D.40.3). The action α_{39} fixes that sign, so pairing with the identity gives the second automorphism in (D.40.5). The normalized conjugators in (D.40.13) pair with the identity conjugators in S_7. Before gauge the S_7 component of x^{-k} is already the identity, because x has order six and $k \equiv 0 \pmod 6$. This proves the simultaneous marking, the gauge and the hexagons in this further extension. Taking the kernels of its explicit fibre-product projections gives (D.40.7).

Finally there is no surjection $\mathcal{V} \to S_{10}$. Factor a proposed map through a finite image

$$J \subset T_0 \times G_1 \times \cdots \times G_j \times S_7. \tag{D.40.16}$$

The T_0-coordinate kernel has image containing A_{10}. The S_7-coordinate kernel also has image containing A_{10}: otherwise it would be trivial, forcing a surjection from S_7 onto the larger group S_{10}. Every perfect-coordinate kernel has image S_{10}, by the same $J = J'N_i$ argument. The iterated commutator of all these kernels is trivial in (D.40.16), but maps onto the nontrivial perfect group A_{10}. This contradiction proves the claim.

Under σ_{10} the kernel $R_T \cap K_{\mathcal{R}} \cap \ker \sigma_7$ has normal image in S_{10}. It is contained in $\widehat{F}' \subset \widehat{B}'$, so this image is contained in A_{10}. The image cannot be trivial, since that would make σ_{10} factor through $\mathcal{V}$. Simplicity therefore makes the image exactly A_{10}. The exact source kernel and the open embedding of $\widehat{F}$ give the final source map in the statement. All previous CM evaluations factor through the unchanged $\mathcal{T}$ component. The nonzero remaining kernel and its actual alternating quotient identify a further full-source test; they do not assert a failure of every possible full lift of the CM pair. □

All marked symmetric quotients at once

D.41. A simultaneous symmetric extension and an explicit remaining projective-linear kernel [Proved]

Let

$$\mathscr{S} = \{M \lhd B : [B : M] < \infty,\ B/M \simeq S_n \text{ for some } n \geq 5\}. \tag{D.41.1}$$

There is one coordinate for each actual marked kernel M. Put $S_M = B/M$, $A_M = S_M'$, and let $\epsilon_M : S_M \to C_2$ be its sign map. The actual simultaneous source image is

$$\mathcal{S} = \left\{ (g_M) \in \prod_{M \in \mathscr{S}} S_M : \epsilon_M(g_M) \text{ is independent of } M \right\}. \tag{D.41.2}$$

The map $\widehat{B} \to \mathcal{S}$ is onto, and

$$\ker(\mathcal{S} \to C_2) = \prod_{M \in \mathscr{S}} A_M, \qquad \mathcal{S} \simeq \left(\prod_{M \in \mathscr{S}} A_M \right) \rtimes C_2. \tag{D.41.3}$$

The section in this semidirect product is the actual tuple of the images of s, and its action on each A_M is the indicated conjugation. Every nonabelian finite quotient of $\mathcal{S}$ has some S_n quotient.

Retain $\mathcal{U} = \mathcal{T} \times_{C_6} \mathcal{R}$ and the exact kernels $R_T, K_{\mathcal{R}}$ of D.40. With $K_{\mathcal{S}} = \bigcap_{M \in \mathscr{S}} \overline{M}$, there is an exact marked isomorphism

$$\mathcal{W} := \widehat{B}/(R_T \cap K_{\mathcal{R}} \cap K_{\mathcal{S}}) \simeq \mathcal{U} \times_{C_2} \mathcal{S}. \tag{D.41.4}$$

The automorphism $(\alpha_{39}, 1_{\mathcal{R}}, 1_{\mathcal{S}})$ is an actual continuous peripheral automorphism of $\mathcal{W}$, with multiplier one and both hexagons. Its simultaneous conjugators and global gauge are specified below. It extends D.40 exactly and acts as the identity on every actual full symmetric quotient in (D.41.1), including both displayed S_7 and S_{10} quotients.

Let M_7 be the kernel of D.40's specified S_7 map and let $\mathcal{V}$ be its source quotient there. The canonical projections have kernels

$$\begin{aligned} \ker(\mathcal{W} \to \mathcal{V}) &= \prod_{M \neq M_7} A_M, \\ \ker(\mathcal{W} \to \mathcal{T}) &= \mathcal{R}' \times \prod_{M} A_M. \end{aligned} \tag{D.41.5}$$

The products carry their profinite topology, and $\mathcal{R}'$ retains its closed-commutator definition from D.40.

An explicit remaining finite source map is

$$\begin{aligned} \sigma : \widehat{B} &\twoheadrightarrow J = \mathrm{PGL}_2(\mathbb{F}_7), \\ s &\longmapsto \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right], \qquad r \longmapsto \left[\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix} \right]. \end{aligned} \tag{D.41.6}$$

The remaining full-source kernel has exact image

$$\sigma(R_T \cap K_{\mathcal{R}} \cap K_{\mathcal{S}}) = \mathrm{PSL}_2(\mathbb{F}_7). \tag{D.41.7}$$

Thus it is nontrivial. The canonical good-prime congruence map $\rho_7 : B \to \mathrm{PSL}_2(\mathbb{F}_7)$ and (D.41.6) have actual joint images

$$\mathrm{im}(\rho_7, \sigma) = \mathrm{PSL}_2(\mathbb{F}_7) \times \mathrm{PGL}_2(\mathbb{F}_7), \quad \mathrm{im}(\rho_7, \sigma)|_{B_{\mathrm{even}}} = \mathrm{PSL}_2(\mathbb{F}_7)^2. \tag{D.41.8}$$

Here B_{even} is the kernel of the original sign map $s \mapsto 1, r \mapsto 0$. In particular the kernel in (D.41.7) kills the canonical ρ_7 coordinate while surjecting under the specified second marked map.

The source $\widehat{F}$ maps onto $\ker(\mathcal{W} \to S_3)$ with the same displayed kernel. D.39's compact-cover matrices and fixed-fibre CM pair are unchanged. D.45 excludes a full profinite lift of this recorded action. Formula (D.41.7) describes the source kernel, while the whole prescribed CM pair remains open for a replacement action.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160.

Proof. The exact simultaneous symmetric image. For every epimorphism $B \to S_n$, $n \geq 5$, the image of r is even because its order divides three. The image of s must be odd, since otherwise the whole image would lie in A_n. Thus all the characters ϵ_M pull back to the same original sign character. In particular $M \subset B_{\mathrm{even}}$ for every $M \in \mathscr{S}$. The restriction $B_{\mathrm{even}} \to A_M$ is onto and has kernel exactly M. Restrictions coming from two different members of $\mathscr{S}$ therefore have different kernels, even when their abstract alternating groups have the same degree.

These restrictions are simultaneously onto every finite product of their simple targets. To prove this, adjoin one target at a time. A subdirect product of a previous full product and a new nonabelian simple group either is the full product or identifies a common simple quotient. Every epimorphism from a product of nonabelian simple groups to a simple group factors through one coordinate: the images of the factors are commuting normal subgroups, and two nontrivial images would make the target abelian. A common simple quotient would therefore identify the new restriction kernel with one of the preceding kernels, which was excluded. This proves finite-product surjectivity. Compactness gives

$$\overline{B_{\mathrm{even}}} \twoheadrightarrow \prod_{M \in \mathscr{S}} A_M. \tag{D.41.9}$$

Multiplication by the actual image of s supplies every tuple with common odd sign. This proves (D.41.2) and (D.41.3), including inverse maps: write a tuple with common sign ϵ as

$$(g_M) = (a_M s_M^{\epsilon}), \qquad a_M = g_M s_M^{-\epsilon} \in A_M, \quad s_M = sM. \tag{D.41.10}$$

The action in the semidirect product is therefore literally conjugation by each s_M.

This construction also explains repeated epimorphisms with the same kernel. Their quotient maps differ by the unique induced automorphism of the quotient, so their image is the graph of that automorphism. Dropping repeated coordinates and reinserting them by these maps are mutually inverse marked isomorphisms. In a finite family with repetitions, the even image is correspondingly a product of diagonal simple blocks. The normalizer of each such block inside the product of its alternating factors is the block itself: a normalizing tuple induces the same conjugation in every identified factor, so its differences centralize the centreless simple group and are trivial. This proves the normalizer assertion directly, and also gives the equality of the even subgroup and the commutator subgroup of each finite simultaneous image.

Finite quotients and the common sign. Consider a finite coordinate image $H = D \rtimes \langle s \rangle$, where $D = \prod_{i=1}^{j} A_{M_i}$. A normal subgroup of D is a product of some of its simple factors. Indeed, a nontrivial projection to a factor, commuted with that factor, gives a nontrivial normal subgroup of that factor and hence contains it; the trivial centre removes every remaining projection.

Let $N \lhd H$. If N contains an odd element as, then on every factor its conjugation is given by an odd permutation. This action cannot be the identity: the centralizer of A_n in S_n is trivial, as proved in D.40. Consequently $[N, A_{M_i}]$ is nontrivial and contains A_{M_i} for every i. Thus N contains D and an odd element, so $N = H$. Every proper normal subgroup of H is therefore a product of some alternating factors. A nonabelian quotient retains at least one factor and maps onto its corresponding S_{n_i}. The only other nontrivial quotient is the common C_2.

Every finite quotient of $\mathcal{S}$ factors through a finite coordinate image. The preceding argument proves the assertion about all its nonabelian finite quotients, and also $\mathcal{S}^{\mathrm{ab}} = C_2$.

The argument of D.40 excluding an S_7 quotient of $\mathcal{U}$ applies to every S_n, $n \geq 5$. In a finite source image $J_0 \subset T_0 \times G_1 \times \cdots \times G_k$, T_0 is solvable and the G_i are perfect. Under a proposed epimorphism to S_n, the T_0-coordinate kernel has image containing A_n. Each perfect-coordinate kernel has image S_n: the derived subgroup maps onto that coordinate, so the kernel maps onto the sign quotient and is normal. Their iterated commutator is trivial in the coordinate product but has image containing the nontrivial perfect group A_n.

It follows that every common finite quotient of $\mathcal{U}$ and $\mathcal{S}$ is a quotient of the common sign C_2. A nonabelian common profinite quotient would have a nonabelian finite quotient and hence an S_n quotient, which has just been excluded. The maximal common marked quotient is exactly C_2. The closed subdirect-product construction of D.40 now proves (D.41.4) in both directions, including its intersection of source kernels.

The actual compatible automorphism. The normalized D.39 conjugators have zero B-abelianization. Pair them with the identity in every $\mathcal{R}$ and $\mathcal{S}$ coordinate. They agree on the common C_6 and C_2 quotients, so are actual elements of the fibre product. Before normalization the coordinate conjugators are x^{-k}, using the actual single $k \in \widehat{\mathbb{Z}}$ of D.39. Their coordinate actions are $\mathrm{Int}(x^{-k})$, which fixes the cusp. The common gauge by x^k changes every additional conjugator to one, and every additional action to the identity. The resulting commutator is $(f_{39}, 1_{\mathcal{R}}, 1_{\mathcal{S}})$.

The source kernel in (D.41.4) is contained in $R_T \subset \widehat{F}'$, so the marked abelianization is still $\widehat{\mathbb{Z}}^2$. The same gauge calculation gives a lift of this commutator in $\widehat{F}'$. The component automorphisms and inverses preserve their common characters, and hence give an actual automorphism of the fibre product. The torsion and cusp equations give both hexagons, compatibly at every finite level. No new multiplier is introduced.

Projection to the M_7 coordinate of $\mathcal{S}$ recovers D.40's action exactly. The kernel of that projection has even sign and consists of the independent remaining alternating factors. Projection to $\mathcal{T}$ additionally sets the C_6 image of $\mathcal{R}$ to zero. These calculations prove both kernels in (D.41.5). In particular the actual S_{10} coordinate now has the identity action; D.40's A_{10} layer is included in the new source quotient.

An explicit remaining marked quotient. The program works in projective matrices over $\mathbb{F}_7$, choosing the representative whose first nonzero entry is one. The two matrices in (D.41.6) satisfy the projective relations $s^2 = r^3 = 1$ and generate all 336 elements. Their $t = sr$ has order eight and $x = t^{-2}$ has order four. The square determinant subgroup $L = \mathrm{PSL}_2(\mathbb{F}_7)$ has order 168, and the determinant square-class map gives $J/L = C_2$. The group B_{even} is generated by r, srs; their images generate all 168 elements of L.

For a complete elementary simplicity check, the program exhausts the conjugacy classes of

L. Their sizes are

$$1, \quad 21, \quad 42, \quad 56, \quad 24, \quad 24. \tag{D.41.11}$$

The subgroup generated by every nonidentity class has order 168. Every nontrivial normal subgroup therefore equals L. The group is nonabelian, so is perfect. The centralizer of L in J has order one, checked on the generating pair r, srs. It follows, by intersecting a normal subgroup with L, that the normal subgroups of J are exactly $1, L, J$. In particular $J' = L$.

The action on the eight points of $\mathbb{P}^1(\mathbb{F}_7)$ is faithful, and its permutation sign equals determinant square class. Its image has order 336, whereas S_8 has order 40320. Thus this specific permutation representation is not an epimorphism to S_8 and is not one of the full symmetric coordinates in (D.41.1). This follows from its explicit action map and image order.

There is no quotient $\mathcal{W} \to J$. A proposed continuous epimorphism factors through a finite marked image

$$H \subset T_0 \times \prod_{i=1}^{a} G_i \times \prod_{j=1}^{b} S_{n_j}, \tag{D.41.12}$$

with T_0 solvable and each G_i perfect. The T_0-coordinate kernel has image containing L, since its quotient is solvable. Each perfect-coordinate kernel has image J: the equality $H = H'N_i$ forces its normal image to meet the nontrivial determinant coset. Each symmetric-coordinate kernel also has image containing L. Otherwise its image would be trivial, and J would be a quotient of S_{n_j}. The only nontrivial proper quotient of that group is C_2, and 336 is not a factorial $n!$ for $n \geq 5$, so that is impossible. The iterated commutator of all coordinate kernels is trivial in (D.41.12), while its image contains the nontrivial perfect group L. This contradiction proves the claim.

The image of the kernel in (D.41.7) is normal in J and lies in $J' = L$, since the kernel is contained in $\widehat{F}' \subset \widehat{B}'$. It cannot be trivial, as that would make σ factor through $\mathcal{W}$. Simplicity of L now proves (D.41.7).

Finally the canonical ρ_7 has the different, previously fixed modular matrices

$$\rho_7(s) = \left[\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\right], \qquad \rho_7(r) = \left[\begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}\right]. \tag{D.41.13}$$

The program evaluates the paired generators from (D.41.6) and (D.41.13), obtaining all $168 \cdot 336 = 56448$ paired elements. Using the actual even-source generators r, srs gives all $168^2 = 28224$ paired elements. These full counts prove (D.41.8) under the displayed maps. The canonical ρ_7 kernel occurs among the subgroups intersected in $K_{\mathcal{R}}$, whereas (D.41.7) remains onto under the second map. The two marked occurrences of the simple group are thus compared explicitly. The source kernel and its open $\widehat{F}$ restriction are exactly those in (D.41.4); full profinite realization has not been inferred from the quotient action. □

Lifting the remaining projective-linear quotient through its full pronilpotent kernel

D.42. An actual simultaneous relative-prime lift and its exact common quotient [Proved]

Retain the marked quotient $\mathcal{W}$ of D.41, its automorphism α_{41}, and its source kernel

$$R_W = R_T \cap K_{\mathcal{R}} \cap K_{\mathcal{S}} \lhd \widehat{B}.$$

Let $\sigma : \widehat{B} \twoheadrightarrow J = \mathrm{PGL}_2(\mathbb{F}_7)$ be the specific map (D.41.6), and put $N_J = \ker(B \to J)$. This is a free group of rank 57. For every prime p, let

$$R_{J,p} = \ker(\widehat{N}_J \longrightarrow N_J^{(p)}), \qquad \mathcal{P}_{J,p} = \widehat{B}/R_{J,p}, \qquad R_{J,\mathrm{nil}} = \bigcap_p R_{J,p}. \tag{D.42.1}$$

Here $N_J^{(p)}$ is the full pro-p completion of the indicated marked free group. These are actual source quotients, and

$$\mathcal{P}_{J,\mathrm{nil}} := \widehat{B}/R_{J,\mathrm{nil}} = \prod_{\substack{J\\p}} \mathcal{P}_{J,p}, \qquad \ker(\mathcal{P}_{J,\mathrm{nil}} \to J) = \prod_{p} N_J^{(p)}. \tag{D.42.2}$$

There is an explicit maximal common marked quotient of $\mathcal{W}$ and $\mathcal{P}_{J,\mathrm{nil}}$. Use D.37's source map $B \to H$, $|H| = 6912$, and set

$$H_{54} = H/O_2(H), \quad |O_2(H)| = 128, \quad L_{54} = \ker(B \to H_{54}), \qquad \mathcal{C}_{54} = \varprojlim_{n\geq 1} B/[L_{54}, L_{54}]L_{54}^{3^n}. \tag{D.42.3}$$

The inverse system uses the actual quotient maps. Its structure is

$$|H_{54}| = 54, \qquad \operatorname{rank} L_{54} = 10, \qquad 1 \longrightarrow \mathbb{Z}_3^{10} \longrightarrow \mathcal{C}_{54} \longrightarrow H_{54} \longrightarrow 1. \tag{D.42.4}$$

Every finite level has order $54 \cdot 3^{10n}$. The cusp $t = sr$ has order $6 \cdot 3^n$ there. The precise finite-base identification, including the images of s, r, is given in the proof.

Consequently the actual simultaneous source image is

$$\mathcal{Z}_{42} := \widehat{B}/(R_W \cap R_{J,\mathrm{nil}}) \simeq \mathcal{W} \times_{\mathcal{C}_{54}} \mathcal{P}_{J,\mathrm{nil}}. \tag{D.42.5}$$

The map $(\alpha_{41}, 1_{\mathcal{P}_{J,\mathrm{nil}}})$ is an actual continuous automorphism of this group. It preserves the indicated peripheral classes with multiplier one, fixes t, and has a commutator parameter satisfying both hexagons. Its conjugators are constructed below from the recorded D.39 conjugators. Thus the specified remaining $\mathrm{PGL}_2(\mathbb{F}_7)$ coordinate is now the identity together with *every* pro-p refinement of its rank-57 free kernel, simultaneously. All D.39–D.41 compact-cover and fixed-fibre CM data are retained.

The new projection kernels are explicit. Put $J^+ = \mathrm{PSL}_2(\mathbb{F}_7)$ and $M = N_J \cap L_{54}$. The joint map to $J \times_{C_2} H_{54}$ is onto, has order 9072, and

$$[N_J : M] = 27, \qquad \operatorname{rank} M = 1513.$$

There is a specified surjection

$$\omega : M^{(3)} \longrightarrow \mathbb{Z}_3^{10}, \qquad \kappa = \ker \omega, \tag{D.42.6}$$

whose 10 by 1513 integral homology matrix is supplied by the program. It is the graph projection induced by $M \subset L_{54}$. With $K_{\mathrm{new}} = \ker(\mathcal{Z}_{42} \to \mathcal{W})$,

$$1 \longrightarrow \left(\prod_{p\neq 3} N_J^{(p)}\right) \times \kappa \longrightarrow K_{\mathrm{new}} \longrightarrow J^+ \longrightarrow 1. \tag{D.42.7}$$

Moreover, with $K_{T,C} = \ker(\mathcal{T} \to \mathcal{C}_{54})$,

$$\ker(\mathcal{Z}_{42} \to \mathcal{P}_{J,\mathrm{nil}}) = K_{T,C} \times \mathcal{R}' \times \prod_{M' \in \mathscr{S}} A_{M'}. \tag{D.42.8}$$

Here $\mathcal{R}'$ is the closed commutator subgroup specified in D.40. The group $K_{T,C}$ has the following two exact descriptions of its successive kernels:

$$\begin{aligned} &1 \longrightarrow \mathcal{K}_2 \longrightarrow K_{T,C} \longrightarrow K_{H,C} \longrightarrow 1,\\ &1 \longrightarrow \left(\prod_{p\neq 3} \mathbb{Z}_p^{1153}\right) \times \ker(P_3) \longrightarrow K_{H,C} \longrightarrow O_2(H) \longrightarrow 1,\\ &P_3 : \mathbb{Z}_3^{1153} \twoheadrightarrow \mathbb{Z}_3^{10}, \qquad \operatorname{rank}_{\mathbb{Z}_3} \ker(P_3) = 1143. \end{aligned} \tag{D.42.9}$$

Here $\mathcal{K}_2 = \ker(T_3 \to C_8)$ is the exact finite group from D.39, of order 2^{111151}, with its indicated source maps. The map P_3 is the stored 10 by 1153 integral graph-projection matrix, completed at 3; $K_{H,C}$ is $\ker(\mathcal{H} \to \mathcal{C}_{54})$.

An explicit nontrivial kernel still remains in the full source. Define the actual marked map

$$\begin{aligned} \sigma_{11} : \widehat{B} &\twoheadrightarrow J_{11} = \mathrm{PGL}_2(\mathbb{F}_{11}), \\ s &\longmapsto \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right], \qquad r \longmapsto \left[\begin{pmatrix} 1 & -1 \\ 1 & 0 \end{pmatrix}\right]. \end{aligned} \tag{D.42.10}$$

Then

$$\sigma_{11}(R_W \cap R_{J,\mathrm{nil}}) = \mathrm{PSL}_2(\mathbb{F}_{11}). \tag{D.42.11}$$

The source $\widehat{F}$ maps onto the inverse image of the identity in S_3 in (D.42.5), with this same source kernel. D.45 excludes a full profinite lift of this recorded action. Formula (D.42.11) records its source kernel; the prescribed CM pair remains open for a replacement.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160.

Proof. The relative completions and their solvable quotients. The images of s and r in J have exact orders two and three. The Bass–Serre graph of the kernel has 336 edges and $168+112 = 280$ vertices, so N_J is free of rank $336-280+1 = 57$. Its profinite completion is the open subgroup $\ker(\widehat{B} \to J)$. Each $R_{J,p}$ is characteristic in that subgroup, hence normal in $\widehat{B}$. The map from $\widehat{N}_J$ to every finite product of the distinct pro-p completions is onto: a common quotient of a pro-p group and a product of groups at other primes is trivial. Finite-product surjectivity and compactness prove (D.42.2), including the literal fibre-product maps. In particular every finite quotient retaining J has a nilpotent normal kernel over J.

Write $B_{\mathrm{even}} = \ker(B \to C_2)$ for the original sign map. Reidemeister–Schreier rewriting with the two representatives $1, s$ gives

$$B_{\mathrm{even}} = \langle r, srs \mid r^3 = (srs)^3 = 1 \rangle = C_3 * C_3. \tag{D.42.12}$$

Let $\mathcal{R}_3$ be its full relative pro-3 completion over the sign quotient:

$$\mathcal{R}_3 = \widehat{B} / \ker(\widehat{B}_{\mathrm{even}} \to B_{\mathrm{even}}^{(3)}). \tag{D.42.13}$$

The action of s interchanges the two displayed order-three generators. The inclusion $N_J \subset B_{\mathrm{even}}$ shows directly that the map to $\mathcal{R}_3$ factors through $\mathcal{P}_{J,3}$.

Here is the exact quotient argument needed for the reverse comparison. Let E be a finite quotient of $\mathcal{P}_{J,\mathrm{nil}}$ retaining J, with nilpotent kernel V. For every quotient E/U, its further quotient by the image of V is

$$(E/U)/\operatorname{im}(V) = J/\operatorname{im}(U). \tag{D.42.14}$$

D.41 proved that the normal subgroups of J are $1, J^+, J$. If $\operatorname{im}(U)$ is J^+ or J, (D.42.14) and the nilpotence of the image of V make E/U solvable. Thus every nonsolvable quotient of E has the actual quotient J, obtained by dividing out the image of V. A finite quotient that does not initially retain J is a quotient of one that does, by adjoining the J coordinate, so the assertion holds for every finite quotient of $\mathcal{P}_{J,\mathrm{nil}}$.

For the solvable images, put $E^+ = \operatorname{im}(B_{\mathrm{even}} \to E)$, the preimage of J^+. Its stable derived subgroup $(E^+)^\infty$ maps onto the perfect group J^+. Therefore

$$E^+ = (E^+)^\infty V.$$

Every solvable image of E^+ kills $(E^+)^\infty$ and is consequently an image of the nilpotent group V. It is generated by the images of r, srs, both of order dividing three. A finite nilpotent group generated by such elements is a 3-group, as its projections to every other Sylow factor are trivial. Thus every solvable source image factors (D.42.13). Conversely $\mathcal{R}_3$ is prosolvable and is an actual quotient of $\mathcal{P}_{J,3}$. This proves

$$(\mathcal{P}_{J,\mathrm{nil}})^{\mathrm{sol}} = \mathcal{R}_3, \qquad (\mathcal{P}_{J,p})^{\mathrm{sol}} = \begin{cases} \mathcal{R}_3, & p = 3, \\ C_2, & p \neq 3. \end{cases} \tag{D.42.15}$$

The superscript here means the quotient by the intersection of all open normal subgroups with solvable quotient; all maps in (D.42.15) are the specified source maps.

*The actual order-*54 *quotient.* The program starts with the literal states (q, a, b) of D.37. Its map is

$$(q, a, b) \longmapsto \big(q\, O_2(Q),\ a \bmod 3,\ b \bmod 3\big). \tag{D.42.16}$$

There are 54 values and 128 elements in each fibre. It checks all 6912 right-s and right-r equations, so (D.42.16) is the actual marked quotient map. The 32 elements of $O_2(Q)$ are verified as a normal subgroup, and their inverse image with $a = b = 0 \bmod 3$ is the indicated normal 2-subgroup of H. The quotient has trivial centre; a normal 2-subgroup in a group of order 54 would have order two and be central. Thus the kernel is exactly $O_2(H)$.

Here is an explicit identification of the finite quotient, rather than only its order. Put $a = r$, $b = srs$, and $c = aba^{-1}b^{-1}$ in H_{54}. Its even subgroup is the order-27 group whose coordinates satisfy

$$(u, v, w)(u', v', w') = (u + u',\ v + v',\ w + w' + uv'), \qquad (u, v, w) \longmapsto b^v a^u c^w, \tag{D.42.17}$$

with all coordinates in $\mathbb{F}_3$. This gives the exact isomorphism from the upper-unitriangular three by three matrices over $\mathbb{F}_3$. The involution s acts by

$$(u, v, w) \longmapsto (v, u, uv - w). \tag{D.42.18}$$

All 27^2 multiplication equations and all 27 involution equations are checked. The source generator r has coordinate $(1, 0, 0)$; s is the displayed involution. The map to Q_0 kills precisely the centre $\langle c \rangle$. In particular its even quotient is the actual abelianization C_3^2 of (D.42.12), with the two indicated generators, so $K = B'_{\mathrm{even}}$ as marked subgroups.

The Bass–Serre graph for L_{54} has 54 edges and $27 + 18 = 45$ vertices. Its ten fundamental integral cycles form the stored basis, with an identity minor at the ten non-tree edges. The same projection on edges sends the 1153 cycles of L to the specified matrix P_3 and sends the path of t^6 to a vector having a unit coordinate at 3. This proves all rank, order and cusp assertions in (D.42.3)–(D.42.4).

The maximal common marked quotient. D.41 excludes J as a quotient of $\mathcal{W}$. By (D.42.14), every common finite quotient of $\mathcal{W}$ and $\mathcal{P}_{J,\mathrm{nil}}$ is therefore solvable. The kernel of $\mathcal{W} \to \mathcal{T}$ is $\mathcal{R}' \times \prod A_{M'}$, whose every finite image is perfect. Hence the maximal prosolvable quotient of $\mathcal{W}$ is precisely $\mathcal{T}$. By (D.42.15) the required common quotient is the maximal relative pro-3 source quotient of $\mathcal{T}$ over the sign.

The projection $\mathcal{T} \to \mathcal{H}$ has kernel $\ker(T_3 \to C_8)$, a finite 2-group of order 2^{11151}. This kernel lies in the even subgroup. Its image in a relative pro-3 source quotient over C_2 is trivial. It remains to compute that quotient of $\mathcal{H} = \widehat{B}/\overline{[L, L]}$. Here $L \subset L_{54}$ and $L_{54}/L = O_2(H)$ has order 128. In every such target the closed images of L_{54} and L coincide: their quotient is both a quotient of this 2-group and a pro-3 group. The image of L is abelian, so the image of L_{54} is abelian. The target therefore factors through $\mathcal{C}_{54}$. Conversely $\mathcal{C}_{54}$ is a quotient of $\mathcal{H}$ and is relative pro-3 over C_2, by (D.42.17)–(D.42.18). This proves maximality with the actual source

maps in both directions. The closed subdirect-product description, or equivalently its finite quotient versions and compactness, now gives (D.42.5).

The retained conjugators and the coprime correction. Let A_W, β_W be D.41's normalized conjugators, so

$$\alpha_{41}(s) = \beta_W s \beta_W^{-1}, \qquad \alpha_{41}(r) = A_W r A_W^{-1}.$$

On the full odd homology component D.39 gave an exact formula over $\mathbb{Z}[1/2]$, with raw action $\mathrm{Int}(x^6)$. Its global gauge has odd component -6. The present program projects those same stored integral paths and applies the same gauge. In the ten-cycle basis its answer is

$$A_W \longmapsto 1, \qquad \beta_W \longmapsto b_C = -\tfrac{1}{2}(e_7 + e_8) \in \mathbb{Z}_3^{10} \subset \mathcal{C}_{54}, \qquad s b_C s^{-1} = b_C. \tag{D.42.19}$$

The notation in the middle is additive in the abelian kernel. In particular the second image is nonzero. The code checks the exact full path vector and its s-invariance, not only a finite reduction. The multiplier of the normalized action is still one; on $\mathcal{C}_{54}$ the action is the identity. Its 3-adic gauge is exactly -6, since x lies in the pro-3 even subgroup of this quotient.

The actual joint map

$$\mathcal{P}_{J,3} \longrightarrow J \times_{C_2} \mathcal{C}_{54} \tag{D.42.20}$$

is onto. Indeed every solvable quotient of J factors through its sign C_2, and the displayed common sign is retained. The kernel of (D.42.20) is pro-3, being contained in $N_J^{(3)}$. The element $(1, b_C)$ centralizes the displayed involution (s_J, s_C). It has a lift $\widetilde{\beta}_3$ that centralizes s in $\mathcal{P}_{J,3}$.

For completeness, this centralizer lifting uses the following elementary coprime conjugacy argument. Lift $(1, b_C)$ arbitrarily to v. The two involutions vsv^{-1} and s have the same image and lie in the preimage of its order-two subgroup, with pro-3 kernel. In a finite 3-group kernel, induction on its centre first conjugates the two involutions modulo the centre. Their remaining ratio is a central element z with $s(z) = z^{-1}$. Conjugation by $z^{-1/2}$ removes it. The square root exists uniquely in this finite abelian odd-order centre. Compactness over finite quotients gives the same conjugacy by an element of a pro-3 kernel. Multiplying v by that element gives $\widetilde{\beta}_3$, without changing either coordinate in (D.42.20).

Its J coordinate is one. Thus

$$\widetilde{\beta}_{\mathrm{nil},p} = \begin{cases} \widetilde{\beta}_3, & p = 3, \\ 1, & p \neq 3 \end{cases} \tag{D.42.21}$$

is a genuine element of the fibre product (D.42.2), centralizes s, and maps to b_C. The pairs

$$A_{42} = (A_W, 1), \qquad \beta_{42} = (\beta_W, \widetilde{\beta}_{\mathrm{nil}}) \tag{D.42.22}$$

therefore belong to the actual group (D.42.5). They give the stated product automorphism on both s and r. It is invertible because both factors are invertible and the common action is the identity. It fixes t.

Both conjugators lie over the identity in the retained S_3 quotient. Their parameter is

$$f_{42} = s\beta_{42} s \beta_{42}^{-1} = (f_{39}, 1_{\mathcal{P}_{J,\mathrm{nil}}}). \tag{D.42.23}$$

This pair is the image of an actual element of $\widehat{F}'$: lift it first through the proved source surjection; its $\mathcal{W}$ component has zero $\widehat{F}^{\mathrm{ab}}$ coordinate, and $R_W \subset \widehat{F}'$ means that $\mathcal{W}$ retains this entire abelianization. Thus every such lift has zero abelianization. The order-two and order-three source relations and the fixed cusp give the two multiplier-one hexagons exactly as in D.25. This also proves that the same global gauge has been retained, rather than replaced at unrecorded primes.

The exact kernels and their completed maps. The joint finite source image $J \times_{C_2} H_{54}$ has order $336 \cdot 54/2 = 9072$. The code enumerates it with the actual two generators. Its kernel M has 9072 graph edges, $4536 + 3024 = 7560$ vertices, and rank 1513. Projection to the H_{54} graph gives the matrix in (D.42.6). Its surjectivity holds even integrally: $L_{54}/M = J^+$ is perfect, hence $L_{54} = M[L_{54}, L_{54}]$. The completed matrix has rank ten over $\mathbb{Z}_3$, also checked directly modulo three.

The subgroup M has 3-power index 27 in N_J. Its full pro-3 topology is induced from N_J: the core of any normal finite 3-index subgroup of M over the 27 cosets is again of 3-power index in N_J. Consequently $M^{(3)}$ is exactly the open subgroup of $N_J^{(3)}$ lying over the identity in the even order-27 group. The kernel of (D.42.20) is precisely

$$\ker(N_J^{(3)} \to (\mathcal{C}_{54})_{\mathrm{even}}) = \ker(\omega) = \kappa. \tag{D.42.24}$$

In particular its further abelian kernel description is $1 \to (M^{(3)})' \to \kappa \to \mathbb{Z}_3^{1503} \to 1$, where the last group is the kernel of the specified completed homology matrix and the prime denotes closed commutators.

The kernel of $\mathcal{P}_{J,\mathrm{nil}} \to \mathcal{C}_{54}$ maps onto J^+, by (D.42.20). Its kernel over J is the product of all the other prime kernels and (D.42.24). This proves (D.42.7). The common quotient retains the full marked abelianization C_6 of B: already $H_{54} \to Q_0$ does so. Hence setting its coordinate to one in the fibre-product formula for $\mathcal{W}$ yields the direct product (D.42.8).

For (D.42.9), first project $\mathcal{T}$ onto $\mathcal{H}$; its exact finite kernel was specified in D.39. The homology kernel of $\mathcal{H} \to H$ is $\widehat{\mathbb{Z}}^{1153}$. Its map to the kernel in (D.42.4) kills every prime other than three and is exactly P_3 at three. The latter is onto. For example, the graph verifies rank ten modulo three; alternatively the index-128 quotient makes coinvariants exact at three. The kernel of the resulting map projects onto $O_2(H)$: every lift of an element of $O_2(H)$ can have its image in $\mathbb{Z}_3^{10}$ removed by this surjection. This proves both exact sequences and the rank $1153 - 10 = 1143$.

An evaluated remaining source kernel. The second program verifies (D.42.10) directly. The two matrices generate all 1320 projective matrices, their even subgroup is the square-determinant group of order 660, and r, srs generate that subgroup. Every conjugacy class in the even subgroup and the normal closure of every nonidentity class representative are enumerated. Each such closure has order 660, proving simplicity. Its centralizer in J_{11} is trivial. Consequently the normal subgroups of J_{11} are $1, \mathrm{PSL}_2(\mathbb{F}_{11}), J_{11}$. The faithful degree-twelve permutation action is recorded; its order is 1320, not 12!. The actual cusp and loop orders are 10 and 5.

The finite-kernel argument of D.41 excludes J_{11} from $\mathcal{W}$ with no new assumption. In a finite subdirect image of its solvable, perfect, and full symmetric coordinates, the solvable-coordinate kernel has image containing the perfect even subgroup under any proposed map to J_{11}. Every perfect-coordinate kernel has image all of J_{11}, because the derived subgroup surjects onto that perfect coordinate and the kernel therefore surjects onto the sign. Every symmetric-coordinate kernel also has image containing the even subgroup; otherwise J_{11} would be a quotient of a full S_n, contradicted by the normal subgroups of S_n and the order 1320. The iterated commutator of these coordinate kernels is trivial, whereas its image contains the nontrivial perfect even subgroup. This is a contradiction.

Likewise J_{11} is not a quotient of $\mathcal{P}_{J,\mathrm{nil}}$: by (D.42.14) it would have a quotient J of order 336, which does not divide 1320. A finite subdirect image of these two groups cannot map onto J_{11} either. Its two coordinate kernels commute, but their images would both contain the same nonabelian simple even subgroup, since neither coordinate admits that quotient. Thus $\mathcal{Z}_{42}$ has no J_{11} quotient.

Its source kernel is normal and is contained in $R_W \subset \widehat{B}'$. Its image under σ_{11} is therefore

contained in the simple even subgroup. That image cannot be trivial, since triviality would factor the indicated epimorphism through $\mathcal{Z}_{42}$. It is the whole even subgroup, proving (D.42.11).

The two independent runnable certificates are `d_cm_relative_pronilpotent.py` and `d_cm_relative_kernel.py`. The former reads the saved D.37 graph and D.39 exact odd paths, without importing or rewriting the preceding executables. It checks all finite source maps, all $1153 + 1513$ integral cycle projections, and the nonzero centralizer vector (D.42.19). The latter uses only the Python standard library for its full finite enumeration. The new source quotient and all its finite characteristic refinements retaining the marked CM data have the stated actual actions. No assertion that these refinements are cofinal among all quotients of $\widehat{F}$ is made. □

All marked simple even bases and their relative pronilpotent kernels

D.43. A simultaneous full-family extension and an evaluated orbit of two simple factors [Proved]

Let $B_{\mathrm{even}} = \ker(B \to C_2)$, with the original sign map, and consider the entire marked family

$$\mathscr{J} = \{N \lhd B : \ N \subset B_{\mathrm{even}},\ B_{\mathrm{even}}/N \text{ is finite nonabelian simple}\}. \tag{D.43.1}$$

There is one coordinate for each actual source kernel. Write $L_N = B_{\mathrm{even}}/N$ and $J_N = B/N$. The actual image s_N gives $J_N = L_N \rtimes \langle s_N \rangle$. For each prime p, put

$$R_{N,p} = \ker(\widehat{N} \to N^{(p)}), \qquad \mathcal{P}_N = \widehat{B}/\bigcap_p R_{N,p}, \qquad R_{\mathscr{J}} = \bigcap_{N\in\mathscr{J},\ p} R_{N,p}. \tag{D.43.2}$$

All completions and intersections are taken with these indicated source maps. Each N is free of rank $1 + |L_N|/3$, as its marked Bass–Serre graph has $2|L_N|$ edges and $|L_N| + 2|L_N|/3$ vertices. Their actual simultaneous images are

$$\begin{aligned} J_{\mathscr{J}} &= \left(\prod_{N\in\mathscr{J}} L_N\right) \rtimes C_2, \\ \mathcal{P}_{\mathscr{J}} := \widehat{B}/R_{\mathscr{J}} &= \prod_{\substack{\mathcal{R}_3 \\ N\in\mathscr{J}}} \mathcal{P}_N, \end{aligned} \tag{D.43.3}$$

where $\mathcal{R}_3$ is the exact relative pro-3 completion (D.42.13). In the first line the involution acts on each coordinate by its actual conjugation action; it does not permute the coordinates. The maximal prosolvable quotient of $\mathcal{P}_{\mathscr{J}}$ is exactly $\mathcal{R}_3$.

Here is an exact common-quotient formula that retains the perfect and symmetric coordinates already present in D.41. Put

$$\begin{aligned} \mathcal{D} = \mathcal{R} \times_{C_2} \mathcal{S} = \widehat{B}/K_D, \qquad K_D = K_{\mathcal{R}} \cap K_{\mathcal{S}}, \\ \mathcal{D}_{\mathscr{J}} = \widehat{B}/(K_D R_{\mathscr{J}}), \qquad \mathcal{C}_{\mathscr{J}} = \mathcal{C}_{54} \times_{C_6} \mathcal{D}_{\mathscr{J}}. \end{aligned} \tag{D.43.4}$$

The products of closed normal subgroups in this display are closed. The maps to C_6 are the original full abelianization of B. Then $\mathcal{C}_{\mathscr{J}}$ is the maximal common marked quotient of $\mathcal{W}$ and $\mathcal{P}_{\mathscr{J}}$, with mutually inverse maps proved below. In particular

$$\mathcal{Z}_{43} := \widehat{B}/(R_W \cap R_{\mathscr{J}}) = \mathcal{W} \times_{\mathcal{C}_{\mathscr{J}}} \mathcal{P}_{\mathscr{J}}. \tag{D.43.5}$$

The automorphism

$$(\alpha_{41}, 1_{\mathcal{P}_{\mathscr{J}}}) \tag{D.43.6}$$

is an actual continuous peripheral automorphism of (D.43.5), with multiplier one, the same global gauge, and both hexagons. One simultaneous centralizer lift supplies its conjugators. It retains D.42 exactly as an action under the canonical projection and acts as the identity on every relative pro-p completion in (D.43.2), for all bases and all primes at once.

All three finite-base types enter this formula. For the inner-action bases, the proof gives the exact isomorphism $J_N \to L_N \times C_2$ and its actual perfect-coordinate map into $\mathcal{R}$. Full symmetric bases use their actual coordinates in $\mathcal{S}$. Let $\mathscr{J}_{\mathrm{new}}$ consist of the remaining bases: the involution has a non-inner action on L_N, and J_N is not a full symmetric group. For this entire subfamily the formula simplifies to

$$\mathcal{D}_{\mathscr{J}_{\mathrm{new}}} = C_6, \qquad \mathcal{C}_{\mathscr{J}_{\mathrm{new}}} = \mathcal{C}_{54}. \tag{D.43.7}$$

It includes both marked projective-linear groups in D.41 and D.42. For the whole family, (D.43.4) specifies its additional properfect common part by its exact source kernel; it has not been replaced by a sign quotient. Its commutator subgroup is properfect, and it maps onto the simultaneous image of every retained inner and symmetric base.

The simultaneous pronilpotent base kernel also has an explicit form. The source inclusions give surjections $N^{(3)} \to B^{(3)}_{\mathrm{even}}$. With $\mathcal{N} = \ker(\mathcal{P}_{\mathscr{J}} \to J_{\mathscr{J}})$,

$$\mathcal{N} = \left(\prod_{\substack{N \in \mathscr{J} \\ p \neq 3}} N^{(p)} \right) \times \left(\prod_{\substack{B^{(3)}_{\mathrm{even}} \\ N \in \mathscr{J}}} N^{(3)} \right). \tag{D.43.8}$$

In particular $\mathcal{N}$ is pronilpotent. The second factor is its pro-3 Sylow subgroup. If $\delta : \widehat{B} \to \mathcal{D}$ is the actual source map, the two comparison kernels in (D.43.5) are

$$\begin{aligned} \ker(\mathcal{Z}_{43} \to \mathcal{P}_{\mathscr{J}}) &= K_{T,C} \times \delta(R_{\mathscr{J}}), \\ \ker(\mathcal{Z}_{43} \to \mathcal{W}) &= \ker(\mathcal{P}_{\mathscr{J}} \to \mathcal{C}_{54}) \cap \ker(\mathcal{P}_{\mathscr{J}} \to \mathcal{D}_{\mathscr{J}}). \end{aligned} \tag{D.43.9}$$

$K_{T,C}$ and all its successive kernels are exactly those of (D.42.9). The second line is an intersection of the displayed actual quotient maps inside $\mathcal{P}_{\mathscr{J}}$.

A further explicit finite source quotient remains. On the symbols $1, \dots, 7$, put

$$a = (1\ 4\ 7), \qquad b = (1\ 2\ 3)(4\ 5\ 6), \qquad J_* = (A_7 \times A_7) \rtimes \langle \iota \rangle, \tag{D.43.10}$$

where ι interchanges the two factors. There is an actual epimorphism

$$\sigma_* : \widehat{B} \twoheadrightarrow J_*, \qquad s \longmapsto \iota, \quad r \longmapsto (a, b). \tag{D.43.11}$$

Its even image has order 6350400, its whole image has order 12700800, and the actual orders of t, x are $14, 7$. The remaining full-source kernel has the exact image

$$\sigma_*(R_W \cap R_{\mathscr{J}}) = A_7 \times A_7. \tag{D.43.12}$$

Thus the new completion still has a nontrivial kernel in $\widehat{B}$, and likewise in the indicated $\widehat{F}$ source. This calculation does not exclude the retained CM pair from a full profinite lift. D.45 excludes the particular recorded action extended here and leaves the CM pair as a target for a replacement.

This addresses Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160.

Proof. The entire marked family and its exact simultaneous images. For each N, the restriction $B_{\mathrm{even}} \to L_N$ has kernel exactly N. Distinct members of (D.43.1) therefore give distinct marked simple quotients of the same even source. The simple subdirect-product argument of D.41 gives surjectivity onto every finite product of them: a failure on adjoining one factor would identify its kernel with that of a preceding factor. Compactness gives the full product. The actual image of s then proves the first line of (D.43.3), with its stated action. Repeated epimorphisms having the same source kernel differ by the unique induced isomorphism of their targets. Their images are its graph; deletion and reinsertion using that isomorphism are inverse marked maps. The same source kernel in (D.43.2) gives the identical relative completion, so this also treats repetitions at every prime and every finite level.

The finite kernel of a simultaneous relative-pronilpotent image over a finite joint base embeds in a product of nilpotent kernels. It is therefore nilpotent. Its even base is a product of nonabelian simple groups and is perfect. The proof of (D.42.15) consequently applies verbatim to the even source: after its perfect residual is killed, every solvable image is an image of that nilpotent kernel and is generated by r, srs. It is a 3-group. The indicated $\mathcal{R}_3$ is a quotient of every $\mathcal{P}_N$, so it is exactly the maximal prosolvable quotient of every finite-family completion.

For precision, the common quotient when adjoining a new member N is also exactly $\mathcal{R}_3$. Suppose a finite common image were nonsolvable. Viewed as a quotient of the new relative completion, its even subgroup, modulo the image of the nilpotent kernel, has the actual quotient L_N. Indeed the only possible even images in the simple base are 1 and L_N; the first would make the whole image solvable. Let $N_I = \bigcap_{M \in I} M$ for the preceding finite family. The simultaneous simple-base surjectivity just proved gives $N_I \twoheadrightarrow L_N$. But in the preceding relative completion the image of N_I lies in its pronilpotent base kernel. Its image in the finite simple group would be nilpotent, a contradiction. Thus every common image is solvable and the maximal common quotient is the stated $\mathcal{R}_3$. Induction and compactness prove the second line of (D.43.3). This argument includes inner and symmetric bases; it never assumes that the involution's action is outer.

The actual joint image of J_N and $\mathcal{R}_3$ is $J_N \times_{C_2} \mathcal{R}_3$: the maximal solvable quotient of J_N is C_2. Consequently its kernel over J_N surjects onto $B_{\mathrm{even}}^{(3)}$. At primes other than three the map to this pro-3 group is trivial. Reordering the literal fibre-product coordinates proves (D.43.8). It also proves that $\mathcal{N}$ projects onto the base kernel of every finite-family completion: extend its common $B_{\mathrm{even}}^{(3)}$ coordinate to each omitted factor using the just-proved surjections. This projection property will be used in the common-quotient and centralizer arguments.

The three base types and their actual existing maps. Let θ_N be conjugation by s_N on L_N. When $\theta_N = \mathrm{Int}(h)$, the element h is unique because L_N is centreless, and $h^2 = 1$ because $\theta_N^2 = 1$. The maps

$$\begin{aligned} L_N \rtimes \langle s_N \rangle &\longrightarrow L_N \times C_2, \\ l s_N^{\epsilon} &\longmapsto (l h^{\epsilon}, \epsilon), \\ (m, \epsilon) &\longmapsto m h^{-\epsilon} s_N^{\epsilon} \end{aligned} \tag{D.43.13}$$

are inverse homomorphisms. The projection to L_N is the actual perfect source map $s \mapsto h,\ r \mapsto r_N$. It is onto, since $B \to J_N$ is onto. Thus this base occurs in $\mathcal{R}$ together with the common sign. When J_N is a full S_n, its actual kernel is one of the coordinates of $\mathscr{S}$, so it occurs in $\mathcal{S}$. These are the marked maps used in (D.43.4).

For the remaining bases, θ_N is non-inner and J_N is not a full symmetric group. Their only normal subgroups are $1, L_N, J_N$. To see this, a normal subgroup contained in L_N is trivial or all of L_N. An odd normal subgroup with trivial intersection with L_N would centralize L_N and make θ_N inner, which was excluded. The finite coordinate-kernel argument of D.41 now excludes each such J_N from $\mathcal{D}$ and from $\mathcal{W}$. Perfect-coordinate kernels have odd image and hence image J_N; symmetric-coordinate kernels have image containing L_N, since J_N is not a

quotient of a full S_n. The solvable coordinate, when present, has kernel image containing L_N. Their iterated commutator is trivial in the subdirect source image but contains the nontrivial perfect L_N in the target.

For a finite joint base in this new subfamily, the even subgroup is a product of its simple factors and the involution acts non-innerly on each. Every proper normal subgroup is a product of even factors. An odd normal element commuted with each factor would otherwise force all those factors, and then the whole group, into the normal subgroup. It follows that every nonsolvable quotient of a relative pronilpotent image retains one of these actual J_N quotients. Thus its common quotient with $\mathcal{D}$ is solvable. The maximal solvable quotient of $\mathcal{D}$ is C_6, and both sides retain that marked quotient. This proves (D.43.7). For the whole family, the maps (D.43.13) and the symmetric coordinates instead show why the properfect part in (D.43.4) must be retained.

The key common-quotient calculation for the whole family. Let Q be any finite common image of $\mathcal{W}$ and $\mathcal{P}_{\mathscr{J}}$. Write

$$P = \operatorname{im}\left(\mathcal{R}' \times \prod A_M \longrightarrow Q\right), \qquad Z = \operatorname{im}\left(\mathcal{T}' \longrightarrow \mathcal{W} \longrightarrow Q\right). \tag{D.43.14}$$

The second map is the *isolated-coordinate* map $t \mapsto (t, 1, 1)$. It is defined on $\mathcal{T}' = \ker(\mathcal{T} \to C_6)$. The group P is perfect, and Q/P is solvable. Hence $P = Q^{\infty}$, the stable derived subgroup. The subgroup Z is normal and centralizes P.

Factor the map to Q through a finite-family relative image retaining its finite joint base. Let V be the image of its nilpotent base kernel in Q. The even subgroup of Q/V is a quotient of a product of nonabelian simple groups, hence is a centreless product of the surviving simple factors. The image of P surjects onto that even subgroup. The image of Z is even and centralizes it, so is trivial. Thus

$$Z \subset V. \tag{D.43.15}$$

In particular Z is nilpotent.

For a prime $p \neq 3$, let Z_p be its characteristic Sylow subgroup. The quotient Q/P is a relative 3-group over the sign, by the first part of the proof. Since Z_p is even, its image there is trivial. Thus $Z_p \subset P$ and Z_p is central in P. For every $q \in Q_{\mathrm{even}}$, choose $a \in P$ with the same image in $(Q/V)_{\mathrm{even}}$. Then $q = av$ for $v \in V$, and conjugation by q on Z_p equals conjugation by v. Because V is nilpotent, this action lies in a p-group: all its other Sylow subgroups commute with Z_p. But Q_{even} is generated by the actual images of r, srs, both of order dividing three. Therefore it acts trivially on Z_p.

Here the final vanishing argument uses no projection from B_{even} to $\mathcal{T}'$. Let $B' = \ker(B \to C_6)$. The actual restriction $B' \to \mathcal{T}'$ is onto after profinite completion, and its image is onto the finite quotient under consideration. Compose this map with (D.43.14) and the characteristic projection $Z \to Z_p$:

$$\phi : B' \longrightarrow Z_p. \tag{D.43.16}$$

It is invariant under conjugation by B_{even}, because Z_p centralizes Q_{even}. Consequently $\ker\phi$ is normal in B_{even}, and

$$1 \longrightarrow \operatorname{im}\phi \longrightarrow B_{\mathrm{even}}/\ker\phi \longrightarrow B_{\mathrm{even}}/B' = C_3 \longrightarrow 1 \tag{D.43.17}$$

is a central extension. A group with central kernel and cyclic quotient is abelian. The group in (D.43.17) is generated by the actual two order-three elements r, srs, so it is abelian of exponent dividing three. Its p-subgroup $\operatorname{im}\phi$ is trivial for $p \neq 3$. The map in (D.43.16) is onto Z_p, so $Z_p = 1$. This proves that Z is a 3-group.

Let $\phi_T : \mathcal{T}' \to Z$ be the map in (D.43.14). Its kernel is normal in $\mathcal{T}$: conjugation can be calculated using any lift to $\mathcal{W}$, and two lifts differ by P, which centralizes Z. The quotient

$\mathcal{T}/\ker\phi_T$ has normal 3-group Z and quotient C_6. Its even subgroup is therefore a 3-group. It is an actual relative pro-3 quotient over the sign, so by D.42 its source map factors through $\mathcal{C}_{54}$. We have proved that every finite common image kills the isolated subgroup $K_{T,C}$.

The exact marked fibre-product formula. Let $K_C = \ker(\widehat{B} \to \mathcal{C}_{54})$. The preceding argument proves that the maximal common quotient of $\mathcal{W}$ and $\mathcal{P}_{\mathscr{J}}$ factors through

$$\mathcal{W}/K_{T,C} = \mathcal{C}_{54} \times_{C_6} \mathcal{D} = \widehat{B}/(K_C \cap K_D). \tag{D.43.18}$$

Conversely both sources retain $\mathcal{C}_{54}$, so $R_{\mathscr{J}} \subset K_C$. Products of closed normal subgroups are closed, and the elementary modular identity gives

$$(K_C \cap K_D)R_{\mathscr{J}} = K_C \cap (K_D R_{\mathscr{J}}). \tag{D.43.19}$$

The common source kernel is $R_W R_{\mathscr{J}}$. The proved inclusion from (D.43.18) and $R_W \subset K_C \cap K_D$ give

$$R_W R_{\mathscr{J}} = (K_C \cap K_D)R_{\mathscr{J}} = K_C \cap (K_D R_{\mathscr{J}}). \tag{D.43.20}$$

These are both directions of the comparison, with the actual source kernels.

The maximal prosolvable quotient of $\mathcal{D}$ is C_6, because its commutator subgroup is the properfect product $\mathcal{R}' \times \prod A_M$. Both $\mathcal{D}$ and $\mathcal{P}_{\mathscr{J}}$ retain C_6, so their common quotient $\mathcal{D}_{\mathscr{J}}$ does also, and its commutator subgroup is properfect. Its maximal common quotient with $\mathcal{C}_{54}$ is therefore C_6. Closed Goursat applied to (D.43.20) proves (D.43.4)–(D.43.5). The inner and symmetric bases are actual common quotients of $\mathcal{D}$ and $\mathcal{P}_{\mathscr{J}}$; their simultaneous image consequently is a quotient of $\mathcal{D}_{\mathscr{J}}$. The description by $K_D R_{\mathscr{J}}$ keeps any further properfect common kernel data as well. It is not an assertion that this group consists only of its finite simple bases.

One simultaneous fixed-element lift. On (D.43.18), the action α_{41} is the identity: it is the identity on $\mathcal{C}_{54}$ by D.42 and on $\mathcal{D}$ by D.40–D.41. It is consequently the identity on $\mathcal{C}_{\mathscr{J}}$. Thus (D.43.6) preserves the actual fibre product and is an automorphism. For the peripheral conjugators one must also retain the nonzero coordinate of D.42. In $\mathcal{C}_{\mathscr{J}}$ the recorded conjugators have images

$$A_C = 1, \qquad \beta_C = (b_C, 1), \qquad b_C = -\tfrac{1}{2}(e_7 + e_8). \tag{D.43.21}$$

This element centralizes s_C.

Set $V = \mathrm{im}(\mathcal{N} \to \mathcal{C}_{\mathscr{J}})$. It is a closed pronilpotent normal subgroup. The projection property following (D.43.8) shows that its image in every finite common quotient is exactly the image of the finite-family base kernel used in (D.43.15). Consequently the image of the isolated $\mathcal{T}'$ coordinate lies in V. It is pro-3 by the same finite argument. In particular $\beta_C \in V_3$, the characteristic pro-3 Sylow subgroup.

The second factor of (D.43.8) maps onto V_3. This map is s-equivariant and has pro-3 kernel. Apply the coprime involution argument proved in D.42 to obtain one element

$$\widetilde{\beta} \in \mathcal{N}_3, \qquad \widetilde{\beta} \longmapsto \beta_C, \qquad s\widetilde{\beta}s^{-1} = \widetilde{\beta}. \tag{D.43.22}$$

Explicitly, use finite quotients of $\mathcal{N}_3$ normal in $\mathcal{N}_3 \rtimes \langle s \rangle$, together with the corresponding finite quotients of V_3. For every finite set of such conditions, their common refinement has a fixed lift by the finite centre-induction argument. The closed sets of solutions in $\mathcal{N}_3$ have the finite intersection property. Compactness gives (D.43.22) at all bases, primes and finite levels simultaneously.

Now the actual compatible pairs

$$(A_W, 1), \qquad (\beta_W, \widetilde{\beta}) \tag{D.43.23}$$

belong to $\mathcal{Z}_{43}$. They conjugate the indicated images of r, s as required, and the product automorphism fixes t. Its parameter is $(f_{39}, 1)$. It is the image of an actual element of $\widehat{F}'$, since the $\mathcal{W}$ coordinate retains the whole abelianization of $\widehat{F}$ and has zero coordinate there. The same order-two, order-three and fixed-cusp equations give both multiplier-one hexagons. No additional finite-level gauge is introduced. The canonical projection to D.42 retains its entire action; the possible choice of a different centralizing representative does not change that action or its commutator parameter.

The formula (D.43.9) follows by taking the two kernels in the now-proved fibre product. For the first, the map $\mathcal{W} = \mathcal{T} \times_{C_6} \mathcal{D}$ to $\mathcal{C}_{54} \times_{C_6} \mathcal{D}_{\mathscr{J}}$ has kernel the direct product of its two displayed kernels. For the second, the two coordinate conditions are exactly those in (D.43.4). This proves the comparison maps and kernels without identifying the additional properfect part with a smaller finite base.

The evaluated orbit of two simple factors. The program enumerates the subgroup generated by a, b in (D.43.10) and obtains all 2520 elements of A_7. Their centralizer orders are respectively 36 and 9, so no automorphism of A_7 interchanges a and b. The two even-source markings

	r	srs
ρ_1	a	b
ρ_2	b	a

(D.43.24)

therefore have different kernels. Simple Goursat gives the full A_7^2 image. There is also a direct word certificate. The program finds a 16-letter word in r, srs and their inverses, equal to a 32-letter word in s, r, r^{-1}, whose paired image is $(1, z)$ with $z \neq 1$. It enumerates the 360 conjugates of z and verifies that they generate all 2520 elements. Thus the paired image contains $1 \times A_7$ and surjects the first factor, proving the same full image directly. Conjugation by s supplies the other factor. This verifies (D.43.11), all stated orders, and the faithful degree-fourteen permutation marking.

The normal subgroups of J_* are $1, A_7^2, J_*$. Indeed a normal subgroup contained in A_7^2 is a product of simple factors, and the involution interchanges them. An odd normal element does not centralize A_7^2; its commutators force the whole even subgroup and then J_* into that normal subgroup. The even subgroup is perfect.

There is no J_* quotient of $\mathcal{W}$, by the finite coordinate-kernel argument already used in D.41: the solvable coordinate kernel has image containing A_7^2, the perfect coordinate kernels have odd image and hence image J_*, and the full symmetric coordinate kernels have image containing A_7^2. The latter assertion uses the normal subgroups of S_n: J_* has order 12700800, is not a full symmetric group, and is not a quotient of one. The iterated commutator of the coordinate kernels would be trivial with nontrivial perfect image.

Nor is J_* a quotient of $\mathcal{P}_{\mathscr{J}}$. In a finite relative image the nilpotent base kernel must have trivial image in J_*: neither of its two nontrivial normal subgroups is nilpotent. The map would therefore factor through a finite joint base in (D.43.3). Each of its simple even factors is normal in the whole base, because its involution preserves that factor. Its image in J_* would be a simple normal subgroup. No such nontrivial subgroup exists in J_*. All even factors are therefore killed, and the remaining C_2 cannot surject onto J_*.

Finally a finite common subdirect image of $\mathcal{W}$ and $\mathcal{P}_{\mathscr{J}}$ cannot map onto J_* either. Its two coordinate kernels commute; their images would both contain A_7^2, since neither coordinate admits that quotient. This contradicts the nontrivial perfectness of A_7^2. Thus $\mathcal{Z}_{43}$ has no such quotient. Its remaining source kernel lies in $\widehat{B}'$ and has normal image in J_*. The image cannot be trivial, as otherwise the displayed epimorphism would factor through $\mathcal{Z}_{43}$. The normal subgroup classification gives exactly (D.43.12).

The independent standard-library certificate is `d_cm_simple_even_family.py`, with its

saved output and JSON word certificate. The remaining kernel in (D.43.12) shows explicitly why no cofinality assertion is being used for the full free profinite source. □

All anabelian even bases, including permuted simple factors

D.44. The entire relative pronilpotent family over anabelian even bases and its exact solvable-top residual [Proved]

Call a finite group anabelian here when all its composition factors are nonabelian simple. This definition includes the trivial group. With the original marked sign map, put

$$\mathscr{A} = \{N \lhd B : N \subset B_{\mathrm{even}},\ B_{\mathrm{even}}/N \text{ is finite anabelian}\}. \tag{D.44.1}$$

For $N \in \mathscr{A}$, write $J_N = B/N$ and use the actual relative completions and source kernels

$$\begin{aligned} R_{N,p} &= \ker(\widehat{N} \longrightarrow N^{(p)}),\\ \mathcal{P}_N &= \widehat{B}/\bigcap_p R_{N,p},\\ R_{\mathscr{A}} &= \bigcap_{N\in\mathscr{A},\ p} R_{N,p},\\ \mathcal{P}_{\mathscr{A}} &= \widehat{B}/R_{\mathscr{A}},\\ J_{\mathscr{A}} &= \varprojlim_{N\in\mathscr{A}} J_N. \end{aligned} \tag{D.44.2}$$

The family is directed under intersections, with all maps induced by the same B. Its actual simultaneous completion and base kernel are

$$\mathcal{P}_{\mathscr{A}} = \varprojlim_N \mathcal{P}_N, \qquad \mathcal{N}_{\mathscr{A}} := \ker(\mathcal{P}_{\mathscr{A}} \to J_{\mathscr{A}}) = \prod_p \varprojlim_N N^{(p)}. \tag{D.44.3}$$

Every displayed transition map on the pro-p kernels is surjective. Thus this kernel is pronilpotent and projects onto the kernel at every finite base and every finite relative level. The maximal prosolvable quotient of $\mathcal{P}_{\mathscr{A}}$ is exactly $\mathcal{R}_3$ of D.42.

This includes every marked semisimple even base, each with its entire relative pronilpotent kernel, and every finite joint base formed from them. In an individual base having a transitive permutation action on its simple factors, that orbit has size one or two. The two-factor source (D.43.11) is included with its actual swapping involution. The larger family also retains successive anabelian even extensions; no factor-permutation action is discarded.

Keep $K_D, \mathcal{D}, \mathcal{W}, K_{T,C}, \mathcal{C}_{54}$ and the actual maps from D.42–D.43. Define

$$\begin{aligned} \mathcal{D}_{\mathscr{A}} &= \widehat{B}/(K_D R_{\mathscr{A}}),\\ \mathcal{C}_{\mathscr{A}} &= \mathcal{C}_{54} \times_{C_6} \mathcal{D}_{\mathscr{A}},\\ \mathcal{Z}_{44} &= \widehat{B}/(R_W \cap R_{\mathscr{A}}) = \mathcal{W} \times_{\mathcal{C}_{\mathscr{A}}} \mathcal{P}_{\mathscr{A}}. \end{aligned} \tag{D.44.4}$$

Here $\mathcal{C}_{\mathscr{A}}$ is the exact maximal common marked quotient, not only a quotient of the common image. There is an actual continuous peripheral automorphism

$$(\alpha_{41}, 1_{\mathcal{P}_{\mathscr{A}}}) \quad \text{of } \mathcal{Z}_{44}. \tag{D.44.5}$$

Its multiplier is one. It retains the D.39 action and every D.40–D.43 coordinate with their actual maps and common gauge. One simultaneous pro-3 fixed-element lift supplies its loop conjugators. If $\delta : \widehat{B} \to \mathcal{D}$ is the original source map, the exact projection kernels are

$$\begin{aligned} \ker(\mathcal{Z}_{44} \to \mathcal{P}_{\mathscr{A}}) &= K_{T,C} \times \delta(R_{\mathscr{A}}),\\ \ker(\mathcal{Z}_{44} \to \mathcal{W}) &= \ker(\mathcal{P}_{\mathscr{A}} \to \mathcal{C}_{54}) \cap \ker(\mathcal{P}_{\mathscr{A}} \to \mathcal{D}_{\mathscr{A}}). \end{aligned} \tag{D.44.6}$$

There is a smaller solvable even quotient at the next unresolved source. Let $a = (1\,2)(3\,4)$, $b = (1\,3\,5)$ in A_5, and use the wreath-product convention

$$(v, q)(w, h) = \big((v_i w_{q^{-1}i})_{i=1}^4, qh\big).$$

The marked homomorphism

$$\begin{aligned} \rho : \widehat{B} &\longrightarrow E = A_5^4 \rtimes S_4, \\ s &\longmapsto ((1, 1, a, 1), (1\,2)), \\ r &\longmapsto ((b, 1, 1, 1), (2\,3\,4)) \end{aligned} \tag{D.44.7}$$

is onto. Its whole image has order 311040000; its even image has order 155520000, and its simple base $M = A_5^4$ has order 12960000. The actual orders of t and x are 20 and 10. The top map is precisely the canonical $B \to \mathrm{SL}_2(\mathbb{Z}/4)/\{\pm I\} \simeq S_4$, with the indicated images of s, r. The remaining source kernel has the exact image

$$\rho(R_W \cap R_{\mathscr{A}}) = A_5^4. \tag{D.44.8}$$

The even quotient above this kernel is A_4, with derived orders $4, 1$. This exhibits an actual nonnilpotent solvable even quotient beyond the anabelian family. D.45 tests its two minimal order-24 relative pro-2 bases and proves nonextension of the recorded action. Neither (D.44.5) nor (D.44.8) decides full realization of its entire prescribed CM pair.

These comparisons address Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160.

Proof. Directed marked bases and the full relative kernels. Normal subgroups and quotients of a finite anabelian group are anabelian, by a composition series. A finite subdirect product of such groups is anabelian: for two factors, its projection kernel is a normal subgroup of the other factor, and induction handles any finite number. Applied to the actual even image of B, this proves that $\mathscr{A}$ is directed under intersections. The actual element s splits every J_N over its even subgroup. Since that subgroup has only nonabelian composition factors, it is perfect and centreless. Every N with nontrivial even base is torsion-free: the order-two factor survives the sign, and killing r would make the nontrivial even image generated by r, srs trivial. For the trivial even base $N = B_{\mathrm{even}} = C_3 * C_3$, the pro-$p$ completion is used literally, without the free-rank formula. All other N are free of rank $1 + [B : N]/6$, by their marked Bass–Serre graphs.

If $N_1 \subset N_2$, the finite quotient N_2/N_1 is a normal subgroup of the even group B_{even}/N_1 and is perfect. In every finite p-group image of N_2, the image of N_1 is the whole image: the quotient would otherwise be a nontrivial p-group quotient of a perfect group. Hence

$$N_1^{(p)} \twoheadrightarrow N_2^{(p)}. \tag{D.44.9}$$

The inclusion of the actual source kernels gives the corresponding relative quotient maps. The finite relative quotients over all finite bases form an inverse system with the original marked source B. Compactness proves that its limit is exactly the quotient by the intersection in (D.44.2). Exactness of inverse limits of these surjective compact systems, followed by reordering the prime coordinates, proves (D.44.3). In particular its kernel projects onto every finite-base kernel, a property needed below.

In a finite relative image let V denote the image of the nilpotent kernel. Its even quotient modulo V is anabelian and perfect. After the perfect residual of the whole image is killed, the remaining even subgroup is an image of V. It is nilpotent and generated by the two actual order-three elements r, srs. It is consequently a 3-group. Conversely the exact relative pro-3

source over the sign is a quotient of every relative completion: $N^{(3)} \to B^{(3)}_{\mathrm{even}}$ is onto by the same perfect-quotient argument. This proves the maximal prosolvable assertion, with its actual marked map.

For a semisimple even base, conjugation by its even subgroup preserves each simple factor. Thus the permutation action of the whole base on these factors factors through its sign C_2. A transitive orbit therefore has size one or two. All these bases, their finite subdirect joint images, and their relative completions belong to the directed system just proved. In particular the actual A_7^2 swapping source in D.43 is retained. This argument uses its permutation action; it does not replace it by two separately normalized simple coordinates.

The common quotient, with both comparison maps. We give the precise extension of the finite common-image argument of D.43. Let Q_* be any finite common quotient of $\mathcal{W}$ and $\mathcal{P}_{\mathscr{A}}$. Let $P = Q_*^{\infty}$, the image of the properfect factor $\mathcal{R}' \times \prod A_n$, and let Z be the image of the isolated coordinate $\mathcal{T}' \to \mathcal{W},\ z \mapsto (z, 1, 1)$. Then Z is normal and centralizes P. Factor through one finite relative base using the directed system, and let V be the image of its nilpotent kernel. The even subgroup of Q_*/V is an anabelian quotient, hence is centreless and perfect. The image of P surjects onto it. The even image of Z, which centralizes that subgroup, is trivial. Therefore $Z \subset V$.

For $p \neq 3$, its characteristic Sylow Z_p has trivial image in Q_*/P, whose even subgroup is a 3-group. Thus $Z_p \subset Z(P)$. For any even $q \in Q_*$, choose $a \in P$ having the same image modulo V; conjugation on Z_p is then that of $a^{-1}q \in V$. This action is a p-group, whereas the even source is generated by the two order-three elements r, srs. The action is trivial. The actual invariant map

$$B' \longrightarrow \mathcal{T}' \longrightarrow Z_p$$

therefore has normal kernel in B_{even}. It yields a central extension of $B_{\mathrm{even}}/B' = C_3$, generated by those two order-three elements. Such an extension is abelian of exponent dividing three. Its p-subgroup Z_p vanishes. Thus Z is a 3-group. This uses a map from B'; it introduces no projection from all of B_{even} onto $\mathcal{T}'$.

The same actual $\mathcal{T}$-quotient as in D.43 now factors through $\mathcal{C}_{54}$. Every common quotient kills the isolated $K_{T,C}$. Put $K_C = \ker(\widehat{B} \to \mathcal{C}_{54})$. Since $R_{\mathscr{A}} \subset K_C$, the two inclusions just proved and the modular identity for normal subgroups give

$$R_W R_{\mathscr{A}} = (K_C \cap K_D) R_{\mathscr{A}} = K_C \cap (K_D R_{\mathscr{A}}). \tag{D.44.10}$$

All products are closed, being images of products of compact groups. The maximal solvable quotient of $\mathcal{D}_{\mathscr{A}}$ is C_6, since its commutator is an image of the properfect commutator of $\mathcal{D}$. Both factors retain the literal C_6. Their maximal common quotient is therefore C_6. Closed Goursat applied to (D.44.10) proves both directions of (D.44.4). Taking its projection kernels gives (D.44.6).

One simultaneous normalized lift. The recorded α_{41} is the identity on $\mathcal{C}_{54}$ and $\mathcal{D}$. Its action on their common quotient is therefore the identity, so the product map (D.44.5) is an automorphism. Its recorded conjugators have common images

$$A_C = 1, \qquad \beta_C = (b_C, 1), \qquad b_C = -\tfrac{1}{2}(e_7 + e_8). \tag{D.44.11}$$

In every finite common image, the preceding proof places the isolated $\mathcal{T}'$ image in the image of the nilpotent base kernel and makes it a 3-group. The surjectivity in (D.44.3) and compactness therefore place β_C in the image of

$$\mathcal{N}_{\mathscr{A},3} = \varprojlim_{N} N^{(3)}.$$

The map onto that image is equivariant for s, with pro-3 kernel. The actual element β_C is s-fixed. Coprime fixed-element lifting provides one s-fixed $\widetilde{\beta} \in \mathcal{N}_{\mathscr{A},3}$ above it. For completeness,

take finite quotients normal in $\mathcal{N}_{\mathscr{A},3} \rtimes \langle s \rangle$. Each finite system has a fixed lift by coprime centre induction; finite intersections are handled by their common refinement. Compactness gives one lift at every base, prime and homology level simultaneously.

The pairs $(A_W, 1)$ and $(\beta_W, \widetilde{\beta})$ are consequently actual elements of (D.44.4). They give the required conjugations of r, s. The automorphism fixes t, and its parameter is $(f_{39}, 1)$. Its actual source preimage lies in $\widehat{F}'$, because the $\mathcal{W}$ coordinate retains the full abelianization of $\widehat{F}$. Both hexagons follow from the same order-two, order-three and fixed-cusp equations as in D.25. The gauge and all projected actions are unchanged. This establishes a quotient automorphism; D.45 supplies an explicit obstruction to extending this recorded action to the full source.

The marked wreath source and its exact residual image. The fixed-block choices in (D.44.7) give $s^2 = r^3 = 1$. The top generators generate S_4. The program `d_cm_anabelian_even.py` enumerates the 24 top elements and all 48 Schreier generators of its kernel. Their first-coordinate images generate all 60 elements of A_5. Top conjugation gives surjectivity to each coordinate. A subdirect product of four simple groups is a product of diagonals over a partition of the coordinates, by the simple Goursat argument. Normality makes that partition S_4-invariant. Its primitive action leaves only the single-block and singleton partitions. The certificate records a literal 22-letter B-word whose first coordinate is 1 and whose other coordinates are nontrivial. This excludes a full diagonal and proves that the kernel is A_5^4. The group orders follow. The program also checks all 576 products for the canonical mod-four top isomorphism and the exact orders of the displayed cusp elements.

The subgroup $M = A_5^4$ is the unique minimal normal subgroup of E. Indeed its factors form one top orbit; any normal subgroup meeting M nontrivially contains M. Moreover $C_E(M) = 1$: a centralizer cannot permute factors, and each simple factor has trivial centre. A normal subgroup disjoint from M centralizes M, so is trivial. Thus every nontrivial normal subgroup of E contains M.

There is no quotient map $\mathcal{W} \twoheadrightarrow E$. Such a map would factor through a finite simultaneous image of its solvable coordinate and finitely many perfect and symmetric coordinates. It cannot factor through any one coordinate: E is nonsolvable and is not perfect; and it is no quotient of any S_n, $n \geq 5$. The latter's nonabelian quotient is S_n itself, whose derived subgroup is simple, whereas E' is not simple. Each coordinate kernel would have a nontrivial normal image containing M. Their iterated commutator is trivial in the subdirect source but contains the perfect M in E, a contradiction.

There is likewise no quotient $\mathcal{P}_{\mathscr{A}} \twoheadrightarrow E$. The image of the pronilpotent base kernel would be a nilpotent normal subgroup. Every nontrivial normal subgroup contains the perfect M, so that image is trivial. The map would then factor through an anabelian even base. But the even image of E has the nontrivial solvable quotient A_4. This is impossible by its composition factors. Finally, an epimorphism from their actual joint image $\mathcal{Z}_{44}$ would give nontrivial images of both coordinate kernels, each containing M. Those kernels commute in the fibre product; the perfect nontrivial M cannot commute with itself. Thus E is not a quotient of $\mathcal{Z}_{44}$.

The image $\rho(R_W \cap R_{\mathscr{A}})$ is consequently a nontrivial normal subgroup and contains M. It is contained in M, because $\mathcal{W}$ already retains the exact canonical S_4 top through Q modulo four. This proves (D.44.8), with both inclusions and the literal marked source map. □

The minimal solvable even bases: a complete obstruction and a corrected elliptic lift

D.45. The recorded D.39 action has no full profinite lift; its elliptic rank-two projection does lift one layer farther [Proved]

The two minimal nonnilpotent solvable even bases of a sign-preserving marked quotient of $B = C_2 * C_3$ have order 24. Their groups and actual source markings can be taken as

	s	r
S_4	$(1\,2)$	$(2\,3\,4)$
$A_4 \times C_2$	$((1\,2)(3\,4), 1)$	$((1\,2\,3), 0)$.

(D.45.1)

Both are already actual quotients of Q. Their full relative pro-2 completions are respectively

$$\begin{aligned} \mathcal{P}_F &= \widehat{B}/\ker(\widehat{F} \to F^{(2)}), \qquad F = \ker(B \to S_3), \\ \mathcal{P}_E &= \widehat{B}/\ker(\widehat{B'} \to (B')^{(2)}), \quad B' = \ker(B \to C_6). \end{aligned} \tag{D.45.2}$$

The exact reductions of (D.45.1) are $S_4/V_4 = S_3$ and $(A_4 \times C_2)/V_4 = C_6$, with these original source maps. The maximal common marked quotient of the two completions is C_2. Consequently their actual joint source image is

$$\mathcal{P}_F \times_{C_2} \mathcal{P}_E. \tag{D.45.3}$$

Over the actual common base $Q_0 = S_3 \times_{C_2} C_6$, its kernel is $F^{(2)} \times (B')^{(2)}$.

The rank-four source $K = \ker(B \to Q_0)$ maps onto that product. In the recorded K-basis, and the rank-two graph bases specified in the proof, its integral homology matrix is

$$P = \begin{pmatrix} -1 & -1 & -2 & 1 \\ -1 & -1 & 1 & -2 \\ -2 & 1 & -1 & -1 \\ 1 & -2 & -1 & -1 \end{pmatrix}, \qquad \det P = -27. \tag{D.45.4}$$

It is an isomorphism over $\mathbb{Z}_2$. The kernel of

$$K^{(2)} \twoheadrightarrow F^{(2)} \times (B')^{(2)} \tag{D.45.5}$$

is the closed normal subgroup generated by the four mixed commutators of any lifts of the two free bases of the two factors. The literal source map and both directions of this kernel description are proved below. The recorded D.39 homology action has the respective blocks, modulo 16,

$$I_2, \qquad \begin{pmatrix} 11 & 8 \\ 8 & 3 \end{pmatrix}. \tag{D.45.6}$$

Write $D_nG = \prod_{i2^j \geq n} \gamma_i(G)^{2^j}$ for the closed mod-two Zassenhaus filtration. The recorded D.39 action has a canonical projection to each of B/D_9F and B/D_9B':

$$\mathcal{T} \longrightarrow T_3 = B/D_9K \longrightarrow B/D_9F,\ B/D_9B'. \tag{D.45.7}$$

The first projection, to the sphere source B/D_9F, *has no peripheral lift with multiplier one to* $B/D_{10}F$. This excludes every conjugator pair inducing the same lower action, not just the recorded pair of words.

In particular there is no genuine full profinite $\widehat{\mathrm{GT}}_0$ element inducing the entire recorded normalized D.39 action. The same conclusion holds for the particular quotient automorphisms

D.40–D.44 extending that action. The retained infinite homology forces the multiplier to be exactly one; thus changing its higher congruence classes cannot remove this obstruction. This conclusion concerns those actual quotient actions. It does not exclude another action having the same prescribed pair of CM coordinates.

Moreover, the sphere obstruction persists for every two-adic multiplier preserving the recorded lower action: the cusp has order 32 in both successive sphere quotients. The actual source inner map $\mathrm{Int}(c_0^{258})$ transports the original D.33 action to this obstruction. Consequently that original action has no full relative pro-two peripheral lift, and its particular outer fibre in (D.35.5a) is empty. The supplement below proves the torsion conjugacy, quotient preservation and full cusp-inner normalization used in this transfer.

The second projection has a different proved outcome. There is an actual peripheral automorphism of $B/D_{10}B'$, with multiplier one, that retains the entire recorded action on B/D_9B'. The certificate specifies and evaluates literal corrected B-words $A_{\mathrm{new}}, \beta_{\mathrm{new}}$ of lengths $3386495, 3109533$. Their cusp equation holds at all six graph vertices through degree nine, and their Frattini action is the identity. This establishes that finite elliptic lift, without asserting simultaneous compatibility with the excluded sphere action or with all D.39 homology coordinates.

The computations address Part D(e), section 51.16, folio 734 verso, scan 174, and D-after-LM2(1), section 51.15, folios 725–727, scans 156–160.

Proof. The minimal bases and their literal reductions. An even image is generated by the images of r, srs, both of order dividing three. A group of order less than 12 generated this way is a 3-group: the only possible orders divisible by three are $3, 6, 9$, and its Sylow 3-subgroup is normal in the order-six case. At order 12, a nonnilpotent such image has four Sylow 3-subgroups. Its action on them is faithful: the kernel lies in a Sylow normalizer of order three, and a nontrivial kernel would make that Sylow subgroup normal. The image is the order-twelve subgroup A_4 of S_4. Thus the least sign-preserving order is 24.

Conjugation by s is an involution of A_4, whose automorphism group is S_4 under its action on the four Sylow 3-subgroups. The nontrivial involutions are inner double transpositions and outer transpositions. The trivial action would give $r = srs$ and would not generate A_4. The inner case is exactly $A_4 \times C_2$: if the action is $\mathrm{Int}(h)$, the mutually inverse marked maps are

$$ls^\epsilon \longmapsto (lh^\epsilon, \epsilon), \qquad (m, \epsilon) \longmapsto mh^{-\epsilon}s^\epsilon.$$

The outer case is S_4. The program `d_cm_solvable_even_boundary.py` enumerates the two actual markings in (D.45.1), not just their group orders. The first is the canonical mod-four quotient, also checked in D.44. The second is the actual mod-three $A_4 = \mathrm{PSL}_2(\mathbb{F}_3)$ coordinate together with the mod-two sign; the program checks all 144 products in the displayed mod-three source isomorphism. Their normal V_4 reductions are precisely the indicated S_3, C_6.

For a normal subgroup of finite 2-power index, the induced pro-2 topology is the full one. Indeed take a subgroup normal of 2-power index in the smaller group and intersect its finitely many conjugates; then take the finite normal 2-power core if necessary. This gives a subgroup normal of 2-power index in the larger group. Applying this to the actual V_4 kernel reductions identifies their relative completions with (D.45.2) in both directions.

Let J be a finite common quotient of the two groups in (D.45.2). The images of their pro-2 kernels are contained in $O_2(J)$. Thus $J/O_2(J)$ is a marked common quotient of S_3 and C_6, and is a quotient of C_2. The group J itself is a 2-group. The original order-three element r is then trivial; s generates the image, so J is a quotient of C_2. The sign is retained on both sides, proving that the maximal common marked quotient is exactly C_2. Closed Goursat proves (D.45.3).

The two graph maps and their exact joint kernel. The marked Bass–Serre graphs each have six edges, three s-vertices, and two r-vertices; their free rank is two. The chosen free bases are

$$\begin{aligned} f_1 &= r^{-1}sr^{-1}s, \quad f_2 = rsrs && \text{for } F, \\ e_1 &= rsr^{-1}s, \qquad e_2 = r^{-1}srs && \text{for } B'. \end{aligned} \tag{D.45.8}$$

In the recorded K-basis $k_1, \dots, k_4$, the literal graph maps are

	F	(B')
k_1	$f_1^{-1}f_2^{-1}$	$e_1^{-1}e_2e_1^{-1}$
k_2	$f_2^{-1}f_1^{-1}$	$e_2^{-1}e_1e_2^{-1}$
k_3	$f_1^{-1}f_2f_1^{-1}$	$e_1^{-1}e_2^{-1}$
k_4	$f_2^{-1}f_1f_2^{-1}$	$e_2^{-1}e_1^{-1}$.

(D.45.9)

The certificate checks these words and all eighteen original Q_0 section words. They give a bijection onto the literal fibre product $S_3 \times_{C_2} C_6$. Abelianizing (D.45.9) gives (D.45.4). Conjugating the recorded D.33/D.39 homology matrix by this P gives the two blocks (D.45.6), with zero off-diagonal blocks.

The kernel of (D.45.3) over Q_0 is exactly the product of the two pro-2 kernels. Thus the image of the actual K is dense in that product, and its map extends to the surjection (D.45.5). Choose lifts a_1, a_2, b_1, b_2 of $(f_1, 1), (f_2, 1), (1, e_1), (1, e_2)$ in $K^{(2)}$. Their Frattini classes are a basis, by the odd determinant in (D.45.4). They are a free pro-2 basis of $K^{(2)}$. Quotienting by the closed normal subgroup generated by $[a_i, b_j]$, $1 \le i, j \le 2$, gives the product of the free pro-2 groups on the a_i and the b_j: the maps from the quotient to that product and back send those four generators to each other. They are inverse continuous homomorphisms. This proves the exact kernel description, with the source map fixed by (D.45.9).

Geometrically the first subgroup has three width-two cusps and genus zero; the second has one width-six cusp and genus one. These statements also follow directly from the regular permutation cycles of $t = sr$ on the two six-point marked quotients and Euler characteristic -1.

The actual lower action and its degree-nine target. Functoriality gives $D_9K \subset D_9F \cap D_9B'$. This proves the two source maps in (D.45.7), and hence the normalization transfer for the recorded action. The program `d_cm_rank2_boundary_test.py` reconstructs the exact D.33 words from the saved integer certificates, checks both original SHA256 hashes, and applies the retained D.39 gauge $k = -438$. It evaluates their full paths at all six vertices of both graphs. The raw cusp word contains 12921196 letters of B. The C++ helper uses exact noncommutative mod-two Magnus series, through degree nine. An independent tuple-monomial Cauchy calculation checks its output on 108 graph paths, including inverse letters and every starting vertex.

Put $X = f_1 - 1$, $Y = f_2 - 1$ in the sphere computation, and use $e_1 - 1, e_2 - 1$ in the elliptic computation. The homogeneous dimensions of the restricted free Lie algebra in degrees $1, \dots, 9$ are

$$2,\ 3,\ 2,\ 6,\ 6,\ 11,\ 18,\ 36,\ 56. \tag{D.45.10}$$

The program constructs a basis by 2-powers of the ordinary Lyndon commutators, and verifies independence in the free associative algebra. In degree nine these are the 56 ordinary Lyndon commutators. The finite kernels have orders $|F/D_9F| = |B'/D_9B'| = 2^{84}$ and the next kernels have order 2^{140}.

Write the recorded action as

$$\alpha(s) = \beta s \beta^{-1}, \qquad \alpha(r) = ArA^{-1}.$$

Both recorded conjugators lie in the indicated free rank-two kernel. The defect $\alpha(t)t^{-1}$ vanishes through degree eight in both graphs. Its degree-nine terms d_F, d_E lie in the degree-nine Lie subspace. Multiplying the two conjugators on the left by arbitrary degree-nine central elements changes that target by

$$(1-s)c_\beta + s(1-r)c_A. \tag{D.45.11}$$

Here s, r act by the actual conjugation substitutions on the source free group. The 112 exact columns of (D.45.11) have ranks 47 and 38 in the sphere and elliptic cases. Both recorded targets are outside those initial images. This calculation alone would only exclude the recorded conjugator images. The next step accounts for all the remaining same-action freedom.

Complete lower centralizers and every same-action correction. In each of the two sources the program `d_cm_rank2_centralizers.py` computes the entire subgroup $C_{G/D_9G}(s)$, not only the s-fixed subspace of the last graded layer. It uses the actual free words sf_is or se_is at every degree. For $1 \to L_n \to G/D_{n+1}G \to G/D_nG \to 1$, the map from a lower centralizer element to the class of $sgs^{-1}g^{-1}$ in $L_n/(1-s)L_n$ is the exact obstruction to lifting that element. The program evaluates this map on actual reduced words. Its kernel supplies all liftable lower elements; the fixed subspace of L_n supplies all new elements in the kernel.

At every step the produced subgroup is elementary abelian. This is checked in the full quotient: each generator has depth at least $\lceil (n+1)/2 \rceil$, so products are additive, squares and commutators vanish, and the displayed truncated Magnus vectors are independent. Induction using the exact central extension therefore proves completeness. The successive ranks in G/D_jG, $j = 2, \dots, 9$, are

j	2	3	4	5	6	7	8	9
F	1	2	3	5	8	13	22	38
B'	2	3	4	7	10	17	28	50.

(D.45.12)

The corresponding connecting ranks are respectively $(0,1,0,2,0,1,0,4)$ and $(0,2,1,3,3,4,7,14)$. The final generators all have depth at least five. Their words and the complete recursive construction are in the certificate.

Replacing β on the right by any lower s-centralizer element preserves its conjugation action modulo D_9G. Its degree-nine defect is precisely the connecting value just computed, up to an inner action of the free kernel, which is trivial on this central layer. Since the final centralizer is elementary and has depth at least five, these defects are additive in the quotient being tested. Thus its whole image is spanned by the 38, respectively 50, additional columns.

No omitted freedom from the r-centralizer can enlarge this obstruction space. The order-three action is coprime to the central 2-group extension. Every lower r-fixed element lifts to an r-fixed element: its defect has zero $1 + r + r^2$ norm, and over $\mathbb{F}_2$ that kernel is the image of $1 - r$. The required central correction is already among (D.45.11). This is also the finite coprime centralizer lifting argument in literal coordinates.

Before concluding nonextension, we identify all possible conjugators of a proposed lift. Its S_3 action equals the identity, as does the recorded action. The image of an s-conjugator lies in $C_{S_3}(s) = \langle s \rangle$. Multiplication on the right by the appropriate s-power puts it in F and changes neither conjugation nor the cusp equation. Likewise an r-conjugator can be put in F using $C_{S_3}(r) = \langle r \rangle$. Two such normalized conjugators inducing the same lower action differ on the right by the complete lower centralizers just computed. Their remaining differences are arbitrary central degree-nine elements. Thus (D.45.11) and the added columns exhaust all lifts of this lower sphere action with multiplier one.

A four-coefficient functional proves the sphere exclusion. Encode a word in X, Y by a binary string, with $0 = X, 1 = Y$, read in the written order. On the homogeneous degree-nine

associative algebra define

$$\Lambda(z) = [000011011]z + [000010111]z + [001001111]z + [000101111]z. \tag{D.45.13}$$

These are the little-endian coefficient positions $432, 464, 484, 488$ in the certificate. All 112 central columns and all 38 additional centralizer columns have $\Lambda = 0$. On the recorded target d_F, the four coefficients are $(0, 0, 1, 0)$, so $\Lambda(d_F) = 1$. The full correction rank stays 47. The exact rational-free bit calculations and the word checks prove that the lower sphere action has no required lift to $B/D_{10}F$.

The normalized D.39 quotient retains all of $\widehat{F}^{\mathrm{ab}}$: its source kernel is contained in $\overline{[L, L]} \subset \widehat{F}'$. The recorded multiplier one action is the identity on this abelianization. A genuine normalized $\widehat{\mathrm{GT}}_0$ element acts there by its full $\lambda \in \widehat{\mathbb{Z}}^*$. The equality on $\widehat{\mathbb{Z}}^2$ therefore forces $\lambda = 1$. The exact braid-source map of D.25 then provides the torsion conjugators and fixed cusp used above. The characteristic pro-2 and Zassenhaus quotients of the actual F give the finite projection of that genuine action. Equation (D.45.7) would make it a lift of the excluded lower action. This contradiction proves nonextension of the entire recorded D.39 action and all its indicated extensions, with no remaining multiplier or conjugator freedom.

The surviving elliptic correction as literal source words. For the elliptic source, adding its 50 exact centralizer columns raises the rank from 38 to the full 56. Number the degree-nine Lyndon commutator words $\ell_0, \ldots, \ell_{55}$ in lexicographic order, with $[a, b] = aba^{-1}b^{-1}$; use the basis e_1, e_2 of (D.45.8). Number the complete lower centralizer words $c_0, \ldots, c_{49}$ in the deterministic recursive certificate order of (D.45.12). The exact solution is

$$\begin{aligned} I_A &= \{0, 1, 2, 3, 4, 6, 10, 11, 12, 13, 14, 15, 17, 22, 23, 24, 25, 30, 31, 34, 36, 41\}, \\ I_\beta &= \{2, 3, 5, 7, 11, 13, 14, 16, 23\}, \\ A_{\mathrm{new}} &= \left(\prod_{i \in I_A} \ell_i\right) A_{39}, \qquad \beta_{\mathrm{new}} = \beta_{39} \left(\prod_{i \in I_\beta} c_i\right). \end{aligned} \tag{D.45.14}$$

Products use increasing indices. There is no left central β-correction. The exact masks are `214c3c2fc5f` and `8168ac`. Every c_i, every substituted B-word, and both corrected full-word hashes are provided in `d_cm_rank2_elliptic_corrected.json`.

Running the boundary program with `-elliptic-correction` constructs (D.45.14) as actual B-words. The six exact degree-nine cusp supports become $(0, 0, 0, 0, 0, 0)$. The two generator substitutions agree with the original ones through degree eight, so this retains the whole lower action. The images of s, r still have their exact orders by conjugation; their free-kernel Frattini action is the identity. They therefore generate the finite quotient, and the resulting endomorphism is an automorphism. This proves the second branch conclusion in its stated finite elliptic source. □

Supplement to D.45: preserving multipliers and the original fibre. The same obstruction also excludes a peripheral lift of the original recorded $\alpha_{33,R}$ to the full relative pro-2 source $\mathcal{P}_2$ of D.34. Consequently the specific fibre over $[\alpha_{33,R}]$ in (D.35.5a) is empty. This strengthens the exclusion of the D.39–D.44 actions without excluding a replacement action having their compact CM coordinate or their prescribed CM pair.

Here are the full source maps and multiplier argument. In the literal sphere graph of D.45, t^2 is conjugate to the free generator f_2. Its eighth power has a nonzero degree-eight mod-two Magnus term, while its sixteenth power has no term below degree sixteen. Thus t has exact order 32 in both B/D_9F and $B/D_{10}F$. The program `d_cm_rank2_multiplier_transport.py` checks this directly by tuple-monomial Cauchy multiplication: the first degrees of t^2, t^4, t^8, t^{16} are $1, 2, 4, 8$, and $t^{32} = 1$ through degree nine. Every λ_2 preserving the recorded lower fixed

cusp therefore satisfies $\lambda_2 \equiv 1 \pmod{32}$ and also fixes that cusp in the next quotient. There is no additional degree-nine multiplier column.

The relative source has the full cusp exponent pair $(\lambda_2, 1)$ from D.34. Its order-three component is one. On the retained Q_0, s, r keep their actual order-two and order-three conjugacy classes; the power of r is one, and the power of s is one because λ_2 is odd. The image of t in Q_0 has order six, and $c_0 = t^6$ lies in $K^{(2)}$. These are the exact conditions used by the D.45 centralizer normalization.

We make explicit that every normalized representative in $\mathrm{Aut}_{Q_0,t}(\mathcal{P}_2)$ has the torsion conjugators required here. Let H be the discrete preimage of $\langle s \rangle < Q_0$. Its nine-coset action has three fixed s-cosets and no fixed r-cosets; the marked Bass–Serre graph has nine edges, six order-two vertices and three order-three vertices, with three nontrivial vertex stabilizers, all of order two. Its graph has first Betti number one. Thus $H = C_2 * C_2 * C_2 * \mathbb{Z}$, with each finite factor a discrete B-conjugate of $\langle s \rangle$. The program `verify_torsion_bridge.py` records the actual cosets, permutations and the three conjugating words. Since $[H : K] = 2$, taking cores in H of normal subgroups of 2-power index in K proves that the induced pro-two topology on K is full. The resulting map of completed extensions identifies the preimage of $\langle s \rangle$ in $\mathcal{P}_2$ with $H^{(2)} = C_2 \amalg_2 C_2 \amalg_2 C_2 \amalg_2 \mathbb{Z}_2$, with precisely the marking in (D.35.15). Herfort–Ribes' torsion theorem [HR85, Theorem A′, pp. 157–159] therefore makes the image of s under any such automorphism conjugate to a finite vertex generator, hence to s in $\mathcal{P}_2$. The subgroups generated by r and its image are complements to K_2 over the same order-three subgroup of Q_0. Profinite coprime complement conjugacy gives a K_2-conjugator; the fixed Q_0 image fixes the exponent as one.

The sphere projection is preserved as well. The preimage $\widetilde{F}$ of $\ker(Q_0 \to S_3)$ is invariant under an automorphism inducing the identity on Q_0. Its maximal pro-two quotient is the marked $F^{(2)}$: every finite two-group quotient of F restricts to one of K, hence factors through $\widetilde{F}$, and the converse follows by restriction to the dense image of F. Its characteristic pro-two residual is consequently preserved. The induced quotient of $\mathcal{P}_2$ is exactly $\mathcal{P}_F$ in (D.45.2), and the characteristic subgroups $D_9F^{(2)}, D_{10}F^{(2)}$ are preserved there. This proves the needed automorphism transfer to both sphere quotients before applying their obstruction.

Now $\mathrm{Int}(c_0^{258})$ is an actual continuous inner automorphism of $\mathcal{P}_2$, with inverse $\mathrm{Int}(c_0^{-258})$. It induces the identity on Q_0 and fixes t as an element, not merely as a conjugacy class. It preserves every peripheral class and commutes with taking the indicated cusp powers. Its reduction on $R = T_3$ is the exact inner map in (D.39.6). In that quotient c_0 has order sixteen, and $258 - 146 = 112$ is divisible by sixteen, proving the two displayed forms agree. A relative lift of $\alpha_{33,R}$, composed on the left with this source inner map, would therefore project to the excluded D.39 sphere action. Its preserving multiplier is invisible in the next sphere quotient by the order calculation above. D.45's exhaustive conjugator argument applies and gives a contradiction.

Finally consider the precise quotient in (D.35.5a). By the exact restriction sequence (D.35.5), an element of the stated outer fibre has a normalized relative representative. Its reduction differs from $\alpha_{33,R}$ by a power of $\mathrm{Int}(c_0)$. The map $\langle c_0 \rangle \simeq \mathbb{Z}_2 \to \langle c_0 \bmod D_9K \rangle = C_{16}$ is onto. Hence that finite discrepancy lifts to an actual c_0^z in the source and can be removed by its inverse inner automorphism. This would give an exact relative lift of $\alpha_{33,R}$, already excluded. It proves emptiness of that particular outer fibre with both directions of the normalization transport retained. □

An arithmetic pro-two replacement and simultaneous relative-prime framing

D.46. A replacement with the prescribed extra CM pair on the full relative pronilpotent source [Proved]

Retain $N = \ker(B \to Q)$, with Q of order 576 and N free of rank 97. For every prime p, define the actual source kernels and relative completions

$$R_p = \ker(\widehat{N} \to N^{(p)}), \qquad \mathcal{P}_p = \widehat{B}/R_p, \qquad R_{\mathrm{nil}} = \bigcap_p R_p, \qquad \mathcal{U} = \widehat{B}/R_{\mathrm{nil}}. \tag{D.46.1}$$

There are exact marked identifications

$$\mathcal{U} = \prod_Q \mathcal{P}_p, \qquad \ker(\mathcal{U} \to Q) = \prod_p N^{(p)}. \tag{D.46.2}$$

In particular this retains the entire nonabelian relative pro-2 kernel, not a finite-degree or class-two quotient.

There is a continuous peripheral automorphism $\alpha^\dagger$ of $\mathcal{U}$, with multiplier one and an actual $\widehat{F}'$ parameter, such that its component at every odd prime is the identity, while its component at two is induced by one actual Galois commutator h. Its finite $G_6 = B/[N,N]N^6$ shadow has

$$V_{12} = I, \qquad T_4 = I, \qquad W_3 = [i], \tag{D.46.3}$$

with the exact global-sign and conjugator conventions of D.25. Thus it realizes the prescribed extra CM pair on this whole relative-pronilpotent quotient. Its finite refinements retaining G_6 give a compatible family of nonarithmetical gentle GT shadows. No assertion of full profinite GT0 realization is made by identifying this quotient family.

The two-component action is particularly explicit. Put $\zeta = \zeta_3$ and

$$a = 1 - 2\zeta^2, \qquad \eta_* = \frac{a}{\bar{a}} = \frac{-3 - 8\zeta^2}{7}. \tag{D.46.4}$$

The normalized action of h on the full sphere group $F^{(2)}$ is the identity. Its actual action on the compact cubic elliptic Tate module is multiplication by η_*. In the elliptic graph basis of D.45 its matrix modulo sixteen is

$$\begin{pmatrix} 11 & 8 \\ 8 & 3 \end{pmatrix}. \tag{D.46.5}$$

The arithmetic element h itself has the CM pair $(-I, [i])$; (D.46.3) is recovered from the new joint relative-prime action through G_2 and G_3. The proof specifies every conjugator map producing that change of framing, while keeping the two-component automorphism unchanged.

There is a canonical identification $\mathcal{P}_2 \simeq \mathcal{P}_2^K$, where $\mathcal{P}_2^K$ is the full rank-four relative source of D.34. The projection of this arithmetic component to (D.45.3) is

$$(1_{\mathcal{P}_F}, \alpha_{h,E}). \tag{D.46.6}$$

Its lift through the rank-four mixed-commutator kernel is the actual restriction of h. Thus this particular replacement solves that entire relative pro-2 lifting step. It replaces the excluded D.33/D.39 lower action: the old sphere action is already nonidentity in degree seven, while (D.46.6) is the identity on the whole sphere factor.

The parameter is the exact compatible tuple

$$f^\dagger = (f_{h,2}, 1, 1, \ldots) \in \mathrm{im}(\widehat{F}' \to \mathcal{U}). \tag{D.46.7}$$

It has trivial image in every pure pro-p completion of F, but it is nontrivial in $\mathcal{U}$, as witnessed by (D.46.4). The exact projection kernels are

$$\ker(\mathcal{U} \to \mathcal{P}_p) = \prod_{q \neq p} N^{(q)}, \qquad \ker(\widehat{B} \to \mathcal{U}) = R_{\mathrm{nil}}. \tag{D.46.8}$$

Under the actual marked epimorphism to S_7 whose permutations, on $0, \dots, 6$, are

$$s \mapsto (1,0,3,2,5,4,6), \qquad r \mapsto (3,6,2,5,1,0,4), \tag{D.46.9}$$

the latter source kernel maps exactly onto A_7. The relative centralizer elements used below do not come from the corresponding full $\widehat{B}$ centralizers. These two exact facts identify the remaining full-source lifting problem.

This continues Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus follow-up to LM2, section 51.15, folios 725–727, scans 156–160.

Proof. The actual arithmetic commutator. Write $k = \mathbb{Q}(\zeta)$ and $\mathcal{O} = \mathbb{Z}_2[\zeta]$. D.35 proves that the actual image of G_k on T_2E_3, for $E_3 : y^2 = x^3 + 1$, is $U = 1 + 2\mathcal{O}$. Its full division field over k is therefore pro-two. The cyclic cubic field $k(\sqrt[3]{4})$ is disjoint from it. Consequently the joint map

$$G_k \longrightarrow U \times C_3, \qquad g \longmapsto (\rho_{E_3,2}(g), \kappa_{4,3}(g))$$

is onto. Choose an actual g with image $(a, 0)$, where a is (D.46.4). This is a use of the proved arithmetic image and disjointness, not an assumed realization of a matrix. The CM norm is $a\bar{a} = 7$, so $\chi_2(g) = 7$. Also $\chi_3(g) \equiv 1 \pmod 3$ and $\kappa_{2,3}(g) \equiv 0 \pmod 3$.

Since $k/\mathbb{Q}$ is ramified at three, choose τ in a fixed inertia group I_3 that induces the nontrivial automorphism of k. Set

$$h = [g, \tau] = g\tau g^{-1}\tau^{-1}. \tag{D.46.10}$$

The entire cyclotomic character of h is one, because the character is abelian. The normalized pro-two action on the thrice-punctured line is unramified at three. The exact statement for the tangential automorphism action appears in [HM03, §8.5, preceding Theorem 8.7], with $N = 1, \ell = 2$; §8.4 states the corresponding outer action for every prime. Thus τ, and hence the commutator h, acts trivially on $F^{(2)}$. The same conclusion follows from the outer statement and the normalization argument of D.23: an inner action fixing x has conjugator in $\langle x \rangle$, and comparison with $y \mapsto f^{-1}yf$, $f \in (F^{(2)})'$, forces both abelian exponents to vanish.

The exact affine Kummer commutator identity is

$$\kappa_b([g, \tau]) = (1 - \chi(\tau))\kappa_b(g) + (\chi(g) - 1)\kappa_b(\tau). \tag{D.46.11}$$

Modulo three, the two conditions imposed on g make $\kappa_{2,3}(h) = 0$. At two, $\chi_2(\tau) = 1$. Also $\kappa_{2,2}(\tau) = 0$: every 2-power root of the 3-adic unit 2 lies in the maximal unramified extension, by Hensel's lemma and its prime-to-three derivative. Therefore $\kappa_{2,2}(h) = 0$. In particular D.18's actual matrix gives

$$V_h = \begin{pmatrix} 1 & -8\kappa_2(h) \\ 0 & 1 \end{pmatrix}, \qquad V_h \bmod 12 = I. \tag{D.46.12}$$

The induced normalized action on Q is the identity.

The compact cubic cover $\mathcal{E}_0$ of D.34 has the exact twist $\rho_{\mathcal{E}_0,2} = \zeta^{-\kappa_{4,3}}\rho_{E_3,2}$. Its value on g is a, and conjugation by the action of τ conjugates the CM coefficient ring. Hence its value on h is exactly $a/\bar{a} = \eta_*$, regardless of the particular semilinear matrix of τ. Equation (D.46.11)

also gives $\kappa_{4,3}(h) = 0$, so the actual E_3 scalar is the same. Its reduction modulo four is $-I$. The Galois comparison of D.25–D.36 then gives the actual arithmetic pair

$$(T_{h,4}, W_{h,3}) = (-I, [i]). \tag{D.46.13}$$

This use of the cover-to-fixed-fibre map is confined to the actual Galois element h.

The full rank-four and two rank-two actions. The subgroup N has 2-power index 32 in K. The full induced pro-two topology argument of D.45 proves equality of the actual source kernels

$$\ker(\widehat{N} \to N^{(2)}) = \ker(\widehat{K} \to K^{(2)}). \tag{D.46.14}$$

Thus the relative completions over Q and Q_0 are canonically the same marked group. The Galois action preserves N because it acts on Q by (D.46.12), and preserves its pro-two residual. It consequently induces an actual automorphism $\alpha_{h,2}$ of this entire relative source. Restriction and the two actual quotient maps of D.45 give (D.46.6). The original arithmetic automorphism is its lift through the exact mixed commutator kernel, so the lifting assertion does not follow merely from the homology determinant.

In the elliptic free basis e_1, e_2 of D.45, the literal graph conjugation by t^2 has matrix

$$Z_E = \begin{pmatrix} 0 & 1 \\ -1 & -1 \end{pmatrix}. \tag{D.46.15}$$

It is the image of ζ^2 under the same CM identification as in D.35. Substitution into $(-3I - 8Z_E)/7$ gives (D.46.5). The arithmetic action preserves each rank-two source map, hence their exact joint kernel as well. The sphere factor is the identity by (D.46.10), whereas D.45 computes a nonzero degree-seven term for the old action. This is an explicit distinction under the same marked quotient map.

The entire relative-prime source and its parameter. For finitely many primes, the actual image of N in a product of finite p-group quotients is the full product, by the coprime-order Goursat argument. Passing to all finite levels and primes proves (D.46.2) with its original source kernels. Its factors have exactly the common quotient Q: in any larger common quotient retaining Q, the kernel would be both a p-group and a prime-to-p group.

Since $\alpha_{h,2}$ is the identity on Q, the product

$$\alpha^{\dagger} = (\alpha_{h,2}, (1_{\mathcal{P}_p})_{p \neq 2}) \tag{D.46.16}$$

preserves the actual fibre product and is a continuous automorphism. It fixes t. Write the canonical normalized Galois conjugators, in the image of $\widehat{F}$, as

$$\alpha_h(s) = \beta_h s \beta_h^{-1}, \qquad \alpha_h(r) = A_h r A_h^{-1}.$$

Put $a_Q = q(A_h)$, $b_Q = q(\beta_h)$. Both lie in $q(F)$; they centralize r_Q, s_Q, respectively, by (D.46.12).

For clarity, the coprime fixed-element statement used next applies to these literal cosets. Let a finite group of order prime to p act on a profinite group with invariant pro-p kernel. Every fixed element of the quotient lifts to a fixed element whenever that acting group is the cyclic order-two or order-three subgroup used here. To prove it, lift the quotient element; its conjugate of the acting cyclic subgroup and the original subgroup are complements in the same pro-p-by-cyclic extension. Coprime complement conjugacy corrects the chosen lift by the kernel. Finite centre induction and compactness give the profinite statement, as in D.42. This is a coset lifting result, not an assumption that the quotient element already lies in the kernel.

Apply it to conjugation by r on $\mathcal{P}_2 \to Q$. There is an actual

$$c_2 \in C_{\mathcal{P}_2}(r), \qquad q(c_2) = a_Q^{-1}. \tag{D.46.17}$$

Set $A_2^\dagger = A_{h,2}c_2$. It has Q-coordinate one and gives exactly the same conjugation of r as before. For every odd prime apply the same result to conjugation by s: choose

$$b_p \in C_{\mathcal{P}_p}(s), \qquad q(b_p) = b_Q. \tag{D.46.18}$$

One choice at each prime is a single compatible element of the full product in (D.46.2). The actual simultaneous conjugators are

$$A^\dagger = (A_{h,2}c_2, (1)_{p\neq 2}), \qquad \beta^\dagger = (\beta_{h,2}, (b_p)_{p\neq 2}). \tag{D.46.19}$$

Their common Q coordinates are 1 and b_Q, so both are actual elements of $\mathcal{U}$. Their images lie in the indicated F source, because their Q coordinates do.

The normalized parameter is recovered from the automorphism of s, not from an arbitrary choice of its conjugator:

$$f^\dagger = s\beta^\dagger s(\beta^\dagger)^{-1} = (f_{h,2}, (1)_{p\neq 2}). \tag{D.46.20}$$

In particular the change (D.46.17) changes neither this parameter nor the two-component automorphism. For each prime p, the source quotient $\mathcal{P}_p$ maps to the actual relative pro-p completion of the original F: the restriction of any such map to N has pro-p image and kills R_p. Therefore $\mathcal{U}$ retains all $\widehat{F}^{\mathrm{ab}} = \prod_p \mathbb{Z}_p^2$. Equation (D.46.20) has zero coordinate there. Every actual F preimage of it consequently lies in $\widehat{F}'$. This proves the parameter claim with its source map, and also shows that its image in every pure $F^{(p)}$ is one.

The order-two and order-three equations, and the fixed t, give the two multiplier-one hexagons by D.25's exact braid-source comparison. Equivalently $\alpha^\dagger(s) = sf^\dagger$ and $\alpha^\dagger(r) = (f^\dagger)^{-1}r$. The automorphism is already bijective by (D.46.16), so the generation condition is also satisfied. Finite characteristic refinements give the stated finite GT shadows: their pullbacks to B_3 are normal subgroups of PB_3, the multiplier is one, and the same actual commutator parameter projects compatibly.

The framing is an invariant of the new joint action. The distinction between (D.46.13) and (D.46.3) is checked using the actual D.25 maps. On G_2 the new automorphism is the arithmetic h action. The first equality in (D.25.6) forces the Q-coordinate of every normalized s-conjugator to be b_Q. On G_3 the new automorphism is the identity. The second equality in (D.25.6) forces the Q coordinate of every normalized r-conjugator to be one. Thus the new finite shadow itself determines the coordinates

$$(A_Q^\dagger, \beta_Q^\dagger, V_Q) = (1, b_Q, 1). \tag{D.46.21}$$

They have not been assigned as independent matrix-centralizer choices. The normalized congruence matrix is $V_Q = 1$: the action on Q is the identity, so this matrix is central in the projective congruence group; its lower-right-one normalization and multiplier one force the scalar matrix I. The actual formulas $T = A^{-1}V$ and $W = \beta^{-1}V$, with the unique global sign fixing the rational order-three point for T_4, now give $T_4 = I$ and retain the arithmetic $W_3 = [i]$.

The finite certificate also checks every possible arithmetic framing used in this argument. The maximal relative-two quotients of Q over the sphere and elliptic bases have orders 24 and 48. Their source kernels have orders $24, 12$, trivial intersection, and product of index two. The first quotient is the actual canonical mod-four S_4, checked on all 576^2 source pairs. Hence

$$Q = Q_F \times_{C_2} Q_E, \qquad |Q_F| = 24,\ |Q_E| = 48. \tag{D.46.22}$$

There are three possible a_Q with arithmetic $T_4 = -I$. There are two possible b_Q with $W_3 = [i]$ and trivial image in Q_F. The latter condition holds here in the full source: the preimage of $\langle s \rangle$ in the sphere relative completion is the free pro-two product $C_2 \amalg \mathbb{Z}_2$. Its marked discrete subgroup has index three in B, one order-two vertex group and one free generator; the full induced pro-two topology identifies the completion. Herfort–Ribes [HR85, Theorem B′] gives $C(s) = \langle s \rangle$ there. Any centralizer element in the whole sphere source projects into $C_{S_3}(s) = \langle s \rangle$, so lies in that subgroup. As β_h belongs to F, the identity sphere action forces its projection to be one. This proves the stated condition on b_Q.

For all six resulting possibilities the certificate explicitly multiplies a_Q by a_Q^{-1}, retains b_Q, and computes the D.25 global-sign bits. They change from $(1, 1)$ to $(0, 1)$ in every case. This verifies the precise framing map without requiring an unidentified choice among the six cases.

The full-source centralizers and the remaining kernel. Herfort–Ribes [HR85, Theorem B′] also gives

$$C_{\widehat{B}}(r) = \langle r \rangle, \qquad C_{\widehat{B}}(s) = \langle s \rangle. \tag{D.46.23}$$

Both cyclic images meet $q(F)$ trivially. Every nontrivial a_Q^{-1} in (D.46.17) lies in $q(F)$, and the two possible b_Q's in (D.46.18) do too. The program verifies these containments and exclusions for all six cases. Thus those relative centralizer lifts do not lift to elements of the corresponding full centralizers. This explains exactly why the construction does not alter the same Galois automorphism on the full source: the odd relative actions in (D.46.16) have been replaced by the identity, and the new parameter is (D.46.20), not the entire f_h. It does not prove nonexistence of a different full automorphism inducing this new parameter.

Taking the projection kernels in (D.46.2) gives (D.46.8). For a concrete test of the remaining source kernel, the program enumerates all 5040 permutations generated by (D.46.9), with odd s and even r. The group Q is solvable: its normal two-subgroup has quotient $S_3 \times C_3$. Every finite image of $\mathcal{U}$ is an extension of a nilpotent group by a quotient of Q, hence is solvable. The quotient of S_7 by the image of R_{nil} is therefore solvable, so that image contains A_7. Conversely $R_{\mathrm{nil}} \subset N \subset B_{\mathrm{even}}$ makes its image even. The image is exactly A_7, as asserted. This is an actual remaining full-source kernel, despite the entire relative-prime lift proved above. □

One coherent replacement over every Fermat level and the full metabelian quotient

D.47. The closed relative-prime source, all Fermat homology maps, and its nonarithmetical metabelian action [Proved]

All subgroups below are closed in $\widehat{B}$, and every commutator subgroup denotes the closed subgroup generated by commutators. Let

$$\begin{aligned} F_m &= \ker(\widehat{F} \to (\mathbb{Z}/m)^2), \quad L_m = \widehat{N} \cap F_m, \\ M &= \widehat{N} \cap \widehat{F}' = \bigcap_{m \geq 1} L_m, \quad \mathcal{H}_\infty = \widehat{B}/M. \end{aligned} \tag{D.47.1}$$

The maps use the original ordered loop abelianization. For every prime p, write $M^{(p)}$ for the maximal continuous pro-p quotient of this compact group, and set

$$\begin{aligned} R_{\infty,p} &= \ker(M \to M^{(p)}), \quad \mathcal{P}_{\infty,p} = \widehat{B}/R_{\infty,p}, \\ R_\infty &= \bigcap_p R_{\infty,p}, \qquad\qquad \mathcal{V} = \widehat{B}/R_\infty. \end{aligned} \tag{D.47.2}$$

There are exact marked identifications

$$\begin{aligned} \mathcal{P}_{\infty,p} &= \varprojlim_{m} \widehat{B}/\ker(L_m \to L_m^{(p)}), \\ \mathcal{V} &= \prod_{\mathcal{H}_\infty} \mathcal{P}_{\infty,p}, \qquad \ker(\mathcal{V} \to \mathcal{H}_\infty) = \prod_p M^{(p)}. \end{aligned} \tag{D.47.3}$$

The first inverse limit is over divisibility, with all maps induced by the original $\widehat{B}$. The maximal common marked quotient of two different prime factors, or of one factor and the joint remaining prime factors, is exactly $\mathcal{H}_\infty$. Both kernel inclusions in the first line, and the actual simultaneous image in the second, are proved below.

The base and all its finite levels are explicit. Put $H = q(\widehat{F}) \subset Q$, and $J = H'$. Then $J \simeq Q_8$, $|J| = 8$, and

$$\mathcal{H}_\infty = (\widehat{B}/\widehat{F}') \times_{Q/J} Q, \qquad |Q/J| = 72. \tag{D.47.4}$$

The image of $\widehat{N}$ in the original loop abelianization is the lattice

$$\begin{aligned} \Lambda &= \{(a,b) \in \widehat{\mathbb{Z}}^2 : a \equiv b \equiv 0 \pmod 2,\ a+b \equiv 0 \pmod 3\} \\ &= \widehat{\mathbb{Z}}(2,-2) + \widehat{\mathbb{Z}}(0,6). \end{aligned} \tag{D.47.5}$$

Thus $\widehat{N}/M = \Lambda$, and, for the actual finite base $H_m = \widehat{B}/L_m$,

$$\begin{aligned} c_m &= \gcd(2,m)^2 \gcd(3,m), \\ |H_m| &= 576\, m^2/c_m, \qquad \operatorname{rank}(L_m) = 1 + 96\, m^2/c_m. \end{aligned} \tag{D.47.6}$$

The rank is that of its original finite-index discrete free subgroup, whose completion is L_m. In particular H_{12} and L_{12} are exactly the order-6912, rank-1153 source of D.37–D.39.

The actual Galois commutator h of D.46 fixes $\mathcal{H}_\infty$ pointwise. There is a continuous peripheral automorphism of $\mathcal{V}$

$$\widetilde{\alpha}^\dagger = (\alpha_{h,\infty,2}, (1_{\mathcal{P}_{\infty,p}})_{p\neq 2}), \tag{D.47.7}$$

with multiplier one and one compatible pair of normalized loop conjugators. Its parameter is the literal tuple

$$\widetilde{f}^\dagger = (f_{h,\infty,2}, 1, 1, \ldots) \in \operatorname{im}(M \to \mathcal{V}). \tag{D.47.8}$$

It projects to the entire D.46 action, including the prescribed finite pair $T_4 = I$, $W_3 = [i]$. It induces compatible actions on every Fermat homology group at every deck level and every coefficient prime: the full two-primary component is the actual h action, and every odd-primary component is identity. The same assertion holds for the congruence/Fermat subgroups L_m. It consequently supplies a replacement on the same underlying all-prime homology source $\mathcal{T}$ used in D.39, with its new arithmetic T_3 component; the excluded D.39 action is not retained.

In particular the exact inclusions

$$R_\infty \subset M' \subset \widehat{F}'' \tag{D.47.9}$$

give a quotient map onto the full metabelian group $\widehat{F}/\widehat{F}''$, and the action descends under this map. Write

$$A = \widehat{F}'/\widehat{F}'' = \prod_p A_p, \qquad \mathcal{B}_{\mathrm{met},p} = \widehat{B}/\ker(\widehat{F}' \to A_p).$$

The exact metabelian action and parameter are

$$\begin{aligned} \widehat{B}/\widehat{F}'' &= \prod_{\widehat{B}/\widehat{F}'} \mathcal{B}_{\mathrm{met},p}, \\ \alpha_{\mathrm{met}}^\dagger &= (\alpha_{h,\mathrm{met},2}, (1)_{p\neq 2}), \\ f_{\mathrm{met}}^\dagger &= (f_{h,\mathrm{met},2}, (1)_{p\neq 2}). \end{aligned} \tag{D.47.10}$$

Here the prime is the coefficient prime; the complete profinite deck group is retained in every component. This metabelian action is nontrivial and its outer class is not in the actual Galois image. In the multiplier-one normalization used here, passage to the outer action is injective. Under the two marked cover maps of D.36 its compact coordinates are

$$\rho_{C_3,2} = \eta_*, \qquad \rho_{C_3,4} = -I, \qquad \rho_{C_4,3} = I, \tag{D.47.11}$$

which violate the proved arithmetic joint graph. This is a proved comparison for the full metabelian quotient and the specified relative source. Whether this metabelian action, or its prescribed CM pair, is induced by a genuine full profinite GT0 element, or this metabelian action by a full loop-preserving automorphism of $\widehat{F}$, remains undecided.

The maps back to D.46 also have exact kernels. Let Λ_p be the p-component of (D.47.5), and put

$$C_p = \ker(N^{(p)} \to \Lambda_p), \qquad D_p = \ker(M^{(p)} \twoheadrightarrow C_p). \tag{D.47.12}$$

These maps are the original subgroup inclusions followed by the loop abelianization. Then

$$\begin{aligned} \ker(\mathcal{U} \to \mathcal{H}_\infty) = \prod_p C_p, \qquad & 1 \longrightarrow \prod_p D_p \longrightarrow \mathcal{V} \longrightarrow \mathcal{U} \longrightarrow 1, \\ \ker(\mathcal{V} \to \mathcal{P}_{\infty,p}) = \prod_{q \neq p} M^{(q)}, \qquad & \ker(\widehat{B} \to \mathcal{V}) = R_\infty. \end{aligned} \tag{D.47.13}$$

Under the actual S_7 source (D.46.9), R_∞ still has exact image A_7. Thus the full-source lifting step has not been replaced by a cofinality claim.

These results continue Part D(e), section 51.16, folio 734 verso, scan 174, and the LM2 two-torus and Fermat-tower follow-ups, section 51.15, folios 725–727, scans 156–160.

Proof. The marked base and its integral lattice.

The program `d_cm_fermat_relative_tower.py` enumerates Q from its original s, r matrices. For $x = (sr)^{-2}$, $y = sxs$, it obtains $|H| = 96$ and $|H'| = 8$. It evaluates the actual conjugates of $[x, y]$, checks the entire derived image, and verifies the quaternion multiplication conditions. The induced abelian quotient has type $C_2 \times C_6$. For all 144 pairs of exponents modulo twelve the certificate proves

$$x^a y^b \in H' \quad \Longleftrightarrow \quad 2 \mid a, \quad 2 \mid b, \quad 3 \mid (a+b). \tag{D.47.14}$$

Right exactness of abelianization identifies the image of N in F^{ab} with precisely this kernel lattice. Its columns $(2, -2), (0, 6)$ have determinant twelve and Smith invariants $2, 6$. Completion gives (D.47.5) with its actual source map.

The source kernels of $\widehat{B} \to Q$ and $\widehat{B} \to \widehat{B}/\widehat{F}'$ have product $\widehat{N}\widehat{F}'$; their maximal common marked quotient is $Q/q(\widehat{F}') = Q/J$. Their intersection is M. Closed Goursat proves both directions of (D.47.4). The image of L_m under $\widehat{N} \to \Lambda$ is $\Lambda \cap m\widehat{\mathbb{Z}}^2$. Its quotient has order

$$|\Lambda/(\Lambda \cap m\widehat{\mathbb{Z}}^2)| = m^2/(\gcd(2,m)\gcd(6,m)) = m^2/c_m.$$

This proves the base orders in (D.47.6). These subgroups are torsion-free finite-index subgroups of $C_2 * C_3$; their actual Bass–Serre graph gives rank $1 + [B : L_m]/6$. The certificate independently enumerates the lattice modulo every $1 \le m \le 60$, including the $m = 4, 12$ comparisons.

Both inclusions in the closed-source inverse limit. Let $R_{m,p} = \ker(L_m \to L_m^{(p)})$. The topology and quotients here are continuous profinite ones. The actual restriction $M \to L_m^{(p)}$ has pro-p image, so

$$R_{\infty,p} \subset R_{m,p} \quad \text{for every } m. \tag{D.47.15}$$

Conversely let C be the kernel of any finite continuous p-group quotient of M. Conjugation by $\widehat{B}$ has a finite orbit on C. Indeed C is clopen in the compact group M; uniform continuity of conjugation gives an open neighbourhood of one in $\widehat{B}$ that preserves its finite coset partition. The normal core formed by the finitely many conjugates is still open in M, normal in $\widehat{B}$, and has p-group quotient. Replace C by that core.

Inside the profinite group $\widehat{B}/C$, the subgroup M/C is finite. There is an open normal subgroup disjoint from its nonidentity elements. Its preimage O satisfies

$$O \cap M = C.$$

In the finite quotient $\widehat{B}/O$, the image of M is a p-group. The quotient by that image is a finite quotient of $\mathcal{H}_\infty$. The subgroups L_m/M are cofinal among its open normal source kernels: they are directed under least common multiples, and their intersection is one. Compactness therefore gives an m with $L_m \subset OM$. The image of L_m in $\widehat{B}/O$ is a p-group. It follows that $R_{m,p} \subset O$.

An element of $\bigcap_m R_{m,p}$ already lies in $\bigcap_m L_m = M$; it consequently lies in $O \cap M = C$. Applying this to every finite p-quotient of M proves

$$\bigcap_m R_{m,p} = R_{\infty,p}. \tag{D.47.16}$$

For $m \mid n$, the restriction to L_n of the pro-p quotient of L_m gives $R_{n,p} \subset R_{m,p}$. The inverse system is therefore an actual system of quotients of $\widehat{B}$. Compactness and (D.47.16) prove the first line of (D.47.3), not just one of its projections.

The maximal common source and all projection kernels. The continuous image of the compact group M in a product of finitely many finite p-group quotients at distinct primes is the full product. Goursat gives this inductively from coprime orders. Compactness over all levels proves

$$M/R_\infty \simeq \prod_p M^{(p)}. \tag{D.47.17}$$

For two distinct primes, the same argument gives $R_{\infty,p}R_{\infty,q} = M$. For one prime and all remaining primes it gives the analogous product-kernel equality. These are exactly the two maximal-common quotient assertions over $\mathcal{H}_\infty$. The marked fibre product in (D.47.3) follows from the same original source map and (D.47.17).

The D.46 group $\mathcal{U}$ retains all of $\widehat{F}^{\mathrm{ab}}$, so its source kernel lies in M, and it maps onto $\mathcal{H}_\infty$. The image of M in $N^{(p)}$ is exactly C_p. To see both directions, its quotient is the maximal pro-p quotient of $\widehat{N}/M = \Lambda$, which is Λ_p. The compact image of M is closed, so this identifies its kernel without a density assumption. The induced $M^{(p)} \to C_p$ is onto. The same restriction $M \to N^{(p)}$ also shows $R_{\infty,p} \subset R_p$; hence $R_\infty \subset R_{\mathrm{nil}}$ and gives the actual source map $\mathcal{V} \to \mathcal{U}$. Applying the distinct-prime product argument again identifies the image of M in $\mathcal{U}$ with $\prod_p C_p$. The source map $\mathcal{V} \to \mathcal{U}$ is the identity on $\mathcal{H}_\infty$ and has kernel $\prod_p D_p$. This proves every map and kernel in (D.47.12)–(D.47.13).

One coherent pair of normalized conjugators. For the actual h of D.46, the normalized action on Q is the identity. Its full multiplier is one and $f_h \in \widehat{F}'$, so the exact equations $\alpha_h(s) = sf_h$, $\alpha_h(r) = f_h^{-1}r$ make its action on $\widehat{B}/\widehat{F}'$ the identity as well. Both coordinates in (D.47.4) are fixed; hence its action on $\mathcal{H}_\infty$ is exactly the identity. It preserves M and every characteristic pro-p residual, giving the arithmetic factor in (D.47.7). The product action is thus an actual continuous automorphism of (D.47.3).

Let A_∞, β_∞ be the images in $\mathcal{H}_\infty$ of the normalized Galois conjugators of r, s. They centralize the respective elements. There is even an exact finite identification of A_∞. On the loop abelianization, the matrix of r has $\det(1 - r) = 3$; the program computes that matrix

from the actual sphere graph. The operator $1 - r$ is injective on $\widehat{\mathbb{Z}}^2$. Since A_∞ centralizes r, its abelian coordinate is zero. It therefore lies in $q(\widehat{F}') = Q_8$. The certificate computes

$$C_{Q_8}(r) = \{I, 5I\}.$$

The arithmetic sign $T_{h,4} = -I$ selects $A_\infty = 5I$ under the exact kernel identification in (D.47.4). In particular this is a nontrivial central element of the marked base.

Apply the pro-two coprime order-three coset lifting proof of D.46 to the map $\mathcal{P}_{\infty,2} \to \mathcal{H}_\infty$. There is one element centralizing r above A_∞^{-1}. Right multiplication of the arithmetic r-conjugator by it leaves the automorphism unchanged and makes its entire $\mathcal{H}_\infty$ coordinate one. At every odd prime, the same coprime order-two proof gives an actual s-centralizing element above β_∞. These are lifts of the whole profinite base elements, not separate incompatible finite choices. The proof by finite complements and compactness works over this infinite base because its kernel is pro-p.

The resulting tuples of conjugators belong to the actual fibre product (D.47.3). They give (D.47.7), fix t, and give exactly (D.47.8). The parameter has trivial base image, so every source preimage belongs to $M \subset \widehat{F}'$. The two hexagons and generation follow as in D.46. Their projections to $\mathcal{U}$ give the same automorphism and parameter as D.46, even when a different centralizer representative was chosen there. The invariant G_2/G_3 framing map therefore retains $T_4 = I$, $W_3 = [i]$. This constructs one simultaneous lift at every prime and every Fermat level.

Every homology quotient and the full metabelian action. An element killed by every pro-p quotient of M is killed by every finite abelian quotient of M: a finite abelian group is a product of its prime-primary factors. It therefore lies in the closed commutator M'. Since $M \subset \widehat{F}'$, we obtain the exact inclusions (D.47.9). Also $M \subset L_m \subset F_m$, so

$$R_\infty \subset M' \subset L'_m \subset F'_m. \tag{D.47.18}$$

This proves every claimed quotient map from the source.

For action descent, the quotient $\mathcal{H}_\infty$ retains the entire loop abelianization, and (D.47.7) fixes that base. Hence the images of F_m, L_m are invariant; so are their closed commutators. Their homology quotients are actual automorphism quotients. For a specified coefficient prime p, the restriction of each such quotient map to M has abelian pro-p image after retaining the relevant finite deck base. It thus kills $R_{\infty,p}$. The action on that coefficient component is exactly the h action at two and identity at every odd prime. The universal property and the finite-level maps in (D.47.3) make these statements compatible for all deck and coefficient levels.

The subgroup $\widehat{F}$ is the preimage of the fixed S_3 quotient. Its image in $\mathcal{V}$ is invariant. Its first and second closed commutators are therefore invariant as well, so (D.47.9) gives action descent to the full metabelian quotient. The map from M to each A_p is pro-p; hence $\mathcal{B}_{\mathrm{met},p}$ is a quotient of $\mathcal{P}_{\infty,p}$. Decomposing the full abelian kernel $\widehat{F}'/\widehat{F}''$ into its prime factors proves the first line of (D.47.10) by its actual source map. The other lines now follow from (D.47.7)–(D.47.8). They give the complete metabelian action, including all prime-to-p deck coordinates, not merely the pure pro-p quotient of F.

For the earlier D.39 underlying source, (D.47.18) at $m = 12$ gives its full L_{12}-homology factor. The map to the actual relative pro-two source gives its T_3 factor. Both are equivariant under the replacement action; their joint source image is the same underlying $\mathcal{T}$ by D.39.2. The two-primary homology and T_3 components are now induced by the actual h; the odd homology components are identity. This specifies the replacement under the exact maps without identifying it with the excluded earlier action.

The actual CM information and nonarithmetical comparison. For the cyclic three and four subgroups K_3, K_4 of D.36, the containment $\widehat{F}' \subset \widehat{K}_n$ gives $\widehat{F}'' \subset \widehat{K}'_n$. Restriction to those

cover homology groups and filling their actual cusp lattices therefore factor through the full metabelian source. Their compact two- and three-adic modules factor through $\mathcal{P}_{\infty,2}$ and $\mathcal{P}_{\infty,3}$, respectively, by the just-proved coefficient-prime argument. The exact maps, cubic twist and quartic tangent convention are those of D.34 and D.36. They give (D.47.11). In particular the metabelian action is nonidentity.

A Galois element inducing this metabelian action would have full cyclotomic character one, from its action on the retained loop abelianization. It would lie over $L = \mathbb{Q}(\zeta_{12})$. The cubic matrix modulo four has trivial order-three factor and normalized sign -1. D.36's exact arithmetic graph then forces projective quartic three-torsion action $[i]$. The actual quartic coordinate in (D.47.11) is I. This contradiction proves that the metabelian action is not in the Galois image. It proves the comparison through specific cover maps and the shared torsion field, and does not impose an unproved full-GT0 cover-to-fixed-fibre identity.

The outer comparison and normalization rigidity. Both the displayed action and a normalized Galois action with multiplier one fix x and send y to $f^{-1}yf$, with $f \in \widehat{F}'$. They are therefore identity on the free nilpotent quotient of class two. An inner conjugator between two such actions acts trivially on that quotient. Its two abelian coordinates vanish: in each actual Heisenberg quotient $U_3(\mathbb{Z}/n)$, commuting with the images of both x and y forces both coordinates to be zero modulo n. Thus the conjugator belongs to $A = \widehat{F}'/\widehat{F}'' \simeq R_A c$, where $R_A = \widehat{\mathbb{Z}}[[\widehat{\mathbb{Z}}^2]]$, by B.8. Fixing x gives $(X-1)a = 0$ in this module.

Here multiplication by $X-1$ is injective in the full profinite completed group ring, including every prime-to-coefficient deck component. For a direct proof, project to coefficients $\mathbb{Z}/m$ and deck group $(\mathbb{Z}/n)^2$, and lift to deck level mn. An X-invariant element at that higher level is a sum of the norm $1 + X + \cdots + X^{mn-1}$ times elements in the other deck coordinate. Its pushforward to level n is multiplied by m, so is zero modulo m. This holds for every m, n, proving injectivity. The inner conjugator is consequently identity. The normalized multiplier-one actions embed in the outer action space. Since an outer Galois comparison already has multiplier one on the retained abelianization, it would give the based comparison excluded above. This proves the claimed nonarithmetical outer class as well.

Finally $\mathcal{H}_\infty$ is prosolvable, by (D.47.4), and the kernel in (D.47.17) is pronilpotent. Every finite quotient of $\mathcal{V}$ is solvable. Under the actual S_7 source of D.46, the quotient by the image of R_∞ is consequently solvable. That image contains A_7; it is contained in A_7 because $R_\infty \subset \widehat{N} \subset B_{\text{even}}$. It is exactly A_7, completing the remaining source-kernel comparison. □

A central double cover and an actual mixed-prime extension

D.48. The all-Fermat replacement extends over a new mixed three-primary kernel [Proved]

Keep the marked groups and the action of D.47. Every subgroup and commutator below is closed. Put

$$\begin{aligned} L &= [\widehat{N}, \widehat{B}], & \mathcal{E} &= \widehat{B}/L, \\ M^+ &= L \cap \widehat{F}' = L \cap M, & \mathcal{H}_\infty^+ &= \widehat{B}/M^+, \\ L_m^+ &= L \cap F_m, & \mathcal{H}_m^+ &= \widehat{B}/L_m^+. \end{aligned} \tag{D.48.1}$$

Then $\mathcal{E} \to Q$ is an actual central extension with kernel C_2, $|\mathcal{E}| = 1152$, and the discrete subgroup whose completion is L is free of rank 193. The image of M in this central kernel is all of C_2. The image of L in the original loop abelianization is exactly the lattice Λ of D.47.5. In particular

$$\begin{aligned} [M : M^+] = 2, \qquad & \mathcal{H}_\infty^+ = (\widehat{B}/\widehat{F}') \times_{Q/J} \mathcal{E}, \qquad J = q(\widehat{F}') \simeq Q_8, \\ \ker(\mathcal{H}_\infty^+ \to \mathcal{H}_\infty) = C_2, \qquad & |\mathcal{H}_m^+| = 1152m^2/c_m, \qquad \operatorname{rank}(L_m^+) = 1 + 192m^2/c_m, \end{aligned} \tag{D.48.2}$$

where $c_m = \gcd(2,m)^2 \gcd(3,m)$. Every displayed map is induced by the same original $\widehat{B}$.

Define the new relative sources by

$$\begin{aligned} R_p^+ &= \ker(M^+ \to (M^+)^{(p)}), \quad \mathcal{P}_p^+ = \widehat{B}/R_p^+, \\ R^+ &= \bigcap_p R_p^+, \qquad\qquad\qquad \mathcal{V}^+ = \widehat{B}/R^+. \end{aligned} \tag{D.48.3}$$

They have exact marked descriptions

$$\begin{aligned} \mathcal{P}_p^+ &= \varprojlim_m \widehat{B}/\ker(L_m^+ \to (L_m^+)^{(p)}), \qquad \mathcal{V}^+ = \prod_{\mathcal{H}_\infty^+} \mathcal{P}_p^+, \\ \ker(\mathcal{V}^+ \to \mathcal{H}_\infty^+) &= \prod_p (M^+)^{(p)}, \qquad R_2^+ = R_{\infty,2}, \quad \mathcal{P}_2^+ = \mathcal{P}_{\infty,2}. \end{aligned} \tag{D.48.4}$$

The base $\mathcal{H}_\infty^+$ is the exact maximal common marked quotient of different prime factors and of one factor with all the remaining ones. For odd p, the actual inclusion $M^+ \subset M$ gives an epimorphism

$$(M^+)^{(p)} \twoheadrightarrow M^{(p)}, \qquad E_p = \ker((M^+)^{(p)} \to M^{(p)}).$$

The comparisons with D.47, including both source kernels, are

$$\begin{aligned} R^+ \subset R_\infty, \qquad & 1 \longrightarrow \prod_{p\neq 2} E_p \longrightarrow \mathcal{V}^+ \longrightarrow \mathcal{V} \longrightarrow 1, \\ \ker(\mathcal{V} \to \mathcal{H}_\infty^+) = (M^+)^{(2)} \times \prod_{p\neq 2} M^{(p)}, \qquad & \ker(\mathcal{V}^+ \to \mathcal{P}_p^+) = \prod_{q\neq p} (M^+)^{(q)}. \end{aligned} \tag{D.48.5}$$

In the second line $(M^+)^{(2)}$ is identified with the actual index-two subgroup $\ker(M^{(2)} \to C_2)$.

The actual arithmetic element h of D.46 fixes $\mathcal{E}$, and hence $\mathcal{H}_\infty^+$, pointwise. Consequently

$$\alpha^+ = (\alpha_{h,\infty,2}, (1_{\mathcal{P}_p^+})_{p\neq 2}) \in \mathrm{Aut}(\mathcal{V}^+) \tag{D.48.6}$$

is a continuous peripheral automorphism with multiplier one. It projects to the entire D.47 action and to its exact parameter. One compatible pair of loop conjugators is constructed below; the required correction to the two-primary r-conjugator centralizes r and has trivial Q-image. Thus all the retained finite CM data, every Fermat homology action, and the nonarithmetical full metabelian outer class of D.47 are preserved.

This extension adds a concrete finite source which does not factor through $\mathcal{V}$. Namely let

$$\mathcal{J}_3 = \widehat{B}/(L'L^3), \qquad V_3 = L/(L'L^3) \simeq (\mathbb{F}_3)^{193}, \qquad G_3 = \widehat{B}/(\widehat{N}'\widehat{N}^3). \tag{D.48.7}$$

Here L^3 is the closed subgroup generated by cubes. The central involution $z \in \mathcal{E}$ acts on V_3, with

$$\dim V_3^+ = 97, \qquad \dim V_3^- = 96. \tag{D.48.8}$$

There are exact source maps and kernel identifications

$$\begin{aligned} \mathcal{P}_3^+ \twoheadrightarrow \mathcal{J}_3, \qquad & \ker(\mathcal{J}_3 \to \mathcal{E} \times_Q G_3) = V_3^-, \\ \mathrm{im}(R_\infty \to \mathcal{J}_3) = V_3^- \simeq C_3^{96}, \qquad & \mathrm{im}(R^+ \to \mathcal{J}_3) = 1. \end{aligned} \tag{D.48.9}$$

Thus the maximal common marked quotient of $\mathcal{V}$ and $\mathcal{J}_3$ is exactly $\mathcal{E} \times_Q G_3$, of order $1152 \cdot 3^{97}$, whereas $|\mathcal{J}_3| = 1152 \cdot 3^{193}$. The new action on $\mathcal{J}_3$ is identity, and it agrees with the D.47 action on this complete common quotient. The actual image of M in $\mathcal{J}_3$ has structure

$$((\mathbb{F}_3)^{95} \oplus (\mathbb{F}_3)^{96}) \rtimes C_2, \tag{D.48.10}$$

where the involution is identity on the first summand and minus identity on the second. This gives the precise mixed prime component killed in D.47 and retained here.

The remaining source kernel R^+ still has exact image A_7 under D.46's displayed S_7 source map. Full profinite GT0 realization of the prescribed CM pair and of the metabelian action remains undecided. The new action is proved on the stated larger source; its finite projections are not substituted for a full-source lift.

The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus and Fermat follow-ups in section 51.15, folios 725–727, scans 156–160.

Proof. The actual central extension and the lattice map. The image of $[\widehat{N}, \widehat{B}]$ in $\widehat{N}^{\mathrm{ab}}$ is the closed augmentation image generated by $(1-s)\widehat{N}^{\mathrm{ab}}$ and $(1-r)\widehat{N}^{\mathrm{ab}}$. The columns $(1-s)b+s(1-r)a$ of D.26 generate the same subgroup: $sv \equiv v$ modulo the first summand. D.26's exact integral Smith form $(1^{96}, 2)$ therefore identifies $\widehat{N}/L = C_2$. This proves the central extension and its order. Its finite-index torsion-free kernel has rank $1 + 1152/6 = 193$, by the actual Bass–Serre graph.

The program `d_cm_central_mixed.py` reconstructs this same quotient. It first forms the oriented graph of $N\backslash T$, its 97 integral fundamental cycles, and the unique nonzero modulo-two coinvariant functional. Let $P(q)$ be the path attached to the original word for $q \in Q$, and extend that functional to the edge module by projection to non-tree edges. The actual central cocycle is

$$\delta(q, q') = \phi(P(q) + qP(q') - P(qq')) \in \mathbb{F}_2. \tag{D.48.11}$$

The marked elements (q, a) multiply by this cocycle. The certificate retains their full right s, r permutations, checks the two torsion relations at all 1152 states, and generates every state. In particular t^{12} maps to the central generator. It also retains an explicit 88-letter word in F' mapping to that generator; its Q-image and both integral loop exponents vanish. Thus the word belongs to M and proves directly that $M \to C_2$ is onto.

There is a second integral check of the same assertion. On the lattice basis $(2, -2), (0, 6)$, the actual r-action is

$$\begin{pmatrix} 1 & -3 \\ 1 & -2 \end{pmatrix}. \tag{D.48.12}$$

Its reduction modulo two has no invariant nonzero covector. A $\widehat{B}$-invariant character $\widehat{N} \to C_2$ vanishing on M would factor through such a covector of Λ, which is impossible. Since $M \to C_2$ is onto, multiplying a preimage of any $\lambda \in \Lambda$ by an appropriate element of M makes it lie in L. Hence $L \to \Lambda$ is onto.

The code computes $|q_{\mathcal{E}}(F)| = 192$ and $|q_{\mathcal{E}}(F')| = 16$ using actual words. The kernel of this last group over J is precisely the central C_2, so $\mathcal{E}/q_{\mathcal{E}}(F') = Q/J$. The two original quotient kernels now give the fibre product in (D.48.2) by closed Goursat, with intersection M^+. Because $M \subset L_m$ already surjects onto this C_2, every L_m^+ has index two in L_m. D.47's index and graph formulas give all finite orders and ranks.

Both relative-prime kernel comparisons. The subgroup M^+ has index two in M. The induced pro-two topology is its full pro-two topology: a normal finite two-power quotient of M^+ is replaced by the intersection of its two M-conjugate kernels; the resulting quotient of M is still a finite two-group. Restriction gives the reverse inclusion of residual kernels. Thus $R_2^+ = R_{\infty,2}$, and $(M^+)^{(2)} = \ker(M^{(2)} \to C_2)$, with the stated maps.

For odd p, the image of M^+ in $M^{(p)}$ is closed and normal, and its quotient is both a pro-p group and a quotient of C_2. That quotient is trivial. Consequently $(M^+)^{(p)} \to M^{(p)}$ is onto and $R_p^+ \subset R_{\infty,p}$. This proves $R^+ \subset R_\infty$ and the actual source map $\mathcal{V}^+ \to \mathcal{V}$.

For the inverse limit in (D.48.4), restriction immediately gives $R_p^+ \subset \ker(L_m^+ \to (L_m^+)^{(p)})$. For the other inclusion, take the kernel C of a finite p-quotient of M^+. Uniform continuity of conjugation on the compact group M^+ makes its $\widehat{B}$-orbit finite. Intersecting that orbit replaces C by a $\widehat{B}$-normal open subgroup with finite p-group quotient. Choose an open normal subgroup O of $\widehat{B}$ with $O \cap M^+ = C$, by separating the finite subgroup M^+/C in $\widehat{B}/C$. The subgroups L_m^+/M^+ form a decreasing divisibility system with intersection one in $\mathcal{H}_\infty^+$. Compactness supplies m with $L_m^+ \subset OM^+$. Its image in $\widehat{B}/O$ is a p-group, so its pro-p residual lies in O. The intersection of these residual kernels already lies in M^+. Intersection over m and then over all such C gives the reverse inclusion. This proves the equality of source kernels before passage to the inverse limit.

Distinct prime quotients of M^+ have no common nontrivial finite quotient. Finite Goursat and compactness therefore give its full product of maximal pro-prime images. They prove the fibre product and the maximal-common-quotient statements in (D.48.4). Under the D.47 map $M \to \prod_p M^{(p)}$, the image of M^+ is $(M^+)^{(2)} \times \prod_{p\neq 2} M^{(p)}$: its sole additional defining condition is the displayed two-primary central character. Applying the same source map to the new product proves exactly (D.48.5), including the product of the odd-prime kernels E_p.

Pointwise base invariance and compatible conjugators. The actual h fixes Q, preserves $\widehat{N}$ and hence preserves L. In $\mathcal{E}$ its images of s, r differ from s, r by central elements z_s, z_r. The relations imply $z_s^2 = z_r^3 = 1$. The exact fixed-cusp equality $\alpha_h(sr) = sr$ gives $z_s z_r = 1$. An element of order dividing both two and three is identity, so both deviations vanish. Thus h fixes $\mathcal{E}$ pointwise. D.47 already proves it fixes $\widehat{B}/\widehat{F}'$; the exact fibre product in (D.48.2) proves pointwise invariance of $\mathcal{H}_\infty^+$. Formula (D.48.6) is consequently an automorphism of the actual product, preserving the cusp and the two torsion conjugacy classes.

Here is the precise conjugator correction, including its projection to the already retained action. Write $A^\dagger, \beta^\dagger$ for a pair constructed in D.47, so their base coordinates there are respectively $1, \beta_\infty$. The two-primary source is unchanged. Let $a \in C_2$ be the image of its $A_2^\dagger$ in the new base. The map $M^{(2)} \to C_2$ has pro-two kernel, and r has order three. Coprime fixed-coset lifting supplies $d \in C_{M^{(2)}}(r)$ above a^{-1}. Replace $A_2^\dagger$ by $A_2^\dagger d$, and take the identity r-conjugator at every odd prime. These have the same new base coordinate, namely one. The change centralizes r, and $q(d) = 1$, so it changes neither the action nor its finite CM coordinate.

Retain $\beta_2^\dagger$ and denote its image in $\mathcal{H}_\infty^+$ by β_+. It centralizes s, because the actual h is identity on this base. At each odd p, coprime order-two fixed-coset lifting through the pro-p kernel supplies a centralizing lift of β_+ in $\mathcal{P}_p^+$. Its image in $\mathcal{P}_{\infty,p}$ and the already chosen $\beta_p^\dagger$ differ by an element of $C_{M^{(p)}}(s)$. The surjection $(M^+)^{(p)} \to M^{(p)}$ has pro-p kernel, so the same coprime fixed-element argument lifts this difference inside $C_{(M^+)^{(p)}}(s)$. Multiplication by that lift makes the projection exactly $\beta_p^\dagger$. These corrections are made once in the complete groups; finite complements and compactness give the stated lifts, as in D.46–D.47.

The resulting tuples belong to the actual fibre product. The whole β-conjugator projects exactly to D.47's $\beta^\dagger$, whereas the A-conjugator projects to $A^\dagger$ times the explicit r-centralizing tuple $(d, 1, 1, \ldots)$. Thus the automorphism and the parameter $f = s\beta s\beta^{-1}$ project exactly to D.47. The new parameter is again $(f_{h,2}, 1, 1, \ldots)$, has trivial $\mathcal{H}_\infty^+$-image, and has an actual source preimage in $M^+ \subset \widehat{F}'$. The two hexagons and generation follow from the actual automorphism, with multiplier one and no further cusp gauge. This proves the compatibility assertions in the statement, rather than merely an unrelated lift of a finite CM matrix pair.

The exact mixed quotient and the newly retained kernel. The quotient $\mathcal{J}_3$ is the actual Bass–Serre homology extension over $\mathcal{E}$, with kernel V_3. Conjugation by a lift of z is independent of that lift on this abelian kernel and has order two. The program builds the full 1152-edge graph, its 193 fundamental cycles and the permutation of edges that toggles the central

coordinate. It verifies the action on every full edge chain, its square, and the two eigenspace ranks in (D.48.8). It also computes the actual graph pushforward $V_3 \to \Lambda/3\Lambda$ and verifies its rank two and its invariance under z.

The map $L \subset \widehat{N}$ induces

$$(V_3)_{C_2} \xrightarrow{\sim} \widehat{N}^{\mathrm{ab}}/3. \tag{D.48.13}$$

For an explicit proof, the two-group quotient in $1 \to L \to \widehat{N} \to C_2 \to 1$ has vanishing positive homology with $\mathbb{F}_3$ coefficients: averaging over it contracts the positive-dimensional resolution. The homology five-term sequence therefore gives the displayed isomorphism with its actual inclusion map. Equivalently the full relative pro-three quotient of $\widehat{N}$ kills the order-two deck action. Because two is invertible in $\mathbb{F}_3$, the coinvariant kernel is exactly V_3^-.

The two source maps from $\mathcal{J}_3$ to $\mathcal{E}$ and G_3 have exact common quotient Q: their kernels over Q have orders two and 3^{97}, respectively. Their joint image is the full fibre product by finite Goursat. Equation (D.48.13) therefore proves its kernel is V_3^-, and proves the two displayed finite orders. This quotient already factors through $\mathcal{V}$. Indeed $\mathcal{E}$ is a quotient of its two-primary factor because $M \to C_2$ is two-primary, and G_3 is a quotient of its three-primary factor. Their actions are both identity: h fixes $\mathcal{E}$, and the three-primary factor has identity action. Hence

$$\mathrm{im}(R_\infty \to \mathcal{J}_3) \subset V_3^-. \tag{D.48.14}$$

For the reverse inclusion, the image of M in $\mathcal{J}_3$ contains an element over z, and its commutators with the image of L give all of $(z-1)V_3 = V_3^-$. This finite image has a normal three-subgroup of index two. A complement exists by the elementary coprime complement theorem, and its involution acts by minus identity on V_3^-. Every nilpotent quotient must kill this subspace: its two- and three-primary subgroups commute, while $z - 1 = -2$ is invertible on the subspace. Conversely, after killing it the image is abelian. Thus its exact nilpotent residual is V_3^-.

The image of the full pronilpotent residual of a compact group in a finite quotient is that finite quotient's nilpotent residual. To verify both inclusions, every nilpotent quotient of the finite image factors through the original maximal pronilpotent quotient, while every finite quotient of that pronilpotent group is nilpotent. Apply this to $M \to \mathrm{im}(M \to \mathcal{J}_3)$. It proves equality in (D.48.14). For the finer structure (D.48.10), note that $L \to \Lambda$ is onto and the image of $L'L^3$ there is exactly 3Λ. The exact maximal common quotient of $\mathrm{im}(\widehat{N} \to \mathcal{J}_3)$ and Λ is consequently $\Lambda/3\Lambda$. Its kernel is the actual image of M, by closed Goursat. The rank-two invariant map removes two dimensions from V_3^+, leaving the dimensions 95 and 96 stated there.

Finally $M^+ \subset L$, so its image in $\mathcal{J}_3$ is an abelian three-group. The actual source map therefore kills R_3^+, giving $\mathcal{P}_3^+ \to \mathcal{J}_3$ and the second kernel image in (D.48.9). Its action is identity by (D.48.6). We have thus extended the retained action across the entire indicated C_3^{96} kernel, including the complete common quotient and the normalized conjugator correction.

The base $\mathcal{H}_\infty^+$ is prosolvable and the new kernel over it is pronilpotent. Thus $\mathcal{V}^+$ is prosolvable. Under the literal S_7 source of D.46 its full source kernel contains A_7, and it is contained in A_7 because $R^+ \subset \widehat{N} \subset B_{\mathrm{even}}$. Its image is exactly A_7. This proves the final residual assertion without a full-source lifting claim. □

The maximal fixed base of the chosen arithmetic parameter

D.49. A simultaneous fixed-base lift, the entire congruence action, and a remaining mixed kernel [Proved]

Keep the actual arithmetic element h, its normalized parameter f_h, and the complete D.48

action. All subgroups and commutators below are closed. Let C_h be the closed normal closure of f_h in $\widehat{B}$, and define

$$\begin{aligned} C_h &= \overline{\langle f_h \rangle}^{\widehat{B}}, \qquad \mathcal{H}_h = \widehat{B}/C_h, \\ \mathfrak{R}_{h,p} &= \ker(C_h \to C_h^{(p)}), \qquad \mathfrak{R}_h = \bigcap_p \mathfrak{R}_{h,p}. \end{aligned} \tag{D.49.1}$$

Then $C_h \subset M^+$. The marked quotient $\mathcal{H}_h$ is exactly the maximal profinite quotient on which this fixed α_h acts pointwise as identity. Put

$$\mathcal{P}_{h,p} = \widehat{B}/\mathfrak{R}_{h,p}, \qquad \mathcal{W}_h = \widehat{B}/\mathfrak{R}_h. \tag{D.49.2}$$

There are actual marked identifications

$$\begin{aligned} \mathcal{P}_{h,p} &= \varprojlim_K \widehat{B}/\ker(K \to K^{(p)}), \qquad \mathcal{W}_h = \prod_{\mathcal{H}_h} \mathcal{P}_{h,p}, \\ \ker(\mathcal{W}_h \to \mathcal{H}_h) &= \prod_p C_h^{(p)}, \qquad \ker(\mathcal{W}_h \to \mathcal{P}_{h,p}) = \prod_{q \neq p} C_h^{(q)}. \end{aligned} \tag{D.49.3}$$

Here K runs through all open normal subgroups of $\widehat{B}$ with $C_h \subset K \subset L$, where $L = [\widehat{N}, \widehat{B}]$ is D.48's actual subgroup. Equivalently, these are all finite h-fixed marked bases refining $\mathcal{E}$. This is one compatible inverse system, with both source-kernel inclusions proved below. The common marked quotient of distinct prime factors is exactly $\mathcal{H}_h$.

On this source there is a continuous peripheral automorphism

$$\begin{aligned} \alpha_h^* &= (\alpha_{h,h,2}, (1_{\mathcal{P}_{h,p}})_{p \neq 2}), \\ f_h^* &= (f_{h,h,2}, (1)_{p \neq 2}) \in \mathrm{im}(C_h \to \mathcal{W}_h). \end{aligned} \tag{D.49.4}$$

The double subscript in $\alpha_{h,h,2}$ means the action of the original arithmetic h on the new source $\mathcal{P}_{h,2}$. The multiplier is one, and one pair of normalized conjugators is constructed on the entire product. The action and parameter project exactly to D.48. In particular the prescribed CM pair, all Fermat homology actions, and the full metabelian nonarithmetical outer class are retained.

The common quotient and projection kernels retain the non-solvable contribution of the new base. Define

$$\begin{aligned} T_h &= M^+/C_h, \\ \mathfrak{r}(T_h) &= \ker(T_h \to \prod_p T_h^{(p)}), \\ \mathcal{D}_h &= \mathcal{H}_h/\mathfrak{r}(T_h) = \widehat{B}/(C_h R^+), \\ D_{h,p} &= \ker((M^+)^{(p)} \to T_h^{(p)}), \\ E_{h,p} &= \ker(C_h^{(p)} \twoheadrightarrow D_{h,p}). \end{aligned} \tag{D.49.5}$$

Then $\mathcal{D}_h$ is the exact maximal common marked quotient of $\mathcal{H}_h$ and $\mathcal{V}^+$, and

$$\begin{aligned} 1 \longrightarrow \prod_p E_{h,p} \longrightarrow \mathcal{W}_h \longrightarrow \mathcal{H}_h \times_{\mathcal{D}_h} \mathcal{V}^+ \longrightarrow 1, \\ 1 \longrightarrow \prod_p E_{h,p} \longrightarrow \ker(\mathcal{W}_h \to \mathcal{V}^+) \longrightarrow \mathfrak{r}(T_h) \longrightarrow 1, \\ \mathfrak{R}_h \subset R^+. \end{aligned} \tag{D.49.6}$$

These are exact sequences with their original source maps; no splitting is asserted. The group $\mathfrak{r}(T_h)$ has an actual A_5 quotient under the marked $\mathrm{PSL}_2(\mathbb{F}_5)$ specialization. Thus the new source is strictly larger than D.48's source.

In fact the full congruence quotient, with its single global sign, factors through the new identity component:

$$\widehat{B} \longrightarrow \mathcal{P}_{h,3} \twoheadrightarrow \mathcal{Q}_{\mathrm{cong}} := \mathrm{SL}_2(\widehat{\mathbb{Z}})/\{\pm I\}. \tag{D.49.7}$$

Consequently α_h^* acts pointwise as identity on $\mathcal{Q}_{\mathrm{cong}}$. For the original arithmetic h, the exact statements used are

$$\kappa_{2,\ell}(h) = 0 \quad (\ell \neq 3), \qquad \kappa_{2,3}(h) \in 3\mathbb{Z}_3. \tag{D.49.8}$$

Full three-adic vanishing for that original h is not asserted. The factorization (D.49.7), with its pro-three kernel calculation below, proves the entire congruence assertion without imposing that additional vanishing.

There is also an explicit remaining finite mixed source. Set

$$\begin{aligned} G = G_3 = \widehat{B}/(\widehat{N}'\widehat{N}^3), \qquad K_3 = \widehat{N}'\widehat{N}^3, \qquad n = |G| = 576 \cdot 3^{97}, \\ \mathcal{J}_{3,2} = \widehat{B}/(K_3'K_3^2), \qquad V_{3,2} = K_3/(K_3'K_3^2) \simeq \mathbb{F}_2^{1+n/6}. \end{aligned} \tag{D.49.9}$$

The actual element $a = f_h \bmod G$ is nonzero of order three in $\widehat{N}/(\widehat{N}'\widehat{N}^3)$. Let $A = \langle a^G \rangle$ be its actual normal closure in that elementary abelian three-group, and define the submodule using the actual deck action:

$$W = [V_{3,2}, A] = \sum_{b \in A} (b-1)V_{3,2}. \tag{D.49.10}$$

The exact common marked quotient of $\mathcal{W}_h, \mathcal{J}_{3,2}$ is $\mathcal{J}_{3,2}/W$, and

$$\begin{aligned} \mathrm{im}(\mathfrak{R}_h \to \mathcal{J}_{3,2}) = W, \qquad \dim_{\mathbb{F}_2} W = \frac{n}{6}\left(1 - \frac{1}{|A|}\right), \\ (a-1)V_{3,2} \subset W, \qquad \dim_{\mathbb{F}_2}(a-1)V_{3,2} = \frac{n}{9} = 64 \cdot 3^{97} > 0. \end{aligned} \tag{D.49.11}$$

The exact formula uses the actual finite subgroup A. Its dimension has not been numerically evaluated; the displayed nonzero cyclic subspace and its dimension are already determined. Thus $\mathfrak{R}_h$ is nontrivial, and the remaining full-source lift is not replaced by a cofinality claim. Genuine full profinite GT0 realization of the CM pair and of the retained metabelian action remains undecided.

Finally the attempted repetition of D.48's central step is completely decided:

$$[L, \widehat{B}] = L, \qquad \widehat{B}/[L, \widehat{B}] \xrightarrow{\sim} \widehat{B}/L. \tag{D.49.12}$$

The exact integral augmentation map on the 193-cycle lattice has Smith form (1^{193}); its Hermite image is the full lattice. This calculation and the maps above continue Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus/Fermat follow-ups in section 51.15, folios 725–727, scans 156–160.

Proof. The exact maximal fixed base. For the original arithmetic element, the normalized identities are $\alpha_h(s) = sf_h$, $\alpha_h(r) = f_h^{-1}r$, and $\alpha_h(t) = t$. A marked quotient map agrees with its composite with α_h exactly when it kills f_h: necessity follows from the first equality, and sufficiency follows from both generator equalities. This proves the maximal property of $\mathcal{H}_h$ and pointwise invariance. It also proves $\alpha_h(C_h) = C_h$, including the reverse inclusion by composing with α_h^{-1}. D.48 proves that h fixes $\mathcal{H}_\infty^+$, so $C_h \subset M^+ \subset \widehat{F}'$.

For the first inverse limit in (D.49.3), restriction $C_h \to K^{(p)}$ proves one residual-kernel inclusion. For the other, start with the kernel of any finite continuous p-quotient of C_h. Uniform continuity on the compact group C_h gives a finite conjugation orbit; its intersection

is a $\widehat{B}$-normal open subgroup C_0 with finite p-group quotient. Choose open normal O in $\widehat{B}$ with $O \cap C_h = C_0$. The open normal subgroups K/C_h with $K \subset L$ are cofinal neighbourhoods of one in $\mathcal{H}_h$. Some such K lies in OC_h; its image modulo O is a p-group. Its residual kernel consequently lies in O. The intersection of all those kernels already lies in C_h, and hence lies in every C_0. This proves the reverse inclusion and the actual inverse-limit equality.

For distinct primes, the maximal pro-prime images of C_h have no common nontrivial finite quotient. Finite Goursat followed by compactness gives their full product as the image of C_h. This proves every product, common-quotient and projection-kernel assertion in (D.49.3), before choosing any automorphism.

One coherent peripheral automorphism and its projection. Because the actual h fixes the entire base $\mathcal{H}_h$, its two-primary action and the odd-primary identity actions define an automorphism of the actual fibre product. Let A_h, β_h be its original r, s-conjugators. Their images in $\mathcal{H}_h$ centralize r, s, respectively. On the two-primary factor, multiply A_h on the right by an r-centralizing lift of its inverse base coordinate. Such a lift exists by the coprime order-three fixed-coset argument through the pro-two kernel. Take the identity r-conjugator on every odd factor. All these base coordinates are now one. For s, retain β_h at two, and on each odd factor choose an s-centralizing lift of the same whole base coordinate. Coprime order-two lifting through the pro-p kernel supplies these elements. The finite complement argument and compactness used in D.46 apply to these entire profinite base elements, so the choices are compatible at every finite level.

The resulting tuples lie in $\mathcal{W}_h$. Their parameter is exactly f_h^* in (D.49.4), with an actual preimage in $C_h \subset \widehat{F}'$. They fix t, give the two hexagons with multiplier one, and generate because the product action is an automorphism. This proves the continuous peripheral construction, not only its finite linear equations.

Restriction $C_h \to (M^+)^{(p)}$ gives $\mathfrak{R}_{h,p} \subset R_p^+$ for every prime. Thus $\mathfrak{R}_h \subset R^+$, with actual maps $\mathcal{P}_{h,p} \to \mathcal{P}_p^+$ and $\mathcal{W}_h \to \mathcal{V}^+$. Each prime component of (D.49.4) maps to the corresponding D.48 component; the action and parameter therefore project exactly. The images of the two new conjugators differ on the right from any already retained D.48 pair by elements of the respective centralizers of r, s. The two actions agree on all of $\mathcal{V}^+$, and D.25's G_2/G_3 rigidity gives trivial Q-image for these differences. This proves retention of the normalized CM coordinates through their actual maps. It also proves all the claimed homology and metabelian comparisons by projection, without replacing the retained arithmetic representative.

The common quotient with D.48, including its full kernel. For a surjection of compact groups $P \to T$, the image of the pronilpotent residual of P is the pronilpotent residual of T. Indeed the quotient by that image is a quotient of the maximal pronilpotent quotient of P, so is pronilpotent; in the other direction the maximal pronilpotent quotient of T is a pronilpotent quotient of P. Compactness makes the image closed. Apply this to $M^+ \to T_h$. The image of R^+ in $\mathcal{H}_h$ is exactly $\mathfrak{r}(T_h)$, giving both descriptions of $\mathcal{D}_h$ in (D.49.5).

The original two quotient kernels C_h and R^+ have product $C_h R^+$. Closed Goursat therefore identifies their full joint source image with $\mathcal{H}_h \times_{\mathcal{D}_h} \mathcal{V}^+$. For each prime the image of C_h in $(M^+)^{(p)}$ is exactly $D_{h,p}$. To prove this, divide by that closed normal image: the resulting group has the universal property of the maximal pro-p quotient of M^+/C_h. The image of C_h in the product of all these prime groups is the full product $\prod_p D_{h,p}$, again by coprime finite Goursat and compactness. Its new product image is $\prod_p C_h^{(p)}$. The kernel of the joint map is consequently $\prod_p E_{h,p}$, proving the first exact sequence in (D.49.6). Projecting the fibre product to $\mathcal{V}^+$ gives kernel $\mathfrak{r}(T_h)$, which proves the second sequence. In particular that kernel is not silently replaced by only the product of the prime kernels.

The full congruence factor and its global sign. Let $\mathcal{Q}^{(3')} = (\prod_{\ell \neq 3} \mathrm{SL}_2(\mathbb{Z}_\ell))/\{\pm I\}$, with one global sign on that product. The original s, r matrices give the actual surjection $\widehat{B} \to \mathcal{Q}_{\mathrm{cong}}$.

At finite level surjectivity follows from elementary row reduction over each local ring $\mathbb{Z}/\ell^k$ and Chinese remainders: upper and lower unipotents generate, and are the actual powers of the marked braid matrices. Passing to the finite inverse system gives the stated map with its single global sign.

For $\tau \in I_3$ and $\ell \neq 3$, every ℓ-power root of unity and every ℓ-power root of the unit two lies in the maximal unramified extension at three. Reduction to the algebraic residue closure followed by Hensel's lemma proves this, since the derivative is prime to three. Hence $\chi_\ell(\tau) = 1$ and $\kappa_{2,\ell}(\tau) = 0$. The actual commutator formula of D.46 gives $\kappa_{2,\ell}(h) = 0$ for every $\ell \neq 3$. At three its exact value is

$$(1 - \chi_3(\tau))\kappa_{2,3}(g) + (\chi_3(g) - 1)\kappa_{2,3}(\tau) \in 3\mathbb{Z}_3, \tag{D.49.13}$$

using the conditions actually imposed in D.46. This proves precisely (D.49.8).

D.18's normalized matrix formula now shows that the original h fixes $\mathcal{Q}^{(3')}$ pointwise. It also fixes $Q_{12} = Q$. Since $\alpha_h(s) = sf_h$, both quotient maps kill f_h, and hence their joint kernel contains the full congruence image of C_h. That joint kernel is exactly

$$\begin{aligned} \ker(\mathcal{Q}_{\mathrm{cong}} \to \mathcal{Q}^{(3')} \times Q_{12}) &\simeq \ker\big(\mathrm{SL}_2(\mathbb{Z}_3) \to \mathrm{SL}_2(\mathbb{F}_3)\big), \\ b &\longmapsto [((I)_{\ell\neq 3}, b)]. \end{aligned} \tag{D.49.14}$$

For both inclusions, an element in the away-from-three kernel has a representative with every away component equal to I, after one global sign. The retained modulo-four coordinate is then I, so identity in Q_{12} forces the modulo-three coordinate also to be I, with positive sign. Conversely every displayed principal-congruence element is in the joint kernel. The map is injective because that away identity representative cannot be changed by a nontrivial global sign. The successive three-adic congruence layers are additive three-groups, so the kernel is pro-three.

Thus the actual full congruence image of C_h is pro-three. Its source map kills $\mathfrak{R}_{h,3}$, proving (D.49.7). The identity action on $\mathcal{P}_{h,3}$ therefore proves the full congruence assertion. The certificate independently checks all 576 Q_{12} states: projection to the independent local projective groups has image of order 288 and kernel $\{I, 5I\}$. Thus the relative-sign factor detected in (D.49.14) is retained, including the finite check that prevents losing it.

The $\mathrm{PSL}_2(\mathbb{F}_5)$ specialization is a quotient of $\mathcal{H}_h$. The program enumerates its 60 matrices and its five Klein-four subgroups. Conjugation gives a faithful even permutation action on those five subgroups, hence the exact marked A_5 map. The actual permutations of s, r, in cycle notation on $\{0, 1, 2, 3, 4\}$, are $(1\,3)(2\,4)$ and $(0\,4\,3)$. Its conjugacy-class sizes $1, 12, 12, 15, 20$ also verify simplicity directly: the only unions containing identity whose orders divide 60 have orders one and 60. The image of M^+ in this simple quotient is all of A_5, because its quotient is a quotient of the prosolvable $\mathcal{H}_\infty^+$. As C_h is killed, T_h maps onto A_5, and its pronilpotent residual does too. This proves the asserted strict enlargement and non-solvable kernel contribution.

An exact nonzero remaining mixed kernel. D.47 proves that the original arithmetic r-conjugator has Q-image $5I$. D.25 proves $q(C_{G_3}(r) \cap \operatorname{im} \widehat{F}) = 1$. Therefore the original h cannot act as identity on G_3. If its actual parameter had trivial image there, its two normalized generator formulas would make that action identity. Hence $a = f_h \bmod G_3$ is nonzero. It lies in the elementary abelian kernel of $G_3 \to Q$, so has exact order three. This establishes the actual source element used in (D.49.10).

The kernel K_3 is a torsion-free free subgroup of index n in B, with rank $1 + n/6$. Its actual modulo-two graph homology gives $\mathcal{J}_{3,2}$ and $V_{3,2}$. Let H be the image of C_h in this finite quotient. It is the normal closure of the actual image of f_h, and projects onto A. Commuting its conjugates with the whole kernel $V_{3,2}$ shows $W \subset H$. The quotient H/W is a

central extension of the elementary abelian three-group A by an elementary abelian two-group. Coprime complements split this extension, so it is abelian.

Conversely averaging over A in characteristic two gives $[W, A] = W$: the augmentation image has zero coinvariant quotient. A three-Sylow complement to $H \cap V_{3,2}$ in H projects isomorphically onto A. In every nilpotent quotient its three-primary image commutes with the two-primary image of W, forcing all of W to vanish. Thus the exact nilpotent residual of H is W. The residual-image argument already proved above, now applied to $C_h \to H$, gives the first equality in (D.49.11). The product of the two actual source kernels then gives the common quotient $\mathcal{J}_{3,2}/W$.

For its dimension, every nonidentity element of A has trivial image in Q, whereas every nonidentity vertex stabilizer in the Bass–Serre graph has a nontrivial order-two or order-three image in Q. Thus A acts freely on all edges and vertices of the graph for K_3. Since its order is odd, the category of $\mathbb{F}_2[A]$-modules is semisimple by averaging. The actual graph chain sequence therefore gives the module decomposition

$$V_{3,2} \simeq \mathbb{F}_2 \oplus \mathbb{F}_2[A]^{\,n/(6|A|)}. \tag{D.49.15}$$

For clarity, this follows by taking the edge permutation module, subtracting the two vertex permutation modules, and adding the constant H_0 module in the split graph sequence: their free-module ranks are respectively $n/|A|$, $n/(2|A|)$, and $n/(3|A|)$. Semisimplicity makes the resulting equality of constituents an actual module isomorphism. The augmentation part has dimension $(|A| - 1)n/(6|A|)$, proving the formula for W. Apply the same actual graph argument to the specific cyclic subgroup $\langle a \rangle$ of order three. Its moving subspace $(a - 1)V_{3,2}$ has dimension $n/9$, and lies in W. This proves the explicit nonzero remaining source-kernel image without evaluating the undetermined dimension of the whole Q-span.

The fully tested central route and its stabilization. The program `d_cm_fixed_parameter.py` reads the actual D.48 marked quotient and rebuilds its integral 193-cycle lattice. It reconstructs the two left generator permutations, verifies that they commute with every right marked generator, and verifies their action on every full integral edge chain. The augmentation matrix has shape 193×386. Its Smith diagonal is (1^{193}), and independently its Hermite image is I_{193}. Thus $L^{\mathrm{ab}}/((1 - s)L^{\mathrm{ab}} + (1 - r)L^{\mathrm{ab}}) = 0$. Because $[L, \widehat{B}]$ contains L', completion gives $[L, \widehat{B}] = L$. This proves the actual quotient isomorphism (D.49.12); a further central repetition cannot produce the new mixed lift constructed above. □

Lifting the remaining moving module through the entire two-adic homology tower

D.50. The actual D.49 action lifts across its moving homology kernel [Proved]

Keep the actual groups, parameter and action of D.49. Thus

$$G = G_3, \quad |G| = n = 576 \cdot 3^{97}, \quad K_3 = \ker(\widehat{B} \to G), \quad a = f_h \bmod G, \quad A = \langle a^G \rangle.$$

The group A is the actual nonzero elementary abelian three-subgroup of G defined there. Set

$$\begin{gathered} U = K_3^{\mathrm{ab},(2)} \simeq \mathbb{Z}_2^{\rho}, \qquad \rho = 1 + n/6, \qquad e_A = |A|^{-1} \sum_{b \in A} b, \\ U^- = (1 - e_A)U, \qquad w = \operatorname{rank}_{\mathbb{Z}_2} U^- = \frac{n}{6}\left(1 - \frac{1}{|A|}\right) \geq \frac{n}{9} > 0. \end{gathered} \tag{D.50.1}$$

The coefficient $|A|^{-1}$ is a unit of $\mathbb{Z}_2$. For $k \geq 1$ define actual marked quotients

$$\begin{gathered} \mathfrak{K}_k = \ker(K_3 \to U/2^k U), \qquad \mathcal{J}_k = \widehat{B}/\mathfrak{K}_k, \qquad W_k = U^-/2^k U^-, \\ \mathfrak{K}_\infty = \ker(K_3 \to U), \qquad \mathcal{J}_\infty = \widehat{B}/\mathfrak{K}_\infty = \varprojlim_k \mathcal{J}_k. \end{gathered} \tag{D.50.2}$$

In particular $\mathcal{J}_1$ is D.49's $\mathcal{J}_{3,2}$, with $W_1 = W$. The exact images of its remaining source kernel are

$$\operatorname{im}(\mathfrak{R}_h \to \mathcal{J}_\infty) = U^-, \qquad \operatorname{im}(\mathfrak{R}_h \to \mathcal{J}_k) = W_k. \tag{D.50.3}$$

Consequently the full common marked quotients with $\mathcal{W}_h$ are $\mathcal{J}_\infty/U^-$ and $\mathcal{J}_k/W_k$, respectively. The D.49 action descends to them and is identity on the retained finite base G.

There is a continuous peripheral automorphism on the strictly larger actual source

$$\mathcal{Z}_h := \widehat{B}/(\mathfrak{R}_h \cap \mathfrak{K}_\infty) = \mathcal{W}_h \times_{\mathcal{J}_\infty/U^-} \mathcal{J}_\infty. \tag{D.50.4}$$

It projects exactly to the complete D.49 action and parameter. Its normalized conjugators are compatible through every level 2^k, its multiplier is one, and it fixes the original cusp $t = sr$. Every previously retained CM, Fermat, full metabelian and full congruence comparison is therefore preserved. The two projection kernels and full source kernel are

$$\begin{gathered} \ker(\mathcal{Z}_h \to \mathcal{W}_h) = U^-, \qquad \ker(\mathcal{Z}_h \to \mathcal{J}_\infty) = \mathfrak{K}_\infty \mathfrak{R}_h/\mathfrak{R}_h, \\ \ker(\widehat{B} \to \mathcal{Z}_h) = \mathfrak{R}_h \cap \mathfrak{K}_\infty. \end{gathered} \tag{D.50.5}$$

The initial lift modulo two is an actual automorphism, with generation proved. A first peripheral endomorphism is obtained from the onto cusp map

$$W_1 \oplus W_1 \longrightarrow W_1, \qquad (b, c) \longmapsto (1 - s)b + s(1 - r)c. \tag{D.50.6}$$

Its possible noninvertibility is corrected by the explicit projective Fitting construction below. After this first correction, every higher step has exact common quotient

$$\begin{gathered} \mathcal{C}_k = \mathcal{J}_k/(2^{k-1}W_k) = \mathcal{J}_{k-1} \times_{\mathcal{J}_{k-1}/W_{k-1}} (\mathcal{J}_k/W_k), \qquad k \geq 2, \\ \ker(\mathcal{J}_k \to \mathcal{C}_k) = 2^{k-1}W_k \simeq W_1. \end{gathered} \tag{D.50.7}$$

This is the maximal common marked quotient of $\mathcal{J}_k$ and the already constructed joint source at level $k - 1$. Cusp corrections in this kernel are onto; the unchanged invertible reduction modulo two proves generation at all higher levels.

The new full source kernel is still nonzero under an explicit next finite map. Put $V = U/2U$, and use the actual class-two, exponent-four quotient

$$\mathcal{T}_2 = \widehat{B}/\overline{\Gamma_3(K_3)(K_3')^2 K_3^4}. \tag{D.50.8}$$

Its kernel over G has abelianization $(\mathbb{Z}/4)^\rho$ and commutator subgroup canonically $\bigwedge^2_{\mathbb{F}_2} V$. In that commutator subgroup,

$$\begin{gathered} W_1 \wedge V \ \subset \ \operatorname{im}(\mathfrak{R}_h \cap \mathfrak{K}_\infty \to \mathcal{T}_2), \\ \dim(W_1 \wedge V) = w(\rho - w) + \binom{w}{2} > 0. \end{gathered} \tag{D.50.9}$$

The inclusion and its source elements are proved by actual commutator maps below. Equality with the entire remaining image, and lifting the retained action through this class-two quotient, are not asserted. Full profinite GT0 realization remains undecided.

The runnable certificate is a separate exact finite verification of the construction. Its literal marked base is $A_5 \times (S_3 \times C_3)$, of order 1080, with a central $A = C_3$. Its kernel ranks are

	full homology	moving part	projective nilpotent part	repaired rank
certificate	181	120	48	181.

(D.50.10)

The initial full kernel map has rank 141. The program verifies its actual replacement, the regular-module retraction, all 181 cycle maps, both torsion relations, the exact cusp identity, and compatible integral lifts through exponents $2, 4, 8, 16$. The actual enormous G_3 application is established by the proof; the certificate does not claim a numerical evaluation of its arithmetic parameter or of $|A|$.

The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus/Fermat follow-ups in section 51.15, folios 725–727, scans 156–160.

Proof. The actual common quotients and their actions. The averaging idempotent e_A is integral over $\mathbb{Z}_2$, is central in the G-action because A is normal, and splits U into its invariant and moving direct summands. Their reductions modulo two are exactly the modules in D.49. The graph calculation there gives the rank in (D.50.1).

Let H be the image of C_h in $\mathcal{J}_\infty$. It projects onto A. Commuting its lifts with the whole abelian kernel U gives $[U, A] = (1 - e_A)U = U^-$, so $U^- \subset H$. The quotient H/U^- is a central extension of the finite odd group A by a pro-two group. Finite coprime complements and compactness split this extension; it is pronilpotent. Conversely $[U^-, A] = U^-$, by averaging. In any pronilpotent quotient of H, the odd image of a complement to its pro-two kernel commutes with the image of U^-, forcing the latter to vanish. Thus the exact pronilpotent residual of H is U^-. D.49's residual-image argument proves (D.50.3); reduction modulo 2^k proves its finite version. The products of the original two source kernels give the asserted maximal common quotients.

The entire G quotient factors through $\mathcal{P}_{h,3}$, since the image of C_h there is the three-group A. It therefore has identity action. The image of K_3 in $\mathcal{W}_h$ is invariant. Taking its maximal abelian pro-two quotient, or its abelian quotient of exponent 2^k, is characteristic. Their actual kernels are the images of $\mathfrak{K}_\infty$ and $\mathfrak{K}_k$. This proves action descent to the common quotients, including their identity action on G.

An actual peripheral endomorphism at the first level. Write $J = \mathcal{J}_1$ and $W = W_1$. Take the images of the already retained normalized conjugators in J/W, and lift these elements to A_J, β_J in J. Their G-images centralize r, s, respectively, since the lower action fixes G. The two conjugates $\beta_J s \beta_J^{-1}$, $A_J r A_J^{-1}$ have the required orders; their product differs from t by an element of W.

Left corrections $b, c \in W$ change that error by exactly (D.50.6). This is an equality in the abelian-kernel group, with no omitted quadratic term. Its image is the augmentation image $(1 - s)W + (1 - r)W$. Since $[W, A] = W$ and $A \subset G = \langle s, r \rangle$, the coinvariants W_G vanish, and this map is onto. Choose a correction killing the error. The resulting map from $\widehat{B}$ fixes the cusp exactly and induces identity on G. It sends K_3 into the abelian exponent-two kernel V, hence kills $K_3' K_3^2$. It is therefore a well-defined peripheral endomorphism $\Phi : J \to J$, whose quotient action on J/W is precisely the retained automorphism. Generation is proved next.

The edge and vertex maps and projective Fitting part. Let $S = \langle s \rangle$, $R = \langle r \rangle$ inside G, and let $\mathbb{F}_2[G]$ act on the left. The actual Bass–Serre graph gives

$$\begin{gathered} 0 \longrightarrow V \longrightarrow C_1 \xrightarrow{\partial} C_0 \longrightarrow \mathbb{F}_2 \longrightarrow 0, \\ C_1 = \mathbb{F}_2[G], \qquad C_0 = \mathbb{F}_2[G/S] \oplus \mathbb{F}_2[G/R], \qquad \partial e_g = e_{gS} + e_{gR}. \end{gathered} \tag{D.50.11}$$

Choose discrete word representatives of the two finite conjugators used for Φ. They define a homomorphism of $C_2 * C_3$ inducing identity on G and the specified map on J. If their G-images are b_0, c_0, its vertex maps are exactly

$$gS \longmapsto g b_0 S, \qquad gR \longmapsto g c_0 R. \tag{D.50.12}$$

They are well-defined bijections: b_0 centralizes s, and c_0 centralizes r. Their inverses use b_0^{-1}, c_0^{-1}. These are G-module automorphisms of both vertex modules.

For an explicit edge map, use the path cocycle $\mathfrak{p}$ with $\mathfrak{p}(s) = 0$, $\mathfrak{p}(r) = e_1 - e_r$. The image of e_1 is the path $\mathfrak{p}(A_J) - \mathfrak{p}(\beta_J) + e_{c_0}$, from $b_0 S$ to $c_0 R$; translate it by g for the image of e_g. This defines a G-linear endomorphism $D : C_1 \to C_1$ commuting with ∂ and the displayed vertex automorphism. Writing ψ for the chosen word homomorphism, the generator formulas give

$$D\mathfrak{p}(g) = \mathfrak{p}(\psi(g)) + (g-1)\mathfrak{p}(\beta_J).$$

For $g \in K_3$ the last term vanishes. Hence the restriction to the actual cycle module V is exactly the induced map $T = \Phi|_V$, including the basepoint transport.

Take the Fitting decomposition of the finite edge module under D: $C_1 = P \oplus C_1^{\mathrm{inv}}$, with D nilpotent on P and invertible on the other summand. Both are G-submodules. If $D^m v = 0$, then the invertible vertex map gives $\partial v = 0$. Consequently $P \subset V$, and it is exactly the nilpotent Fitting part of T. The quotient map of T on V/W is invertible, so $P \subset W$. Most importantly, P is a direct summand of the regular module C_1, and is therefore projective over $\mathbb{F}_2[G]$. Let

$$e : C_1 \longrightarrow P \tag{D.50.13}$$

be the G-linear Fitting projection. Its restriction to V is the exact projection onto the possible noninvertible part of T.

The explicit derivation and the cyclic norm correction. The path function $\mathfrak{p}$ is a crossed homomorphism to C_1: its given values satisfy the order-two and order-three relations, the latter by telescoping. Its restriction to K_3, after abelianization modulo two, is the actual inclusion $V \hookrightarrow C_1$. Thus

$$\delta_0 = e\mathfrak{p} : \widehat{B} \longrightarrow P \tag{D.50.14}$$

is an explicit derivation restricting to $e|_V$. It factors through J, since its restriction to K_3 is additive of exponent two.

This also gives the full cohomological comparison. Projective $\mathbb{F}_2[G]$-modules are summands of finite sums of the regular coinduced module. Its positive cohomology vanishes by the contracting cochain resolution, and hence $H^1(G, P) = H^2(G, P) = 0$. The actual inflation–restriction sequence therefore identifies

$$H^1(\widehat{B}, P) \xrightarrow{\sim} \mathrm{Hom}_G(V, P). \tag{D.50.15}$$

The concrete cocycle (D.50.14) realizes the particular homomorphism needed here; existence is not left as an unspecified cohomology class.

The order of the image of t in $G = G_3$ is $m = 36$, by D.26's exact cusp computation. Since $\Phi(t) = t$, the cycle $c = [t^m] \in V$ is fixed by T, and hence is killed by its nilpotent projection e. The crossed-homomorphism identity gives

$$(1 + t + \cdots + t^{m-1})\,\delta_0(t) = \delta_0(t^m) = e(c) = 0. \tag{D.50.16}$$

Restriction of P to $\langle t \rangle$ is still projective: the regular G-module restricts to a free module for this subgroup. For a free cyclic-group module the kernel of the norm is the image of $1 - t$, directly from its coefficient sums. This exactness persists under direct summands. Consequently there is $b \in P$ with $(1 - t)b = \delta_0(t)$; no division by the even cusp order is used.

Subtract the inner derivation $(1 - g)b$, and write $\delta(g) = \delta_0(g) - (1 - g)b$. Then $\delta(t) = 0$, and, in characteristic two,

$$\delta(s) = (1 - s)b, \qquad \delta(r) = (1 - r)(e(e_1) + b), \qquad \delta|_V = e|_V. \tag{D.50.17}$$

Define $\Phi'(g) = \delta(g)\Phi(g)$ in the abelian-kernel extension. It is a homomorphism because Φ is identity on G, the group controlling this module action. The two displayed values change the

conjugators by left multiplication by b and $e(e_1)+b$, respectively. Thus the two torsion images remain conjugates of the original torsion generators, and the cusp stays exactly fixed. All corrections lie in $P \subset W$, preserving the complete quotient action and conjugators modulo W.

On V the new map is $T+e$. It is T on the invertible Fitting summand and $T+1$ on the nilpotent summand, so is invertible on both. With identity on the finite quotient G, this proves Φ' is an automorphism of J. This completes the initial generation argument.

The exact higher common quotients and compatible lifts. Suppose the constructed level is $k-1$. Its joint source has full kernel $\mathfrak{R}_h \cap \mathfrak{K}_{k-1}$. Because $\mathfrak{K}_k \subset \mathfrak{K}_{k-1}$, the image of that intersection in $\mathcal{J}_k$ is exactly

$$W_k \cap 2^{k-1}(U/2^kU) = 2^{k-1}W_k. \tag{D.50.18}$$

Here equality of images follows from the actual preimage property: an element of $\mathfrak{R}_h$ whose image in $\mathcal{J}_k$ reduces to identity already lies in $\mathfrak{K}_{k-1}$. The last equality uses the integral direct summand $U^- \subset U$. This proves that the maximal common quotient is the first description of $\mathcal{C}_k$. The two original quotient kernels in $\mathcal{J}_k$ have intersection $2^{k-1}W_k$ and sum $W_k + 2^{k-1}(U/2^kU)$. Finite Goursat gives precisely the fibre product in (D.50.7).

The already constructed automorphism on $\mathcal{J}_{k-1}$ and the D.49 action on $\mathcal{J}_k/W_k$ agree on the indicated common quotient. Their conjugators agree there as actual elements, not just conjugacy classes. They therefore give an automorphism and conjugators on $\mathcal{C}_k$. Lift those two elements to $\mathcal{J}_k$. The resulting cusp error is in $2^{k-1}W_k \simeq W_1$; the same onto map (D.50.6) supplies left corrections in this kernel. This preserves both previous conjugators and their full D.49 common-coordinate values.

The corrected map sends K_3 into the abelian kernel of exponent 2^k, so defines an endomorphism of $\mathcal{J}_k$. Its kernel matrix reduces to the already invertible matrix of Φ' modulo two. Its determinant is therefore a unit modulo 2^k, proving it is an automorphism. This is the finite-module form of Nakayama's criterion; no repeated generation assumption enters the induction.

Choose these lifts successively. Their actual conjugators, automorphisms and inverses are compatible, so their limits give an automorphism of $\mathcal{J}_\infty$ and one pair of complete conjugators. Closed Goursat and (D.50.3) identify its joint source with (D.50.4). The two component automorphisms agree on the complete common quotient, so define an automorphism of that actual fibre product.

The chosen conjugators form elements of the fibre product because their reductions agree at every level. Its parameter therefore projects exactly to the D.49 parameter. That source retains the entire loop abelianization, and its full source kernel lies in $\widehat{F}''$. Every source preimage of the new parameter thus belongs to $\widehat{F}'$. This proves the normalization, the two hexagons, and retention of the prescribed CM data through the original maps. The two fibre-product projections give exactly (D.50.5).

The next class-two residual map is nonzero. The class-two exponent-four quotient of the free group K_3 has collected normal form

$$g_1^{a_1}\cdots g_\rho^{a_\rho}\prod_{i<j}[g_i,g_j]^{b_{ij}}, \qquad a_i \in \mathbb{Z}/4, \quad b_{ij} \in \mathbb{F}_2. \tag{D.50.19}$$

The commutator convention is $[u,v]=uvu^{-1}v^{-1}$. These coordinates are independent: multiplication on $(\mathbb{Z}/4)^\rho \oplus \mathbb{F}_2^{\binom{\rho}{2}}$ with the usual bilinear reordering term gives a group realizing all of them. It proves that the actual quotient in (D.50.8) has the stated abelianization and commutator module $\bigwedge^2 V$. The defining subgroup is characteristic in K_3, hence normal in $\widehat{B}$; this is an actual finite source quotient.

Choose a basis u_i of the integral direct summand U^-. By (D.50.3), each has an actual preimage $z_i \in \mathfrak{R}_h$. Its image in G is identity, so $z_i \in K_3$. For a basis vector v_j of U, choose an actual preimage $x_j \in K_3$. The commutator $[z_i, x_j]$ lies in $\mathfrak{R}_h$, by normality, and in $\mathfrak{K}_\infty$, because its abelian pro-two image is zero. Its image in the displayed finite commutator module is exactly $\bar{u}_i \wedge \bar{v}_j$. These elements span $W_1 \wedge V$. As W_1 is a direct summand of V, its dimension is $w(\rho - w) + \binom{w}{2}$, proving (D.50.9) and nontriviality of the remaining source kernel by specified maps and elements.

The separate finite verification. The program `d_cm_projective_repair.py` generates the literal marked product $A_5 \times (S_3 \times C_3)$ from the A_5 permutations of D.49 and the original order-two/order-three permutations and cyclic coordinate in $S_3 \times C_3$. It enumerates all 1080 states and builds the full 181-cycle graph. The central three-deck moving summand has dimension 120 and the exact cusp map has rank 120.

The first fixed-seed test produces a peripheral endomorphism of full kernel rank 141, whose nilpotent Fitting summand has dimension 48. The certificate constructs its regular-module retraction, checks equivariance and idempotence, and uses the actual cyclic norm equation to correct the two conjugators. All 181 cycle images are checked; the new matrix has rank 181 and is nonidentity. The faithful edge-cocycle group model verifies both torsion relations and the exact cusp product before and after the correction.

Finally the program uses the integral idempotent $1 - (1 + z + z^2)/3$, with three inverted modulo sixteen, and lifts the same corrected conjugators through exponents two, four, eight and sixteen. It verifies their unchanged original parity, membership in the actual moving summand, and every torsion and cusp identity. This is a finite verification of the symbolic construction; the actual G_3, h, and A in the statement are the ones proved above. □

A further class-two lift with the complete homology action retained

D.51. The moving class-two kernel lifts and the invariant residual is exact [Proved]

Keep the actual D.49 groups and the chosen complete D.50 automorphism, including its two conjugators. Put

$$\begin{gathered} n = |G_3|, \qquad q = |A|, \qquad m = \frac{n}{6q}, \qquad \rho = 1 + mq, \qquad w = m(q-1), \\ \mathfrak{R}_{50} = \mathfrak{R}_h \cap \mathfrak{K}_\infty, \qquad V = U/2U, \qquad W = U^-/2U^-. \end{gathered} \tag{D.51.1}$$

Here q is the order of the same actual nonzero elementary three-subgroup A; no numerical choice of that arithmetic subgroup is substituted. Let $\mathcal{T}_2$ be D.50's actual class-two, exponent-four quotient, with kernel $\mathcal{D}$ over G_3. Thus

$$\mathcal{D}^{\mathrm{ab}} = (\mathbb{Z}/4)^\rho, \qquad \mathcal{D}' = \bigwedge\nolimits^2_{\mathbb{F}_2} V.$$

The exact image, strengthening D.50's inclusion, is

$$\mathrm{im}(\mathfrak{R}_{50} \to \mathcal{T}_2) = Z := W \wedge V. \tag{D.51.2}$$

Conjugation on $\mathcal{D}'$ factors through G_3. The normal odd subgroup A therefore gives canonical G_3-submodules

$$\begin{gathered} Z_0 = e_A Z = Z^A, \qquad Z_- = (1 - e_A)Z, \qquad Z = Z_0 \oplus Z_-, \\ \dim Z_0 = \frac{m^2(q-1)}{2} > 0, \qquad \dim Z_- = \frac{w\rho}{2} > 0. \end{gathered} \tag{D.51.3}$$

All dimensions here are over $\mathbb{F}_2$.

Define the actual further marked quotient and its source kernel by

$$\mathcal{T}_- = \mathcal{T}_2/Z_0, \qquad \mathfrak{K}_- = \ker(\widehat{B} \to \mathcal{T}_-). \tag{D.51.4}$$

The maximal common marked quotient of the D.50 source $\mathcal{Z}_h$ and $\mathcal{T}_-$ is exactly $\mathcal{T}_2/Z$. The fixed D.50 action and its complete conjugator pair extend to a peripheral automorphism of the strictly larger source

$$\mathcal{Y}_h := \widehat{B}/(\mathfrak{R}_{50} \cap \mathfrak{K}_-) = \mathcal{Z}_h \times_{\mathcal{T}_2/Z} \mathcal{T}_-. \tag{D.51.5}$$

Its multiplier is one and it fixes $t = sr$. It preserves every D.50 homology level and the entire retained D.49 action, parameter, CM pair, full metabelian action and global-sign congruence action. The new projection and full-source kernels are

$$\begin{aligned} \ker(\mathcal{Y}_h \to \mathcal{Z}_h) = Z_-, \qquad & \ker(\mathcal{Y}_h \to \mathcal{T}_-) = \mathfrak{K}_-\mathfrak{R}_{50}/\mathfrak{R}_{50}, \\ \mathfrak{R}_{51} := \ker(\widehat{B} \to \mathcal{Y}_h) = \mathfrak{R}_{50} \cap \mathfrak{K}_-, \qquad & \operatorname{im}(\mathfrak{R}_{51} \to \mathcal{T}_2) = Z_0. \end{aligned} \tag{D.51.6}$$

In particular the next remaining image is exact and nonzero. The orders of the two finite comparison quotients are

$$\begin{aligned} |\mathcal{T}_2/Z| &= n\, 2^{\,2\rho+\binom{m+1}{2}}, \\ |\mathcal{T}_-| &= n\, 2^{\,2\rho+\binom{\rho}{2}-m^2(q-1)/2}. \end{aligned} \tag{D.51.7}$$

The lift uses actual central corrections in Z_-. The induced exterior action preserves Z_0, so the map factors through $\mathcal{T}_-$; its unchanged invertible Frattini action proves generation. These two points are included in the proof.

There are two separate runnable finite calculations. They retain the exact order-1080 D.50 certificate, with $\rho = 181$, $q = 3$, $m = 60$, and $w = 120$. The class-two lift has exact dimensions

$$\dim Z = 14460, \qquad \dim Z_0 = 3600, \qquad \dim Z_- = 10860. \tag{D.51.8}$$

All 181 actual free-generator words and the unchanged homology matrix are checked. The corrected cusp word has 4 159 444 letters in B; direct collection verifies its equality to an element of Z_0, hence identity in $\mathcal{T}_-$. Both exact torsion relations are also checked.

The second calculation exhibits the reason for retaining the invariant residual. For this particular finite D.50 homology action, the ribbon intersection form has rank 146, but its preservation defect has rank 72, with a nonzero entry at $(0, 6)$. Consequently that specified finite homology action has no multiplier-one peripheral lift to the unquotiented class-two source. This is an exclusion for the separate order-1080 certificate. It does not evaluate or exclude the actual arithmetic G_3 action in (D.51.1). Full profinite GT0 realization remains undecided.

The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus/Fermat follow-ups in section 51.15, folios 725–727, scans 156–160.

Proof. The exact class-two residual. Write $D = \mathcal{D}$ in this paragraph. Its abelianization is the free $\mathbb{Z}/4$-module $U/4U$, and the alternating commutator pairing is the stated identification $D' = \bigwedge^2 V$. Let H be the image of C_h in $\mathcal{T}_2$. It is normal, projects onto A, and has a finite two-group kernel. Choose an odd-order complement $\widetilde{A}$ in H. Its conjugation action on D^{ab} and D' is the original A-action: different lifts differ by D, which acts trivially on those two modules.

Put $E = [D, \widetilde{A}]$. It is normal in D and stable under $\widetilde{A}$. For example, the product identity for $[d_1 d_2, a]$ expresses the d_1-conjugate of $[d_2, a]$ as a product of two such commutators. Normality of H gives $E \subset H$. Its abelian image is

$$\operatorname{im}(E \to U/4U) = (1 - e_A)(U/4U) = U^-/4U^-. \tag{D.51.9}$$

Commuting this image with all of D proves $Z = W \wedge V \subset E \cap D'$.

We prove the reverse inclusion by an actual quotient map. Let D_+ be the free class-two exponent-four group on $\rho - w$ generators. Choose a basis of $e_A(U/4U)$. The projection

$$p : U/4U \longrightarrow e_A(U/4U)$$

lifts to a homomorphism $D \to D_+$: choose images of the free generators lifting its basis coordinates. The homomorphisms with this prescribed abelian map form an affine space for $\mathrm{Hom}_{\mathbb{F}_2}(V, D'_+)$. Indeed the images of each generator can differ independently by a central element of exponent two. Precomposition by $\widetilde{A}$ preserves this affine space, because p is A-invariant. Averaging any affine point over the odd group $\widetilde{A}$ gives a fixed point. This produces an actual A-invariant surjection

$$\pi : D \longrightarrow D_+. \tag{D.51.10}$$

Surjectivity follows from its onto Frattini map. It kills E, and on commutators its map is $\bigwedge^2(V \to V^A)$, whose kernel is exactly $W \wedge V$. Hence

$$E \cap D' = Z. \tag{D.51.11}$$

For completeness, the needed coprime commutator identity in this group follows directly from (D.51.9)–(D.51.11). Put $E_1 = [E, \widetilde{A}]$. Its abelian image is still $U^-/4U^-$, by odd averaging. Its own commutators therefore contain $\bigwedge^2 W$. The other summand $W \wedge V^A$ of Z has zero A-invariants, so it equals its commutators with A and lies in E_1. Thus $Z \subset E_1$, and the equality of abelian images gives $E_1 = E$. The quotient H/E is the product of its two-group kernel and its odd complement, since that complement centralizes D/E. It is nilpotent. Every nilpotent quotient of H kills $[E, \widetilde{A}] = E$, because its two- and three-parts commute. Thus E is the exact pronilpotent residual of H. D.49's proved residual-image comparison gives

$$\mathrm{im}(\mathfrak{R}_h \to \mathcal{T}_2) = E. \tag{D.51.12}$$

Now take the actual intersection with $\mathfrak{K}_\infty$. Its image lies in $E \cap D' = Z$. Conversely, take $z \in Z$ and a preimage $b \in \mathfrak{R}_h$. Its image in U lies both in U^- and in $4U$, hence in $4U^-$. By D.50, choose $c \in \mathfrak{R}_h$ whose image in U is one quarter of that image. The element bc^{-4} then lies in $\mathfrak{R}_h \cap \mathfrak{K}_\infty$. It still maps to z, because D has exponent four. This proves (D.51.2) using the actual source preimages.

The invariant and moving dimensions. The actual graph calculation of D.49 gives, as an A-module,

$$V \simeq \mathbb{F}_2 \oplus \mathbb{F}_2[A]^m. \tag{D.51.13}$$

Consequently V^A has dimension $m + 1$, and each nontrivial character of the elementary three-group A has multiplicity m after extension to $\mathbb{F}_4$. The nontrivial characters occur in distinct inverse pairs. In $\bigwedge^2 W$, each inverse pair contributes exactly m^2 invariant dimensions. The summand $W \wedge V^A$ has no invariants. Extension of the coefficient field preserves dimensions, so

$$\begin{aligned} \dim Z_0 &= \frac{q-1}{2}\, m^2, \\ \dim Z_- &= w(m+1) + \binom{w}{2} - \frac{q-1}{2}\, m^2 = \frac{w\rho}{2}. \end{aligned} \tag{D.51.14}$$

The central idempotent e_A commutes with the G_3-action, because A is normal. Thus both subspaces are actual normal subgroups of $\mathcal{T}_2$. They are nonzero since $q \geq 3$ and $m > 0$. The collected order of D is $2^{2\rho + \binom{\rho}{2}}$; subtracting the indicated central dimensions gives (D.51.7).

The exact common action and its central lift. By (D.51.2), the image of $\mathfrak{R}_{50}$ in $\mathcal{T}_-$ is exactly $Z/Z_0 = Z_-$. Products of the original source kernels therefore give the maximal common quotient $\mathcal{T}_2/Z$. Its action is also determined, not presumed. The image of K_3 in $\mathcal{Z}_h$ is invariant under the fixed D.50 action, since the latter is identity on G_3. Its maximal class-two exponent-four quotient is characteristic. The kernel in the original source is precisely the product of $\mathfrak{R}_{50}$ with $\overline{\Gamma_3(K_3)(K_3')^2K_3^4}$. Thus the resulting quotient is $\mathcal{T}_2/Z$, and the actual D.50 action and conjugators descend to it.

Lift these two conjugator elements to $\mathcal{T}_-$. Their torsion conjugates have the required orders; their cusp error is in Z_-. Left multiplication of the conjugators by $b, c \in Z_-$ changes that error by exactly

$$(b, c) \longmapsto (1 - s)b + s(1 - r)c. \tag{D.51.15}$$

This uses the abelian normal kernel Z_-; conjugation on it depends only on G_3. There are no omitted quadratic corrections. The image is the augmentation image for $\langle s, r\rangle = G_3$. Since $(Z_-)_A = 0$, it is all of Z_-. Choose a correction killing the error. It preserves the complete common-quotient conjugators and fixes the cusp exactly.

We check both factorization and generation. The corrected map from $\widehat{B}$ sends K_3 into a class-two exponent-four group, so first factors from $\mathcal{T}_2$. Its induced map T on V is the fixed D.50 map and is G_3-linear, since the base action is identity. Therefore it preserves $W = [V, A]$, and its exterior map preserves $Z^A = Z_0$. The map consequently kills the additional relation Z_0 in the target, and is an endomorphism of $\mathcal{T}_-$. Its action on the Frattini quotient of D/Z_0 is the unchanged invertible T. For a finite two-group an onto Frattini map is onto: a proper image would be contained in a maximal subgroup, whose index is two and which contains the Frattini subgroup. Thus the endomorphism is an automorphism.

The two actual component automorphisms agree on $\mathcal{T}_2/Z$, as do their specified conjugators. The closed fibre product gives exactly (D.51.5), with one conjugator pair on that source. Its parameter projects to the entire D.50 parameter. Since the latter retains the full loop abelianization and has source kernel inside $\widehat{F}''$, the new parameter has an actual $\widehat{F}'$ preimage. This preserves the normalized hexagons and all the earlier comparison maps.

The fibre-product projections give the first two kernel formulas in (D.51.6). Finally an element of $\mathfrak{R}_{50}$ maps into the kernel of $\mathcal{T}_2 \to \mathcal{T}_-$ exactly when its image is in Z_0. Every element of Z_0 has such a preimage by (D.51.2). This proves the final exact image in (D.51.6).

The exact finite moving lift. The program `d_cm_class2_moving.py` reads the frozen D.50 certificate without modifying it. It reconstructs the literal order-1080 base, all 181 free graph generators as words in s, r, r^{-1}, and their collected class-two coordinates. For a closed path, appending a free generator $g_j^{\pm 1}$ changes its $\mathbb{Z}/4$ coordinate and adds the commutators with the higher-index odd coordinates. This is the faithful normal form of D.50, so the word checks take place in the actual quotient.

Here $A = \langle z\rangle$ is central of order three. The program obtains an actual order-three lift by taking the fourth power of a chosen z-word; its cube is checked in the full class-two group. On the moving abelian summand, $(1 - z)^{-1} = 2 - z$ modulo four. The recorded D.50 conjugator coordinates therefore lift to actual commutators with this order-three element. Their abelian coordinates remain exactly the recorded modulo-four reductions of the D.50 modulo-sixteen conjugators.

Their cusp error lies in Z, verified in an adapted A-basis consisting of 61 fixed vectors and 60 nontrivial two-dimensional blocks. The program splits the error by $1 + z + z^2$. On the moving central part the inverse of $1 + z$ is z. Writing the literal z-word in s, r telescopes $(1 - z)u$ into the two columns of (D.51.15), giving explicit central corrections. Each central coordinate is then realized by its actual free-generator commutator word.

The resulting s- and r-conjugates are checked to have exact orders two and three. Direct

collection of all 4 159 444 letters in their cusp word gives exactly the invariant component of the original error. It vanishes in $\mathcal{T}_-$. Every column of the old 181-dimensional Frattini matrix is unchanged and has full rank. Thus the finite verification checks a compatible automorphism, not merely a central equation without generation.

The separate invariant obstruction in the finite certificate.

The program `d_class2_diagnostic.py` independently reconstructs the same marked graph and reads only the frozen D.50 matrix. The associated torsion-free cover has 36 cusps of width 30. Euler characteristic therefore gives genus 73 and compact homology rank 146.

For two integral graph cycles a, b, its oriented ribbon intersection form is

$$\Omega(a,b) = \frac{1}{2}\sum_{g\langle r\rangle}(a_g b_{gr} - b_g a_{gr}). \tag{D.51.16}$$

This formula follows by averaging the two small transverse push-offs of each cycle in the thickened graph. The trivalent vertices contribute the displayed determinant; the bivalent vertices contribute zero. Each crossing is then counted twice. The cycle condition makes the expression independent of the representative of each cyclically ordered r-vertex. The program checks integrality in the entire integral cycle basis and obtains rank 146 modulo two.

A multiplier-one peripheral automorphism inducing identity on this marked base fixes each cusp conjugacy class. To see the resulting class-two constraint directly, use a surface presentation with the single boundary relation $\prod_i [a_i, b_i] \prod_j c_j = 1$. After taking its class-two image and quotienting homology by the span of the cusp classes, conjugating any c_j contributes a wedge with c_j and hence vanishes. The compact symplectic tensor is therefore preserved. Equivalently its homology matrix satisfies $T^{\mathrm{t}}\Omega T = \Omega$ modulo two. This constraint holds for every choice of conjugators with that same homology action.

The exact D.50 finite matrix violates this identity: the defect has rank 72 and entry $(0,6)$ equal to one. It cannot lift peripherally to the full class-two source. The moving lift above retains the same homology action and succeeds in the quotient specified in (D.51.4). This computation addresses only the literal finite certificate; the actual arithmetic G_3 and its chosen D.50 action are covered by the symbolic lift and exact residual calculations above. □

Solving the entire invariant boundary equation in the marked finite test

D.52. An evaluated full class-two replacement and its complete homology continuation [Proved]

This statement uses the precise marked finite source of the D.50–D.51 certificates. Write

$$\begin{aligned} G_* &= A_5 \times (S_3 \times C_3), \qquad |G_*| = 1080, \\ s &\longmapsto ((0,3,4,1,2),((1,0,2),0)), \\ r &\longmapsto ((4,1,2,0,3),((1,2,0),1)). \end{aligned} \tag{D.52.1}$$

The permutations are in zero-based one-line notation. Put $K_* = \ker(\widehat{B} \to G_*)$, of free rank 181, and let $A_* = C_3$ be the central factor. Let $U_* = K_*^{\mathrm{ab},(2)}$, $V_* = U_*/2U_*$, and $W_* = (1 - e_{A_*})V_*$, of dimension 120. The full class-two exponent-four source is

$$\mathcal{T}_* = \widehat{B}/\overline{\Gamma_3(K_*)(K_*')^2 K_*^4}. \tag{D.52.2}$$

There is an explicitly evaluated multiplier-one peripheral automorphism of $\mathcal{T}_*$ which fixes $t = sr$, induces identity on G_*, and has nonidentity homology action T_*, with

$$\operatorname{rank} T_* = 181, \qquad \operatorname{rank}(T_* - 1) = 96, \qquad T_*^{\mathrm{t}}\Omega T_* = \Omega. \tag{D.52.3}$$

Here T_* acts on V_* over $\mathbb{F}_2$; both ranks and the pairing identity are over $\mathbb{F}_2$. The matrix Ω is the reduction modulo two of the retained integral ribbon intersection form. It replaces the particular finite D.50 homology action excluded in D.51. The two matrices differ; this is not an extension of that excluded action. Both actions and both conjugator pairs project to identity on their common invariant quotient $\mathcal{T}_*/[D, \widetilde{A}_*]$, where $D = \ker(\mathcal{T}_* \to G_*)$ and $\widetilde{A}_*$ is the actual order-three complement specified below. Their homology actions also project to identity on the marked invariant quotient by $(1 - e_{A_*})U_*$.

The allowable moving parameters are evaluated as actual peripheral conjugator parameters. This pair space has dimension 120; each pair induces the displayed derivation. The full remaining boundary equation has 13 coordinates, obtained from a 3600-dimensional invariant central module whose augmentation image has rank 3587. In the program's specified moving basis, a solved parameter is

$$\texttt{0x2fc2bc437a2c2a199b32d59580fd1f}. \tag{D.52.4}$$

The least significant bit is the first basis coefficient. Its weight is 60. All linear and quadratic coefficients, and the actual central corrections, are included in the certificate. Direct collection of the complete 6 352 056-letter corrected cusp word gives identity in $\mathcal{T}_*$. Both torsion relations and generation are checked.

The new finite action has a compatible complete two-adic homology continuation. Define

$$\begin{aligned} \mathfrak{L}_\infty = \ker(K_* \to U_*), \qquad \mathcal{J}^*_\infty = \widehat{B}/\mathfrak{L}_\infty, \qquad \mathcal{J}^*_2 = \widehat{B}/\ker(K_* \to U_*/4U_*), \\ \mathfrak{L}_c = \overline{\Gamma_3(K_*)(K'_*)^2 K_*^4}, \\ \mathcal{Y}_* = \widehat{B}/(\mathfrak{L}_\infty \cap \mathfrak{L}_c) = \mathcal{J}^*_\infty \times_{\mathcal{J}^*_2} \mathcal{T}_*. \end{aligned} \tag{D.52.5}$$

One actual conjugator pair on this fibre product retains the full evaluated $\mathcal{T}_*$-action. The two projection kernels are

$$\begin{aligned} \ker(\mathcal{Y}_* \to \mathcal{J}^*_\infty) = \bigwedge^2_{\mathbb{F}_2} V_*, \qquad \dim = 16290, \\ \ker(\mathcal{Y}_* \to \mathcal{T}_*) = 4U_*. \end{aligned} \tag{D.52.6}$$

The program verifies the common conjugator coordinates and cusp identities through homology exponents $4, 8, 16$.

The full source kernel remains nonzero. Its exact image in

$$\mathcal{T}^*_8 = \widehat{B}/\overline{\Gamma_3(K_*)(K'_*)^4 K_*^8}$$

is

$$\operatorname{im}(\mathfrak{L}_\infty \cap \mathfrak{L}_c \to \mathcal{T}^*_8) = 2\bigwedge^2_{\mathbb{Z}/4}(U_*/4U_*). \tag{D.52.7}$$

For the first two actual free graph generators g_0, g_1, the 272-letter word $[g_0, g_1]^2$ is in this source kernel and has nonzero coordinate 2 at the corresponding commutator in $\mathcal{T}^*_8$.

The actual arithmetic G_3, the complete D.50–D.51 arithmetic actions, and their remaining invariant residual are unchanged by this calculation. No map transferring the finite solution (D.52.4) to that arithmetic residual is asserted. The evaluated finite repair supplies exact allowable-correction and boundary-equation data for that next comparison. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The actual parameter space and its peripheral maps. All group operations, the 1080 marked states, the 181 free graph cycles, and the central three-deck action are rebuilt from

(D.52.1). The program checks that $1-t$ has rank 120 on W_*. For $w \in W_*$, set

$$\begin{aligned} b(w) = (1-t)^{-1}s(1-r)w, \qquad a(w) = w + b(w), \\ (1-s)b(w) + s(1-r)a(w) = 0. \end{aligned} \tag{D.52.8}$$

Conversely every pair in the kernel of this cusp map is of this form: take $w = a + b$ and solve the displayed invertible equation. This proves that these are all 120 allowable moving parameters at the first level.

The corrections give the actual derivation values

$$\delta(s) = (1-s)b, \qquad \delta(r) = (1-r)a, \qquad \delta(t) = 0. \tag{D.52.9}$$

They satisfy the two torsion relations and the cusp relation as crossed homomorphisms. Their restrictions to the free kernel are therefore actual G_*-linear homology maps, with zero transgression. This construction does not replace them by all abstract $\mathrm{Hom}_{G_*}(V_*, W_*)$. The edge map sends the identity edge to itself plus $w = a + b$, giving every matrix column used in the calculation.

Each chosen pair lifts integrally modulo four in $(1 - e_{A_*})(U_*/4U_*)$. The odd averaging idempotent first gives moving lifts; the onto modulo-two cusp map then corrects their exact modulo-four cusp error. An actual order-three lift $\widetilde{z}$ of the central deck generator is obtained, and checked, as in D.51. On the moving abelian summand, $(1-z)^{-1} = 2 - z$ modulo four. Thus the two abelian coordinates lift to literal commutators with $\widetilde{z}$. The resulting words A_i, B_i lie in $E = [D, \widetilde{z}]$, have the prescribed homology coordinates, and have central cusp errors. All these assertions are checked in the faithful collected graph model.

The exact invariant quotient and quadratic boundary map. The central residual is $Z = W_* \wedge V_*$. Its invariant part Z_0 has dimension 3600, and its moving part has dimension 10860, by D.51. An adapted basis consists of 61 fixed homology vectors and 60 two-dimensional blocks u_i, v_i, with

$$zu_i = v_i, \qquad zv_i = u_i + v_i.$$

A basis of Z_0 is

$$\begin{aligned} u_i \wedge v_i, \\ u_i \wedge u_j + u_i \wedge v_j + v_i \wedge v_j, \qquad u_i \wedge v_j + v_i \wedge u_j \quad (i < j). \end{aligned} \tag{D.52.10}$$

This lists $60 + 2\binom{60}{2} = 3600$ independent invariant coordinates. The actual exterior actions of s, r give the full augmentation matrix

$$(1-s)Z_0 + (1-r)Z_0 \longrightarrow Z_0. \tag{D.52.11}$$

Its rank is 3587. Exact elimination constructs 13 dual coordinate functionals for its cokernel; they are retained in the certificate and annihilate every column. The moving central cusp equation is already onto. Thus the remaining error is evaluated in these actual 13 coordinates.

Here is the quadratic formula, including its source in the word relation. For a pair of kernel elements A, B, write its cusp error as

$$C(A, B) = B\, s(B)^{-1} s(A)\, t(A)^{-1}, \tag{D.52.12}$$

where s, t denote their conjugation actions. For the chosen pairs this is central, since the abelian cusp equation was solved modulo four. Let a, b and a', b' be their modulo-two abelian coordinates. Collecting the product of two pairs in the class-two group gives

$$\begin{aligned} C(AA', BB') = C(A, B) + C(A', B') \\ + (1+s)b' \wedge s(a+b). \end{aligned} \tag{D.52.13}$$

Central coordinates are written additively. To verify the cross term, order the four factors in (D.52.12) for each pair and move the primed factors into their product-pair positions. The cross commutators sum to

$$(b' + sb') \wedge (sb + sa + ta) + (sa' + ta') \wedge ta = (1 + s)b' \wedge s(a + b),$$

using the two abelian cusp equations. This proves the formula with no linearization of the group relation.

Choose the program's ordered moving basis and the corresponding actual pairs A_i, B_i. For a binary parameter vector v, use the ordered products of the selected pairs. The exact remaining obstruction is

$$\omega(v) = \sum_i \lambda_i v_i + \sum_{i<j} \gamma_{ij} v_i v_j \ \in \mathbb{F}_2^{13}. \tag{D.52.14}$$

The coefficients are obtained by applying the invariant projection and the 13 cokernel functionals to the actual individual errors and the cross wedges in (D.52.13). The program stores every coefficient and checks the polar identity again on 36 literal two-parameter word products. Thus the evaluated equations retain the torsion, cusp and transgression constraints.

The unsuccessful pairing-only route is also evaluated. The earlier attempted correction solves only the intersection-form equations. Their linearized rank is 9, leaving 111 directions. A selected nonidentity direction has homology derivation rank 16, gives an invertible pairing-preserving matrix, and satisfies the exact abelian cusp equation. Its full 13-coordinate obstruction is nevertheless nonzero.

The program also evaluates all moving modulo-four homology variations with that same modulo-two conjugator pair. They are the 120 pairs in (D.52.8), realized by moving projections of actual squares. They lie in the central Frattini subgroup of the class-two group, so their effects are exactly linear, and are checked there. Their additional image in the 13-coordinate cokernel has rank zero. This records the precise failure of that attempted repair; it is not silently discarded.

The solved parameter and full word verification. Evaluation of all equations (D.52.14) gives zero at the parameter (D.52.4). The fixed deterministic search first reaches it at trial 5784. The selected product has 60 basis factors. Its full cusp word is collected directly, and its invariant error lies in the actual augmentation image (D.52.11). Exact elimination supplies invariant central corrections. The moving correction is obtained by the literal odd-word telescoping used in D.51.

Each central correction is realized by its product of actual free-generator commutator words. The program then collects all 6 352 056 letters in the corrected cusp word in the full class-two quotient, and obtains zero, including the invariant coordinates. It also checks the squares and cubes of the two torsion conjugates. The full homology matrix has rank 181 and deviation rank 96; it preserves the exact intersection form. An onto Frattini action generates the finite two-group kernel, so the resulting endomorphism is an automorphism of $\mathcal{T}_*$.

All conjugators lie in E, including both central corrections. Hence their images in $\mathcal{T}_*/E$ are identity. This is an actual quotient: the subgroup $E = [D, \widetilde{A}_*]$ is normal in D; two odd complements above A_* are D-conjugate and give the same E. Since A_* is normal in G_*, conjugation by the full group preserves E. This proves the stated comparison to the earlier finite actions. The new modulo-two matrix differs from the excluded one: the latter has pairing-defect rank 72, whereas the new defect is zero. The replacement is therefore explicit at the common homology quotient.

The complete homology continuation and exact common source. The maximal common quotient of $\mathcal{J}_\infty^*$ and $\mathcal{T}_*$ is exactly $\mathcal{J}_2^*$. Indeed the image of $\mathfrak{L}_c$ in U_* is $4U_*$: its commutator and third lower-central generators have zero abelian image, and its fourth powers give all of

$4U_*$. The image of $\mathfrak{L}_\infty$ in $\mathcal{T}_*$ is exactly D': it contains the actual commutator generators, and its abelian image is zero. These two original kernel computations give (D.52.5)–(D.52.6).

The evaluated conjugator coordinates modulo four lie in the integral moving summand. Lift them successively through homology exponents 2^k. At each step the cusp error is in the next copy of W_*. The onto map in (D.52.8) supplies corrections. Every integral correction is first projected by the full odd averaging idempotent, so it remains in the moving summand at all later levels. For $k \geq 3$ the changes are multiples of four and preserve the common $\mathcal{J}_2^*$ coordinates. The unchanged invertible reduction modulo two proves generation at every level.

The resulting automorphisms, their inverses, and both conjugators are compatible. Their limits define an automorphism on $\mathcal{J}_\infty^*$, whose action and conjugators agree with the full evaluated $\mathcal{T}_*$ component on $\mathcal{J}_2^*$. The fibre product gives one pair and one automorphism on $\mathcal{Y}_*$. The program checks the successive $4, 8, 16$ coordinates, the exact cusp identities, and their unchanged common coordinates.

The remaining full-source kernel has an exact finite image. In $\mathcal{T}_8^*$, the kernel over G_* has generator exponents modulo eight and independent commutator coordinates modulo four. Its class-two collection law realizes the commutator module $\bigwedge^2_{\mathbb{Z}/4}(U_*/4U_*)$. The image of $\mathfrak{L}_c$ is generated by fourth powers of generators and squares of commutators. It has abelian coordinates divisible by four and even commutator coordinates: collecting a fourth power adds $\binom{4}{2} = 6$, which is even modulo four.

An element of $\mathfrak{L}_c \cap \mathfrak{L}_\infty$ has zero abelian coordinate, so its image is in twice that commutator module. Conversely every commutator square has an actual preimage in both original kernels. This proves equality in (D.52.7). The program evaluates $[g_0, g_1]^2$ in both faithful normal forms. It is identity in $\mathcal{T}_*$ and has coordinate 2 at $(0, 1)$ in $\mathcal{T}_8^*$. Its actual 272-letter B-word is retained in the certificate.

This completes the finite replacement, its complete homology continuation, and its remaining kernel computation. The arithmetic G_3 has a different specified source map, whose actual correction and transgression comparison is still being determined; the present proof makes no transfer between those two marked sources. □

The actual arithmetic correction space and its integrated automorphisms

D.53. Exact local and transgression maps for the arithmetic moving corrections [Proved]

Return to the actual arithmetic groups of D.49–D.51:

$$\begin{gathered} G = G_3, \qquad n = 576 \cdot 3^{97}, \qquad K_3 = \ker(\widehat{B} \to G), \\ U = K_3^{\mathrm{ab},(2)}, \qquad V = U/2U, \qquad \mathscr{W} = (1 - e_A)U, \qquad W = \mathscr{W}/2\mathscr{W}, \\ q = |A| \geq 3, \qquad w = \dim_{\mathbb{F}_2} W = \frac{n}{6}(1 - q^{-1}). \end{gathered} \tag{D.53.1}$$

Here A is the same actual normal elementary three-subgroup generated by $f_h \bmod G_3$. For $g = s, r, t$, write $W^g = \ker(1 - g : W \to W)$.

The complete allowable modulo-two conjugator-pair space is

$$\mathcal{P} = \{(b, c) \in W \oplus W : (1 - s)b + s(1 - r)c = 0\}, \qquad \dim \mathcal{P} = w. \tag{D.53.2}$$

Every pair induces the actual continuous derivation

$$\delta_{b,c}(s) = (1 - s)b, \qquad \delta_{b,c}(r) = (1 - r)c, \qquad \delta_{b,c}(t) = 0. \tag{D.53.3}$$

Restriction to K_3 gives a specified linear map

$$\mathcal{P} \longrightarrow \mathrm{Hom}_G(V, W). \tag{D.53.4}$$

Its kernel is exactly the injective image

$$\begin{aligned} W^s \oplus W^r \oplus W^t &\longrightarrow \mathcal{P}, \\ (u_s, u_r, d) &\longmapsto (u_s + d, u_r + d). \end{aligned} \tag{D.53.5}$$

Thus the actual image $\mathscr{D}$ has dimension

$$d = w - \dim W^s - \dim W^r - \dim W^t. \tag{D.53.6}$$

The local and transgression restrictions are computed by the exact sequences

$$\begin{gathered} H^i(G, W) = 0 \quad (i \geq 0), \qquad H^1(\widehat{B}, W) \xrightarrow{\sim} \mathrm{Hom}_G(V, W), \\ 0 \longrightarrow W/(W^s + W^r) \longrightarrow \mathrm{Hom}_G(V, W) \longrightarrow H^1(\langle s \rangle, W) \longrightarrow 0, \\ 0 \longrightarrow \mathscr{D} \longrightarrow W/(W^s + W^r) \xrightarrow{\kappa} W/(1-t)W \longrightarrow 0, \\ \kappa([v]) = [s(1-r)v], \qquad H^1(\langle s \rangle, W) = \ker(1+s)/(1-s)W. \end{gathered} \tag{D.53.7}$$

In particular vanishing transgression does not remove the displayed local and cusp restrictions.

These are nonzero correction directions for the actual arithmetic source. Their dimension satisfies the explicit positive bound

$$d \;\geq\; 80 \cdot 3^{95} - 4 \cdot 3^{45} - 4 \cdot 3^{34} \;> 0. \tag{D.53.8}$$

The proof uses the actual 576-state marked graph and its rank-97 three-homology extension. Its exact centralizer and Jordan data are

$$\begin{gathered} |C_Q(s)| = 16, \qquad |C_Q(r)| = 18, \\ \dim(N^{\mathrm{ab}}/3)^s = 45, \qquad (N^{\mathrm{ab}}/3)|_{\langle r \rangle} = J_3^{29} \oplus J_2^5, \\ |C_G(s)| = 16 \cdot 3^{45}, \qquad |C_G(r)| = 3^{35}. \end{gathered} \tag{D.53.9}$$

Here J_j is the length-j unipotent block over $\mathbb{F}_3$.

The correction directions integrate on the complete actual D.50 source. Define

$$\mathscr{P} = \ker\bigl(\mathscr{W} \oplus \mathscr{W} \xrightarrow{(1-s)b + s(1-r)c} \mathscr{W}\bigr). \tag{D.53.10}$$

This is a free $\mathbb{Z}_2$-module of rank w; its reduction modulo two maps onto $\mathcal{P}$. For the chosen D.50 automorphism α_0, every pair $2(b, c)$, with $(b, c) \in \mathscr{P}$, gives an actual continuous peripheral automorphism $\alpha_{2(b,c)}$ of $\mathcal{Z}_h$. It fixes the cusp, retains the entire D.49 action and parameter, and has the same action modulo two as α_0.

The closed subgroup generated by $\alpha_{2(b,c)}\alpha_0^{-1}$ has a continuous elementary abelian quotient of dimension d, detected by its homology action modulo four:

$$\alpha \longmapsto \frac{\alpha|_{U/4U} - 1}{2} \bmod 2. \tag{D.53.11}$$

This is an integrated automorphism quotient of the specified actual D.50 source. It is not a claim that these automorphisms lift to $\widehat{B}$ or solve the remaining invariant class-two equation.

The separate order-1080 source provides an evaluated check of the local ambiguities. There

$$\begin{gathered} \dim W^s = 64, \qquad \dim W^r = 38, \qquad \dim W^t = 0, \\ \dim \mathcal{P} = 120, \qquad \dim \ker(\mathcal{P} \to \mathscr{D}) = 102, \qquad \dim \mathscr{D} = 18. \end{gathered} \tag{D.53.12}$$

The same-action kernel has rank 4 in D.52's 13-coordinate invariant obstruction. It repairs the previously unsuccessful pairing-preserving action while keeping that entire modulo-two homology matrix unchanged. Its full corrected cusp word has 4 220 100 letters and is checked directly in the unquotiented class-two target; generation has rank 181, and the homology deviation still has rank 16. This finite check does not evaluate the arithmetic invariant obstruction. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The actual moving block and the transgression. The averaging element e_A is a central idempotent of $\mathbb{F}_2[G]$, since A is normal and has odd order. It acts as identity on the trivial module and as zero on W. A projective resolution of the trivial module can be taken in the e_A block: apply that idempotent to a free resolution. Every homomorphism from that resolution to W is zero. This proves $H^i(G, W) = 0$ in every degree, including degree zero.

The actual inflation–restriction sequence for $K_3 \subset \widehat{B}$ now gives the isomorphism in (D.53.7). Since the action on W factors through G, restriction on K_3 factors through its elementary two-homology V. Thus every G-equivariant homomorphism has zero transgression and represents a unique cohomology class of continuous derivations. The peripheral restrictions are computed next.

The edge map and the exact local restriction. Use the actual graph sequence

$$0 \longrightarrow V \longrightarrow \mathbb{F}_2[G] \longrightarrow \mathbb{F}_2[G/\langle s\rangle] \oplus \mathbb{F}_2[G/\langle r\rangle] \longrightarrow \mathbb{F}_2 \longrightarrow 0. \tag{D.53.13}$$

Its boundary sends the identity edge to its two incident vertices. Apply $\mathrm{Hom}_G(-, W)$, separating the image of the boundary into a short exact sequence. The just-proved vanishing of all $\mathrm{Ext}^i_G(\mathbb{F}_2, W)$ removes the augmentation terms. The edge module is projective, and the r-vertex module is projective because three is invertible in the coefficient field. The remaining vertex extension group is $H^1(\langle s\rangle, W)$. This gives the first short exact sequence in (D.53.7).

More explicitly, its map from W is the restriction to the cycle module of the edge map $e_g \mapsto gv$. Its kernel is $W^s + W^r$, the two vertex values. Their intersection is $W^G = 0$. The corresponding path derivation has values

$$\delta_v(s) = 0, \qquad \delta_v(r) = (1 - r)v.$$

It is locally a coboundary at both finite peripheral subgroups. Conversely all such cohomology classes lie in this edge-map image, by the exact sequence. This proves the claimed local restriction without assuming that every abstract equivariant map is peripheral.

The cusp value of this derivation is $s(1-r)v$. Changing it by an inner derivation changes that value by $(1 - t)W$. Thus the remaining map is exactly κ. It is well-defined on $W/(W^s + W^r)$: an r-fixed vector contributes zero, and an s-fixed vector contributes $(1-t)v$. It is onto because

$$(1 - t)W + s(1 - r)W = (1 - s)W + s(1 - r)W = W. \tag{D.53.14}$$

The last equality is the augmentation identity $W_G = 0$, already implied by $W_A = 0$. Its kernel consists exactly of the cusp-fixed peripheral restrictions. This proves the second short exact sequence.

Conjugator parameters and their full kernel. The same augmentation identity proves that the map defining $\mathcal{P}$ is onto. Its kernel therefore has dimension w. Each pair gives (D.53.3); the two torsion relations hold by telescoping, and the defining equation makes its cusp value zero.

Its restriction to K_3 is zero exactly when the derivation inflates from G. Vanishing $H^1(G, W)$ makes it the inner derivation $g \mapsto (1 - g)d$; d is unique because $W^G = 0$. The zero cusp value gives $d \in W^t$. The two generator equations then say

$$b - d \in W^s, \qquad c - d \in W^r. \tag{D.53.15}$$

This is precisely (D.53.5). That map is injective: an element in its kernel would have $d \in W^s \cap W^r = W^G = 0$. Taking dimensions proves (D.53.6).

The actual arithmetic centralizer data. The program `d_arithmetic_correction_graph.py` enumerates the actual Q matrices modulo twelve and checks the marked order 576. It reconstructs all 97 free graph cycles and both complete $\mathbb{F}_3$ action matrices. Exact elimination gives

$$\operatorname{rank}(s-1) = 52, \qquad \operatorname{rank}(r-1) = 63, \qquad \operatorname{rank}(r-1)^2 = 29. \tag{D.53.16}$$

These are the fixed dimension 45 and the Jordan decomposition in (D.53.9).

The centralizer maps are checked in the actual elementary-three extension, not by enumerating a smaller substitute. For each of the 16 or 18 matrices centralizing s or r in Q, choose its recorded B-word. Its commutator with the torsion generator is an actual vector in $N^{\mathrm{ab}}/3$. A kernel correction centralizes the generator exactly when

$$(1-g)v = -[\widetilde{c}, g] \quad \text{in } N^{\mathrm{ab}}/3. \tag{D.53.17}$$

The certificate retains every such equation and its solution or failure. All 16 images lift for s. For r, the liftable images are exactly $1, r, r^2$. The two centralizer kernels have orders 3^{45} and 3^{34}, respectively. This proves the exact centralizer orders in (D.53.9).

For later use, every quotient of the $\mathbb{F}_3[\langle r\rangle]$-module $N^{\mathrm{ab}}/3$ is generated by at most $29+5 = 34$ elements over that local cyclic-group algebra. Its invariant dimension equals its coinvariant dimension, and is therefore at most 34. Applying this to the actual quotient by A gives

$$|C_{G/A}(\bar{r})| \le 18 \cdot 3^{34}. \tag{D.53.18}$$

The factor 18 is the maximal centralizer image in Q; the kernel is the just-bounded fixed subgroup.

Fixed dimensions and the positive lower bound. Write

$$q_s = |A^s|, \qquad a_t = |A \cap \langle t\rangle|, \qquad \bar{c}_r = |C_{G/A}(\bar{r})|.$$

The actual fixed dimensions are

$$\begin{aligned} \dim W^s &= \frac{w}{2} + 4 \cdot 3^{45}(1 - q_s^{-1}), \\ \dim W^r &= \frac{w}{3} - \frac{2}{9}(3^{35} - \bar{c}_r), \\ \dim W^t &= \frac{n}{216}(1 - a_t/q), \qquad a_t \in \{1, 3\}. \end{aligned} \tag{D.53.19}$$

We justify each formula from its actual map.

Apply $1 - e_A$ to the graph sequence. The augmentation disappears, leaving a short exact edge–vertex sequence. The moving edge module and the moving r-vertex module are projective. For the order-two subgroup, the periodic resolution with maps $1 - s, 1 + s$ therefore identifies $H^1(\langle s\rangle, W)$ with the degree-zero Tate group of the moving s-vertex module. A permutation module contributes one such coordinate for each fixed vertex. There are $|C_G(s)|/2$ fixed vertices before odd averaging, and $|C_{G/A}(\bar{s})|/2$ afterwards. The centralizer map onto the latter group is onto: an involution discrepancy in the odd elementary kernel is corrected by solving $1 - s$ on its anti-invariant part. Its kernel is A^s. Consequently

$$\dim H^1(\langle s\rangle, W) = \frac{|C_G(s)|}{2}(1 - q_s^{-1}).$$

The left side is $2 \dim W^s - w$. This gives the first formula.

For r, odd-order invariants are exact. There are $n/3$ edge orbits, $n/6$ s-vertex orbits, and $n/9 + 2|C_G(r)|/9$ r-vertex orbits. The last count uses $|C_G(r)|/3$ fixed r-vertices. The classes of r, r^{-1} are distinct, as is already checked in Q. The graph Euler formula thus gives

$$\dim V^r = 1 + n/18 - 2|C_G(r)|/9.$$

The identical computation in G/A computes $(V^A)^r$. Their difference is the second formula in (D.53.19).

For the cusp, the certificate checks that $\langle t \rangle$ in Q has order 12 and meets no conjugate of either nontrivial elliptic subgroup. It also checks that t^{12} is a nonzero vector in $N^{\mathrm{ab}}/3$; the order in G is 36. Thus a_t is one or three, and the cusp order in G/A is $36/a_t$. Both cyclic subgroups act freely on the vertices and edges of their graphs, because any elliptic intersection would already project to one in Q.

The preimage in $\widehat{B}$ of such a cyclic subgroup of order h is free of rank $1 + |G|/(6h)$. The homology five-term sequence gives the same dimension for $V_{\langle t \rangle}$: the free source has zero H_2, and the even cyclic quotient has one-dimensional H_1 and H_2 over $\mathbb{F}_2$. For a cyclic operator, invariants and coinvariants have equal dimensions. Apply this first with $h = 36$ and then with $h = 36/a_t$ in G/A. Subtracting gives the third formula.

Using (D.53.18) and $a_t \geq 1$, these formulas imply

$$\dim W^s \leq w/2 + 4 \cdot 3^{45}, \qquad \dim W^r \leq w/3 + 4 \cdot 3^{34}, \qquad \dim W^t \leq w/36.$$

Since $q \geq 3$, we also have $w \geq n/9$. Substitution in (D.53.6) gives

$$d \geq \frac{5w}{36} - 4 \cdot 3^{45} - 4 \cdot 3^{34} \geq 80 \cdot 3^{95} - 4 \cdot 3^{45} - 4 \cdot 3^{34} > 0,$$

as asserted.

Integral lifting and actual source automorphisms. The augmentation identity holds integrally: $\mathscr{W}_A = 0$, because the odd averaging idempotent is integral over $\mathbb{Z}_2$. Hence the map defining $\mathscr{P}$ is onto $\mathscr{W}$. It splits as a map of finite free $\mathbb{Z}_2$-modules. Its kernel is free of rank w, and reduction gives

$$\mathscr{P}/2\mathscr{P} \xrightarrow{\sim} \mathcal{P}. \tag{D.53.20}$$

This proves the required integrality and reduction surjectivity.

D.50 identifies the actual kernel of $\mathcal{Z}_h \to \mathcal{W}_h$ with the additive group $\mathscr{W}$. Left-multiply the two chosen conjugators of α_0 by $2b, 2c$ in this kernel. The old conjugates project to s, r in G, respectively; hence their actions on this abelian kernel are exactly the displayed s, r module actions. Their exact cusp change is $2((1-s)b + s(1-r)c) = 0$. The torsion images remain conjugates. On $\mathcal{W}_h$ the action and conjugators are unchanged. On the complete homology component the construction is an endomorphism, since it sends K_3 into its abelian pro-two kernel and induces identity on G.

Write T_0 for the invertible integral homology matrix of α_0. The new matrix is $T_0 + 2D_{b,c}$, where $D_{b,c}$ is the restriction of the displayed integral derivation. Its reduction modulo two is unchanged, so it is invertible over $\mathbb{Z}_2$. It is therefore an automorphism of the homology component. The unchanged common-quotient action and literal conjugator coordinates give an automorphism on the actual fibre product $\mathcal{Z}_h$. This proves the asserted integrated family.

For $\alpha_{2(b,c)}\alpha_0^{-1}$, the matrix modulo four is

$$1 + 2\,\overline{D}_{b,c}\,\overline{T}_0^{-1}. \tag{D.53.21}$$

All bars here denote reduction modulo two. Multiplication of such matrices adds their divided differences modulo two. Their images are exactly the vector space $\mathscr{D}\,\overline{T}_0^{-1}$, of dimension d.

Thus the closed subgroup they generate maps onto its additive elementary abelian group, proving (D.53.11). Every generator preserves the projection to $\mathcal{J}_\infty$, induces identity on G, and acts identically on $U/2U$. These properties persist under products, inverses and closure. Consequently the displayed modulo-four action is defined on that whole closed subgroup, and its kernel is precisely the elements acting identically on $U/4U$. The map is continuous because it is detected on the finite homology quotient modulo four. The entire D.49 action is retained; its full loop abelianization also retains the normalized commutator parameter condition.

The separate finite same-action check. The program `d_cm_same_action_obstruction.py` rebuilds the order-1080 marked graph and computes the exact fixed spaces. It obtains ranks $56, 82, 120$ for $1-s, 1-r, 1-t$ on its moving module. Thus (D.53.5) has kernel dimension $64 + 38 = 102$, with derivation image 18. This checks explicitly that the 120 pair parameters in D.52 were not 120 distinct derivations.

The full D.52 quadratic coefficients are then evaluated on this kernel. All kernel polar terms against all 120 parameters vanish in the recorded 13-coordinate quotient. The resulting same-action correction image has rank 4. It contains the actual obstruction of the earlier pairing-preserving rank-16 action. An exact solution gives the new conjugator parameter

$$\texttt{0x40000000000000000000002004840}. \tag{D.53.22}$$

The program verifies equality of every modulo-two homology column before and after this change.

Finally `d_cm_same_action_word.py` uses that prescribed parameter in the full D.52 word construction. The entire 4 220 100-letter corrected cusp word is identity in the unquotiented class-two target; both torsion relations and rank-181 generation are checked. The homology deviation remains rank 16, and its matrix is exactly the earlier pairing-preserving one. The integral continuation through exponents $4, 8, 16$ retains this new full class-two component. These finite checks illustrate the explicit local ambiguities. They do not assign values to the arithmetic invariant obstruction or lift the automorphisms above to $\widehat{B}$. □

Actual doubled corrections across the full class-two source

D.54. Integrated even corrections, the reduced arithmetic boundary map, and the complete finite comparison [Proved]

Keep the actual arithmetic source and the fixed normalized conjugators of D.50. Write

$$\begin{gathered} R_{50} = \mathfrak{R}_h \cap \mathfrak{K}_\infty, \qquad L_2 = \overline{\Gamma_3(K_3)(K_3')^2 K_3^4}, \\ \mathcal{T}_2 = \widehat{B}/L_2, \qquad D = \ker(\mathcal{T}_2 \to G_3), \qquad C = D' = \textstyle\bigwedge^2_{\mathbb{F}_2} V, \\ Z = W \wedge V, \qquad Z^0 = Z^A, \qquad Z^- = (1 - e_A)Z. \end{gathered} \tag{D.54.1}$$

D.51 proved that the exact image of R_{50} in $\mathcal{T}_2$ is Z. Consequently the full joint source and its two specified projections are

$$\begin{gathered} \mathcal{X}_{54} = \widehat{B}/(R_{50} \cap L_2) = \mathcal{Z}_h \times_{\mathcal{T}_2/Z} \mathcal{T}_2, \\ \ker(\mathcal{X}_{54} \to \mathcal{Z}_h) = Z, \qquad \ker(\mathcal{X}_{54} \to \mathcal{T}_2) = L_2 R_{50}/R_{50}. \end{gathered} \tag{D.54.2}$$

The map to D.51's joint source has kernel Z^0.

There is an actual peripheral automorphism of $\mathcal{X}_{54}$ for every doubled integral pair $2(b, c)$, $(b, c) \in \mathscr{P}$ of D.53.10. These automorphisms act identically on $\mathcal{W}_h$, fix $t = sr$, and have complete homology action

$$1 + 2D_{b,c} \quad \text{on } U. \tag{D.54.3}$$

Here $D_{b,c}$ is the actual restriction of the integral derivation with generator values $(1-s)b, (1-r)c$. On the $\mathcal{T}_2$ component every element of the full central module C, including Z^0, is fixed. The closed subgroup generated by these automorphisms has the continuous elementary abelian quotient $\mathscr{D}$ of D.53, detected by its action modulo four. Thus its dimension is the positive arithmetic number d bounded in D.53.8.

These are automorphisms of the identity-marked joint source. The explicit parameter transport below lifts every difference $\alpha_{2(b,c)}\alpha_0^{-1}$ from D.53 through this construction. The chosen D.50 automorphism α_0 itself is not asserted to have a cusp-fixed extension. The exact comparison with that fixed automorphism is the following. Its fixed two conjugators lift to $\mathcal{X}_{54}$; their two torsion conjugates define an actual automorphism $\widetilde{\alpha}_0$, with

$$\widetilde{\alpha}_0(t) = z_0 t, \qquad z_0 \in Z. \tag{D.54.4}$$

The cusp defect class

$$\omega_0 = [z_0] \in Z_G = (Z^0)_{G/A} \simeq \left(\bigwedge\nolimits^2 W\right)_G \tag{D.54.5}$$

is independent of these lifts of the fixed conjugators. Composition with any of the constructed doubled corrections leaves the literal z_0 unchanged. In particular those higher even homology corrections do not change the invariant boundary obstruction. No value or vanishing assertion for the actual arithmetic ω_0 is made.

Throughout the following reduction the two D.49 conjugators themselves are retained on $\mathcal{W}_h$. Within this fixed framing, the complete modulo-two pair space $\mathcal{P}$ of D.53 has a well-defined actual affine quadratic boundary map

$$q_{\alpha_0} : \mathcal{P} \longrightarrow Z_G, \qquad q_{\alpha_0}(0) = \omega_0. \tag{D.54.6}$$

Its polar map, for $p = (b, c)$ and $p' = (b', c')$, is

$$q_{\alpha_0}(p + p') + q_{\alpha_0}(p) + q_{\alpha_0}(p') + q_{\alpha_0}(0) = [(1 + s)b' \wedge s(c + b)]. \tag{D.54.7}$$

The right side is interpreted in Z_G. It vanishes whenever one input belongs to the exact kernel $\mathcal{K}$ in D.53.5. The map

$$\ell : \mathcal{K} \longrightarrow Z_G, \qquad \ell(k) = q_{\alpha_0}(k) + q_{\alpha_0}(0) \tag{D.54.8}$$

is therefore linear, and there is a specified affine quadratic descent

$$\overline{q}_{\alpha_0} : \mathscr{D} \longrightarrow Z_G/\ell(\mathcal{K}). \tag{D.54.9}$$

This removes every doubled integral ambiguity and every same-modulo-two-action conjugator ambiguity from the displayed arithmetic boundary map. Its arithmetic zero locus remains unevaluated in this work; invertibility of the corresponding $\overline{T}_0 + D_p$ is also retained as part of the arithmetic calculation.

In the separate literal order-1080 source, the calculation within the retained 120-parameter moving family is complete:

pair parameters	120
same-action kernel	102
actual homology parameters	18
same-action obstruction image	4
remaining obstruction coordinates	9
homology parameters tested	262 144
full class-two homology images	864.

(D.54.10)

All 864 images are invertible and form an exactly verified group on homology. They are exactly the points preserving the retained modulo-two intersection form. For every point the certificate gives a full 120-coordinate conjugator parameter solving all 13 original invariant equations. The comparison is of zero loci: the two nine-coordinate polynomial maps do not have identical coefficient spans. This evaluated finite classification is not a numerical evaluation of $\overline{q}_{\alpha_0}$ for G_3.

The source kernel in (D.54.2) remains nonzero under the explicit next map

$$\mathcal{T}_8 = \widehat{B}/\overline{\Gamma_3(K_3)(K_3')^4 K_3^8}. \tag{D.54.11}$$

For the actual elements $z_i \in \mathfrak{R}_h$ and $x_j \in K_3$ constructed in D.50, the elements $[z_i, x_j]^2$ belong to $R_{50} \cap L_2$, and their images include the nonzero module

$$2\big((U^-/4U^-) \wedge (U/4U)\big) \;\subset\; \bigwedge\nolimits^2_{\mathbb{Z}/4}(U/4U). \tag{D.54.12}$$

Full profinite realization remains unresolved. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The full joint source and its actual kernels. The equality $\mathrm{im}(R_{50} \to \mathcal{T}_2) = Z$ was proved by the actual preimage and fourth-power argument in D.51. The product of the two source kernels therefore gives the maximal common marked quotient $\mathcal{T}_2/Z$. Closed Goursat gives (D.54.2), including both projection kernels. Replacing $\mathcal{T}_2$ by $\mathcal{T}_2/Z^0$ gives exactly D.51's source, so that projection has kernel Z^0.

The square module and its moving part. The subgroup

$$P_D = D^2 D' = D^2 \tag{D.54.13}$$

is central in D, abelian of exponent two, and hence a module over $\mathbb{F}_2[G]$. Indeed $[u^2, v] = [u, v]^2 = 1$, and $(uv)^2 u^{-2} v^{-2}$ is a commutator. Collected class-two coordinates identify its exact sequence as

$$0 \longrightarrow \bigwedge\nolimits^2 V \longrightarrow P_D \xrightarrow{\pi} V \longrightarrow 0. \tag{D.54.14}$$

The last map sends an even abelian coordinate $2v$ modulo four to v modulo two. Squares of free generators give every basis vector; its kernel is exactly the commutator module. This description includes the square cross terms and does not choose an equivariant splitting of the sequence.

Apply the actual central idempotent $1 - e_A$. Averaging is exact because $|A|$ is odd, so

$$0 \longrightarrow Z^- \longrightarrow P_D^- \longrightarrow W \longrightarrow 0, \qquad P_D^- = (1 - e_A)P_D. \tag{D.54.15}$$

For the kernel, decompose $V = V^A \oplus W$. The moving part of $\bigwedge^2 V$ is exactly Z^-: the summand $\bigwedge^2(V^A)$ is fixed, and every other moving summand lies in $W \wedge V$.

Let E denote the image of $\mathfrak{R}_h$ in $\mathcal{T}_2$. D.51 proved $E \cap C = Z$ and $E/Z \simeq U^-/4U^-$ by its actual abelianization map. Since

$$P_D^- = [P_D, A] \subset [D, A] \subset E, \tag{D.54.16}$$

the image of P_D^- in $\mathcal{T}_2/Z$ is precisely the even subgroup $2(U^-/4U^-)$. The map is the one in (D.54.14). This proves the needed agreement with the D.50 homology coordinate.

The exact compatible correction module. Inside $\mathcal{X}_{54}$, form the actual fibre product

$$\mathscr{E} = 2U^- \times_W P_D^-, \qquad 2U^- \longrightarrow W, \quad 2u \longmapsto u \bmod 2. \tag{D.54.17}$$

Its $\mathcal{W}_h$ coordinate is identity. The two remaining coordinates agree in $\mathcal{T}_2/Z$ by (D.54.16), so this is an embedded subgroup of the displayed joint source. It is abelian, centralizes the image of K_3, and has its G-action by the literal conjugation maps. Its exact sequence is

$$0 \longrightarrow Z^- \longrightarrow \mathscr{E} \longrightarrow 2U^- \longrightarrow 0. \tag{D.54.18}$$

Choose the integral pair (b, c) in D.53.10 and lift $2b, 2c$ to elements B_0, A_0 of $\mathscr{E}$. Their cusp defect under conjugation of s, r is exactly

$$(1-s)B_0 + s(1-r)A_0. \tag{D.54.19}$$

There are no unrecorded quadratic terms: $\mathscr{E}$ is abelian and centralizes the kernel. The projection of this error in $2U^-$ is zero. It therefore lies in Z^-. The map $(1-s)Z^- + s(1-r)Z^-$ is onto Z^-, since $Z_A^- = 0$. Correct B_0, A_0 by elements of this exact kernel to kill (D.54.19).

The resulting conjugates define an endomorphism of the homology component: its kernel matrix is $1 + 2D_{b,c}$, which is invertible over $\mathbb{Z}_2$. They also define an endomorphism of $\mathcal{T}_2$, because the image of K_3 is still class two with the stated exponent relations. Its induced map on $D/\Phi(D) = V$ is identity. The finite two-group Frattini criterion makes its restriction to D an automorphism; it is identity on the finite quotient G. Thus it is an automorphism of $\mathcal{T}_2$. The two component automorphisms agree on the actual common quotient through the displayed conjugators. They give the claimed automorphism of $\mathcal{X}_{54}$.

Its effect on C is the exterior square of its effect on V, hence identity. The parameter has trivial $\mathcal{W}_h$ coordinate. Its preimage lies in $\mathfrak{R}_h \subset \widehat{F}'$; equivalently the reconstruction from the two conjugators retains the full normalized commutator condition. All loop, Fermat, CM and congruence maps retained by $\mathcal{W}_h$ therefore have identity action.

Modulo four the homology matrices are $1 + 2\overline{D}_{b,c}$. Their divided differences add under composition. D.53.20 proves that the integral pairs reduce onto all of $\mathcal{P}$, and D.53.4 identifies their restriction image as $\mathscr{D}$. The generated closed subgroup thus maps continuously onto its additive elementary abelian group. The kernel of this detected quotient is exactly the identity action on $U/4U$. This proves the actual integrated claim without extending α_0.

The exact transport of the D.53 differences. Write β_0, A_0 for the fixed D.50 conjugators on $\mathcal{Z}_h$, and set

$$\begin{aligned} v_s &= \alpha_0^{-1}(\beta_0^{-1}), & v_r &= \alpha_0^{-1}(A_0^{-1}), \\ b' &= \delta_{b,c}(v_s) + \bar{v}_s b, & c' &= \delta_{b,c}(v_r) + \bar{v}_r c. \end{aligned} \tag{D.54.23}$$

Bars here are the actual G-images, acting on U^-. The integral derivation $\delta_{b,c}$ factors through $\mathcal{J}_\infty$, since its restriction to K_3 is abelian and pro-two. Hence it is defined on these elements of $\mathcal{Z}_h$. The two inverse generator images are $v_s s v_s^{-1}$ and $v_r r v_r^{-1}$. The crossed-homomorphism identity therefore gives

$$(\delta_{b,c} \circ \alpha_0^{-1})(s) = (1-s)b', \qquad (\delta_{b,c} \circ \alpha_0^{-1})(r) = (1-r)c'.$$

Its cusp value is zero because $\alpha_0^{-1}(t) = t$. Thus $(b', c') \in \mathscr{P}$. On the complete source $\mathcal{Z}_h$, the D.53 difference is exactly $g \mapsto 2(\delta_{b,c} \circ \alpha_0^{-1})(g)\, g$, with additive notation in the kernel. It is the identity-source correction for (b', c'), which has just been lifted to $\mathcal{X}_{54}$. Its homology matrix is the D.53 matrix $1 + 2D_{b,c}T_0^{-1}$. This proves the stated transport as an equality of actual maps.

The fixed arithmetic cusp defect. Lift the two specified D.50 conjugators through $\mathcal{X}_{54} \to \mathcal{Z}_h$. Their torsion conjugates give a homomorphism from $\widehat{B}$. Its homology component is the original α_0, and its $\mathcal{T}_2$ component descends through the characteristic class-two exponent-four relations of K_3. Its modulo-two homology map is the invertible matrix of α_0. The Frattini criterion again proves that this component is an automorphism. The components agree on the common

quotient by construction, giving $\widetilde{\alpha}_0$ on $\mathcal{X}_{54}$. Since α_0 fixes the cusp, its error belongs to the actual projection kernel Z, proving (D.54.4).

Changing either lifted conjugator by an element of Z changes its cusp error by $(1-s)z_\beta + s(1-r)z_A$. The image is the augmentation submodule of Z, so its cokernel is Z_G. This proves independence in (D.54.5). Odd averaging kills Z^- in these coinvariants. The cross summand $W \otimes V^A$ also has zero A-coinvariants, which proves the last isomorphism in (D.54.5). Finally every constructed doubled automorphism fixes t and all of Z pointwise. It therefore sends $z_0 t$ to the same $z_0 t$. Composition on the other side also leaves that value unchanged because the doubled automorphism fixes t.

The reduced actual arithmetic quadratic map. For a pair in $\mathcal{P}$, choose an integral lift in $\mathscr{P}$, which exists by D.53.20. Use it to change the two D.50 conjugators in the kernel U^-. These give a cusp-fixed peripheral endomorphism on $\mathcal{Z}_h$; invertibility is not needed to define its boundary error. Lift its conjugators to $\mathcal{X}_{54}$ and take the class of the cusp error in Z_G.

Changing those lifts contributes only the augmentation image of Z. Two integral lifts differ by a doubled pair. The construction (D.54.17)–(D.54.19) gives its exact full class-two correction. Since the two base actions on G are identity, the same calculation applies to left corrections of the chosen endomorphism: their action on $\mathscr{E}$ is exactly the displayed s, r, t action. It contributes zero in Z_G. Thus the error defines the actual map q_{α_0} on $\mathcal{P}$, independently of all these choices.

For completeness its polar formula is the class-two collected product of the two correction pairs. Write A, B for the corrections of the r, s conjugators, and write a, b for their abelian coordinates modulo two. The correction cusp is

$$B\, s(B)^{-1} s(A) t(A)^{-1}. \tag{D.54.20}$$

Replacing A, B by AA', BB', collecting the central commutators, and using the two zero abelian cusp values gives the cross term $(1+s)b' \wedge s(a+b)$. The actions of the original lifted conjugates have the same actions on V and C as s, r, t; their additional terms are linear. Reducing in Z_G gives (D.54.7). This calculation includes the square terms of the exponent-four kernel; they are part of the linear coefficients over $\mathbb{F}_2$.

Now let $p' \in \mathcal{K}$. D.53.5 writes it as $(u_s + d, u_r + d)$ with $u_s \in W^s, u_r \in W^r, d \in W^t$. Its cross term reduces to $[(1+s)d \wedge s(c+b)]$. Coinvariance under s identifies this with $[d \wedge (1+s)(c+b)]$. The defining cusp equation for p gives

$$(1+s)(c+b) = (1+t)c. \tag{D.54.21}$$

Since d is fixed by t, its wedge with this last vector vanishes in the coinvariants. The polar map is symmetric, so it vanishes when either input lies in $\mathcal{K}$. Consequently its restriction to $\mathcal{K}$, after subtracting the constant value, is linear. Quotienting its image gives exactly (D.54.9). This is the actual reduction of the remaining arithmetic equation; no zero of that unevaluated map is asserted.

The complete separate finite calculation. The program `d_cm_doubled_class2.py` first verifies the square-module construction on this same literal finite source. It checks all 543 actual conjugations of free-generator squares by s, r, z. This verifies the full action, including the central square cross terms, against the marked B-words. The square module has dimension 16 471, and its moving part has dimension 10 980. All 120 basis pairs are lifted through the exact moving square module; every abelian projection and full cusp module equation is checked. An explicit nonzero combination has divided homology action of rank 16 modulo four. Its entire 4 031 900-letter cusp word and both torsion relations are evaluated directly in the full class-two group. This finite verification is separate from the symbolic arithmetic proof above.

The program `d_cm_reduced_invariant_equations.py` reconstructs the literal marked order-1080 graph. It verifies the 102-dimensional kernel and chooses an explicit 18-dimensional

complement in the original 120-dimensional pair space. The full retained D.52 quadratic coefficients are then reduced by the four-dimensional same-action obstruction image proved in D.53. This gives all linear and quadratic coefficients of the remaining nine-coordinate map.

For the same eighteen basis parameters, the program computes every coefficient of $T^{\mathrm{t}}\Omega T-\Omega$, where Ω is the retained modulo-two intersection form. Its coefficient rank is also nine. An exact comparison of the coefficient relations records that the two polynomial spans differ; no equality of those displayed maps is used. The separate program `d_cm_invariant_zero_loci.py` evaluates both maps on all 2^{18} parameters. Their zero sets coincide and have exactly 864 elements.

For every such parameter the first program solves the exact linear equation in the 102-dimensional same-action kernel. It records the resulting full 120-coordinate parameter and checks all original thirteen invariant equations. The D.52 central correction construction then gives full class-two cusp-fixed conjugators for every point. Every one of the 864 full homology matrices is explicitly checked to have rank 181. The finite two-group Frattini criterion proves generation in the full class-two quotient.

Finally the program multiplies the actual homology matrices to build the subgroup generated by the five recorded elements

$$\texttt{0x368},\quad \texttt{0x37e},\quad \texttt{0x3c9},\quad \texttt{0x635},\quad \texttt{0x80a}. \tag{D.54.22}$$

Each product is reduced in the exact eighteen-dimensional matrix space and is checked to belong to the listed solution set. The generated set is the full set of 864 invertible matrices. This proves the asserted group statement. These calculations concern their literal finite source throughout.

A nonzero remaining source-kernel map. Choose the actual z_i, x_j of D.50's final commutator construction. The commutator $[z_i, x_j]$ belongs to R_{50}, and maps to $\bar{u}_i \wedge \bar{v}_j$ in the commutator module of $\mathcal{T}_2$. Its square is therefore killed in $\mathcal{T}_2$, so belongs to $R_{50} \cap L_2$. The collected class-two coordinates in $\mathcal{T}_8$ have commutator module $\bigwedge^2_{\mathbb{Z}/4}(U/4U)$. The same square maps there to $2\bar{u}_i \wedge \bar{v}_j$. Because U^- is a nonzero integral direct summand and U has rank greater than one, these images include a nonzero basis coordinate. Their span is precisely the displayed submodule in (D.54.12). This proves the explicit nontriviality claim without asserting an equality with the entire remaining image. □

The arithmetic boundary norm and the compact orientation class

D.55. An exact morphism from the arithmetic boundary obstruction to the intersection form [Proved]

Use the actual marked quotient $G = G_3$ and the arithmetic boundary map of D.54. Let $n = 576 \cdot 3^{97}$. The subgroup K_3 is the fundamental group of the marked oriented surface with

$$\begin{gathered} c = n/36 \quad \text{cusps}, \qquad 2g = 2 + 5n/36, \qquad \dim V = 1 + n/6, \\ \mathcal{R} = \langle [gt^{36}g^{-1}] : g \in G/\langle t \rangle \rangle \subset V, \qquad \dim \mathcal{R} = c - 1, \\ H = V/\mathcal{R}, \qquad \dim H = 2g. \end{gathered} \tag{D.55.1}$$

Here $V = H_1(K_3, \mathbb{F}_2)$, and the projection $\pi : V \to H$ is the actual homology map of compactification. The oriented surface determines its nondegenerate intersection bivector $\Theta \in \bigwedge^2 H$, the inverse of its alternating intersection form.

There is a specified linear map

$$\begin{aligned} \mathcal{N} : Z_G &\longrightarrow (\bigwedge^2 H)^G, \\ [z] &\longmapsto \bigwedge^2 \pi \left(\sum_{g \in G} gz \right), \qquad Z = W \wedge V. \end{aligned} \tag{D.55.2}$$

Every modulo-two pair parameter $p \in \mathcal{P}$ of D.54 gives the following equality of actual maps:

$$\boxed{\mathcal{N}(q_{\alpha_0}(p)) = (\bigwedge^2 T_{p,c})(\Theta) - \Theta.} \tag{D.55.3}$$

The map $T_p = \overline{T}_0 + D_p$ fixes $\mathcal{R}$ pointwise and induces $T_{p,c}$ on H. Neither its invertibility nor a zero of the arithmetic boundary map is assumed in this equality.

The complete local image $\mathcal{L} = \ell(\mathcal{K})$ of D.54 lies in $\ker \mathcal{N}$. Consequently the exact remaining comparison is

$$\begin{gathered} \overline{\mathcal{N}} : Z_G/\mathcal{L} \longrightarrow \operatorname{im} \mathcal{N}, \\ 0 \longrightarrow \mathcal{E}_{\mathcal{N}} \longrightarrow Z_G/\mathcal{L} \xrightarrow{\overline{\mathcal{N}}} \operatorname{im} \mathcal{N} \longrightarrow 0, \qquad \mathcal{E}_{\mathcal{N}} = \ker \mathcal{N}/\mathcal{L}. \end{gathered} \tag{D.55.4}$$

No vanishing assertion for the actual $\mathcal{E}_{\mathcal{N}}$ is made. Its two explicit sources are identified by the exact sequence

$$\begin{gathered} \nu : Z_G \longrightarrow Z^G, \qquad [z] \longmapsto \sum_{g \in G} gz, \qquad p_C = (\bigwedge^2 \pi)|_Z, \\ 0 \longrightarrow \widehat{H}^{-1}(G, Z) \longrightarrow \ker \mathcal{N} \xrightarrow{\nu} \operatorname{im} \nu \cap \ker p_C \longrightarrow 0. \end{gathered} \tag{D.55.5}$$

Thus the norm kernel and the boundary projection kernel remain in the comparison, with their literal maps.

The actual zero fiber of $\overline{q}_{\alpha_0}$ consists entirely of invertible homology actions. Generation imposes no additional restriction on that fiber: the compact map preserves a nondegenerate bivector and the cusp radical is already fixed. The zero fiber itself remains unevaluated, as does the full profinite realization.

There is also a complete ordinary relative-homology calculation for the actual arithmetic base. With trivial $\mathbb{F}_2$ coefficients and the marked peripheral collections

$$\mathcal{S}_G = (\langle s \rangle, \langle r \rangle, \langle t \rangle), \qquad \mathcal{S}_B = (\langle s \rangle, \langle r \rangle, \overline{\langle t \rangle}), \tag{D.55.6}$$

the cusp orders are 36 and $\widehat{\mathbb{Z}}$, respectively. The exact maps are

$$\begin{gathered} H_2(Q, \mathbb{Z}) = C_2, \qquad Q^{\mathrm{ab}} = C_6, \qquad H_2(G, \mathbb{F}_2) \simeq \mathbb{F}_2^2, \\ H_2(\langle s \rangle, \mathbb{F}_2) \oplus H_2(\langle t \rangle, \mathbb{F}_2) \xrightarrow{\sim} H_2(G, \mathbb{F}_2), \\ H_2(\widehat{B}, \mathcal{S}_B; \mathbb{F}_2) \xrightarrow{\sim} H_2(G, \mathcal{S}_G; \mathbb{F}_2) \simeq \mathbb{F}_2. \end{gathered} \tag{D.55.7}$$

The two actual central double covers $\mathrm{SL}_2(\mathbb{Z}/12) \to Q$ and D.48's $\mathcal{E} \to Q$ give the dual restriction matrix

	$\langle s \rangle$	$\langle t \rangle$
$\mathrm{SL}_2(\mathbb{Z}/12)$	1	0
$\mathcal{E}$	0	1

. (D.55.8)

Their pulled-back lift orders on G are $(4, 36)$ and $(2, 72)$. The relative orientation map is therefore computed using the original markings, rather than only abstract dimensions.

For the separate order-1080 family, the norm in (D.55.2) has rank 9 on the 13-coordinate obstruction space. Its kernel is exactly the four-dimensional local image. All 18 linear and all 153 quadratic coefficients in (D.55.3) are checked through that actual norm. This proves the finite 864-point intersection comparison via the specified morphism. It does not assert equality of the arithmetic norm kernel with its local image. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The marked compactification and its homology map. The regular G-cover of the oriented modular orbifold has one edge for every element of G, with vertices indexed by $G/\langle s\rangle$ and $G/\langle r\rangle$. The rotations are the actual right multiplications by s, r. Their product gives the cusp cycles. D.53 checked that t has exact order 36 in G; hence there are $c = n/36$ cusp cycles. The torsion-free kernel has rank $\rho = 1 + n/6$. Capping these oriented boundary components gives $2g = \rho - c + 1 = 2 + 5n/36$.

The integral homology exact sequence of this compactification identifies the boundary sublattice with the free group on the c oriented cusp classes modulo their one sum. It is a primitive sublattice. Reduction modulo two proves the dimension and quotient in (D.55.1); in particular there is no additional modulo-two boundary relation. An oriented handle and boundary presentation identifies the intersection bivector as

$$\Theta = \sum_{i=1}^{g} a_i \wedge b_i. \tag{D.55.9}$$

It is independent of that presentation, since it is the inverse of the intersection form on compact homology. All deck maps preserve the orientation, so the quotient map and Θ are G-equivariant and G-invariant, respectively.

The full cusp norm. Use the actual class-two lift of the chosen pair parameter from D.54. Write φ for its map from the marked source to $\mathcal{T}_2$. It induces identity on G, sends the two torsion generators to their specified conjugates, and has $\varphi(t) = zt$, $z \in Z$. These are actual endomorphisms of the class-two target; no generation assumption enters the argument.

Because Z is central in the kernel D, its conjugates by powers of t commute. Consequently

$$(zt)^{36} = \left(\sum_{j=0}^{35} t^j z\right) t^{36}. \tag{D.55.10}$$

Choose one marked coset representative g_i for each cusp. The original cusp loop is $c_i = g_i t^{36} g_i^{-1}$. The element $\varphi(g_i)g_i^{-1}$ belongs to D, since the base action is identity. Thus the image of c_i differs from its original conjugacy class by the central term

$$g_i \sum_{j=0}^{35} t^j z. \tag{D.55.11}$$

The remaining conjugation correction has exterior coordinate $[u_i] \wedge [c_i]$ for a kernel element u_i. It maps to zero under $\bigwedge^2 \pi$, because the cusp class is in $\mathcal{R}$. In abelian homology the central error also disappears. Hence every cusp class is fixed, proving the asserted pointwise action on $\mathcal{R}$.

Sum (D.55.11) over the cusp cosets. The representatives followed by the thirty-six powers of t enumerate each element of G once. The total central error is therefore exactly $\sum_{g\in G} gz$. This proves the literal full-group norm in (D.55.2). The map is well-defined on coinvariants since the sum is unchanged by translating z. Its image is invariant after the equivariant compact projection.

The orientation identity. Choose the integral punctured-surface presentation with handles a_i, b_i and boundary loops $c_1, \dots, c_c$:

$$\prod_{i=1}^{g} [a_i, b_i]\, c_1 \cdots c_c = 1. \tag{D.55.12}$$

The free basis consists of the handles and the first $c-1$ boundary generators. There is an actual homomorphism of its free class-two exponent-four quotient which kills these $c-1$ generators and retains the handles. On central coordinates it is exactly $\bigwedge^2 \pi$. The last boundary word then maps to the inverse of the handle commutator product. This construction specifies the map used in the following computation; it does not assume that φ descends through a compact surface relation.

Apply φ to (D.55.12) and then use that homomorphism. The handle commutators contribute $(\bigwedge^2 T_{p,c})(\Theta)$. The original boundary words contribute $-\Theta$. Their additional conjugation terms vanish by the preceding paragraph, and their central errors give $\mathcal{N}([z])$. All surviving terms are central. The resulting equality, in characteristic two, is precisely (D.55.3). Since $[z]$ is the actual boundary map of D.54, this also proves its compatibility with all integral and local choices.

The exact residual and generation. A pair in $\mathcal{K}$ has zero homology restriction. Adding it changes the boundary value by its linear local image and leaves $T_{p,c}$ unchanged. Subtracting the two orientation identities shows $\mathcal{L} \subset \ker \mathcal{N}$. The kernel and image of the induced linear map give (D.55.4).

For (D.55.5), the finite Tate group in degree minus one is, by its norm definition,

$$\widehat{H}^{-1}(G, Z) = \ker\left(\sum_{g \in G} g : Z \to Z\right) / I_G Z. \tag{D.55.13}$$

It is exactly the kernel of ν. The compact norm is the composition $p_C \nu$. Its kernel maps onto $\operatorname{im} \nu \cap \ker p_C$, with the preceding Tate kernel. This proves the claimed exact sequence and retains both possible kernel contributions.

On the zero fiber of the reduced boundary map, (D.55.3) gives $(\bigwedge^2 T_{p,c})(\Theta) = \Theta$. The support of the nondegenerate bivector on the right is all of H. The support on the left is contained in $\operatorname{im} T_{p,c}$. Thus $T_{p,c}$ is onto, and hence invertible in this finite-dimensional space. Its full map T_p fixes $\mathcal{R}$ and induces this isomorphism on the quotient; it is therefore invertible on V. The complete homology lift is invertible over $\mathbb{Z}_2$, and the finite class-two lift generates by the Frattini criterion. This proves the generation assertion about the exact zero fiber without asserting its nonemptiness.

The ordinary relative-homology computation. The discrete kernel N of $B = C_2 * C_3 \to Q$ has free rank 97. D.26's integral lattice calculation, reconstructed in `d_arithmetic_relative_h2.py`, gives

$$(N^{\mathrm{ab}})_Q = C_2. \tag{D.55.14}$$

The exact Smith factors are $(1^{96}, 2)$; appending the actual cusp column $[t^{12}]$ gives (1^{97}). The marked abelianization map $Q \to C_6$ sends s to 3 and r to 2. It is onto and equals the abelianization, because $B^{\mathrm{ab}} = C_6$. The homology five-term sequence, with $H_2(B, \mathbb{Z}) = 0$, therefore identifies $H_2(Q, \mathbb{Z})$ with (D.55.14).

The universal coefficient sequence has the two one-dimensional terms $H_2(Q, \mathbb{Z}) \otimes \mathbb{F}_2$ and $\operatorname{Tor}(C_6, \mathbb{F}_2)$. Hence $H_2(Q, \mathbb{F}_2)$ has dimension two. The kernel of $G \to Q$ is the actual odd group of order 3^{97}. Averaging its bar resolution contracts all positive-degree homology over $\mathbb{F}_2$. The induced map on modulo-two homology is therefore an isomorphism, giving the first line of (D.55.7).

To compute the peripheral map, use the two stated central double covers. In $\mathrm{SL}_2(\mathbb{Z}/12)$ the marked lifts of s, r, t have orders $4, 3, 12$. In the actual central cover $\mathcal{E}$ they have orders $2, 3, 24$. The program checks the full marked state spaces and their projection maps, including t^{12} as the nontrivial central element of $\mathcal{E}$. Pulling both extensions back to G gives orders $4, 36$ and $2, 72$ on s, t. A central double cover of an even cyclic group has nonzero restriction class exactly when its generator lift has doubled order. Thus the dual restriction matrix is (D.55.8). It is invertible, so the displayed map on H_2 is an isomorphism.

The r-subgroup has odd order, so its positive modulo-two homology vanishes. In the relative homology sequence, the peripheral H_2 map is the just-proved isomorphism. The next kernel is the one-dimensional kernel of

$$H_1(\langle s\rangle, \mathbb{F}_2) \oplus H_1(\langle t\rangle, \mathbb{F}_2) \longrightarrow H_1(G, \mathbb{F}_2), \qquad (a, b) \longmapsto a + b. \tag{D.55.15}$$

This proves the finite relative dimension and specifies its boundary. For $\widehat{B}$, the free-product homology sequence identifies its H_2 with that of C_2. The procyclic cusp has zero H_2 and one-dimensional H_1. The same relative sequence has the same boundary kernel (D.55.15). The original quotient preserves both generators, so induces identity on that kernel. This proves the marked relative orientation isomorphism in (D.55.7).

The exact separate finite norm certificate. The program `d_cm_boundary_norm.py` reconstructs the order-1080 marked graph and the actual intersection form of D.51. It constructs the 35-dimensional cusp radical and the rank-146 compact quotient. The chosen complement, projection matrix and inverse-form bivector are all retained. The code verifies the actual s, r deck actions, the compact projection and orientation invariance.

It then reconstructs the exact 3600-dimensional invariant exterior basis and the thirteen original obstruction functionals. Exact elimination supplies thirteen representatives dual to those functionals. The full group norm is evaluated through literal subgroup transversals of orders $3, 4, 5, 3, 2$, with the central order-three average separately checked. These are the actual A_5, S_3, and central C_3 factors of this finite source. They enumerate the full group norm, not an unrelated averaging operator.

After compact projection the norm has rank nine. It kills the four independent local-image vectors, so its kernel is exactly their span. Every one of the eighteen linear coefficients and the one hundred fifty-three quadratic coefficients equals the corresponding compact bivector defect through this norm. This proves the stated finite morphism and explains the zero-locus equality of D.54. The arithmetic maps above are proved symbolically on their actual source; the finite kernel equality is not transferred to G_3. □

An evaluated arithmetic parameter and its smaller obstruction map

D.56. The actual order-3888 quotient, its unique arithmetic parameter, and the forty-dimensional correction space [Proved]

Retain the actual $h, f_h, G_3, A, R_h, \alpha_0$ of D.46–D.54. All source kernels below are closed. Put $A_* = \mathbb{F}_3^4$, with column-vector convention, and set

$$S_* = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \qquad R_* = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 1 \end{pmatrix}, \tag{D.56.1}$$

$$v_* = (1, 1, 1, 0)^{\mathsf{t}}, \qquad Q_* = \langle S_*, R_*\rangle, \qquad G_* = A_* \rtimes Q_*.$$

The following is a specified epimorphism from the original marked source:

$$\varphi_* : \widehat{B} \longrightarrow G_*, \quad s \longmapsto (0, S_*), \quad r \longmapsto (v_*, R_*). \tag{D.56.2}$$

It factors through the actual $G_3 = B/(N'N^3)$. The induced linear quotient and its kernel satisfy

$$\begin{gathered} Q \twoheadrightarrow Q_*, \qquad |Q_*| = 48, \qquad \ker(Q \to Q_*) \simeq A_4, \qquad 5I \longmapsto -I_4, \\ |G_*| = 3888, \qquad 1 \longrightarrow C_3^{93} \longrightarrow P_* \longrightarrow A_4 \longrightarrow 1, \quad P_* = \ker(G_3 \to G_*), \\ |P_*| = 12 \cdot 3^{93}, \qquad H_1(P_*, \mathbb{F}_2) = 0. \end{gathered} \tag{D.56.3}$$

The Q_*-module A_* is irreducible, with $\mathrm{End}_{Q_*}(A_*) = \mathbb{F}_9$. The image of the actual arithmetic parameter is determined exactly:

$$\begin{gathered} a_* = \varphi_*(f_h) = (2,1,2,1)^{\mathsf{t}}, \qquad \varphi_*(A) = A_*, \qquad \varphi_*(C_h) = A_*, \\ t_* = \varphi_*(sr), \qquad |t_*| = 12, \qquad t_*^6 = ((0,2,1,0)^{\mathsf{t}}, -I_4), \\ \varphi_* \alpha_h = \mathrm{Int}(t_*^6)\varphi_*. \end{gathered} \tag{D.56.4}$$

Here $\mathrm{Int}(k)(g) = kgk^{-1}$. In particular the normalized arithmetic action preserves this specific quotient and acts as $-I_4$ on its translation module. The map from the spliced D.49 source factors through its identity three-primary component:

$$\mathcal{W}_h \longrightarrow P_{h,3} \longrightarrow G_*, \qquad \alpha_0|_{G_*} = \mathrm{id}\,. \tag{D.56.5}$$

These are the two actual comparison maps for the original action and its retained splice.

Write $K_* = \ker(\widehat{B} \to G_*)$, and put

$$U_* = K_*^{\mathrm{ab},(2)}, \quad V_* = U_*/2, \quad e_* = \frac{1}{81}\sum_{a \in A_*} a, \quad U_*^- = (1-e_*)U_*, \quad W_* = U_*^-/2. \tag{D.56.6}$$

The actual marked graph and its compactification give

quantity	value
$\mathrm{rank}_{\mathbb{Z}_2} U_*$	649
$\mathrm{rank}_{\mathbb{Z}_2} e_* U_*$	9
$\dim W_*$	640
cusp count	324
cusp radical dimension	323
compact homology dimension	326
moving cusp radical dimension	320
moving compact homology dimension	320

. (D.56.7)

Every dimension in the last six rows is over $\mathbb{F}_2$. The genus is 163. The peripheral maps are the original right multiplications by $\varphi_*(s), \varphi_*(r)$; the cusp loops are the conjugates of t^{12}.

The actual correction and same-action parameter maps are

$$\begin{gathered} \mathcal{P}_* = \ker\Big[W_*^2 \xrightarrow{L_*} W_*\Big], \qquad L_*(b,c) = (1-s)b + s(1-r)c, \\ \mathcal{K}_* = \{(u_s + d, u_r + d) : u_s \in W_*^s,\ u_r \in W_*^r,\ d \in W_*^t\}, \qquad \mathscr{D}_* = \mathcal{P}_*/\mathcal{K}_*, \end{gathered} \tag{D.56.8}$$

	W_*^s	W_*^r	W_*^t	$\mathcal{P}_*$	$\mathcal{K}_*$	$\mathscr{D}_*$
dim	336	210	54	640	600	40

.

The map defining $\mathcal{K}_*$ is injective. Thus 640 counts actual peripheral conjugator-pair parameters, and 40 counts their induced homology derivations.

There are surjective maps from the full arithmetic modules and parameters, all induced by $K_3 \hookrightarrow K_*$:

$$\begin{gathered} u : U \twoheadrightarrow U_*, \quad j : V \twoheadrightarrow V_*, \quad j_W : W \twoheadrightarrow W_*, \quad \xi : \mathcal{P} \twoheadrightarrow \mathcal{P}_*, \quad \mathscr{D} \twoheadrightarrow \mathscr{D}_*, \\ \theta : Z = W \wedge V \twoheadrightarrow Z_* = W_* \wedge V_*, \qquad \tau : Z_{G_3} \twoheadrightarrow (Z_*)_{G_*}, \\ \dim Z_* = 210240, \qquad \dim Z_*^{A_*} = 2560, \qquad \dim(1-e_*)Z_* = 207680. \end{gathered} \tag{D.56.9}$$

For example $\xi(b,c)=(j_W b, j_W c)$, and $\theta(w\wedge v)=j_W w\wedge jv$. The original chosen D.50 action and its conjugators induce a specific actual action $\alpha_{0,*}$ on the smaller joint homology source. With these choices the actual boundary maps commute:

$$\begin{gathered}\tau(q_{\alpha_0}(p))=q_{\alpha_{0,*}}(\xi(p)),\\ \overline{\tau}: Z_{G_3}/\ell(\mathcal{K})\twoheadrightarrow (Z_*)_{G_*}/\ell_*(\mathcal{K}_*),\\ \overline{\tau}\,\overline{q}_{\alpha_0}=\overline{q}_{\alpha_{0,*}}(\mathscr{D}\longrightarrow\mathscr{D}_*).\end{gathered}\tag{D.56.10}$$

The right-hand quadratic map is an actual forty-parameter arithmetic obstruction. Its coefficients and zero locus have not been evaluated here.

For completeness, its source and projection kernels are explicit. Set

$$\begin{gathered}M_*=\ker(K_*\to U_*),\quad R_{50,*}=R_h\cap M_*,\quad L_{2,*}=\overline{\Gamma_3 K_*(K_*')^2K_*^4},\\ \mathcal{Z}_{h,*}=\widehat{B}/R_{50,*},\qquad \mathcal{T}_{2,*}=\widehat{B}/L_{2,*},\qquad \mathcal{X}_*=\widehat{B}/(R_{50,*}\cap L_{2,*}).\end{gathered}\tag{D.56.11}$$

The image of R_h in $\widehat{B}/M_*$ is exactly U_*^-. The image of $R_{50,*}$ in $\mathcal{T}_{2,*}$ is exactly Z_*. Consequently

$$\begin{gathered}\mathcal{X}_*=\mathcal{Z}_{h,*}\times_{\mathcal{T}_{2,*}/Z_*}\mathcal{T}_{2,*},\qquad \ker(\mathcal{Z}_{h,*}\to\mathcal{W}_h)=U_*^-,\\ \ker(\mathcal{X}_*\to\mathcal{Z}_{h,*})=Z_*,\qquad \ker(\mathcal{X}_*\to\mathcal{T}_{2,*})=L_{2,*}R_{50,*}/R_{50,*},\\ \ker(\mathcal{Z}_h\to\mathcal{Z}_{h,*})=\frac{R_h\cap M_*}{R_h\cap M_\infty},\qquad \ker(\mathcal{X}_{54}\to\mathcal{X}_*)=\frac{R_{50,*}\cap L_{2,*}}{R_{50}\cap L_2}.\end{gathered}\tag{D.56.12}$$

Here $M_\infty=\ker(K_3\to U)$ is D.50's $\mathfrak{K}_\infty$. All maps retain the D.49 parameter and its original conjugator framing. The ordinary group norms have the further exact comparison

$$\theta\left(\sum_{g\in G_3} gz\right)=|P_*|\sum_{g_*\in G_*} g_*\theta(z)=0\quad(z\in Z).\tag{D.56.13}$$

The smaller boundary equation in (D.56.10) therefore remains part of the comparison, even after the compact norm of D.55 has been calculated.

The certificates use the actual Q-graph and the actual arithmetic f_h. They do not substitute the separate order-1080 family for this quotient. The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The literal finite source. Multiplication in $A_*\rtimes Q_*$ is $(v,M)(w,N)=(v+Mw,MN)$. The displayed matrices have orders $2,3$, and $(1+R_*+R_*^2)v_*=0$. Thus (D.56.2) is already a homomorphism from C_2*C_3, and it extends continuously to its profinite completion.

There is a particularly short complete certificate that the linear map factors through the marked Q. Starting at its identity, assign to a Q-state the product of S_*,R_* along its shortest marked word. The program checks consistency on every right-s and right-r edge of all 576 states. Every word with the same Q-endpoint therefore has the same linear matrix. This proves the claimed quotient map, rather than only a match of generator orders. Its complete image has 48 matrices. Its kernel has twelve states: the four marked states

$$0,\ 572,\ 574,\ 575\tag{D.56.14}$$

form a Klein four group, and the marked state 321 has order three and cyclically permutes its three nonidentity elements. The twelve products are all distinct. This gives the explicit semidirect-product identification of that kernel with A_4. The numbering is the original Q-matrix ordering in the D.53 certificate; both the matrices and the state maps are retained, so the identification has a literal inverse.

Since N has trivial linear image, its affine image is contained in A_*. Reconstruct the 97 marked free generators of N from the same spanning tree as D.53. Their affine images are the columns of the retained rank-four map

$$\phi : N^{\mathrm{ab}}/3 \twoheadrightarrow A_*. \tag{D.56.15}$$

Every one of the 97 word images is checked against the corresponding collapsed graph cycle. This proves surjectivity and the exact coordinate map. In particular $N'N^3$ is killed. The full affine map retaining Q has image $A_* \rtimes Q$, of order 46656; projection of its linear factor gives (D.56.2). The 3888 actual source states are all generated by the two displayed images.

The kernel of ϕ has dimension 93. For any of the twelve linear kernel states, lift it to G_3 and remove its translation using $\phi(N)$. This gives the surjection $P_* \to A_4$, whose kernel is precisely $\ker \phi$. It proves the exact sequence in (D.56.3), including its maps. The odd kernel has no positive-degree homology with $\mathbb{F}_2$ coefficients, so $H_1(P_*, \mathbb{F}_2) = H_1(A_4, \mathbb{F}_2) = 0$. Equivalently, A_4 is generated by its elements of order three, which all map trivially to an abelian two-group.

The forty projective lines of A_* are tested directly: the S_*, R_*-span of each has dimension four. This proves irreducibility. Solving the sixteen-entry matrix commutation equations gives a two-dimensional $\mathbb{F}_3$-algebra. All eight nonzero elements have inverses in the same algebra, proving that it is $\mathbb{F}_9$. The dual character pairs have orbit sizes $8, 12, 12, 8$, with stabilizer orders $6, 4, 4, 6$; their complete maps are also retained for later coefficient calculations.

Determining the actual arithmetic parameter. D.49 supplies the original, normalized identities

$$\alpha_h(s) = sf_h, \qquad \alpha_h(r) = f_h^{-1} r, \qquad \alpha_h(t) = t, \qquad f_h \in N, \tag{D.56.16}$$

and D.47 identifies the original r-conjugator's Q-coordinate as $5I$. In our module this coordinate is $-I_4$. Write $a = \varphi_*(f_h)$. The order-two relation for sf_h gives

$$S_* a = -a. \tag{D.56.17}$$

Every affine lift of the original r-conjugator has form $(b, -I_4)$, and its square is the identity. Its conjugation of (v_*, R_*) has translation $(1 - R_*)b - v_*$. Comparison with $f_h^{-1} r$ consequently gives

$$a - 2v_* \in \mathrm{im}(1 - R_*). \tag{D.56.18}$$

These two conditions determine a without a choice of its remaining conjugator coordinate. Indeed $\mathrm{im}(1 - R_*)$ is exactly the span of the second and fourth coordinate vectors. Thus the first and third coordinates of a equal 2, and (D.56.17) makes the other two equal 1. This proves the value in (D.56.4). It also proves directly why no lift of $5I$ centralizes the actual r-image: that would require $2v_* \in \mathrm{im}(1 - R_*)$.

The normal span of this nonzero vector is all of the irreducible A_*. This is the image of both the actual A and the actual closed normal subgroup C_h. Direct multiplication gives the displayed value of t_*^6, and conjugation by that value sends the two marked generators to

$$s_* a_*, \qquad a_*^{-1} r_*. \tag{D.56.19}$$

Equation (D.56.16) proves the equality of homomorphisms in (D.56.4). It proves preservation of the quotient kernel as well as the resulting automorphism. The certificate independently checks the map on all 7776 directed marked edges and all 3888 states.

Finally C_h has three-group image A_* in G_*. The restriction of this finite map to C_h factors through $C_h^{(3)}$. Therefore $R_{h,3}$ lies in its source kernel, which proves the actual factorization through $P_{h,3}$. The retained splice is the identity on that factor, proving (D.56.5). No change of the marked source or of the original arithmetic parameter is used.

The smaller homology and its peripheral parameter space. The original s, r orders remain $2, 3$, so K_* is torsion-free. Its Bass–Serre graph has 3888 edges and $1944 + 1296$ vertices. Its rank is 649. The t_*-cycles have length 12, hence there are 324 cusps and genus $(649 - 324 + 1)/2 = 163$. The integral cusp sublattice is primitive, with its single sum relation. Its modulo-two rank is 323.

The coefficient-two averaging idempotent e_* is integral because 81 is odd. The quotient graph for Q_* has rank $1 + 48/6 = 9$. Its cusp count is $48/12 = 4$. The covering to that graph has group A_*; both cusp orders are 12, so no additional cusp ramification occurs. Averaging identifies invariant homology with the rank-nine quotient homology. It identifies the invariant cusp radical with its rank-three radical. The complementary ranks are therefore 640 and 320, and the moving compact rank is 320. This proves all the maps and numbers in (D.56.7).

The certificate reconstructs all 649 free words and their integral cycles. It computes the integral ribbon pairing as one half of the sum of the oriented trivalent-vertex determinants. Every entry is integral; its modulo-two rank is 326, and its radical is exactly the span of the 324 marked cusp loops. The source permutations give the homology matrices. All deck equivariances of this pairing are checked. The A_*-norm of each cycle gives the actual invariant projection; subtracting it gives W_*. The fixed-space nullities of its s, r, t matrices are $336, 210, 54$, respectively. Every matrix and each fixed-space basis is included in the certificate.

Here $(W_*)_{G_*} = 0$, because already $(W_*)_{A_*} = 0$. The image of L_* is the augmentation subspace, so L_* is onto. Its kernel has dimension 640. The cohomological and peripheral argument of D.53 applies with the literal smaller coefficient map: the averaging idempotent kills all cohomology of the G_*-module W_*, and the two marked cyclic restrictions identify the kernel of the pair-to-derivation map with

$$W_*^s \oplus W_*^r \oplus W_*^t \longrightarrow W_*^2, \quad (u_s, u_r, d) \longmapsto (u_s + d, u_r + d). \tag{D.56.20}$$

For its injectivity, a vector in the kernel must be fixed by s, r, t, and hence by G_*; its A_*-invariant part in W_* is zero. The dimensions in (D.56.8) now follow.

Surjective maps from the full arithmetic obstruction. The inclusion $K_3 \hookrightarrow K_*$ has quotient P_*. The image of $U \to U_*$ has quotient equal to the maximal abelian pro-two quotient of P_*. That quotient is trivial by (D.56.3); thus u and its reduction j are onto. Conjugation by $\widehat{B}$ commutes with these maps. The action on the smaller target factors through G_*. Since $A \to A_*$ is onto and has odd kernel, their averaging idempotents satisfy

$$ue_A = e_* u. \tag{D.56.21}$$

It follows that $U^- \to U_*^-$ and j_W are onto. Taking exterior products gives θ; taking coinvariants gives τ, with

$$\ker \tau = \operatorname{im}\big((\ker \theta)_{G_3} \longrightarrow Z_{G_3}\big). \tag{D.56.22}$$

This is right exactness of the specified coinvariant sequence, with the action on Z_* inflated along $G_3 \to G_*$.

For surjectivity on peripheral pairs, put $J = \ker j_W$. Exactness of averaging implies $J_A = 0$. Therefore

$$(1 - s)J + s(1 - r)J = J. \tag{D.56.23}$$

Lift any pair in $\mathcal{P}_*$ to W^2. Its cusp defect belongs to J; (D.56.23) removes it using a pair in J^2. This proves surjectivity of ξ. The image of $\mathcal{K}$ lies in $\mathcal{K}_*$, so it also proves surjectivity $\mathscr{D} \to \mathscr{D}_*$, with literal kernel $\xi^{-1}(\mathcal{K}_*)/\mathcal{K}$.

As an A_*-module, W_* consists of eight copies of its augmentation module. Indeed the graph chain modules are free $\mathbb{F}_2[A_*]$-modules, since A_* has trivial intersection with the two vertex stabilizers. Their moving chain difference has multiplicity eight; semisimplicity over the odd

group A_* gives the stated isomorphism. The eighty nontrivial characters occur in forty inverse pairs, each with multiplicity eight. Thus $\dim(\bigwedge^2 W_*)^{A_*} = 40 \cdot 8^2 = 2560$. The cross term with $V_*^{A_*}$ has no invariant part. Also

$$\dim(W_* \wedge V_*) = \binom{640}{2} + 640 \cdot 9 = 210240. \tag{D.56.24}$$

Subtracting the invariant dimension gives 207680. Exactness of odd averaging also proves that $Z^A \to Z_*^{A_*}$ is onto.

The actual joint source and the boundary square. The D.50 action is an automorphism of $\mathcal{Z}_h = \widehat{B}/R_{50}$. Since it induces identity on G_3, it preserves the actual subgroup K_*/R_{50}, the preimage of P_* in this source. The inclusions $R_{50} \subset M_\infty \subset M_*$ identify its maximal abelian pro-two kernel as M_*/R_{50}. This characteristic subgroup is therefore preserved as well. Separately, the retained map to $\mathcal{W}_h$ makes R_h/R_{50} invariant. Thus their intersection $(R_h \cap M_*)/R_{50}$ is invariant, and the action descends to $\mathcal{Z}_{h,*} = \widehat{B}/(R_h \cap M_*)$. The quotient map $\mathcal{Z}_{h,*} \to \mathcal{W}_h$ remains equivariant. The action also descends to $\widehat{B}/M_*$. These are the actual subgroup and quotient maps defining $\alpha_{0,*}$ and its inherited conjugators in (D.56.10).

To identify the image of R_h, consider the image of C_h in $\widehat{B}/M_*$. It is normal, projects onto A_*, and contains $[A_*, U_*] = U_*^-$. Modulo U_*^-, its pro-two kernel commutes with its odd A_*-quotient, and the extension splits by the coprime averaging argument. Its image there is pronilpotent. The universal pronilpotent quotient of C_h therefore kills exactly the moving part in this image. This proves $\mathrm{im}(R_h) = U_*^-$. One can verify the reverse inclusion directly at every finite level: in a pronilpotent image the three-primary image of C_h commutes with the normal two-primary kernel, whereas their commutators generate U_*^-.

Apply the class-two construction of D.51 to these actual subgroups. The odd averaging gives an A_*-equivariant section in the affine space of class-two lifts. Its full moving abelian image is W_*; commutators with the whole kernel give exactly $W_* \wedge V_*$. The mixed invariant quotient is zero, and fourth-power corrections remove its abelian coordinate. This gives $\mathrm{im}(R_{50,*}) = Z_*$. It proves both the common quotient and the two small projection kernels in (D.56.12).

The inclusions $K_3 \subset K_*$ imply $M_\infty \subset M_*$ and $L_2 \subset L_{2,*}$. Taking the displayed intersections with R_h gives the two full-source projection kernels in (D.56.12). Thus these are quotient maps of the same marked $\widehat{B}$, with the literal joint-source maps.

Choose the original class-two lift of the D.50 conjugators used to define q_{α_0}. Its image defines the smaller lift. Every pair correction and every central cusp error maps by j_W and θ, respectively. After taking the actual central correction cokernels this gives the first equality in (D.56.10). The same equality on local same-action pairs gives $\tau\ell(\mathcal{K}) \subset \ell_*(\mathcal{K}_*)$. It proves the entire reduced boundary square. In particular

$$\ker \overline{\tau} = \frac{\tau^{-1}(\ell_*(\mathcal{K}_*))}{\ell(\mathcal{K})} \tag{D.56.25}$$

is its exact comparison kernel. All of this retains the actual arithmetic seed α_0; no value of that seed's remaining invariant obstruction is inserted.

Finally, each G_*-element has $|P_*|$ lifts to G_3. Equivariance of θ therefore proves (D.56.13). Since $|P_*|$ is even, that comparison is zero over $\mathbb{F}_2$. The boundary square, rather than a division by this norm factor, supplies the evaluated smaller arithmetic test. □

The marked genus-three source and an arithmetic class-two repair

D.57. Both arithmetic branches have a full class-two repair over the actual order-3888 quotient [Proved]

Use the actual arithmetic element h, the group $G_* = \mathbb{F}_3^4 \rtimes Q_*$, and the maps of D.46, D.49 and D.56. In particular $|Q_*| = 48$, $|G_*| = 3888$, and the marked epimorphism sends s, r to the elements specified in D.56. Put

$$K_Q = \ker(\widehat{B} \longrightarrow Q_*), \qquad U_Q = K_Q^{\mathrm{ab},(2)}, \qquad L_{Q,2} = \overline{\Gamma_3 K_Q (K_Q')^2 K_Q^4}, \qquad J_{Q,2} = \widehat{B}/L_{Q,2}. \tag{D.57.1}$$

The subgroup K_Q has free rank nine. Its actual marked compactification is

$$C: \ y^4 = x^3 + 1, \qquad b(x,y) = \frac{(x^6 + 20x^3 - 8)^2}{64(x^3+1)^3}. \tag{D.57.2}$$

The four points above infinity are removed. The ordered deck generators are

$$s : (x,y) \longmapsto \left(\frac{2-x}{x+1}, \frac{\sqrt{3}\, y}{x+1} \right), \qquad r : (x,y) \longmapsto (\zeta_3 x, y). \tag{D.57.3}$$

These formulas identify the actual 48-state marked quotient, including both generator maps, with the deck group of (D.57.2).

Write D_t for the action of $t = sr$ on U_Q, $J = D_t^3$, and $N = \sum_{q \in [Q_*, Q_*]} q$. The derived group in this sum is the marked quaternion group of order eight. The actual arithmetic action of h on U_Q is one of the two integral matrices

$$T_1 = J + \frac{N/2 - D_t^2 N}{7}, \qquad T_3 = J^3 + \frac{N/2 - D_t^2 N}{7}. \tag{D.57.4}$$

Both matrices preserve the integral ribbon intersection form over $\mathbb{Z}_2$ and are invertible modulo two. Their difference modulo two has rank four. The proof below treats both matrices and the complete integral conjugator ambiguity; no choice between them is required for the conclusion.

Let K_*, M_*, $L_{*,2}$, U_*, V_*, W_*, and $Z_* = W_* \wedge V_*$ be the actual objects of D.56. There is an actual continuous peripheral automorphism of

$$X_* = \widehat{B}/(\mathfrak{R}_h \cap M_* \cap L_{*,2}) \tag{D.57.5}$$

which fixes t, has multiplier one, and induces exactly D.49's prescribed automorphism on $\mathcal{W}_h = \widehat{B}/\mathfrak{R}_h$. It induces the identity on G_* and an automorphism of the full class-two exponent-four quotient $\mathcal{T}_{*,2} = \widehat{B}/L_{*,2}$. Its action on the full integral homology quotient $\widehat{B}/M_*$ is a repaired action; the particular homology representative chosen in D.50 is not asserted to remain fixed. The exact underlying source and all projection kernels are those of D.56.12. The resulting parameter belongs to the image of $\widehat{F}'$ and maps to the same literal parameter f_h^* in $\mathcal{W}_h$. Consequently the previous metabelian and prescribed CM comparisons are retained through those quotient maps.

Here are the evaluated dimensions in the actual arithmetic source:

$\dim V_*$	649
$\dim W_*$	640
$\dim(Z_*^{A_*})$	2560
$\dim(Z_*)_{G_*}$	74
rank(full group norm)	40
rank(compact group norm)	6
rank(local same-homology correction)	64
dim(reduced obstruction space)	10
dim(compact-norm kernel/local image)	4.

(D.57.6)

The complete forty-variable equations retain all ten output coordinates. In the recorded coordinates their coefficients span a nine-dimensional subspace, with the explicit relation that output coordinate zero equals output coordinate two. The full lists are retained, rather than replacing the ten-coordinate map by this smaller image. For the two matrices in (D.57.4), the recorded zeros are

branch	forty-parameter mask	fixed-seed trial	reduced constant
J	`ba190a8d71`	682	`040`
J^3	`dfd51862a6`	18	`04d.`

(D.57.7)

Each row has all eight integral s-conjugator framing cases, with an explicit correction in the 600-dimensional local parameter space. All sixteen corrected literal cusp words have zero abelian error and zero in every one of the 74 moving central coinvariant coordinates. Their lengths range from 114712 to 119132 letters in s, r, r^{-1}. Every resulting marked homology map has rank 649, preserves the intersection pairing, and projects to its specified matrix in (D.57.4).

This proves the actual automorphism of (D.57.5). The next full-source map still has the explicit kernel

$$\widehat{B} \longrightarrow X_* : \mathfrak{R}_h \cap M_* \cap L_{*,2}. \tag{D.57.8}$$

No surjectivity from genuine full profinite GT_0 is inferred. The exact map from the original full G_3 class-two source of D.54 to (D.57.5) is induced by

$$\mathfrak{R}_h \cap M_\infty \cap L_2 \ \subset \ \mathfrak{R}_h \cap M_* \cap L_{*,2}. \tag{D.57.9}$$

Its kernel is the quotient of the right-hand subgroup by the left-hand subgroup. The original full G_3 invariant equation remains to be lifted across this map.

The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and the two-torus/Fermat follow-ups in section 51.15, folios 725–727, scans 156–160.

Proof. The exact marked curve. Use homogeneous coordinates (X, Z), with $Y^4 = X^3Z + Z^4$. The matrices for the two maps in (D.57.3) are

$$S = \frac{1}{\sqrt{3}} \begin{pmatrix} -1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} \zeta_3 & 0 \\ 0 & 1 \end{pmatrix}, \tag{D.57.10}$$

and Y is unchanged. Both matrices preserve the displayed quartic, $S^2 = R^3 = I$, and $(SR)^3 = iI$. The latter scalar acts on the affine curve by $(x, y) \mapsto (x, -iy)$, so it is exactly the operator J. Direct substitution shows that b is invariant under both maps. The useful second identity is

$$b - 1 = \frac{x^3(x^3 - 8)^3}{64(x^3 + 1)^3}. \tag{D.57.11}$$

The branch permutations are therefore 2^{24}, 3^{16}, and 12^4 above zero, one, and infinity. Their degree is 48. The exact generated matrix group also has 48 elements. The program constructs its bijection with the 48 matrices of D.56 and verifies all 96 ordered s, r edges in both directions. Thus this comparison identifies the marked epimorphism, not only the abstract group order. On the positive real interval, the s-fixed point has $x = \sqrt{3} - 1$, while the r-fixed point has $x = 0$; the local multipliers are respectively -1 and ζ_3. These fix the branch convention in (D.57.3). The genus is three and there are four cusps, giving free rank $2 \cdot 3 + 4 - 1 = 9$.

The quotient by $y \mapsto -y$ is the elliptic curve $E_3 : Y^2 = x^3 + 1$, through $Y = y^2$. The natural index-eight map to the elliptic source of D.45 is this map followed by multiplication by two. Explicitly it is

$$u = \frac{x(x^3 - 8)}{4(x^3 + 1)}, \qquad v = \frac{x^6 + 20x^3 - 8}{8y^6}, \qquad v^2 = u^3 + 1, \quad b = v^2. \tag{D.57.12}$$

Substitution gives $s : (u, v) \mapsto (u, -v)$ and $r : (u, v) \mapsto (\zeta_3 u, v)$. Its deck kernel is therefore exactly the marked quaternion group, since these images generate the abelian quotient of order six. In particular t^2 acts by $u \mapsto \zeta_3^2 u$, which proves the precise CM evaluation convention used in (D.57.4). This also identifies the rational Tate map with D.46's elliptic source; no unidentified elliptic twist is introduced. The two further degree-two quotient maps can also be written explicitly. For $a = \sqrt{3}$ or $-\sqrt{3}$, put

$$X_a = \frac{x + 1 - a}{x + 1 + a}, \qquad u_a = \frac{y}{x + 1 + a}, \qquad z_a = \frac{(9 + 6a)X_a^2 - 9}{6a}.$$

Then

$$z_a^2 = 1 - (12 + 8a)u_a^4, \qquad E_a : \; v^2 = w^3 + (3 + 2a)w, \tag{D.57.13}$$

where the rational map to E_a is obtained first by $w_0 = 2(1 + z_a)/u_a^2$, $v_0 = 4(1 + z_a)/u_a^3$, then by $w = w_0/4$, $v = v_0/8$. All identities are verified in the function field of (D.57.2). The pulled-back differentials for the two quartics are

$$\frac{du_a}{z_a} = \frac{-x - 1 + a}{4} \frac{dx}{y^3}. \tag{D.57.14}$$

Together with dx/y^2 they are independent. The induced homomorphism of Jacobians is consequently an isogeny $\mathrm{Jac}(C) \to E_3 \times E_{\sqrt{3}} \times E_{-\sqrt{3}}$: its differential is an isomorphism, hence its image has full dimension and its kernel is finite. Each E_a becomes isomorphic over $\overline{\mathbb{Q}}$ to $E_i : v^2 = w^3 - w$ by scaling by a fourth root of $-(3 + 2a)$. This gives the actual comparison of the four-dimensional complementary Tate module with two copies of the Tate module of E_i.

Why there are exactly two integral arithmetic branches. On the nine-dimensional rational homology the simultaneous centralizer of the two marked deck matrices has dimension five. Here this is an exact calculation: the 81-variable commutator matrix has rank 76 modulo five, and the five independent rational operators $I, J, J^2, J^3, D_t^2 N$ commute with both deck matrices. The three mutually orthogonal projectors are

$$e_3 = N/8, \qquad e_i = (I - J^2)/2, \qquad e_c = (I + J^2)/2 - N/8, \tag{D.57.15}$$

of ranks two, four, and three. The last is the cusp subspace; the first is the marked E_3 subspace. On the middle subspace the centralizer is exactly $\mathbb{Q}_2(i)$, with i represented by J.

There is an actual finite-action argument for the middle arithmetic scalar. The curve E_i has good reduction at three: its displayed integral model has discriminant 64. Its 2^n-torsion is finite etale at three for every n, so the inertia element τ used in D.46 acts trivially on $T_2 E_i$. Therefore the actual commutator $h = [g, \tau]$ also acts trivially there. The isogeny just constructed supplies the evaluation map

$$\mathrm{Hom}_{\overline{\mathbb{Q}}}(E_i, \mathrm{Jac}(C)^-) \otimes_{\mathbb{Z}[i]} T_2 E_i \longrightarrow T_2 \,\mathrm{Jac}(C)^- \tag{D.57.16}$$

as an isomorphism after tensoring with $\mathbb{Q}_2$. Its Hom lattice is finitely generated: under a complex embedding it embeds in the integral homomorphisms between the first homology lattices, because an abelian-variety homomorphism is determined by its analytic lattice map. Its Galois action is therefore finite: choose a finite integral generating set of morphisms, each defined over a finite extension, and take their common field of definition. All deck maps are defined over $\mathbb{Q}(\zeta_{12})$. The commutator h fixes this field. Passing from the arithmetic action on the curve to the marked source action introduces a deck element; since the latter action is the identity on Q_*, that deck element is central. Equivalently the finite Galois image on the Hom lattice normalizes the finite deck image, so their combined image is finite. It follows from (D.57.16) that the marked scalar on the middle summand has finite order. The only roots

of unity in $\mathbb{Q}_2(i)$ are $1, -1, i, -i$: the residue field has order two, and its ramified quadratic extension contains μ_4 but not μ_8; indeed $\Phi_8(X+1)$ is Eisenstein at two and has degree four.

On the first summand, D.46 gives the actual marked scalar $\eta_* = (-3 - 8\zeta_3^2)/7$, with ζ_3^2 represented by D_t^2. On the cusp summand the action is the identity, because the source action fixes the marked deck quotient and has multiplier one. Consequently the four rational candidates are exactly

$$J^k + \left(\frac{-3I - 8D_t^2}{7} - (-1)^k I\right) N/8, \qquad 0 \le k < 4. \tag{D.57.17}$$

The marked integral graph lattice excludes $k = 0, 2$. For both of those candidates the entry in row one and column zero, with zero-based indices in the certificate, is $-1/14$. The other two candidates have integral $\mathbb{Z}_2$ entries and are precisely (D.57.4). The certificate checks their exact rational pairing identities, ranks modulo two, and all marked deck commutations. This argument uses the two-adic Hom and Tate comparison (D.57.16); it does not identify a two-adic scalar from the ramified three-adic representation.

The actual common quotient and the remaining framing. The exact sequence $1 \to K_* \to K_Q \to A_* \to 1$, with $|A_*| = 81$, gives the integral onto map $U_* \to U_Q$. Odd averaging identifies its kernel with U_*^-. Thus, already on full homology, the marked common quotient is

$$(\widehat{B}/M_*)/U_*^- \simeq G_* \times_{Q_*} J_{Q,1}, \qquad J_{Q,1} = \widehat{B}/\ker(K_Q \to U_Q). \tag{D.57.18}$$

Both projections are the original source projections. For the class-two quotient let $D_* = \ker(\mathcal{T}_{*,2} \to G_*)$ and let E_* be the image of $\mathfrak{R}_h$ in $\mathcal{T}_{*,2}$. D.51's averaged quotient construction, applied through D.56's actual maps, gives $E_* = [D_*, \widetilde{A}_*]$, $E_* \cap D'_* = Z_*$, and

$$\mathcal{T}_{*,2}/E_* \simeq G_* \times_{Q_*} J_{Q,2}. \tag{D.57.19}$$

Here are both directions of this identification. In the preimage of A_*, an odd complement exists and the maximal two-group quotient kills it and its commutators with D_*. The resulting group is the free class-two exponent-four quotient of $K_Q^{(2)}$, of rank nine. Conversely its defining laws already hold on the image of K_*, so its quotient map factors through $\mathcal{T}_{*,2}$. Its kernel on D_* is exactly E_*, by the averaged quotient and commutator calculation of D.51. The two remaining kernels over Q_* have orders a power of two and a power of three; finite Goursat therefore gives the full displayed fibre product.

The actual $\mathcal{W}_h$ action on (D.57.19) is (id, α_h). The first projection factors through the identity three-primary component. For the second, C_h maps into the two-group kernel over Q_*, so its map factors through the two-primary component, where the action is exactly h. This proves the common action on the full class-two quotient, before using its homology matrices.

There is also an exact way to remove the homology-level based ambiguity without discarding it. The integral graph sequence gives $U_Q^{Q_*} = 0$ and $H^1(Q_*, U_Q) = 0$. For the latter assertion the invariant vertex kernel is generated by $(2, -3)$, since the vertex orbit sizes are 24 and 16; this is exactly the boundary of the invariant edge norm. The two vertex permutation lattices and the edge lattice have zero first cohomology, as their stabilizers are finite and the coefficients are torsion free. Hence two automorphisms of $J_{Q,1}$ with the same action on U_Q and the same identity action on Q_* differ by a unique inner element of U_Q. When both fix t, that element is in U_Q^t. The exact marked graph calculation gives

$$U_Q^t = \mathbb{Z}_2[t^{12}], \tag{D.57.20}$$

including primitivity of the displayed cusp class. Thus the comparison is the actual map $\alpha \mapsto \mathrm{Int}(t^{-12z})\alpha$ for one $z \in \mathbb{Z}_2$. This inner automorphism exists in every source under

discussion and fixes t. We construct the repair after this comparison, then compose with $\mathrm{Int}(t^{12z})$ to recover the original prescribed $\mathcal{W}_h$ action.

After this exact comparison, conjugators for the same $J_{Q,1}$ action differ by the fixed lattices U_Q^s and U_Q^r. The Smith decompositions of $s-1$ and $r-1$ are respectively

$$(1,1,1,2,2,2,0,0,0), \qquad (1,1,1,1,1,3,3,3,0). \tag{D.57.21}$$

Their unimodular change-of-basis matrices are retained; the resulting saturated integral fixed ranks are three and one. In particular these are the integral fixed lattices, not the larger fixed spaces obtained by reducing the two-primary action modulo two.

The finite s-conjugator coordinate in Q_* is one of the two order-four elements 26 and 33 recorded in D.56's ordered matrix list. Indeed a normalized conjugator lies in F, centralizes s, and its finite CM coordinate excludes the central elements. The centralizer in Q_* is a two-group, so its even part lies in the quaternion derived group. Exact integral equations pair state 26 with T_1, and state 33 with T_3. The other two pairings have no integral solution. The recorded row reductions have only denominators $1, 3, 6$, whose two-primary parts divide two; the program explicitly asserts that they divide four before enumerating the five free variables modulo four. Changing any free variable by an element of $4\mathbb{Z}_2$ therefore changes every solved coordinate by an integral element. The final scaling by seven is a unit. Thus the 1024 tested free residue choices exhaust all integral possibilities in each rejected pairing. The normalized r-conjugator has quotient coordinate one, by D.49's coprime centralizer correction. Above each of the two s-coordinates there are nine elements of $C_{G_*}(s)$. Any prescribed one can be used: the actual joint quotient of the three-primary factor with $\mathcal{H}_h$ is their fibre product over Q_*, and coprime order-two fixed-coset lifting through its pro-three kernel lifts the chosen centralizing pair. This only changes a conjugator within its centralizer and leaves the prescribed automorphism and parameter unchanged.

The complete arithmetic central module. All calculations now use the 3888-state marked graph. There are 649 explicit free words. The 40 nonzero character pairs of $A_* \simeq \mathbb{F}_3^4$ split W_* into 40 summands of dimension 16 over $\mathbb{F}_2$. For a character ℓ, the projector is $e_{\ker \ell} + e_{A_*}$; a generator of $A_*/\ker \ell$ acts by $(u,v) \mapsto (v, u+v)$ on each of eight pairs. The invariant exterior square has the following 64-element basis:

$$u_i \wedge v_i, \quad u_i \wedge u_j + u_i \wedge v_j + v_i \wedge v_j, \quad u_i \wedge v_j + v_i \wedge u_j \quad (i<j). \tag{D.57.22}$$

Cross terms between different character pairs and terms with $V_*^{A_*}$ have zero A_*-invariants. Thus the actual invariant module has dimension $40 \cdot 64 = 2560$. The program verifies every projection column and both marked generator actions, then reduces the columns of $1-s$ and $1-r$. Their rank is 2486, giving the actual 74-dimensional coinvariant module. It also evaluates all 48 terms of the remaining deck norm on every coinvariant representative. The full norm rank is 40, while compactification lowers the rank to six. This gives all norm dimensions in (D.57.6) by actual maps.

Write $L(b,a) = (1-s)b + s(1-r)a$ on $W_* \oplus W_*$. It is onto. Its kernel P_* has dimension 640, and the exact same-homology kernel is

$$K_*^{\mathrm{par}} = \{(u_s + d, u_r + d) : u_s \in W_*^s,\ u_r \in W_*^r,\ d \in W_*^t\}. \tag{D.57.23}$$

The three dimensions are 336, 210, and 54; their displayed injection gives total dimension 600. An exact complementary basis of dimension 40 in P_* is retained in the certificate. Each parameter is lifted into the integral moving summand modulo four by the actual 81-term averaging operator. Its even cusp error is removed with the onto map L, yielding an actual integral peripheral pair. This verifies the entire reduction map rather than assigning formal mod-two parameters.

The class-two collection model uses coordinates (a, c), with $a \in (\mathbb{Z}/4)^{649}$ and $c \in \bigwedge^2 \mathbb{F}_2^{649}$. Appending generator j adds the coefficients a_i with $i > j$ to its central coordinates. The program pulls back all 74 coinvariant functionals through (D.57.22), and constructs the exact s, t actions on this collected group from the 649 literal conjugated words. Its resulting local map has rank 64 for each arithmetic seed. Every local column is killed by the compact norm. Since that norm has kernel dimension 68, the four-dimensional residual in (D.57.6) is retained.

The forty-variable equations and their literal zeros. For a fixed homology seed, let d_s be the abelian coordinate of its image of s, multiplied by s^{-1}. If $q_{\mathrm{id}}(b, a)$ is the collected cusp error for the identity seed, the actual affine equation is

$$q_0 + q_{\mathrm{id}}(b, a) + [d_s \wedge s(a - b)] \quad \text{in } (Z_*)_{G_*}. \tag{D.57.24}$$

To verify the formula, the corrected cusp product in the kernel is $b\, s_0(b^{-1}a)\, t(a^{-1})$, multiplied by the original central error. On central coordinates the action of s_0 differs from that of s by exactly the commutator with d_s. The collected-product polar term is

$$[(1 + s)b' \wedge s(a + b)]. \tag{D.57.25}$$

Both formulas are evaluated on the actual integral pair lifts. In particular changing central representatives adds only augmentation columns and cannot change (D.57.24).

The exact local image in (D.57.24) has rank 64. Quotienting by this image gives the stated ten coordinates. The certificate contains the 74-to-10 quotient matrix, all 40 linear coefficients, all 780 quadratic coefficients, and the constant, separately for both branches. The saturated s-fixed lattice of (D.57.21) has three generators. Each has zero obstruction in all 74 coordinates, as does its double. The integral r-fixed generator and its double also have zero obstruction. Cross terms with moving parameters vanish: the plus and moving summands pair to zero under the odd norm, and (D.57.25) gives the same conclusion directly. Thus its complete reduction consists of the eight explicitly retained s-framing cases, with no unrecorded higher two-adic choice.

The search evaluates the full ten-coordinate polynomial and stops at the actual trials in (D.57.7). These are existence searches, not an assertion that all 2^{40} inputs were tested. For each of the sixteen cases, Gaussian elimination then supplies the actual 600-parameter local correction. The corrected words are evaluated anew in the full collected group. They have zero abelian error and zero under all 74 functionals. Independently the literal edge cycle map is evaluated on all 649 generators; it has full rank, preserves the actual ribbon form, commutes with the two marked deck maps, and has the precise rank-nine projection (D.57.4). These are the tests reported in the statement, including their recorded word lengths.

Alignment to the actual arithmetic class-two action. It remains to check that these numerical zeros lift the prescribed source action. The common quotient is the full quotient (D.57.19), on which the actual arithmetic action already exists. Its conjugators have one of the two rank-nine matrices and one of the saturated integral framing classes just exhausted. Choose their actual lifts to $\mathcal{T}_{*,2}$. The onto moving homology map permits their abelian coordinates to be the corresponding ones in the certificate. If the selected literal words differ in their plus central coordinates, correct this difference by the actual onto map

$$\bigwedge^2 V_* \longrightarrow \bigwedge^2 V_*^{A_*}. \tag{D.57.26}$$

It is the commutator map of the averaged quotient in D.51, not a new arbitrary quotient. All 74 functionals kill this plus summand, and changes of central conjugators give only augmentation columns. Hence the computed 74-coordinate zero is unchanged.

After this alignment the cusp error maps to one in (D.57.19), since the prescribed action there fixes t. It is central with zero abelianization, so it belongs to $E_* \cap D'_* = Z_*$. Its zero

image in $(Z_*)_{G_*}$ says exactly that it lies in the augmentation image $(1-s)Z_* + s(1-r)Z_*$. Choose those two central corrections and add them to the conjugators. The cusp identity is now exact in $\mathcal{T}_{*,2}$. The two finite-order relations are preserved because the images remain conjugates of s and r. The rank-649 calculation proves generation of the finite two-group kernel by Frattini; the map on G_* is the identity, so this is an automorphism of the whole finite quotient.

The same correction continues through the full integral homology tower. The moving integral cusp map is onto by odd averaging and the augmentation formula for L. At stage 2^k, with $k \geq 2$, correct the next cusp error by 2^k times a moving integral pair. Such corrections can be chosen as fourth powers of actual $\mathfrak{R}_h$ lifts, and therefore are one in $\mathcal{T}_{*,2}$. They retain the already constructed finite class-two action and the entire $\mathcal{W}_h$ action. Compactness gives compatible conjugators at all levels. The map is invertible on the full two-adic homology kernel because its reduction modulo two is invertible. The exact fibre-product description of D.56 therefore gives the claimed continuous automorphism of (D.57.5). Finally undo the explicitly recorded cusp-inner comparison (D.57.20).

After the initial common-quotient alignment, all changes used to repair the prescribed action lift in $\mathfrak{R}_h \subset \widehat{F}'$. Thus the final normalized parameter still belongs to the image of $\widehat{F}'$ and maps exactly to f_h^*. This proves the retained metabelian and CM statements through their existing source maps.

For clarity, the remaining source kernel in (D.57.8) is nontrivial. Choose an actual element $u \in \mathfrak{R}_h$ whose image in U_*^- is a primitive moving vector, using D.50/D.56's onto map, and a free generator $v \in K_*$ independent of it modulo two. Then $[u,v]^2$ belongs to $\mathfrak{R}_h \cap M_* \cap L_{*,2}$. In the free class-two quotient with abelianization $(\mathbb{Z}/8)^{649}$ and commutator module $\bigwedge^2(\mathbb{Z}/4)^{649}$, its image is $2\bar{u} \wedge \bar{v} \neq 0$. This supplies an actual surviving source element and the explicit comparison detecting it. The inclusion (D.57.9) follows from $M_\infty \subset M_*$ and functoriality of the defining class-two exponent-four laws for $K_3 \subset K_*$. It records the next unsolved lifting map without asserting a full profinite realization. □

Restoring full arithmetic homology while retaining the repaired invariant block

D.58. The D.57 repair extends through the full arithmetic homology tower and the remaining moving class-two module [Proved]

Retain the actual groups, maps and automorphism of D.57. Use $G = G_3$, K_3, $U = K_3^{\mathrm{ab},(2)}$, U^-, $V = U/2U$, $W = U^-/2U^-$, and $Z = W \wedge V$ from D.49–D.54. Let $\theta_U : U \twoheadrightarrow U_*$ and $\theta_Z : Z \twoheadrightarrow Z_*$ be the actual onto maps of D.56. Set

$$\mathcal{L} = \ker(U^- \longrightarrow U_*^-), \qquad S_{57} = \mathfrak{R}_h \cap M_* \cap L_{*,2}, \qquad R_{50} = \mathfrak{R}_h \cap M_\infty. \tag{D.58.1}$$

Then $\mathcal{L}$ is a saturated, free $\mathbb{Z}_2$-module of rank $w - 640$, and

$$\mathrm{im}(S_{57} \longrightarrow U) = \mathcal{L}. \tag{D.58.2}$$

There is an actual continuous peripheral automorphism of

$$Y = \widehat{B}/(\mathfrak{R}_h \cap M_\infty \cap L_{*,2}) \tag{D.58.3}$$

which projects to the complete D.57 automorphism of $X_* = \widehat{B}/S_{57}$, including one compatible pair of its normalized conjugators. It has multiplier one, fixes t, and induces identity on G. It restores the full integral homology source U; its particular homology action is newly repaired.

The exact comparison kernels are

$$\begin{aligned} &\ker(Y \longrightarrow X_*) \simeq \mathcal{L}, \\ &\ker(Y \longrightarrow \mathcal{Z}_h) \simeq Z_*, \qquad\qquad \mathcal{Z}_h = \widehat{B}/R_{50}, \\ &\qquad\qquad Y \simeq \mathcal{Z}_h \times_{\mathcal{T}_{*,2}/Z_*} \mathcal{T}_{*,2}. \end{aligned} \tag{D.58.4}$$

All arrows in this display are induced by the same source $\widehat{B}$. In particular the full small class-two action of D.57 and the entire prescribed $\mathcal{W}_h$ action are retained simultaneously.

There is also a further actual extension. Let

$$X = \widehat{B}/(R_{50} \cap L_2), \qquad \mathcal{K} = \ker(\theta_Z : Z \to Z_*), \qquad \mathcal{K}_0 = \mathcal{K}^A, \quad \mathcal{K}_1 = (1 - e_A)\mathcal{K}. \tag{D.58.5}$$

The identification of Z with $\ker(X \to \mathcal{Z}_h)$ in D.54 is used in this definition. Thus $\mathcal{K}_0, \mathcal{K}_1$ are actual normal subgroups of X, with their displayed G-module structures. The source map is exactly

$$Y = X/\mathcal{K}. \tag{D.58.6}$$

The constructed automorphism of Y extends to an actual continuous peripheral automorphism of

$$X_{58} = X/\mathcal{K}_0. \tag{D.58.7}$$

The projection $X_{58} \to Y$ has the exact kernel $\mathcal{K}_1$. The full remaining class-two map is $X \to X_{58}$, with exact kernel $\mathcal{K}_0$.

For the actual finite arithmetic group put

$$n = 576 \cdot 3^{97}, \qquad q = |A|, \qquad m = n/(6q), \qquad \rho = 1 + n/6, \qquad w = m(q-1).$$

These are the actual quantities of D.49, with $3^4 \le q \le 3^{97}$. The exact dimensions are

$$\begin{aligned} \dim \mathcal{K} &= w\rho - \frac{w(w+1)}{2} - 210240, \\ \dim \mathcal{K}_0 &= \frac{m^2(q-1)}{2} - 2560, \\ \dim \mathcal{K}_1 &= \frac{w\rho}{2} - 207680. \end{aligned} \tag{D.58.8}$$

Thus the extension retains all 2560 invariant central coordinates of the solved actual order-3888 quotient, in addition to the full original moving class-two module. Its remaining invariant kernel is nonzero: already $m \ge 96$ and $q \ge 81$ give $\dim \mathcal{K}_0 \ge 366080$. The order q is the order of the actual normal span of f_h; no new evaluation of that entire span is asserted in this statement.

The normalized parameter of X_{58} still maps to the literal f_h^* in $\mathcal{W}_h$ and hence retains its established full metabelian and prescribed CM comparisons. The actual source kernel of (D.58.7) is

$$R_{58} = \{b \in R_{50} : \text{the image of } b \text{ in } \mathcal{T}_2 \text{ lies in } \mathcal{K}_0\}. \tag{D.58.9}$$

In particular $R_{58}/(R_{50} \cap L_2) = \mathcal{K}_0$. The original full profinite realization and the remaining invariant class-two lifting problem are not asserted to be complete.

The manuscript passages are Part D(e), section 51.16, folio 734 verso, scan 174, and section 51.15, folios 725–727, scans 156–160.

Proof. The exact image of the new source kernel. The onto map $U^- \to U_*^-$ is the integral map proved in D.56, with the original marked source maps. Its target is free over $\mathbb{Z}_2$. Its kernel $\mathcal{L}$ is consequently saturated and free, and its rank is $w - 640$. Odd averaging gives $e_A\mathcal{L} = 0$.

We first evaluate an actual image of an intersection, rather than intersecting two independently computed images. Every element of S_{57} maps into U^-, because it lies in $\mathfrak{R}_h$, and has zero image in U_*, because it lies in M_*. Its image therefore belongs to $\mathcal{L}$. Conversely take $u \in \mathcal{L}$. By D.50's onto map choose an actual $b \in \mathfrak{R}_h$ mapping to u. Then $b \in M_*$. In the actual small class-two quotient its image has zero abelianization and belongs to the image of $\mathfrak{R}_h$, so it lies in Z_*, by D.51 and D.56.

The map

$$R_{50} \longrightarrow Z_* \tag{D.58.10}$$

is onto. Indeed D.51 gives the onto map $R_{50} \to Z$, and D.56 gives the onto marked exterior map $Z \to Z_*$. Choose $c \in R_{50}$ with the same small class-two image as b. Then $bc^{-1} \in S_{57}$, while its full integral homology image is still u. This proves (D.58.2) using compatible source preimages. Its kernel is $S_{57} \cap M_\infty$. Because $M_\infty \subset M_*$, this intersection is exactly the source kernel in (D.58.3), proving the first kernel formula of (D.58.4).

The second formula in (D.58.4) follows from the same onto map (D.58.10): the kernel of $Y \to \mathcal{Z}_h$ is $R_{50}/(R_{50} \cap L_{*,2})$, identified with its actual image Z_* in $\mathcal{T}_{*,2}$. The product of the two original source kernels gives the maximal common quotient $\mathcal{T}_{*,2}/Z_*$. Closed Goursat then gives the displayed fibre product, including both original projection maps.

The common quotient at every integral homology level. For $k \geq 1$, put

$$M_k = \overline{K_3' K_3^{2^k}}, \qquad J_k = \widehat{B}/M_k, \qquad U_k = U/2^k U, \qquad \mathcal{L}_k = \mathcal{L}/2^k \mathcal{L}. \tag{D.58.11}$$

Saturation identifies $\mathcal{L}_k$ with its actual image in U_k. Equation (D.58.2) proves that the maximal common marked quotient of X_* and J_k is exactly $J_k/\mathcal{L}_k$.

The D.57 automorphism descends to this common quotient. To verify the assertion directly, X_* maps to $G = G_3$ through its retained $\mathcal{W}_h$ projection; that action is the identity. The image of K_3 in X_* is therefore invariant. Taking its abelian quotient of exponent 2^k is characteristic. Its actual kernel is the image of M_k. Hence the resulting quotient is $\widehat{B}/(S_{57}M_k) = J_k/\mathcal{L}_k$, with the induced automorphism and the actual images of the fixed D.57 conjugators. This descent does not assume a lift of any centralizer element from $\mathcal{W}_h$ to the old D.50 homology action.

An actual first-level peripheral endomorphism. Write $J = J_1$, $V = U_1$, and $L = \mathcal{L}_1$. Lift the two prescribed conjugators from J/L to J. Their images in G centralize s, r, respectively, since the lower action fixes G. Their two conjugates have the required orders, and their product differs from t by an element of L.

Left corrections $b, a \in L$ change this error by exactly

$$(1 - s)b + s(1 - r)a. \tag{D.58.12}$$

Its image is the augmentation image $I_G L$. Since $e_A L = 0$, odd averaging gives $[L, A] = L$, and therefore $L_G = 0$. Thus (D.58.12) is onto. Kill the cusp error with an actual pair of these corrections. The resulting homomorphism from $\widehat{B}$ is identity on G, fixes t, and sends K_3 to the elementary abelian kernel V. It kills M_1 and hence defines an actual peripheral endomorphism $\Phi : J \to J$. It retains both conjugators and the automorphism on the whole quotient J/L.

Projective generation repair inside the retained kernel. We spell out the repair because it must remain inside L. The actual marked Bass–Serre graph gives

$$0 \longrightarrow V \longrightarrow \mathbb{F}_2[G] \xrightarrow{\partial} \mathbb{F}_2[G/\langle s \rangle] \oplus \mathbb{F}_2[G/\langle r \rangle] \longrightarrow \mathbb{F}_2 \longrightarrow 0. \tag{D.58.13}$$

The chosen conjugators define the edge chain map D of D.50. Its vertex maps are the actual permutations given by their G-coordinates, which centralize the corresponding stabilizers. They are invertible. The exact path identity $D\mathfrak{p}(g) = \mathfrak{p}(\Phi(g)) + (g - 1)\mathfrak{p}(\beta)$ proves that the restriction of D to V is the actual homology map T.

Let P be the nilpotent Fitting summand of D on the regular edge module. Invertibility on the vertex module forces $P \subset V$. The induced map on V/L is invertible, since the quotient action on J/L is an automorphism. Consequently $P \subset L$. The Fitting decomposition is G-linear, so P is a direct summand of the regular module and is projective. Let e be its G-linear projection.

The exact path cocycle has values $\mathfrak{p}(s) = 0$ and $\mathfrak{p}(r) = e_1 - e_r$. The derivation $\delta_0 = e\mathfrak{p}$ restricts to $e|_V$. The cusp has order 36 in G, and $\Phi(t) = t$, so the cycle $[t^{36}]$ is fixed by T and killed by e. Therefore

$$(1 + t + \cdots + t^{35})\delta_0(t) = 0. \tag{D.58.14}$$

Restriction of a projective $\mathbb{F}_2[G]$-module to the cyclic subgroup generated by t remains projective. In a free cyclic-group module the norm kernel is the image of $1 - t$, by its coefficient sums; direct summands retain this exactness. Choose $b \in P$ with $(1 - t)b = \delta_0(t)$, and subtract the inner derivation $(1 - g)b$. The resulting δ fixes the cusp and has

$$\delta(s) = (1 - s)b, \qquad \delta(r) = (1 - r)(e(e_1) + b), \qquad \delta|_V = e|_V. \tag{D.58.15}$$

These are actual left corrections to the two conjugators inside $P \subset L$. The corrected map has homology map $T + e$, invertible on both Fitting summands, and remains identity on G. Thus it is an automorphism of J, preserving the entire prescribed common quotient and the actual conjugators there. No division by the even cusp order occurs in this construction.

All higher levels and compatible conjugators. Suppose level $k - 1$ has been constructed together with its X_* component. Its source kernel is $S_{57} \cap M_{k-1}$. Its image in J_k is exactly

$$\mathcal{L}_k \cap 2^{k-1}U_k = 2^{k-1}\mathcal{L}_k. \tag{D.58.16}$$

For the first equality, an element of S_{57} whose image in J_k reduces to one already belongs to M_{k-1}; this is the actual preimage property. The second uses saturation of $\mathcal{L} \subset U$.

Thus the next maximal common quotient is $J_k/2^{k-1}\mathcal{L}_k$. Its two quotient maps identify it with the fibre product of J_{k-1} and $J_k/\mathcal{L}_k$ over their original common quotient. The previously constructed action and the D.57 action agree there, and their conjugators agree as elements. Lift those two elements to J_k. Their cusp error lies in $2^{k-1}\mathcal{L}_k \simeq L$, where (D.58.12) is onto. These corrections preserve both previous conjugators and the entire X_* component. The resulting map is an automorphism: its homology matrix reduces to the already invertible matrix at level one.

Successive compatible choices give automorphisms, inverses and one pair of conjugators in the inverse limit. Closed Goursat and (D.58.2) identify this limit with (D.58.3). This proves the first actual extension and all its comparison assertions.

The exact full class-two kernel. The source inclusion $L_2 \subset L_{*,2}$ comes from the inclusion $K_3 \subset K_*$ and the defining class-two exponent-four laws. Thus there is an actual projection $X \to Y$. Its kernel is the image in $\mathcal{T}_2$ of $R_{50} \cap L_{*,2}$. The image of R_{50} is exactly Z. An element of this image is represented by a source element in $L_{*,2}$ exactly when its image in $\mathcal{T}_{*,2}$ is one. Hence the kernel is exactly $\ker \theta_Z = \mathcal{K}$, with both inclusions furnished by those same source representatives. This proves (D.58.6).

The constructed Y action has identity G-action and full integral homology automorphism T. Its map to the retained small class-two action gives $\theta_V T = T_* \theta_V$. Therefore $\bigwedge^2 T$ preserves $\mathcal{K}$, and it preserves $\mathcal{K}_0, \mathcal{K}_1$ because it commutes with the G-action and the central averaging operator e_A.

Lift the two actual Y conjugators to $X/\mathcal{K}_0$. Their product differs from t by an element of $\mathcal{K}/\mathcal{K}_0 = \mathcal{K}_1$. This is an abelian module with action through G. Its A-coinvariants vanish, so

$$(1 - s)\mathcal{K}_1 + s(1 - r)\mathcal{K}_1 = \mathcal{K}_1. \tag{D.58.17}$$

Kill the error by these actual central conjugator corrections. The finite-order relations remain exact and the cusp is fixed.

To verify factorization, the resulting source map already projects to the constructed automorphism of $\mathcal{Z}_h$. Its finite class-two projection is identity on G and sends K_3 into a class-two exponent-four kernel, so it kills L_2. On its commutator module the map is $\bigwedge^2 T$, which preserves $\mathcal{K}_0$, as just proved. It therefore induces a homomorphism of $\mathcal{T}_2/\mathcal{K}_0$. It is an automorphism by its invertible Frattini map and identity quotient map. These two automorphisms agree on their original common quotient. Their fibre product is exactly $X/\mathcal{K}_0$, so the corrected source map is the claimed continuous automorphism of (D.58.7). The conjugators project to the fixed Y pair, and hence to the fixed D.57 pair, throughout this construction.

Dimensions and the remaining actual source kernel. The map θ_Z is onto. Odd averaging is exact, so it also gives the onto map $Z^A \to Z_*^{A_*}$, whose kernel is $\mathcal{K}_0$; the action on the target factors through the actual quotient $A \twoheadrightarrow A_*$. D.51 gives $\dim Z = w\rho - w(w+1)/2$, $\dim Z^A = m^2(q-1)/2$, and $\dim Z^- = w\rho/2$. D.56–D.57 give the respective target dimensions 210240, 2560 and 207680. Their differences prove (D.58.8) with the exact module maps, and (D.58.9) is the preimage of the remaining $\mathcal{K}_0$ under $R_{50} \to Z$.

For an actual surviving source element, choose a nonzero $z \in \mathcal{K}_0$, which exists by the proved positive dimension, and an actual source preimage $b \in R_{50}$ under the onto map to Z. Then $b \in R_{58}$, while its image in X is the specified nonzero z. This detects the source kernel through an explicit quotient map. For the constructed X_{58} action, the next invariant boundary is the actual class of the lifted cusp error in $(\mathcal{K}_0)_G$; changing its two central lifts changes that error by the corresponding augmentation columns. That class has not been evaluated here. All parameter and CM comparisons claimed in the statement follow from the retained $\mathcal{W}_h$ projection and $\mathfrak{R}_h \subset \widehat{F}'$. □

E. The next approximations and exactness on coverings

We write $F = \langle x, y \rangle$ for the discrete group, $F^{(i)}$ for its derived series, and $\gamma_i F$ for its lower central series, with $F^{(0)} = \gamma_1 F = F$. All commutators in this section are $[a, b] = aba^{-1}b^{-1}$. The puncture used to define depth is the puncture labelled 1; it is additional data. The manuscript anchors are §51.16, folios 733v–736v, scans 172–178, especially (371)–(379). The arithmetic model in those passages is $X_n : x_0^n + x_1^n + x_\infty^n = 0$, with U_n obtained by deleting the coordinate divisors. We retain that model over $\mathbb{Q}$.

Derived quotients and the depth filtration

E.1. The third derived quotient and its exact relation module [Proved]

Put $Q = F/F''$ and $R = \mathbb{Z}[Q]$. There is an exact sequence of discrete groups and an exact sequence of left R-modules

$$1 \longrightarrow F''/F''' \longrightarrow F/F''' \longrightarrow Q \longrightarrow 1, \qquad 0 \longrightarrow F''/F''' \longrightarrow R^2 \xrightarrow{d} R \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0,$$

where $d(a, b) = a(\bar{x} - 1) + b(\bar{y} - 1)$, with right factors in this formula and the regular left action on each term. This is an explicit module description over the noncommutative ring R.

Proof. The cover of the two-loop graph corresponding to the normal subgroup F'' has vertices Q and oriented edges $(q, x), (q, y)$ from q to $q\bar{x}, q\bar{y}$. Its cellular differential is the displayed map. The graph is connected, so its reduced zeroth homology vanishes. Its fundamental group is

F'', and its first homology is $F''/[F'',F''] = F''/F'''$. Lifting paths identifies conjugation by F with the left deck action of Q. These identifications give every arrow, its kernel and its image. The group sequence is the quotient sequence for the nested normal subgroups $F''' \subset F''$.

For the algebraic constructions below we use the rational relative unipotent completion over a specified finite quotient, constructed in E.14. For an unbased finite-dimensional representation problem one can also use the affine proalgebraic group with coordinate Hopf algebra the matrix coefficients of all finite-dimensional rational representations of F/F'''. Evaluation is the canonical group map into its rational points, and factoring matrix coefficients proves its universal property. These constructions specify an envelope and its comparison maps; they do not require a finite-index abelian subgroup of F/F'''. □

E.2. The depth-two Lie algebra and its comparison with the derived quotient [Proved]

Let $L = \mathbb{L}_{\mathbb{Q}}(x,y)$, graded by total degree, and let $D^r L$ be the span of Lie monomials with at least r occurrences of y. Put $v_i = \mathrm{ad}_x^i(y)$ and $V = \bigoplus_{i\geq 0} \mathbb{Q} v_i$. There is an isomorphism of graded Lie algebras

$$L/D^3L \simeq \mathbb{Q}x \ltimes (V \oplus \Lambda^2 V).$$

Here $[v_i, v_j] = v_i \wedge v_j$, the bracket of V with $\Lambda^2 V$ vanishes, and $[x, v_i] = v_{i+1}$; the action of x on $\Lambda^2 V$ is the induced derivation. In weight $n \geq 2$ the depth-two dimension is $\lfloor (n-1)/2 \rfloor$. The identity on x, y induces the surjection $L/L''' \twoheadrightarrow L/D^3L$ with kernel D^3L/L'''.

Proof. We give the elimination argument. Jacobi moves every occurrence of x in a bracket in the ideal generated by y onto a bracket entry. Consequently that ideal is generated as a Lie algebra by the v_i. In the tensor algebra, the leading words of the v_i are $x^i y$. The words obtained by concatenating these blocks have unique factorizations at the occurrences of y. Thus the homomorphism from the tensor algebra on the symbols v_i into $\mathbb{Q}\langle x, y\rangle$ is injective, by its triangular leading words. Its primitive Lie subalgebra is the free Lie algebra on the v_i. Keeping at most two such symbols yields $V \oplus \Lambda^2 V$. The derivation formula follows from Jacobi. A basis in weight n is $v_i \wedge v_j$ with $i < j$ and $i + j = n - 2$, which has the stated cardinality.

Every element of L' has depth at least one, hence $L'' \subset D^2 L$ and $L''' \subset D^4 L \subset D^3 L$. This proves the displayed quotient map and its kernel. The element $[y, [y, [y, x]]]$ has weight four and belongs to $D^3 L$; its tensor expansion is nonzero, whereas L''' has no weight-four part. Thus the kernel is nonzero. Also $[[x, y], [x, [x, y]]]$ survives in $\Lambda^2 V$, so the identity on x, y does not descend from L/L'' to $L/D^3 L$: its defining kernel L'' contains this explicitly nonzero image. Completion for the total-degree filtration gives the same maps of pronilpotent Lie algebras, and exponentiation gives maps of rational prounipotent groups. No integral exponential is used. □

E.3. The rational Galois calculation in depth two through weight fourteen [Known]

Normalize the depth-one generators by $\bar{\sigma}_{2m+1} = \mathrm{ad}_x^{2m}(y)$. For the positive-weight Lie algebra of the outer pro-ℓ Galois image, tensored with $\mathbb{Q}_\ell$, the depth-one dimensions are one in weights $3, 5, 7, 9, 11, 13$ and zero in the other weights at most fourteen. Its depth-two dimensions in weights $1, \ldots, 14$ are

$$0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 2.$$

The first relation among the depth-one brackets is

$$\{\bar{\sigma}_3, \bar{\sigma}_9\} - 3\{\bar{\sigma}_5, \bar{\sigma}_7\} = 0 \quad \text{in depth two.}$$

This is a statement after tensoring with $\mathbb{Q}_\ell$ and taking the specified filtrations; it asserts no integral saturation index.

Proof. We use Hain–Matsumoto, Theorem 8.4, its application in §8.4, Lemma 8.5 and Remark 8.6 of the opened version [HM03], and Brown's depth-two bracket formula and exact sequence (7.8), Example 7.2 [Bro21]. Here U_1 has $H^1_{\mathrm{et}}(U_{1,\overline{\mathbb{Q}}}, \mathbb{Q}_\ell(1)) = \mathbb{Q}_\ell^2$ with trivial Galois action: the two Kummer covers are those of t and $1-t$. Its outer pro-ℓ action is unramified away from ℓ. The indicated theorem therefore identifies the weight-graded Lie algebra of its Zariski closure with the rationalized graded Galois image. The weight-one generator maps to zero by Lemma 8.5; the odd generators in weights at least three have nonzero depth-one coefficient, as recorded in Remark 8.6. This supplies actual Galois elements after rationalization, rather than only solutions of a linear equation.

The polynomial form of the depth-two bracket is

$$\begin{aligned} B_{a,b}(X,Y) = {} & X^a\bigl(Y^b - (Y-X)^b\bigr) + (Y-X)^a\bigl(X^b - Y^b\bigr) \\ & + Y^a\bigl((Y-X)^b - X^b\bigr), \qquad a, b \text{ positive and even.} \end{aligned}$$

One can derive this formula directly by substituting $D_f(x) = 0$, $D_f(y) = [y, f]$ into $[f, g] + D_f(g) - D_g(f)$ in the tensor algebra, and reading the two y-blocks from the right. The program `e_depth.py` does both expansions independently over the integers and compares every coefficient. The only pairs in weights $8, 10, 12, 14$ are, respectively, $(3,5)$, $(3,7)$, $(3,9), (5,7)$, and $(3,11), (5,9)$. Their exact ranks are $1, 1, 1, 2$. The other weights contain no pair.

For completeness of this span in depth two, every bracket word of length at least three in the Galois generators has depth at least three. A generator itself has a nonzero depth-one image in its weight; different generator weights cannot cancel in a weight-graded piece. Thus the generation theorem reduces the calculation exactly to the displayed pairs. The polynomial

$$X^2Y^2(X-Y)^3(X+Y)^3 = X^8Y^2 - 3X^6Y^4 + 3X^4Y^6 - X^2Y^8$$

satisfies the antisymmetry and the three-term period equation. Substitution in $B_{a,b}$ gives the displayed relation, also checked word by word in the free tensor algebra. Brown's exact sequence identifies its line with the weight-twelve cusp-form period line. This verifies the requested modular relation without making a freeness assertion for an integral Galois Lie algebra. □

E.4. What the metabelian representation forgets about these brackets [Proved]

Under the canonical Lie homomorphism $L \to L/L''$, every Ihara bracket of two elements of L' maps to zero. The depth-two image of $\{\bar\sigma_3, \bar\sigma_5\}$ is nonzero. Thus the specified comparison maps exhibit information already added in weight eight; the first dependence between these new bracket directions occurs in weight twelve as in E.3.

Proof. Write the derived module of L/L'' as $\mathbb{Q}[X,Y]c$, where $c = [x, y]$ and X, Y act by $\mathrm{ad}_x, \mathrm{ad}_y$. For $f = pc$ one has $D_f(c) = [x, [y, pc]] = XYpc$. The derivation D_f vanishes on the operators X, Y when applied to the derived module: its correction there is a bracket of two derived elements. Hence $D_f(qc) = XYpq\,c = D_g(pc)$ for $g = qc$, while $[f, g] = 0$. The Ihara bracket therefore vanishes. In E.3 the polynomial $B_{2,4} = 2X^5Y - 5X^4Y^2 + 5X^2Y^4 - 2XY^5$ is explicitly nonzero. Both claims concern the images of the same derivations through the two quotient maps just constructed. □

Localizers and finite coverings

E.5. Two-out-of-three and relative Theorem A [Proved]

For each of abelianization, the metabelian quotient, a fixed nilpotent quotient $G/\gamma_{c+1}G$ with $c \geq 1$, pro-p completion, and rational prounipotent completion, the class $W_{\mathfrak{a}}$ in the question satisfies two-out-of-three, retract closure, the terminal-object axiom and relative Theorem A for all small categories.

Proof. We first specify the tests determining each class. They are the classifying groupoids BH for, respectively, all abelian groups, all metabelian groups, all groups of nilpotence class at most c, all finite p-groups, and all groups $U(\mathbb{Q})$ of rational points of finite-dimensional unipotent rational algebraic groups. We include all discrete sets as tests to detect components. For a category C, functors $C \to BH$ invert every arrow and therefore factor uniquely through its fundamental groupoid. For a connected pointed category the homotopy fiber of evaluation $\mathrm{Fun}(C, BH) \to BH$ is equivalent to the discrete groupoid $\mathrm{Hom}(\pi_1 C, H)$. The universal properties of the five approximations, followed by Yoneda, prove that a component-bijective functor is in $W_{\mathfrak{a}}$ precisely when it induces an equivalence of these functor groupoids for every indicated test. Conjugations cause no loss here: the evaluation fiber restores the pointed homomorphisms. Two-out-of-three and retract closure follow from equivalences of groupoids. A category with a terminal object has contractible nerve and passes every test.

Here is the relative argument. For $C \to S$, the Grothendieck construction of $s \mapsto C/s$ has objects $(s, c, u : f(c) \to s)$. Its forgetful functor to C has a section $c \mapsto (f(c), c, 1)$ and a natural transformation from section followed by forgetful to the identity, furnished by u. It therefore has the same fundamental groupoid as C. Functors from this construction to a groupoid T consist exactly of functors $C/s \to T$ with invertible coherent transition transformations along the arrows of S. This is the pseudolimit of the groupoids $\mathrm{Fun}(C/s, T)$. A pointwise equivalence of such diagrams induces an equivalence of pseudolimits: choose objectwise lifts, transport the transitions through full faithfulness, and use faithfulness to verify their coherence; the same argument applies to morphisms. A map $C \to D$ over S which passes every test on every comma category consequently passes the test on $C \to D$. This is the stated relative Theorem A. □

E.6. The exact comparison maps for quotient approximations [Proved]

Let $q : F \twoheadrightarrow G$ be any finite quotient, $N = \ker q$, and $V(H)$ one of H', H'', or $\gamma_{c+1}H$. There is a natural short exact sequence

$$1 \longrightarrow \frac{N \cap V(F)}{V(N)} \longrightarrow \frac{N}{V(N)} \longrightarrow \ker\left(\frac{F}{V(F)} \longrightarrow \frac{G}{V(G)}\right) \longrightarrow 1.$$

The comparison requested with target G itself is the specialization $V(G) = 1$ of this exact sequence. The map $F/V(F) \to G$ with composite q exists exactly when $V(G) = 1$.

Proof. Surjectivity of q gives $q(V(F)) = V(G)$, by the definitions of the verbal subgroups. The inverse image of $V(G)$ is $NV(F)$, so the kernel in the display is $NV(F)/V(F)$. The map from N onto it has kernel $N \cap V(F)$ and kills $V(N)$. This proves exactness and identifies every map. A factorization of q through $F/V(F)$ kills $q(V(F)) = V(G)$; the converse is the same calculation. The formulas thus also give an explicit replacement target for every quotient for which the arrow to G written in the question does not descend. □

E.7. All nontrivial finite quotients have a nonzero comparison kernel [Proved]

For every finite quotient $F \twoheadrightarrow G$ of order $d > 1$, each of the three comparison kernels in E.6 is nonzero. For abelianization it is a free abelian group of rank $d-1$; the image is a rank-two lattice of index $|G^{\mathrm{ab}}|$ in F^{ab}. For the nilpotent quotient of class c its rational Hirsch rank is

$$\sum_{j=1}^{c}(w_j(d+1)-w_j(2)), \qquad w_j(r)=\frac{1}{j}\sum_{e\mid j}\mu(e)r^{j/e}.$$

Proof. The connected Cayley graph of G has d vertices and $2d$ edges. A maximal tree gives $d+1$ free generators of N, hence $N^{\mathrm{ab}}=\mathbb{Z}^{d+1}$. The abelian image and its index follow from E.6. A subgroup of a free abelian group is free, and the rank of the kernel is $(d+1)-2=d-1$.

For the nilpotent case the domain is the free nilpotent group on $d+1$ generators and the image is a finite-index subgroup of the free nilpotent group on two generators. Both are torsion-free. Their lower central ranks are the Witt numbers: the tensor algebra identity $1/(1-rt)=\prod_{j\geq 1}(1-t^j)^{-w_j(r)}$, obtained by ordering a homogeneous Lie basis in the universal enveloping algebra, gives the displayed formula by logarithms and Möbius inversion. Hirsch rank is additive in a short exact sequence of torsion-free nilpotent groups. The first summand is already $d-1$.

For the metabelian case choose $a \in N \cap F'$ with nonzero image A in N^{ab}; such an element exists by the abelian calculation. Choose $t \in N$ whose abelian class T is linearly independent from A over $\mathbb{Q}$. In the free metabelian quotient of N the module N'/N'' injects into $\mathbb{Z}[N^{\mathrm{ab}}]^{d+1}$ by the covering-graph differential. It is therefore torsion-free over this integral domain. The image of $[t,a]$ is nonzero: its degree-two term is $T \wedge A$. The element

$$w=[a,tat^{-1}] \in N \cap F''$$

equals $[a,[t,a]]$ and has image $(A-1)[t,a]$ in that module for our convention $[u,v]=uvu^{-1}v^{-1}$. This image is nonzero. Thus $w \notin N''$, giving a specific construction of a nontrivial element of $(N \cap F'')/N''$ for every such quotient. For $d=1$, $N=F$ and every comparison is the identity. □

E.8. The pro-p comparison and the precise exactness criterion [Proved]

Write $G^{(p)}$ for the maximal p-group quotient of the finite group G. There is a continuous surjection

$$\widehat{N}_p \longrightarrow \ker(\widehat{F}_p \longrightarrow G^{(p)}).$$

Its source is free pro-p of rank $|G|+1$ and its target is free pro-p of rank $|G^{(p)}|+1$. It is an isomorphism exactly for p-groups G. For p-groups the target is the open subgroup corresponding to the given cover. More explicitly, put $r=|G|+1$ and $s=|G^{(p)}|+1$. There are free pro-p bases $a_1,\dots,a_s,b_1,\dots,b_{r-s}$ in the source and $\bar{a}_1,\dots,\bar{a}_s$ in the target for which the comparison sends $a_i \mapsto \bar{a}_i$ and $b_j \mapsto 1$. Its kernel is exactly the closed normal subgroup generated by those $r-s$ elements b_j. This description is noncanonical; the original continuous comparison is canonical.

Proof. The image is the closure of N. The quotient by that closure is the maximal pro-p quotient of the finite group F/N, namely $G^{(p)}$. This proves surjectivity. For a normal subgroup of p-power index the induced pro-p topology is its full topology: the core in the ambient group of a normal subgroup of finite p-power index is the intersection of finitely many conjugates, and its quotient is an extension of finite p-groups. Completing therefore identifies $\widehat{N}_p$ with the open subgroup when G is a p-group. The Schreier rank calculation applied to finite p-quotients, or

the covering tree, gives the two ranks. An isomorphism would identify their Frattini quotients, whose dimensions are these ranks, and hence forces $|G| = |G^{(p)}|$. This equality says precisely that G is a p-group. All arrows used here are continuous homomorphisms of compact pro-p groups.

For the kernel description, lift a target basis to elements $a_1, \dots, a_s$. Their classes in the source Frattini quotient are independent. Extend them to a basis of that vector space and choose lifts b_j. The pro-p basis theorem and Hopficity show that the resulting r elements form a free basis. The subgroup generated by the a_i maps isomorphically onto the target: the induced surjection of free groups of rank s is an isomorphism by Hopficity. For each b_j, choose the unique word $w_j(a_1, \dots, a_s)$ with its target image and replace b_j by $b_j w_j^{-1}$. The Frattini classes remain a basis, so these are again free generators. The quotient by their closed normal closure is now the free group on the a_i, proving the stated kernel and image, not merely their ranks. □

E.9. Ordinary rational completion of a finite-index subgroup [Proved]

For $|G| = d$, the canonical morphism of rational prounipotent groups $N^{\mathrm{un}} \to F^{\mathrm{un}}$ is surjective. Its map on abelianized Lie algebras is a surjection $\mathbb{Q}^{d+1} \to \mathbb{Q}^2$ with kernel of dimension $d - 1$. It is an isomorphism exactly for $d = 1$. In suitable noncanonical free pronilpotent coordinates, its Lie algebra map sends X, Y to the two target generators and $z_1, \dots, z_{d-1}$ to zero. Its kernel is the closed ideal generated by the z_j; by free-Lie elimination it is the completed free Lie algebra on $\mathbb{Q}\langle X, Y\rangle \otimes \langle z_1, \dots, z_{d-1}\rangle$, where a word acts by iterated adjoints. Exponentiation is the exact kernel group. Every morphism from a connected unipotent rational algebraic group to the constant finite group G is trivial; thus the specified map to G in a kernel formula descends only for the trivial quotient. The preceding surjection is the comparison obtained from the trivial completion of G.

Proof. A homomorphism from F to a unipotent group has the same Zariski closure on N as on F: the quotient of the closures has finite order, while a unipotent group in characteristic zero is connected and admits no nontrivial finite quotient. Equivalently, every $f \in F$ has a positive power in N, and taking roots in the one-parameter unipotent subgroup containing its image recovers that image. The resulting map of completions is surjective. Both Lie algebras are completed free Lie algebras on their abelianizations, of ranks $d + 1$ and two. The abelian comparison is the rationalization of E.7. A morphism from a connected scheme to a constant finite scheme is constant, which proves the last assertion and types the replacement map.

For the coordinate assertion, lift the target free Lie generators and complete their independent linear terms to a basis of the source abelianization. The homomorphism of completed free Lie algebras determined by these lifts is invertible: its map in each graded degree is invertible, and degreewise inversion converges. Its first two generators give a section of the target free Lie algebra. Subtract from every remaining generator the section of its image. This triangular change is again invertible and puts that image equal to zero. The kernel is consequently the closed ideal generated by the remaining z_j. Jacobi writes this ideal as the Lie algebra generated by the iterated X, Y adjoints of the z_j. In the free associative algebra their leading words have unique cuts at the letters z_j, proving freeness exactly as in E.2. This also computes the whole kernel for every finite quotient, including those for which the proposed map to the original G has no descent. □

E.10. Which finite regular coverings preserve the localizers [Proved]

For a regular covering with monodromy $q : F \twoheadrightarrow G$, membership in $\mathrm{Fib}_{W_{\mathfrak{a}}}$ holds only for $G = 1$ for abelianization, the metabelian quotient, every fixed positive nilpotent class, and

rational prounipotent completion. For pro-p completion it holds exactly for finite p-groups G. These are exactly the cases of isomorphism in E.7–E.9, with the properly descended target of E.6 or E.8.

Proof. We first give uniform counterexamples over BF, so the universal base-change requirement is tested, not merely the given fiber. For a nontrivial finite G, choose a prime $r \nmid |G|$ and let V be the augmentation-zero summand of the regular $\mathbb{F}_r[G]$-module. Averaging shows $V_G = 0$, and $V \neq 0$. Let $A = V \rtimes F$ for the action through q, with retraction $A \to F$. The inclusion $F \to A$ is an abelian and every nilpotent equivalence: $[F, V] = V$ puts V in every term $\gamma_j A$, $j \geq 2$. It is also a rational unipotent equivalence, since the finite normal subgroup V maps trivially to every rational unipotent group. Pullback along the G-cover replaces the inclusion by $N \to V \times N$. Its abelian and nilpotent quotients retain V, so they fail those tests. For the rational test use instead the integral augmentation ideal $V = \ker(\mathbb{Z}[G] \to \mathbb{Z})$. Every homomorphism $A = V \rtimes F \to U(\mathbb{Q})$, with U a rational unipotent group, kills V. To prove this, take the Zariski closure of the image of V. It is an abelian unipotent group. Conjugation by each image of an element of F acts unipotently on its Lie algebra and has finite order, since the action on V factors through G. A finite-order unipotent linear operator in characteristic zero is the identity. Hence the image factors through V_G, which is finite because its rationalization is zero. A unipotent group in characteristic zero has no nontrivial finite subgroup, so that image is trivial. The universal property therefore identifies the rational prounipotent completions of F and A. After pullback, $N \to V \times N$ retains the nonzero vector quotient $V \otimes \mathbb{Q}$, of dimension $d - 1$.

For the metabelian test form a finite quotient $H = F/(N'N^r)$, with r any prime. The inclusion $N \cap F' \hookrightarrow N$ has nonzero primitive abelian image of rank $d-1$, by E.7. Its reduction modulo r remains nonzero. Thus H' is nontrivial and the image of N' is trivial. Choose a prime $s \nmid |H|$ and put $V = (1 - e_{H'})\mathbb{F}_s[H]$, where $e_{H'} = |H'|^{-1} \sum_{h \in H'} h$. This is a nonzero H-module with $[H', V] = V$. For $A = V \rtimes F$ one has $A'' = V \rtimes F''$, so $F \to A$ induces an isomorphism of metabelian quotients. On the pulled-back group $V \rtimes N$, N' acts trivially on V; its second derived subgroup is $\{0\} \rtimes N''$. Consequently the metabelian quotient retains V, and $N \to V \rtimes N$ fails the test. These are explicit finitely presented group constructions. Their standard finite presentation complexes and mapping cylinders give finite simplicial-category models with the same fundamental-group calculation.

For a finite p-group cover, E.8 applies to every group occurring after any base change, not just F. An isomorphism of pro-p completions identifies the finite orbit sets and the appropriate open stabilizer subgroups, whose completions are those open subgroups. Hence pullback preserves W_p.

For a regular cover with G not a p-group, let P be a Sylow p-subgroup. The image of $F \to G \times (F/F'F^p)$ is the fiber product over $G/G'G^p$: this follows by taking the two projections and their kernels, or directly from $q(F'F^p) = G'G^p$. The map $P \to G/G'G^p$ is surjective, because its cokernel has order both a power of p and prime to p. Choose words u, v mapping into P whose images in $F/F'F^p$ are x, y. The endomorphism $x \mapsto u, y \mapsto v$ induces a pro-p automorphism: it is onto the Frattini quotient, hence onto the finitely generated free pro-p group, and that group is Hopfian. Its pulled-back regular G-set has image contained in the proper subgroup P, and is therefore disconnected. This violates the component condition in W_p. Finally, a trivial covering is a product with a discrete set and preserves all five classes. □

E.11. The explicit three-letter obstruction in every nilpotent approximation [Proved]

The endomorphism $\alpha(x) = x$, $\alpha(y) = y[x, y]$ induces the identity on every graded lower central quotient and an automorphism on every nilpotent quotient, every pro-p completion

and the rational prounipotent completion of F. Pullback of the action $x \mapsto (12), y \mapsto (13)$ changes its orbit set from one orbit to $\{1,2\},\{3\}$. It consequently refutes intrinsic-fibration membership for this cover for each of these approximations, and for every functor of their inverse system which sends isomorphisms to isomorphisms.

Proof. Substituting a generator times a commutator in an iterated commutator changes it only in the next lower central degree. Thus the graded action is the identity. Induction in the central extensions gives an automorphism on each nilpotent quotient. The Frattini argument of E.10 proves the pro-p statement; the completed free rational Lie algebra gives the unipotent statement by degreewise inversion. The inverses commute with the transition maps, so the map of inverse systems is an isomorphism. For $a = (12), b = (13)$, multiplication gives $b[a,b] = a$. Both pulled-back generators therefore act by (12). The program `e_obstructions.py` verifies these permutations and the orbit sets. The ordinary metabelian quotient retains the map to S_3, since $S_3'' = 1$; the map of E.11 is not substituted for a metabelian equivalence. □

E.12. Exhaustive small-quotient and Schreier verification [Proved]

For all ordered generating pairs in every group model of order at most twenty-four, the program `e_smallgroups.py` constructs an integral cycle basis of N^{ab}, the comparison to F^{ab}, and the deck character; it computes the free-metabelian relation-module generic rank and the first four lower central ranks. Every generated kernel has rank $|G|+1$, the comparison lattice has index $|G^{\mathrm{ab}}|$, and the rational deck character is $1+\mathrm{reg}_G$. All checks pass.

Proof. The input is the complete order-at-most-twenty-four list in the opened primary GroupNames database [Gro]. It contains seventy-four groups. The accompanying input file records the permutation generators and exact source URLs. The program closes each set of generators, checks its order, constructs its multiplication table, and enumerates every ordered pair. It tests generated subgroup size without a presentation heuristic. For every generating pair it constructs the $2|G|$ edges and a spanning tree, and checks that every fundamental cycle has zero boundary. The non-tree edge coefficients form an integral basis, so this proves the integral calculation. The gcd of all two-by-two minors of the pushforward matrix checks the Smith index. Deck traces are calculated for every group element. Independent tests exhaust the images of a generating pair in every other two-generated model of the same order to verify pairwise nonisomorphism of the models used.

The lower central ranks are calculated from the proved Witt formula in E.7. For the free group N on $r = |G|+1$ generators, the graph over N^{ab} gives $N'/N'' = \ker(R_N^r \to R_N)$, $R_N = \mathbb{Z}[N^{\mathrm{ab}}]$; localization to its fraction field has dimension $r-1 = |G|$. Thus the printed free-metabelian invariant has an exact module presentation behind it. The output is accompanied by `e_all_surjections.csv`, with a row for every ordered epimorphism, and integral cycle witnesses for every group. The formulas E.6–E.10 prove the assertions for all finite quotients, so the bounded computation also has a general proof. □

E.13. Specialization to the Fermat coverings [Proved]

For $N_n = \ker(F \to (\mathbb{Z}/n)^2)$ the abelian comparison is

$$0 \longrightarrow \mathbb{Z}^{n^2-1} \longrightarrow N_n^{\mathrm{ab}} \longrightarrow n\mathbb{Z}^2 \longrightarrow 0.$$

The metabelian and class-c comparisons are the exact sequences of E.6 with kernels F''/N_n'' and $\gamma_{c+1}F/\gamma_{c+1}N_n$, respectively; both are nonzero for $n > 1$. Pro-p completion is exact exactly when n is a power of p. With $n = p^a m$, $(m,p) = 1$, its replacement target is the open subgroup

of $\widehat{F}_p$ of index p^{2a}, and the source has rank n^2+1. The rational unipotent comparison is the surjection of E.9, with abelian kernel dimension n^2-1.

Proof. The abelian covering map sends a closed edge cycle to its two exponent sums. Each sum is divisible by n, and the loops x^n, y^n realize the two generators of $n\mathbb{Z}^2$. The source rank is n^2+1 by the graph calculation. Since N_n contains F', it contains both F'' and every $\gamma_{c+1}F$ for $c \geq 1$, giving exactly the two kernels in the statement. E.7 proves their nonvanishing. The maximal p-quotient of $(\mathbb{Z}/n)^2$ is $(\mathbb{Z}/p^a)^2$, so E.8 gives the asserted ranks and the exactness criterion. E.9 gives the last assertion. These are the maps induced by the actual covering inclusion, and coincide with the Fox matrices computed in section B. □

E.14. Relative completion retains the entire covering kernel [Proved]

The rational relative unipotent completion $\mathcal{R}_q$ of $F \twoheadrightarrow G$, for finite G, fits into

$$1 \longrightarrow N^{\mathrm{un}} \longrightarrow \mathcal{R}_q \longrightarrow G_{\mathbb{Q}} \longrightarrow 1.$$

It splits over $\mathbb{Q}$, with splittings conjugate by $N^{\mathrm{un}}(\mathbb{Q})$. Its quotient with abelian kernel has vector kernel $H_1(N,\mathbb{Q})$, of dimension $|G|+1$ and character $1+\mathrm{reg}_G$. For $q_2 : F \to (\mathbb{Z}/2)^2$ its four character multiplicities are $2,1,1,1$.

Proof. Choose a transversal of G in F. Its conjugation maps on N extend uniquely to N^{un}, and its products give a factor set with values in $N \to N^{\mathrm{un}}$. The associated multiplication on the disjoint union of $|G|$ copies of N^{un} is associative by the factor-set identities. This constructs $\mathcal{R}_q$ and the displayed exact sequence as rational proalgebraic group schemes. Every map from F into an extension of G by a unipotent group extends uniquely on N and then on the chosen components. It respects multiplication by the same identities. This proves the relative universal property.

At each successive central vector quotient of the unipotent kernel, a section lifts because the cohomology in positive degree of the finite group G with rational coefficients vanishes. To see the vanishing, average a cochain contraction over G and divide by $|G|$ in the standard bar resolution. The same averaging in degree one conjugates two sections. Induction gives compatible sections and conjugators through the central quotients, and their inverse limits give the assertions for N^{un}. This specifies the conjugacy ambiguity; uniqueness of the conjugating element is not claimed. Finally, the chain modules of the finite Cayley graph are $\mathbb{Q}[G]^2$ and $\mathbb{Q}[G]$, with zeroth homology $\mathbb{Q}$. Semisimplicity of $\mathbb{Q}[G]$ gives $H_1(N,\mathbb{Q}) \simeq \mathbb{Q} \oplus \mathbb{Q}[G]$. For $G=(\mathbb{Z}/2)^2$, expanding its regular character gives the four stated multiplicities. This action has a -1 eigenvalue; it survives in the relative completion, whereas the map of E.9 takes coinvariants and loses these characters. □

E.15. The genuine action on a relative completion [Proved]

Let $\mathrm{Aut}(F,q)$ consist of the automorphisms of F carrying N to itself. Its homomorphism to $\mathrm{Aut}_{\mathbb{Q}}(\mathcal{R}_q)$ is injective, and its image is exactly the subgroup preserving the embedded discrete subgroup $F \subset \mathcal{R}_q(\mathbb{Q})$ setwise. The automorphism $\mathrm{Int}([x,y])$, at $q=q_2$, acts nontrivially on the relative completion, although its action on F^{ab} and the image of its conjugator there are both the identity.

Proof. The universal property in E.14 extends an automorphism and its inverse, with their induced automorphism of G, so the extension is an automorphism. The map $N \to N^{\mathrm{un}}$ is injective. Write a nontrivial reduced word in maximal nonzero-power blocks $z_{i_1}^{a_1}\cdots z_{i_k}^{a_k}$, with adjacent i_j different. Under $z_i \mapsto 1+Z_i$, the coefficient of $Z_{i_1}\cdots Z_{i_k}$ with these k

alternating blocks is $a_1 \cdots a_k \neq 0$: omitting a block cannot yield that sequence of k blocks, and each retained block must contribute its linear term. A finite truncation of the power-series ring detects the word in its unipotent group $1 + (Z_i)$. Since components detect F/N, the map $F \to \mathcal{R}_q(\mathbb{Q})$ is injective as well. This proves faithfulness. An algebraic automorphism preserving this subgroup restricts to an automorphism of F preserving its identity component N; the universal property identifies its extension with the given automorphism. This proves the image description.

At q_2 the commutator $w = [x, y]$ has a nonzero cycle class in $H_1(N_2, \mathbb{Q})$, as can be read from the unit-square cycle in the four-vertex graph. Choose another cycle with independent class. Their bracket is nonzero in the degree-two free Lie algebra of N_2^{un}, so conjugation by w is nontrivial. Passing from a genuine action to its outer class kills this particular inner action. Thus this example also specifies the actual quotient map responsible for the ambiguity, rather than identifying an outer action with a genuine one. □

Sections and arithmetic obstructions

E.16. All sections of the rank-two toric quotient [Proved]

Let $T = \mathbb{G}_m^{\{0,1,\infty\}}/\mathbb{G}_m$ over $\mathbb{Q}$, with its rational identity. Its arithmetic fundamental-group extension has geometric kernel $\widehat{\mathbb{Z}}(1)^2$. Its continuous sections modulo conjugacy by this kernel form the abelian group

$$\left(\varprojlim_m \mathbb{Q}^*/\mathbb{Q}^{*m}\right)^2.$$

One factor has the explicit description

$$\{\pm 1\} \times \left\{(a_p)_p \in \prod_{p \text{ prime}} \widehat{\mathbb{Z}} : \forall m,\ a_p \in m\widehat{\mathbb{Z}} \text{ for all but finitely many } p\right\}.$$

This computes the cohomology in (379) for its specified rank-two quotient at every level n.

Proof. The section at the identity gives the semidirect product $\widehat{\mathbb{Z}}(1)^2 \rtimes G_{\mathbb{Q}}$. A section has form $g \mapsto (c(g), g)$; multiplication says $c(gh) = c(g) + gc(h)$. Conjugation changes c by a coboundary. At finite level the Kummer sequence and Hilbert 90 identify $H^1(G_{\mathbb{Q}}, \mu_m)$ with $\mathbb{Q}^*/\mathbb{Q}^{*m}$. To verify the reduction, a multiplicative cocycle of a finite Galois extension is a coboundary by choosing b for which $\sum_\tau c(\tau)\tau(b) \neq 0$ and translating the sum; linear independence of field embeddings supplies such a b. Every algebraic cocycle is defined over a finite extension. Taking limits gives the formula. The possible $\varprojlim^1 H^0(\mu_m)$ vanishes because these groups are finite and their descending images stabilize.

Unique prime factorization gives $\mathbb{Q}^* = \{\pm 1\} \times \bigoplus_p p^{\mathbb{Z}}$. An element modulo m has finitely many nonzero valuation residues. Compatible residues are exactly the families in the second display. Even m retain the sign and odd m kill it; their compatible limit is $\{\pm 1\}$.

The map $U_n \to T$ is inclusion on the open with all coordinates nonzero. On geometric groups it is the covering pushforward of E.13 divided by n, hence surjective onto $\widehat{\mathbb{Z}}(1)^2$. The map to the base torus in coordinates is $(u, v) \mapsto (-u^n, -v^n)$, with $t = -u^n$, $1 - t = -v^n$. It induces multiplication by n on geometric kernels and retains the arithmetic translation $(-1, -1)$. Thus the quotient extension is the arithmetic fundamental group of T even when U_n has no rational point. This supplies the comparison underlying (373)–(379). □

E.17. Pairwise separation of cusp sections by the minimum quotient [Refuted]

For every odd n, the rational cusps $P_0 = [0 : 1 : -1]$ and $P_\infty = [1 : -1 : 0]$ admit rational tangent choices whose sections become conjugate in the rank-two quotient of (373). Both toric Kummer classes are $(1, -1)$ in E.16. For the three punctures of U_1 the complete families of toric images are $(w, 1)$, $(1, -w)$ and $(w^{-1}, -w^{-1})$, with $w \in \mathbb{Q}^*$, in the conventional coordinates $(t, 1 - t)$ and the tangent coordinates specified below.

Proof. Use $u = x_0/x_\infty, v = x_1/x_\infty$. Near P_0 take $s = u$; the branch is $v = -(1 + s^n)^{1/n}$ with constant coefficient -1. The unit tangent has leading torus coefficients $(1, -1)$. Near P_∞ take $s = x_\infty/x_0$; then $(u, v) = (s^{-1}, -s^{-1}(1 + s^n)^{1/n})$, giving the same leading coefficients. Adjoining compatible roots of these units computes the tangential cocycle: coefficientwise Galois action fixes the formal parameter and acts on roots of the leading coefficients. Both classes therefore coincide. This also derives the applicable case of Wickelgren's tangential formula, Lemma 4 [Wic12b].

For U_1 in coordinate t, at 0 with tangent w the units $t, 1-t$ give $(w, 1)$; at 1 with coordinate $t - 1$ they give $(1, -w)$; at infinity with coordinate $1/t$ they give $(w^{-1}, -w^{-1})$. These exhaust rational tangential choices. E.16 determines equality of their images. The unit tangent at 0 and tangent -1 at 1, for example, both give $(1, 1)$. The exhibited coincidences are recorded in `e_obstructions.py`. They concern precisely this quotient, with the quotient map specified. □

E.18. The real obstruction for the manuscript quartic [Proved]

The geometrically abelianized arithmetic fundamental group of $U_4/\mathbb{Q}$ has no continuous section. The section-class sets are also empty for its geometrically metabelian, full profinite and geometric pro-2 quotients. The obstruction already lies in a group of order two in $H^2(C_2, H_1(U_4(\mathbb{C}), \mathbb{Z}_2))$.

Proof. The plus quartic has no real points, by positivity of fourth powers. Complex conjugation acts freely on $U_4(\mathbb{C})$. Its quotient is a connected nonorientable punctured surface of Euler characteristic -8, since the double cover has Euler characteristic -16. It retracts to a graph, so its fundamental group J is free of rank nine. The geometric subgroup N of index two is free of rank seventeen. Finite complex topological covers with conjugation descent data identify the real arithmetic extension with the profinite completion of $1 \to N \to J \to C_2 \to 1$: descend the complex cover with its antilinear involution; in the other direction analytify a finite etale cover. This is the equivariant finite-cover comparison, and finite-index completion preserves this extension.

Free-generator changes put $J \to C_2$ in the form $a \mapsto \bar{1}$, $b_i \mapsto \bar{0}$ for $1 \le i \le 8$. A Schreier basis of N is

$$a^2, \quad b_i, \quad ab_ia^{-1} \quad (1 \le i \le 8).$$

Conjugation by a fixes the first abelian basis vector and exchanges each remaining pair. The factor set at $(\bar{1}, \bar{1})$ is a^2. Replacing the lift by na changes its square by $(1 + \operatorname{Ad} a)[n]$. Its first coordinate is $1 + 2n_0$, which cannot vanish in $\mathbb{Z}_2$. Each exchanged pair has invariant diagonal equal to the image of the norm. Hence the indicated cohomology group is $\mathbb{Z}/2$, and the extension is its nonzero class. The finite program tests the factor-set equation for every modulus dividing twenty-four and for additional odd powers; precisely the even moduli are obstructed.

A rational section restricts to a real decomposition group, and each quotient in the statement projects to this 2-adic abelian quotient. All are therefore obstructed. Its image under $N^{\mathrm{ab}} \to 4\widehat{\mathbb{Z}}(1)^2$ is zero in H^2: conjugation acts as -1 on that torsion-free rank-two module,

whose invariants vanish. This explicit pushout explains the splitting of (373) together with its failure to lift. The odd-adic and rational pushouts of the first-coordinate obstruction also vanish because two is invertible there; this cohomology calculation makes no assertion of a global section there. □

E.19. The cubic cusps are separated by their own 3-adic abelianization [Proved]

The three rational cusps $[0:1:-1]$, $[1:0:-1]$, $[1:-1:0]$ determine pairwise nonconjugate sections of the geometric pro-3 abelianization of $\pi_1(U_3/\mathbb{Q})$, for every rational tangent choice. They remain distinct in the geometric abelian, metabelian and full profinite quotients.

Proof. Filling the cusps maps the affine fundamental group to the Tate module of the elliptic curve X_3, based at one cusp. A cusp section maps to the point section there, independently of its tangent. Each difference of two distinct cusps has exact order three: the ratio of their tangent lines has divisor three times their difference, and a genus-one curve has no function with a single simple pole. Section C gives the same exact-order calculation.

The cubic has good reduction at two. Its equation over $\mathbb{F}_2$ has exactly the three projective points with two nonzero coordinates, as the program enumerates. The kernel of good reduction over $\mathbb{Q}_2$ is pro-2, from its formal-parameter congruence filtration. Odd torsion therefore injects into this group of three points. A rational point of exact order three cannot be divisible by three: a preimage would have order nine. Its Kummer class modulo three is nonzero. The difference of the two point sections is this class in $H^1(G_{\mathbb{Q}}, X_3[3])$. Projection from affine 3-adic homology detects it and proves the assertion. This map also identifies information lost in the quotient of E.17. □

E.20. A new obstruction to lifting a toric Kummer section [Proved]

For U_1 based at the tangent from 0 towards 1, the abelian section with Kummer coordinates $(2,3)$ does not lift to the class-two quotient with geometric group

$$\langle x, y, z \mid x^2 = y^2 = z^2 = 1, [x,y] = z,\ [x,z] = [y,z] = 1 \rangle.$$

Its obstruction is the nonzero class $\{2\} \cup \{3\}$ in $H^2(G_{\mathbb{Q}}, \mu_2)$.

Proof. The eight normal forms $x^a y^b z^c$ have central multiplication term ba'. The code verifies this group law and presentation. The tangential Galois action is trivial on it: $x \mapsto x^\chi = x$ and $y \mapsto f^{-1} y^\chi f = y$, since χ is odd and the commutator f is central here. Pulling the central factor set back along the two Kummer characters gives their cup product (the order of factors does not change its class modulo two).

Over $\mathbb{Q}_3$, the quadratic extension generated by $\sqrt{2}$ is unramified because 2 is a nonsquare modulo three. Norms have even valuation, so 3 is not a norm. The crossed-product description of the cup product identifies its vanishing with the norm equation $3 = u^2 - 2v^2$ over $\mathbb{Q}_3$: represent $\sqrt{2}$ by its multiplication matrix, and the matrix equation for the second quaternion generator is exactly this norm equation. Thus the local and the global class are nonzero. A lifted section would give a cochain whose coboundary is this factor set, a contradiction. □

E.21. The unmarked depth assignment fails to descend [Refuted]

The quotient of $\mathbb{L}(x,y)$ by D_y^3, with its canonical projection, is not functorial for all automorphisms of the unmarked free group. It is functorial for maps preserving the ideal of the marked puncture and its lower central filtration. Thus E.5 tests the five specified ordinary group approximations; the depth construction additionally uses this marking and these morphisms.

Proof. The swap of x, y sends $[y,[y,[y,x]]] \in D_y^3$ to $[x,[x,[x,y]]]$, a nonzero depth-one element of E.2. It therefore fails to preserve the kernel of the canonical projection, which proves the non-descent. A map preserving the marked ideal preserves its lower central terms and does induce the quotient map. No LM8 return specifying an additional unmarked functor is present in the supplied archive. □

E.22. The scheme coordinates do not carry the requested relative action [Refuted]

Put $q_2 : F \to (\mathbb{Z}/2)^2$ and extend the relative completion of E.14 to $\mathbb{Q}_\ell$. There is no homomorphism from the normalized diagonal arithmetic image in $\mathcal{M}(0)(\widehat{\mathbb{Z}})$ to $\mathrm{Out}(\mathcal{R}_{q_2,\mathbb{Q}_\ell})$ whose composite with $G_{\mathbb{Q}}$ is its actual outer action. In particular no such genuine action descends from those coordinates. The actual action is defined on the continuous geometric fundamental group, and its relative universal property gives its action on $\mathcal{R}_{q_2,\mathbb{Q}_\ell}$. E.15 gives the faithful discrete version and its exact image.

Proof. First we type the comparison with the arithmetic action. The continuous unipotent completion of the free profinite group $\widehat{N}_2$ over $\mathbb{Q}_\ell$ is the free prounipotent group on its five Schreier generators. Indeed each finite tuple in a unipotent $\mathbb{Q}_\ell$-group generates a relatively compact subgroup, as follows from upper-triangular coordinates and their polynomial group law. Its homomorphism from the discrete free group consequently extends continuously to the free profinite group. The two universal properties identify this completion with $N_2^{\mathrm{un}} \otimes \mathbb{Q}_\ell$. The transversal construction in E.14 then gives the continuous relative completion and its functorial arithmetic action. The normalized Galois action is the identity on $(\mathbb{Z}/2)^2$, because its abelian multiplier is $\chi \equiv 1 \pmod 2$.

The homomorphism $F \to G \times F^{\mathrm{un}}$, $g \mapsto (q_2(g), g)$, and the relative universal property give the canonical surjection

$$\mathcal{R}_{q_2,\mathbb{Q}_\ell} \longrightarrow F^{\mathrm{un}}_{\mathbb{Q}_\ell}.$$

It is the maximal unipotent quotient, so its kernel is characteristic. Every genuine automorphism descends through this map and inner automorphisms descend to inner automorphisms. This gives the needed map of outer automorphism groups. A nontrivial outer action on its target is therefore a nontrivial outer action on its source.

A.19 proves that the normalized arithmetic image is contained in $\mathcal{M}(0) \simeq \Gamma$. On $G_{\mathbb{Q}^{\mathrm{cyc}}}$ its multiplier-scheme coordinate is the identity, and the exact law is that of $C_2 \times \widehat{\mathbb{Z}}$, an abelian group. Every commutator of two elements of $G_{\mathbb{Q}^{\mathrm{cyc}}}$ thus maps to the identity in these scheme coordinates.

We now exhibit a commutator whose actual outer unipotent action is nontrivial. E.3 gives nonzero actual graded Galois classes in weights three and five, with depth-one coefficients proportional to $\bar\sigma_3, \bar\sigma_5$. Choose representatives g_3, g_5 of nonzero such classes and choose $\tau \in G_{\mathbb{Q}}$ with $\chi_\ell(\tau) = 1 + \ell$. The cyclotomic action on weight w is multiplication by χ_ℓ^w. Hence $h_w = [\tau, g_w]$, for $w = 3, 5$, has leading class $((1+\ell)^w - 1)$ times that of g_w, which is nonzero over $\mathbb{Q}_\ell$. Both h_w lie in $G_{\mathbb{Q}^{\mathrm{ab}}} = G_{\mathbb{Q}^{\mathrm{cyc}}}$, since they are commutators. Their commutator has nonzero weight-eight, depth-two outer class: its polynomial is a nonzero scalar multiple of $B_{2,4}$ computed in E.3. This assertion uses actual graded Galois classes, not arbitrary special derivations.

Consequently $[h_3, h_5]$ is killed by $\mathcal{M}(0)$ and acts nontrivially on $\mathrm{Out}(F^{\mathrm{un}}_{\mathbb{Q}_\ell})$, and thus on the relative completion. A homomorphism agreeing with both actions would send the identity to this nonidentity element, which proves the claimed non-descent. For a separate genuine geometric witness, $w = [x, y] \in \Pi_0'$ has trivial final coordinates $(a, b, U) = (1, 1, \bar w)$ but its inner action on the relative completion is nontrivial by E.15. The arithmetic witness above also survives passage to outer automorphisms. □

E.23. The remaining cubic section calculation [Open]

The complete set of continuous section classes for the geometric metabelian and marked depth-two arithmetic quotients of $U_3/\mathbb{Q}$ is not determined in this return. The remaining calculation is the image and every fiber of their section maps to E.16. E.37 determines the full intrinsic pro-3 class-two section map over every abelian class. The next pro-3 map whose entire image remains unevaluated is

$$\mathrm{Sect}(N/\gamma_4 N \rtimes G_{\mathbb{Q}})/N \longrightarrow \mathrm{Sect}(N/\gamma_3 N \rtimes G_{\mathbb{Q}})/N, \qquad N = \pi_1(U_{3,\overline{\mathbb{Q}}})^{(3)}.$$

E.38 proves that this map excludes an explicit nonempty class-two fiber. E.44 gives its exact local-error and convergence criterion, with the genuine higher Frobenius action retained as arithmetic input. E.41 computes all unramified-outside-three classes of the full, metabelian and single-cusp marked groups, and E.43 excludes an infinite family of ramified class-two fibers at the further stages. The other prime components and the mixed finite quotients also remain part of the requested full profinite metabelian and marked-depth classification.

E.84 evaluates the complete ordinary class transfer in the first complex base-change step. Its kernel and image both have order three. Actual field normalizers and their certified class matrices force all six ramified prime cubes and the three-prime product over three to be principal. The complete remaining S-divisor kernel lies on one explicit order-three line; the earlier twenty-seven candidates reduce to three literal ideal classes. It also proves capitulation of the original L_2 prime class in B_2 and identifies the exact ordinary-unit norm coset left by the corrected target ξc_1.

E.86 completes these S-class and capitulation calculations at every level, in actual coherent prime bases. The groups are $\mathrm{Cl}_S(B_d)\{3\} = C_3$ and $\mathrm{Cl}_S(L_d)\{3\} = C_3^2$; every adjacent norm is the identity matrix and every extension matrix is zero. It also gives the complete ordinary groups $\mathrm{Cl}(B_2)\{3\} = C_9 \oplus C_3^2$ and $\mathrm{Cl}(L_2)\{3\} = C_9 \oplus C_3$, with all genus restriction, norm and extension maps and their inverse coordinates. In particular, all twenty-seven E.81 ideals are principal, both E.84 residual classes vanish, and the full principal S-divisor lattice in B_2 has index 27. The cyclic cubic and ninth-degree B-unit norm defects are C_3, with full quotients $C_{3^r}^5 \oplus C_3$; the adjacent original L-unit defects are C_3^2, with full quotient C_3^3. Every horizontal norm congruence and every finite completed norm fibre is evaluated through the actual Hilbert–90 and ideal-class maps. The higher ordinary prime orders determining h, and the particular unit-norm coset of ξc_1, remain unevaluated.

E.85 proves the existence converse for the compatible joint wild family and computes its full intrinsic central fibres over a fixed actual abelian section. A normalized rational cusp gives its full origin; the actual coefficient map has image $\mathbb{Z}_3 \oplus 3\mathbb{Z}_3 \oplus 0$ in $\mathbb{Z}_3^2 \oplus C_3$, with cokernel C_3^2. Every nonempty fibre over that intrinsic abelian class is $\mathbb{Z}_3^{44} \oplus C_3^9 \oplus C_9^5$, with complete representatives and inverse coordinates. Other intrinsic abelian preimages are further evaluated in E.87.

E.87 computes every abelian preimage of the joint local first stage: the cohomology map is onto with kernel $\mathbb{Z}_3^{10} \oplus C_3^3$, and all inverse coordinates retain the finite carry. On the entire locus with the same B_0 inertia coordinate and the E.85 joint abelian image, the full class-two obstruction has image C_3^3 and zero locus $\mathbb{Z}_3^4$. Each point of the retained index-nine joint image has fibre $\mathbb{Z}_3^{48} \oplus C_3^9 \oplus C_9^5$ in this enlarged locus. The actual quotient filling eight cusps, with its rational based involution and explicit central coordinate morphism, evaluates the free obstruction on every intrinsic abelian input as $-7t(U_1 + 21t/2)$. Its two zero branches are $t = 0$ and $U_1 = -21t/2$, with complete filled fibres $\mathbb{Z}_3^2$. The fifteen finite intrinsic coordinates for general relative inputs, followed by their global and higher compatibilities, remain to be evaluated.

E.88 evaluates the entire complex multiplicative norm system. The outside-$\{3, 17\}$ divisor image is the kernel of two congruences modulo three in the fixed certified degree-six field

L_1. Compact completed S-unit fibres prove their sufficiency for an actual compatible lift through every level. The actual primes $7, 73$ supply a complete restricted $\mathbb{Z}_3$ basis and inverse coordinates; convergent products give the continuous splitting with its exact coherent-unit ambiguity. The ordinary L_2 class map evaluates the normalized base valuation phase modulo three. For pure positional component one, the total-norm cube condition and scaled toric map are thereby evaluated on the entire complex component, with every coordinate and inverse. The eventual integer h, the residual phase modulo 3^h, the pure radical component and the intrinsic preimages remain part of the calculation.

E.89 allows every horizontal intrinsic value over the actual B_0 relative value. Its complete obstruction has image $C_3^2 \oplus C_9$, with three exact congruences modulo $3, 3, 9$. The full zero locus modulo its fixed geometric gauge is $\mathbb{Z}_3^6$, with literal representatives and inverse coordinates. Its joint image has four free coordinates: over the varying first-stage coordinates L, m, the relative central mean lies in $3\mathbb{Z}_3$, the horizontal central mean has residue m modulo three, and the horizontal difference is arbitrary. The full source locus is $\mathbb{Z}_3^{52} \oplus C_3^9 \oplus C_9^5$, and every joint-image point has fibre $\mathbb{Z}_3^{48} \oplus C_3^9 \oplus C_9^5$. All these coordinates use the actual Hall image, its proved finite character and the complete geometric gauge. Other relative values and their higher and global compatibilities remain part of the unfinished calculation.

E.90 evaluates every relative multiple re_{B_0}, including zero, with every horizontal intrinsic input. Its exact obstruction is r times the map of E.89; the three image orders are $81, 3, 1$. Integral particular corrections and the full nonlinear geometric gauge give every joint image and all source fibres. The rational tangent swap and centre-free path comparison construct the global cusp family; the actual map $-y/x$ is a Kummer left inverse. Its global-to-local map is onto, with the literal logarithmic prime-exponent formula, the restricted completion topology and every inverse fibre. The full unramified intrinsic class-two source is $\mathbb{Z}_3^{30} \times C_3^{12} \times C_9^5$, its joint image is $\mathbb{Z}_3^3 \times C_3^2$, and each fibre is $\mathbb{Z}_3^{27} \times C_3^{11} \times C_9^5$. The entire full-group fibre appends the actual recursive Hall coordinates of E.41. These exact comparisons retain the actual handle phase and Kummer basis constants. Other relative vectors, higher wild obstructions and the general ramified global fibres remain to be evaluated.

E.91 proves triviality of every adjacent ordinary B-class action and surjectivity of the actual ordinary norms. Each original ordinary L-class group has the exact type $C_{3^{\alpha_d+1}} \oplus C_3$, with its specified prime above three generating its subgroup of triples. The full ordinary norm tower stabilizes with the already proved finite exponent h. At B3 the actual ninth-genus Artin map has an order-at-most-three kernel generated by the prime73 transvection. The full action proves exponent nine, three explicit group-type alternatives, and the identity $[Q_1] + [Q_2] + [Q_3] = 3[Q_i]$. The original horizontal norm has its exact index-three Artin-kernel image. The complete unit-lifting attempt has an evaluated rank-three symbol image. The actual transvection and prime triple, the numerical value of h, and the particular unit-norm coset remain to be determined.

E.92 computes the entire global intrinsic abelian localization image: it is exactly the subgroup $\theta = 0$, of type $\mathbb{Z}_3^{14} \times C_3^3$. The actual compact filling and toric Shapiro maps prove both inclusions. Every fibre, the three literal quadratic prime pivots, all logarithmic and μ_3 coordinates, and the nontrivial elliptic carry have explicit inverse maps. They give a continuous linear section of the actual localization map. Thus every global intrinsic input has zero relative elliptic coordinate, leaving eight toric relative directions for the local lifting calculation.

E.93 computes every central localization fibre with the unchanged marked Kummer constants. Its image is $\mathbb{Z}_3^{25} \times C_3^9 \times C_9^5$, its kernel is the restricted prime module outside $\{2, 3\}$, and its cokernel is $\mathbb{Z}_3^{21}$. The complete integral quotient maps retain the actual primitive relative Tate line and its horizontal Smith coordinates. Its genuine joint coefficient image vanishes on every ramified central cocycle. The only remaining global joint variation is horizontal ρ, with kernel the restricted prime module outside three times $\mathbb{Z}_3^{23} \times C_3^9 \times C_9^5$. The actual vanishing stabilizers

transfer these maps to every nonempty global class-two fibre. The particular joint origins for all global abelian inputs and the higher nonabelian lifts remain to be evaluated.

E.94 evaluates the actual extension obtained by adjoining μ_{27} to B_3. The prime-three prolongations are totally ramified and the prime-seventeen prolongations are unramified of residue degree three, with their exact ideal norm and extension coefficients. The relative ordinary and S-split genus quotients over B_1 are $C_9^3 \oplus C_3$ and $C_9 \oplus C_3$. All ray restriction, Kummer, prime-removal, normalizer, norm and genuine ideal-extension maps are evaluated. The ordinary genus norm has kernel C_3^2, and extension kills the actual old transvection kernel, with $JN = NJ = 3$. The S-genus norm is $-x \bmod 3$. The noncyclic quotient of the actual deck coinvariants proves the precise failure of the earlier cyclic-module premise after this base change. The full upper class groups, original transvection and prime triple, eventual h, and particular unit-norm coset remain to be evaluated.

E.95 evaluates the complete wild class-two obstruction on the entire global abelian localization image. All eight actual framed-cusp cochains and the full marking transport are explicit. Eleven modulo-three and four modulo-nine coordinates have a completely evaluated relative image: 16,713 residues modulo $9W$, with every inverse and conjugacy fibre. The maximal all-horizontal relative lattice is $9W + 3W^J$. The vector $A_0 + B_0$ has a constant nonzero obstruction and actual global abelian preimages, giving a global nonlifting family.

E.96 evaluates the full global slice with zero free compact elliptic coordinate. An explicit integral forty-four-coordinate polynomial gives the exact local correction $-\mathcal{N}/[3(q-1)]$, including every restricted convergence exponent. Its exact modulo-three inertia image is the single rational-cusp ray corresponding to the finite elliptic residue, and actual tangential sections attain every finite family of residues. The nonzero free elliptic slice, general joint origins, higher geometric stages, the quartic curve, and full other-primary or mixed sources remain to be evaluated through their actual maps.

E.97 evaluates the previously undecided unit-norm coset: $\omega_B(\xi c_1) = +1$ in the coherent prime73 basis. The actual ordinary-unit defect and complete divisor map prove this value; every completed power-line fibre is computed, with preimages exactly for exponents in$3\mathbb{Z}_3$. The eventual exponent$h$ and the old ordinary transvection remain active.

E.99 computes every intrinsic class-two section unramified outside$\{2, 3\}$. The actual source, all81 Frobenius strata, all261 admissible residue pairs, every inertia lattice and every geometric fibre have exact inverse coordinates. The full abelian gauge and its finite carry are retained. A displayed global section over $9\kappa_3(2)(e_{A_0} + e_{B_0})$ has no metabelian or full lift. Unrestricted global ramification, the full and metabelian sources beyond their already evaluated loci, higher geometric stages, the quartic curve and other-primary or mixed sources remain on the requested list.

E.98 computes the entire$\{3, 7\}$ global ramification locus for the full, intrinsic metabelian and class-two geometric pro-three groups of the cubic. The class-two relative image is the union of nine cusp lines modulo$9W$, with the exact horizontal test in the quotient $C_3^5 \oplus C_9^2$ at each cusp. Every fixed-abelian and fixed-inertia fibre has complete inverse coordinates. Every full and metabelian ramified section has an actual cusp packet with its complete residual geometric quotient. The full-to-metabelian map is onto on these packets, with every global fibre computed by actual Hall/Chen coordinates; its first new factor has rank504 in degree4. An explicit class-two inertia$9(e_{A_0} + e_{B_0})$ has no full or intrinsic metabelian lift. The unrestricted global and other-primary or mixed-source questions remain active outside the stated computed loci.

E.100 computes all full and intrinsic-metabelian cubic sections unramified outside$\{2, 3\}$. Its three rational-cusp strata have unique representatives and complete compatible Hall coordinates. Every full-to-metabelian fibre is evaluated, including the nonlinear zero-inertia

comparisons and the actual boundary topology. The unrestricted global source and the other requested prime components remain active. For the plus quartic, E.18 and E.25 already prove emptiness for the full, metabelian, pro-two and every quotient retaining its two-adic geometric abelianization. The remaining quartic work concerns the specified maps that remove that obstruction, including intrinsic odd-primary and rational relative targets.

E.102 evaluates the complete first integral unramified full-to-metabelian coefficient kernel: $\mathbb{Z}_3^{504} \oplus C_3^{144} \oplus C_9^{56} \oplus C_{27}^{30} \oplus C_{81}^{9}$. Its actual integral basis maps and inverses are supplied; the nonlinear comparison of E.100 gives every corresponding degree-four section fibre.

E.101 evaluates all first ordinary-unit and S-unit norm images, including the scalar previously left on the product of the two lower free units. Its actual field, divisor, Artin and horizontal norm maps give every norm fibre. The next ordinary tower layer and eventual exponent remain active.

E.103 computes all unrestricted global odd-primary and prime-to-two abelian quartic sections, their onto toric norm map and every fibre. It also computes every rational abelian pushout. The degree-four coset section and exact integral isogeny inverses are supplied. The higher odd-primary and rational nonabelian targets retain their lifting questions.

E.104 computes the entire full and intrinsic-metabelian cubic pro-three section spaces for both infinite two-prime families of E.55 and E.56. All original-root source maps, ramified cusp coordinates, unramified strata, comparison fibres and boundary topologies are explicit. General global ramification and the other geometric prime components remain outside those completed arithmetic sources.

E.106 determines the complete third ordinary layer: $\mathrm{Cl}(B_3)\{3\} = C_9^2 \oplus C_3$ and $\mathrm{Cl}(L_3)\{3\} = C_9 \oplus C_3$. The ninth-to-cubic character square on the complete ordinary-unit norm lattice has rank four, so the previous transvection vanishes. Every actual class norm and extension, all six S-prime columns, the entire principal S-divisor lattice and their inverses are evaluated. The horizontal ordinary-unit norm is onto. The vertical ordinary norm quotients are C_3^5 and C_3^3, and both norm kernels are explicit. Thus the third finite exponent is one; the later ordinary layers and eventual exponent remain active at this stage. These conclusions supersede the third-layer alternatives recorded in the earlier milestones.

E.105 lifts the actual quartic four-coset origin through class two for every odd integral and every rational coefficient ring. The entire two-coordinate toric image is evaluated by the exact continuous correction criterion, including its nonclosed denominator. The compact-handle arithmetic action and general abelian inputs continue in the subsequent calculation.

E.107 now determines the previously retained all-level ordinary class maps and coherent exponent: $h = 1$. The actual cyclic-nine Artin square preserves the fifth ordinary-unit character, and the named-prime class norm maps stabilize as isomorphisms. The full complex divisor phase and every scaled toric fibre with pure positional component one are evaluated explicitly. These conclusions supersede the later-layer and phase questions recorded in the earlier milestones. The separate pure-field norm factor, general coupled arithmetic sources and higher intrinsic lifting questions remain active.

Work and obstruction. E.18 determines the empty set for the plus quartic and all quotients retaining its 2-adic geometric abelianization; E.25 extends this to every even plus-Fermat level. E.24 computes the entire cubic abelian section set and every fiber over the minimum torus. E.19 separates the three rational cubic cusps inside that set, while E.17 proves their collision under the stated toric map.

The relation-module presentation E.1, the rank-two factor-set calculation, local cup products, the exact generalized-Jacobian extension, and a new three-punctured-line map on the cubic were successive attempts to compute the nonabelian fibers. They give the following actual refinements. E.28 exhibits a whole nonempty abelian toric fiber with no class-two lifts. E.29

classifies every rational toric pair for the inherited cubic class-two quotient. E.32 completes that calculation for every completed pro-3 Kummer pair, including the exact convergence condition and all three classes in each nonempty fiber; it also exhibits an actual obstruction invisible at every finite level. E.31 computes a genuine intrinsic quotient with its arithmetic origin term: in one toric fiber the zero abelian class lifts, whereas an explicit other class and its entire congruence coset have no intrinsic class-two lift. Thus the two minimum coordinates do not determine intrinsic lifting, as witnessed by this exact same-fiber comparison. E.34 computes the entire intrinsic abelian family unramified outside three and proves that every one of its classes lifts to the full geometric pro-3 group. E.35 evaluates every section of the intrinsic finite Heisenberg quotient by a cubic norm equation and gives all of its nonempty fibers in explicit Kummer coordinates. E.36 proves that every abelian class lifts through the cuspidally central quotient retaining all ten abelian coordinates; each fiber is the specified completed group of rational units. Its quotient map from the full intrinsic class-two group has a central kernel of rank 44.

E.37 then evaluates that full intrinsic pro-3 obstruction. It constructs the integral generalized-Jacobian lattice, computes every peripheral path term from a framed cusp point, and gives the necessary and sufficient integral restricted-support matrix criterion. Every nonempty fiber is evaluated, including its torsion and its specified coordinate map. E.38 supplies a further obstruction: the class with toric coordinates $(2, 19^{27})$ has a full intrinsic class-two lift, but its entire fiber is excluded at class three, already in an arithmetic quotient of order 3^{30}. The same quotient receives the intrinsic metabelian and marked-depth groups, so excludes those lifts as well. E.39 constructs the arithmetic free-inertia comparison and proves that its local Frobenius correction words cannot all be removed, with a cyclic cubic witness at nineteen. E.40 evaluates every nonempty class-three lifting fiber and supplies finite matrix presentations for the higher central fiber groups. E.41 constructs compatible coordinates for the entire unramified family at every degree in all three pro-three targets. E.42 computes the exact local equation in the actual arithmetic module quotient. E.43 gives the exact class-two local exponent and an infinite family with nonempty class-two fibers and no further metabelian or marked lift. E.44 proves the general higher image criterion, retaining both the actual higher Frobenius action and the corrected Frobenius value of E.39. E.45 computes all local conjugacy classes in the actual inherited module quotient, with unique explicit representatives in both affine-orbit cases. E.46 supplies actual norm-system coordinates for every global first-coordinate fiber of that quotient and constructs explicit ramified families. E.47 evaluates its entire zero-first-coordinate family by norm-coherent cyclotomic units and an independent free-arithmetic-group calculation. E.48 constructs a compact product of independent ramified classes by exact polynomial norms. E.49 evaluates the entire first-coordinate fiber at $\kappa_3(3)$, and E.51 extends that calculation to every first coordinate unramified outside three. E.50 supplies the actual multiplication-by-three elliptic comparison, including its abelian kernel of rank 332 and the outer Tate-image factorization of the specified quotient. E.52 evaluates the entire nonzero first-coordinate line generated by $\kappa_3(2)$, with certified fundamental-unit norm matrices. E.54 gives an actual class-three quotient with its fully evaluated Frobenius action and every local lifting fiber at two. E.53 joins this action to the elliptic comparison over their exact common arithmetic quotient, with abelian kernel of rank 329. E.55 computes every mixed first coordinate supported on three and a prime congruent to two or five modulo nine, using an exact arithmetic free-product presentation and integral cocycle normal forms. E.56 computes the corresponding complete mixed families at every split prime ℓ with $v_3(\ell - 1) = 1$ and three a noncube modulo ℓ, using its two actual decomposition factors and an explicitly solved index-three parameter lattice. Their precise geometric kernels retain the intrinsic lifting problem. E.57 evaluates the first complex norm transition over $3\kappa_3(17)$ with certified class-group and divisor coordinates. E.62 proves that the explicit class $(1, \zeta_3)$ lifts through

that transition and fails at the next one; it computes every local norm stage at seventeen, its full inverse limit and each toric fiber. E.64 and E.65 evaluate every adjacent complex and pure norm in that same positional tower, including their complete norm images and fibers. The pure factor has an explicit finite wild character at three; its first fundamental unit has character two and no next norm preimage. E.69 proves finite-index stable projections for the entire coherent complex three-unit subsystem, gives its actual integral product coordinates and every toric fiber, and evaluates its first two base norm images. Its eventual finite base index remains to be evaluated. E.72 gives every surviving complex divisor chain and proves sufficiency of the exact principal-class equations for a coherent multiplicative lift. Each divisor fiber is the evaluated ordinary-unit product. The remaining higher class matrices are retained explicitly; the actual prime-above-three class has eventual order 3^h. The pure and complex surviving prime sets are disjoint away from three and seventeen, and their positional divisor norm has the displayed rank-three kernel at those two primes. E.58 evaluates the actual rank-thirteen elliptic quotient action at two, and E.60 computes every local section and every central lifting fiber there. E.59 evaluates the first nonabelian Frobenius word for every inert prime congruent to eight modulo nine, including the exact prime-seventeen Frattini orbit. E.61 classifies all ramified local sections of the full cubic pro-three group at every K-place away from three. It also proves that the abelian inertia image of all local intrinsic metabelian sections is exactly the union of the nine integral cusp lines. Its actual quotient section $\nu = c^{e_{00}+e_{01}+V}$, $z = \eta^2$ has no intrinsic metabelian lift: the explicit Frattini-line cover of degree 19683 supplies the comparison to its compact Jacobian Tate module. E.66 completes the ramified local metabelian source: all its sections are the same nine cusp families, and each has exactly one full pro-three class above it. Every unramified local metabelian class also has a full lift. E.67 classifies every marked-depth local section whose inertia dies after filling the marked cusp, with its exact central Frobenius parameter and full-source image. Its explicit primitive-ray section has no full lift, detected by the evaluated first-moment quotient. E.68 computes every unramified class and every full lifting fiber, with actual nonlinear comparison coordinates, completing the entire local metabelian map. E.70 evaluates the marked central parameter as a countable product of $\mathbb{Z}_3$ through its finite orbit maps. E.71 completes the other eight marked cusp loci: their actual compact Kummer paths exclude periodic cosets, and the exact two-step measure argument gives a unique full lift. Every local metabelian and marked class and every full-source fiber away from three is therefore computed. E.73 proves the entire convergent global relation set and sufficiency of its coupled local packet equations, with every actual Frobenius word and rational descent map retained. Its finite Kummer witness over nineteen detects the failure of independent local Frobenius choices. E.74 computes the complete wild-local central coefficient groups and every nonempty central fiber, with numerical obstruction groups in degrees one through three and a proved finite precision bound in every higher degree. Its source comparison, local duality and central-difference maps are explicit. E.75 identifies the actual relative relation module uniformly, with its equivariant source change and inverse. Evaluation on one actual relative generator gives the explicit matrix kernel $\ker((I - J_r)(T_r - 7I))$, with exact saturated bases and inverse coordinates in degrees one through three. The local three-term cochain model retains the factor $1/2$ in its local-invariant comparison and every finite coefficient transition. E.76 computes the complete six-place unit-norm characters after adjoining a ninth root of unity, the first two genus quotients and the actual descent of their prime classes. The original third-level prime class still has unevaluated order three or nine, and the eventual norm index remains to be determined. E.77 constructs the actual equivariant nonlinear source presentation over the fixed free quotient. It gives every wild section as an explicit pair satisfying its one relation, and its exact central defect with the factor $1/2$ in the invariant coordinate. The actual geometric operators remain in those equations. Evaluating their remaining arithmetic values and solving the coupled global equations, the other-primary and mixed sources, and the remaining global principal-class equations and indices are the specified unfinished steps.

E.78 evaluates the genuine wild arithmetic operators on the actual rank-two class-three quotient induced by h over $\mathbb{Q}_3(\zeta_3)$. It computes both lower lifting images, every geometric-conjugacy fibre and the exact two central errors. An actual cusp path and a central commutator change construct an intrinsic class-two family with equal abelian coordinates: one specified member lifts to the full geometric group and another has no intrinsic class-three lift. The general intrinsic image through the displayed kernel and rational descent remain among the unfinished steps.

E.79 proves the full six-place S-unit field-norm image at every cubic and cyclic ninth-degree step after the specified cyclotomic base change. Actual ideal norm identifies the S-class coinvariants with the base S-class group. The first-level divisor lattices, ordinary-unit corrections and a compatible prime above 73 are explicit. The two Hilbert–90 maps identify the remaining S-unit norm defect with an invariant-class quotient whose order equals the capitulation kernel. E.86 evaluates its actual annihilator and every S-capitulation map; the original eventual ordinary-prime exponent remains active.

E.80 computes the actual joint image of h and its conjugate, including the connected degree-three cover, its rank-seven intermediate group and both filling kernels. The complete wild section set over $\mathbb{Q}_3$ injects into the first-factor set over $\mathbb{Q}_3(\zeta_3)$ with four explicit character conditions and an explicit inverse gauge. Both lower lifting images and all their fibres are evaluated. The intrinsic third-degree map has image of index nine in the product central kernel; the actual quotient kernels retain the remaining intrinsic problem.

E.81 proves that the actual prime above 73 retains ordinary class order three after extension from B_1 to B_2. The complete ramified Artin matrix and literal principal S-divisors reduce its S-capitulation to twenty-seven explicit ordinary principality tests, with inverse lattice coordinates and all principal cube generators. The ambiguous genus kernel has order at most three, so the number of principal candidates is zero, nine or twenty-seven. E.86 proves all twenty-seven ideals principal and evaluates the resulting S-unit norm defect as C_3.

E.82 evaluates all forty-five intrinsic second-commutator variations of an actual local section at each of the nine cusps. It computes the full geometric gauge, the exact intrinsic third-degree obstruction, its image and kernel, every nonempty fibre and a convergent particular central correction. The elliptic commutator gives a literal nonlifting parameter at every cusp. The nine evaluated images generate a subgroup whose exact remaining quotient in the intrinsic obstruction group is $C_3^{11} \oplus C_9$. The arbitrary wild inputs and their higher and global compatibilities remain to be evaluated.

E.83 computes the actual intrinsic-to-joint central obstruction map, including its full integral Hall matrix and its literal Smith-coordinate character. Its image is the diagonal C_3 in C_3^2. The natural obstruction square excludes every intrinsic class-two preimage of the specified incompatible joint-quotient family. All nine cusp-family obstruction images lie in its kernel, and the remaining quotient of that kernel by their sum is exactly $C_3^{10} \oplus C_9$, with explicit coordinates and inverse. E.85 and E.87 evaluate the compatible joint image and fibres on their specified intrinsic abelian loci; the other relative inputs and their higher compatibilities remain active.

All these transfers use the actual quotient homomorphisms of E.25, E.36–E.38. Wickelgren's opened formulas apply to the ambient rank-two group; the full intrinsic calculation retains its separately computed integral action and its eight cusp path classes. E.22 settles the proposed descent from the manuscript scheme by an actual Galois counterexample. The remaining task is to evaluate the higher obstructions over all the class-two sections, continue through the metabelian and marked-depth quotients, and include the other prime and mixed finite quotients. The computed pro-3 stages are retained as such; they do not assert the full profinite classification.

The complete cubic abelian calculation and the precise depth maps

In this subsection $E = X_3^+$ is the cubic $x^3 + y^3 + z^3 = 0$, $D = E \setminus U_3$, and $G = J_{E,D}$ is the generalized Jacobian for the reduced divisor D. We use the origin $O = [1 : -1 : 0]$, and set $P = [0 : -1 : 1]$, $Q = [-1 : 0 : 1]$. For an abelian group A we write $A^\wedge = \varprojlim_m A/mA$, with m ordered by divisibility. This is the completion for multiplication by m; the quotients need not be finite. All inverse limits of cohomology groups below carry their inverse-limit prodiscrete topology. For a countable set $\mathcal{I}$ define

$$\mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) = \left\{ (a_j) \in \prod_{j \in \mathcal{I}} \widehat{\mathbb{Z}} : \text{for every } m,\ a_j \in m\widehat{\mathbb{Z}} \text{ for all but finitely many } j \right\}.$$

Its topology is the one coming from $\varprojlim_m \bigoplus_{\mathcal{I}} \mathbb{Z}/m$. In particular this space is not asserted to be compact.

E.24. All cubic abelian sections and every fiber over the toric minimum [Proved]

Put $K = \mathbb{Q}(\zeta)$, $\zeta^2 + \zeta + 1 = 0$, and $\lambda = 1 - \zeta$. There is a retraction of algebraic groups over $\mathbb{Q}$

$$r : G \longrightarrow \mathbb{G}_m^2, \qquad G \simeq \mathbb{G}_m^2 \times G', \qquad 1 \longrightarrow (\mathrm{Res}_{K/\mathbb{Q}}\, \mathbb{G}_m)^3 \longrightarrow G' \longrightarrow E \longrightarrow 0.$$

The map r is the map induced by the two units $x/z, y/z$ on U_3. The group $E(\mathbb{Q})$ is cyclic of order three, and

$$G'(\mathbb{Q}) \simeq \mu_6^3 \oplus \bigoplus_{j \in \mathcal{I}} \mathbb{Z}.$$

Here one may take $\mathcal{I}$ to consist of three copies of the nonzero prime ideals of $\mathbb{Z}[\zeta]$, with the first copy of (λ) replaced by one new generator. The displayed abstract isomorphism includes explicit choices of prime generators.

Choose the unit rational tangent at O, with local parameter z/x, as the origin of the set of sections of the geometrically abelian arithmetic fundamental group. Up to geometric conjugation, that entire set is then the group

$$H^1(G_{\mathbb{Q}}, T_\bullet G) \simeq (\mathbb{Q}^{\times,\wedge})^2 \times \mu_6^3 \times \mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}. \tag{E24.1}$$

The last factor and its splitting are noncanonical. The actual map to the toric minimum of E.16 is the projection onto the first two factors followed by translation by the Kummer class of $(1, -1)$. Consequently it is surjective, and every fiber is a torsor under

$$\mu_6^3 \times \mathcal{R}_{\mathcal{I}}(\widehat{\mathbb{Z}}) \times \widehat{\mathbb{Z}}. \tag{E24.2}$$

For the pro-3 geometric abelianization the same formulas hold with $\mathbb{Q}^{\times,\wedge}$ replaced by $\varprojlim_a \mathbb{Q}^\times/\mathbb{Q}^{\times 3^a}$, μ_6^3 replaced by μ_3^3, and each $\widehat{\mathbb{Z}}$ replaced by $\mathbb{Z}_3$. The translation by -1 is zero in this pro-3 version.

Proof. The rational elliptic-curve calculation. The maps

$$X = -\frac{3z}{x+y}, \qquad Y = \frac{4x - 5y}{x+y}, \qquad [x : y : z] = [Y + 5 : 4 - Y : -3X]$$

identify the smooth cubic with $Y^2 + Y = X^3 - 7$, with O at infinity. Indeed the exact identity is

$$Y^2 + Y - X^3 + 7 = \frac{27(x^3 + y^3 + z^3)}{(x+y)^3}.$$

The inverse formula proves the birational assertion, which extends to an isomorphism of the smooth projective curves.

For the rank calculation we use the explicit identification over $\mathbb{Q}$ of the Fermat cubic with $X_0(27)$ in Li–Long–Tu, §2.4 and Remark 4.2 [LLT18]. The corresponding normalized differential has newform $f(\tau) = \eta(3\tau)^2\eta(9\tau)^2$. The Mellin formula therefore gives

$$L(E,1) = 2\pi \int_0^\infty \eta(3it)^2\eta(9it)^2\, dt > 0.$$

The product formula for η makes the integrand strictly positive for $t > 0$, and cuspidality gives convergence at both ends. The automorphism induced by multiplication of one Fermat coordinate by ζ gives complex multiplication by $\mathbb{Z}[\zeta]$; this is the maximal order of the norm-Euclidean field K. Coates–Wiles, Theorem 1 on p. 223 [CW77], now gives rank zero. The smooth cubic has exactly three points over $\mathbb{F}_2$, so the prime-to-2 rational torsion injects into a group of order three. An order-two point on the displayed Weierstrass model would have $Y = -1/2$ and $X^3 = 27/4$, which has no rational solution. Thus there is no 2-primary torsion. The distinct points O, P, Q already give three rational points, so $E(\mathbb{Q}) = \langle P \rangle \simeq \mathbb{Z}/3$.

The retraction and the exact rational-point presentation. The cusp scheme is the disjoint union of three rational points and three copies of $\operatorname{Spec} K$. Taking the frame at O as one identifies the torus of G with

$$T_D = \mathbb{G}_m^2 \times (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3.$$

We write its coordinates as (a, b, A, B, C), at P, Q and the three quadratic cusp orbits, respectively. The convention is that a frame is a nonzero vector in the fiber of the line bundle.

Here is an explicit construction of the characters used in r. For a unit h on U_3, write $\operatorname{div}(h) = \sum_{d \in D} m_d d$. For a framed line bundle $(L, (e_d))$, choose a rational section s nonvanishing along D, and set

$$\chi_h(L, (e_d)) = h(\operatorname{div} s) \prod_{d \in D} (s(d)/e_d)^{-m_d}. \tag{E24.3}$$

Products over nonrational points mean the corresponding norms. Replacing s by fs leaves this expression unchanged, since Weil reciprocity gives $h(\operatorname{div} f) = f(\operatorname{div} h)$. Tensor products multiply (E24.3). The construction in families, or the divisor presentation of the generalized Jacobian, consequently gives a morphism of algebraic groups. On the torus it is $\prod e_d^{m_d}$, and on a divisor $u - u_0$ in U_3 it is $h(u)/h(u_0)$. Thus it is precisely the character induced by h.

For $h = x/z, y/z$ their divisors are the respective coordinate-cusp orbits minus the orbit at infinity. Hence

$$r_T(a, b, A, B, C) = \big(aN(A)/N(C),\ bN(B)/N(C)\big). \tag{E24.4}$$

The map $i(a, b) = (a, b, 1, 1, 1)$ is an algebraic section of r. The isomorphism with the direct product is explicitly $g \mapsto (r(g),\ g\, i(r(g))^{-1})$. The kernel torus is parametrized by

$$(A, B, C) \longmapsto \big(N(C)/N(A),\ N(C)/N(B),\ A, B, C\big). \tag{E24.5}$$

Hilbert 90 gives $H^1(\mathbb{Q}, T_D) = 0$ and $H^1(\mathbb{Q}, (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3) = 0$.

To calculate the extension on rational points, let $z_0 \in G(\mathbb{Q})$ be the framed line bundle $\mathcal{O}(P - O)$ with the local frames specified next. Put $f = (y + z)/(x + y)$, so $\operatorname{div}(f) = 3P - 3O$. At P use $s = x/z$ and the frame s^{-1} for $\mathcal{O}(P - O)$; at O use $s = z/x$ and the frame s. At the other cusps use the canonical section 1. Multiplication by f trivializes its third tensor power. The two leading values of f are

$$\lim_P f/s^3 = 1/3, \qquad \lim_O fs^3 = 3, \qquad f(Q) = -1.$$

For example $y/z = -(1+s^3)^{1/3}$ at P, which proves the first equality; the same expansion at O proves the second. At the nonrational cusps use the representatives $[0:-\zeta:1]$, $[-\zeta:0:1]$, $[1:-\zeta:0]$. Dividing every frame value by the value 3 at O gives

$$3z_0 = \left(3^{-2}, -3^{-1}, \lambda^{-1}, \lambda^{-2}, \zeta^2\lambda^{-3}\right) \in \mathbb{Q}^{\times 2} \times K^{\times 3}. \tag{E24.6}$$

This calculation also follows directly from $3 = -\zeta^2\lambda^2$. Equation (E24.4), with $N(\lambda) = 3$, sends (E24.6) to $(1,-1)$. The rational cube roots of 1 and -1 are unique, so $r(z_0) = (1,-1)$. For $z' = z_0 - i(1,-1)$ the presentation becomes

$$G'(\mathbb{Q}) = (K^{\times 3} \oplus \mathbb{Z}z') / \left\langle 3z' - (\lambda^{-1}, \lambda^{-2}, \zeta^2\lambda^{-3}) \right\rangle. \tag{E24.7}$$

There are no additional relations: the quotient onto $E(\mathbb{Q}) = \mathbb{Z}/3$ is surjective by Hilbert 90, the class of z' generates it, and (E24.7) specifies its third multiple in the kernel.

The norm-Euclidean ring $\mathbb{Z}[\zeta]$ has units μ_6, so prime factorization gives $K^\times = \mu_6 \oplus \bigoplus_{\mathfrak{p}} \mathbb{Z}$ after choosing one generator for each prime ideal. In (E24.7) the coefficient of the first λ generator is -1. Eliminate that generator and retain z' instead. This operation changes no torsion relation and proves the asserted description of $G'(\mathbb{Q})$. As an independent finite-support check, `e_cubic_abelian.py` obtains the Smith diagonal $1, 6, 6, 6$ for the relation involving the three μ_6 generators, the three λ generators and z'.

The full cohomology calculation. Write $B = (\mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m)^3$. The extension of E by B in G' is killed by three: the extension of E by T_D is classified by the cusp-difference map, all of whose values are killed by three by C.5, and (E24.5) and the direct product exhibit the component defining G'. It follows that the connecting map

$$\delta : H^1(\mathbb{Q}, E) \longrightarrow H^2(\mathbb{Q}, B)$$

is killed by three. Since $H^1(\mathbb{Q}, B) = 0$, $H^1(\mathbb{Q}, G')$ injects into $H^1(\mathbb{Q}, E)$. Every element of the maximal divisible subgroup of the latter is three times an element of that subgroup; hence its image under δ is zero. Consequently the injection identifies maximal divisible subgroups:

$$\mathrm{Div}\, H^1(\mathbb{Q}, G') = \mathrm{Div}\, H^1(\mathbb{Q}, E) \simeq \mathbb{Q}/\mathbb{Z}. \tag{E24.8}$$

The last equality is exactly Çiperiani–Stix, Theorem 18(2), printed p. 18 [ÇS13], applied to the analytic-rank-zero curve already established above. Their additional restrictions on primes concern the separate assertion about the Tate–Shafarevich intersection. They impose no restriction on (E24.8).

The Kummer sequences and inverse limit give

$$0 \longrightarrow G'(\mathbb{Q})^\wedge \longrightarrow H^1(G_\mathbb{Q}, T_\bullet G') \longrightarrow \mathrm{Hom}(\mathbb{Q}/\mathbb{Z}, H^1(\mathbb{Q}, G')) \longrightarrow 0. \tag{E24.9}$$

For precision, take the countable cofinal subsystem $m = a!$. The transitions $G'(\mathbb{Q})/m \to G'(\mathbb{Q})/d$, $d \mid m$, are surjective, so their first derived inverse limit vanishes. The finite groups $H^0(\mathbb{Q}, G'[m])$ satisfy the Mittag–Leffler condition, which identifies continuous H^1 of $T_\bullet G'$ with the inverse limit of the finite-coefficient H^1 groups. No finiteness of $G'(\mathbb{Q})/m$ or $H^1(\mathbb{Q}, G'[m])$ is used. The last term in (E24.9) is the limit of $H^1(\mathbb{Q}, G')[m]$ under multiplication transitions. Every homomorphic image of $\mathbb{Q}/\mathbb{Z}$ is divisible, so (E24.8) identifies it with $\mathrm{End}(\mathbb{Q}/\mathbb{Z}) = \widehat{\mathbb{Z}}$.

Choose a lift of 1 in (E24.9). Multiplication by $\widehat{\mathbb{Z}}$ then gives a continuous splitting: modulo m its value depends only on the scalar modulo m. The description (E24.7) gives

$$G'(\mathbb{Q})^\wedge = \mu_6^3 \times \mathcal{R}_\mathcal{I}(\widehat{\mathbb{Z}}). \tag{E24.10}$$

Indeed the image at each m has finite support in $\mathcal{I}$, and compatibility is exactly the restricted-sequence condition in the definition. This proves both the algebraic and topological assertions in (E24.9). The identical argument with the subsystem $m = 3^a$ gives the pro-3 version, whose divisible quotient is $\mathrm{Hom}(\mathbb{Q}_3/\mathbb{Z}_3, \mathbb{Q}_3/\mathbb{Z}_3) = \mathbb{Z}_3$.

Finally, $T_\bullet G = H_1(U_{3,\overline{\mathbb{Q}}}, \widehat{\mathbb{Z}})$ by the generalized-Jacobian realization in C.1. A rational tangential section splits the corresponding arithmetic extension; its conjugacy classes of splittings are exactly continuous one-cocycles modulo coboundaries. Thus (E24.9), (E24.10), Hilbert 90 and the algebraic retraction give (E24.1). At O, with $s = z/x$, the leading terms of $(x/z, y/z)$ are $(s^{-1}, -s^{-1})$. This makes the toric origin the Kummer class of $(1, -1)$ and proves the claimed affine map. The covering pushforward q_{3*} is multiplication by three after r_* on geometric homology; the map used here is the actual coordinate-unit map r_*, as in E.16. This fixes the comparison and the translation, and proves the fiber assertion. □

E.25. Intrinsic depth, inherited depth, and the even-level obstruction [Proved]

Let $N_n = \pi_1(U_{n,\overline{\mathbb{Q}}})$, embedded in the geometric profinite group $\widehat{F}$ of U_1 by the covering. In this statement write $F = \widehat{F}$ in the profinite formulas. Use y for inertia at the base puncture 1. For an intrinsic depth filtration choose a Galois-stable set of cusps above 1, and let I_n be the closed normal subgroup generated by their inertia. Put $D^r_{\mathrm{intr}} = \gamma_r I_n$. One allowed choice for $n = 3$ is the rational cusp Q alone; for $n = 4$ take the entire four-point coordinate-cusp orbit. There are actual quotient maps

$$N_n/D^3_{\mathrm{intr}} \longrightarrow \mathrm{im}(N_n \longrightarrow F/D^3_y), \qquad \ker = \frac{N_n \cap D^3_y}{D^3_{\mathrm{intr}}}, \tag{E25.1}$$

where $D^3_y = \gamma_3\langle\langle y \rangle\rangle$. The same formulas apply to the corresponding Lie ideals and their completed rational Lie algebras.

Over each field $\mathbb{Q}_\ell$ the inherited prounipotent image is all of $F^{\mathrm{un}}_{\mathbb{Q}_\ell}/D^3_y$. The corresponding inherited arithmetic extension is the base arithmetic extension and has a rational tangential section, for every n. On abelianized Lie algebras the map from the intrinsic quotient has a kernel of dimension $n^2 - 1$.

For every even n, the inherited pro-2 depth quotient, the inherited pro-2 class-two quotient, and every intrinsic quotient retaining the curve's own pro-2 abelianization have empty section sets over $\mathbb{Q}$. The obstruction is the explicit central extension with class $\kappa_2(-1) \cup \kappa_2(-1)$ described below.

Proof. Inertia above 1 maps to a conjugate of a power of y. Consequently I_n maps into $\langle\langle y \rangle\rangle$, and lower central series give (E25.1), including its exact kernel. The chosen sets of cusps are Galois stable, so their normal closures and lower central terms are preserved by the full arithmetic fundamental group. These are therefore arithmetic quotients, not just quotients of unmarked geometric groups.

By E.9 the map $N^{\mathrm{un}}_{n,\mathbb{Q}_\ell} \to F^{\mathrm{un}}_{\mathbb{Q}_\ell}$ is surjective: positive powers of x, y lie in N_n, and unique roots in characteristic-zero unipotent groups recover x, y. Its inherited quotient is thus the entire base quotient. The map on arithmetic extensions induces this surjection on geometric kernels and the identity on $G_{\mathbb{Q}}$. Quotienting by the geometric kernel of that surjection identifies the inherited arithmetic extension with the base one, which is split by the unit tangent at 0 towards 1. Equivalently its fibers are the full fibers of $(F^{\mathrm{un}}_{\mathbb{Q}_\ell}/D^3_y) \rtimes G_{\mathbb{Q}}$. Since both depth ideals lie in the respective derived Lie algebras, their abelianizations are the original ones. The kernel dimension is therefore exactly the $n^2 - 1$ of B.5 and E.13.

The integral obstruction uses the quotient

$$F \longrightarrow D_8 = \langle x, y, c \mid x^2 = y^2 = c^2 = 1,\ c \text{ central},\ [x, y] = c \rangle. \tag{E25.2}$$

With the standard rational tangential section, the Galois action on this quotient is trivial: in the normalized action of E.20 the cyclotomic exponent is odd, while the conjugating element lies in F' and hence has central image in D_8. The arithmetic base quotient is $D_8 \times G_{\mathbb{Q}}$.

In the plus model, with $t = -x^n/z^n$, the two covering roots are

$$(x/z)^n = -t, \qquad (y/z)^n = -(1-t). \tag{E25.3}$$

At the unit tangent $t = 0$, their constant root factors are nth roots of -1. For even n, their reductions to quadratic root factors both give $\epsilon = \kappa_2(-1) : G_{\mathbb{Q}} \to \mathbb{Z}/2$. The sign in the stabilizer condition has no effect modulo two. The geometric image of N_n in D_8 is exactly $\langle c \rangle$: its two exponent sums are even, and $[x, y] \in N_n$ maps to c. The arithmetic image is consequently the explicit pullback

$$\mathcal{P} = \{(d, g) \in D_8 \times G_{\mathbb{Q}} : d^{\mathrm{ab}} = (\epsilon(g), \epsilon(g))\}. \tag{E25.4}$$

For each g there is a lift in $\pi_1(U_n)$, and multiplication by the geometric element c gives both lifts in (E25.4); this proves equality of the image, not just its containment. Using $(xy)^{\epsilon(g)}$ as a set-theoretic lift gives the factor set $c^{\epsilon(g)\epsilon(h)}$. Thus (E25.4) represents $\kappa_2(-1) \cup \kappa_2(-1)$.

Complex conjugation has $\epsilon = 1$. Both possible lifts, xy and xyc, square to $c \neq 1$. The extension cannot have a section even after restriction to $G_{\mathbb{R}}$, hence cannot have one over $\mathbb{Q}$. The group D_8 kills $\gamma_3 F$ and hence D_y^3. Its geometric image from N_n is abelian, so it also kills N_n'. These two explicit factorizations prove all three nonexistence assertions in the statement. They recover E.18 at level four and prove the assertion for all even levels. Pushing to the rank-two toric lattice kills the central element c, exactly as in E.18; extending scalars to $\mathbb{Q}_\ell$ gives the inherited comparison already proved. Thus neither operation is silently identified with the integral intrinsic quotient.

For clarity the intrinsic marked-depth quotient also has a precise comparison with the third derived quotient. Their common quotient is $N_n/(N_n''' D^3_{\mathrm{intr}})$. For the markings just specified neither kernel contains the other. Indeed filling the marked cusps gives free groups of ranks 9 and 13, respectively. On six distinct remaining generators use the word $w = [[[a, b], [a, c]], [[d, e], [d, f]]]$. Its degree-eight Magnus term has the monomial *abacdedf* with coefficient one, and is nonzero. This third-derived word survives the filling map, so N_n''' is not contained in D^3_{intr}. In the other direction choose a marked cusp generator z_0 and an unmarked free generator a. The element $[z_0, [z_0, [z_0, a]]]$ lies in $\gamma_3 I_n$. Its degree-four Magnus term is nonzero, whereas every element of N_n''' has zero Magnus terms below degree eight. Thus it does not lie in N_n'''. The corresponding maps from N_n, and the common quotient, state the relation without imposing the base rank-two containment from E.2 on a different free group. □

The first obstruction beyond the splitting of §51.12

E.26. The class-two splitting obstruction is the generator modulo three [Proved]

Let

$$P = \langle \rho, \sigma \mid \rho^3 = \sigma^2 = 1 \rangle = \mathrm{PSL}_2(\mathbb{Z}), \qquad q : P \longrightarrow S_3,$$

where $q(\rho)$ is a three-cycle and $q(\sigma)$ a transposition. The kernel is the free group

$$F = \langle x, y \rangle, \qquad x = (\sigma\rho)^2, \quad y = (\rho\sigma)^2.$$

The extension of S_3 by F/F' splits by B.11. Its class-two refinement

$$1 \longrightarrow F/\gamma_3 F \longrightarrow P/\gamma_3 F \longrightarrow S_3 \longrightarrow 1 \tag{E.26a}$$

does not split. Neither does its profinite completion, its relative pro-3 completion, or any of the discrete or profinite extensions $P/F^{(j)} \to S_3$ for $j \geq 2$.

Write $c = [x, y]$ and $r' = x\rho$. The inverse image in (E.26a) of the chosen abelian complement is

$$1 \longrightarrow \mathbb{Z}(\mathrm{sign}) \xrightarrow{1 \mapsto c} \langle r', \sigma \mid \sigma^2 = 1,\ \sigma r' \sigma = (r')^{-1} \rangle \longrightarrow S_3 \longrightarrow 1, \qquad (r')^3 = c. \tag{E.26b}$$

Its cohomology class is 1 in $H^2(S_3, \mathbb{Z}(\mathrm{sign})) \simeq \mathbb{Z}/3$. The finite quotient

$$P/(\gamma_3 F\, F^3) \longrightarrow S_3 \tag{E.26c}$$

has order 162, Heisenberg kernel of order 27, and no section. The inverse image of the chosen abelian complement is the dihedral group of order 18.

For the class-two relative pro-p extension, the set of continuous sections modulo conjugation by its kernel is empty at $p = 3$, has two elements at $p = 2$, and has one element at every $p \geq 5$. Over $\mathbb{Q}$ or $\mathbb{Q}_\ell$, the rational class-two relative extension does split, with one conjugacy class of sections. An explicit section in every one of these cases where it exists is

$$\bar{\rho} \longmapsto c^{-1/3} x\rho, \qquad \bar{\sigma} \longmapsto \sigma. \tag{E.26d}$$

Here c^a denotes the actual central coordinate a of the class-two kernel; it is defined over $\mathbb{Z}_p$ for $p \neq 3$ and over the indicated fields.

Manuscript: §51.12, folios 700–704 and 715–716, scans 105–114 and 135–138; in particular the abelian splitting on scan 136. This answers the splitting part of E(c) with its integral, profinite, and rational comparison maps.

Proof. The free-kernel identification and the matrices of its conjugation action were proved in A.2 and B.9. In the present central quotient of the modular group the exact free-group substitutions are

$$\rho x \rho^{-1} = y, \qquad \rho y \rho^{-1} = x^{-1} y^{-1}, \qquad \sigma x \sigma = y, \quad \sigma y \sigma = x. \tag{E.26e}$$

They also follow immediately by substituting the displayed words for x, y and using $\rho^3 = \sigma^2 = 1$. Consequently, already in P,

$$(x\rho)^3 = xyx^{-1}y^{-1} = c, \qquad ((x\rho)\sigma)^2 = 1. \tag{E.26f}$$

The second identity follows alternatively by reducing $(\sigma\rho\sigma\rho^2\sigma)^2$.

We specify the integral class-two group rather than replacing it by a vector space. Every element of $F/\gamma_3 F$ has a unique normal form $x^a y^b c^k$, $a, b, k \in \mathbb{Z}$, with law

$$(a, b, k)(a', b', k') = (a + a', b + b', k + k' - a'b). \tag{E.26g}$$

Existence is collection using the central commutator c. Uniqueness follows from the injective matrix model

$$x^a y^b c^k \longmapsto \begin{pmatrix} 1 & a & ab + k \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus c has infinite order. Conjugation by r' fixes c and conjugation by σ sends c to c^{-1}. The inverse image of the complement generated by $\overline{x\rho}, \bar{\sigma}$ in P/F' has elements $c^k (r')^i \sigma^j$, with $0 \leq i < 3$, $0 \leq j < 2$. Since $c = (r')^3$, these are exactly the normal forms of the infinite dihedral group in (E.26b). There are no additional relations: the image in S_3 first determines i, j, and $c^k = 1$ then forces $k = 0$.

For clarity we compute the cohomology class. Write elements of S_3 as $(i,j) = \bar{\rho}^i\bar{\sigma}^j$, with $0 \le i < 3$ and $0 \le j < 2$. The section $(i,j) \mapsto (r')^i\sigma^j$ gives the normalized integral two-cocycle

$$f((i,j),(i',j')) = \left\lfloor \frac{i + (-1)^j i'}{3} \right\rfloor, \qquad (i,j)\cdot a = (-1)^j a. \tag{E.26h}$$

Its restriction to the cyclic subgroup has carry 1, namely $(r')^3 = c$. We can compute the whole cohomology group directly. In an arbitrary extension by $A(\mathrm{sign})$, for $A = \mathbb{Z}$ or $\mathbb{Z}_3$, choose lifts r, s. The elements s^2 and $(rs)^2$ lie in A and are fixed by conjugation by s and rs, respectively. Those conjugations act as -1 on A; because A has no 2-torsion, both elements vanish. The only remaining parameter is $r^3 = a \in A$. Replacing r by br, with additive notation on A, changes a to $a + 3b$. Conversely every $a \in A$ defines an extension: take $(A \oplus \mathbb{Z}r)/(3r - a)$ and adjoin the involution acting by negation. This proves $H^2(S_3, A(\mathrm{sign})) = A/3A$ and identifies (E.26b) with 1. Equivalently, the one-cochain $(i,j) \mapsto i$ has coboundary $3f$.

To exclude sections of the whole group, we must also exclude other abelian complements. B.10 proves, over every coefficient group used here,

$$H^0(S_3, M) = H^1(S_3, M) = 0, \qquad M = F^{\mathrm{ab}}.$$

Thus conjugation by a unique element of M carries any abelian section to the B.11 section. The surjection $F/\gamma_3 F \to M$ lifts this conjugating element. After this conjugation any proposed section of (E.26a) must send $\bar{\rho}$ to $c^a r'$ and $\bar{\sigma}$ to $c^b\sigma$. Its order-three relation is

$$(c^a r')^3 = c^{1+3a} = 1. \tag{E.26i}$$

There is no solution $a \in \mathbb{Z}$ or $a \in \mathbb{Z}_3$. The other two defining relations hold for all a, b, because σ and $r'\sigma$ invert c and have order two. This proves the integral and relative pro-3 assertions.

Here are the exact completion and quotient comparisons. Since F is normal in P and $\gamma_3 F$, F^3, and $F^{(j)}$ are characteristic in F, every displayed quotient is a group extension over the same S_3. The relative pro-p class-two kernel is obtained by completing the normal form (E.26g) to coordinates in $\mathbb{Z}_p$. Conjugation (E.26e) extends continuously: its formulas use addition, multiplication, and $\binom{a}{2}$, the continuous integer-valued polynomial on $\mathbb{Z}_p$. The finite cosets of F give the corresponding continuous extension over S_3. The quotient (E.26c) receives (E.26a) by reducing its three kernel coordinates modulo three. It therefore also receives the profinite completion of (E.26a), continuously. A section of that completion would give a section of (E.26c).

To construct (E.26c) completely, over $\mathbb{F}_3$ the law (E.26g) is the Heisenberg group of order 27 and exponent three. It is exactly $F/(\gamma_3 F F^3)$: collection bounds its order by 27, and the matrix model realizes all 27 normal forms. The quotient has six such cosets, hence order 162. With states $x^a y^b c^k \rho^i \sigma^j$, right multiplication by σ toggles j. Right multiplication by ρ increments i when $j = 0$, and when $j = 1$ it multiplies the kernel coordinate by $\rho^i x \rho^{-i}$ and replaces i by $i + 2$ modulo three. The three kernel factors are $x, y, x^{-1}y^{-1}$. These rules construct its regular permutation representation without a presentation-order guess. The program `code/e_splitting.py` checks closure of the 162 normal-form permutations under both generators, all 4 251 528 associativity triples, and all $27^2 = 729$ pairs of possible section-generator images. There are no sections. Its abelian quotient of order 54 has exactly nine sections; all 81 pairs lifting these have their $\bar{\rho}$ image of order nine. The chosen complement has preimage $\langle r', \sigma\rangle = D_{18}$, with $(r')^3 = c$. The code also checks all 216 two-cocycle identities and all 243 normalized one-cochains modulo three: none has coboundary f.

For $j \ge 2$, the commutator identities $[\gamma_a F, \gamma_b F] \subset \gamma_{a+b} F$ give $F^{(j)} \subset \gamma_{2^j} F \subset \gamma_3 F$. Consequently the identity on generators gives the exact quotient map

$$P/F^{(j)} \twoheadrightarrow P/\gamma_3 F, \quad \ker = \gamma_3 F/F^{(j)}. \tag{E.26j}$$

A section of the source extension would compose to a section of (E.26a). For the profinite version, use the continuous map to the finite group (E.26c); its kernel contains the closed jth derived subgroup. The same composition gives the contradiction. This establishes the transfer to the requested third-derived quotient by an actual surjection.

Finally, the normalized lift calculation (E.26i) works with central coordinates in $\mathbb{Z}_p$, $\mathbb{Q}$, or $\mathbb{Q}_\ell$. It has the unique solution $a = -1/3$ exactly in the coefficient cases specified in the statement. The parameter b remains arbitrary. Any kernel conjugation preserving the normalized abelian complement has zero image in M, by $H^0(S_3, M) = 0$, and is therefore c^u. It changes b to $b + 2u$. Thus the conjugacy classes are $\mathbb{Z}_p/2\mathbb{Z}_p$ for $p \neq 3$, and a singleton over the indicated characteristic-zero fields. Formula (E.26d) gives their specified representative. In the cochain description, $-i/3$ cancels (E.26h), so the comparison to rational coefficients explicitly kills this integral obstruction. The rational group here is the class-two relative unipotent quotient over S_3: its six cosets are copies of the Heisenberg group (E.26g) over the field. To verify the universal comparison, a homomorphism from F to a class-two unipotent group, with images X, Y, extends uniquely by $x^a y^b c^k \mapsto X^a Y^b [X, Y]^k$. Rational powers are given by the finite exponential and logarithm polynomials. The finite-coset relations then give the unique extension of the original homomorphism from P over S_3. The program also checks the normalized central equations $1 + 3a = 0$ and the conjugacy relation $b \sim b + 2u$ modulo p^e for $p = 2, 3, 5, 7$ and $1 \leq e \leq 4$.

We also record precisely how the marked-depth construction in E.2 compares. There is a surjective Lie homomorphism

$$L/D^3L \longrightarrow L/\gamma_3 L, \qquad \ker = \gamma_3 L/D^3 L,$$

because a monomial with at least three occurrences of y has total degree at least three. The specific full permutation action does not descend to that marked quotient: the swap $x \leftrightarrow y$ sends $[y, [y, [y, x]]] \in D^3L$ to $[x, [x, [x, y]]]$, a nonzero depth-one element modulo D^3L. Thus (E.26j) is an S_3-extension comparison for the derived quotients, whereas the displayed marked-depth map has the stated Lie source and target. The rational section (E.26d) and the integral obstruction (E.26i) retain these exact domains and coefficient rings. □

Boundary types and the scope of the covering criterion

E.27. The four boundary types and their actual comparison maps [Proved]

Over an algebraically closed field of characteristic zero the geometric fundamental groups of types $(0,0), (0,1), (0,2), (1,0)$ are, respectively,

$$1, \qquad 1, \qquad \widehat{\mathbb{Z}}(1), \qquad TE \simeq \widehat{\mathbb{Z}}^{,2},$$

where a chosen origin gives the elliptic curve E in the last case. The last two groups are their own centers and their own abelian and metabelian quotients. The assertions about nontrivial comparison kernels in E.7–E.9 have the stated rank-two free-group source. For the discrete group $F_1 = \mathbb{Z}$ and its subgroup $d\mathbb{Z}$, the abelian, metabelian and fixed nilpotent comparisons to their correctly descended kernels are isomorphisms for every d. The rational completion map is multiplication by d on $\mathbb{Q}$, hence is also an isomorphism. For $d = p^a m$ with $(m, p) = 1$, the pro-p comparison is the isomorphism

$$\mathbb{Z}_p \longrightarrow p^a \mathbb{Z}_p, \qquad u \longmapsto du.$$

The same statements hold for a finite-index lattice in $\mathbb{Z}^2$, with its inclusion matrix in place of d.

For the corresponding nontrivial finite covers of $B\mathbb{Z}$ or $B\mathbb{Z}^2$, intrinsic-fibration membership for the five localizers of E.5 still holds only for the p-group covers in the pro-p case. Thus the equivalence between the two criteria in E.10 is proved for its specified source F_2; the displayed lattice comparisons compute what happens at these boundary sources.

Proof. For the projective line a connected degree-d finite etale cover has $2g-2=-2d$ by Riemann–Hurwitz, forcing $d=1$. For the affine line, extend a finite etale cover to smooth projective curves. Ramification can occur only over infinity, and its total contribution is at most $d-1$. Hence $2g-2\le -d-1$, again forcing $d=1$. For the twice-punctured line, ramification is allowed at two points. The same formula forces $g=0$ and total ramification over each of them. Choosing coordinates at their unique preimages makes the covering $z\mapsto z^d$. The cofinal inverse system of these maps has deck groups μ_d, which proves the asserted Tate-module description and its action.

A finite etale cover of a genus-one curve has genus one by the same formula. Choose a point over the fixed origin. The resulting pointed map of elliptic curves is an isogeny: its map to the target Jacobian is the pointed curve map under the degree-zero divisor identification. Its kernel is finite; its dual isogeny shows that a multiplication map dominates it. The multiplication coverings are therefore cofinal, and their inverse deck system is TE. In characteristic zero $E[n]\simeq(\mathbb{Z}/n)^2$, giving the displayed geometric group. For the cubic over $\mathbb{Q}$ the specified origin is a rational coordinate cusp; E.24 computes its required arithmetic abelian cohomology. We impose no rational-point conclusion on a genus-one torsor without such an origin.

For the rank-one discrete group, every verbal commutator subgroup is zero, so the covering inclusion itself is exactly the displayed kernel map. Rationalization gives multiplication by d. In $\mathbb{Z}_p$, multiplication by m is a unit and multiplication by p^a maps bijectively onto the closed subgroup $p^a\mathbb{Z}_p$. For a rank-two lattice, integral Smith normal form gives the same calculation in two coordinates. These arguments compute both the image and zero kernel, rather than applying the free-group rank formula outside its hypotheses. The program `e_boundary.py` checks the rank-one formulas for $d\le 24$ and the inverse unit coordinates modulo p^k, for $p=2,3,5,7$ and $k\le 4$.

Intrinsic-fibration membership is preserved by pullback, directly from its universal base-change definition. Pull a nontrivial lattice cover back by $BF_2\to B\mathbb{Z}$ or $BF_2\to B\mathbb{Z}^2$, using a surjection onto the lattice. Its monodromy is still the given finite quotient, so E.10 excludes the pulled-back cover in all the claimed cases. The general p-group sufficiency in E.10 proves the other cases. This supplies the comparison between the two criteria at every boundary source under discussion.

Real descent also retains its arithmetic data. The plus conic $X_2: x_0^2+x_1^2+x_\infty^2=0$ has no real point, but its geometric projective fundamental group is trivial, so its arithmetic fundamental group maps isomorphically to $G_{\mathbb{Q}}$ and has its unique section. Deleting the coordinate divisors gives the nontrivial real obstruction treated by E.25. The exact open-to-proper quotient kills that obstruction. The elliptic and multiplicative centers above, the retained central C_2 in A.14, and the real orientation-reversing extension in E.18 are consequently part of the calculations rather than omitted exceptional cases. □

A cubic Kummer obstruction at the prime 109

E.28. A nonempty cubic abelian fiber with no class-two lifts [Proved]

In the pro-3 version of E.24, consider the toric section class

$$(\kappa(6),\kappa(109))\in H^1(\mathbb{Q},\mathbb{Z}_3(1)^2). \tag{E28.1}$$

Its fiber of geometrically abelian sections is nonempty and is a torsor under $\mu_3^3 \times \mathcal{R}_{\mathcal{I}}(\mathbb{Z}_3) \times \mathbb{Z}_3$. No point of this fiber lifts to a section of the curve's own class-two pro-3 quotient. No such lift exists for the intrinsic metabelian quotient, the intrinsic marked-depth quotient of E.25, or the inherited class-two pro-3 quotient either.

Proof. Let F_3 be the free pro-3 group on the base loops x, y, and put $\mathcal{H} = F_3/\gamma_3 F_3$. In its class-two Lie coordinates over $\mathbb{Z}_3$ write $X = \log x$, $Y = \log y$, $C = [X, Y]$. The Baker–Campbell–Hausdorff formula has only the denominator two, which is a unit in $\mathbb{Z}_3$. The image H of N_3 consists exactly of

$$\exp(3aX + 3bY + cC), \qquad a, b, c \in \mathbb{Z}_3. \tag{E28.2}$$

The exponent-sum condition proves containment, and the elements $x^3, y^3, [x, y] \in N_3$ generate the full lattice in (E28.2). In coordinates (a, b, c) its multiplication is

$$(a, b, c)(a', b', c') = \left(a + a', b + b', c + c' + \frac{9}{2}(ab' - ba')\right). \tag{E28.3}$$

In particular the commutator is $(0, 0, 9(ab' - ba'))$. The coordinates a, b are the two coordinate-unit homology coordinates of U_3; the covering pushforward to base homology is $(3a, 3b)$.

The normalized Galois action sends x to x^χ and y to $f_g^{-1} y^\chi f_g$, with $f_g \in F_3'$. In $\mathcal{H}$ this conjugator is central, so the action on (E28.2) is exactly

$$(a, b, c) \longmapsto (\chi(g)a, \chi(g)b, \chi(g)^2 c). \tag{E28.4}$$

The unit tangent at $t = 0$ lifts to the plus cubic by $w = -x/z$, $t = w^3$ and $y/z = -(1 - w^3)^{1/3}$. Its toric leading coefficients are $(-1, -1)$. Thus the base rational tangential section lifts to the cubic and the inherited extension is $H \rtimes G_{\mathbb{Q}}$. The pro-3 Kummer classes of both these signs, and of the sign at the origin used in E.24, are zero. Changing between these two tangential origins therefore fixes the toric coordinates of every abelian class, so it leaves (E28.1) and its entire fiber unchanged.

Reduce all three coordinates in (E28.3) modulo 27. This gives a Galois-stable finite quotient H_{27} of order 27^3, of exponent 27, whose commutator of lifts of $(1, 0)$ and $(0, 1)$ is $(0, 0, 9) \neq 1$. At the local field $\mathbb{Q}_{109}$ the action (E28.4) on H_{27} is trivial: the 3-adic cyclotomic character is unramified there, with Frobenius $109 \equiv 1 \pmod{27}$. The integer 6 has exact order 108 in $\mathbb{F}_{109}^\times$. Consequently its Kummer class modulo 27 is unramified with Frobenius value a generator of μ_{27}, while the Kummer class of 109 has tame-inertia value a generator. Choose the common generator of μ_{27} so that the Frobenius value for 6 is 1; a generator of tame inertia can then be chosen with the value for 109 also equal to 1. We may choose Frobenius fixing the selected roots of the uniformizer. The two projected values are thus $(1, 0)$ and $(0, 1)$.

A section would therefore give a homomorphism $G_{\mathbb{Q}_{109}} \to H_{27}$ with those two projections. Wild inertia is pro-109 and maps trivially to this finite 3-group. In the tame quotient, choose Frobenius σ and tame inertia τ using roots of the uniformizer. Frobenius acts on their roots of unity by the 109th power, so $\sigma\tau\sigma^{-1} = \tau^{109}$. Since H_{27} has exponent 27, this relation forces the two images to commute. Equation (E28.3) gives their commutator as $(0, 0, 9)$, independently of their central coordinates, a contradiction. The program `e_cubic_class2.py` checks the order of 6 modulo 109, the exponent and the full multiplication formula, and all 27^2 choices of the two central lifts.

Finally the map $N_3 \to H_{27}$ kills $\gamma_3 N_3$, N_3'' and $\gamma_3 I_3$. These are the actual factorizations through the quotients in the statement. The explicit toric projection is (a, b), so every hypothetical lift of any point of the E.24 fiber would give the forbidden homomorphism. E.24 proves the asserted nonemptiness and size of the abelian fiber independently of this obstruction.

There is no reduction map from the rational group $H_{\mathbb{Q}_3}$ to H_{27} extending this reduction on its integral lattice: every element of $H_{\mathbb{Q}_3}$ is a 27th power, whereas every 27th power in H_{27} is one. Thus any such homomorphism is trivial. The finite obstruction is used only for the indicated integral pro-3 quotients. The result does not classify the remaining cubic class-two sections whose toric pairs pass this local test. □

Every rational toric pair in the inherited cubic class-two quotient

E.29. The exact tame-residue criterion and the three classes in each nonempty fibre [Proved]

Let F be the free pro-3 geometric fundamental group of $\mathbb{P}^1 - \{0, 1, \infty\}$, based at the rational tangent $t = 0$, $dt = 1$, and let $N_3 = \ker(F \to (\mathbb{Z}/3)^2)$. The image of N_3 in $F/\overline{\gamma_3 F}$ is the pro-3 group H with coordinates $(u, v, w) \in \mathbb{Z}_3^3$, law

$$(u, v, w)(u', v', w') = \left(u + u', v + v', w + w' + \frac{9}{2}(uv' - vu')\right), \tag{E.29a}$$

and arithmetic action

$$g(u, v, w) = (\chi_3(g)u, \chi_3(g)v, \chi_3(g)^2 w).$$

It is the inherited class-two quotient of the plus cubic; its arithmetic extension, with the lifted tangent, is $H \rtimes G_{\mathbb{Q}}$.

For $a, b \in \mathbb{Q}^*$ let $\mathscr{F}(a, b)$ be the fibre over (κ_a, κ_b) of the continuous section-class map

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, H) \longrightarrow H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(1)^2).$$

For a rational prime q, put

$$\theta_q(a, b) = (-1)^{v_q(a)v_q(b)} \overline{a^{v_q(b)}/b^{v_q(a)}} \ \in \mathbb{F}_q^*. \tag{E.29b}$$

The expression under the bar is a q-adic unit. Let $o_3(\theta)$ denote the largest power of three dividing its multiplicative order. Then the complete answer on rational toric pairs is

$$|\mathscr{F}(a, b)| = \begin{cases} 3, & o_3(\theta_q(a, b)) \le 9 \text{ for every prime } q, \\ 0, & o_3(\theta_q(a, b)) > 9 \text{ for at least one prime } q. \end{cases} \tag{E.29c}$$

Only primes dividing a numerator or denominator of a or b need to be tested. For each nonempty fibre, choose one continuous cocycle $z(g) \in H$. Its three distinct classes are represented by

$$z_j(g) = \left(0, 0, j\frac{\chi_3(g)^2 - 1}{3}\right) z(g), \qquad j = 0, 1, 2. \tag{E.29d}$$

The choice of z is the stated ambiguity in this parametrization. The obstruction itself is exactly $9\kappa_a \cup \kappa_b \in H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2))$.

For example, $(6, 109)$ and $(2, 163)$ have empty fibres, whereas $(6^3, 109)$ and $(2^9, 163)$ each have three classes. The source of this classification is the displayed inherited quotient. The intrinsic rank-ten class-two quotient maps onto it by

$$N_3/\overline{\gamma_3 N_3} \twoheadrightarrow H, \qquad \ker = (N_3 \cap \overline{\gamma_3 F})/\overline{\gamma_3 N_3}. \tag{E.29e}$$

Thus its section classes map to the fibres computed here.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); the Kummer coordinates of the minimum quotient are retained, and their next inherited integral obstruction is computed. This statement treats every rational toric pair.

Proof. We first fix the comparison and its factors of three. Put $X = \log x$, $Y = \log y$, and $C = [X, Y]$ in the integral class-two pro-3 group; the denominator 2 in its Baker–Campbell–Hausdorff law is a unit in $\mathbb{Z}_3$. The image of N_3 has the lattice

$$A = 3X, \qquad B = 3Y, \qquad C = [X, Y], \qquad [A, B] = 9C.$$

Indeed its abelian coordinates are divisible by three, and $x^3, y^3, [x, y] \in N_3$ generate all these coordinates. This proves (E.29a) and the exact surjection (E.29e). For the normalized arithmetic action, $x \mapsto x^{\chi_3(g)}$ and $y \mapsto f_g^{-1} y^{\chi_3(g)} f_g$, where $f_g \in F'$. The element f_g is central after quotienting by $\overline{\gamma_3 F}$, giving the displayed diagonal action.

On the plus cubic put $u = x_0/x_\infty$, $v = x_1/x_\infty$. The covering is $t = -u^3$, $1 - t = -v^3$. At its rational cusp above $t = 0$, the parameter $s = -u$ gives $t = s^3$ and $v = -(1 - s^3)^{1/3}$ with constant coefficient -1. The tangential Galois section at $s = 0$, $ds = 1$ maps to the chosen base section: the corresponding embedding of Puiseux fields sends $t^{1/m}$ to $s^{3/m}$ and commutes with coefficientwise $G_{\mathbb{Q}}$-action. This specifies the comparison at the ramified cusp. Both leading toric units are -1, whose pro-3 Kummer classes are zero, since -1 has compatible rational 3^nth roots. Thus a toric cocycle (κ_a, κ_b) has inherited coordinates $\kappa_a A + \kappa_b B$ exactly. The comparison to base toric coordinates is multiplication by three, coming from $t = -u^3$, $1 - t = -v^3$.

Write $V = \mathbb{Z}_3(1)^2$, $Z = \mathbb{Z}_3(2)$, so that $1 \to Z \to H \to V \to 1$ is a central extension of pro-3 groups with $G_{\mathbb{Q}}$-action. For a continuous V-cocycle $v_g = (\alpha_g, \beta_g)$, use its continuous set-theoretic lift $(\alpha_g, \beta_g, 0)$. Its obstruction two-cocycle is

$$\omega(g, h) = \frac{9}{2} \left(\alpha_g \chi_3(g) \beta_h - \beta_g \chi_3(g) \alpha_h \right). \tag{E.29f}$$

The symmetric sum of the two cup cocycles is the negative coboundary of $g \mapsto \alpha_g \beta_g$. Hence the class of (E.29f) is $9[\alpha \cup \beta]$. Vanishing of this class means precisely that a continuous central one-cochain cancels (E.29f), which is the cocycle condition for a lift to H.

We now prove the criterion in continuous cohomology itself. The pairing

$$(a, b) \longmapsto \kappa_a \cup \kappa_b \quad \text{in } H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, Z)$$

is bilinear and satisfies the integral Steinberg relation. For the latter, the rational point $t = a$ of $\mathbb{P}^1 - \{0, 1, \infty\}$, for $a \neq 0, 1$, supplies an actual arithmetic section. Its class-two pro-3 quotient has toric cocycle (κ_a, κ_{1-a}) and commutator coefficient one. The calculation (E.29f) with 9 replaced by 1 therefore gives $\kappa_a \cup \kappa_{1-a} = 0$ in continuous integral cohomology. Bilinearity now defines a homomorphism

$$\beta : K_2^M(\mathbb{Q}) \longrightarrow H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, Z), \qquad \{a, b\} \longmapsto \kappa_a \cup \kappa_b. \tag{E.29g}$$

This constructs (E.29g) before taking any inverse limit of degree-two cohomology groups.

We use Matsumoto's presentation and the localization calculation in the opened source [Wei13, III.6.1, III.6.5–III.6.5.1, III.5.2.2]. For $\mathbb{Q}$ these give the exact sequence

$$0 \longrightarrow \mathbb{Z}/2 \longrightarrow K_2^M(\mathbb{Q}) \xrightarrow{(\partial_q)_q} \bigoplus_q \mathbb{F}_q^* \longrightarrow 0. \tag{E.29h}$$

The kernel is generated by $\{-1, -1\}$. In the convention of that source, $\partial_q\{a, b\} = \theta_q(a, b)^{-1}$; coordinatewise inversion is the explicit automorphism of the right-hand side relating the two conventions. Every element of the middle group is torsion: its finitely supported image has finite order, and its remaining kernel is killed by two. In particular (E.29h) identifies its

3-primary subgroup with $\bigoplus_q(\mathbb{F}_q^*)\{3\}$. The target of (E.29g) is a $\mathbb{Z}_3$-module, so β kills all primary components at primes different from three.

We give the local detection argument to prove injectivity on the remaining 3-primary subgroup. Let $q \neq 3$. For $T_n = \mu_{3^n}^{\otimes 2}$, wild inertia has no positive cohomology with these coefficients, and the 3-part of tame inertia is $\mathbb{Z}_3(1)$. The quotient Galois group of the finite residue field and this tame 3-part have 3-cohomological dimension one. Their two-term procyclic resolutions, followed by Hochschild–Serre, give

$$H^2(\mathbb{Q}_q, T_n) \simeq H^1(\mathbb{F}_q, \mu_{3^n}) \simeq (\mathbb{Z}/3^n)/(q-1). \tag{E.29i}$$

In particular the degree-one finite-coefficient local groups are finite, so their first derived inverse limit vanishes. Passing to the inverse limit of (E.29i) therefore gives an actual continuous-cohomology isomorphism

$$\mathrm{res}_q^{\mathrm{tame}} : H^2_{\mathrm{cts}}(\mathbb{Q}_q, Z) \xrightarrow{\sim} H^1_{\mathrm{cts}}(\mathbb{F}_q, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3/(q-1). \tag{E.29j}$$

Here is the cup-product normalization of this map. Choose arithmetic Frobenius ϕ and a tame generator τ with $\phi\tau\phi^{-1} = \tau^q$; these are the operators on compatible roots of a uniformizer. The conjugation formula is also given in the opened source [BC09, (8.2.2)]. For a unit u, its Kummer class is unramified. Write $a = \kappa_u(\phi)$ and choose ϕ fixing the chosen uniformizer roots, so $\kappa_q(\phi) = 0$ and $\kappa_q(\tau) = 1$. The extension two-cocycle $\kappa_u \cup \kappa_q$ has the relation defect qa between $\phi\tau\phi^{-1}$ and τ^q. Changing the central lift of τ by b changes that defect by $q(q-1)b$. Thus (E.29j), which divides the defect by q, sends this cup product to the unit's residue-field Kummer class. This specifies the map and its sign, and verifies the edge-map calculation directly. Two unit Kummer classes have zero cup product because they factor through the procyclic residue Galois group. Also $\kappa_q \cup \kappa_q = 0$, since graded commutativity kills it by two and two is a unit in $\mathbb{Z}_3$. Writing $a = q^\alpha u$, $b = q^\beta v$ now yields

$$\mathrm{res}_q^{\mathrm{tame}}(\kappa_a \cup \kappa_b) = \kappa_{\overline{u}^\beta/\overline{v}^\alpha} = \kappa_{\theta_q(a,b)}. \tag{E.29k}$$

The sign in (E.29b) has zero pro-3 Kummer class. Finite-field Kummer theory identifies the image of an element of $\mathbb{F}_q^*$ in (E.29j) with its 3-primary component, preserving its order.

Every nonzero 3-primary element of $K_2^M(\mathbb{Q})$ has a nonzero tame component at some $q \neq 3$, by (E.29h), since $\mathbb{F}_3^*$ has order two. Equation (E.29k) detects that component under β. Consequently β is injective on this subgroup. It follows that

$$9\kappa_a \cup \kappa_b = 0 \iff 9\{a,b\}_{\{3\}} = 0 \iff o_3(\theta_q(a,b)) \leq 9 \quad \text{for every } q. \tag{E.29l}$$

This proves both necessity and sufficiency in continuous cohomology, including sufficiency for the existence of a continuous central cochain. No global inverse-limit injectivity assertion was used.

It remains to compute each nonempty fibre. We first compute

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) \simeq \mathbb{Z}/3. \tag{E.29m}$$

A continuous cocycle with values in $\mathbb{Q}_3(2)$ restricts on inertia at $q \neq 3$ to a homomorphism. Wild inertia contributes zero. If a is its value on a tame 3-generator, Frobenius equivariance gives $q^2a = qa$, so $(q-1)a = 0$, hence $a = 0$. It therefore vanishes on the closed normal subgroup generated by these inertia groups and factors through $G_{\mathbb{Q},\{3\}}$. Theorem 7.2 and Theorem B.1 of the opened Hain–Matsumoto paper [HM03] give

$$\dim_{\mathbb{Q}_3} H^1_{\mathrm{cts}}(G_{\mathbb{Q},\{3\}}, \mathbb{Q}_3(2)) = r_2(\mathbb{Q}) = 0.$$

This proves the vanishing over the full $G_{\mathbb{Q}}$. The exact coefficient sequence $0 \to \mathbb{Z}_3(2) \to \mathbb{Q}_3(2) \to \mathbb{Q}_3/\mathbb{Z}_3(2) \to 0$ thus identifies (E.29m) with $H^0(G_{\mathbb{Q}}, \mathbb{Q}_3/\mathbb{Z}_3(2))$. The image of χ_3 is $\mathbb{Z}_3^*$; the ideal generated by $u^2 - 1$ for all units u is $3\mathbb{Z}_3$. The invariants are consequently $(\frac{1}{3}\mathbb{Z}_3)/\mathbb{Z}_3$, and the boundary of $j/3$ is exactly the central cocycle in (E.29d).

To see that this calculation gives the whole fibre, choose one lifted cocycle. The ratio of two lifts with the same toric cocycle is a central one-cocycle, and every such cocycle gives another lift. A kernel conjugation which preserves the exact toric cocycle has projection in $H^0(G_{\mathbb{Q}}, V) = 0$: complex conjugation acts by -1 on the torsion-free group V. Such a conjugation is therefore central and changes the ratio only by a central coboundary. When the toric cocycles initially differ by a coboundary, the surjection $H \to V$ supplies the conjugation which makes them equal. Thus the fibre is a torsor under the explicitly computed group (E.29m), with exactly the representatives (E.29d).

The program `code/e_rational_pairs.py` computes (E.29b) by exact rational arithmetic. Among all 40 000 ordered positive integer pairs up to 200, it finds 39 378 nonempty and 622 empty fibres. The histogram of the largest 3-primary residue order is

order	1	3	9	27	81
number of pairs	19036	15198	5144	364	258.

It also checks all 101 124 ordered pairs of the 318 nonzero signed reduced rationals of numerator and denominator height at most 16, and all 317 applicable Steinberg pairs in that set. It checks (E.29d) as a cocycle modulo 3^n for $1 \le n \le 4$, and the three twist-two invariants and absence of twist-one invariants at those finite levels. All output and every obstructed integer pair are retained in the accompanying JSON file. □

Arithmetic completion over the proconstant base

E.30. Arithmetic completion over the proconstant base kills the geometric group [Proved]

Put $U = \mathbb{P}^1_{\mathbb{Q}} - \{0, 1, \infty\}$ and $\mathcal{E} = \pi_1([U/S_3])$, with the positive rational tangential base point lifted to U. The quotient by its geometric free subgroup $\widehat{F}_2$ is $S_3 \times G_{\mathbb{Q}}$. For $m \ge 1$, let $K_m = \mathbb{Q}(\mu_m)$, $C_m = \mathrm{Gal}(K_m/\mathbb{Q})$ and $B_m = S_3 \times C_m$. There are two precise versions of the following construction: abstract relative completion over $k = \mathbb{Q}$, and continuous relative completion over $k = \mathbb{Q}_\ell$. Use the corresponding version throughout each formula, and let $\mathcal{A}_{m,k}$ be the relative completion of $G_{\mathbb{Q}} \to C_m$. The relative completion of $\mathcal{E} \to B_m$ is exactly

$$\mathcal{R}_{m,k} = S_3 \times \mathcal{A}_{m,k}, \qquad 1 \longrightarrow (G_{K_m})^{\mathrm{un}}_k \longrightarrow \mathcal{A}_{m,k} \longrightarrow C_m \longrightarrow 1. \tag{E.30.1}$$

For $m \mid n$ the transition maps are induced by the identity on $G_{\mathbb{Q}}$; on the displayed kernels they are induced by $G_{K_n} \hookrightarrow G_{K_m}$. Consequently the completion over the proconstant group scheme $B = S_3 \times \widehat{\mathbb{Z}}^\times_k$ is the specified inverse limit

$$\mathcal{R}_{B,k} = \varprojlim_m \mathcal{R}_{m,k}, \qquad 1 \longrightarrow \varprojlim_m (G_{K_m})^{\mathrm{un}}_k \longrightarrow \mathcal{R}_{B,k} \longrightarrow B \longrightarrow 1. \tag{E.30.2}$$

The natural homomorphism $\mathcal{E} \to \mathcal{R}_{B,k}(k)$ kills the whole geometric subgroup $\widehat{F}_2$. In particular its restriction to every Fermat subgroup is trivial; it retains none of the geometric homology computed in E.14.

There is a different, explicit completion retaining that geometry. Let $\mathcal{A}_\ell$ be the negatively weighted completion of $G_{\mathbb{Q}} \xrightarrow{\chi_\ell} \mathbb{G}_m(\mathbb{Q}_\ell)$, with central cocharacter $t \mapsto t^{-2}$. For the arithmetic

group $\pi_1(U) = \widehat{F}_2 \rtimes G_{\mathbb{Q}}$ with the stated tangential section, its negatively weighted completion is

$$(\widehat{F}_2)^{\mathrm{un}}_{\mathbb{Q}_\ell} \rtimes \mathcal{A}_\ell. \tag{E.30.3}$$

The geometric factor is the free prounipotent group on two generators. No algebraic morphism from the proconstant B to $\mathbb{G}_m$ induces χ_ℓ on its cyclotomic factor.

Proof. The morphism $[U/S_3] \to [\operatorname{Spec}\mathbb{Q}/S_3]$ gives the stated quotient $S_3 \times G_{\mathbb{Q}}$, since the permutation action is defined over $\mathbb{Q}$. Its geometric kernel is $\widehat{F}_2$. Thus the kernel N_m of $\mathcal{E} \to B_m$ has the tangentially split sequence

$$1 \longrightarrow \widehat{F}_2 \longrightarrow N_m \longrightarrow G_{K_m} \longrightarrow 1.$$

On $\widehat{F}_2^{\mathrm{ab}} = \widehat{\mathbb{Z}}^2$ the arithmetic action is scalar multiplication by the full cyclotomic character.

First work continuously over $\mathbb{Q}_\ell$. Let $r : N_m \to U_0(\mathbb{Q}_\ell)$ be any continuous homomorphism to a unipotent algebraic group. Let A be the Zariski closure of $r(\widehat{F}_2)$. It is a connected unipotent normal subgroup of the Zariski closure of $r(N_m)$; unipotent groups in characteristic zero have no nontrivial finite algebraic quotients. The map to A^{ab} factors through $\mathbb{Z}_\ell^2$ and spans its Lie algebra over $\mathbb{Q}_\ell$. Choose $g \in G_{K_m}$ with $\chi_\ell(g) \neq 1$, possible because the cyclotomic image of this finite-index subgroup is infinite. Conjugation by $r(g)$ on $\mathrm{Lie}(A^{\mathrm{ab}})$ is both multiplication by $\chi_\ell(g)$ and a unipotent linear operator. The latter assertion follows, for example, by putting U_0 in upper triangular unipotent matrices and considering its conjugation action on its Lie algebra and invariant subquotients. It follows that $A^{\mathrm{ab}} = 1$. A nonzero nilpotent Lie algebra has nonzero abelianization, so $A = 1$. Every such r therefore kills $\widehat{F}_2$.

For the abstract rational version, the required abelianization is still exactly $\widehat{\mathbb{Z}}^2$: the abstract commutator subgroup of a finitely generated profinite group is closed by [NS07, Theorem 1.4]. Choose $u \in \widehat{\mathbb{Z}}^\times$ with $u \equiv 1 \pmod m$ and $u - 1 = dv$, where $d \neq 0$ is an integer and $v \in \widehat{\mathbb{Z}}^\times$, as in D.3. Explicitly use $u_p = 2$ for odd $p \nmid m$, $u_p = 1 + p^{v_p(m)}$ for odd $p \mid m$, and $u_2 = 1 + 2^{\max(1, v_2(m))}$. Cyclotomic surjectivity supplies $g \in G_{K_m}$ with $\chi(g) = u$. For an abstract homomorphism $r : N_m \to U_0(\mathbb{Q})$, there is an equivariant surjection

$$\widehat{\mathbb{Z}}^2 \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \mathrm{Lie}(A^{\mathrm{ab}}).$$

On the source $u - 1 = dv$ is invertible. Therefore $\mathrm{Ad}(r(g)) - 1$ on the target is surjective. It is also nilpotent, since $r(g)$ is unipotent. The target is zero, and the preceding nilpotent Lie algebra argument again gives $A = 1$. This proves the rational assertion without assuming continuity of a homomorphism to $U_0(\mathbb{Q})$.

For a group mapping onto a finite group C, its relative completion is the extension of C by the unipotent completion of the kernel. One can construct it directly by choosing coset representatives: multiplication on the finite disjoint union of copies of the completed kernel is given by the induced conjugation automorphisms and the original factor-set cocycle. This is the construction and universal-property proof used in E.14; it applies to the abstract and continuous categories just considered. The result above identifies N_m^{un} with $(G_{K_m})^{\mathrm{un}}$. Hence every representation defining $\mathcal{R}_{m,k}$ kills $\widehat{F}_2$ and factors uniquely through $S_3 \times G_{\mathbb{Q}}$. Completing this direct product gives $S_3 \times \mathcal{A}_{m,k}$: the finite S_3 image commutes with the arithmetic image, and these two homomorphisms have exactly the product universal property. This proves (E.30.1) and the specified transition maps.

An algebraic homomorphism from a proconstant profinite group scheme to a finite-type affine group scheme factors through a finite quotient. Indeed its finitely many coordinate generators belong to one finite stage of the coordinate-ring direct limit; the multiplication identities also hold at a sufficiently large stage. Every finite-dimensional algebraic representation of B therefore factors through some B_m. Defining relative completion by representations

whose semisimplifications are algebraic representations of B gives precisely the inverse system (E.30.2): every such finite-dimensional representation occurs at one of its stages. This also proves its universal property. The transition maps on the unipotent kernels are faithfully flat: a finite-index subgroup has Zariski-dense image in the connected unipotent closure of the larger group's image. Inverse limits therefore give the exact proalgebraic sequence displayed in (E.30.2). The geometric image was already killed at every stage.

For (E.30.3), the genuine action of $G_{\mathbb{Q}}$ on the geometric prounipotent group has scalar χ_ℓ on its abelianization. Its kernel on that abelianization is prounipotent, with negative weights for $t \mapsto t^{-2}$: the degree-d part of its IA Lie algebra maps a degree-one generator to a degree-d word and has weight $-2(d-1) < 0$ for $d \geq 2$. Thus the universal property of weighted completion extends this action uniquely to $\mathcal{A}_\ell$; the weighted conventions and construction are those of [HM03, §§2–4]. The semidirect product in (E.30.3) receives the actual arithmetic group. Given any lift of that group into a negatively weighted extension of $\mathbb{G}_m$, its restriction to $\widehat{F}_2$ extends uniquely by continuous unipotent completion, and its restriction to the tangential $G_{\mathbb{Q}}$ extends uniquely by weighted completion. Their equivariance holds on the dense original images, hence algebraically. Thus these two extensions define a unique homomorphism from the semidirect product, proving its weighted universal property. Its geometric kernel is visibly the free prounipotent group, proving its injection as well. Finally an algebraic map $B \to \mathbb{G}_m$ has finite image by the finite-stage argument, whereas $\chi_\ell(G_{\mathbb{Q}})$ is infinite. This proves the claimed obstruction to that specific base-change comparison. We do not identify an abstract rational completion after scalar extension with a continuous ℓ-adic completion; the two constructions above specify their source categories independently. □

An intrinsic cubic obstruction inside the zero toric fiber

E.31. Two abelian classes in one toric fiber have different liftability [Proved]

Use the pro-3 version of E.24, with its unit rational tangent at $O = [1 : -1 : 0]$. Put $K = \mathbb{Q}(\zeta)$, $\lambda = 1 - \zeta$ and $\pi = 3 + \zeta$, so $N_{K/\mathbb{Q}}(\pi) = 7$. In the frame coordinates of (E24.4) take the rational torus point

$$t = (7, 7, 1, 1, \pi) \in T_D(\mathbb{Q}) \subset G(\mathbb{Q}). \tag{E31.1}$$

Its Kummer class $\alpha \in H^1(G_{\mathbb{Q}}, T_3G)$ has zero image in the toric minimum. Let $\mathcal{K}$ denote the kernel of that toric map on section classes. Every class in the nonempty coset $\alpha + 3\mathcal{K}$ fails to lift to the cubic's own class-two pro-3 arithmetic quotient, its metabelian quotient, and its marked-depth quotient of E.25. The zero class in the same toric fiber lifts to a full tangential section.

The obstruction is visible in an explicit quotient of the intrinsic based Galois action over K. Let H_3 be the group of order 27 with coordinates $(a, b, c) \in \mathbb{F}_3^3$ and multiplication

$$(a, b, c)(a', b', c') = \left(a + a', b + b', c + c' + \tfrac{1}{2}(ab' - ba')\right). \tag{E31.2}$$

There is a surjection from the cubic geometric group to H_3 whose Galois action, with the tangent at O as origin, is

$$g : (a, b, c) \longmapsto (a, b, c + \kappa_3(\zeta)(g)b), \qquad g \in G_K. \tag{E31.3}$$

Thus (E31.3) gives the arithmetic cross term explicitly, as well as a finite obstruction to the indicated section classes.

Proof. Equation (E24.4) sends (E31.1) to $(7/N(\pi), 7/N(\pi)) = (1, 1)$. Kummer functoriality therefore puts α in $\mathcal{K}$. This is a genuine geometrically abelian section class by E.24. It is

nonzero modulo three: the valuation at the prime (π) in the third $K^\times$ factor of (E24.7) is zero on its displayed λ relation, and takes the value one on t. It extends to an integral homomorphism on that rational-point presentation, so t is not three times a rational point of G'.

Over K use the actual morphism

$$h = \frac{x+y}{x+\zeta y} : U_3 \longrightarrow \mathbb{P}^1 \setminus \{0,1,\infty\}, \qquad 1-h = \frac{(\zeta-1)y}{x+\zeta y}. \tag{E31.4}$$

The equations $h=0$ and $h=\infty$ force $z=0$ on the cubic; $h=1$ forces $y=0$. All these points have been removed from U_3, so the displayed target and morphism are exact. Write $C_i = [1 : -\zeta^i : 0]$ and $B_i = [-\zeta^i : 0 : 1]$. The numerator and denominator tangent lines give

$$\operatorname{div}(h) = 3O - 3C_2, \qquad \operatorname{div}(1-h) = B_0 + B_1 + B_2 - 3C_2. \tag{E31.5}$$

The frame at C_2 is $\bar\pi$, not π. Formula (E24.3) consequently gives the exact character values

$$\chi_h(t) = \bar\pi^{-3}, \qquad \chi_{1-h}(t) = 7\bar\pi^{-3}. \tag{E31.6}$$

The based arithmetic map is obtained by applying (E31.4) to the tangential fiber functor at O and choosing a geometric path to the standard unit tangent at zero in the target. Its affine effect can be computed without a path convention. At O put $s = z/x$; then $y/x = -(1+s^3)^{1/3}$, and hence

$$h = -\frac{s^3}{3(1-\zeta)} + O(s^6) = \zeta\lambda^{-3}s^3 + O(s^6), \qquad 1-h = 1 + O(s^3). \tag{E31.7}$$

The identity in the second expression is $\lambda^2 = -3\zeta$. The map of Puiseux fiber functors under (E31.7) has the constant Kummer factors $(\zeta\lambda^{-3}, 1)$. Thus the image of the tangential section at O has, modulo three, abelian coordinates $(\kappa_3(\zeta), 0)$ relative to the standard target origin. Combining (E31.6) and (E31.7), the abelian section with class α has target coordinates

$$\big(\kappa_3(\zeta\lambda^{-3}\bar\pi^{-3}), \kappa_3(7\bar\pi^{-3})\big) = (\kappa_3(\zeta), \kappa_3(7)). \tag{E31.8}$$

This retains the nonzero affine origin term explicitly.

Take the quotient of the target free pro-3 group in which the two generators and their central commutator have order three; it is (E31.2). Its standard based Galois action over K is trivial. Indeed $\mu_3 \subset K$, and the normalized action of E.20 has cyclotomic exponent one modulo three and a conjugator with central image. Let $\phi_h : N_3 \to H_3$ be the geometric map induced by (E31.4). It is surjective. The two geometric Kummer characters $h, 1-h$ are independent modulo cubes: the valuation at a B_i forces the coefficient of $1-h$ to be zero modulo three; h is not a cube, since such a cube root would have divisor $O - C_2$, a degree-one map from an elliptic curve to $\mathbb{P}^1$. Their independent abelian images generate $H_3/[H_3, H_3]$, and their commutator generates its central subgroup, proving surjectivity.

Let $q_g \in H_3$ be the first component of the image of the tangential section at O in $H_3 \times G_K$. Equivariance of the arithmetic map gives $\phi_h(gng^{-1}) = q_g \phi_h(n) q_g^{-1}$. Its abelian coordinate is $(\kappa_3(\zeta)(g), 0)$ by (E31.7); its central coordinate does not affect conjugation. Conjugation in (E31.2) is therefore exactly (E31.3). The character $\kappa_3(\zeta)$ is nonzero: a cube root of ζ would be a unit in $\mathbb{Z}[\zeta]$, whose unit group μ_6 has only ± 1 as cubes. This proves the asserted nonzero arithmetic cross term.

Now pass to the prime (π) over 7. Its residue field is $\mathbb{F}_7$ and $\zeta \equiv -3 \equiv 4 \pmod{\pi}$, so $K_{(\pi)} = \mathbb{Q}_7$. The character $\kappa_3(\zeta)$ is unramified and nonzero: choose $\zeta_9^3 = \zeta$ in the unramified cubic extension. Arithmetic Frobenius sends ζ_9 to ζ_9^7, whose ratio to ζ_9 is ζ^2. The Kummer

character of the uniformizer 7 has tame-inertia value ζ after a choice of generator, and Frobenius can be chosen to fix its selected roots. Consequently (E31.8) sends Frobenius and tame inertia to $(2,0)$ and $(0,1)$ in H_3^{ab}. Every pair of lifts has commutator $(0,0,2) \neq 1$.

An arithmetic section lifting α would yield a homomorphism $G_{\mathbb{Q}_7} \to H_3$, because the standard target action is trivial. Wild inertia is pro-7 and maps trivially to this finite 3-group. The tame relation is $\sigma\tau\sigma^{-1} = \tau^7$. Every element of H_3 has exponent three, so this relation forces its two images to commute, a contradiction. The exact field identities, the local leading constants, all 27 group elements and all nine choices of the two central lifts are checked by `e_intrinsic_obstruction.py`.

The map ϕ_h kills $\gamma_3 N_3$, N_3'' and $\gamma_3 I_3$. Thus it factors through each intrinsic quotient named in the statement. These factorizations also apply to the arithmetic maps after restriction to G_K, which suffices to rule out a section over $\mathbb{Q}$. There is also a single map over $\mathbb{Q}$ displaying both conjugate comparisons: the adjoint of (E31.4) is

$$U_3 \longrightarrow \mathrm{Res}_{K/\mathbb{Q}}(\mathbb{P}^1 \setminus \{0,1,\infty\}). \tag{E31.9}$$

After base change to $\overline{\mathbb{Q}}$ it is $(h, \bar{h})$, giving the finite geometric image inside $H_3 \times H_3$. The arithmetic map to this image is defined over $\mathbb{Q}$ and its geometric kernel contains $\gamma_3 N_3$. Its restriction has the first projection used above. This supplies an actual global finite quotient comparison as well.

Adding any element of $3\mathcal{K}$ to α leaves (E31.8) unchanged, since the target abelian coefficient group has exponent three. The same obstruction applies to the whole coset $\alpha + 3\mathcal{K}$. The zero class is the original rational tangential section and lifts to the full arithmetic fundamental group. Both lie over the zero pro-3 toric class, as asserted. In particular this proof distinguishes liftability inside one specified fiber, using a quotient that detects an intrinsic direction beyond the inherited toric comparison of E.28. □

All completed toric pairs in the inherited cubic class-two quotient

E.32. The complete convergence criterion, including the invisible inverse-limit obstruction [Proved]

Retain the arithmetic group $H \rtimes G_{\mathbb{Q}}$ of E.29. For a set I write

$$\mathcal{R}_I = \{(z_i) \in \textstyle\prod_{i\in I} \mathbb{Z}_3 : \text{for every } n,\ z_i \in 3^n\mathbb{Z}_3 \text{ outside a finite subset of } I\}.$$

Let $\mathcal{P}$ be the rational primes and $\mathcal{P}^* = \mathcal{P} - \{3\}$. The Kummer isomorphism is

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(1)) = \mathcal{R}_{\mathcal{P}}, \qquad (a_p)_p \longmapsto \sum_p a_p \kappa_p. \tag{E.32a}$$

Choose a compatible system of roots for each κ_p; the sum in (E.32a) converges uniformly as a cocycle.

Here is an explicit criterion for every pair $a = (a_p), b = (b_p) \in \mathcal{R}_{\mathcal{P}}$. For $q \neq 3$ put $e_q = v_3(q-1)$ and choose a primitive 3^{e_q}th root $\eta_q \in \mathbb{F}_q^*$ when $e_q > 0$. For $p \neq q$ let $r_{q,p} \in \{0, \dots, 3^{e_q} - 1\}$ be specified by

$$\overline{p}^{(q-1)/3^{e_q}} = \eta_q^{r_{q,p}}.$$

When $e_q = 0$, put $r_{q,p} = 0$. Define convergent sums and a central correction by

$$A_q = \sum_{p\neq q} a_p r_{q,p}, \qquad B_q = \sum_{p\neq q} b_p r_{q,p}, \qquad c_q = -\frac{9(A_q b_q - B_q a_q)}{q-1} \in \mathbb{Q}_3. \tag{E.32b}$$

The fibre of $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, H)$ over the pair in (E.32a) has exactly three classes precisely when

$$c_q \in \mathbb{Z}_3 \quad (q \neq 3), \qquad (c_q)_{q\neq 3} \in \mathcal{R}_{\mathcal{P}^*}; \tag{E.32c}$$

in all other cases that fibre is empty. The criterion is independent of the roots and generators used above. For a nonempty fibre its three representatives are given by (E.29d), starting with any one lift.

The continuous degree-two group, and the morphism used to compute this obstruction, are

$$\begin{aligned} H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_3(2)) &\xrightarrow[\sim]{\mathrm{loc}_{\mathrm{cts}}} \mathcal{R}_{\mathcal{P}^*}/d\mathcal{R}_{\mathcal{P}^*}, \\ 9\Big(\sum_p a_p\kappa_p\Big) \cup \Big(\sum_p b_p\kappa_p\Big) &\longmapsto [\,9(A_q b_q - B_q a_q)\,]_q. \end{aligned} \tag{E.32d}$$

Here $d(z)_q = (q-1)z_q$. These are isomorphisms of $\mathbb{Z}_3$-modules; the denominator is the actual image of d, with no closure added. The map from this quotient to the product of local groups is coordinatewise reduction modulo $q-1$. Its exact kernel is

$$\frac{\{z \in \mathcal{R}_{\mathcal{P}^*} : z_q \in (q-1)\mathbb{Z}_3 \text{ for all } q\}}{d\mathcal{R}_{\mathcal{P}^*}}. \tag{E.32e}$$

This kernel contains an obstruction of an actual completed toric pair. Choose recursively increasing primes q_j, $j \geq 3$, with

$$q_j \equiv 1 \pmod{3^j}, \qquad 3^{(q_j-1)/3} \not\equiv 1 \pmod{q_j},$$

and write $r_j = r_{q_j,3} \in \mathbb{Z}_3^*$. Take $a_3 = 1$, $a_p = 0$ for $p \neq 3$, and

$$b_{q_j} = \frac{q_j - 1}{9r_j}, \qquad b_p = 0 \text{ at the other primes.} \tag{E.32f}$$

Then $b \in \mathcal{R}_{\mathcal{P}}$ and every local obstruction and every finite-coefficient global obstruction vanishes. There is nevertheless no continuous section with this completed toric pair: its corrections are $c_{q_j} = -1$ for all j. Every finite-support partial pair obtained from (E.32f) does have three continuous section classes.

The classification concerns the inherited quotient H. The map from the cubic's intrinsic quotient is exactly (E.29e); the additional intrinsic obstruction E.31 remains present. *Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379), with its completed Kummer parameters retained.*

Proof. We first calculate the continuous cohomology group and its localization morphism, retaining the derived inverse limit. Put

$$U = \operatorname{Spec}\mathbb{Z}[1/3], \qquad G_S = G_{\mathbb{Q},\{3,\infty\}}, \qquad T_n = \mathbb{Z}/3^n(2), \qquad T = \mathbb{Z}_3(2).$$

Milne's comparison for locally constant sheaves [Mil06, II.2.9] identifies $R\Gamma(U, T_n)$ with $R\Gamma(G_S, T_n)$. We need the following actual vanishing:

$$H^2(G_S, T_n) = H^3(G_S, T_n) = 0, \quad H^1(G_S, T_n) \text{ finite}, \qquad n \geq 1. \tag{E.32g}$$

For $n = 1$, $T_1 = \mathbb{F}_3$ with trivial action, since $\chi_3^2 \equiv 1 \pmod 3$. The Kronecker–Weber theorem [Mil20b, I.4.16] gives $H^1(G_S, \mathbb{F}_3) = \mathbb{F}_3$. More explicitly, an abelian extension unramified away from three is contained in a cyclotomic field, and its quotient kills the inertia factors at all other primes. Its cubic quotient therefore factors through $\mathrm{Gal}(\mathbb{Q}(\zeta_{3^m})/\mathbb{Q})$, which has a unique quotient of order three. The nontrivial quotient already occurs in $\mathbb{Q}(\zeta_9)^+$. Both $H^0(G_S, \mathbb{F}_3)$

and $H^1(G_S, \mathbb{F}_3)$ have order three. The global Euler characteristic [Mil06, I.5.1] equals one here: its real factor is $|H^0(G_{\mathbb{R}}, \mathbb{F}_3)|/3 = 1$. Consequently $H^2(G_S, \mathbb{F}_3) = 0$. Filter T_n by its powers of three. Each successive quotient is the same trivial $\mathbb{F}_3$, so the long exact cohomology sequence proves $H^2(G_S, T_n) = 0$ for all n. Milne I.4.10(c) identifies every degree at least three with real-place cohomology, which is zero for a module of odd order. Finiteness is I.4.15. This proves (E.32g).

For a finite set $J \subset \mathcal{P}^*$, apply the localization triangle to $U_J = U - \{q : q \in J\}$, using [Mil06, II.0.1(f), II.1.7(b)]. Here the local purity identification can also be checked directly: over the strict Henselization at q, a locally constant T_n is constant; the fraction field has tame 3-inertia $\mathbb{Z}_3(1)$, while its wild inertia is pro-q. Its cohomology is T_n in degree zero, $T_n(-1)$ in degree one, and zero above degree one. The local support complex is therefore $T_n(-1)[-2]$, with its residue-field action. Thus there is the compatible distinguished triangle

$$R\Gamma(U, T_n) \longrightarrow R\Gamma(U_J, T_n) \longrightarrow \bigoplus_{q \in J} R\Gamma(\mathbb{F}_q, \mathbb{Z}/3^n(1))[-1] \longrightarrow R\Gamma(U, T_n)[1]. \tag{E.32h}$$

First take the filtered direct limit over finite J. Filtered direct limits of abelian groups are exact. On the generic-field term this gives $R\Gamma(G_{\mathbb{Q}}, T_n)$: every continuous cochain with finite coefficients factors through a finite quotient of $G_{\mathbb{Q}}$, whose number field is ramified at only finitely many primes. In particular the third term becomes a direct sum at each fixed coefficient level n; no infinite product is substituted.

Next take $R\varprojlim_n$ of these triangles. The comparison with continuous cochains is Jannsen's theorem [Jan88, (2.2), together with (1.6)]. For the first term (E.32g), the finiteness of the degree-one groups kills their first derived inverse limit. It follows that

$$H^2(R\varprojlim_n R\Gamma(U, T_n)) = H^3(R\varprojlim_n R\Gamma(U, T_n)) = 0. \tag{E.32i}$$

For the support term, the procyclic residue Galois group has the compatible two-term cochain model

$$R\Gamma(\mathbb{F}_q, \mathbb{Z}/3^n(1)) \simeq [\, \mathbb{Z}/3^n \xrightarrow{q-1} \mathbb{Z}/3^n \,] \quad \text{in degrees } 0, 1. \tag{E.32j}$$

Choose arithmetic Frobenius for this model. After taking the direct sum over $q \neq 3$, the transition in each of its two terms is the surjective reduction $\bigoplus_q \mathbb{Z}/3^{n+1} \to \bigoplus_q \mathbb{Z}/3^n$. The inverse systems of terms are therefore acyclic for $\varprojlim$. Their derived inverse limit is the literal complex

$$[\, \mathcal{R}_{\mathcal{P}^*} \xrightarrow{d} \mathcal{R}_{\mathcal{P}^*} \,]. \tag{E.32k}$$

Indeed a compatible sequence of finitely supported residues is exactly a family of 3-adic integers tending to zero. The degree-two segment of the resulting triangle, using (E.32i), proves the first isomorphism in (E.32d). This calculation takes the inverse limit of complexes. It does not replace it by the inverse limit of their cohomology.

We next compute the image of the actual obstruction under this morphism. The Kummer sequence and $\mathbb{Q}^* = \{\pm 1\} \times \bigoplus_p p^{\mathbb{Z}}$ give (E.32a); the sign factor has trivial pro-3 completion. Multiplying a fixed cocycle κ_p by a_p and summing is legitimate on continuous cochains: for each n, all but finitely many summands vanish modulo 3^n, uniformly on $G_{\mathbb{Q}}$. Write these sums as α, β. Their obstruction is the continuous cocycle

$$\omega(g, h) = \frac{9}{2}\{\alpha(g)\chi_3(g)\beta(h) - \beta(g)\chi_3(g)\alpha(h)\}, \tag{E.32l}$$

representing $9\alpha \cup \beta$ as in E.29.

Fix $q \neq 3$ and a decomposition group. Choose tame inertia τ_q and arithmetic Frobenius ϕ_q such that

$$\phi_q \tau_q \phi_q^{-1} = \tau_q^q, \qquad \kappa_q(\tau_q) = 1, \qquad \kappa_q(\phi_q) = 0. \tag{E.32m}$$

These choices use compatible roots of a uniformizer, as in E.29. Put $\alpha(\tau_q) = a_q$, $\beta(\tau_q) = b_q$ and $\alpha(\phi_q) = A'_q$, $\beta(\phi_q) = B'_q$. On the full inertia group the two cocycles are proportional to the same tame character; wild inertia maps to zero. Consequently (E.32l) is identically zero on two inertia arguments, giving its canonical central-zero inertia splitting. In the pullback of $H \rtimes G_{\mathbb{Q}}$, let the central coordinates of the two lifts be w at τ_q and z at ϕ_q. Direct multiplication in (E.29a) gives the defect between the two sides of (E.32m) as

$$q\{(q-1)w + 9(A'_q b_q - B'_q a_q)\}. \tag{E.32n}$$

It is independent of z. In particular the central-zero inertia splitting represents the residue cochain with value $9(A'_q b_q - B'_q a_q)$ at Frobenius. The division by q in (E.32n) is the identification $\mathrm{Hom}(\mathbb{Z}_3(1), T) = T(-1)$: the conjugation defect at τ_q^q is pulled back to its value at τ_q. Changing the inertia splitting by w adds precisely $(q-1)w$, the differential in (E.32j). This is the chain-level localization morphism in (E.32h): it records the failure of an inertia splitting to be invariant under residue-field Frobenius.

At each finite coefficient level these inertia splittings agree with an unramified splitting outside finitely many primes, since α, β factor through a finite quotient modulo 3^n. They therefore form a compatible representative in the direct-sum complexes used in (E.32h)–(E.32k). The resulting family tends to zero, since it contains a factor a_q or b_q in each summand. This establishes the localization representative itself, including its class in the quotient (E.32k).

For a unit p at q, finite-field Kummer theory gives

$$\kappa_p(\phi_q) \equiv r_{q,p} \pmod{q-1}.$$

Choose the place over q to send the selected global 3^{e_q}th root of unity to η_q. The formula follows by applying Frobenius to a 3^{e_q}th root of p: its quotient is $\overline{p}^{(q-1)/3^{e_q}}$. Write $\kappa_p(\phi_q) - r_{q,p} = (q-1)t_{q,p}$ with $t_{q,p} \in \mathbb{Z}_3$. The sums of $a_p t_{q,p}$ and $b_p t_{q,p}$ converge and are uniformly bounded in $\mathbb{Z}_3$. Hence replacing A'_q, B'_q by A_q, B_q changes the representative in (E.32d) by d of a family tending to zero: the change divided by $q-1$ has a factor a_q or b_q. This proves the second formula in (E.32d). The same argument proves independence of root choices. Changing a residue generator scales the corresponding coordinate by a 3-adic unit and adds a term of this form, so it also preserves (E.32c).

For completeness, changing global cocycles by $\alpha \mapsto \alpha + du$, $\beta \mapsto \beta + dv$, with $du(g) = (\chi_3(g) - 1)u$, changes the residue representative by

$$(q-1)\, 9(ub_q - va_q).$$

The family in the second factor lies in $\mathcal{R}_{\mathcal{P}^*}$. The corresponding central inertia correction changes by $-9(ub_q - va_q)$, exactly as the change $z_g \mapsto h^{-1} z_g g(h)$ with $h = (u, v, 0)$ gives. Also replacing ϕ_q by $\tau_q^r \phi_q$ leaves $A'_q b_q - B'_q a_q$ unchanged. These formulas specify both the coboundary sign and the actual coefficient comparisons.

The obstruction vanishes precisely when its representative lies in $d\mathcal{R}_{\mathcal{P}^*}$. Since $q - 1 \neq 0$ in $\mathbb{Z}_3$, its unique possible central correction is (E.32b). This proves (E.32c) and the empty-fibre assertion. The central group has $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, T) = \mathbb{Z}/3$ by E.29, and $H^0(G_{\mathbb{Q}}, \mathbb{Z}_3(1)^2) = 0$. The exact nonabelian central-extension sequence therefore makes each nonempty fibre a torsor under this group. Its three explicit twists are (E.29d).

We finally verify the inverse-limit example. The field $\mathbb{Q}(\sqrt[3]{3})$ cannot be contained in a cyclotomic field: a subfield of an abelian Galois extension is Galois, whereas this irreducible real cubic does not contain the two nonreal roots of $X^3 - 3$. Thus $\mathbb{Q}(\zeta_{3^j}, \sqrt[3]{3})/\mathbb{Q}(\zeta_{3^j})$ has degree three. A nonidentity element of this subgroup has a conjugacy class restricting trivially to the cyclotomic field. Chebotarev [Mil20b, V.3.23] provides infinitely many rational primes with

that class, giving the two requirements on q_j. Choose each q_j to be the least such prime larger than the previous choice. Noncubicity says r_j is a 3-adic unit, so $v_3(b_{q_j}) = e_{q_j} - 2 \geq j - 2$. This proves that (E.32f) is an actual completed Kummer pair. Its representative in (E.32d) is

$$z_{q_j} = q_j - 1, \qquad z_q = 0 \text{ otherwise.} \tag{E.32o}$$

It is a restricted family, but its only preimage under coordinatewise multiplication by $q-1$ is the family with value one at every q_j, which is not restricted. Thus its class is nonzero in (E.32d), despite lying in the kernel (E.32e). At every finite level its class is zero in the cohomology of the direct-sum complex: only finitely many entries of (E.32o) survive modulo 3^n, and they are obtained by applying d to the finitely supported family with value one at those entries. The finite-level global obstruction is zero by (E.32g)–(E.32h). At $q \neq 3$ the local obstruction is zero by coordinatewise divisibility. At three, finite local duality [Mil06, I.2.3] gives $H^2(\mathbb{Q}_3, T_n) = H^0(\mathbb{Q}_3, \mathbb{Z}/3^n(-1))^* = 0$: the local cyclotomic character contains -1, whose action has no fixed vector on this odd-order module. Finite local degree-one groups then give $H^2_{\mathrm{cts}}(\mathbb{Q}_3, T) = 0$ as well. Real cohomology with these coefficients is zero. Every finite-support partial pair in (E.32f) has finitely many corrections equal to -1, so it satisfies (E.32c). All assertions follow. □

The continuous arithmetic kernel and its trace maps

E.33. The continuous arithmetic kernels are free, with explicit generators as a Galois module [Proved]

Fix a prime ℓ and put $K_m = \mathbb{Q}(\mu_m)$, $d_m = [K_m : \mathbb{Q}]$ and $r_m = r_2(K_m) + 1$. The continuous unipotent completion occurring in E.30 is

$$(G_{K_m})^{\mathrm{un}}_{\mathbb{Q}_\ell} \simeq \exp \widehat{\mathbb{L}}(V_m), \qquad V_m = \frac{K_m \otimes_{\mathbb{Q}} \mathbb{Q}_\ell}{\log_\ell(\mathcal{O}^\times_{K_m} \otimes_{\mathbb{Z}} \mathbb{Q}_\ell)}. \tag{E.33.1}$$

Here $\widehat{\mathbb{L}}$ is completion of the free Lie algebra by bracket length. The vector space V_m is the canonical abelianization of this Lie algebra; choosing lifts of a basis gives the displayed noncanonical free presentation. It has dimension r_m. For $m \geq 3$, complex conjugation c gives the explicit $\mathrm{Gal}(K_m/\mathbb{Q})$-equivariant identification

$$V_m \xrightarrow{\sim} \mathbb{Q}_\ell \oplus (K_m \otimes \mathbb{Q}_\ell)^-, \qquad [z] \longmapsto \left(\frac{\mathrm{Tr}_{K_m/\mathbb{Q}}(z)}{d_m}, \frac{z - cz}{2} \right). \tag{E.33.2}$$

For $m = 1, 2$ it is $V_m = \mathbb{Q}_\ell$. For $m \mid n$, the actual homomorphism of completed groups is induced by $G_{K_n} \hookrightarrow G_{K_m}$, as in E.30, and its map on abelianizations is

$$[z] \longmapsto [\mathrm{Tr}_{K_n/K_m} z]. \tag{E.33.3}$$

In (E.33.2) this is $(a, w) \mapsto ([K_n : K_m]a, \mathrm{Tr}_{K_n/K_m} w)$. Thus every transition is faithfully flat. These groups and these specified homomorphisms compute the continuous arithmetic kernel in (E.30.2); no identification of the abstract rational completion with this continuous completion is used. More explicitly, along the cofinal conductors $m_j = 4j!$, free generators can be chosen compatibly so that each transition keeps the first r_{m_j} generators and sends every additional generator to 1. Thus that kernel is an inverse limit of free prounipotent groups of the explicit ranks r_{m_j} with these deletion maps.

Proof. Write $K = K_m$, and let S consist of the places above ℓ and the infinite places. Every continuous homomorphism $\rho : G_K \to U(\mathbb{Q}_\ell)$, for U unipotent algebraic, is unramified outside

S. Indeed its compact image is pro-ℓ: use a central series of U with additive quotients and the fact that a compact additive subgroup of a finite-dimensional $\mathbb{Q}_\ell$-space is pro-ℓ. At a finite place $v \nmid \ell$, wild inertia therefore dies, and tame inertia maps through its $\mathbb{Z}_\ell(1)$ quotient. If N is the logarithm of the image of a tame generator and F_v is arithmetic Frobenius, the exact relation $F_v \tau F_v^{-1} = \tau^{q_v}$ gives

$$\mathrm{Ad}(\rho(F_v))N = q_v N.$$

The left-hand operator is unipotent and $q_v \neq 1$ in $\mathbb{Q}_\ell$. Hence $N = 0$. The normal closed subgroup generated by these inertia groups is the kernel of $G_K \to G_{K,S}$, which proves the claimed factorization, with its universal property.

The class-field-theoretic exact sequence of local principal units, global units and the maximal abelian pro-ℓ extension unramified outside S gives

$$H_1(G_{K,S}, \mathbb{Q}_\ell) = \mathrm{coker}\left(\mathcal{O}_K^\times \otimes \mathbb{Q}_\ell \xrightarrow{\log_\ell} \bigoplus_{v|\ell} K_v\right). \tag{E.33.4}$$

The finite ideal class group disappears after tensoring. The exact sequence, including its proof from reciprocity, is [Shaa, Theorem 1.5.4 and Theorem 1.5.7]. The diagonal logarithm is injective here: $K_m/\mathbb{Q}$ is abelian, so the proved abelian-field case of Leopoldt applies [Shaa, Theorems1.5.17 and 1.5.21]. To specify the reduction, the unit representation of the maximal real subfield is its augmentation representation. A rational normal generator gives conjugate units whose logarithms have just the norm relation over $\mathbb{Q}$; Brumer's linear-independence theorem retains independence over algebraic coefficients. Diagonalizing the abelian group matrix by characters makes every nontrivial character eigenvalue nonzero. Thus the logarithm has the full Dirichlet rank. This is precisely the proof of the cited Theorem 1.5.21, and applies to every ℓ, including 2. Consequently (E.33.4) has dimension $[K : \mathbb{Q}] - (r_1(K) + r_2(K) - 1) = r_2(K) + 1$.

We also need degree two, rather than infer freeness from this dimension. Apply the finite-coefficient Euler characteristic [Mil06, TheoremI.5.1] to $\mathbb{Z}/\ell^a$ with trivial action. Its formula is exactly

$$\frac{|H^0(G_{K,S}, \mathbb{Z}/\ell^a)||H^2(G_{K,S}, \mathbb{Z}/\ell^a)|}{|H^1(G_{K,S}, \mathbb{Z}/\ell^a)|} = \ell^{-ar_2(K)}. \tag{E.33.5}$$

All these finite-coefficient groups are finite, by [Mil06, CorollaryI.4.15]. The class-field computation says that H^1 has order $\ell^{a(r_2+1)}$ times a bounded factor, because its abelian pro-ℓ source is finitely generated with finite torsion. Since $|H^0| = \ell^a$, (E.33.5) bounds the orders of H^2 independently of a. Inverse limits are exact on these finite systems; the resulting $H^2(G_{K,S}, \mathbb{Z}_\ell)$ is torsion, and tensoring with $\mathbb{Q}_\ell$ gives

$$H^2_{\mathrm{cts}}(G_{K,S}, \mathbb{Q}_\ell) = 0. \tag{E.33.6}$$

Continuous cochains with $\mathbb{Q}_\ell$ coefficients are the union of the cochain complexes with values in $\ell^{-b}\mathbb{Z}_\ell$, by compactness of their domain; this justifies the last tensoring step. The real places at $\ell = 2$ cause no exception: (E.33.5) is exactly Milne's degree-zero-through-two formula, whose footnote explicitly treats this case.

For completeness, the universal unipotent completion has the same H^1 and injects on H^2 into continuous group cohomology. The first assertion follows from homomorphisms to the additive group. For the second, an algebraic continuous Lie-cohomology class factors through a finite-dimensional quotient. Represent it by a central extension by $\mathbb{G}_a$; exponential coordinates give a continuous set-theoretic section of the projection. Its pullback along the dense continuous group homomorphism is a continuous group extension. A zero pullback class supplies a continuous lift; the universal property of unipotent completion extends that

lift algebraically. Its composite with the projection is the identity on the dense original group, hence the identity, so it splits the original extension. This proves injectivity, with no comparison theorem left implicit. Equation (E.33.6) therefore makes the second cohomology of the completed Lie algebra zero.

Choose lifts of a basis of its abelianization V_m. They give a surjection from $\widehat{\mathbb{L}}(V_m)$, with a closed kernel R contained in its commutator ideal. The free-Lie five-term sequence identifies the continuous dual of $R/\overline{[\widehat{\mathbb{L}}(V_m), R]}$ with degree-two cohomology. A nonzero R has a nonzero term of least bracket degree, which cannot lie in $\overline{[\widehat{\mathbb{L}}(V_m), R]}$. Thus $R = 0$, proving (E.33.1).

For the more explicit module in (E.33.2), a unit in a CM field has unit/conjugate-unit a root of unity. Its logarithm is therefore fixed by c and has trace zero. The injectivity already proved and the equal dimensions $d_m/2 - 1$ show that the logarithmic image is exactly the trace-zero subspace of $(K_m \otimes \mathbb{Q}_\ell)^+$. The kernel and image of the displayed formula are now as asserted. The scalar $1/2$ is legitimate also over $\mathbb{Q}_2$.

Finally reciprocity sends the inclusion $G_{K_n} \to G_{K_m}$ to the local norm maps. Logarithms send norms to traces and send norms of global units to the logarithms of global units. This proves (E.33.3), including its compatibility with every tower. Trace is onto on the displayed quotient: on an embedded element of K_m it is multiplication by the nonzero degree. A morphism of prounipotent groups surjective on this finite-dimensional abelianization is surjective, successively on the lower central quotients. The transition maps are consequently faithfully flat. To obtain the final presentation, choose generators at m_1. At the next stage lift their logarithms through the surjection. Choose a basis of the remaining kernel on abelianizations and lift it into the Lie kernel: an arbitrary lift can be corrected by a commutator, because the map on commutator ideals is onto. Together these elements form a basis modulo commutators, and hence a free set of topological Lie generators by the freeness already proved. Iterating gives exactly the claimed deletion maps.

The accompanying exact cyclotomic-field program verifies the conjugation projections, all divisor trace maps and their tower compositions through conductor 24, and prints the corresponding free-Lie ranks through degree 8. This checks the explicit maps, not merely their dimensions. □

A complete intrinsic family with full geometric pro-3 lifts

E.34. Every cubic abelian class unramified outside three lifts fully [Proved]

Let N be the free pro-3 geometric fundamental group of the plus Fermat cubic with its nine cusps removed, of rank ten. Use the rational tangential section above the base tangent $t = 0$, $dt = 1$, specified by $s = -x/z$, $t = s^3$, $y/z = -(1 - s^3)^{1/3}$. Write $V = N^{\mathrm{ab}} = T_3 J_{E,D}$ and $G_S = G_{\mathbb{Q},\{3,\infty\}}$. The genuine action on N factors through G_S. Here a section is called unramified outside three when its geometric cocycle relative to this tangent factors through G_S.

The entire group of such geometrically abelian section classes is

$$H^1_{\mathrm{cts}}(G_S, V) \simeq \mathbb{Z}_3^6 \oplus (\mathbb{Z}/3)^3. \tag{E34.1}$$

Every one of these classes lifts to a section of $N \rtimes G_{\mathbb{Q}}$ whose geometric cocycle also factors through G_S. In particular all of them lift to the intrinsic metabelian, intrinsic class-two and marked-depth quotients. The actual coordinate-unit map sends (E34.1) onto

$$\{(a\kappa_3, b\kappa_3) : a, b \in \mathbb{Z}_3\} \simeq \mathbb{Z}_3^2 \tag{E34.2}$$

in the toric minimum. Its kernel is $\mathbb{Z}_3^4 \oplus (\mathbb{Z}/3)^3$, and every abelian class in each of these fibers

has a full geometric pro-3 lift. These are assertions about the specified unramified family, not about all the ramified classes of E.24.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); this is an intrinsic rank-ten geometric pro-3 family compared through the actual coordinate-unit map.

Proof. The actual action and its relative pro-3 quotient. The projective Fermat cubic over $\mathbb{Z}[1/3]$ is smooth: its three partial derivatives cannot vanish simultaneously. Each of its three coordinate-cusp schemes is finite etale of degree three over that ring, and they are pairwise disjoint. The specified cusp and parameter are integral away from three. For the based comparison, the displayed Puiseux formulas send $t^{1/m}$ to $s^{3/m}$ and commute with coefficientwise Galois action. The constant cubic roots are -1 and have compatible rational 3-power roots. Thus N is the actual index-nine subgroup of the base free pro-3 group F and its action is the restriction of the standard based action on F, with no additional pro-3 Kummer twist.

The based unramifiedness used here is stated explicitly in Hain–Matsumoto, §8.5, printed p. 28 of the opened version [HM03]: take $X_N = \mathbb{P}^1 - \{0, \mu_N, \infty\}$, $N = 1$ and $\ell = 3$ there. Smooth base change gives the *genuine* action on the fundamental group with tangent $\overrightarrow{01}$ over G_S. Restricting to the above subgroup proves the asserted unramifiedness for N.

Put $K = \mathbb{Q}(\zeta)$ and let P be the maximal pro-3 quotient of $G_{K,\{3\}}$. Its defining kernel is stable under G_S, so the relative quotient

$$\Gamma = G_S / \ker(G_{K,\{3\}} \longrightarrow P) \quad \text{fits into} \quad 1 \longrightarrow P \longrightarrow \Gamma \longrightarrow C_2 \longrightarrow 1. \tag{E34.3}$$

Complex conjugation splits this sequence. Over K the action on $F/\Phi(F) \simeq \mathbb{F}_3^2$ is trivial, because $\mu_3 \subset K$. The kernel of $\mathrm{Aut}(F) \to \mathrm{GL}(F/\Phi(F))$ is pro-3. One way to see this is to use the characteristic lower exponent-3 series: successive automorphism kernels embed in homomorphism groups between its finite $\mathbb{F}_3$-graded pieces and are 3-groups; the finite Burnside basis theorem starts the induction. The inverse limit is the asserted pro-3 kernel. The action of $G_{K,\{3\}}$ on F, and hence on N, therefore factors through P. The full action factors through (E34.3).

The same factorization holds for every V-cocycle on G_S. Indeed the map $g \mapsto (\alpha(g), \rho(g))$ takes $G_{K,\{3\}}$ into the pro-3 group $V \rtimes \rho(G_{K,\{3\}})$. By the defining universal property of P it kills the kernel in (E34.3). This proves that both the action and all the cocycles under consideration factor through the same actual quotient.

The free Galois kernel and vanishing of obstructions. The norm-Euclidean ring $\mathbb{Z}[\zeta]$ is a PID, its units are μ_6, and it has the single prime (λ), $\lambda = 1 - \zeta$, above three. Kummer theory gives

$$H^1(G_{K,\{3\}}, \mathbb{F}_3) = \langle \zeta, \lambda \rangle \simeq \mathbb{F}_3^2. \tag{E34.4}$$

For completeness, an element defining a cubic extension unramified away from three has valuation divisible by three at every other prime. Unique factorization then writes its class as $\zeta^a \lambda^b$ modulo cubes. Conversely these classes define extensions unramified away from three. Their independence follows first from the λ valuation and then from the unit group.

The global Euler characteristic of Milne, I.5.1 [Mil06], for this imaginary quadratic field and the trivial module $\mathbb{F}_3$ is $1/3$. The groups in degrees zero and one have orders three and nine, respectively, by (E34.4). Therefore

$$H^2(G_{K,\{3\}}, \mathbb{F}_3) = 0. \tag{E34.5}$$

Inflation embeds $H^2(P, \mathbb{F}_3)$ into this group. To verify that point, let J be the kernel defining P. An invariant nonzero character $J \to \mathbb{F}_3$ would produce a quotient of $G_{K,\{3\}}$ which is

an extension of P by a central group of order three. That quotient is pro-3, contradicting maximality of P. The five-term sequence therefore proves the claimed injection.

These calculations prove that P is free pro-3 of rank two. Here is a direct explanation of the criterion being used. The dimension in (E34.4) gives a minimal surjection from a free pro-3 group on two generators to P. The vanishing of $H^2(P, \mathbb{F}_3)$ gives vanishing in degree two for every finite 3-primary P-module: each such module has a filtration with trivial $\mathbb{F}_3$ factors, since its action image is a finite 3-group. Every finite embedding problem with 3-group kernel is then soluble. Indeed use a nonzero characteristic elementary abelian subgroup of the center of its kernel, solve the quotient problem by induction, and split the remaining abelian extension using this H^2 vanishing. For a characteristic open subgroup U of the minimal free group F_0 and its surjection $f : F_0 \to P$, the precise finite-kernel problem is the pullback of $F_0/U \to P/f(U)$ along $P \to P/f(U)$. Its solution gives the required homomorphism $P \to F_0/U$. Apply this to all those finite quotients. There are finitely many homomorphisms to each finite target, because P has two generators; compactness gives a compatible section of the minimal surjection. Its resulting idempotent endomorphism of the free group is the identity on the Frattini quotient, hence is an automorphism by the pro-3 basis theorem. An idempotent automorphism is the identity, so the minimal surjection is an isomorphism.

In particular P has 3-cohomological dimension one. The Hochschild–Serre sequence for (E34.3), with averaging over C_2 (two is invertible on every 3-primary module), gives

$$H^2(\Gamma, M) = 0 \quad \text{for every finite 3-primary } \Gamma\text{-module } M. \tag{E34.6}$$

The same conclusion holds over G_S for every finite module whose restriction to $G_{K,\{3\}}$ has 3-group action image: filter the restriction by trivial $\mathbb{F}_3$ factors, apply (E34.5), and again take C_2 invariants.

Now take any cocycle $\alpha : \Gamma \to V$. Choose a decreasing characteristic series of N with finite 3-group quotients N_j and intersection one. At each level, pull the surjection $N_j \rtimes \Gamma \to N_j^{\mathrm{ab}} \rtimes \Gamma$ back along the reduction of α. Its kernel is a finite 3-group. The same central-subgroup induction using (E34.6) supplies a section. There are finitely many choices at every level: Γ has the two generators of P and complex conjugation, and the geometric target is finite. The compatible-section sets form a nonempty inverse limit. Explicitly, every finite collection of levels is supplied by a section at their largest common refinement; compactness of the product of the finite sets gives a coherent choice. The limit is a continuous N-cocycle lifting α. Inflating along $G_{\mathbb{Q}} \to G_S \to \Gamma$ gives the asserted section of $N \rtimes G_{\mathbb{Q}}$. This constructs lifts in the full geometric pro-3 group, not merely compatible names for degree-two obstruction classes.

Counting the entire unramified abelian family. The genuine action on $V/3^nV$ has 3-group image over $G_{K,\{3\}}$, by the action factorization already proved. The last assertion following (E34.6) gives $H^2(G_S, V/3^nV) = 0$ for every n. These finite-coefficient degree-one groups are finite (Milne I.4.15); hence the inverse-limit sequence also gives $H^2_{\mathrm{cts}}(G_S, V) = 0$. Consequently

$$H^1_{\mathrm{cts}}(G_S, V)/3^n \simeq H^1(G_S, V/3^nV). \tag{E34.7}$$

Complex conjugation has four plus eigenvalues and six minus eigenvalues on V. To compute them, it fixes the three rational cusps and interchanges three pairs among the other six cusps. Its permutation action on degree-zero cusp divisors therefore has multiplicities five and three. The toric Tate twist reverses their signs, giving multiplicities three and five on the rank-eight toric part. On T_3E, its order is two and its determinant is -1 by the Weil pairing, so its two eigenvalues are $1, -1$. Adding the exact toric and elliptic pieces gives the claimed multiplicities. Since two is a unit in $\mathbb{Z}_3$, the same ranks hold for the integral eigensublattices.

By E.24 the rational 3-primary torsion of $G = J_{E,D}$ is $(\mathbb{Z}/3)^3$. Thus $H^0(G_S, V/3^nV) = G(\mathbb{Q})[3^n]$ has order 27 for every $n \geq 1$. Milne's Euler formula I.5.1, now over $\mathbb{Q}$, has real factor

$3^{4n}/3^{10n} = 3^{-6n}$. Together with the preceding H^2 vanishing it yields

$$|H^1(G_S, V/3^nV)| = 3^{6n+3}. \tag{E34.8}$$

The continuous degree-one group is a finitely generated $\mathbb{Z}_3$-module: its cocycles factor through the three-generated group Γ and form a closed submodule of V^3, modulo the closed coboundary image. Equations (E34.7)–(E34.8) give rank six and torsion of order 27 and exponent three. Equivalently that torsion is $H^0(G_{\mathbb{Q}}, V \otimes \mathbb{Q}_3/V) = G(\mathbb{Q})[3^\infty]$. This proves (E34.1).

Finally the algebraic retraction of E.24 gives a split map of these Galois modules to $\mathbb{Z}_3(1)^2$. Kummer theory over $\mathbb{Z}[1/3]$ gives $H^1(G_S, \mathbb{Z}_3(1)) = \mathbb{Z}_3\kappa_3$. Its rank-two target is torsion-free, and the splitting therefore proves (E34.2) and the stated rank-four kernel with its three torsion factors. The change from the origin at O in E.24 to the present rational cusp is also unramified outside three. The framed cusp-difference relation (E24.6) involves only the units λ, powers of 3 and signs, so this framed point extends over $\mathbb{Z}[1/3]$ and its Kummer cocycle is unramified there. Equivalently both integral tangential fiber functors give the same unramified comparison. Both origins have zero toric pro-3 class. Thus changing origins preserves the unramified family and the displayed linear toric map. The matrix and order calculations are recorded in `e_intrinsic_positive.py`; the Galois vanishing and lifting arguments above establish the infinite assertions. □

All sections of the intrinsic Heisenberg quotient

E.35. A norm equation and every conjugacy class in the finite intrinsic quotient [Proved]

Let $K = \mathbb{Q}(\zeta)$ and retain the quotient H_3 with the based G_K-action (E31.3). Write $\mathcal{A} = K^\times/K^{\times 3}$ multiplicatively. For an abelian section pair $(a, b) \in \mathcal{A}^2$ put $A = \zeta a$, choose nonzero representatives $A, b \in K$, and form the etale cubic algebra $L_A = K[t]/(t^3 - A)$. The fiber of section conjugacy classes of $H_3 \rtimes G_K$ over (a, b) is nonempty exactly when the explicit equation

$$X^3 + AY^3 + A^2Z^3 - 3AXYZ = b \quad \text{has a solution } X, Y, Z \in K. \tag{E35.1}$$

Every nonempty fiber is a torsor under the explicitly specified group

$$\mathcal{A}/\langle A, b\rangle = K^\times/(K^{\times 3}\langle A, b\rangle). \tag{E35.2}$$

The subgroup in the denominator has rank zero, one or two over $\mathbb{F}_3$, according to the span of the two indicated cube classes. There are concrete coordinates for this group. Choose one generator $\pi_{\mathfrak{p}}$ for each nonzero prime ideal of $\mathbb{Z}[\zeta]$, and put $J = \{\zeta\} \amalg \{\mathfrak{p}\}$. Unique factorization gives the explicit isomorphism

$$\mathcal{A} \longrightarrow \bigoplus_{j \in J} \mathbb{F}_3, \qquad [\zeta^e \prod_{\mathfrak{p}} \pi_{\mathfrak{p}}^{e_{\mathfrak{p}}}] \longmapsto (e, (e_{\mathfrak{p}})) \pmod 3. \tag{E35.2a}$$

The sign in the factorization is a cube and is omitted. Thus (E35.2) is obtained by Gaussian elimination of the two finite-support vectors $[A], [b]$: choose zero, one or two pivot coordinates according to their rank, subtract the corresponding row-reduced basis vectors, and retain all remaining coordinates. This is an explicit parameterization by the direct sum of $\mathbb{F}_3$ over the coordinates in J outside those pivots. After choosing the lift n_0 constructed from a norm solution, all classes are represented by $n_d(g) = (0, 0, \kappa_3(d)(g))n_0(g)$ with $d \in K^\times/(K^{\times 3}\langle A, b\rangle)$.

All dependent cases are included. If A is a cube, (E35.1) is always soluble and the fiber group is $\mathcal{A}/\langle b\rangle$. If A is not a cube and b is in its span, (E35.1) is always soluble and the fiber

group is $\mathcal{A}/\langle A\rangle$. If both classes are trivial the group is $\mathcal{A}$. For two independent classes, a solution of (E35.1) constructs an actual Heisenberg Galois extension as in the proof.

The quotient is the one obtained from the actual map (E31.4) and the based arithmetic comparison of E.31. For example the pair $(a,b)=(1,7)$ has empty fiber, while $(a,b)=(2\zeta^2,3)$ has nonempty fiber, with norm solution $1+\sqrt[3]{2}$. No assertion is made that every pair here is the image of a class in the full intrinsic rank-ten abelian section group.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); this computes a specified finite intrinsic quotient reached by the additional morphism (E31.4).

Proof. Choose the path in E.31 within the formal punctured disc at the target zero. Equation (E31.7) then gives the image of the source tangential section as

$$q_g=(\kappa_3(\zeta)(g),0,0)\in H_3. \tag{E35.3}$$

To see the zero central coordinate directly, write $h=\zeta\lambda^{-3}s^3u(s)$ with $u(0)=1$. Its chosen cube root is $(\zeta_9/\lambda)s\,u(s)^{1/3}$. The last factor has coefficients in K, so the only Galois change is the inertia power supplied by ζ_9. The formal puncture has only that one geometric generator and gives no commutator term. Thus $q:G_K\to H_3$ is the homomorphism in (E35.3), and the intrinsic action is $\mathrm{Ad}(q_g)$.

There is an explicit isomorphism of arithmetic groups

$$H_3\rtimes_{\mathrm{Ad}\,q}G_K\longrightarrow H_3\times G_K,\qquad (n,g)\longmapsto(nq_g,g). \tag{E35.4}$$

Its inverse sends (h,g) to (hq_g^{-1},g). It is the identity on the geometric group. Consequently it identifies section classes with continuous homomorphisms $G_K\to H_3$ modulo conjugation by H_3, and sends the intrinsic abelian pair (a,b) to the ordinary Kummer pair $(A,b)=(\zeta a,b)$.

Write these latter Kummer characters as $u,v:G_K\to\mathbb{F}_3$. A homomorphism with those abelian coordinates has the form (u_g,v_g,w_g), where its central coordinate solves

$$dw(g,h)=-\tfrac{1}{2}(u_gv_h-v_gu_h). \tag{E35.5}$$

The two-cocycle on the right, with sign reversed, differs from u_gv_h by $d(u_gv_g/2)$; hence its obstruction is exactly $\kappa_3(A)\cup\kappa_3(b)$. The Kummer exact sequence and Hilbert 90 embed this group of classes into $\mathrm{Br}(K)[3]$.

We spell out the resulting norm condition. Suppose first that A is not a cube, let $\alpha^3=A$ and let $\sigma(\alpha)=\zeta\alpha$ generate $\mathrm{Gal}(L_A/K)$. The cyclic algebra is

$$\mathcal{C}=L_A\oplus L_AU\oplus L_AU^2,\qquad U\ell=\sigma(\ell)U,\quad U^3=b. \tag{E35.6}$$

Its factor set is $b^{\lfloor(i+j)/3\rfloor}$ on σ^i,σ^j. Choose $\beta^3=b$ and represent the values of u by $0,1,2$. The multiplicative coboundary of $g\mapsto\beta^{u_g}$ is

$$b^{\lfloor(u_g+u_h)/3\rfloor}\zeta^{v_gu_h}. \tag{E35.7}$$

The identity for $d(u_gv_g)$ in (E35.5) shows that the class of this factor set is precisely $\kappa_3(A)\cup\kappa_3(b)$. The crossed-product comparison with the Brauer group is Milne, *Class Field Theory*, IV.3.14 [Mil20b]; its construction uses exactly the multiplication and factor set just displayed.

The algebra (E35.6) is split exactly when b is a norm. This equivalence also has a direct proof here. A norm solution $b=N_{L_A/K}(w)$ represents $\mathcal{C}$ on the three-dimensional K-space L_A, with ℓ acting by multiplication and U acting as $w\sigma$. The relations hold and the resulting homomorphism is an isomorphism with $\mathrm{End}_K(L_A)$, by central simplicity and the equal dimension nine. Conversely, a splitting realizes $\mathcal{C}$ on a three-dimensional K-space which, by

its embedded degree-three field L_A, is one-dimensional over L_A. The operator U is necessarily $w\sigma$ in a choice of L_A-basis. Its cube is $N(w)$, so $b = N(w)$. The norm of $X + Y\alpha + Z\alpha^2$, by its multiplication matrix on this same basis, is $X^3 + AY^3 + A^2Z^3 - 3AXYZ$. This proves (E35.1).

For independent cube classes A, b the norm solution also yields an explicit finite Galois realization. Write $w = X + Y\alpha + Z\alpha^2$ with $N(w) = b$ and put

$$\beta^3 = b, \qquad v = w^2\sigma(w), \qquad \gamma^3 = v. \tag{E35.8}$$

The field $M = L_A(\beta, \gamma)$ has the automorphisms

	α	β	γ
$\widetilde{\sigma}$	$\zeta\alpha$	β	$\beta\gamma/w$
τ	α	$\zeta\beta$	γ.

(E35.9)

Indeed $\sigma(v)/v = b/w^3$, so these formulas preserve the defining equations, and $\widetilde{\sigma}^3(\gamma) = b\gamma/N(w) = \gamma$. Both displayed automorphisms have order three. The root γ does not already lie in $L_A(\beta)$. Otherwise the fact that its cube is in L_A would write $\gamma = c\beta^j$ for $c \in L_A$. The equation $\sigma(v)/v = b/w^3$ would then imply that b is a cube in L_A, contradicting the independence of A, b. Thus $[M : K] = 27$, the formulas make M/K Galois, and their commutator acts on γ as multiplication by ζ^{-1}. Taking that automorphism as the central generator identifies its Galois group with (E31.2), with the two stated Kummer characters. Composing the Galois restriction map with this identification and then (E35.4)$^{-1}$ gives an actual arithmetic section.

The dependent cases require no degree-27 field. If A is a cube, replace its representative by one. Then $L_A \simeq K^3$ and the norm is the product of its three coordinates, so it is surjective. In (E35.1) the choice $X = (b+2)/3$, $Y = Z = (b-1)/3$ gives norm b. The homomorphism $g \mapsto (0, \kappa_3(b)(g), 0)$ provides a lift. If A is nontrivial and $b = A^j c^3$ for $j = 0, 1, 2$, the element $w = c\alpha^j$ has norm b. The homomorphism $g \mapsto (\kappa_3(A)(g), j\kappa_3(A)(g), 0)$ has the required coordinates; its alternating term is zero. This includes $b = 1$ and all other dependent cases. The construction is independent of the selected representatives modulo cubes.

Finally choose one lift h_g of (u, v). Every other lift is uniquely $z_g h_g$ with $z \in H^1(G_K, \mathbb{F}_3) = \mathcal{A}$. Conjugating by an element with abelian coordinates (r, s) adds the central character $rv - su$. Central conjugations do nothing. The exact subgroup of differences generated by all geometric conjugations is therefore $\langle \kappa_3(A), \kappa_3(b) \rangle$. This proves the entire torsor description (E35.2), including each of the dependent cases just treated. For completeness, the coordinate computation (E35.2a) uses the Euclidean norm $N(a + b\zeta) = a^2 - ab + b^2$ on $\mathbb{Z}[\zeta]$. Rounding the two rational coefficients of any quotient to nearest integers leaves errors r, s of absolute value at most $1/2$, and $r^2 - rs + s^2 \le 3/4 < 1$. The division algorithm proves that this ring is a principal ideal domain and a unique factorization domain. Its norm-one units are $\{\pm 1, \pm\zeta, \pm\zeta^2\}$, so their quotient by cubes is generated freely over $\mathbb{F}_3$ by ζ. Factoring the numerator and denominator of an element of $K^\times$ now proves both the injectivity and surjectivity of (E35.2a). Row reduction gives an actual isomorphism from its quotient by $[A], [b]$ to the stated direct sum: normalize the pivot entries to one and all other pivot entries to zero, then send a vector to its unique representative with every pivot entry zero. This also describes the inverse by inclusion of those nonpivot coordinates.

At $(a, b) = (1, 7)$, equation (E35.1) would assert that 7 is a norm from $K(\zeta_9)$, contradicted by the local calculation at $(3 + \zeta)$ in E.31. For $A = 2, b = 3$, the element $1 + \alpha$ has norm $1 + \alpha^3 = 3$, proving the positive example. The program `e_intrinsic_finite.py` checks the norm polynomial, this solution and the relations in (E35.8)–(E35.9), and enumerates every conjugation orbit for all two-generator assignments in H_3. The finite intrinsic quotient map of

E.31 remains the comparison to the full cubic problem; no inverse to that quotient map is asserted. □

Every fiber of the cuspidally central cubic quotient

E.36. A class-two quotient retaining all ten abelian coordinates, with every fiber evaluated [Proved]

Let $E = X_3^+$, let D be its nine coordinate cusps, and put $O = [1 : -1 : 0]$. Use the tangential base point at O with parameter $s = z/x$ and tangent one, both on $U_3 = E \setminus D$ and on $E \setminus \{O\}$. Write

$$N = \pi_1(U_{3,\overline{\mathbb{Q}}})^{(3)}, \qquad V = N^{\mathrm{ab}} = T_3 J_{E,D}, \qquad T = T_3 E, \qquad I = \ker(N \longrightarrow T).$$

Closures are understood in commutator subgroups. Set

$$N^{\mathrm{cc}} = N/[N, I], \qquad C = \pi_1((E \setminus \{O\})_{\overline{\mathbb{Q}}})^{(3)}/\gamma_3.$$

The maps induced by the open immersion $U_3 \hookrightarrow E \setminus \{O\}$ and by abelianization give an actual, based, $G_{\mathbb{Q}}$-equivariant isomorphism

$$N^{\mathrm{cc}} \xrightarrow{\sim} C \times_T V, \qquad n \longmapsto (q_O(n), n^{\mathrm{ab}}). \tag{E36.1}$$

In particular the quotient retains all ten coordinates of V and its nonzero elliptic commutator. The exact quotient from the full intrinsic class-two group is

$$1 \longrightarrow \ker\Big(\bigwedge\nolimits^2 V \longrightarrow \bigwedge\nolimits^2 T\Big) \longrightarrow N/\gamma_3 N \longrightarrow N^{\mathrm{cc}} \longrightarrow 1. \tag{E36.2}$$

The displayed kernel is a free $\mathbb{Z}_3$-module of rank 44, with the Galois action induced on the exterior square; the remaining central commutator module is $\bigwedge^2 T \simeq \mathbb{Z}_3(1)$.

Every geometric-abelian section class $\alpha \in H^1(\mathbb{Q}, V)$ lifts to a section of $N^{\mathrm{cc}} \rtimes G_{\mathbb{Q}}$. For every such α, the complete fiber of geometric conjugacy classes is a torsor under

$$\mathcal{R}_{\mathbb{P}}(\mathbb{Z}_3) = \left\{ (a_p)_{p\ mrational\ prime} \in \prod_p \mathbb{Z}_3 : \begin{smallmatrix} \text{for every } m,\ a_p \in 3^m\mathbb{Z}_3 \\ \text{for all but finitely many } p \end{smallmatrix} \right\}. \tag{E36.3}$$

The parameter $a = (a_p)$ acts by the central Kummer cocycle of the compatible classes $\prod_p p^{a_p}$ modulo $\mathbb{Q}^{\times 3^m}$. These finite products modulo each 3^m define the indicated element of the completion for multiplication by powers of three. Thus this is a computation of every fiber over the entire rank-ten abelian section group evaluated in E.24, including its ramified classes. The classes obstructed in E.31 also lift to (E36.1), while their further lifting through (E36.2) is obstructed by the explicit quotient of E.31.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379). The quotient maps (E36.1)–(E36.2) specify exactly the intrinsic cubic quotient evaluated here.

Proof. First compute the geometric maps integrally. An oriented presentation of the punctured complex elliptic curve is

$$N = \langle a, b, c_O, c_1, \dots, c_8 \mid [a, b] c_O c_1 \cdots c_8 = 1 \rangle_{\text{pro-3}}. \tag{E36.4}$$

Here a, b map to a basis of T; the c_d are the peripheral loops. Eliminating c_O gives the free pro-three group of rank ten. The kernel I is normally generated by these peripheral loops.

Passing to $N/[N,I]$ makes all c_d central and leaves only the indicated relation. Consequently the geometric cuspidal kernel is

$$I/[N,I] \simeq \mathbb{Z}_3(1)[D], \qquad [a,b] = -\sum_{d\in D} c_d, \tag{E36.5}$$

with its actual Galois action: the cusps are permuted and each inertia generator has the cyclotomic twist. This action can also be obtained from the torus $\mathrm{Res}_{D/\mathbb{Q}}\mathbb{G}_m$. The central quotient is the one defined in Borne–Emsalem–Stix, §3.2; their Proposition 3.5 identifies it with the fundamental group of their torus torsor by the canonical map from the open curve [BES18].

For an entirely explicit check of (E36.5), take the two-step Lie module with basis $a, b, c_1, \dots, c_8, k$ and only bracket $[a,b] = k$. The law $u * v = u + v + [u,v]/2$ is integral over $\mathbb{Z}_3$. Sending c_O to $-k - \sum_{i=1}^{8} c_i$ satisfies (E36.4). Conversely its generators and brackets give precisely the relations of $N/[N,I]$. The two maps are inverse, so there are no further relations or torsion in the nine-dimensional central lattice of (E36.5).

The map q_O fills the other eight punctures. On (E36.5) it is projection to the O-coordinate. Abelianization quotients (E36.5) by the diagonal element $\delta = \sum_{d\in D} c_d$. The maps on the two geometric cuspidal kernels are therefore

$$\mathbb{Z}_3(1)[D] \longrightarrow \mathbb{Z}_3(1) \oplus \big(\mathbb{Z}_3(1)[D]/\mathbb{Z}_3(1)\delta\big), \quad t \longmapsto (t_O, \bar{t}). \tag{E36.6}$$

This is an isomorphism: in coordinates it sends $(t_O, t_1, \dots, t_8)$ to $(t_O, t_1 - t_O, \dots, t_8 - t_O)$, with inverse $(u, v_1, \dots, v_8) \mapsto (u, v_1 + u, \dots, v_8 + u)$. It is $G_{\mathbb{Q}}$-equivariant because O is rational and the diagonal element is invariant apart from its cyclotomic twist. Both sides of (E36.1) map onto T, and (E36.6) identifies their kernels. This proves (E36.1). The isomorphism is induced by the stated morphisms of curves and abelianization, with the same tangent throughout. It therefore identifies the actual arithmetic extensions, not just their abstract geometric groups. There is no change of section origin in (E36.1).

The commutator module of the free class-two pro-three group on V is $\bigwedge^2 V$. The bracket surviving in (E36.5) is the exterior-square map induced by $V \to T$. In the basis from (E36.4) it sends $a \wedge b$ to one generator and sends the other 44 basic wedges to zero. Its kernel is therefore exactly $[N,I]/\gamma_3 N$, free of rank $\binom{10}{2} - 1 = 44$; its image has index one. The orientation identifies the image with $\bigwedge^2 T = \mathbb{Z}_3(1)$, compatibly with the Weil pairing. Functoriality proves the asserted arithmetic action and (E36.2).

We next establish the existence of all the sections in the statement. Use the rational point O as origin on the proper elliptic curve. Its full geometric fundamental group is $T_\bullet E = \prod_\ell T_\ell E$, and its arithmetic fundamental group is the corresponding semidirect product with $G_{\mathbb{Q}}$. For a given α, let $\beta \in H^1(\mathbb{Q}, T_3 E)$ be its image, and choose compatible cocycle representatives. There is a specified full proper-curve section

$$g \longmapsto \big((\beta(g), 0, 0, \dots), g\big), \tag{E36.7}$$

where every component at a prime other than three is zero. The inclusion of the three-primary factor is continuous and $G_{\mathbb{Q}}$-equivariant, so (E36.7) is an actual section of the full arithmetic fundamental group of E.

Borne–Emsalem–Stix, Theorem 5.10, states that for a smooth projective positive-genus curve over $\mathbb{Q}$ and a boundary which is a union of torsion sub-packets, every section lifts to its cuspidally central quotient [BES18]. Apply the theorem to E and the singleton boundary $\{O\}$. This is a torsion sub-packet by their Definition 2.3: the only difference of two points in its support is zero in the Picard group. All hypotheses are now satisfied. The cuspidal kernel for a once-punctured elliptic curve is the geometric commutator subgroup; making it

geometrically central is precisely quotienting by γ_3. Projecting the resulting full lift of (E36.7) to its geometric pro-three quotient gives a section c_β of $C \rtimes G_{\mathbb{Q}}$ above β.

If this section projects to a representative conjugate to the chosen β, conjugate it by a geometric lift of the relevant element of T. Its projection then equals the chosen representative. Pairing it with the given abelian section α gives a section of the fiber product (E36.1). This proves existence for every α, without imposing any ramification condition.

Finally the kernel of $N^{\mathrm{cc}} \to V$ is the central copy of $\mathbb{Z}_3(1)$ generated by the elliptic commutator. After choosing one lift of a fixed representative of α, the other lifts differ by continuous one-cocycles with values in this module. Central geometric conjugation changes the cocycle by an ordinary coboundary. There is no additional identification from geometric elements with nonzero image in V: such an element could fix the prescribed abelian cocycle only if its image lay in $H^0(\mathbb{Q}, V)$, and this group is zero. Here is a direct verification. In the exact sequence $0 \to T_D \to J_{E,D} \to E \to 0$, the rational torsion of $T_D \simeq \mathbb{G}_m^2 \times (\mathrm{Res}_{K/\mathbb{Q}}\, \mathbb{G}_m)^3$ is finite, being a product of groups of roots of unity. The group $E(\mathbb{Q})$ has order three by E.24. Thus $J_{E,D}(\mathbb{Q})[3^\infty]$ is finite. An invariant in $T_3 J_{E,D}$ is a compatible sequence in this finite three-primary group under multiplication by three. Every such sequence is zero. Hence $H^0(\mathbb{Q}, V) = 0$ as asserted.

The exact fiber is consequently a torsor under $H^1(\mathbb{Q}, \mathbb{Z}_3(1))$. Kummer theory identifies this group with $\varprojlim_m \mathbb{Q}^\times/\mathbb{Q}^{\times 3^m}$. Unique factorization in $\mathbb{Q}^\times$ gives the isomorphism with (E36.3): the sign is a cube, each prime valuation is reduced modulo 3^m, and every such residue vector has finite support. The inverse constructs the compatible Kummer classes from precisely these finite-support residue vectors. This proves the complete coordinate description of all fibers.

The matrix in (E36.6) has determinant one, and the exterior-square map in (E36.2) has Smith diagonal one and kernel rank 44. The program `e_cuspidally_central.py` verifies these integral lattice computations. The existence theorem and the arithmetic fiber calculation above establish the infinite section assertions. □

The full intrinsic cubic class-two section set

E.37. A complete local matrix criterion and every nonempty intrinsic fiber [Proved]

Use the same curve, tangent and group N as in E.36, and put

$$V = N^{\mathrm{ab}}, \quad M = \bigwedge^2 V, \quad K = \mathbb{Q}(\zeta), \quad \lambda = 1 - \zeta.$$

The following data give the entire image and every fiber of

$$\mathrm{Sect}(N/\gamma_3 N \rtimes G_{\mathbb{Q}})/\text{geometric conjugacy} \longrightarrow H^1(\mathbb{Q}, V). \tag{E37.1}$$

All coordinates in the criterion belong to the full rank-ten module V.

Write the eight cusps other than $O = C_0$ in the order

$$\mathcal{D} = (A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2), \quad A_i = [0 : -\zeta^i : 1],\ B_i = [-\zeta^i : 0 : 1],\ C_i = [1 : -\zeta^i : 0].$$

In the basis $(P, Q) = (A_0, A_2)$ of $E[3]$ their coordinates are the rows of

$$\begin{pmatrix} 1 & 0 \\ 2 & 2 \\ 0 & 1 \\ 2 & 0 \\ 1 & 1 \\ 0 & 2 \\ 2 & 1 \\ 1 & 2 \end{pmatrix}. \tag{E37.2}$$

Let e_d be the positive peripheral cocharacters of the split torus T_D over K, and write $V = \mathbb{Z}_3 f_1 \oplus \mathbb{Z}_3 f_2 \oplus \bigoplus_{d\in\mathcal{D}} \mathbb{Z}_3 e_d$. Set

$$R = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C_E = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}, \quad U_d = (v_d, -u_d), \qquad (u_d, v_d) \text{ from (E37.2).} \tag{E37.3}$$

These are integral coordinates in which the unique homomorphism $j : E \to J_{E,D}$ with $p\circ j = [3]$ has Tate matrix $j_* = \binom{3I_2}{U}$. For a prime $\mathfrak{q} \nmid 3$ of K, let $q = N\mathfrak{q}$ and let $\psi_\mathfrak{q} = a + b\zeta$ be the elliptic Frobenius scalar, equivalently the Jacobi sum of D.6. It is the primary generator of $\mathfrak{q}$, congruent to one modulo three; at an inert rational prime p it is $-p$. The actual Frobenius matrix on V is

$$A_\mathfrak{q} = \begin{pmatrix} A_E & 0 \\ U(A_E - qI_2)/3 & qI_8 \end{pmatrix}, \qquad A_E = aI_2 + bR. \tag{E37.4}$$

Write $A(\psi, q)$ for this displayed block matrix. Let $B_\mathfrak{q} = \bigwedge^2 A_\mathfrak{q}$. The matrix of complex conjugation on V is the same block formula with $A_E = C_E$ and qI_8 replaced by C_T, where C_T is minus the permutation interchanging the subscripts one and two in each cusp orbit.

Put

$$w = (3f_1 + U_1) \wedge (3f_2 + U_2), \quad L = \mathbb{Z}_3 w \simeq \mathbb{Z}_3(1), \quad M_0 = \ker(M \to \bigwedge^2 T_3 E). \tag{E37.5}$$

The coefficient of w at $e_{A_0} \wedge e_{A_2}$ is one. Use as a basis of M/L the other 44 basic wedges, and as a basis of M_0 the 44 basic wedges other than $f_1 \wedge f_2$. The map

$$d_\mathfrak{q} : M/L \longrightarrow M_0, \qquad \overline{m} \longmapsto (q^{-1}B_\mathfrak{q} - 1)m \tag{E37.6}$$

is therefore a specified 44 by 44 matrix over $\mathbb{Z}_3$. It is invertible over $\mathbb{Q}_3$.

Here is the arithmetic input for its right-hand side. For each $d \neq O$, define $z_d \in J_{E,D}(K)$ to be $\mathcal{O}(d-O)$, framed by s_d^{-1} at d, by s_O at O, and by one at every other cusp. The unit tangent parameters are $s_{A_i} = x/z$, $s_{B_i} = y/z$, and $s_{C_i} = z/x$. Their third multiples are completely specified by

$$3z_d = (t_{d,r})_{r\in\mathcal{D}}, \qquad t_{d,r} = \frac{\mathrm{lead}_r(f_d)}{\mathrm{lead}_O(f_d)}, \quad f_d = \frac{L_d}{x+y}, \tag{E37.7}$$

where $L_{A_i} = y + \zeta^i z$, $L_{B_i} = x + \zeta^i z$, $L_{C_i} = x + \zeta^{2i} y$. At d and O, the leading values mean respectively f_d/s_d^3 and $f_d s_O^3$; elsewhere they mean ordinary evaluation. Explicitly these exceptional values are

d	$\mathrm{lead}_d(f_d)$	$\mathrm{lead}_O(f_d)$
A_i	$1/3$	3
B_i	$1/3$	-3
C_i $(i = 1, 2)$	$-1/(3(1-\zeta^i))$	$-3(1-\zeta^{2i})$.

(E37.8)

Thus every entry in (E37.7) is an explicitly given unit away from λ; the program also lists all 64 entries as powers of $-\zeta^2$ and λ.

Represent $\alpha \in H^1(\mathbb{Q}, V)$ by a continuous cocycle and restrict to K. At each $\mathfrak{q} \nmid 3$ choose arithmetic Frobenius $F_\mathfrak{q}$ and a tame pro-three inertia generator $\tau_\mathfrak{q}$ with $F\tau F^{-1} = \tau^q$. Put

$$a_\mathfrak{q} = \alpha(F_\mathfrak{q}), \qquad v_\mathfrak{q} = \alpha(\tau_\mathfrak{q}) = \sum_{d\in\mathcal{D}} b_{\mathfrak{q},d} e_d. \tag{E37.9}$$

The assertion that $v_\mathfrak{q}$ lies in this toric submodule is part of the theorem. Let $k_{\mathfrak{q},d}$ be any integral representative of the unramified Kummer class of z_d in $V/(A_\mathfrak{q} - 1)V$. It is computable

from the framed point $\overline{z}_d$ of $J_{E,D}(\mathbb{F}_q)$: take its successive 3^m-division points and apply $F_{\mathfrak{q}} - 1$. Define the following vector in the stated basis of M_0:

$$D_{\mathfrak{q}} = a_{\mathfrak{q}} \wedge v_{\mathfrak{q}} - \sum_{d \in \mathcal{D}} b_{\mathfrak{q},d} k_{\mathfrak{q},d} \wedge e_d, \qquad c_{\mathfrak{q}} = -d_{\mathfrak{q}}^{-1} D_{\mathfrak{q}}. \tag{E37.10}$$

Then the fiber of (E37.1) over α is nonempty *exactly* when

$$c_{\mathfrak{q}} \in \mathbb{Z}_3^{44} extfor every \mathfrak{q} \nmid 3, \quad \text{and for every } m, \quad c_{\mathfrak{q}} \in 3^m \mathbb{Z}_3^{44} \text{ for all but finitely many } \mathfrak{q}. \tag{E37.11}$$

All choices of the displayed Kummer representatives give the same answer. The convergence requirement is included in the criterion.

Every nonempty fiber is a torsor under the explicitly evaluated group

$$\mathcal{R}_{\{p \neq 3\}}(\mathbb{Z}_3) \oplus \mathbb{Z}_3^{24} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/9)^5. \tag{E37.12}$$

Here the restricted prime coordinates act by the actual central cocycles $\sum_{p \neq 3} a_p \kappa_3(p) w$. For the remaining factor, put $A_\tau = A(1 + 3\zeta, 7)$, $B_\tau = \bigwedge^2 A_\tau$, and $B_c = \bigwedge^2 C_V$. Its exact presentation, and hence a specified coordinate map, is

$$H^1(G_{\mathbb{Q},\{3,\infty\}}, M) = M/(B_\tau - 1)M^c = \operatorname{coker}\big((B_\tau - 1)(1 + B_c) : M \to M\big). \tag{E37.13}$$

The supplied exact Smith matrices have seven unit factors, nine factors three, five factors nine, and 24 zero factors. Multiplication by the supplied left Smith matrix gives the coordinates in (E37.12).

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379). Equations (E37.1) and (E36.2) specify the full intrinsic class-two group, including the 44 *directions discarded by the cuspidally central quotient.*

Proof. The integral Tate lattice and its arithmetic action. The nine cusps are precisely $E[3]$ with origin O. The tangent-line divisors prove that each is killed by three, and they are nine distinct points. The collinear cusp triples and the automorphism $z \mapsto \zeta z$ give (E37.2). On the Weierstrass model of E.24, $P = (3, -5)$ and $Q = (3\zeta, -5)$. The tangent functions $y + 3x - 4$ and $y + 3\zeta^2 x - 4$ have divisors $3P - 3O$ and $3Q - 3O$, with equal leading coefficients at O. The normalized Weil formula gives

$$e_3(P, Q) = -\frac{9(\zeta - 1)}{9(\zeta^2 - 1)} = \zeta. \tag{E37.14}$$

The sign in this formula is included in [Sut19, Corollary 24.27].

The framed-line-bundle description of the generalized Jacobian identifies its extension characters with $\mathcal{O}(d - O)$. Pulling back by [3] trivializes these characters, and gives the unique homomorphism j with $pj = [3]$; uniqueness follows from $\operatorname{Hom}(E, T_D) = 0$. For precision about its sign, if $\operatorname{div}(h_d) = [3]^*(d - O)$, the vector frame of the trivialized pullback is $1/h_d$. Translation by $t \in E[3]$ multiplies this frame by $h_d(X)/h_d(X + t) = e_3(d, t)^{-1}$. Here e_3 is the divisor pairing used in (E37.14). The equivalence with this translation formula, and the contrary argument ordering in one presentation of the pairing, are proved and discussed in [Eng15, §3, Remark 9]. Thus the character at d of $j(t)$ is the inverse divisor pairing. In the basis (E37.14), this gives exactly $U_d = (v_d, -u_d)$.

Equivalently, the actual semiabelian variety is the quotient of $T_D \times E$ by the graph $\{(j(t)^{-1}, t) : t \in E[3]\}$, with projection to E given by multiplication by three on the second factor. The pairing computation proves that its extension characters are the specified framed line bundles, so this is an isomorphism of the given extensions. Its Tate lattice is generated

by the eight torus generators and two lifts f_1, f_2 with $j_*(t_i) = 3f_i + U_i$. This proves (E37.3). Nondegeneracy of (E37.14) shows that j has trivial kernel. Its image is an elliptic subvariety, and the quotient is a torus; hence its Tate sublattice is primitive. The wedge w is also primitive, as witnessed by its coefficient one at the indicated toric wedge.

The $\mathbb{Z}_3[\zeta]$-module T_3E is torsion free of rank one over this discrete valuation ring. All its three-torsion is rational over K, so the image of G_K lies in $1 + 3\mathbb{Z}_3[\zeta]$. Its determinant is the cyclotomic character by the Weil pairing. The Jacobi-sum comparison D.6 gives the scalar $\psi_{\mathfrak{q}}$ used in (E37.4). Its primary congruence can also be checked directly: the transformations $t \mapsto 1 - t$ and $t \mapsto 1/t$ preserve the cubic character of $t(1 - t)$. Their orbits have sizes six, three, or two; the size-two orbit satisfies $t^2 - t + 1 = 0$ and has character one. Thus the character counts are $2, 0, 0$ modulo three, and the negative character sum is one modulo three. This works also in characteristic two. Its reduction at $\mathfrak{q}$ vanishes, because the corresponding power sum has degree $2(q - 1)/3 < q - 1$; its norm is q by D.6. These facts give the primary generator at split primes; at inert primes the trace-fiber calculation in D.7 gives $-p$. Only one Eisenstein unit is one modulo three, proving uniqueness.

Equation $A_V j_* = j_* A_E$, together with the actual torus action qI_8, now forces (E37.4). To choose the stated complex-conjugation basis, first lift the rational point P to an $\mathbb{Z}_3[\zeta]$-basis t of T_3E. It is a basis because $(\zeta - 1)P \neq 0$. Write $c(t) = \epsilon t$. Then $\epsilon\overline{\epsilon} = 1$ and $\epsilon \equiv 1$ modulo three. Its unique square root v in $1 + 3\mathbb{Z}_3[\zeta]$ satisfies $\overline{v} = v^{-1}$, so vt is fixed by c and still lifts P. Thus c is conjugation on the Eisenstein coefficients, with matrix C_E. The toric action is C_T, and equivariance of j_* gives the claimed remaining block. These arguments determine all the displayed arithmetic matrices on the integral lattice.

The rational decomposition induced by the isogeny $T_D \times E \to J_{E,D}$ has Frobenius eigenvalues q eight times and the two elliptic eigenvalues $\alpha_{\mathfrak{q}}, \beta_{\mathfrak{q}}$. On M the only eigenvalue equal to q is their product, on the line L. The other eigenvalues have complex absolute values $q^{3/2}$ or q^2. Consequently (E37.6) is injective on M/L and invertible over $\mathbb{Q}_3$ onto M_0. The index $[M : (L + M_0)]$ is nine, since the projection of w to $\bigwedge^2 T_3E$ is nine. This explains why the integral domain in (E37.6) must be M/L. For example its determinant, up to a unit sign, is

$$\det d_{\mathfrak{q}} = \#E(\mathbb{F}_q)^8 (q - 1)^{28}/9. \tag{E37.15}$$

Tangential paths and the complete peripheral term. Use a geometric path p_d from the base tangent to the chosen tangent at d. Its abelian path cocycle is $k_d(g) = p_d^{-1} g(p_d)$ in V. It is the Kummer class of the framed point z_d above. One can see the comparison on the actual covers as follows. Choose an ordinary geometric base point and the Abel–Jacobi map to the generalized Jacobian. Pullbacks of its multiplication-by-3^m covers are the maximal abelian exponent-3^m covers of the curve. At a chosen tangent parameter, separate the power of that parameter from the unit part of the local framing. The coefficientwise Galois section fixes all chosen powers of the parameter. Roots of the remaining power series with constant term one are uniquely normalized. Therefore the comparison of the two tangential fibers is exactly the torsor of divisions of the ratio of their constant vector frames. There is an explicit local sign in that ratio. In the family $\mathcal{O}(u_d - u_O)$, the canonical frame at d is $-t_d$ times $(s_d - t_d)^{-1}$, and at O it is $(-t_O)^{-1}$ times $(s_O - t_O)$. Removing the tangent powers therefore gives frames $-s_d^{-1}$ and $-s_O$, with one elsewhere. Their ratio to the frames defining z_d is the torus point ϵ_d with value -1 at d and O and one elsewhere, modulo common scaling. This point has order two. Its compatible 3^m-division points are ϵ_d itself, all rational, so its pro-three Kummer torsor has the specified canonical trivialization. Tensoring by that trivialization constructs an isomorphism of the path torsor with the division torsor of z_d at every level, compatible with every transition map. It proves the integral Kummer comparison, including the section origin; the path-cocycle convention is the one in [Wic12a, (12.4)–(12.10)].

Multiplication by f_d trivializes the third tensor power of this framed line bundle. Evaluating

the frames gives (E37.7). Expanding the appropriate branch $(1+s^3)^{1/3}$ gives (E37.8) directly. These formulas prove that the points and their division torsors are unramified away from λ; the cubic and its distinct cusp divisor have good reduction there. Over $\mathbb{F}_q$, the Kummer construction identifies the three-primary part of $J_{E,D}(\mathbb{F}_q)$ with $V/(A_{\mathfrak{q}}-1)V$. Explicitly a compatible division sequence y_m of a point gives $((F-1)y_m)_m$. Changing the divisions changes this by $(A_{\mathfrak{q}}-1)V$. Conversely $F-1$ is surjective on the geometric semiabelian group, so the construction gives every such class. It is therefore a finite-field division computation of each required class, and does not require any unknown class-two action.

Choose lifts of f_1, f_2 together with the eight peripheral generators; they are a free pro-three basis by the Frattini criterion. Since two is a unit in $\mathbb{Z}_3$, the full class-two group in this basis has the Baker–Campbell–Hausdorff law

$$(v,m)(v',m') = (v+v', m+m'+\tfrac{1}{2}v\wedge v'). \tag{E37.16}$$

Write its arithmetic action as $g(v,m) = (A_g v, B_g m + \eta_g(v))$. Let δ_d be the local positive inertia loop. Since the based peripheral loop is $p_d^{-1}\delta_d p_d$, its Galois transform is

$$k_d(g)^{-1}(p_d^{-1}\delta_d p_d)^{\chi(g)}k_d(g).$$

Thus, for $g \in G_K$, the exact term on the toric submodule is

$$\eta_g(e_d) = -\chi(g)k_d(g)\wedge e_d. \tag{E37.17}$$

This retains all eight cusp-path classes, with their elliptic as well as toric components.

Local obstruction and the inverse limit. At $\mathfrak{q}\nmid 3$ the genuine based action on the geometric pro-three group is unramified, as established in E.34. A cocycle restricted to tame inertia is therefore a homomorphism. Its value v satisfies $A_{\mathfrak{q}}v = qv$. Projecting to T_3E gives zero, because the elliptic Frobenius has no eigenvalue q. The toric sublattice is primitive, so v belongs to it integrally. Wild inertia has no nonzero homomorphism to this pro-three group. This proves (E37.9).

Lift the inertia generator in the pulled-back extension by (v,m) and Frobenius by (a,n). The relation $F\tau F^{-1} = \tau^q$, computed with (E37.16), gives the exact central equation

$$(B_{\mathfrak{q}}-q)m = -\eta_F(v) - q\,a\wedge v. \tag{E37.18}$$

The central Frobenius coordinate n cancels. Dividing by q and substituting (E37.17) gives (E37.10). Inertia itself has the canonical zero-central splitting $\tau^u \mapsto (uv,0)$, because the wedge of two multiples of v is zero. Thus (E37.18) is the actual localization representative, including its choice of inertia splitting, of the global central extension obstruction.

Here is the continuous localization argument, with the required support condition. Put $S = \operatorname{Spec}\mathcal{O}_K[1/3]$ and $M_n = M/3^nM$. The proof of E.34 gives $H^2(G_{K,S},\mathbb{F}_3)=0$. The mod-three action on M has three-group image, so each finite M_n has a filtration with trivial $\mathbb{F}_3$ factors. It follows that $H^2(S,M_n)=0$; global three-cohomological dimension gives $H^3(S,M_n)=0$. The finite-coefficient $H^1(S,M_n)$ are finite. Localization and tame purity therefore give, exactly as in E.32,

$$H^2_{\mathrm{cts}}(K,M) \simeq \operatorname{coker}\left(\mathcal{R}_{\mathfrak{q}\nmid 3}(M(-1)) \xrightarrow{F_{\mathfrak{q}}-1} \mathcal{R}_{\mathfrak{q}\nmid 3}(M(-1))\right). \tag{E37.19}$$

Here $\mathcal{R}$ means sequences tending to zero modulo every 3^n. To justify this formula, first take the direct limit over the finite sets of removed primes in the finite-coefficient localization triangles. Their local two-term complexes have terms $\bigoplus_{\mathfrak{q}} M_n(-1)$ and transition maps which are surjective. Their derived inverse limit is consequently the literal two-term complex of restricted sequences. The two unramified groups in degrees two and three vanish also after

derived inverse limit, by the preceding vanishing and finiteness. This proves (E37.19) without discarding a derived-limit class. The comparison and localization results used here are those already opened in E.32, [Mil06, Jan88].

Continuity of α implies $v_{\mathfrak{q}} \to 0$: at every finite level it ramifies at only finitely many primes. All $a_{\mathfrak{q}}$ and $k_{\mathfrak{q},d}$ are integral, so (E37.10) is a restricted sequence. Its values lie in M_0 because every summand has a toric wedge factor. The kernel of each local differential is exactly L, and the fixed integral section $M/L \to M$ obtained by deleting the specified pivot wedge lifts restricted sequences to restricted sequences. Therefore the support representative is in the image in (E37.19) exactly when its unique quotient corrections satisfy (E37.11).

Changing $k_{\mathfrak{q},d}$ by $(A_{\mathfrak{q}} - 1)t_{\mathfrak{q},d}$ changes $D_{\mathfrak{q}}$ by the differential of the integral vector $-\sum_d b_{\mathfrak{q},d} t_{\mathfrak{q},d} \wedge e_d$. These vectors form a restricted sequence since $b_{\mathfrak{q},d} \to 0$. Changing the cocycle representative of α similarly changes D by the differential of an integral restricted sequence. Frobenius and tame-generator choices are the changes of the displayed actual local extension, so give the same localization class. This proves the asserted independence of all representatives. Finally restriction from $\mathbb{Q}$ to K is injective in the obstruction group: corestriction composed with restriction is multiplication by two, a unit on these three-primary modules. Thus the criterion over K is necessary and sufficient for a section over $\mathbb{Q}$ itself.

All the fibers in explicit coordinates. E.36 proved $H^0(\mathbb{Q}, V) = 0$. Hence the fiber over a fixed abelian section class, once nonempty, is exactly a torsor under $H^1(\mathbb{Q}, M)$, without a further stabilizer quotient. We now compute this group. At a rational prime $p \neq 3$, the only Frobenius eigenline of M with eigenvalue p is L. The toric permutation eigenvalues do not alter this conclusion: all the other complex absolute values are $p^{3/2}$ or p^2. Its integral fixed lattice after Tate twist is exactly $\mathbb{Z}_3 w$ because w is primitive. Localization, with $H^2(G_{\mathbb{Q},\{3,\infty\}}, M_n) = 0$ proved by the same filtration and quadratic descent, gives

$$0 \to H^1(G_{\mathbb{Q},\{3,\infty\}}, M) \to H^1(\mathbb{Q}, M) \to \mathcal{R}_{\{p\neq 3\}}(\mathbb{Z}_3) \to 0. \tag{E37.20}$$

The map $(a_p) \mapsto \sum_{p\neq 3} a_p \kappa_3(p) w$ is an actual splitting: the sum converges uniformly as a continuous cocycle and has precisely those inertia residues.

It remains to evaluate the unramified term, not just to name it. Let $P_S = G^{(3)}_{K,S}$, free pro-three of rank two by E.34. The actual CM image is all $1 + 3\mathbb{Z}_3[\zeta]$: Frobenius at the prime over two is -2, and Frobenius at a prime over seven is $1+3\zeta$ or its conjugate. Their logarithms divided by three reduce to a basis modulo three, so they topologically generate this group. Choose the prime over seven giving $1 + 3\zeta$, and a Frobenius lift τ in P_S. Then $\tau' = c\tau c$ has scalar $1 + 3\zeta^2$. The logarithms of these latter two elements also give a basis modulo three. Since P_S is free of rank two, the Frattini criterion shows that τ, τ' freely generate it. Complex conjugation interchanges them exactly.

The based arithmetic action and all the cocycles under consideration factor through $P_S \rtimes \langle c \rangle$ as in E.34. Every abelian cocycle may be normalized to vanish on c, since two is invertible. In that normalization it is determined freely by its value $x \in M$ on τ, and its value on τ' must be $B_c x$. The coboundaries preserving the normalization are exactly $(B_\tau - 1)y$ with $y \in M^c$. This proves the actual evaluation isomorphism (E37.13). The map $1 + B_c$ has image M^c over $\mathbb{Z}_3$, again because two is a unit.

The same description makes all abelian inputs of (E37.10) concrete. Every proper elliptic cocycle is unramified away from three: its inertia value would otherwise give a Frobenius eigenvector of eigenvalue p in T_3E, which the good-reduction eigenvalues exclude. Evaluation at τ gives

$$H^1(\mathbb{Q}, T_3E) = T_3E/(A_E(\tau) - 1)(T_3E)^c = \mathbb{Z}_3 t_1 \oplus (\mathbb{Z}/3) t_2, \tag{E37.22}$$

since $(T_3E)^c = \mathbb{Z}_3 t_1$ and $(A_E(\tau) - 1)t_1 = 3t_2$. For $b \in \mathbb{Z}_3$ and $r \in \{0, 1, 2\}$, the unramified V-cocycle with value $bf_1 + rf_2$ at τ, its conjugate value at τ', and zero at c is a specified lift of

this elliptic class. Subtracting that cocycle from an arbitrary α leaves a unique toric Kummer class. Uniqueness follows from $H^0(\mathbb{Q}, T_3E) = 0$, and the torus is $\mathbb{G}_m^2 \times (\mathrm{Res}_{K/\mathbb{Q}}\,\mathbb{G}_m)^3$. Thus all inputs can equivalently be supplied by (b, r) and two completed rational multiplicative classes and three completed K-multiplicative classes. This is a bijection of sets of inputs; it does not assert that this choice of the representatives r is a group splitting. The Kummer classes have the prime and unit coordinates computed in E.24, and the unramified cocycle is determined on the two actual free generators just specified. This provides evaluations of (E37.9) without assigning a toric approximation to the elliptic or peripheral-path coordinates.

The matrices (E37.2)–(E37.4) now compute the cohomology by ordinary linear algebra over $\mathbb{Z}_3$. The program performs exact rational elementary operations whose denominators are prime to three and saves both transformation matrices. Their product with $(B_\tau - 1)(1 + B_c)$ is exactly

$$\mathrm{diag}(1^{\times 7}, 3^{\times 9}, 9^{\times 5}, 0^{\times 24}). \tag{E37.21}$$

This is an exact certificate for (E37.12), with the coordinate map given by the saved left matrix. For an independent check, the simultaneous invariants of $M \otimes \mathbb{Q}_3/M$ under the matrices for -2, $1 + 3\zeta$ and c have elementary divisors $3^{\times 9}, 9^{\times 5}$, exactly the torsion just computed. Complex conjugation has minus-space dimension $4 \cdot 6 = 24$, agreeing with the free rank.

The same program verifies all cusp frames, the Weil sign, the primitive determinant line, and several rank-44 determinant identities (E37.15). Formula (E37.17), applied to the morphism of E.31 at the prime above seven, gives the central residue two computed there; this checks the peripheral minus sign against an independent nonzero intrinsic obstruction. The next maps for further lifting are the actual surjections $N/N'' \to N/\gamma_3 N$ and $N/\gamma_3 I_{\mathrm{marked}} \to N/\gamma_3 N$, with respective kernels $\gamma_3 N/N''$ and $\gamma_3 N/\gamma_3 I_{\mathrm{marked}}$. Those further section-lifting problems are not asserted to vanish by the class-two calculation. □

An explicit cubic class that stops at intrinsic class two

E.38. A finite class-three obstruction after a full class-two lift [Proved]

Let $N = \pi_1(U_{3,\overline{\mathbb{Q}}})^{(3)}$ and $V = N^{\mathrm{ab}}$, with the cubic, cusp labels and arithmetic origin of E.37. Let $\alpha_P \in H^1(\mathbb{Q}, V)$ be the abelian class of the actual rational tangential section at $P = A_0 = [0 : -1 : 1]$, with tangent coefficient two for the parameter $s = -x/z$. Write $e_Q \in V$ for positive inertia at $Q = B_0 = [-1 : 0 : 1]$. This is a $G_{\mathbb{Q}}$-equivariant copy of $\mathbb{Z}_3(1)$ in V. The class

$$\alpha = \alpha_P + 27\kappa_3(19)e_Q \quad \text{has toric coordinates} \quad (2, 19^{27}). \tag{E38.1}$$

Its fiber in the full intrinsic class-two section map (E37.1) is nonempty, and is the torsor (E37.12). There is no section above α in any of the following three actual arithmetic quotients:

$$N/\gamma_4 N, \qquad N/N'', \qquad N/\gamma_3 I_{\mathrm{marked}}. \tag{E38.2}$$

Here the marked cusp subgroup is the closed normal subgroup generated by Q alone, or by any Galois-stable subset of the cusps above the base puncture 1 that contains Q. The obstruction is detected at the rational prime 19 by a finite $G_{\mathbb{Q}}$-equivariant quotient of N of order 3^{30} and nilpotency class three. Thus every class-two section above (E38.1) fails to lift through each of (E38.2).

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379); Part E(c),(d) and the fourth exactness follow-up. The quotient maps below specify the precise transfer from the full intrinsic cubic group.

Proof. The positive full class-two calculation. The tangential section defining α_P is a section of the full arithmetic fundamental group, so it has a lift at every quotient in question. Relative to the origin at O its path class is the framed Kummer class of E.37, with the tangent coefficient two added at P. Its coordinates are unramified away from 2 and 3: all the unscaled framed values (E37.7) are units away from 3.

We apply the full 44-dimensional criterion (E37.11), over $K = \mathbb{Q}(\zeta)$. At the prime over 2 the Kummer class of 19 is zero. Indeed $19 \in 1 + 2\mathcal{O}_{K,2}$, and taking the third power is an automorphism of this pro-two unit group. The coherent third-power roots give the zero local pro-three Kummer class. Consequently α is locally cohomologous to α_P there and has zero local obstruction.

At either prime over 19 the class α_P is unramified. Choose tame inertia so that $\kappa_3(19)(\tau) = 1$. The inertia value in (E37.9) is then $27e_Q$. Writing $a_P = \alpha_P(F)$ and $u = \kappa_3(19)(F)$ gives the exact local numerator

$$D = (a_P + 27ue_Q) \wedge 27e_Q - 27k_Q \wedge e_Q = 27(a_P - k_Q) \wedge e_Q \in 27M_0. \tag{E38.3}$$

For these two primes the CM scalars are $-2 + 3\zeta$ and $-5 - 3\zeta$. The exact integral matrices (E37.6), evaluated by `e_intrinsic_19.py`, have the same Smith valuations

$$0^{\times 8}, \quad 1^{\times 6}, \quad 2^{\times 18}, \quad 3^{\times 12}. \tag{E38.4}$$

The sum is 78, also obtained from the determinant formula (E37.15), and the largest is three. Thus $27M_0 \subseteq d_{\mathfrak{q}}(M/L)$ at each prime over 19. Equation (E38.3) has an integral correction. At every other prime away from 3 there is no inertia in (E38.1), so its support representative is zero. The corrections therefore have finite support and satisfy the convergence condition of E.37. This proves the full intrinsic class-two lift and its stated fiber.

A continuous arithmetic quotient with its exact map. Use the rational unit tangent $s = -x/z$ at P for this calculation. The covering coordinate is $t = s^3$. The other covering root is $y/z = -(1 - s^3)^{1/3}$, with rational constant term -1. Taking coefficientwise roots of these power series identifies this tangential fiber with the fiber over the unit tangent at $t = 0$. It gives a continuous $G_{\mathbb{Q}}$-equivariant inclusion

$$N \longrightarrow F = \langle x, y \rangle^{\text{pro-3}}, \qquad N = \ker(F \longrightarrow (\mathbb{Z}/3)^2). \tag{E38.5}$$

Here x, y are the positive base inertia loops at $0, 1$. A geometric path from this tangent to the tangent at O identifies the two geometric groups. Changing that path conjugates the identification. On V this conjugation is trivial, and the change of arithmetic origin is the actual tangential cocycle already used to define α_P. Thus (E38.5) acts on precisely the class (E38.1), with its origin accounted for.

Put $I_y = \ker(F \to \mathbb{Z}_3)$, where $x \mapsto 1$ and $y \mapsto 0$, and let $T = 1 + U$. The continuous quotient by the closed commutator subgroup of I_y is

$$R = F/[I_y, I_y] = \mathbb{Z}_3[[U]] \rtimes \mathbb{Z}_3, \quad (a, v)(b, w) = (a + b, v + T^a w), \quad x = (1, 0),\ y = (0, 1). \tag{E38.6}$$

For an explicit inverse construction, I_y is the closed normal subgroup generated by y. Its abelianization is therefore generated, as a continuous $\mathbb{Z}_3[[\mathbb{Z}_3]]$-module, by the image of y, with T acting by conjugation by x. This gives a continuous surjection from the completed group ring onto I_y^{ab}. In the other direction the free pro-three property sends x, y to $(1, 0), (0, 1)$ in the displayed semidirect product. Its restriction to I_y is abelian and sends the first module generator back to one. The composite on the completed group ring is the identity, so the first surjection is an isomorphism. The section generated by x now gives exactly (E38.6), including its quotient map and topology. Its restriction to N is onto

$$H = \{(a, v) \in R : a \in 3\mathbb{Z}_3,\ v(1) \in 3\mathbb{Z}_3\} = 3\mathbb{Z}_3 \ltimes J, \qquad J = (3, U)\mathbb{Z}_3[[U]]. \tag{E38.7}$$

Indeed $F \to R$ is onto and induces the identity on $F^{\mathrm{ab}} = \mathbb{Z}_3^2$, with second coordinate the augmentation $v(1)$, so the inverse image of H is exactly N.

The normalized genuine Galois action used in E.20 is $x \mapsto x^{\chi(g)}$, $y \mapsto f_g^{-1} y^{\chi(g)} f_g$, with $f_g \in F'$. Its image in R has $f_g \in I_y/[I_y, I_y]$, which commutes with y. It follows on the actual generators and then by continuity that

$$g(a, v(T)) = (\chi(g)a,\ \chi(g)v(T^{\chi(g)})). \tag{E38.8}$$

This proves the arithmetic action and preservation of H without discarding an unknown conjugating term.

The group R is metabelian, so N'' maps to one. Every inertia loop above 1 maps to a conjugate of a power of y, hence into the abelian normal module in (E38.6). The subgroup I_{marked} maps into that same module, and $\gamma_3 I_{\mathrm{marked}}$ maps to one. Thus there are the actual continuous arithmetic surjections

$$N/N'' \longrightarrow H, \qquad N/\gamma_3 I_{\mathrm{marked}} \longrightarrow H. \tag{E38.9}$$

More precisely, put $J_y = [I_y, I_y]$, taking the closed commutator subgroup. Since $J_y \subset F' \subset N$, the kernel of $N \to H$ is exactly J_y. The kernels of the two maps in (E38.9) are therefore J_y/N'' and $J_y/\gamma_3 I_{\mathrm{marked}}$, respectively. Indeed the whole subgroup $[I_{\mathrm{marked}}, I_{\mathrm{marked}}]$ already maps to one.

The local equation, including normalization of the section. At 19 fix a compatible root-of-unity identification with ζ_9 reducing to $4 \in \mathbb{F}_{19}$. Since 4 has order nine and $2^{(19-1)/9} = 4$, the unramified Kummer value is $\kappa_3(2)(F) \equiv 1 \pmod 9$. In (E38.5) the section at P with tangent coefficient two has first coordinate $3\kappa_3(2)$; changing the tangent sends $t = s^3$ to a leading coefficient 2^3. The added cusp class has first coordinate zero and second augmentation $3 \cdot 27\kappa_3(19)$, because inertia at Q maps to a conjugate of y^3.

For every possible section above α, its Frobenius first coordinate can consequently be set equal to $a = 3$ by geometric conjugation by x^{3c}: the change is $3(1-19)c$, and $(\kappa_3(2)(F)-1)/18 \in \mathbb{Z}_3$. This conjugation preserves the inertia augmentation. The tame inertia value has first coordinate zero and is $(0, v(T))$ with

$$v(1) = 81. \tag{E38.10}$$

The cyclotomic action is 19 at Frobenius and one on tame inertia. In (E38.6)–(E38.8), the relation $F\tau F^{-1} = \tau^{19}$ now becomes

$$T^3 v(T^{19}) = v(T). \tag{E38.11}$$

The module coordinate of Frobenius cancels because the normal module is abelian. This equation therefore tests every possible Frobenius lift.

For a quick exact obstruction over $\mathbb{Q}_3$, set $z = \log T$. Equation (E38.11) becomes $e^{3z} h(19z) = h(z)$, $h(0) = 81$. Multiplying by $e^{z/6}$ shows that every positive coefficient of the resulting series vanishes, since $19^n - 1 \neq 0$. Thus its unique formal solution is

$$v(T) = 81 T^{-1/6}. \tag{E38.12}$$

For $r = -1/6$, every $r - j$, $j \in \mathbb{Z}$, has three-adic valuation -1. Hence

$$v_3\left(81 \binom{-1/6}{n}\right) = 4 - n - v_3(n!).$$

At $n = 4$ this equals -1, excluding an element of $\mathbb{Z}_3[[U]]$. The finite certificate below also detects the failure after an intrinsic class-three quotient, and makes this nonexistence independently checkable by integer matrix multiplication.

A finite class-three certificate. Reduce the module to $A = (\mathbb{Z}/729)[U]/(U^5)$ and the first coordinate modulo 2187. The binomial coefficients of $(1+U)^{2187} - 1$ through degree four are divisible by 729, so these choices define a finite quotient R_6 of (E38.6). Formula (E38.8) descends to it: substitution and its inverse are defined over A, and changing χ by 2187 changes neither the substitution nor the scalar action. Its subgroup H_6 is the image of (E38.7), of order 3^{35}.

Write $v(U) = 81 + \sum_{i=1}^{4} v_i U^i$. The four positive-degree coefficients of (E38.11) are $Kv = b \pmod{729}$, with

$$K = \begin{pmatrix} 18 & 0 & 0 & 0 \\ 228 & 360 & 0 & 0 \\ 1539 & 7581 & 6858 & 0 \\ 7315 & 86640 & 205770 & 130320 \end{pmatrix}, \qquad b = \begin{pmatrix} -243 \\ -243 \\ -81 \\ 0 \end{pmatrix}. \tag{E38.13}$$

The single integral row

$$\ell = (1, 660, 558, 162) \quad \text{satisfies} \quad \ell K = 0 \pmod{729}, \qquad \ell b = 486 \pmod{729}. \tag{E38.14}$$

It excludes all four-tuples v_i.

To obtain the stronger class-three assertion, put $\delta = T^3 - 1$. Direct commutators in (E38.7) give $\gamma_{c+1} H_6 = \delta^c J$ for $c \geq 1$: the module is abelian, and conjugation by the generator x^3 acts by T^3. In the present ring the subgroup $\gamma_4 H_6$ is generated by

$$3\delta^3 = 81U^3 + 243U^4, \qquad U\delta^3 = 27U^4. \tag{E38.15}$$

These elements generate a subgroup of order 3^5, so $H_6/\gamma_4 H_6$ has order 3^{30}. Its class is exactly three: $\delta^2 U = 9U^3 + 18U^4$ is in $\gamma_3 H_6$ and has U^3 coefficient nine, whereas every element of (E38.15) has that coefficient divisible by 81. The row ℓ also annihilates their coefficient columns $(0, 0, 81, 243)^{\mathrm{t}}$ and $(0, 0, 0, 27)^{\mathrm{t}}$ modulo 729. Consequently (E38.14) still excludes the local tame equation after quotienting by $\gamma_4 H_6$. The composite

$$N \longrightarrow H \longrightarrow H_6 \longrightarrow H_6/\gamma_4 H_6 \tag{E38.16}$$

is continuous, onto and $G_{\mathbb{Q}}$-equivariant. Functoriality of the lower central series makes it factor through $N/\gamma_4 N$. Its factorizations through the other two groups in (E38.2) follow from (E38.9). Any section of any one of those arithmetic quotients above α would therefore give the excluded local section. This proves all the claimed nonlifting assertions.

The program `e_next_depth.py` constructs the matrix from the power series, checks (E38.14) and (E38.15), and tests membership in the full integer column lattice $[K \mid 3^m I]$ by Hermite normal form. The first failed modulus is 3^6. Adding the columns of $\delta^c J$ tests the same finite quotient at each lower central stage: the class-one and class-two systems have solutions, whereas the class-three system has the explicit annihilating row (E38.14). Its saved output and JSON certificate include every matrix and residue used here. □

The arithmetic free-inertia comparison and its correction words

E.39. Riemann existence gives the exact kernel, with unavoidable Frobenius corrections [Proved]

Put $K = \mathbb{Q}(\zeta)$, $S = \{(1-\zeta)\}$, $\mathcal{G} = G_K^{(3)}$, and $P = G_{K,S}^{(3)} = \langle \tau, \tau' \rangle$ as in E.37. Let $L = \ker(\mathcal{G} \to P)$. There is an actual continuous free-product isomorphism

$$\underset{w \nmid 3 \text{ in } K_S(3)}{*} I_w \xrightarrow{\sim} L, \qquad I_w \simeq \mathbb{Z}_3(1), \tag{E39.1}$$

induced by the inclusions of the chosen inertia groups. The indexing and free product have the continuous-bundle topology; finite quotients retain only finitely many base primes and locally constant data on their prolongations. Choose lifts of the two Frobenii over seven, related by complex conjugation. They define a conjugation-equivariant splitting $s : P \to \mathcal{G}$, and hence $\mathcal{G} = L \rtimes s(P)$.

For an actual local Frobenius $F_{\mathfrak{q}}$ put

$$\sigma_{\mathfrak{q}} = \mathrm{pr}(F_{\mathfrak{q}}), \qquad h_{\mathfrak{q}} = F_{\mathfrak{q}} s(\sigma_{\mathfrak{q}})^{-1} \in L. \tag{E39.2}$$

No splitting makes every $h_{\mathfrak{q}}$ belong to the corresponding local inertia group. For the specified splitting, at $\mathfrak{q} = (-2 + 3\zeta)$ above nineteen, the actual cyclic cubic quotient already detects

$$\kappa_{2,3}(h_{\mathfrak{q}}) = 2 \in \mathbb{F}_3, \tag{E39.3}$$

where $\kappa_{2,3}$ is the Kummer character of 2. This character is zero on $I_{\mathfrak{q}}$.

The precise section equations supplied by (E39.1) are as follows. Let W be N, N/N'', or one of the Galois-stable marked quotients in E.25, with its actual based action. A section over K is specified by a continuous homomorphism $\lambda : L \to W$ and a cocycle $u : P \to W$ satisfying

$$\lambda(s(g)\ell s(g)^{-1}) = u(g)\, g(\lambda(\ell))\, u(g)^{-1}. \tag{E39.4}$$

Here u is freely specified by its two values on τ, τ', and λ is specified by its continuous family of values on the factors in (E39.1). The section cocycle is reconstructed by the exact formula

$$a(\ell s(g)) = \lambda(\ell) u(g). \tag{E39.5}$$

For tame inertia $t_{\mathfrak{q}}$, writing $n_{\mathfrak{q}} = \lambda(t_{\mathfrak{q}})$ and $q = N\mathfrak{q}$, (E39.4) gives the necessary local equation

$$[\lambda(h_{\mathfrak{q}})u(\sigma_{\mathfrak{q}})]\sigma_{\mathfrak{q}}(n_{\mathfrak{q}})[\lambda(h_{\mathfrak{q}})u(\sigma_{\mathfrak{q}})]^{-1} = n_{\mathfrak{q}}^{q}. \tag{E39.6}$$

The correction word is part of this formula, including for metabelian W.

Manuscript: §51.16, folios 733v–736v, scans 172–178; Part E(c),(d). These are comparisons with the actual arithmetic Galois group and intrinsic geometric quotients.

Proof. The precise primary result used here is Wingberg, [Win07, Theorem 2.4, pp. 13–14]; its forward implication explicitly includes infinite sets of primes. The $K(\pi, 1)$ hypothesis holds for $\operatorname{Spec} \mathcal{O}_K[1/3]$, as stated in the remark after Proposition 2.1 of that paper. It also follows from the finite-coefficient calculation in E.34: its degree-two cohomology vanishes, and induction along finite three-covers preserves that vanishing. The universal three-cover kills degree-one cohomology. The local nontriviality hypothesis holds at every $\mathfrak{q} \nmid 3$ because the cyclotomic $\mathbb{Z}_3$-extension is inside $K_S(3)$ and its arithmetic Frobenius is $q \neq 1$ in $1 + 3\mathbb{Z}_3$. The theorem therefore gives (E39.1) for the set of all finite primes. At these residue characteristics, pro-three inertia is $\mathbb{Z}_3(1)$.

The quotient P is the free rank-two group already proved in E.34 and E.37. Select an actual Frobenius lift above $(1 + 3\zeta)$ in $\mathcal{G}$, and use its complex conjugate above the other prime over seven. The free property produces s with these values. Its composite with $\mathcal{G} \to P$ fixes both free generators and is the identity. Conjugation interchanges the two chosen lifts, so s also commutes with conjugation.

For completeness, the failure of simultaneous local normalization holds for every splitting, not only this one. Let $\chi = \kappa_{2,3} : \mathcal{G} \to \mathbb{F}_3$, and set $\beta = \chi \circ s : P \to \mathbb{F}_3$. The character χ ramifies at the prime (2): the local polynomial $X^3 - 2$ is Eisenstein, and K contains μ_3. The character β is unramified away from three. Were all $h_{\mathfrak{q}}$ local-inertia elements, for every prime away

from six we would have $\chi(F_{\mathfrak{q}}) = \beta(\sigma_{\mathfrak{q}})$. Apply the abelian Chebotarev theorem to the finite extension cut out by these two characters [Mil20b, V, Theorem 3.23]. It makes the characters equal on all of $\mathcal{G}$, contradicting their different inertia values at two. This proves the asserted failure by actual character maps.

There is also the finite witness (E39.3), requiring no unspecified prime. At the primes over seven, the images of ζ in the residue fields are two and four. For a unit b, the exponent of ζ in $b^{(q-1)/3}$ is its cubic Kummer Frobenius value. These exponents give

	$\kappa_{2,3}$	$\kappa_{\zeta,3}$	$\kappa_{\lambda,3}$
$\zeta \equiv 2 \pmod 7$	2	2	0
$\zeta \equiv 4 \pmod 7$	1	2	2.

(E39.7)

The last two characters are the two unramified Kummer generators, and evaluation on the free generators shows

$$\kappa_{2,3} \circ s = \kappa_{\zeta\lambda,3} \quad \text{as characters of } P. \tag{E39.8}$$

At $\mathfrak{q} = (-2 + 3\zeta)$ the residue of ζ is seven, that of λ is thirteen, and $2^6 = 7$, $(\zeta\lambda)^6 = 11 = 7^2$ modulo nineteen. Thus (E39.2) has character value $1 - 2 = 2$ modulo three. The program `e_riemann_correction.py` verifies all residues and the two-character calculation.

The based action on W factors through P over K, by E.34. A section cocycle factors through $\mathcal{G}$: the homomorphism into W semidirect the arithmetic image has pro-three target. Its restriction to L is a homomorphism because L acts trivially on W. Restriction to $s(P)$ is the cocycle u. Applying the cocycle identity to $s(g)\ell s(g)^{-1}$ gives (E39.4). Conversely this identity verifies the cocycle law for (E39.5), using $\mathcal{G} = L \rtimes s(P)$, so the displayed description is a bijection of actual cocycles. The free-product universal property gives exactly the continuity condition stated for λ; arbitrary discontinuous families are excluded.

For sections over $\mathbb{Q}$, first conjugate so that the value at complex conjugation c is one. This is possible because the complement of the pro-three kernel in its extension by $\langle c \rangle$ is conjugate to the standard complement, by the finite Schur–Zassenhaus theorem and inverse limits. Then add the two exact equations

$$u(\tau') = c(u(\tau)), \qquad \lambda(c\ell c) = c(\lambda(\ell)). \tag{E39.9}$$

The remaining geometric conjugations are $w \in W^c$; they send (λ, u) to $(w\lambda w^{-1}, g \mapsto wu(g)g(w)^{-1})$. This specifies the rational descent and its conjugacy quotient rather than changing the arithmetic base field.

Finally $F_{\mathfrak{q}} = h_{\mathfrak{q}} s(\sigma_{\mathfrak{q}})$ and the actual local relation is $F_{\mathfrak{q}} t_{\mathfrak{q}} F_{\mathfrak{q}}^{-1} = t_{\mathfrak{q}}^{q}$. Apply (E39.5) to obtain (E39.6). In a metabelian target the value $\lambda(h_{\mathfrak{q}})$ still uses the image of that word in L/L''; (E39.3) shows that even its first abelian coordinate cannot be set to zero in general. Equations (E39.4)–(E39.9) therefore record the precise further arithmetic input needed for this approach. They do not replace the evaluation of those words, or of their conjugation action, by the two unramified Frobenius matrices. □

Every higher central lifting fiber and the complete class-three fiber group

E.40. Higher intrinsic fibers have finite matrix presentations; the next one is evaluated [Proved]

For the full cubic pro-three group N put

$$Q_r = N/\gamma_{r+1}N, \qquad L_r = \gamma_r N/\gamma_{r+1}N = \mathrm{Lie}_r(V), \qquad r \geq 3. \tag{E40.1}$$

The equality is the actual graded commutator isomorphism over $\mathbb{Z}_3$. Set $A_r = \mathrm{Lie}_r(A_\tau)$ and $C_r = \mathrm{Lie}_r(C_V)$, using the integral matrices of E.37. Here c denotes complex conjugation; $L_r^{\langle c\rangle}$ means its fixed sublattice. Every nonempty geometric-conjugacy fiber of

$$\mathrm{Sect}(Q_r \rtimes G_{\mathbb{Q}}) \longrightarrow \mathrm{Sect}(Q_{r-1} \rtimes G_{\mathbb{Q}}) \tag{E40.2}$$

is a torsor under the explicitly presented compact group

$$L_r/(A_r - 1)L_r^{\langle c\rangle} = \mathrm{coker}\big((A_r - 1)(1 + C_r) : L_r \to L_r\big). \tag{E40.3}$$

There is no further stabilizer quotient. The central cocycles acting in these fibers are all unramified away from three, including when the lower section is ramified.

For the first subsequent step, namely $Q_3 \to Q_2$, every nonempty fiber is exactly a torsor under

$$\mathbb{Z}_3^{166} \ \oplus\ (\mathbb{Z}/3)^7 \ \oplus\ (\mathbb{Z}/9)^{56} \ \oplus\ (\mathbb{Z}/27)^{25} \ \oplus\ (\mathbb{Z}/81)^3. \tag{E40.4}$$

Thus the nonempty fibers at this stage have a fully computed integral parameter group, while E.38 supplies an entire abelian class whose class-two fibers all have empty preimage at this stage.

The presentation (E40.3) is effectively finite in every degree. Its free rank is

$$\frac{n_r - t_r}{2}, \qquad n_r = \frac{1}{r}\sum_{d\mid r} \mu(d) 10^{r/d}, \qquad t_r = \frac{1}{r}\sum_{d\mid r} \mu(d)\big(\mathrm{tr}(c^d \mid V)\big)^{r/d}, \tag{E40.5}$$

where the last trace is 10 for even d and -2 for odd d. Its torsion is killed by

$$3^{r+b_r}, \qquad b_r = \max_{r\le j\le 2r}(1 + v_3(j)). \tag{E40.6}$$

Consequently a Hall-basis matrix calculation modulo 3^{r+b_r+1} determines all its elementary divisors.

Manuscript: §51.16, folios 733v–736v, scans 172–178, Part E(c),(d). The comparison is with the full intrinsic lower-central quotients; the precise further maps from the metabelian and single-cusp marked quotients are given below.

Proof. The free pro-three group N has rank ten. The graded commutator map identifies its lower-central graded algebra with the free $\mathbb{Z}_3$-Lie algebra on V; the exact pro-p statement is [KKL98, Theorem 2.1, pp. 4–5]. In Magnus coordinates this sends the degree-r leading term of a group commutator to the same Lie bracket in the tensor algebra. It is functorial for automorphisms, so the arithmetic action on L_r is precisely Lie_r of the action on V. This uses the integral graded quotient, including at $r = 3$; it does not use a class-three exponential with denominator three.

Conjugation by an element of N acts trivially on L_r, since $[N, \gamma_r N] \subset \gamma_{r+1}N$. Therefore the action on this central kernel after pulling back along any lower section is the same displayed arithmetic action. Twisting the lower section cannot change that module.

At a rational prime $p \neq 3$ the original action is unramified. The eigenvalues on $V \otimes \mathbb{Q}_3$ have complex absolute values $p^{1/2}$ or p: the actual isogeny $T_D \times E \to J_{E,D}$ in E.24 gives these toric and elliptic factors. The eigenvalues on $L_r \otimes \mathbb{Q}_3$, as a subspace of $V^{\otimes r}$, therefore have absolute values $p^{j/2}$ with $r \le j \le 2r$. For $r \ge 3$ none is p and none is one.

Let a be a continuous L_r-valued cocycle. Wild inertia at p has zero image because it is pro-p. On tame pro-three inertia, the cocycle is a homomorphism; its value v on a generator satisfies $Fv = pv$ by the tame relation. The absence of the eigenvalue p makes $v = 0$ in the torsion-free

lattice. Thus a is zero on every inertia group outside three, and factors through $G_{\mathbb{Q},\{3,\infty\}}$. This is an equality of cocycle groups, so there is no inverse-limit ramification assumption in this step.

The same weight calculation excludes a geometric stabilizer of a lower section. If a nonidentity element of Q_{r-1} were fixed by its section-twisted Galois action, take its first nonzero lower-central coordinate. This coordinate lies in some L_i, $1 \le i < r$, and is fixed by the original action: inner twists act trivially on the graded coordinate. Every Frobenius eigenvalue there has absolute value strictly greater than one, contradicting the nonzero fixed coordinate. Hence

$$H^0(G_{\mathbb{Q}}, Q_{r-1}^{\mathrm{section}}) = 1. \tag{E40.7}$$

For two lifts of a fixed lower section, their pointwise ratio is an L_r-valued cocycle. Central conjugation changes it by a coboundary. Equation (E40.7) means that no additional conjugation in the lower group identifies two such cocycle classes. This proves that the full geometric-conjugacy fiber is a torsor under $H^1(\mathbb{Q}, L_r)$.

This cohomology group is computed directly, rather than left as its name. Over K, the unramified group $P = \langle \tau, \tau' \rangle$ is free and $c\tau c = \tau'$. As in E.34, its action and every cocycle factor through $P \rtimes \langle c \rangle$. Normalize an additive cocycle to be zero at c by division by two. It is then freely determined by $x = a(\tau) \in L_r$; its value at τ' is $C_r x$. The coboundaries preserving this normalization are exactly $(A_r - 1)y$ with $y \in L_r^{\langle c \rangle}$. Since two is a unit, the image of $1 + C_r$ is exactly that fixed sublattice. This gives (E40.3) by an actual evaluation map. Frobenius at seven has no eigenvalue one, so $A_r - 1$ is rationally injective and the free rank of its cokernel on the fixed sublattice is the minus-space dimension. Witt's character formula [ABRdS15, Theorem A.1], evaluated at the ten eigenvalues of c, gives (E40.5).

We give a uniform integral bound that makes the finite matrix calculation conclusive. In V consider the stable sublattice

$$V_0 = T_3 T_D + j(T_3 E). \tag{E40.8}$$

Its matrix in the basis of E.37 is $B = \left(\begin{smallmatrix} 3I_2 & 0 \\ U & I_8 \end{smallmatrix}\right)$, and $B \left(\begin{smallmatrix} I_2 & 0 \\ -U & 3I_8 \end{smallmatrix}\right) = 3I_{10}$. Thus $3V \subset V_0 \subset V$. By multilinearity $3^r L_r \subset \mathrm{Lie}_r(V_0) \subset L_r$. At the unique K-prime above two, the elliptic scalar is -2 and the toric scalar is four. A word with a elliptic letters has scalar $(-2)^a 4^{r-a} = (-2)^{2r-a}$ on the split lattice. For $j \ge 1$, elementary binomial induction gives $v_3((-2)^j - 1) = 1 + v_3(j)$. Hence $3^{b_r}(F-1)^{-1}$ preserves $\mathrm{Lie}_r(V_0)$, and

$$3^{r+b_r}(F-1)^{-1} L_r \subset L_r. \tag{E40.9}$$

The torsion of $H^1(\mathbb{Q}, L_r)$ is exactly $H^0(\mathbb{Q}, L_r \otimes \mathbb{Q}_3/L_r)$: this follows from the long exact sequence and $H^0(\mathbb{Q}, L_r \otimes \mathbb{Q}_3) = 0$. An invariant class has $(F-1)v \in L_r$, so (E40.9) kills it by 3^{r+b_r}. This proves (E40.6).

Here are full coordinates and the certificate in degree three. Use the 330 brackets

$$[[x_i, x_j], x_k], \qquad 0 \le j < i \le 9, \quad j \le k \le 9, \tag{E40.10}$$

ordered lexicographically in (i, j, k), with the x_i in the order of the V-basis of E.37. Antisymmetry and the identity

$$[[x_i, x_j], x_k] = [[x_i, x_k], x_j] - [[x_j, x_k], x_i] \quad (k < j < i) \tag{E40.11}$$

reduce every bracket to these coordinates. They span; their number is $(10^3 - 10)/3 = 330$, the free rank of L_3, so they are a basis. Applying the two known ten-dimensional matrices to all three letters and then (E40.11) constructs A_3, C_3 by integer operations.

Their cocycle matrix $P_3 = (A_3-1)(1+C_3)$ has rank 164: the trace of C_3 is $((-2)^3-(-2))/3 = -2$, giving plus and minus dimensions 164 and 166. The program `e_lie3_fibers.py` records invertible modular row and column transformations S, T and verifies

$$SP_3T \equiv \operatorname{diag}(1^{\times 73}, 3^{\times 7}, 9^{\times 56}, 27^{\times 25}, 81^{\times 3}, 0^{\times 166}) \pmod{729}. \tag{E40.12}$$

Every scaling operation is a unit modulo three, and the output includes every operation and both complete transformation matrices. Here $b_3 = 2$, so (E40.6) bounds the torsion exponent by 3^5. Therefore the modulo-3^6 certificate determines the integral elementary divisors exactly; none can be hidden above the tested modulus. The independent stacked matrix $\binom{A_3-1}{C_3-1}$ computes the invariants of $L_3 \otimes \mathbb{Q}_3/L_3$ and gives the same torsion factors. The program also verifies the exact split-lattice identities used in (E40.9), with scalar values $64, -32, 16, -8$. This proves (E40.4). The exact quotient presentation (E40.3) supplies full $\mathbb{Z}_3$ coordinates; (E40.12) gives their Smith coordinates modulo 729, and the same elementary operations can be continued at any precision.

We finally specify the scope of this calculation for the requested further quotients. Since $N'' \subset \gamma_4 N$, there is the actual arithmetic surjection

$$N/N'' \longrightarrow Q_3, \qquad \ker = \gamma_4 N/N''. \tag{E40.13}$$

For the single rational cusp marking I_Q, the image of I_Q in V has rank one. Hence its commutator subgroup maps to zero in $\gamma_2 N/\gamma_3 N = \bigwedge^2 V$, and $[I_Q, I_Q] \subset \gamma_3 N$. It follows that $\gamma_3 I_Q \subset \gamma_4 N$, giving

$$N/\gamma_3 I_Q \longrightarrow Q_3, \qquad \ker = \gamma_4 N/\gamma_3 I_Q. \tag{E40.14}$$

The maps (E40.13)–(E40.14), followed by $Q_3 \to Q_2$, identify exactly the common stage whose fibers were computed. They do not identify either full kernel with the degree-three central lattice. □

All unramified sections of the full, metabelian and single-cusp marked groups

E.41. Complete compatible coordinates for all sections unramified outside three [Proved]

Use the actual based cubic action and the meaning of unramified outside three in E.34. Let $Q = B_0$ be the rational cusp over one, let q be its positive peripheral generator, and let I_Q be its closed normal closure in N. All commutator subgroups below are closed. Consider the three actual arithmetic quotients

$$W_{\mathrm{free}} = N, \qquad W_{\mathrm{met}} = N/N'', \qquad W_{\mathrm{mark}} = N/\gamma_3 I_Q. \tag{E41.1}$$

For each of these W, put

$$W_r = W/\gamma_{r+1}W, \qquad \mathcal{A}_r(W) = \gamma_r W/\gamma_{r+1}W. \tag{E41.2}$$

These graded modules are finite free over $\mathbb{Z}_3$. Let $T_r(W)$ and $J_r(W)$ be the matrices induced on them by the two explicit matrices A_τ, C_V of E.37. Define the following explicitly computable compact abelian group:

$$\mathcal{C}_r(W) = \operatorname{coker}\big((T_r(W)-1)(1+J_r(W)) : \mathcal{A}_r(W) \longrightarrow \mathcal{A}_r(W)\big). \tag{E41.3}$$

There are compatible noncanonical homeomorphisms of pointed profinite spaces

$$\begin{aligned} \operatorname{Sect}_{\{3\}}(W_r)/W_r &\simeq \prod_{i=1}^{r} \mathcal{C}_i(W), \\ \operatorname{Sect}_{\{3\}}(W)/W &\simeq \varprojlim_r \operatorname{Sect}_{\{3\}}(W_r)/W_r \simeq \prod_{i\geq 1} \mathcal{C}_i(W). \end{aligned} \tag{E41.4}$$

Here $\mathrm{Sect}_{\{3\}}$ consists of sections whose geometric cocycle, relative to the specified tangent, factors through $G_{\mathbb{Q},\{3,\infty\}}$. The statement computes this entire section family, including all its higher coordinates. The product structures arise from specified Hall lifts and Smith representatives; no group structure on the section set is asserted.

The integral graded modules and their matrix bases are

W	$\mathcal{A}_r(W)$	basis for the finite calculation
W_{free}	$\mathrm{Lie}_r(V)$	Hall commutators
W_{met}	$(\mathrm{Lie}(V)/\mathrm{Lie}(V)'')_r$	$[x_i, x_j, x_{k_3}, \dots, x_{k_r}],\ i > j \le k_3 \le \cdots \le k_r$
W_{mark}	$(\mathrm{Lie}(V)/\gamma_3\mathfrak{i}_Q)_r$	Hall commutators with at most two occurrences of q.

(E41.5)

In the last row $\mathfrak{i}_Q$ is the Lie ideal generated by the actual vector e_Q. Choose the other nine generators to complete q to a free pro-three basis inducing the integral V-basis of E.37. The marking has the single rational cusp Q throughout this statement.

The first three factors in (E41.4) agree for all three quotients and are, respectively,

$$\begin{aligned} \mathcal{C}_1 &= \mathbb{Z}_3^6 \oplus (\mathbb{Z}/3)^3, \\ \mathcal{C}_2 &= \mathbb{Z}_3^{24} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/9)^5, \\ \mathcal{C}_3 &= \mathbb{Z}_3^{166} \oplus (\mathbb{Z}/3)^7 \oplus (\mathbb{Z}/9)^{56} \oplus (\mathbb{Z}/27)^{25} \oplus (\mathbb{Z}/81)^3. \end{aligned} \tag{E41.6}$$

For every degree the ranks and traces below, together with the matrix (E41.3), evaluate all remaining factors. Writing $n_r = \operatorname{rank} \mathcal{A}_r$ and $t_r = \operatorname{tr}(J_r)$, the free rank is $(n_r - t_r)/2$, and the torsion is killed by

$$3^{r+b_r}, \qquad b_r = \max_{r \le j \le 2r} (1 + v_3(j)). \tag{E41.7}$$

Thus a finite Smith calculation modulo 3^{r+b_r+1} determines all elementary divisors; there is a proved bound on the precision required. In degree four the actual matrices have the following sizes and free ranks:

W	n_4	t_4	$\operatorname{rank}_{\mathbb{Z}_3} \mathcal{C}_4$
W_{free}	2475	-21	1248
W_{mark}	2466	-22	1244
W_{met}	1485	-3	744

(E41.8)

The supplied program constructs these integral matrices and verifies the split-lattice certificates behind (E41.7).

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379), Part E(c),(d). These are complete coordinates for the family unramified outside three. The ramified classes excluded in E.31 and E.38 remain outside this family; their exact quotient maps are retained.

Proof. The based arithmetic quotient and an actual section parameter. The action on N factors through $\Gamma = P \rtimes \langle c \rangle$ by E.34, where P is free pro-three on τ, τ' and $c\tau c = \tau'$. E.37 proves that the two actual conjugate Frobenii above seven form this free basis. The same action descends to each group in (E41.1): derived subgroups are characteristic, and the rational cusp makes I_Q stable under the based Galois action.

Every cocycle of $G_{\mathbb{Q},\{3,\infty\}}$ with values in any of these W factors through Γ. Indeed, over K, the map $g \mapsto (a(g), \rho(g))$ has image in the pro-three semidirect product $W \rtimes \rho(G_{K,\{3\}})$. The universal property defining P kills the same kernel as in E.34. This supplies the actual comparison from the arithmetic section problem to the following explicit group problem.

Normalize a cocycle by geometric conjugation so that $a(c) = 1$. Coprime conjugacy supplies this normalization: in each central graded layer the required correction is obtained by division by two. The corrections lift compatibly through the lower-central quotients, giving

the normalization in W itself. The completeness and torsion-freeness used here are proved below. A normalized cocycle is freely specified by a single element

$$x = a(\tau) \in W, \qquad a(\tau') = c(x), \qquad a(c) = 1. \tag{E41.9}$$

To verify existence, map the two free generators of P to (x, τ) and $(c(x), \tau')$ in $W \rtimes P$. Freeness gives a continuous homomorphism whose projection to P is the identity. Conjugation by $(1, c)$ interchanges these generator images, so the homomorphism extends to a section over Γ. Conversely the cocycle identities force these formulas. The remaining geometric conjugations are exactly

$$W^{\langle c \rangle}, \qquad x \longmapsto wx\tau(w)^{-1}. \tag{E41.10}$$

In particular the normalized cocycle space is the compact space W, and its section-class space is the Hausdorff compact orbit space for this displayed continuous compact-group action.

The integral metabelian comparison, including its topology. For the full free group, the integral graded commutator isomorphism is the one in E.40 and [KKL98, Theorem 2.1]. We prove the metabelian passage directly in the pro-three setting. Set

$$R = \mathbb{Z}_3[[N]], \quad \Lambda = \mathbb{Z}_3[[N/N']] = \mathbb{Z}_3[[X_1, \dots, X_{10}]], \quad \mathfrak{m} = (X_1, \dots, X_{10}), \tag{E41.11}$$

where $X_i = x_i - 1$. The ring R is the noncommutative power-series ring in these ten variables with its $(3, X_1, \dots, X_{10})$ topology. Here is a direct topological justification. The units $1 + X_i$ in that complete ring define a homomorphism from the free pro-three group N, because their images at each finite coefficient-and-degree level generate a finite three-group. In the reverse direction, $X_i \mapsto x_i - 1$ defines a continuous power-series map into R: in the group algebra of every finite three-group over $\mathbb{F}_3$ the augmentation ideal is nilpotent, and lifting to coefficients modulo 3^a preserves nilpotence of the combined ideal. The resulting maps are inverse on the generators and therefore inverse on the complete rings. The same argument after imposing commutativity gives the topology on Λ in (E41.11).

Unique decomposition of a nonempty word by its last letter shows that the augmentation ideal of R is free as a left R-module on the X_i. Thus

$$0 \longrightarrow R^{10} \xrightarrow{(a_i) \mapsto \sum a_i X_i} R \longrightarrow \mathbb{Z}_3 \longrightarrow 0 \tag{E41.12}$$

is an actual completed free resolution. Completed tensoring with the right R-module Λ gives the exact identification

$$N'/N'' \xrightarrow{d} \mathcal{K} := \ker\bigl(\Lambda^{10} \xrightarrow{(X_i)} \Lambda\bigr). \tag{E41.13}$$

For clarity, the continuous homology identification in this assertion is the induced-module comparison $\Lambda = \mathbb{Z}_3 \widehat{\otimes}_{\mathbb{Z}_3[[N']]} R$. There is a continuous coset section $(a_1, \dots, a_{10}) \mapsto x_1^{a_1} \cdots x_{10}^{a_{10}}$; the usual bar contraction along this transversal identifies the first homology after tensoring (E41.12) with $H_1(N', \mathbb{Z}_3) = N'/N''$. On a loop its chain map is precisely the continuous Fox derivative

$$d(x_i) = e_i, \qquad d(uv) = d(u) + u^{\mathrm{ab}} d(v), \qquad \sum X_i d_i(u) = u^{\mathrm{ab}} - 1. \tag{E41.14}$$

This describes the isomorphism, rather than only its two objects. Both sides of (E41.13) are compact and the target is a closed kernel. Its continuous bijection is consequently a homeomorphism. In particular the topology on N'/N'' is exactly the induced $(3, \mathfrak{m})$ topology in (E41.13).

The elements X_i form a regular sequence in Λ. Its Koszul complex presents $\mathcal{K}$ by the vectors $X_i e_j - X_j e_i$ and their Jacobi relations. Commutation with x_i in the group corresponds under (E41.14) to multiplication by $-X_i$. It follows, with closed images throughout, that

$$\gamma_r(W_{\mathrm{met}}) = \mathfrak{m}^{r-2}\mathcal{K} \quad (r \geq 2), \qquad \mathcal{A}_r(W_{\mathrm{met}}) = \ker\bigl(\mathrm{Sym}^{r-1} V \otimes V \longrightarrow \mathrm{Sym}^r V\bigr). \tag{E41.15}$$

The filtration comparison used in the second equality is, explicitly,

$$\mathcal{K} \cap \mathfrak{m}^{k+1}\Lambda^{10} = \mathfrak{m}^k\mathcal{K} \qquad (k \geq 0).$$

One inclusion follows from the degree-one generators. For the other, decompose a syzygy into its homogeneous parts of degrees at least $k+1$. Homogeneous Koszul exactness expresses each part by $X_i e_j - X_j e_i$ with coefficients of degree at least k. Assemble these coefficients into power series; their convergent sum is the required element of $\mathfrak{m}^k\mathcal{K}$. Thus the second equality of (E41.15) is exactly the homogeneous degree-$r-1$ syzygy map. This also proves that no extra initial relations or completion torsion occur. This is the integral free metabelian Lie quotient; its commutator-to-Koszul map is the one in [PS03, Theorem 6.2, including (6.3)–(6.4)]. Its stated Chen basis is the integral basis in [Sat10, §2.4, equation(2), p. 5]. The preceding resolution proves explicitly the passage to the pro-three completion used here.

The exact single-cusp marked comparison. Write the other nine degree-one generators as X. Integral Lazard elimination gives

$$\mathrm{Lie}(X \oplus \mathbb{Z}_3 q) = \mathrm{Lie}(X) \ltimes \mathrm{Lie}\big(\mathbb{Z}_3 q \otimes T(X)\big), \tag{E41.16}$$

where the free generators of the ideal are $[q, x_{i_1}, \ldots, x_{i_s}]$, $s \geq 0$. The version over an arbitrary commutative ring is opened in [Stö08, §2, Lazard Elimination Theorem, p. 3]. Each ideal generator contains one q. Its third lower-central term is therefore exactly the direct sum of the multihomogeneous components containing at least three q's.

We verify that this is the initial ideal of the actual group subgroup $\gamma_3 I_Q$. In the Magnus power-series expansion, $u-1$ for $u \in I_Q$ has at least one occurrence of the variable $q-1$ in every monomial: I_Q is the kernel of the retraction setting $q=1$. The expansion of an element of $\gamma_3 I_Q$ consequently has at least three occurrences in each monomial. Taking the first nonzero ordinary degree gives one inclusion in the proposed initial ideal. In the reverse direction each generator of the ideal in (E41.16) is the leading term of the corresponding group commutator in I_Q. Every bracket of at least three such generators is the leading term of an element of $\gamma_3 I_Q$. Products and $\mathbb{Z}_3$ powers realize their sums and scalar multiples. This proves the reverse inclusion in every degree, including the closure. Hence the graded quotient is exactly the last row of (E41.5). The Hall basis is multihomogeneous; discarding its elements with at least three q's leaves a free integral basis. In particular the marked quotient has no extra graded torsion.

All three descriptions are functorial for the relevant based arithmetic action. In the marked case the line $\mathbb{Z}_3 e_Q$ is stable; the chosen complement need not be stable, and the quotient matrix is obtained by actual substitution followed by the just-proved ideal reduction. Thus the displayed matrices use the whole rank-ten action.

Every layer, its free rank and a finite torsion computation. The graded modules in (E41.5) are quotients of degree-r Lie powers with the actual induced arithmetic action. Their rational Frobenius eigenvalues have absolute values $p^{j/2}$ with $r \leq j \leq 2r$. In particular none is one. An element fixed by a section-twisted action on W_r would have a first nonzero graded coordinate fixed by the original action: inner conjugation acts trivially on the associated graded module. The preceding eigenvalue calculation excludes such a coordinate. We have therefore proved

$$H^0(\Gamma, W_r^{\mathrm{section}}) = 1 \quad \text{for every section and every } r. \tag{E41.17}$$

The central action on $\mathcal{A}_r$ is unaffected by a lower section. A normalized additive cocycle is freely specified by its value at τ, and the normalized coboundaries are exactly $(T_r - 1)\mathcal{A}_r^{\langle c \rangle}$. The image of $1 + J_r$ equals that fixed lattice because two is a unit. This proves (E41.3) as an evaluation map and shows that it is the complete parameter group for each nonempty central fiber. Frobenius at seven makes $T_r - 1$ rationally injective, so the free rank is the minus-space dimension $(n_r - t_r)/2$.

Here are formulas evaluating those dimensions in every degree. For the full group they are (E40.5). For the metabelian group and $r \geq 2$, put $h_s = [t^s](1-t)^{-4}(1+t)^{-6}$. Taking the trace in the exact kernel (E41.15) gives

$$n_r^{\mathrm{met}} = (r-1)\binom{r+8}{r}, \qquad t_r^{\mathrm{met}} = -2h_{r-1} - h_r. \tag{E41.18}$$

For the marked group define

$$w_r(9) = \frac{1}{r}\sum_{d|r} \mu(d) 9^{r/d}, \qquad u_r = \frac{1}{r}\sum_{d|r} \mu(d)(4 + (-1)^d 5)^{r/d}.$$

Complex conjugation on $V/\mathbb{Z}_3 e_Q$ has plus and minus multiplicities four and five; on q it is minus one. Averaging a projection by c supplies an invariant integral complement for computing this trace. Elimination separates zero, one and two q occurrences. Their dimensions and traces give, for $r \geq 2$,

$$\begin{aligned} n_r^{\mathrm{mark}} &= w_r(9) + 9^{r-1} + \frac{(r-1)9^{r-2} - \mathbf{1}_{2|r} 9^{r/2-1}}{2}, \\ t_r^{\mathrm{mark}} &= u_r + (-1)^r + \frac{(r-1)(-1)^r - \mathbf{1}_{2|r} 9^{r/2-1}}{2}. \end{aligned} \tag{E41.19}$$

For example the last terms are obtained from $\bigwedge^2(\mathbb{Z}_3 q \otimes T(X))$; its degree-$r$ trace is one half of the trace on the tensor square minus the diagonal trace of c^2. Witt's formula handles the zero-q part. In degree one each group has rank ten and trace minus two. These formulas prove (E41.8).

The bound (E41.7) passes through these exact integral quotients. Namely, the actual split sublattice $V_0 = T_3 T_D + j(T_3 E)$ contains $3V$. The image of the corresponding degree-r lattice contains $3^r \mathcal{A}_r(W)$. It is a direct sum by the number of elliptic letters: the metabelian relations are homogeneous for this count, and the marked letter is toric. Frobenius at the K-prime above two acts on each summand by $(-2)^j$ with $r \leq j \leq 2r$. The identity $v_3((-2)^j - 1) = 1 + v_3(j)$ proves

$$3^{r+b_r}(F-1)^{-1}\mathcal{A}_r(W) \subset \mathcal{A}_r(W). \tag{E41.20}$$

Since the rational invariant space is zero, the torsion of the cokernel (E41.3) is precisely $H^0(\Gamma, \mathcal{A}_r \otimes \mathbb{Q}_3/\mathcal{A}_r)$. Equation (E41.20) kills this group by the stated power of three. Substituting the two known ten-dimensional matrices into the explicit bases (E41.5), reducing by antisymmetry, Jacobi and the displayed ideal, and taking Smith form at the stated precision therefore evaluates every factor by a finite procedure. Exact Smith representatives may be chosen over $\mathbb{Z}_{(3)}$ since the input matrices are integral; unit denominators define continuous maps on the resulting $\mathbb{Z}_3$ coordinates.

The first three evaluations are E.34, the unramified summand of E.37, and E.40. The two defining group kernels in (E41.1) lie in $\gamma_4 N$, as proved in (E40.13)–(E40.14), so their actual graded comparisons identify these three layers. This proves (E41.6). The program `e_unramified_sections.py` also performs the first distinct degree, four. It constructs exact Hall and Chen action matrices, the marked matrix by actual quotient reduction, and the matrices of the maps B, C with $BC = 3I$ from E.40. Its saved integer identities verify $B_r C_r = 3^r I$, conjugation squared equal to the identity, the traces in (E41.8), and Frobenius conjugacy to the split letter-count blocks. Every operation uses exact Python integers. The output includes the sparse matrices and rank tables through degree ten.

Compatible normal forms and the inverse limit. We now construct (E41.4); central liftability alone would not supply these coordinates. Choose the ordered commutator representatives

for each basis in (E41.5). Their $\mathbb{Z}_3$ powers, ordered by degree and then by the chosen basis order, give continuous unique coordinates on each W_r. This follows inductively from its central kernel $\mathcal{A}_r$, and supplies a continuous set section

$$h_r : W_{r-1} \longrightarrow W_r$$

by setting the last-degree coordinates to zero. Choose Smith-coordinate representatives $s_r : \mathcal{C}_r(W) \to \mathcal{A}_r(W)$, continuous and taking zero to zero: retain $\mathbb{Z}_3$ in a free coordinate and the digits $0, \dots, 3^e - 1$ in a factor $\mathbb{Z}/3^e$. Starting with $x_0 = 1$, define

$$x_r = h_r(x_{r-1})\, s_r(z_r), \qquad z_r \in \mathcal{C}_r(W). \tag{E41.21}$$

The additive element $s_r(z_r)$ is viewed here in the central kernel. Formula (E41.9) makes each x_r an actual normalized section. In particular every lower section class has a lift in the unramified family.

We prove uniqueness of these representatives inductively. A normalized section at level r has a lower representative at level $r-1$ already classified by the induction. There is a unique element of $W_{r-1}^{\langle c\rangle}$ aligning it with that representative: two aligning elements differ by the stabilizer (E41.17). This element has a lift in $W_r^{\langle c\rangle}$. To see the latter surjectivity explicitly, lift it arbitrarily to $\widetilde{w}$ and write $\delta = c(\widetilde{w})\widetilde{w}^{-1}$ additively in the central kernel. Then $c\delta = -\delta$, and $(\delta/2)\widetilde{w}$ is fixed by c. After this alignment the remaining difference lies in $\mathcal{A}_r$. Its changes under the remaining central fixed conjugators are exactly $(T_r - 1)\mathcal{A}_r^{\langle c\rangle}$. The chosen Smith representative therefore gives one and only one z_r. Equation (E41.17) again shows uniqueness of the full aligning conjugator. This proves the finite product bijection, with the asserted compatibility under projection.

For completeness, every pro-three group here is its inverse limit of lower-central quotients. An element in every $\gamma_r W$ dies in every finite three-group quotient, since such a quotient is nilpotent; it is therefore the identity. Compactness proves surjectivity onto the inverse limit by the nested cosets of the closed subgroups $\gamma_r W$. Hence

$$W \xrightarrow{\sim} \varprojlim_r W_r. \tag{E41.22}$$

The words x_r in (E41.21) have a compatible limit $x \in W$, and (E41.9) gives an actual continuous full section. For any other normalized section the unique finite-level aligning conjugators are compatible; their limit lies in $W^{\langle c\rangle}$. This proves the full bijection in (E41.4), including injectivity on geometric-conjugacy classes. The construction is continuous because every coordinate at a finite stage uses continuous Hall lifts and Smith representatives. A continuous bijection from the compact product to the Hausdorff orbit space (E41.10) is a homeomorphism.

Finally the comparisons among the three families are the induced maps from the actual epimorphisms

$$N \longrightarrow N/N'', \quad \ker = N'', \qquad N \longrightarrow N/\gamma_3 I_Q, \quad \ker = \gamma_3 I_Q. \tag{E41.23}$$

At level r their kernels are respectively $N''\gamma_{r+1}N/\gamma_{r+1}N$ and $\gamma_3 I_Q\, \gamma_{r+1}N/\gamma_{r+1}N$. On the section coordinates, apply the displayed quotient to the word x_r and then the just-proved unique normalization in its target. These formulas specify the comparison even when the independently chosen Smith coordinates do not make it a coordinate projection. They also retain the common quotient $N/(N''\gamma_3 I_Q)$ and its two quotient maps. The full ramified section problem is not replaced by this unramified family: E.39 gives its additional inertia coordinates and their actual correction words, while E.38 exhibits a ramified central obstruction. □

Exact local lifting in the arithmetic module quotient

E.42. A tame existence test and finite coordinates for nonzero first inertia [Proved]

Use the actual arithmetic quotient of E.38, with

$$\mathscr{A} = \mathbb{Z}_3[[U]], \quad T = 1 + U, \quad J = (3, U)\mathscr{A}, \quad H = 3\mathbb{Z}_3 \ltimes J. \tag{E42.1}$$

Its law is $(a, v)(b, w) = (a+b, v+T^a w)$, and the based Galois action is $(a, v(T)) \mapsto (\chi a, \chi v(T^\chi))$. Let $\mathfrak{q} \nmid 3$ be a prime of $K = \mathbb{Q}(\zeta)$, put $q = N\mathfrak{q}$, and choose local Frobenius F and tame pro-three inertia τ with $F\tau F^{-1} = \tau^q$. Here $q \equiv 1 \pmod 3$. Fix four actual coordinates $a, b, c, d \in 3\mathbb{Z}_3$. We compute the continuous section cocycles with

$$s(\tau) = (a, v), \qquad s(F) = (b, w), \qquad v(1) = c, \quad w(1) = d. \tag{E42.2}$$

The notation in (E42.2) records the geometric coordinates of the section, with the arithmetic generator understood.

For $a = 0$ and $c = 0$, all solutions have $v = 0$, and w is an arbitrary element of J of augmentation d. For $a = 0$ and $c \neq 0$, solutions exist exactly when

$$\frac{b}{q-1} \in \mathbb{Z}_3. \tag{E42.3}$$

Every solution then has $v = cT^{-b/(q-1)}$, and again w is arbitrary with the specified augmentation. For any negative valuation of $b/(q-1)$, the obstruction holds for every nonzero c, regardless of its three-adic divisibility.

For $a \neq 0$, set $m = 3^{v_3(a)}$ and list the orbits $O_1, \dots, O_e$ of the actual affine permutation

$$\mathbb{Z}/m \longrightarrow \mathbb{Z}/m, \quad k \longmapsto qk + b. \tag{E42.4}$$

Use representatives $0 \le k < m$, and define

$$\begin{aligned} P_i(T) &= \sum_{k \in O_i} T^k, \quad \ell_i = |O_i|, \quad \sigma_i = \sum_{k \in O_i} k, \\ \delta_i &= \frac{(q-1)\sigma_i + (b - a(q-1)/2)\ell_i}{a} \in \mathbb{Z}_3. \end{aligned} \tag{E42.5}$$

The displayed integrality is part of the assertion. Solutions exist exactly when the following finite two-row integral system has a solution $(z_1, \dots, z_e, h_0) \in \mathbb{Z}_3^{e+1}$:

$$\sum_i \ell_i z_i = c, \qquad \sum_i \delta_i z_i + (q-1)h_0 = d. \tag{E42.6}$$

This is an explicit Smith test over $\mathbb{Z}_3$. For every solution, put

$$\begin{aligned} n &= T^a - 1, \qquad S = \sum_{j=0}^{q-1} T^{aj}, \\ P &= \sum_i z_i P_i, \qquad h = h_0 + Uh_1, \quad h_1 \in \mathscr{A}, \\ D(P) &= \frac{qT^b P(T^q) - SP(T)}{nS} \in \mathscr{A}, \\ v &= P + nh, \qquad w = D(P) + qT^b h(T^q) - h(T). \end{aligned} \tag{E42.7}$$

These are all the actual section cocycles in (E42.2). Modulo geometric conjugations preserving all four fixed coordinates, their classes are in bijection with the solutions of (E42.6). Every

nonempty such set is a torsor for the free module $\mathbb{Z}_3^{e-1}$, with coordinates supplied by that finite matrix.

For $a = 0$ the conjugations preserving the four coordinates are also explicit: they leave v fixed and send w to $w + u - qT^b u(T^q)$, with $u \in U\mathscr{A}$. Thus the formulas specify both the actual cocycles and their equivalence relation in this case as well.

Manuscript: §51.16, folios 733v–736v, scans 172–178; Part E(c),(d) and the fourth exactness follow-up. The arithmetic quotient maps $N/N'' \to H$ and $N/\gamma_3 I_{\mathrm{marked}} \to H$, with their kernels, are precisely (E38.5)–(E38.9).

Proof. The action on H is the generator computation of E.38. Over $K_{\mathfrak{q}}$ its cyclotomic image is pro-three. Consequently a section cocycle factors through the maximal pro-three quotient of the local Galois group: its graph maps into the pro-three group formed by H and that cyclotomic image. Killing wild inertia, which is pro-p for $p \neq 3$, and the prime-to-three tame factors gives the actual presentation

$$\langle F, \tau \mid F\tau F^{-1} = \tau^q \rangle^{\text{pro-3}}. \tag{E42.8}$$

The unramified factor and the remaining inertia are each $\mathbb{Z}_3$, with conjugation multiplication by q. Conversely images of these two generators satisfying the relation define a continuous homomorphism by this presentation. Thus the equation below tests actual continuous local sections.

Compute the geometric coordinates of $F\tau$ and $\tau^q F$ in H semidirect the arithmetic action. Their first coordinates are both $b + qa$. Their second coordinates give exactly

$$(T^{qa} - 1)w(T) = qT^b v(T^q) - S_q(T^a)v(T), \quad S_q(T^a) = \sum_{j=0}^{q-1} T^{aj}. \tag{E42.9}$$

The scalar q in the first term and the translation T^b both come from the genuine action.

For $a = 0$ this reduces to $T^b v(T^q) = v(T)$. Work first in $\mathbb{Q}_3[[U]]$ and substitute $T = \exp z$. Multiplication by $\exp(bz/(q-1))$ converts the equation to invariance under $z \mapsto qz$. Since $q^j - 1 \neq 0$ for every $j > 0$, the resulting series is constant. Hence $v = cT^{-b/(q-1)}$, including $v = 0$ for $c = 0$. For an exponent $r \in \mathbb{Z}_3$, the coefficients $\binom{r}{j}$ are integral, by continuity from integer exponents. For $v_3(r) = -s < 0$, every factor $r - i$, $i \in \mathbb{Z}$, has that same valuation. Therefore

$$v_3\left(c\binom{r}{j}\right) = v_3(c) - sj - v_3(j!), \tag{E42.10}$$

which is negative for large j whenever $c \neq 0$. This proves the exact integral alternative (E42.3). The term containing w vanishes in (E42.9), so there is no further restriction on w.

Now let $a = mu$ with $u \in \mathbb{Z}_3^{\times}$. The quotient $((T^m)^u - 1)/(T^m - 1)$ is an element of $\mathscr{A}$ with constant term u, hence is a unit. The ideals (n) and $(T^m - 1)$ are consequently equal. The actual quotient map $T \mapsto \overline{T}$ gives

$$\mathscr{A}/(n) \xrightarrow{\sim} \mathbb{Z}_3[\mathbb{Z}/m]. \tag{E42.11}$$

For example, division by the monic distinguished polynomial $(1 + U)^m - 1$ gives the unique representatives of degree less than m; this also proves the claimed topological isomorphism. The factor $S_q(T^a)$ has augmentation $q \in \mathbb{Z}_3^{\times}$, so it too is a unit in $\mathscr{A}$.

Modulo n, equation (E42.9) becomes $T^b v(T^q) = v(T)$. In the displayed group-ring basis this is precisely invariance under the permutation (E42.4). A vector is invariant exactly when its coefficients are constant along each orbit. Thus its invariant submodule has the free basis

P_i, and every permissible v has a unique expression $v = P + nh$ as in (E42.7). Invariance makes the numerator defining $D(P)$ divisible by n; division by the unit S keeps it integral. Substituting $v = P + nh$ in (E42.9) and using $n(T^q) = n(T)S(T)$ gives the exact formula for w in (E42.7). Since $\mathscr{A}$ is an integral domain, this w is unique for the chosen v.

Differentiate the numerator and denominator of $D(P_i)$ at $T = 1$. The numerator has value zero, the derivative of nS is qa, and the numerator derivative is

$$q\big((q-1)\sigma_i + (b - a(q-1)/2)\ell_i\big).$$

It follows that $D(P_i)(1) = \delta_i$. The integrality of $D(P_i)$ proves that of this formula even when a is divisible by a large power of three. Taking augmentations in (E42.7) now gives exactly (E42.6). The required values $c, d \in 3\mathbb{Z}_3$ ensure $v, w \in J$, so these are sections of H itself.

We check the conjugacy assertion without changing the specified coordinates. Conjugation by an element $(s, u) \in H$ changes the first Frobenius coordinate b by $(1 - q)s$. Equality with the fixed b forces $s = 0$. With $s = 0$, the inertia and Frobenius module coordinates change respectively by

$$v \longmapsto v - nu, \qquad w \longmapsto w + u - qT^b u(T^q). \tag{E42.12}$$

Preserving d forces $(1 - q)u(1) = 0$, hence $u \in U\mathscr{A}$. For $a \neq 0$, (E42.12) changes h to $h - u$, fixes every z_i and h_0, and removes the arbitrary Uh_1. These are all conjugations preserving the four coordinates. The matrix in (E42.6) has rank two: its last column is $(0, q-1)^{\mathsf{T}}$ and each ℓ_i is nonzero. Its kernel is therefore a free $\mathbb{Z}_3$-module of rank $e - 1$. This proves the full assertion about those classes. For $a = 0$ the same computation gives the equivalence relation stated after (E42.7). Its linear image is closed, since $U\mathscr{A}$ is compact and the map is continuous.

The program `e_tame_module.py` checks 96 finite orbit systems, with $q = 4, 7, 19, 37$, $m = 3, 9, 27$, and both units $1, 2$ multiplying m. It verifies the two-row Smith valuations, both augmentations, and the original relation through degree 18 by exact rational coefficients integral at three. It also checks 28 zero-first-inertia cases, including the forced coefficient 1729/384 of E.38. The all-degree conclusions follow from the ideal, permutation and power-series arguments above; no finite precision is used to infer them. □

An exact local exponent and infinitely many obstructed cubic fibers

E.43. Every local class-two exponent and an infinite family excluded at higher stages [Proved]

For every prime $\mathfrak{q} \nmid 3$ of $K = \mathbb{Q}(\zeta)$, put $q = N\mathfrak{q}$ and $s = v_3(q - 1)$. The local rank-44 map of E.37 has the exact cokernel exponent

$$\exp \operatorname{coker} d_{\mathfrak{q}} = 3^{s+1}. \tag{E43.1}$$

This is the exponent in the actual integral source M/L and target M_0 of (E37.6).

Let $\mathcal{P}$ be the following set of rational primes:

$$\mathcal{P} = \{p : p \equiv 1 \pmod 9, \quad 2^{(p-1)/3} \not\equiv 1 \pmod p\}. \tag{E43.2}$$

It has natural density $1/9$, in particular it is infinite. Use the actual rational tangential class α_P and rational peripheral vector e_Q of E.38. For $p \in \mathcal{P}$ and every integer $k \geq v_3(p - 1) + 1$, the class

$$\alpha_{p,k} = \alpha_P + 3^k \kappa_3(p) e_Q, \qquad \text{toric coordinates } (2, p^{3^k}), \tag{E43.3}$$

has a nonempty full intrinsic class-two fiber, with the complete parameter group (E37.12). Every section in that fiber fails to lift to N/N'', to N, or to any marked quotient in (E38.2). In

fact the local H obstruction excludes those lifts for every $k \geq 0$, independently of the positive class-two bound. The classes for distinct primes are distinct already in the minimum torus quotient.

The primes in (E43.2) through 1000 are

$$\begin{gathered} 19, 37, 73, 163, 181, 199, 271, 379, 487, 523, 541, 577, \\ 613, 631, 757, 829, 883, 937, 991. \end{gathered} \tag{E43.4}$$

All 38 local rank-44 matrices at their two primes in K have been computed and agree with (E43.1). For $p = 19$ the bound is $k = 3$, recovering the finite class-three obstruction of E.38.

Manuscript: §51.16, folios 733v–736v, scans 172–178; Part E(c),(d) and the fourth exactness follow-up. The positive calculation uses the full intrinsic matrix, and the negative calculation uses the actual quotient maps (E38.9).

Proof. We first prove the exact exponent. Write $\mathcal{O} = \mathbb{Z}_3[\zeta]$ and $z = \psi_{\mathfrak{q}} - 1$. The scalar is primary, so $z \in 3\mathcal{O}$, and $z \neq 0$ since its Frobenius norm is $q > 1$. The two Smith exponents of multiplication by z on $\mathcal{O}$ are

$$m = \lfloor v_{1-\zeta}(z)/2 \rfloor, \qquad h = \lceil v_{1-\zeta}(z)/2 \rceil. \tag{E43.5}$$

Indeed $\mathcal{O}$ is a discrete valuation ring, 3 is a unit times $(1-\zeta)^2$, multiplication by a unit is an integral automorphism, and multiplication by $1-\zeta$ has Smith factors $1, 3$. These facts give (E43.5) by separating the even and odd powers of the uniformizer.

The trace of $\mathcal{O}$ is $\mathbb{Z}_3$ and the trace of $(1-\zeta)\mathcal{O}$ is $3\mathbb{Z}_3$: use the bases $1, \zeta$ and $1-\zeta, \zeta-\zeta^2$. Consequently $\operatorname{Tr}(z) \in 3^h\mathbb{Z}_3$, and $N(z) \in 3^{m+h}\mathbb{Z}_3$. The identity

$$q - 1 = N(1+z) - 1 = \operatorname{Tr}(z) + N(z)$$

therefore gives $s \geq h$.

Inside M_0 consider the explicit stable sublattice

$$S_0 = \bigwedge^2 T_3 T_D + (T_3 T_D \wedge j(T_3 E)). \tag{E43.6}$$

For the elliptic basis t_i, the columns $j(t_i) = 3f_i + U_i$ in E.37 show that $3M_0 \subset S_0$: the toric wedges are already in S_0, and three times each mixed wedge is a mixed j-wedge minus toric wedges. The inclusion $S_0 \to M/L$ is injective, since L has nonzero $f_1 \wedge f_2$ coordinate while S_0 has none. On its toric wedges, $d_{\mathfrak{q}}$ is multiplication by $q-1$. On its mixed wedges it is the actual matrix $A_E - 1$, whose largest Smith exponent is $h \leq s$. Hence $3^s d_{\mathfrak{q}}^{-1} S_0 \subset S_0$. Together with $3M_0 \subset S_0$ this proves

$$3^{s+1} M_0 \subset d_{\mathfrak{q}}(M/L). \tag{E43.7}$$

We establish equality of the exponent by an integral direct summand. The two columns of U span a primitive rank-two sublattice of $\mathbb{Z}_3^8$: their minor at A_0, A_2 is a unit. A change of torus basis sends U to $\binom{I_2}{0}$. In this basis the actual Frobenius module decomposes as

$$V = V_4 \oplus T_6, \qquad A_{V_4} = \begin{pmatrix} A_E & 0 \\ C & qI_2 \end{pmatrix}, \quad C = (A_E - qI_2)/3, \tag{E43.8}$$

where T_6 is toric of rank six and $L \subset \bigwedge^2 V_4$. Thus $V_4 \otimes T_6$ is an actual direct summand of both M/L and M_0, and $d_{\mathfrak{q}}$ acts there as six copies of $A_{V_4} - 1$.

Identify its two-dimensional blocks with $\mathcal{O}$. The element represented by C is $-\psi_{\mathfrak{q}}\overline{z}/3$. Since conjugation preserves the uniformizer valuation, the matrix

$$(A_E - 1)C^{-1} = -3\psi_{\mathfrak{q}}^{-1} z/\overline{z}$$

is three times an integral automorphism. Also $(q-1)C^{-1}$ is integral: its uniformizer valuation is $2s + 2 - v_{1-\zeta}(z) \geq 2$. Integral block row and column operations on

$$A_{V_4} - 1 = \begin{pmatrix} A_E - 1 & 0 \\ C & (q-1)I_2 \end{pmatrix}$$

therefore give a block diagonal matrix with blocks C and three times $q-1$ times an integral automorphism. Its Smith exponents are

$$m - 1,\ h - 1,\ s + 1,\ s + 1. \tag{E43.9}$$

This direct summand has exponent 3^{s+1}. Combined with (E43.7) it proves (E43.1) for every prime under consideration.

We next verify the positive global lifting assertion. The class α_P comes from a section of the full arithmetic group and is unramified outside $2, 3$. At the prime of K above two, an odd rational prime p lies in $1 + 2\mathcal{O}_{K,2}$. Cubing is an automorphism of this pro-two unit group, so $\kappa_3(p)$ vanishes there. The local class in (E43.3) thus has the same lifting property as α_P at two.

At either prime above p, normalize the tame valuation of p to one. With the notation of E.37 the full local numerator is

$$D_{\mathfrak{p}} = 3^k(a_P - k_{\mathfrak{p},Q}) \wedge e_Q \in 3^k M_0. \tag{E43.10}$$

The self-wedge term from the unramified value of $\kappa_3(p)$ is zero. Since $k \geq s+1$, (E43.1) makes the unique correction integral at both primes. All other primes away from three have zero inertia value, apart from the already liftable prime above two. The correction family has finite support and satisfies the full continuity condition (E37.11). E.37 proves the claimed global class-two lift and identifies every class in its fiber.

For the negative assertion, use the unit tangent at P and the quotient $N \to H$ of E.38. At $p \in \mathcal{P}$ the first inertia coordinate is $a = 0$, and the inertia augmentation is $c = 3^{k+1} \neq 0$. The first Frobenius coordinate is $b = 3\kappa_3(2)(F)$, up to addition of $3(p-1)\mathbb{Z}_3$ by geometric conjugation. The residue condition in (E43.2) says precisely that $\kappa_3(2)(F)$ is a unit modulo three. Thus $v_3(b) = 1$, while $v_3(p-1) \geq 2$. Condition (E42.3) fails for every representative and every nonzero c. This excludes a local H-section above the prescribed abelian class. Any global section in any of the stated higher quotients would restrict and map to that excluded local section. The exact maps are (E38.9), and N maps onto each of its quotients. This proves the nonlifting statements through those morphisms.

Finally let $L_9 = \mathbb{Q}(\zeta_9, 2^{1/9})$. At a prime above two, the cyclotomic field $\mathbb{Q}(\zeta_9)$ is unramified and $X^9 - 2$ is Eisenstein. The Kummer degree is therefore nine, and $[L_9 : \mathbb{Q}] = 54$. An automorphism is the actual pair

$$(u, v):\quad \zeta_9 \mapsto \zeta_9^u,\quad 2^{1/9} \mapsto \zeta_9^v 2^{1/9},\qquad (u, v)(u', v') = (uu', v + uv').$$

The conjugacy class of $(1, 1)$ consists of the six pairs $(1, v)$ with $v \in (\mathbb{Z}/9)^\times$. At an unramified rational prime, this class is exactly the two residue requirements in (E43.2). Chebotarev's theorem [Mil20b, V, Theorem 3.23], applied to this explicit finite Galois extension, gives density $6/54 = 1/9$. The toric valuation at p in (E43.3) is 3^k, whereas the same coordinate for

a different prime has valuation zero at p. This proves distinctness through the actual minimum torus quotient.

The program `e_infinite_obstructions.py` enumerates every eligible prime through 1000, counts $E(\mathbb{F}_p)$ on the model $y^2 + y = x^3 - 7$, finds both primary CM scalars, and computes both complete rank-44 Smith lists. The exact matrices and lists are saved. It checks (E43.1) in all 38 cases and records the first forced nonintegral coefficient from (E42.10). The uniform exponent, infinitude, and nonlifting at every k are supplied by the preceding proofs. □

The exact higher central lifting image, including convergence and Frobenius corrections

E.44. A global higher lift is equivalent to integral restricted tame corrections [Proved]

Let W be any of the three actual cubic quotients in (E41.1), and retain W_r and $\mathcal{A}_r(W)$ from (E41.2). Fix $r \geq 3$, put $A = \mathcal{A}_r(W)$, and let $a : G_{\mathbb{Q}} \to W_{r-1}$ be an actual continuous cocycle for the based arithmetic action. It may be ramified outside three. Fix the continuous Hall set section $h : W_{r-1} \to W_r$ with $h(1) = 1$ from E.41, and write $\widehat{a} = h \circ a$.

For every prime $\mathfrak{q} \nmid 3$ of $K = \mathbb{Q}(\zeta)$ fix one actual prolongation, a tame pro-three inertia generator $t_{\mathfrak{q}}$, and an arithmetic Frobenius $\phi_{\mathfrak{q}}$. Write $q = N\mathfrak{q}$; the tame relation is $\phi_{\mathfrak{q}} t_{\mathfrak{q}} \phi_{\mathfrak{q}}^{-1} = t_{\mathfrak{q}}^{q}$ after killing wild inertia. Set

$$\begin{aligned} n_{\mathfrak{q}} &= \widehat{a}(t_{\mathfrak{q}}), \qquad x_{\mathfrak{q}} = \widehat{a}(\phi_{\mathfrak{q}}), \\ \epsilon_{\mathfrak{q}} &= x_{\mathfrak{q}}\, \rho_r(\phi_{\mathfrak{q}})(n_{\mathfrak{q}})\, x_{\mathfrak{q}}^{-1} n_{\mathfrak{q}}^{-q} \ \in A, \\ z_{\mathfrak{q}} &= -(\Phi_{\mathfrak{q}} - qI)^{-1} \epsilon_{\mathfrak{q}} \ \in A \otimes \mathbb{Q}_3. \end{aligned} \tag{E44.1}$$

Here ρ_r is the genuine action on W_r and $\Phi_{\mathfrak{q}}$ is its induced linear action on A. The central element $\epsilon_{\mathfrak{q}}$ is written additively in the last line. The inverse exists for every prime in (E44.1).

The geometric-conjugacy fiber of the actual map

$$\mathrm{Sect}(W_r \rtimes G_{\mathbb{Q}})/W_r \longrightarrow \mathrm{Sect}(W_{r-1} \rtimes G_{\mathbb{Q}})/W_{r-1} \tag{E44.2}$$

over $[a]$ is nonempty exactly for the following explicit condition:

$$z_{\mathfrak{q}} \in A \quad (\mathfrak{q} \nmid 3), \qquad \text{for every } m, \quad z_{\mathfrak{q}} \in 3^m A \text{ outside finitely many } \mathfrak{q}. \tag{E44.3}$$

Every nonempty fiber is a torsor under the evaluated finite-matrix group $\mathcal{C}_r(W)$ in (E41.3). Thus the coefficient lattice and all parameters within a nonempty fiber are already explicit.

More precisely, write

$$\mathcal{R}(A) = \{(v_{\mathfrak{q}}) \in \prod_{\mathfrak{q} \nmid 3} A : v_{\mathfrak{q}} \longrightarrow 0 \text{ in the three-adic topology}\}, \quad d(v)_{\mathfrak{q}} = (q^{-1}\Phi_{\mathfrak{q}} - 1)v_{\mathfrak{q}}.$$

The exact continuous localization map sends the actual obstruction to

$$H^2_{\mathrm{cts}}(K, A) \xrightarrow{\sim} \mathcal{R}(A)/d\mathcal{R}(A), \qquad \mathrm{ob}_r(a|_{G_K}) \longmapsto [(q^{-1}\epsilon_{\mathfrak{q}})_{\mathfrak{q}}]. \tag{E44.4}$$

The denominator is the image, with no closure added. The restricted sequence condition in (E44.3) consequently retains the possible inverse-limit obstruction.

One local target in this criterion is completely evaluated in the next intrinsic degree. At the K-prime (2), with $q = 4$, the common degree-three module of all three quotients has

$$\mathrm{coker}(\Phi_{(2)} - 4I) = (\mathbb{Z}/3)^{70} \oplus (\mathbb{Z}/9)^{24} \oplus (\mathbb{Z}/27)^{76} \oplus (\mathbb{Z}/81)^{28}. \tag{E44.5}$$

The group has order 3^{458}. At the rational prime two the corresponding actual target is

$$H^2_{\mathrm{cts}}(\mathbb{Q}_2, \mathrm{Lie}_3\, V) = \mathrm{coker}(\Phi_{2,\mathbb{Q}} - 2I) = (\mathbb{Z}/3)^{36} \oplus (\mathbb{Z}/9)^{12} \oplus (\mathbb{Z}/27)^{38} \oplus (\mathbb{Z}/81)^{14}, \quad \text{(E44.5a)}$$

of order 3^{230}. Their Smith certificates and matrices are included in `e_higher_images.py`, its output and JSON file.

The error in (E44.1) uses the actual higher Frobenius action. With the splitting in E.39, its lower Frobenius value is exactly

$$a(\phi_{\mathfrak{q}}) = \lambda(h_{\mathfrak{q}})u(\sigma_{\mathfrak{q}}). \tag{E44.6}$$

Formula (E44.4) proves the higher lifting image for these supplied arithmetic data; evaluation of all the higher words in (E44.1) and (E44.6) for all ramified inputs remains the further step in the full metabelian and marked section classification. The rank-ten linear matrices determine $\Phi_{\mathfrak{q}}$, while $\rho_r(\phi_{\mathfrak{q}})$ is the genuine action on the specified group quotient.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379), Part E(c),(d). The comparison uses the intrinsic quotients and the exact maps in (E41.23); it applies above ramified as well as unramified lower sections.

Proof. The actual central extension and its local error. The kernel of $W_r \to W_{r-1}$ is the finite free lattice A proved in E.41. It is central in W_r, so pullback along any lower section has its original arithmetic action on A. The continuous set lift $\widehat{a}$ defines the central continuous two-cocycle

$$\omega(g,h) = \widehat{a}(g)\, g(\widehat{a}(h))\, \widehat{a}(gh)^{-1} \in A. \tag{E44.7}$$

Associativity gives its cocycle identity, and changing the set lift changes it by a continuous coboundary. A continuous A-valued correcting cochain b with $\delta b = -\omega$ gives the actual lifted cocycle $\widetilde{a}(g) = b(g)\widehat{a}(g)$. Conversely every lift has this form. Thus the vanishing of the continuous class of (E44.7) is exactly the lifting assertion.

At $\mathfrak{q} \nmid 3$ the based arithmetic action on every W_r is unramified. Wild inertia has trivial cocycle image: it is pro-p for $p \neq 3$ and the geometric target is pro-three. On tame inertia the lower cocycle is therefore a homomorphism, and $a(t_{\mathfrak{q}}^s) = a(t_{\mathfrak{q}})^s$ for $s \in \mathbb{Z}_3$. The corresponding lifted inertia homomorphism is

$$t_{\mathfrak{q}}^s \longmapsto n_{\mathfrak{q}}^s. \tag{E44.8}$$

Its difference from $\widehat{a}(t_{\mathfrak{q}}^s)$ is an explicit local null-homotopy of (E44.7). The lower section satisfies the tame relation, so the error in (E44.1) belongs to the central kernel A.

Replacing $n_{\mathfrak{q}}$ by $zn_{\mathfrak{q}}$ with $z \in A$ changes this error by exactly

$$\epsilon_{\mathfrak{q}} \longmapsto \epsilon_{\mathfrak{q}} + (\Phi_{\mathfrak{q}} - qI)z. \tag{E44.9}$$

This follows by moving the central factors through the displayed word: Frobenius contributes $\Phi_{\mathfrak{q}} z$ and the qth power contributes qz. Changing $x_{\mathfrak{q}}$ by a central element has no effect at all. Thus (E44.1) is the unique rational central correction of the chosen inertia lift.

The weights on $A \otimes \mathbb{Q}_3$ are among the degree-r tensor weights of V, namely Frobenius absolute values $q^{j/2}$ with $r \leq j \leq 2r$. This follows from each actual graded quotient in (E41.5). For $r \geq 3$ none is q, so $\Phi_{\mathfrak{q}} - qI$ is invertible over $\mathbb{Q}_3$. There is no choice of a rational kernel coordinate in (E44.1).

Localization at finite coefficients, followed by the derived limit. Put $U = \mathrm{Spec}\, \mathcal{O}_K[1/3]$ and $A_m = A/3^m A$. The action on this finite module over $G_{K,\{3\}}$ has three-group image by E.34. A finite three-group module has a filtration by trivial $\mathbb{F}_3$ modules. The vanishing $H^2(G_{K,\{3\}}, \mathbb{F}_3) = 0$ proved in E.34 consequently gives

$$H^2(U, A_m) = 0, \qquad H^3(U, A_m) = 0, \qquad H^1(U, A_m) \text{ finite.} \tag{E44.10}$$

Here the comparison is [Mil06, II.2.9]; the degree-three vanishing and finiteness are [Mil06, I.4.10(c), I.4.15]. The field K has no real places and three is inverted on U, as these statements require.

For a finite set J of primes of U, the actual localization triangle is

$$R\Gamma(U, A_m) \longrightarrow R\Gamma(U - J, A_m) \longrightarrow \bigoplus_{\mathfrak{q}\in J} R\Gamma(k_{\mathfrak{q}}, A_m(-1))[-1] \longrightarrow R\Gamma(U, A_m)[1]. \quad \text{(E44.11)}$$

It follows from [Mil06, II.0.1(f), II.1.7(b)]. The purity term can also be computed directly over the strict Henselization: wild inertia contributes no three-primary cohomology, tame inertia is $\mathbb{Z}_3(1)$, and its degree-one cohomology is $A_m(-1)$. Taking the filtered direct limit over finite J gives the generic field G_K term. The support term remains a direct sum at each fixed coefficient level.

Now take $R\varprojlim_m$. Its comparison with continuous cochains and its derived limit sequence are [Jan88, Theorem 2.2 and Proposition 1.6]. Equation (E44.10) and the finiteness of $H^1(U, A_m)$ give zero in degrees two and three for the unramified inverse-limit complex. For the residue-field term use the compatible two-term complex

$$[\, A_m \xrightarrow{q^{-1}\Phi_{\mathfrak{q}}-1} A_m \,] \quad \text{in degrees } 0, 1. \quad \text{(E44.12)}$$

The direct-sum transitions in both terms are surjective. Their derived inverse limit is therefore the literal complex

$$[\, \mathcal{R}(A) \xrightarrow{d} \mathcal{R}(A) \,]. \quad \text{(E44.13)}$$

The resulting degree-two isomorphism is the first part of (E44.4). In particular the image in its denominator is retained as an image; this calculation has not replaced inverse limits of complexes by inverse limits of cohomology groups.

The chain-level representative and uniform continuity. The support map in (E44.11) records the failure of an inertia splitting to be Frobenius invariant. Use exactly the splitting (E44.8). Transporting it by $(x_{\mathfrak{q}}, \phi_{\mathfrak{q}})$ and evaluating at $t_{\mathfrak{q}}$ gives the error at $t_{\mathfrak{q}}^q$ pulled back by its qth-root map. In formulas,

$$x_{\mathfrak{q}}\phi_{\mathfrak{q}}(n_{\mathfrak{q}})x_{\mathfrak{q}}^{-1} = \epsilon_{\mathfrak{q}} n_{\mathfrak{q}}^q \quad \Longrightarrow \quad x_{\mathfrak{q}}\phi_{\mathfrak{q}}(n_{\mathfrak{q}}^{q^{-1}})x_{\mathfrak{q}}^{-1}n_{\mathfrak{q}}^{-1} = q^{-1}\epsilon_{\mathfrak{q}}. \quad \text{(E44.14)}$$

This is the twist in $\mathrm{Hom}(\mathbb{Z}_3(1), A) = A(-1)$. Equation (E44.9) divided by q is exactly the differential in (E44.12). Thus the representative is $D_{\mathfrak{q}} = q^{-1}\epsilon_{\mathfrak{q}}$ at the cochain level, with the sign in (E44.1) forced by (E44.9).

We verify that these local null-homotopies are admissible in the direct-sum complexes, including their inverse limit. For every finite characteristic quotient of W_{r-1}, the map $g \mapsto (a(g), \rho(g))$ modulo that quotient has finite image. Its kernel defines a finite extension of K, ramified at finitely many primes. Hence $a(t)$ approaches 1 uniformly on every inertia group outside a finite set, including all prolongations of those primes. Continuity of h at 1 gives the same assertion for $h(a(t))$ in every finite quotient of W_r. The map $(n, s) \mapsto n^s$ is continuous on the compact space $W_r \times \mathbb{Z}_3$; indeed a closed subgroup containing n contains every n^s. It follows that the difference between $n_{\mathfrak{q}}^s$ and $h(a(t_{\mathfrak{q}}^s))$ is zero modulo $3^m A$, uniformly in s, outside finitely many base primes.

The conjugation and arithmetic action are continuous and preserve sufficiently small characteristic open subgroups. The same argument proves $\epsilon_{\mathfrak{q}} \in 3^m A$ outside finitely many primes. On the prolongations of any fixed prime, these finite-coefficient cochains depend continuously, and hence locally constantly, on the conjugating Galois element. In (E44.4) we use the one fixed actual prolongation per base prime specified in the statement. These facts justify the direct-sum representative before taking the derived limit and prove $(D_{\mathfrak{q}}) \in \mathcal{R}(A)$. Equations

(E44.11)–(E44.14) now give the second part of (E44.4), including its actual representative in the restricted quotient.

The component maps $d_{\mathfrak{q}}$ are rationally injective. Consequently D is in $d\mathcal{R}(A)$ precisely when its unique negative inverse is the integral restricted sequence in (E44.3). This proves the criterion over K. Restriction from $H^2_{\mathrm{cts}}(\mathbb{Q}, A)$ to $H^2_{\mathrm{cts}}(K, A)$ is injective: the composite with corestriction is multiplication by two, which is an automorphism of this three-primary cohomology module. Thus the criterion also proves the asserted global lift over $\mathbb{Q}$.

Actual gluing and independence of choices. The vanishing of the continuous class in (E44.7) supplies a continuous global correcting cochain, hence an actual lift. The gluing can also be performed at finite coefficients. The reductions of the corrected inertia lifts $z_{\mathfrak{q}} n_{\mathfrak{q}}$ are compatible and have finite support in their central corrections at each level. Finite-coefficient localization first gives a global lift modulo $3^m A$. Its difference from the prescribed inertia lifts is a finite family in $\bigoplus_{\mathfrak{q}} H^0(k_{\mathfrak{q}}, A_m(-1))$. The preceding degree of (E44.11) is surjective onto this family because $H^2(U, A_m) = 0$. Subtract the resulting global one-cocycle to align every prescribed inertia lift.

Two such aligned global correcting cochains differ by a cocycle zero on all these inertia groups, hence by an unramified cocycle. The group of such finite-coefficient cocycles is finite: its cohomology is finite by (E44.10), and its coboundaries are images of the finite module A_m. The nonempty finite sets of aligned cochains therefore have a compatible inverse limit. Each component is continuous, so the resulting A-valued cochain is continuous. This construction also shows that the continuous inertia-bundle equivariance of E.39 is fulfilled by the global lift; no independent choice of incoherent inertia values has been introduced.

Changing the Hall lift changes $n_{\mathfrak{q}}$ by an integral restricted central sequence, by the uniform continuity just proved. Equation (E44.9) changes D by its image under d, preserving (E44.3). Changing Frobenius by an inertia power replaces its geometric lift by $n_{\mathfrak{q}}^s x_{\mathfrak{q}}$ up to a central factor; this leaves $\epsilon_{\mathfrak{q}}$ unchanged. A change of tame generator first transports the inertia lift to $n_{\mathfrak{q}}^u$, $u \in \mathbb{Z}_3^*$, and rescales its additive coordinate by that unit in $\mathbb{Z}_3$, as does the identification of the corresponding Tate line. Such rescalings preserve integrality and the restricted condition. Returning from this transported lift to $h(a(t_{\mathfrak{q}}^u))$ changes the representative by a restricted central boundary, by the same uniform continuity argument. Changing prolongations similarly gives the corresponding Galois-module coordinate isomorphisms followed by a restricted central boundary when returning to the Hall lift. Their integral automorphisms preserve every $3^m A$. Finally, geometric conjugation of a can be lifted to W_r; it conjugates the error word and acts trivially on its central value. Returning to Hall lifts again adds a restricted boundary. Thus the criterion is well defined on the fiber (E44.2).

The differences between two global lifts are A-valued one-cocycles. At every good prime their inertia value v satisfies $\Phi v = qv$. The same weight calculation forces $v = 0$, so every such cocycle is unramified outside three. The lowest nonzero graded coordinate argument of E.41 excludes every section-twisted lower stabilizer, including for a ramified section. The full geometric-conjugacy fiber is therefore the torsor $\mathcal{C}_r(W)$, with no further quotient. This supplies the claimed parameters whenever (E44.3) holds.

An evaluated degree-three local obstruction module. At the K-prime (2) the actual elliptic scalar is -2 and the toric scalar is four. In degree three the split-lattice eigenvalues and multiplicities are

eigenvalue	64	−32	16	−8
multiplicity	168	128	32	2.

(E44.15)

They come from zero, one, two and three elliptic letters, respectively. The actual rank-330 integral matrix is constructed from the matrix A_2 of E.37 in Hall coordinates. The split-lattice maps satisfy $B_3 C_3 = 27I$, and $9/(\lambda - 4)$ is three-integral for each eigenvalue λ in (E44.15).

Hence

$$3^5(\Phi_{(2)} - 4I)^{-1}A \subset A. \tag{E44.16}$$

The local cokernel is killed by 3^5 and has full rational rank zero. The saved modular row and column transformations in `e_higher_images.py` verify modulo 729 the exact Smith valuation counts

$$0^{\times 132}, \quad 1^{\times 70}, \quad 2^{\times 24}, \quad 3^{\times 76}, \quad 4^{\times 28}. \tag{E44.17}$$

All transformations are invertible modulo three. The proven bound (E44.16) makes this finite certificate conclusive. As an independent check, its valuation sum is $70 + 48 + 228 + 112 = 458$, which equals the valuation of the determinant from (E44.15), namely $168 + 2 \cdot 128 + 32 + 2 = 458$. The three actual group quotients have identical degree-three kernels by E.40–E.41, so this proves (E44.5) for all of them.

We also compute the rational-prime target, retaining the integral arithmetic comparison. Let $\mathcal{O} = \mathbb{Z}_3[\zeta]$. The rational Frobenius at two acts on the rank-one $\mathcal{O}$-module T_3E as uc for $u \in 1 + 3\mathcal{O}$: all $E[3]$ is defined over K, and its nontrivial residual Galois element is conjugation. The determinant is two by the Weil pairing, while $\det(c) = -1$, so $u\bar{u} = -2$. The point count on $Y^2 + Y = X^3 - 7$ over $\mathbb{F}_2$ gives three points and trace zero, in agreement with its square -2 over K.

Let $s = \sqrt{-2} \in 1 + 3\mathbb{Z}_3$ and set $v = (s/u)^{1/2}$ in $1 + 3\mathcal{O}$. Unique square roots exist there; $v\bar{v} = 1$ and $vu/\bar{v} = s$. The full CM image proved in E.37 realizes v by an actual G_K action whose toric action is the identity. Its entire integral V-matrix is

$$\begin{pmatrix} v & 0 \\ U(v - I)/3 & I_8 \end{pmatrix}. \tag{E44.18}$$

Thus it conjugates the actual rational Frobenius representation to the matrix with elliptic block sC_E, toric block $2P_D$, and lower block $(UsC_E - 2P_DU)/3$. Here $P_D = -C_T$ is the cusp permutation, and U, C_E, C_T are exactly the E.37 matrices. The comparison (E44.18) acts on the whole generalized-Jacobian lattice and is integral.

The square root modulo 2187 determines this divided block modulo 729. The program constructs its degree-three matrix, verifies that its square is the K-Frobenius matrix modulo 729, and gives Smith valuations

$$0^{\times 230}, \quad 1^{\times 36}, \quad 2^{\times 12}, \quad 3^{\times 38}, \quad 4^{\times 14}. \tag{E44.19}$$

The exact exponent bound follows from (E44.16) and

$$(\Phi_{2,\mathbb{Q}} - 2I)^{-1} = (\Phi_{2,\mathbb{Q}} + 2I)(\Phi_{(2)} - 4I)^{-1}.$$

The matrix on the left is therefore determined for Smith purposes at the computed precision. An independent determinant check uses the split degree-one eigenvalues $s, -s, 2^{\times 5}, (-2)^{\times 3}$ and all degree-three Hall monomials; its valuation is $36 + 24 + 114 + 56 = 230$. The local support sequence has zero degree-two and degree-three unramified residue terms, so its degree-two field cohomology is precisely the displayed cokernel. This proves (E44.5a).

The inverse corrections included in the code are lattice checks; the theorem does not assert that every arbitrary lattice error is realized by a global lower section.

Lastly (E44.6) is the exact decomposition (E39.5) applied to the actual local Frobenius. Substituting it into (E44.1) retains its correction word and the genuine higher arithmetic action. E.38 and E.43 evaluate particular ramified inputs; (E44.1)–(E44.4) give the proved general lifting test with its remaining arithmetic data explicitly specified. □

All tame-local conjugacy classes in the actual inherited module quotient

E.45. Complete local section coordinates, including zero first inertia [Proved]

Use the actual arithmetic group H and local field $K_{\mathfrak{q}}$ of E.42. Put

$$\mathscr{A} = \mathbb{Z}_3[[U]], \quad T = 1 + U, \quad \sigma = v_3(q-1), \quad \mathcal{B}_\sigma = \{0, 3, 6, \dots, 3^{\sigma+1} - 3\}. \tag{E45.1}$$

Every continuous local section, modulo geometric conjugacy by H, has uniquely normalized four coordinates

$$a, c \in 3\mathbb{Z}_3, \qquad b, d \in \mathcal{B}_\sigma, \qquad s(\tau) = (a, v), \quad s(F) = (b, w), \quad v(1) = c, \quad w(1) = d. \tag{E45.2}$$

After this normalization the following formulas give all the classes, with no additional identifications.

For $a \neq 0$, the classes are exactly the solutions of the finite two-row Smith system (E42.6), with the representatives (E42.7) and $h = h_0$. Every nonempty such set is the explicit $\mathbb{Z}_3^{e-1}$ torsor of E.42, where e is the number of the affine orbits (E42.4).

For $a = 0$, put $r = b/(q-1)$. When $r \in \mathbb{Z}_3$, every $c \in 3\mathbb{Z}_3$ is allowed. The classes have the following unique representatives, indexed by arbitrary $z_{j,u} \in \mathbb{Z}_3$:

$$\begin{aligned} v &= cT^{-r}, \\ w &= d + T^{-r} \sum_{\substack{j \geq 0 \\ u \in (\mathbb{Z}/3^\sigma)^\times}} z_{j,u}(T^{3^j u} - 1). \end{aligned} \tag{E45.3}$$

Choose the representatives $1 \leq u < 3^\sigma$. The series converges in the actual profinite completed-group-ring topology, equivalently the $(3, U)$-adic topology. Its parameter space is the full compact product

$$\prod_{j \geq 0} \prod_{u \in (\mathbb{Z}/3^\sigma)^\times} \mathbb{Z}_3. \tag{E45.4}$$

When $r \notin \mathbb{Z}_3$, put $t = v_3(b) < \sigma$. A class exists exactly for $c = 0$. Its unique representatives, indexed by $z_0, \dots, z_{3^t-1} \in \mathbb{Z}_3$, are

$$v = 0, \qquad w = d + z_0 q(1 - T^b) + \sum_{i=1}^{3^t-1} z_i(T^i - 1). \tag{E45.5}$$

Thus this fiber is explicitly $\mathbb{Z}_3^{3^t}$. Here $1 \leq t < \sigma$, since $b \in 3\mathbb{Z}_3$. The fixed-coordinate fibers in (E45.3) and (E45.5) have their product and three-adic topologies, respectively.

The precise linear maps underlying these coordinates are also explicit. Set

$$D_b(f) = qT^b f(T^q) - f(T), \qquad f \in U\mathscr{A}. \tag{E45.6}$$

For integral r, translation by T^r gives

$$\operatorname{coker}(D_b : U\mathscr{A} \to U\mathscr{A}) \xrightarrow{\sim} \prod_{j,u} \mathbb{Z}_3. \tag{E45.7}$$

For nonintegral r, the corresponding map identifies this cokernel with the actual lattice

$$\left\{ (I_0, \dots, I_{3^t-1}) \in \mathbb{Z}_3^{3^t} : \sum I_i \in (q-1)\mathbb{Z}_3 \right\}, \tag{E45.8}$$

whose basis is $(q-1)e_0$ and $e_i - e_0$, $1 \leq i < 3^t$. The proof defines the maps by weighted integrals of integral measures and proves their kernels and surjectivity.

Manuscript: §51.16, folios733v–736v, scans172–178, Part E(c),(d). This completes all local section classes of the actual quotient H in E.38 and E.42. The maps from the intrinsic metabelian and marked groups, with their geometric kernels, remain exactly (E38.5)–(E38.9).

Proof. The full conjugation action and normalization. All local cocycles are the actual generator images satisfying the tame presentation (E42.8). Conjugate their section by $(s,u)\in H$, with $s\in 3\mathbb{Z}_3$ and $u(1)\in 3\mathbb{Z}_3$. Direct multiplication in the given semidirect product gives

$$\begin{aligned} a' &= a, & v' &= T^s v + (1-T^a)u,\\ b' &= b+(1-q)s, & w' &= T^s w + u - qT^{b'}u(T^q),\\ c' &= c, & d' &= d+(1-q)u(1). \end{aligned} \tag{E45.9}$$

In particular a,c are actual invariants. The two remaining augmentations vary independently by $3(q-1)\mathbb{Z}_3 = 3^{\sigma+1}\mathbb{Z}_3$. Their representatives in $\mathcal{B}_\sigma$ are unique. They are achieved simultaneously by choosing the unique required s and taking u to be the required constant with that augmentation. Once b,d have their normalized values, (E45.9) preserves them exactly when $s=0$ and $u(1)=0$. The remaining conjugations are therefore precisely $u\in U\mathscr{A}$. Formula (E42.12) then proves the asserted classification for $a\neq 0$.

For $a=0$, E.42 gives $v=cT^{-r}$ when r is integral, and gives $v=0, c=0$ when r is nonintegral. The remaining w is arbitrary of augmentation d, and the residual action is $w\mapsto w-D_b(u)$. The constant $w=d$ is an actual solution. Thus it remains to compute the exact compact quotient $U\mathscr{A}/D_b(U\mathscr{A})$ and supply representatives. We do that over the integral completed group ring itself.

Integral measures and the free orbit calculation. Identify $\mathscr{A}$ with $\mathbb{Z}_3[[\mathbb{Z}_3]]$ by $T^x\leftrightarrow\delta_x$. An element is an integral measure: its coefficients on the finite quotients $\mathbb{Z}/3^k$ give compatible $\mathbb{Z}_3$ values on clopen subsets. A continuous $\mathbb{Z}_3$-valued function is integrated by uniform limits of its locally constant approximations. The resulting integral exists because all coefficients have absolute value at most one. Augmentation is total mass. Under this identification, $T^b\psi_q$, with $\psi_q(f)=f(T^q)$, is pushforward by the actual affine map

$$\varphi(x)=qx+b. \tag{E45.10}$$

On a compact free orbit $\varphi^n(x_0)$, $n\in\mathbb{Z}_3$, the parameter n identifies its measure module with $\mathbb{Z}_3[[S-1]]$. The operator D_b becomes multiplication by $qS-1$. Evaluation at $S=q^{-1}$ gives an exact continuous sequence

$$0\longrightarrow \mathbb{Z}_3[[S-1]]\xrightarrow{qS-1}\mathbb{Z}_3[[S-1]]\xrightarrow{I}\mathbb{Z}_3\longrightarrow 0, \qquad I(\mu)=\int q^{-n}\,d\mu(n). \tag{E45.11}$$

Indeed the power series may be evaluated at $S-1=q^{-1}-1\in 3\mathbb{Z}_3$; division by the monic linear factor proves that the kernel is precisely $(qS-1)$. The first map is injective in the power-series domain. Dirac mass at $n=0$ proves surjectivity. This computes the orbit cokernel as an actual map of compact modules.

We shall use the elementary valuation identity

$$v_3(q^n-1)=\sigma+v_3(n)\quad(0\neq n\in\mathbb{Z}_3). \tag{E45.12}$$

For positive integers n it follows by binomial expansion: a unit exponent preserves the first valuation, and cubing an element of $1+3\mathbb{Z}_3$ raises it by exactly one. The identity for all n follows by continuity on each annulus. In particular $n\mapsto q^n$ is a homeomorphism from $\mathbb{Z}_3$ onto $1+3^\sigma\mathbb{Z}_3$. Surjectivity can be checked without a logarithm: (E45.12) makes the normalized map a bijection at every finite three-power residue level, and compactness passes to the inverse limit.

Integral fixed point and infinitely many orbit coordinates. Suppose $r\in\mathbb{Z}_3$. Multiplication by T^r is an automorphism of $U\mathscr{A}$, and a direct substitution gives

$$T^r D_b(f)=D_0(T^r f). \tag{E45.13}$$

We therefore compute D_0. The punctured compact space $\mathbb{Z}_3 - \{0\}$ is the disjoint union of clopen sets

$$A_{j,u} = 3^j(u + 3^\sigma \mathbb{Z}_3), \quad j \geq 0, \quad u \in (\mathbb{Z}/3^\sigma)^\times. \tag{E45.14}$$

By (E45.12) each is a single free compact orbit, parametrized by $x = 3^j u q^n$. Its map (E45.11) is concretely

$$I_{j,u}(\mu) = \int_{A_{j,u}} \frac{3^j u}{x}\, d\mu(x). \tag{E45.15}$$

The integrand takes integral unit values and is continuous on that clopen set.

There is an exact topological module isomorphism

$$U\mathscr{A} \xrightarrow{\sim} \prod_{j,u} \mathscr{A}(A_{j,u}), \qquad \mu \longmapsto (\mu|_{A_{j,u}})_{j,u}. \tag{E45.16}$$

Its inverse is the centered measure series

$$(\mu_{j,u}) \longmapsto \sum_{j,u} (\mu_{j,u} - \operatorname{aug}(\mu_{j,u})\delta_0). \tag{E45.17}$$

To check convergence, project to the group ring of $\mathbb{Z}/3^k$. Every term with $j \geq k$ becomes zero there, since its support and its compensating mass both reduce to zero. Only finitely many terms remain. This proves convergence and continuity in the actual profinite topology. It is the $(3, U)$-adic topology; no convergence in the U-adic topology alone is needed. The resulting measure has total mass zero and the prescribed restrictions. For uniqueness, a measure whose restrictions all vanish is supported at zero: testing any locally constant function shows it equals its total mass times δ_0. Total mass zero then makes it zero.

The isomorphism (E45.16) intertwines D_0 with its orbit operators. Indeed

$$D_0(\nu - m\delta_0) = D_0\nu - (q-1)m\delta_0, \qquad \operatorname{aug}(D_0\nu) = (q-1)m.$$

The product of the exact sequences (E45.11) therefore proves (E45.7), with no closure or extra kernel. The representative of the coordinate $z_{j,u}$ is $z_{j,u}(\delta_{3^j u} - \delta_0)$. Transporting it back by T^{-r} and adding d gives exactly (E45.3). Each orbit integral of this representative is its stated coordinate. This proves both surjectivity and uniqueness of the section representatives.

No integral fixed point and the finite-rank lattice. Suppose $r \notin \mathbb{Z}_3$, and put $t = v_3(b) < \sigma$. The compact affine orbits are now exactly

$$A_i = i + 3^t \mathbb{Z}_3, \qquad 0 \leq i < 3^t. \tag{E45.18}$$

For a proof, the continuous iterate is

$$\varphi^n(i) = q^n i + b\frac{q^n - 1}{q - 1}, \qquad \varphi^n(i) + r = q^n(i + r).$$

Here $v_3(i + r) = t - \sigma < 0$. Equation (E45.12) yields

$$v_3(\varphi^n(i) - \varphi^m(i)) = t + v_3(n - m).$$

Thus the normalized parameter is an isometry on every finite residue quotient and a homeomorphism onto A_i. Each orbit is free.

The finite partition into these orbits and (E45.11) give the explicit map

$$\mathscr{A}/D_b\mathscr{A} \xrightarrow{\sim} \mathbb{Z}_3^{3^t}, \qquad I_i(\mu) = \int_{A_i} \frac{i + r}{x + r}\, d\mu(x). \tag{E45.19}$$

The ratio is q^{-n} under the orbit parameter. It is congruent to one modulo $q-1$, so

$$\sum_i I_i(\mu) \equiv \operatorname{aug}(\mu) \pmod{q-1}. \tag{E45.20}$$

This sends $U\mathscr{A}$ into the lattice (E45.8).

We prove the exact kernel and image on $U\mathscr{A}$. For any lattice vector $I=(I_i)$, the measure

$$\mu_I = \sum_i I_i\delta_i - D_b\left(\frac{\sum_i I_i}{q-1}\delta_0\right) \tag{E45.21}$$

has augmentation zero and has weighted integrals I_i. It proves surjectivity. If all integrals of a measure $\mu \in U\mathscr{A}$ vanish, (E45.19) gives $\mu = D_b\nu$ in $\mathscr{A}$. Taking augmentations yields $0=(q-1)\operatorname{aug}(\nu)$, so $\nu \in U\mathscr{A}$. This is exactly the required kernel, proving (E45.8).

The lattice basis $(q-1)e_0, e_i-e_0$ has representatives

$$q(1-T^b), \qquad T^i-1, \quad 1\le i<3^t.$$

For the first, $\varphi(0)=b$ and its weighted integral is $q(1-q^{-1})=q-1$ on orbit zero; for the others the integrals are e_i-e_0. This proves (E45.5), including its uniqueness. All quotient maps above are continuous surjections between compact Hausdorff modules, and the displayed representative maps are continuous. Hence the asserted topologies also follow from these actual maps.

Finite certificates and their transition maps. The program `e_tame_classes.py` works on the exact finite group-ring lattices $\mathbb{Z}_3[\mathbb{Z}/3^k]$ in the augmentation basis $\delta_j-\delta_0$. For an affine cycle of length ℓ, the full coinvariant is $\mathbb{Z}_3/(q^\ell-1)$ by the weighted coordinate $\sum_n q^{-n}a_n$. If the cycle exponents $e_i=\sigma+v_3(\ell_i)$ are ordered increasingly, the augmentation-kernel coinvariants have exponents

$$e_1-\sigma, e_2, \dots, e_f, \tag{E45.22}$$

with a zero exponent omitted. This follows directly from the kernel of the sum map to $\mathbb{Z}_3/(q-1)$: use the first coordinate multiplied by $q-1$ and the differences from that coordinate. The program independently verifies the Smith matrices and this formula in 48 cases.

It also verifies 36 actual transition maps from level $k+1$ to k, rather than only comparing invariant factors. If a higher cycle starts at a point whose projection has lower orbit parameter n_0, its weighted coordinate contributes q^{-n_0} to that lower coordinate, modulo the lower cycle's $q^\ell-1$. The program checks this formula on every Dirac mass, and checks that reduction of the group ring commutes with D_b. The proposed representatives in (E45.3) and (E45.5) are independently verified to generate every finite augmentation coinvariant. Three complete modular Smith transformation certificates are retained in the JSON output. The measure proofs above establish the all-level isomorphisms themselves.

Finally the actual quotient maps of E.38 send every intrinsic local section to one of the classes just computed. Their specified geometric kernels remain part of this comparison. The additional intrinsic lifting tests are E.37 and E.44, with their genuine arithmetic actions; the computation here completes exactly the local section set of H. □

Global section coordinates by actual norm systems

E.46. Every global fiber for the inherited module quotient, and explicit ramified families [Proved]

Use the actual arithmetic quotient of E.38,

$$\Lambda=\mathbb{Z}_3[[U]], \quad T=1+U, \quad J=(3,U)\Lambda, \quad H=3\mathbb{Z}_3\ltimes J, \qquad g(a,v)=(\chi_g a, \chi_g v(T^{\chi_g})). \tag{E46.1}$$

Fix a continuous first-coordinate cocycle $A \in Z^1(G_{\mathbb{Q}}, 3\mathbb{Z}_3(1))$. For $n \geq 0$ let $X_{A,n} = \mathbb{Z}/3^n$ with the actual action

$$gi = A_g + \chi_g i \pmod{3^n}, \qquad E_{A,n} = \mathrm{Map}(X_{A,n}, \overline{\mathbb{Q}})^{G_{\mathbb{Q}}}. \tag{E46.2}$$

The action on functions is $(gf)(i) = g(f(g^{-1}i))$. The reduction $X_{A,n+1} \to X_{A,n}$ induces an inclusion of finite etale algebras $E_{A,n} \to E_{A,n+1}$ of rank three, with its usual multiplicative norm. In particular $E_{A,0} = \mathbb{Q}$. For a finite product of number fields E write

$$\widehat{E^\times}_3 = \varprojlim_m E^\times/(E^\times)^{3^m}, \qquad \mathcal{N}_A = \left\{(u_n)_n : u_n \in \widehat{E^\times_{A,n3}},\ N_{E_{A,n+1}/E_{A,n}} u_{n+1} = u_n\right\}. \tag{E46.3}$$

These completions carry the inverse-limit topology of the discrete power-class groups. They are not assumed compact. All the transition maps in (E46.3) are the indicated norms, including when either algebra is a product.

The global sections of $H \rtimes G_{\mathbb{Q}}$ whose first coordinate has class $[A]$, modulo conjugation by H, are exactly

$$\boxed{\mathcal{N}_A^{(3)} = \{(u_n)_n \in \mathcal{N}_A : u_0 \in (\widehat{\mathbb{Q}^\times}_3)^3\}.} \tag{E46.4}$$

This is an isomorphism of topological abelian groups after taking the section $g \mapsto (A_g, 0)$ as origin. The proof gives the map and its inverse, the exact change-of-origin maps, and the continuous-cochain topology. No surjectivity of the norm maps is needed. Every first-coordinate class occurs.

For a rational $a \in \mathbb{Q}^\times$, take $A = 3\kappa_3(a)$ with compatible Kummer roots. Then

$$E_{A,n} = \mathbb{Q}[X_n]/(X_n^{3^n} - a^3), \qquad X_n \longmapsto X_{n+1}^3. \tag{E46.5}$$

For $a^3 \neq 1$ and every $r, z \in \mathbb{Z}_3$, the explicit tuple

$$u_{n,r,z} = X_n^r (1 - X_n)^{3z} \tag{E46.6}$$

defines a section class in (E46.4). Its image under the actual homomorphism

$$H \longrightarrow \mathbb{Z}_3(1)^2, \qquad (a, v) \longmapsto \big(a/3, \mathrm{aug}(v)/3\big) \tag{E46.7}$$

is

$$\big(\kappa_3(a), r\kappa_3(a) + z\kappa_3(1 - a^3)\big). \tag{E46.8}$$

The parameter z distinguishes the subfamily $r = 0$. For $a = 2$, both parameters are distinguished by the valuations at two and seven, giving an embedded $\mathbb{Z}_3^2$ family in this fiber. In particular $a = 2, r = 0, z = 1$ gives the rational toric Kummer pair $(2, -7)$ in this quotient.

For this same first coordinate $a = 2$, the entire norm system in (E46.4), including its field factors, has the following exact description. For $n \geq 1$ choose

$$\alpha_n = 2^{1/3^{n-1}}, \qquad \beta_n = \alpha_n \zeta_{3^n}, \qquad F_n = \mathbb{Q}(\alpha_n), \qquad L_n = \mathbb{Q}(\beta_n). \tag{E46.9}$$

Then

$$\begin{gathered} E_{A,n} = F_n \times L_n, \quad [F_n : \mathbb{Q}] = 3^{n-1}, \quad [L_n : \mathbb{Q}] = 2 \cdot 3^{n-1}, \quad L_1 = \mathbb{Q}(\zeta_3), \\ \alpha_{n+1}^3 = \alpha_n, \qquad \beta_{n+1}^3 = \beta_n. \end{gathered} \tag{E46.10}$$

Thus (E46.4) consists precisely of two sequences $v_n \in \widehat{F_n^\times}_3$ and $w_n \in \widehat{L_n^\times}_3$, each compatible under the displayed degree-three field norms, with

$$v_1 N_{L_1/\mathbb{Q}}(w_1) \in (\widehat{\mathbb{Q}^\times}_3)^3. \tag{E46.11}$$

Formula (E46.6) is the pair $(\alpha_n^r(1-\alpha_n)^{3z}, \beta_n^r(1-\beta_n)^{3z})$ in these coordinates.

Manuscript: §51.16, folios733v–736v, scans172–178, Part E(c),(d). The arithmetic group here is the quotient H in E.38. The actual maps from the intrinsic metabelian and marked groups have exactly the kernels (E38.9); (E46.4) classifies their target section set. It does not supply a lift through either of those kernels.

Proof. The twisted module and geometric conjugation. Write a section cocycle as (A_g, b_g). The multiplication in (E46.1) gives

$$b_{gh} = b_g + \chi_g T^{A_g} b_h(T^{\chi_g}). \tag{E46.12}$$

Define Λ_A to be the compact additive group Λ with action

$$g \cdot_A f = \chi_g T^{A_g} f(T^{\chi_g}), \tag{E46.13}$$

and let J_A be its stable submodule J. The cocycle identity for A verifies the action law in (E46.13). Thus sections with their first cocycle equal to A have precisely the second cocycles $Z^1(G_{\mathbb{Q}}, J_A)$.

For completeness, conjugating by an actual element $(s,u) \in H$ gives

$$\begin{aligned} A'_g &= A_g + (1-\chi_g)s, \\ b'_g &= T^s b_g + u - \chi_g T^{A'_g} u(T^{\chi_g}). \end{aligned} \tag{E46.14}$$

The first coordinate is unchanged exactly when $s = 0$, since complex conjugation has $\chi = -1$ and $3\mathbb{Z}_3$ is torsion-free. The remaining identifications in (E46.12) are exactly the J_A coboundaries. Every cocycle whose first class is $[A]$ can be brought to first cocycle A by the unique required s. This change is continuous: evaluate the first-coordinate difference at complex conjugation and divide by two. Consequently the fiber in the statement is precisely $H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, J_A)$, with its quotient topology and its origin $b = 0$.

Equation (E46.14) also supplies the exact comparison between representatives of a first-coordinate class. If $A'_g = A_g + (1-\chi_g)s$, the map $i \mapsto i + s$ is an equivariant bijection $X_{A,n} \to X_{A',n}$. Multiplication by T^s intertwines Λ_A and $\Lambda_{A'}$ and induces these bijections at every positional level. The corresponding algebra isomorphisms commute with the norms. Thus the description of the fiber is transported by explicit maps, rather than by an unspecified normalization.

Finite positional modules and Shapiro's map. Set

$$M_{A,n} = \mathbb{Z}_3(1) \otimes_{\mathbb{Z}_3} \mathbb{Z}_3[X_{A,n}], \qquad g(\delta_i) = \chi_g \delta_{A_g + \chi_g i}. \tag{E46.15}$$

The coefficient transition sends δ_i to $\delta_{i \bmod 3^n}$. The actual group-ring identification is

$$\Lambda_A = \varprojlim_n M_{A,n}, \qquad T^i \longmapsto \delta_{i \bmod 3^n}. \tag{E46.16}$$

Its topology is the inverse limit over both the positional level and the coefficient modulus 3^m, equivalently the $(3, U)$-adic topology. The coefficient projections are surjective and split continuously as maps of $\mathbb{Z}_3$-modules; equivariance of a chosen splitting is not required.

Decompose $X_{A,n}$ into orbits and choose a point i_0 in each. Its stabilizer is G_F, where F is the corresponding field factor of $E_{A,n}$. For a cocycle $c : G_{\mathbb{Q}} \to M_{A,n}$, Shapiro's map is explicitly

$$h \longmapsto c(h)_{i_0} \quad (h \in G_F). \tag{E46.17}$$

Here the subscript denotes the coefficient of δ_{i_0}. To verify the map and its inverse directly, choose g_i with $g_i i_0 = i$ and $g_{i_0} = 1$. A continuous cocycle $z : G_F \to \mathbb{Z}_3(1)$ induces

$$c_z(g)_i = \chi(g_i)\, z(g_i^{-1} g g_{g^{-1}i}). \tag{E46.18}$$

The argument of z fixes i_0. Its product identity proves the cocycle equation, and (E46.17) recovers z. Conversely, applying (E46.18) to (E46.17) gives $c_z = c + \delta m$, where $m_i = c(g_i)_i$ and $\delta m(g) = gm - m$. These formulas are continuous and show the precise degree-one Shapiro isomorphism. They agree with [Mil20b, II, Proposition1.11 and Example1.27(a)].

The Kummer sequence and Hilbert's theorem 90 [Mil20b, II, Proposition1.22 and its proof] give

$$H^1(G_F, \mu_{3^m}) = F^\times/(F^\times)^{3^m}.$$

Passing to $\mathbb{Z}_3(1)$ gives $H^1_{\mathrm{cts}}(G_F, \mathbb{Z}_3(1)) = \widehat{F^\times}_3$. Indeed the finite groups $\mu_{3^m}(F)$ satisfy the Mittag–Leffler condition, so their first inverse-limit derivative is zero. The continuous-cochain comparison is [Jan88, Proposition1.6 and Theorem2.2]. The same conclusion follows by aligning the finite Kummer cocycles and taking their continuous limit. Thus (E46.17) supplies actual isomorphisms

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, M_{A,n}) \simeq \widehat{E^\times_{A,n}}_3. \tag{E46.19}$$

Under these maps the positional pushforward is corestriction, and therefore the multiplicative norm. Here is a direct check of this identification. For a field inclusion $F \subset L$, choose a 3^m-th root of $u \in L^\times$ and multiply its conjugates over the embeddings of L into $\overline{\mathbb{Q}}$ over F. Their product is a 3^m-th root of $N_{L/F}u$. The Galois ratios of this product are exactly the sum of the transferred Kummer cocycles, with the coefficient twists in (E46.18). A different choice of coset representatives changes the root by the corresponding coboundary. For products of fields, multiply the norms of the factors. This proves the assertion at every finite coefficient level and hence after completion. It is the explicit transfer construction of [Mil20b, II, Example1.29].

The direction of the residual deck action can also be read from these maps. When $A = 0$, write $D_\gamma(\delta_i) = \delta_{\gamma i}$ on a unit orbit, and choose g with $\chi_g = \gamma$. For h fixing the chosen unit root, the cocycle identity gives

$$\chi_g c(g^{-1}hg)_1 = c(h)_\gamma + (\chi_h - 1)c(g)_\gamma.$$

The left side is the usual Galois action on its Kummer cohomology class. The right side is Shapiro evaluation of $D_{\gamma^{-1}}c$, up to the displayed coboundary. Consequently the usual field action corresponds to inverse deck translation. This explicitly fixes the convention for the cyclotomic norm system in E.47.

The positional inverse limit without a hidden term. For every number field F, $H^0(G_F, \mathbb{Z}_3(1)) = 0$: its group of roots of unity is finite, so it contains no compatible nonzero Tate-module element. By the degree-zero version of (E46.17),

$$H^0(G_{\mathbb{Q}}, M_{A,n}) = 0 \quad \text{for every } n. \tag{E46.20}$$

We now prove the needed inverse-limit statement on actual cocycles. Given compatible classes $[c_n] \in H^1(G_{\mathbb{Q}}, M_{A,n})$, choose a representative c_0. Inductively, the projection of a representative at level $n+1$ differs from c_n by a coboundary δb_n. Lift b_n through the surjective coefficient projection and subtract its coboundary. This makes the cocycles compatible. Their limit is continuous by (E46.16), and is a cocycle in Λ_A.

If a compatible cocycle is a coboundary at every level, write $c_n = \delta b_n$. The differences between the projection of b_{n+1} and b_n are invariant and hence zero by (E46.20). Thus the b_n themselves have a compatible limit, which bounds the original cocycle. This proves

$$H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, \Lambda_A) \xrightarrow{\sim} \varprojlim_n H^1_{\mathrm{cts}}(G_{\mathbb{Q}}, M_{A,n}) \xrightarrow{\sim} \mathcal{N}_A. \tag{E46.21}$$

In particular no first inverse-limit derivative of degree-zero cohomology has been discarded, and the proof makes no assertion that a norm map on multiplicative classes is surjective.

These maps also prove the claimed topology. Equip continuous cocycles with uniform convergence in finite coefficient-and-positional quotients, then take the quotient by continuous coboundaries. If a cocycle becomes a coboundary in one such finite quotient, its bounding element lifts to Λ_A. Subtracting that lifted coboundary gives a representative zero in that quotient. Therefore the kernels of the finite cohomology maps give exactly the quotient neighborhoods. Under (E46.19) these are the power-class neighborhoods in (E46.3). The algebraic isomorphism (E46.21) is consequently a homeomorphism. This argument permits the noncompact Kummer completions of number fields.

The exact total-norm condition. Augmentation modulo three gives the exact sequence of continuous modules

$$0 \longrightarrow J_A \longrightarrow \Lambda_A \xrightarrow{\text{aug mod } 3} \mathbb{F}_3(1) \longrightarrow 0. \tag{E46.22}$$

The last action is the cyclotomic action; the translation in (E46.13) has augmentation one. Since $\mu_3(\mathbb{Q}) = \{1\}$, $H^0(G_{\mathbb{Q}}, \mathbb{F}_3(1)) = 0$. Thus $H^1(J_A)$ injects into $H^1(\Lambda_A)$, with image the kernel of

$$H^1(\Lambda_A) \longrightarrow H^1(\mathbb{F}_3(1)) = \mathbb{Q}^\times/(\mathbb{Q}^\times)^3. \tag{E46.23}$$

Under (E46.21), this is exactly u_0 modulo cubes: augmentation is the positional map to $n = 0$, which is the total norm from every $E_{A,n}$. The kernel of $\widehat{\mathbb{Q}^\times}_3 \to \mathbb{Q}^\times/(\mathbb{Q}^\times)^3$ is $(\widehat{\mathbb{Q}^\times}_3)^3$. One can check this without an exactness assumption on completion: the prime valuations identify $\widehat{\mathbb{Q}^\times}_3$ with the sequences $(v_p)_p \in \prod_p \mathbb{Z}_3$ tending to zero in the three-adic sense. Reduction modulo three has finite support, and a sequence reducing to zero can be divided by three and still tends to zero. The rational sign disappears because cubing is an automorphism of $\{\pm 1\}$. This proves (E46.4).

The injection in (E46.23) is also a topological identification with its kernel. A cocycle whose reduction is a coboundary has a unique bounding element in $\mathbb{F}_3(1)$, since its invariant group is zero. Choose a fixed lift of each of the three possible elements. Subtracting its coboundary continuously turns the cocycle into a J_A cocycle. This is an explicit continuous inverse on the kernel.

Rational first coordinates and actual ramified sections. Choose compatible 3^n-th roots a_n of a and compatible ζ_{3^n}. The roots $a_n^3\zeta_{3^n}^i$ of $X^{3^n} - a^3$ transform by

$$g(a_n^3\zeta_{3^n}^i) = a_n^3\zeta_{3^n}^{3\kappa_3(a)(g)+\chi_g i}.$$

Their cube map is the reduction in (E46.2). This proves precisely the finite-etale algebra and its transition in (E46.5).

The element $1 - X_n$ is invertible in that algebra because its total norm is $1 - a^3 \neq 0$. In the rank-three extension given by $Y^3 = X_n$, the determinant of multiplication by Y is X_n, and that of multiplication by $1 - Y$, in the basis $1, Y, Y^2$, is

$$\det \begin{pmatrix} 1 & 0 & -X_n \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} = 1 - X_n. \tag{E46.24}$$

This determinant proves the norm identity even when the algebras have several factors. Taking continuous $\mathbb{Z}_3$ powers and products proves compatibility of (E46.6), with $u_{0,r,z} = a^{3r}(1 - a^3)^{3z}$. The construction (E46.21)–(E46.23) therefore gives an actual continuous H section class. Augmentation and (E46.7) give (E46.8). For rational $a \neq 0$ with $a^3 \neq 1$, the number $1 - a^3$ is neither 1 nor -1: these alternatives would give $a = 0$ or $a^3 = 2$, respectively. Its nonzero rational prime-valuation vector therefore makes $z \mapsto z\kappa_3(1 - a^3)$ injective. For $a = 2$, the

valuations at two and seven distinguish $r\kappa_3(2) + z\kappa_3(-7)$ for all pairs (r, z). This distinguishes the stated families through an actual quotient homomorphism. No identification with an unspecified rational-point path cocycle is used.

For $a = 2$, the factor $X^{3^{n-1}} - 2$ is Eisenstein at two. The complementary roots are the conjugates of β_n in (E46.9). This element satisfies $\beta_n^{3^{n-1}} = 2\zeta_3$ and contains $\mathbb{Q}(\zeta_3)$, since $\beta_n^{3^{n-1}}/2 = \zeta_3$. The prime two is unramified in $\mathbb{Q}(\zeta_3)$; in fact $X^2 + X + 1$ is irreducible modulo two. Over that field $X^{3^{n-1}} - 2\zeta_3$ is Eisenstein at its prime above two. It gives exactly the second degree in (E46.10). The sum of the two degrees is 3^n, proving the full factorization. Cubing respects both factors and gives the displayed tower maps. At $n = 1$, the total norm is $v_1 N_{L_1/\mathbb{Q}} w_1$, proving (E46.11). In particular $N_{\mathbb{Q}(\zeta_3)/\mathbb{Q}}(1 - 2\zeta_3) = 7$ and $1 - 2 = -1$, so the explicit family has total norm $2^{3r}(-7)^{3z}$, as asserted.

Finite verification. The program `e_norm_sections.py` verifies the actual affine positional actions, their two orbits for $a = 2$, and the intertwining of all tested coefficient pushforwards. It implements (E46.18) in the explicit finite splitting group of the ninth-root and twenty-seventh-root-of-unity tower, checks its cocycle identities, and checks the corestriction-to-norm formula on the nontrivial Kummer class of $2\zeta_3$ and on the rational class of 2. Its 78732 cocycle identities and 13122 inverse-deck comparisons all pass. It also verifies (E46.24) and the total-norm and factorization identities at the listed finite levels. These computations retain the scalar cyclotomic factor and the direction of the root-cubing maps. The proofs above give the asserted all-level comparison. □

The entire global zero-first-coordinate family in the inherited module quotient

E.47. Global section coordinates from explicit norm-coherent cyclotomic units [Proved]

Use the actual arithmetic group H and the unit tangential origin of (E38.5)–(E38.8). Write

$$\mathscr{A} = \mathbb{Z}_3[[\mathbb{Z}_3]] = \mathbb{Z}_3[[U]], \qquad T = 1 + U, \qquad J = (3, U)\mathscr{A}, \qquad H = 3\mathbb{Z}_3 \ltimes J. \tag{E47.1}$$

Here $T^x = \delta_x$, multiplication by T translates the measure variable, and $g(a, v) = (\chi(g)a, \chi(g)v(T^{\chi(g)}))$. Let $\mathrm{Sec}_0(H)$ denote the geometric conjugacy classes of global sections whose image in $H^1(\mathbb{Q}, 3\mathbb{Z}_3(1))$ is zero. There is an explicit isomorphism of pointed topological spaces, and a group structure specified by its right side,

$$\mathrm{Sec}_0(H) \xrightarrow{\sim} 3\widehat{\mathbb{Q}^\times}_3 \times \prod_{j \geq 0} (\mathbb{Z}_3(1) \oplus \mathbb{Z}_3[[\Gamma]]), \qquad \Gamma = 1 + 3\mathbb{Z}_3. \tag{E47.2}$$

The completion means exactly $\widehat{F^\times}_3 = \varprojlim_m F^\times/(F^\times)^{3^m}$. In particular the first factor is the group of restricted prime-valuation sequences

$$3\widehat{\mathbb{Q}^\times}_3 = \left\{ (3b_p)_p \in \prod_{p \text{ prime}} 3\mathbb{Z}_3 : \text{for every } m,\ b_p \in 3^m\mathbb{Z}_3 \text{ for all but finitely many } p \right\}, \tag{E47.3}$$

with its inverse-limit topology, not an asserted compact topology. The factors indexed by j in (E47.2) are compact; their displayed product is the full product.

The coordinate maps are as follows. First conjugate a section so its first cocycle is identically zero. This normalization is unique in the first geometric coordinate. Its second coordinate is a cocycle $v : G_\mathbb{Q} \to J$. Take its augmentation $\nu(g) = v(g)(1)$, and for each $j \geq 0$ restrict its measure to $3^j\mathbb{Z}_3^\times$ and scale that set by 3^{-j}. Under the exact Shapiro and norm maps of E.46, the latter class is the compatible system

$$(u_{j,n})_{n \geq 1} \in \varprojlim_{N_{K_{n+1}/K_n}} \widehat{K_n^\times}_3, \qquad K_n = \mathbb{Q}(\zeta_n), \quad \zeta_{n+1}^3 = \zeta_n, \quad \mathrm{ord}(\zeta_n) = 3^n. \tag{E47.4}$$

Every such system has a unique expression

$$\begin{aligned} u_{j,n} &= \zeta_n^{\,a_j \bmod 3^n} \prod_{\gamma\in\Gamma_n} \gamma(\epsilon_n)^{\,b_{j,n,\gamma}}, \\ \epsilon_n &= (1-\zeta_n)(1-\zeta_n^{-1}), \qquad \Gamma_n = \Gamma/(1+3^n\mathbb{Z}_3), \\ a_j &\in \mathbb{Z}_3, \qquad b_j = (\sum_{\gamma\in\Gamma_n} b_{j,n,\gamma}[\gamma])_n \in \mathbb{Z}_3[[\Gamma]]. \end{aligned} \tag{E47.5}$$

Three-adic powers are taken in the indicated multiplication-by-3^m completion. Equations (E47.4)–(E47.5), together with ν, are the coordinates in (E47.2). The root-of-unity line has its usual Tate action under the natural field action of Γ. On measures this action is the inverse deck action $\delta_x \mapsto \delta_{\gamma^{-1}x}$; the proof specifies the resulting involution on the group ring.

The two-coordinate augmentation map $H \to (3\mathbb{Z}_3(1))^2$, $(a,v)\mapsto(a,\operatorname{aug} v)$, and the scaled toric map (E46.7) have, respectively, the following exact maps on the displayed parameters:

$$\begin{aligned} (\nu,(a_j,b_j)_{j\ge0}) &\longmapsto (0,\nu) \in H^1(\mathbb{Q},(3\mathbb{Z}_3(1))^2), \\ (\nu,(a_j,b_j)_{j\ge0}) &\longmapsto (0,\nu/3) \in H^1(\mathbb{Q},\mathbb{Z}_3(1)^2). \end{aligned} \tag{E47.6}$$

Consequently every such toric coordinate occurs, and each fiber is the displayed compact product in (E47.2). Every annular coordinate is unramified outside 3. Its independent arithmetic evaluation is the explicit free-group presentation

$$\begin{aligned} H^1(\mathbb{Q},\mathbb{Z}_3(1)\otimes\mathbb{Z}_3[[\mathbb{Z}_3^\times]]) \simeq \frac{B\oplus B}{\{(dz,dz): z\in B\}} &\xrightarrow{\sim} B\oplus\mathbb{Z}_3, \\ B=\mathbb{Z}_3[[\Gamma]], \qquad d=7S_7-1, & \\ [(a,b)] \longmapsto \big(a-b,\ b\big|_{S_7=1/7}\big). & \end{aligned} \tag{E47.7}$$

In this formula S_γ denotes direct positional translation $\delta_x\mapsto\delta_{\gamma x}$. Thus the natural field action on the last line is S_γ^{-1}, acting as multiplication by γ.

The actual geometric maps being evaluated are $N \twoheadrightarrow H$ with kernel $J_y=[I_y,I_y]$, and

$$N/N'' \twoheadrightarrow H, \quad \ker = J_y/N'', \qquad N/\gamma_3 I_{\text{marked}} \twoheadrightarrow H, \quad \ker = J_y/\gamma_3 I_{\text{marked}}, \tag{E47.8}$$

for the markings in (E38.9). The parameterization (E47.2) evaluates the target of these precise maps.

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d). The finite program `e_global_zero.py` *checks the cyclotomic elements, their actual norm maps, and the compatible finite presentations underlying (E47.7).*

Proof. Normalization, the centered decomposition, and topology. If the first cocycle has zero class, it equals $g\mapsto(\chi(g)-1)s$ for some $s\in3\mathbb{Z}_3$. Conjugation by $(s,0)$ makes it zero. This s is unique: complex conjugation acts by -1, and $3\mathbb{Z}_3$ has no two-torsion. In fact $s=-a(c)/2$, so this normalization is continuous. Once the first cocycle is zero, a geometric conjugation preserving it has first coordinate zero and changes v by the ordinary coboundary of an element of J. Thus the normalized section set is exactly the abelian cohomology quotient $H^1(\mathbb{Q},J)$, including its uniform continuous-cochain quotient topology.

There is an actual continuous $G_\mathbb{Q}$-module isomorphism

$$\begin{aligned} J &\longrightarrow 3\mathbb{Z}_3(1)\delta_0 \oplus \prod_{j\ge0} M_j, \\ \mu &\longmapsto (\operatorname{aug}\mu,\ (\mu|_{3^j\mathbb{Z}_3^\times})_j), \\ M_j &= \mathbb{Z}_3(1)\otimes\mathbb{Z}_3[[3^j\mathbb{Z}_3^\times]]. \end{aligned} \tag{E47.9}$$

Its inverse is the centered series

$$(\nu, (\mu_j)_j) \longmapsto \nu\delta_0 + \sum_{j\geq 0}(\mu_j - \mathrm{aug}(\mu_j)\delta_0). \tag{E47.10}$$

In the positional quotient $\mathbb{Z}/3^k$ every summand with $j \geq k$ is zero. Hence this series converges in the completed group-ring topology for arbitrary choices of its factors; no sum of their uncentered augmentations is required. Restricting the sum to each annulus recovers that factor, and its augmentation is ν. Conversely a measure of augmentation zero whose restrictions to all the annuli vanish has support at zero and is zero. These checks prove both inverse identities and continuity. Multiplication of the variable by a cyclotomic unit preserves every annulus, so the maps are equivariant, including their Tate twist. Scaling identifies each M_j with the module M in (E47.7).

Continuous cochains into a product are precisely products of continuous cochains, and the uniform cochain topology into that product is the product of the uniform topologies. Cocycle equations and coboundaries are componentwise; the quotient maps are open. Therefore (E47.9) induces an isomorphism of topological cohomology groups with the product of the cohomology groups of its factors. The first factor is $3\widehat{\mathbb{Q}^\times}_3$ by Kummer theory. Indeed $\mathbb{Q}^\times/\{\pm 1\}$ is the free abelian group on the rational primes, and completion gives precisely (E47.3). The inclusion $3\mathbb{Z}_3(1) \hookrightarrow \mathbb{Z}_3(1)$ induces multiplication by three on this group.

The exact annular Shapiro maps. At positional level $n \geq 1$, the unit positions are $(\mathbb{Z}/3^n)^\times$. Their stabilizer at one is G_{K_n}. Shapiro evaluation at that position gives

$$H^1(\mathbb{Q}, \mathbb{Z}_3[(\mathbb{Z}/3^n)^\times](1)) \xrightarrow{\sim} \widehat{K_n^\times}_3. \tag{E47.11}$$

Projection of unit positions induces corestriction; the Kummer formula for corestriction is the product of conjugate roots, so the map is exactly the field norm. E.46 proves the inverse-limit isomorphism by aligning continuous cocycles and coboundaries. The relevant degree-zero groups vanish because no number field contains a compatible sequence of roots of unity of unbounded three-power order. That proof does not require compactness of $\widehat{K_n^\times}_3$.

Here is the action convention explicitly. For a cocycle c at positional level n, a lift g with $\chi(g) = \gamma$, and $h \in G_{K_n}$, the cocycle identity gives

$$\gamma\, c(g^{-1}hg)_1 = c(h)_\gamma + (\chi(h) - 1)c(g)_\gamma. \tag{E47.12}$$

The last term is a coboundary. Consequently the natural field action by γ corresponds to the inverse deck translation, whose coefficient at one is $c(h)_\gamma$. The comparison between the direct positional ring and the group ring in (E47.5) is the explicit involution $S_\gamma \mapsto [\gamma^{-1}]$.

The three-primary class groups in this tower vanish. The ring $\mathbb{Z}[\zeta_1]$ is Euclidean for its norm, so K_1 has class number one. For K_{n+1}/K_n there is one ramified finite prime, the unique prime above three, and it is totally ramified. There is no infinite-place ramification. These facts follow directly from $\Phi_{3^n}(X) = X^{2\cdot 3^{n-1}} + X^{3^{n-1}} + 1$: $\Phi_{3^n}(X+1)$ is Eisenstein at three, and the polynomial discriminant is, up to sign, $3^{(2n-1)3^{n-1}}$. The latter identity follows by taking norms of $\Phi'_{3^n}(\zeta_n) = 3^n\zeta_n^{-1}/(\zeta_n^{3^{n-1}} - 1)$. The field discriminant divides it, excluding ramification at every other finite prime.

Apply the exact ambiguous class number formula of [Lem13, Theorem 1, p. 1 of the opened preprint] to this cyclic extension of degree three. Its ramification factor is $3^{1-1} = 1$, so

$$|\,\mathrm{Cl}(K_{n+1})^{C_3}| = \frac{h(K_n)}{[\mathcal{O}_{K_n}^\times : \mathcal{O}_{K_n}^\times \cap N_{K_{n+1}/K_n}K_{n+1}^\times]}. \tag{E47.13}$$

If $3 \nmid h(K_n)$, the integer on the left is prime to three. A nonzero finite three-group acted on by C_3 has a nonzero fixed element: nonfixed orbits have size three, so the number of fixed

elements is divisible by three and already includes zero. Applied to the three-primary class group, this proves it is zero. Induction proves $3 \nmid h(K_n)$ for every n. This is also the one-prime class-group argument proved in [Shaa, Lemma 1.3.13].

Put $F_n = K_n^+$. Extension of ideals followed by norm multiplies $\mathrm{Cl}(F_n)$ by two. It is therefore injective on its three-primary subgroup, whose image in $\mathrm{Cl}(K_n)$ is zero. Thus $3 \nmid h(F_n)$ as well.

Every norm-coherent class is a three-unit. Fix a rational prime $p \neq 3$. Its decomposition subgroup in Γ is open: its positive cyclotomic component has infinite order, since no positive integral power of p is 1 or -1 as a rational, hence as a three-adic number. After a finite level the number of primes over p is constant and their residue degrees in successive layers are three. For a compatible system u_n, norm compatibility therefore makes the valuation at each fixed finite-level prime divisible by arbitrarily large powers of three. That valuation is in $\mathbb{Z}_3$, so it is zero.

This valuation argument applies to the completed multiplicative groups themselves. To verify the kernel, let $\mathcal{U}_n = \mathcal{O}_{K_n}[1/3]^\times$. The image of the away-from-three divisor map $K_n^\times \to \bigoplus_{\mathfrak{p}\nmid 3} \mathbb{Z}$ is a free abelian subgroup with cokernel the finite S-class group, of order prime to three. The sequence with kernel $\mathcal{U}_n$ splits as a sequence of abstract abelian groups because its image is free. Completion is exact on this split sequence; the inclusion of that image into the divisor group induces an isomorphism on three-completions, since its finite index is a three-adic unit. Hence the kernel of all the completed valuations is exactly $\widehat{\mathcal{U}_{n3}}$. We have proved the actual equality of inverse limits

$$\varprojlim_N \widehat{K_n^\times}_3 = \varprojlim_N \widehat{\mathcal{U}_{n3}}. \tag{E47.14}$$

A basis of the completed three-units and its norms. At level n the real field has degree $r_n = 3^{n-1}$. The element ϵ_n generates its unique prime above three: its valuation there is one, and its absolute norm is three. Let $\epsilon_{n,a} = (1-\zeta_n^a)(1-\zeta_n^{-a})$ for the r_n unit classes modulo signs. The normalized real circular units are

$$\eta_{n,a} = \zeta_n^{(1-a)/2} \frac{1-\zeta_n^a}{1-\zeta_n}, \qquad \frac{\epsilon_{n,a}}{\epsilon_n} = \eta_{n,a}^2. \tag{E47.15}$$

The half exponent is read modulo 3^n. The opened proof of the Hasse index theorem, [Shab, Theorem 4.3.3 and its proof, pp. 108–109], states that these $r_n - 1$ units form a basis of the real circular units modulo $\{\pm 1\}$ and that their index in the full real units is $h(F_n)$. Thus the subgroup generated by $\{\pm 1\}$ and the $\epsilon_{n,a}$ has the exact index

$$2^{r_n-1} h(F_n) \quad \text{in } \mathcal{O}_{F_n}[1/3]^\times. \tag{E47.16}$$

Indeed ϵ_n accounts for the valuation at three, while the ratios in (E47.15) account for the unit kernel. This index is prime to three. Both the number of generators and the rank of the real three-units are r_n, so

$$\mathbb{Z}_3[\Gamma_n] \longrightarrow \widehat{\mathcal{O}_{F_n}[1/3]^\times}_3, \qquad \sum b_\gamma[\gamma] \longmapsto \prod \gamma(\epsilon_n)^{b_\gamma} \tag{E47.17}$$

is an isomorphism of compact $\mathbb{Z}_3$-modules.

Complex conjugation gives integral plus and minus projectors $(1 \pm c)/2$ on the three-completion. Its plus part is precisely the completion in (E47.17): norm followed by inclusion acts as two on invariants. For $u \in \mathcal{U}_n$, the quotient $u/\bar{u}$ has valuation zero at every finite prime, because the only allowed prime is fixed by conjugation, and every conjugate has complex absolute value one. It is a root of unity. For completeness this last assertion follows by applying boundedness of the coefficients of minimal polynomials to all its powers: only finitely many

such integral polynomials occur, so two powers coincide. The minus part of the completion is consequently μ_{3^n}. We have proved the actual direct-sum decomposition

$$\widehat{\mathcal{U}_{n3}} = \mu_{3^n} \oplus \langle \gamma(\epsilon_n) : \gamma \in \Gamma_n \rangle_{\mathbb{Z}_3} . \tag{E47.18}$$

The relative Galois group has exponents $1, 1+3^n, 1+2\cdot 3^n$. Multiplication gives

$$N\zeta_{n+1} = \zeta_n, \qquad N(1-\zeta_{n+1}) = 1-\zeta_n, \qquad N\epsilon_{n+1} = \epsilon_n. \tag{E47.19}$$

The middle identity is also proved as [Shab, Lemma 4.3.6]. Norm commutes with the abelian field action, so on (E47.18) it is reduction $\mathbb{Z}/3^{n+1} \to \mathbb{Z}/3^n$ on the root exponents and the group-ring projection $\mathbb{Z}_3[\Gamma_{n+1}] \to \mathbb{Z}_3[\Gamma_n]$ on the real unit exponents. There is no extra factor of three in this latter map: each conjugate generator maps to its conjugate generator downstairs. Taking inverse limits proves (E47.5), its uniqueness, and

$$\varprojlim_N \widehat{K_n^\times}_3 \simeq \mathbb{Z}_3(1) \oplus \mathbb{Z}_3[[\Gamma]]. \tag{E47.20}$$

This also constructs the inverse section map. Use the compatible Kummer cocycles and the inverse Shapiro map of E.46 for each system (E47.5), scale their measures to their annuli, and form (E47.10) together with a Kummer cocycle representing ν. Every finite positional quotient sees only finitely many summands, so this is a continuous J-valued cocycle. The section is $g \mapsto ((0, v_g), g)$. Changes of the chosen representatives are exactly the componentwise coboundaries in (E47.9), hence geometric conjugations.

The independent free-inertia calculation and the topology of the limit. The annular module M is the finite-index induced module from G_K, $K = \mathbb{Q}(\zeta_1)$, of the module $\mathcal{B}$ whose underlying ring is $B = \mathbb{Z}_3[[\Gamma]]$ and whose action is $g \cdot z = \chi(g) S_{\chi(g)} z$. The two cosets of G_K correspond to the positive and negative residue classes of unit positions modulo three; on the second coset complex conjugation also contributes its Tate sign. This specifies the induction map and proves the continuous Shapiro comparison with $H^1(K, \mathcal{B})$.

For a prime $\mathfrak{q} \nmid 3$ of K, the coefficient action is unramified. The restriction of a cocycle to inertia is therefore an additive homomorphism. Wild inertia and prime-to-three tame inertia have zero image in this pro-three module. Put $q = N\mathfrak{q}$ and let v be its value at a pro-three tame generator. The relation $F\tau F^{-1} = \tau^q$ gives

$$qS_q v = qv, \qquad (S_q - 1)v = 0. \tag{E47.21}$$

The ring B is the domain $\mathbb{Z}_3[[S_7 - 1]]$; $S_q - 1$ is nonzero since $q \neq 1$ in $1 + 3\mathbb{Z}_3$. Thus $v = 0$. The cocycle kills all inertia away from three. Equivalently it kills every factor of the actual free-inertia kernel (E39.1), so it factors through $G_{K,\{3\}}$. Its graph has pro-three target, and it factors further through the free group $P = \langle \tau, \tau' \rangle$ of E.37.

Both actual Frobenius generators have cyclotomic value seven. A B-valued cocycle is freely specified by the pair of its values on them, and the coboundaries are the diagonal pairs $((7S_7 - 1)z, (7S_7 - 1)z)$. Evaluation at $S_7 = 1/7$ is continuous because $1/7 - 1 \in 3\mathbb{Z}_3$. Its kernel is the principal ideal $(7S_7 - 1)$: formal division by $S_7 - 1 - (1/7 - 1)$ converges integrally. The kernel and inverse of the last map of (E47.7) now follow directly; a representative of (x, y) is $(x + y, y)$, where y is the constant series. This proves that presentation and also shows its cohomology quotient is compact Hausdorff. Finite-index Shapiro is continuous, so the annular cohomology has this compact topology. The natural map to (E47.20) is a continuous bijection onto an inverse limit of compact three-unit groups. It is therefore a homeomorphism. This establishes the claimed topology without treating the larger groups $\widehat{K_n^\times}_3$ as compact.

Finally, (E47.9) sends augmentation exactly to its first summand, proving (E47.6). The geometric kernels in (E47.8) follow from the actual quotient $F \to F/[I_y, I_y]$ and its restriction to N in E.38; they retain the arithmetic origin used throughout.

The exact program verifies forty identities (E47.15) in the integral cyclotomic rings of conductors $3, 9, 27, 81$, all three norm identities (E47.19) on their conjugates, and the finite centered isomorphisms at five positional levels. At level n the finite version of $7S_7 - 1$ has cyclic cokernel $\mathbb{Z}_3/(7^{3^{n-1}} - 1)$, of order 3^n; its evaluation map and every transition through $n = 5$ are checked. These are exact algebraic checks of the maps used above. The all-level class-number and generator-index assertions follow from (E47.13)–(E47.18). □

Infinitely many ramified directions from exact polynomial norm systems

E.48. Cyclotomic-polynomial norms and a compact product of independent section classes [Proved]

Use E.46 with a rational $a \neq 0$, the first cocycle $A = 3\kappa_3(a)$, and the actual algebras $E_{A,n} = \mathbb{Q}[X_n]/(X_n^{3^n} - a^3)$. For every $m \geq 1$ prime to three the rank-three algebra norm has the exact identity

$$N_{E_{A,n+1}/E_{A,n}} \Phi_m(X_{n+1}) = \Phi_m(X_n). \tag{E48.1}$$

There is no omitted sign or unit. Whenever $\Phi_m(a^3) \neq 0$, these elements are units of the finite etale algebras and

$$u_{n,m,z} = \Phi_m(X_n)^{3z}, \qquad z \in \mathbb{Z}_3, \tag{E48.2}$$

is an actual norm system in (E46.4). Its scaled toric coordinates are $(\kappa_3(a), z\kappa_3(\Phi_m(a^3)))$.

In particular fix an integer $a \geq 2$, and enumerate the primes at least five as $\ell_0, \ell_1, \ldots$. Put $B_j = \Phi_{\ell_j}(a^3)$. The formulas

$$\begin{aligned} u_{n,\mathbf{z}} &= \prod_{j \geq 0} \Phi_{\ell_j}(X_n)^{\,3^{j+1} z_j}, \qquad \mathbf{z} = (z_j)_j \in \prod_{j \geq 0} \mathbb{Z}_3, \\ (\kappa_3(a), \beta_{\mathbf{z}}) &= \left(\kappa_3(a), \sum_{j \geq 0} 3^j z_j \kappa_3(B_j) \right) \end{aligned} \tag{E48.3}$$

give a closed topological embedding of the full compact product $\prod_{j \geq 0} \mathbb{Z}_3$ into this global first-coordinate section fiber of H. The series and products converge in the multiplication-by-3^r completions specified in E.46. The image is torsion-free. The toric projection itself is injective on this family.

The integers B_j are greater than one, pairwise coprime, and prime to $3a$. For each j, choose any prime $p_j \mid B_j$. Its toric inertia coordinate is exactly

$$v_{p_j}(\beta_{\mathbf{z}}) = 3^j z_j \, v_{p_j}(B_j). \tag{E48.4}$$

Thus these are independent ramified directions at distinct primes away from $3a$. The choice $z_j = 1$ for all j gives a continuous section with nonzero toric inertia at infinitely many primes.

For clarity, the universal rank-three norm for the remaining cyclotomic indices is also determined:

$$N_{Y^3 = X} \Phi_m(Y) = \begin{cases} \Phi_m(X), & 3 \nmid m, \\ \Phi_{m/3}(X)^2, & 3 \mid m,\ 9 \nmid m, \\ \Phi_{m/3}(X)^3, & 9 \mid m. \end{cases} \tag{E48.5}$$

The geometric target and its comparison kernels are the actual H and (E47.8). All section classes in (E48.2)–(E48.3) are obtained through the proved norm-to-section isomorphism (E46.4).

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d). The program `e_cyclotomic_norms.py` *retains the exact resultant and convergence certificates.*

Proof. Work first over the polynomial ring $\mathbb{Q}[X]$ with the free rank-three algebra $\mathbb{Q}[X,Y]/(Y^3-X)$. Over a ring containing a primitive cube root of unity, the three embeddings send Y to $Y, \omega Y, \omega^2 Y$. For every root of unity ξ their product is

$$N(Y-\xi) = (Y-\xi)(\omega Y-\xi)(\omega^2 Y-\xi) = X-\xi^3. \tag{E48.6}$$

The leading product ω^3 is one, and the constant term is $-\xi^3$, fixing the sign. Multiplicativity of the determinant norm gives

$$N\Phi_m(Y) = \prod_{\operatorname{ord}(\xi)=m} (X-\xi^3). \tag{E48.7}$$

When $3 \nmid m$, cubing is a permutation of the primitive mth roots. For $m = 3s$ with $3 \nmid s$, its image consists of the primitive sth roots, each with multiplicity two. When $9 \mid m$, its image is the primitive $(m/3)$th roots, each with multiplicity three. This proves every case of (E48.5) as an identity over $\mathbb{Q}[X]$. Specialization commutes with the determinant norm, including for products of fields, so (E48.1) holds in the exact algebras of E.46.

Iteration down to level zero gives the total norm $\Phi_m(a^3)$. Its nonvanishing is exactly invertibility of $\Phi_m(X_n)$ in the finite etale algebra. Taking continuous three-adic powers proves norm compatibility in (E48.2). The level-zero element is the cube $\Phi_m(a^3)^{3z}$; the map (E46.7) divides this Kummer coordinate by three, giving the asserted toric coordinates.

Let $B = a^3 > 1$. For distinct primes ℓ, r, a common prime divisor p of $\Phi_\ell(B)$ and $\Phi_r(B)$ cannot divide B. The order of B modulo p then divides both ℓ and r, so $B \equiv 1 \pmod p$. But $\Phi_\ell(B) \equiv \ell$ and $\Phi_r(B) \equiv r \pmod p$, forcing p to divide both distinct primes. This proves pairwise coprimality. Each sum $\Phi_\ell(B) = 1 + B + \cdots + B^{\ell-1}$ is greater than one and prime to a. For $\ell \geq 5$ prime, its value modulo three is 1, ℓ, or 1 as B is respectively 0, 1, or -1 modulo three. It is therefore also prime to three.

At coefficient exponent r, every factor of the product (E48.3) with $j+1 \geq r$ is trivial. Thus the product exists in every completed algebra, varies continuously with the product topology on $\mathbf{z}$, and is compatible under all norms. Its level-zero element is again a cube. This proves that it defines actual global section classes by E.46, including when infinitely many coordinates are nonzero.

The Kummer valuation at p_j receives a contribution only from B_j, giving (E48.4). The positive integer $v_{p_j}(B_j)$ is a nonzero element of $\mathbb{Z}_3$, so equality of toric coordinates forces equality of every z_j. This proves injectivity and torsion-freeness. The target cohomology quotient is Hausdorff: coboundaries are the continuous image of the compact module J_A in the Hausdorff uniform cocycle space, hence form a closed subgroup. A continuous injection from the compact product is consequently a homeomorphism onto a closed image. This proves the topological assertion, as well as the claimed infinite ramification.

The program checks (E48.5) by exact resultants for $1 \leq m \leq 45$, verifies 108 total norms in the actual radical etale algebras, and checks 84 coprimality identities with explicit supporting prime factorizations. It also records the exact coefficient bounds for the infinite product. The universal determinant identity and the preceding convergence proof establish all levels. □

A complete nonzero-first-coordinate global section calculation

E.49. Every global section of the inherited quotient with first Kummer class three [Proved]

Use the actual arithmetic quotient H and origin of E.38 and E.46. Choose positive

compatible real roots of 3, put $\kappa = \kappa_3(3)$, and fix the first cocycle $A = 3\kappa$. Define the explicit set of rational primes

$$\mathscr{P} = \{p \neq 3 : \ 3 \nmid \operatorname{ord}_{\mathbb{F}_p^\times}(27 \bmod p)\}. \tag{E49.1}$$

For $p = 2$ the order is one. Equivalently the three-primary part of the order of 3 mod p has order at most three. In particular every prime $p \equiv 2 \pmod 3$, including two, belongs to $\mathscr{P}$.

Let

$$\mathcal{R}(\mathscr{P}) = \{(b_p)_p \in \prod_{p\in\mathscr{P}} \mathbb{Z}_3 : \text{for each } r,\ b_p \in 3^r\mathbb{Z}_3 \text{ for all but finitely many } p\}. \tag{E49.2}$$

It carries its multiplication-by-3^r inverse-limit topology. The global section classes of H with first class $[3\kappa]$ have complete coordinates

$$\boxed{(b_3, (b_p)_{p\in\mathscr{P}}, z, f) \in \mathbb{Z}_3 \times \mathcal{R}(\mathscr{P}) \times \mathbb{Z}_3 \times \mathbb{Z}_3[[Z]].} \tag{E49.3}$$

The proof constructs a homeomorphism after fixing one specified norm lift at each prime of $\mathscr{P}$. The toric map scaled as in (E46.7) is exactly

$$(b_3, (b_p), z, f) \longmapsto \left(\kappa,\, b_3\kappa + \sum_{p\in\mathscr{P}} b_p\kappa_3(p)\right). \tag{E49.4}$$

Thus its image consists exactly of completed rational Kummer classes whose valuations at primes outside $\{3\} \cup \mathscr{P}$ vanish. Every fiber has the explicit compact coordinates $\mathbb{Z}_3 \times \mathbb{Z}_3[[Z]]$. The entire section group, with the origin $g \mapsto (A_g, 0)$, is torsion-free.

Here are formulas for the compact parameters. Write

$$\Lambda = \mathbb{Z}_3[[U]], \quad T = 1 + U, \quad W = (T - T^{-1})/2, \quad \eta = 1 - (T^{-1} + T^{-2})/2. \tag{E49.5}$$

The proof selects actual conjugate Frobenii τ, τ' over seven, still a free basis of $P = G^{(3)}_{\mathbb{Q}(\zeta_3),\{3\}}$, with $\kappa(\tau) = 1$ and $\kappa(\tau') = -1$. Every section class unramified outside three has a unique second-cocycle representative with

$$\begin{aligned} v(c) &= 0, \\ v(\tau) &= 3b_3 + z\eta + T^3 W f(W^2), \\ v(\tau') &= -v(\tau)(T^{-1}). \end{aligned} \tag{E49.6}$$

These values define the cocycle on the actual free group and its extension by complex conjugation. All coefficients of the power series f are arbitrary elements of $\mathbb{Z}_3$.

There is also a complete unit and valuation interpretation of (E49.3). For $n \geq 1$ put

$$\begin{aligned} \alpha_n = 3^{1/3^{n-1}}, \qquad \beta_n = \alpha_n\zeta_{3^n}, \\ F_n = \mathbb{Q}(\alpha_n), \qquad L_n = \mathbb{Q}(\beta_n), \qquad E_{A,n} = F_n \times L_n. \end{aligned} \tag{E49.7}$$

The fields have degrees 3^{n-1} and $2 \cdot 3^{n-1}$, and their tower maps are $\alpha_{n+1}^3 = \alpha_n$ and $\beta_{n+1}^3 = \beta_n$. The inverse norm limit of their completed ordinary global units is explicitly

$$\varprojlim_N \left(\widehat{\mathcal{O}^\times_{F_n 3}} \times \widehat{\mathcal{O}^\times_{L_n 3}}\right) \ \xrightarrow{\sim}\ \mathbb{Z}_3[[Z]]. \tag{E49.8}$$

The inverse map $\epsilon(f)$ is the exact Kummer–Shapiro norm system of the unramified cocycle with τ-value $T^3 W f(W^2)$.

Define two norm-coherent three-unit systems ξ_F, ξ_L by the unramified cocycles with τ-values 1 and $1-\eta$, respectively, in Λ_A. The combination (E49.9) will satisfy the additional J_A condition. They have respective three-prime valuation pairs $(1,0)$ and $(0,1)$ in F_n, L_n, and both have level-zero norm 3. For each $p \in \mathscr{P}$, fix a norm system π_p whose divisor at each level is the unique compatible degree-one prime above p, with zero valuations at every other prime, including three. The proof constructs all these systems using surjective unit norm maps. Their level-zero norms are exactly p. The complete norm-to-section map in (E49.3) is then

$$\epsilon(f)\, \xi_F^{3b_3+z} \xi_L^{-z} \prod_{p \in \mathscr{P}} \pi_p^{3b_p}. \tag{E49.9}$$

Every product and power is taken in the completed multiplicative groups of E.46. No hypothesis on an unevaluated class group or unit norm map remains.

The geometric map is precisely $N \to H$ with kernel $[I_y, I_y]$; its factorizations through the intrinsic metabelian and marked groups have the kernels (E47.8). Statement (E49.3) evaluates the entire indicated global section fiber of this target.

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d). The calculation retains the actual arithmetic origin, first-coordinate cocycle, and completed-group-ring topology.

Proof. The actual arithmetic basis and its normalization. Let $K = \mathbb{Q}(\zeta_3)$ and retain the positive real roots defining κ. Then $\kappa(c) = 0$. Start with the actual Frobenii $\tau_0, \tau_0' = c\tau_0 c$ of E.37, taking the prime over seven at which ζ_3 reduces to two. The cubic residue calculation $3^{(7-1)/3} = 2 = \zeta_3$ gives $t = \kappa(\tau_0) \equiv 1 \pmod 3$. The cocycle identity gives $\kappa(\tau_0') = -t$.

Put $\rho_0 = \tau_0' \tau_0^{-1}$. Its cyclotomic character is one and $A(\rho_0) = -6t$. The three-adic power

$$h = \rho_0^k, \qquad k = (t^{-1} - 1)/12 \in \mathbb{Z}_3 \tag{E49.10}$$

is therefore defined, and $A(h) = -6tk$. Set $\tau = h\tau_0 h^{-1}$ and $\tau' = c\tau c$. The exact cocycle formula for conjugation gives

$$A(\tau) = 3t + (1-7)(-6tk) = 3, \qquad A(\tau') = -3. \tag{E49.11}$$

Both elements remain actual Frobenii at conjugate prolongations over seven. Their Frattini classes are the unchanged classes of τ_0, τ_0'; the Frattini criterion in the free rank-two group therefore proves that they remain a free basis. This supplies the arithmetic map behind the normalization.

Use the Nielsen basis $\sigma = \tau$, $\rho = \tau' \tau^{-1}$. It satisfies

$$c\sigma c = \rho\sigma, \qquad c\rho c = \rho^{-1}. \tag{E49.12}$$

On the twisted coefficient module Λ_A of E.46, the actions are the explicit operators

$$\sigma u = 7T^3 u(T^7), \qquad \rho u = T^{-6} u, \qquad Cu = cu = -u(T^{-1}). \tag{E49.13}$$

Their compositions verify (E49.12) on the coefficient module as well.

Finite coefficients and the full arithmetic group. We record the exact comparison used below. For a finite three-primary module M whose action over K factors through P, inflation induces an isomorphism

$$H^i(P, M) \longrightarrow H^i(G_{K,\{3\}}, M) \qquad (i \geq 0). \tag{E49.14}$$

For degree one, a cocycle gives a homomorphism into M semidirect its finite three-group of arithmetic actions. This target is a three-group, so the homomorphism factors through the

actual maximal pro-three quotient P. Coboundaries use the same elements of M on both sides, proving injectivity as well as surjectivity. Degree zero is immediate.

For higher degrees, every simple factor of M for its finite three-group action is the trivial module $\mathbb{F}_3$. The established calculation of E.34 gives $H^2(G_{K,\{3\}},\mathbb{F}_3)=0$; the arithmetic cohomological dimension bound there gives zero in higher degrees. Induction along a finite composition series gives the same vanishing for M. The free group P has three-cohomological dimension one. This proves the asserted comparison in every degree for precisely these coefficient modules.

In particular take the finite images Λ_r, J_r in $(\mathbb{Z}/3^r)[\mathbb{Z}/3^r]$. Their action over K has three-power image: the positions transform by $i\mapsto A_g+\chi_g i$ with $A_g\in 3\mathbb{Z}_3$ and $\chi_g\in 1+3\mathbb{Z}_3$, and the scalar twist has three-power image as well. Thus (E49.14) applies. Restriction to K followed by corestriction multiplies by two, so the corresponding degree-two and degree-three groups over $G_{\mathbb{Q},\{3\}}$ also vanish. Its degree-one groups with these finite coefficients are finite: over P a cocycle is specified by two values in a finite module, and descent adds only the finite group of order two.

Taking the cofinal system $r\to\infty$ therefore retains vanishing in degree two. In particular the inverse systems of unramified degree-one groups are Mittag–Leffler. The continuous-cochain comparison is the finite-coefficient comparison proved in E.44 and [Jan88, Proposition 1.6 and Theorem 2.2]. No comparison with a pro-three quotient for an unrelated coefficient action is being assumed.

An explicit normal form for all unramified classes. Normalize a second cocycle at c to zero using division by two. It is then freely specified by $u=v(\tau)\in\Lambda$ and $v(\tau')=Cu$. The remaining coboundaries are

$$Dm=7T^3m(T^7)-m,\qquad m\in\Lambda^C=\{m : m(T^{-1})=-m(T)\}. \tag{E49.15}$$

The first-degree comparison in (E49.14) and the free property of P prove that these are exactly the global classes unramified outside three. For J_A one takes $u\in J$; the residual fixed lattice is the same, because an element of Λ^C has augmentation zero.

Here is a reduction algorithm entirely inside the integral completed ring. Put $n=T^{-6}-1$. The exact identity

$$n=-T^{-6}(T^3+1)(T^3-1) \tag{E49.16}$$

identifies its ideal with (T^3-1): the extra factor is a unit. Consequently $\Lambda/(n)=\mathbb{Z}_3[C_3]$, with the specified representatives $1,T,T^2$. For $u\in\Lambda$, let its image be $u_0+u_1T+u_2T^2$ and set

$$\begin{aligned} a&=\operatorname{aug} u, & z&=-(u_1+u_2),\\ u_*&=u-a-z\eta, & m&=\frac{Cu_*-T^{-6}u_*}{T^{-6}-1}. \end{aligned} \tag{E49.17}$$

The numerator has zero image in $\mathbb{Z}_3[C_3]$: the image of u_* is antisymmetric under inversion. Thus the division is integral. Multiplication by n is injective in the domain Λ. Applying C to the numerator shows $m=Cm$; explicitly $C(nm)=-T^6nCm$ and the numerator equals $-T^{-6}$ times its image under C. Therefore (E49.15) permits subtracting Dm.

The resulting $v=u_*-Dm$ satisfies $Cv=T^{-6}v$. Thus $T^{-3}v$ is antisymmetric under $T\mapsto T^{-1}$. The variable W in (E49.5) has linear coefficient one in U. Its inverse substitution is $T=W+\sqrt{1+W^2}$, with the unique square root of constant term one. Hence $\Lambda=\mathbb{Z}_3[[W]]$, inversion sends W to $-W$, and

$$v=T^3Wf(W^2)\quad\text{for a unique } f\in\mathbb{Z}_3[[Z]]. \tag{E49.18}$$

This gives the normal form $a+z\eta+T^3Wf(W^2)$.

It is unique modulo (E49.15). Augmentation of Dm is zero, so it cannot change a. The ρ-value of a cocycle with τ-value u is $Cu - T^{-6}u$. Its image modulo n detects z and is unchanged by the coboundary, whose ρ-value is nm. Once a, z are zero, a representative of the remaining form has ρ-value zero; a coboundary with this property has $nm = 0$, hence $m = 0$. The power series is consequently unchanged. All the division maps just used are continuous on their closed principal ideals, since they invert continuous bijections from a compact ring. This also proves the normal-form topology. For $u \in J$ its augmentation is $a = 3b_3$, giving (E49.6).

The radical fields and surjectivity of their unit norms. The rational polynomial factors as

$$X^{3^n} - 27 = (X^{3^{n-1}} - 3)(X^{2\cdot 3^{n-1}} + 3X^{3^{n-1}} + 9). \tag{E49.19}$$

The first factor is Eisenstein at three. For the second, $\beta_n^{3^{n-1}} = 3\zeta_3$, so β_n contains K. Normalize the valuation of K at $\lambda = 1 - \zeta_3$ by $v_\lambda(3) = 2$. The valuation $2/3^{n-1}$ of β_n forces ramification degree at least 3^{n-1} over K, since its numerator is prime to three. This proves the full degree, irreducibility of the second factor, and the unique totally ramified prime above three. The same argument applies to $M_n = KF_n$.

Both L_{n+1}/L_n and M_{n+1}/M_n are cyclic cubic: the base contains μ_3 and the next generator has cube equal to the preceding one. They are unramified away from three, as seen from the polynomial discriminants, and have exactly one totally ramified prime. Starting from $L_1 = M_1 = K$, the ambiguous class-number argument of E.47 therefore gives $3 \nmid h(L_n)h(M_n)$ at every level. Ideal extension followed by norm from M_n multiplies $\mathrm{Cl}(F_n)$ by two, so $3 \nmid h(F_n)$ also.

The strongly ambiguous formula in [Lem13, Theorem 1, formula (2)] has the especially useful consequence, for each of these cyclic cubic layers L'/L,

$$|\operatorname{Am}_{\mathrm{st}}(L'/L)| = \frac{h(L)}{[\mathcal{O}_L^\times : N\mathcal{O}_{L'}^\times]}. \tag{E49.20}$$

The denominator is a power of three, because the cube of each base unit is a norm. It divides the prime-to-three integer $h(L)$, and is therefore one. Actual ordinary unit norms in the L_n and M_n towers are onto. For the F_n tower, project the latter norm map to the fixed submodule of complex conjugation by $(1 + c)/2$. The fixed submodule of the completed units of M_n is exactly the completed units of F_n: norm followed by inclusion is $1 + c$, and two is invertible. This proves surjectivity of the completed ordinary unit norms for both factors of (E49.7).

The ordinary-unit norm limit in exact cochain coordinates. Apply finite Shapiro over $G_{\mathbb{Q},\{3\}}$ to the positional modules. Every field in (E49.7) is unramified outside three. Kummer theory identifies $H^1(G_{F,\{3\}}, \mathbb{Z}_3(1))$ with the completed three-units: the inverse Tate module of the finite S-class group is zero. In this tower its three-primary class group is already zero by the preceding paragraph. The norm comparison of E.46 therefore identifies the unramified cohomology normal form with the entire inverse norm limit of the completed three-units.

To obtain the ordinary units, compute the two three-prime valuations at level one. There $E_{A,1} = \mathbb{Q} \times K$. Write $u \bmod (T^3 - 1) = u_0 + u_1T + u_2T^2$. Shapiro at the rational root gives valuation u_0, because $\kappa(\tau) = 1$. At the unit-root orbit, the two K-cocycle values are u_1 and $-u_2$. The torsion Kummer class of ζ_3 has values $2, 2$, since it is represented over K by $(\chi - 1)/3$. The relation $\lambda^2 = -3\zeta_3$ gives $2[\kappa_3(\lambda)] = [\kappa] + [\kappa_3(\zeta_3)]$. Thus the two values of its class have difference one. A coboundary changes both values by the same $6m$. The valuation at λ of the unit-orbit class is therefore exactly $u_1 + u_2$.

For the normal form just proved, these two valuations are

$$(a + z, -z). \tag{E49.21}$$

Indeed $T^3Wf(W^2)$ reduces to an antisymmetric vector modulo $T^3 - 1$, while η reduces to $1 - (T + T^2)/2$. The residue degrees at the unique three-prime in each tower layer are one, so

norm compatibility makes these valuations constant in the tower. Consequently the ordinary-unit norm limit is exactly $a = z = 0$. This proves (E49.8), including its explicit map and compact topology. It also proves the asserted valuations and total norms of ξ_F, ξ_L.

The surviving primes and the exact residue lattice. At $p \neq 3$ the coefficient action is unramified. For a chosen actual arithmetic Frobenius F_p it acts, after removing the Tate twist, by pushforward under

$$\phi_p(x) = A_{F_p} + px \quad (x \in \mathbb{Z}_3). \tag{E49.22}$$

Its only possible fixed point is $x_p = -A_{F_p}/(p-1)$. An integral fixed point is exactly a compatible system of 3^nth roots of 27 in $\mathbb{Q}_p$, using the root labels of E.46. Cubing is an automorphism on the principal units at $p \neq 3$. Thus those compatible roots exist exactly when the residue of 27 has order prime to three, proving (E49.1). For $p \equiv 2 \pmod 3$, $p-1$ is a three-adic unit and the fixed point is always integral.

The invariant integral measure lattice for ϕ_p is exactly $\mathbb{Z}_3\delta_{x_p}$ when this point is integral, and is zero otherwise. Here is the integral argument. With an integral center, translate it to zero and use multiplication by p^2. On every nonzero annulus its clopen orbits are free $\mathbb{Z}_3$ orbits, as in E.45. An invariant integral measure on a free orbit has its mass divisible by arbitrarily large powers of three, hence vanishes; equivalently $S-1$ is injective on $\mathbb{Z}_3[[S-1]]$. Only the atom at the fixed point remains. When the center is not integral, necessarily $p \equiv 1 \pmod 3$. Put $s = v_3(p-1)$ and $t = v_3(A_{F_p}) < s$. The identity $v_3(\phi_p^n(x) - x) = t + v_3(n)$ shows that the clopen sets $i + 3^t\mathbb{Z}_3$ are free orbits. The same measure argument gives zero. Intersecting the invariant lattice with J, whose masses are divisible by three, gives precisely

$$(J(-1))^{F_p} = \begin{cases} 3\mathbb{Z}_3\delta_{x_p}, & p \in \mathscr{P}, \\ 0, & p \notin \mathscr{P},\ p \neq 3. \end{cases} \tag{E49.23}$$

This also computes the valuation inverse system on the norm side. At each finite positional level, the Frobenius orbits correspond to the primes of $E_{A,n}$ above p. The invariant vector $\sum_{i \in O} \delta_i$ represents valuation one at that prime. Positional pushforward multiplies it by the relative residue degree, exactly as does the ideal norm. The fixed point x_p therefore gives the unique compatible degree-one prime chain. Every other valuation is forced to zero by its unbounded three-power residue degrees. At three there are the two totally ramified chains of residue degree one in (E49.7). The complete valuation inverse limit is consequently $\mathbb{Z}_3^2 \times \mathcal{R}(\mathscr{P})$. Restrictedness is exact: the mass at level zero forces the coefficients at distinct p to tend to zero modulo every power of three, and that condition is sufficient in every positional level.

Residue surjectivity over the actual arithmetic group. For clarity the same surjectivity can be proved before choosing elements of the radical fields. Localization, purity, and the group-cohomology comparison used in E.44 give, for each J_r, the exact sequence

$$\begin{aligned} 0 \longrightarrow H^1(G_{\mathbb{Q},\{3\}}, J_r) &\longrightarrow H^1(G_{\mathbb{Q}}, J_r) \\ &\longrightarrow \bigoplus_{p \neq 3} (J_r(-1))^{F_p} \longrightarrow 0. \end{aligned} \tag{E49.24}$$

The last zero is the finite-coefficient degree-two vanishing proved above. The primary maps are [Mil06, II.0.1(f), II.1.7(b), II.2.9]. The left groups are finite, so their first inverse-limit derivative vanishes. The degree-zero finite groups needed to compare continuous degree-one cohomology also satisfy Mittag–Leffler. Taking the inverse limit therefore preserves surjectivity and gives exactly the restricted fixed measures in (E49.23), without taking a closure or discarding an inverse-limit term. The tame generator can be normalized by $\kappa_3(p)(\tau_p) = 1$; the residue $3b_p\delta_{x_p}$ then has scaled toric valuation b_p.

Constructing all norm systems and all section coordinates. There is an equally explicit construction using the fields. At each finite level, the completed divisor map is onto: the

class groups of F_n, L_n have order prime to three. If an ideal has class of order h prime to three, choose a generator of its hth power and take its h^{-1} power in the three-completion. This produces its valuation vector exactly. The kernel of the completed divisor map is the completed ordinary unit group. For example this follows by splitting the sequence onto the free principal-divisor image, then using its prime-to-three index in the full divisor group.

For $p \in \mathscr{P}$, choose a completed element with divisor the prime at each level of its degree-one chain. If its norm differs from the element already chosen downstairs, the ratio is an ordinary unit. The surjective unit norms proved in (E49.20) correct this ratio. Induction constructs π_p with exactly the specified divisors. At level zero its norm is p, because the completion of $\mathbb{Q}$'s ordinary unit group is zero. This construction uses actual norm equations, not an assumed saturation of the divisor image.

Any compatible norm system has unique valuations (d_F, d_L) at three and $(c_p)_p$ on these other chains. After dividing by $\xi_F^{d_F}\xi_L^{d_L}\prod \pi_p^{c_p}$, it is an ordinary-unit system, and has the unique form $\epsilon(f)$ by (E49.8). The infinite product converges because $c_p \to 0$ three-adically; at every coefficient exponent only finitely many factors contribute. Its level-zero norm has valuations $d_F + d_L$ at three and c_p at p. The cube condition of E.46 is exactly

$$d_F + d_L \in 3\mathbb{Z}_3, \qquad c_p \in 3\mathbb{Z}_3 \quad (p \in \mathscr{P}). \tag{E49.25}$$

Set $d_F = 3b_3 + z$, $d_L = -z$, and $c_p = 3b_p$. This proves (E49.9), its uniqueness, and the complete coordinates (E49.3)–(E49.4).

All valuation maps are continuous in the power-class topologies. The products just constructed are continuous for those topologies, and the remaining unit map and its inverse are continuous by the compact normal-form calculation. Thus the classification is a homeomorphism. Changes of the chosen systems π_p multiply them by explicit ordinary-unit norm systems, and alter the f coordinate by the corresponding convergent linear sum. They do not change the toric criterion or any fiber. Every parameter group in (E49.3) is three-torsion-free, proving the stated torsion assertion. □

The exact once-punctured elliptic comparison in intrinsic class three

E.50. The multiplication-by-three cover and its integral ambient-metabelian kernel [Proved]

Let E be the cubic elliptic curve used in E.37, with origin O, and put $U = E \setminus E[3]$ and $E^\circ = E \setminus \{O\}$. Choose the actual rational tangent v at O used for U, and use $3v$ as the tangent on E°. The morphism

$$[3] : (U, v) \longrightarrow (E^\circ, 3v) \tag{E50.1}$$

induces an inclusion of geometric pro-three groups

$$N = \pi_1(U_{\overline{\mathbb{Q}}}, v)^{(3)} \hookrightarrow F = \pi_1(E^\circ_{\overline{\mathbb{Q}}}, 3v)^{(3)}, \qquad N = \ker(F \longrightarrow F^{\mathrm{ab}}/3), \tag{E50.2}$$

equivariant for the resulting based $G_{\mathbb{Q}}$ actions. Here F is free pro-three of rank two and N is free pro-three of rank ten. All derived and lower central subgroups below are closed.

Write $C = N/\gamma_4 N$. The actual map to the image in the class-three quotient of the ambient metabelian group is

$$C \longrightarrow \overline{C} = (N/F'')/\gamma_4(N/F'') = N/(F''\gamma_4 N), \qquad K_0 = \ker(C \longrightarrow \overline{C}) = F''\gamma_4 N/\gamma_4 N. \tag{E50.3}$$

Let $V = N^{\mathrm{ab}}$ and let $p : V \to T_3 E$ be the map induced by filling the nine punctures. The map $V \to F^{\mathrm{ab}} = T_3 E$ in (E50.2) is exactly $3p$. Thus

$$T = \ker(p) = \ker(V \longrightarrow F^{\mathrm{ab}}) \tag{E50.4}$$

is the primitive rank-eight cusp lattice of E.37. Choose the free basis described in the proof, and write $V = L \oplus T$, where L is spanned by its two handle generators. This splitting is a choice of integral coordinates. Then

	gr_1	gr_2	gr_3
C	V	$\Lambda^2 V$	$\mathrm{Lie}_3(V)$
K_0	0	$\Lambda^2 T$	$[T, \mathrm{Lie}_2(V)]$
$\overline{C}$	V	$\Lambda^2 L \oplus (L \otimes T)$	$\mathrm{Lie}_3(L) \oplus (\mathrm{Sym}^2 L \otimes T)$

(E50.5)

is an exact description of the induced lower central filtrations over $\mathbb{Z}_3$. The last row has ranks $10, 17, 26$; the kernel row has ranks $0, 28, 304$. In particular K_0 is an abelian pro-three group isomorphic to $\mathbb{Z}_3^{332}$, and all the displayed graded quotients are torsion-free. The maps and kernel in (E50.3) are $G_{\mathbb{Q}}$-equivariant; the direct sums involving the chosen L record the stated coordinate splitting.

There is also a direct ambient route retaining the whole intrinsic class-three group:

$$F''' \subset \gamma_4 N, \qquad F/F''' \longrightarrow F/\gamma_4 N, \qquad \ker(F/\gamma_4 N \longrightarrow F/N) = C. \tag{E50.6}$$

Thus an actual based action on the indicated ambient derived-length-three quotient induces the actual action on C through these maps.

The comparison has a concrete automorphism kernel. There are based automorphisms of F preserving its peripheral conjugacy class which act identically on F/F'' and nontrivially on C. One is conjugation by $w = [t_1, t_2] \in F''$, in the basis below:

$$\mathrm{Ad}_w(e_1) e_1^{-1} = [[t_1, t_2], e_1] \neq 1 \quad \text{in } C. \tag{E50.7}$$

This is an assertion about these specified based automorphism maps.

Two further transfers retain the exact geometric and arithmetic origins. The full peripheral kernel of $N \to T_3 E$ is $I = F'$, and the quotient in E.36 fits into the actual equivariant sequence

$$1 \longrightarrow \mathbb{Z}_3^{42} \longrightarrow \overline{C} \longrightarrow N^{\mathrm{cc}} = N/[N, I] \longrightarrow 1. \tag{E50.8}$$

The kernel has induced graded ranks $0, 16, 26$ and is abelian; the displayed isomorphism with $\mathbb{Z}_3^{42}$ is an assertion about its underlying group.

Let $\rho_E : G_{\mathbb{Q}} \to \mathrm{Aut}(T_3 E)$ be the actual elliptic Tate representation. The based action on $\overline{C}$ induces a continuous homomorphism

$$\rho_E(G_{\mathbb{Q}}) \longrightarrow \mathrm{Out}(\overline{C}) \tag{E50.9}$$

whose composition with ρ_E is the genuine outer action on this exact quotient. For $g \in \ker \rho_E$ the proof gives its actual inner conjugator in $F'/F'' \subset N/F''$.

The exact integer Hall certificates are saved in `a_elliptic_kernel.py`, its output and JSON file. They verify the cover relation, 1,000 tensor identities, all 304 integral kernel coordinates, and 280 normality identities.

Manuscript: §51.16, folios 733v–736v, scans 172–178, (371)–(379), Part E(c),(d). This comparison supplies the precise quotient needed when applying once-punctured elliptic formulae to the intrinsic cubic group.

Proof. The actual cover and the based action. Multiplication by three is an etale isogeny in characteristic zero. Its inverse image of O is $E[3]$, so (E50.1) is a finite etale cover of degree nine. Its differential at O sends v to $3v$; these are actual tangential basepoints, defined over $\mathbb{Q}$. The geometric covering subgroup is the kernel of the homology reduction modulo three. Its

induced pro-three topology is the full one: for a normal subgroup of three-power index in the covering group, intersect its nine conjugates. The resulting quotient of the covering group is a three-group and the quotient of the ambient group is an extension of it by $(\mathbb{Z}/3)^2$. Taking these quotients proves the asserted injection of pro-three completions. The map of tangential arithmetic fundamental groups gives the equivariance, including the chosen sections of $G_{\mathbb{Q}}$.

Cut a complex torus along a symplectic pair of loops x_1, x_2 based just before the origin, and let $z = [x_1, x_2]^{-1}$ be its positive peripheral loop. In the nine-square cover choose the peripheral paths in the following order of pairs:

$$(0,0), (0,1), (1,1), (1,0), (0,2), (1,2), (2,2), (2,1), (2,0). \tag{E50.10}$$

For a pair (a, b) take $x_{ab} = x_1^a x_2^b$ for $a \geq b$ and $x_{ab} = x_1^b x_2^b x_1^{a-b}$ for $a < b$, and put $z_{ab} = x_{ab} z x_{ab}^{-1}$. The boundary of the nine-square polygon is the exact relation

$$[x_1^3, x_2^3] z_0 z_1 \cdots z_8 = 1. \tag{E50.11}$$

These paths and the same relation are given in Nakamura, §2.4.1–2.4.2, p.205 [Nak95]. The program also checks (E50.11) directly by free reduction in x_1, x_2. The punctured surface has genus one and nine punctures, hence its group is free of rank ten, with basis

$$e_1 = x_1^3, \quad e_2 = x_2^3, \quad t_i = z_i \ (1 \leq i \leq 8), \tag{E50.12}$$

after eliminating z_0 with (E50.11). The same is true of the pro-three group. Transporting the complex basepoint to the tangent gives this geometric basis of the already specified based group; it does not alter the arithmetic morphism (E50.1).

Filling the punctures sends the two handle classes to a basis of $T_3 E$ and each t_i to zero. The proper map in the commutative filling square is $[3] : E \to E$, whose Tate module map is multiplication by three. This proves (E50.4) and the precise factor $3p$. The image of N in F^{ab} is $3F^{\mathrm{ab}}$ and its kernel is F'. Consequently, inside N,

$$F' = \ker(N \longrightarrow 3F^{\mathrm{ab}}) = \langle\langle t_1, \ldots, t_8, [e_1, e_2] \rangle\rangle_N. \tag{E50.13}$$

Indeed, quotienting by the displayed normal generators gives precisely the abelian free pro-three group on e_1, e_2, and the latter maps isomorphically onto $3F^{\mathrm{ab}}$.

Passage to the actual class-three group. Integral Hall collection gives unique $\mathbb{Z}_3$ coordinates for C on its ten weight-one generators, its 45 basic commutators of weight two, and its 330 Hall commutators of weight three. These are group coordinates, not rational Campbell–Hausdorff coordinates. For clarity, order the ten generators by the indices $0, 1, 2, \ldots, 9$, with the first two the handles. The weight-three basis is

$$[[v_i, v_j], v_k], \qquad i > j, \quad k \geq j. \tag{E50.14}$$

Unique collection for integral free nilpotent groups, followed by the inverse limit of their three-power coordinate quotients, gives these coordinates in the free pro-three quotient. Equivalently, the successive central extensions have the integral Hall lattices constructed in E.41. In particular this argument uses no division by three.

Let J be the image of F' in C, let H be the closed subgroup generated by the t_i, and put $A = \gamma_2 C$. From (E50.13) and the abelianization of C,

$$J = HA, \qquad [A, A] = 1, \qquad [H, A] \subset \gamma_3 C. \tag{E50.15}$$

In the last inclusion the target is central. Let B be the closed subgroup generated by the 28 actual basic group commutators $[t_i, t_j]$, $i > j$, and let L_3 be the closed subgroup of $\gamma_3 C$ generated by $[t_i, [v_a, v_b]]$. Commutator collection in (E50.15) gives

$$K_0 = J' = B L_3. \tag{E50.16}$$

Here $[H,H]$ is generated by B and triple commutators on the t_i, while $[H,A]=L_3$. Those triple commutators already belong to L_3. Conversely, both B and L_3 consist of commutators of elements of J, since $A\subset J$. This proves equality of actual closed subgroups, before considering their associated graded modules.

All generators of B lie in $\gamma_2 C$, so they commute with each other, and L_3 is central. Their degree-two Hall coordinates are 28 distinct basis vectors. The continuous map $\mathbb{Z}_3^{28}\to B$ given by their ordered powers is therefore injective, is surjective by the definition of B, and satisfies $B\cap\gamma_3 C=1$. Thus

$$K_0 = B\times L_3, \qquad B\simeq \mathbb{Z}_3^{28}, \qquad K_0\cap\gamma_3 C = L_3. \tag{E50.17}$$

This supplies the required passage from graded information to the whole kernel at the prime three.

The saturated degree-three lattice. In the central group $\gamma_3 C$, a commutator $[t,[v,w]]$ is exactly the corresponding integral Lie bracket. Hence $L_3=[T,\mathrm{Lie}_2(V)]$. The degree-two quotient is obtained by deleting the basis $\Lambda^2 T$. For degree three, use the decomposition $V=L\oplus T$. Jacobi shows that $[V,[T,T]]\subset[T,\mathrm{Lie}_2(V)]$. Every Lie monomial with at least two T letters therefore vanishes in the quotient. For a fixed $t\in T$, the remaining relation with one T letter is $[t,[e_1,e_2]]=0$. Consequently

$$[e_i,[e_j,t]] = [e_j,[e_i,t]], \tag{E50.18}$$

and the surviving one-T coordinates are the three symmetric pairs of the two handles. In the Hall basis (E50.14), retain exactly the two pure handle triples and

$$[[t,e_1],e_1], \quad [[t,e_1],e_2], \quad [[t,e_2],e_2] \quad (t=t_1,\dots,t_8). \tag{E50.19}$$

Projection to these 26 coordinates annihilates all generators of L_3. Conversely, the 304 discarded Hall coordinates are integral combinations of those generators. One direct certificate uses the identity

$$[[v_i,v_j],v_k] = [[v_i,v_k],v_j] - [[v_j,v_k],v_i] \quad (k<j<i). \tag{E50.20}$$

The included program forms the 360 columns $[t_i,[v_a,v_b]]$ and saves an integer right inverse onto every discarded Hall coordinate. All its nonzero coefficients are 1 or -1; it verifies the right-inverse matrix identity over $\mathbb{Z}$. This is also an explicit certificate of the integral generation argument just given. The decomposition by the number of T letters is

number of T letters	0	1	2	3
$\mathrm{Lie}_3(V)$	2	32	128	168
L_3	0	8	128	168
$\mathrm{Lie}_3(V)/L_3$	2	24	0	0.

(E50.21)

Thus L_3 is saturated of rank 304. Equations (E50.17) and (E50.21) prove the actual abelian kernel rank 332. They also give (E50.5) with free integral quotients. Normality follows either from (E50.16) or from the integral Jacobi inclusions; the latter are checked for every handle and cusp generator in the program.

The further ambient map and the automorphism comparison. Since $F'\subset N$, one has $F''\subset\gamma_2 N$ and then $F'''=[F'',F'']\subset\gamma_4 N$. All these subgroups are preserved by every automorphism of F, since $N=\ker(F\to F^{\mathrm{ab}}/3)$ is characteristic. This proves (E50.6) and its compatibility with based actions. For (E50.7), $w\in F''$ acts trivially on F/F'', whereas its commutator with e_1 is a nonzero Hall combination in L_3. Inner automorphisms preserve the peripheral conjugacy class. This proves the specified based-automorphism claim.

For outer actions, changing an ambient representative by Ad_f changes its restriction by the action of f on N. Elements of N give inner automorphisms of N; the remaining ambiguity is the deck action of $F/N = (\mathbb{Z}/3)^2$. The tangent map in (E50.1) fixes the based representatives used here. This describes the precise transfer without choosing an unspecified outer lift.

The transfer to the cuspidally central quotient. The filling map has kernel I and its composite with multiplication by three has kernel F' by (E50.13). Multiplication by three is injective on T_3E, so $I = F'$. Consequently

$$F'' = [I, I] \subset [N, I], \qquad \gamma_3 N = [N, N'] \subset [N, I], \tag{E50.22}$$

and (E50.8) is the resulting quotient map. In the class-three Hall coordinates, $[N, I]$ is generated by the weight-two commutators with at least one T letter and all of $\gamma_3 C$. Modulo K_0 its weight-two lattice is $L \otimes T$ of rank sixteen and its central weight-three lattice is the rank-twenty-six quotient in (E50.5). These are subsets of the remaining Hall coordinates. Their actual group commutators vanish, because they have weight at least four. As in (E50.17), independent weight-two coordinates split off the central part. This proves the abelian kernel and its rank forty-two. The target and its arithmetic origin are exactly the group $N/[N, I]$ of E.36.

The actual outer factorization. Use $z = [x_1, x_2]^{-1}$ in F and let

$$\mathcal{A} = \mathbb{Z}_3[[F^{\mathrm{ab}}]] = \mathbb{Z}_3[[T_1, T_2]], \qquad X_i = 1 + T_i. \tag{E50.23}$$

The two-generator instance of the completed Fox resolution proved in E.41 identifies F'/F'' with the free $\mathcal{A}$-module on $\bar{z}$: the syzygy kernel of $(a, b) \mapsto aT_1 + bT_2$ is generated by $(T_2, -T_1)$, the Fox image of $\bar{z}$.

Fix $g \in \ker \rho_E$ and let f_g be its actual tangential action reduced on F/F''. The Tate determinant is χ_g, so $\chi_g = 1$. Tangential inertia therefore gives $f_g(\bar{z}) = \bar{z}$ exactly, while f_g acts trivially on the abelianization. Write

$$f_g(\bar{x}_i)\bar{x}_i^{-1} = G_i \cdot \bar{z} \quad (i = 1, 2). \tag{E50.24}$$

Computing the commutator of these two images in the metabelian group gives $f_g(\bar{z}) = (1 + T_2G_1 - T_1G_2) \cdot \bar{z}$. Thus $T_2G_1 = T_1G_2$. Reduction modulo T_1 shows $G_1 \in (T_1)$, because T_2 is a non-zero-divisor in $\mathcal{A}/(T_1)$. Write $G_1 = T_1H_g$ and cancel T_1 to obtain $G_2 = T_2H_g$. This determines H_g uniquely in the actual integral ring. Conjugation by

$$w_g = -H_g \cdot \bar{z} \in F'/F'' \subset N/F'' \tag{E50.25}$$

changes $\bar{x}_i$ by $(1 - X_i)w_g = T_iH_g \cdot \bar{z}$. It therefore equals f_g on the two generators and hence on the whole group. This is the direct calculation in Nakamura's Proposition4.7–§4.8 [Nak95], applied to the actual tangential lift. Its restriction and quotient on $\overline{C}$ is inner by the image of precisely w_g. The genuine outer action thus kills $\ker \rho_E$. Since $G_{\mathbb{Q}}$ is compact and its Tate image is Hausdorff, descent gives the continuous homomorphism (E50.9). This proves the factorization in its stated outer automorphism group.

Finally, in Nakamura's §6.3 $I_m = \ker(\mathbb{Z}_3[[F]] \to \mathbb{Z}_3[[F/\gamma_m F]])$ differs from the augmentation power I^m. The Magnus formula in Theorem6.6 and Corollary6.7 is modulo I_2; its full noncommutative matrix is defined in §6.5. The maps (E50.3) and (E50.6) identify the respective intrinsic targets for that formula and for an actual ambient derived-length-three action [Nak95]. □

Every first coordinate unramified outside three

E.51. All nonzero three-power Kummer first coordinates in the inherited quotient [Proved]

Retain the based arithmetic quotient H of E.46 and the positive real root system defining $\kappa = \kappa_3(3)$ in E.49. Let $r \in \mathbb{Z}_3 \setminus \{0\}$, and write

$$r = 3^k e, \qquad k = v_3(r), \quad e \in \mathbb{Z}_3^\times, \qquad M = 3^{k+1}, \quad h = (M-1)/2. \tag{E51.1}$$

Define a set of rational primes and its restricted three-adic product by

$$\begin{aligned} \mathscr{P}_k &= \{p \neq 3 : v_3(\mathrm{ord}_{\mathbb{F}_p^\times}(3)) \leq k+1\}, \\ \mathcal{R}(\mathscr{P}_k) &= \{(b_p) \in \prod_{p \in \mathscr{P}_k} \mathbb{Z}_3 : \text{for every } a,\ b_p \in 3^a\mathbb{Z}_3 \text{ for almost all } p\}. \end{aligned} \tag{E51.2}$$

The order at $p = 2$ is one. After the choices of prime norm lifts specified below, the complete global section fiber of H with first class $[3r\kappa]$ has the following additive topological coordinates:

$$\boxed{\mathbb{Z}_3 \times \mathcal{R}(\mathscr{P}_k) \times \mathbb{Z}_3^h \times \mathbb{Z}_3[[Z]].} \tag{E51.3}$$

Write its coordinates as $(b_3, (b_p), (z_i)_{i=1}^h, f)$. The scaled toric map of E.46 is exactly

$$(b_3, (b_p), (z_i), f) \longmapsto \left(r\kappa,\ b_3\kappa + \sum_{p \in \mathscr{P}_k} b_p \kappa_3(p) \right). \tag{E51.4}$$

Thus every toric fiber over an allowed second coordinate is explicitly $\mathbb{Z}_3^h \times \mathbb{Z}_3[[Z]]$. All the displayed groups are three-torsion-free.

Here are actual cocycle coordinates for the unramified part. Use the Frobenius basis $\tau, \tau' = c\tau c$ of E.49, for which $\chi(\tau) = 7$ and $\kappa(\tau) = 1$. Put $T = 1 + U$, $W = (T - T^{-1})/2$, and

$$\eta_{M,i} = 1 - \frac{T^{-i} + T^{i-M}}{2} \qquad (1 \leq i \leq h). \tag{E51.5}$$

For the normalized first cocycle $A = M\kappa$, every second-cocycle class unramified outside three has exactly one representative of the form

$$\begin{aligned} v(c) &= 0, \\ v(\tau) &= 3b_3 + \sum_{i=1}^{h} z_i \eta_{M,i} + T^M W f(W^2), \\ v(\tau') &= -v(\tau)(T^{-1}). \end{aligned} \tag{E51.6}$$

For the prescribed cocycle $3r\kappa = eM\kappa$, replace every occurrence of T in the module coordinate by T^e. The exact map accomplishing this change is the continuous arithmetic automorphism

$$\Phi_e : H \xrightarrow{\sim} H, \qquad (a, v(T)) \longmapsto (ea, v(T^e)). \tag{E51.7}$$

Its inverse is $\Phi_{e^{-1}}$. It preserves $J = (3, U)$, commutes with the based Galois action, and preserves the second scaled toric coordinate. On every finite root set the corresponding equivariant bijection is $i \mapsto ei$. Pullback gives an isomorphism from the algebra for $eM\kappa$ to the one for $M\kappa$; its inverse transports the chosen norm systems below. These isomorphisms commute with every tower map and norm.

For each $p \in \mathscr{P}_k$, choose the norm system $\pi_{k,p}$ constructed in the proof. Its divisor is the compatible degree-one prime above p, and all other valuations, including those at three, are zero. Its level-zero norm is p. The ramified factor in the norm coordinates of E.46 is $\prod_p \pi_{k,p}^{3b_p}$. Together with the Kummer–Shapiro image of (E51.6), this specifies the map in (E51.3), including all infinite products.

The first classes unramified outside three are exactly $[3r\kappa]$ for $r \in \mathbb{Z}_3$. E.47 supplies the fiber at zero, and (E51.3) supplies every nonzero fiber. These are section classes of the specified target H; the maps from the intrinsic groups retain the kernels of E.46–E.47.

The program `e_unramified_first.py` checks the exact coordinate maps, finite affine orbits, field factors and norm transitions used below.

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d) and the fourth exactness follow-up.

Proof. The change of first cocycle. The multiplication in H is $(a, v)(b, w) = (a + b, v + T^a w)$. Substituting T^e in this identity proves that (E51.7) is a group homomorphism. Its inverse follows by substituting $T^{e^{-1}}$. The series $T^e - 1$ has unit linear coefficient, hence the substitution is a continuous automorphism of $\mathbb{Z}_3[[U]]$ carrying J to itself. It commutes with $T \mapsto T^{\chi_g}$ and the scalar factor χ_g. Thus it is an arithmetic automorphism, induces a bijection on the stated section fibers, and changes their first coordinate by e. Augmentation is unchanged, which proves the assertion about the second toric coordinate. It suffices to compute the fiber for the actual cocycle $A = M\kappa$.

Integral computation of the unramified group. The free arithmetic basis and the full-to-pro-three cohomology comparison are those proved in E.49. They apply to every finite coefficient quotient here: over $K = \mathbb{Q}(\zeta_3)$ its arithmetic image is a three-group. Its filtration by trivial $\mathbb{F}_3$ modules gives vanishing of unramified H^2 and H^3; its H^1 is finite. These facts also justify the inverse-limit comparisons below.

After normalizing $v(c) = 0$, write $u = v(\tau)$. The other free value is $Cu = -u(T^{-1})$. For $\rho = \tau'\tau^{-1}$ the coefficient operators are

$$Su = 7T^M u(T^7), \qquad Ru = T^{-2M}u, \qquad Cu = -u(T^{-1}). \tag{E51.8}$$

They satisfy $CSC = RS$ and $CRC = R^{-1}$, matching the actual conjugation relations of c, τ, ρ. The remaining coboundaries change u by $Dm = (S - 1)m$, where $Cm = m$. Such m has augmentation zero and belongs to J.

Let $n = T^{-2M} - 1$. The exact factorization

$$n = -T^{-2M}(T^M + 1)(T^M - 1) \tag{E51.9}$$

gives $(n) = (T^M - 1)$, since $T^M + 1$ has constant term two. The quotient is the finite free ring $R_M = \mathbb{Z}_3[C_M]$. Write $\bar{u}_j$ for the coefficient of T^j in the image of u, where $0 \le j < M$. The involution $T \mapsto T^{-1}$ has a fixed submodule of rank $h+1$, with basis given by 1 and the images of $\eta_{M,i}$. Its projection $(1 + \iota)/2$ annihilates every $\overline{Dm}$: substitution by T^7 commutes with ι, and $\bar{m}$ is anti-invariant.

We evaluate the whole quotient by these coboundaries. Set

$$\begin{aligned} a &= \operatorname{aug} u, \qquad z_i = -(\bar{u}_i + \bar{u}_{M-i}), \\ u_* &= u - a - \sum_{i=1}^{h} z_i \eta_{M,i}, \\ m &= \frac{Cu_* - Ru_*}{n}, \qquad v = u_* - Dm. \end{aligned} \tag{E51.10}$$

The image of u_* in R_M is anti-invariant under inversion. Thus the numerator defining m vanishes in R_M and the division is integral. The domain $\Lambda = \mathbb{Z}_3[[U]]$ has no n-torsion. Applying C to the numerator gives $-T^{2M}$ times that numerator, while $C(nm) = -T^{2M} nCm$. Therefore $Cm = m$. The operator identities in (E51.8) now give

$$C(Dm) - R(Dm) = nm, \qquad Cv = Rv. \tag{E51.11}$$

Consequently $T^{-M}v$ is anti-invariant under inversion. The integral change of variable $W = (T - T^{-1})/2$ has inverse $T = W + \sqrt{1 + W^2}$, with constant term one. It identifies Λ with $\mathbb{Z}_3[[W]]$ and inversion with $W \mapsto -W$. This proves $v = T^M W f(W^2)$ for a unique $f \in \mathbb{Z}_3[[Z]]$.

The coordinates are unique modulo coboundaries. The fixed projection in R_M detects every a, z_i. For two representatives with the same finite coordinates, their difference has zero ρ-value. A coboundary with this property satisfies $nm = 0$, hence $m = 0$. Their f coordinates therefore coincide as well. All divisions above are continuous on their closed principal ideals: multiplication by a nonzero series is a continuous bijection from the compact ring onto its image. This proves the topological normal form. Restricting from Λ to J imposes exactly $a \in 3\mathbb{Z}_3$, yielding (E51.6).

The exact allowable residues. At $p \neq 3$, the twisted module is unramified. After removing its Tate twist, an actual arithmetic Frobenius acts on the positions by $x \mapsto M\kappa(F_p) + px$. Its possible fixed point is unique. An integral fixed point is, by the root labels in E.46, exactly a compatible system of 3^nth roots of 3^M in $\mathbb{Q}_p$. Cubing is an automorphism on the pro-p principal units; in the finite residue-unit group the compatible divisibility criterion is

$$3 \nmid \operatorname{ord}_{\mathbb{F}_p^\times}(3^M) \quad \Longleftrightarrow \quad v_3(\operatorname{ord}_{\mathbb{F}_p^\times}(3)) \leq k + 1. \tag{E51.12}$$

This proves the prime set in (E51.2).

The integral fixed-measure argument of E.49 applies verbatim to this affine map: translate an integral fixed point to zero, and use multiplication by p^2 on every other annulus. Each of its nonzero clopen orbits is a free $\mathbb{Z}_3$ orbit. A compatible invariant integral measure there has mass divisible by arbitrarily high powers of three, so is zero. With a nonintegral center, the valuation of the displacement is $t + v_3(n)$ on every orbit, with $t < v_3(p-1)$, and the same argument applies on its finitely many residue balls. Intersecting with J therefore gives

$$(J_A(-1))^{F_p} = \begin{cases} 3\mathbb{Z}_3 \delta_{x_p}, & p \in \mathscr{P}_k, \\ 0, & p \notin \mathscr{P}_k, \end{cases} \tag{E51.13}$$

where x_p denotes the unique fixed point.

The finite-coefficient localization sequence used in E.49 is exact with zero right obstruction because of the unramified H^2 vanishing already checked. The finite unramified H^1 groups satisfy Mittag–Leffler. The continuous inverse-limit comparison therefore gives the exact sequence

$$0 \longrightarrow H^1(G_{\mathbb{Q},\{3\}}, J_A) \longrightarrow H^1(G_{\mathbb{Q}}, J_A) \longrightarrow 3\mathcal{R}(\mathscr{P}_k) \longrightarrow 0. \tag{E51.14}$$

Here the residue at p is its coefficient of δ_{x_p}. Positional pushforward is ideal norm, so its total mass at level zero forces precisely the restrictedness in (E51.2). Conversely that restrictedness is sufficient at every finite coefficient and positional level. This is the actual inverse limit of the direct sums, without replacing an image by its closure.

Explicit norm lifts of the residues. We give a field construction of the splitting of (E51.14). A cofinal range is $n = d + k + 1$ with $d \geq 0$. The finite etale algebra of E.46 factors as

$$E_{A,n} = \prod_{j=0}^{k+1} L_{d,j}, \qquad L_{d,0} = \mathbb{Q}(3^{1/3^d}), \quad L_{d,j} = \mathbb{Q}(3^{1/3^d} \zeta_{3^{d+j}}) \quad (j \geq 1). \tag{E51.15}$$

The degrees are 3^d and $3^d\varphi(3^j)$. The first polynomial is Eisenstein at three. For the others, the element $\rho = \tau'\tau^{-1}$ in the actual unramified arithmetic group has $\chi(\rho) = 1$ and $\kappa(\rho) = -2$. Its powers fix every three-power root of unity and permute all 3^d roots above a prescribed $3\zeta_{3^j}$. These are actual arithmetic actions on the root torsor: its Galois action factors through that group, since the splitting fields are unramified outside three. Moreover the generator in (E51.15) contains ζ_{3^j}. This proves the asserted degree and irreducibility. Equivalently, the polynomial identity is

$$X^{3^{d+k+1}} - 3^{3^{k+1}} = (X^{3^d} - 3) \prod_{j=1}^{k+1} 3^{\varphi(3^j)} \Phi_{3^j}(X^{3^d}/3). \tag{E51.16}$$

Cubing the chosen generator at $d+1$ gives the one at d, so these are the actual field maps in the norm tower.

For $j \geq 1$ each vertical layer is cyclic cubic and is unramified away from three. At $d = 0$ it starts at $\mathbb{Q}(\zeta_{3^j})$, whose three-primary class group is zero by E.47. There is a unique prime above three. A nontrivial cyclic cubic layer unramified also at this prime would be everywhere unramified, contradicting that class-group calculation. Thus this prime is totally ramified, remains unique, and has residue degree one. The ambiguous class-number formula proves inductively that the next class group again has order prime to three: its fixed three-primary subgroup is zero, while a nonzero finite three-group acted on by a group of order three has nonzero fixed points. The strongly ambiguous formula [Lem13, Theorem1, formula(2)] also shows that the ordinary unit norm is onto, because its index is a power of three dividing the base class number.

For the $j = 0$ factor, first adjoin $K = \mathbb{Q}(\zeta_3)$. The resulting vertical layers are cyclic cubic; the valuation $2/3^d$ of the radical proves their degree and their unique totally ramified prime. The same argument starts at K and proves both claims there. Extension and norm over the quadratic map remove the three-primary class group downstairs, and the integral projector $(1+c)/2$ proves surjectivity of the completed ordinary unit norms in the pure radical tower. All factors of (E51.15) therefore have surjective completed divisor maps and surjective completed unit norms.

For $p \in \mathscr{P}_k$, its fixed position gives a unique compatible degree-one prime chain in these algebras. At each level, take a generator of a prime-to-three power of this ideal and raise it to the inverse of that power in the three-completion. This gives precisely the desired divisor. Correct each succeeding norm discrepancy by an ordinary unit using the surjective unit norm just proved. Induction constructs $\pi_{k,p}$ with no other valuation, including at three. Its norm to level zero is exactly p, since the three-completion of the ordinary rational units is zero. This supplies the claimed explicit choice of residue lift.

Every global norm system satisfying the cube condition of E.46 has unique residue coefficients $3b_p$. Divide by $\prod_p \pi_{k,p}^{3b_p}$; the remaining class is unramified outside three and therefore has the unique coordinates in (E51.6). The product converges in every power-class and positional quotient because b_p tends to zero three-adically. The construction and its inverse are continuous. This proves (E51.3), its group law and topology, and (E51.4). Changing a chosen $\pi_{k,p}$ multiplies it by an ordinary-unit norm system and changes the compact coordinates by the corresponding convergent additive sum.

Finally, the completed rational Kummer group has the restricted prime-valuation coordinates of E.16. Being unramified outside three sets every coordinate except the one at three to zero. Thus its first-coordinate classes are exactly the stated $3r\kappa$. The zero fiber and the nonzero fibers are all accounted for by their explicit maps. No lift through the intrinsic geometric kernels is asserted by this quotient calculation. □

The entire nonzero first-coordinate line generated by two

E.52. Complete global sections for every nonzero multiple of the Kummer class of two [Proved]

Use the based arithmetic quotient $H = 3\mathbb{Z}_3 \ltimes J$ of E.38 and E.46, where $J = (3, T-1)\mathbb{Z}_3[[T-1]]$. Choose positive compatible real roots of two and put $\kappa = \kappa_3(2)$. For a nonzero $r \in \mathbb{Z}_3$, write uniquely

$$r = 3^{m-1}e, \qquad m \geq 1, \quad e \in \mathbb{Z}_3^{\times}, \qquad h_m = (3^m - 1)/2. \tag{E52.1}$$

Define

$$\begin{aligned} \mathscr{P}_m^{(2)} &= \{p \neq 2, 3 : \ v_3(\mathrm{ord}_{\mathbb{F}_p^{\times}}(2)) \leq m\}, \\ \mathcal{R}_m^{(2)} &= \{(b_p)_{p \in \mathscr{P}_m^{(2)}} : b_p \in \mathbb{Z}_3, \ b_p \longrightarrow 0 \text{ modulo every } 3^a\}, \\ \mathcal{U}_m &= \mathbb{Z}_3^{h_m - m} \times \prod_{n \geq m} \mathbb{Z}_3^{3^n}. \end{aligned} \tag{E52.2}$$

The topology on $\mathcal{R}_m^{(2)}$ is the inverse-limit topology of its finite-exponent direct sums. All factors of $\mathcal{U}_m$ have their usual compact topology and the displayed product has the product topology.

The entire set of global section classes of H whose first class is $[3r\kappa]$ has the following complete coordinates:

$$\boxed{(b_2, b_3, (b_p), \boldsymbol{z}_2, \boldsymbol{z}_3, \boldsymbol{u}) \in \mathbb{Z}_3^2 \times \mathcal{R}_m^{(2)} \times \mathbb{Z}_3^m \times \mathbb{Z}_3^m \times \mathcal{U}_m.} \tag{E52.3}$$

The proof specifies norm generators, finite fundamental-unit matrices, and right inverses that give an actual homeomorphism. In the scaled toric target of (E46.7), its value is exactly

$$\left(r\kappa, \ b_2\kappa_3(2) + b_3\kappa_3(3) + \sum_{p \in \mathscr{P}_m^{(2)}} b_p \kappa_3(p) \right). \tag{E52.4}$$

Thus every second toric class supported on $\{2, 3\} \cup \mathscr{P}_m^{(2)}$ occurs, and its fiber has the explicit compact coordinates

$$\mathbb{Z}_3^{h_m + m} \times \prod_{n \geq m} \mathbb{Z}_3^{3^n}. \tag{E52.5}$$

The finite block in (E52.5) is obtained by the displayed coordinate concatenation $(\boldsymbol{z}_2, \boldsymbol{z}_3, \boldsymbol{u}_0)$; no arithmetic information is discarded. With origin $g \mapsto (3r\kappa_g, 0)$ the entire fiber (E52.3) is an abelian torsion-free group. The case $r = 0$ is the explicitly computed fiber (E47.2), related here by the common projection $H \to 3\mathbb{Z}_3(1)$.

Here are the fields and maps that construct these coordinates. First take $e = 1$, and use the cofinal levels $n = m + d$, $d \geq 0$. Put

$$\begin{aligned} \alpha_d &= 2^{1/3^d}, \qquad F_d = \mathbb{Q}(\alpha_d), \\ \beta_{d,j} &= \alpha_d \zeta_{3^{d+j}}, \qquad L_{d,j} = \mathbb{Q}(\beta_{d,j}) \quad (1 \leq j \leq m), \\ E_{A,m+d} &= F_d \times \prod_{j=1}^{m} L_{d,j}, \qquad A = 3^m\kappa. \end{aligned} \tag{E52.6}$$

The transition maps are cubing of the indicated generators. Each field has one prime above two and one above three. At two, their absolute residue degrees are

$$w_0 = 1, \qquad w_j = \varphi(3^j) \quad (1 \leq j \leq m); \tag{E52.7}$$

at three all residue degrees are one. Every class number in (E52.6) is prime to three.

In the jth complex factor let $\mathcal{E}^1_{d,j}$ be the kernel of the three-primary residue map at two on the completed ordinary unit group. Then

$$N : \mathcal{E}^1_{d+1,j} \twoheadrightarrow \mathcal{E}^1_{d,j}, \qquad \mathcal{E}^1_{d,j} \simeq \mathbb{Z}_3^{3^{d+j-1}-1}. \tag{E52.8}$$

The real-factor completed unit norm is likewise onto, with rank $(3^d-1)/2$ at level d. The inverse norm limit of all completed ordinary units in (E52.6) is precisely $\mathcal{U}_m$, by the following explicit matrix prescription.

Choose fundamental units in each field. Multiply each complex-factor unit by the unique power of ζ_{3^j} which removes its three-primary residue at two. Their images are a basis of $\mathcal{E}^1_{d,j}$. Together with a fundamental-unit basis in F_d, they give a basis $\boldsymbol{\epsilon}_d$ of rank

$$R_d = (3^{m+d}-1)/2 - m. \tag{E52.9}$$

Let $B_d \in M_{R_d,R_{d+1}}(\mathbb{Z})$ be the matrix of the actual relative norms in these bases, after removing the prime-to-three torsion. Every Smith invariant of B_d is a three-adic unit. Choose a right inverse S_d and a kernel basis K_d over $\mathbb{Z}_{(3)}$, so that

$$B_dS_d = I, \qquad B_dK_d = 0, \qquad [S_d\ K_d] \in \mathrm{GL}_{R_{d+1}}(\mathbb{Z}_3). \tag{E52.10}$$

For $\boldsymbol{u} = (\boldsymbol{u}_0, \boldsymbol{t}_m, \boldsymbol{t}_{m+1}, \ldots) \in \mathcal{U}_m$, set

$$\boldsymbol{a}_0 = \boldsymbol{u}_0, \qquad \boldsymbol{a}_{d+1} = S_d\boldsymbol{a}_d + K_d\boldsymbol{t}_{m+d}, \qquad \epsilon(\boldsymbol{u})_d = \boldsymbol{\epsilon}_d^{\boldsymbol{a}_d}. \tag{E52.11}$$

This is the promised norm-coherent unit system, with a unique inverse given by the same finite matrices.

Choose norm-coherent systems $\xi_{3,j}$ with valuation one along the jth three-prime chain and zero elsewhere, $0 \le j \le m$. The proof constructs them. The two-prime systems require no choice:

$$\xi_{2,0,d} = \alpha_d, \qquad \xi_{2,j,d} = \beta_{d,j}, \tag{E52.12}$$

placed in the indicated factor and with one in every other factor. They have total norms 2^{w_j}. One may take, explicitly, $\xi_{3,0,d} = 1+\alpha_d$ and $\xi_{3,1,d} = 1+\beta_{d,1}$, again supported in their individual factors; their total norms are three. For every $p \in \mathscr{P}^{(2)}_m$, choose the constructed norm system π_p with divisor the unique compatible degree-one prime chain above p. It has total norm p.

Write $\boldsymbol{z}_i = (z_{i,1}, \ldots, z_{i,m})$. The complete norm-to-section formula for (E52.3), with $e = 1$, is

$$\epsilon(\boldsymbol{u})\, \xi_{2,0}^{3b_2 - \sum_{j=1}^m w_j z_{2,j}} \prod_{j=1}^m \xi_{2,j}^{z_{2,j}}\, \xi_{3,0}^{3b_3 - \sum_{j=1}^m z_{3,j}} \prod_{j=1}^m \xi_{3,j}^{z_{3,j}} \prod_{p \in \mathscr{P}^{(2)}_m} \pi_p^{3b_p}. \tag{E52.13}$$

The last product converges in every finite-exponent quotient. For general e, apply the actual arithmetic automorphism

$$\Phi_e : H \longrightarrow H, \qquad (a, v(T)) \longmapsto (ea, v(T^e)). \tag{E52.14}$$

It preserves augmentation and gives the first and second coordinates stated in (E52.4).

The maps from the intrinsic geometric groups remain the actual quotient maps

$$N \twoheadrightarrow H, \qquad N/N'' \twoheadrightarrow H, \qquad N/\overline{\gamma_3(I_Q)} \twoheadrightarrow H \tag{E52.15}$$

with kernels respectively $[I_y, I_y]$, $[I_y, I_y]/N''$, and $[I_y, I_y]/\overline{\gamma_3(I_Q)}$ as in (E47.8). Formula (E52.3) computes the target section fibers of these maps. The source lifting obstructions are retained by those exact kernels. This addresses the arithmetic section problem arising from §51.16, fols. 733v–736v, scans 172–178, with the full first-coordinate line and every continuous norm-compatible second coordinate specified.

Proof. The actual factorization and its ramification. For $A = 3^m\kappa$, E.46 gives the finite etale algebra $\mathbb{Q}[X]/(X^{3^{m+d}} - 2^{3^m})$. The exact factorization

$$X^{3^{m+d}} - 2^{3^m} = (X^{3^d} - 2) \prod_{j=1}^{m} 2^{\varphi(3^j)} \Phi_{3^j}(X^{3^d}/2) \tag{E52.16}$$

gives (E52.6). The real factor is Eisenstein at two. For $j \geq 1$, $\beta_{d,j}^{3^d} = 2\zeta_{3^j}$ shows that $L_{d,j}$ contains $K_j = \mathbb{Q}(\zeta_{3^j})$. Two is a primitive root modulo 3^j: the assertion for $j = 1$ and $v_3(2^{2\cdot 3^{j-1}} - 1) = j$ prove this by induction. Thus K_j has one prime over two, of residue degree $\varphi(3^j)$. The valuation of $2\zeta_{3^j}$ there is one. The equation $X^{3^d} = 2\zeta_{3^j}$ is Eisenstein at that prime, proving the full degree $[L_{d,j} : K_j] = 3^d$ and total ramification at two. This proves irreducibility of every factor in (E52.16) and all its specified field embeddings.

The vertical extension $L_{d+1,j}/L_{d,j}$ is cyclic cubic: it has degree three by its two-adic ramification, and the base contains μ_3. Its polynomial $X^3 - \beta_{d,j}$ shows that only two and three can ramify. We now prove simultaneously that three has one prime, totally ramified in every vertical step, and that every class number is prime to three. At $d = 0$ the field is K_j, whose three-class group vanishes by the one-prime induction proved in E.47.

Suppose the assertions hold in the base $B = L_{d,j}$ and put $C = L_{d+1,j}$. The residue field at two remains $\mathbb{F}_{2^{\varphi(3^j)}}$. In a totally ramified cubic extension the residue of a unit norm is the cube of its residue. The reduction of ζ_{3^j} has order 3^j, and the three-primary order of this residue multiplicative group is exactly 3^j. It is therefore not a cube. A field element whose norm is a unit has valuation zero at this unique prime, so the same obstruction excludes ζ_{3^j} from $N_{C/B}C^\times$.

Apply the two exact formulas in Lemmermeyer's Theorem 1 [Lem13]:

$$\begin{aligned} \#\operatorname{Am}(C/B) &= \frac{h_B 3^{t-1}}{[\mathcal{O}_B^\times : \mathcal{O}_B^\times \cap N_{C/B}C^\times]}, \\ \#\operatorname{Am}_{\mathrm{st}}(C/B) &= \frac{h_B 3^{t-1}}{[\mathcal{O}_B^\times : N_{C/B}\mathcal{O}_C^\times]}. \end{aligned} \tag{E52.17}$$

Here $t \leq 2$, since the only possible ramified primes are the unique primes over two and three, and there is no real-place ramification. Every base unit cubed is a norm, so both displayed indices are powers of three. The local obstruction just proved makes both indices at least three. Because $3 \nmid h_B$, integrality of the first formula forces $t = 2$ and forces its index to equal three. The three-prime is therefore ramified and, in this cubic extension, totally ramified with one prime above it. The ambiguous class group has order prime to three. If the three-class group of C were nonzero, its subgroup of elements killed by three would be a nonzero vector space in characteristic three acted on by a cyclic group of order three. The nilpotence of $\sigma - 1$ gives a nonzero fixed vector. That would give an ambiguous three-class, which was excluded. This proves the induction. The second formula in (E52.17) also forces its index to equal three. Both the element-norm and the actual unit-norm image are consequently exactly the kernel of the nontrivial residue-cube character at two.

The identical induction applies to $M_d = K_1F_d$, starting from $M_0 = K_1$, with cubic equation $X^3 - \alpha_d$. Extension of ideals $\mathrm{Cl}(F_d) \to \mathrm{Cl}(M_d)$ followed by norm is multiplication by two. It is injective on the three-primary subgroup, so F_d has no three-class group either. The unique three-prime of M_d, with absolute ramification index $2 \cdot 3^d$, lies over a unique prime of F_d; the intermediate quadratic extension forces its ramification index in $F_d/\mathbb{Q}$ to be 3^d. This proves every assertion about class numbers and primes in (E52.6)–(E52.7).

Actual unit norms, residues, and the free coordinates. For a complex factor write $\mathcal{E}_{d,j} = \widehat{\mathcal{O}^\times_{L_{d,j}\,3}}$. Reduction at two followed by projection to its three-primary part gives the split

sequence

$$0 \longrightarrow \mathcal{E}^1_{d,j} \longrightarrow \mathcal{E}_{d,j} \longrightarrow \mu_{3^j} \longrightarrow 0. \tag{E52.18}$$

The splitting is the actual inclusion of μ_{3^j} into the field. There are no additional three-power roots of unity: reduction at the two-prime injects such roots into the fixed residue field, whose three-primary multiplicative subgroup has order 3^j. It follows that $\mathcal{E}^1_{d,j}$ is torsion-free. Dirichlet's unit rank and the degree already proved give its rank $3^{d+j-1} - 1$.

The actual norm index three in (E52.17) identifies its image with the residue-cube subgroup. Every element of $\mathcal{E}^1_{d,j}$ therefore has a unit preimage. The three-primary residue of any such preimage has cube one. Multiplication by the suitable element of μ_3 removes that residue and leaves its norm unchanged, since $N_{C/B}\zeta_3 = \zeta_3^3 = 1$. This proves (E52.8). These deductions hold on three-completions as stated: the actual index-three unit quotient survives completion exactly, and all groups of units modulo torsion are finitely generated free abelian.

For the real factor, the three-completed units of F_d identify with the complex-conjugation invariants in those of M_d. Inclusion followed by the quadratic norm is multiplication by two, and the projector is $(1 + c)/2$. A unit from F_d has residue one at its two-prime, whose residue field is $\mathbb{F}_2$. It has a norm preimage in M_{d+1} by the already proved unit-norm criterion. Averaging this preimage by $(1 + c)/2$ gives a preimage in the completed units of F_{d+1}. Thus these norm maps are onto as well.

Under (E52.18), the map on μ_{3^j} is cubing, while its map on the residue-one units is onto. An inverse norm system in the finite residue group is zero: iterating j cubings is already zero. Hence the entire ordinary-unit norm limit is exactly the limit of the real completed units and the groups $\mathcal{E}^1_{d,j}$. Their total rank is

$$\frac{3^d - 1}{2} + \sum_{j=1}^{m} (3^{d+j-1} - 1) = R_d, \qquad R_{d+1} - R_d = 3^{m+d}. \tag{E52.19}$$

The normalized fundamental units in the statement are explicit bases of these completed groups. Their relative norms are products of the next base's normalized fundamental units, times a root of unity of order prime to three. Indeed the remaining three-power root would have nontrivial residue. This defines the integer matrix B_d. Surjectivity of the norm proves that all its nonzero Smith entries are prime to three. For an entirely specified construction of (E52.10), take the lexicographically first maximal minor whose determinant is a three-adic unit. Its inverse gives S_d in those pivot rows, with zero in the other rows. For each remaining column indexed by i, take the kernel vector $e_i - S_d B_d e_i$. These are the columns of K_d. The resulting square matrix $[S_d\ K_d]$ has determinant a three-adic unit; its inverse extracts $B_d \boldsymbol{a}$ and the nonpivot coordinates. This proves (E52.10) and the bijection (E52.11), including its continuous inverse. All these factors are compact, so the inverse-limit identification has exactly the product topology asserted for $\mathcal{U}_m$.

Every surviving valuation, and actual lifting of divisors. Outside two and three the coefficient action in E.46 is unramified. For an arithmetic Frobenius F_p, the action on the position space $\mathbb{Z}_3$ is the affine map

$$x \longmapsto px + A_{F_p}. \tag{E52.20}$$

The inertia Tate twist cancels the coefficient cyclotomic scalar. There is an integral fixed point exactly when $x_p = -A_{F_p}/(p - 1)$ lies in $\mathbb{Z}_3$. For the actual Kummer torsor of 2^{3^m}, this is equivalent to having compatible roots of all three-power orders in $\mathbb{Q}_p$. Cubing is an automorphism on the principal units for $p \neq 3$; the finite residue group therefore gives precisely $p \in \mathscr{P}^{(2)}_m$. This argument includes primes $p \equiv 2 \pmod 3$, where the fixed point always exists.

Here is the full integral measure calculation behind this criterion. When the fixed point is integral, translate it to zero and use the square of the Frobenius action. Its action off zero is

multiplication by p^2, and every nonzero point belongs to a clopen orbit homeomorphic to $\mathbb{Z}_3$ under the powers of p^2. A translation-invariant $\mathbb{Z}_3$-valued measure on this orbit is zero: the common mass of a level-a coset must be divisible by every power of three upon refinement. If no integral fixed point exists, then $p \equiv 1 \pmod 3$. Writing $s = v_3(p-1)$ and $t = v_3(A_{F_p}) < s$, the affine map acts freely on the clopen cosets modulo 3^t; the identity

$$v_3(\phi^a(x) - x) = t + v_3(a) \quad (a \in \mathbb{Z}_3,\ a \neq 0) \tag{E52.21}$$

gives the same orbit description. Thus the fixed measure module is $\mathbb{Z}_3\delta_{x_p}$ at an eligible prime and zero otherwise. At the finite positional levels the orbit sum is the valuation-one prime divisor, and pushforward multiplies it by the relative residue degree, which is the actual ideal norm. Consequently the unique fixed point is exactly the unique compatible degree-one prime chain. All other valuation directions have zero inverse norm limit.

At two and three the field calculation above supplies all $2(m+1)$ chains, with relative residue degree one. The complete inverse limit of completed divisor groups is therefore

$$\mathbb{Z}_3^{m+1} \times \mathbb{Z}_3^{m+1} \times \mathcal{R}_m^{(2)}. \tag{E52.22}$$

The restricted convergence condition is necessary already at positional level zero and sufficient at every level: all fixed-point measures have integral unit mass. This proves the assertion in the actual norm system, including its inverse-limit topology.

At every finite field, its class group has order prime to three. The completed principal-divisor map is therefore onto the completed divisor group, with kernel the completed ordinary units. Explicitly, a prime ideal raised to the class number is principal, and the inverse of that class number in $\mathbb{Z}_3$ gives a completed element with valuation one. The kernel assertion follows by splitting the original map onto its free abelian principal ideal image and then taking the specified multiplication-by-3^a completions.

For completeness, the finite residue unit obstruction is also removed by an explicit formula. Take arbitrary completed elements z_d with prescribed compatible divisors in one complex factor, and write the norm defects Nz_{d+1}/z_d as units. Let r_d be their μ_{3^j} component in (E52.18). Define

$$\rho_d = \prod_{a=0}^{j-1} r_{d+a}^{3^a} \in \mu_{3^j}. \tag{E52.23}$$

Then $\rho_d = r_d \rho_{d+1}^3$. Replacing z_d by $z_d\rho_d$ makes every norm defect have three-primary residue one. The remaining defects are removed recursively using the onto maps (E52.8), with the chosen matrix right inverses. The real factor uses its onto unit norm maps directly. This constructs an actual norm lift of every divisor system in (E52.22). It also verifies the inverse-limit exactness: the images of ordinary unit norms stabilize after at most m steps to their residue-one subgroups, and these subgroups have onto norm transitions. No uncomputed $\varprojlim^1$ term is omitted.

The specified $\xi_{3,j}$ and π_p now follow by applying this construction to their individual prime chains. For the other generators use the identities

$$\begin{aligned} N_{d+1/d}(\alpha_{d+1}) &= \alpha_d, & N_{d+1/d}(\beta_{d+1,j}) &= \beta_{d,j}, \\ N_{d+1/d}(1+\alpha_{d+1}) &= 1+\alpha_d, & N_{d+1/d}(1+\beta_{d+1,1}) &= 1+\beta_{d,1}. \end{aligned} \tag{E52.24}$$

Each equality is the exact norm in an odd cubic extension, computed from $X^3 - a$. The total norms of $1+\alpha_d$ and $1+\beta_{d,1}$ are respectively 3 and $N_{K_1/\mathbb{Q}}(1+2\zeta_3) = 3$. Their only nonzero valuation is therefore one at the indicated three-prime. This proves all the literal generator claims.

The full section coordinates and their topology. Divide an arbitrary norm system by the systems just constructed with its actual prime valuations. The quotient is an ordinary-unit system and has the unique form $\epsilon(\boldsymbol{u})$ by (E52.11). Thus the valuation and unit coordinates are both exhaustive and unique. If $d_{2,j}$ and $d_{3,j}$ denote the bad-prime valuations, its level-zero norm has valuation $\sum_j w_j d_{2,j}$ at two, $\sum_j d_{3,j}$ at three, and c_p at an eligible prime. The exact cube condition of E.46 is therefore

$$\sum_{j=0}^{m} w_j d_{2,j} \in 3\mathbb{Z}_3, \qquad \sum_{j=0}^{m} d_{3,j} \in 3\mathbb{Z}_3, \qquad c_p \in 3\mathbb{Z}_3. \tag{E52.25}$$

Solving these three displayed conditions with the coefficient one of $d_{2,0}$ and $d_{3,0}$ gives exactly (E52.13). The infinite product converges because at every coefficient exponent only finitely many b_p are nonzero. Valuations give a continuous inverse to these coordinates; the remaining unit inverse is the compact continuous matrix inverse (E52.11). This proves the homeomorphism, its toric formula, and all its stated fibers. Changes of the chosen prime norm lifts multiply them by definite ordinary-unit norm systems, and change $\boldsymbol{u}$ by the corresponding convergent linear series. They preserve the displayed toric map.

Finally, substitution $T \mapsto T^e$ is a continuous automorphism of Λ, with inverse $T \mapsto T^{e^{-1}}$. It preserves J and intertwines T^a with T^{ea}. This verifies the group law in (E52.14). The arithmetic action is $g(a, v) = (\chi_g a, \chi_g v(T^{\chi_g}))$; these substitutions commute with Φ_e. Augmentation is unchanged. Hence Φ_e gives the exact transport from the computed first cocycle $3^m\kappa$ to $3r\kappa$, and changes no second toric coordinate. All groups in (E52.3) are torsion-free, which proves the final group assertion.

Executed finite certificates. The retained programs `e_two_units.gp` and `e_two_line.py` record the actual integral bases, fundamental units, normalized units, relative norms, and matrix splittings. The PARI computation uses `bnfinit(P,1)` followed by `bnfcertify`; the latter returns one for each of the eight fields

$$F_0, F_1, F_2, \quad L_{0,1}, L_{1,1}, L_{2,1}, \quad L_{0,2}, L_{1,2}. \tag{E52.26}$$

Thus every finite unit basis and class group used in these checks is certified without a hypothesis on zeros of zeta functions [The26]. All eight certified class numbers are one. In the recorded normalized bases the three norm matrices with nonzero target rank are

$$\begin{aligned} B_{F_2/F_1} &= \begin{pmatrix} 3 & 2 & -1 & 1 \end{pmatrix}, \\ B_{L_{2,1}/L_{1,1}} &= \begin{pmatrix} 0 & -3 & 0 & 0 & 0 & 0 & 1 & 1 \\ 3 & -3 & 1 & 1 & 0 & 0 & 1 & 1 \end{pmatrix}, \\ B_{L_{1,2}/L_{0,2}} &= \begin{pmatrix} 0 & -3 & 0 & 1 & 1 & 0 & 1 & -1 \\ 3 & -3 & 1 & 0 & 0 & 1 & 0 & -1 \end{pmatrix}. \end{aligned} \tag{E52.27}$$

Their Smith entries are respectively (1), $(1, 1)$, and $(1, 1)$. The exact rational right inverses, kernel bases, and inverse coordinate matrices are retained in the JSON certificate. The Python program independently computes all twenty-three relative unit norms as multiplication determinants in the actual cubic algebras, verifies their unit relations and all matrix identities, and checks the rank formulas, prime-chain criteria, residue correction, and complete valuation coordinates. These finite certificates verify the specified maps; the preceding induction and norm-limit arguments prove the assertions at every level. □

An evaluated class-three Frobenius quotient of the cubic

E.54. The genuine class-three action, its tangential transport, and every local lifting fiber at two [Proved]

Put $K=\mathbb{Q}(\zeta)$, $\zeta=\zeta_3$, $\lambda=1-\zeta$, and $a=\zeta\lambda^{-3}$. Use the cubic U_3, its tangent $s=z/x$ of value one at $O=[1:-1:0]$, and the actual map

$$h=\frac{x+y}{x+\zeta y}:U_3\longrightarrow\mathbb{P}^1\setminus\{0,1,\infty\}. \tag{E54.1}$$

Let $N=\pi_1^{(3)}(U_{3,\overline{K}},\vec{O})$ and let F be the free pro-three group on the standard positive peripheral generators x,y of the target. Choose the comparison path in the punctured formal disc at zero, with compatible roots of a. Then $h_*:N\twoheadrightarrow F$ is surjective. In particular there is an actual arithmetic quotient

$$N/\gamma_4N\twoheadrightarrow C=F/\gamma_4F,\qquad \ker=\frac{h_*^{-1}(\gamma_4F)}{\gamma_4N}. \tag{E54.2}$$

The maps from N/N'' and from $N/\overline{\gamma_3(I_Q)}$ to C factor through the corresponding maps to N/γ_4N in E.41.

Write

$$c=[x,y],\qquad d=[x,c],\qquad e=[y,c],\qquad [u,v]=uvu^{-1}v^{-1}. \tag{E54.3}$$

Every element of C has unique integral Hall coordinates $x^Ay^Bc^nd^re^s$, with all five exponents in $\mathbb{Z}_3$. For $g\in G_K$, put $\chi=\chi_3(g)$ and $k=\kappa_a(g)$ using the chosen roots. The genuine based action on C is

$$\begin{aligned}&\alpha_{\chi,k}(x)=x^{\chi},\\&\alpha_{\chi,k}(y)=x^kc^{-b(\chi)}y^{\chi}c^{b(\chi)}x^{-k},\qquad b(\chi)=\frac{\chi^2-1}{24}\in\mathbb{Z}_3.\end{aligned} \tag{E54.4}$$

Equivalently, all its Hall images are

$$\begin{aligned}&x\longmapsto x^{\chi},\\&y\longmapsto y^{\chi}c^{k\chi}d^{\chi k(k-1)/2}e^{\chi b(\chi)-k\chi(\chi+1)/2},\\&c\longmapsto c^{\chi^2}d^{\chi^2(\chi-1)/2+k\chi^2}e^{\chi^2(\chi-1)/2},\\&d\longmapsto d^{\chi^3},\qquad e\longmapsto e^{\chi^3}.\end{aligned} \tag{E54.5}$$

The composition law is $\alpha_{\chi,k}\alpha_{\mu,l}=\alpha_{\chi\mu,k+\chi l}$. For every compatible root choice its full image over K is the faithful image of the explicit group

$$\Gamma_a=\left\{(\chi,k)\in(1+3\mathbb{Z}_3)\ltimes\mathbb{Z}_3:k\equiv\frac{\chi-1}{3}\pmod 3\right\}. \tag{E54.6}$$

Its outer image is the cyclotomic quotient, with the factorization supplied by $\alpha_{\chi,k}=\mathrm{Ad}(x^k)\alpha_{\chi,0}$.

Here is an exact local root choice and Frobenius evaluation. At the inert prime of K over two the residue field is $\mathbb{F}_4$ and λ has Teichmuller part ζ^2. Take

$$a_n=\zeta_{3^{n+1}}(a/\zeta)^{1/3^n}, \tag{E54.7}$$

where the second factor is the unique 3^nth root in the principal pro-two units. Arithmetic Frobenius F_2 then has $\chi(F_2)=4$ and $\kappa_a(F_2)=1$ exactly. Thus

$$\begin{aligned}&\alpha_{F_2}(x)=x^4,\qquad &&\alpha_{F_2}(y)=y^4c^4e^{-15/2},\\&\alpha_{F_2}(c)=c^{16}d^{40}e^{24},\quad &&\alpha_{F_2}(d)=d^{64},\qquad &&\alpha_{F_2}(e)=e^{64}.\end{aligned} \tag{E54.8}$$

The alternate assembled roots $a_n^{\mathrm{old}} = \zeta_{3^{n+1}}\lambda_n^{-3}$, with $\lambda_n = \zeta_{3^{n+1}}^{2}(\lambda/\zeta^2)^{1/3^n}$, give $\kappa_a(F_2) = -5$. The actual transport between these two actions is

$$\frac{a_n}{a_n^{\mathrm{old}}} = \zeta_{3^n}^{2}, \qquad k_{\mathrm{new}} = k_{\mathrm{old}} + 2(\chi - 1), \qquad \alpha_{\mathrm{new}} = \mathrm{Ad}(x^{-2})\alpha_{\mathrm{old}}\,\mathrm{Ad}(x^2). \tag{E54.9}$$

Put $C_2 = C/\gamma_3 C$. All local C_2-valued cocycles at K_2 are described by their values

$$\nu = x^a y^b c^n, \qquad z = x^A y^B c^N \tag{E54.10}$$

on a positive tame inertia generator and arithmetic Frobenius, respectively. The exact condition on the six coordinates is

$$L = 4Ab - 4Ba + 6ab + 4b + 12n = 0. \tag{E54.11}$$

Use the same words as Hall lifts to C, with zero d, e coordinates, and define

$$\begin{aligned} R &= 2A^2b - 4ABa + 16Aab + 2Ab + 16An - 8Ba^2 + 2Ba - 4Na \\ &\quad + 17a^2b + 13ab + 54an + 40n, \\ S &= \big(16Ab^2 - 4Ab - 4B^2a + 4Ba + 8Bb + 32Bn - 8Nb \\ &\quad + 14ab^2 - 6ab + 16b^2 + 108bn + b + 48n\big)/2. \end{aligned} \tag{E54.12}$$

The image of the map on local section classes $H^1(K_2, C) \to H^1(K_2, C_2)$ is exactly the set of classes whose representatives satisfy

$$R, S \in 3\mathbb{Z}_3. \tag{E54.13}$$

For each such representative, one lift is specified by

$$t \longmapsto \nu d^{-R/60} e^{-S/60}, \qquad F_2 \longmapsto z. \tag{E54.14}$$

Every nonempty geometric-conjugacy fiber has exactly 81 elements. Its actual coordinates are the central changes $z \mapsto z d^u e^v$, with $u, v \in \{0, \dots, 8\}$; its translation group is $(\mathbb{Z}/9)^2$.

For a small explicit family take inertia y and Frobenius $x^{-1}c^N$ in C_2. These satisfy (E54.11), and their class-three error is

$$e^{\,5/2 - 4N}. \tag{E54.15}$$

The members with $N \equiv 1 \pmod 3$ lift. The member $N = 0$ has empty fiber, whereas the member $N = 1$ has the exact lift with inertia $ye^{1/40}$ and Frobenius $x^{-1}c$. They have the same abelian coordinates.

All the obstruction maps above are attached to (E54.2). More precisely, for the local maps $N/\gamma_4 N \to C$ and $N/\gamma_3 N \to C_2$, the diagram of section sets commutes, and the central map is $\mathrm{Lie}_3(N^{\mathrm{ab}}) \to \mathrm{Lie}_3(F^{\mathrm{ab}}) = \mathbb{Z}_3(3)^2$. This gives an evaluated component of the intrinsic obstruction in E.44. The displayed local C_2 classes are classes of this specified quotient; their preimages in the full intrinsic section set are governed by the kernel in (E54.2). The further quotient of C obtained by killing $\gamma_3 C$ and the cubes of x, y, c is exactly the order-27 quotient of E.31 and E.35, with its stated based arithmetic action. The manuscript passage is §51.16, folios 733v–736v, scans 172–178, equations (371)–(379).

Proof. The geometric quotient and its based action. The divisors computed directly in E.31 are

$$\mathrm{div}(h) = 3O - 3C_2, \qquad \mathrm{div}(1 - h) = B_0 + B_1 + B_2 - 3C_2. \tag{E54.16}$$

The two Kummer characters are independent modulo three. Indeed the valuation at any B_i first kills the coefficient of $1 - h$. A cube root of h would have divisor $O - C_2$, which would give

a degree-one rational function on the elliptic curve. The latter is impossible: a degree-one finite map between smooth projective curves is an isomorphism, and the genera are one and zero. Thus the image of N in $F/\Phi(F)$ is all of $\mathbb{F}_3^2$. Its image in F is closed, being the continuous image of a compact group. A proper closed subgroup of a finitely generated pro-three group is contained in a maximal open subgroup of index three. Such a subgroup cannot surject onto the Frattini quotient. This proves the surjectivity and the exact kernel in (E54.2). Surjectivity also identifies the image of each lower central subgroup with the corresponding subgroup of F, which proves the asserted central map.

The complete local expansion at O is $h = as^3u(s)$, with $u(s) \in K[[s]]$ and $u(0) = 1$. Every compatible 3^nth root of $u(s)$ with constant term one belongs to $K[[s]]$, by formal binomial expansion. The source tangent fixes the roots of s. In the punctured target disc the only remaining Galois change is therefore the power $q_g = x^{\kappa_a(g)}$. This statement precedes passage to any nilpotent quotient: the geometric group of the punctured formal disc is procyclic. Relative to the standard target tangential action ρ, the induced action is exactly $\mathrm{Ad}(q_g)\rho_g$. The arithmetic comparison is the continuous isomorphism

$$(C \rtimes_\alpha G_K) \longrightarrow (C \rtimes_\rho G_K), \qquad (w, g) \longmapsto (wq_g, g), \tag{E54.17}$$

whose inverse is $(w, g) \mapsto (wq_g^{-1}, g)$. The cocycle identity $q_{gh} = q_g\rho_g(q_h)$ proves the group law.

In the standard positive-path convention of D.23, $\rho_g(x) = x^\chi$ and $\rho_g(y) = f_g^{-1}y^\chi f_g$, with $f_g \in F'$. Write $f_g = c^\beta$ modulo γ_3F. Its genuine three-cycle relation, in the convention of [Fur10, Theorem 2, equation (5), and §3], has $m = (\chi - 1)/2$ and $xyz = 1$. In F/γ_3F the three commutators $[z, x], [y, z], [x, y]$ all equal c. The remaining product $z^m y^m x^m$ equals $c^{-m(m+1)/2}$, by the class-two collection rule. Consequently $3\beta = m(m + 1)/2$ and $\beta = (\chi^2 - 1)/24$ with the positive sign. Multiplication by three is injective on $\mathbb{Z}_3$, and $\chi^2 - 1$ is divisible by three, so this is an integral identity. The part of f_g in γ_3F is central in C and drops out of its conjugation action. This proves (E54.4), without choosing unknown higher coefficients of the Galois conjugator. The identity $b(\chi\mu) = b(\chi) + \chi^2 b(\mu)$ proves its composition law, either on the generators or by (E54.5).

Integral Hall coordinates and the actual arithmetic image. The integral free nilpotent group of rank two and class three has the Hall basis x, y, c, d, e. Collection gives a group structure on $\mathbb{Z}_3^5$. If $g = (a, b, n, r, s)$ and $g' = (A, B, N, R, S)$, its exact law is

$$\begin{aligned} gg' = (&a + A,\ b + B,\ n + N - Ab,\\ &r + R + bA(A + 1)/2 - An,\\ &s + S + Ab(b + 1)/2 - Bn + AbB). \end{aligned} \tag{E54.18}$$

All denominators are units at three. The law is first verified on integral powers using $xy = cyx$, $xc = dcx$ and $yc = ecy$, with d, e central, and then extends continuously to $\mathbb{Z}_3$. Uniqueness follows successively from the free integral graded lattices of ranks $2, 1, 2$. Reduction modulo 3^v in all five coordinates gives finite three-groups. Their inverse limit is generated by x, y, and the successive Hall coordinates show that its kernel from F is precisely γ_4F. This proves the asserted integral model at the prime three; no division by three in a Baker–Campbell–Hausdorff formula is used. Applying (E54.18) to (E54.4) proves every entry of (E54.5).

With roots assembled from λ and ζ,

$$k = \frac{\chi - 1}{3} - 3\kappa_\lambda. \tag{E54.19}$$

Changing compatible roots adds $v(\chi - 1)$ for some $v \in \mathbb{Z}_3$, so it preserves the congruence in (E54.6). The cyclotomic projection of G_K is all of $1 + 3\mathbb{Z}_3$. Over $L = K(\mu_{3^\infty})$, the character κ_λ has image $\mathbb{Z}_3$. To prove this, λ is not a cube in L: $\lambda^2 = -3\zeta$ and the existence of every

three-power root of ζ in L would make a cube root of three belong to L. But $L/\mathbb{Q}$ is abelian, whereas $\mathbb{Q}(\sqrt[3]{3})/\mathbb{Q}$ is not Galois. The Kummer character is therefore nonzero modulo three. Its closed image is a subgroup of $\mathbb{Z}_3$ containing a unit and hence is $\mathbb{Z}_3$. For a fixed cyclotomic value, (E54.19) now gives precisely one coset of $3\mathbb{Z}_3$. This proves equality with (E54.6). Finally, the action on C^{ab} detects χ. For $\chi = 1$ the action is $\mathrm{Ad}(x^k)$; its effect on y modulo $\gamma_3 C$ detects k. This proves faithfulness and the asserted outer image.

Frobenius and the change of roots. At the prime over two, reduction gives $1 - \zeta = \zeta^2$ in $\mathbb{F}_4$. The map of taking a 3^nth power is an automorphism of the principal pro-two units. Thus the principal-unit factors in (E54.7) and (E54.9) already lie in K_2 and are fixed by local Galois. Arithmetic Frobenius acts on $\zeta_{3^{n+1}}$ by the fourth power. It follows directly that $\kappa_\lambda(F_2) = 2$, $\kappa_{a,\mathrm{old}}(F_2) = 1 - 3 \cdot 2 = -5$, and $\kappa_{a,\mathrm{new}}(F_2) = 1$. Moreover $\zeta_{3^{n+1}}^6 = \zeta_{3^n}^2$, so the ratio of the two root systems is exactly (E54.9). Conjugating an action $\mathrm{Ad}(x^k)\rho_g$ on the left by $\mathrm{Ad}(x^{-2})$ and on the right by $\mathrm{Ad}(x^2)$ changes k to $k + 2(\chi - 1)$, proving the transport with its sign. Substitution of $(\chi, k) = (4, 1)$ into (E54.5) gives (E54.8). In general Hall coordinates that automorphism is

$$(a, b, n, r, s) \longmapsto \left(4a, 4b, 4b + 16n,\ 40n + 64r,\ 24n + 64s - 16\binom{b}{2} - 15b/2\right). \tag{E54.20}$$

All local lifts and their conjugacy fibers. The action on C at K_2 is unramified: in the formula (E54.4), both χ and the chosen unit Kummer character are unramified. Its image is pro-three, as follows either from (E54.6) or from its trivial action on the Frattini quotient. A cocycle together with this action is a homomorphism into a pro-three semidirect product. Wild inertia, which is pro-two, therefore has zero image. The actual maximal pro-three tame quotient is

$$\langle F_2, t \mid F_2 t F_2^{-1} = t^4 \rangle_{\text{pro-3}}. \tag{E54.21}$$

It is $\mathbb{Z}_3 \rtimes \mathbb{Z}_3$ with action $t \mapsto t^4$; the unit 4 lies in $1 + 3\mathbb{Z}_3$, so this action is continuous for the stated exponent group. The tame generators and relation thus give both the necessity and sufficiency of the cocycle relation $z\alpha_{F_2}(\nu)z^{-1} = \nu^4$.

Use (E54.18) and (E54.20) for the two zero-central Hall lifts in (E54.10). Direct collection gives

$$z\alpha_{F_2}(\nu)z^{-1}\nu^{-4} = (0, 0, L, R, S) \tag{E54.22}$$

in Hall coordinates, with exactly (E54.11)–(E54.12). In particular the lower relation is $L = 0$. The central group is $Z = \langle d, e \rangle = \mathbb{Z}_3(3)^2$, and F_2 acts on it by 64. Changing the inertia lift by $d^r e^s$ changes (E54.22) by $(64 - 4)(r, s) = 60(r, s)$. Changing the Frobenius lift by a central element does not change that error. Since $60\mathbb{Z}_3 = 3\mathbb{Z}_3$, this proves the exact image condition and the explicit correction (E54.14).

The differences between two successful lifts are central one-cocycles with values in Z. The tame relation forces their inertia value v to satisfy $(64 - 4)v = 0$, hence $v = 0$. Their Frobenius value is arbitrary in Z, and a central conjugation changes it by $(1 - 64)w$. Thus the central cohomology is exactly $Z/63Z = (\mathbb{Z}/9)^2$. There is no further quotient by a stabilizer of the lower section. An element fixed by its twisted action has, in its first nonzero lower central graded coordinate, a vector fixed by the scalar 4 or 16. Both scalar-minus-one operators are injective on the corresponding integral lattice. Its two coordinates therefore vanish successively. This proves the fiber cardinality and the representatives in the statement. Substitution $(a, b, n, A, B, N) = (0, 1, 0, -1, 0, N)$ in (E54.22) gives (E54.15), and $60/40 = 3/2$ verifies the displayed positive lift.

Finally, all lower central quotients and their obstruction cocycles are functorial for the based arithmetic map (E54.2); applying h_* to the actual relation defect is precisely (E54.22).

The order-27 projection has $\chi \equiv 1$ and $k \equiv (\chi - 1)/3 = \kappa_3(\zeta)$ modulo three, so (E54.4) reduces to the conjugation action in E.31. This proves both comparison claims.

The exact programs `a_class3_frobenius.py` and `a_class3_local.py` verify the calculations in an independent truncated noncommutative Magnus algebra. They check the hexagon sign through degree three, all Hall images, the affine composition law, the actual inner root transport, the collection law, and 639 exact lower-section examples with their central correction signs. The local example and its 1/40 lift are checked as literal group identities. The polynomial derivations and the integral arguments above establish the formulas for all three-adic coordinates. □

Joining the elliptic comparison to the evaluated Frobenius quotient

E.53. An exact common quotient and a class-three refinement with kernel of rank 329 [Proved]

Use the actual based maps of E.50 and E.54 over $K = \mathbb{Q}(\zeta_3)$. To distinguish their targets, put

$$D = N/\gamma_4 N, \qquad B = N/(F''_{\mathrm{ell}}\gamma_4 N), \qquad C = F_h/\gamma_4 F_h. \tag{E53.1}$$

Here F_{ell} is the group of the once-punctured elliptic curve reached by [3], and F_h is the free rank-two group of the thrice-punctured line reached by h. The maps $D \twoheadrightarrow B$ and $D \twoheadrightarrow C$ have their actual tangential origins and arithmetic actions from those two results. Let $I = \ker(N \to T_3E)$ be the full peripheral kernel, let $K_0 = I'\gamma_4 N/\gamma_4 N$, and let $K_h = \ker(D \to C)$.

Write x, y, c, d, e for the Hall generators of C in (E54.3). The image of K_0 in C is the exact closed normal subgroup

$$P' = \langle c^3, d^3, e\rangle \simeq \mathbb{Z}_3^3. \tag{E53.2}$$

Consequently the maximal common quotient of B and C induced by their maps from D is

$$Q = D/(K_0K_h) = C/\langle c^3, d^3, e\rangle. \tag{E53.3}$$

In particular every element of Q has unique coordinates $x^a y^b c^{\bar n} d^{\bar r}$ with $a, b \in \mathbb{Z}_3$ and $\bar n, \bar r \in \mathbb{Z}/3$. These are group coordinates with the quotient of the Hall law (E54.18).

There is an exact based arithmetic sequence

$$1 \longrightarrow J \longrightarrow D \longrightarrow B \times_Q C \longrightarrow 1, \qquad J = K_0 \cap K_h \simeq \mathbb{Z}_3^{329}. \tag{E53.4}$$

The fiber product uses the actual two maps to (E53.3); the map on the right is surjective. All the maps and kernels in (E53.2)–(E53.4) are G_K-equivariant.

Put $V = N^{\mathrm{ab}}$, $T = \ker(V \to T_3E)$, and $W = F_h^{\mathrm{ab}} = \mathbb{Z}_3(1)^2$. Write $f : V \to W$ for the map induced by h. Then f is onto and its cusp image is exactly

$$f(T) = 3\mathbb{Z}_3 x + \mathbb{Z}_3 y. \tag{E53.5}$$

The induced lower central filtration on J is given by the integral kernels

$$\begin{aligned} &\mathrm{gr}_1 J = 0, \\ &\mathrm{gr}_2 J = \ker\bigl(\Lambda^2 T \longrightarrow \Lambda^2 W\bigr), \\ &\mathrm{gr}_3 J = \ker\bigl([T, \mathrm{Lie}_2(V)] \longrightarrow \mathrm{Lie}_3(W)\bigr). \end{aligned} \tag{E53.6}$$

Their ranks are $0, 27, 302$. The quotient in (E53.4) has torsion-free graded lattices of ranks

$$10, \quad 18, \quad 28. \tag{E53.7}$$

The map on the degree-two part of K_0 has image $3\mathbb{Z}_3 c$; the map on its degree-three part has image $3\mathbb{Z}_3 d + \mathbb{Z}_3 e$. These precise integral image lattices are retained in (E53.3). In chosen integral coordinates the projection $B \times_Q C \to B$ has kernel (E53.2); its three coordinates have degrees two, three, three.

The actual based action on the common quotient Q factors through the cyclotomic character itself. Put $r = (\chi - 1)/3$ for $\chi \in 1 + 3\mathbb{Z}_3$. The action is

$$\begin{aligned} x &\longmapsto x^{\chi}, \quad y \longmapsto y^{\chi} c^{r} d^{r(r-1)/2}, \\ c &\longmapsto c d^{r}, \quad d \longmapsto d, \end{aligned} \tag{E53.8}$$

where the exponents of c, d are taken modulo three. At the actual Frobenius of E.54, these images are $x^4, y^4 c, cd, d$, respectively. This supplies the literal compatibility map between the elliptic quotient and the newly evaluated class-three Frobenius quotient.

The integral certificate `a_combined_quotient.py` records the actual cusp matrix, a unimodular cusp basis change, the two graded kernel bases, and integer right inverses for the quotient maps of ranks 18 and 28. It also verifies the group identities generating (E53.2), 375 tensor substitutions, 270 normality identities and 72 common-action identities. The manuscript passage is §51.16, folios 733v–736v, scans 172–178, equations (371)–(379).

Proof. The cusp lattice and its actual subgroup image. Use the eight positive peripheral classes away from O, ordered as $A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2$. They are a basis of the primitive cusp lattice T. The valuation of h gives the first coordinate of f, and that of $1 - h$ gives the second. Thus (E31.5) gives the literal matrix

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & -3 \end{pmatrix}. \tag{E53.9}$$

For example the new cusp basis

$$-C_2 - 3B_0,\ B_0,\ A_0,\ A_1,\ A_2,\ B_1 - B_0,\ B_2 - B_0,\ C_1 \tag{E53.10}$$

is unimodular, and its image is $3x, y, 0, \ldots, 0$. This proves (E53.5) with its factor three.

Let P be the image of I in C. The map $h_* : N \to F_h$ is surjective by E.54, so $h_*(N') = F_h'$. As $N' \subset I$, the group P contains $\gamma_2 C$. Its image in C^{ab} is exactly (E53.5). It follows that

$$P = \{ w \in C :\ w^{\mathrm{ab}} \in 3\mathbb{Z}_3 x + \mathbb{Z}_3 y \} = \langle x^3, y, c, d, e \rangle. \tag{E53.11}$$

This is equality of closed subgroups. Derived subgroups commute with a continuous surjection of compact groups, so the image of K_0 is P'. The Hall law gives the actual identities

$$[x^3, y] = c^3 d^3, \qquad [x^3, c] = d^3, \qquad [y, c] = e. \tag{E53.12}$$

The elements c^3, d^3, e commute and are independent in the integral Hall coordinates. Their subgroup is normal in C: conjugation by x changes c^3 by d^3, conjugation by y changes it by e^3, and d, e are central. Equations (E53.12) put this subgroup in P'; the generators in (E53.11) commute modulo it. This proves the exact equality (E53.2), including closedness, normality and its underlying free three-adic group.

The common quotient and the integral kernel. The product $K_0 K_h$ of the two compact normal subgroups is closed. Taking its quotient gives (E53.3), with the displayed Hall coordinates. The image of D in $B \times C$ lies in the fiber product over Q. Given a compatible pair, lift its B coordinate to D. The discrepancy in C belongs to the image of K_0 by (E53.3); lift that discrepancy to K_0 and correct the chosen element of D. This proves surjectivity to the actual

fiber product and the kernel formula in (E53.4). Any common quotient obtained from the two maps out of D kills both K_0 and K_h, hence factors through their product quotient. This proves the stated maximal common quotient.

E.50 proves that K_0 is actually abelian $\mathbb{Z}_3^{332}$ and that its induced graded pieces are the saturated lattices $\Lambda^2 T$ and $[T, \mathrm{Lie}_2(V)]$. The map $K_0 \to P'$ is onto a free group $\mathbb{Z}_3^3$ by (E53.12). Its kernel is therefore an actual free abelian group of rank 329. The strict filtered assertion requires the following additional check. The degree-three map has image

$$[\,3\mathbb{Z}_3 x + \mathbb{Z}_3 y,\ \mathrm{Lie}_2(W)\,] = 3\mathbb{Z}_3 d + \mathbb{Z}_3 e. \tag{E53.13}$$

This is also $P' \cap \gamma_3 C$. There are actual central preimages: choose $a, b \in I$ with images x^3, y and $w \in N'$ with image c. Then $[a, w]$ and $[b, w]$ belong to $I' \cap \gamma_3 N$ and map to d^3, e. Such choices exist by (E53.11) and the surjectivity on derived subgroups. Consequently a degree-two kernel class in K_0 can be corrected in its degree-three coordinates to have zero image in C. This proves exactness on both graded pieces in (E53.6). The respective image ranks are one and two, giving 27 and 302. Both kernels are saturated in the respective free integral lattices: their quotients are the free image lattices $3\mathbb{Z}_3 c$ and $3\mathbb{Z}_3 d + \mathbb{Z}_3 e$. Together with the saturation proved in E.50, this proves torsion-freeness and the ranks in (E53.7).

For completeness, these saturation arguments have an explicit integer coordinate certificate. Choose a splitting $V = L \oplus T$ and then adjust the two handle lifts by elements of T. Since f is onto and (E53.5) has index three in W, a unit change of handle basis and the adjustments in T give the following linear map in a basis $u, v, t_1, \ldots, t_8$:

$$f(u) = x, \qquad f(v) = 0, \qquad f(t_1) = 3x, \qquad f(t_2) = y, \qquad f(t_i) = 0 \quad (i \geq 3). \tag{E53.14}$$

Here one first chooses a handle with unit x coefficient, scales it to coefficient one, and subtracts its y coefficient using t_2. Subtracting a multiple of this handle from the other kills its x coefficient; another t_2 adjustment kills its y coefficient. The resulting handle images under p still form an integral basis. This is a chosen coordinate splitting; the maps and kernel in (E53.4) remain the previously specified equivariant ones.

For the Hall basis $[[v_i, v_j], v_k]$ with $i > j$ and $k \geq j$, write $H_i = f(v_i)$ and $\Delta_{ij} = \det(H_i, H_j)$. The actual induced graded maps are

$$[v_i, v_j] \longmapsto \Delta_{ij} c, \qquad [[v_i, v_j], v_k] \longmapsto -\Delta_{ij}(H_{k,x} d + H_{k,y} e). \tag{E53.15}$$

On the 28 cusp-cusp rows, division of the image coordinate by three leaves a row with one entry -1 and all other entries zero. On the 304 degree-three kernel rows, use coordinates relative to $3d, e$. The two rows $(i, j, k) = (3, 0, 2), (3, 0, 3)$ map to the two unit vectors. Subtracting their multiples from the other 302 rows gives an integral kernel basis and an integral direct-sum inverse. Adding the 17 and 26 retained rows of E.50 gives integer quotient maps with integer right inverses of ranks 18 and 28. The saved certificate checks both products of each claimed inverse and all brackets of a degree-two kernel vector with a weight-one generator. Thus it verifies the integral normal subgroup calculation.

The common based arithmetic action. The equalities of closed subgroups and all quotient maps above start with the two based arithmetic maps of E.50 and E.54. They therefore give exactly the asserted G_K actions and equivariance. In Q, (E53.12) shows that x^3 is central. The action $\alpha_{\chi,k}$ of E.54 therefore depends on k only modulo three. Its genuine image satisfies $k \equiv (\chi - 1)/3 \pmod 3$. Substituting this congruence into the literal Hall images (E54.5), and using $c^3 = d^3 = e = 1$, gives (E53.8). Equivalently, the homomorphism $\chi \mapsto \alpha_{\chi,(\chi-1)/3}$ gives this action, since $(\chi\mu - 1)/3 = (\chi - 1)/3 + \chi(\mu - 1)/3$. This proves the factorization and the stated actual Frobenius images. □

Mixed first coordinates at three and an inert prime

E.55. A free-product arithmetic presentation and every mixed first-coordinate fiber [Proved]

Let ℓ be a rational prime congruent to 2 or 5 modulo 9, and put $q = \ell^2$. In particular $\ell = 2$ is allowed. Retain the actual based arithmetic quotient H of E.46. For $a \in \mathbb{Z}_3 \setminus \{0\}$ and $b \in \mathbb{Z}_3$, fix the first Kummer class

$$[A] = [3a\kappa_3(\ell) + 3b\kappa_3(3)], \qquad A_0 = 3a = 3^s\epsilon, \quad s = v_3(3a), \quad \epsilon \in \mathbb{Z}_3^\times, \quad B_0 = 3b. \tag{E55.1}$$

For a rational prime $p \notin \{\ell, 3\}$, set $h_p = v_3(p-1)$. Choose integer representatives a_p, b_p of a, b modulo $3^{\max(h_p-1,0)}$, and define

$$\mathscr{P}^\ell_{a,b} = \left\{ p \notin \{\ell, 3\} : \left(\ell^{3a_p} 3^{3b_p}\right)^{(p-1)/3^{h_p}} \equiv 1 \pmod{p} \right\}. \tag{E55.2}$$

This condition is independent of the representatives. Write $\mathcal{R}(\mathscr{P})$ for the restricted three-adic product defined in (E49.2). After the explicit arithmetic choices below, the complete global section fiber of H over $[A]$ has additive topological coordinates

$$\boxed{\mathbb{Z}_3^2 \times \mathcal{R}(\mathscr{P}^\ell_{a,b}) \times \mathbb{Z}_3^s \times \mathbb{Z}_3[[Z]].} \tag{E55.3}$$

Writing the coordinates as $(\beta_\ell, \beta_3, (\beta_p), (z_d)_{d=1}^s, f)$, the scaled toric map is exactly

$$\left(a\kappa_3(\ell) + b\kappa_3(3), \quad \beta_\ell\kappa_3(\ell) + \beta_3\kappa_3(3) + \sum_{p \in \mathscr{P}^\ell_{a,b}} \beta_p\kappa_3(p) \right). \tag{E55.4}$$

Every fiber over an allowed second coordinate is therefore $\mathbb{Z}_3^s \times \mathbb{Z}_3[[Z]]$, with the displayed integral coordinate maps. The classes with $a = 0$ are those computed in E.47 and E.51. Thus these statements together compute every first class supported on $\{\ell, 3\}$.

The arithmetic input is an actual continuous pro-three isomorphism. With $K = \mathbb{Q}(\zeta_3)$ and $S = \{\text{primes of } K \text{ over } \ell, 3\}$, there are generators t, σ, ρ for which

$$G^{(3)}_{K,S} \simeq \langle t, \sigma \mid \sigma t \sigma^{-1} = t^q \rangle_3 \underset{3}{*} \langle \rho \rangle, \qquad \langle \rho \rangle \simeq \mathbb{Z}_3. \tag{E55.5}$$

Here t is actual tame inertia, and σ is actual arithmetic Frobenius at the unique K-prime over ℓ. In the relative pro-three quotient over $\mathrm{Gal}(K/\mathbb{Q}) = \langle c \rangle$ one has

	t	σ	ρ
χ_3	1	q	1
$\kappa_3(\ell)$	1	0	0
$\kappa_3(3)$	0	0	1
$c(\,\cdot\,)c$	t^{-1}	σ	ρ^{-1}.

(E55.6)

The projection from the full relative arithmetic group onto this group has a continuous c-equivariant section retaining the actual decomposition group at ℓ.

For the part unramified outside $\{\ell, 3\}$, all second cocycles have unique representatives with

$v(c)=0$ and

$$\begin{aligned} u=v(t)&=T^{A_0/2}\left(3\beta_\ell+\sum_{d=1}^{s} z_d\bigl(Q_d(T)-\varphi(3^d)\bigr)\right),\\ Q_d(T)&=\sum_{\substack{-(3^d-1)/2\le j\le (3^d-1)/2\\ 3\nmid j}} T^{\epsilon 3^{s-d}j},\\ v(\sigma)&=\frac{q\,u(T^q)-(1+T^{A_0}+\cdots+T^{(q-1)A_0})u(T)}{T^{qA_0}-1},\\ v(\rho)&=T^{B_0/2}\bigl(3\beta_3+Vf(V)\bigr),\qquad V=T+T^{-1}-2. \end{aligned} \tag{E55.7}$$

The quotient in this formula belongs to $\mathbb{Z}_3[[T-1]]$, is anti-invariant under inversion, and has augmentation zero. No division by a nonunit is left unspecified. The proof constructs the extra global classes by actual homomorphisms on the arithmetic inertia kernel and gives their precise toric normalization.

The program `e_two_prime_first.py` verifies the coordinate equations and integral divisions, recovers arbitrary tested representatives from cyclic group rings, and checks the residue criterion by four independent finite descriptions.

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d) and the fourth exactness follow-up.

Proof. The actual arithmetic generators. All groups in this paragraph are relative pro-three Galois groups over K, together with the indicated order-two action. The local group at ℓ has tame inertia $\mathbb{Z}_3(1)$ and unramified quotient $\mathbb{Z}_3\times C_2$ over $\mathbb{Q}_\ell$. Its injection into the full relative global group follows from the inertia injection in E.39 and the injective unramified cyclotomic character. Write its Frobenius as F. The order-two component d of its procyclic unramified subgroup acts on inertia by inversion. The element $\sigma=F^2$ commutes with d and acts on inertia by q. Choose the Frobenius complement to fix compatible roots of the uniformizer ℓ. Cubing is an automorphism of $\mathbb{Z}_\ell^\times$: its residue group has order $\ell-1$ prime to three, and its principal units are pro-ℓ. Consequently the compatible roots of 3 are in $\mathbb{Q}_\ell$. With these root choices the two Kummer coordinates vanish on d and σ.

The finite Schur–Zassenhaus conjugacy theorem, followed by compact inverse limits, conjugates the image of a complex conjugation to d. We use this conjugate and denote it by c. Choose the compatible roots of positive rational primes fixed by this c. The roots of ℓ and 3 just chosen are such roots. This is a permitted change of representatives of the same rational Kummer classes. More explicitly, replacing roots by $\zeta_{3^n}^{e_i}$ times the old roots changes their Kummer cocycles by $(\chi_3-1)e_i$. For (E55.1) the change is $(\chi_3-1)3(ae_\ell+be_3)$; conjugation by the element $(-3(ae_\ell+be_3),0)\in H$ effects exactly this change of first cocycle, using the multiplication law of E.46. Thus the based section fibers are transported by an explicit geometric conjugation. The coefficient involution after normalization is still $C(f)=-f(T^{-1})$.

Put $P=G^{(3)}_{K,\{3\}}$. E.34 and E.37 prove that P is free pro-three of rank two. The characters given by the cyclotomic character modulo nine and by $\kappa_3(3)$ modulo three are independent on its Frattini quotient. Since $v_3(q-1)=1$, the image of σ is primitive in that quotient, has nonzero first coordinate and zero second coordinate. Choose an element γ of the full group whose image has second coordinate a unit. Multiply it by $t^{-\kappa_3(\ell)(\gamma)}$ to obtain γ_0 with $\kappa_3(\ell)(\gamma_0)=0$. The exact element

$$r=\gamma_0c\gamma_0^{-1}c\quad\text{satisfies}\quad crc=r^{-1},\quad \chi_3(r)=1,\quad \kappa_3(\ell)(r)=0,\quad \kappa_3(3)(r)=2\kappa_3(3)(\gamma_0). \tag{E55.8}$$

Raise r to the inverse of the last, unit, coordinate to obtain ρ. Its image and σ are a free basis of P: their Frattini images form a basis, and the resulting surjective endomorphism of a free pro-three group of finite rank is an automorphism. This proves all the Kummer and involution identities in (E55.6) on actual arithmetic elements.

The free-product comparison and its section. Apply the opened arithmetic Riemann-existence theorem [Win07, Theorem2.4] to $\{3\} \subset S$. Its hypotheses are exactly those checked in E.39: the base has the $K(\pi,1)$ property, and the cyclotomic tower gives nontrivial unramified completion at ℓ. Its kernel is the continuous free pro-three product of the inertia groups over the coset space $P/\langle\sigma\rangle$. The actual decomposition group $D = \langle t, \sigma \mid \sigma t \sigma^{-1} = t^q \rangle_3$ and ρ therefore induce a continuous surjection $D *_3 \langle\rho\rangle \to G^{(3)}_{K,S}$ over P.

Its map on kernels is an isomorphism. One can see the kernel of the source directly in its pro-three coset tree: the vertices are the cosets of D and $\langle\rho\rangle$, and the edges are the group itself. Quotienting by that kernel gives the coset tree for $P = \langle\sigma\rangle *_3 \langle\rho\rangle$. The vertex stabilizers over the first kind of vertex are exactly the conjugate copies of $\langle t\rangle$, indexed by $P/\langle\sigma\rangle$; the other vertex and edge stabilizers are trivial. The tree's universal property identifies its kernel with their continuous free product. The map in question sends each such factor isomorphically to the corresponding actual inertia factor of Wingberg's isomorphism. The free-product universal property gives its inverse on the kernel, and the map on the quotient P is the identity. This proves (E55.5), including injectivity.

We made the choices of D and ρ in the full arithmetic group. Its same elements now define a map in the reverse direction from (E55.5). Projection back fixes all three generators, so this map is a continuous section. Equation (E55.6) makes it c-equivariant. Applying the same opened theorem to S and the set of all primes identifies the kernel of this projection with the continuous free inertia product over primes outside S. The $K(\pi,1)$ property for S is also a conclusion of that theorem for $\{3\} \subset S$. These statements specify the splitting and its exact arithmetic domain; they do not discard the correction words of E.39 for a different splitting.

Solving every unramified cocycle equation. For the fixed first cocycle, the coefficient module is $J = (3, T-1)\mathbb{Z}_3[[T-1]]$ with operators

$$t(f) = T^{A_0} f, \quad \sigma(f) = q f(T^q), \quad \rho(f) = T^{B_0} f, \quad C(f) = -f(T^{-1}). \tag{E55.9}$$

Normalize $v(c) = 0$ by the order-two averaging argument. Writing its other values as (u, v_0, w), the complete relations of (E55.5)–(E55.6) give exactly

$$\begin{aligned} u(T^{-1}) &= T^{-A_0} u(T), \\ v_0(T^{-1}) &= -v_0(T), \\ w(T^{-1}) &= T^{-B_0} w(T), \\ q u(T^q) - (1 + T^{A_0} + \cdots + T^{(q-1)A_0}) u(T) &= (T^{qA_0} - 1) v_0(T). \end{aligned} \tag{E55.10}$$

The remaining geometric conjugations are precisely $m \in J$ with $C(m) = m$, that is, anti-invariant series. They send

$$(u, v_0, w) \longmapsto (u - (T^{A_0} - 1)m,\ v_0 - (q\psi_q - 1)m,\ w - (T^{B_0} - 1)m), \quad \psi_q(f) = f(T^q). \tag{E55.11}$$

The first geometric coordinate must be zero here: preserving the normalized first value at c would otherwise change it by twice that coordinate. Thus no conjugations have been lost.

Set $M = 3^s$. The substitution $T \mapsto T^\epsilon$ is an integral automorphism, and

$$(T^{A_0} - 1) = (T^M - 1), \qquad (T^{qA_0} - 1) = (T^{A_0} - 1) \tag{E55.12}$$

as ideals in the completed ring. The first equality follows by the unit-exponent substitution, or by factoring its ratio as a power series with constant term ϵ in $T^M - 1$; the ratio in the second

equality has constant term q. The quotient is $\mathbb{Z}_3[C_M]$. Reducing (E55.10) in it shows that $\bar{u}$ is invariant under both inversion and position multiplication by q. Since $v_3(q-1)=1$, the closed subgroup generated by q is $1+3\mathbb{Z}_3$; together with -1 it acts transitively on positions of each fixed three-power order. There are precisely $s+1$ orbits: zero and the orders $3,\dots,3^s$.

The augmentation condition on $\bar{u}$ is exactly that its total mass is in $3\mathbb{Z}_3$. This lattice has the basis given by the images of 3 and $Q_d-\varphi(3^d)$ for $1\le d\le s$. The factor $T^{A_0/2}$ is one in this quotient, so it supplies a symmetric lift about $A_0/2$ without changing any of these finite coordinates. This gives precisely the first line of (E55.7).

For any such lift the numerator in its σ-value vanishes modulo $T^{A_0}-1$, by these finite invariants. Both ideal equalities in (E55.12) prove integral divisibility by the displayed denominator. Inversion of its numerator multiplies it by T^{-qA_0}; inversion of its denominator multiplies it by $-T^{-qA_0}$. The quotient is therefore anti-invariant, has augmentation zero, and lies in J. This also proves all of (E55.10) for the proposed values.

Conversely take any solution of (E55.10). Its finite coordinates determine a unique proposed u_0 with the same image in $\mathbb{Z}_3[C_M]$. There is a unique integral series $m=(u-u_0)/(T^{A_0}-1)$. Inverting the numerator and denominator shows $m(T^{-1})=-m(T)$, so $m\in J$ is an allowed conjugation. It sends u to u_0. The last equation in (E55.10), in a domain, then fixes v_0 uniquely. No further nonzero conjugation fixes u_0. Finally $T^{-B_0/2}w$ is invariant under inversion. The change of coordinate $W=(T-T^{-1})/2$ identifies the completed ring with $\mathbb{Z}_3[[W]]$, with inversion $W\mapsto -W$. Its even subring is also $\mathbb{Z}_3[[V]]$, since $V=2(\sqrt{1+W^2}-1)$ has unit linear coefficient in W^2. The augmentation restriction on w gives exactly $3\beta_3+Vf(V)$. This proves existence and uniqueness of all coordinates in (E55.7). The division maps used here are continuous on closed principal ideals, by compactness and the domain property. All remaining maps are continuous substitutions or finite coefficient extractions. Thus this normal form is a homeomorphism, including its stated integral topology.

All other primes and the exact global splitting. Let L be the arithmetic kernel of the split projection onto (E55.5). It acts trivially on J_A. Restriction to L of a second cocycle is therefore an additive continuous homomorphism $\lambda: L\to J_A$. The splitting gives the explicit reconstruction

$$v(l\,i(g))=\lambda(l)+v_S(g),\qquad \lambda(i(g)li(g)^{-1})=g\lambda(l), \tag{E55.13}$$

where i is the section just proved. This is a bijection between actual cocycles, with c-equivariance imposed. On the abelianized free inertia product, a different prolongation of the same prime of $K_S(3)$ only conjugates a factor by an element of L; its image under λ is unchanged. Consequently its equivariant homomorphisms are exactly the compatible local inertia values, with no unevaluated nonabelian correction.

At a rational prime $p\notin\{\ell,3\}$ the Tate-untwisted action on positions is the affine map $x\mapsto A(F_p)+px$. E.49 and E.51 prove directly that its invariant integral measures in $J_A(-1)$ are zero without an integral fixed point, and are precisely $3\mathbb{Z}_3\delta_{x_p}$ with one. The fixed-point condition is divisibility of the local Kummer class of $\ell^{3a}3^{3b}$ by every power of three. Principal units at p are uniquely three-divisible; in its residue group the condition says that the three-primary component is one. Computing that component gives exactly (E55.2). The representatives there are sufficient because changing an exponent by 3^{h_p-1} changes its triple by a multiple of 3^{h_p}. In particular every p with $h_p\le 1$ is allowed.

Normalize tame inertia at an allowed p by $\kappa_3(p)(t_p)=1$. Assign it $\lambda_p(t_p)=3\delta_{x_p}$ and assign every other base prime zero. Extend to all prolongations by the equivariance in (E55.13), and then by the free-product universal property. The fixed-point condition proves independence of the chosen coset representative. At a split rational prime the conjugate factor is prescribed by c; at an inert prime the local rational Frobenius condition already gives that descent. Thus λ_p is an actual rational equivariant homomorphism. Continuity of a family of such maps is

exactly $\beta_p \to 0$ three-adically: augmentation forces this condition at every finite coefficient exponent, and the same condition makes the family uniformly zero at all but finitely many primes in every finite quotient of J. This is the continuous-bundle topology of E.39.

There is a small toric normalization to record. The cocycle reconstructed with $v_S = 0$ from λ_p has scaled second coordinate

$$\kappa_3(p) - \kappa_3(p)(\rho)\kappa_3(3). \tag{E55.14}$$

Indeed it has valuation one at p, zero at every other prime except possibly three, and vanishes on ρ. At ℓ its whole decomposition restriction is zero. Also $\kappa_3(p)(\sigma) = 0$: the chosen c fixes σ, whereas its Kummer action is negation and its value at c is zero. The rational completed Kummer coordinates of E.16 now prove (E55.14), including the coefficient at three. Add the unramified cocycle (E55.7) with $\beta_3 = \kappa_3(p)(\rho)$ and every other coordinate zero. Denote the result by v_p. It has scaled second coordinate exactly $\kappa_3(p)$.

Every full global cocycle has unique residues $3\beta_p\delta_{x_p}$. Subtract the convergent sum $\sum_p \beta_p v_p$; (E55.13) shows that the remainder is unramified outside $\{\ell, 3\}$, and (E55.7) gives its unique remaining coordinates. The terms used to normalize v_p have coefficients in $\mathbb{Z}_3$, so the same restrictedness proves convergence of that normalization as well. All constructions and their inverses are continuous at every finite coefficient and positional level. This proves (E55.3) and (E55.4) as actual topological group coordinates. It also proves three-torsion-freeness.

Finally a global section of H with the specified first class is exactly this twisted additive cocycle modulo the conjugations used above, by the multiplication law of E.46. Its finite images over K are three-groups, so passing to the relative pro-three arithmetic quotient preserves all these sections and all their conjugacies. The maps from the intrinsic Fermat groups to H retain their previously computed kernels. The classification here applies to this specified target and first family.

For $\ell = 2$ and $b = 0$, the comparison with the norm coordinates of E.52 is explicit through their common section cocycle. Apply the finite Kummer–Shapiro maps of E.46 to a norm system, evaluate its cocycle on t, σ, ρ, c, and normalize its value at c. Its image in $\mathbb{Z}_3[C_{3^s}]$ supplies the coefficients of 3 and $Q_d - \varphi(3^d)$; division by $T^{A_0} - 1$ then gives the unique conjugator in (E55.11), and the remaining ρ-value gives f. For the inverse, reconstruct that actual cocycle by (E55.7) and (E55.13), apply the inverse Shapiro maps, and extract the valuations and successive unit coordinates by (E52.10)–(E52.13). Both composites are the identity because each of those component maps was proved bijective. This specifies the arithmetic homeomorphism between the two displayed parameter groups, including their finite coordinate blocks and their norm transitions. □

Mixed first coordinates at three and a split prime

E.56. Every mixed first-coordinate fiber at an explicitly specified split prime [Proved]

Let ℓ be a rational prime satisfying

$$v_3(\ell - 1) = 1, \qquad 3^{(\ell-1)/3} \not\equiv 1 \pmod{\ell}. \tag{E56.1}$$

For example, $\ell = 7, 13, 31, 43, 79, 97$ satisfy these conditions. Use the actual arithmetic quotient H and the scaled toric map of E.46. Fix $a \neq 0$ and b in $\mathbb{Z}_3$, and put

$$\begin{aligned} A = 3a = 3^s\epsilon, \quad s \geq 1, \quad \epsilon \in \mathbb{Z}_3^\times, \qquad B = 3b, \qquad h = (\ell-1)/3 \in \mathbb{Z}_3^\times, \\ r = B/(\ell-1) = b/h \in \mathbb{Z}_3, \qquad [\mathcal{A}] = [3a\kappa_3(\ell) + 3b\kappa_3(3)]. \end{aligned} \tag{E56.2}$$

For $p \notin \{\ell, 3\}$ let $h_p = v_3(p-1)$, choose integer representatives a_p, b_p modulo $3^{\max(h_p-1,0)}$, and define

$$\mathscr{P} = \left\{ p \notin \{\ell, 3\} : \left(\ell^{3a_p} 3^{3b_p}\right)^{(p-1)/3^{h_p}} \equiv 1 \pmod{p} \right\}. \tag{E56.3}$$

The condition is independent of the representatives. Let $\mathcal{R}(\mathscr{P})$ have the restricted inverse-limit topology of (E49.2).

All global section classes of H with first class $[\mathcal{A}]$ have complete additive topological coordinates

$$\boxed{\mathbb{Z}_3^2 \times \mathcal{R}(\mathscr{P}) \times \mathbb{Z}_3^{2s} \times \mathbb{Z}_3[[Z]].} \tag{E56.4}$$

The first two coordinates are denoted β_ℓ, β_3, and the restricted ones β_p. The scaled toric map is exactly

$$\left(a\kappa_3(\ell) + b\kappa_3(3),\ \beta_\ell\kappa_3(\ell) + \beta_3\kappa_3(3) + \sum_{p \in \mathscr{P}} \beta_p \kappa_3(p) \right). \tag{E56.5}$$

Every allowed second coordinate has the explicitly computed compact fiber $\mathbb{Z}_3^{2s} \times \mathbb{Z}_3[[Z]]$. The entire group in (E56.4), with the zero second cocycle as origin, is torsion-free. The first classes with $a = 0$ are already computed in E.47 and E.51.

Here are actual arithmetic generators and the complete normal form. Put $K = \mathbb{Q}(\zeta_3)$ and let S contain the primes over ℓ and three. There are actual local generators t, σ at one prime over ℓ, and $t' = ct^{-1}c$, $\sigma' = c\sigma c$ at its conjugate, such that

$$G_{K,S}^{(3)} \simeq \langle t, \sigma \mid \sigma t \sigma^{-1} = t^\ell \rangle_3 \underset{3}{*} \langle t', \sigma' \mid \sigma' t' \sigma'^{-1} = t'^\ell \rangle_3,$$

$$\begin{array}{r|cccc} & t & t' & \sigma & \sigma' \\ \hline \chi_3 & 1 & 1 & \ell & \ell \\ \kappa_3(\ell) & 1 & 1 & 0 & 0 \\ \kappa_3(3) & 0 & 0 & 1 & -1. \end{array} \tag{E56.6}$$

The full relative arithmetic group over C_2 projects onto this group by a map with an actual c-equivariant section retaining both decomposition groups.

For $1 \le d \le s$ and $\delta \in \{+1, -1\}$ define the centered orbit sums

$$Q_{d,\delta}(T) = \sum_{\substack{-(3^d-1)/2 \le j \le (3^d-1)/2 \\ j \equiv \delta \pmod 3}} T^{\epsilon 3^{s-d} j}. \tag{E56.7}$$

The finite parameter lattice has the explicit basis given by the free coordinates

$$\eta,\ z_{1,-},\ (z_{d,+}, z_{d,-})_{2 \le d \le s} \in \mathbb{Z}_3, \qquad z_{1,+} = 3\eta + \sum_{d=1}^{s} z_{d,-} - \sum_{d=2}^{s} z_{d,+}. \tag{E56.8}$$

There are exactly $2s$ free coordinates. Put

$$\begin{aligned} f_0(T) &= 3\beta_\ell + \sum_{d=1}^{s} \sum_{\delta=\pm 1} z_{d,\delta}(Q_{d,\delta}(T) - 3^{d-1}), \\ u_0(T) &= T^{-r} f_0(T), \qquad S_\ell(T) = \sum_{i=0}^{\ell-1} T^{iA}, \\ v_0(T) &= \frac{\ell T^B u_0(T^\ell) - S_\ell(T) u_0(T)}{T^{\ell A} - 1}, \\ V &= T + T^{-1} - 2, \qquad z(T) = \frac{\beta_3}{h} - \eta + \frac{3\beta_\ell}{2} + V f(V). \end{aligned} \tag{E56.9}$$

The quotient defining v_0 is integral. For an arbitrary series $f \in \mathbb{Z}_3[[Z]]$, the unique normal representative of the class unramified outside S is

$$\begin{aligned} v(c) &= 0, \\ u = v(t) &= u_0 + (T^A - 1)z, \\ v(\sigma) &= v_0 + (\ell T^B \psi_\ell - 1)z, \qquad \psi_\ell(g) = g(T^\ell), \\ v(t') &= T^A u(T^{-1}), \qquad v(\sigma') = -v(\sigma)(T^{-1}). \end{aligned} \tag{E56.10}$$

Its augmentations on t, σ are exactly $3\beta_\ell, 3\beta_3$. The proof constructs the additional global residue cocycles and their exact toric normalization. It also gives the forward and inverse maps to the actual norm systems in E.46, rather than identifying the parameter products without a comparison map.

The maps from the intrinsic Fermat groups to this H retain the kernels displayed in (E47.8). Equations (E56.4)–(E56.10) evaluate the target fibers for the stated first-coordinate family. They leave the source lifting conditions attached to those exact kernels. The manuscript passage is §51.16, folios 733v–736v, scans 172–178, equations (371)–(379), Part E(c),(d).

Proof. The two actual decomposition factors. Choose compatible roots of the positive rational numbers fixed by complex conjugation c. Since $\ell \equiv 1 \pmod 3$, it splits in K. Choose the prime at which the cubic Kummer Frobenius value of three is one modulo three. Such a choice exists by (E56.1); conjugation changes the value to its negative. Choose an actual Frobenius σ_0 fixing the compatible roots of the uniformizer ℓ, and put $\sigma_0' = c\sigma_0 c$. Write $t_0 = \kappa_3(3)(\sigma_0) \equiv 1 \pmod 3$. The exact Kummer values are

$$\chi(\sigma_0) = \chi(\sigma_0') = \ell, \qquad \kappa_\ell(\sigma_0) = \kappa_\ell(\sigma_0') = 0, \qquad \kappa_3(\sigma_0) = t_0, \quad \kappa_3(\sigma_0') = -t_0. \tag{E56.11}$$

Here $\kappa_\ell = \kappa_3(\ell)$ and $\kappa_3 = \kappa_3(3)$ in this calculation. The cyclotomic character modulo nine and κ_3 modulo three give the two independent Frattini characters of the free rank-two group $P = G^{(3)}_{K,\{3\}}$ proved in E.34. The two Frobenii have determinant $-2((\ell-1)/3)t_0$, a three-adic unit, in these coordinates. Their images are therefore a free basis of P.

An explicit conjugation makes the displayed values exact, not just values modulo three. Put

$$\rho_0 = \sigma_0'\sigma_0^{-1}, \qquad k = \frac{1 - t_0}{2t_0(\ell - 1)} \in \mathbb{Z}_3, \qquad g = \rho_0^k, \qquad \sigma = g\sigma_0 g^{-1}, \quad \sigma' = c\sigma c. \tag{E56.12}$$

The integrality of k follows from $t_0 \equiv 1 \pmod 3$ and $v_3(\ell - 1) = 1$. Since $\chi(\rho_0) = 1$, $\kappa_\ell(\rho_0) = 0$, and $\kappa_3(\rho_0) = -2t_0$, the cocycle identity gives

$$\kappa_3(\sigma) = t_0 + 2t_0(\ell - 1)k = 1. \tag{E56.13}$$

All other entries in (E56.11) are preserved. Conjugating each of a free pair does not change its Frattini images, so the new images remain a free basis. Conjugate the first tame inertia generator by g as well, normalize its uniformizer Kummer value to one, and call it t. Set $t' = ct^{-1}c$. This proves the entire table (E56.6), with actual local generators.

At each prime the local pro-three tame group injects into the full pro-three arithmetic group: its inertia injects by the free-inertia theorem, and its unramified part injects through the nontrivial cyclotomic character. Applying the opened Wingberg Theorem 2.4 [Win07] to $\{3\} \subset S$ identifies the kernel over P with the continuous free inertia bundle over the two coset spaces $P/\langle\sigma\rangle$ and $P/\langle\sigma'\rangle$. The $K(\pi, 1)$ hypothesis and nontrivial local unramified completions are the ones verified in E.39 and E.55.

There is an explicit inverse for the resulting free-product comparison. For either generator $\tau = \sigma, \sigma'$, form the induced inertia bundle with $(p\tau^z, t_\tau^u)$ identified with $(p, t_\tau^{\ell^z u})$. Take

the continuous free pro-three product of these two bundles and its semidirect product with $P = \langle\sigma\rangle *_3 \langle\sigma'\rangle$. The inclusions of t, σ, t', σ' give maps in both directions between this semidirect product and the right side of (E56.6). They are inverse on those generators, by the free-product universal property. On the arithmetic kernel, Wingberg's isomorphism sends every factor to its actual inertia group, and the map on P is the identity. This proves the isomorphism (E56.6) including its kernel map. We chose the generators in the full arithmetic group, so their same inclusions define a reverse section of its projection onto (E56.6). Its projection fixes every generator, and it commutes with c. Enlarging S to all primes in the same theorem identifies the remaining kernel as the continuous free inertia product outside S.

The exact two-generator cocycle calculation. For the first cocycle $\mathcal{A}$, the coefficient module is $J = (3, T-1)\Lambda$, with $\Lambda = \mathbb{Z}_3[[T-1]]$. On the first decomposition group its operators are

$$t(f) = T^A f, \qquad \sigma(f) = \ell T^B f(T^\ell), \qquad C(f) = -f(T^{-1}). \tag{E56.14}$$

Normalize the second cocycle so that $v(c) = 0$, using division by two. Since c exchanges the decomposition factors, its values on the second factor are then exactly the last line of (E56.10). There are no further relations between the two factors. The single condition on $u = v(t)$ and $v = v(\sigma)$ is

$$\ell T^B u(T^\ell) - S_\ell(T) u(T) = (T^{\ell A} - 1) v(T). \tag{E56.15}$$

This follows directly from the cocycle identity for $\sigma t \sigma^{-1} = t^\ell$, and is sufficient by the actual presentation.

The remaining geometric conjugations have zero first coordinate: a first coordinate preserving the value at c is killed by multiplication by two. Their second coordinates are exactly the series m with $m(T^{-1}) = -m(T)$. They act by

$$(u, v) \longmapsto \big(u - (T^A - 1)m,\ v - (\ell T^B \psi_\ell - 1)m\big). \tag{E56.16}$$

Every such m has augmentation zero and hence belongs to J.

Put $M = 3^s$. The ideals $(T^A - 1)$ and $(T^{\ell A} - 1)$ both equal $(T^M - 1)$. Their ratios are power series with constant terms ϵ and ℓ, respectively, and are units in Λ. Thus the quotient is the actual cyclic group ring $\mathbb{Z}_3[C_M]$. Because $B = (\ell - 1)r$, multiplication by T^r turns the affine operator $T^B \psi_\ell$ into ψ_ℓ. Reducing (E56.15) consequently says exactly that

$$\bar{f} = \overline{T^r u} \quad \text{is fixed by multiplication by } \ell \text{ in } \mathbb{Z}_3[C_M]. \tag{E56.17}$$

The closed subgroup generated by ℓ is $1 + 3\mathbb{Z}_3$. At each nonzero order 3^d there are precisely two orbits, represented by $\epsilon 3^{s-d}$ and $-\epsilon 3^{s-d}$. Their orbit sums are the images of (E56.7). Together with zero, these are the $2s + 1$ orbits of the complete finite permutation module.

Augmentation of u and f agrees. Thus all invariant elements with augmentation in $3\mathbb{Z}_3$ have the unique lift f_0 in (E56.9), with β_ℓ and all $z_{d,\delta}$ initially arbitrary. The numerator defining v_0 vanishes modulo $T^A - 1$ by (E56.17), and therefore is divisible by $T^{\ell A} - 1$ in Λ. This proves the claimed integral division.

There is one additional, exactly computable condition for v_0 to belong to J. Let $\partial = T\, d/dT$ and evaluate at $T = 1$. The centered representatives satisfy

$$\begin{aligned} \operatorname{aug} Q_{d,\delta} &= 3^{d-1}, & (\partial Q_{d,\delta})(1) &= \delta A/3, \\ \operatorname{aug} v_0 &= \frac{\ell - 1}{A}\left((\partial f_0)(1) - \frac{A}{2}\operatorname{aug} f_0\right) \\ &= (\ell - 1)\left(\frac{1}{3}\sum_{d=1}^{s}(z_{d,+} - z_{d,-}) - \frac{3\beta_\ell}{2}\right). \end{aligned} \tag{E56.18}$$

For the first-moment identity, the 3^{d-1} centered integers congruent to $+1$ modulo three have mean one; the negative orbit has mean minus one. The derivative formula follows by differentiating the actual numerator and denominator of (E56.9); the r terms cancel because $B=(\ell-1)r$. As $(\ell-1)/3$ is a unit, $\operatorname{aug} v_0 \in 3\mathbb{Z}_3$ is equivalent to

$$\sum_{d=1}^{s}(z_{d,+}-z_{d,-}) \in 3\mathbb{Z}_3. \tag{E56.19}$$

This is exactly the lattice with the explicit basis (E56.8). It has index three in the free $2s$-coordinate lattice; the stated formula solves that condition integrally rather than dividing a free coordinate by three.

Every solution with these finite coordinates has a unique expression

$$u=u_0+(T^A-1)z, \qquad v=v_0+(\ell T^B\psi_\ell-1)z, \qquad z\in\Lambda. \tag{E56.20}$$

Indeed the first difference is divisible by T^A-1, and the second follows from (E56.15) in the integral domain Λ. If an arbitrary solution starts with $u,v\in J$, then $\operatorname{aug} v-\operatorname{aug} v_0=(\ell-1)\operatorname{aug} z$ is already divisible by three. It follows that (E56.19) is necessary for every solution, not only for the proposed lift.

Split z into its even and odd parts under $T\mapsto T^{-1}$. The unique allowed conjugation in (E56.16) removes its odd part. The even ring is $\mathbb{Z}_3[[V]]$: with $W=(T-T^{-1})/2$, the ring is $\mathbb{Z}_3[[W]]$, inversion sends W to $-W$, and $V=2(\sqrt{1+W^2}-1)$ has unit coefficient of W^2. The desired augmentation $\operatorname{aug} v=3\beta_3$, together with (E56.18) and $\sum(z_+-z_-)=3\eta$, fixes the constant part of z to be exactly

$$\operatorname{aug} z=\beta_3/h-\eta+3\beta_\ell/2. \tag{E56.21}$$

Its remaining even part is uniquely $Vf(V)$. This proves every formula in (E56.9)–(E56.10), its existence and its uniqueness. No further nonzero geometric conjugation fixes u, since $(T^A-1)m=0$ in a domain forces $m=0$. The specified substitutions, finite coefficient extraction, division on closed principal ideals, and even/odd projectors are continuous. This proves the full compact normal-form homeomorphism.

The entire global section fiber. Let L be the kernel of the actual split arithmetic projection onto (E56.6). Its action on $J_{\mathcal{A}}$ is trivial. As in the exact reconstruction (E55.13), a global second cocycle is equivalent to its cocycle on (E56.6) and an equivariant continuous additive homomorphism $\lambda: L\to J_{\mathcal{A}}$. The free inertia bundle computes these homomorphisms. At $p\notin\{\ell,3\}$ their possible normalized inertia values are zero or $3\mathbb{Z}_3\delta_{x_p}$, where $x_p=-\mathcal{A}(F_p)/(p-1)$. The fixed-measure proof in E.49 and E.55 shows that this Dirac line occurs exactly when $x_p\in\mathbb{Z}_3$. For the actual Kummer cocycle this is precisely the three-primary residue condition (E56.3). At a split prime its conjugate value is determined by c; at an inert prime the rational Frobenius condition supplies the same descent. The family of coefficients is exactly the restricted sequence module $\mathcal{R}(\mathscr{P})$.

For clarity, the rational toric normalization differs from the one used for the inert decomposition factor in E.55 by an explicitly computed evaluation map. Choose roots of every rational prime fixed by c. A residue homomorphism supported at p, with inertia value $3\delta_{x_p}$, and zero cocycle on the split arithmetic subgroup, has scaled second Kummer class exactly

$$\kappa_3(p)-\kappa_3(p)(\sigma)\kappa_3(3). \tag{E56.22}$$

Its valuation is one at p, zero at every other prime except possibly three, and zero at ℓ. It vanishes on σ and σ'; the two evaluation identities $\kappa_3(3)(\sigma)=1$ and $\kappa_3(p)(\sigma')=-\kappa_3(p)(\sigma)$

fix the coefficient at three exactly as displayed. These facts determine the rational completed Kummer class by E.16.

The normal form (E56.10) with $\beta_\ell = 0$, all finite coordinates and f zero, and $\beta_3 = \kappa_3(p)(\sigma)$ has scaled toric class $\kappa_3(p)(\sigma)\kappa_3(3)$. Adding it to the residue cocycle produces an actual global cocycle v_p with toric class $\kappa_3(p)$. Every global cocycle has unique residues $3\beta_p\delta_{x_p}$. Subtracting $\sum_p \beta_p v_p$ leaves precisely a cocycle on the subgroup (E56.6), for which the normal form is unique. The series converges in every finite coefficient and positional quotient: $\beta_p \to 0$, and the normalizing coefficients $\kappa_3(p)(\sigma)$ are integral. Valuation extraction, the normal form, and this reconstruction are continuous in the uniform cochain quotient topology. This proves (E56.4)–(E56.5) as a full global homeomorphism.

Comparison with the norm calculation and verification. The map to E.46 is fully specified. Starting with a norm-compatible system, apply its finite Kummer–Shapiro inverse, align the compatible cocycles as proved there, and fix the first representative $\mathcal{A}$. Normalize $v(c) = 0$. Extract the fixed inertia measures at primes outside S and subtract the just specified v_p terms. For the remaining cocycle, take $\overline{T^r v(t)}$ in $\mathbb{Z}_3[C_{3^s}]$ and read its orbit coefficients. Divide $v(t) - u_0$ by $T^A - 1$, project to its even part, and use (E56.21) to extract the final power series. This is the forward map to (E56.4). Conversely the actual arithmetic presentation constructs the cocycle (E56.10), the additive inertia reconstruction adds the global v_p terms, and E.46 sends it to the norm system by the stated finite Kummer–Shapiro maps. The equations above and the inverse theorem in E.46 prove that these two maps are mutual inverses at every finite level and hence in the stated topologies.

The retained exact program `e_split_prime_first.py` checks the actual Kummer normalization and the Frattini determinants, the complete finite orbit systems, both first-moment identities, the index-three lattice, the integral Laurent division and its unit-denominator series, and recovery of every tested finite, toric, odd, and even coordinate. It also checks the eligible-prime criterion by four finite descriptions. The proof establishes the infinite arithmetic and every three-adic parameter; the finite certificates verify the explicit formulas and their signs. □

The first complex norm transition at seventeen

E.57. Actual class-group corrections and the complete first complex norm map [Proved]

For the first cocycle $3\kappa_3(17)$ in E.46, the first two positional algebras and their cubing map are

$$\begin{gathered} R_1 = \mathbb{Q}[X]/(X^3 - 17^3) = \mathbb{Q} \times K, \qquad K = \mathbb{Q}(\zeta), \quad \zeta^2 + \zeta + 1 = 0, \\ R_2 = \mathbb{Q}[X]/(X^9 - 17^3) = F \times L, \\ F = \mathbb{Q}(\alpha), \quad \alpha^3 = 17, \qquad L = K(\beta), \quad \beta^3 = 17\zeta, \\ R_1 \longrightarrow R_2, \quad X \longmapsto X^3, \qquad N_{R_2/R_1} = (N_{F/\mathbb{Q}}, N_{L/K}). \end{gathered} \tag{E57.1}$$

In particular, $L = \mathbb{Q}(\beta)$ has defining polynomial $\beta^6 + 17\beta^3 + 289$. The following formulas evaluate its entire norm map, including the integral ideal-class corrections. Every completion in this statement is $A^{\wedge_3} = \varprojlim_n A/3^n A$; restricted infinite coordinates carry that inverse-limit topology.

Put $\lambda = 1 - \zeta$, $\pi = 2 - \zeta$, $\bar{\pi} = 3 + \zeta$, and

$$\begin{gathered} \mathfrak{p}_1 = (7, \beta - 3), \quad \mathfrak{p}_2 = (7, \beta + 1), \quad \mathfrak{p}_3 = (7, \beta + 2), \\ c = \beta + 3\zeta + 1, \qquad g_1 = -2 - 3\zeta - \zeta\beta, \qquad g_2 = -3 - \zeta + \zeta\beta. \end{gathered} \tag{E57.2}$$

The class group and its specified generators are

$$\mathrm{Cl}(L) = (\mathbb{Z}/3)[\mathfrak{p}_1] \oplus (\mathbb{Z}/3)[\mathfrak{p}_2], \qquad [\mathfrak{p}_3] = -[\mathfrak{p}_1] - [\mathfrak{p}_2]. \tag{E57.3}$$

There is one prime $\mathfrak{q}_3$ above three, one $\mathfrak{q}_{17}$ above seventeen, and one degree-three prime $\mathfrak{q}_{\bar\pi}$ above $\bar\pi$. The literal principal-ideal identities are

$$\begin{gathered}(c) = \mathfrak{q}_3, \quad (\beta) = \mathfrak{q}_{17}, \quad (\bar\pi) = \mathfrak{q}_{\bar\pi}, \quad (\pi) = \mathfrak{p}_1\mathfrak{p}_2\mathfrak{p}_3, \\ (g_1) = \mathfrak{p}_1^3, \qquad (g_2) = \mathfrak{p}_2^3.\end{gathered} \tag{E57.4}$$

Thus a divisor supported above $3, 7, 17$, with coefficients n_i at $\mathfrak{p}_i$, is principal precisely when $n_1 \equiv n_2 \equiv n_3 \pmod 3$.

Let S_K consist of the primes above $3, 7, 17$, and let S_L be its inverse image. Two actual fundamental units of L are

$$\begin{aligned}e_1 &= -18 + 3\beta + 3\beta^2 - 18\beta^3/17 - 5\beta^4/17 + 2\beta^5/17, \\ e_2 &= -18 - 5\beta + 2\beta^2 - \beta^3/17 - 8\beta^4/17 - \beta^5/17.\end{aligned} \tag{E57.5}$$

The source S-unit completion has the specified basis

$$\mathcal{O}_{L,S_L}^{\times,\wedge_3} = \langle e_1, e_2, c, \beta, \bar\pi, \pi, g_1, g_2\rangle_{\mathbb{Z}_3} \oplus \langle\zeta\rangle_{\mathbb{Z}/3}. \tag{E57.6}$$

The target basis is $\lambda, 17, \pi, \bar\pi$ over $\mathbb{Z}_3$ and ζ of order three. Relative norms are exactly

v	e_1	e_2	c	β	$\bar\pi$	π	g_1	g_2	ζ
$N(v)$	1	1	λ	17ζ	$\bar\pi^3$	π^3	$-\zeta^2\pi^3$	$\zeta^2\pi^3$	$1.$

(E57.7)

The sign is killed by the explicitly defined three-primary completion. In these coordinates the norm matrix has its last row read modulo three:

$$\begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 3 & 3 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 2 & 2 & 0 \end{pmatrix}. \tag{E57.8}$$

Consequently its full S-unit kernel and cokernel are

$$\begin{aligned}\ker N &= \langle e_1, e_2, -g_1/g_2, (g_2/\pi)^3\rangle_{\mathbb{Z}_3} \oplus \langle\zeta\rangle_{\mathbb{Z}/3} \simeq \mathbb{Z}_3^4 \oplus \mathbb{Z}/3, \\ \operatorname{coker} N &\simeq (\mathbb{Z}/3)^2, \qquad \text{coordinates } (v_\pi \bmod 3, v_{\bar\pi} \bmod 3).\end{aligned} \tag{E57.9}$$

There is a concrete correction with zero relative divisor norm:

$$w = (g_1/\pi)^2, \qquad N(w) = \zeta, \qquad \operatorname{div}(w) = 4\mathfrak{p}_1 - 2\mathfrak{p}_2 - 2\mathfrak{p}_3. \tag{E57.10}$$

By contrast, the ordinary-unit norm image is exactly $\{1, -1\}$. Formula (E57.10) specifies the extra valuations which supply this root norm.

Here are full global coordinates, so that the calculation also specifies every column outside S. For every prime $\mathfrak{P} \notin S_L$, compute its unique class coordinates $a_\mathfrak{P}, b_\mathfrak{P} \in \{0, 1, 2\}$ in (E57.3), and choose the actual principal generator $q_\mathfrak{P}$ of

$$(q_\mathfrak{P}) = \mathfrak{P}\mathfrak{p}_1^{-a_\mathfrak{P}}\mathfrak{p}_2^{-b_\mathfrak{P}}. \tag{E57.11}$$

Class logarithms and generators are computed by the certified ideal algorithm retained with this statement. Fix a generator $p_\mathfrak{p}$ for each prime $\mathfrak{p} \notin S_K$ in the Eisenstein principal ideal

domain. If $\mathfrak{P}$ lies over $\mathfrak{p}$, let $f_{\mathfrak{P}}$ be its relative residue degree. The exact element norm determines $\varepsilon_{\mathfrak{P}} \in \mathbb{Z}/3$ by

$$N(q_{\mathfrak{P}}) = \pm\zeta^{\varepsilon_{\mathfrak{P}}}\, p_{\mathfrak{p}}^{f_{\mathfrak{P}}}\, \pi^{-a_{\mathfrak{P}}-b_{\mathfrak{P}}}. \tag{E57.12}$$

These are evaluated number-field operations, with the defining field, basis and class generators fixed above.

Every element of $L^{\times,\wedge_3}$ has unique coordinates

$$e_1^{t_1} e_2^{t_2} c^{\alpha} \beta^{b_0} \bar{\pi}^{\delta} \pi^{\gamma} g_1^{\eta} g_2^{\theta} \zeta^{e} \prod_{\mathfrak{P} \notin S_L} q_{\mathfrak{P}}^{n_{\mathfrak{P}}}, \quad n_{\mathfrak{P}} \longrightarrow 0 \text{ modulo every } 3^m. \tag{E57.13}$$

The complete global norm is obtained from (E57.8) by the following explicit additions:

$$\begin{aligned} v_{\lambda} &= \alpha, & v_{17} &= b_0, \\ v_{\pi} &= 3(\gamma+\eta+\theta) - \sum_{\mathfrak{P}} (a_{\mathfrak{P}} + b_{\mathfrak{P}}) n_{\mathfrak{P}}, & v_{\bar{\pi}} &= 3\delta, \\ v_{\mathfrak{p}} &= \sum_{\mathfrak{P} \mid \mathfrak{p}} f_{\mathfrak{P}} n_{\mathfrak{P}}, & e_{\mathrm{root}} &= b_0 + 2\eta + 2\theta + \sum_{\mathfrak{P}} \varepsilon_{\mathfrak{P}} n_{\mathfrak{P}} \pmod 3. \end{aligned} \tag{E57.14}$$

All sums have their stated restricted convergence. This is the actual first complex transition of E.46. It imposes no conclusion about compatibility with every subsequent norm transition, or about lifting through the intrinsic kernels of (E47.8). The manuscript anchor is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379), Part E(c),(d).

Proof. Factoring the two polynomials in (E57.1) and substituting X^3 gives all four stated algebra maps. On the complex factor the substitution is literally $\beta^3 = 17\zeta$, so the associated corestriction in E.46 is $N_{L/K}$ with this precise embedding.

The retained PARI/GP computation uses the officially specified unconditional `bnfcertify` procedure [The26]. It returns one for both K and L. It certifies the complete class group $(3,3)$, the maximal-order basis

$$1,\ \zeta+1,\ \beta,\ \zeta\beta,\ \beta^2,\ \zeta\beta^2, \tag{E57.15}$$

the two complete fundamental units in (E57.5), and the torsion group μ_6. The class logarithms of the displayed $\mathfrak{p}_1, \mathfrak{p}_2$ form an invertible two-by-two matrix modulo three on the certified class basis. Changing by that actual matrix gives coordinates $(1,0), (0,1), (2,2)$ for the three primes. This proves (E57.3) with its specified generators, not just its abstract order.

All remaining displayed ideal and element equalities are checked independently of PARI's class-group arithmetic. The Python certificate constructs the exact multiplication matrices in the integral basis (E57.15), reduces modulo $\beta^6 + 17\beta^3 + 289$, and compares their Hermite lattices. It verifies (E57.4) and the divisor in (E57.10), including its negative exponents. Each relative norm is also the independently computed determinant

$$N_{L/K}(a + b\beta + d\beta^2) = a^3 + (17\zeta)b^3 + (17\zeta)^2 d^3 - 3(17\zeta)abd. \tag{E57.16}$$

This gives every entry in (E57.7). The official specification of `bnfisprincipal` supplies both the class logarithm and the principal generator, and the code verifies the resulting ideals and norms exactly.

For divisors supported on S_L, (E57.3) reduces principality to $n_1 - n_3 = n_2 - n_3 = 0$ modulo three. The lattice of these triples has basis $(1,1,1), (3,0,0), (0,3,0)$, of index nine in $\mathbb{Z}^3$. The other three prime coordinates are independently principal by (E57.4). Combining these six divisor generators with the two fundamental units and the torsion generator proves the full

basis (E57.6). No missing unit or class index is absorbed into this basis. The analogous target basis follows from the Eisenstein principal ideal domain and its units μ_6.

Writing the source exponents in the order of (E57.6), formula (E57.7) gives (E57.8). For a norm-one vector its first, second, and fourth target coordinates force $\alpha = b_0 = \delta = 0$. The third says $\gamma+\eta+\theta = 0$, and its root coordinate says $\eta+\theta \in 3\mathbb{Z}_3$. Write $\eta+\theta = 3t$ and then $\theta = 3t-\eta$, $\gamma = -3t$. These are exactly the four free kernel generators in (E57.9), together with the order-three root. The root coordinate is attained by (E57.10); the two remaining obstructions are precisely the valuations at $\pi, \bar{\pi}$ modulo three. Equivalently, adjoining the target relation $3e_{\mathrm{root}} = 0$ to the integer norm matrix gives Smith entries $1, 1, 1, 3, 3$, as independently verified. For ordinary units the certified basis has norms $1, 1$, while $N(-1) = -1$ and $N(\zeta) = 1$. This proves the exact ordinary-unit assertion and the actual comparison (E57.10).

For the global formula, the primes in S_L generate the full class group. Thus (E57.11) exists for every outside prime. The resulting divisor sequence has the explicit splitting

$$0 \longrightarrow \mathcal{O}_{L,S_L}^{\times} \longrightarrow L^{\times} \xrightarrow{(v_{\mathfrak{P}})} \bigoplus_{\mathfrak{P}\notin S_L} \mathbb{Z} \longrightarrow 0, \qquad 1_{\mathfrak{P}} \longmapsto q_{\mathfrak{P}}. \tag{E57.17}$$

This proves (E57.13), first as a literal group decomposition and then after completion for multiplication by 3^n. The completion of its direct sum is exactly the restricted sequence group stated there. This argument does not use compactness of those infinite quotients.

Taking the ideal norm of (E57.11) gives $\mathfrak{p}^{f_{\mathfrak{P}}}(\pi)^{-a_{\mathfrak{P}}-b_{\mathfrak{P}}}$. The difference between its two displayed generators is a unit of K, hence lies in μ_6. This proves the exact and computable formula (E57.12). Taking its coefficients gives (E57.14). Since each class coordinate is in $\{0, 1, 2\}$, the sums contributing to the S coordinates converge for every restricted input. There are at most three primes over each target prime, and their residue degrees are at most three, so the outside target coordinates again form a restricted family. This proves continuity and the entire asserted norm map.

The retained computation evaluates all 31 outside-prime columns above rational primes at most 61 other than $3, 7, 17$. Each corrected principal divisor, its exact relative norm, and its root coordinate is retained. The Python certificate independently checks these columns, 41 cubic multiplication determinants, nine ideal-lattice identities, and the full Smith calculation. The formulas (E57.11)–(E57.17) prove the construction for every prime; the finite computation checks its concrete normalizations. □

An evaluated elliptic Frobenius retaining the ten abelian coordinates

E.58. Two actual elliptic quotients and their based Frobenius at two [Proved]

Use the actual degree-nine elliptic covering of E.50, with $F = \pi_1^{(3)}(E \setminus \{O\}, \vec{w})$ and $N = \ker(F \to F^{\mathrm{ab}}/3)$. The specified cubic tangent $s = z/x$ maps to $\vec{w}$ under this cover, by the exact morphism and tangent calculation below. Choose the modular path with the principal compatible complex roots of the Weierstrass discriminant, as specified in the proof. Let x, y be its free pro-three generators, put

$$c = [x, y], \quad d = [x, c], \quad e = [y, c], \qquad \Lambda = \mathbb{Z}_3[[X-1, Y-1]], \quad \mathfrak{m} = (X-1, Y-1), \tag{E58.1}$$

and identify $F'/F'' = \Lambda c$, with X, Y acting by conjugation by x, y. Write

$$U = X^3-1, \quad V = Y^3-1, \quad J = (U, V), \quad P = (1+X+X^2)(1+Y+Y^2), \quad I_B = J^3 + PJ^2. \tag{E58.2}$$

For an ideal $L \subset \Lambda$, let $\widetilde{L}$ denote the inverse image of Lc in F'. The following are actual quotient maps:

$$\begin{gathered} B = N/(F''\gamma_4 N) = N/\widetilde{I}_B, \qquad R = N/\widetilde{J\mathfrak{m}}, \qquad P_3 = N/\gamma_4 F, \\ B \twoheadrightarrow R \twoheadrightarrow P_3, \qquad B \twoheadrightarrow R \twoheadrightarrow C_{\mathrm{cc}} = N/[N, F']. \end{gathered} \tag{E58.3}$$

Here C_{cc} is exactly the cuspidally central quotient of E.36. The comparison is the fiber product

$$R \;\simeq\; C_{\mathrm{cc}} \underset{Q_*}{\times} P_3, \qquad Q_* = N/\widetilde{J + \mathfrak{m}^2}. \tag{E58.4}$$

The commutator-module quotients in B, R, P_3, Q_* are, respectively,

$$\Lambda/I_B, \qquad M = \Lambda/(J\mathfrak{m}), \qquad \Lambda/\mathfrak{m}^2, \qquad \Lambda/(J + \mathfrak{m}^2) \simeq \mathbb{Z}_3 \oplus (\mathbb{Z}/3)^2. \tag{E58.5}$$

The first three are free of ranks $51, 11, 3$ over $\mathbb{Z}_3$. Thus the kernels of $B \to R$, $B \to P_3$, $R \to P_3$ and $R \to C_{\mathrm{cc}}$ are abelian and free of ranks $40, 48, 8, 2$. The abelianization of R is the full N^{ab}, of rank ten.

Use the following integral basis of M:

$$e_{ij} = X^i Y^j \quad (0 \le i, j \le 2), \qquad U, V. \tag{E58.6}$$

Its multiplication is determined by

$$\begin{gathered} e_{ij} e_{kl} = e_{\overline{i+k}, \overline{j+l}} + \left\lfloor \frac{i+k}{3} \right\rfloor U + \left\lfloor \frac{j+l}{3} \right\rfloor V, \\ e_{ij} U = U, \qquad e_{ij} V = V, \qquad U^2 = UV = V^2 = 0, \end{gathered} \tag{E58.7}$$

where the barred indices belong to $\{0, 1, 2\}$. Every element of R has unique coordinates $\xi^a \eta^b c^H$, with $\xi = x^3$, $\eta = y^3$, $a, b \in \mathbb{Z}_3$ and $H \in M$. It has the exact lower-central lattices

$$\gamma_2 R = \mathbb{Z}_3 Pc \oplus \mathbb{Z}_3 Uc \oplus \mathbb{Z}_3 Vc, \qquad \gamma_3 R = 9\mathbb{Z}_3 Uc \oplus 9\mathbb{Z}_3 Vc, \qquad \gamma_4 R = 1. \tag{E58.8}$$

In particular $\gamma_2 R/\gamma_3 R \simeq \mathbb{Z}_3 \oplus (\mathbb{Z}/9)^2$. These finite summands remain part of the integral quotient.

Choose the actual prolongation at the prime of $K = \mathbb{Q}(\zeta_3)$ over two for which the compatible positive roots of three lie in $\mathbb{Q}_2$. Let φ be arithmetic Frobenius there. Its genuine based action on R is the following fully evaluated automorphism. Put

$$\begin{aligned} H_x &= (1 + X + X^2)(3 + Y - Y^2) + \frac{31}{4} U + \frac{3}{2} V, \\ H_y &= (X^2 - X - 3)(1 + Y + Y^2) - \frac{3}{2} U - \frac{23}{4} V, \\ \Phi(e_{ij}) &= e_{ij} - iU - jV, \qquad \Phi(U) = -2U, \quad \Phi(V) = -2V. \end{aligned} \tag{E58.9}$$

Then

$$\varphi(\xi) = \xi^{-2} c^{H_x}, \qquad \varphi(\eta) = \eta^{-2} c^{H_y}, \qquad \varphi(c^H) = c^{4\Phi(H)}. \tag{E58.10}$$

Equivalently its map on all thirteen coordinates is

$$\begin{aligned} (a, b, H) \longmapsto \big(&- 2a, -2b, 4\Phi(H) + aH_x + bH_y + 9a(a-1)U \\ &+ [18ab - 9b(b-1)]V \big). \end{aligned} \tag{E58.11}$$

The induced ambient class-three action on $F/\gamma_4 F$ is

$$\begin{array}{c|l} x & x^{-2} c^3 d^{19/4} e^{1/2} \\ y & y^{-2} c^{-3} d^{-1/2} e^{-11/4} \\ c & c^4 \\ d & d^{-8} \\ e & e^{-8}. \end{array} \tag{E58.12}$$

Restriction to P_3 gives

$$\xi \longmapsto \xi^{-2}c^{9}d^{129/4}e^{3/2}, \qquad \eta \longmapsto \eta^{-2}c^{-9}d^{-3/2}e^{-105/4}, \tag{E58.13}$$

with the last three images from (E58.12).

All these coordinates include an exact change from the $\mathbb{Q}_2$ discriminant-root convention. Define $T_1(a) : x \mapsto xy^{-a},\ y \mapsto y$. On R and on $F/\gamma_4 F$ the two actual coordinate actions satisfy

$$\varphi_{\text{principal}} = T_1(1/2)\varphi_{\mathbb{Q}_2}T_1(-1/2) = \mathrm{Int}(c^{1/8})\varphi_{\mathbb{Q}_2}. \tag{E58.14}$$

The second equality compares the two actions; the first is the actual integral change of generators. In (E58.12), the $\mathbb{Q}_2$ version replaces $19/4, -11/4$ by $9/2, -3$. In (E58.9), it replaces $31/4, -23/4$ by $15/2, -6$.

The factorization proved below is through the actual image of the pair consisting of the Tate representation and the specified discriminant Kummer cocycle. It does not construct a section from an unrestricted matrix group. Equations (E58.3)–(E58.5) retain the remaining rank-forty kernel in the larger quotient B; its full based Frobenius action is not evaluated here. The manuscript anchor is §51.16, scans 172–178, folios 733v–736v, equations (371)–(379), with Part E(c),(d).

Proof. The based elliptic morphism. On the compact cubic $x_0^3 + y_0^3 + z_0^3 = 0$, the formulas

$$X_0 = -\frac{3z_0}{x_0 + y_0}, \qquad Y_0 = \frac{9(x_0 - y_0)}{x_0 + y_0} \tag{E58.15}$$

give $Y_0^2 = 4X_0^3 - 27$, with inverse $(x_0 : y_0 : z_0) = (9 + Y_0 : 9 - Y_0 : -6X_0)$. They extend across the displayed poles as inverse morphisms of the smooth compact curves, carrying $O = (1 : -1 : 0)$ to the origin. For $s = z_0/x_0$, the branch through O has $y_0/x_0 = -(1 + s^3)^{1/3}$. Consequently the Weierstrass parameter is

$$t = -2X_0/Y_0 = \frac{2s}{3(1 + (1 + s^3)^{1/3})} = s/3 + O(s^4). \tag{E58.16}$$

Multiplication by three has derivative three at the origin. Thus the actual covering [3] takes the original s-tangent to the Weierstrass t-tangent of coefficient one. Its remaining formal unit has constant term one, whose compatible roots also have constant term one. This proves the based comparison used here without introducing an additional tangent Kummer character. The discriminant for dX_0/Y_0 is $\Delta = -3^9$.

The integral quotient rings and groups. The free metabelian module in E.50 gives $F'/F'' = \Lambda c$. In N/F'', the two handle generators are x^3, y^3, and the whole module Λc is present. Their commutators give Pc, Uc, Vc. Since N is normal in F, its derived subgroup is stable under X, Y; hence

$$N'/F'' = (P, J)c, \qquad \gamma_r N\, F''/F'' = J^{r-2}(P, J)c \quad (r \geq 2). \tag{E58.17}$$

This proves the ideal I_B and every quotient in (E58.3). Similarly $[N, F']/F'' = Jc$ and $\gamma_4 F/F'' = \mathfrak{m}^2 c$.

Here is the exact intersection, over $\mathbb{Z}_3$:

$$J \cap \mathfrak{m}^2 = J\mathfrak{m}. \tag{E58.18}$$

Write $u = X - 1$, $v = Y - 1$. For $Ua + Vb \in \mathfrak{m}^2$, its linear term is $3a(0)u + 3b(0)v$. Since $\mathbb{Z}_3$ has no three-torsion, $a(0) = b(0) = 0$, proving the nontrivial containment. The usual explicit lifting argument for two normal quotients now gives (E58.4), including surjectivity onto the

fiber product. The sum ideal gives $3u = 3v = u^2 = uv = v^2 = 0$, which is the last module in (E58.5).

The ring defined by (E58.7) is free on its eleven displayed vectors. Its multiplication is associative: the two integer carries in each coordinate add independently. It satisfies $X^3 = 1 + U$, $Y^3 = 1 + V$ and $\mathfrak{m}U = \mathfrak{m}V = 0$. Conversely these relations reduce every formal series to the displayed basis. They define continuous maps in both directions between this ring and $\Lambda/(J\mathfrak{m})$, inverse on X, Y; the powers of $\mathfrak{m}$ tend to zero three-adically in the finite ring. This proves its integral normal form and topology.

For the rank in B, use U, V as formal variables. Λ is free of rank nine over $\mathbb{Z}_3[[U, V]]$, with basis e_{ij}, by distinguished-polynomial division. Modulo J^3 it has rank $9(1 + 2 + 3) = 54$. The additional relations PU^2, PUV, PV^2 are independent primitive vectors, one sum of all nine coset vectors in each degree-two component. Multiplication by X or Y adds only terms in J^3, so these are all the relations from PJ^2. The rank is therefore 51. The maps to the free modules M and $\Lambda/\mathfrak{m}^2$ are surjective, proving the kernel ranks and their freeness. The same reasoning proves the two smaller kernel ranks.

Modulo $J\mathfrak{m}$, the ideal (P, J) is exactly $\mathbb{Z}_3P + \mathbb{Z}_3U + \mathbb{Z}_3V$: $(X - 1)P = 3U$, $(Y - 1)P = 3V$ there. Furthermore $UP = 9U$, $VP = 9V$ and $J^2 = 0$. These equalities prove (E58.8). The three vectors P, U, V are primitive and independent in M, so the abelianization has rank $2 + (11 - 3) = 10$ with no torsion. The subgroup $\widetilde{J\mathfrak{m}}$ lies in N', so its quotient identifies this abelianization with the actual N^{ab}.

The universal arithmetic factorization with its sign. We use the original modular-path construction of [Nak13, §§3.6,5.9 and Theorem 6.10.3]. In its convention $z = [x, y]^{-1} = c^{-1}$. On the kernel of the Tate representation, its equation (3.6.3) identifies the action on F/F'' with conjugation by $z^{-E_\sigma} = c^{E_\sigma}$. The augmentation of this integral measure is $\kappa_\Delta(\sigma)/12$. In particular the kernel of the pair (ρ_E, κ_Δ) has $E_\sigma \in \mathfrak{m}$. Conjugation by such a module element changes x^3, y^3 by elements of $J\mathfrak{m}c$ and fixes F'/F''. It consequently acts trivially on R. It also acts trivially on F/γ_4F, since that commutator change lies in $\mathfrak{m}^2c$. This proves the stated factorization through the actual image of the cocycle pair, with product $(A, k)(B, l) = (AB, k + \det(A)l)$. It specifies the sign of the cusp correction as well.

An explicit reference element. Nakamura's Weierstrass-section formulas [Nak13, Proposition 5.7.3] give the actual twists

$$T_1(a) : x \mapsto xy^{-a},\ y \mapsto y, \qquad T_2(a) : x \mapsto x,\ y \mapsto yx^a. \tag{E58.19}$$

Their discriminant winding is $-a$. Direct multiplication gives $(T_1(1)T_2(1))^6 = \mathrm{Int}(c^{-1})$. Define

$$D = T_2(-2)T_1(-1/2)T_2(-3)T_1(-1)T_2(-1). \tag{E58.20}$$

It has Tate matrix $\mathrm{diag}(-2, -1/2)$ and discriminant winding $15/2$. These three-adic twist powers are actual continuous automorphisms, and are images of the corresponding profinite braid powers.

On the Tate tangential splitting, an actual Galois element whose three-adic cyclotomic character is four has Tate matrix $\mathrm{diag}(1, 4)$ and discriminant cocycle zero. The explicit formula of [Nak13, Theorem 5.2.4], with the genuine coefficient $(4^2 - 1)/24 = 5/8$ verified in E.54, has ambient class-three images

$$x \longmapsto xc^{3/2}d^{-3/2}e^{-5/4}, \qquad y \longmapsto y^4. \tag{E58.21}$$

There is also a convenient exact metabelian reference map

$$A_4 : x \longmapsto c^{h(Y)}x,\ y \longmapsto y^4, \qquad h(Y) = \frac{Y^2 + 2Y + 3}{(Y + 1)(Y^2 + 1)}. \tag{E58.22}$$

The denominator is a unit in Λ. Writing $S_4 = 1 + Y + Y^2 + Y^3$, one has $h = (S_4 - 4)/(Y^4 - 1)$, which proves directly that $A_4(c) = c^4$. Its invertible Frattini map proves that it is an automorphism of the free metabelian pro-three group. Expansion to first order in $Y - 1$ gives $h = 3/2 - (5/4)(Y - 1) + \cdots$, so it has exactly (E58.21). The ratio of this map and the genuine Tate map is a Tate-trivial, cusp-fixing automorphism; the regular-sequence calculation in E.50 writes it as conjugation by c^H. Equality of the class-three actions gives $H \in \mathfrak{m}$. Thus (E58.22) and the genuine Tate map agree on R. This last comparison is asserted on R and on $F/\gamma_4 F$, with their displayed kernels.

The composite DA_4 therefore represents the actual pair $(-2I, 15/2)$ on these two quotients. Left composition by $\mathrm{Int}(c^{-1/2})$ gives the pair $(-2I, 3/2)$. Left composition instead by $\mathrm{Int}(c^{-5/8})$ gives $(-2I, 0)$. Collecting these explicit words proves (E58.9)–(E58.13). For example the substitutions on the module are $X \mapsto X^{-2}$, $Y \mapsto Y^{-2}$; in (E58.7) these are precisely Φ. Powering the two handle images gives (E58.11).

The actual local Frobenius and the root transport. The good integral model is $Y_1^2 + Y_1 = X_0^3 - 7$. It has three points over $\mathbb{F}_2$, hence Frobenius polynomial $T^2 + 2$. The prime of K over two has residue field $\mathbb{F}_4$, so its Tate action is exactly $-2I$. All three-power roots of three can be embedded compatibly in $\mathbb{Q}_2$, where the odd-power map on units is bijective. For every n, put

$$\Delta_n^{\mathrm{loc}} = -3^{9/3^n}, \qquad \Delta_n^{\mathrm{pr}} = e^{\pi i/3^n} 3^{9/3^n}. \quad \frac{\Delta_n^{\mathrm{pr}}}{\Delta_n^{\mathrm{loc}}} = \zeta_{3^n}^{(3^n+1)/2}. \tag{E58.23}$$

These are compatible systems; the ratio has coherent exponent $1/2 \in \mathbb{Z}_3$. Frobenius fixes the first system and has three-adic cyclotomic character four. Its discriminant cocycles are therefore 0 and $3/2$, respectively.

A modular path can be chosen to give the first system: starting with any path, its root ratios give a compatible three-adic exponent, and a twist (E58.19), whose discriminant winding is surjective, corrects that exponent. Changing that path by $T_1(1/2)$ changes the discriminant cocycle by $(\chi - 1)/2$, since its winding is $-1/2$. Its matrix commutes with the scalar $-2I$, so the resulting local Tate matrix remains exactly the one used above. The actual change of coordinates is the first equality in (E58.14); direct collection verifies it on the two ambient class-three generators and on every generator of R. The second equality follows by the computed difference $\kappa_\Delta/12 = (3/2)/12 = 1/8$. Every exponent used in this transport is integral over $\mathbb{Z}_3$.

The retained programs verify the cubic morphism, its tangent, the point count, the exact braid, Tate, and root-change formulas, and all thirteen generator images. They check the finite ring's 1,331 associativity identities, 144 exact group/action comparisons and 156 power identities. The projection to the ambient three-coordinate commutator module has an explicit integer right inverse. These checks accompany the integral ideal and group proofs above. □

The first nonabelian inert Frobenius and its exact Frattini orbit

E.59. A uniform cubic-residue formula and the rank-five Frobenius orbit at seventeen [Proved]

Let $K = \mathbb{Q}(\zeta_3)$ and $P = G^{(3)}_{K,\{3\}}$. Use the actual free basis of E.55, written x, y, with

$$\chi(x) = 4, \quad \kappa_3(3)(x) = 0, \qquad \chi(y) = 1, \quad \kappa_3(3)(y) = 1, \qquad cxc = x, \quad cyc = y^{-1}. \tag{E59.1}$$

Here x is the specified arithmetic Frobenius at the prime over two, and c is the specified complex conjugation in $P \rtimes C_2$. Put $\Phi = P^3[P, P]$, $\Phi_2 = \Phi^3[\Phi, \Phi]$, and let $D_n(P)$ be the three-Zassenhaus filtration. The Frattini field is

$$K_1 = K(\zeta_9, \sqrt[3]{3}), \qquad \mathrm{Gal}(K_1/K) = (\mathbb{Z}/3)^2, \qquad \dim_{\mathbb{F}_3} \Phi/\Phi_2 = 10. \tag{E59.2}$$

Every inert rational prime $\ell \equiv 8 \pmod 9$ has arithmetic Frobenius $\sigma_\ell \in D_3(P)$. The following explicit formula determines its whole image in $D_3(P)/D_4(P)$.

Set $z = \zeta_9$, $\omega = z^3$, $r = \sqrt[3]{3}$, and define actual elements of K_1 by

$$u = -z^2(1-z)^2, \quad u_1 = -z^8(1-z^4)^2, \quad u_2 = -z^{14}(1-z^7)^2,$$
$$h = \frac{(r-u)(\omega^2 r - u_1)}{(\omega^2 r - u)(r-u_1)}. \tag{E59.3}$$

The denominators have no prime divisor away from three. In $\mathbb{F}_{\ell^2}$ choose a primitive ninth root $\bar z$, put $\bar\omega = \bar z^3$, and take the unique cube root $\bar r$ of 3 in $\mathbb{F}_\ell$. Define $\alpha_\ell, \beta_\ell \in \{0,1,2\}$ by

$$\alpha_\ell = \frac{\ell^2-1}{9} \pmod 3, \qquad \bar h^{\,(\ell^2-1)/3} = \bar\omega^{\,\beta_\ell}. \tag{E59.4}$$

Then the actual comparison is

$$\boxed{\sigma_\ell = x^{3\alpha_\ell}[y,[x,y]]^{\beta_\ell} \quad \text{in } P/D_4(P).} \tag{E59.5}$$

Changing the prolongation, or either of the compatible residue-field roots in (E59.4), gives the same two coefficients. For example,

$$\sigma_{17} = x^6[y,[x,y]]^2 \pmod{D_4(P)}, \qquad \sigma_{17} \in \Phi \setminus \Phi_2, \qquad \sigma_{17} \in D_3(P) \setminus D_4(P). \tag{E59.6}$$

Here is an integral-source comparison which also determines the exact span of its conjugates. Write

$$R = \mathbb{F}_3[X,Y]/(X^3-1, Y^3-1) = \mathbb{F}_3[s,t]/(s^3,t^3), \qquad s = X-1, \quad t = Y-1, \quad I = (s,t). \tag{E59.7}$$

Continuous Fox derivatives give an actual R-module isomorphism

$$\Phi/\Phi_2 \xrightarrow{\sim} M = \ker\bigl(R^2 \xrightarrow{(s,t)} R\bigr). \tag{E59.8}$$

The involution induced by c is

$$(A,B) \longmapsto \bigl(A(X,Y^{-1}), -Y^{-1}B(X,Y^{-1})\bigr). \tag{E59.9}$$

Its fixed subspace has the exact basis

$$a = (s^2, 0), \qquad b = (-t^2, st - st^2), \qquad sb, \qquad s^2 b. \tag{E59.10}$$

For $w = Aa + (B + Cs + Ds^2)b$, its entire cyclic R-module has dimension

coordinates	$\dim_{\mathbb{F}_3} Rw$	one basis
$B \neq 0$	5	w, sw, tw, s^2w, stw
$B = 0,\ A \neq 0$	3	w, tw, t^2w
$A = B = 0,\ C \neq 0$	3	w, sw, tw
$A = B = C = 0,\ D \neq 0$	1	w
$A = B = C = D = 0$	0	$\varnothing$.

(E59.11)

A prolongation with $c\sigma_\ell c = \sigma_\ell$ exists, and its coordinates in (E59.10) satisfy $A = \alpha_\ell$, $B = \beta_\ell$. Thus the nine conjugate Frobenius classes have rank five when $\beta_\ell \neq 0$, and rank three when $\beta_\ell = 0$, $\alpha_\ell \neq 0$. In particular, their rank at seventeen is *exactly five*. Each individual class at seventeen is nonzero, but the nine classes cannot be part of a free generating set of Φ. This last conclusion follows from the displayed map on its Frattini quotient.

For completeness, here is a separate full arithmetic certificate of the rank at seventeen. In the ten-dimensional Kummer basis $e_1, \dots, e_8, \pi_3, z$ specified by the retained exact coefficient file, the nine Frobenius columns are

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 2 & 2 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 2 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 0 & 2 & 0 & 0 & 0 & 2 & 1 & 0 \\ 2 & 0 & 0 & 2 & 2 & 1 & 0 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}. \tag{E59.12}$$

Its rank is five; its kernel has dimension four and contains the all-one vector. The residue procedure is evaluated at every one of the fifty primes $\ell \le 2000$ in this congruence class. The first coefficients include

ℓ	17	53	71	89	107	179	197	233	251	269
α_ℓ	2	0	2	1	0	2	1	2	1	0
β_ℓ	2	1	1	0	2	1	2	2	2	0.

(E59.13)

The three primes in this range with both coefficients zero are $269, 809, 1187$. Their full ten-unit residue matrices, also retained, have respective ranks $3, 1, 3$. These ranks use the entire Frattini quotient, including the coordinates beyond (E59.5). There is also an exact distribution theorem. For ranks $0, 1, 3, 5$, respectively, the relative Dirichlet densities among rational primes $\ell \equiv 8 \pmod 9$ are

$$\frac{1}{81}, \qquad \frac{2}{81}, \qquad \frac{8}{27}, \qquad \frac{2}{3}. \tag{E59.13a}$$

Their absolute Dirichlet densities are $1/486, 1/243, 4/81, 1/9$. They follow from the actual finite Galois quotient below.

The maps in this statement are the actual maps $P \to P/\Phi$, $\Phi \to \Phi/\Phi_2$ and $P \to P/D_4(P)$, together with their Kummer duals. The correction words in the arithmetic inertia presentation of E.39 are retained. The calculation does not replace an actual local Frobenius by the image of a chosen section of an arithmetic quotient. The manuscript anchor is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379), and Part E(c),(d).

Proof. The arithmetic field and the two filtrations. E.34 and the actual basis construction in E.55 give (E59.1). The two independent mod-three characters are the cyclotomic character on ζ_9 and the Kummer character of 3. Their fixed field is exactly K_1. The subgroup Φ has index nine in the free pro-three group of rank two, so its Schreier rank is $1 + 9(2 - 1) = 10$.

For $\ell \equiv 8 \pmod 9$, cubing is an automorphism of $\mathbb{Q}_\ell^\times$ on its unit subgroup: both its residue-unit group and its principal-unit group have order prime to three at every finite quotient. All compatible cube-power roots of 3 can therefore be taken in $\mathbb{Q}_\ell$. The local Frobenius has $\chi = \ell^2$ and $\kappa_3(3) = 0$. It is trivial on both generators of K_1/K, so it lies in $\Phi = D_2(P)$.

There is also a precise normalization by complex conjugation. The image of the local decomposition group in $P \rtimes C_2$ is procyclic, since the extension is unramified at ℓ. Its cyclotomic image identifies it with $C_2 \times \mathbb{Z}_3$. Its unique involution is conjugate to c by an element of P, by the coprime complement theorem. After making this actual conjugation, its Frobenius commutes with c. Both involutions have zero Kummer character on 3; the

conjugating element has zero such character because the conjugation formula gives twice that character. Hence the normalization also preserves $\kappa_3(3)(\sigma_\ell) = 0$.

The quotient D_2/D_3 is generated by $[x, y]$ and c acts on it by minus one. Consequently the c-fixed Frobenius lies in D_3. Collection, or the truncated Magnus expansion over $\mathbb{F}_3$, gives the four independent generators

$$x^3, \quad y^3, \quad [x,[x,y]], \quad [y,[x,y]] \quad \text{of } D_3/D_4. \tag{E59.14}$$

Their signs under c are $+, -, -, +$. Thus only the first and the last can occur. On x^3 the cyclotomic character is 64, and $(64-1)/9 \equiv 1 \pmod 3$. This proves the formula for α_ℓ. At seventeen $\chi(\sigma_{17}) = 289$ is one modulo nine but not modulo twenty-seven. Both Φ_2 and D_4 have cyclotomic image contained in $1 + 27\mathbb{Z}_3$. This proves the two strict assertions in (E59.6).

The Fox map, including the fixed lattice. The degree-nine covering of the two-circle graph with deck group P/Φ has chain complex $R^2 \to R$ with boundary (s, t). Its first homology is the Frattini quotient of its free pro-three fundamental group. This gives (E59.8), and also proves its compatibility with conjugation by x, y. The Fox derivatives of x and y^{-1} are respectively 1 and $-Y^{-1}$. They give (E59.9) without a change of coefficient convention.

Write a polynomial in t to degree two. The first coordinate of a fixed vector has the form $A_0(s) + A_2(s)t^2$, while the second has the form $B_1(s)(t - t^2)$. The boundary equation is $sA_0 = 0$ and $sA_2 + B_1 = 0$. It follows that $A_0 = As^2$ and $A_2 = -(B + Cs + Ds^2)$. This proves the complete basis (E59.10). The leading Fox coordinates of x^3 and $[y,[x,y]]$ are respectively $(s^2, 0)$ and $(-t^2, st)$. Thus the first two fixed coordinates are exactly the two coefficients in (E59.5).

Here are all the cyclic-module calculations in (E59.11). For $B \neq 0$, the five displayed vectors have independent lowest homogeneous terms. The ideal I^3 and the polynomial $As^2 + Bt^2$ annihilate w; their quotient in R has dimension five. For $B = 0, A \neq 0$, $s + (C/A)t^2$ annihilates w, and w, tw, t^2w are independent. For $A = B = 0, C \neq 0$, I^2 annihilates w, and w, sw, tw are independent. The remaining two cases follow by multiplying s^2b by s and t. These identities prove the complete table, including all degenerate cases. The retained finite program verifies all eighty-one fixed vectors as a second certificate.

There is a universal relation as well. For a cycle (A, B), reducing $sA + tB = 0$ modulo I^2 shows $A, B \in I$. The group norm of $(\mathbb{Z}/3)^2$ is s^2t^2; it annihilates I. Thus the sum of all nine conjugates of every element of Φ/Φ_2 is zero. The rank-five assertion at seventeen is stronger than this one relation.

The finite arithmetic quotient and the density count. Let $G_0 = P/\Phi_2$. The exact Fox map embeds it as the following entire set of pairs:

$$\begin{aligned} G_0 &= \{(q, D) : q \in \langle X, Y\rangle, \quad D \in R^2, \quad (s,t)D = q - 1\}, \\ (q, D)(q', D') &= (qq', D + qD'). \end{aligned} \tag{E59.13b}$$

Indeed, the Fox identity gives the boundary condition. For each fixed q, a chosen word over q supplies one derivative, and (E59.8) supplies every difference in M. The kernel is exactly Φ_2. Thus $|G_0| = 3^{12}$. The characteristic subgroup Φ_2 is stable under c, so $G = G_0 \rtimes \langle c\rangle$ is an actual finite Galois quotient over $\mathbb{Q}$, unramified outside three.

The cyclotomic condition $\chi_9 = -1$ is precisely the subset $g = (Y^b, D, c)$, with $b = 0, 1, 2$ and $(s,t)D = Y^b - 1$. Its size is 3^{11}. Write $D = D_b + m$, where D_b is the Fox derivative of y^b and $m \in M$. The square g^2 is in M and is

$$T_b(D) = D + Y^b c(D), \qquad T_b(D_b) = 0, \qquad Y^b T_b(D) = (1 + c)Y^b m. \tag{E59.13c}$$

The last equality uses $Y^3 = 1$ and the actual semilinear involution (E59.9). Multiplication by Y^b is an invertible R-module map. Since two is invertible in $\mathbb{F}_3$, $(1 + c)Y^b : M \to M^c$ is

surjective, with kernel of dimension six. Every one of the eighty-one fixed vectors therefore has 3^7 preimages in the complete cyclotomic subset.

The five cases in (E59.11) give exactly $1, 2, 24, 54$ fixed vectors of ranks $0, 1, 3, 5$. Their corresponding finite Galois subsets have sizes $2187, 4374, 52488, 118098$. The rank is invariant under all conjugations in G: conjugation by P multiplies the cycle by an invertible deck element, and c acts semilinearly. For a rational prime in this cyclotomic class, g^2 is its actual arithmetic Frobenius over K. Chebotarev's theorem for the resulting conjugacy-stable subsets [Mil20b, VIII, Theorem 7.4] therefore gives (E59.13a), upon division by $|G| = 2 \cdot 3^{12}$. The retained program checks the ten-generator Schreier basis, all three maps in (E59.13c), and these exact finite counts.

An explicit character dual to the triple bracket. Let X act on K_1 by $z \mapsto z^4$, $r \mapsto r$, and let Y act by $z \mapsto z$, $r \mapsto \omega r$. These are exactly the restrictions of x, y. Direct calculation in $\mathbb{Q}[z]/(z^6 + z^3 + 1)$ gives

$$uu_1u_2 = 3, \qquad \gamma = u_1u_2^2, \qquad q = \frac{-2z^5 + 2z^4 - z^2 + z}{3}, \qquad \frac{3 - u^3}{3 - u_1^3}\,\gamma = 9q^3. \tag{E59.15}$$

The resultant of $3 - u^3$ and $z^6 + z^3 + 1$ is 3^{10}. Consequently each factor $r - u_i$ in (E59.3), and each of its Y-conjugates, is a unit away from three. In particular h defines a Kummer character of Φ unramified outside three.

Set $\phi = (r - u)/(r - u_1)$, so that $h = \phi/(Y^{-1}\phi)$. The exact norm identity (E59.15), with $r^6 = 9$, gives the literal equality

$$\frac{Y^{-1}h}{h} = \gamma \left(\frac{Y^{-1}\phi}{r^2 q} \right)^3. \tag{E59.16}$$

Also $X\gamma/\gamma = (u/r)^3$ and $Y\gamma = \gamma$. These are actual element identities. In the Kummer group they imply that h is killed by $(X - 1)^2, (X - 1)(Y - 1), (Y - 1)^2$: the first follows also from $N_X\phi = 1$, and the last from $N_Y h = 1$. Furthermore $c(u_i) = \omega u_i$, so $c(\phi) = Y^{-1}\phi$ and $c(h) = h^{-1}$.

Here is the exact filtration comparison behind the character calculation. The Fox module has dimension ten and I^2M has dimension five, by multiplying the explicit cycles $(-t, s), (s^2, 0), (0, t^2)$. We have $\Phi_2 \subset D_4$ and $I^2M \subset D_4/\Phi_2$. The five independent collected classes $[x, y], x^3, y^3, [x, [x, y]], [y, [x, y]]$ give $\dim \Phi/D_4 = 5$. Hence

$$D_4/\Phi_2 = I^2M. \tag{E59.17}$$

The Kummer character of h therefore factors through Φ/D_4. The sign $c(h) = h^{-1}$ combines with $c(\omega) = \omega^{-1}$ to make that character fixed under conjugation by c.

The element γ is not a cube in K_1. Indeed put $F = \mathbb{Q}(z)$. A cube root in $F(r)$ would be an eigenvector for Y, hence would have the form $r^k v$ with $v \in F$ and $k \in \{0, 1, 2\}$. It would follow that $\gamma = 3^k v^3$. Then $X\gamma/\gamma = u^3/3$ would be a cube in F, forcing 3 to be a cube there. But every intermediate field of the abelian extension $F/\mathbb{Q}$ is Galois over $\mathbb{Q}$, whereas the real pure cubic $\mathbb{Q}(\sqrt[3]{3})$ omits the two nonreal conjugates. Thus the extension $K_1(\delta)$, $\delta^3 = \gamma$, has lifts

$$X(\delta) = (u/r)\delta, \qquad Y(\delta) = \delta. \tag{E59.18}$$

Their commutator $[X, Y]$ sends δ to $\omega\delta$; their third powers are the identity. Thus the pairing of $[x, y]$ with γ is exactly one, with this commutator convention. For any Kummer character κ_h on Φ, conjugation gives

$$\kappa_h([y, g]) = \kappa_{(Y^{-1}h)/h}(g) \quad (g \in \Phi). \tag{E59.19}$$

Applying (E59.16) proves $\kappa_h([y,[x,y]]) = 1$.

The other required value is $\kappa_h(x^3) = 0$. At the chosen prime over two, r reduces to one, and the residue field in K_1 is $\mathbb{F}_{64} = \mathbb{F}_2[z]/(z^6+z^3+1)$. Substitution in (E59.3) gives $h = 1+z^3 = \omega^2$ there. Its cubic residue is $h^{21} = 1$. By comparison, the character of z on x^3 is one, and it vanishes on every commutator. Thus z, h are the exact dual characters to $x^3, [y,[x,y]]$ on the fixed part of D_3/D_4. The ordinary cubic-residue formula for arithmetic Frobenius now proves (E59.4)–(E59.5). Conjugation by P is trivial on D_3/D_4, proving independence of prolongation and of the associated roots.

At seventeen one can take $\bar{r} = 7$, $\bar{z} = 2+8\bar{\omega}$, where $\bar{\omega}^2+\bar{\omega}+1 = 0$. Then $\bar{u} = 6\bar{\omega}$, $\bar{u}_1 = 12\bar{\omega}$, $\bar{h} = 15+9\bar{\omega}$, and $\bar{h}^{96} = \bar{\omega}^2$. Together with $(17^2-1)/9 \equiv 2 \pmod 3$ this proves the explicit word in (E59.6).

The complete residue certificates. The retained PARI source reconstructs K_1 from z^6+z^3+1 and $r^3 = 3$, with the actual primitive element $\theta = r - z$. Its absolute polynomial is

$$\theta^{18} - 21\theta^{15} + 105\theta^{12} + 326\theta^9 + 3633\theta^6 + 3660\theta^3 + 2197. \tag{E59.20}$$

The unconditional certification procedure of [The26] returns one. It certifies class number one, discriminant -3^{39}, root group μ_{18}, the eight complete fundamental units, and the unique prime above three. The coefficient file supplies the literal polynomials for these units, a generator π_3 of that prime, z, r, the integral basis, and every prime ideal used in (E59.12). Thus the ten displayed Kummer coordinates are specified actual elements. Their classes form a basis because valuations outside three are divisible by three and the class group is trivial. The subgroup Φ is the maximal pro-three group unramified outside three over K_1: taking a normal closure of a finite three-extension over the finite three-extension K_1/K preserves this property.

At each of the nine degree-two primes over seventeen, reduction of these ten elements and raising to the ninety-sixth power gives (E59.12), relative to z^3. An independent finite-field implementation matches the prime-ideal Hermite lattices and verifies all ninety entries. It also checks thirty residue entries at two, the exact field actions, the low Kummer subspace and the Heisenberg sign. It independently evaluates (E59.4) for all fifty relevant primes through two thousand and for all nine prolongations, giving four hundred fifty comparisons. For $269, 809, 1187$ it evaluates all ten Kummer characters, giving another two hundred seventy residue values and the exact ranks $3, 1, 3$. These finite checks accompany the separate finite-Galois proof of the density theorem. The calculation evaluates the displayed quotient maps; it does not determine the full Frobenius word in P. □

All local sections of the evaluated elliptic quotient

E.60. Complete local coordinates and the exact integral class-three image at two [Proved]

Let R, M, ξ, η, U, V and the genuine Frobenius φ be those of E.58. Work over the actual local field $K_v = \mathbb{Q}_2(\zeta_3)$, with residue field $\mathbb{F}_4$, at the prime of $K = \mathbb{Q}(\zeta_3)$ above two. Use the same actual prolongation and discriminant-root convention. All geometric-conjugacy classes of local sections of R are the following explicit disjoint union:

$$\boxed{\coprod_{A,B\in\{0,1,2\}} \mathcal{L}_{A,B} \times ((\mathbb{Z}/3)^7 \oplus (\mathbb{Z}/27)^2),} \tag{E60.1}$$

where

$$\begin{gathered} h = (h_{ij})_{0\le i,j\le 2} \in \mathbb{Z}_3^9, \qquad h_0 = \sum h_{ij}, \quad I = \sum i h_{ij}, \quad J_0 = \sum j h_{ij}, \\ \mathcal{L}_{A,B} = \{h : I - Ah_0 \in 3\mathbb{Z}_3, \quad J_0 - Bh_0 \in 3\mathbb{Z}_3\}. \end{gathered} \tag{E60.2}$$

Each $\mathcal{L}_{A,B}$ is a specified index-nine lattice, free of rank nine. The equations and representatives below give the homeomorphism in (E60.1).

Let t be tame three-primary inertia and F arithmetic Frobenius, so $FtF^{-1} = t^4$. A representative section has values

$$\begin{aligned} \nu = v(t) = c^H, \qquad & H = \sum h_{ij} e_{ij} + uU + vV, \\ z = v(F) = \xi^A \eta^B c^K, \qquad & u = \frac{Ah_0 - I}{3}, \quad v = \frac{Bh_0 - J_0}{3}. \end{aligned} \tag{E60.3}$$

The allowed K are exactly the classes in $M/D_{A,B}M$, where

$$\begin{aligned} D_{A,B} e_{ij} &= -3e_{ij} + (4i - A)U + (4j - B)V, \\ D_{A,B} U &= 9U, \qquad D_{A,B} V = 9V. \end{aligned} \tag{E60.4}$$

This quotient has the finite coordinates in (E60.1) by the following actual maps. Put

$$p = (\overline{A+1}, B), \quad q = (A, \overline{B+1}), \quad C_{ij} = \begin{pmatrix} 4i - A \\ 4j - B \end{pmatrix}, \quad G = (C_p \; C_q) \in \mathrm{GL}_2(\mathbb{Z}_3), \tag{E60.5}$$

where the bars denote representatives in $\{0, 1, 2\}$. Write $K = \sum k_{ij} e_{ij} + k_U U + k_V V$. Its seven order-three coordinates are k_{ij} mod 3 for $(i, j) \neq p, q$. Its two order-twenty-seven coordinates are

$$\begin{pmatrix} k_p \\ k_q \end{pmatrix} + \sum_{(i,j) \neq p,q} k_{ij} G^{-1} C_{ij} + 3G^{-1} \begin{pmatrix} k_U \\ k_V \end{pmatrix} \pmod{27}. \tag{E60.6}$$

For the inverse, choose the seven residues in $\{0, 1, 2\}$, put $k_U = k_V = 0$, and choose k_p, k_q by subtracting the sum in (E60.6) from the two residues in $\{0, \ldots, 26\}$. These are three-adically integral representatives in M.

There is also a complete answer to the next central lifting question within this quotient. Put $R_2 = R/\gamma_3 R$, where $\gamma_3 R = 9\mathbb{Z}_3 Uc \oplus 9\mathbb{Z}_3 Vc$. For a local R_2 section, its inertia handle coordinates are zero. Write its remaining inertia coordinates as in (E60.3), with $u, v \in \mathbb{Z}_3/9$ and arbitrary representatives in $\mathbb{Z}_3$, and its Frobenius handle coordinates as $A, B \in \mathbb{Z}_3$. The lower tame relation is exactly

$$Ah_0 - I - 3u \in 9\mathbb{Z}_3, \qquad Bh_0 - J_0 - 3v \in 9\mathbb{Z}_3. \tag{E60.7}$$

The actual image of local section classes of R consists exactly of those lower classes satisfying the stronger tests

$$\boxed{Ah_0 - I - 3u \in 27\mathbb{Z}_3, \qquad Bh_0 - J_0 - 3v \in 27\mathbb{Z}_3.} \tag{E60.8}$$

The tests are independent of the representatives of u, v. Every nonempty fiber of the map of *full geometric-conjugacy classes* is a single point. For a fixed lower representative, the central lift classes modulo $\gamma_3 R$-conjugation form an eighty-one-element torsor $(\mathbb{Z}/9)^2$; the lower stabilizer acts simply transitively on it. This specifies the exact comparison between the two equivalence relations.

For instance the lower data $z = 1$, $\nu = c^{3U}$ define an R_2 section which fails (E60.8), whereas $z = 1$, $\nu = 1$ lifts. These two sections have identical rank-ten abelian data. All statements concern the actual quotient R of E.58; the intrinsic source still has the exact kernels in E.58.3. The manuscript anchor is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379), Part E(c),(d).

Proof. The punctured elliptic curve and its three-torsion punctures have good reduction at two, with the specified integral tangent. Its pro-three geometric action is unramified there. The induced action of Frobenius on $R^{\mathrm{ab}}/3$ is unipotent: it is the identity on the two handle and eight cusp diagonal blocks, with only the handle-to-cusp block remaining. The cube of this action is the identity. The kernel of the automorphism action on the Frattini quotient of a finitely generated pro-three group is pro-three, as follows from its characteristic exponent-three central series. Thus the closed Frobenius-action group is pro-three. Every cocycle consequently factors through the actual maximal pro-three tame quotient

$$\mathcal{D} = \langle t, F \mid FtF^{-1} = t^{4} \rangle_{3}. \tag{E60.9}$$

Here wild inertia is pro-two and contributes no map to the resulting pro-three semidirect product. The remaining tame components of prime-to-three order disappear in the same way. The displayed local quotient and its relation retain arithmetic Frobenius of residue cardinality four. Conversely any two values satisfying that relation define a continuous cocycle, by the presentation of $\mathcal{D}$.

In the coordinates $(a, b, H) = \xi^{a}\eta^{b}c^{H}$, an exact group law follows from the commutator module:

$$\begin{aligned}(a,b,H)(A,B,K) = (a+A,\, b+B,\, H+K-AbP \\ + [9Ab(A+1)/2 - A\operatorname{aug} H]U \\ + [9Ab(b+2B+1)/2 - B\operatorname{aug} H]V).\end{aligned} \tag{E60.10}$$

For example it follows by moving the commutator module past $x^{3A}y^{3B}$ and using $X^{3} = 1 + U$, $Y^{3} = 1 + V$, $U\mathfrak{m} = V\mathfrak{m} = 0$. All its denominators are units in $\mathbb{Z}_{3}$. This law also proves that the thirteen-coordinate parametrization is a homeomorphism with its product topology.

Solving the complete tame equation. The equation for the two cocycle values is

$$z\varphi(\nu)z^{-1} = \nu^{4}. \tag{E60.11}$$

On the two handle coordinates it reads $-2a = 4a$, $-2b = 4b$. They are torsion-free, so both vanish. Inertia therefore lies in the abelian module, and we can write it as in (E60.3). If $z = \xi^{A}\eta^{B}c^{K}$, then conjugation by z on the module is multiplication by $X^{3A}Y^{3B} = 1 + AU + BV$. Using E58.9 gives the exact error

$$z\varphi(\nu)z^{-1}\nu^{-4} = c^{4(Ah_{0} - I - 3u)U + 4(Bh_{0} - J_{0} - 3v)V}. \tag{E60.12}$$

This proves (E60.2)–(E60.3), including necessity and sufficiency. It puts no condition on the Frobenius module coordinate K.

Every geometric conjugation and the finite coordinates. A conjugating element with handle coordinates (s, t) sends the Frobenius handle coordinates to $(A + 3s, B + 3t)$, and the inertia central coordinates to $(u + sh_{0}, v + th_{0})$. Its nine coordinates h_{ij} are unchanged. There is therefore a unique choice of handle coordinates of a conjugator which puts A, B in $\{0, 1, 2\}$. After doing so, the remaining conjugators are precisely c^{L} with $L \in M$. They fix the inertia and send the Frobenius coordinate to

$$K + (1 - AU - BV)L - 4\Phi(L) = K + D_{A,B}L. \tag{E60.13}$$

This is (E60.4), with its sign and its full integral domain.

The determinant of $D_{A,B}$ is $(-3)^{9} \cdot 9^{2}$, so it has no kernel. To compute its cokernel with actual representatives, use the two columns in (E60.5). Modulo three they are the two nonzero coordinate-axis vectors, so G is invertible over $\mathbb{Z}_{3}$. The corresponding two relations express

$$\begin{pmatrix} U \\ V \end{pmatrix} = 3(G^{\mathsf{T}})^{-1} \begin{pmatrix} e_{p} \\ e_{q} \end{pmatrix}. \tag{E60.14}$$

Every other relation says that $e_{ij} - (G^{-1}C_{ij})_1 e_p - (G^{-1}C_{ij})_2 e_q$ has order dividing three. The two remaining relations $9U = 9V = 0$ are exactly $27e_p = 27e_q = 0$ after this invertible change. No further relations remain. This proves the complete invariant factors, (E60.6), and its explicit inverse. In particular the quotient has order 3^{13}.

For completeness the rank-nine inertia lattice also has explicit free coordinates. With p, q as in (E60.5), put $\alpha_A = \overline{A+1} - A$ and $\alpha_B = \overline{B+1} - B$, each either 1 or -2. Choose the seven h_{ij} away from p, q freely, and choose $r, s \in \mathbb{Z}_3$. Set

$$\begin{aligned} h_p &= \alpha_A^{-1}\left(3r - \sum_{(i,j)\neq p,q} (i-A)h_{ij}\right), \\ h_q &= \alpha_B^{-1}\left(3s - \sum_{(i,j)\neq p,q} (j-B)h_{ij}\right). \end{aligned} \tag{E60.15}$$

These solve both congruences and are unique for their free coordinates. All extraction, division on the closed principal ideals, residue choices and finite-coordinate maps are continuous. The explicit construction and its inverse therefore prove the homeomorphism (E60.1).

The exact class-three image and stabilizer action. In R_2, the coordinates u, v are taken modulo nine. The handle equations still force inertia handles to zero, and (E60.12) becomes exactly (E60.7). Changing an inertia lift by $c^{9rU+9sV}$ changes the error by $c^{-108rU-108sV}$, since Frobenius acts as -8 on U, V and the tame exponent is four. Thus the error can be killed precisely when its two parentheses lie in $27\mathbb{Z}_3$. Their quotients by 27 give the unique correction of the inertia central coordinates. The Frobenius central coordinates have no further relation, which proves existence as well as necessity in (E60.8).

The central module $\gamma_3 R$ has Frobenius -8 and trivial inertia. An additive cocycle into it has inertia value killed by $-8 - 4 = -12$, hence zero, and an arbitrary Frobenius value. Central conjugation changes the latter by multiplication by $1 - (-8) = 9$. The framed lift classes consequently form $(\gamma_3 R)/9(\gamma_3 R) \simeq (\mathbb{Z}/9)^2$.

The lower quotient has a central subgroup generated by the images of c^U, c^V, of order 9^2. They are fixed by Frobenius because $-8 \equiv 1 \pmod 9$, and are fixed by every twist of the lower section because they are central. They are therefore actual stabilizers of every lower section. They form its entire lower stabilizer: a fixed Frobenius value first forces $3s = 3t = 0$ on the torsion-free handle coordinates of a conjugator, and then (E60.13) forces $-3l_{ij} = 0$ on each of its nine free module coordinates. Only the two central coordinates modulo nine remain. Conjugating a lift by c^{rU+sV} changes its Frobenius coordinate by

$$9rU + 9sV. \tag{E60.16}$$

Modulo $9\gamma_3 R = 81\mathbb{Z}_3 U + 81\mathbb{Z}_3 V$, this is a bijection from that lower stabilizer to the framed lift torsor. Hence all framed lifts give the same full geometric-conjugacy class, proving the single-point fiber assertion. Finally $\nu = c^{3U}$, $z = 1$ has error c^{-36U}, which is zero in R_2 and is not removable by a multiple of $108U$. It gives the stated explicit obstruction.

The exact program checks all nine local Smith systems, their displayed finite coordinate maps with 108 full coordinate round trips, 54 tame sections, 108 exact conjugation and normalization identities, 486 lower representatives and their lift tests, and all 81 stabilizer values. The proofs above apply to all three-adic inputs, with these calculations checking the integral constants and the actual maps. □

Every ramified local section of the full cubic pro-three group

E.61. The nine full local cusp families and the abelian inertia image of metabelian sections [Proved]

Let $K = \mathbb{Q}(\zeta_3)$, let C be the smooth cubic $X^3 + Y^3 + Z^3 = 0$, and remove its nine points on the coordinate axes to obtain U. Take $(1 : -1 : 0)$ as origin on C. Use the marked [3]-cover identification of E.50 and E.58, so that

$$\Pi = \pi_1(U_{\overline{K}})^{(3)} = N \quad \text{is free pro-three of rank ten.} \tag{E61.1}$$

Fix a finite place $v \nmid 3$, put $q = |k_v|$ and $h = v_3(q-1) \geq 1$, and use an integral nonzero tangent at the base cusp. The resulting genuine geometric action of G_{K_v} on Π is unramified. Fix compatible roots defining a tame three-primary inertia generator t and arithmetic Frobenius F. For each of the nine cusps, choose an integral tangent and an inertia-invariant geometric path from the base tangent. Write $I_i = \langle d_i \rangle \simeq \mathbb{Z}_3(1)$ for its oriented geometric inertia and $F_i = z_i F$ for its actual tangential Frobenius lift. In particular, $F_i d_i F_i^{-1} = d_i^q$.

The continuous sections with nontrivial geometric inertia, modulo conjugation by Π, form exactly the following space:

$$\coprod_{i=1}^{9} (\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1)\mathbb{Z}_3. \tag{E61.2}$$

The actual representative attached to $(i, a, \bar{b})$ is

$$t \longmapsto d_i^a t, \qquad F \longmapsto d_i^b F_i. \tag{E61.3}$$

Changing b by $(1-q)k$ is precisely conjugation by d_i^k. The cusp and the nonzero exponent a are unchanged by any geometric conjugation between representatives. Formula (E61.2) is a homeomorphism for the quotient topology on continuous cocycles and the indicated three-adic and finite topologies.

Put $M = \Pi/\Pi''$, with its induced genuine arithmetic action, and $V = \Pi^{\mathrm{ab}}$. The image of the continuous map taking a local M-section class to its abelian inertia is exactly

$$\mathrm{im}\big(\mathrm{Sect}(M \rtimes G_{K_v})/M \longrightarrow V\big) = \bigcup_{i=1}^{9} \mathbb{Z}_3[d_i]. \tag{E61.14}$$

The map uses the fixed tame generator t above; geometric conjugation acts trivially on its value in V. Each vector on the right is realized by a full cuspidal section, including the zero vector. This statement computes the abelian inertia image and retains the further commutator and Frobenius fibers.

At $K_v = \mathbb{Q}_2(\zeta_3)$, let $\Pi \twoheadrightarrow R$ be the actual arithmetic-equivariant quotient of E.58. Number the cusps by $(i, j) \in \{0, 1, 2\}^2$ so that their images are $c^{e_{ij}}$. In the normalized local coordinates of E.60, the image of every full ramified section has

$$(A, B) = (i, j), \qquad (h_{kl}) = a e_{ij}, \qquad u = v = 0, \quad a \neq 0. \tag{E61.4}$$

The continuous R-section

$$\nu = c^{e_{00}+e_{01}+V}, \qquad z = \eta^2 \tag{E61.5}$$

has no lift to a continuous section with geometric kernel M, and hence has no full Π-lift, even after full geometric conjugation. The map $\Pi \twoheadrightarrow R$ factors through $M \twoheadrightarrow R$, since the

commutator subgroup of R lies in its displayed abelian module. Its inertia has an actual word preimage $c(ycy^{-1})[y^3, c]$. The obstruction is already detected by its image $e_{00} + e_{01}$ in

$$\bigoplus_{i,j} \mathbb{F}_3 e_{ij}/\mathbb{F}_3 \sum_{i,j} e_{ij} \;\hookrightarrow\; \Pi^{\mathrm{ab}}/3. \tag{E61.6}$$

In the associated finite-cover proof the detecting cover has degree 19683, has 59049 cusps, and its compactification has genus 59050. These integers and all twenty-seven possible cuspidal-line comparisons in (E61.6) are checked by the retained program. This settles the entire ramified local source at these good places and gives an explicit reverse-lifting failure for the computed quotient. The global and unramified source questions remain at their stated maps in E.23. The manuscript anchor is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379), Part E(c),(d).

Proof. The actual specialization and local source. The Fermat equation is smooth away from three: its three partial derivatives have no common projective zero there. The nine cusp sections are disjoint since the third roots of unity remain distinct. With the displayed origin they are $C[3]$; the identity over K extends over the good integral model because [3] is etale there and all nine sections remain distinct. Thus the special fibre has genus one, nine rational punctures, and rational three-torsion.

Apply [GR71, XIII, Corollary 2.9 and construction 2.10] to this smooth proper model with its disjoint cusp divisor, using finite three-groups. The specialization functor identifies every finite three-group torsor on the two geometric fibres, compatibly with its fibre functor and the cusp inertia [GR71, XIII, Lemma 2.11]. It induces a continuous isomorphism of the two geometric pro-three groups. The integral tangential fibre functor makes this isomorphism equivariant for $G_{K_v} \twoheadrightarrow G_{k_v}$ and makes the based action unramified. We therefore work with the actual special-fibre arithmetic extension

$$\Pi \rtimes G_{k_v}, \qquad G_{k_v} = \widehat{\mathbb{Z}}. \tag{E61.7}$$

The genus-one, nine-puncture presentation follows as well from [GR71, XIII, Corollary 2.12]; eliminating any one cusp generator gives ten free pro-three generators.

On $\Pi^{\mathrm{ab}}/3$, Frobenius preserves the exact sequence whose subspace is the eight-dimensional cusp space and whose quotient is $C[3]$. It acts as the identity on each, so its matrix is unipotent. The kernel of $\mathrm{Aut}(\Pi) \to \mathrm{Aut}(\Pi^{\mathrm{ab}}/3)$ is pro-three, as seen from the characteristic exponent-three central series. Hence the closed Frobenius action group is pro-three. Every cocycle together with this action is a homomorphism into a pro-three semidirect product and therefore factors through $G^{(3)}_{K_v}$.

The latter quotient is the tame group

$$D = \langle t, F \mid FtF^{-1} = t^q \rangle_{(3)}. \tag{E61.8}$$

Indeed wild inertia is pro-residue-characteristic, its image in a pro-three group is trivial, and the prime-to-three parts of tame inertia and of the unramified quotient have trivial image. The usual tame relation survives with this exact q; it defines the semidirect product $\mathbb{Z}_3 \rtimes \mathbb{Z}_3$ because $q \in 1 + 3\mathbb{Z}_3$. The same argument supplies the universal map in the reverse direction from the tame quotient to every such pro-three image. Thus a pair $(\nu, z) \in \Pi^2$ defines a continuous section exactly when

$$zF(\nu)z^{-1} = \nu^q. \tag{E61.9}$$

Recognition of the nonzero inertia in the full group. We give the pro-three, finite-field form of the argument in [Sti12, Theorem 30], including the cover and its comparison map. The cited

paper works with full profinite groups in characteristic zero; the following argument verifies the precise variant used here.

Every cusp group I_i is a free pro-three factor of Π. Any two cusp generators occur simultaneously in a free basis, by eliminating a third cusp in the surface presentation. Theorem B' of [HR85] applies to the extension- closed class of finite three-groups: a nontrivial element of a free factor has its centralizer in that factor, and the factor has trivial intersection with any different conjugate of itself. Different cusp factors have disjoint nonzero conjugates as well: an equality $d_i^a = g d_j^b g^{-1}$ in Π gives $a[d_i] = b[d_j]$ in the free integral abelianization, whose two cusp vectors are independent. This forces $a = b = 0$ for $i \neq j$. These facts also show that every procyclic subgroup meeting a cusp group nontrivially is contained in that group: its generator centralizes the nonzero intersection. Every nonzero procyclic subgroup is isomorphic to $\mathbb{Z}_3$, since the free pro-three group is torsion-free (Theorem A' of the same source applied to its free pro-three cyclic factors).

Take a solution of (E61.9) with $\nu \neq 1$, and put $I = \overline{\langle \nu \rangle} \simeq \mathbb{Z}_3$ and $\psi = zF$. Then ψ normalizes I by its q-power action. Let W be the union of all conjugates of all nine cusp groups. It is compact, being the image of $\Pi \times \coprod_i I_i$ under conjugation. Suppose I is contained in no cusp group. The preceding centralizer calculation gives $I \cap W = \{1\}$. The compact set $I \setminus I^3$ is disjoint from W. Compact separation in a profinite group supplies an open normal subgroup for which their images are disjoint. Intersect its finitely many automorphic images to obtain a characteristic open subgroup N_0 with the same property; there are finitely many subgroups of any fixed index because Π is finitely generated. In particular, $I \cap N_0 \subset I^3$.

The subgroup $H = N_0 I$ is open and ψ-stable. Its exact quotient map

$$H \longrightarrow I/(I \cap N_0) \longrightarrow I/I^3 \simeq \mathbb{F}_3 \tag{E61.10}$$

is nonzero on I and kills every cusp subgroup of H. For otherwise some element of $W \cap H$ would have, modulo N_0, a representative in $I \setminus I^3$, contrary to the choice of N_0.

The continuous map $\widehat{\mathbb{Z}} \to \Pi \rtimes \widehat{\mathbb{Z}}$ taking 1 to ψ is a section: its composition with the projection is the identity. Consequently $H \rtimes \langle \psi \rangle$ is an open arithmetic subgroup and defines an actual geometrically connected finite etale cover $U' \to U_{k_v}$ over k_v. Its geometric group is H. More explicitly, any finite three-cover of its geometric fibre has a three-group Galois closure over $U_{\overline{k_v}}$, by the finite wreath-product construction; thus taking pro-three completion loses none of the covering subgroups of H. Let C' be the smooth compactification of U'. Filling the cusp inertia gives the continuous surjection

$$H \twoheadrightarrow \pi_1(C'_{\overline{k_v}})^{(3)} \twoheadrightarrow T_3 \operatorname{Jac}(C'). \tag{E61.11}$$

The last arrow is abelianization, with its canonical identification from [Mil08, III, Proposition 9.1]. It is independent of the Abel–Jacobi base point after abelianization and is Frobenius-equivariant. The kernel of the first arrow is precisely the closed normal subgroup generated by the puncture inertia, since an etale cover of U' extends etale to C' exactly when its inertia is trivial. Hence (E61.10) factors through (E61.11). The image of I in the free $\mathbb{Z}_3$-module $T_3 \operatorname{Jac}(C')$ is nonzero modulo three.

Conjugation by ψ acts as arithmetic Frobenius on this Tate module and by multiplication by q on that nonzero image. Thus q would be an eigenvalue. The actual Frobenius eigenvalues on the Tate module of an abelian variety over $\mathbb{F}_q$ have complex absolute value $\sqrt{q}$ [Mil08, II, Theorem 1.1(b), Lemmas 1.2–1.3]. Since $q > 1$, the asserted eigenvalue is impossible. This proves that I lies in a unique conjugate of a cusp group. Notice that the weight argument has been applied to the finite-field Jacobian in (E61.11), with its constructed Frobenius action.

Evaluation of the full ramified section space. Conjugate the unique containing cusp group to the fixed I_i. Then $\nu = d_i^a$ for one nonzero $a \in \mathbb{Z}_3$. The element ψ preserves I_i, since it

preserves its nonzero subgroup I. Both ψ and the chosen F_i normalize I_i and project to F. Their ratio lies in the geometric normalizer, which is I_i. It follows that $\psi = d_i^b F_i$ for a unique $b \in \mathbb{Z}_3$. Conversely these elements satisfy (E61.9) directly, and (E61.8) proves continuity and existence of the entire section.

A geometric conjugation between two such representatives must take the unique containing cusp group to the other one. The disjointness already proved forces the same cusp and places the conjugator in I_i. Conjugation by d_i^k fixes a and sends b to $b + (1-q)k$. This proves (E61.2)–(E61.3), including every equivalence relation. Continuity follows from three-adic exponentiation. For the inverse, the primitive integral cusp vectors in Π^{ab} determine the cusp and the nonzero exponent continuously. On each clopen set $v_3(a) = r$, the parameter space is compact and its continuous bijection to the corresponding Hausdorff conjugacy quotient is a homeomorphism. These sets exhaust the nonzero-inertia space; their clopen property is detected by the same abelianization. This proves the topological assertion.

The exact abelian inertia image of metabelian sections. Let a continuous M-section have tame coordinates $(\bar\nu, \bar z)$, and let w be the image of $\bar\nu$ in V. Its abelian tame relation is $F(w) = qw$. We prove that w lies on one of the nine integral cusp lines. Suppose it does not. The compact primitive annulus $\mathbb{Z}_3 w \setminus 3\mathbb{Z}_3 w$ is disjoint from $\bigcup_i \mathbb{Z}_3[d_i]$. Indeed each $[d_i]$ is primitive in the free lattice V; a nonzero intersection of its rational line with $\mathbb{Q}_3 w$ would put w on its integral line. Choose n so that these compact sets remain disjoint in $V/3^nV$. This also gives $\mathbb{Z}_3 w \cap 3^n V \subset 3\mathbb{Z}_3 w$.

Let H be the inverse image of $3^nV + \mathbb{Z}_3 w$ under $\Pi \twoheadrightarrow V$. It is open, contains Π', and is preserved by zF for any lift $z \in \Pi$ of $\bar z$. The exact quotient character

$$H \longrightarrow (3^nV + \mathbb{Z}_3 w)/3^nV \longrightarrow \mathbb{Z}_3 w/3\mathbb{Z}_3 w \tag{E61.15}$$

detects the image of $\bar\nu$ and kills every cusp group of H, by the chosen separation. Because $\Pi' \subset H$, we have the actual closed-subgroup inclusion $\Pi'' \subset [H,H]$. There is therefore a continuous quotient map

$$H/\Pi'' \longrightarrow H^{\mathrm{ab}} \longrightarrow T_3 \operatorname{Jac}(C'), \tag{E61.16}$$

where C' is the compactification of the finite-field cover defined by $H \rtimes \langle zF \rangle$ exactly as in (E61.11). The relation of the proposed section holds modulo Π'', so becomes the exact eigenvalue-q relation in (E61.16). Character (E61.15) factors through this same Tate module and detects a nonzero image modulo three. The finite-field Weil bound again excludes that eigenvalue. This proves the inclusion in (E61.14). Every vector on its right occurs in (E61.3), including $a = 0$ for the zero vector, which proves equality with the stated image.

The actual quotient comparison and its failed lift. The images of the marked cusp words $x^i y^j c y^{-j} x^{-i}$ in R are $c^{e_{ij}}$. The full tangential Frobenius at this cusp has handle coordinates (i,j) in the normalization of E.60: in the ambient two-handle abelianization, Frobenius is $-2I$, so $pF(p)^{-1}$ for $p = x^i y^j$ has coordinates $(3i, 3j)$. Inner conjugation in $\Pi = N$ changes these normalized coordinates by multiples of three. The nonzero cusp inertia becomes exactly $c^{ae_{ij}}$, proving (E61.4). The additional finite Frobenius coordinates depend on the chosen actual paths and tangents. This statement does not assign them from a reference automorphism of the ambient free group: the remaining inner parameter described in E.58 can change those origins while acting trivially on R.

For (E61.5) the moment data are

$$A = 0, \quad B = 2, \quad h_0 = 2, \quad I = 0, \quad J_0 = 1, \quad u = 0, \quad v = 1. \tag{E61.12}$$

The exact E.60 defect is $4[(Ah_0 - I - 3u)U + (Bh_0 - J_0 - 3v)V] = 0$. The retained program also evaluates the actual eleven-module Frobenius and the group law, obtaining $z\varphi(\nu)z^{-1}\nu^{-4} = 1$. It evaluates the displayed word preimage using the same group law. Thus this is a continuous

section of the actual quotient. Its image under the isomorphism $R^{\mathrm{ab}} \simeq \Pi^{\mathrm{ab}}$ of E.58 has cusp vector $e_{00} + e_{01}$. For every cusp label k and every $a \in \mathbb{F}_3$, the vector $e_{00} + e_{01} - ae_k$ has unequal coordinates and so cannot belong to the diagonal line. This checks all twenty-seven possibilities in (E61.6). Geometric conjugation acts trivially on this abelianization. The exact metabelian image (E61.14) therefore forbids every metabelian lift of (E61.5), and the full classification gives the same obstruction for Π.

There is a finite geometric cover detecting exactly this obstruction. Put $\Phi\Pi = \Pi^3[\Pi, \Pi]$ and let w be the nonzero vector (E61.6). The inverse image H of the line $\langle w \rangle$ in $\Pi/\Phi\Pi \simeq \mathbb{F}_3^{10}$ has index $3^9 = 19683$. Each cusp vector is outside this line, so each cusp has ramification index three in the cover; the number of lifted cusps is $9 \cdot 3^8 = 59049$. The punctured Euler characteristic is multiplied by 3^9, and hence the compact genus is

$$1 + \frac{9 \cdot 3^9 - 59049}{2} = 59050. \tag{E61.13}$$

For a proposed metabelian lift, choose any representative $z \in \Pi$ of its Frobenius coordinate. Its zF stabilizes this exact H, by (E61.9). The quotient character $H \to H/\Phi\Pi = \langle w \rangle$ kills all its cusps and detects its inertia. Here $\Pi' \subset \Phi\Pi \subset H$ gives $\Pi'' \subset [H, H]$, so its relation descends along the literal map (E61.16). Thus (E61.11) gives the same impossible Frobenius eigenvalue four on the compact Jacobian of this specific cover. The finite certificate verifies the line separation, the cover indices and Euler characteristic, and the cusp parameter formulas modulo $3, 9, 27, 81$. □

A positional class which lifts once and fails at the next norm

E.62. The complete local norm tower at seventeen and an actual second-step obstruction [Proved]

Use the genuine arithmetic quotient of E.38 and E.46,

$$\begin{gathered} \Lambda = \mathbb{Z}_3[[T-1]], \quad J = (3, T-1)\Lambda, \quad H = 3\mathbb{Z}_3 \ltimes J, \\ (a, v)(b, w) = (a + b, v + T^a w), \qquad g(a, v) = \big(\chi(g)a, \chi(g)v(T^{\chi(g)})\big). \end{gathered} \tag{E62.1}$$

For $n \geq 1$ put

$$\begin{gathered} \Lambda_n = \Lambda/(T^{3^n} - 1), \qquad J_n = \ker(\Lambda_n \xrightarrow{\mathrm{aug}} \mathbb{F}_3), \\ H_n = H/\big(0 \times (T^{3^n} - 1)\Lambda\big) = 3\mathbb{Z}_3 \ltimes J_n. \end{gathered} \tag{E62.2}$$

Fix the actual first cocycle $A = 3\kappa_3(17)$. Under the Kummer–Shapiro comparison of E.46, the positional algebras and their transition maps are exactly

$$\begin{gathered} R_n = \mathbb{Q}[X]/(X^{3^n} - 17^3) = F_{n-1} \times L_{n-1}, \\ F_d = \mathbb{Q}(\alpha_d), \quad \alpha_d^{3^d} = 17, \qquad L_d = K(\beta_d), \quad \beta_d^{3^d} = 17\zeta_3, \quad K = \mathbb{Q}(\zeta_3), \\ \alpha_{d+1}^3 = \alpha_d, \quad \beta_{d+1}^3 = \beta_d, \qquad \alpha_0 = 17, \quad \beta_0 = 17\zeta_3. \end{gathered} \tag{E62.3}$$

The transition on section coordinates is the actual norm in these two field factors.

There is a global section class of H_1 over the fixed first cocycle whose positional Kummer coordinates are

$$(1, \zeta_3) \in \mathbb{Q}^{\times, \wedge_3} \times K^{\times, \wedge_3}. \tag{E62.4}$$

It lifts to a section class of H_2. It has no lift to a section class of H_3, and hence no lift to a section class of H. This failure is already detected by a finite residue-field quotient at seventeen.

The entire local section tower over $\mathbb{Q}_{17}$ can be evaluated. Choose the primitive ninth root $\iota \in K_{17} = \mathbb{Q}_{17}(\zeta_3)$ whose residue is $2 + 8\bar{\zeta}_3$; thus $\iota^3 = \zeta_3$. Put

$$\mathcal{L} = \{(e, f) \in \mathbb{Z}_3^2 : e + 2f \in 3\mathbb{Z}_3\}. \tag{E62.5}$$

For every $n \geq 1$, the full local fiber of section conjugacy classes of H_n over $[A]$ is

$$\mathcal{L} \oplus \mathbb{Z}/9, \qquad (e, f, t) \longleftrightarrow (\alpha_{n-1}^e, \beta_{n-1}^f \iota^t). \tag{E62.6}$$

Here the displayed field elements denote their completions for multiplication by 3^m. The transition is precisely

$$(e, f, t) \longmapsto (e, f, 3t). \tag{E62.7}$$

Consequently the full local fiber for H is the lattice $\mathcal{L}$, and its projection to the first positional level is

$$(e, f) \longmapsto (17^e, (17\zeta_3)^f). \tag{E62.8}$$

These are homeomorphisms for the natural continuous-cohomology and three-adic topologies. The common toric quotient of E.46 is $(a, v) \mapsto (a/3, \mathrm{aug}(v)/3)$. On these local classes it is the explicit map

$$(e, f) \longmapsto \left(\kappa_3(17), \frac{e + 2f}{3}\kappa_3(17)\right). \tag{E62.9}$$

It is onto the entire second local toric coordinate. Every such toric fiber is a copy of $\mathbb{Z}_3$, parametrized by f, with $e = 3b - 2f$ for second coordinate $b\kappa_3(17)$.

The global class (E62.4) restricts to $(e, f, t) = (0, 0, 3)$ in (E62.6). Its first lifts have $t = 1, 4, 7$; none has a further lift. Its total norm to $\mathbb{Q}$ is one, so its second toric class is zero. The positional class $(1, 1)$ has the same toric pair and does extend: the literal global section is $g \mapsto (A(g), 0, g)$ of $H \rtimes G_{\mathbb{Q}}$. Thus the failed positional lift and the surjective toric map are related by the actual maps (E62.6)–(E62.9). The manuscript anchor is §51.16, folios 733v–736v, scans 172–178, formulas (371)–(379), Part E(c),(d).

Proof. The exact section coordinates. For a section with first cocycle A, its second coordinate is a cocycle for the actual module action

$$g \cdot v = \chi(g) T^{A(g)} v(T^{\chi(g)}). \tag{E62.10}$$

This action preserves J_n. The Kummer–Shapiro map of E.46 identifies $H^1(G_{\mathbb{Q}}, \Lambda_{A,n})$ with $R_n^{\times,\wedge_3}$. The augmentation is the total norm, and the transition is induced by root cubing, hence by the field norms in (E62.3). The two polynomial factors there are $X^{3^{n-1}} - 17$ and its complementary factor. The first is Eisenstein at seventeen; the complementary factor over K is $X^{3^{n-1}} - 17\zeta_3$, also Eisenstein at the unique prime over seventeen. These facts prove all degrees and the stated field embeddings.

Both globally and over $\mathbb{Q}_{17}$, $H^0(\mathbb{F}_3(1)) = 0$. The exact sequence $0 \to J_{A,n} \to \Lambda_{A,n} \to \mathbb{F}_3(1) \to 0$ therefore identifies $H^1(J_{A,n})$ with the kernel of the total norm modulo cubes. Moreover $H^0(\mathbb{Z}_3(1)) = 0$, so fixing a representative A of its first class leaves no first-coordinate stabilizer. Thus these are the full section conjugacy classes over $[A]$, with the actual geometric conjugations accounted for.

The literal first lift. Use the field L_1 and elements of E.57:

$$\beta_1^3 = 17\zeta_3, \qquad \pi = 2 - \zeta_3, \qquad g_1 = -2 - 3\zeta_3 - \zeta_3\beta_1, \qquad w = (g_1/\pi)^2. \tag{E62.11}$$

The actual norm is $N_{L_1/K}(w) = \zeta_3$. Hence $(1, w) \in R_2^{\times,\wedge_3}$ maps to $(1, \zeta_3)$. The total norm of each is one, so the preceding exact sequence places both in the asserted J-cohomology classes.

This proves the global first lift with an explicit element. The identity pair at every positional level is represented by the zero second cocycle. Since A is a genuine $\mathbb{Z}_3(1)$-cocycle, the group law in (E62.1) verifies directly that $g \mapsto (A(g), 0, g)$ is a full section. Its common toric pair is $(\kappa_3(17), 0)$, which is also the toric pair of (E62.4).

The local unit norm and the obstruction. The prime seventeen is inert in K. Its completion has residue field $\mathbb{F}_{289}$, whose unit group has order $288 = 32 \cdot 9$. The local extensions

$$F_{d,17} = \mathbb{Q}_{17}(\alpha_d), \qquad L_{d,17} = K_{17}(\beta_d) \tag{E62.12}$$

are totally ramified of degree 3^d, by their Eisenstein polynomials. They have residue fields $\mathbb{F}_{17}$ and $\mathbb{F}_{289}$, respectively. The norm of a unit in a totally ramified extension of degree 3^d reduces to the 3^dth power of its residue. This follows either by taking the determinant of multiplication on the local integral algebra and reducing its length filtration, or by multiplying its conjugates in a normal closure: all residue embeddings are the identity and occur with ramification multiplicity 3^d.

For $d = 2$ the residue norm image is the ninth-power subgroup of order thirty-two. The residue of ζ_3 has order three, so it is not in this subgroup; explicitly $\bar{\zeta}_3^{32} = \bar{\zeta}_3^{2} \neq 1$. Any element whose norm has valuation zero must itself have valuation zero, since the residue degree is one. It follows that ζ_3 is not a norm from $L_{2,17}$. The same obstruction holds for the completed multiplicative groups: it factors through the finite quotient $\mathbb{F}_{289}^{\times}/(\mathbb{F}_{289}^{\times})^9$. Thus it also forbids a norm equal to the completed class of ζ_3. The complex component of the actual map $R_3 \to R_1$ is $N_{L_2/K}$. Local restriction of any purported lift would contradict this calculation, proving the global failure at H_3.

One can see the first lift in the same finite field. With $\bar{\omega}^2 + \bar{\omega} + 1 = 0$, the element (E62.11) has residue

$$\bar{w} = 2 + 8\bar{\omega}, \qquad \bar{w}^3 = \bar{\omega}, \qquad \bar{w}^{96} = \bar{\omega}^2. \tag{E62.13}$$

Thus it is a primitive ninth root whose cubic residue is nontrivial. Every possible first preimage of ζ_3 encounters the same second-step obstruction, since the composite norm has ninth-power residue image.

Evaluation of every local level and the limit. Principal units in a finite extension of $\mathbb{Q}_{17}$ form a pro-seventeen group. Raising to 3^m is an automorphism on that group, by Hensel's lemma, so it contributes zero to the completion for multiplication by 3^m. The Teichmuller units in $F_{d,17}$ have order sixteen and also contribute zero. The three-primary Teichmuller units in $L_{d,17}$ are exactly the same $\mu_9 \subset K_{17}$ at every level: a totally ramified extension adds no unramified roots of order prime to seventeen. Consequently the actual coordinate maps are

$$\begin{aligned} F_{d,17}^{\times,\wedge_3} &\xrightarrow{\sim} \mathbb{Z}_3, & \alpha_d^e &\longleftrightarrow e, \\ L_{d,17}^{\times,\wedge_3} &\xrightarrow{\sim} \mathbb{Z}_3 \oplus \mathbb{Z}/9, & \beta_d^f \iota^t &\longleftrightarrow (f,t). \end{aligned} \tag{E62.14}$$

The uniformizer norms are $N(\alpha_{d+1}) = \alpha_d$ and $N(\beta_{d+1}) = \beta_d$, from their odd-degree monic polynomials. The norm of the common root ι is ι^3. This proves the transition $(e, f, t) \mapsto (e, f, 3t)$. The inverse limit of $\mathbb{Z}/9 \xleftarrow{3} \mathbb{Z}/9 \xleftarrow{3} \cdots$ is zero, since any coordinate is divisible by nine after two further transitions. The other two coordinates are unchanged.

The total local norm of the two uniformizer coordinates is 17^{e+2f}. The ninth root has trivial norm to $\mathbb{Q}_{17}$, since the unramified quadratic norm on it raises it to $17 + 1 = 18$. The condition for belonging to $H^1(J_{A,n})$ is therefore exactly $e + 2f \in 3\mathbb{Z}_3$. This proves (E62.6)–(E62.8). The continuous inverse-limit comparison is the actual cocycle-alignment map of E.46: the positional module maps are surjective, and every positional Tate module has zero invariant subgroup. Compatible classes have compatible cocycle representatives, uniquely up to a compatible

bounding element. All the groups in (E62.14) are compact, and the displayed coordinate maps and their inverses are continuous. This also proves the stated topology.

Finally the toric map divides both raw coordinates by three. Its second class is the Kummer class of 17^{e+2f} divided by three, giving (E62.9). The integral map $(b, f) \mapsto (3b - 2f, f)$ is a bijection from $\mathbb{Z}_3^2$ to $\mathcal{L}$, with inverse $((e + 2f)/3, f)$. This explicitly evaluates every toric fiber and proves its surjectivity. The finite-field certificate checks all residue norm images used above, the literal element (E62.13), the torsion transitions, and the lattice coordinates modulo $3, 9, 27, 81$. □

Every adjacent norm in the complex seventeen tower

E.64. Explicit global norm images, completed Hilbert 90, and the next fixed class group [Proved]

Keep the actual first cocycle $A = 3\kappa_3(17)$ of E.62. Put $K = \mathbb{Q}(\zeta)$, $\zeta = \zeta_3$, and choose

$$L_d = K(\beta_d), \qquad \beta_0 = 17\zeta, \qquad \beta_{d+1}^3 = \beta_d \quad (d \geq 0). \tag{E64.1}$$

Thus $L_0 = K$ and $\beta_d^{3^d} = 17\zeta$. Write $\widehat{L_{d\,3}^{\times}} = \varprojlim_m L_d^{\times}/(L_d^{\times})^{3^m}$, with its prodiscrete topology. The following formulas evaluate the image and every nonempty fiber of each actual adjacent norm

$$N_d : \widehat{L_{d+1\,3}^{\times}} \longrightarrow \widehat{L_{d\,3}^{\times}}. \tag{E64.2}$$

There is exactly one prime $\mathfrak{t}_d$ over 17 and exactly one prime $\mathfrak{l}_d$ over 3 in L_d. Their residue fields are $\mathbb{F}_{289}$ and $\mathbb{F}_3$, respectively. The extension L_{d+1}/L_d is cyclic of degree three, totally ramified at these two primes and unramified elsewhere. Its specified generator is $\sigma_d(\beta_{d+1}) = \zeta\beta_{d+1}$. For a prime $\mathfrak{q} \nmid 3 \cdot 17$ of L_d, define $\gamma_{d,\mathfrak{q}} \in \mathbb{F}_3$ by the actual finite-field equation

$$\overline{\beta_d}^{(\mathrm{N}\mathfrak{q}-1)/3} = \overline{\zeta}^{\gamma_{d,\mathfrak{q}}}, \qquad \mathcal{I}_d = \{\mathfrak{q} \nmid 3 \cdot 17 : \gamma_{d,\mathfrak{q}} \neq 0\}. \tag{E64.3}$$

Precisely the primes in $\mathcal{I}_d$ are inert in L_{d+1}/L_d; every other prime away from $3, 17$ splits completely.

For $z \in L_d^{\times}$, let $v_{\mathfrak{t}_d}(\beta_d) = 1$ and set

$$\begin{aligned} \delta_{d,17}(z) &= \log_{\overline{\zeta}}\left(\overline{z\beta_d^{-v_{\mathfrak{t}_d}(z)}}^{96}\right) \in \mathbb{F}_3, \\ \delta_{d,\mathfrak{q}}(z) &= \gamma_{d,\mathfrak{q}} v_{\mathfrak{q}}(z) \pmod 3 \quad (\mathfrak{q} \in \mathcal{I}_d), \\ \Theta_d(z) &= (\delta_{d,17}(z), (\delta_{d,\mathfrak{q}}(z))_{\mathfrak{q}\in\mathcal{I}_d}). \end{aligned} \tag{E64.4}$$

The bar in the first line is reduction of the displayed unit at $\mathfrak{t}_d$. Every coordinate kills cubes, so the same map on the completion is evaluated on any representative of its class modulo cubes. The sequence of coordinates away from 17 has finite support. There is an exact sequence of topological abelian groups

$$\widehat{L_{d+1\,3}^{\times}} \xrightarrow{N_d} \widehat{L_{d\,3}^{\times}} \xrightarrow{\Theta_d} \mathbb{F}_3 \oplus \bigoplus_{\mathfrak{q}\in\mathcal{I}_d} \mathbb{F}_3 \longrightarrow 0. \tag{E64.5}$$

The right-hand group has the discrete topology. In particular the complete global norm test uses only the explicit residue computations and valuation congruences in (E64.4); its image is an open subgroup. The omitted local norm condition at the single prime over 3 follows from global reciprocity, as proved below.

Both exact completed comparisons are

$$\begin{aligned} \ker N_d &= (\sigma_d - 1)\widehat{L^{\times}_{d+1\,3}}, \\ (\widehat{L^{\times}_{d+1\,3}})^{\sigma_d} &= \widehat{L^{\times}_{d\,3}} \quad \text{under the actual field inclusion}, \\ \widehat{L^{\times}_{d+1\,3}}/\widehat{L^{\times}_{d\,3}} &\xrightarrow{\sim} \ker N_d, \qquad u \longmapsto \sigma_d(u)/u. \end{aligned} \tag{E64.6}$$

These are equalities and a topological isomorphism, rather than a replacement of ordinary multiplicative groups by their completions without an exactness proof. For completeness, each nonempty fiber of (E64.2) has the explicit noncanonical topological form

$$\mu_3 \times \left\{ (a_i)_{i\ge 1} \in \prod_{i\ge 1} \mathbb{Z}_3 : \text{for every } m,\ a_i \in 3^m\mathbb{Z}_3 \text{ for all but finitely many } i \right\}. \tag{E64.7}$$

Choose a lift y_0 of the target and a free basis u_i of the ordinary norm-one group modulo its roots of unity; the map is $(\zeta^e, (a_i)) \mapsto y_0\zeta^e \prod_i u_i^{a_i}$. The proof gives the precise basis choices and explains convergence.

At the first degree-eighteen extension there is also the exact class-group calculation

$$\left|\mathrm{Cl}(L_2)^{\mathrm{Gal}(L_2/L_1)}\right| = 27, \qquad E_{L_1} \cap N_{L_2/L_1} L_2^{\times} = E_{L_1}, \tag{E64.8}$$

where E_{L_i} denotes the ordinary unit group. The actual comparison with unit norms is the exact sequence

$$1 \longrightarrow \mathrm{Am}_{\mathrm{st}}(L_2/L_1) \longrightarrow \mathrm{Cl}(L_2)^{\mathrm{Gal}(L_2/L_1)} \longrightarrow E_{L_1}/N_{L_2/L_1}E_{L_2} \longrightarrow 1. \tag{E64.9}$$

Thus (E64.8) proves element-norm surjectivity on base units. Formula (E64.9) retains the additional quotient needed to determine which of those norms have an ordinary unit as preimage.

Under the exact Kummer–Shapiro map E.46, (E64.2) is the complex-factor component of the actual positional transition

$$\widehat{F^{\times}_{d+1\,3}} \times \widehat{L^{\times}_{d+1\,3}} \quad \xrightarrow{(N,N_d)} \quad \widehat{F^{\times}_{d\,3}} \times \widehat{L^{\times}_{d\,3}}, \qquad F_d = \mathbb{Q}(17^{1/3^d}) \tag{E64.10}$$

from level $d+2$ to level $d+1$ of E.62. Total norms to $\mathbb{Q}$ commute with this arrow; consequently, once both component preimages have been chosen, the target's actual J-condition modulo cubes also holds for the source. The next global question is the existence and evaluation of compatible choices through all these arrows, together with the pure-factor norm conditions. An assertion that membership in each individual norm image already supplies compatible choices is not used. The manuscript anchor remains §51.16, scans172–178, folios733v–736v, equations(371)–(379), and Part E(c),(d).

Proof. Fields and the two ramified primes. The polynomial $X^{3^d} - 17\zeta$ is Eisenstein at the prime of K over 17. It proves $[L_d : K] = 3^d$, uniqueness of $\mathfrak{t}_d$, and the assertions about its uniformizer and residue field. For $d \ge 1$ the absolute minimal polynomial is

$$P_d(X) = X^{2\cdot 3^d} + 17X^{3^d} + 289. \tag{E64.11}$$

Modulo three it is $(X+1)^{2\cdot 3^d}$, whereas $P_d(-1) = 273 = 3 \cdot 91$. Hence $P_d(Y-1)$ is Eisenstein at three. It proves that L_d has exactly one prime over three, with absolute ramification index $2\cdot 3^d$ and residue degree one. The statement for $d = 0$ is the elementary decomposition in $\mathbb{Q}(\zeta)$.

These indices show that both primes are ramified of degree three in each adjacent extension. Since $\zeta \in L_d$, adjoining β_{d+1} gives the specified cyclic extension and automorphism. The relative polynomial $X^3 - \beta_d$ has discriminant $-27\beta_d^2$; away from the two displayed primes it defines an etale algebra. There is no infinite ramification because L_d is totally imaginary. This proves the ramification assertions.

The exact local tests and their global sufficiency. At $\mathfrak{q} \nmid 3 \cdot 17$, the extension is unramified. The polynomial $X^3 - \bar{\beta}_d$ either splits into three distinct linear factors or is irreducible, since the residue field contains μ_3. Its splitting is exactly the zero condition in (E64.3). For an unramified cubic extension the unit norm is onto and the norm of a uniformizer has valuation three. One may prove the unit assertion by first taking the surjective finite-field norm and then the surjective trace on each successive principal-unit quotient. Thus the local norm condition is precisely $\delta_{d,\mathfrak{q}} = 0$ at inert primes.

At $\mathfrak{t}_d$ the extension is tamely and totally ramified of degree three. Its norm on residue units is cubing. Cubing is an automorphism of the principal-unit pro-seventeen group, and the norm of an element of the base field is its cube. These facts prove sufficiency as well as necessity of the residue-unit test. The literal identity

$$N_d(\beta_{d+1}) = \beta_d \tag{E64.12}$$

allows every valuation, and leaves exactly the first line of (E64.4).

Now let $z \in L_d^\times$ have all displayed coordinates zero. It is a norm at every place other than possibly $\mathfrak{t}_d$. Global reciprocity makes the product of its local Artin symbols one. Since this is the only remaining place, its Artin symbol there is also one. The local reciprocity kernel is the local norm group. The Hasse norm theorem for a cyclic extension then gives $z = N_d(y)$ for an actual $y \in L_{d+1}^\times$. Here the exact theorem is [Mil20b, VIII, Theorem3.1], including the proof by the injection $L_d^\times / N_d L_{d+1}^\times \to \bigoplus_v L_{d,v}^\times / N_d L_{d+1,v}^\times$. Necessity follows from the local calculations.

We also need surjectivity of the entire last map of (E64.5), rather than just the zero criterion. For a cyclic extension the relative Brauer exact sequence in [Mil20b, VIII, §4, the exact sequence preceding Lemma4.1 and its proof; IV, Theorem3.14] identifies the global norm quotient with the tuples of local norm classes having total invariant zero. All nonzero local norm quotients here have order three. Arbitrary finitely supported prescribed classes outside $\mathfrak{t}_d$ can be supplemented by the unique class at $\mathfrak{t}_d$ whose invariant makes their sum zero. This is permitted because its local degree is three. The relative exact sequence supplies a global class, and the cyclic crossed-product description represents it by an element of $L_d^\times$. Each coordinate in (E64.4) is an explicit isomorphism from the corresponding local norm quotient to $\mathbb{F}_3$; a possible change of sign of a local invariant is an invertible change of that one coordinate. This proves the asserted surjectivity with exactly the displayed coordinates.

Exactness of completion, with both kernels checked. We give the required abelian-group argument in additive notation. If

$$0 \longrightarrow C \longrightarrow B \longrightarrow D \longrightarrow 0 \tag{E64.13}$$

is exact and the three-primary torsion of D is killed by 3^r, then

$$C \cap 3^{n+r}B \subseteq 3^n C \subseteq C \cap 3^n B. \tag{E64.14}$$

Indeed $c = 3^{n+r}b \in C$ implies that the image of b in D is killed by 3^{n+r}, hence by 3^r; therefore $3^r b \in C$. The kernels $C/(C \cap 3^n B)$ in the quotient exact sequences have surjective transition maps. Taking inverse limits is consequently exact, and (E64.14) identifies the induced topology with completion for multiplication by 3^n. This proves exactness of the completed version of (E64.13), including its injective left-hand map.

Apply this first to

$$0 \longrightarrow K_d \longrightarrow L_{d+1}^{\times} \xrightarrow{N_d} I_d^{\mathrm{norm}} \longrightarrow 0, \quad K_d = \ker(N_d : L_{d+1}^{\times} \to L_d^{\times}), \quad I_d^{\mathrm{norm}} = N_d L_{d+1}^{\times}. \tag{E64.15}$$

The torsion of I_d^{norm} is contained in the finite roots of unity of L_d, so (E64.14) applies. To compare its completion with the target completion, note separately that $3L_d^{\times} \subset I_d^{\mathrm{norm}}$ in additive notation, because the norm of a base element is its cube. Consequently

$$I_d^{\mathrm{norm}} \cap 3^{n+1} L_d^{\times} \subseteq 3^n I_d^{\mathrm{norm}}, \qquad 3^n I_d^{\mathrm{norm}} \subseteq I_d^{\mathrm{norm}} \cap 3^n L_d^{\times}. \tag{E64.16}$$

Its induced and intrinsic filtrations are cofinal. The exact quotient sequences then identify $\widehat{I}_d^{\mathrm{norm}}$ with the kernel of the completed map to $L_d^{\times}/I_d^{\mathrm{norm}}$. This quotient has exponent three and is exactly the discrete group evaluated above. It proves (E64.5) on completions and identifies the completed norm kernel with $\widehat{K}_d$.

For the second direction use ordinary Hilbert90, in its exact form [Mil20b, II, Proposition1.22 and Corollary1.23]:

$$0 \longrightarrow L_d^{\times} \longrightarrow L_{d+1}^{\times} \xrightarrow{\sigma_d - 1} K_d \longrightarrow 0. \tag{E64.17}$$

The torsion of K_d is contained in the finite roots of unity of L_{d+1}, so (E64.14) applies to this different exact sequence too. Its completed sequence identifies the kernel of $\sigma_d - 1$ with the actual included $\widehat{L_{d\,3}^{\times}}$, and its image with $\widehat{K}_d$. The latter has already been identified with the norm kernel by (E64.15)–(E64.16). This proves every assertion in (E64.6). The quotient maps are open: on the intrinsic power filtrations their images of 3^n times the completed source are 3^n times the completed image. Together with the cofinal inclusions this proves the claimed topological statements.

There is also an explicit way to produce a completed norm preimage once the test has passed. Represent the target a modulo cubes by $a_0 \in L_d^{\times}$. The established ordinary criterion supplies y_0 with $N_d y_0 = a_0$. For example, an enumeration of the elements of the explicit number field, followed by exact norm computation, finds such a y_0 and terminates by the theorem. Write $a/a_0 = u^3$ in $\widehat{L_{d\,3}^{\times}}$. This division exists: using units and a basis of the principal-divisor group identifies the multiplicative completion with a finite root-of-unity group and restricted three-adic sequences; its kernel modulo cubes is exactly its subgroup of cubes. Then $y_0 u$, with u included from the base field, is the required preimage. This construction takes no inverse limit over noncompact nonempty fibers.

The fibers as explicit topological groups. The three-primary roots of unity in every L_d are exactly μ_3. Indeed, for any cyclotomic three-power extension C/K, the polynomial $X^{3^d} - 17\zeta$ remains Eisenstein at a prime over 17, because C/K is unramified there. Its degree remains 3^d, which proves that L_d and C are linearly disjoint over K. In particular $\zeta_9 \notin L_d$. Since $N_d(\zeta) = 1$, the three-primary torsion of K_d is exactly μ_3.

The quotient of a number field's multiplicative group by its roots of unity is free abelian of countable rank: use the ordinary unit group and the subgroup of the free divisor group consisting of principal divisors. Hence K_d modulo its torsion is free abelian of at most countable rank. Its rank is infinite. For infinitely many primes of L_d split in L_{d+1}, choose a prime $\mathfrak{P}$ above each and an element v with $(v) = \mathfrak{P}^h$, where h is the class number of L_{d+1}. The elements $\sigma_d(v)/v$ have nonzero divisors $h(\sigma_d\mathfrak{P} - \mathfrak{P})$ on disjoint prime orbits; they are independent. Infinitude of split primes follows from the proved Chebotarev theorem [Mil20b, VIII, Theorem7.4]. Choose a basis of this free norm-one group modulo its torsion and lifts $u_i \in K_d$. Since the quotient is free, these choices split its ordinary torsion subgroup. Hilbert90 gives actual v_i with $u_i = \sigma_d(v_i)/v_i$. Completing this direct sum gives exactly (E64.7): the restrictedness condition ensures that the displayed product converges modulo every power

of three and that its coordinates are unique. The infinite quotients in this completion are discrete and need not be finite; the resulting global group is not being treated as compact.

The fixed class group in the second complex layer. E.57 certified $\mathrm{Cl}(L_1) \simeq C_3^2$ and the fundamental units

$$\begin{aligned}\epsilon_1 &= -18 + 3\beta_1 + 3\beta_1^2 - 18\beta_1^3/17 - 5\beta_1^4/17 + 2\beta_1^5/17,\\ \epsilon_2 &= -18 - 5\beta_1 + 2\beta_1^2 - \beta_1^3/17 - 8\beta_1^4/17 - \beta_1^5/17.\end{aligned} \tag{E64.18}$$

At the unique prime over 17, their residues are both $-1 - \bar{\zeta} = \bar{\zeta}^2$. The roots of unity are generated by -1 and ζ; each of these residues is a cube in $\mathbb{F}_{289}^{\times}$. The unit valuations at all unramified primes vanish. The norm criterion therefore proves that every ordinary unit of L_1 is the norm of an element of $L_2^{\times}$. There are exactly two ramified finite primes and no ramified infinite primes. The ambiguous class-number formula [Lem13, Theorem1, formula(1)] now gives

$$|\mathrm{Cl}(L_2)^{\mathrm{Gal}(L_2/L_1)}| = 9\,\frac{3^{2-1}}{[E_{L_1} : E_{L_1} \cap N_d L_2^{\times}]} = 27. \tag{E64.19}$$

The same source's Proposition1 gives (E64.9). Its actual map sends an ambiguous ideal class represented by $\mathfrak{a}$, with $\sigma_d\mathfrak{a} = (\alpha)\mathfrak{a}$, to $N_d\alpha$ modulo norms of units. This formula specifies exactly the additional unit-norm comparison. No full class-group or fundamental-unit calculation in the degree-eighteen field is needed for (E64.19).

The retained certificate verifies the exact prime decompositions and field inclusions at degrees two, six and eighteen, certifies the degree-six class group and units, and compares (E64.3) with actual prime contraction at all 58 primes tested. Its independent Python part checks 1,508 finite-field norm identities, 124 characteristic-zero norm identities and six Eisenstein expansions. The theorem's statements for every d follow from the proofs above; finite tests are accompanied by the exact field and norm maps. □

Every adjacent norm in the pure seventeen tower

E.65. Pure norm images, an explicit wild character, and the certified first unit norm [Proved]

Let

$$F_d = \mathbb{Q}(\alpha_d), \qquad \alpha_0 = 17, \qquad \alpha_{d+1}^3 = \alpha_d, \qquad M_d = F_d(\zeta), \quad \zeta = \zeta_3, \tag{E65.1}$$

where the α_d are positive real roots. Complex conjugation c fixes F_d and sends ζ to ζ^{-1}. Write $\widehat{F_d^{\times}}_3 = \varprojlim_m F_d^{\times}/(F_d^{\times})^{3^m}$, with the multiplication-completion topology. The following formulas evaluate the image and every fiber of the actual adjacent norm

$$N_d^{\mathrm{p}} : \widehat{F_{d+1}^{\times}}_3 \longrightarrow \widehat{F_d^{\times}}_3. \tag{E65.2}$$

For a prime $v \nmid 3 \cdot 17$ of F_d, let $q_v = \mathrm{N}v$. When $q_v \equiv 1 \pmod 3$, choose a primitive third root $\zeta_v \in \mathbb{F}_{q_v}$, and define

$$\overline{\alpha_d}^{(q_v-1)/3} = \zeta_v^{\gamma_{d,v}}, \qquad \mathcal{I}_d^{\mathrm{p}} = \{v \nmid 3 \cdot 17 : q_v \equiv 1 \pmod 3,\ \gamma_{d,v} \neq 0\}. \tag{E65.3}$$

The choice of ζ_v specifies one of the two primes of M_d over v. Changing that choice multiplies this one coordinate by -1. For $v \in \mathcal{I}_d^{\mathrm{p}}$ set $\delta_{d,v}(z) = \gamma_{d,v}v(z) \pmod 3$.

For $d \geq 1$ there is one further, explicitly computable character $\delta_{d,3} : F_d^{\times} \to \mathbb{F}_3$. Its finite matrix definition below uses the completion of F_d at its unique prime of ramification index

$2 \cdot 3^{d-1}$ over three. There is no such coordinate for $d = 0$. With that convention there is an exact sequence of topological groups

$$\widehat{F^{\times}_{d+1\,3}} \xrightarrow{N^{\mathrm{p}}_d} \widehat{F^{\times}_{d\ 3}} \xrightarrow{\Theta^{\mathrm{p}}_d} \begin{cases} \bigoplus_{v \in \mathcal{I}^{\mathrm{p}}_0} \mathbb{F}_3, & d = 0, \\ \mathbb{F}_3 \oplus \bigoplus_{v \in \mathcal{I}^{\mathrm{p}}_d} \mathbb{F}_3, & d \geq 1, \end{cases} \longrightarrow 0, \tag{E65.4}$$

where $\Theta^{\mathrm{p}}_d(z)$ consists of $\delta_{d,3}(z)$, when present, and the $\delta_{d,v}(z)$. The right-hand side is discrete. Every coordinate kills cubes, so the map on the completion is evaluated on any actual representative modulo cubes. Its prime coordinates have finite support. Thus (E65.4) is a sufficient as well as necessary global norm test, and its norm image is open.

Here is the finite definition of the wild character, including all arithmetic choices. There is a unique $\rho \in \mathbb{Q}_3$ with $\rho^3 = 17$; it satisfies $\rho \equiv 23 \pmod{27}$. For $d \geq 1$ put

$$m = 3^{d-1}, \qquad e = 2m, \qquad n = 2e + 1, \qquad Q_d(Z) = (Z-1)^{2m} + \rho(Z-1)^m + \rho^2. \tag{E65.5}$$

At the specified prime, $\pi = \alpha_d + 1$ is a uniformizer, $Q_d(\pi) = 0$, and its integer ring is $\mathbb{Z}_3[\pi]$. Its quotient modulo π^n is exactly the finite ring

$$R_d = (\mathbb{Z}/27)[Z]/(Q_d(Z) \bmod 27,\ Z^n). \tag{E65.6}$$

Let $g_i = 1 + \pi^i$, $1 \leq i < n$. Every unit of R_d has a unique expression $\epsilon \prod_{i=1}^{n-1} g_i^{a_i}$ with $\epsilon \in \{1, -1\}$ and $a_i \in \{0, 1, 2\}$. Write $\mathbf{a}(u) = (a_1, \ldots, a_{n-1})$. In the finite free cubic algebra

$$R'_d = R_d[\Pi]/(\Pi^3 - 3\Pi^2 + 3\Pi - \pi), \tag{E65.7}$$

compute norms as determinants of multiplication in the basis $(1, \Pi, \Pi^2)$. Form the matrix over $\mathbb{F}_3$ whose rows are

$$\mathbf{a}(g_i^3) \quad (1 \leq i < n), \qquad \mathbf{a}\big(\det(1 + \Pi^j \mid R'_d)\big) \quad (1 \leq j < 3n). \tag{E65.8}$$

Its nullspace has dimension one. Let $\mathbf{w}_d$ be its unique generator whose first nonzero coordinate is 1. The specified wild character is

$$\delta_{d,3}(z) = \mathbf{w}_d \cdot \mathbf{a}\big(z\pi^{-v_3(z)} \bmod \pi^n\big) \quad \text{in } \mathbb{F}_3, \tag{E65.9}$$

where $v_3(\pi) = 1$ at this one prime. The notation $\det(1 + \Pi^j \mid R'_d)$ in (E65.8) means the determinant of multiplication by that element, not a determinant of a scalar matrix. Formulas (E65.5)–(E65.9) give a finite algorithm for every d, using only binomial coefficients, integral lattice reduction and 3-by-3 determinants.

Each nonempty fiber of (E65.2) is a torsor under the group

$$\mathcal{R}_3 = \left\{ (a_i)_{i \geq 1} \in \prod_{i \geq 1} \mathbb{Z}_3 : \text{for every } r,\ a_i \in 3^r\mathbb{Z}_3 \text{ for all but finitely many } i \right\}, \tag{E65.10}$$

with its multiplication-completion topology. Explicitly, choose an actual free basis u_i of the ordinary norm-one group and a lift y_0 of the target. The isomorphism is $(a_i) \mapsto y_0 \prod_i u_i^{a_i}$. In particular the ordinary norm-one group has no torsion and is free abelian of countably infinite rank; (E65.10) retains its infinite discrete quotients.

The first wild matrix and its certified unit witness are

$$\mathbf{w}_1 = (1, 2, 0, 0), \qquad \varepsilon = 7\alpha_1 - 18, \qquad \mathbf{a}(\varepsilon) = (2, 0, 2, 2), \qquad \delta_{1,3}(\varepsilon) = 2. \tag{E65.11}$$

Thus ε is not in $N_{F_2/F_1}(F_2^\times)$, nor in the image of its completed norm. Both F_1 and F_2 have class number one. The following actual fundamental units of F_2, written with $x = \alpha_2$, have norm row $(0,3,0,0)$ in the fundamental-unit basis ε of F_1:

$$\begin{aligned} u_1 &= (20 + 18x - 5x^2 - 8x^3 + 5x^5 + 2x^6 - 3x^7 - 2x^8)/3, \\ u_2 &= -18 + 7x^3, \\ u_3 &= -1 - 4x + 6x^2 - 6x^4 + 2x^5 - 2x^7 + 2x^8, \\ u_4 &= \tfrac{1}{3}(-235 + 148x + 182x^2 - 50x^3 - 97x^4 \\ &\qquad - 5x^5 + 53x^6 + 16x^7 - 25x^8), \\ (Nu_1, Nu_2, Nu_3, Nu_4) &= (1, \varepsilon^3, -1, 1). \end{aligned} \tag{E65.12}$$

Consequently the ordinary unit norm image is $\langle -1, \varepsilon^3 \rangle$, of index three, and its kernel is free on $u_1, -u_3, u_4$. This specifies the preimages and signs in addition to the index.

Under E.46, the actual positional transition of E.62 is the product of (E65.2) and (E64.2):

$$\widehat{F^\times_{d+1\,3}} \times \widehat{L^\times_{d+1\,3}} \xrightarrow{(N_d^{\mathrm{p}}, N_d)} \widehat{F^\times_{d\ 3}} \times \widehat{L^\times_{d\ 3}}. \tag{E65.13}$$

Its total norm to $\mathbb{Q}$ is preserved. Therefore the $J = (3, T-1)\mathbb{Z}_3[[T-1]]$ condition from E.46 is preserved by every preimage supplied by these two norm tests. The two tests and their fiber coordinates evaluate every adjacent positional arrow. The compatible inverse limit of these arrows is retained as the further global section problem; an independent selection of preimages at each finite level is not asserted to be a compatible selection. The manuscript anchor is §51.16, scans172–178, folios733v–736v, equations(371)–(379), and Part E(c),(d).

Proof. The exact descent map. The polynomial $X^{3^d} - 17$ is Eisenstein at seventeen. Hence $[F_d : \mathbb{Q}] = 3^d$, $[M_d : F_d] = 2$, and M_{d+1}/M_d is cyclic of degree three, generated by $\sigma(\alpha_{d+1}) = \zeta\alpha_{d+1}$. The identity $c\sigma c = \sigma^{-1}$ holds on both generators. Let $z \in F_d^\times$ and let $b \in M_{d+1}^\times$ satisfy $N_{M_{d+1}/M_d} b = z$. Then

$$w = \frac{z}{b\,c(b)} \in F_{d+1}^\times, \qquad N_{F_{d+1}/F_d} w = \frac{z^3}{z^2} = z. \tag{E65.14}$$

Conversely, an actual pure norm remains a norm after adjoining ζ. This proves the equivalence of the two norm tests by an explicit preimage map.

The same formulas hold in the multiplication completions. For clarity, the actual field inclusion gives

$$(\widehat{M_d^\times}_{\,3})^c = \widehat{F_d^\times}_{\,3}, \qquad v \longmapsto N_{M_d/F_d}(v)^{1/2} \tag{E65.15}$$

as inverse on the fixed subgroup. Indeed the inclusion followed by norm is squaring, which is an automorphism of a pro-3 multiplication completion; for a fixed v, inclusion of its norm is v^2. This proves injectivity, surjectivity onto the stated fixed subgroup, and continuity of both maps. The corresponding assertion for norm quotients also holds before completion. Given a c-fixed class $[a]$ in the cyclic norm quotient, whose exponent divides three, the actual element $(a\,c(a))^2 \in F_d^\times$ has class $4[a] = [a]$. Injectivity follows from (E65.14).

Conjugation on the local norm coordinates. Identify the cyclic norm quotient with the relative Brauer group by the crossed product $a \mapsto (M_{d+1}/M_d, \sigma, a)$. The actual conjugation map sends this algebra to $(M_{d+1}/M_d, \sigma^{-1}, c(a))$, which is the inverse Brauer class of $(M_{d+1}/M_d, \sigma, c(a))$. Thus, when the generator σ is kept fixed, c acts on local norm coordinates by negative permutation of the places. The cyclic crossed-product identification, the exact relative Brauer sequence and the Hasse norm theorem are the opened results [Mil20b, IV, Theorem3.14; VIII,

§4, the exact sequence preceding Lemma4.1 and its proof; VIII, Theorem3.1]. They identify the global cyclic norm quotient with the direct sum of its nontrivial local norm groups, subject to the one sum-of-invariants relation.

All these nontrivial groups have order three. On a place fixed by c, a c-fixed tuple satisfies $a = -a$, hence has value zero. On a pair of conjugate places its values are $a, -a$; their sum is zero. It follows that the fixed quotient is exactly one independent $\mathbb{F}_3$ for every conjugate pair at which the cyclic local extension is nontrivial. Every finite assignment to those pairs is realized: its sum of invariants is zero, the relative Brauer sequence supplies a global cyclic norm class, and the averaging formula after (E65.15) supplies an actual representative in $F_d^\times$. This proves both the global sufficiency and the cokernel surjectivity that will be used in (E65.4).

Identifying every relevant local pair. Away from three and seventeen, the relative polynomial $X^3 - \alpha_d$ is etale. If $q_v \equiv 2 \pmod 3$, adjoining ζ gives the unramified quadratic completion, fixed by c. Cubing is a bijection on $\mathbb{F}_{q_v}^\times$, so α_d already has a cube root there. The pure local algebra is the product of the base field and an unramified quadratic field. Its norm is onto because of the first factor. If $q_v \equiv 1 \pmod 3$, the quadratic algebra splits into two conjugate copies. On either copy, (E65.3) is the Kummer splitting test for the unramified cubic extension. A nonzero symbol gives an unramified cubic field, whose unit norm is onto and whose uniformizer norm has valuation three. A zero symbol gives three copies of the base field. This yields exactly the coordinates $\gamma_{d,v} v(z)$ in (E65.4).

At seventeen, $X^{3^d} - 17$ is Eisenstein both over $\mathbb{Q}_{17}$ and over its unramified quadratic extension K_{17}. There is one prime in M_d over seventeen, fixed by c. Directly in the pure fields, the relative extension is tamely and totally ramified of degree three, its residue field is $\mathbb{F}_{17}$, and $N(\alpha_{d+1}) = \alpha_d$. Cubing is a bijection on $\mathbb{F}_{17}^\times$ and on the principal-unit pro-seventeen group. Hence all local valuations and units are norms; this gives the same absence of a seventeen coordinate by an explicit local map. Infinite places contribute no 3-primary norm quotient.

For three, Hensel's lemma applied at 23 gives a root ρ of $X^3 - 17$ with $\rho \equiv 23 \pmod{27}$: $v_3(23^3 - 17) = 5 > 2v_3(3 \cdot 23^2) = 2$. There is only one such root in $\mathbb{Q}_3$, because this field has no primitive third root of unity. For $d \geq 1$ one has the exact factorization

$$X^{3^d} - 17 = (X^m - \rho)(X^{2m} + \rho X^m + \rho^2), \quad m = 3^{d-1}. \tag{E65.16}$$

After $X = Z - 1$, both factors are Eisenstein at three. Their reductions are Z^m and Z^{2m}; their constant terms are $-1 - \rho$ and $1 - \rho + \rho^2$, each of valuation one. Consequently F_d has precisely two primes over three, with ramification indices m and $2m$, both with residue field $\mathbb{F}_3$. The first completion has odd degree and therefore does not contain K_3. The second contains it, since α_d^m/ρ is a primitive third root of unity. After adjoining ζ, the first prime remains a single place fixed by c, while the second becomes a conjugate pair. Choose one of the latter pair by $\alpha_d^m = \rho\zeta$. At the next level the two degrees in (E65.16) are multiplied by three. On the chosen pair, adjoining α_{d+1} is the cyclic totally ramified cubic extension given by (E65.7). Thus there is exactly one nontrivial conjugate pair over three for every $d \geq 1$. At $d = 0$, the local algebra after adjoining ζ splits completely, because $17 = \rho^3$ in $\mathbb{Q}_3$; there is no pair contributing a norm condition. This proves the list of all coordinates in (E65.4).

A finite presentation of the wild local norm group. The Eisenstein polynomial Q_d proves that π is a uniformizer, $v_3(3) = e$, and the integer ring is $\mathbb{Z}_3[\pi]$. For example, the last assertion follows directly by writing an element in the basis $1, \pi, \ldots, \pi^{e-1}$: the valuations of its nonzero terms are distinct modulo e, so integrality forces all coefficients to be 3-adic integers. Since $3e \geq 2e + 1 = n$, reduction modulo π^n kills 27. It therefore depends only on ρ mod 27 and is exactly (E65.6). Each successive unit quotient $(1 + \pi^i\mathcal{O})/(1 + \pi^{i+1}\mathcal{O})$ is the additive $\mathbb{F}_3$ generated by $1 + \pi^i$. Successively removing the leading term gives the unique ordered unit coordinates used in (E65.8).

The group of principal units modulo π^n has a presentation with commuting generators g_i and the relations expressing g_i^3 in the generators of higher index. These relations reduce every word to exponents $0, 1, 2$, so the presented group has at most 3^{n-1} elements; the actual unit filtration has exactly that many. The presentation is therefore exact. An $\mathbb{F}_3$-valued character of the unit group is precisely a vector annihilating the first set of rows in (E65.8).

In the actual next completion, $\Pi = \alpha_{d+1} + 1$ is a uniformizer and its minimal polynomial over the base is the polynomial in (E65.7). It is Eisenstein at π and has norm π. The target quotient of its integer ring by π^n is exactly R'_d, and $\Pi^{3n} = 0$ there. Its units are generated by -1 and $1 + \Pi^j$ for $1 \le j < 3n$. Norms on this finite free ring are the displayed determinants. Units congruent to one modulo π^n in the source have norm congruent to one modulo π^n in the target, so these rows compute the whole unit norm image in the finite quotient.

Finally $1 + \pi^n\mathcal{O}$ is contained in the cube subgroup of the base field: for u in this group, Hensel's lemma applied to $X^3 - u$ at 1 uses $v(1-u) \ge 2e + 1 > 2v(3)$. A cube in the base is the norm of its base-field cube root, viewed in the cubic extension. Thus this deep unit group is already in the actual norm image. Local reciprocity gives index three for the norm of a cyclic cubic local extension, and $N\Pi = \pi$ removes the valuation quotient [Mil20b, III, Theorem3.4 and its norm-index proof]. The unit quotient computed by (E65.8) therefore has exactly order three. This proves that its character nullspace is one-dimensional, and proves necessity and sufficiency of the explicit character (E65.9) at the wild prime.

Completion and all norm fibers. Let $B = F_{d+1}^{\times}$, let $I = NB \subset F_d^{\times}$, and let $C = \ker(B \to I)$. In additive notation, $0 \to C \to B \to I \to 0$ has 3-torsion-free quotient I, because F_d is real and has only the roots of unity $\{1, -1\}$. Thus $C \cap 3^r B = 3^r C$ for every r. The exact finite-power quotient sequences have surjective transition maps on the kernel quotients. Their inverse limit is exact and identifies $\widehat{C}_3$ with the kernel of $\widehat{B}_3 \to \widehat{I}_3$. Separately, $3F_d^{\times} \subset I$, because the norm of an included base element is its cube. Hence

$$I \cap 3^{r+1}F_d^{\times} \subset 3^r I \subset I \cap 3^r F_d^{\times}. \tag{E65.17}$$

These cofinal filtrations embed $\widehat{I}_3$ as the actual kernel of the discrete exponent-three quotient in (E65.4). This proves the completed norm test and the claimed kernel without using compactness of the infinite multiplicative groups.

The group C has no torsion, since the norm of -1 in an odd-degree extension is -1. The quotient $F_{d+1}^{\times}/\{1, -1\}$ is free abelian of countable rank: its unit subgroup is finitely generated free, and its quotient is the group of principal divisors, a subgroup of the free abelian group on the finite primes. The latter is free, so the exact sequence splits as abelian groups. The subgroup C is therefore free abelian of at most countable rank. Its rank is infinite. Take infinitely many rational primes splitting completely in the actual normal closure $\mathbb{Q}(\mu_{3^{d+1}}, \alpha_{d+1})/\mathbb{Q}$, using [Mil20b, VIII, Theorem7.4]. This field contains $M_{d+1} = \mathbb{Q}(\zeta, \alpha_{d+1})$ under the specified root choices, so those primes split completely in F_{d+1} as well. For each choose distinct primes P, Q of F_{d+1} above the same prime of F_d. A class-number multiple of $P - Q$ is the divisor of an element y. These elements have disjoint prime supports and are independent. Their norms are units of F_d, a group of finite rank. The kernel of the resulting map from a countably infinite free abelian group to this finite-rank unit group has infinite rank and lies in C. Choose a free basis u_i of C. Completing $\bigoplus_i \mathbb{Z}u_i$ for multiplication by 3^r gives exactly (E65.10), with the displayed product formula. This also specifies the topology of each nonempty fiber.

Exact finite certificates and the positional comparison. The retained local program constructs (E65.6) by integral Hermite normal form and checks the index of every ideal π^k, $0 \le k \le n$. It computes all determinant norms in (E65.8), and independently checks character additivity with the actual ring multiplication. For $d = 1, 2, 3$ the nullspace is one-dimensional as proved

above; the first two normalized vectors are

$$\mathbf{w}_1 = (1,2,0,0), \qquad \mathbf{w}_2 = (1,2,1,1,2,2,2,2,0,0,0,0). \tag{E65.18}$$

The full degree-eighteen local vector, matrices and relations are retained in the output. Substitution of $\varepsilon = 7\pi - 25$ gives (E65.11). The official PARI program separately certifies the two number fields with `bnfcertify=1`, records their actual integer rings and prime ideals, and computes the relative norms in (E65.12) through the embedding $\alpha_1 \mapsto \alpha_2^3$. The Python verification recomputes the four norm determinants in characteristic zero. Thus the unit basis, the sign row, and the non-norm witness have independent exact certificates. The actual prime contractions away from $3, 17$ are also checked against (E65.3), including the linear–quadratic local algebras when $q_v \equiv 2 \pmod 3$.

Finally E.62 factors the positional algebra as $F_d \times L_d$. The root-cubing map acts on these factors by the actual relative field norms in E.46. Hence (E65.13) is that same positional arrow, with both components now evaluated. Norm transitivity gives

$$N_{F_d/\mathbb{Q}}(N_d^{\mathrm{p}} u)\, N_{L_d/\mathbb{Q}}(N_d v) = N_{F_{d+1}/\mathbb{Q}}(u)\, N_{L_{d+1}/\mathbb{Q}}(v). \tag{E65.19}$$

The congruence modulo cubes defining J is therefore preserved literally, as claimed. □

The full ramified local metabelian section space

E.66. Every ramified local metabelian section is cuspidal, with its unique full lift [Proved]

Retain the curve, field, good place, integral tangents, actual paths and tame generators of E.61. In particular $K = \mathbb{Q}(\zeta_3)$, $v \nmid 3$, $q = |k_v|$, and $\Pi = \pi_1(U_{\overline{K}})^{(3)}$ is free pro-three of rank ten. All derived subgroups below are closed. Put

$$M = \Pi/\Pi'', \qquad V = \Pi^{\mathrm{ab}}, \qquad A = \Pi'/\Pi'', \qquad v_i = [d_i] \in V. \tag{E66.1}$$

These are actual continuous quotients with the genuine unramified action fixed in E.61. The space of local M-section classes with nontrivial geometric inertia is exactly

$$\coprod_{i=1}^{9} (\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1)\mathbb{Z}_3. \tag{E66.2}$$

The representative for $(i, a, \bar{b})$ is the image in M of the actual full tangential section

$$t \longmapsto d_i^a t, \qquad F \longmapsto d_i^b F_i. \tag{E66.3}$$

Every equivalence in (E66.2) is conjugation by d_i^k, sending b to $b + (1-q)k$. The cusp and the nonzero exponent are unchanged. The parametrization is a homeomorphism for the quotient topology.

The actual map

$$\mathrm{Sect}(\Pi \rtimes G_{K_v})/\Pi \longrightarrow \mathrm{Sect}(M \rtimes G_{K_v})/M \tag{E66.4}$$

is a continuous surjection. Its restriction over the ramified locus is a homeomorphism; in particular every ramified metabelian class has exactly one full section class above it. Its unramified locus also has full lifts, by lifting its arbitrary Frobenius coordinate. The further classification of those unramified conjugacy fibers is retained in the specified remaining local and global maps of E.23.

For the actual quotient $M \twoheadrightarrow R$ at $K_v = \mathbb{Q}_2(\zeta_3)$, the ramified image therefore has precisely the cusp representatives of E.61. The displayed R-section of E61.5 lies outside that image.

All nine cusp inertia lines, its twenty-seven finite comparisons and the detecting degree-19683 cover retain their literal maps and certificates. The theorem settles the complete ramified local metabelian source at every specified good place, with no theta-value normalization required. The manuscript anchor is §51.16, folios 733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d).

Proof. The arithmetic relation and its finite covers. E.61 identifies the actual good-reduction action with the special fibre action and proves its pro-three factorization. A local section with geometric kernel M is a pair $(\bar\nu, \bar z) \in M^2$ satisfying

$$\bar z F(\bar\nu)\bar z^{-1} = \bar\nu^q. \tag{E66.5}$$

Its abelian inertia $w \in V$ satisfies $F(w) = qw$. E61.14 proves that w belongs to one of the nine integral cusp lines. We must determine the entire inertia element, including the case $w = 0$.

Choose any lift $z \in \Pi$ of $\bar z$, and write $\psi = zF$. For an F-stable open lattice $W_n \subset V$, let H_n be its inverse image in Π. The group H_n contains Π', hence $\Pi'' \subset [H_n, H_n]$. It is ψ-stable. As in E61.7–E61.11, the actual open arithmetic subgroup $H_n \rtimes \langle\psi\rangle$ defines a finite etale cover $U_n \to U_{k_v}$ over k_v, with smooth compactification C_n. All cover maps used below are the inclusion maps between these subgroups, with this same Frobenius lift ψ.

When $w \in W_n$, the image $\nu_n \in H_n^{\mathrm{ab}}$ of a lift of $\bar\nu$ is independent of that lift. Relation (E66.5) gives the exact equality

$$\psi(\nu_n) = q\nu_n \quad \text{in } H_n^{\mathrm{ab}}. \tag{E66.6}$$

The continuous, Frobenius-equivariant filling sequence is

$$0 \longrightarrow B_n \longrightarrow H_n^{\mathrm{ab}} \longrightarrow T_3\,\mathrm{Jac}(C_n) \longrightarrow 0, \tag{E66.7}$$

where B_n is the free cusp lattice modulo its single diagonal relation. The integral exactness follows from the punctured-surface presentation, with its oriented cusp generators; its specialization and Jacobian comparison are those proved in E.61 from [GR71, XIII, 2.9–2.12] and [Mil08, III, Proposition9.1]. The finite-field Frobenius eigenvalues on the final Tate module have complex absolute value $\sqrt{q}$ [Mil08, II, Theorem1.1(b)]. It follows from (E66.6) that

$$\nu_n \in B_n. \tag{E66.8}$$

This application uses the actual finite-field cover C_n; no weight statement about an arbitrary local-field Jacobian is substituted for that comparison.

Two exact inverse-limit comparisons. We record the maps needed to use all the finite covers at once. For a decreasing sequence of closed subgroups H_n of the compact group Π, put $H_\infty = \bigcap_n H_n$. The natural inclusion maps give an isomorphism

$$H_\infty^{\mathrm{ab}} \xrightarrow{\sim} \varprojlim_n H_n^{\mathrm{ab}}. \tag{E66.9}$$

Here abelianization is the maximal abelian pro-three quotient. To prove this, take a continuous character of H_∞ with finite three-primary image. Its kernel contains $H_\infty \cap U$ for some open normal subgroup U of Π. The decreasing images of the H_n in the finite group Π/U eventually equal the image of H_∞: each element in their intersection lifts to the intersection by compactness. The character therefore extends to all sufficiently deep H_n. If such a character restricts to zero on H_∞, choose U inside its kernel on the relevant H_n and use the same finite image argument to see that it is zero on every sufficiently deep H_m. Thus restriction identifies the direct limit of the finite-character groups with the character group of H_∞. Taking their compact Pontryagin duals proves (E66.9), with exactly the displayed inclusion maps.

We also take inverse limits of compact short exact sequences

$$0 \longrightarrow \Delta_n \longrightarrow P_n \longrightarrow B_n \longrightarrow 0. \tag{E66.10}$$

Exactness at the right can be proved directly. For a compatible sequence in the B_n, the set of lifts satisfying any finite collection of compatibility equations is nonempty: lift its coordinate at the largest index and map that lift down. These are closed subsets of the compact product of the P_n. Their finite intersection property supplies a compatible lift. The kernel is the inverse limit of the displayed Δ_n. This proves exactness here without a surjectivity assumption on the transition maps of the diagonal lattices.

Vanishing of metabelian inertia with zero abelian image. Suppose first that $w = 0$, and set $W_n = 3^n V$. Then $H_\infty = \Pi'$. The geometric deck group of the cover is $Q_n = V/3^n V$. Write $\bar{I}_{i,n}$ for the subgroup generated by v_i in this finite group. Since every v_i is primitive, it has order 3^n. The exact cusp modules in (E66.10) are

$$P_n = \bigoplus_{i=1}^{9} \mathbb{Z}_3[Q_n/\bar{I}_{i,n}], \qquad \Delta_n = \mathbb{Z}_3 \sum_{i,g} e_{i,g}. \tag{E66.11}$$

The basis vector $e_{i,g}$ is the actual oriented inertia class of $g d_i^{3^n} g^{-1}$. Under $H_{n+1} \hookrightarrow H_n$, every such cusp class maps to three times the corresponding class below. Consequently the transition on each summand of P_n is three times the trace along its finite coset map. Every fixed coordinate of a compatible inverse sequence is therefore divisible by every power of three; thus $\varprojlim P_n = 0$. Exactness of (E66.10) gives $\varprojlim B_n = 0$ as well. By (E66.8)–(E66.9), the class of $\bar{\nu}$ in Π'/Π'' is zero. Hence $\bar{\nu} = 1$ in M.

The surviving cusp module for nonzero abelian inertia. Now let $w = a v_i \neq 0$, put $r = v_3(a)$ and $u = a/3^r \in \mathbb{Z}_3^\times$, and use

$$W_n = \mathbb{Z}_3 w + 3^n V \quad (n \geq r), \qquad Y = V/\mathbb{Z}_3 v_i. \tag{E66.12}$$

We have $H_\infty = \Pi' \langle d_i^{3^r} \rangle$. Any two distinct cusp vectors occur together in a basis of V, by the punctured-surface basis used in E.61. In V/W_n, cusp i has order 3^r, whereas every other cusp has order 3^n. Thus its own boundary summand has transition equal to the trace

$$\mathbb{Z}_3[Y/3^{n+1}Y] \longrightarrow \mathbb{Z}_3[Y/3^n Y], \tag{E66.13}$$

and each of the other eight transitions is three times a trace. Their inverse limits are zero by the preceding divisibility argument. The covering degree between consecutive levels is 3^9, so the transition on Δ_n is multiplication by 3^9 and its inverse limit is zero. The exact compact comparison (E66.10) therefore identifies the full boundary limit as

$$\varprojlim_n B_n \simeq \mathbb{Z}_3[[Y]]. \tag{E66.14}$$

The right side is the module of integral three-adic measures on the compact additive group Y. This identification sends the inertia class of $g d_i^{3^r} g^{-1}$ to the Dirac measure at the image of g in Y. Thus it specifies the actual geometric origin of every basis coordinate.

The Frobenius action on this module is q times the permutation of measures induced by the affine map

$$T : Y \longrightarrow Y, \qquad \bar{g} \longmapsto F(\bar{g}) + \overline{[z] - [z_i]}. \tag{E66.15}$$

Indeed $F_i = z_i F$ sends d_i to d_i^q; conjugating the displayed cusp word by zF gives precisely the label and factor in (E66.15). This verifies both the translation term and the orientation factor, without an arbitrary choice of a nonbase cusp Frobenius origin.

Integral invariant measures for this actual affine action. The eigenvalues of F on $Y \otimes \mathbb{Q}_3$ are q on the remaining seven-dimensional cusp space and the two finite-field elliptic eigenvalues of complex absolute value $\sqrt{q}$ on the compact quotient. In particular $F^m - I$ is invertible over $\mathbb{Q}_3$ for every positive integer m. The affine map (E66.15) has one fixed point in $Y \otimes \mathbb{Q}_3$, obtained by applying $(I-F)^{-1}$ to its translation. Every affine periodic point equals this fixed point: the equation for period m has a unique solution and that same fixed point is a solution.

Moreover the closure of the permutation group generated by T is pro-three. The linear action is pro-three by E.61, and translations form the pro-three group Y. Thus every orbit on every finite quotient $Y/3^nY$ has three-power length.

Let $\mu \in \mathbb{Z}_3[[Y]]$ be T-invariant. Take an orbit O on a fixed finite quotient whose full inverse image does not contain the rational fixed point. For each integer b, all the orbits above O on a sufficiently deep quotient have length at least $3^b|O|$. To see this, the closed sets on that inverse image defined by $T^{3^c}x \equiv x \pmod{3^nY}$ decrease, as n grows, to its fixed-point set for T^{3^c}. This set is empty. Compactness gives a uniform depth where they are empty, for each prescribed c.

At a deep enough level, the contribution of each refined orbit to a fixed coarse coset is its orbit-length ratio times its common integral coefficient. It is divisible by 3^b. Trace compatibility therefore forces the coefficient of every coset of O to be divisible by every power of three, hence to vanish. All clopen sets avoiding the fixed point have measure zero. It follows that

$$(\mathbb{Z}_3[[Y]])^T = \begin{cases} \mathbb{Z}_3\delta_{g_0}, & g_0 \in Y, \\ 0, & g_0 \notin Y, \end{cases} \tag{E66.16}$$

where $g_0 = (I-F)^{-1}\overline{[z]-[z_i]}$ is the stated rational fixed point. When it lies outside Y, every coarse coset avoids it, so the same argument kills all coefficients.

Apply this calculation to the compatible boundary class of $\bar{\nu}$ supplied by (E66.8). Relation (E66.6), together with the factor q in (E66.15), says that its measure is T-invariant. Its total mass is u: under the inclusion in V, each selected cusp generator maps to $3^r v_i$, whereas $\bar{\nu}$ maps to av_i. Thus $g_0 \in Y$ and the measure is exactly $u\delta_{g_0}$. Choose $g \in \Pi$ with this image in Y. The elements $\bar{\nu}$ and $gd_i^a g^{-1}$ have the same image in H_∞^{ab}, by (E66.9) and (E66.14).

The exact conjugating correction. In M the subgroup H_∞/Π'' is the split extension $A \rtimes \langle d_i^{3^r} \rangle$. Its derived group is the closed image

$$[H_\infty, H_\infty]/\Pi'' = (d_i^{3^r} - 1)A = (d_i^a - 1)A. \tag{E66.17}$$

The first equality follows by multiplying commutators in this split extension, with A abelian. Its displayed image is closed because A is compact. For the second equality, the formal power series $((1+T)^u - 1)/T$ is a unit in $\mathbb{Z}_3[[T]]$, since its constant coefficient is $u \in \mathbb{Z}_3^\times$. Evaluate it at the action of $d_i^{3^r} - 1$ on A. Conjugation of $gd_i^a g^{-1}$ by elements of A changes its A coordinate by exactly $(1 - d_i^a)A$. The agreement in H_∞^{ab} and (E66.17) therefore supply an actual A-conjugator carrying it to $\bar{\nu}$. We have proved that every nonzero metabelian inertia is conjugate to one of the actual d_i^a.

Centralizers and all equivalences. We verify the required centralizer in the metabelian group, including nonunit exponents. Choose a free pro-three basis $d_i = b_1, b_2, \ldots, b_{10}$ of Π, put $\Lambda = \mathbb{Z}_3[[V]]$, and write X_j for its basis elements. The continuous Magnus map is

$$M \longrightarrow V \ltimes \Lambda^{10}, \qquad b_j \longmapsto (X_j, e_j), \qquad (X,c)(Y,d) = (XY, c + Xd). \tag{E66.18}$$

It is injective. Here is the precise comparison proving injectivity. For the open subgroups above 3^nV, the finite covering graph of the ten-petal rose identifies H_n^{ab} with the kernel of its integral edge-boundary map $\mathbb{Z}_3[V/3^nV]^{10} \to \mathbb{Z}_3[V/3^nV]$. The maps between these graphs are

the actual edge-trace maps. Taking their compact inverse limit and using (E66.9) embeds A in Λ^{10}. The Magnus map restricts to this embedding on A, while its first coordinate is the identity on V; hence its full kernel is trivial. Its path-boundary identity is

$$\sum_j (X_j - 1)c_j = X - 1. \tag{E66.19}$$

For an element (X, c) commuting with d_i^a, multiplication in (E66.18) gives

$$(X_1^a - 1)c = (X - 1)\frac{X_1^a - 1}{X_1 - 1}e_1. \tag{E66.20}$$

The ring $\Lambda \simeq \mathbb{Z}_3[[T_1, \dots, T_{10}]]$ is a domain, and $a \neq 0$ makes the first factor nonzero. Thus $c_j = 0$ for $j > 1$. Identity (E66.19), reduced at $X_1 = 1$, forces $X = X_1^b$ for some $b \in \mathbb{Z}_3$. It then forces $c_1 = (X_1^b - 1)/(X_1 - 1)$. Injectivity of (E66.18) proves

$$C_M(d_i^a) = \langle d_i \rangle. \tag{E66.21}$$

This calculation never divides by the possibly nonunit integer a.

Conjugate the section so that $\bar{\nu} = d_i^a$. Both its Frobenius lift and F_i conjugate d_i^a to d_i^{qa}. Their ratio centralizes this nonzero power; (E66.21) makes the ratio d_i^b. Thus the section is exactly (E66.3). Any geometric conjugation between two such representatives preserves the cusp and a, since their abelian cusp vectors do. It centralizes that inertia, hence lies in $\langle d_i \rangle$, and changes b by precisely $(1 - q)k$. This proves all the classes and equivalences in (E66.2). The same compact-annulus argument as in E.61 proves the homeomorphism: on each clopen locus $v_3(a) = r$ the parametrization is a continuous bijection from a compact space to a Hausdorff geometric-conjugacy quotient.

Finally the full ramified section classification E.61 has these identical parameters and representatives, so (E66.4) is a homeomorphism on that locus and each fiber there is a single class. A section with $\bar{\nu} = 1$ has an arbitrary Frobenius coordinate $\bar{z} \in M$. Choose any lift $z \in \Pi$. The pair $(1, z)$ satisfies the full tame relation and defines a continuous full section by E61.8. This proves surjectivity everywhere. A full ramified section has nonzero abelian cusp inertia, so its image in M remains ramified; the unramified lifting assertion retains precisely the unramified conjugacy fibers stated in the theorem. □

The marked-cusp inertia locus and its surviving central Frobenius classes

E.67. Complete marked-cusp inertia classes and an explicit ramified class without a full lift [Proved]

Retain the actual cubic curve, the good place $v \nmid 3$ of $K = \mathbb{Q}(\zeta_3)$, and the local arithmetic group of E.61. Write $q = |k_v|$ and let $Q = B_0$ be the single rational cusp marked in E.41. Its actual positive peripheral generator is $d = d_Q$. Set

$$I = \overline{\langle\langle d \rangle\rangle}_\Pi, \qquad W = \Pi/\gamma_3 I, \qquad B = \Pi/I, \qquad K_I = I/\gamma_3 I. \tag{E67.1}$$

Here Π is the free pro-three group of rank ten and every subgroup and commutator image is closed. Thus W is exactly the single-cusp marked-depth quotient in E41.1; the kernels of $\Pi \to W \to B$ are respectively $\gamma_3 I$ and K_I. The quotient B is the actual pro-three group of the cubic with this one cusp filled, and is free of rank nine.

Use the actual cusp Frobenius $F_Q = z_Q F$ of E.61, so that $F_Q(d) = d^q$. Replacing the generator F by F_Q is the literal change of arithmetic coordinates $(w, F_Q) \mapsto (w z_Q, F)$; it

retains the given arithmetic extension and the same inertia generator. Denote the induced automorphism of B by θ. The two actual normal-kernel modules and their actions are

$$\begin{aligned} D &= I/I' = \mathbb{Z}_3[[B]], \\ C &= \gamma_2 I/\gamma_3 I = \widehat{\bigwedge}^2_{\mathbb{Z}_3} D, \\ F_Q|_D &= q\theta_*, \qquad F_Q|_C = q^2\theta_2, \qquad \theta_2 = \widehat{\bigwedge}^2 \theta_*. \end{aligned} \tag{E67.2}$$

The element bdb^{-1} maps to $\delta_b \in D$. The completed exterior module in this formula means the compact inverse limit

$$C = \varprojlim_{U} \bigwedge^2_{\mathbb{Z}_3} \mathbb{Z}_3[B/U], \qquad \mathcal{Z}_Q = C/(q^2\theta_2 - 1)C, \tag{E67.3}$$

where U runs through characteristic open subgroups of B. All coefficient maps are the actual trace maps on the displayed finite coset sets; the denominator is its compact image and hence is already closed.

Consider the precise locus of local W-sections whose geometric inertia becomes the identity in B. Its ramified part, up to full geometric conjugacy in W, is

$$(\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1)\mathbb{Z}_3 \times \mathcal{Z}_Q. \tag{E67.4}$$

For $(a, \bar{b}, \bar{w})$ choose any $w \in C$ over $\bar{w}$. The representative is the actual section

$$t \longmapsto d^a t, \qquad F \longmapsto d^b c_w F_Q, \tag{E67.5}$$

where c_w denotes the element of the central subgroup $C \subset K_I$. The parametrization is a homeomorphism. Its complete equivalence relation is

$$b \longmapsto b + (1-q)k, \qquad w \longmapsto w + (1-q^2\theta_2)v \quad (k \in \mathbb{Z}_3,\ v \in C). \tag{E67.6}$$

The remaining part of the specified locus has trivial geometric inertia. It consists exactly of the pairs $(1, z)$ for arbitrary $z \in W$, modulo $z \mapsto gzF_Q(g)^{-1}$. Every such unramified class has a full Π-lift obtained by lifting z. There is no nontrivial inertia with zero image in D in this locus.

The actual full-to-marked map on the ramified locus whose inertia dies in B has image precisely

$$(\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1)\mathbb{Z}_3 \times \{0\}. \tag{E67.7}$$

It is a homeomorphism onto this image, and each of these classes has exactly one full section class above it. This assertion specifies the selected locus; it does not identify the inertia classes whose image in B is nontrivial.

At $K_v = \mathbb{Q}_2(\zeta_3)$, so $q = 4$, the finite arithmetic quotient of $\mathcal{Z}_Q$ supplied by the first moment map is

$$\mathcal{Z}_Q \twoheadrightarrow B^{\mathrm{ab}}/(16F_B - 1)B^{\mathrm{ab}} \simeq (\mathbb{Z}/9)^5 \oplus (\mathbb{Z}/27)^2. \tag{E67.8}$$

Here F_B is the actual quotient of the E.37 Frobenius matrix by the cusp vector v_Q. In particular set $b_0 = d_{A_0}$ and use the literal word

$$c_* = [d, b_0 d b_0^{-1}] \in \gamma_2 I/\gamma_3 I. \tag{E67.9}$$

Then

$$t \longmapsto dt, \qquad F \longmapsto c_* F_Q \tag{E67.10}$$

is a ramified local W-section with no full Π-lift. Its abelian inertia is the single primitive vector v_Q. The finite moment character and the integral matrix certificate below detect its nonzero third coordinate in (E67.4).

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d); the intrinsic single marked cusp and its exact kernel are those of E.25 and E.41.

Proof. The actual normal-kernel model. Complete d to a free pro-three basis of Π. Its other nine generators give a geometric splitting of $\Pi \twoheadrightarrow B$. This splitting is only used to label conjugates and is not asserted to be arithmetic. The normal closure I is the free pro-three group on the compact generator space

$$B \longrightarrow I, b \longmapsto bdb^{-1}. \tag{E67.11}$$

One can prove this universal description directly. Form the free pro-three group on the profinite space B, with its continuous B-action by left translation. Its semidirect product with B is generated by the generator at 1 and the chosen nine generators of B. The universal property of Π gives a map to this semidirect product. Conversely (E67.11) gives a B-equivariant map back to I, hence to Π. The two composites fix every indicated generator, so are inverse. The free group on the profinite space is constructed as the inverse limit of the free pro-three groups on its finite clopen partitions; this also verifies continuity of both maps.

At nilpotence class two, this gives the explicit group

$$K_I = D \times C, \qquad (h, w)(h', w') = (h + h', w + w' + \tfrac{1}{2}h \wedge h'). \tag{E67.12}$$

The generator at b is $(\delta_b, 0)$, the commutator of $(h, 0)$ and $(h', 0)$ is $(0, h \wedge h')$, and the central group is C. Division by two is integral over $\mathbb{Z}_3$. For each finite coset set B/U, this formula is the free class-two pro-three group on that set: collection gives its unique coordinates and proves the universal property. Taking the compact inverse limit proves (E67.12) with kernel exactly $\gamma_3 I$. Equivalently C is the alternating part of $\mathbb{Z}_3[[B \times B]]$: the map sends $\delta_g \wedge \delta_h$ to $\delta_{(g,h)} - \delta_{(h,g)}$, and the inverse divides by two after the natural exterior projection. This describes its topology, including the diagonal, exactly.

Since $F_Q(d) = d^q$, applying F_Q to bdb^{-1} gives $q\delta_{\theta(b)}$ in D. Its unknown inner part in I has no effect on that abelianization. Applying commutators then gives $q^2\theta_2$ on C. Thus (E67.2) follows from the actual action, with no choice of an arithmetic splitting of $W \to B$.

Two integral measure facts for this Frobenius. The group B is the good-reduction group of the curve with Q filled. Its abelianization has seven cusp Tate directions and the rank-two compact elliptic quotient. Frobenius eigenvalues have complex absolute values q and $\sqrt{q}$, respectively, by the same actual filling and specialization maps as in E.61. On every lower-central graded piece of the free pro-three group B, all eigenvalues are products of one or more of these eigenvalues and have absolute value greater than one. The integral free Lie comparison and separation of the lower-central filtration follow by the Magnus argument used in E.41, applied to the nine free generators. Consequently

$$B^{\theta^m} = 1 \qquad (m \geq 1). \tag{E67.13}$$

An element fixed by θ^m would have a lowest nonzero graded coordinate fixed by that matrix, which the eigenvalue calculation excludes.

For $z_B \in B$, consider the affine permutation $T(b) = z_B\theta(b)$. Each power of T has at most one fixed point: the quotient of two such points would be fixed by a power of θ. A periodic point of T is therefore a fixed point. The closure of T is pro-three, because both the actual Frobenius action and the group of translations have this property by E.61. On characteristic

finite quotients of B, orbit lengths are consequently three-powers. Compactness, applied to the fixed-point equations for each bounded power of T, forces these lengths to grow uniformly away from its possible fixed point. The trace argument of E66.16 gives

$$D^T = \begin{cases} \mathbb{Z}_3\delta_{b_*}, & T(b_*) = b_*, \\ 0, & \text{there is no such point.} \end{cases} \tag{E67.14}$$

We also need the finite-coefficient version with a scalar twist. Set $D_0 = \ker(D \xrightarrow{\text{aug}} \mathbb{Z}_3)$; then $D = \mathbb{Z}_3\delta_1 \oplus D_0$ is θ_*-stable. For every nonzero $a \in \mathbb{Z}_3$,

$$(D_0/aD_0)^{q\theta_*=1} = 0. \tag{E67.15}$$

For a unit a there is nothing to prove. Otherwise work over $\mathbb{Z}/3^r$, $r = v_3(a)$, and let a measure satisfy $q\theta_*\mu = \mu$. All nonidentity θ-orbits are infinite by (E67.13). On a refined cycle its coefficients are related by powers of q. The contribution to a fixed coarser cycle is a unit multiple of a coefficient times the geometric sum $\sum_{j=0}^{L-1} q^{j\ell}$, where L is the ratio of the cycle lengths. Since $q \equiv 1 \pmod 3$, this sum has three-adic valuation $v_3(L)$; this follows by induction from $1 + x + x^2 = 3 + 3(x-1) + (x-1)^2$ for $x \equiv 1 \pmod 3$. Orbit-length growth therefore kills every coefficient away from 1 modulo 3^r. The measure is supported at 1, and its zero augmentation makes it zero. Finite-coefficient trace systems have surjective maps, so they identify exactly with D/aD; the centered splitting identifies the stated submodule. This proves (E67.15) with the actual compact topology.

The inertia element, including its zero case. Use the arithmetic coordinates based on F_Q. A section in the specified locus has $\nu \in K_I$ and $z \in W$, with

$$zF_Q(\nu)z^{-1} = \nu^q. \tag{E67.16}$$

Its image $h \in D$ is fixed by the affine permutation $T(b) = z_B\theta(b)$, since the action on D is qT. If $h = 0$, then $\nu \in C$. The action on C is $q^2 \bigwedge^2 T$, so (E67.16) would make $(q \bigwedge^2 T - 1)\nu = 0$. On each finite coset quotient the exterior permutation has finite order; $q^n - 1 \neq 0$ in $\mathbb{Z}_3$ for every $n > 0$. Thus this operator is injective at each finite integral level, and hence on C. It follows that $\nu = 1$. This proves the full zero-inertia assertion, rather than only a statement about its abelian image.

For $h \neq 0$, (E67.14) gives $h = a\delta_{b_*}$ for a unique $a \neq 0$ and a fixed point b_*. Conjugating by a lift of b_*^{-1} sends z_B to 1 and h to $a\delta_1$. We now have

$$\nu = d^a c_w, \qquad z = (h_z, w_z) \in K_I. \tag{E67.17}$$

The genuine action fixes d by its q-power and acts on C by $q^2\theta_2$. Formula (E67.12) turns the relation into the exact central equation

$$(q\theta_2 - 1)w = a\,\delta_1 \wedge h_z. \tag{E67.18}$$

Decompose $C = (\delta_1 \wedge D_0) \oplus \widehat{\bigwedge}^2 D_0$. On the second summand, $q\theta_2 - 1$ is injective by the same finite-permutation argument. Hence $w = \delta_1 \wedge r$ for a unique $r \in D_0$. Equation (E67.18) becomes

$$(q\theta_* - 1)r = a\big(h_z - \mathrm{aug}(h_z)\delta_1\big). \tag{E67.19}$$

By (E67.15), $r = av$ with $v \in D_0$. Conjugating by the actual element $(v, 0) \in K_I$ subtracts $a\delta_1 \wedge v$ from the central coordinate of ν, so sends it to d^a. Its effect on the first coordinate of z is $h_z + (1 - q\theta_*)v = \mathrm{aug}(h_z)\delta_1$. The remaining central coordinate is retained as an arbitrary element of C; no unknown higher action has been dropped from that conjugation.

Centralizers, representatives and topology. In W, conjugation of the first coordinate $a\delta_1$ by a nonidentity element of B moves its support. Thus an element centralizing d^a lies in K_I. There (E67.12) says that its first coordinate h must satisfy $a\delta_1 \wedge h = 0$. The integral exterior module is torsion-free and its centered decomposition makes this equivalent to $h \in \mathbb{Z}_3\delta_1$. Consequently

$$C_W(d^a) = \langle d \rangle \times C \qquad (a \neq 0). \tag{E67.20}$$

After inertia has been made d^a, the ratio of its Frobenius lift and F_Q centralizes that inertia, so is exactly $d^b c_w$. Conversely every such element gives (E67.5), because C commutes with d. All geometric equivalences are in the same centralizer and give precisely (E67.6). Different nonzero a have different abelian inertia. The compact annuli $v_3(a) = r$, together with the compact groups in the other two coordinates, map continuously and bijectively to the corresponding Hausdorff orbit spaces. They therefore give homeomorphisms, and these clopen strata exhaust the ramified locus.

By E.61, a full ramified section is at one of the nine cusps. The other eight cusp vectors remain nonzero in B^{ab}, so only Q occurs in the selected locus. Its full representatives are $d^b F_Q$ with no central factor. This proves (E67.7), including uniqueness of the full class. For $\nu = 1$, every lift of z satisfies the full tame relation; this proves the stated unramified lifting map.

The finite moment quotient and the explicit failed lift. There is a continuous, surjective homomorphism

$$\begin{aligned} &\mathfrak{m}_1 : C \longrightarrow B^{\mathrm{ab}}, \\ &\delta_g \wedge \delta_h \longmapsto [h] - [g]. \end{aligned} \tag{E67.21}$$

Under the alternating-measure identification following (E67.12), it is one half of the integral of the continuous function $(g, h) \mapsto [h] - [g]$. Equivalently apply the first moment to $\mathrm{aug}(x)y - \mathrm{aug}(y)x$. These formulas agree on the dense finite sums and show continuity at every finite coefficient level. It is equivariant for simultaneous left translation, which acts trivially on its target, and for θ. Thus its exact Frobenius action is $q^2 F_B$ and it induces the surjection in (E67.8). For any chosen free basis $b_1, \ldots, b_9$ of B, the vectors $\delta_1 \wedge \delta_{b_j}$ give an integral right inverse on its abelian basis.

At the place above two, the E.37 basis is ordered $f_1, f_2, e_{A_0}, e_{A_1}, e_{A_2}, e_{B_0}, e_{B_1}, e_{B_2}, e_{C_1}, e_{C_2}$. The marked quotient removes the sixth vector. In the remaining order its actual matrix is

$$F_B = \begin{pmatrix} -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 4 & 0 & 0 & 0 & 0 & 0 & 0 \\ -4 & 4 & 0 & 4 & 0 & 0 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 \\ -2 & 2 & 0 & 0 & 0 & 4 & 0 & 0 & 0 \\ -4 & 0 & 0 & 0 & 0 & 0 & 4 & 0 & 0 \\ -2 & 4 & 0 & 0 & 0 & 0 & 0 & 4 & 0 \\ -4 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 4 \end{pmatrix}. \tag{E67.22}$$

The independent exact Smith calculation has integer diagonal entries

$$1, 1, 63, 63, 63, 63, 63, 2079, 2079.$$

Their three-primary valuations are $0, 0, 2, 2, 2, 2, 2, 3, 3$, proving (E67.8). The supplied certificate retains an integral three-adic left and right inverse transformation for the same matrix.

More economically, the explicit character

$$\ell = e^*_{A_0} + e^*_{A_1} - e^*_{B_2} \pmod 3 : B^{\mathrm{ab}} \longrightarrow \mathbb{F}_3 \tag{E67.23}$$

kills $(16F_B - 1)B^{\mathrm{ab}}$. On the two columns of the matrix U in E.37 its value is $(0, -3)$. The actual Frobenius gluing block is $-2U$, on which its value is $(0, 6)$. The remaining columns of $16F_B - 1$ are divisible by three. But $\mathfrak{m}_1(c_*) = [b_0] = e_{A_0}$ and $\ell(e_{A_0}) = 1$. Hence c_* represents a nonzero class in $\mathcal{Z}_Q$. Its commutator with d lies in the actual subgroup $\gamma_3 I$, so (E67.10) satisfies the literal tame relation. Formula (E67.7) excludes every full lift, including all possible conjugate representatives.

The program `a_marked_local.py` checks the actual quotient matrix, both exact Smith transforms, the moment and character identities, the class-two word and tame equation, and the weighted trace divisibility used above. Every reported computation is an exact integral or finite-coefficient calculation. □

All unramified local coordinates and the exact full-to-quotient fibres

E.68. Complete unramified local normal forms, triangular comparison maps and every full lifting fibre [Proved]

Retain the actual good-reduction local arithmetic action of E.61, with $K = \mathbb{Q}(\zeta_3)$, $v \nmid 3$, residue cardinality q, and its specified based Frobenius F. Put

$$\begin{aligned} P = \Pi, \qquad M = P/P'', \qquad W = P/\gamma_3 I_Q, \\ G_r = G/\gamma_{r+1}G, \qquad L_r(G) = \gamma_r G/\gamma_{r+1}G \quad (G = P, M, W). \end{aligned} \tag{E68.1}$$

These are the actual free, metabelian and single-cusp marked groups of E.41. Let $F_r(G)$ be the matrix induced by the actual abelian Frobenius on the integral Hall, Chen or retained-Hall module $L_r(G)$. Define the finite abelian groups

$$\mathcal{U}_r(G) = \operatorname{coker}(F_r(G) - 1). \tag{E68.2}$$

After the explicit representative choices described below, there are homeomorphisms of pointed profinite spaces

$$\begin{aligned} \mathrm{Sect}_{\mathrm{unr}}(G_r)/G_r &\simeq \prod_{j=1}^{r} \mathcal{U}_j(G), \\ \mathrm{Sect}_{\mathrm{unr}}(G)/G &\simeq \prod_{j\geq 1} \mathcal{U}_j(G). \end{aligned} \tag{E68.3}$$

Here an unramified section has tame coordinates $(1, z)$: its geometric inertia is the identity and its actual Frobenius image is zF. The quotient relation is exactly $z \mapsto gzF(g)^{-1}$.

For either actual quotient $\pi : P \twoheadrightarrow Q$, $Q = M$ or W, write $R = \ker \pi$ and give R the induced filtration $R \cap \gamma_r P$. Its finite free graded kernel is

$$\begin{aligned} K_r(R) &= (R \cap \gamma_r P)/(R \cap \gamma_{r+1} P) \\ &= \ker\big(L_r(P) \longrightarrow L_r(Q)\big), \\ \mathcal{V}_r(R) &= \operatorname{coker}(F_r|_{K_r(R)} - 1). \end{aligned} \tag{E68.4}$$

The integral maps give short exact sequences of finite groups

$$0 \longrightarrow \mathcal{V}_r(R) \xrightarrow{\iota_r} \mathcal{U}_r(P) \xrightarrow{\bar{\pi}_r} \mathcal{U}_r(Q) \longrightarrow 0. \tag{E68.5}$$

Every unramified target class has full lifts. For a fixed target represented by $z_Q F$, choose any actual lift $z_0 \in P$ of z_Q. Its full geometric-conjugacy fibre is canonically the following nonabelian cohomology set, and has the displayed compatible coordinates:

$$\begin{aligned} \pi^{-1}([z_Q F]) &\simeq H^1_{\mathrm{cts}}(\mathbb{Z}_3, R, \mathrm{Ad}(z_0)F) \\ &\simeq \prod_{r\geq 1} \mathcal{V}_r(R) = \prod_{r\geq 4} \mathcal{V}_r(R). \end{aligned} \tag{E68.6}$$

The second homeomorphism uses Hall lifts and finite Smith representatives; the first is the actual map $r \mapsto rz_0F$. The fibre in (E68.6) includes every full local section over that target: E.61 excludes a ramified full preimage of an unramified target.

In the product coordinates (E68.3), the actual comparison is the triangular map

$$b_r = \bar{\pi}_r(a_r) + \varepsilon_r(a_1, \dots, a_{r-1}). \tag{E68.7}$$

Formula (E68.15) below defines each correction from the unique lower conjugator, its specified Hall lift and the genuine nonlinear Frobenius on that lift. Formula (E68.17) gives every point of every fibre recursively, using exactly the maps in (E68.5). Thus (E68.6) retains the actual comparison, including all its correction coordinates.

At $K_v = \mathbb{Q}_2(\zeta_3)$, the following table evaluates the first four local factors. An entry in column e is the multiplicity of $\mathbb{Z}/3^e$; omitted unit Smith factors contribute the zero group.

factor	lattice rank	$e=1$	$e=2$	$e=3$	$e=4$	$e=5$
$\mathcal{U}_1(P) = \mathcal{U}_1(M) = \mathcal{U}_1(W)$	10	6	2	0	0	0
$\mathcal{U}_2(P) = \mathcal{U}_2(M) = \mathcal{U}_2(W)$	45	15	3	12	1	0
$\mathcal{U}_3(P) = \mathcal{U}_3(M) = \mathcal{U}_3(W)$	330	24	74	74	24	2
$\mathcal{U}_4(P)$	2475	537	435	51	319	3
$\mathcal{U}_4(M)$	1485	309	252	15	210	3
$\mathcal{U}_4(W)$	2466	532	433	51	319	3
$\mathcal{V}_4(P'')$	990	228	183	36	109	0
$\mathcal{V}_4(\gamma_3 I_Q)$	9	5	2	0	0	0

(E68.8)

At this place every remaining factor is determined by a finite integral Hall/Chen calculation modulo 3^{r+b_r+1}, where $b_r = \max_{r \le j \le 2r}(1 + v_3(j))$. This is a proved precision bound for the kernel factors as well as the three quotient factors.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d). Together with E.66, this describes every fibre of the local full-to-metabelian map. For the marked map it describes all unramified fibres; E.67 retains its separate exact ramified marked-cusp locus.

Proof. Integral strictness and the actual kernel filtration. The free pro-three lower-central graded algebra is the free integral Lie algebra on $V = P^{\mathrm{ab}}$, with its Hall basis. E.41 proves, through the completed Fox resolution and the actual commutator map, that $\mathrm{gr}(M)$ is the integral free metabelian Lie algebra. In particular its graded pieces are free and the map from the full Lie algebra is the natural commutator-to-syzygy map, with kernel $\mathrm{Lie}(V)''$. For the marked quotient, E.41 proves that the initial ideal of the actual closed subgroup $\gamma_3 I_Q$ is $\gamma_3 \mathfrak{i}_Q$, where $\mathfrak{i}_Q$ is the Lie ideal generated by the actual cusp vector. Its multihomogeneous Hall basis consists of the words with at least three occurrences of that vector. The Magnus inclusion and the lifting of those bracket generators prove both directions of this equality over $\mathbb{Z}_3$, including passage to the closed groups.

Consequently the exact degree-r sequence is

$$0 \longrightarrow K_r(R) \longrightarrow L_r(P) \xrightarrow{\pi_r} L_r(Q) \longrightarrow 0, \qquad L_r(Q) \text{ finite free over } \mathbb{Z}_3. \tag{E68.9}$$

For clarity, the group calculation behind its first term is

$$\frac{(R\gamma_{r+1}P) \cap \gamma_r P}{\gamma_{r+1}P} = \frac{(R \cap \gamma_r P)\gamma_{r+1}P}{\gamma_{r+1}P} \simeq \frac{R \cap \gamma_r P}{R \cap \gamma_{r+1}P}.$$

The products are closed products of compact subgroups. Thus the kernel is saturated and is exactly the induced filtration on R, not its intrinsic lower-central filtration. Every quotient in

(E68.9) is torsion-free. At degrees below four the kernels vanish. At degree four, $K_4(P'')$ has the Hall basis of brackets of two degree-two Hall words and rank $\binom{45}{2} = 990$; $K_4(\gamma_3 I_Q)$ has basis $[d,[d,[d,b_j]]]$, $1 \le j \le 9$, and rank nine.

All three groups equal the inverse limits of the indicated lower-central quotients, by the Magnus and completed Fox descriptions in E.41. Since R is a closed subgroup of P, it likewise equals $\varprojlim R/(R \cap \gamma_{r+1}P)$. These statements supply the integral groups and topologies to which the following central induction applies.

Weights, unique conjugators and unramified sections. The local action is the genuine good-reduction action of E.61 and factors through its pro-three arithmetic quotient. An arbitrary $z \in G$ defines an unramified section: the closure of zF in $G \rtimes \mathbb{Z}_3$ is procyclic, and projection sends its specified generator to the specified generator of $\mathbb{Z}_3$. The corresponding continuous homomorphism from $\mathbb{Z}_3$ is therefore a section. This proves the actual parametrization by z, before conjugacy.

On V the finite-field weights are the rank-two compact elliptic weights and the cusp Tate weights. On every degree-r Lie piece their products have complex absolute values $q^{j/2}$ with $r \le j \le 2r$. They survive as subquotients on all the lattices in (E68.9), so none has eigenvalue one. In particular every $F_r - 1$ in (E68.2) and (E68.4) is rationally invertible. Its integral cokernel is finite.

Conjugating the Frobenius action by an arbitrary element of P changes no associated-graded matrix. A nonidentity fixed element in any of the finite nilpotent quotients, or in their separated inverse limits, would have a first nonzero graded coordinate fixed by F_r. Hence

$$H^0(\mathbb{Z}_3, G, \mathrm{Ad}(z)F) = 1, \quad H^0(\mathbb{Z}_3, R, \mathrm{Ad}(z_0)F) = 1, \tag{E68.10}$$

with the same conclusion at every filtered finite stage. For R use the actual induced graded lattice $K_r(R)$. Thus any conjugator between two equivalent Frobenius representatives is unique.

Actual recursive representatives. Choose ordered integral Hall lifts, or the Chen and retained-Hall lifts of E.41, giving continuous set sections $s_r^G : G_{r-1} \to G_r$ with zero last coordinates. Central induction on the free integral graded bases proves existence and uniqueness of these collected coordinates; it uses group commutators and three-adic powers, without a rational exponential. Choose a Smith representative map $\sigma_r^G : \mathcal{U}_r(G) \to L_r(G)$, with $\sigma_r^G(0) = 0$. The representative associated to a finite coordinate sequence is defined literally by

$$z_0 = 1, \qquad z_r = s_r^G(z_{r-1})\, \sigma_r^G(a_r). \tag{E68.11}$$

The last factor is viewed in the central kernel of $G_r \to G_{r-1}$.

To normalize an arbitrary $z \in G_r$, first use the unique lower conjugator and lift it by s_r^G. The resulting element differs from the chosen lift of the lower representative by a unique central coordinate $e \in L_r(G)$. Its class is $a_r = [e] \in \mathcal{U}_r(G)$. There is a unique integral solution

$$v = (F_r(G) - 1)^{-1}(e - \sigma_r^G(a_r)) \in L_r(G). \tag{E68.12}$$

The displayed inverse is only applied to an element of its integral image, as certified by the Smith coordinates. Conjugation by the central element v subtracts $(F_r - 1)v$, so supplies exactly (E68.11). This constructs the unique full conjugator and proves both surjectivity and injectivity of the finite-stage normal form.

All normalizations are continuous: Smith reduction has finite residue targets, and the inverse of an injective lattice map is continuous on its compact image. The representatives (E68.11) are compatible. If two inverse-limit sections are equivalent at each finite stage, the unique conjugators are compatible and give one element of G. Conversely a compatible

sequence of finite normal forms has the actual representative $\varprojlim z_r$. This proves (E68.3) as a homeomorphism. The same argument, using the induced filtration on R, gives the second map in (E68.6).

The exact fibre as a twisted kernel section set. Fix z_Q and z_0 as in (E68.6). For a source class over $[z_Q F]$, its quotient representative is carried to z_Q by a unique element of Q, by (E68.10). Lift that element to P and apply the actual twisted conjugation. The resulting representative has the form rz_0 with $r \in R$. Any two choices of the lifted conjugator differ by an element of R. Conversely a conjugacy between two such representatives projects to a stabilizer of $z_Q F$, hence projects to the identity. Their exact equivalence relation is

$$r \longmapsto ur\,(\mathrm{Ad}(z_0)F)(u)^{-1} \qquad (u \in R). \tag{E68.13}$$

This proves the first homeomorphism in (E68.6): it is a continuous bijection from a compact orbit space to the Hausdorff fibre. Every target has a lift because $P \to Q$ is onto. Changing the chosen lift to $z_0' = r_0 z_0$ changes the representative by the explicit map $r \mapsto r r_0^{-1}$ and changes the action to $\mathrm{Ad}(r_0)\,\mathrm{Ad}(z_0)F$. Thus all choices are related by the actual inner cocycle transport.

Both kernels R lie in P', and both quotient maps induce the identity on V. A full ramified section has nonzero cusp inertia in V by E.61. It therefore cannot map to an unramified target. This proves that (E68.6) is the whole full section fibre, not merely its intersection with a chosen subclass.

The finite comparison sequence and its nonlinear correction. Apply $F_r - 1$ to (E68.9). Its three kernels are zero, so the snake lemma gives exactly (E68.5). There is also a direct integral inverse description. Choose any lattice section $s : L_r(Q) \to L_r(P)$. For $x \in L_r(P)$ whose quotient lies in $(F_r(Q) - 1)L_r(Q)$, let $y = (F_r(Q) - 1)^{-1}\pi_r(x)$. Then

$$x - (F_r(P) - 1)s(y) \in K_r(R) \tag{E68.14}$$

represents its unique class in $\mathcal{V}_r(R)$. Changing x by a full coboundary changes this element by a kernel coboundary. This specifies every map and the inverse on the kernel in (E68.5).

To compute the actual section map, take a source coordinate prefix and its representative z_{r-1}^P. Let z_{r-1}^Q be the already computed target normal form. Let $h_{r-1} \in Q_{r-1}$ be the unique element aligning their quotient representatives. Put $\widehat{h} = s_r^Q(h_{r-1})$ and define $e_r \in L_r(Q)$ by the exact group identity

$$\widehat{h}\,\pi\big(s_r^P(z_{r-1}^P)\big)F(\widehat{h})^{-1} = s_r^Q(z_{r-1}^Q)\,e_r, \qquad \varepsilon_r = [e_r]. \tag{E68.15}$$

Both sides have the same lower image, so this central element exists uniquely. In particular $F(\widehat{h})$ is the genuine Frobenius value, including its higher coordinates. Centrality then gives (E68.7). To continue the comparison, set

$$\begin{aligned} v_r &= (F_r(Q) - 1)^{-1}\big(e_r + \pi_r\sigma_r^P(a_r) - \sigma_r^Q(b_r)\big),\\ h_r &= v_r\,\widehat{h}. \end{aligned} \tag{E68.16}$$

The cokernel equality makes v_r integral. This is the unique actual aligning conjugator at the new stage. Equations (E68.15)–(E68.16) retain all nonlinear corrections and their signs.

Choose for each surjection in (E68.5) a section of finite sets τ_r on the specified Smith residues. For a target coordinate sequence (b_r), every source in its fibre is obtained uniquely by the recursion

$$a_r = \tau_r\big(b_r - \varepsilon_r(a_{<r})\big) + \iota_r(k_r), \qquad k_r \in \mathcal{V}_r(R). \tag{E68.17}$$

All maps at any finite stage are maps of finite sets, so this recursion and its inverse are continuous on the inverse products. It is the comparison-coordinate version of the twisted-kernel homeomorphism (E68.6), with every chosen representative and correction specified. No homomorphic splitting of (E68.5) is required.

Finite exact calculations at the place above two. The actual rank-ten matrix in E.37 is

$$A_F = \begin{pmatrix} -2I_2 & 0 \\ -2U & 4I_8 \end{pmatrix}, \qquad U_{d,\bullet} = (v_d, -u_d). \tag{E68.18}$$

The two cusp rows A_2, A_0 give a unit minor of U. Integral row operations make them I_2 and clear the remaining rows. After permuting coordinates, (E68.18) is integrally conjugate to two blocks $\left(\begin{smallmatrix} -2 & 0 \\ -2 & 4 \end{smallmatrix}\right)$ and six scalar blocks 4. This change fixes the actual marked vector e_{B_0} and has an integral inverse, both retained in the certificate. An actual compatible free pro-three basis is obtained by taking the ordered products of the original generators with exponent columns of that inverse matrix. Their images modulo the Frattini subgroup are a basis, so they give a free basis; the marked generator itself is unchanged. All nonlinear Frobenius values in these new words are retained in (E68.15)–(E68.16). Hall multidegrees in these eight invariant blocks are preserved by Frobenius. Through degree four their largest block has size24. The program computes every full, quotient and kernel matrix in these blocks and saves both Smith transformations and their inverses.

At degree four the metabelian kernel consists of the Hall brackets of pairs of degree-two words, so is exactly $\bigwedge^2(\bigwedge^2 V)$ with its integral basis. The marked kernel is the nine-word basis in the proof of (E68.9); its action is 4^3F_B. These supply independent checks on the kernel maps. The determinant valuations of the full and the two quotient degree-four operators are $2851, 1713, 2842$; the kernel valuations are $1138, 9$. Their differences agree with every exact sequence in (E68.5). The actual factors are those in (E68.8).

For completeness, the precision bound follows on all the lattices, not only the finite computed degrees. The split sublattice $T \oplus jT_3E$ contains $3V$. Its degree-r functorial maps B_r, C_r satisfy $B_rC_r = 3^rI$, as in E.41. They preserve the metabelian kernel and the marked ideal, so restrict to the kernels in (E68.9) as well as descend to the quotients. In the split lattice every multihomogeneous piece has Frobenius scalar $(-2)^j$, $r \le j \le 2r$. The valuations $v_3((-2)^j - 1) = 1 + v_3(j)$ imply

$$3^{\,r+b_r}(F_r - 1)^{-1}L_r \subset L_r \quad \text{and} \quad 3^{\,r+b_r}(F_r|_{K_r} - 1)^{-1}K_r \subset K_r. \tag{E68.19}$$

Thus all finite factors are killed by that power, and the stated extra digit suffices to determine their Smith invariants.

Finally the program checks the nonlinear normalization separately in the actual rank-two arithmetic quotient of E.54, where the Hall coordinates are (a, b, n, r, s) and

$$F(a, b, n, r, s) = (4a, 4b, 4b + 16n, 40n + 64r, 24n + 64s - 8b(b-1) - 15b/2). \tag{E68.20}$$

Its unique normal representatives have $a, b, n \in \{0, 1, 2\}$ and $r, s \in \{0, \dots, 8\}$. Five hundred exact tests recover the complete conjugator, including nonintegral rational coordinates with three-adic unit denominators. For the separate actual class-two projection test, deliberately choose the alternative nonpointed target degree-one Hall lift $x^{a+3}y^{b+3}$. The main normal forms (E68.3) retain their pointed Smith representatives. In this illustrative test the identity source has target coordinates $(0, 0, 1)$. The resulting correction in (E68.15) is $\varepsilon_2(a, b) = 1 + a - b \pmod 3$. All2187 source representatives give exactly 27 target classes with81 representatives in each fibre. This verifies the actual nonlinear correction alongside the952 exact graded sequence certificates; it does not substitute this smaller quotient for the rank-ten matrices of (E68.8).

All completed-group and section maps in the theorem are therefore supplied by the actual quotient, the unique conjugators, and the displayed integral coordinate procedures. □

An evaluated coherent-unit family at every positional level

E.69. Kato's norm theorem in the actual complex tower and a product of global section fibers [Proved]

Keep $L_d = K(\beta_d)$ and the specified roots of E.64, where $K = \mathbb{Q}(\zeta)$ and $\beta_d^{3^d} = 17\zeta$. Put

$$\mathcal{U}_d = \mathcal{O}_{L_d}[1/3]^\times \otimes_{\mathbb{Z}} \mathbb{Z}_3, \qquad \mathcal{U}_\infty = \varprojlim_{d,N} \mathcal{U}_d, \qquad I_d = \mathrm{pr}_d(\mathcal{U}_\infty). \tag{E69.1}$$

These are the usual finitely generated 3-unit completions and their compact inverse limit, with the actual adjacent field norms. For every d, I_d is a finite-index, torsion-free subgroup of $\mathcal{U}_d$, free over $\mathbb{Z}_3$ of rank 3^d. The actual restricted norms give split exact sequences of compact $\mathbb{Z}_3$-modules

$$0 \longrightarrow K_d \longrightarrow I_{d+1} \xrightarrow{N} I_d \longrightarrow 0, \qquad K_d \simeq \mathbb{Z}_3^{2\cdot 3^d}. \tag{E69.2}$$

Here the splittings are choices of lifts of an integral basis of I_d; they are not asserted to extend to all of $\mathcal{U}_d$.

Let $\lambda = 1 - \zeta$. There is a unique finite integer $h \geq 1$ such that

$$I_0 = \lambda^{3^h \mathbb{Z}_3}. \tag{E69.3}$$

Its exact value, and the higher indices $[\mathcal{U}_d : I_d]$, are retained as unevaluated finite norm-index data. The assertions that they are finite and that $h \geq 1$ are proved below from the actual norm maps. The first two finite projections are completely evaluated:

$$N_{L_1/K}\mathcal{U}_1 = \lambda^{\mathbb{Z}_3}, \qquad N_{L_2/K}\mathcal{U}_2 = \lambda^{3\mathbb{Z}_3}. \tag{E69.4}$$

For the second map an actual preimage of λ^3 is the 3-unit $c_1 = \beta_1 + 3\zeta + 1$, included into L_2. The second image has index nine in $\mathcal{U}_0 = \mu_3 \oplus \lambda^{\mathbb{Z}_3}$. More generally define the finite projection exponent a_n by $N_{L_n/K}\mathcal{U}_n = \lambda^{3^{a_n}\mathbb{Z}_3}$ for $n \geq 1$. Then

$$a_1 = 0, \quad a_2 = 1, \quad a_n \leq a_{n+1} \leq a_n + 1, \qquad a_n = h \text{ for all sufficiently large } n. \tag{E69.5}$$

Choose $s_d : I_d \to I_{d+1}$ with $Ns_d = \mathrm{id}$, and choose a basis $k_{d,1}, \ldots, k_{d,2\cdot 3^d}$ of K_d. The literal recursive maps

$$\begin{aligned} u_0 &= \lambda^{3^h z}, \\ u_{d+1} &= s_d(u_d) \prod_{j=1}^{2\cdot 3^d} k_{d,j}^{a_{d,j}}, \\ (z, (a_{d,j})) &\in \mathbb{Z}_3 \times \prod_{d\geq 0} \mathbb{Z}_3^{2\cdot 3^d} \end{aligned} \tag{E69.6}$$

give a topological group isomorphism onto the entire group $\mathcal{U}_\infty$. The inverse reads z from u_0, and reads $(a_{d,j})_j$ as the basis coordinates of $u_{d+1}s_d(u_d)^{-1}$ in K_d. In particular

$$\ker(\mathrm{pr}_0) = \prod_{d\geq 0} \mathbb{Z}_3^{2\cdot 3^d} \tag{E69.7}$$

under these specified maps. More generally, every fiber of $\mathrm{pr}_r : \mathcal{U}_\infty \to I_r$ is a torsor under $\prod_{d\geq r} \mathbb{Z}_3^{2\cdot 3^d}$. The product in (E69.6) is an unrestricted compact product; its finite-level coordinates are given recursively by the finite products displayed there.

This gives actual global section classes through the following closed embedding. Fix $A = 3\kappa_3(17)$ and the origin $g \mapsto (A_g, 0)$ in E.46. For $u = (u_d)_d \in \mathcal{U}_\infty$ define the positional norm tuple

$$t_{d+1} = (1, u_d) \in \widehat{F_d^\times}_3 \times \widehat{L_d^\times}_3, \qquad t_0 = N_{K/\mathbb{Q}} u_0, \quad F_d = \mathbb{Q}(17^{1/3^d}). \tag{E69.8}$$

All these tuples satisfy the total-norm cube condition defining J in E.46. Kummer–Shapiro therefore identifies them with precisely the global H-section classes having pure component 1 and complex components in the 3-unit groups at every level. Their actual toric map (E46.7), in the coordinates (E69.6), is

$$(z, (a_{d,j})) \longmapsto (\kappa_3(17),\, 3^{h-1} z\, \kappa_3(3)). \tag{E69.9}$$

Every nonempty fiber of this displayed family has the fully specified product coordinates (E69.7). In particular the fiber over $(\kappa_3(17), 0)$ consists exactly of the tuples with $u_0 = 1$ and is the full product in (E69.7).

The indices in (E69.3) have a further exact finite-level comparison:

$$I_d = \bigcap_{n \geq d} N_{L_n/L_d} \mathcal{U}_n, \qquad I_d = N_{L_n/L_d} \mathcal{U}_n \quad \text{for every sufficiently large } n. \tag{E69.10}$$

The proof supplies the actual maps and proves stabilization. It does not compute the stabilization heights or identify a single observed equality of consecutive images with permanent stabilization. The remaining numerical indices and the sections with other positional divisors remain separate data in the full norm system of E.46. The maps from intrinsic metabelian and marked sections to these H-sections remain the actual quotient maps of E.38; (E69.8) supplies section classes in their target. The manuscript anchor is §51.16, scans172–178, folios733v–736v, equations(371)–(379), and Part E(c),(d).

Proof. The exact use of a coherent universal-norm theorem. Choose compatible roots of unity with $\zeta_3 = \zeta$ and choose the roots of seventeen so that

$$\beta_d = 17^{1/3^d} \zeta_{3^{d+1}}. \quad \mathcal{L} = K(\mu_{3^\infty}, 17^{1/3^\infty}) = L_d(\mu_{3^\infty}, \beta_d^{1/3^\infty}). \tag{E69.11}$$

Both equalities are literal equalities inside the chosen algebraic closure. For example, dividing β_{d+j} by $\zeta_{3^{d+j+1}}$ recovers $17^{1/3^{d+j}}$; multiplying these two factors recovers β_{d+j}. Thus the two sets of generators in (E69.11) give mutual field inclusions.

Kato defines universal norms as the image of the inverse limit of completed 3-units over all finite intermediate fields, with norm transitions. His Theorem1.1(2) states that this image has finite index in the base 3-unit group when the base has exactly one prime over three, for the radical–cyclotomic extensions in (E69.11) [Kat06, pp.551–552, definition and Theorem1.1; Corollary5.2(2)]. The complete proof was checked through the cohomological norm comparison in §2.2–2.6, the abelian case Proposition3.1, the Tor-module argument Proposition4.2, and §5.4–5.5. In particular the source theorem concerns compatible systems, not just separate membership in finite norm images.

By E.64, L_d has exactly one prime over three. Apply this theorem with base L_d and radical β_d. There is the following actual restriction map of inverse systems:

$$\varprojlim_{\substack{L_d \subset E \subset \mathcal{L} \\ [E:L_d] < \infty}} (\mathcal{O}_E[1/3]^\times \otimes \mathbb{Z}_3) \longrightarrow \mathcal{U}_\infty. \tag{E69.12}$$

For $n \geq d$ its nth coordinate is the given coordinate at L_n; for $n < d$ it is N_{L_d/L_n} of the coordinate at L_d. Norm transitivity proves compatibility at every adjacent arrow, including

the arrow across d. The projection of (E69.12) to $\mathcal{U}_d$ is exactly Kato's universal-norm image. Consequently this finite-index subgroup is contained in I_d. The latter is closed, since it is the continuous image of the compact group $\mathcal{U}_\infty$. It follows that I_d also has finite index in $\mathcal{U}_d$. This proves the claimed finite-index assertion through a map of the actual norm systems.

The finite ranks and the local torsion exclusion. The field L_d has degree $2 \cdot 3^d$, has no real embeddings, and has one prime over three. Dirichlet's unit theorem gives

$$\mathcal{U}_d \simeq \mu_3 \oplus \mathbb{Z}_3^{3^d}. \tag{E69.13}$$

For the roots of unity, E.64 proves the exact intersection with the 3-power cyclotomic fields: over any such field containing K, $X^{3^d} - 17\zeta$ remains Eisenstein at a prime over seventeen. Hence L_d is linearly disjoint from that field over K and has no additional 3-power roots of unity. Its 3-primary torsion is precisely the displayed μ_3.

Let $\mathfrak{t}_d$ be the unique prime over seventeen. Reduction of 3-units gives a continuous map

$$r_d : \mathcal{U}_d \longrightarrow (\mathbb{F}_{289}^\times)^{\wedge_3} \simeq \mu_9. \tag{E69.14}$$

The extension L_{d+2}/L_d at these primes is totally ramified of degree nine, with the same residue field. Its residue unit norm is the ninth power. Thus $r_d(N_{L_{d+2}/L_d} v) = r_{d+2}(v)^9 = 1$ for all $v \in \mathcal{U}_{d+2}$. Every $u_d \in I_d$ comes from that actual coordinate of a compatible system, so

$$I_d \subset \ker r_d. \tag{E69.15}$$

Reduction is injective on μ_3 because the residue characteristic is seventeen. Equations (E69.13)–(E69.15) therefore prove that I_d has no torsion. A finite-index torsion-free submodule of the finitely generated module (E69.13) is free of rank 3^d. This proves the precise assertion needed for (E69.2).

The actual base coordinate and the total-norm condition. The certified E.57 calculation gives the entire 3-unit group at the first complex level: its free basis is $e_1, e_2, c_1 = \beta_1 + 3\zeta + 1$, its torsion basis is ζ, and their norms to K are $1, 1, \lambda, 1$. Indeed (c_1) is the single prime over three, so adjoining c_1 to the ordinary unit basis gives all 3-units. It follows that

$$N_{L_1/K}\mathcal{U}_1 = \lambda^{\mathbb{Z}_3}, \qquad I_0 \subset \lambda^{\mathbb{Z}_3}. \tag{E69.16}$$

In $\mathbb{F}_{289}$ one has $\lambda^2 = -3\zeta$ and $\lambda^{32} = (-3)^{16}\zeta^{16} = \zeta$. The projection to the 3-primary factor of this group of order $288 = 32 \cdot 9$ is the power 64; hence $r_0(\lambda) = \zeta^2$. Combining (E69.15) with (E69.16) shows that an exponent occurring in I_0 is divisible by three. The subgroup I_0 is closed and of finite index in $\mu_3 \oplus \lambda^{\mathbb{Z}_3}$, so it is exactly $\lambda^{3^h\mathbb{Z}_3}$ for one finite $h \geq 1$. This proves (E69.3) with both the finite-index and lower-bound assertions justified. It leaves the value of h as the stated finite arithmetic datum.

The same local argument already puts $N_{L_2/K}\mathcal{U}_2$ in $\lambda^{3\mathbb{Z}_3}$. For the opposite inclusion, the included element c_1 has $N_{L_2/L_1} c_1 = c_1^3$, and E.57 gives $N_{L_1/K} c_1 = \lambda$. Thus its total norm is λ^3, proving the second equality of (E69.4) with an actual preimage. The identity for its first norm can also be checked directly: $N_{L_1/K}(\beta_1 + 1 + 3\zeta) = (1 + 3\zeta)^3 + 17\zeta = 1 - \zeta$. At every finite height the included c_1 supplies the nonzero image $\lambda^{3^{n-1}}$; hence that closed image in $\lambda^{\mathbb{Z}_3}$ has the unique exponent a_n. For an element of $\mathcal{U}_n$ included into $\mathcal{U}_{n+1}$, the adjacent norm is its cube. Therefore the next projection image contains the cube of the current image and is contained in the current image. This proves the inequalities in (E69.5). Their eventual value is h by (E69.10), proved next.

Surjective stable-image transitions and recursive coordinates. For $u_d \in I_d$, choose a compatible sequence that projects to it. Its next coordinate lies in I_{d+1} and norms to u_d. Thus the actual map $N : I_{d+1} \to I_d$ is onto. Choose any integral basis of I_d, lift its basis vectors to

I_{d+1}, and extend $\mathbb{Z}_3$-linearly. This gives the continuous splitting s_d. Its kernel is free of rank $3^{d+1} - 3^d = 2 \cdot 3^d$, proving (E69.2).

For the product on the right of (E69.6), the displayed recursion uses a finite product at each level and has norm u_d by construction. It therefore gives an element of $\mathcal{U}_\infty$ for every unrestricted set of product coordinates. Conversely a compatible sequence has $u_0 \in I_0$ and $u_{d+1} s_d(u_d)^{-1} \in K_d$, so the inverse formulas determine all its coordinates uniquely. Each finite output coordinate depends continuously on finitely many input coordinates, and the inverse basis maps are continuous between finite free compact $\mathbb{Z}_3$-modules. This proves the topological isomorphism, (E69.7), and the analogous fiber formula at any level r.

To prove (E69.10), the inclusion from left to right is norm compatibility. If an element lies in every norm image on the right, then every finite set of the norm equations above level d has a solution: choose a preimage at its highest level and norm it to the lower levels. These equations define closed subsets of the product of the compact 3-unit groups. The finite intersection property gives a compatible infinite system. Hence their intersection is I_d. Every norm image contains the finite-index subgroup I_d; after quotienting by I_d, these images form a descending chain of subgroups in the finite group $\mathcal{U}_d/I_d$. The chain stabilizes, proving the second assertion of (E69.10). This argument uses compactness only for the finitely generated 3-unit completions in (E69.1).

The closed map to global sections and its toric coordinate. The natural map $\mathcal{U}_d \to \widehat{L_d^\times}_3$ is a closed embedding. For an elementary verification, a 3^rth power in $L_d^\times$ which is a 3-unit has a 3-unit root, since all valuations away from three of that root must vanish. Thus the induced power filtration on the 3-unit group is its own multiplication filtration. The quotient is a subgroup of the free divisor group away from three and is 3-torsion-free; the power quotient sequences and their inverse limits preserve the asserted injection. Compactness of $\mathcal{U}_d$ and the Hausdorff target make its image closed.

The positional algebra at level $d+1$ is $F_d \times L_d$, with the root-cubing norms proved in E.46 and E.62. Consequently (E69.8) is a continuous injection into the actual positional norm limit. The norm of its base coordinate is

$$t_0 = N_{K/\mathbb{Q}}(\lambda^{3^h z}) = 3^{3^h z}. \tag{E69.17}$$

Since $h \geq 1$, this is a cube in $\widehat{\mathbb{Q}^\times}_3$. It is exactly the condition in E46.4 for the J-cocycle, with no additional lift chosen at any positional level. The E.46 Kummer–Shapiro map and its inverse identify this tuple with an actual global section class and preserve the given topology. The domain is compact and the target norm limit is Hausdorff, so this is a closed embedding. Conversely the specified pure and complex coordinates of any section in this image give exactly a sequence in $\mathcal{U}_\infty$; hence the image is the full family described in the statement. Finally E46.7 divides the two raw augmentation coordinates by three. Applying Kummer to (E69.17) gives (E69.9). The class $\kappa_3(3)$ has infinite order, as detected by its valuation at three, so its zero fiber is exactly $z = 0$, and the other fibers follow from (E69.7). □

The complete central parameter of the ramified marked-cusp family

E.70. Finite orbit transitions and full product coordinates for the marked central parameter [Proved]

Retain the actual groups and Frobenius of E.67: $B = \Pi/I_Q$, $D = \mathbb{Z}_3[[B]]$, $C = \widehat{\bigwedge}^2 D$, $\theta = F_Q|_B$, and $q = |k_v|$. Put $s = v_3(q-1) \geq 1$ and

$$\mathcal{Z}_Q = C/(q^2\theta_2 - 1)C, \qquad \theta_2 = \widehat{\bigwedge}^2 \theta_*. \tag{E70.1}$$

There is a homeomorphism of compact $\mathbb{Z}_3$-modules

$$\mathcal{Z}_Q \simeq \prod_{n\geq 1} \mathbb{Z}_3. \tag{E70.2}$$

The actual noncanonical coordinates and their inverse are constructed in (E70.10)–(E70.13) from finite orbit characters and coherent division, retaining the Frobenius and every transition map.

More precisely, for a characteristic open subgroup $U \subset B$, let Ω_U be the unordered pairs of distinct elements of B/U. For each θ-orbit $O \subset \Omega_U$, choose an oriented representative (g_O, h_O). Its orbit length L_O is a power of three. Transport this orientation by θ; it returns with the same orientation. Then the actual finite quotient is

$$\begin{aligned} Z_U &= \operatorname{coker}\left(q^2\theta_2 - 1 : \bigwedge^2 \mathbb{Z}_3[B/U] \longrightarrow \bigwedge^2 \mathbb{Z}_3[B/U]\right) \\ &\simeq \bigoplus_{O\in\Omega_U/\theta} \mathbb{Z}_3/(q^{2L_O}-1) \simeq \bigoplus_{O} \mathbb{Z}/3^{\,s+v_3(L_O)}, \\ \mathcal{Z}_Q &\xrightarrow{\sim} \varprojlim_{U} Z_U. \end{aligned} \tag{E70.3}$$

For $U' \subset U$, an oriented finer orbit generator maps to zero when its endpoints coincide modulo U. Otherwise its image is

$$e_{O'} \longmapsto \epsilon q^{-2j} e_O, \qquad \epsilon \in \{1,-1\}, \tag{E70.4}$$

where its reduced ordered pair is ϵ times the jth transported orientation of the parent orbit. This is a homomorphism between the displayed finite cyclic factors; all finite maps in (E70.3) are surjective. The same formula applies to every θ-stable open normal subgroup; its finite quotient is dominated by a characteristic one.

Consequently the ramified marked-cusp inertia locus of E.67 has the complete coordinates

$$(\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1)\mathbb{Z}_3 \times \prod_{n\geq 1} \mathbb{Z}_3. \tag{E70.5}$$

The actual full image is precisely the zero sequence in the last factor, with the representatives and unique full lifts of E67.5–E67.7. The word E67.9 is detected by its specified finite moment character; its finite torsion quotient is compatible with the torsion-free compact module (E70.2) through that actual quotient map.

At $q = 4$, the actual rank-nine quotient $B/B'B^3$ gives

$$Z_{B'B^3} \simeq (\mathbb{Z}/3)^{2390391} \oplus (\mathbb{Z}/9)^{63770004}. \tag{E70.6}$$

There is also an explicit primitive arithmetic quotient $B \twoheadrightarrow \mathbb{Z}_3$ on which θ is multiplication by four. Its positional level 3^n gives the fully evaluated finite system

$$Z_n^{\mathrm{line}} \simeq \bigoplus_{e=0}^{n-1} (\mathbb{Z}/3^{e+1})^{\,4\cdot 3^e - 1}, \tag{E70.7}$$

with the literal signed phase transitions (E70.4). The program evaluates all of these transitions through $n = 6$, as well as coherent character lifts. These are actual quotients of (E70.3).

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d); the marked subgroup and its actual quotient maps are exactly those in E.41 and E.67.

Proof. The exact finite orbit system. By E.67 the normal-kernel centre is the compact module $C = \varprojlim_U \bigwedge^2 \mathbb{Z}_3[B/U]$. The closure of θ is pro-three. Therefore its permutation of B/U has three-power order, and every orbit in Ω_U has odd length. An ordered pair cannot return with its endpoints exchanged: that would give a nontrivial permutation of order two induced by a group of odd order. Thus, on each orbit, the transported vectors

$$e_j = \delta_{\theta^j g_O} \wedge \delta_{\theta^j h_O} \quad (0 \le j < L_O)$$

give an integral basis with $\theta_2 e_j = e_{j+1}$, indices read modulo L_O. The relation $q^2 e_{j+1} - e_j$ gives $[e_j] = q^{-2j}[e_0]$, and the final relation is $(q^{2L_O} - 1)[e_0] = 0$. Eliminating the intermediate basis vectors uses unit integral coefficients. This proves the first two lines of (E70.3), including its literal generator. The lifting-the-exponent identity gives $v_3(q^{2L_O} - 1) = s + v_3(L_O)$.

The finite operators $q^2\theta_2 - 1$ are injective: a power of the permutation is the identity, while $q^{2m} - 1$ is a nonzero integer for $m > 0$. Their short exact cokernel sequences are compact. The same finite-intersection lifting argument as in E66.10 makes their inverse limit exact. It yields exactly the final map in (E70.3). Its denominator is the compact image of the continuous operator on C.

Under reduction of finite coset sets a wedge becomes zero when its two endpoints coalesce. Otherwise its orientation and position in the parent orbit give (E70.4). The parent orbit length divides the finer length, so its cyclic relation annihilates the image of the finer relation. Each parent wedge lifts to a finer wedge; the resulting column is a unit times its parent generator. Hence each displayed finite cokernel map is surjective.

No three-torsion survives in the actual compact quotient. The exterior module has the exact alternating realization

$$C \xrightarrow{\sim} \{\mu \in \mathbb{Z}_3[[B \times B]] : \mathrm{swap}_* \mu = -\mu\}, \quad \delta_g \wedge \delta_h \longmapsto \delta_{(g,h)} - \delta_{(h,g)}. \tag{E70.8}$$

The inverse is one half of the natural exterior projection. The idempotent $(1 - \mathrm{swap})/2$ is integral. At each finite positional level its image is a direct summand of a finite free lattice. Compact exact limits of the multiplication-by-three sequences show that reducing (E70.8) modulo three gives exactly the alternating direct summand of $\mathbb{F}_3[[B \times B]]$.

E67.13 proves $B^{\theta^m} = 1$ for every $m > 0$, using the actual finite-field weights and the separated integral lower-central filtration. The only periodic point of $\theta \times \theta$ on $B \times B$ is therefore $(1, 1)$. An invariant $\mathbb{F}_3$-measure on this compact space is supported at that point. Indeed, above any coarse orbit whose inverse image avoids $(1, 1)$, compactness forces all refining orbit lengths to grow beyond every fixed three-power multiple. Trace compatibility sums equal coefficients around each refined cycle, and eventually makes the coefficient of the coarse orbit a multiple of three. It is therefore zero. This proves the asserted support statement on every clopen set.

Since $q^2 \equiv 1 \pmod 3$, an element of the kernel of $q^2\theta_2 - 1$ on $C/3C$ would be such an invariant alternating measure. The Dirac measure at $(1, 1)$ is symmetric, and two is invertible, so that kernel is zero. Suppose $3x = (q^2\theta_2 - 1)y$ in C. Reduction modulo three gives $y = 3y_1$; the torsion-freeness of C then gives $x = (q^2\theta_2 - 1)y_1$. We have proved

$$\mathcal{Z}_Q[3] = 0. \tag{E70.9}$$

The same argument applies to all powers of three. Thus the compact pro-three module is torsion-free.

Countably infinitely many independent directions. The finitely generated pro-three group B has a countable cofinal system of characteristic open subgroups. The positional and coefficient presentations make C and $\mathcal{Z}_Q$ metrizable. Their continuous character groups are countable unions of finite groups.

Choose a strictly decreasing cofinal sequence $U_1 \supset U_2 \supset \cdots$ of characteristic open subgroups, and set $A_n = U_n \setminus U_{n+1}$. There are continuous alternating characters with values in $\mathbb{F}_3$ defined by

$$\lambda_n(\delta_g \wedge \delta_h) = 1_{A_n}(g)1_{U_{n+1}}(h) - 1_{A_n}(h)1_{U_{n+1}}(g). \tag{E70.10}$$

They are θ_2-invariant and hence annihilate $q^2\theta_2 - 1$ modulo three. For $g_m \in A_m$, their values on $\delta_{g_m} \wedge \delta_1$ are exactly δ_{nm}. Thus the order-three continuous character space of $\mathcal{Z}_Q$ has countably infinite dimension. This argument uses actual Frobenius-stable clopen sets, including their centred limit at the identity.

Coherent finite characters and the actual product map. Write $\mathcal{A} = \varinjlim_U Z_U^\vee$, where $Z_U^\vee = \mathrm{Hom}(Z_U, \mathbb{Q}_3/\mathbb{Z}_3)$. Every transition in this direct system is the dual injection of (E70.4). The order-three characters of each finite cyclic orbit factor enumerate $\mathcal{A}[3]$. Choose a basis $\alpha_1, \alpha_2, \ldots$ of that space by successive finite linear algebra in this enumeration. To compare any finite list of characters, pull them to one common finite positional level; injectivity makes that a literal finite computation.

Every finite character has an explicit division by three at a sufficiently fine level. For the finitely many parent orbits on which it is nonzero, choose a finer characteristic subgroup so that each surviving finer orbit above them has length divisible by three times the parent length. Such a subgroup exists uniformly: the inverse image of a parent unordered distinct pair is compact, contains no diagonal point, and contains no periodic point of $\theta \times \theta$. The orbit-length argument used above applies. Put $t_O = s + v_3(L_O)$. For a character with value $a_O/3^{t_O}$ on the parent generator, define its divided character on a finer generator by

$$e_{O'} \longmapsto \frac{\epsilon q^{-2j} a_O}{3^{t_O+1}} \pmod{\mathbb{Z}_3}; \tag{E70.11}$$

set the value to zero on generators mapping to zero or to a zero parent coordinate. The finer exponent is at least $t_O + 1$, so these are legitimate characters of the actual finite cyclic factors. Multiplying them by three gives the exact pullback of the original character. The unit in the numerator is the same phase unit as in (E70.4), now evaluated to the required higher precision.

Starting with $\alpha_{i,1} = \alpha_i$, this procedure chooses coherent characters $\alpha_{i,n}$ of order 3^n with $3\alpha_{i,n+1} = \alpha_{i,n}$. For $z \in \mathcal{Z}_Q$, define

$$\Phi(z)_i = (a_{i,n})_{n\geq 1}, \qquad \alpha_{i,n}(z) = a_{i,n}/3^n \pmod{\mathbb{Z}_3}, \quad a_{i,n} \in \mathbb{Z}/3^n. \tag{E70.12}$$

Here $a_{i,n}$ is the unique residue with the displayed character value. Coherence makes this a continuous $\mathbb{Z}_3$-linear map to $\prod_i \mathbb{Z}_3$.

The character chains form a direct sum of Pruefer three-groups and span $\mathcal{A}$. For independence, take a finite relation at common order 3^n and multiply by 3^{n-1}. Independence of the α_i makes every coefficient divisible by three. Repeating proves that every coefficient is zero at the original order. For spanning, take a character of order 3^n, express its 3^{n-1} multiple in the basis $\{\alpha_i\}$, and subtract the corresponding finite sum of $\alpha_{i,n}$. The remainder has smaller order; induction finishes the expression.

This also gives an explicit inverse to (E70.12). For a prescribed sequence $(u_i) \in \prod_i \mathbb{Z}_3$, evaluate $\alpha_{i,n}$ as $u_i/3^n$ modulo $\mathbb{Z}_3$. The just-proved direct sum makes this a well-defined homomorphism on $\mathcal{A}$. Restrict it to the character group of each Z_U and use the elementary duality of its displayed finite cyclic factors. The resulting $z_U \in Z_U$ are compatible with (E70.4), so give

$$(u_i) \longmapsto (z_U)_U \in \varprojlim_U Z_U. \tag{E70.13}$$

All finite evaluations use only the character expressions obtained by the order induction and the finite linear algebra above. Equations (E70.12)–(E70.13) are inverse continuous maps of compact modules. This proves (E70.2) with actual product coordinates and their inverse.

Arithmetic finite examples and all scalar transitions. For $q = 4$, E67.22 gives the actual matrix F_B. Modulo three, $F_B - I$ has rank two and square zero. Thus its permutation of $\mathbb{F}_3^9$ has order three and exactly 3^7 fixed points. An unordered pair is fixed exactly when both points are fixed, because a permutation of odd order cannot exchange just those two points. Therefore the numbers of pair orbits of lengths one and three are

$$\binom{3^7}{2} = 2390391, \qquad \frac{\binom{3^9}{2} - \binom{3^7}{2}}{3} = 63770004. \tag{E70.14}$$

The finite presentation (E70.3) proves (E70.6), without enumerating a much larger matrix than these orbit counts require.

In the E.37 basis, the primitive functional

$$f_2^* + e_{A_0}^* + e_{A_1}^* - e_{B_2}^* : B^{\mathrm{ab}} \longrightarrow \mathbb{Z}_3 \tag{E70.15}$$

annihilates the filled cusp vector and satisfies $\ell F_B = 4\ell$. It takes value one on e_{A_0}, so is surjective. Composing with $B \to B^{\mathrm{ab}}$ gives the actual arithmetic quotient to $\mathbb{Z}_3$ claimed in the statement. Pushforward of measures and exterior powers gives corresponding surjections from C and $\mathcal{Z}_Q$ to the scalar orbit systems.

On $\mathbb{Z}/3^n$, multiplication by four has period 3^{k-1} on a pair whose minimum valuation is $n - k$. The number of unordered distinct pairs of that valuation is $4 \cdot 3^{2k-2} - 3^{k-1}$: subtract the diagonal from the two-coordinate valuation stratum, then divide by two. Consequently there are exactly $4 \cdot 3^{k-1} - 1$ orbits of that length. This proves (E70.7) for every n. Each surviving finer orbit has three times the parent length; those whose endpoints coalesce map to zero. Hence the character division (E70.11) can be carried out at the next level in this scalar system.

The exact program `a_central_orbits.py` checks the actual primitive functional and the rank-nine mod-three matrix, every oriented scalar pair through level729, all finite coboundaries and transitions, the finite order-three basis extensions, 378 coherent character divisions, and the independent annular characters. Its retained output distinguishes the full rank-nine finite quotient from the explicit scalar arithmetic quotient throughout. No uncomputed higher Frobenius word is assigned a numerical value. □

The complete marked local section map at every good place

E.71. All ramified marked classes and every fibre of the full-to-marked local map [Proved]

Use the actual good-reduction action, oriented cusp generators and integral tangential Frobenius lifts of E.61. Fix the marked cusp $Q = B_0$ and put

$$P = \Pi, \qquad I = \overline{\langle\langle d_Q \rangle\rangle}, \qquad W = P/\gamma_3 I, \qquad B = P/I, \qquad K = I/\gamma_3 I. \tag{E71.1}$$

Let $v \nmid 3$, $q = |k_v|$, and let $F_Q = z_Q F$ be the actual marked tangential Frobenius. Use E.67's centre $C = \widehat{\bigwedge}^2 \mathbb{Z}_3[[B]]$ and its exact module

$$Z_Q = C/(q^2\theta_Q^{[2]} - 1)C, \qquad \theta_Q = F_Q|_B. \tag{E71.2}$$

E.70 gives its finite orbit presentations and explicit choices of coordinates identifying it with $\prod_{n\geq 1} \mathbb{Z}_3$.

The entire ramified local W-section space, modulo W, has the following homeomorphism:

$$\begin{aligned}\mathrm{Sect}_{\mathrm{ram}}(W)/W \simeq\ & ((\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1) \times Z_Q) \\ & \sqcup \coprod_{i \neq Q} ((\mathbb{Z}_3 \setminus \{0\}) \times \mathbb{Z}_3/(q-1))\,.\end{aligned} \tag{E71.3}$$

The first summand is the actual marked-cusp family of E.67: $t \mapsto d_Q^a t$, $F \mapsto d_Q^b c_w F_Q$. In each of the eight other summands the representative is

$$t \longmapsto d_i^a t, \qquad F \longmapsto d_i^b F_i, \qquad a \neq 0. \tag{E71.4}$$

Every class in (E71.4) has exactly one full P-section class above it. In the first summand of (E71.3), the full-source image is exactly $w = 0$, with exactly one full class above each such point. All other points have empty full fibre.

The unramified W-classes and every full fibre over them are exactly E.68.3 and E.68.6 for $R = \gamma_3 I$. Thus these results compute every local W-class and every fibre of

$$\mathrm{Sect}(P)/P \longrightarrow \mathrm{Sect}(W)/W \tag{E71.5}$$

at every K-place away from three. Formula (E71.3) describes the ramified subspace; the unramified subspace retains its subspace topology in the complete section space.

There is a stronger framed assertion underlying (E71.4). Fix an actual B-cusp section at $i \neq Q$, including its representative and its nonzero exponent a. Its actual lift to W in (E71.4) makes K a group with a continuous local action. Every K-valued cocycle for that action is conjugate to the zero cocycle by exactly one element of K. The comparison uses the actual path term from Q to P_i; it applies to every nonzero $a \in \mathbb{Z}_3$, including nonunits, and to every $b \in \mathbb{Z}_3$.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d). The wild place, the global compatibility problem and the other primary and mixed finite quotients retain their stated source maps.

Proof. The actual filled curve and its ramified source. Let $X = C_{k_v} \setminus \{P_i : i \neq Q\}$; in this paragraph C_{k_v} denotes the smooth compact Fermat cubic. The geometric group of X is B, free pro-three of rank nine. The map $P \to B$ is precisely the filling map for the marked cusp, by E.41 and the specialization in E.61. It carries all other peripheral generators to their actual positive peripheral generators, denoted c_i. Its base point at Q becomes an ordinary rational point.

The ramified-source argument of E.61 applies to X with the following same-property comparison, which we spell out. Its eight rational cusp vectors have the single diagonal relation, and any two are independent. Eliminating a third cusp in the genus-one presentation includes any chosen pair in a free basis. The free-factor centralizer and intersection statements used in E.61 therefore hold for these eight inertia subgroups in B [HR85, Theorems A' and B'].

For a nonzero inertia element $\nu \in B$ normalized by $\psi = zF$, put $J = \overline{\langle \nu \rangle}$. Suppose that J lies in no conjugate of those cusp groups. The compact-separation construction of E.61 chooses a characteristic open $N_0 \subset B$ for which the image of $J \setminus J^3$ is disjoint from every cusp image. The actual subgroup $H = N_0 J$ is ψ-stable, and

$$H \longrightarrow J/(J \cap N_0) \longrightarrow J/J^3 \simeq \mathbb{F}_3 \tag{E71.6}$$

is nonzero on J and kills every cusp subgroup of H. The arithmetic subgroup defined by H and ψ gives an actual finite cover $X' \to X$. Filling its cusps gives the equivariant map $H \to T_3 \,\mathrm{Jac}(\overline{X}')$ through which (E71.6) factors. Thus ν has nonzero image in this free integral

Tate module. The relation $\psi(\nu) = \nu^q$ makes that image a q-eigenvector. Every eigenvalue of this compact Jacobian Frobenius has complex absolute value $\sqrt{q}$, by the same specialization and Tate-module maps already checked in E.61 [Mil08, II, Theorem1.1(b); III, Proposition9.1]. This is a contradiction. This proves the required cuspidality for this actual filled curve.

Its free-factor centralizer is $C_B(c_i^a) = \langle c_i \rangle$ for every $a \neq 0$. Comparing with the actual lift F_i now gives exactly $c_i^b F_i$, and conjugating by c_i^k changes b by $(1-q)k$. Distinct cusps and distinct nonzero exponents remain distinct in the integral abelianization. This proves that the ramified B-section space consists of its eight actual cusp families. Each has the displayed full P lift, so the comparison needed below is onto.

The exact affine action on the marked kernel. Set $D = I/I' = \mathbb{Z}_3[[B]]$; E.67 supplies the actual universal maps and the compact central sequence

$$0 \longrightarrow C \longrightarrow K \longrightarrow D \longrightarrow 0. \tag{E71.7}$$

For $i \neq Q$, put $b_i = \overline{z_i z_Q^{-1}} \in B$. Since $F_i = (z_i z_Q^{-1}) F_Q$, the actual action of F_i on D is q times the affine permutation $g \mapsto b_i \theta_Q(g)$. For the reference section (E71.4), its inertia action on D is the permutation $T_0 : g \mapsto c_i^a g$, and its Frobenius action is qT, where

$$T(g) = c_i^b b_i \theta_Q(g). \tag{E71.8}$$

Its actions on C are the corresponding diagonal permutation $T_0^{[2]}$ and $q^2 T^{[2]}$. These formulas follow directly on the generators $g d_Q g^{-1}$ of I and on their commutators. Possible central terms in the action on K are retained by (E71.7); they do not change these two induced actions.

Put $H_a = \langle c_i^a \rangle$. The relation $T T_0 T^{-1} = T_0^q$ shows that T acts on the compact left coset space $H_a \backslash B$ and on $H_a \backslash (B \times B)$, with diagonal left translation in the latter. The closure of this permutation action is pro-three: it comes from the actual pro-three semidirect product of the geometric group with the local arithmetic quotient of E.61. All its finite permutation orbits therefore have three-power length.

No periodic coset: the actual compact Kummer map. Use Q as origin on the compact elliptic curve E. Filling all the remaining cusps gives the actual equivariant surjection

$$p : B \longrightarrow T_3 E. \tag{E71.9}$$

Every $p(c_i)$ is zero. The nine original coordinate cusps are the nine distinct three-torsion points on E, also after good specialization. Indeed each coordinate cusp is a flex, whose tangent cuts out three times that point. Comparing with the tangent at Q gives $3(P_i - Q) = 0$, and distinct points on the genus-one curve have distinct Abel–Jacobi images. The disjoint integral sections of E.61 preserve this nonzero three-torsion at every residue characteristic other than three.

With the path direction used for $F_i = z_i F$, the cocycle $p(b_i)$ is the Kummer cocycle of $P_i - Q$. This assertion follows on every actual division cover $[3^n] : E \to E$: transporting the zero fibre from Q to P_i chooses a compatible division point R_n with $[3^n]R_n = P_i - Q$, and the Frobenius translation is $F_E(R_n) - R_n$. Changing the chosen path adds the corresponding Tate-module coboundary. Thus the comparison retains both its actual representative and its precisely specified ambiguity.

Suppose $H_a g$ were periodic for T, of period $m > 0$. Then $T^m(g) = c_i^{ak} g$ for some $k \in \mathbb{Z}_3$. Projecting by (E71.9) gives

$$\sum_{j=0}^{m-1} F_E^j p(b_i) = (1 - F_E^m) p(g). \tag{E71.10}$$

The terms from c_i^b vanish under p. Equation (E71.10) would make the restriction of that actual Kummer cocycle to $k_{v,m} = \mathbb{F}_{q^m}$ a coboundary. More precisely, (E71.10) makes $R_n + p(g)_n$

fixed by F_E^m, where $p(g)_n$ is the 3^n-torsion coordinate of the actual Tate element $p(g)$. These are compatible $k_{v,m}$-rational divisions of $P_i - Q$ at every level. The finite group $E(k_{v,m})$ has no nonzero three-primary element divisible by arbitrarily high powers of three. Since $P_i - Q$ is nonzero of order three, this is impossible. Hence T has no periodic point on $H_a\backslash B$. This proof includes every nonunit a and every b; it uses the actual affine term b_i in (E71.8).

The compact measure and coinvariant maps. For any continuous permutation with pro-three closure on a compact profinite space and with no periodic points, its invariant integral measures are zero. Here is the exact trace argument needed. Choose a cofinal system of invariant finite partitions. Bounded orbit lengths along all refinements would, by compactness, produce a point fixed by some power T^{3^e}. Thus the minimum orbit length tends to infinity. At a fixed coarse orbit, refining to orbit lengths divisible by 3^n times its length makes the coarse coefficient of every invariant measure divisible by 3^n. Every such coefficient is therefore zero.

The actual augmentation to cosets gives

$$\begin{aligned} D/(T_0-1)D &\simeq \mathbb{Z}_3[[H_a\backslash B]], \\ C/(T_0^{[2]}-1)C &\simeq e^-\mathbb{Z}_3[[H_a\backslash(B\times B)]], \qquad e^- = (1-\mathrm{swap})/2. \end{aligned} \tag{E71.11}$$

For the first map, δ_g goes to $\delta_{H_a g}$. For the second, use the alternating-measure embedding of E.67, then pass to diagonal orbits. The idempotent e^- commutes with diagonal translation and with all quotient maps. At a finite quotient these are the ordinary exact coinvariant maps of finite permutation modules. Passing to the compact inverse limits is exact: the finite systems of lift equations have the finite intersection property. The images of $T_0 - 1$ and $T_0^{[2]} - 1$ are already compact and closed. This proves (E71.11), without choosing a coset transversal.

The no-periodic-point result and the trace argument show

$$(D/(T_0-1)D)^T = 0. \tag{E71.12}$$

Also $D^{T_0} = 0$ and $C^{T_0^{[2]}} = 0$. For these two assertions, left translation by c_i^a acts freely on B and diagonally on $B \times B$, and its finite orbit lengths tend to infinity. The same trace argument applies; C is an invariant direct summand of the second measure module. The assertions also hold for T_0^q, which generates the same closed translation group.

Finally, on the second coinvariant module of (E71.11) the operator $qT^{[2]} - 1$ is injective. In each finite integral orbit module $T^{[2]}$ is a permutation P of finite order N. A vector in its kernel is killed by $q^N - 1$. That nonzero integer acts injectively on the free $\mathbb{Z}_3$-module, and hence on its alternating direct summand. The compact limit therefore has zero kernel as well.

Eliminating the entire cocycle, including its central part. Let a K-valued cocycle for the reference local action have inertia and Frobenius values whose images in D are x, u. The tame relation gives the exact additive equation

$$(qT - N_q(T_0))x = (T_0^q - 1)u, \qquad N_q(T_0) = 1 + T_0 + \cdots + T_0^{q-1}. \tag{E71.13}$$

Modulo $(T_0 - 1)D$ this reads $q(T-1)\bar{x} = 0$. Since q is a three-adic unit, (E71.12) gives $x = (T_0 - 1)v$ for one and only one $v \in D$. Lift v to an actual element of K and conjugate the section by it. This replaces x by $x - (T_0 - 1)v = 0$. The new equation gives $(T_0^q - 1)u = 0$, so the new Frobenius image in D is zero. The entire new cocycle consequently takes values in C.

Write these central values as x_C, u_C. Their exact relation is

$$(q^2T^{[2]} - N_q(T_0^{[2]}))x_C = ((T_0^{[2]})^q - 1)u_C. \tag{E71.14}$$

On its inertia coinvariant module the equation is $q(qT^{[2]} - 1)\bar{x}_C = 0$. The proved injectivity gives $x_C = (T_0^{[2]} - 1)v_C$ for a unique $v_C \in C$. Conjugating by v_C kills this inertia value; the

remaining Frobenius value is zero because $C^{(T_0^{[2]})^q} = 0$. Thus every cocycle is conjugate to zero. An element fixing the zero cocycle has T_0-fixed image in D, hence has image zero there; its remaining central value is also $T_0^{[2]}$-fixed and is zero. This proves uniqueness of the full conjugator. All higher central corrections in the original action on K have been handled by the actual first conjugation and the subsequent central equation; none was replaced by a graded action on the whole group.

Every geometric orbit and every full fibre. Take a W-section whose inertia has nontrivial image in B. Normalize its projected section to its unique cusp parameters $(i, a, \bar{b})$. Lift its conjugator through $W \to B$. The remaining section is a K-valued cocycle over the actual reference (E71.4), so the preceding argument conjugates it to that reference. Conversely two references equivalent in W have equivalent B images. The B classification forces the same cusp, the same exponent, and exactly the stated change of b; that change is realized in W by the actual lift d_i^k.

The other ramified W-sections have inertia image one in B, so E.67 gives exactly the first summand of (E71.3), including its full-source image. The map $P \to W$ preserves the full abelianization. E.61 therefore shows that an unramified target can have only unramified full preimages, and that every class in (E71.4) has precisely its displayed full cusp class above it. This proves all set-theoretic assertions and the framed assertion.

For topology, each fixed nonzero valuation annulus for a has compact parameter space, and its actual representative map to the Hausdorff geometric orbit space is continuous and bijective. It is therefore a homeomorphism. Such annuli are open in and cover the ramified space, as detected by the valuation of its nonzero abelian inertia vector. The nonzero cusp vectors in the free integral abelianization distinguish its nine branches; their intersections occur only at zero, outside this ramified space. This gives the homeomorphism (E71.3). The unramified topology and all its full fibres are those already proved with the actual unique conjugators in E.68. The retained program independently enumerates the entire projective cubic over $\mathbb{F}_4$ and checks the literal tangent polynomial at all nine flexes. Using the actual compact Frobenius -2 of E.37, it also checks the Kummer obstruction for every one of the eight nonzero three-torsion coordinates over all residue extension degrees from one through 81. All 648 nonzero residues lie outside the actual Frobenius coboundary image; the argument above proves that assertion for every positive degree. This completes (E71.5). □

All complex divisor coordinates and exact norm compatibility

E.72. Surviving prime chains, principal-class equations, and saturation of the positional norm system [Proved]

Keep the actual fields and roots

$$\begin{gathered} K = \mathbb{Q}(\zeta), \qquad L_d = K(\beta_d), \quad \beta_d^{3^d} = 17\zeta, \qquad F_d = \mathbb{Q}(\alpha_d), \quad \alpha_d^{3^d} = 17, \\ \beta_{d+1}^3 = \beta_d, \qquad \alpha_{d+1}^3 = \alpha_d. \end{gathered} \tag{E72.1}$$

Thus $L_0 = K$ and $F_0 = \mathbb{Q}$. For a number field E, write $\widehat{E^\times}_3 = \varprojlim_r E^\times/(E^\times)^{3^r}$ and

$$\widehat{\mathrm{Div}(E)}_3 = \varprojlim_r \bigoplus_{\mathfrak{p}} \mathbb{Z}/3^r, \qquad \mathscr{R}(J) = \{(a_j) \in \prod_{j\in J} \mathbb{Z}_3 : a_j \longrightarrow 0 \pmod{3^r} \text{ for every } r\}. \tag{E72.2}$$

These divisor groups carry their displayed prodiscrete topology. In particular, the infinite restricted sequence factor is not asserted compact.

Here is the entire complex divisor norm limit. For each rational prime $p \neq 3, 17$ such that $v_3(\mathrm{ord}_p(17)) = 1$, there is exactly one prime $\mathfrak{p} \mid p$ of K for which the residue of 17ζ has order

prime to three. Call the set of these primes $\mathcal{P}_1$. Each p is split in K. For $\mathfrak{p} \in \mathcal{P}_1$, let $\mathfrak{P}_{d,\mathfrak{p}}$ be the degree-one prime of L_d specified by

$$\beta_d \equiv r_{d,\mathfrak{p}} \pmod{\mathfrak{P}_{d,\mathfrak{p}}}, \qquad r_{d,\mathfrak{p}}^{3^d} = \overline{17\zeta}, \qquad r_{d,\mathfrak{p}} \text{ has order prime to three.} \tag{E72.3}$$

The root in (E72.3) is unique in the prime-to-three subgroup of $\mathbb{F}_p^\times$; its cube is the corresponding root at the preceding level. Let $\mathfrak{q}_{3,d}$ and $\mathfrak{q}_{17,d}$ be the unique primes of L_d above 3 and 17. The explicit map

$$\begin{aligned} \mathcal{D} = \mathbb{Z}_3^2 \oplus \mathscr{R}(\mathcal{P}_1) \;&\xrightarrow{\sim}\; \varprojlim_{d,N} \widehat{\mathrm{Div}(L_d)}_3, \\ a = (a_3, a_{17}, (a_\mathfrak{p})) \;&\longmapsto\; D_d(a) = a_3\mathfrak{q}_{3,d} + a_{17}\mathfrak{q}_{17,d} + \sum_{\mathfrak{p}\in\mathcal{P}_1} a_\mathfrak{p}\mathfrak{P}_{d,\mathfrak{p}} \end{aligned} \tag{E72.4}$$

is a topological isomorphism. Its inverse reads the coefficients of the divisor at K. No other prime chain contributes a nonzero compatible completed divisor. The further rational norm sends these coordinates to

$$a_3[3] + 2a_{17}[17] + \sum_{\mathfrak{p}\in\mathcal{P}_1} a_\mathfrak{p}[p]. \tag{E72.5}$$

Consequently this rational divisor map is injective.

For every d, let $C_d = \mathrm{Cl}(L_d)\{3\}$, and use the actual classes of the displayed ideals to define

$$\begin{aligned} \mathcal{D}_{\mathrm{pr}} &= \left\{ a \in \mathcal{D} : a_3[\mathfrak{q}_{3,d}] + \sum_{\mathfrak{p}\in\mathcal{P}_1} a_\mathfrak{p}[\mathfrak{P}_{d,\mathfrak{p}}] = 0 \text{ in } C_d \text{ for every } d \right\}, \\ \mathcal{X} &= \varprojlim_{d,N} \widehat{L_d^\times}_3, \qquad \mathcal{E}_\infty = \varprojlim_{d,N} (\mathcal{O}_{L_d}^\times \otimes \mathbb{Z}_3). \end{aligned} \tag{E72.6}$$

The term at 17 is absent from the class equation because $(\beta_d) = \mathfrak{q}_{17,d}$. The restricted sums in the finite groups C_d are well defined. These are literal principal-class equations in the fields $X^{2\cdot 3^d} + 17X^{3^d} + 289$, with the ideals specified in (E72.3)–(E72.4).

The entire completed multiplicative norm system fits into the exact sequence

$$0 \longrightarrow \mathcal{E}_\infty \longrightarrow \mathcal{X} \xrightarrow{\mathrm{div}} \mathcal{D}_{\mathrm{pr}} \longrightarrow 0, \qquad \mathcal{E}_\infty \simeq \prod_{d\ge 0} \mathbb{Z}_3^{2\cdot 3^d}. \tag{E72.7}$$

In particular the class equations in (E72.6) are sufficient for one coherent multiplicative lift, including compatibility at all levels. Every nonempty fiber of the divisor map is a compact torsor under the displayed product. Its product coordinates are the actual recursive norm-kernel coordinates of E.69.

The base element, including its root-of-unity factor, has a finite-level test. For $a \in \mathcal{D}_{\mathrm{pr}}$, choose a completed principal generator g_1 with $\mathrm{div}(g_1) = D_1(a)$, using E.57's specified ideal coordinates, and put

$$\nu(a) = N_{L_1/K}(g_1). \tag{E72.8}$$

This is independent of the generator, because the completed ordinary-unit norm from L_1 is 1. It is an injective map with divisor $D_0(a)$, and the base projection of (E72.7) is exactly $\nu \circ \mathrm{div}$. Thus, for an actual $z \in \widehat{K^\times}_3$, membership in $\mathrm{pr}_0(\mathcal{X})$ is equivalent to the following three explicit requirements: its divisor has the support in (E72.4); that divisor satisfies every class equation (E72.6); and $z = \nu(\mathrm{div}\, z)$. The last equality tests the remaining μ_3 factor by E.57's norm formulas. The higher class matrices and the integer h of E.69 remain unevaluated arithmetic data; this statement is an exact membership theorem, not a numerical evaluation of those matrices.

The prime above three gives a particularly precise interpretation of that remaining integer. If $N_{L_d/K}\mathcal{U}_d = \lambda^{3^{a_d}\mathbb{Z}_3}$ as in E.69, then

$$\operatorname{ord}[\mathfrak{q}_{3,d}] = 3^{a_d}, \qquad (\beta_1 + 3\zeta + 1)\mathcal{O}_{L_d} = \mathfrak{q}_{3,d}^{3^{d-1}} \quad (d \geq 1). \tag{E72.9}$$

The order here is the order in the full ideal class group; it is a power of three. In particular $[\mathfrak{q}_{3,2}]$ has exact order three. The stable integer h is the eventual exponent of these specified prime classes. There is no assertion that equality at two adjacent levels proves stabilization.

There are also exact saturation and positional comparison maps. Every member of $\mathcal{X}$ whose rational base norm is supported on $\{3, 17\}$ consists of $\{3, 17\}$-units at every level. The entire such subgroup is

$$\begin{aligned} \mathcal{X}_{\{3,17\}} &= \{(\beta_d^b u_d)_d : b \in \mathbb{Z}_3,\ (u_d) \in \mathcal{U}_\infty\}, \\ \operatorname{pr}_0(\beta^b u) &= (17\zeta)^b \lambda^{3^h z}, \qquad N_{K/\mathbb{Q}} \operatorname{pr}_0(\beta^b u) = 17^{2b} 3^{3^h z}. \end{aligned} \tag{E72.10}$$

The inverse removes the common valuation b at the unique prime above 17. In particular rational norm supported only at 3 forces $b = 0$ and gives exactly the whole family of E.69. For positional tuples with pure component 1, the total-norm cube condition is $b \in 3\mathbb{Z}_3$. Writing $b = 3r$, their scaled toric coordinates from E.46 are

$$\Big(\kappa_3(17),\ 2r\,\kappa_3(17) + 3^{h-1} z\,\kappa_3(3)\Big). \tag{E72.11}$$

For the full positional algebra $R_{d+1} = F_d \times L_d$, put $\mathcal{P}_0 = \{p \neq 3, 17 : v_3(\operatorname{ord}_p 17) = 0\}$. The pure divisor limit away $3, 17$ has exactly one degree-one prime chain for each $p \in \mathcal{P}_0$. Its coefficient norm is the identity. The sets $\mathcal{P}_0$ and the rational primes underlying $\mathcal{P}_1$ are disjoint. At 3 the pure chain has two persistent branches, both of residue degree one; the complex chain has one. At 17 the pure and complex chains have residue degrees one and two. In these actual branch coordinates the rational divisor norm is

prime	norm on the compatible coefficients
3	$(b_3, c_3, a_3) \longmapsto b_3 + c_3 + a_3$
17	$(e_{17}, f_{17}) \longmapsto e_{17} + 2f_{17}$
$p \in \mathcal{P}_0 \cup \mathcal{P}_1$	$a_p \longmapsto a_p$.

(E72.12)

Its kernel is exactly $\mathbb{Z}_3^3$, supported at 3 and 17. Therefore every full positional norm tuple whose scaled second toric coordinate is supported on $\{3, 17\}$ consists of actual $\{3, 17\}$-unit components in both field towers. This is the map to the quotient H of E.46; it makes no assertion of a lift through the intrinsic metabelian or marked comparison kernels. The manuscript anchor is §51.16, fols. 733v–736v, scans 172–178, through the actual covering and toric maps of E.25, E.27 and E.46.

Proof. First fix a prime $\mathfrak{p} \nmid 3 \cdot 17$ of K, with residue size q. The roots $\beta_d \mu_{3^d}$ identify the root set with $\mathbb{Z}/3^d$. The actual arithmetic Frobenius acts by

$$x \longmapsto qx + B \pmod{3^d}, \tag{E72.13}$$

where the compatible $B \in \mathbb{Z}_3$ is the local Kummer cocycle of 17ζ at the chosen Frobenius. A different root system changes B by $(q-1)t$, transporting all the following prime chains by the same translation. Frobenius orbits are precisely the primes above $\mathfrak{p}$, and their lengths are their relative residue degrees. If a higher orbit lies over a lower orbit, the ideal norm multiplies its coefficient by the ratio of those lengths.

The affine transformation in (E72.13) has an integral fixed point exactly when $-B/(q-1) \in \mathbb{Z}_3$. This is equivalent to the residue of 17ζ belonging to the prime-to-three subgroup of $\mathbb{F}_q^\times$: the

local principal-unit group has bijective cubing, and the finite residue group supplies precisely the remaining Kummer class. When it exists, the fixed point is unique, since the integer $q > 1$ is not 1 in $\mathbb{Z}_3$.

Every other orbit has unbounded three-part length. For a direct verification, write the fixed point in $\mathbb{Q}_3$ as $x_* = -B/(q-1)$. Iteration gives $\phi^j(x) - x = (q^j - 1)(x - x_*)$. Since $q \equiv 1 \pmod 3$, $v_3(q^{3^k} - 1) = v_3(q-1) + k$. On a fixed clopen residue class which does not contain an integral fixed point, $v_3(x - x_*)$ is bounded. Thus the orbit lengths of every refinement have three-parts tending uniformly to infinity. The norm coefficient of a compatible divisor on that lower orbit is divisible by every power of three and is zero. This reasoning uses a finite sum at each higher level, so it also applies to completed divisors.

Only the singleton fixed-point branch can remain. Its residue degrees are all one and its norm coefficient is constant. Conversely any coefficient in $\mathbb{Z}_3$ defines a compatible divisor on that branch. This proves the asserted local inverse limit, together with its literal coefficient map.

For an inert rational $p \neq 3, 17$, the element 17 lies in $\mathbb{F}_p^\times$, of order prime to three, whereas ζ has order three in $\mathbb{F}_{p^2}^\times$. There is no complex fixed branch. For split p, the two residues of the radicand are $17\zeta_{\mathfrak{p}}$ and $17\zeta_{\mathfrak{p}}^{-1}$. Their ratio has order three, so at most one has order prime to three. Such a residue exists precisely when the three-primary component of 17 has exact order three; its inverse determines the required value of $\zeta_{\mathfrak{p}}$. This proves both the stated set $\mathcal{P}_1$ and formula (E72.3).

At 3 and 17 the complex tower has the unique primes and total ramification established in E.64; their relative residue degrees are one. The completed divisor coefficients at all other primes have just been determined. At each level the condition that they tend to zero modulo every 3^r is exactly the same condition on the surviving base coefficients. This proves (E72.4) as an isomorphism of the specified prodiscrete groups. The residue degrees from K to $\mathbb{Q}$ are respectively $1, 2, 1$, giving (E72.5) and its injectivity.

For clarity, the completion and the inverse limit in (E72.6) are handled separately. For any fixed number field E, the discrete divisor sequence gives

$$0 \longrightarrow \mathcal{O}_E^\times \otimes \mathbb{Z}_3 \longrightarrow \widehat{E^\times}_3 \xrightarrow{\operatorname{div}} \widehat{\operatorname{Div}(E)}_3 \longrightarrow \operatorname{Cl}(E)\{3\} \longrightarrow 0. \tag{E72.14}$$

Indeed the group of principal divisors is a subgroup of the free divisor group and is itself free. The sequence defining it therefore splits as a sequence of abstract abelian groups before completion. Its induced topology inside the divisor group is the multiplication topology: if the three-primary class group has exponent 3^e and a principal divisor is $3^{r+e}D$, the prime-to-three part of $[D]$ is already zero, and $3^e[D] = 0$. Hence the intersection with $3^{r+e}\operatorname{Div}(E)$ is contained in 3^r times the principal-divisor group. This proves the exact image and the stated topology in (E72.14). The kernel is compact because the ordinary-unit group is finitely generated.

Necessity of every class equation is immediate from (E72.14). For sufficiency, fix a satisfying all of them. At each d let T_d be the fiber of the divisor map in (E72.14) over $D_d(a)$. It is a nonempty compact torsor under the ordinary-unit completion. Norms send T_{d+1} to T_d. In the compact product $\prod_d T_d$, impose the closed equations $N(x_{d+1}) = x_d$. Every finite collection of them has a solution: choose an element in the highest required fiber, norm it down to all preceding fibers, and choose arbitrary elements in the other fibers. The finite intersection property gives a simultaneous solution. This proves the surjectivity in (E72.7), without assuming surjectivity of any individual ordinary-unit norm or compactness of the whole multiplicative completion.

The kernel is precisely $\mathcal{E}_\infty$. An ordinary-unit norm tuple is a 3-unit tuple in E.69. Its base lies in μ_3 and also in $I_0 = \lambda^{3^h\mathbb{Z}_3}$, so it is 1. Conversely, in a coherent 3-unit tuple with base 1, the coefficient at the unique prime above 3 is constant under norm and therefore zero at every

level. Such a tuple consists of ordinary units. Consequently $\mathcal{E}_\infty = \ker(\mathcal{U}_\infty \to I_0)$, and E.69 gives exactly the product in (E72.7) with its recursive coordinates. Translation by any one lift proves the asserted compact topology and coordinates on each fiber.

Any two choices of g_1 in (E72.8) differ by a completed ordinary unit. E.57 computes its norm as 1, including the torsion generator. Thus ν is well defined. Every coherent lift of a has first component in this same fiber, so its base is $\nu(a)$. Conversely every a in $\mathcal{D}_{\mathrm{pr}}$ has such a lift, already proved. Its divisor at K is exactly the base coordinate a. This proves injectivity, the entire base-image criterion, and the claimed finite test for the root factor. E.57's principal generators and norm matrix specify the calculation; no assertion that the divisor alone determines an arbitrary element of $\widehat{K^\times}_3$ has been used.

Let m_d be the full order of $[\mathfrak{q}_{3,d}]$. The included element $c_1 = \beta_1 + 3\zeta + 1$ has ideal $\mathfrak{q}_{3,d}^{3^{d-1}}$, since its ideal at L_1 is $\mathfrak{q}_{3,1}$ and every adjacent extension is totally ramified. Hence m_d is a power of three. The image of the valuation $\mathcal{O}_{L_d}[1/3]^\times \to \mathbb{Z}$ is exactly $m_d\mathbb{Z}$: this is the definition of the order of that ideal class. Its completed image is $m_d\mathbb{Z}_3$. The relative residue degree at 3 is one, and E.69 places every total norm in $\lambda^{\mathbb{Z}_3}$. Reading its λ-valuation therefore gives $m_d = 3^{a_d}$ and (E72.9). E.69 supplies $a_2 = 1$ and eventual equality $a_d = h$.

If the rational base norm of a coherent complex tuple has no valuation outside $\{3, 17\}$, the injectivity and explicit support in (E72.5) force every other compatible divisor coefficient to vanish. Exactness in (E72.14), or its version with these two primes inverted, then places each component in the completed $\{3, 17\}$-unit group. Its common coefficient b at 17 is removed by the literal coherent tuple $(\beta_d^b)_d$, leaving exactly a member of $\mathcal{U}_\infty$. This proves (E72.10). The rational norm is as displayed because $N_{K/\mathbb{Q}}\lambda = 3$ and $N_{K/\mathbb{Q}}(17\zeta) = 17^2$. The cube condition and the division by three in E.46 then give (E72.11).

For the pure comparison use the actual root set $\alpha_d\mu_{3^d}$ over $\mathbb{Q}$ at $p \neq 3, 17$. Its affine Frobenius is $x \mapsto px + B$. If $p \equiv 1 \pmod 3$, the argument above applies unchanged, with radicand 17 and fixed branch exactly when $v_3(\mathrm{ord}_p\, 17) = 0$. If $p \equiv 2 \pmod 3$, $p - 1$ is a unit in $\mathbb{Z}_3$, so there is always a unique integral fixed point. For the other orbits apply the same length argument to Frobenius squared: $p^2 \equiv 1 \pmod 3$ and $v_3(p^{2\cdot 3^k} - 1) = v_3(p^2 - 1) + k$. This proves the pure assertion and disjointness of the surviving prime sets.

Finally the local factors at 3 are the actual factors of E.65. If $\rho^3 = 17$ is the root in $\mathbb{Q}_3$ and $m = 3^{d-1}$, then for $d \geq 1$

$$X^{3^d} - 17 = (X^m - \rho)(X^{2m} + \rho X^m + \rho^2). \tag{E72.15}$$

Both factors become Eisenstein at 3 after $X \mapsto X - 1$. Cubing carries each factor to its predecessor, and each successive extension on either branch is totally ramified of degree three. Both residue degrees are one, including their maps to $\mathbb{Q}_3$. At 17, the pure polynomial is Eisenstein with residue degree one, whereas the complex branch has the residue degree two already used above. This proves (E72.12). A basis of its kernel is $(1, -1, 0)$ and $(1, 0, -1)$ at 3, and $(-2, 1)$ at 17. There is no kernel away these primes. The total-norm map of E.46 is precisely this rational norm on divisor coordinates, so the final saturation assertion follows.

There is a concrete finite obstruction to prolonging the first norm correction in E.57. At $\pi = 2 - \zeta$ above 7, the roots of $\beta_1^3 = 17\zeta$ are $3, 5, 6$ in $\mathbb{F}_7$. The only root of order prime to three is $6 = -1$. The two other primes are inert cubic in L_2/L_1, since 3 and 5 are not cubes in $\mathbb{F}_7^\times$. E.57's element $w = (g_1/\pi)^2$ has the actual divisor

$$\mathrm{div}(w) = 4\mathfrak{p}_{\beta=3} - 2\mathfrak{p}_{\beta=6} - 2\mathfrak{p}_{\beta=5}. \tag{E72.16}$$

The coefficients at the two inert primes are not divisible by three. Consequently w is not a norm from $\widehat{L_2^\times}_3$, even though its first norm is the root ζ. The obstruction is the actual

ideal-norm map at 7; it requires no inference from a nonprincipal ideal or an uncomputed higher class matrix.

The retained finite certificate compares the actual factorizations of both defining polynomials with the affine-Frobenius orbit formulas. It also checks the weighted norm transition matrices, the eligible-prime sets, and the displayed saturated kernel. These finite checks support the formulas; the uniform orbit-length proof above supplies the all-level assertion. □

The actual global presentation and the coupled local packets

E.73. The complete relation set for global local-packet gluing [Proved]

Put $K = \mathbb{Q}(\zeta_3)$, $S = \{(1-\zeta_3)\}$ and

$$\mathcal{G} = G_K^{(3)}, \qquad P = G_{K,S}^{(3)} = \langle \tau, \tau' \rangle, \qquad L = \ker(\mathcal{G} \to P). \tag{E73.1}$$

Use the actual splitting $s : P \to \mathcal{G}$ of E.39. For each finite prime $\mathfrak{q} \notin S$, choose an actual prolongation, a positive tame generator $t_\mathfrak{q}$ and an arithmetic Frobenius $F_\mathfrak{q}$ in its decomposition group. Write $q = N\mathfrak{q}$. Let $\mathcal{F}$ be the free pro-three group on τ, τ' and the pointed convergent family $(T_\mathfrak{q})_{\mathfrak{q}\notin S}$, with $T_\mathfrak{q} \to 1$. The continuous map

$$\pi : \mathcal{F} \longrightarrow \mathcal{G}, \qquad \tau \mapsto s(\tau), \quad \tau' \mapsto s(\tau'), \quad T_\mathfrak{q} \mapsto t_\mathfrak{q} \tag{E73.2}$$

is a minimal epimorphism. Choose actual preimages $Y_\mathfrak{q} \in \mathcal{F}$ with $\pi(Y_\mathfrak{q}) = F_\mathfrak{q}$. Then its entire kernel is

$$\ker \pi = \overline{\langle\langle Y_\mathfrak{q} T_\mathfrak{q} Y_\mathfrak{q}^{-1} T_\mathfrak{q}^{-q} : \mathfrak{q} \notin S \rangle\rangle}. \tag{E73.3}$$

This is a presentation with actual pro-three words $Y_\mathfrak{q}$, not prescribed independent Frobenius symbols. Its cohomology and relation module have the exact topologies

$$\begin{aligned} &H^2_{\mathrm{cts}}(\mathcal{G}, \mathbb{F}_3) \xrightarrow[\text{restriction}]{\sim} \bigoplus_{\mathfrak{q}\notin S} H^2_{\mathrm{cts}}(D_\mathfrak{q}, \mathbb{F}_3) \simeq \bigoplus_{\mathfrak{q}\notin S} \mathbb{F}_3, \\ &R/\overline{R^3[R,\mathcal{F}]} \simeq \prod_{\mathfrak{q}\notin S} \mathbb{F}_3, \qquad R = \ker \pi. \end{aligned} \tag{E73.4}$$

The relators displayed in (E73.3) are the coordinate vectors of the compact product in (E73.4). No product replaces the continuous-cohomology direct sum.

Let V be one of the actual geometric pro-three groups Π, Π/Π'', or $\Pi/\gamma_3 I_Q$, with its given based action $\rho : P \to \mathrm{Aut}(V)$. Define

$$c_0(\mathcal{P}_K \setminus S, V) = \{(n_\mathfrak{q}) : n_\mathfrak{q} \to 1\}. \tag{E73.5}$$

For $A, A' \in V$ and such a family, the universal property gives the unique homomorphism

$$\widetilde{a} : \mathcal{F} \longrightarrow V \rtimes \mathcal{G}, \quad \tau \mapsto (A, s\tau), \quad \tau' \mapsto (A', s\tau'), \quad T_\mathfrak{q} \mapsto (n_\mathfrak{q}, t_\mathfrak{q}). \tag{E73.6}$$

Define $v_\mathfrak{q}$ by the actual word evaluation $\widetilde{a}(Y_\mathfrak{q}) = (v_\mathfrak{q}, F_\mathfrak{q})$. These data give a global section exactly when, for every $\mathfrak{q} \notin S$,

$$v_\mathfrak{q}\, \rho(\sigma_\mathfrak{q})(n_\mathfrak{q}) v_\mathfrak{q}^{-1} = n_\mathfrak{q}^q, \qquad \sigma_\mathfrak{q} = \mathrm{pr}(F_\mathfrak{q}). \tag{E73.7}$$

Thus the actual local equations are sufficient as well as necessary; there is no further relation after all the evaluated words in (E73.7) have been retained. Their local solution packets are

exactly E.61, E.66, E.68 and E.71. In particular a ramified full or metabelian packet has, for one of the nine actual cusps, the form

$$\begin{aligned} n_{\mathfrak{q}} &= g_{\mathfrak{q}} d_i^{a} g_{\mathfrak{q}}^{-1}, \\ v_{\mathfrak{q}} &= g_{\mathfrak{q}} d_i^{b} z_{i,\mathfrak{q}} \rho(\sigma_{\mathfrak{q}})(g_{\mathfrak{q}})^{-1}, \qquad a \neq 0, \end{aligned} \tag{E73.8}$$

where $F_{i,\mathfrak{q}} = z_{i,\mathfrak{q}} F_{\mathfrak{q}}$ is the actual tangential lift. For the marked quotient, the Q packet additionally has the central coordinate of E.67 and E.70. The unramified packet has $n_{\mathfrak{q}} = 1$ and arbitrary $v_{\mathfrak{q}}$.

Let $\mathcal{X}_V$ denote the subset of $V^2 \times c_0(\mathcal{P}_K \setminus S, V)$ defined by (E73.7). Restriction to the displayed generators gives the homeomorphism

$$\mathrm{Sect}_K(V)/V \simeq \mathcal{X}_V/V, \tag{E73.9}$$

with its full conjugation action

$$(A, A', n_{\mathfrak{q}}) \longmapsto (wA\rho(\tau)(w)^{-1},\, wA'\rho(\tau')(w)^{-1},\, wn_{\mathfrak{q}}w^{-1}). \tag{E73.10}$$

The topology on c_0 is uniform congruence at every coordinate, equivalently the inverse limit of the discrete finite-support configuration spaces for the finite quotients of V. The quotient maps from Π to the two approximations induce exactly the componentwise maps in (E73.9), with their word evaluations and local packet maps retained.

For sections over $\mathbb{Q}$, there is an equally exact descent condition. Fix the actual complex conjugation c. Choose words $C_{\mathfrak{q}} \in \mathcal{F}$ lifting $ct_{\mathfrak{q}}c$, tending to one, and take $C_\tau = \tau'$, $C_{\tau'} = \tau$. Write $\widetilde{a}(C_{\mathfrak{q}}) = (a_{\mathfrak{q}}^{c}, ct_{\mathfrak{q}}c)$. After conjugating the value at c to one, the precise additional equations and remaining conjugations are

$$A' = c(A), \qquad a_{\mathfrak{q}}^{c} = c(n_{\mathfrak{q}}) \quad (\mathfrak{q} \notin S), \qquad w \in V^{c}. \tag{E73.11}$$

No chosen inertia prolongation is presumed fixed by c.

Independent local packets do not give a Cartesian product of global parameters. In the actual cubic Kummer quotient of (E.39), the prescribed values on $s(\tau), s(\tau')$ and inertia at two are $(2,1,1) \in \mathbb{F}_3^3$, with zero values on all other inertia generators. They force the value at $(-2+3\zeta)$ over nineteen to be one. Its quotient-section Frobenius value is two. Both values define valid unramified local packets, and only the former extends with those global generator values. This is an actual finite obstruction to omitting the word evaluation in (E73.7), verified by the retained program. The theorem leaves the evaluation of all $Y_{\mathfrak{q}}$, and the solution of the resulting coupled global equations, as the stated arithmetic input; it does not identify that unevaluated system with an independent product of the local classes.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379); Part E(c),(d) and follow-up(4).

Proof. The actual inertia bundle and its support. E.39, using Wingberg's Theorem2.4, gives the continuous free product of the inertia groups over the primes of $K_S(3)$. The preceding definition and Lemma1.1 of that same original give its cohomology with the continuous direct-sum topology [Win07, pp. 4–5, Lemma1.1; pp. 13–14, Theorem2.4]. Here is the specialization, including all support maps. For a chosen prime over $\mathfrak{q}$ let $\Delta_{\mathfrak{q}} \subset P$ be its decomposition subgroup. It is $\mathbb{Z}_3$: the unramified cyclotomic character sends its Frobenius to $q \neq 1$ in $1+3\mathbb{Z}_3$. Its actual local group in $\mathcal{G}$ is the full tame pro-three group

$$D_{\mathfrak{q}} = \langle t_{\mathfrak{q}} \rangle \rtimes \langle F_{\mathfrak{q}} \rangle, \qquad F_{\mathfrak{q}} t_{\mathfrak{q}} F_{\mathfrak{q}}^{-1} = t_{\mathfrak{q}}^{q}. \tag{E73.12}$$

Indeed its inertia injection is supplied by the actual free product; its unramified quotient injects because the displayed cyclotomic character has infinite order.

Put $M = H^1_{\mathrm{cts}}(L, \mathbb{F}_3)$, a discrete P-module. Restriction to the inertia factors gives

$$M = \bigoplus_{\mathfrak{q} \notin S} \mathrm{Coind}_{\Delta_{\mathfrak{q}}}^{P} \mathbb{F}_3. \tag{E73.13}$$

The summand is the locally constant functions on $P/\Delta_{\mathfrak{q}}$, with the contragredient permutation action. The mod-three orientation is trivial because $\mu_3 \subset K$. A continuous homomorphism from L to the finite group $\mathbb{F}_3$ is nonzero over only finitely many base primes and is locally constant on each of their prolongation spaces. Conversely those data give the continuous homomorphism by the free-product universal property. This proves (E73.13), including the direct-sum assertion. It also identifies its P-invariants with the direct sum of the constant functions on those spaces.

Both P and L have cohomological dimension one. The split extension $L \rtimes P$ consequently gives

$$H^2(\mathcal{G}, \mathbb{F}_3) = H^1(P, M). \tag{E73.14}$$

We describe this edge map to retain its naturality. A central extension of $\mathcal{G}$ by $\mathbb{F}_3$ splits continuously on L and on $s(P)$. Comparing the two lifted conjugation actions gives a one-cocycle $z : P \to \mathrm{Hom}_{\mathrm{cts}}(L, \mathbb{F}_3)$. Changing the splitting on L changes z by a coboundary. Conversely such a z gives the explicit two-cocycle, in the coordinates $(\ell, p) \in L \rtimes P$,

$$\omega_z((\ell, p), (\ell', p')) = z(p)(p\ell' p^{-1}). \tag{E73.15}$$

The cocycle equation for z and additivity on L verify the two-cocycle equation directly. The two constructions are inverse. They are continuous: the conjugation discrepancy is a continuous function on the compact product $P \times L$ with finite range, so it is locally constant uniformly in its L coordinate. This proves (E73.14) as well as its compatibility with actual subgroup restrictions.

Every continuous P-cocycle with values in (E73.13) has finite support in the base primes. For example its values on the two free generators lie in a finite sum of summands, which is P-stable and hence contains all its values. The same assertion holds for coboundaries. Thus $H^1(P, -)$ commutes with this particular direct sum. The Shapiro map on each summand is restriction to $\Delta_{\mathfrak{q}}$ followed by evaluation at its base coset. For precision, a continuous transversal of $P \to P/\Delta_{\mathfrak{q}}$ is constructed by compatible choices in the finite quotients of P. Each prescribed finite representative lifts to the next level after multiplying by an element of $\Delta_{\mathfrak{q}}$. If $j(x)$ is this transversal, with $j(\Delta) = 1$, the inverse for a character $\psi : \Delta \to \mathbb{F}_3$ is

$$z_\psi(p)(x) = \psi\big(j(x)^{-1} p\, j(p^{-1}x)\big). \tag{E73.16}$$

The argument lies in Δ. Multiplication verifies the cocycle equation. For any cocycle z, setting $m(x) = z(j(x))(x)$ gives $z = z_\psi + m - pm$, proving both directions. Compactness of P/Δ gives the uniform local constancy needed for a discrete coinduced module.

The local analogue of (E73.14), applied to (E73.12), identifies $H^2(D_{\mathfrak{q}}, \mathbb{F}_3)$ with $H^1(\Delta_{\mathfrak{q}}, \mathbb{F}_3) = \mathbb{F}_3$. In the central extension described above, lifting $F_{\mathfrak{q}} = h_{\mathfrak{q}} s(\sigma_{\mathfrak{q}})$ as the product of its two chosen lifts shows that its local relation error is

$$z(\sigma_{\mathfrak{q}})(t_{\mathfrak{q}}). \tag{E73.17}$$

Inner conjugation by $h_{\mathfrak{q}} \in L$ disappears on the character of L, and the orientation factor q is one in $\mathbb{F}_3$. Thus (E73.17) is precisely the Shapiro evaluation, with the positive sign for the

relator in (E73.3). This proves the restriction isomorphism in (E73.4), with finite support, directly.

Minimal generators and the complete relation set. The analogous degree-one restriction sequence, split by s, gives the exact evaluation isomorphism

$$H^1(\mathcal{G},\mathbb{F}_3)\longrightarrow \mathbb{F}_3^2\oplus\bigoplus_{\mathfrak{q}\notin S}\mathbb{F}_3,\quad \chi\longmapsto(\chi(s\tau),\chi(s\tau'),\chi(t_{\mathfrak{q}})). \tag{E73.18}$$

Every finite extension of K ramifies at finitely many primes, so the chosen $t_{\mathfrak{q}}$ tend to one in $\mathcal{G}$, independently of the chosen prolongations. This proves continuity of (E73.2). If its compact image were proper, a finite three-group quotient would contain that image in a proper subgroup and then in a subgroup of index three. The resulting nonzero character would vanish on every displayed generator, contradicting (E73.18). Thus π is onto, and (E73.18) proves that it induces an isomorphism on Frattini quotients.

For a central extension of $\mathcal{G}$ by $\mathbb{F}_3$, lift the free group $\mathcal{F}$ to that extension. These lifts respect the convergent basis: an open subgroup of the extension avoiding its finite kernel maps isomorphically onto an open subgroup of $\mathcal{G}$, and contains the unique small lifts of all sufficiently late inertia generators. Restriction of the lift to R is an $\mathcal{F}$-invariant character. Different lifts change it by the restriction of a character of $\mathcal{F}$, which is zero on R by minimality. Conversely pushout by such an invariant character constructs the central extension. Consequently

$$H^2(\mathcal{G},\mathbb{F}_3)=\mathrm{Hom}_{\mathrm{cts}}(R/\overline{R^3[R,\mathcal{F}]},\mathbb{F}_3). \tag{E73.19}$$

This is the actual relation-module map: its value on a relator is the central coordinate obtained by evaluating that word in any lift of the generators. Pulling back along the two-generator local free group, with its generators sent to $T_{\mathfrak{q}},Y_{\mathfrak{q}}$, identifies this evaluation with the local coordinate (E73.17). The tame group (E73.12) itself has the one-relator presentation in (E73.3): its relation makes the closed cyclic inertia normal, and collection gives t^aF^b, whose two coordinates are detected in $\mathbb{Z}_3\rtimes_q\mathbb{Z}_3$.

It follows from (E73.4) and (E73.19) that the displayed relators span the compact relation module densely, and that their evaluations are the coordinate basis. They tend to one in $\mathcal{F}$: for every open normal subgroup, eventually $T_{\mathfrak{q}}$ belongs to it, as do its conjugate and every integral power. This argument needs no convergence of the $Y_{\mathfrak{q}}$ themselves.

Let R_0 be their closed normal closure. Were $R_0\neq R$, the nontrivial normal subgroup $H=R/R_0$ of $\mathcal{F}/R_0$ would have a nontrivial finite three-group image J in a finite quotient. The finite module $J/\Phi(J)$ has nonzero coinvariants under that finite three-group: the augmentation ideal in its group algebra over $\mathbb{F}_3$ is nilpotent. It therefore gives a nonzero invariant character of H, hence of R, vanishing on R_0. This contradicts dense spanning in (E73.19). Thus $R_0=R$, proving (E73.3).

The same proof applies with any finite set of additional primes. Deleting the other inertia generators sends $Y_{\mathfrak{q}}$ to actual Frobenius preimages in that finite-ramification presentation. All omitted relators become one. The inverse limit of those actual quotient maps recovers (E73.2)–(E73.3). This also proves the stated product topology on the relation module.

Sufficiency and the nonlinear arithmetic correction. The actual action on each V factors through P by E.34, so L acts trivially. Its semidirect product with $\mathcal{G}$ is pro-three. More explicitly, a cocycle on the full arithmetic group G_K defines a homomorphism to $V\rtimes\rho(G_K)$, whose target is pro-three. It therefore factors through $\mathcal{G}$, and conversely every constructed cocycle inflates to G_K. These are inverse maps on the actual based sections and their conjugacies. The convergence condition (E73.5) is exactly what makes (E73.6) continuous on the pointed generating space. Evaluation of a relator in (E73.3) has geometric component

$$v_{\mathfrak{q}}\rho(\sigma_{\mathfrak{q}})(n_{\mathfrak{q}})v_{\mathfrak{q}}^{-1}n_{\mathfrak{q}}^{-q}. \tag{E73.20}$$

Thus all (E73.7) hold exactly when the homomorphism (E73.6) kills the entire kernel, proving sufficiency. Its descended homomorphism has second projection the identity on $\mathcal{G}$, so it is the desired section. Restriction is the inverse construction.

Let $u : P \to V$ be the free cocycle specified by A, A'. In $\mathcal{F}$ the word $Y_{\mathfrak{q}} s_{\mathcal{F}}(\sigma_{\mathfrak{q}})^{-1}$ lies in the kernel of $\mathcal{F} \to P$ and maps to $h_{\mathfrak{q}}$ of E.39. On a descended solution, its evaluation is $\lambda(h_{\mathfrak{q}})$. Hence the actual equality is

$$v_{\mathfrak{q}} = \lambda(h_{\mathfrak{q}}) u(\sigma_{\mathfrak{q}}). \tag{E73.21}$$

Before descent, (E73.6) evaluates the displayed free word itself. This specifies the correction without presupposing a homomorphism λ or its equivariance. The entire global relation problem is consequently the coupled equations (E73.7), not independently chosen local Frobenius coordinates. E.61, E.66 and E.71 applied to the actual pair $(n_{\mathfrak{q}}, v_{\mathfrak{q}})$ give exactly the packets in the statement.

Choose cofinal characteristic open subgroups V_m of V, so the action preserves them. Two cocycles are uniformly equal modulo V_m exactly when their values agree on all the displayed generators. The implication from generators follows by their images in the finite semidirect product using the actual finite arithmetic action on V/V_m. Thus restriction gives the claimed homeomorphism before conjugacy. Word evaluation is continuous for these topologies, so $\mathcal{X}_V$ is closed in $V^2 \times c_0$. Formula (E73.10) is the literal section-conjugation formula; taking its quotient proves (E73.9). The compact group V acts continuously on a Hausdorff space, so this orbit space is Hausdorff. The actual geometric quotient maps commute with every generator and word evaluation, proving their componentwise description.

Rational descent with all conjugators. The actual arithmetic quotient over $\mathbb{Q}$ is $\mathcal{G} \rtimes \langle c \rangle$. Every section can be conjugated so that its value at c is one, by coprime complement conjugacy in the extension by the pro-three group V. The remaining conjugations are exactly V^c. The elements $c t_{\mathfrak{q}} c$ tend to one. Since the epimorphism π is open, compatible choices in successively smaller open subgroups of $\mathcal{F}$ give preimages $C_{\mathfrak{q}} \to 1$. For the two free generators, the actual splitting is equivariant, so the indicated preimages are exactly τ' and τ.

For a descended section, (E73.11) says that the two continuous homomorphisms obtained by conjugating the arithmetic source, or by conjugating both factors of the semidirect target, agree on every generator. They therefore agree everywhere. This is precisely the extension condition over $\mathbb{Q}$. Conversely every normalized rational section has these equations. Changing any preimage $C_{\mathfrak{q}}$ or $Y_{\mathfrak{q}}$ by an element of R changes none of the descended evaluations. The coordinate description is therefore independent of these choices by the identity map on its generator values; the actual prolongations and words remain part of its typed construction.

An actual finite failure of independent Frobenius packets. We use the explicit Kummer extension

$$K(\sqrt[3]{2}, \sqrt[3]{\zeta}, \sqrt[3]{1-\zeta}).$$

The three Kummer classes are independent: the valuations at two and at $1-\zeta$ separate the first and third, while ζ is not a cube in K. Its Galois group is consequently $H = \mathbb{F}_3^3$, with coordinates $(\kappa_2, \kappa_\zeta, \kappa_{1-\zeta})$. The actual Frobenii and the inertia generator give

	κ_2	κ_ζ	$\kappa_{1-\zeta}$
$s\tau$	2	2	0
$s\tau'$	1	2	2
$t_{(2)}$	1	0	0
$F_{(-2+3\zeta)}$	1	0	2.

(E73.22)

The first two rows are the exact E.39 choice of the Frobenii above seven. The last is obtained by reduction with $\zeta = 7$ in $\mathbb{F}_{19}$. The projection of that last row to $s(P)$ is $2s\tau + s\tau' = (2, 0, 2)$,

and their difference is $2t_{(2)}$. The unique character taking the prescribed values $(2,1,1)$ on the first three rows is κ_2, so its value on the last row is one, not two.

This finite obstruction is detected by an actual geometric quotient. With the original tangent at $O = C_0$, the unit z/x on the affine cubic induces the based map $\Pi \to \mathbb{Z}_3(1)$; its order at O is one. Reduction modulo three is equivariant over K and factors through both approximations. The cusp cocycle $g \mapsto d_O^{\kappa_2(g)}$ realizes the first three rows and the value one at nineteen. Keeping its generator data and replacing only that unramified local value by two remains a valid local packet but violates (E73.22). Conjugacy cannot change this finite character because its coefficient action over K is trivial. The program recomputes all residues, enumerates all27 characters of H, and verifies the forced value and the actual correction. It also checks the explicit Shapiro cochains and the positive local-relator transgression in finite groups. This proves every asserted comparison while retaining the unevaluated higher Frobenius words in (E73.21). □

The wild local central groups and their actual lifting fibers

E.74. Integral cohomology and every nonempty central fiber at three [Proved]

Retain the based cubic pro-three groups W and their integral graded lattices $\mathcal{A}_r(W)$ from E.41. Fix an embedding of the algebraic closure of $\mathbb{Q}$ into an algebraic closure of $\mathbb{Q}_3$, and put

$$k = \mathbb{Q}_3(\zeta_3), \quad \mathcal{D} = \mathrm{Gal}(k(3)/\mathbb{Q}_3), \quad D = \mathrm{Gal}(k(3)/k), \quad \Gamma = P \rtimes \langle c \rangle. \tag{E74.1}$$

Here $k(3)$ is the maximal pro-three extension of k, and $P = \langle \tau, \tau' \rangle$ is the actual global free quotient from E.37, with $c\tau c = \tau'$. The localization map $\pi_3 : \mathcal{D} \to \Gamma$ is onto. It has a continuous group section j constructed below. Write $R_3 = \ker \pi_3$ and use the resulting actual coordinates $\mathcal{D} = R_3 \rtimes j(\Gamma)$.

For $A = \mathcal{A}_r(W)$ write $n_r = \mathrm{rank}\, A$, $t_r = \mathrm{tr}\, J_r$, where T_r, J_r are the actual matrices of E.41. Define

$$\begin{aligned} \mathcal{B}_r(W) &= \mathrm{coker}\big([\,T_r - 7I \mid J_r + I\,] : A^2 \longrightarrow A\big), \\ b_1 &= 5, \qquad b_2 = 1, \qquad b_r = 0 \quad (r \geq 3), \\ d_r &= \frac{n_r + t_r}{2} + b_r. \end{aligned} \tag{E74.2}$$

All cokernels have their compact quotient topology. There are isomorphisms of compact $\mathbb{Z}_3$-modules

$$\begin{aligned} H^2_{\mathrm{cts}}(G_{\mathbb{Q}_3}, A) &\simeq \mathcal{B}_r(W), \\ H^1_{\mathrm{cts}}(G_{\mathbb{Q}_3}, A) &\simeq \mathcal{C}_r(W) \oplus \mathrm{Hom}_{\Gamma,\mathrm{cts}}(R_3^{\mathrm{ab}}, A) \\ &\simeq \mathcal{C}_r(W) \oplus \mathbb{Z}_3^{d_r}. \end{aligned} \tag{E74.3}$$

The first isomorphism is the local invariant pairing with the invariant Tate dual, after a choice of compatible three-power roots of unity. The second is restriction to $j(\Gamma)$ and R_3; its inverse is the literal cocycle

$$a(rj(g)) = \theta(r) + u(g), \qquad r \in R_3, \quad g \in \Gamma. \tag{E74.4}$$

The last isomorphism uses a chosen basis of the actual equivariant homomorphism lattice. Two normal generators of R_3, explicit evaluation coordinates, and the inverse coordinate map are specified in the proof. The resulting coordinate choice is recorded separately from numerical evaluation of a local relator.

For the first three degrees, all three geometric quotients have the same answer. With $C_m = \mathbb{Z}/m\mathbb{Z}$, the complete obstruction groups are

$$\begin{aligned} \mathcal{B}_1 &= \mathbb{Z}_3^{5}, \\ \mathcal{B}_2 &= \mathbb{Z}_3 \oplus C_3^{11} \oplus C_9^{4}, \\ \mathcal{B}_3 &= C_3^{63} \oplus C_9^{23} \oplus C_{27}^{7}. \end{aligned} \tag{E74.5}$$

The complete central parameter groups are

$$\begin{aligned} H^1(G_{\mathbb{Q}_3}, \mathcal{A}_1) &= \mathbb{Z}_3^{15} \oplus C_3^{3}, \\ H^1(G_{\mathbb{Q}_3}, \mathcal{A}_2) &= \mathbb{Z}_3^{46} \oplus C_3^{9} \oplus C_9^{5}, \\ H^1(G_{\mathbb{Q}_3}, \mathcal{A}_3) &= \mathbb{Z}_3^{330} \oplus C_3^{7} \oplus C_9^{56} \oplus C_{27}^{25} \oplus C_{81}^{3}. \end{aligned} \tag{E74.6}$$

In particular the additional free ranks in (E74.3) are $9, 22, 164$. In degree four the full, marked and metabelian groups have additional free ranks $1227, 1222, 741$, respectively; their total H^1 ranks are $2475, 2466, 1485$. Their torsion is precisely the corresponding E.41 torsion.

For every $r \geq 3$, $\mathcal{B}_r(W)$ is finite and is killed by

$$3^{r+e_r}, \qquad e_r = \max_{r-2 \leq j \leq 2r-2} (1 + v_3(j)). \tag{E74.7}$$

Thus the actual matrix in (E74.2) modulo 3^{r+e_r+1} evaluates it by a finite Smith calculation. The retained certificate gives the integer transformation and inverse maps in degree one, and the modular transformation and inverse maps, with proved sufficient precision, in degrees two and three.

Let $W_r = W/\gamma_{r+1}W$. For each actual section $\sigma : G_{\mathbb{Q}_3} \to W_r \rtimes G_{\mathbb{Q}_3}$, write $\bar{\sigma}$ for its projection. The map

$$H^1(G_{\mathbb{Q}_3}, \mathcal{A}_r(W)) \longrightarrow \mathrm{Fib}_{[\bar{\sigma}]}\big(\mathrm{Sect}_{\mathbb{Q}_3}(W_r)/W_r \longrightarrow \mathrm{Sect}_{\mathbb{Q}_3}(W_{r-1})/W_{r-1}\big), \quad [z] \longmapsto [z\sigma] \tag{E74.8}$$

is a homeomorphism of pointed spaces. Its inverse aligns the lower sections by their unique geometric conjugator and takes their central difference. Consequently (E74.3) computes every nonempty central fiber at the wild place. The obstruction of an arbitrary lower section takes values in the explicitly computed group $\mathcal{B}_r(W)$. Its individual nonlinear values remain part of E.23.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379); Part E(c),(d) and follow-up(4). The coefficient prime, geometric groups and arithmetic source in this statement are exactly those displayed above.

Proof. The actual local source and an equivariant splitting. Restriction to the completion supplies the continuous map $D \to P$. A proper closed image is contained in a subgroup of index three: choose a finite three-group quotient in which the image is proper, and then a maximal subgroup of that finite group. The corresponding cyclic cubic extension of $K = \mathbb{Q}(\zeta_3)$ is unramified outside three and has trivial decomposition group at its unique prime above three. It is therefore unramified at every finite prime. The class number one computation in E.34 excludes this extension. This proves surjectivity.

The relative field $k(3)$ is stable under the automorphisms of $k/\mathbb{Q}_3$, so (E74.1) gives an exact sequence with quotient C_2. Its localization map onto Γ is compatible with that quotient and is surjective by the preceding argument. In the preimage of the specified subgroup $\langle c \rangle \subset \Gamma$, choose a complement to the normal pro-three kernel. The profinite coprime complement theorem follows here by the finite coprime complement theorem and compactness of compatible finite complements. It gives an element $\widetilde{c}$ of order two mapping to c.

Choose any $\widetilde{\tau} \in D$ mapping to τ and put $\widetilde{\tau}' = \widetilde{c}\widetilde{\tau}\widetilde{c}$. The free property of P extends these two choices to a continuous group homomorphism $P \to D$. Its composite with localization is the identity on the two free generators and hence on P. Conjugation by $\widetilde{c}$ interchanges the two chosen lifts, so this homomorphism extends to the section $j : \Gamma \to \mathcal{D}$. Every involution just used lies in the stated relative quotient, with its map from the absolute local group.

By E.34 the based action on each W factors through Γ, and R_3 consequently acts trivially on W and A. The local cyclotomic character also factors through Γ: all three-power roots of unity occur in the global extension unramified outside three. In particular

$$\chi_3(j\tau) = 7, \qquad \chi_3(jc) = -1. \tag{E74.9}$$

These are the values of the specified lifted arithmetic elements. All matrices below refer to those elements.

We prove the cohomological comparison with the full absolute local group directly. The actual local/global quotient is also recorded in the opened original [Win, pp. 1–2; §3]. Let $N_0 = G_{k(3)}$. Kummer theory gives $H^1(N_0, \mathbb{F}_3) = 0$: every $a \in k(3)^\times$ lies in a finite Galois three-extension E/k. Adjoining cube roots of all its k-conjugates gives an elementary three-extension of E, since $\mu_3 \subset E$. The resulting splitting field is Galois of three-power degree over k, so it already lies in $k(3)$.

Continuous cohomology of the closed subgroup N_0 is the direct limit over these finite fields E of $H^i(G_E, \mathbb{F}_3)$. This comparison is induced by restriction: continuous finite-valued cochains extend to a sufficiently small open neighborhood of the closed subgroup, and their cocycle and coboundary identities hold after shrinking that neighborhood. For degree two, identifying $\mathbb{F}_3$ with μ_3 identifies this system with $\mathrm{Br}(E)[3]$. Its local invariant is multiplied by $[E' : E]$ under restriction to E'. A degree-three step of the unramified tower beyond the finite residue degree of E gives $[E' : E] = 3$ inside $k(3)$, and kills that group. Thus $H^2(N_0, \mathbb{F}_3) = 0$. All higher groups vanish by finite local duality before passage to the same direct limit. These are the restriction and invariant maps in [Mil06, I, Theorem2.1 and Corollary2.3].

For every finite three-primary D-module M, the N_0 action is trivial. A filtration of its underlying finite abelian group with trivial $\mathbb{F}_3$ factors therefore proves $H^i(N_0, M) = 0$ for $i > 0$. The Hochschild–Serre spectral sequence now gives inflation isomorphisms $H^i(D, M) \to H^i(G_k, M)$ in every degree, with their actual restriction maps. Exact C_2 invariants give the comparison over $\mathbb{Q}_3$. The four Kummer classes below give $\dim H^1(D, \mathbb{F}_3) = 4$, so the pro-three Frattini criterion supplies four topological local generators. Local duality and the proved comparison also recover its dimension-two Demuškin pairing directly. At the coefficient levels $A/3^m A$ all local cohomology groups are finite, by local duality. Their inverse systems are Mittag–Leffler, so every relevant derived-limit term vanishes. Passing to the inverse limit gives the continuous compact-coefficient comparisons

$$H^i_{\mathrm{cts}}(G_{\mathbb{Q}_3}, A) \xleftarrow[\text{inflation}]{\sim} H^i_{\mathrm{cts}}(\mathcal{D}, A), \qquad i = 1, 2, \tag{E74.10}$$

and vanishing above degree two. For degree one and for the nonabelian section sets, factorization is also direct: over k the map $g \mapsto (a(g), \rho(g))$ has a pro-three semidirect target. It kills N_0, so every such cocycle factors through $\mathcal{D}$. Inflation and factorization are inverse on cocycles and conjugations.

The integral degree-one splitting and its coordinates. For an additive cocycle a on $\mathcal{D}$, restriction to R_3 is a continuous homomorphism θ. The cocycle identity and the trivial R_3 action give

$$\theta(j(g)rj(g)^{-1}) = g\theta(r). \tag{E74.11}$$

Restriction to $j(\Gamma)$ is a cocycle u. Conversely, the multiplication law of $R_3 \rtimes j(\Gamma)$ and (E74.11) verify (E74.4). A coboundary vanishes on R_3 and restricts to the ordinary coboundary on Γ.

Thus these restriction maps and (E74.4) give the middle isomorphism in (E74.3), integrally. E.41 computes the first factor by normalizing the value at c to zero and evaluating at τ:

$$H^1(\Gamma, A) = A/(T_r - I)A^c = \mathcal{C}_r(W), \qquad A^c = (I + J_r)A. \tag{E74.12}$$

Division by two proves the last equality.

Here are actual coordinates for the second factor. Set $\lambda = 1-\zeta_3$, $\mathcal{O} = \mathbb{Z}_3[\zeta_3]$ and $U_i = 1+\lambda^i\mathcal{O}$. The unit filtration gives $U_1 = \mu_3 \times U_2$: the quotient U_1/U_2 has order three, and $\zeta_3 \notin U_2$. The logarithm and exponential converge inversely on U_2 and $\lambda^2\mathcal{O}$, since $v_3(\lambda^2) = 1 > 1/2$. Their valuations follow term by term from the factorial and integer denominators. Put $u_2 = 1 + \lambda^2$, $u_3 = 1 + \lambda^3$. Their logarithms are respectively λ^2, λ^3 modulo $\lambda^4\mathcal{O}$. For example the denominator-three term of $\log(1 + \lambda^2)$ already has λ-valuation four; every later term has at least that valuation. Since $3\lambda^2\mathcal{O} = \lambda^4\mathcal{O}$ up to a unit, the four Kummer classes

$$\lambda, \quad \zeta_3, \quad u_2, \quad u_3 \quad \text{form a basis of } k^\times/k^{\times 3}. \tag{E74.13}$$

The first two span precisely the restrictions of $H^1(P, \mathbb{F}_3)$, by the explicit global S-unit calculation of E.34. The split five-term sequence and $H^2(P, \mathbb{F}_3) = 0$ show that the restrictions of $\kappa_{u_2}, \kappa_{u_3}$ are a basis of $H^1(R_3, \mathbb{F}_3)^P$. Choose elements $r_1, r_2 \in R_3$ dual to that basis. They normally generate R_3 in D. Indeed, a nontrivial quotient by their closed normal closure would have a nontrivial finite normal three-group image. Its Frattini quotient has nonzero coinvariants under the finite acting three-group, and hence a nonzero invariant character. That character vanishes on r_1, r_2, contrary to their chosen dual coordinates.

Let $\Lambda = \mathbb{Z}_3[[P]]$. Conjugation through j and the two generators give an actual continuous module surjection $\Lambda^2 \to R_3^{\mathrm{ab}}$. Evaluation therefore embeds the second factor of (E74.3) as

$$E : \mathrm{Hom}_{\Gamma,\mathrm{cts}}(R_3^{\mathrm{ab}}, A) \hookrightarrow A^2, \qquad E(\theta) = (\theta(r_1), \theta(r_2)). \tag{E74.14}$$

Its image $\mathcal{E}_A$ consists of pairs whose Λ-linear map kills the actual relation submodule and respects the actual c action. These are closed integral linear equations. The image is closed and saturated: when three times a pair satisfies all those equations, torsion-freeness of A implies that the pair satisfies them. Thus $\mathcal{E}_A$ is a finite free direct summand of A^2.

Choose a unimodular map $S_A : A^2 \to A^2$ carrying this actual submodule to the first d_r coordinates, whose rank is proved below. The extra coordinates of $[a]$ are the first d_r entries of $S_A(a(r_1), a(r_2))$. Conversely append zeros, apply S_A^{-1}, and take the unique equivariant homomorphism having that pair of values. Formula (E74.4) gives the inverse class together with its E.41 coordinate. All choices and both maps have now been specified. The numerical entries of S_A depend on the actual relation submodule; these entries are not among the Smith computations in (E74.5). The evaluated module type and rank below require no unevaluated relator value.

The topological assertions follow from the same maps. Cocycles on the finitely generated pro-three group D form a closed submodule of the finite product of copies of A obtained by evaluating on its generators. The extra C_2 equations are closed. Coboundaries are compact images of A. Thus all the degree-one groups are compact Hausdorff, and the displayed continuous bijections are homeomorphisms.

Local duality, with its exact twist. At each finite level the cup product followed by the local Brauer invariant gives the perfect duality

$$H^2(G_{\mathbb{Q}_3}, A/3^mA) \times H^0(G_{\mathbb{Q}_3}, \mathrm{Hom}(A/3^mA, \mu_{3^m})) \longrightarrow \tfrac{1}{3^m}\mathbb{Z}/\mathbb{Z}. \tag{E74.15}$$

This is the specialization of [Mil06, I, Theorem2.1 and Corollary2.3, pp. 26–28]. The maps under coefficient reduction are dual to the inclusions of the Tate-dual coefficient groups. Their

inverse limit consequently identifies H^2 with the compact coinvariants of $A(-1)$. Explicitly, a coinvariant is paired with an invariant element of $A^*(1) \otimes \mathbb{Q}_3/\mathbb{Z}_3$ by evaluation; the inverse is the unique cohomology class giving that functional under (E74.15). This defines both directions of the first isomorphism in (E74.3).

The localization map is onto Γ, so these coinvariants are precisely the Γ coinvariants. On $A(-1)$ the two topological generators τ, c act as $7^{-1}T_r, -J_r$. Their relation submodule is therefore $(T_r - 7I)A + (J_r + I)A$. Relations for products and inverses follow from those for the generators; continuity gives the relations for their closure. This image is already compact and closed. This proves the exact matrix formula (E74.2).

Its rational free rank is b_r. To compute it, use the actual split rational lattice from E.40–E.41. On degree one the Frobenius at the K-prime above two acts as -2 on the elliptic part and as 4 on the toric part. The toric character module is the degree-zero cusp module on nine cusps; c has three fixed cusps and three pairs. After the Tate twist its coinvariant dimension is $6 - 1 = 5$. The elliptic part has no eigenvalue four, proving $b_1 = 5$. In degree two the only four-eigenspace is $\bigwedge^2 T_3E \otimes \mathbb{Q}_3 = \mathbb{Q}_3(1)$, with its actual Weil-pairing action. It contributes one dimension, proving $b_2 = 1$. All degrees at least three have eigenvalues $(-2)^j$, $r \le j \le 2r$; none is four. The exact graded quotients of E.41 retain these weight spaces and hence have $b_r = 0$.

The integral rank and all degree-one torsion. The element $j(\tau)$ acts by the positive-weight Frobenius matrix T_r, which has no eigenvalue one. Thus $H^0(G_{\mathbb{Q}_3}, A) = 0$. Write R_A for the free rank of $H^1(G_{\mathbb{Q}_3}, A)$; finite generation was proved in (E74.12)–(E74.14). For m beyond the exponents of its torsion and of $\mathcal{B}_r$, the coefficient exact sequence gives

$$\begin{aligned} &H^0(G_{\mathbb{Q}_3}, A/3^m) = H^1(G_{\mathbb{Q}_3}, A)[3^m], \\ &0 \longrightarrow H^1(A)/3^m \longrightarrow H^1(A/3^m) \longrightarrow H^2(A)[3^m] \longrightarrow 0, \\ &H^2(G_{\mathbb{Q}_3}, A/3^m) = H^2(G_{\mathbb{Q}_3}, A)/3^m. \end{aligned} \tag{E74.16}$$

In the middle row the arithmetic group is the same $G_{\mathbb{Q}_3}$ throughout. The finite local Euler characteristic theorem [Mil06, I, Theorem2.8, pp. 31–34] says that the alternating product of these orders is 3^{-mn_r}. The two finite torsion orders cancel in (E74.16), giving $3^{m(b_r - R_A)} = 3^{-mn_r}$. Therefore

$$R_A = n_r + b_r, \qquad \operatorname{rank} \mathcal{E}_A = n_r + b_r - \frac{n_r - t_r}{2} = d_r. \tag{E74.17}$$

The direct summand $\mathcal{E}_A$ is torsion-free, so every torsion summand of local H^1 is exactly the E.41 torsion. This proves (E74.6) and all stated degree-four ranks.

Finite precision and the retained Smith maps. Let F denote the actual Frobenius at the K-prime above two. The split lattice of E.41 has maps B_r, C_r with $B_rC_r = 3^rI$ and $FB_r = B_rF_{\text{split}}$. Its diagonal eigenvalues are $(-2)^j$, $r \le j \le 2r$. For $r \ge 3$,

$$v_3((-2)^j - 4) = 1 + v_3(j - 2). \tag{E74.18}$$

The scalar four is a three-adic unit. It follows that $3^{e_r}(F_{\text{split}} - 4I)^{-1}$ is integral. Multiplying by B_r, C_r shows that $3^{r+e_r}(F - 4I)^{-1}A \subset A$. The image $(F - 4I)A$ is zero in the coinvariants (E74.2), because $\chi_3(F) = 4$. This proves the annihilator bound (E74.7), including its passage through each actual geometric quotient.

For degree two the free coordinate is the first Hall coordinate $[x_1, x_0]$, the projection to $\bigwedge^2 T_3E$. The entire first row of the matrix in (E74.2) is zero. On its coordinate-zero lattice $K_2 = \langle e_1, \dots, e_{44} \rangle$, the supplied integer matrices verify

$$(F - 4I)B_2D_2C_2\big|_{K_2} = 108I_{K_2}, \qquad D_2\big|_{\text{eigenvalue } e} = \begin{cases} 0 & e = 4, \\ 12/(e-4) & e = -8, 16. \end{cases} \tag{E74.19}$$

The elliptic coordinate of C_2K_2 is zero, which verifies the identity without a rational projection error. Hence the torsion is killed by 3^3, and precision 3^4 suffices.

The program `e_wild_layers.py` constructs the actual matrices by integral Hall substitution from E.41. It checks equality with both quotient matrices in degrees one through three. In degree one it retains an integer Smith decomposition and both inverse transformations. In degrees two and three it retains every elementary operation, the left and right transformations, both inverses, and the exact modular Smith identities at moduli 81 and 729. The nonzero Smith valuations and free ranks are

r	n_r	$v=0$	$v=1$	$v=2$	$v=3$	free rank
1	10	5	0	0	0	5
2	45	29	11	4	0	1
3	330	237	63	23	7	0.

(E74.20)

These are (E74.5). The annihilator bounds exclude undetected elementary divisors beyond the precision. For a finite factor, multiply by the stored left transformation and reduce each coordinate modulo its displayed power of three. For the inverse, lift those digits, apply the stored inverse, and pass to the cokernel. Differences between lifts are killed by the proved annihilator. In degree two first subtract the exact free coordinate times e_0 and apply this construction to K_2; retain the first coordinate in $\mathbb{Z}_3$. This supplies the actual coordinate maps, including the free part.

Section fibers, stabilizers, and the comparison maps. A geometric element fixed by a section-twisted action on W_{r-1} has a first nonzero graded coordinate. Inner conjugation by the section's geometric part acts trivially on that grade, so this coordinate is fixed by T_i. Its absence of eigenvalue one is a contradiction. Thus every such stabilizer is trivial.

Fix the actual section σ in (E74.8). Multiplying it by an A-cocycle gives a section because A is central in W_r and has the original arithmetic action. Another section whose lower class is $[\bar{\sigma}]$ has a unique lower geometric conjugator aligning the two lower sections. Lift that conjugator to W_r and align. Their difference is then an A-cocycle. Changing the lifted conjugator changes this cocycle by an A-coboundary. Conversely two central differences yielding conjugate sections have a conjugator projecting to a lower stabilizer; that projection is the identity, so their difference is an A-coboundary. This proves both directions and injectivity of (E74.8). All section spaces factor through the finitely generated relative source $\mathcal{D}$ and are compact; their geometric orbit spaces are Hausdorff. The continuous bijection in (E74.8) is consequently a homeomorphism.

For completeness, the precise obstruction map of a lower geometric cocycle a is obtained with the continuous Hall lift $h_r : W_{r-1} \to W_r$ of E.41:

$$(g,h) \longmapsto h_r(a(g))\, g(h_r(a(h)))\, h_r(a(gh))^{-1} \in A. \tag{E74.21}$$

Associativity gives its additive two-cocycle identity; changing the Hall lift changes it by a coboundary. Its local invariant functional (E74.15), followed by the proved Smith map, is its value in $\mathcal{B}_r$. Vanishing means that an actual central correction of $h_r \circ a$ is a cocycle, with inverse provided by the same identity. This specifies the obstruction and its map while leaving its unevaluated nonlinear values in the outstanding classification.

Finally the actual epimorphisms from the full group to its metabelian and marked quotients induce the graded maps $q_r : A \to A'$. These commute with T_r, J_r and with the local source. On (E74.3) the map is $([u],\theta) \mapsto ([q_r u], q_r\theta)$; on (E74.2) it is the map of cokernels induced by q_r. Formula (E74.21) commutes with these maps at the cohomology-class level, since changing a chosen lift only changes a coboundary. Thus the computed wild layers retain the original geometric comparison maps and every stated ambiguity. □

A uniform presentation of the actual wild relative module

E.75. The relative relation module and explicit wild kernel coordinates [Proved]

Use the actual split local source of E.74,

$$\mathcal{D} = R_3 \rtimes j(\Gamma), \qquad D = R_3 \rtimes j(P), \qquad P = \langle \tau, \tau' \rangle, \quad c\tau c = \tau', \tag{E75.1}$$

and the cyclotomic character $\chi(\tau) = \chi(\tau') = 7$. Put $\Lambda = \mathbb{Z}_3[[P]]$, with its completed group-ring topology. Conjugation by c induces an involution ι of Λ. Write $U = \tau - 7$ and $V = \tau' - 7$. All modules in the following presentation are *left* compact Λ-modules; their coefficient vectors are rows. There is an actual continuous Γ-module isomorphism

$$\Phi : R_3^{\mathrm{ab}} \xrightarrow{\sim} \mathcal{M}_\chi := \Lambda^2/\{\lambda(U, -V) : \lambda \in \Lambda\}, \qquad c(\lambda_1, \lambda_2) = (\iota(\lambda_2), \iota(\lambda_1)). \tag{E75.2}$$

The displayed relation submodule is closed, and the map $\lambda \mapsto \lambda(U, -V)$ is injective. The proof constructs Φ from the actual completed Fox row of the local source, including its inverse and its c-equivariant change of basis.

Choose an actual element $r_* \in R_3$ whose image under Φ is $(1, 0)$, and put $r'_* = cr_*c$. These two elements normally generate R_3 in D. For every compact lattice $A = \mathcal{A}_r(W)$ of E.41, set $T = T_r$, $J = J_r$ and $A^{\pm} = \ker(J \mp I)$. Evaluation on this single, uniformly chosen source element gives

$$\begin{aligned} \mathrm{Hom}_{\Gamma,\mathrm{cts}}(R_3^{\mathrm{ab}}, A) &\xrightarrow{\sim} \mathcal{K}_A, \qquad \theta \longmapsto \theta(r_*), \\ \mathcal{K}_A &= \ker\big((I - J)(T - 7I) : A \longrightarrow A^-\big). \end{aligned} \tag{E75.3}$$

For $a \in \mathcal{K}_A$ the inverse is the literal map

$$\theta_a(r) = \ell_a(\Phi(\bar{r})), \qquad \ell_a(\lambda_1, \lambda_2) = \lambda_1 a + \lambda_2 J a. \tag{E75.4}$$

Thus the wild degree-one group, its coordinates and their inverse are

$$\begin{aligned} H^1(G_{\mathbb{Q}_3}, A) &\xrightarrow{\sim} \mathcal{C}_r(W) \oplus \mathcal{K}_A, \\ [z] &\longmapsto ([z \circ j], z(r_*)), \\ ([u], a) &\longmapsto [\, rj(g) \longmapsto \theta_a(r) + u(g)\,]. \end{aligned} \tag{E75.5}$$

All arrows are homeomorphisms for the compact topologies. The source element r_* and Φ are fixed once for all coefficient lattices. Their construction uses the actual local Fox row; it does not assign a numerical pro-three word to that row.

More precisely, the local cochain groups in degrees zero through two have the following actual finite model, also valid after reduction modulo 3^m:

$$A^+ \xrightarrow{d^0} A \oplus A \xrightarrow{d^1} A^-, \qquad d^0(w) = ((T - I)w, 0), \quad d^1(u, a) = (I - J)(T - 7I)a. \tag{E75.6}$$

With the top-cell orientation specified below, its degree-two comparison with the local invariant coordinate of E.74 is

$$\begin{aligned} \frac{A^-}{(I - J)(T - 7I)A} &\xrightarrow{\sim} \frac{A}{(T - 7I)A + (J + I)A}, \\ [y] \longmapsto [y/2], \qquad [x] &\longmapsto [(I - J)x] \quad \text{for the inverse.} \end{aligned} \tag{E75.7}$$

The retained program constructs exact saturated kernel bases and inverse coordinate maps in degrees one, two and three. Their ranks are $9, 22, 164$. The nonzero Smith valuations of the map in (E75.3), using a basis of A^-, are respectively

$$\{0^1\}, \qquad \{0^8, 1^{11}, 2^4\}, \qquad \{0^{73}, 1^{63}, 2^{23}, 3^7\}. \tag{E75.8}$$

These give exactly the degree-two obstruction groups of E.74, including their free coordinates. The finite model also verifies their coefficient transitions and the eighteen finite-cohomology order identities at $1 \le m \le 6$.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379); Part E(c),(d) and follow-up(4). This computes the previously implicit saturated Hom lattice in E.74 by one uniform source normalization. Individual nonlinear obstruction values remain part of E.23.

Proof. An actual pair of relative normal generators. Use the local units $u_2 = 1 + \lambda^2$ and $u_3 = 1 + \lambda^3$ from E.74, where $\lambda = 1 - \zeta_3$. We use the identification $\mu_3 = \mathbb{F}_3$ given by the fixed root ζ_3. As elements of $k^\times/k^{\times 3}$,

$$cu_2 = u_2 u_3^{-1}, \qquad cu_3 = u_3^{-1}. \tag{E75.9}$$

Indeed $c\lambda = -\zeta_3^2\lambda$; the logarithms modulo λ^4 give respectively $\lambda^2 - \lambda^3$ and $-\lambda^3$. The logarithm identifies cubing on these units with $3\lambda^2\mathcal{O} = \lambda^4\mathcal{O}$, up to a unit, so these congruences prove the asserted Kummer identities.

On characters with *trivial* $\mathbb{F}_3$ coefficients the involution is the negative of this Kummer action, because $c\zeta_3 = \zeta_3^{-1}$. Thus the character values on crc are $(-a+b, b)$ when the values on r are (a, b). The existence of these prescribed values is the actual relative kernel calculation in E.74: $H^1(D, \mathbb{F}_3)$ has Kummer basis $\lambda, \zeta_3, u_2, u_3$, and the first two are exactly the inflated basis of $H^1(P, \mathbb{F}_3)$. Since P is free, the five-term sequence identifies $H^1(R_3, \mathbb{F}_3)^P$ with the quotient by those first two classes. The restrictions of $\kappa_{u_2}, \kappa_{u_3}$ are therefore independent. Their joint evaluation map $R_3 \to \mathbb{F}_3^2$ is onto: a proper image would give a nonzero linear relation between those characters. Choose $r \in R_3$ with values $(0, 1)$ and put $r' = crc$; its values are $(1, 1)$. These two columns form a basis over $\mathbb{F}_3$. The relative Frattini argument in E.74 proves that r, r' normally generate R_3 in D. Together with $t = j(\tau)$ and $t' = j(\tau')$ they give a minimal system of four generators of D. Their involution is exactly $t \leftrightarrow t'$ and $r \leftrightarrow r'$.

The minimal completed Fox resolution. We prove the needed resolution directly from the local cohomology comparison of E.74. Put $\Omega = \mathbb{Z}_3[[D]]$ and let $\mathfrak{n}$ be its maximal ideal, generated by 3 and the four elements $t-1, t'-1, r-1, r'-1$. All maps below are continuous maps of compact left modules. The augmentation ideal has the epimorphism

$$\Omega^4 \longrightarrow I_D, \qquad e_t \mapsto t-1, \quad e_{t'} \mapsto t'-1, \quad e_r \mapsto r-1, \quad e_{r'} \mapsto r'-1. \tag{E75.10}$$

It is onto because the identity $gh - 1 = (g-1) + g(h-1)$ expresses every group word in its image, and that image is compact. Let Z be its kernel.

Here compact Nakayama has an elementary proof. For any compact module M, a set spanning $M/\mathfrak{n}M$ topologically generates M: otherwise a nonzero finite quotient of the remaining compact quotient would equal its maximal-ideal multiple. The maximal ideal of the finite acting three-group ring is nilpotent, a contradiction. The products in this argument are closed, since the maximal ideal has the displayed finite right-generating set. This proof also shows that $\mathrm{Hom}_\Omega(M, \mathbb{F}_3) = 0$ implies $M = 0$.

Applying continuous Hom to (E75.10) and to the augmentation sequence gives

$$\mathrm{Hom}_\Omega(Z, \mathbb{F}_3) = H^2(D, \mathbb{F}_3) = \mathbb{F}_3. \tag{E75.11}$$

The maps on the four degree-one generators are an isomorphism because this is a minimal system. The bar resolution identifies the continuous Ext groups used here with group cohomology: evaluation on a bar basis is precisely the continuous inhomogeneous cochain complex. Choose a lift $e \in Z$ of a generator modulo $\mathfrak{n}$. Compact Nakayama gives a surjection $\Omega e \to Z$. If Z_2 is its kernel, the same Hom sequence and dimension shifting give

$$\mathrm{Hom}_\Omega(Z_2, \mathbb{F}_3) = \mathrm{Ext}^1_\Omega(Z, \mathbb{F}_3) = H^3(D, \mathbb{F}_3) = 0. \tag{E75.12}$$

The last equality is the proved local comparison and local duality. Compact Nakayama gives $Z_2 = 0$. We have proved the exact free resolution

$$0 \longrightarrow \Omega \xrightarrow{d_2} \Omega^4 \xrightarrow{d_1} \Omega \longrightarrow \mathbb{Z}_3 \longrightarrow 0. \tag{E75.13}$$

This is the completed Fox resolution with the chosen generators. To identify its top generator with an actual relation, let the free pro-three group on those four letters map to D, with kernel N. Its complete group algebra is the noncommutative power-series ring, as proved in E.41. The unique last-letter decomposition gives its free augmentation resolution. Completed base change to Ω has first homology N^{ab} and gives the exact Fox sequence $0 \to N^{\mathrm{ab}} \to \Omega^4 \to I_D \to 0$. The identification sends a relation to its four continuous Fox derivatives. It can also be checked on the completed bar complex: choose compatible coset representatives in the finite quotients of the free group, and use the usual coset decomposition in each bar factor. These maps and their contracting homotopies are compatible under refinement, proving the completed Shapiro comparison. Thus $Z = N^{\mathrm{ab}}$, and the generator e has an actual relation lift in N.

The involution permutes the four basis elements of Ω^4 in the two indicated pairs. On its rank-one kernel write $c(e) = we$. Then $\iota(w)w = 1$, where here ι denotes the involution on Ω. Modulo $\mathfrak{n}$ the scalar is -1. Indeed local duality pairs $H^2(D, \mathbb{F}_3)$ with μ_3, and the field involution preserves the Brauer invariant and inverts μ_3. It therefore acts by -1 on this one-dimensional cohomology group and on its dual top generator. The explicit unit

$$v = (1 - w)/2 \in \Omega^{\times} \quad \text{satisfies} \quad \iota(v)w = -v. \tag{E75.14}$$

Replace e by ve. Its involution is now exactly $c(e) = -e$.

Projection to the free quotient. Project the four Fox coefficients to Λ. The identity $d_1 d_2 = 0$ says that the first two coefficients $a_t, a_{t'}$ satisfy $a_t(\tau - 1) + a_{t'}(\tau' - 1) = 0$. The augmentation ideal of the free power-series ring is a free *left* module on these two elements, by the unique last letter. Thus they are both zero. The exact top involution gives the projected row

$$\bar{d}_2(1) = (0, 0, a, -\iota(a)). \tag{E75.15}$$

Completed base change of (E75.13) along $\Omega \to \Lambda$ computes the homology of R_3. Here the section j gives the actual coset decomposition $D = R_3 j(P)$, so the completed Shapiro maps just described use this specified section. Since the first two coordinates form the free augmentation resolution of P, we obtain the actual map

$$R_3^{\mathrm{ab}} = \Lambda^2 / \{\lambda(a, -\iota(a)) : \lambda \in \Lambda\}. \tag{E75.16}$$

It sends the two coordinate classes to the images of the actual elements r, r'.

Finite regular coefficients and the right ideal. The coefficients $a, -\iota(a)$ generate the following *right* ideal:

$$a\Lambda + \iota(a)\Lambda = I_\chi, \qquad I_\chi = \ker\bigl(\epsilon_\chi : \Lambda \to \mathbb{Z}_3,\ p \mapsto \chi(p)\bigr). \tag{E75.17}$$

We prove equality, including its topology and orientation. Choose a cofinal sequence of pairs (m, H), with $H \triangleleft P$ open and $\chi(H) \equiv 1 \pmod{3^m}$, and put $\Lambda_{m,H} = (\mathbb{Z}/3^m)[P/H]$. Use left regular P action and commuting right multiplication. Finite local duality [Mil06, I, Theorem2.1 and Corollary2.3] gives a natural right-module isomorphism

$$H^2(D, \Lambda_{m,H}) = \bigl(\Lambda_{m,H}(-1)\bigr)_P \xrightarrow{\sim} \mathbb{Z}/3^m, \quad \sum b_p[p] \longmapsto \sum b_p \chi(p). \tag{E75.18}$$

The inverse sends 1 to [1] in the coinvariants. The right action of p on the target is multiplication by $\chi(p)$. The transitions are the ordinary reductions on that target. The finite degree-one cohomology groups are finite, so their inverse system is Mittag–Leffler. Consequently the

inverse limit of (E75.18) is the continuous cohomology with coefficient Λ. The finite free resolution (E75.13) also commutes with this inverse limit and gives

$$\Lambda/(a\Lambda + \iota(a)\Lambda) = H^2(D, \Lambda) \simeq \mathbb{Z}_3(\chi) \quad \text{as right } \Lambda\text{-modules.} \tag{E75.19}$$

The image of 1 is a unit of $\mathbb{Z}_3$, since it generates the right-hand module. The annihilator of that element is exactly I_χ. This proves (E75.17); the ideal is closed because it is a finite sum of compact images. Rescale the top generator e by the same scalar unit so that the local invariant in (E75.19) sends 1 exactly to 1. This scalar rescaling preserves $c(e) = -e$. It is the top-cell orientation used below.

The actual equivariant change of basis. The continuous substitution $p \mapsto \chi(p)^{-1}p$ is an automorphism of the completed free ring. In power-series variables it sends X to $(X-6)/7$ and Y to $(Y-6)/7$; its inverse sends them to $6+7X, 6+7Y$. Both substitutions preserve the maximal-ideal topology. The ordinary augmentation ideal is right-free on X, Y, by unique first letter. It follows that I_χ is right-free on U, V. The unique twisted first-letter decomposition defines a matrix M by

$$(a, -\iota(a)) = (U, V)M. \tag{E75.20}$$

Equality (E75.17) says that this endomorphism of a free right module of rank two is onto. Its reduction modulo the maximal ideal is therefore invertible. Lifting that inverse and using the convergent matrix geometric series proves $M \in \mathrm{GL}_2(\Lambda)$. Put $S = M^{-1}\,\mathrm{diag}(1, -1)$. Then

$$(a, -\iota(a))S = (U, -V), \qquad S_0\iota(S)S_0 = S, \quad S_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{E75.21}$$

For the second identity apply the involution and swap to the first identity. Both displayed rows change sign; uniqueness follows because the entries of $(a, -\iota(a))$ are a right basis of I_χ.

Right multiplication of row coordinates by S gives Φ in (E75.2); its inverse is right multiplication by S^{-1}. Equation (E75.21) proves the exact involution comparison. The relation image is compact. It is injective, since $\lambda U = 0$ forces $\lambda = 0$: the lowest nonzero word-length coefficient of a nonzero power series is multiplied by the nonzero constant -6 of $U = X - 6$. This also proves the exact sequence $0 \to \Lambda \to \Lambda^2 \to R_3^{\mathrm{ab}} \to 0$.

Lift $\Phi^{-1}(1, 0)$ to an actual $r_* \in R_3$ and set $r'_* = cr_*c$. They form a basis of the relative Frattini quotient, because S is invertible and both U, V lie in the maximal ideal. The relative Frattini proof again gives normal generation. This constructs the source pair in the statement. Its lift is a genuine choice in R_3, shared by all coefficients. For instance, to evaluate it in the original relative basis one uses the first row of S^{-1}; conversely the first row of S expresses the image of r in the new module. The numerical Fox coefficients have not been replaced by a guessed local presentation.

Cochains, inverse maps and the index-two factor. For a Γ-equivariant homomorphism, the values on the two new generators are a, Ja. Their one relation gives

$$(T - 7I)a - (JTJ - 7I)Ja = (I - J)(T - 7I)a = 0. \tag{E75.22}$$

This proves (E75.3)–(E75.4), in both directions. The map is saturated since the target A^- is torsion-free. The cocycle restriction and reconstruction of E.74 now give (E75.5).

One can obtain all three cochain degrees directly from (E75.13). Lift the matrix S back to Ω using the actual ring section induced by j. Use the coordinate change $(x_1, x_2) \mapsto (x_1, x_2)S$ on the last two coordinates of Ω^4; its new basis vectors have the rows of S^{-1} in the old basis. This is invertible and c-equivariant. For every coefficient module on which R_3 acts trivially, the projected Fox row is now $(0, 0, U, -V)$. The degree-zero fixed cochains are A^+. The paired degree-one generators give A^2; the sign on the top generator gives A^-. Taking the exact

C_2-invariant functor gives exactly (E75.6). The same argument works for each finite module $A/3^mA$ and its reduction maps. The top-cell scalar has been fixed by the k-local invariant in (E75.19).

Restriction from $\mathbb{Q}_3$ to k has degree two. Under local invariant duality it is dual to corestriction on degree-zero Tate duals. Thus on the coinvariants of $A(-1)$ it is the actual norm map $I - J$. On the invariant cochain complex it sends $y \in A^-$ to the same y in the D complex. Therefore its $\mathbb{Q}_3$ invariant coordinate is $[y/2]$. This proves (E75.7) with the stated sign and scalar. Directly, $(I - J)(J + I) = 0$, and

$$\frac{(I-J)x}{2} - x = -\frac{(I+J)x}{2}; \qquad (I-J)(y/2) = y \quad (y \in A^-). \tag{E75.23}$$

The relation maps in the two cokernels also agree by these identities.

Exact numerical coordinates and topology. The program `a_wild_kernel.py` selects columns of $I + J$ and $I - J$ whose reductions form the two eigenspace bases modulo three. Their union gives an invertible matrix over $\mathbb{Z}_3$; its exact rational inverse has denominators prime to three. In those coordinates the target of (E75.3) is a specified copy of $\mathbb{Z}_3^{\operatorname{rank} A^-}$.

Exact elimination always chooses an entry of least three-adic valuation in the remaining rectangle. Every subsequent elimination factor is consequently three-integral. The program retains the domain transformation Q, its exact inverse, the codomain transformation and the entire diagonal identity over $\mathbb{Q} \cap \mathbb{Z}_3$. If the rank is s, the last $n - s$ columns of Q give the saturated kernel basis, and the last $n - s$ rows of Q^{-1} give its inverse coordinate map. Their product is the identity and the original matrix annihilates those columns exactly. These are full integral coordinates, not finite-precision kernel guesses.

The same computation on $(T - I)A^+$ retains the degree-zero map. All index-two identities in (E75.23) and both original coefficient maps are checked exactly. The eighteen finite-level checks verify the actual identity

$$\#H^1(G_{\mathbb{Q}_3}, A/3^mA) = \#(H^1(G_{\mathbb{Q}_3}, A)/3^m) \cdot \#H^2(G_{\mathbb{Q}_3}, A)[3^m] \quad (1 \le m \le 6) \tag{E75.24}$$

from the two independent differentials. The resulting kernel ranks and invariant factors are (E75.8), matching the previously computed groups through the literal map (E75.7). All these modules are compact and all basis maps are continuous, so their continuous inverse maps also prove the topological assertions.

Finally, for any of the actual nonlinear geometric groups W, restriction to the split source has the exact form

$$z(rj(g)) = \vartheta(r)u(g), \quad \vartheta(j(g)rj(g)^{-1}) = u(g)\, g(\vartheta(r))\, u(g)^{-1}, \tag{E75.25}$$

where u is a Γ-cocycle and $\vartheta : R_3 \to W$ a continuous homomorphism. Multiplication in (E75.1) proves necessity and sufficiency. Geometric conjugation acts on u by its cocycle formula and on ϑ by the same inner conjugation. The module computation above is its exact additive specialization through $R_3 \to R_3^{\mathrm{ab}}$. For the actual geometric quotient maps $q_r : A \to A'$, (E75.3) sends a to $q_r a$; every displayed cocycle and cochain map commutes with this operation. No nonlinear relation value is inferred from its abelian image. □

A certified cyclotomic base change and its six local norm characters

E.76. The first two genus quotients after adjoining a ninth root of unity [Proved]

Keep K, L_d, β_d, and $\mathfrak{q}_{3,d}$ from E.72. Choose $\xi^3 = \zeta$, with ξ of order nine, and put

$$B_d = L_d(\xi), \qquad B = B_1 = \mathbb{Q}(\xi, a), \qquad a^3 = 17, \quad \beta_1 = a\xi. \tag{E76.1}$$

For every $d \geq 1$, B_d/L_d is cyclic of degree three, unramified at every finite prime, and split completely at the unique primes above 3 and 17. Thus B_d has three primes $P_{d,j}$ above 3 and three primes $Q_{d,j}$ above 17. The adjacent extensions B_{d+1}/B_d are totally ramified at exactly these six primes. The extensions B_{d+2}/B_d are cyclic of degree nine and are totally ramified at the same six primes.

The full ideal class group of B, with the certified coordinates specified below, is

$$\mathrm{Cl}(B) = (\mathbb{Z}/3)^2, \qquad [P_{1,j}] = (1,0), \quad [Q_{1,j}] = (2,0) \quad (1 \leq j \leq 3). \tag{E76.2}$$

The second class generator is the explicit prime of norm 109 in the certificate. In particular the six ramified prime classes generate precisely the first cyclic summand.

Let $U = \mathcal{O}_B^\times$, and order the six local places as $P_{1,1}, P_{1,2}, P_{1,3}, Q_{1,1}, Q_{1,2}, Q_{1,3}$. Let $(\xi, u_1, \ldots, u_8)$ be the complete certified unit generators in `e_basechange_field.gp`; the omitted generator -1 is a norm in both odd-degree extensions. The six local unit-norm quotient characters for B_2/B, in the explicitly recorded local generators, have matrix

$$M_3 = \begin{pmatrix} 2 & 1 & 0 & 1 & 0 & 0 & 0 & 2 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 2 & 0 & 2 & 0 & 0 & 0 & 2 & 2 \end{pmatrix} \pmod 3. \tag{E76.3}$$

Its rank is three. For B_3/B, the corresponding six characters take values in $\mathbb{Z}/9$ and have matrix

$$M_9 = \begin{pmatrix} 1 & 5 & 3 & 2 & 6 & 0 & 3 & 4 & 2 \\ 1 & 5 & 3 & 8 & 3 & 6 & 0 & 3 & 6 \\ 1 & 5 & 3 & 5 & 0 & 3 & 6 & 8 & 7 \\ 5 & 7 & 6 & 4 & 0 & 0 & 3 & 1 & 4 \\ 8 & 4 & 6 & 7 & 3 & 3 & 0 & 1 & 1 \\ 1 & 5 & 3 & 5 & 3 & 3 & 3 & 5 & 8 \end{pmatrix} \pmod 9. \tag{E76.4}$$

Here $\mathrm{coker}(M_9 : (\mathbb{Z}/9)^9 \to (\mathbb{Z}/9)^6) \simeq (\mathbb{Z}/9)^2 \oplus \mathbb{Z}/3$. The displayed maps are actual local norm-quotient coordinates; they do not identify the six quotient generators with a single common Hilbert-symbol root. Their kernels, the simultaneous kernel, and the following indices are consequently fixed without a Hilbert-symbol sign convention:

$$[U : U \cap N_{B_2/B} B_2^\times] = 27, \qquad [U : U \cap N_{B_3/B} B_3^\times] = 2187. \tag{E76.5}$$

For the actual cyclic groups $G_n = \mathrm{Gal}(B_{1+\log_3 n}/B)$, $n = 3, 9$, these give

$$|\,\mathrm{Cl}(B_2)^{G_3}| = 81, \qquad |\,\mathrm{Cl}(B_3)^{G_9}| = 243. \tag{E76.6}$$

There are more precise finite quotient maps. Let $\mathcal{G}_n/B$ be the largest abelian extension containing $B_{1+\log_3 n}$ that is unramified over that field. This is a finite subfield of its Hilbert class field. Write $\Gamma_n = \mathrm{Gal}(\mathcal{G}_n/B)$. The ray-class presentations and their full integral Smith coordinate maps give

$$\Gamma_3 \simeq \mathbb{Z}/9 \oplus (\mathbb{Z}/3)^3, \qquad \Gamma_9 \simeq (\mathbb{Z}/9)^3 \oplus \mathbb{Z}/3. \tag{E76.7}$$

For each j, the Frobenius of $P_{2,j}$ in $\mathcal{G}_3/B_2$ has Smith coordinates $(3,1,0,1)$. The Frobenii of $P_{3,1}, P_{3,2}, P_{3,3}$ in $\mathcal{G}_9/B_3$ have respective coordinates

$$(3,6,0,2), \quad (6,6,6,2), \quad (0,6,3,2). \tag{E76.8}$$

Every one of these six elements has exact order three. The inclusion $\mathcal{G}_3 \subset \mathcal{G}_9$ induces the explicit surjective matrix in the certificate; it sends each prime Frobenius in (E76.8) to its counterpart in (E76.7).

The actual automorphism

$$\delta(\xi) = \xi^4, \qquad \delta(a) = a\xi^{-3}, \qquad \delta(\beta_1) = \beta_1 \tag{E76.9}$$

acts on Γ_9. The coordinate calculation gives $\Gamma_9/(\delta-1)\Gamma_9 \simeq (\mathbb{Z}/3)^3$. Let $A_9 = \mathrm{Gal}(\mathcal{G}_9/B_3)$, the kernel of the actual Kummer character $\Gamma_9 \to \mathbb{Z}/9$. Then

$$A_9 \simeq (\mathbb{Z}/9)^2 \oplus \mathbb{Z}/3, \qquad A_9/(\delta-1)A_9 \simeq (\mathbb{Z}/3)^2. \tag{E76.10}$$

In the latter Smith coordinates all three elements of (E76.8) have image $(2,0)$.

The comparison with E.72 uses the literal ideal maps

$$\begin{aligned} N_{B_d/L_d}[P_{d,j}] &= [\mathfrak{q}_{3,d}], \\ j_d[\mathfrak{q}_{3,d}] &= \sum_{j=1}^{3} [P_{d,j}], \\ N_{B_d/L_d} j_d &= 3, \qquad j_d N_{B_d/L_d} = 1+\delta+\delta^2. \end{aligned} \tag{E76.11}$$

Moreover $\mathcal{G}_9/L_3$ is unramified and Galois. The subgroup $A_9/(\delta-1)A_9$ injects into $\mathrm{Gal}(\mathcal{G}_9/L_3)^{\mathrm{ab}}$; the class $[\mathfrak{q}_{3,3}]$ maps to $(2,0)$ in this subgroup. This gives the explicit order-three quotient of that original prime class. E.72 still gives the full bound $3 \mid \mathrm{ord}[\mathfrak{q}_{3,3}] \mid 9$. The finite quotients here evaluate the stated maps, without evaluating the remaining order in $\mathrm{Cl}(L_3)$ or the eventual integer h.

Finally $|\mathrm{Cl}(B_2)\{3\}| \geq 81 > 9 = |\mathrm{Cl}(B)\{3\}|$. Thus the initial norm map on three-class groups is surjective with nontrivial kernel. This is an exact failed initial equality for the stability test of Li–Ouyang–Xu–Zhang, Proposition 3.2, applied to $B_3/B_2/B$; all its cyclicity and ramification requirements hold. The computation does not supply a later equality. The source motivation remains LM §51.16, fols. 733v–736v, scans 172–178: these are concrete arithmetic calculations inside the norm systems of E.46 and E.72.

Proof. First fix all the field maps. The element 17 has a cube root in $\mathbb{Q}_3$: the cube map from $1+3\mathbb{Z}_3$ to $1+9\mathbb{Z}_3$ is bijective by the three-adic logarithm, and $-17 \in 1+9\mathbb{Z}_3$. Hence $\xi = \beta_1/a$ belongs to the completion of L_1 at three. At seventeen, the completion of K is the unramified quadratic extension of $\mathbb{Q}_{17}$ and its residue field has order 289. It contains μ_9 by Teichmuller lifting. The extension $K(\xi)/K$ is unramified away from three. These local maps show that its base change to every L_d is everywhere unramified and split at three and seventeen. The degree remains three: L_d/K is totally ramified at its unique prime above seventeen, whereas seventeen splits into three primes in $K(\xi)/K$, excluding an inclusion $K(\xi) \subset L_d$.

The element $\beta_d + 1$ is a uniformizer at three. Indeed the polynomial $(-1+Y)^{3^d} - 17\zeta$ is Eisenstein over K_3: its constant coefficient $-1-17\zeta$ has valuation one and its intermediate coefficients are divisible by three. The corresponding assertion at seventeen follows directly from $Y^{3^d} - 17\zeta$. The split completions above prove the stated total ramification in the B_d chain. The polynomial $Y^9 - \beta_d$ defines B_{d+2}/B_d, and $\mu_9 \subset B_d$ makes this extension cyclic.

Here is the unconditional class and unit certificate for B. The chosen primitive element is $t = a - \xi$, with minimal polynomial

$$t^{18} - 105t^{15} + 4137t^{12} - 81182t^9 + 1128687t^6 - 6822222t^3 + 28934443. \tag{E76.12}$$

The exact maps from ξ, a to this polynomial field, the integral basis, both class-generator ideals and all fundamental units are retained. PARI 2.19.0 computes the discriminant $-3^{27}17^{12}$ and

returns `bnfcertify`=1 for the full class group and unit basis. That return value certifies the calculation without a generalized Riemann hypothesis. The six `bnfisprincipal` coordinates give (E76.2); the first generator is $P_{1,1}$ and the other is the retained prime above 109. These are exact ideals in the displayed integral basis, not classes inferred from an unsuccessful principal-element search. The primary software descriptions and the complete fresh execution are part of the certificate [The26].

We explain why the finite local computations determine the entire six characters. At a place $P_{1,j}$ the field $B_{P_{1,j}}$ is $\mathbb{Q}_3(\xi)$, of absolute ramification degree six. Write $U_k = 1 + P_{1,j}^k$. Logarithm and exponential are inverse on U_4, so

$$U_{10} = U_4^3, \qquad U_{16} = U_4^9. \tag{E76.13}$$

Thus reduction modulo P^{10}, respectively P^{16}, followed by quotienting by third, respectively ninth, powers computes the actual local unit quotient by powers. Its elementary divisors are seven copies of 3, respectively 9. At the tame places the corresponding quotients are the third- and ninth-power quotients of $\mathbb{F}_{289}^\times$, each cyclic of the indicated order.

For $n = 3, 9$ put $Y^n = \beta_1$. The element $Y+1$ is a uniformizer of the local degree-n extension. For each k the program computes the literal multiplication determinant

$$N_k = N(1 + (Y+1)^k) \tag{E76.14}$$

in $B[Y]/(Y^n - \beta_1)$. All these elements are local units at the three wild places. For $n = 3$, the columns with $k = 1, 2, 4, 5, 7, 10$ have rank six modulo three. Their annihilator in the recorded seven local coordinates is $(2, 0, 2, 1, 2, 1, 0)$. For $n = 9$, the first ten columns together with $9I_7$ have Smith quotient $\mathbb{Z}/9$ at each wild place. The respective quotient rows are

$$\begin{gathered}\begin{pmatrix}1 & -8 & -4 & -7 & -1 & -6 & -3\end{pmatrix}, \qquad \begin{pmatrix}1 & -8 & -7 & -1 & -4 & -6 & -3\end{pmatrix}, \\ \begin{pmatrix}1 & -8 & -1 & -4 & -7 & -6 & -3\end{pmatrix}.\end{gathered} \tag{E76.15}$$

Local reciprocity identifies the unit norm quotient of a totally ramified cyclic degree-n extension with $\mathbb{Z}/n$. Consequently each computed subgroup of norm units, already of index n, is the entire unit norm kernel. At seventeen the tame unit norm kernel is exactly the subgroup with nth-power residue. This proves the characters, including their precise coordinate meanings [Mil20b, III, Theorems 3.1, 3.4].

Applying these six quotient rows to the complete global generators gives (E76.3)–(E76.4). The integer Smith calculations include their unimodular transformations and inverse maps. A global unit is automatically a local norm at every other place, since the extension there is unramified. The Hasse norm theorem identifies the simultaneous kernel with $U \cap NB^\times_{1+\log_3 n}$ [Mil20b, VIII, Theorem 3.1]. Its index is 3^3 for $n = 3$ and $9^6/(9^2 \cdot 3) = 3^7$ for $n = 9$. Chevalley's exact ambiguous class formula now reads

$$|\operatorname{Cl}(B_{1+\log_3 n})^{G_n}| = \frac{9\,n^6}{n\,[U : U \cap NB^\times_{1+\log_3 n}]}, \tag{E76.16}$$

which gives (E76.6). The degree-three formula and its complete ideal/unit proof are [Lem13, Theorem 1]; the general cyclic formula is also stated explicitly in [LOXZ22, Theorem 2.6, Remark 2.7(2), Lemma 2.8].

We next identify the finite ray quotients with the claimed fields, including the relevant quotient maps. Let $\mathfrak{m}_n$ have exponent ten at the three wild primes for $n = 3$, exponent sixteen for $n = 9$, and exponent one at the three tame primes. In $\operatorname{Cl}_{\mathfrak{m}_n}(B)$ quotient by all local norm-unit subgroups. The local maps are realized by the Chinese remainder theorem: put a

chosen local norm unit at one place and put 1 at the other five, and take the ray class of its principal ideal. These are exactly the columns in the retained presentation, together with the ray cyclic relations. The finite idelic description is

$$\Gamma_n = \mathbb{A}_{B,f}^{\times}/B^{\times}\left(\prod_{v\in S} N\mathcal{O}_{B_{1+\log_3 n},w}^{\times}\prod_{v\notin S}\mathcal{O}_{B,v}^{\times}\right). \tag{E76.17}$$

The global norm subgroup for the cyclic extension contains the denominator, so its class field contains $B_{1+\log_3 n}$. At each ramified place, the inertia quotient has size at most n and maps onto the inertia group of that cyclic field, of size n. It follows that the resulting extension over that field is unramified. Conversely local norm functoriality kills precisely the displayed norm units in every abelian extension of B unramified over $B_{1+\log_3 n}$. This proves the maximal property defining $\mathcal{G}_n$ by the actual Artin map [Mil20b, III, Lemma 3.2; V, Proposition 5.2].

The norm compatibility used here follows from the proved cup-product comparison in [Mil20b, III, Theorem 3.1 and its restriction/corestriction proof]. For each actual finite local tower $L/E/K$ above, put $G = \mathrm{Gal}(L/K)$ and $H = \mathrm{Gal}(L/E)$. Corestriction in Tate degree -2 is the inclusion-induced map $H^{\mathrm{ab}} \to G^{\mathrm{ab}}$, and in degree zero it is $N_{E/K}$ on the multiplicative norm quotients. The fundamental classes satisfy $\mathrm{Res}_H^G u_{L/K} = u_{L/E}$. The projection formula

$$\mathrm{Cor}_H^G(x \smile u_{L/E}) = \mathrm{Cor}_H^G(x) \smile u_{L/K}$$

therefore says that the inverse Artin maps commute with these two maps. Inverting those isomorphisms gives $\mathrm{Art}_K(N_{E/K}z) = \iota\,\mathrm{Art}_E(z)$, where $\iota : H^{\mathrm{ab}} \to G^{\mathrm{ab}}$ is induced by inclusion. This proves precisely the norm and local-unit compatibilities used in the genus and prime-Frobenius maps.

This also gives a direct proof of the ambiguous-order formula in the two cases used here. The kernel of the surjection from (E76.17) to $\mathrm{Cl}(B)$ is the quotient of the six local unit norm quotients by the image of U. Consequently its order is $n^6/[U : U \cap NB_{1+\log_3 n}^{\times}]$. Let H be the full Hilbert class field of $E = B_{1+\log_3 n}$. The group $\mathrm{Gal}(H/B)$ is generated by the abelian subgroup $\mathrm{Cl}(E)$ and one lift of a generator σ of $\mathrm{Gal}(E/B)$. Its commutator subgroup is exactly $(\sigma - 1)\,\mathrm{Cl}(E)$. Its maximal abelian quotient is Γ_n, with kernel over $\mathrm{Gal}(E/B)$ equal to $\mathrm{Cl}(E)/(\sigma - 1)\,\mathrm{Cl}(E)$. For a finite abelian group the kernel and cokernel of $\sigma - 1$ have equal order. Dividing $|\Gamma_n|$ by n therefore proves (E76.16) and (E76.6) directly from these exact maps.

The quotient in (E76.17) is an extension of the ordinary class group $(\mathbb{Z}/3)^2$ by a quotient of $(\mathbb{Z}/n)^6$. Its exponent therefore divides $3n$. This proves that using the ray class group modulo $3n$th powers in the certificate loses no element of Γ_n. Its relation matrices have exact integer Smith forms (E76.7). The local norm groups for $n = 9$ are contained in those for $n = 3$. This proves $\mathcal{G}_3 \subset \mathcal{G}_9$ and the surjection between the displayed ray presentations. The program computes this map on actual ray ideals; its change of ray generators is recorded with an inverse modulo the cyclic relation lattices.

For clarity, the prime Frobenii are computed by a local uniformizer norm, not by assigning an unramified symbol to a ramified prime of B. For both values of n,

$$N(Y + 1) = \beta_1 + 1 = \pi. \tag{E76.18}$$

For the chosen wild prime P, take $x \equiv \pi$ modulo $P^{1+v_P(\mathfrak{m}_n)}$ and $x \equiv 1$ at the other modulus factors. The ideal $I = (x)P^{-1}$ is coprime to the modulus. Global reciprocity identifies the local Artin class of π with $-[I]$ in additive ray coordinates. Norm compatibility identifies it with the Frobenius of the upper prime in $\mathcal{G}_n/B_{1+\log_3 n}$. Applying the Smith maps gives exactly (E76.8) and its cubic counterpart.

The final descent uses the actual automorphism, not an abstract permutation of Smith factors. In the primitive element (E76.12), it is $t \mapsto a\xi^{-3} - \xi^4$. The program verifies the polynomial relation, $\delta(\beta_1) = \beta_1$, and $\delta(\xi) = \xi^4$. Its action matrix preserves the ray relation lattice and the local norm-unit lattice, giving the stated action on Γ_9. The Kummer character $\Gamma_9 \to \mathbb{Z}/9$ is evaluated on actual ideal generators: at an unramified prime $\mathfrak{l}$, its coordinate is the unique $e \pmod 9$ with

$$\overline{\beta_1}^{(N\mathfrak{l}-1)/9} = \overline{\xi}^{\,e}. \tag{E76.19}$$

In the retained sixteen ray coordinates the row is

$$(8, 7, 5, 8, 4, 5, 2, 8, 3, 2, 8, 5, 3, 8, 1, 0) \pmod 9. \tag{E76.20}$$

The exact lattice kernel of this row, the restriction of δ to it, and both Smith coordinate maps prove (E76.10). They give image $(2, 0)$ for each of the three prime Frobenii.

The extension B_3/L_3 is generated by the restriction of δ, extending (E76.9) by $\delta(\beta_3) = \beta_3$. Its conjugation on $\mathrm{Gal}(B_3/B)$ raises a generator to the fourth power, so $\mathcal{G}_9$ is stable under it by the maximal property already proved. Hence $\mathcal{G}_9/L_3$ is Galois and unramified. For the exact sequence with abelian kernel

$$1 \longrightarrow A_9 \longrightarrow \mathrm{Gal}(\mathcal{G}_9/L_3) \longrightarrow \mathbb{Z}/3 \longrightarrow 1, \tag{E76.21}$$

the commutator subgroup is exactly $(\delta - 1)A_9$: the group is generated by A_9 and one lift of the cyclic quotient, and its quotient by these commutators is abelian. This proves the claimed injection into the abelianization. Since $\mathfrak{q}_{3,3}$ splits in B_3/L_3, its Frobenius belongs to this kernel and is the computed upper Frobenius. This establishes the precise transfer back to the original prime.

Finally (E76.11) follows directly from residue degree one and complete splitting, and holds on full class groups and on their three-primary parts. The norm from B_2 to B is surjective: its cyclic degree-three extension is ramified, so it has no nontrivial unramified subextension; write H_B for the Hilbert class field of B. Then $H_B \cap B_2 = B$, and $H_B B_2/B_2$ is unramified and abelian. Its Artin quotient gives the surjection

$$\mathrm{Cl}(B_2) \twoheadrightarrow \mathrm{Gal}(H_B B_2/B_2) \xrightarrow{\mathrm{res}} \mathrm{Gal}(H_B/B) \simeq \mathrm{Cl}(B).$$

Norm functoriality identifies this actual composite with $N_{B_2/B}$. Equation (E76.6) then shows that the norm kernel has order at least nine. Thus the equality premise in the opened stability theorem fails at this actual first step [LOXZ22, Proposition 3.2 and its full proof]. The classes in the kernels of the displayed genus maps have not been evaluated here; the statement retains exactly the remaining prime-class orders from E.72. □

The actual wild source over its fixed free quotient

E.77. A relative equivariant presentation and its nonlinear obstruction [Proved]

Retain the actual local groups, quotient and involution of E.74–E.75. There are actual generators t, t', r, r' of $D = G_{\mathbb{Q}_3(\zeta_3)}(3)$ for which

$$\begin{aligned} ctc = t', \qquad crc = r', \qquad \pi(t) = \tau, \quad \pi(t') = \tau', \quad \pi(r) = \pi(r') = 1, \\ D \;\simeq\; \left\langle t, t', r, r' \;\middle|\; (trt^{-1}r^{-7})(t'r't'^{-1}r'^{-7})^{-1} = 1 \right\rangle_{\text{pro-3}}. \end{aligned} \tag{E77.1}$$

The displayed isomorphism commutes with the fixed map to P and with c. Its construction retains the actual local relation and a convergent sequence of changes of generators over P. In

particular it does not replace that quotient by another free quotient with the same rank or signature.

The generators t, t' define a new actual section $j_1 : \Gamma \to \mathcal{D}$ of the same quotient. Let j_0 be the section fixed in E.75. Its exact comparison with j_1 is the continuous cocycle

$$h(g) = j_1(g)j_0(g)^{-1} \in R_3, \qquad h(c) = 1, \qquad h(gg') = h(g)\, j_0(g)h(g')j_0(g)^{-1}. \tag{E77.2}$$

The final r, r' have the same classes in R_3^{ab} as E.75's r_*, r'_*. All choices in (E77.1)–(E77.2) are made once in the actual source.

Let W be any of the actual geometric pro-three quotients in E.41, with its actual Γ action, and write $T(w) = \tau(w)$ and $J(w) = c(w)$. The full continuous local section set, modulo geometric conjugacy, has the following exact compact coordinates:

$$\frac{\{(u, a) \in W^2 : J\big(uT(a)u^{-1}a^{-7}\big) = uT(a)u^{-1}a^{-7}\}}{W^c}. \tag{E77.3}$$

The residual action is

$$w : (u, a) \longmapsto \big(wuT(w)^{-1},\, waw^{-1}\big), \qquad w \in W^c. \tag{E77.4}$$

The inverse sends c to the prescribed c-lift and gives the four actual source generators the section values

$$z(t) = u, \quad z(t') = J(u), \quad z(r) = a, \quad z(r') = J(a). \tag{E77.5}$$

Equations (E77.1) and (E77.3) prove directly that these values extend to a continuous section. The quotient in (E77.3) has its compact orbit topology; the correspondence and its inverse are homeomorphisms.

This also evaluates each wild central obstruction from actual lower section coordinates. For the central extension $W_r \to W_{r-1}$ with kernel $A = \mathcal{A}_r(W)$, choose the retained Hall lifts u_0, a_0 of a pair in (E77.3) at level $r - 1$ and put

$$v_0 = u_0T(a_0)u_0^{-1}a_0^{-7}, \qquad \omega = v_0J(v_0)^{-1} \in A^-. \tag{E77.6}$$

The local obstruction in the invariant coordinate of E.74 is

$$[\omega/2] \;\in\; A/((T - 7I)A + (J + I)A). \tag{E77.7}$$

For the actual lifts $u = u_0b$, $a = a_0d$, $b, d \in A$, the exact new defect is

$$\omega + (I - J)(T - 7I)d. \tag{E77.8}$$

Thus the image consists precisely of the lower pairs whose displayed defect is in the negative of that explicit image. Each nonempty fibre is the compact group $\mathcal{C}_r(W) \oplus \mathcal{K}_A$ of E.75, with its actual correction coordinates. In particular the source relation in this criterion is fully specified. Its evaluation retains the actual geometric operators T, J.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379); Part E(c),(d) and follow-up(4). This supplies the nonlinear local source and exact obstruction map following E.74–E.75. It does not replace the remaining global arithmetic evaluations in E.23.

Proof. An actual initial relative presentation. Start with the generators and minimal relation module used to prove E.75. Let

$$F = \langle t_0, t'_0, r_0, r'_0\rangle_{\text{pro-3}}, \qquad N = \ker(F \to D), \qquad K = \ker(F \to P). \tag{E77.9}$$

The involution swaps the two pairs and the horizontal generators initially lift j_0. All normal subgroups and commutator subgroups in this proof are closed. The universal relative abelian quotient is the actual split group

$$F/K' \simeq \Lambda^2 \rtimes P, \qquad K^{\mathrm{ab}} \simeq \Lambda^2, \qquad \Lambda = \mathbb{Z}_3[[P]]. \tag{E77.10}$$

Here a row (λ_1, λ_2) represents the corresponding completed sum of conjugates of r_0, r_0'. For an explicit proof, assign t_0, t_0' their marked images in P and assign r_0, r_0' the two module basis vectors. The free universal property gives the map in (E77.10). Its inverse sends a point mass $\delta_p e_i$ to pr_ip^{-1} modulo K'. The continuous action extends this map to the completed group ring. The two maps are inverse on all marked generators and then on their closed generated groups.

The equivariant matrix used in E.75 is an actual change of generators in this free relative presentation. Indeed every prescribed row of Λ^2 lifts to an element of K. Lift the first row of the inverse basis matrix and take its c-conjugate for the second row. Together with t_0, t_0' these elements are a free basis of F: their reduction on the Frattini quotient is an invertible block matrix. The resulting endomorphism is onto by the Frattini criterion and is injective because a finitely generated profinite group is Hopfian. One direct proof of the latter fact uses the finite set of open normal subgroups of any bounded index: a surjective endomorphism permutes that set, and hence is injective on every associated finite characteristic quotient. This change fixes the quotient P and commutes with c.

After this change the oriented top generator $e \in N^{\mathrm{ab}}$ has relative Fox row $(\tau - 7, -\tau' + 7)$ and satisfies $ce = -e$, exactly as in E.75. Lift $e/2$ to an actual $b \in N$ and take

$$\rho = b\, c(b)^{-1}. \tag{E77.11}$$

Its class is e, it normally generates N by the relative Frattini criterion, and $c(\rho) = \rho^{-1}$ holds in the free group itself. A lift of a chosen r_* in E.75 may be used: any discrepancy of its relative module row is in the image of N^{ab} and can be corrected by an element of N. Thus the initial actual images of the two vertical generators can be exactly r_*, r_*'.

In this new basis put

$$\rho_0 = (trt^{-1}r^{-7})(t'r't'^{-1}r'^{-7})^{-1}. \tag{E77.12}$$

It too is exactly inverted by c. Its row in K^{ab} is the same oriented row $(\tau - 7, -\tau' + 7)$. Consequently $\rho_0^{-1}\rho \in K'$.

The integral filtration and its relative kernel. Write $F_1 = F$ and $F_{n+1} = F_n^3[F_n, F]$. We use the commutator $[x, y] = xyx^{-1}y^{-1}$. The associated graded object is

$$\operatorname{gr} F = \mathfrak{L}_{\mathbb{F}_3[\pi]}(T, T', R, R'), \qquad \deg \pi = 1. \tag{E77.13}$$

Cubing induces multiplication by π. This is also the odd-prime case of the original lower-central calculation in [Lab67, §2, Propositions1–2, pp.112–113]. We give the needed integral justification. Use the Magnus map into the completed noncommutative power-series algebra over $\mathbb{Z}_3$, assigning 3 and every variable weight one. Its graded coefficient ring is $\mathbb{F}_3[\pi]$. For $g \in F_n$, the leading term of g^3 is three times the leading term of g: the other two terms in $(1 + X)^3$ have strictly higher weight, also for $n = 1$. The commutator leading term is the Lie bracket. The same formulas show additivity of the cube map on each layer and $\pi[x, y] = [\pi x, y] = [x, \pi y]$. Induction on the defining filtration therefore spans each layer by the elements $\pi^a H$ for integral Hall commutators H. Their Magnus leading polynomials are independent: Hall polynomials of each ordinary degree have distinct triangular leading words, and the powers of π separate the remaining weights. This proves (E77.13), with no division by three. The filtration tends to one, since its image in every finite three-group does.

Let $\mathfrak{J}$ be the Lie ideal generated by R, R' in (E77.13). The actual induced filtrations satisfy

$$\operatorname{gr} K = \mathfrak{J}, \qquad \operatorname{gr} K' = \mathfrak{J}'. \tag{E77.14}$$

For clarity, the second equality is for $K' \cap F_n$, not a reindexed intrinsic filtration on K'. Here is the strict comparison. Put $\mathfrak{m} = (3, \tau - 1, \tau' - 1) \subset \Lambda$. The relative Fox map $K \to \Lambda^2$ in (E77.10) is filtered from $K \cap F_n$ to $\mathfrak{m}^{n-1}\Lambda^2$. To check this without an intersection assumption, filter $\Lambda^2 \rtimes P$ by $\mathfrak{m}^{n-1}\Lambda^2 \rtimes P_n$. The formula for a commutator and $(1 + p + p^2)v$ for a cube show that this is a lower three-central filtration. Its graded relative module is $\mathbb{F}_3[\pi]\langle T, T'\rangle^2$.

The ideal $\mathfrak{J}$ is the free Lie algebra on the free module with basis

$$\operatorname{ad}_{X_1} \cdots \operatorname{ad}_{X_d} R_i, \qquad X_j \in \{T, T'\}, \quad R_i \in \{R, R'\}. \tag{E77.15}$$

An elementary universal-property proof constructs this free Lie algebra and lets T, T' act by prepending their adjoints as derivations. Its semidirect product with the free horizontal Lie algebra receives and sends back all four free generators. The two maps are inverse. Thus its abelianization is precisely $\mathbb{F}_3[\pi]\langle T, T'\rangle^2$, and the graded relative Fox map has kernel $\mathfrak{J}'$. If $k \in K' \cap F_n$, its Fox value is zero, so its initial form lies in $\mathfrak{J}'$. Conversely every bracket and every π multiple defining a homogeneous element of $\mathfrak{J}'$ lifts to a commutator or cube in K'. This proves both inclusions in the second equality of (E77.14). The first follows either from the same argument or from the actual section $P \to F$, which makes $F_n \to P_n$ surjective with the stated graded kernel.

The relative correction map. Suppose $\rho = \rho_0 e_n$ with $e_n \in K' \cap F_n$. At degree two the error is a multiple of $[R, R']$. The paired shear

$$t \longmapsto t(r')^a, \qquad t' \longmapsto t' r^a \tag{E77.16}$$

changes the initial form of ρ_0 by $-2a[R, R']$. Choose a modulo three to remove the error. This is an invertible triangular change of the Frattini basis, and it preserves the actual quotient and the two vertical module classes. Thereafter $n \geq 3$.

Change t, t' by right factors in $K \cap F_{n-1}$ and r, r' by right factors in $K' \cap F_{n-1}$. For their initial forms $(x, x', y, y') \in \mathfrak{J}_{n-1}^2 \oplus (\mathfrak{J}'_{n-1})^2$, the initial change of ρ_0 is

$$\mathcal{L}(x, x', y, y') = [x, R] - [x', R'] + (\operatorname{ad} T - 2\pi)y - (\operatorname{ad} T' - 2\pi)y'. \tag{E77.17}$$

In particular the coefficient is 2π because $7 - 1 = 2 \cdot 3$. The ordinary commutator expansion proves this formula; terms from $\binom{7}{2} = 21$ have an extra factor of three and are above the displayed weight. It is the relative version of the variation on pp.116–118 of [Lab67].

The image of (E77.17) is all of $\mathfrak{J}'_n$. This assertion is integral in the graded coefficient ring and has a direct proof. Using Jacobi, every bracket in the free Lie algebra (E77.15) is a sum of brackets $[v, b]$ where v is one of its displayed free generators and $b \in \mathfrak{J}$. This follows by repeatedly expanding $[[v_1, v_2], b] = [v_1, [v_2, b]] - [v_2, [v_1, b]]$; coefficients and powers of π may be placed on b. Modulo the image of (E77.17),

$$[\operatorname{ad} T(v), b] \equiv 2\pi[v, b] - [v, \operatorname{ad} T(b)], \tag{E77.18}$$

and the same identity holds for T'. Indeed their difference is $(\operatorname{ad} T - 2\pi)[v, b]$, an allowed term. Induction on the horizontal word in v reduces its first entry to R or R'. The resulting bracket is in $[\mathfrak{J}_{n-1}, R] + [\mathfrak{J}_{n-1}, R']$, also in the image. All operations preserve the total weight, which proves surjectivity in every degree.

This calculation preserves the actual involution. Let c_{dom} interchange the paired correction entries and apply c to each. Then $\mathcal{L} c_{\mathrm{dom}} = -c\mathcal{L}$. The initial error is anti-invariant: the exact

identities $c(\rho)=\rho^{-1}$ and $c(\rho_0)=\rho_0^{-1}$ give this after passing to the central layer. Average a solution with its c_{dom} transform, dividing by two. Lift its first and third entries to the stated actual closed subgroups and use their c-conjugates for the other entries. This is an exactly paired change of generators. Choose its sign to cancel the error.

All these changes are over P. Horizontal changes by K act on K^{ab} by the same marked P action, and vertical changes by K' do not change its basis. The two relative rows therefore continue to agree exactly, so the next error remains in K' as well as in F_{n+1}. For $n\geq 3$ the changes are Frattini automorphisms. Their depths tend to infinity; the four generators converge to a basis, because their limiting Frattini matrix remains invertible. The limiting relation is exactly $\rho=\rho_0$ in that basis. This proves (E77.1) as an isomorphism over the fixed P, with c retained. It also proves the asserted R_3^{ab} comparison.

Change of section and nonlinear coordinates. These are the full local section sets for $G_{\mathbb{Q}_3}$. Indeed the kernel of its map to $\mathcal{D}$ acts trivially on W, and E.74's Kummer comparison proves that this kernel has no quotient C_3. Any nontrivial continuous homomorphism from it to a pro-three group would have a nontrivial finite three-group image, and hence a quotient C_3. Its restriction under every cocycle is therefore trivial. All the cocycles in question factor through the displayed actual local source.

Since π maps the new horizontal pair to the free basis τ,τ', their map from P is injective and is a section. Together with the unchanged c it gives j_1. Multiplication proves (E77.2). For any actual local section write, as in E.75, $z(rj_0(g))=\theta(r)u_0(g)$. The exact new coordinates are

$$u_1(g)=\theta(h(g))u_0(g),\qquad z(rj_1(g))=\theta(r)u_1(g). \tag{E77.19}$$

Conversely $u_0(g)=\theta(h(g))^{-1}u_1(g)$. The equivariance of θ changes by the inner conjugation by $\theta(h(g))$, as is seen by applying θ to $j_1(g)rj_1(g)^{-1}$. These formulas retain the actual change of splitting.

Coprime conjugacy for C_2 normalizes the section on c. For completeness this follows at finite three-group quotients by the conjugacy of order-two complements: induction on the nontrivial invariant centre reduces it to an abelian group, where division by two kills the cocycle. Compactness gives compatible conjugators in their inverse limit. The residual conjugators are exactly W^c. The source relation (E77.1), evaluated in $W\rtimes\Gamma$, is $vJ(v)^{-1}$, where $v=uT(a)u^{-1}a^{-7}$. This proves (E77.3)–(E77.5) in both directions. The generators identify the continuous Hom space with a closed subspace of W^4, and all conjugating groups are compact. The obtained continuous bijections between compact Hausdorff orbit spaces are therefore homeomorphisms.

Finally, in a central layer ω in (E77.6) lies in A^- because $J(\omega)=\omega^{-1}$. Replacing a_0 by a_0d changes v_0 by $(T-7I)d$; replacing u_0 by u_0b has no effect. This proves (E77.8) exactly, including its sign. The oriented relator was retained throughout, so the index-two comparison in E.75 is still $y\mapsto y/2$, giving (E77.7). When the defect is removed, the differences are $b\in A$ and $d\in\mathcal{K}_A$; central residual conjugation by $w\in A^+$ changes b by $(I-T)w$ and leaves d fixed. There is no additional twisted lower stabilizer by E.74's positive-weight argument. The resulting fibre is exactly $A/(T-I)A^+\oplus\mathcal{K}_A=\mathcal{C}_r(W)\oplus\mathcal{K}_A$, with the retained Smith representatives and inverse maps of E.41 and E.75. □

An evaluated wild class-three arithmetic quotient

E.78. Actual wild operators, both lifting images, and complete class-three fibres [Proved]

Put $K=\mathbb{Q}(\zeta_3)$, $k=K_3=\mathbb{Q}_3(\zeta_3)$, $\lambda=1-\zeta_3$, and $a=\zeta_3\lambda^{-3}$. Retain E.54's actual map $h=(x+y)/(x+\zeta_3y)$ from the nine-punctured cubic, its source tangent $s=z/x$ of value

one, and its induced surjection $N \twoheadrightarrow F$ onto the free pro-three group on the positive target generators x, y. Write

$$C = F/\gamma_4 F, \qquad C_2 = F/\gamma_3 F, \qquad C_1 = F^{\mathrm{ab}}, \qquad c = [x,y], \quad d = [x,c], \quad e = [y,c]. \tag{E78.1}$$

Commutators are $[u,v] = uvu^{-1}v^{-1}$ and Hall words are $x^A y^B c^n d^r e^s$.

Fix positive coherent roots $3_n = 3^{1/3^n}$ and $\zeta_{3^{n+1}}$. There is a unique compatible choice of λ_n satisfying

$$\lambda_n^{3^n} = \lambda, \quad \lambda_n^2 = -3_n \zeta_{3^{n+1}}, \quad \lambda_{n+1}^3 = \lambda_n. \qquad a_n = \zeta_{3^{n+1}} \lambda_n^{-3}. \tag{E78.2}$$

Use the punctured-disc comparison determined by these roots in E.54. If $\kappa = \kappa_3(3)$, its exact path character is

$$\kappa_a = -\frac{\chi - 1}{6} - \frac{3}{2}\kappa. \tag{E78.3}$$

Here and below every divided expression is integral in $\mathbb{Z}_3$ on its stated domain.

The source marking is the following actual transport of E.37's Frobenii. Start with $\tau_0, \tau_0' = c_{\mathrm{ar}} \tau_0 c_{\mathrm{ar}}$, where $\chi(\tau_0) = 7$ and $t_0 = \kappa(\tau_0) \in 1 + 3\mathbb{Z}_3$. The symbol c_{ar} denotes arithmetic complex conjugation, distinct from the geometric Hall word c. Set

$$\begin{aligned} \rho_0 &= \tau_0' \tau_0^{-1}, \qquad & h_0 &= \rho_0^{(t_0^{-1}-1)/12}, \\ \tau &= h_0 \tau_0 h_0^{-1}, \qquad & \tau' &= c_{\mathrm{ar}} \tau c_{\mathrm{ar}}. \end{aligned} \tag{E78.4}$$

These are the E.49 free basis of the same actual group $P = G^{(3)}_{K,\{3\}}$, and

$$(\chi(\tau), \kappa_a(\tau)) = (7, -5/2), \qquad (\chi(\tau'), \kappa_a(\tau')) = (7, 1/2). \tag{E78.5}$$

The map from the old to the new free basis and the root/path conjugations are retained in the proof.

For $\mu \in \{-5/2, 1/2\}$ define T_μ on C by

$$\begin{aligned} T_\mu(x) &= x^7, \\ T_\mu(y) &= y^7 c^{7\mu} d^{7\mu(\mu-1)/2} e^{14-28\mu}, \\ T_\mu(c) &= c^{49} d^{147+49\mu} e^{147}, \\ T_\mu(d) &= d^{343}, \qquad T_\mu(e) = e^{343}. \end{aligned} \tag{E78.6}$$

These are the actual based operators of τ, τ'. Put $T_- = T_{-5/2}$ and $T_+ = T_{1/2}$. Apply the source construction of E.77 with the explicitly transported marking (E78.4). It gives actual generators t_s, t_s', r_s, r_s' of $D = G_k(3)$, mapping to $\tau, \tau', 1, 1$, with

$$D = \left\langle t_s, t_s', r_s, r_s' \;\middle|\; f(t_s, r_s) f(t_s', r_s')^{-1} = 1 \right\rangle_{\text{pro-3}}, \qquad f(t,r) = trt^{-1}r^{-7}. \tag{E78.7}$$

All choices are made in this actual source over its fixed quotient $D \to P$.

For $v_\mu(u, a_0) = u T_\mu(a_0) u^{-1} a_0^{-7}$, the full local section set $H^1(k, C)$ has the following exact compact coordinates:

$$\begin{gathered} \{(u, u', a_0, a_0') \in C^4 : v_{-5/2}(u, a_0) = v_{1/2}(u', a_0')\}/C, \\ w : (u, u', a_0, a_0') \longmapsto (wuT_-(w)^{-1},\ wu'T_+(w)^{-1},\ wa_0 w^{-1},\ wa_0' w^{-1}). \end{gathered} \tag{E78.8}$$

The four entries are the values on t_s, t_s', r_s, r_s', respectively.

Both lower central lifting images are evaluated as follows. A C_2-valued section is represented by

$$\begin{aligned} a_0 &= x^A y^B c^n, \qquad u = x^U y^V c^N, \\ a_0' &= x^{A'} y^{B'} c^{n'}, \qquad u' = x^{U'} y^{V'} c^{N'}, \\ L_\mu &= 3AB - AV + BU + B\mu + 6n. \end{aligned} \tag{E78.9}$$

Its exact relation is $L_{-5/2} = L'_{1/2}$, where a prime means substitution of all six primed variables. For the same words with zero d, e coordinates put

$$\begin{aligned} R_\mu = \frac{7}{2}\big(&29A^2B - 7A^2V + 14ABU + 14AB\mu - 3AB \\ &- 2AN - 2AUV + AV + 90An + BU^2 + 2BU\mu - BU \\ &+ B\mu^2 - B\mu + 14Un + 14\mu n + 42n\big), \end{aligned} \tag{E78.10}$$

and

$$\begin{aligned} S_\mu = \frac{7}{2}\big(&13AB^2 - 3AB - AV^2 + AV + 7B^2U + 7B^2\mu \\ &- 2BN - BU + 2BV\mu - B\mu + 90Bn + 4B + 14Vn + 42n\big). \end{aligned} \tag{E78.11}$$

The image of $H^1(k, C) \to H^1(k, C_2)$ is exactly

$$\Delta R = R_{-5/2} - R'_{1/2} \in 3\mathbb{Z}_3, \qquad \Delta S = S_{-5/2} - S'_{1/2} \in 3\mathbb{Z}_3. \tag{E78.12}$$

A particular lift keeps u, u', a_0' unchanged and replaces

$$a_0 \longmapsto a_0 d^{-\Delta R/336} e^{-\Delta S/336}. \tag{E78.13}$$

Every nonempty fibre, modulo full geometric conjugacy, has the explicit compact coordinates

$$\mathbb{Z}_3^4 \times (\mathbb{Z}/9)^2. \tag{E78.14}$$

More precisely, after the particular lift write $[z] = d^{z_1} e^{z_2}$. Every other lift has the following changes, and its class has the indicated unique coordinates:

$$\begin{aligned} &a_0 \mapsto a_0[z], \qquad a_0' \mapsto a_0'[z], \qquad u \mapsto u[b], \qquad u' \mapsto u'[b'], \\ &\qquad\qquad (z,\ b - b',\ b' \bmod 342\mathbb{Z}_3^2). \end{aligned} \tag{E78.15}$$

For the inverse choose the two residues of b' in $\{0, \dots, 8\}$ and add $b - b'$ to obtain b. The exact remaining conjugation is $(b, b') \mapsto (b - 342w, b' - 342w)$.

At the preceding stage, every abelian section has four arbitrary vector values on the same source generators. For their coordinates in (E78.9) define

$$\begin{aligned} D_2 = 3(AB - A'B') - \tfrac{5}{2}B - \tfrac{1}{2}B' \\ + UB - VA - U'B' + V'A'. \end{aligned} \tag{E78.16}$$

Its class is in the image of $H^1(k, C_2) \to H^1(k, C_1)$ exactly when $D_2 \in 3\mathbb{Z}_3$. A particular lift sets $n = -D_2/6$, $n' = N = N' = 0$. Every nonempty fibre has coordinates $\mathbb{Z}_3^2 \times \mathbb{Z}/3$, given by the scalar version of (E78.15) with c in place of (d, e) and 48 in place of 342. The abelian section group itself has the explicit coordinates

$$H^1(k, C_1) \simeq \mathbb{Z}_3^6 \times (\mathbb{Z}/3)^2 : \quad (u - u', a_0, a_0',\ u' \bmod 6\mathbb{Z}_3^2). \tag{E78.17}$$

All inverse maps retain the actual conjugating element.

For an explicit family take, in C_2,

$$a_0 = y, \qquad u = x^{5/2}c^N, \qquad a_0' = u' = 1. \tag{E78.18}$$

Every member satisfies the source relation at class two. Its exact class-three error is $e^{7(2-N)}$. Its fibre is nonempty exactly for $N \equiv 2 \pmod 3$; one lift sends r_s to $ye^{(N-2)/48}$. Thus $N = 0$ has empty fibre and $N = 2$ already lifts with the same displayed words. They have the same abelian coordinates.

Every member of (E78.18) has an actual intrinsic $N/\gamma_3 N$ preimage. The proof constructs them from a rational cusp path and two chosen geometric lifts of x, y. The preimage constructed for $N = 2$ comes from a full N-valued local section. The preimage constructed for $N = 0$ has no $N/\gamma_4 N$-valued lift. The two constructed intrinsic sections have the same intrinsic abelian coordinates.

The arithmetic comparison with the cubic is the restriction to G_k followed by the actual maps

$$\begin{aligned} N/\gamma_4 N \twoheadrightarrow C, \qquad N/\gamma_3 N \twoheadrightarrow C_2, \\ \ker(N/\gamma_j N \to F/\gamma_j F) = h_*^{-1}(\gamma_j F)/\gamma_j N, \qquad j = 3, 4. \end{aligned} \tag{E78.19}$$

It gives a commutative diagram of section sets and the central map $\mathrm{Lie}_3(N^{\mathrm{ab}}) \to \mathrm{Lie}_3(F^{\mathrm{ab}}) = \mathbb{Z}_3(3)^2$. Consequently (E78.12) is an evaluated component of the intrinsic wild obstruction. In particular every constructed preimage of (E78.18) with $N \not\equiv 2 \pmod 3$ has empty intrinsic class-three fibre. For an arbitrary quotient section the kernel in (E78.19) is the retained intrinsic lifting problem.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). The actual operators, source marking and tangent transport in this quotient are all evaluated.

Proof. Coherent roots and the literal source transport. The identity $\lambda^2 = -3\zeta_3$ gives (E78.2): of the two square roots of $-3_n\zeta_{3^{n+1}}$, exactly one has 3^nth power λ. The same uniqueness makes its cube equal to the preceding root. Hence these are coherent roots, rather than separate choices at finite levels. Since the coherent roots of -1 can all be taken equal to -1, their Kummer character is zero. Taking Galois ratios in (E78.2) therefore gives

$$2\kappa_\lambda = \kappa + \frac{\chi - 1}{3}, \qquad \kappa_a = \frac{\chi - 1}{3} - 3\kappa_\lambda, \tag{E78.20}$$

which proves (E78.3). All these identities hold on the actual G_K and after restriction to G_k.

For the E.37 pair, the cubic residue of three at the prime over seven with $\zeta_3 \bmod 7 = 2$ gives $t_0 \equiv 1 \pmod 3$. Positive real roots give $\kappa(c_{\mathrm{ar}}) = 0$, so $\kappa(\tau_0') = -t_0$. The cocycle law now gives the exact identities

$$\begin{aligned} \chi(\rho_0) = 1, \quad \kappa(\rho_0) = -2t_0, \quad \kappa(h_0) = (t_0 - 1)/6, \\ \kappa(\tau) = 1, \quad \kappa(\tau') = -1, \quad \kappa_a(h_0) = (1 - t_0)/4. \end{aligned} \tag{E78.21}$$

The exponent in (E78.4) is integral because $t_0 - 1 \in 3\mathbb{Z}_3$. The endomorphism of the free group P sending its original basis to (τ, τ') induces the identity on its Frattini quotient. It is consequently a continuous automorphism. It commutes with c_{ar} and preserves χ. Formula (E78.4), together with the inverse of this actual free-group automorphism, is the source marking map.

This transport also has an exact geometric description. Before it, the two operators of E.54 are $\alpha_{7,-1-3t_0/2}$ and $\alpha_{7,-1+3t_0/2}$. The operator of h_0 is $\alpha_{1,(1-t_0)/4} = \mathrm{Ad}(x^{(1-t_0)/4})$. The composition law in E.54 conjugates the first operator to $\alpha_{7,-5/2}$. For the second one, $c_{\mathrm{ar}}h_0c_{\mathrm{ar}}$

has the opposite Kummer coordinate and conjugates it to $\alpha_{7,1/2}$. These are literal equalities of based automorphisms of C. Thus the change of Frobenius prolongations and the induced inner path operators have both been specified.

For completeness, implement the new marking in the source construction as follows. In E.75's initial splitting replace the horizontal lifts by $j_0(h_0)t_{s,0}j_0(h_0)^{-1}$ and its c_{ar} conjugate, leaving the initial vertical pair unchanged. In the free relative presentation these substitutions form a Frattini-invertible automorphism and induce exactly (E78.4) on P. Run the oriented Fox change and the convergent relative corrections of E.75–E.77 with these horizontal generators. Their proofs use the actual free basis, its swapped involution and its two cyclotomic values seven, all retained by (E78.4). Every later correction is in the kernel of the same map to P. This gives (E78.7) in the actual local group. The change from the old source splitting to the new one is the actual cocycle $j_{\mathrm{new}}(g)j_{\mathrm{old}}(g)^{-1}$ in that kernel, with the change-of-section formula of (E77.2) and (E77.19). In particular no equality of the original and transported Frobenius lifts is used.

The genuine operators and the source equation. E.54 gives

$$\alpha_{\chi,k}(y) = x^k c^{-(\chi^2-1)/24} y^\chi c^{(\chi^2-1)/24} x^{-k}.$$

Its degree-two coefficient follows by integral collection in the genuine three-cycle relation, with the positive commutator convention checked there; the original relation and coefficient are also in [Fur10, Theorem2, equation(5), §3 and Corollary12]. The higher part of the conjugator is central in C and acts trivially by conjugation there. Substitution of (E78.5) in E.54's full Hall images gives (E78.6), including its c, d, e terms.

The action on C factors through P: the two characters in (E78.3) are unramified outside three, and their action is E.54's explicit pro-three image. For a continuous C-valued cocycle, its graph in the semidirect product with that pro-three image is a homomorphism into a pro-three group. It therefore factors through $D = G_k(3)$. Assigning arbitrary four values to the free generators of the presentation (E78.7) defines a homomorphism into $C \rtimes P$. The relator maps precisely to $v_{-5/2}(u, a_0)v_{1/2}(u', a_0')^{-1}$. It vanishes exactly under (E78.8), proving both existence and exhaustiveness of the section coordinates. Geometric conjugation gives the four formulas displayed there.

Integral collection and both images. The Hall law from E.54 is

$$\begin{aligned}(a,b,n,r,s)&(A,B,N,R,S)\\ &= (a+A,\ b+B,\ n+N-Ab,\\ &\quad r+R+bA(A+1)/2-An,\\ &\quad s+S+Ab(b+1)/2-Bn+AbB).\end{aligned} \tag{E78.22}$$

Every denominator is a unit at three. It is an identity for integral powers by collection, and hence for all five three-adic coordinates by density and continuity. In these coordinates the actual operator has the complete expression

$$\begin{aligned}T_\mu(A,B,n,r,s) = (&7A,\ 7B,\ 7\mu B+49n,\\ &7\mu(\mu-1)B/2+(147+49\mu)n+343r,\\ &(14-28\mu)B-49\mu B(B-1)/2+147n+343s).\end{aligned} \tag{E78.23}$$

Substitute (E78.22)–(E78.23) into the four factors defining v_μ. Collection yields the exact identity

$$v_\mu(u, a_0) = c^{7L_\mu} d^{R_\mu} e^{S_\mu} \tag{E78.24}$$

for the zero-central Hall lifts in (E78.9). The explicit polynomials are (E78.10)–(E78.11). They may alternatively be verified coefficient by coefficient in the truncated integral Magnus algebra; that independent computation is part of the certificate below.

In C_2 the relator is therefore $7(L_{-5/2} - L'_{1/2})$. Its part independent of n, n' is $7D_2$, and its remaining part is $42(n - n')$. This proves (E78.16) and its actual particular lift. At class three the common c coordinate cancels: the subgroup generated by c, d, e is abelian, so the exact remaining error is $d^{\Delta R} e^{\Delta S}$. Changing the central coordinates of a_0, a'_0 by $p, p' \in \mathbb{Z}_3^2$ changes this error by

$$(343 - 7)(p - p') = 336(p - p'). \tag{E78.25}$$

Changing the central coordinates of u, u' does not change it. Since $336\mathbb{Z}_3 = 3\mathbb{Z}_3$, this proves the necessity and sufficiency in (E78.12) and the particular lift (E78.13).

The full conjugacy fibres and their topology. At either step let the central kernel be A_j, where $A_2 = \mathbb{Z}_3(2)c$ and $A_3 = \mathbb{Z}_3(3)d \oplus \mathbb{Z}_3(3)e$. After one solution, the differences on r_s, r'_s satisfy $(7^j - 7)(p - p') = 0$. These lattices are torsion-free and $j = 2, 3$, so $p = p'$. The differences on t_s, t'_s are arbitrary b, b', modulo exactly

$$(b, b') \longmapsto (b - (7^j - 1)w,\ b' - (7^j - 1)w). \tag{E78.26}$$

This proves the stated coordinates and their inverse with $7^2 - 1 = 48$ and $7^3 - 1 = 342$. More explicitly the normalizing conjugator is $c^{(b' - \epsilon)/48}$ at class two and $[(b' - \epsilon)/342]$ at class three, where ϵ is the chosen finite residue.

There is no additional quotient from a lower stabilizer. Indeed, a nonidentity element fixed by a lower twisted section has a first nonzero Hall graded coordinate. The operator $\mathrm{Ad}(u)T_-$ on degree j is multiplication by 7^j. The equation $(7^j - 1)v = 0$ has no nonzero solution in the torsion-free graded lattice. Thus the lower fixed subgroup is trivial. Every geometric conjugacy of lower sections consequently has a unique lower conjugator. Lifting it gives precisely the central freedom already divided out in (E78.26). At the abelian stage the same calculation uses $7 - 1 = 6$ and gives (E78.17) directly.

All coordinate spaces and the relator-zero sets are compact. Their group actions are continuous and their orbit spaces are Hausdorff. The orbit map for a fixed lower section has trivial stabilizer, hence is a homeomorphism from the compact lower group to that orbit. Division by 48 or 342 on its image is continuous, and selection of the indicated finite residues is continuous. These facts prove the asserted homeomorphisms, including the inverses with their actual normalizing conjugators.

Substituting (E78.18) in (E78.24) gives $L = R = 0$ and $S = 7(2 - N)$. Equation (E78.25) proves every assertion about its quotient fibres. The maps in (E78.19) and their kernels are induced by the actual surjection h_* of E.54. Its path comparison is the group isomorphism (E54.17), with the specific roots (E78.2), so it also intertwines the arithmetic semidirect products and all section and central obstruction maps.

Actual intrinsic preimages and a wild nonlifting section. For the full based action used here, the rational coordinate isomorphism $[x : y : z] \mapsto [z : y : x]$ sends E.34's cusp $[0 : -1 : 1]$ to $[1 : -1 : 0]$. It sends the source parameter $s_0 = -x/z$ to the parameter $s = z/x$ with $s = -s_0$. Choose the compatible 3^n-th roots of -1 all equal to -1. These roots are rational and compatible, so the resulting punctured-disc comparison is Galois invariant. The coordinate isomorphism followed by this tangential comparison therefore transports E.34's genuine based N-action to the present one. In particular it still factors through the same P over K, as required when using the actual source generators in (E78.7).

Choose any of the three K-rational cusps B_i where $1 - h$ has order one, as in (E54.16). An actual geometric path from the original tangent to a positive tangent at this cusp gives an inertia generator $d_* \in N$ and an element $u_* \in N$ with

$$u_* T(d_*) u_*^{-1} = d_*^7. \tag{E78.27}$$

Here T is the genuine operator of τ on N. Explicitly, for the path p the element $p\,\tau(p)^{-1}$ is such a conjugator, because the local cusp action raises its positive inertia generator to the seventh power. The image of d_* under h_* is conjugate to y, with exponent one. If it is gyg^{-1}, choose $v \in N$ with $h_*(v) = g^{-1}$ and replace

$$d_* \longmapsto v d_* v^{-1}, \qquad u_* \longmapsto v u_* T(v)^{-1}. \tag{E78.28}$$

Surjectivity of the actual h_* supplies v, and (E78.28) is the exact morphism preserving (E78.27). We now have $h_*(d_*) = y$.

Write the image of u_* in C in Hall coordinates. The identity $v_{-5/2}(h_*(u_*), y) = 1$ forces its x coordinate to be $5/2$. Equations (E78.10)–(E78.11) then force its c coordinate to be $2 - 5V/2$, where V is its y coordinate; the d, e coordinates are retained without evaluation. Replace the actual generator value u_* by $d_*^{-V} u_*$. Left multiplication by this cusp power preserves (E78.27). The exact Hall law gives

$$\begin{aligned} h_*(d_*^{-V} u_*) \bmod \gamma_3 F &= x^{5/2} c^2, \\ (u, u', a_0, a_0') &= (d_*^{-V} u_*,\ 1,\ d_*,\ 1). \end{aligned} \tag{E78.29}$$

These four actual N-values satisfy both sides of the relator in (E78.7) separately with value one. They therefore define a full N-valued local section. Formula (E78.29) specifies a choice of section values, rather than identifying distinct original sections without a map.

Choose actual $x_*, y_* \in N$ with images x, y and put $w_* = [x_*, y_*]$. The class of w_* is central in $N/\gamma_3 N$ and its image is c. In that quotient replace only the t_s value in (E78.29) by

$$d_*^{-V} u_* \, w_*^{N-2}. \tag{E78.30}$$

The relator remains exactly one: its changed factor conjugates $T(d_*)$ by a central element. Thus (E78.30) constructs an actual intrinsic class-two section for every $N \in \mathbb{Z}_3$, with image (E78.18). All their intrinsic abelian values agree. At $N = 2$ the original full section is an intrinsic lift. At $N = 0$, the nonzero class-three obstruction already computed in its actual quotient excludes every intrinsic class-three lift. This proves both intrinsic assertions by explicit source and geometric maps.

The self-contained program `a_wild_class3.py` proves the full symbolic identities for L_μ, R_μ, S_μ and for the central change 336. It checks 640 identities in an independent associative Magnus model, all five Hall images for both actual operators, eight exact source/root transports, 67 class-two lifting cases, seven class-three lifting cases, 180 abelian and 201 class-two fibre inverses, and 35 complete central-fibre coordinates and inverses. Its symbolic family calculation retains the variable N, and its symbolic cusp-path normalization retains V and both unevaluated central path coordinates. All final checks pass; the all-input conclusions follow from the polynomial identities and the proved actual source presentation. □

Uniform S-unit field norms and the exact capitulation obstruction

E.79. All cubic and ninth-degree S-class norm quotients in the base-changed tower [Proved]

Use the actual fields $B_d = L_d(\xi)$ of E.76, with $\xi^3 = \zeta$, $\beta_{d+1}^3 = \beta_d$, and $\beta_1 = a\xi$, $a^3 = 17$. Let S_d be the six primes above 3 and 17, and put

$$\mathscr{U}_d = \mathcal{O}_{B_d}[1/3, 1/17]^\times, \qquad A_d = \mathrm{Cl}_{S_d}(B_d)\{3\}. \tag{E79.1}$$

For $r = 1, 2$, write $n = 3^r$, $E = B_{d+r}$, $F = B_d$, $G = \langle\sigma\rangle = \mathrm{Gal}(E/F)$, and

$$K_{d,r} = \mathscr{U}_d \cap N_{E/F} E^\times. \tag{E79.2}$$

In every one of these actual cyclic extensions, the six Hilbert characters, with the first argument in $\mathscr{U}_d$ and the second argument β_d, map onto the entire product-one subgroup:

$$\rho_{d,r}(u) = \big((u,\beta_d)_{F_v,n}\big)_{v\in S_d}, \qquad \rho_{d,r}(\mathscr{U}_d) = \ker\left(\mu_n^6 \xrightarrow{\prod} \mu_n\right), \quad \ker\rho_{d,r} = K_{d,r}. \tag{E79.3}$$

Consequently, $[\mathscr{U}_d : K_{d,r}] = n^5$ at every level. The actual ideal norm induces the isomorphism

$$N_{E/F} : (A_{d+r})_G \xrightarrow{\sim} A_d, \qquad \ker(N : A_{d+r} \to A_d) = (\sigma-1)A_{d+r}. \tag{E79.4}$$

In particular, $|(A_{d+r})^G| = |A_d|$. At $d = 1$, the two displayed coinvariant groups are both $\mathbb{Z}/3$.

Here are complete finite coordinates for the norm calculation at $B = B_1$. Use the certified ordinary-unit generators $(\xi, u_1, \dots, u_8)$ of E.76 and the six literal S-unit generators $(s_1, \dots, s_6)$ in the accompanying certificate. Together with -1, they generate $\mathscr{U}_1$; their valuation matrix, in the order of E.76, is

$$V = \begin{pmatrix} 3 & 2 & 2 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E79.5}$$

The six full field-norm characters on these fifteen generators are $[U_n \mid C_n]$ modulo n. The ordinary-unit blocks satisfy $U_3 = M_3$ and $U_9 = \mathrm{diag}(1,1,1,8,8,7)M_9$, with the M_n of E.76. Thus the change of local quotient generators is explicit. The new blocks are

$$C_3 = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 2 & 2 & 0 & 0 & 1 \\ 2 & 2 & 0 & 1 & 0 & 1 \\ 2 & 2 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad C_9 = \begin{pmatrix} 2 & 2 & 6 & 6 & 6 & 1 \\ 8 & 3 & 5 & 8 & 2 & 8 \\ 2 & 4 & 1 & 0 & 6 & 2 \\ 1 & 1 & 6 & 2 & 0 & 5 \\ 7 & 1 & 3 & 0 & 5 & 8 \\ 4 & 7 & 3 & 0 & 0 & 1 \end{pmatrix}. \tag{E79.6}$$

The two image groups are $(\mathbb{Z}/n)^5$. Their full fifteen-coordinate kernel lattices and all integral Smith maps and inverses are retained. The allowed exponent vectors of the s_i, after multiplication by an ordinary unit, are respectively $K_n\mathbb{Z}^6$, where

$$K_3 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad K_9 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 1 & 2 & 2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E79.7}$$

Hence $\mathbb{Z}^6/K_3\mathbb{Z}^6 = (\mathbb{Z}/3)^2$ and $\mathbb{Z}^6/K_9\mathbb{Z}^6 = \mathbb{Z}/9 \oplus \mathbb{Z}/3$. In the actual six valuation coordinates w, the full criterion is

$$\begin{aligned} w_1 + w_2 + w_3 + 2(w_4 + w_5 + w_6) &\equiv 0 \pmod 3, \\ w_4 \equiv w_5 \equiv w_6 &\pmod 3, \\ 3w_2 - 3w_3 - 2w_4 - 2w_5 + 4w_6 &\equiv 0 \pmod 9 \quad (n = 9). \end{aligned} \tag{E79.8}$$

The first line is exactly the principal S-divisor condition. For each n, the retained 9 by 6 matrix R_n satisfies $U_nR_n + C_nK_n = 0$ modulo n. It gives the literal element

$$z \longmapsto \prod_{i=1}^{9} (\xi, u_1, \dots, u_8)_i^{(R_nz)_i} \prod_{i=1}^{6} s_i^{(K_nz)_i} \in K_{1,r}, \qquad \mathrm{div}_{S_1} = VK_nz. \tag{E79.9}$$

The same included elements have the same field-norm characters in every B_{d+r}/B_d.

An actual coherent prime chain makes (E79.4) concrete. In the polynomial model $t = a - \xi$ of E.76, choose the degree-one prime $\mathfrak{p}_1$ above 73 with residues

$$t \mapsto 33, \quad \xi \mapsto 55, \quad a \mapsto 15, \quad \beta_1 \mapsto 22, \qquad [\mathfrak{p}_1] = (2,1) \text{ in } \mathrm{Cl}(B). \tag{E79.10}$$

For every d, extend this residue map by $\beta_d \mapsto 22$ for odd d and $\beta_d \mapsto 63$ for even d. This defines a degree-one prime $\mathfrak{p}_d$ of B_d; $N_{B_{d+1}/B_d}\mathfrak{p}_{d+1} = \mathfrak{p}_d$. The three-primary components $[\mathfrak{p}_2]_3$ and $[\mathfrak{p}_3]_3$ of their ideal classes generate A_2 and A_3 as modules over $\mathbb{Z}_3[\mathrm{Gal}(B_2/B)]$ and $\mathbb{Z}_3[\mathrm{Gal}(B_3/B)]$, respectively. Their annihilator ideals are not evaluated here. If I_r is the kernel of the actual surjection $\mathbb{Z}_3[G] \to A_{1+r}$, $1 \mapsto [\mathfrak{p}_{1+r}]_3$, then the extension of the base generator is the explicit element $j[\mathfrak{p}_1]_3 = (1 + \sigma + \cdots + \sigma^{n-1})[\mathfrak{p}_{1+r}]_3$. Thus its remaining capitulation test is membership of this norm polynomial in the actual ideal I_r.

The exact remaining obstruction to a norm preimage being an S-unit is as follows. Let $j : A_d \to A_{d+r}$ extend ideals. There is a surjective homomorphism, with its construction in the proof,

$$\begin{aligned} \omega_{d,r} : K_{d,r} &\longrightarrow (A_{d+r})^G / jA_d, \\ \ker \omega_{d,r} &= N_{E/F}\mathscr{U}_{d+r}, \\ |K_{d,r}/N\mathscr{U}_{d+r}| &= |\ker j|, \qquad [\mathscr{U}_d : N\mathscr{U}_{d+r}] = n^5 |\ker j|. \end{aligned} \tag{E79.11}$$

At $d = 1$ the obstruction group has order dividing three. In particular every element of $K_{1,r}$ has its cube in $N\mathscr{U}_{1+r}$. One explicit family of field norms uses $c_1 = \beta_1 + 3\zeta + 1$:

$$\xi c_1 \in K_{d,1}, \qquad \xi u_4 u_5 u_7^6 u_8^6 c_1 \in K_{d,2} \qquad (d \geq 1), \tag{E79.12}$$

where these base elements are included in B_d. The uncorrected c_1 has residue $(1+3\zeta)^{96} = \zeta \neq 1$ at every prime above 17, and therefore is not a norm in the cubic step. This has been corrected in (E79.12) by an actual unit, before applying (E79.11).

All finite-index lattice and obstruction sequences above remain exact after completion for multiplication by 3^m; the completed S-unit groups are compact. The norm-coherent subgroup of E.69 still requires compatible actual S-unit norms. Its original eventual exponent h, the higher full S-class groups, and the capitulation maps j remain unevaluated. The literal norm and extension maps in E76.11 and the positional maps of E.46 and E.72 are the comparisons to those original systems. The manuscript anchor is LM §51.16, fols. 733v–736v, scans 172–178.

Proof. We first prove the uniform assertion with its exact local normalization. The field B_d contains μ_9, and $B_{d+r} = B_d(\beta_{d+r})$ with $\beta_{d+r}^n = \beta_d$. At each of the three primes above 17, β_d is a uniformizer, so this polynomial is Eisenstein. At a prime above 3, use $(Y-1)^n - \beta_d$: the constant term $-1 - \beta_d$ has valuation one and each intermediate coefficient is divisible by 3. The uniformizer assertion for $\beta_d + 1$ follows from the Eisenstein polynomial over $\mathbb{Q}_3(\zeta)$ in E.76. Thus every one of the six local extensions has degree n and is totally ramified. There are no other ramified primes, since the discriminant of $Y^n - \beta_d$ is supported above 3 and 17. All infinite places are complex. The inclusions $B_1 \subset B_d$ have, at these six places, the full local degree 3^{d-1} and a single prime over each base prime.

Fix the arithmetic local Artin maps. Our Hilbert symbol is

$$(x,y)_{F_v,n} = \frac{\mathrm{Art}_{F_v}(x)(y^{1/n})}{y^{1/n}}, \qquad (x,y)_{F_v,n} = (y,x)_{F_v,n}^{-1}. \tag{E79.13}$$

For a finite extension of local fields M/F_v, reciprocity gives the square

$$\begin{array}{ccc} M^\times & \xrightarrow{\mathrm{Art}_M} & G_M^{\mathrm{ab}} \\ N_{M/F_v} \downarrow & & \downarrow \iota^{\mathrm{ab}} \\ F_v^\times & \xrightarrow{\mathrm{Art}_{F_v}} & G_{F_v}^{\mathrm{ab}}. \end{array} \tag{E79.14}$$

Here the right arrow is induced by inclusion of absolute Galois groups. On a finite Galois overfield, this square is precisely corestriction compatibility for the Tate cup-product isomorphism $\widehat{H}^{-2}(G,\mathbb{Z}) \to \widehat{H}^{0}(G,L^{\times})$: corestriction on $\widehat{H}^{-2}$ is the inclusion-induced map on abelianizations, and on $\widehat{H}^{0}$ it is norm. The fundamental class restricts to the fundamental class, and the projection formula $\mathrm{cor}(z \cup \mathrm{res}\, u) = \mathrm{cor}(z) \cup u$ proves the square. These are the proved compatibility formulas immediately before and after Milne's III Theorem 3.1; III Lemma 3.2 also states the square but does not print a separate proof [Mil20b]. Passing to finite Galois overfields and then their inverse limit gives (E79.14) without a Galois assumption on M/F_v.

For $x \in F_v^{\times}$ and $y \in M^{\times}$, applying that square to the root $x^{1/n}$ first proves $(y,x)_{M,n} = (N_{M/F_v}y, x)_{F_v,n}$. Taking inverses in (E79.13) therefore gives the second-argument formula

$$(x,y)_{M,n} = (x, N_{M/F_v}y)_{F_v,n}. \tag{E79.15}$$

This fixes both arguments and the orientation. In the present local tower the minimal polynomial of β_d over B_1 is $Y^{3^{d-1}} - \beta_1$, and its odd degree gives the exact norm $N\beta_d = \beta_1$. Hence, for every included $u \in \mathscr{U}_1$ and each of the six compatible places,

$$(u,\beta_d)_{(B_d)_v,n} = (u,\beta_1)_{(B_1)_{v_1},n}. \tag{E79.16}$$

The same chosen $\mu_n \subset B_1$ occurs on both sides.

We next verify the finite input to this identity. The full unconditional class and unit certificate in E.76 gives $\mathrm{Cl}(B) = (\mathbb{Z}/3)^2$ and the six ramified-prime coordinates $(1,0)$ and $(2,0)$. Thus $A_1 = \mathbb{Z}/3$. The certified S-unit algorithm gives six additional generators, whose actual ideal identities are the columns of (E79.5). Their determinant is three, and their image is exactly the kernel of

$$\mathbb{Z}^6 \longrightarrow \mathrm{Cl}(B), \qquad w \longmapsto (w_1+w_2+w_3+2w_4+2w_5+2w_6, 0) \pmod 3. \tag{E79.17}$$

Consequently these six generators, the complete ordinary unit basis, and -1 generate all S-units. The source and output include their literal polynomials and verify every principal ideal identity [The26].

To extend the local unit characters of E.76 to the new S-units, divide by an actual norm uniformizer before using the recorded unit logarithm. At the wild places use $\pi = \beta_1 + 1 = N(Y+1)$; at the tame places use $\beta_1 = N(Y)$, where $Y^n = \beta_1$. Both identities have positive sign because n is odd. For a local input z, the unit to be evaluated is $z/\pi^{v(z)}$, respectively $z/\beta_1^{v(z)}$. These are exactly the evaluations in (E79.6). The tame quotient generators in the ninth-degree run differ from those of E.76 by the displayed diagonal unit matrix, which the program checks on all nine ordinary unit generators. No common-root identification is imposed on the six computational rows.

The HNF lattices $n\mathbb{Z}^6 + [U_n \mid C_n]\mathbb{Z}^{15}$ have index n in $\mathbb{Z}^6$. Their image modulo n therefore has order n^5; the complete Smith maps identify it with $(\mathbb{Z}/n)^5$. Multiplying each row by the unit that changes its local quotient generator to (E79.13) preserves this order. Global reciprocity puts the common-root image in the product-one subgroup, whose order is n^5. Thus equality holds. Formula (E79.16) includes this entire group in $\rho_{d,r}(\mathscr{U}_d)$ at every higher level, and the same reciprocity bound proves (E79.3). At places outside S_d, an S-unit is a local norm because the extension is unramified and norm is onto the unit group. The cyclic Hasse norm theorem then identifies the simultaneous kernel with (E79.2). We use the exact statement and full proof in Milne VIII Theorem 3.1; the same character-kernel argument is written in Li–Ouyang–Xu–Zhang, Lemma 2.8 [Mil20b, LOXZ22].

For completeness, here is a direct class-field proof of the actual norm isomorphism. Let $\mathcal{H}_E^S/E$ be the maximal abelian extension unramified everywhere and split at all primes of

S_{d+r}. Let $\mathcal{G}/F$ be its maximal subfield abelian over F, and set $\Gamma = \mathrm{Gal}(\mathcal{G}/F)$. The extension $\mathcal{H}_E^S/F$ is Galois. Its decomposition group at any prime above S_d maps isomorphically onto G, because that prime is totally ramified in E/F and split in $\mathcal{H}_E^S/E$. This gives a splitting of the finite group extension by $\mathrm{Cl}_S(E)$. Its commutator subgroup is consequently $(\sigma-1)\,\mathrm{Cl}_S(E)$, and

$$\ker(\Gamma \to G) = \big(\mathrm{Cl}_S(E)\big)_G. \tag{E79.18}$$

Reciprocity describes Γ as the quotient of finite ideles of F by

$$F^\times \left(\prod_{v\in S_d} N_{E_w/F_v} E_w^\times \prod_{v\notin S_d} \mathcal{O}_{F_v}^\times \right). \tag{E79.19}$$

This description can be checked in both directions. Its class field contains E. At $v \in S_d$, its local decomposition group has order at most n and maps onto the degree-n local extension E_w/F_v, so its local extension over E is trivial. Outside S_d it is unramified. Conversely every abelian subextension of $\mathcal{H}_E^S/F$ kills the local norms in (E79.19), by (E79.14), and kills the unit groups outside S_d. This proves the asserted maximality and the actual reciprocity quotient.

Mapping a finite idele to its divisor outside S_d gives an exact sequence

$$\mathscr{U}_d \longrightarrow \prod_{v\in S_d} F_v^\times / N E_w^\times \longrightarrow \Gamma \longrightarrow \mathrm{Cl}_{S_d}(F) \longrightarrow 0. \tag{E79.20}$$

The product has order n^6, and (E79.3) says that the first arrow has image order n^5. Thus the kernel of the last arrow has order n. The local Artin image at any one prime in S_d maps onto G; it is contained in this kernel, so that the kernel maps isomorphically to G. It follows that the restriction $\ker(\Gamma \to G) \to \mathrm{Cl}_{S_d}(F)$ is an isomorphism. For a prime w of E outside S, its Frobenius restricts to the $f(w/v)$th power of the Frobenius at v; its image in the last class group is $f(w/v)[v] = N[w]$. Thus this is the actual ideal norm on (E79.18). Taking three-primary parts proves (E79.4). Equivalently, norm surjectivity follows by base changing the S-split Hilbert class field of F: its intersection with E is F because every nontrivial subextension of E/F ramifies totally at S_d. The resulting S-split unramified extension of E has restriction map on Frobenii equal to ideal norm. For a finite group and an endomorphism $\sigma - 1$, kernel and cokernel have equal order. This proves the invariant-order assertion as well. It agrees with Gras' formula in [LOXZ22, Theorem 2.6], with D generated by the six ramified primes; here $\Lambda_D = \mathscr{U}_d$ and the actual norm of D is the full base S-divisor group.

We give the full lattice argument behind (E79.7)–(E79.9). Put $H_n = n\mathbb{Z}^6 + U_n\mathbb{Z}^9$. A vector $t \in \mathbb{Z}^6$ allows an ordinary-unit correction precisely when $C_n t \in H_n$. The retained integer kernel computation gives $\{t : C_n t \in H_n\} = K_n\mathbb{Z}^6$. Both inclusions are certified: $H_n^{-1} C_n K_n$ is integral, and

$$|\det K_n| = \frac{[\mathbb{Z}^6 : H_n]}{[\mathbb{Z}^6 : H_n + C_n\mathbb{Z}^6]}. \tag{E79.21}$$

The indices are 9 and 27. Integral Smith transformations and their inverses give exactly the quotients stated after (E79.7); they also give $\mathbb{Z}^6/VK_n\mathbb{Z}^6 = (\mathbb{Z}/3)^3$ for $n = 3$, and $\mathbb{Z}/9 \oplus (\mathbb{Z}/3)^2$ for $n = 9$. Applying V^{-1} and then K_n^{-1} gives (E79.8), including its integer divisibility conditions. The computed R_n gives (E79.9) directly. The complete fifteen-coordinate kernel has determinant n^5, so it contains every S-unit field norm, with the relation $\xi^9 = 1$ already lying in that kernel. The matrix identities and all their inverse coordinate maps are independently checked by `e_sclass_norms.py`.

The prime certificate in (E79.10) is an actual ideal of norm 73 in the retained integral basis. Reducing that basis gives its quotient map to $\mathbb{F}_{73}$; the ideal is exactly the kernel, as follows from its determinant 73. The certified class coordinates are $(2, 1)$, which maps to a

generator of A_1. The two identities $63^3 = 22$ and $22^3 = 63$ in $\mathbb{F}_{73}$ prove compatibility of all the displayed roots. Their order is eight. At 73 the polynomials $Y^{3^{d-1}} - \beta_1$ are etale, so these residue maps define actual degree-one primes of the maximal orders. Ideal norms send each to its predecessor. At the first two levels, (E79.4) shows that the three-primary component of the prime class generates the coinvariants. Projection to that component is the canonical projection in the primary decomposition of the finite ideal class group, and commutes with ideal norm and extension. To see the asserted cyclic-module generation without an extra assumption, let Q be its quotient in the finite three-group A_{1+r}. It satisfies $Q = (3, \sigma - 1)Q$. Modulo 3 the augmentation ideal is nilpotent since $(\sigma - 1)^n = 0$ in $\mathbb{F}_3[G]$. Iterating therefore gives $Q = 3Q$, and finiteness gives $Q = 0$. This proves the stated surjection from $\mathbb{Z}_3[G]$ with the literal prime component as image of one. The base prime splits completely in both extensions, since its displayed root and all its nth-root-of-unity multiples lie in $\mathbb{F}_{73}$. Extension of its ideal is therefore the product of these n conjugate upper primes. This proves the stated norm-polynomial formula for j.

We finally prove the norm-defect and capitulation maps. Let $\mathrm{Div}_{E,S}$ denote divisors outside S_{d+r}. On each orbit of unramified primes, norm is residue degree times the sum of coefficients. Therefore

$$\ker(N : \mathrm{Div}_{E,S} \to \mathrm{Div}_{F,S}) = (\sigma - 1)\,\mathrm{Div}_{E,S}, \qquad (\mathrm{Div}_{E,S})^G = j\,\mathrm{Div}_{F,S}\,. \tag{E79.22}$$

The second identity uses exactly that all ramification lies in S: extension of a base prime is the sum of its conjugate upper primes, each with coefficient one. For $u \in K_{d,r}$ choose $x \in E^\times$ with $Nx = u$. Then $\mathrm{div}_{E,S}(x)$ has norm zero, so choose D with $\mathrm{div}_{E,S}(x) = (\sigma - 1)D$. The class $[D]$ is G-invariant. Its image modulo $j\,\mathrm{Cl}_{S_d}(F)$ is independent of D by the second identity in (E79.22). If x changes to another norm preimage, Hilbert's theorem 90 writes the quotient as $\sigma(z)/z$. Replacing D by $D + \mathrm{div}_{E,S}(z)$ shows independence of this choice too. This defines ω.

Every invariant ideal class has a representative D with $(\sigma - 1)D = \mathrm{div}_{E,S}(x)$ for some x, and then Nx is an S-unit; thus ω is onto. Its vanishing says that $D = \mathrm{div}_{E,S}(z) + jD_0$. The element $x/(\sigma z/z)$ then has divisor zero outside S and the same norm u, proving the kernel formula. Conversely an S-unit norm has the zero representative $D = 0$. The quotient is killed by n because every $u \in \mathscr{U}_d$ has $u^n = N(u)$ after inclusion in E. Thus the quotient of full class groups just constructed is its three-primary part, giving exactly the groups in (E79.11).

There is a second exact map which records capitulation. For a class $[D_0] \in \ker j$, choose x with $\mathrm{div}_{E,S}(x) = jD_0$ and send it to the cocycle $g \mapsto g(x)/x$ in $\mathscr{U}_{d+r}$. Principal changes and changing x by an S-unit change this by a coboundary. Hilbert 90 and the invariant divisor identity in (E79.22) give the inverse. Hence

$$\ker(j : A_d \to A_{d+r}) \simeq H^1(G, \mathscr{U}_{d+r}). \tag{E79.23}$$

All these groups are three-primary since $Nj = n$. The S-unit Herbrand quotient is n^5: in the logarithmic permutation representation, the six finite S-places contribute six trivial lattices, the complex places contribute regular lattices, and the product formula removes one trivial lattice. The resulting quotient is n^6/n; finite-index lattice changes and torsion have Herbrand quotient one. This is also Milne VII Proposition 3.1, whose full lattice proof applies with these exact local degrees [Mil20b]. Therefore

$$\frac{[\mathscr{U}_d : N\mathscr{U}_{d+r}]}{|\ker j|} = n^5. \tag{E79.24}$$

Together with (E79.3) this proves the norm-defect size in (E79.11). It also follows directly from $|(A_{d+r})^G| = |A_d|$ and the actual image of j. At $d = 1$, $A_1 = \mathbb{Z}/3$, so the defect has order dividing three and every cube assertion follows.

The literal corrections in (E79.12) are checked on all six local characters; (E79.16) transports them to every higher step. At the tame residue field $\mathbb{F}_{17}[\zeta]/(\zeta^2+\zeta+1)$, direct exponentiation gives $(1+3\zeta)^{96}=\zeta$. Norm on residues in the totally ramified cubic extension is cubing. This proves the asserted failure for the uncorrected c_1. Multiplication by ξ cancels that residue and the three wild characters as well, as the complete matrix check records. The ninth-degree correction uses the same certified ordinary-unit basis.

For the completion assertion, all S-unit groups and all their displayed subgroups are finitely generated abelian groups. The kernels of the finite characters have finite index. Also $N\mathscr{U}_{d+r}$ contains $\mathscr{U}_d^{\,n}$, so it has finite index in $\mathscr{U}_d$. Completing these exact sequences therefore retains their kernels, images and finite three-primary quotients; explicitly the groups are their finite three-torsion summands plus finite products of $\mathbb{Z}_3$. In particular the completed version of (E79.11) gives the same obstruction group and the same zero criterion. This proves an exact membership theorem for each step. The all-level compatible norm condition remains the actual inverse system of these maps, with the capitulation maps and upper annihilator ideals retained as unevaluated arithmetic data. □

The conjugate pair and its actual wild descent

E.80. The joint cubic quotient, its integral filtration, and complete descended local fibres [Proved]

Put $K=\mathbb{Q}(\zeta)$, $\zeta=\zeta_3$, and $k=\mathbb{Q}_3(\zeta)$. Retain the cubic, tangent and actual source marking of E.78. In particular $O=[1:-1:0]$, the source parameter is $s=z/x$ of value one, and N is the based geometric free pro-three group of $U=\{x^3+y^3+z^3=0,\ xyz\neq 0\}$. Use conjugate paths for the two maps

$$\begin{aligned} h&=\frac{x+y}{x+\zeta y}, & \bar h&=\frac{x+y}{x+\zeta^2 y}, & v&=-\frac{x+y}{z},\\ \bar h&=-\frac{\zeta^2 h}{h+\zeta}, & \frac{1-h}{1-\bar h}&=-\zeta\frac{h}{\bar h}, & v^3&=h\bar h. \end{aligned} \tag{E80.1}$$

Both target groups are the same marked free pro-three group $F=\langle x,y\rangle$ of the three-punctured line. Write $X(f),Y(f)$ for its abelian coordinates and $\ell=X-Y$. The same notation denotes the characters of $C_r=F/\gamma_{r+1}F$ for $1\le r\le 3$.

There is the following exact intermediate cover of schemes over K:

$$\begin{gathered} U\longrightarrow X_0=E\setminus\{yz=0\}\longrightarrow S=\mathbb{P}^1_h\setminus\{0,1,\infty,-\zeta\},\\ X_0\simeq\left\{(h,v):v^3=-\frac{\zeta^2h^2}{h+\zeta}\right\}\longrightarrow S. \end{gathered} \tag{E80.2}$$

The first arrow fills the three cusps with $x=0$. The second is connected, cyclic, finite étale of degree three. Let $H=\pi_1^{(3)}(S)$, let $N_0=\pi_1^{(3)}(X_0)\subset H$, and let $K_\infty,K_{-\zeta}$ be the closed normal subgroups generated by the indicated peripheral generators of H. Then H is free of rank three, N_0 is its index-three subgroup of rank seven, and the exact joint image is

$$\begin{gathered} \mathcal{J}=\operatorname{im}(N\longrightarrow F\times F)=\{(f,g):\ell(f)=\ell(g),\ X(f)+X(g)\in 3\mathbb{Z}_3\},\\ \ker(N\longrightarrow\mathcal{J})=(N\longrightarrow N_0)^{-1}(K_\infty\cap K_{-\zeta}),\\ K_\infty\cap K_{-\zeta}=\overline{[K_\infty,K_{-\zeta}]}\subset N_0. \end{gathered} \tag{E80.3}$$

In particular the two arithmetic factors are retained with their actual joint image. For each

$r = 1, 2, 3$ put

$$\begin{aligned} \mathcal{J}_r &= \{(f,g) \in C_r^2 : \ell(f) = \ell(g),\ X(f) + X(g) \in 3\mathbb{Z}_3\}, \\ N/\gamma_{r+1}N &\twoheadrightarrow \mathcal{J}_r, \qquad \ker = \frac{h_*^{-1}(\gamma_{r+1}F) \cap \bar{h}_*^{-1}(\gamma_{r+1}F)}{\gamma_{r+1}N}. \end{aligned} \tag{E80.4}$$

These are actual $G_{\mathbb{Q}_3}$-equivariant quotient maps with the action below.

Let $3_n, \zeta_n = \zeta_{3^{n+1}}, \lambda_n, a_n$ be the coherent roots in E.78, and use their complex conjugates for $\bar{h}$. They satisfy the literal identities

$$\begin{gathered} \lambda_n/\bar{\lambda}_n = -\zeta_n, \qquad \lambda_n\bar{\lambda}_n = 3_n, \qquad a_n/\bar{a}_n = -\zeta_n^{-1}, \qquad a_n\bar{a}_n = 3_n^{-3}, \\ \kappa_a = -\frac{\chi-1}{6} - \frac{3}{2}\kappa_3(3), \qquad \kappa_{\bar{a}} = \frac{\chi-1}{6} - \frac{3}{2}\kappa_3(3). \end{gathered} \tag{E80.5}$$

The product of the chosen cube roots of $h, \bar{h}$ is exactly v, whose leading term is $s^2/3$. Thus (E80.2) uses the actual sheet selected by the source tangent.

Write $c = [x,y]$, $d = [x,c]$, $e = [y,c]$ and $x^A y^B c^n d^r e^s$ for the Hall convention of E.78. Let T_μ be the polynomial operator (E78.6), now evaluated at all four values below, and let $I(x) = x^{-1}, I(y) = y^{-1}$. The actual arithmetic operators of τ, τ' and the order-two arithmetic generator in the actual relative source of E.74 and E.77 are respectively

$$\begin{aligned} \mathcal{T}(f,g) &= (T_{-5/2}f, T_{-1/2}g), \\ \mathcal{T}'(f,g) &= (T_{1/2}f, T_{5/2}g), \\ \jmath(f,g) &= (Ig, If), \qquad \jmath\mathcal{T}\jmath = \mathcal{T}'. \end{aligned} \tag{E80.6}$$

The operators descend through every map in (E80.4). Their source generators are the actual E.77 generators after the E.78 Frobenius transport.

Normalize a local cocycle on that relative order-two subgroup to one. Its complete section set with coefficients in $\mathcal{J}_r$ is

$$\begin{gathered} \{(u_0,a_0) \in \mathcal{J}_r^2 : V = u_0\mathcal{T}(a_0)u_0^{-1}a_0^{-7}, \quad \jmath(V) = V\}/\mathcal{J}_r^\jmath, \\ w_0 : (u_0,a_0) \longmapsto (w_0u_0\mathcal{T}(w_0)^{-1}, w_0a_0w_0^{-1}), \\ \mathcal{J}_r^\jmath = \{(w, Iw) : \ell(w) = 0\}. \end{gathered} \tag{E80.7}$$

Restriction to G_k followed by the first target projection induces an injection

$$H^1(\mathbb{Q}_3, \mathcal{J}_r) \hookrightarrow H^1(k, C_r). \tag{E80.8}$$

Its exact image is given on any E.78 representative (u, u', a, a') by the four conditions

$$\begin{aligned} \ell(a) + \ell(a') = 0, \qquad & X(a) - X(a') \in 3\mathbb{Z}_3, \\ H_u = \ell(u) + \ell(u') \in 3\mathbb{Z}_3, \qquad & X(u) - X(u') \in 3\mathbb{Z}_3. \end{aligned} \tag{E80.9}$$

Here every representative still satisfies E.78's actual local source relation. For an explicit inverse, apply its gauge by $w = x^{H_u/12}$, so the new horizontal ℓ-sum is zero, and set

$$u_0 = (u, Iu'), \qquad a_0 = (a, Ia'). \tag{E80.10}$$

The exact remaining gauge is $\ker \ell$, as in (E80.7). Thus (E80.8) includes the full geometric-conjugacy quotient, and its inverse retains the actual normalizing element.

The abelian joint lattice has the integral coordinates

$$\begin{gathered} (L,p,m) \longmapsto ((p, p-L), (3m-p, 3m-p-L)), \\ [\jmath]_{L,p,m} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & -1 \end{pmatrix}, \qquad [\mathcal{T}] = 7I_3. \end{gathered} \tag{E80.11}$$

Consequently the complete first-stage coordinates and the exact next lifting image are

$$\begin{aligned} H^1(\mathbb{Q}_3,\mathcal{J}_1) &\simeq \mathbb{Z}_3^5\times\mathbb{Z}/3,\\ (L_a,p_a,m_a,L_u,m_u,\ p_u \bmod 3),&\\ \operatorname{im}\bigl(H^1(\mathbb{Q}_3,\mathcal{J}_2)\to H^1(\mathbb{Q}_3,\mathcal{J}_1)\bigr):&\quad L_a(p_u-1)-L_up_a\equiv 0 \pmod 3. \end{aligned} \tag{E80.12}$$

Choose $p_u\in\{0,1,2\}$ for the inverse and use the fixed gauge with $p=(p_u^{\mathrm{old}}-p_u)/6$. For a particular second-stage lift use the E.78 substitution $n=-D_2/6$, $n'=N=N'=0$ after (E80.10). Every nonempty fibre at this stage is $\mathbb{Z}_3^2\times\mathbb{Z}/3$, with exactly the scalar representative and inverse maps of E.78.

At the next stage apply (E78.10)–(E78.11) to the restricted quadruple. The complete image and fibres are

$$\begin{aligned} \operatorname{im}\bigl(H^1(\mathbb{Q}_3,\mathcal{J}_3)\to H^1(\mathbb{Q}_3,\mathcal{J}_2)\bigr):&\quad \Delta R,\Delta S\in 3\mathbb{Z}_3,\\ \text{each nonempty fibre}&\simeq \mathbb{Z}_3^4\times(\mathbb{Z}/9)^2,\\ a\longmapsto ad^{-\Delta R/336}e^{-\Delta S/336}&\quad\text{for a particular lift.} \end{aligned} \tag{E80.13}$$

All other representatives and their inverses are (E78.15), followed by (E80.10). These are homeomorphisms of compact section fibres.

For example the actual descended second-stage family

$$\begin{gathered} a=y,\qquad a'=y^{-1},\qquad u=x^{5/2}c^N,\qquad u'=x^{-1/2}y^2c^M,\\ \text{class-three error}=e^{7(5-N-M)} \end{gathered} \tag{E80.14}$$

has a third-stage lift exactly for $N+M\equiv 2 \pmod 3$. A particular lift changes a to $ye^{(N+M-5)/48}$. Thus the member $N=M=0$ has empty fibre and $N=2,M=3$ already lifts with its displayed words. These are sections of the actual joint quotient over $\mathbb{Q}_3$; their intrinsic preimages are governed by the explicit kernels in (E80.4).

Finally the integral lower central comparison is as follows. The kernel of $\mathcal{J}_2\to\mathcal{J}_1$ is all of $\mathbb{Z}_3^2$ in the two c coordinates, while

$$\begin{aligned} \gamma_2\mathcal{J}_2&=\{(c^n,c^{n'}):n+n'\in 3\mathbb{Z}_3\},\\ \gamma_2\mathcal{J}_3&=\{(c^nd^re^s,c^{n'}d^{r'}e^{s'}):n+n'\in 3\mathbb{Z}_3\},\\ \gamma_3\mathcal{J}_3&=\{(d^re^s,d^{r'}e^{s'}):r'-r\in 3\mathbb{Z}_3,\quad s+s'-2r\in 3\mathbb{Z}_3\}. \end{aligned} \tag{E80.15}$$

In particular the actual comparison maps are

$$\begin{gathered} 0\longrightarrow\mathbb{Z}/3\longrightarrow\mathcal{J}_2^{\mathrm{ab}}\longrightarrow\mathcal{J}_1\longrightarrow 0,\qquad (n,n')\longmapsto n+n' \bmod 3,\\ 1\longrightarrow(\mathbb{Z}/3)^2\longrightarrow\mathcal{J}_3/\gamma_3\mathcal{J}_3\longrightarrow\mathcal{J}_2\longrightarrow 1,\\ (r,s,r',s')\longmapsto(r'-r,\ s+s'-2r) \bmod 3. \end{gathered} \tag{E80.16}$$

The two coordinate lattice Smith lists are $(1,3)$ and $(1,1,3,3)$. As compact $\mathbb{Z}_3$-modules, $\mathcal{J}_3^{\mathrm{ab}}\simeq\mathbb{Z}_3^3\oplus\mathbb{Z}/3$ and $\gamma_2\mathcal{J}_3/\gamma_3\mathcal{J}_3\simeq\mathbb{Z}_3^2\oplus(\mathbb{Z}/3)^2$. The image of the intrinsic central map $\mathrm{Lie}_3(N^{\mathrm{ab}})\to\ker(\mathcal{J}_3\to\mathcal{J}_2)=\mathbb{Z}_3^4$ is exactly the last lattice in (E80.15), of index nine. These exact sequences specify how the product truncation used for (E80.12)–(E80.13) compares with the actual lower central filtration.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). The result evaluates the conjugate joint quotient and its full wild descent; (E80.4) retains the intrinsic kernel.

Proof. **The exact schemes and their groups.** Writing $t = x/y$ gives $h = (t+1)/(t+\zeta)$ and $\bar{h} = (t+1)/(t+\zeta^2)$. Substitution and $1+\zeta+\zeta^2 = 0$ prove all three identities in (E80.1). The inverse to the second line of (E80.2) is the literal morphism

$$t = \frac{1-\zeta h}{h-1}, \qquad (h, v) \longmapsto [t : 1 : -(t+1)/v]. \tag{E80.17}$$

Every denominator is invertible on the indicated schemes. The equation for v proves that this point lies on the cubic; substitution proves both compositions are the identity. The Kummer function has orders $2, -1, -1$ at $0, -\zeta, \infty$, respectively. Its three-valuation at $-\zeta$ proves that it is not a cube. Removing these points and 1 makes the degree-three cover finite étale and connected. The filled points $x = 0$ map to $h = \zeta^2$; this is an unramified fibre of three points.

Apply the finite-cover comparison in [GR71, Exposé XII, Theorem 5.1 and Corollary 5.2] to these actual morphisms. The topological punctured curves give the free pro-three presentations just stated. For completeness, the pro-three topology on the index-three subgroup is its full one: intersect the three conjugates of any normal subgroup of three-power index in that subgroup. The intersection is normal in the whole group, and the resulting extension has three-power order. Thus the subgroup comparison preserves every finite three-cover, and its rank is $1 + 3(3-1) = 7$. Filling the three specified punctures gives the surjection $N \to N_0$ with the closed normal closure of their peripheral generators as its kernel.

On S the first map fills $-\zeta$, and the second fills ∞ before applying the displayed Möbius transformation. Their common quotient fills both points. It is the pro-three group of $\mathbb{P}^1 \setminus \{0, 1\}$, hence $\mathbb{Z}_3$. The two maps to this quotient are exactly ℓ: the Kummer functions $h/(1-h)$ and $\bar{h}/(1-\bar{h})$ have ratio $-\zeta^{-1}$, so their geometric characters agree. Given a pair with equal common character, lift its first coordinate to H and correct its second coordinate by the first kernel. This proves surjectivity onto the fibre product, since the image of that kernel is exactly the kernel of the second common-character map. All these images and products are closed, being images of compact groups.

Here is also a direct verification of the asserted intersection kernel. Choose a free peripheral basis of H consisting of one surviving generator and the two separately killed generators. Let A, B be the kernels of the retractions from the respective free rank-two factors onto the surviving $\mathbb{Z}_3$. The fibre product is $(A \times B) \rtimes \mathbb{Z}_3$. In $H/\overline{[K_\infty, K_{-\zeta}]}$, conjugation of the first killed generator by the other kernel does nothing. Its normal closure therefore comes from A; the other normal closure comes from B. The two commute, and the surviving generator acts on both. This constructs the reverse continuous map from $(A \times B) \rtimes \mathbb{Z}_3$. The two compositions fix the three free generators, so are the identity. Hence the intersection of the original kernels is exactly their closed mutual commutator.

The character of the cover (E80.2) is $X(f) + X(g) \bmod 3$, since its actual defining root is $(h\bar{h})^{1/3}$. It vanishes on the intersection kernel. Restricting the already proved fibre-product surjection to this character kernel proves (E80.3). For (E80.4), lift each prescribed C_r coordinate arbitrarily to F. Both required conditions are abelian, so the pair of lifts lies in $\mathcal{J}$. The stated kernel is then the literal inverse image of $\gamma_{r+1}F$ under both maps, divided by $\gamma_{r+1}N$.

Roots, paths and the descended action. The chosen square roots satisfy $(\lambda_n/\bar{\lambda}_n)^2 = \zeta_n^2$. At level zero their ratio is $-\zeta$; compatibility and the odd cube power give $-\zeta_n$ at every level. Their product is the positive root 3_n. The remaining multiplicative identities in (E80.5) follow directly. Taking Kummer characters gives

$$\kappa_a + \kappa_{\bar{a}} = -3\kappa_3(3), \qquad \kappa_a - \kappa_{\bar{a}} = -\frac{\chi - 1}{3}, \tag{E80.18}$$

which proves the two individual formulas with their actual coherent roots.

Near O, the cubic equation gives $y/x = -1 - s^3/3 + s^6/9 + \cdots$ and $v = s^2/3 + \cdots$. The leading terms of $h, \bar{h}$ are $as^3, \bar{a}s^3$. The product of the two chosen cube-root leading coefficients

is $1/3$. The product cube root therefore has the same leading term as v; their quotient is a cube root of one with leading coefficient one, so it is one. At every higher level the same argument identifies the product roots with the corresponding coherent roots of v. This proves the actual sheet and tangential compatibility, including its constant.

E.78 transports the actual source marking so that $\chi(\tau) = \chi(\tau') = 7$ and $\kappa_3(3)(\tau) = 1$, $\kappa_3(3)(\tau') = -1$. Substitution in (E80.5) gives the four parameters in (E80.6). E.54's genuine operator, whose class-three coefficient is $(\chi^2-1)/24$, gives exactly E.78's polynomial T_μ at these values. All Hall terms are therefore those of the actual arithmetic action. For conjugate paths functoriality is the exact identity $\bar{h}_* \circ c_{\mathrm{ar}} = I \circ h_*$. On the target the positive real interval is fixed by complex conjugation and its peripheral circles are reversed; thus $I(x) = x^{-1}, I(y) = y^{-1}$. The Hall formula is

$$I(A, B, n, r, s) = (-A, -B, n, -n - r, -n - s). \tag{E80.19}$$

It proves both the stated involution and $\jmath\mathcal{T}\jmath = \mathcal{T}'$ by direct substitution. Each T_μ multiplies both abelian characters by seven; $\jmath$ negates and swaps them. Hence all preserve the actual joint image. E.78's rational origin transport and E.34 identify the based N action with the same actual $P \rtimes C_2$ action. This proves equivariance of the quotient maps, rather than inferring a descent from the first factor alone.

Full local conjugacy and its inverse. The cohomological source comparison of E.74 and the actual relative presentation of E.77 apply to this pro-three target. Coprime complement conjugacy normalizes the cocycle on C_2 to one. E.77 then gives exactly (E80.7), with all residual conjugators fixed by $\jmath$. A fixed pair has the form (w, Iw); its common-character condition says $\ell(w) = -\ell(w)$, hence $\ell(w) = 0$. Its index-three condition is automatic. This proves the final line of (E80.7).

Write $u_0 = (u, Iu')$, $a_0 = (a, Ia')$. The first component of the relation in (E80.7) is exactly $v_{-5/2}(u, a) = v_{1/2}(u', a')$, because of $\jmath\mathcal{T}\jmath = \mathcal{T}'$. Thus restriction gives E.78's actual source equation. Membership of these two pairs is exactly (E80.9) with the stronger equality $H_u = 0$.

For an arbitrary E.78 gauge w, the abelian coordinates transform by

$$\begin{aligned} \ell(u) + \ell(u') &\longmapsto H_u - 12\ell(w), \\ X(u) - X(u') &\longmapsto X(u) - X(u'), \\ (X(a), Y(a), X(a'), Y(a')) &\longmapsto (X(a), Y(a), X(a'), Y(a')). \end{aligned} \tag{E80.20}$$

The existence of an integral normalizing gauge is therefore exactly $H_u \in 3\mathbb{Z}_3$, and the displayed choice $w = x^{H_u/12}$ supplies it. This proves necessity, sufficiency and the inverse map in (E80.9)–(E80.10).

If two normalized joint representatives have the same image in $H^1(k, C_r)$, an actual $w \in C_r$ relates their first quadruples. Both horizontal ℓ-sums are zero, so (E80.20) forces $12\ell(w) = 0$. The lattice $\mathbb{Z}_3$ is torsion-free, hence $\ell(w) = 0$. The pair (w, Iw) is precisely a residual conjugator in (E80.7). This proves injectivity on full geometric-conjugacy classes, including the normalization which removes the larger ambient target gauge. All spaces of solutions are closed in finite products of compact groups; quotienting a compact group action gives a compact Hausdorff space. The displayed maps are continuous, including division by twelve on its closed image. The bijections and inverses are thus homeomorphisms.

Evaluated local coordinates and fibres. The map in (E80.11) is an integral parametrization of the joint abelian lattice: $p = X(f)$, $L = \ell(f)$, $m = (X(f) + X(g))/3$ recover its inverse. The involution follows by negating and swapping the two factors. Its fixed lattice is exactly the p-axis. E.77's relation is identically zero on this abelian target since $\mathcal{T} = 7$. The fixed gauge changes u by minus six times that axis and fixes a, proving the complete coordinates and inverse in (E80.12).

Under (E80.10), the first-factor abelian values are

$$\begin{aligned}(A,B)&=(p_a,p_a-L_a),\quad (A',B')=(p_a-3m_a,p_a-3m_a+L_a),\\(U,V)&=(p_u,p_u-L_u),\quad (U',V')=(p_u-3m_u,p_u-3m_u+L_u).\end{aligned}\tag{E80.21}$$

Their horizontal ℓ-sum is already zero. Substituting them into E.78's exact second-stage defect gives

$$\begin{aligned}D_2=\;&9L_am_a+3L_am_u-6L_ap_a-2L_ap_u+2L_a\\&-3L_um_a+2L_up_a-27m_a^2+18m_ap_a+\tfrac{3}{2}m_a-3p_a.\end{aligned}\tag{E80.22}$$

Reduction modulo three is precisely the congruence in (E80.12). E.78 proves that divisibility is sufficient and gives its particular lift.

All the joint-image constraints depend only on abelian coordinates. Consequently every E.78 lift of the restricted normalized quadruple, at either of the next two stages, still satisfies those constraints and returns by (E80.10). Conversely every joint lift restricts to that E.78 lift. Injectivity already proved identifies the full fibres, not only selected representatives. The explicit central changes, stabilizer calculation and inverse gauges in E.78 therefore give exactly (E80.12)–(E80.13). In particular the coefficient for a vertical third-degree change is 336, and the remaining horizontal diagonal gauge is 342; their three-adic valuations are one and two. At the preceding step the corresponding coefficients are 6 and 48, with valuations one and one. The lower stabilizers vanish as in E.78: their first nonzero Hall coordinate would be fixed by 7^j for a positive integer j, impossible in a torsion-free $\mathbb{Z}_3$ lattice. Substitution of (E80.14) into the full E.78 polynomials gives $\Delta R=0$ and $\Delta S=7(5-N-M)$, proving every assertion about that descended family.

Integral group-level lower central comparison. The following four actual elements of $\mathcal{J}_3$ will suffice:

$$\begin{aligned}g_1&=(y^{-1},y^{-1}),\qquad g_2=(xy,x^{-1}y^{-1}),\\g_3&=(1,x^3y^3),\qquad g_4=(c,1).\end{aligned}\tag{E80.23}$$

The first three abelian columns are the three columns of (E80.11). For any two elements their second-degree commutator coordinates are the two abelian determinants. On this integral lattice these span the columns

$$\begin{pmatrix}-1&0&0\\1&-3&0\end{pmatrix},\qquad \text{Smith list } (1,3).\tag{E80.24}$$

The resulting image is exactly $\{(n,n'):n+n'\in 3\mathbb{Z}_3\}$.

For the full class-three calculation use, in order, $[g_4,g_1]$, $[g_4,g_2]$, $[[g_1,g_2],g_1]$, $[[g_1,g_2],g_2]$. They have zero first- and second-degree coordinates. Their four central coordinate columns and their integer inverse are

$$\begin{aligned}B&=\begin{pmatrix}0&-1&0&-1\\1&-1&1&-1\\0&0&0&-1\\0&0&-1&-1\end{pmatrix},\qquad \det B=-1,\\B^{-1}&=\begin{pmatrix}-1&1&-1&1\\-1&0&1&0\\0&0&1&-1\\0&0&-1&0\end{pmatrix}.\end{aligned}\tag{E80.25}$$

Thus all four central directions actually belong to $\gamma_2\mathcal{J}_3$. Combining these exact commutators with (E80.24) proves both claims for γ_2 in (E80.15). It also proves that (E80.23) topologically

generates $\mathcal{J}_3$: the first three generators cover the abelian pair lattice, g_4 and $[g_1, g_2]$ cover both remaining c coordinates, and (E80.25) covers all central coordinates. Continuous $\mathbb{Z}_3$ powers give the stated integral coordinates, so this proves the group-level generation and not only its rational graded image.

For γ_3 only the abelian coordinates of the first input and the c coordinates of the second input enter. Bracketing the three columns of (E80.11) with the two lattice basis vectors $(1,-1),(0,3)$ gives the exact matrix

$$M = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -3 & -3 & 9 \\ 1 & -3 & 1 & -3 & -3 & 9 \end{pmatrix}, \qquad \operatorname{Smith}(M) = (1,1,3,3). \tag{E80.26}$$

The two rows $(-1,0,1,0)$ and $(-2,1,0,1)$ kill its columns modulo three and are independent there. Their kernel has index nine, equal to the Smith index, so it is exactly the image of M. This proves the last line of (E80.15) integrally. The stored certificate includes the complete unimodular left and right Smith maps and both inverses, as well as (E80.25).

It follows that the product-central kernel has the two finite quotients in (E80.16). On the second quotient the actual involution and τ act by

$$[\jmath] = \begin{pmatrix} 1 & 0 \\ 2 & -1 \end{pmatrix} \pmod 3, \qquad [\tau] = I_2. \tag{E80.27}$$

Indeed the central involution is minus the factor swap, and τ is multiplication by 343. The abstract compact-module decompositions follow by splitting their free $\mathbb{Z}_3$ quotients; the exact maps and action are the preceding ones. Finally surjectivity in (E80.4) implies $\operatorname{im}(\gamma_3 N) = \gamma_3 \mathcal{J}_3$. Since $\gamma_4 N$ is killed, this proves the stated index-nine image of the intrinsic third-degree lattice.

The self-contained program `a_joint_wild.py` verifies the eleven scheme and tangent identities, the actual Hall operators, all integral Smith maps and their inverses. Separate finite Hall arithmetic enumerates the complete joint images at class one modulo 27, class two modulo 9, and class three modulo 3, of orders 6561, 19683, and 6561. It checks 1200 arithmetic action identities, 240 exact restriction/gauge normalizations and inverses, 62 descended second-stage sections, 13 third-stage lifts, and all 36 displayed small family cases. The accompanying JSON retains all matrices and the complete finite-image hashes. These checks verify the formulas used in the proofs above. □

An exact ordinary noncapitulation and the twenty-seven S-corrections

E.81. The prime-seventy-three norm polynomial and a finite S-capitulation reduction [Proved]

Put $F = B_1$, $E = B_2$ in E.76–E.79, with $\beta_2^3 = \beta_1$, and let $G = \langle\sigma\rangle = \operatorname{Gal}(E/F)$. Order the primes of F above three and seventeen as in E.79. Write $Q_1, Q_2, Q_3, T_1, T_2, T_3$ for the respective unique primes of E above them. Let $\mathfrak{p}$ be the actual degree-one prime above 73 with $t = a - \xi \mapsto 33$ and $\beta_1 \mapsto 22$. Its certified class in $\operatorname{Cl}(F) = (\mathbb{Z}/3)^2$ is $(2,1)$. Set $I = \mathfrak{p}\mathcal{O}_E$.

Use the actual genus Artin map from E.76,

$$\varphi : \operatorname{Cl}(E)\{3\} \longrightarrow \Gamma = \mathbb{Z}/9 \oplus (\mathbb{Z}/3)^3, \tag{E81.1}$$

where Γ is the Galois group over F of the maximal abelian subfield of the three-primary Hilbert class field of E. The image of φ is the subgroup fixing E. In the specified coordinates

its values on the six upper ramified primes and on I are

$$C = \begin{pmatrix} 3 & 3 & 3 & 0 & 3 & 6 \\ 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 2 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \varphi[I] = (6,0,0,0). \tag{E81.2}$$

The rows are reduced modulo $9, 3, 3, 3$, respectively. Consequently $[I]$ has order exactly three in the ordinary class group. More generally, every class of F whose second coordinate is nonzero retains order three after extension to E; the ordinary capitulation kernel is contained in the first coordinate subgroup.

There is an explicit comparison to the S-class question in E.79. Choose the prime $\mathfrak{p}_2$ above $\mathfrak{p}$ with $\beta_2 \mapsto 63$ in $\mathbb{F}_{73}$. Then

$$[I] = (1+\sigma+\sigma^2)[\mathfrak{p}_2] \quad \text{in } \mathrm{Cl}(E), \qquad \pi_S[I] = j_S \pi_S[\mathfrak{p}], \tag{E81.3}$$

where π_S is the quotient by the classes of primes above 3 and 17, and j_S is actual ideal extension on S-classes. The first element in (E81.3) is nonzero. Its image under the second displayed quotient is evaluated by the following finite set of ordinary principal-ideal tests.

For $a, b, c \in \{-1, 0, 1\}$, define the actual fractional ideals

$$J_{a,b,c} = I Q_1^{3a+2b+2c} Q_2^b Q_3^c T_1^{-1} T_2. \tag{E81.4}$$

Then $j_S \pi_S[\mathfrak{p}] = 0$ precisely when at least one of these twenty-seven ideals is principal. Every $J_{a,b,c}$ is G-invariant, has a principal cube, and has zero image under φ. The certificate retains an actual $f_{a,b,c} \in F$ with

$$(f_{a,b,c}) = \mathfrak{p}^3 \prod_{i=1}^{6} \mathfrak{s}_i^{e_i(a,b,c)}, \qquad J_{a,b,c}^3 = f_{a,b,c}\mathcal{O}_E, \quad e(a,b,c) = (3a{+}2b{+}2c, b, c, -1, 1, 0). \tag{E81.5}$$

Here $\mathfrak{s}_i$ are the six base primes in the prescribed order. Both ideal identities are checked in the actual maximal orders, including negative exponents.

The following lattices and literal principal functions make the reduction to (E81.4) exact. With the valuation matrix V of E.79, the seven proved S-principal relations in E are the columns of

$$M = \left(3V \mid (0,0,0,1,1,1)^{\mathsf{t}}\right), \qquad H = \begin{pmatrix} 9 & 6 & 6 & 3 & 3 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad H\mathbb{Z}^6 = M\mathbb{Z}^7. \tag{E81.6}$$

The six columns of H are the divisors of the included $s_1, \dots, s_5$ from E.79 and of β_2, respectively. Thus an element Hq is the actual principal divisor of $s_1^{q_1} \cdots s_5^{q_5} \beta_2^{q_6}$. The quotient by these proved relations is $\mathbb{Z}/9 \oplus (\mathbb{Z}/3)^4$, of order 729. No assertion that these are all S-principal relations is needed.

For an arbitrary correction exponent $e \in \mathbb{Z}^6$, the genus obstruction vanishes exactly on the affine lattice

$$e \in p + K\mathbb{Z}^6, \qquad p = (0,0,0,-1,1,0)^{\mathsf{t}},$$
$$K = \begin{pmatrix} 3 & 2 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad K\mathbb{Z}^6/H\mathbb{Z}^6 \simeq (\mathbb{Z}/3)^3. \tag{E81.7}$$

To recover its representative (E81.4), compute $z = K^{-1}(e - p)$ and take the centered residues of $(z_1 - z_4 - z_5, z_2, z_3)$ modulo three as (a, b, c). The vector $q = H^{-1}(e - e(a, b, c))$ is integral, and the principal function following (E81.6) is the explicit comparison of the two ideals. All these matrices, their Smith transformations and their inverses are retained.

There is a further numerical bound on these remaining tests. Put

$$W = \ker\Big(\mathrm{Cl}(E)\{3\}^G \xrightarrow{\varphi} \Gamma\Big). \tag{E81.8}$$

Then $|W|$ divides three, and all twenty-seven classes in (E81.4) lie in W. Their class map is an affine map from $(\mathbb{Z}/3)^3$ to this actual group. Accordingly the number of principal ideals in the displayed list is one of $0, 9, 27$. The subgroup generated by the six S-prime classes has order 27 or 81. Equivalently, the kernel of the actual map $\mathbb{Z}^6 \to \mathrm{Cl}(E)$, $e \mapsto [\prod_i \mathfrak{S}_i^{e_i}]$, contains $H\mathbb{Z}^6$ with index 27 or 9, respectively.

The remaining S-capitulation value, the higher annihilator ideals of E.79, and the eventual exponent h of E.69 are not evaluated here. The bounded reduction searches retained with the certificate found no positive generator; no nonprincipality conclusion is drawn from those searches. The ordinary noncapitulation above follows from (E81.2). The exact positional comparisons remain E.46, E.72 and E76.11. The manuscript anchor is LM §51.16, fols. 733v–736v, scans 172–178.

Proof. We describe the actual number-field maps first. The certified base has primitive element $t = a - \xi$, $a^3 = 17$, $\xi^3 = \zeta$, and $\beta_1 = a\xi$. The ideal $(73, t - 33)$ has norm 73, and reduction gives $\xi = 55$, $a = 15$, $\beta_1 = 22$ in its residue field. Its class is $(2, 1)$ in the certified class basis. The program retains $f \in F$ and verifies $(f) = \mathfrak{p}^3$ and $N_{F/\mathbb{Q}} f = 73^3$. It constructs $E = F[Y]/(Y^3 - \beta_1)$ by an actual relative maximal order. The absolute degree is 54, the discriminant is $-3^{135}17^{48}$, and the relative power-basis index is one. At every base prime in S it checks

$$\mathfrak{s}_i \mathcal{O}_E = \mathfrak{S}_i^3, \qquad (e(\mathfrak{S}_i/\mathbb{Q}), f(\mathfrak{S}_i/\mathbb{Q})) = \begin{cases} (18, 1) & i \le 3, \\ (9, 2) & i \ge 4. \end{cases} \tag{E81.9}$$

These identities fix the prime ordering, rather than matching two unordered lists. No class-group computation in degree 54 is used. The official maximal-order, ideal-extension and principal-generator semantics are those in [The26]; the complete base BNF is certified with default flag zero and return value one.

Let $\mathcal{G}/F$ be the genus field defining Γ in E.76. It is unramified over E, so Frobenius gives the homomorphism (E81.1) on every ordinary ideal class of E. At a prime outside S, restriction of Frobenius to $\mathcal{G}/F$ is the Artin symbol of the ideal norm down to F. In particular, $\mathfrak{p}$ splits in E: the root 63 satisfies $63^3 = 22$ in $\mathbb{F}_{73}$, and μ_3 is contained there. Thus I is the product of its three conjugate upper primes. This proves (E81.3) and

$$\varphi[I] = 3\,\mathrm{Art}_{\mathcal{G}/F}(\mathfrak{p}). \tag{E81.10}$$

The local normalization at the six ramified primes is also explicit. At a three-prime use the upper uniformizer $\beta_2 + 1$, whose norm is $\pi = \beta_1 + 1$; at a seventeen-prime use β_2, whose norm is β_1. These are exact positive norm identities because the relative degree is odd. Arithmetic local reciprocity and its norm/inclusion square therefore compute the upper Frobenius as the base local Artin symbol of that specified norm [Mil20b, III Theorems 3.1 and 3.4].

For clarity, these local symbols are represented by actual coprime ideals in the certificate. Take the modulus with exponents $(10, 10, 10, 1, 1, 1)$ of E.76. For its ith prime, choose by the Chinese remainder map an element x_i congruent to the specified uniformizer one precision step beyond that modulus, and congruent to one at its other primes. Then $(x_i)\mathfrak{s}_i^{-1}$ is coprime

to S, and the desired local symbol is the negative of its ray Artin class. The full integer quotient map of E.76 is then applied to that class. This gives exactly the six columns in (E81.2). Applying it to (E81.10) gives $(6,0,0,0)$. All integer Smith maps and inverse maps are checked again in the new certificate.

The ideal I^3 is generated by the included f, so its class has order dividing three. Its nonzero Artin image has order three, proving the claimed exact order. The first base class generator is a prime above three, with class $(1,0)$. Its extension is Q_1^3, whose image is three times the first column of (E81.2), hence zero. Since $(2,1)$ has image $(6,0,0,0)$, extension followed by φ on the whole base class group is

$$(a,b) \longmapsto (6b,0,0,0). \tag{E81.11}$$

This proves the assertion about all six classes with nonzero second coordinate.

We next prove that the finite correction list is complete through actual principal functions. Include the six complete base S-unit generators of E.79 into E. Formula (E81.9) multiplies each valuation vector by three, giving $3V$. The divisor of β_2 is $T_1+T_2+T_3$, giving the last column of M. The columns of H in (E81.6) are columns $1,2,3,4,5,7$ of M. The omitted sixth column equals the first minus the fourth and fifth plus three times the seventh. This proves equality of the two relation lattices and supplies the explicit function for each column, in both directions. The complete Smith transformation gives $\mathbb{Z}^6/H\mathbb{Z}^6 = \mathbb{Z}/9 \oplus (\mathbb{Z}/3)^4$.

For a correction e, the Artin image of $I\prod \mathfrak{S}_i^{e_i}$ is $(6,0,0,0)+Ce$. Reducing the four rows of (E81.2) gives exactly

$$e_1+e_2+e_3 \equiv 0, \qquad e_4-e_6 \equiv -1, \qquad e_5-e_6 \equiv 1 \pmod 3. \tag{E81.12}$$

The solution lattice is $p+K\mathbb{Z}^6$. Direct integer multiplication gives

$$K^{-1}H = \begin{pmatrix} 3 & 0 & 0 & 1 & 1 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E81.13}$$

This proves the quotient and the inverse coordinate rule in (E81.7). It gives exactly (E81.4) as the twenty-seven representatives. If an S-principalization is given with any integer exponents, its Artin image vanishes, so (E81.12) applies. Reducing those exponents by (E81.13) and multiplying by the explicit principal function associated to Hq gives a principal ideal in (E81.4). Conversely any principal ideal in that list exhibits $[I]$ as a combination of S-prime classes, so its S-class is zero. This proves the two directions with the actual ideal comparison.

For the cube identities, the six base prime classes are $(1,0)$ three times and $(2,0)$ three times. For $e=e(a,b,c)$ their weighted sum is zero modulo three. Consequently $\mathfrak{p}^3\prod \mathfrak{s}_i^{e_i}$ is principal in F. Its certified principal generator is $f_{a,b,c}$ in (E81.5). Cubing (E81.4) and using (E81.9) proves the upper identity. The program checks that identity in the upper maximal order, while an independent rational-matrix computation checks every lower principal-ideal identity in the actual integral basis of F. The twenty-seven exponent classes are pairwise distinct modulo H.

Finally E.76 computes the entire ordinary ambiguous class group order as

$$|\operatorname{Cl}(E)\{3\}^G| = \frac{9\cdot 3^6}{3\cdot 27} = 81. \tag{E81.14}$$

This uses the certified base class number nine, the six ramification indices three, and the ordinary-unit field-norm index twenty-seven proved there. All six ramified prime classes are

G-invariant. Their Artin columns span a subgroup of order twenty-seven, as is verified both by the retained Smith maps and by the three independent columns consisting of the first, fourth, and fifth minus fourth columns. Therefore the kernel (E81.8) has order dividing three. The subgroup generated by the six S-prime classes has Artin image of order twenty-seven and kernel a subgroup of W, so its order is twenty-seven or eighty-one. The kernel of its actual divisor-class map has that index in $\mathbb{Z}^6$; comparison with $[\mathbb{Z}^6 : H\mathbb{Z}^6] = 729$ proves the stated index of additional principal S-relations. Each candidate ideal is G-invariant and satisfies (E81.12), so its class lies in W. The principal functions for H identify differences of candidate classes with a homomorphism $(\mathbb{Z}/3)^3 \to W$. Translation by $[J_{0,0,0}]$ gives their affine class map. Such a map has either zero, nine, or twenty-seven zero values, including the case $W = 0$. This proves the final bound.

For reproducibility, the degree-fifty-four computation constructs only a maximal order and exact ideals. The complete class-and-unit computation is the already certified degree-eighteen base. The separate unfiltered search was stopped after the genus constraint was evaluated; its real interrupted output is retained. The subsequent 108 reductions tested all twenty-seven necessary correction classes in four directions each and found no generator. Those outcomes are recorded as attempts. They do not evaluate the zero set of the affine class map just proved, and do not change the remaining S-capitulation or universal-norm index in E.79. □

All intrinsic second-commutator variations at the nine cusps

E.82. The nine complete cusp-family obstruction maps, their kernels, and every third-stage fibre [Proved]

Retain the based plus Fermat cubic of E.36–E.37, with $O = [1 : -1 : 0]$, parameter $s = z/x$ and tangent one. Let N be its geometric free pro-three group of rank ten, and use the actual local source and action of E.74–E.77. Write

$$\begin{aligned} W_2 = N/\gamma_3 N, \qquad W_3 = N/\gamma_4 N, \qquad V = N^{\mathrm{ab}}, \\ A_2 = \gamma_2 N/\gamma_3 N = \mathrm{Lie}_2 V, \qquad A_3 = \gamma_3 N/\gamma_4 N = \mathrm{Lie}_3 V, \\ T_r = \mathrm{Lie}_r(T_1), \qquad J_r = \mathrm{Lie}_r(J_1), \qquad T_1 = A(1 + 3\zeta, 7) \end{aligned} \tag{E82.1}$$

in the integral basis of (E37.2)–(E37.4). Thus τ is the specified actual generator with elliptic block $\left(\begin{smallmatrix} 1 & -3 \\ 3 & -2 \end{smallmatrix}\right)$, and J_1 is E.37's actual conjugation matrix. The bases of A_2, A_3 are the integral Hall bases of E.41, of ranks $45, 330$.

For every cusp $i \in \{A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2, O\}$, fix its unit tangent over $K = \mathbb{Q}(\zeta)$ and a geometric path p_i to it from the stated base point. Let ι_i be its positive pro-three inertia generator, and set

$$d_i = p_i \iota_i p_i^{-1}, \qquad u_i = p_i T(p_i)^{-1} \in N, \qquad u_i T(d_i) u_i^{-1} = d_i^7. \tag{E82.2}$$

The conventions for a path mean that $p_i \iota_i p_i^{-1}$ is based at O. The section families below are labelled by these fixed paths. Their abelian inertia vectors in V are

$$\bar{d}_i = e_i \quad (i \neq O), \qquad \bar{d}_O = -\sum_{i \neq O} e_i, \tag{E82.3}$$

where the eight e_i occur after f_1, f_2 in E.37's basis.

For $z \in A_2$, choose the continuous Hall representative $w_z \in \gamma_2 N$ described below. The pair

$$(u_i w_z, d_i) \quad \text{in } W_2^2 \tag{E82.4}$$

gives an actual local section over $\mathbb{Q}_3$ by E.77. For fixed i its complete geometric-conjugacy parameter space is the compact abelian group

$$\begin{gathered}\mathcal{P} = A_2/(I-T_2)A_2^{J_2} = \operatorname{coker} G, \qquad G = (I-T_2)(I+J_2),\\ \mathcal{P} \simeq \mathbb{Z}_3^{24} \oplus C_3^9 \oplus C_9^5, \qquad C_m = \mathbb{Z}/m\mathbb{Z}.\end{gathered} \tag{E82.5}$$

The nine resulting subsets of $H^1(\mathbb{Q}_3, W_2)$ are pairwise disjoint.

Let $\mathcal{B}_3$ be the actual local obstruction group of E.74. On each entire family (E82.4), the obstruction map is the following continuous homomorphism of compact groups:

$$\begin{gathered}\delta_i : \mathcal{P} \longrightarrow \mathcal{B}_3 = \frac{A_3}{(T_3-7I)A_3 + (J_3+I)A_3} \simeq C_3^{63} \oplus C_9^{23} \oplus C_{27}^7,\\ \delta_i([z]) = [7[z, \bar{d}_i]].\end{gathered} \tag{E82.6}$$

In particular every value uses the full intrinsic rank-330 central lattice. The image, its exact kernel, and its cokernel are as follows:

i	$\operatorname{im}\delta_i$	$\ker\delta_i$
A_0, B_0, O	$C_3^{11} \oplus C_9^3$	$\mathbb{Z}_3^{24} \oplus C_3^3 \oplus C_9^4$
$A_1, A_2, B_1, B_2, C_1, C_2$	$C_3^{22} \oplus C_9^7 \oplus C_{27}$	$\mathbb{Z}_3^{24} \oplus C_3^6$

(E82.7)

i	$\operatorname{coker}\delta_i$	$[\mathcal{P} : \ker\delta_i]$
A_0, B_0, O	$C_3^{52} \oplus C_9^{20} \oplus C_{27}^7$	3^{17}
$A_1, A_2, B_1, B_2, C_1, C_2$	$C_3^{41} \oplus C_9^{16} \oplus C_{27}^6$	3^{39}

(E82.8)

The certificate supplies a separate matrix and both coordinate inverses for each of the nine entries; equality of the displayed invariants does not remove their marked maps.

The image of the third-stage section map over (E82.4) is exactly $\ker\delta_i$. Every nonempty fibre, with a particular lift chosen by (E82.13) below, is homeomorphic to

$$\mathbb{Z}_3^{330} \oplus C_3^7 \oplus C_9^{56} \oplus C_{27}^{25} \oplus C_{81}^3. \tag{E82.9}$$

The zero parameter in each family has the actual full N-section (E82.2); the parameters outside the stated kernel have no W_3-section above them. In particular the single Hall vector $z = [f_2, f_1]$ has nonzero obstruction at every one of the nine cusps; its full 93 coordinates and exact order are saved. At A_0 its first obstruction coordinate is 2 mod 3. Thus the literal pair $(u_{A_0}[q_2, q_1], d_{A_0})$, where q_j is the chosen lift of f_j, has no intrinsic third-stage lift. The actual maps from N to its intrinsic metabelian and single-cusp marked quotients identify their first three nilpotent stages with those used here. Thus their third-stage lifting maps on these families have the same images and fibres.

Here are exact coordinate maps for (E82.6)–(E82.9). Let $M = [\,T_3 - 7I \mid J_3 + I\,]$, and take the retained matrices L, R of E.74, with

$$LMR \equiv [D \mid 0] \pmod{729}, \qquad D = \operatorname{diag}(3^{v_1}, \dots, 3^{v_{330}}). \tag{E82.10}$$

The multiplicities of $v = 0, 1, 2, 3$ are $237, 63, 23, 7$. Both L and R have their recorded inverse modulo 729, and hence define invertible matrices over $\mathbb{Z}_3$. Let A_i be the 330 by 45 integral matrix $z \mapsto [z, \bar{d}_i]$. Selecting the 93 rows with $v_j > 0$ gives the actual finite homomorphism

$$\begin{gathered}F_i : A_2 \longrightarrow \bigoplus_{v_j>0} \mathbb{Z}/3^{v_j}, \qquad (F_i z)_j = (L\,7A_i z)_j \bmod 3^{v_j},\\ \Lambda_i = \ker F_i = K_i \mathbb{Z}_3^{45}, \qquad \ker\delta_i = \Lambda_i / G\mathbb{Z}_3^{45}.\end{gathered} \tag{E82.11}$$

Every K_i is a saved integer Hermite basis, with exact rational inverse, determinant 3^{17} or 3^{39} as in (E82.8), and $27K_i^{-1}$ integral. The matrices $K_i^{-1}G$ are integral. Exact Smith maps over $\mathbb{Z}_{(3)}$ and both their inverses are retained for G, all K_i, and all $K_i^{-1}G$. In detail, write

$$\begin{aligned} L_iK_iR_i = D_i, \qquad P_i(K_i^{-1}G)Q_i = E_i. \\ \operatorname{im}\delta_i: \quad z \longmapsto L_iz \pmod{D_i}, \qquad \eta \longmapsto F_iL_i^{-1}\eta; \\ \ker\delta_i: \quad z \longmapsto P_iK_i^{-1}z \pmod{E_i}, \qquad \eta \longmapsto K_iP_i^{-1}\eta \pmod{G}. \end{aligned} \tag{E82.12}$$

The first line of coordinates identifies an obstruction value using any parameter preimage z; the saved generators $F_iL_i^{-1}\eta$ give its inverse. For $\mathcal{P}$ itself the corresponding coordinates use the saved Smith maps of G. All nonzero Smith entries have been scaled to powers of three by actual units in $\mathbb{Z}_{(3)}$.

For a particular geometric third-stage lift, put $v = 7A_iz$, take the first 330 columns R_* of R, and set

$$\begin{aligned} C = D^{-1}LMR_*, \qquad C \equiv I \pmod{27}, \qquad C^{-1} = \sum_{n\ge 0}(I-C)^n, \\ \begin{pmatrix} d \\ e \end{pmatrix} = -R_*C^{-1}D^{-1}Lv, \qquad (u_iw_z,\ d_i\,d) \quad \text{in } W_3^2. \end{aligned} \tag{E82.13}$$

Here $z \in \Lambda_i$, and d in the last pair denotes the central group element with its displayed coordinate in A_3; e is an auxiliary relation coordinate. This is a convergent formula in $\mathbb{Z}_3$, with no unevaluated higher Frobenius coefficient. The certificate verifies the actual corrections modulo 729 on all 405 columns of the nine kernel bases.

Finally let $D_V \subset V$ be the eight-dimensional cusp lattice generated by the e_i, and define the homomorphism

$$\begin{aligned} A_2^8 \longrightarrow A_3, \qquad (z_i)_{i\neq O} \longmapsto \sum_{i\neq O}[z_i, e_i], \\ \mathscr{I} = \sum_i \operatorname{im}\delta_i \subset \mathcal{B}_3. \end{aligned} \tag{E82.14}$$

The first image is a saturated direct summand of rank 304, with a specified Hall-coordinate complement of rank 26. The sum in the second line is the image of the induced homomorphism $\mathcal{P}^8 \to \mathcal{B}_3$, and has exact groups

$$\mathscr{I} \simeq C_3^{61} \oplus C_9^{25} \oplus C_{27}^2, \qquad \mathcal{B}_3/\mathscr{I} \simeq C_3^{11} \oplus C_9. \tag{E82.15}$$

The quotient map in (E82.15), its finite coordinates and their inverse are given by the saved 93 by 93 relation-lattice Smith maps. Thus the nine families evaluate a specified subgroup of the full obstruction group, with its remaining quotient explicitly retained.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). These are full intrinsic local families and all their third-stage fibres; the remaining global and higher intrinsic equations stay in E.23.

Proof. Actual paths and sections. The rational coordinate permutation $[x : y : z] \mapsto [z : y : x]$ carries the base tangent of E.34 to the stated point O and changes its parameter by the rational factor -1. The compatible three-power roots of this factor are all -1. This is the actual Galois-invariant tangent comparison used in E.78, so the based action still factors through the same actual $P \rtimes \langle c\rangle$ of E.74. We use its generator with the specified E.37 matrix in (E82.1).

Every cusp and chosen unit tangent is defined over K. On its local inertia the actual τ acts by the seventh power. Applying T to the path expression for d_i and cancelling $T(p_i)^{-1}T(p_i)$

gives (E82.2) as an identity in the full group N. In particular the relator value $u_i T(d_i) u_i^{-1} d_i^{-7}$ is one. E.77 now gives an actual section over $\mathbb{Q}_3$, with values $u_i, J(u_i), d_i, J(d_i)$ on its four specified generators and trivial value on its relative order-two subgroup. The latter subgroup lies in the actual relative source of E.74, with its proved inflation map from the absolute local group.

Choose lifts of the two elliptic basis vectors of V and use the eight fixed d_i, $i \neq O$, for the other eight generators. Their classes form a basis of $V/3V$. They generate N by the pro-three Frattini criterion. The induced surjection from a free pro-three group of rank ten to N is an isomorphism, by the Hopfian property used in E.12. They are therefore an actual free basis. For $z = (z_b)$ in its 45 Hall coordinates define w_z to be the ordered product of the corresponding group commutators raised to their $\mathbb{Z}_3$ powers z_b. This is a continuous map into $\gamma_2 N$ whose image modulo $\gamma_3 N$ is exactly z. The relation among all peripheral cocharacters gives (E82.3). A change of the lift w_z by $\gamma_3 N$ has no effect on (E82.4).

In W_2 the subgroup A_2 is central, so multiplication of u_i on the right by w_z leaves its relator value one. This proves the section assertion for every parameter, not just for parameters in the later lifting kernel.

The entire parameter gauge. By E.77 two pairs (E82.4) are geometrically conjugate precisely by its residual J-fixed geometric conjugators. Let $g \in W_2^J$ carry the parameter z to z'. Comparing the abelian coordinates of their first entries gives $(I - T_1)\bar{g} = 0$. The elliptic block has characteristic polynomial $X^2 + X + 7$, whose value at one is nine, and the cusp block is $7I_8$. Consequently $I - T_1$ is invertible over $\mathbb{Q}_3$, and its kernel on the torsion-free integral lattice is zero. Thus g belongs to A_2 and is a vector $b \in A_2^{J_2}$. The exact gauge formula of E.77 becomes

$$z' - z = (I - T_2)b. \tag{E82.16}$$

Conversely every such central b fixes d_i and gives exactly this change. This proves the full orbit relation. Since two is a unit in $\mathbb{Z}_3$, the image of $I + J_2$ is precisely $A_2^{J_2}$: the inverse on the fixed lattice is $b \mapsto b/2$. This proves (E82.5) with the actual matrix G. Its Smith computation has seven unit factors, nine factors three, five factors nine and 24 zero factors, in agreement with E.41. Conjugation leaves the second entry's abelian vector unchanged. The nine vectors (E82.3) are distinct, so the nine families have disjoint geometric-conjugacy images.

The exact intrinsic third-degree error. Compute in W_3. Its third subgroup is central, and $[w_z, T(d_i)]$ has coordinate $7[z, \bar{d}_i]$. The group identity $waw^{-1} = [w, a]a$ and the full identity (E82.2) give

$$\begin{aligned} (u_i w_z) T(d_i) (u_i w_z)^{-1} d_i^{-7} &= u_i [w_z, T(d_i)] u_i^{-1} u_i T(d_i) u_i^{-1} d_i^{-7} \\ &= [w_z, T(d_i)] = 7[z, \bar{d}_i] \quad \text{in } A_3. \end{aligned} \tag{E82.17}$$

All higher terms of the original path and Frobenius cancel in this equality. Changing w_z by a third-degree element leaves the displayed commutator unchanged. Writing v for (E82.17), E.77 gives the defect $(I - J_3)v$ and the local invariant class $[(I - J_3)v/2]$. In the quotient by $(J_3 + I)A_3$ this class is $[v]$, which proves (E82.6), with its factor seven and its sign.

The operator identity needed for the parameter quotient is an identity of integral Lie maps:

$$T_3[z, \bar{d}_i] = [T_2 z, 7\bar{d}_i], \qquad 7[(I - T_2)b, \bar{d}_i] = -(T_3 - 7I)[b, \bar{d}_i]. \tag{E82.18}$$

Here $T_1 \bar{d}_i = 7\bar{d}_i$ is the actual inertia character. The second equality shows that every gauge relation maps to zero in $\mathcal{B}_3$, and proves that δ_i is a homomorphism on the full group $\mathcal{P}$. The matrices A_i and both identities are checked integrally in the certificate.

All finite coordinates and their inverse maps. E.74 proves that the obstruction cokernel is killed by 3^5 before making its finite computation modulo 3^6. Its actual Smith calculation (E82.10) has maximum valuation three. Therefore multiplication by L and reduction in its

positive rows identifies the exact completed cokernel with the displayed sum of 93 cyclic groups. This gives the map F_i in (E82.11).

We describe the complete kernel computation, including the passage from its integer certificate to the compact lattice. Start with $K = I_{45}$. At stage $j = 0, 1, 2$ use the rows of F_i having exponent greater than j. All entries of their product with the current K are divisible by 3^j. Divide by that power and row-reduce over $\mathbb{F}_3$. For each pivot column take its unit vector; for each free column subtract the pivot-vector combination prescribed by that reduced column. These vectors form an integer unimodular matrix: a permutation of its rows puts it in block triangular form with identity diagonal blocks. Multiply its pivot columns by three, multiply by K, and take the integer Hermite basis of the resulting full lattice.

At this step a vector satisfies the next congruence exactly when its pivot coordinates in that unimodular basis are multiples of three. Induction therefore proves equality with the solution lattice after each stage, over both $\mathbb{Z}$ and $\mathbb{Z}_3$. After three stages all 93 congruences have been imposed. The resulting matrix is the claimed K_i. The successive pivot counts are $(14, 3, 0)$ at A_0, B_0, O and $(30, 8, 1)$ at the other six cusps. They give the determinants in (E82.8). The direct checks $F_iK_i = 0$ in each target cyclic group and $27K_i^{-1}$ integral also retain the two containments and the exponent bound.

For completeness the exact Smith operation used on these bases works in the localization $\mathbb{Z}_{(3)}$. Choose an entry of minimum three-adic valuation in the remaining block, move it to the next diagonal position, and multiply its row by the inverse of its unit part. The pivot is now exactly 3^a. Every other entry in that block is divisible by 3^a in $\mathbb{Z}_{(3)}$, so row and column additions clear its column and row with coefficients in that ring. Earlier pivots remain unchanged. The remaining entries still have valuation at least a, proving that the resulting diagonal is a Smith diagonal. The program tracks the left and right operations and their inverse operations, and separately multiplies all full matrices to verify $LAR = D$, $LL^{-1} = L^{-1}L = I$, and $RR^{-1} = R^{-1}R = I$. Every denominator in these four matrices is prime to three. Extending these exact maps to $\mathbb{Z}_3$ proves the stated continuous coordinate isomorphisms.

For K_i the positive valuations are $(1^{11}, 2^3)$ or $(1^{22}, 2^7, 3)$, respectively. For $K_i^{-1}G$ they are $(1^3, 2^4)$ or (1^6), with 24 zero rows in both cases. Their cokernels prove (E82.7) and the explicit two-sided coordinate maps (E82.12). The actual matrices $F_iL_i^{-1}$ are saved modulo the target cyclic orders, using the inverse of every rational unit denominator. Thus the obstruction generators themselves, and not just their orders, are retained. The first degree-two Hall vector is $[f_2, f_1]$. Its first matrix column is nonzero in each of the nine actual cyclic targets, as recorded separately in the certificate. At A_0 its first row is two modulo three, proving the stated concrete nonlifting pair.

There is also an independent finite check of every image invariant. For $k = 0, 1, 2, 3$, compute the cokernel of the actual matrix

$$[\,\mathrm{diag}(3^{v_j}) \mid 3^k F_i\,] \quad (v_j > 0) \tag{E82.19}$$

modulo 81. These cokernels are killed by 27 because the diagonal columns already impose that bound. Minimum-valuation elimination modulo 81 therefore gives their exact Smith valuations. Subtracting their order exponents from 130 gives $\log_3 |3^k \operatorname{im} F_i|$. Successive differences recover the multiplicities of cyclic factors of orders three, nine and twenty-seven. They agree with the exact K_i computations and also give the cokernels in (E82.8).

Particular lifts and all full conjugacy fibres. Equation (E82.10), with its integer representatives, implies $LMR_* = D + 729E$ for an integer matrix E. Since every diagonal entry of D divides 27, $C = D^{-1}LMR_*$ is integral and congruent to the identity modulo 27. Its inverse series in (E82.13) converges uniformly on the compact lattice. For $z \in \Lambda_i$, the vector $D^{-1}Lv$

is integral by (E82.11). Thus both d, e in (E82.13) belong to A_3, and multiplication gives the exact identity

$$(T_3 - 7I)d + (J_3 + I)e = -v. \tag{E82.20}$$

Apply $I - J_3$ to this equality. The second summand vanishes, and the first cancels the exact defect $(I - J_3)v$ from (E82.17). E.77 therefore proves that the displayed corrected pair is an actual W_3-section. The stored correction matrix checks this equality on every kernel basis column modulo 729. At that precision the inverse series is $I - (C - I)$, since $(C - I)^2$ is divisible by 729. The convergent formula proves the lift at every precision.

Conversely a third-stage section above (E82.4) has zero local obstruction, so its parameter lies in Λ_i. This establishes the entire image. E.74's actual central-difference map now identifies each nonempty full geometric-conjugacy fibre with $H^1(\mathbb{Q}_3, A_3)$, giving (E82.9). More explicitly one multiplies the particular lift by a central cocycle; the inverse first aligns its lower section by the unique lower geometric conjugator and then takes the central difference. The absence of a lower stabilizer is E.74's positive weight argument for these actual modules. E.75 and E.77 give the cocycle coordinate inverse $A_3/(T_3 - I)A_3^{J_3} \oplus \ker((I - J_3)(T_3 - 7I))$. Every map just used is continuous; the orbit spaces are compact Hausdorff. Their specified continuous inverse maps prove the homeomorphism assertions.

The quotient maps to the intrinsic metabelian and marked groups are the actual maps of E.41. They induce isomorphisms on the first three integral graded groups. In the finite lower-central filtration of $N/\gamma_4 N$, a kernel element therefore maps trivially in grade one, then in grade two, then in grade three, and is the identity. The maps onto the corresponding third-stage groups are surjective by their definition, hence are isomorphisms. They retain the arithmetic action and the source markings, which proves the stated transfer of section images and fibres.

The sum of the nine evaluated images. The chosen Hall order first places the two elliptic letters and then the eight cusp letters. A degree-three Hall word is $[[a, b], c]$ with $a > b$ and $b \le c$. Every such word whose outer letter c is a cusp letter is itself a column of (E82.14). The remaining Hall words number 26: there are two involving only the elliptic letters and three for each of the eight cusp letters. They are the words with outer elliptic letter and with its allowed inner smaller elliptic letter. A column $[[a, b], e_i]$ that is not already Hall has $b > e_i$. Then a, b, e_i are all cusp letters, and its Jacobi rewrite has outer letters among a, b, again cusp letters. This proves that every column has zero coordinates in the 26 complementary rows. Independent associative tensor expansions in the program check this integral rewrite. The other 304 rows have distinct unit columns. Selecting those columns is an actual integer inverse on this summand, proving both its saturation and the stated complement.

The O-map is the negative sum of the eight others by (E82.3). Hence the subgroup $\mathscr{I}$ is exactly the image of the saved concatenated matrix $[F_i]_{i \neq O}$. Applying (E82.19) to this matrix gives

$$\log_3 |3^k \mathscr{I}| = 117, 29, 2, 0 \quad (k = 0, 1, 2, 3). \tag{E82.21}$$

These values give the first decomposition in (E82.15), and its presented cokernel has eleven factors three and one factor nine. For actual quotient coordinates form the integer column lattice of $[\mathrm{diag}(3^{v_j}) \mid F_{A_0} \mid \cdots \mid F_{C_2}]$ in $\mathbb{Z}^{93}$. Its saved Hermite basis has determinant 3^{13}. Every presentation column lies in it; the finite index 3^{13} just computed proves equality of the lattices. Its exact local Smith maps and both inverses therefore give the quotient coordinates claimed in (E82.15).

The programs `e_cusp_family_obstruction.py` and `e_cusp_local_smith.py` reconstruct all of these data. They check all 405 Hall brackets against independent associative tensor expansions, the nine T and nine J intertwiners, all original E.74 inverse matrices, every complete parameter kernel and gauge map, the 405 particular central corrections, and the joint

direct-summand and quotient inverses. The saved JSON contains the entire matrices, kernel derivations, exact rational Smith transformations and inverse maps; it has no unspecified arithmetic coefficient in the nine obstruction homomorphisms. □

The intrinsic image in the joint wild obstruction

E.83. The diagonal obstruction image and empty intrinsic preimages [Proved]

Retain the actual maps, source marking and conjugate paths of E.80. Put

$$\begin{gathered} N_2 = N/\gamma_3 N, \qquad N_3 = N/\gamma_4 N, \qquad A_3 = \gamma_3 N/\gamma_4 N = \mathrm{Lie}_3(N^{\mathrm{ab}}), \\ A = \ker(\mathcal{J}_3 \to \mathcal{J}_2) = \mathbb{Z}_3^4, \quad (r,s,r',s') \longleftrightarrow (d^r e^s, d^{r'} e^{s'}). \end{gathered} \tag{E83.1}$$

The central coefficient map induced by $N_3 \to \mathcal{J}_3$ factors as the actual equivariant surjection and inclusion

$$\begin{gathered} A_3 \twoheadrightarrow L \hookrightarrow A, \\ L = \{(r,s,r',s') : r'-r \in 3\mathbb{Z}_3, \quad s+s'-2r \in 3\mathbb{Z}_3\}. \end{gathered} \tag{E83.2}$$

In particular these are the image and inclusion from the integral group computation in E.80.

Use the actual local cohomology coordinates of E.74 and E.77. They give the natural maps

$$\begin{gathered} \mathcal{B}_3 = H^2(\mathbb{Q}_3, A_3) \simeq C_3^{63} \oplus C_9^{23} \oplus C_{27}^{7}, \\ \mathcal{B}_3 \twoheadrightarrow H^2(\mathbb{Q}_3, L) \longrightarrow H^2(\mathbb{Q}_3, A). \end{gathered} \tag{E83.3}$$

Both groups after $\mathcal{B}_3$ have the following fully specified coordinates. A basis for L is

$$\begin{gathered} (r,s,a,b) \longmapsto (r,s,r+3a,2r-s+3b), \\ K_L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 \\ 2 & -1 & 0 & 3 \end{pmatrix}, \qquad \det K_L = 9. \end{gathered} \tag{E83.4}$$

In this basis and the displayed basis of A, the involutions and Frobenius operators are

$$\begin{gathered} J_L = \begin{pmatrix} -1 & 0 & -3 & 0 \\ -2 & 1 & 0 & -3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{pmatrix}, \qquad J_A = -\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \\ T_L = T_A = 343 I_4, \qquad J_A K_L = K_L J_L. \end{gathered} \tag{E83.5}$$

Consequently

$$\begin{aligned} H^2(\mathbb{Q}_3, L) &\xrightarrow{\sim} \mathbb{F}_3^2, \quad (r,s,a,b) \longmapsto (r, b-a), \\ H^2(\mathbb{Q}_3, A) &\xrightarrow{\sim} \mathbb{F}_3^2, \quad (r,s,r',s') \longmapsto (r+r', s+s'), \\ H^2(K_L) : \mathbb{F}_3^2 &\longrightarrow \mathbb{F}_3^2, \qquad (r,t) \longmapsto (2r, 2r). \end{aligned} \tag{E83.6}$$

All coordinates on the right are reduced modulo three. The image of the full intrinsic map $\Phi : \mathcal{B}_3 \to H^2(\mathbb{Q}_3, A)$ is therefore exactly the diagonal C_3. Its exact remaining quotient is the character

$$\mu : \mathbb{F}_3^2 \twoheadrightarrow \mathbb{F}_3, \qquad \mu(\xi,\eta) = \eta - \xi. \tag{E83.7}$$

Write $\delta_N, \delta_{\mathcal{J}}$ for the actual central obstruction maps for $N_3 \to N_2$ and $\mathcal{J}_3 \to \mathcal{J}_2$. The exact naturality square is

$$\begin{array}{ccc} H^1(\mathbb{Q}_3, N_2) & \longrightarrow & H^1(\mathbb{Q}_3, \mathcal{J}_2) \\ \delta_N \downarrow & & \downarrow \delta_{\mathcal{J}} \\ \mathcal{B}_3 & \xrightarrow{\Phi} & \mathbb{F}_3^2. \end{array} \tag{E83.8}$$

For the normalized representative of E.80,

$$\delta_{\mathcal{J}} = (\Delta R, \Delta S) \bmod 3. \tag{E83.9}$$

Thus the actual intrinsic class-two image is contained in the explicitly computed locus $\Delta R \equiv \Delta S \pmod 3$.

In particular let $[\sigma_{N,M}]$ be E.80's descended second-stage section with $a = y$, $a' = y^{-1}$, $u = x^{5/2}c^N$, $u' = x^{-1/2}y^2c^M$. For every $N, M \in \mathbb{Z}_3$ with $N + M \not\equiv 2 \pmod 3$ its full intrinsic class-two preimage is empty:

$$\big(H^1(\mathbb{Q}_3, N_2) \longrightarrow H^1(\mathbb{Q}_3, \mathcal{J}_2)\big)^{-1}([\sigma_{N,M}]) = \varnothing. \tag{E83.10}$$

This evaluates a specific obstruction within the kernel retained in (E80.4), before an intrinsic class-three lifting question is formed. The intrinsic preimages of the congruence-compatible members are retained by that same exact map; no sufficiency assertion for those preimages is included in (E83.10).

The whole integral coefficient map is also evaluated in the retained cubic basis. In the column order $(f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2)$, the four geometric unit characters $(X_h, Y_h, X_{\bar{h}}, Y_{\bar{h}})$ form the matrix

$$Q = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & -3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 & 1 & 1 & -3 & 0 \end{pmatrix}. \tag{E83.11}$$

For a degree-three Hall word $[[q_a, q_b], q_c]$, the actual 4 by 330 matrix $F_3 : A_3 \to A$ has the following two coordinate pairs, using $\epsilon = 0, 2$ for the two pairs of rows of Q:

$$-(Q_{\epsilon,a}Q_{\epsilon+1,b} - Q_{\epsilon+1,a}Q_{\epsilon,b})(Q_{\epsilon,c}, Q_{\epsilon+1,c}). \tag{E83.12}$$

Here row and column indices of Q start at zero; the Hall ordering is exactly E.41's retained ordering. The complete matrix is saved, together with its actual arithmetic intertwiners and image basis.

Let $b_1, \dots, b_{93}$ be E.74's positive Smith coordinates, in their retained order. The first 63 have order three, the next 23 have order nine and the final seven have order twenty-seven. Then the full intrinsic obstruction map and its kernel have the explicit coordinates

$$\begin{gathered} \Phi(b) = (\lambda(b), \lambda(b)), \qquad \lambda(b) = b_{89} + b_{91} + 2b_{92} \pmod 3, \\ \ker \Phi \simeq C_3^{63} \oplus C_9^{24} \oplus C_{27}^{6}, \\ k = (b_{89} + b_{91} + 2b_{92})/3 \in C_9, \qquad b_{89} = 3k - b_{91} - 2b_{92} \in C_{27}. \end{gathered} \tag{E83.13}$$

For these inverse kernel coordinates retain all b_j except b_{89} and replace it by k. Division by three in the displayed subgroup of C_{27} means its unique inverse identification with C_9.

Let $\mathscr{I}$ be E.82's sum of the nine actual cusp-family obstruction images. The entire subgroup $\mathscr{I}$ lies in $\ker \Phi$. In E.82's retained residual coordinates $(\eta_1, \dots, \eta_{12})$ on $\mathcal{B}_3/\mathscr{I} = C_3^{11} \oplus C_9$, the exact induced map and its kernel are

$$\begin{gathered} \bar{\lambda}(\eta) = \eta_9 + \eta_{10} \pmod 3, \\ (\ker \Phi)/\mathscr{I} \simeq C_3^{10} \oplus C_9, \qquad \eta_9 = -\eta_{10}. \end{gathered} \tag{E83.14}$$

Its inverse retains the other eleven residual coordinates and inserts the displayed ninth one. Thus the cohomological comparison also computes one exact quotient of the remaining E.82 obstruction coordinates.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). The result computes the exact cohomological image of the integral central map and its effect on actual descended quotient sections.

Proof. The maps in (E83.2) are E.80's actual central maps. Their image is $L = \gamma_3\mathcal{J}_3$, since $N_3 \to \mathcal{J}_3$ is onto. The lattice is free of rank four; it has the basis (E83.4), obtained by writing $r' = r + 3a$ and $s' = 2r - s + 3b$. The first map is onto this lattice as an integral module. Its kernel is a saturated free submodule of A_3, because the quotient L is torsion-free. The inclusion $L \to A$ has the separately retained index nine.

On a pure third-degree target coordinate, both actual E.80 operators of τ and τ' are multiplication by 7^3. The involution is minus the factor swap, by (E80.19). This gives T_A, J_A. Substitution of the basis (E83.4) gives the integer matrix $J_L = K_L^{-1} J_A K_L$ displayed in (E83.5). In particular $J_L^2 = I$; no rational identification of L with all of A is used.

The local-duality proof in (E74.15) identifies $H^2(\mathbb{Q}_3, M)$ naturally with the compact coinvariants of $M(-1)$ for each of these finite free modules. Coefficient homomorphisms preserve the finite-level evaluation pairing, and the inverse-limit identifications are compatible with them. Equivalently E.75's actual relative source resolution gives the same coefficient maps. Since the local image is the actual Γ, this natural presentation is

$$H^2(\mathbb{Q}_3, M) = M/\big((T_M - 7I)M + (J_M + I)M\big), \qquad M = A_3, L, A. \tag{E83.15}$$

The same proof applies to L and A: their actions factor through that actual quotient, and all lattices and finite coefficient systems are finite free before reduction. The images in the denominator are compact. This also proves the surjectivity of the first arrow in (E83.3) directly: lift a representative in L through the coefficient surjection $A_3 \to L$ and pass to (E83.15).

For $M = A$ or L, $T_M - 7I = 336I = 3 \cdot 112I$, and 112 is a three-adic unit. Reduce the involution relations modulo three. For A they identify the two factors; the complete quotient coordinates are $(r + r', s + s')$. For L, the image of $J_L + I$ modulo three is spanned by the s basis vector and the sum of the a, b basis vectors. Its quotient coordinates are $(r, b - a)$. These observations prove both isomorphisms in (E83.6), including their surjectivity and exact kernels. The actual inclusion sends

$$(r, s, a, b) \longmapsto (r, s, r + 3a, 2r - s + 3b) \longmapsto (2r, 2r) \pmod 3. \tag{E83.16}$$

This proves the displayed natural map, its diagonal image, and the residual quotient (E83.7). In particular the intermediate exact sequence is $0 \to \mathbb{F}_3 \to H^2(\mathbb{Q}_3, L) \to H^2(\mathbb{Q}_3, A) \to \mathbb{F}_3 \to 0$, with first map $t \mapsto (0, t)$ and last map μ.

There is also an explicit nonzero image witness in the actual intrinsic lattice. Choose geometric lifts $n_1, n_2 \in N$ of g_1, g_2 in (E80.23), using the proved geometric surjection. The word $[n_2, [n_1, n_2]]$ maps to the central vector $(1, 1, 1, 1)$; this is the third column of (E80.26). Its cohomological image is $(2, 2)$, a generator of the diagonal. This is consistent with, and also witnesses, the surjectivity argument above.

To prove the obstruction naturality with its normalization, use the literal equivariant diagram of central extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & A_3 & \longrightarrow & N_3 & \longrightarrow & N_2 & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & A & \longrightarrow & \mathcal{J}_3 & \longrightarrow & \mathcal{J}_2 & \longrightarrow & 1. \end{array} \tag{E83.17}$$

The left arrow is the composite (E83.2), rather than a surjection onto the full bottom kernel. Choose continuous Hall lifts of a lower cocycle to N_3 and apply the middle arrow. Its multiplication defect is the image of the A_3-valued defect. Changing to the target's chosen Hall lifts adds an A-valued coboundary. Centrality makes the coefficient action independent of the lower geometric conjugator. This proves (E83.8) on the full section sets. It also proves that its assertion is independent of the retained representatives and gauges.

In E.80's normalized source notation let $v_- = v_{-5/2}(u, a)$ and $v_+ = v_{1/2}(u', a')$. The second-stage relation makes $v_- v_+^{-1}$ central in C_3, with coordinates $\Delta = (\Delta R, \Delta S)$. The joint lifted word is (v_-, Iv_+). Since I is multiplication by minus one in third degree, its anti-invariant source defect and the E.77 invariant are exactly

$$\omega = (\Delta, \Delta), \qquad [\omega/2] = [(\Delta/2, \Delta/2)] \in H^2(\mathbb{Q}_3, A). \tag{E83.18}$$

Applying the target coordinate sum gives precisely Δ mod 3, which proves (E83.9) without a factor-of-two ambiguity. The map μ vanishes on the image of Φ, so (E83.8) proves the claimed necessary locus for every actual intrinsic class-two section.

For the family in (E80.14), the computed target defect is $(0, 7(5 - N - M))$. Its residual value is $7(5 - N - M)$ mod 3, which is nonzero exactly for the parameters in (E83.10). An intrinsic class-two preimage would contradict the exact commutative square and the diagonal image. This proves the empty-preimage assertion.

The full coefficient and residual maps. The positive peripheral coefficient of a unit character is its order at that cusp. E.54 gives $\mathrm{div}(h) = 3O - 3C_2$ and $\mathrm{div}(1-h) = \sum_j B_j - 3C_2$. Conjugating swaps C_1, C_2. These are exactly the last eight columns of (E83.11), in E.37's positive peripheral convention. Its actual integral gluing has columns $j_* = [3I; U]$, with $U_d = (v_d, -u_d)$ in that section's table. The composite of the unit homomorphism from the generalized Jacobian with $j : E \to J_{E,D}$ is trivial: a group morphism from the proper connected elliptic curve to $\mathbb{G}_m$ is constant, and its value at the identity is one. Thus a unit character with toric row m_T has its exact elliptic row determined by

$$3m_f + m_T U = 0, \qquad m_f = -m_T U/3. \tag{E83.19}$$

This proves all four first blocks of (E83.11) integrally, with their signs. The rows for ℓ, X_h, $(X_h + X_{\bar{h}})/3$ have an integer determinant-one minor in columns (f_2, B_0, C_1). They therefore also recover the surjection onto E.80's joint abelian lattice.

The precise E.78 marking transport also preserves the retained coefficient matrices. It replaces τ_0 by $h_0 \tau_0 h_0^{-1}$. Over K the E.37 matrix on V is $\left(\begin{smallmatrix} A_E & 0 \\ (UA_E - \chi U)/3 & \chi I \end{smallmatrix}\right)$. Over $\mathbb{Q}_3$ this is conjugate, by the fixed matrix $[3I, 0; U, I]$, to $\mathrm{diag}(A_E, \chi I)$. The CM matrices A_E commute, and therefore these actual integral matrices commute as well. Conjugation by the matrix of h_0 leaves T_1 unchanged; the arithmetic involution is the retained one. Thus T_3, J_3 in the full coefficient calculation are exactly E.74's matrices after the actual source-word transport, with no change of the coefficient coordinate system.

A homomorphism of free pro-three groups induces the free Lie homomorphism on its lower central graded groups. In the target convention $c = [x, y]$, $d = [x, c]$, $e = [y, c]$, a second-degree determinant is the coefficient of c, and $[c, Ax + By] = -Ad - Be$. This proves (E83.12). The resulting image has the basis of (E83.4) up to an integer unimodular change, as verified by its determinant-nine Hermite basis. The arithmetic equalities are the exact identities $F_3 T_3 = 343 F_3$ and $F_3 J_3 = J_A F_3$.

Let L_* and L_*^{-1} be E.74's retained left Smith matrix and inverse modulo 729, and let I_+ be its ordered set of positive rows. The complete coordinate map on the source obstruction group is

$$\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} F_3 (L_*^{-1})_{*, I_+} \pmod 3. \tag{E83.20}$$

The columns belonging to the omitted unit rows vanish modulo three. Both rows of (E83.20) equal the character in (E83.13). The three nonzero entries occur at the specified order-twenty-seven coordinates, with coefficients $1, 1, 2$. These statements are exact modulo three, so the retained inverse precision gives the actual map. The formula for k and its inverse then proves the kernel type and both coordinate maps in (E83.13).

There is a geometric reason that every E.82 cusp image is killed. Modulo three, (E83.11) sends each B_j to (y, y) and every other cusp cocharacter to zero. A second-degree pair has coordinates (n, n') with $n + n' = 0$ modulo three, by (E80.15). Its bracket with (y, y) has central coordinates $(0, -n, 0, -n')$, whose image under the target sum in (E83.6) is zero. This proves $\Phi\delta_i = 0$ for all nine actual maps, hence $\mathscr{I} \subset \ker \Phi$.

For the numerical residual coordinate map, let P_* be E.82's exact left Smith matrix of its 93 by 93 relation lattice and let I'_+ be its twelve positive positions. Using the full row λ in (E83.13), the quotient character is

$$\lambda(P_*^{-1})_{*,I'_+} \pmod 3 = (0,0,0,0,0,0,0,0,1,1,0,0). \tag{E83.21}$$

All rational denominators in this inverse matrix are prime to three. The unit positions vanish, and all relation-lattice columns are killed, so the displayed row is the literal induced homomorphism. This proves (E83.14), including its surjectivity, kernel and inverse.

The self-contained program `a_joint_intrinsic_image.py` reconstructs the actual Hall image matrix from the two integral abelian lattices. It saves its integer Smith maps and inverses, those for both local cohomology relation matrices, the integral involution on L, and the explicit natural cohomology map. It verifies all 81 ambient residue vectors, all 729 inputs to the six actual Hall image columns, all nine index-two normalizations, and all 81 family residues with parameters modulo nine. Every check passes; the program and its actual output and JSON are retained. The second program `a_joint_intrinsic_coordinates.py` derives the four unit characters, checks all 660 target tensor expansions independently, reconstructs the entire 4 by 330 map, checks both exact arithmetic intertwiners, and saves both actual cohomology matrices. It verifies the vanishing of all 405 E.82 cusp columns and computes the complete residual row and its coordinate inverse. The 6561 triples for the changed kernel coordinates in (E83.13) and all three values of the residual coordinate in (E83.14) are also checked in both directions. Its retained input hashes specify the exact E.37/E.41, E.74 and E.82 coordinate systems used in (E83.13)–(E83.14). □

The exact ordinary capitulation kernel and three remaining S-class tests

E.84. Normalizer action evaluates the ordinary transfer and reduces the S-unit problem [Proved]

Use the actual fields and prime orderings of E.81:

$$\begin{gathered} K = \mathbb{Q}(\zeta), \quad \xi^3 = \zeta, \quad a^3 = 17, \quad \beta_1 = a\xi, \quad \beta_2^3 = \beta_1, \\ F = B_1 = K(\xi, a), \qquad E = B_2 = F(\beta_2), \qquad G = \mathrm{Gal}(E/F) = \langle\sigma\rangle. \end{gathered} \tag{E84.1}$$

Write Q_1, Q_2, Q_3 for the upper primes above three and T_1, T_2, T_3 for those above seventeen. Their corresponding base classes are respectively $(1, 0)$ and $(2, 0)$ in the certified group $\mathrm{Cl}(F) = (\mathbb{Z}/3)^2$. Let $P = (73, t - 33)$, $t = a - \xi$, and $I = P\mathcal{O}_E$; then $[P] = (2, 1)$.

The ordinary ideal-extension map is completely evaluated:

$$\begin{gathered} j : \mathrm{Cl}(F) \longrightarrow \mathrm{Cl}(E)\{3\}, \qquad j(u, v) = v[I], \\ \ker j = \langle(1, 0)\rangle \simeq \mathbb{Z}/3, \qquad \mathrm{im}\, j = \langle[I]\rangle \simeq \mathbb{Z}/3. \end{gathered} \tag{E84.2}$$

Every one of the six ramified prime classes has exact order three. In addition,

$$[Q_1] + [Q_2] + [Q_3] = 0, \qquad [T_1] + [T_2] + [T_3] = 0. \tag{E84.3}$$

Thus all the ideals $Q_i^3, T_i^3, Q_1Q_2Q_3$ are principal; the second product in (E84.3) is literally (β_2).

Here is the exact remaining S-divisor map. Put

$$\begin{gathered} S=(Q_1,Q_2,Q_3,T_1,T_2,T_3),\quad \Delta=(-1,1,0,0,0,0)^{\mathsf t},\\ L_0=3\mathbb{Z}^6+\mathbb{Z}(1,1,1,0,0,0)^{\mathsf t}+\mathbb{Z}(0,0,0,1,1,1)^{\mathsf t},\\ \delta=[Q_2]-[Q_1],\qquad w=[I]+[T_2]-[T_1]. \end{gathered} \tag{E84.4}$$

The lattice L_0 consists of principal S-divisors and has index 81. Its HNF and quotient are

$$H_0=\begin{pmatrix} 3&0&1&0&0&0\\ 0&3&1&0&0&0\\ 0&0&1&0&0&0\\ 0&0&0&3&0&1\\ 0&0&0&0&3&1\\ 0&0&0&0&0&1 \end{pmatrix},\qquad \mathbb{Z}^6/L_0\simeq(\mathbb{Z}/3)^4. \tag{E84.5}$$

Let ϕ be the actual genus Artin map of E.81 and

$$W=\ker\Big(\phi:\mathrm{Cl}(E)\{3\}^G\longrightarrow\Gamma\Big). \tag{E84.6}$$

Both w and δ belong to W, whose order divides three. On $\mathbb{Z}^6/L_0$, the kernel of the genus map is exactly the line generated by Δ. Consequently the full lattice L of principal S-divisors satisfies the unconditional exact identity

$$L/L_0=\ker\Big(\mathbb{F}_3\xrightarrow{t\mapsto t\delta}W\Big)\quad\text{inside}\quad\mathbb{Z}^6/L_0,\qquad t\longmapsto t\Delta. \tag{E84.7}$$

The image of the six S-prime classes still has order 27 or 81; (E84.7) identifies its entire remaining kernel.

The twenty-seven ideals of E.81 now have the exact class formula

$$[J_{a,b,c}]=w+(b+2c)\delta. \tag{E84.8}$$

Thus the S-class of I vanishes exactly when one of the three literal ideals

$$I\,T_2T_1^{-1}(Q_2Q_1^{-1})^t,\qquad t=-1,0,1, \tag{E84.9}$$

is principal. Each of these three class values occurs nine times in the old list. The values of w,δ in W are not evaluated here.

There is also an evaluated comparison with the original complex tower. The fixed field of the automorphism h below is $L_2=\mathbb{Q}(\beta_2)$; its discriminant is $-3^{45}17^{16}$, and E/L_2 is unramified everywhere and split at the two primes above $3,17$. The exact ordinary transfer gives

$$\mathrm{ord}[\mathfrak{q}_{3,2}]=3,\qquad j_{E/L_2}[\mathfrak{q}_{3,2}]=[Q_1Q_2Q_3]=0. \tag{E84.10}$$

Finally retain the exact distinction between the two norm maps that arise in the proposed unit calculation. For $c_1=\beta_1+1+3\zeta$,

$$N_{F/K(\xi)}c_1=1-\zeta,\qquad c_1\notin N_{E/F}E^\times,\qquad \xi c_1\in N_{E/F}E^\times. \tag{E84.11}$$

The last membership is the proved field-norm assertion of E.79. Every S-unit preimage of ξc_1 has the now proved principal ideal $Q_1Q_2Q_3$. Choose a generator γ of that ideal. The entire remaining norm fiber is the explicitly translated ordinary-unit fiber

$$\{x\in\mathcal{O}_E[1/3,1/17]^\times:N_{E/F}x=\xi c_1\}=\gamma\,\{u\in\mathcal{O}_E^\times:N_{E/F}u=\xi c_1/N_{E/F}\gamma\}. \tag{E84.12}$$

The quotient on the right is a unit of F, and changing γ changes it by an actual unit norm. Neither this unit-norm coset nor the eventual exponent h of E.69 is evaluated by the ordinary capitulation calculation. The manuscript anchor is LM §51.16, fols. 733v–736v, scans 172–178.

Proof. The essential extra input is an actual normalizer action, including its action on the certified base class group. Define

	ξ	β_2	a	β_1
h	$\zeta\xi$	β_2	$\zeta^2 a$	β_1
g	ξ	$\xi\beta_2$	ζa	$\zeta\beta_1$

$$h^3 = 1, \quad g^3 = \sigma, \quad g^9 = 1, \quad hgh^{-1} = g^4. \tag{E84.13}$$

These are automorphisms of the displayed field, not formal actions on a class group. For example $\beta_2^9 = 17\xi^3$; both substitutions preserve this equation and $\xi^6 + \xi^3 + 1 = 0$, and their inverse substitutions are powers of themselves. Their 27 distinct products $g^i h^j$, $0 \le i < 9, 0 \le j < 3$, are all the K-automorphisms, since $[E : K] = 27$. Both normalize F and commute with $\sigma = g^3$.

In the actual six-prime ordering the permutations are

$$h = (Q_1\, Q_2\, Q_3)(T_1\, T_2\, T_3), \qquad g = (Q_1\, Q_3\, Q_2), \qquad g(T_i) = T_i. \tag{E84.14}$$

The certificate checks these as equalities of ideals in the maximal order of E, using the substitutions (E84.13). In the primitive base coordinate $t = a - \xi$, the two substitutions are $t \mapsto \zeta^2 a - \zeta\xi$ and $t \mapsto \zeta a - \xi$. Their actions on the certified class basis of E.76 are

$$h_F = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad g_F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad \text{over } \mathbb{F}_3. \tag{E84.15}$$

For each basis ideal and for P, the program retains the actual image ideal, its class coordinates and a literal principal generator of its quotient by the specified class-basis ideals. A separate rational matrix calculation verifies these six principal-ideal identities in the complete integral basis of F. Thus (E84.15) uses the exact certified class map, with the arithmetic automorphisms fixed by (E84.13) [The26, entries `bnfisprincipal` and `nfgaloisapply`].

We spell out why these automorphisms preserve the particular kernel W. Let $\mathcal{H}_E$ be the maximal unramified abelian three-extension of E, and let $\mathcal{G}$ be its maximal subfield which is abelian over F, as constructed in E.76. An extension to an algebraic closure of an automorphism normalizing E and F carries $\mathcal{H}_E$ to itself: unramifiedness and abelianity are preserved by conjugation, and the inverse gives the reverse inclusion. It carries $\mathcal{G}$ to itself for the same reason and its maximal property. The Artin map is equivariant: on an unramified prime this is the equality

$$\mathrm{Frob}_{\nu\mathfrak{P}} = \nu\, \mathrm{Frob}_{\mathfrak{P}}\, \nu^{-1}, \qquad \nu = g, h, \tag{E84.16}$$

and multiplication gives the assertion on every ideal class. Since g, h commute with σ, they also preserve the subgroup of σ-fixed classes. Hence both preserve the kernel in (E84.6). The retained ray matrices additionally verify that they preserve the exact relation lattice of E.76, and that their genus actions carry each of its six Artin columns to the column specified by (E84.14).

E.81 proves $|W| \mid 3$. An automorphism of a group of order three has order dividing two. The actions of g, h have orders dividing nine and three, respectively, so both are the identity on W. This argument also includes $W = 0$.

Put $k = j(1, 0)$, $i = [I]$, and $t_j = [T_j]$. The actual Artin matrix of E.81 gives $\phi(i + t_2 - t_1) = 0$, so $w = i + t_2 - t_1 \in W$. By (E84.15),

$$g(i) - i = j\big((0, 1) - (2, 1)\big) = j(1, 0) = k. \tag{E84.17}$$

Equation (E84.14) fixes both t_1, t_2. Since g acts trivially on W, (E84.17) implies $k = (g-1)w = 0$. On the other hand E.81 proves that i has order exactly three, through its nonzero Artin

value $(6,0,0,0)$ in $\mathbb{Z}/9 \oplus (\mathbb{Z}/3)^3$. Because $(2,1) = 2(1,0) + (0,1)$, this proves (E84.2) on the entire certified base class group.

Extension of the six base primes gives exactly their upper cubes. Thus

$$3[Q_j] = j(1,0) = 0, \qquad 3[T_j] = j(2,0) = 0. \tag{E84.18}$$

Every one of their Artin columns in E.81 is nonzero; their orders are therefore exactly three. The three Q-columns are equal, so $\delta = [Q_2] - [Q_1]$ belongs to W. The identity $h\delta = \delta$, together with the actual cycle (E84.14), gives $[Q_3] - [Q_2] = \delta$. Therefore $[Q_3] = [Q_1] + 2\delta$, and their sum is $3[Q_1] + 3\delta = 0$. The T-product is (β_2), since this element is a uniformizer at each of the three primes over seventeen and a unit at every other finite prime. This proves (E84.3) and all the principal relations defining L_0.

The displayed HNF has determinant 81; its full integral Smith maps and inverse maps are retained. The genus images of the six S-primes have order 27 by E.81. Since $\Delta \notin L_0$ has order three and has zero Artin image, it is the entire genus kernel on $\mathbb{Z}^6/L_0$. Every principal divisor has zero Artin image. On that remaining line the actual class map is exactly $t\Delta \mapsto t\delta$, proving (E84.7) in both directions.

For the comparison of the old list with (E84.9), substitute $[Q_2] = [Q_1]+\delta$ and $[Q_3] = [Q_1]+2\delta$ into its literal exponent vector $(3a + 2b + 2c, b, c, -1, 1, 0)$. This gives (E84.8). There is an exact ideal comparison behind the reduction. Choose the centered representative t of $b + 2c$ modulo three and write $m = (b + 2c - t)/3$. The difference of its exponent vector from the vector of (E84.9) is

$$c(1,1,1,0,0,0) + 3(a + b + c - m)(1,0,0,0,0,0) + 3(m - c)(0,1,0,0,0,0). \tag{E84.19}$$

All three displayed factors have just been proved principal. Thus multiplying by their principal functions gives the actual ideal-class comparison; no identification of abstract vector spaces is substituted for it. The linear map $(a, b, c) \mapsto b + 2c$ has kernel of order nine, which proves the multiplicity assertion.

The fixed field of h contains β_2 and has degree eighteen. This is exactly $L_2 = \mathbb{Q}(\beta_2)$, with polynomial $X^{18} + 17X^9 + 289$. The maximal-order calculation gives its discriminant $-3^{45}17^{16}$; its cube equals the independently retained discriminant $-3^{135}17^{48}$ of E. The relative discriminant ideal therefore has norm one. All infinite places are complex, so the extension is unramified everywhere. The two three-cycles in (E84.14) give complete splitting at $3, 17$, also agreeing with the all-level comparison in E.76. The original prime extends to $Q_1Q_2Q_3$, which is principal by (E84.3). E.72 proves that its class in L_2 has exact order three. This establishes the concrete capitulating class (E84.10).

For the norm comparison, put $u = 1 + 3\zeta$. The literal identities $u^3 = 1 - 18\zeta$ and $\beta_1^3 = 17\zeta$ give $N_{F/K(\xi)}(\beta_1 + u) = 1 - \zeta$. At each base prime over seventeen the residue field has order 289, and the cubic relative extension is totally ramified. The residue of a unit norm is therefore a cube. The certificate checks, in each of the three actual residue maps,

$$c_1^{96} = \zeta, \qquad (\xi c_1)^{96} = 1. \tag{E84.20}$$

Since $\zeta \neq 1$, the first identity excludes the uncorrected norm. The second is the tame part of the corrected field-norm calculation; E.79 supplies its full six-place verification and the Hasse norm theorem, yielding (E84.11).

The actual base valuations of ξc_1 are $(1,1,1,0,0,0)$. Each of the six relative extensions has residue degree one and a single upper prime, so ideal norm preserves these valuation coefficients. An S-unit preimage therefore has divisor precisely $Q_1 + Q_2 + Q_3$. All elements with that divisor are $\gamma\mathcal{O}_E^\times$, and taking their literal norms gives (E84.12). The ratio there is a

unit because its divisor is zero. Changing γ by an ordinary unit multiplies the ratio by the inverse of its norm, proving the stated independence of the remaining coset.

The searches retained with this result found no small generator or norm preimage in their stated finite search sets. They are not used to infer nonprincipality or failure of an S-unit norm. In particular the principality conclusions above follow from the exact normalizer and class-group argument, even where those searches did not produce a principal function. □

Compatible joint sections and their full intrinsic central fibres

E.85. The compatible joint family and an exact index-nine central comparison [Proved]

Retain the actual local source, conjugate maps and tangent of E.77–E.80. Write

$$V = N^{\mathrm{ab}}, \qquad A_2 = \gamma_2 N/\gamma_3 N, \qquad N_2 = N/\gamma_3 N, \qquad A = \ker(\mathcal{J}_2 \to \mathcal{J}_1) = \mathbb{Z}_3^2. \tag{E85.1}$$

The two coordinates of A are the exponents of $(c,1)$ and $(1,c)$ in E.80. Its actual operators are $49I$ and the factor swap. Use the unchanged integral cubic basis $(f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2)$.

There is an actual full intrinsic local section with E.77 values (u_0, d) such that

$$\begin{gathered} J(d) = d^{-1}, \qquad u_0 T(d) u_0^{-1} = d^7, \qquad (h_*, \bar{h}_*)(d) = (y, y), \\ (h_*, \bar{h}_*)(u_0) \bmod \gamma_3 F^2 = (x^{5/2}c^2,\ x^{1/2}y^{-2}c^3). \end{gathered} \tag{E85.2}$$

Here d is the positive inertia of the rational cusp B_0. The proof constructs a fixed path and its actual normalizing conjugators; the constants two and three are forced by the full cusp identity. Let α_0 be the abelian class of this section in $H^1(\mathbb{Q}_3, V)$ and β_0 its image in $H^1(\mathbb{Q}_3, \mathcal{J}_1)$.

For E.80's whole family

$$a = y, \qquad a' = y^{-1}, \qquad u = x^{5/2}c^N, \qquad u' = x^{-1/2}y^2c^M, \qquad N, M \in \mathbb{Z}_3, \tag{E85.3}$$

the actual intrinsic class-two preimage is nonempty exactly for

$$N + M \equiv 2 \pmod 3. \tag{E85.4}$$

Every compatible member already has a preimage over the single intrinsic abelian class α_0. The complete geometric-conjugacy classes of the compatible target family have the coordinate and inverse

$$t = N - M + 1 \in \mathbb{Z}_3, \qquad N = 2 + t/2, \quad M = 3 - t/2. \tag{E85.5}$$

In particular this is an exact existence converse to E.83 for this family.

The entire fibre over α_0, including variations of both relative source generators, is evaluated as follows. Put

$$\begin{aligned} \mathcal{P} &= A_2/(I - T_2)A_2^{J_2} \simeq \mathbb{Z}_3^{24} \oplus C_3^9 \oplus C_9^5, \\ \mathcal{K}_2 &= \ker((I - J_2)(T_2 - 7I) : A_2 \to A_2) \simeq \mathbb{Z}_3^{22}, \\ \mathfrak{F}_{\alpha_0} &:= \big(H^1(\mathbb{Q}_3, N_2) \to H^1(\mathbb{Q}_3, V)\big)^{-1}(\alpha_0) \xrightarrow{\sim} \mathcal{P} \oplus \mathcal{K}_2. \end{aligned} \tag{E85.6}$$

The inverse of the last arrow is the actual section with values

$$(u_0 w_z,\ dw_a), \qquad ([z], a) \in \mathcal{P} \oplus \mathcal{K}_2, \tag{E85.7}$$

where w_z, w_a are the ordered Hall representatives modulo $\gamma_3 N$. All four source values and the trivial value on the order-two source subgroup are then determined by E.77. The residual gauge is exactly $z \mapsto z + (I - T_2)b$, $b \in A_2^{J_2}$; it leaves a unchanged.

The target fibre over β_0 is

$$\begin{gathered}\mathfrak{G}_{\beta_0} := \left(H^1(\mathbb{Q}_3, \mathcal{J}_2) \to H^1(\mathbb{Q}_3, \mathcal{J}_1)\right)^{-1}(\beta_0) \xrightarrow{\sim} \mathbb{Z}_3 \oplus \mathbb{Z}_3 \oplus C_3, \\ (\rho, \sigma, \epsilon) = (n - m,\ s,\ (n + m)/2 \bmod 3).\end{gathered} \tag{E85.8}$$

In the last line the central changes from the origin (E85.2) are (n, m) on its horizontal value and (s, s) on its relative value. For $\epsilon \in \{0, 1, 2\}$ an actual inverse is

$$u_0^{\mathcal{J}}(c^{\rho/2+\epsilon}, c^{-\rho/2+\epsilon}), \qquad a_0^{\mathcal{J}}(c^{\sigma}, c^{\sigma}), \qquad a_0^{\mathcal{J}} = (y, y). \tag{E85.9}$$

Let $F_2 : A_2 \to \mathbb{Z}_3^2$ be the two-by-forty-five matrix obtained from the four rows of (E83.11): on a Hall vector $[q_a, q_b]$, its two entries are

$$Q_{\epsilon,a}Q_{\epsilon+1,b} - Q_{\epsilon+1,a}Q_{\epsilon,b}, \qquad \epsilon = 0, 2. \tag{E85.10}$$

Its exact integral image and arithmetic comparisons are

$$\begin{gathered}L_2 = \{(n, m) : n + m \in 3\mathbb{Z}_3\}, \qquad F_2T_2 = 49F_2, \quad F_2J_2 = \operatorname{swap} F_2, \\ F_2(\mathcal{K}_2) = \{(3s, 3s) : s \in \mathbb{Z}_3\}.\end{gathered} \tag{E85.11}$$

Consequently the complete natural map on the two fibres in (E85.6) and (E85.8) is

$$\begin{gathered}([z], a) \longmapsto ((F_{2,1} - F_{2,2})z,\ F_{2,1}a,\ 0), \\ \operatorname{im}(\mathfrak{F}_{\alpha_0} \to \mathfrak{G}_{\beta_0}) = \mathbb{Z}_3 \oplus 3\mathbb{Z}_3 \oplus 0, \\ \operatorname{coker}(H^1(\mathbb{Q}_3, A_2) \to H^1(\mathbb{Q}_3, A)) \simeq C_3^2, \\ \ker(H^1(\mathbb{Q}_3, A_2) \to H^1(\mathbb{Q}_3, A)) \simeq \mathbb{Z}_3^{44} \oplus C_3^9 \oplus C_9^5.\end{gathered} \tag{E85.12}$$

Every point in the displayed image has exactly this compact fibre, with the explicit coordinates below. Thus each compatible member (E85.3) has a complete preimage *over* α_0 of this type.

There are small literal splitting vectors. With brackets in the retained Hall convention put

$$\begin{gathered}z_* = -\tfrac{1}{2}[f_2, f_1], \qquad a_* = -[B_0, f_1] + [B_0, f_2] - [B_0, A_1], \\ F_2z_* = (1/2, -1/2), \qquad a_* \in \mathcal{K}_2, \qquad F_2a_* = (3, 3).\end{gathered} \tag{E85.13}$$

The inverse to the image coordinates $(\rho, 3s, 0)$ can therefore be taken to be the section (E85.7) with $z = \rho z_*$, $a = sa_*$. Adding the exact kernel coordinates gives all its preimages over α_0.

For completeness these kernel coordinates use the actual retained Smith and relative-kernel maps. Let $L_G G R_G = D_G$ be E.82's exact Smith system for $G = (I - T_2)(I + J_2)$, where

$$D_G = \operatorname{diag}(1^7, 3^9, 9^5, 0^{24}). \tag{E85.14}$$

Write $p = L_G z$. Set its first seven entries to zero, take $p_8, \ldots, p_{16}$ in C_3 and $p_{17}, \ldots, p_{21}$ in C_9, and retain its final twenty-four entries in $\mathbb{Z}_3$. In that order the row for ρ on the final twenty-four entries is

$$r = (-1, -1, -1, 0, 0, 0, 0, -3, 3, 0, -1, 0, -1, 0, 0, 3, -3, 0, 0, -2, -3, -3, -1, 0). \tag{E85.15}$$

Let B and B^{ℓ} be E.75's retained saturated kernel basis and left inverse for $\mathcal{K}_2$; write $a = B\eta$, $\eta = B^{\ell}a \in \mathbb{Z}_3^{22}$. The exact row for $\sigma/3$ is

$$s = (0, 0, 0, 0, -3, 0, 0, 0, 0, 0, 1, 0, -3, 2, -2, 0, 0, 0, -1, 2, -1, -2). \tag{E85.16}$$

For any image value $(\rho, 3v, 0)$ retain all the fourteen cyclic coordinates, the twenty-three entries $p_{23}, \dots, p_{45}$, and the twenty-one entries η_j with $j \neq 11$. The inverse inserts

$$\begin{aligned} p_{22} &= \sum_{j=2}^{24} r_j p_{21+j} - \rho, \qquad \eta_{11} = v - \sum_{j \neq 11} s_j \eta_j, \\ z &= L_G^{-1} p, \qquad a = B\eta, \end{aligned} \tag{E85.17}$$

and then applies (E85.7). These are mutually inverse homeomorphisms on the full geometric-conjugacy fibres. The source generators are E.77's actual generators; B is used as a basis of its displayed coefficient kernel, without identifying distinct earlier choices of relative source elements.

Restricting to $a = 0$ in (E85.7) recovers the embedded forty-five-parameter cusp family. Its map onto the compatible target family has exact fibres

$$\mathbb{Z}_3^{23} \oplus C_3^9 \oplus C_9^5. \tag{E85.18}$$

The entire intrinsic preimage can also contain sections over other abelian preimages of β_0. The source of the complete fibre assertion in (E85.12) is explicitly $\mathfrak{F}_{\alpha_0}$; no other intrinsic abelian class is identified with α_0.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). This evaluates actual compatible joint preimages and their full central fibres over a specified intrinsic abelian section.

Proof. The actual rational cusp origin. For the path construction use actual global complex conjugation $c_\infty \in G_{\mathbb{Q}}$, which fixes the two rational unit tangents. Its image in the global quotient Γ is the marked c, and its based action on N is exactly J by E.34.3 and E.78's rational tangent transport. The local involution of E.77 has the same image under $\mathcal{D} \to \Gamma$, as constructed in E.74. Choose an actual global lift $\tilde{\tau}$ of the marked $\tau \in P$; its based action is T and its three-adic cyclotomic value is seven. Thus the global path actions used below induce the prescribed based operators. The element $p\tilde{\tau}(p)^{-1}$ supplies the displayed norm identity, and E.77 constructs the local section from that identity and $J(d) = d^{-1}$. This is the exact passage from the global path groupoid to the local section presentation.

Choose a rational unit tangent at B_0 fixed by the order-two action, and start with the torsor of geometric pro-three paths from the stated base tangent to it. This torsor has a fixed point. Here the needed coprime fixed-point fact has a direct proof: for a finite three-group, every order-two cocycle is a coboundary. Induct on the group through an invariant elementary abelian central subgroup; on that subgroup averaging divides by two. For a pro-three group, the nonempty solution sets in finite invariant quotients give a nonempty compact inverse limit. The same argument proves that an equivariant surjection of pro-three groups is onto on fixed points: the defect of a lift is such a cocycle in its kernel.

Use a fixed path p, and put $d = p\iota_{B_0}p^{-1}$. Thus $J(d) = d^{-1}$. The maps $h, \bar{h}$ have order one at the respective target cusp 1, so the first image is $\alpha y \alpha^{-1}$ for an actual $\alpha \in F$. The equality $J(d) = d^{-1}$ and the literal factor-swap involution of E.80 then make its second image exactly $I(\alpha) y I(\alpha)^{-1}$. Set

$$w = y^{-\ell(\alpha)} \alpha^{-1}. \tag{E85.19}$$

Then $\ell(w) = 0$. Hence (w, Iw) belongs to $\mathcal{J}^J$ and conjugates this inertia pair to (y, y). The actual equivariant surjection $N \to \mathcal{J}$ in E.80 lifts it to a fixed element of N, by the preceding fixed-point argument. Change p by that element. This retains a fixed path and proves the exact inertia assertion in (E85.2).

Initially put $u = pT(p)^{-1}$. The actual tangent action gives $uT(d)u^{-1} = d^7$ in the full group. Project this identity to each class-three target, retaining all five Hall coordinates. The class-two coefficient first forces the two x-coordinates of u to be $5/2, 1/2$. Membership of its pair in $\mathcal{J}$ makes their y-coordinates $V, V-2$ for one $V \in \mathbb{Z}_3$. The actual E.78 polynomial identity, with central coordinates r_i, s_i retained, is

$$\begin{aligned} T_{-5/2}: &\quad x^{5/2}y^V c^{2-5V/2} d^{r_1} e^{s_1} \quad \text{conjugates } T_{-5/2}(y) \text{ to } y^7, \\ T_{-1/2}: &\quad x^{1/2}y^{V-2} c^{3-V/2} d^{r_2} e^{s_2} \quad \text{conjugates } T_{-1/2}(y) \text{ to } y^7. \end{aligned} \tag{E85.20}$$

Conversely the two c-coordinates displayed are forced by those identities: their errors are respectively $e^{7(2-N-5V/2)}$ and $e^{7(3-M-V/2)}$. Replace u in the full group by $u_0 = d^{-V}u$. This preserves the norm identity and gives exactly the two class-two words in (E85.2). No value of r_i, s_i is assigned or needed. E.77 now supplies the actual full intrinsic section with values $u_0, J(u_0), d, J(d)$.

Every compatible family member. Let $e = [f_2, f_1]$ and $f = [B_0, f_1]$ in A_2. Their actual images, computed from (E83.11), are

$$F_2 e = (-1, 1), \qquad F_2 f = (-2, -1). \tag{E85.21}$$

They form an integral basis for L_2. For $n = N-2$, $m = M-3$ and $\ell = (n+m)/3$, the literal Hall vector

$$z = (2\ell - n)e - \ell f \tag{E85.22}$$

has image (n, m) whenever (E85.4) holds. Since A_2 is central in N_2, the pair $(u_0 w_z, d)$ still has norm defect one. Its joint image is precisely (E85.3), with the conjugate source values prescribed in E.80. This proves existence. E.83 proves emptiness for all incompatible members, completing the equivalence (E85.4).

For the complete target gauge, a normalized geometric conjugator is an element of $\mathcal{J}_2^J$. Comparing the fixed abelian horizontal coordinates of two members forces its abelian vector to be zero, since $7-1$ is injective on that lattice. Therefore it is (c^b, c^b), $b \in \mathbb{Z}_3$. Its exact change is

$$(N, M) \longmapsto (N - 48b, M - 48b). \tag{E85.23}$$

For compatible members the actual normalizing value $b = (N + M - 5)/96$ is integral. It gives (E85.5), and no nonzero remaining b fixes the resulting pair. Conversely equality of t for compatible members makes their common coordinate difference divisible by three, so (E85.23) supplies the required integral conjugator. This proves the complete orbit statement, including the nonunit factor in the normalizing denominator.

The full fibre over the chosen intrinsic abelian class. The abelian section α_0 has no geometric stabilizer. Indeed the actual T_1 on V has eight eigenvalues seven and an elliptic block with characteristic polynomial $X^2 + X + 7$; neither has eigenvalue one. Thus its fixed lattice is zero. An intrinsic section with abelian class α_0 can consequently be aligned with its actual abelian representative by a unique abelian conjugator. After normalizing the order-two source value to one, that conjugator is J_1-fixed. Fixed-point lifting through $N_2 \to V$ gives an actual fixed lift. All remaining conjugators lie in $A_2^{J_2}$.

The aligned values therefore have the unique form (E85.7). Centrality and the full identity in (E85.2) give its literal norm value

$$(u_0 w_z) T(dw_a)(u_0 w_z)^{-1} (dw_a)^{-7} = (T_2 - 7I)a \quad \text{in } A_2. \tag{E85.24}$$

E.77 makes the source relation precisely $(I - J_2)(T_2 - 7I)a = 0$. Conjugation by a remaining $b \in A_2^{J_2}$ changes z by $(I - T_2)b$ and does not change a. This proves every direction of

(E85.6)–(E85.7). The Smith and saturated kernel maps used are exactly those of E.82 and E.75 for these actual coefficient operators. The E.78 Frobenius transport preserves them by the explicit CM and gluing comparison in E.83.

Apply the same argument to the target abelian lattice. Its fixed group is zero because its operator is $7I$. For its central lattice A the condition in (E85.24) is $42(I - \mathrm{swap})(s_1, s_2) = 0$. It is equivalent to $s_1 = s_2 = s$. The remaining central conjugator is (b, b), changing (n, m) by $(-48b, -48b)$. Thus its coordinates are exactly (E85.8). To check the inverse, take ϵ to be the residue of $(n + m)/2$ and use

$$b = \frac{(n + m)/2 - \epsilon}{48} \in \mathbb{Z}_3. \tag{E85.25}$$

This is the actual conjugator producing (E85.9), and proves that these coordinates already include full geometric conjugacy.

The coefficient map and all inverse fibre coordinates. The coefficient map of the central extensions is the alternating map (E85.10), since both targets use $c = [x, y]$. Its two-by-two image minor (E85.21) has determinant three. Every column has sum divisible by three, proving the exact image L_2. The two arithmetic intertwiners in (E85.11) follow from the actual four unit characters and are also checked on all forty-five Hall columns.

For $a \in \mathcal{K}_2$ those intertwiners give

$$42(I - \mathrm{swap})F_2 a = 0. \tag{E85.26}$$

Thus $F_2 a$ is diagonal. Its membership in L_2 makes its common entry divisible by three. The retained vector a_* in (E85.13) satisfies the defining integral kernel equation and has image $(3, 3)$. This proves equality in (E85.11), not merely a finite-level bound. Similarly the vector z_* has the displayed image and gives all values of ρ. The third target coordinate vanishes because every $F_2 z$ has coordinate sum divisible by three. These facts prove the exact map and image (E85.12) with the explicit section (E85.13).

For the complete coordinate inverse use the literal matrices in (E85.14). Multiplying $(F_{2,1} - F_{2,2})L_G^{-1}$ gives zero on the first twenty-one columns and the row (E85.15) on the last twenty-four. Thus the map kills all fourteen torsion coordinates, and its first free coefficient is minus one. The exact product $F_2 B$ has two identical rows, equal to three times (E85.16); its eleventh coefficient after division by three is one. Solving these two unit equations gives precisely (E85.17). The retained matrices include both inverse pairs over $\mathbb{Z}_{(3)}$, and $B^\ell B = I$. Their ranks and these unit coefficients therefore give the asserted kernel type, with no unproved index or saturation step.

Restricting the relative central coordinate a to zero removes its twenty-two free coordinates. The remaining kernel of the surjective row $\rho : \mathcal{P} \to \mathbb{Z}_3$ is (E85.18), with inverse obtained by the same first equation in (E85.17). By contrast the full fibre over α_0 retains the twenty-one free relative coordinates in that formula. This is the exact inclusion between the two source fibres.

Every space used here is compact, the displayed groups are Hausdorff, and all Hall representatives, matrix maps and unit divisions are continuous. The explicit inverse formulas prove the homeomorphism assertions directly. In particular no closure is silently added to a gauge image; these images are already compact images of the indicated finite free lattices.

The executed certificate independently expands all ninety alternating tensor images, checks the exact image lattice, both arithmetic intertwiners, all inverse Smith and kernel maps, and the two literal splitting vectors. It verifies four symbolic class-three cusp norm identities while retaining the unknown central path entries. It also checks all 729 sampled target family pairs, $1,296$ exact coordinate inverse identities, and the index-nine finite coordinate quotients through modulus 81. The complete input hashes, matrices and actual output are retained. □

Complete S-class towers and the first ordinary class groups

E.86. All S-class norms, the ordinary class groups at the second level, and the exact unit-norm defects [Proved]

Keep the literal roots of E.76 and E.79:

$$K=\mathbb{Q}(\zeta),\quad \xi^3=\zeta,\quad \beta_d^{3^d}=17\zeta,\quad \beta_{d+1}^3=\beta_d,\quad L_d=K(\beta_d),\quad B_d=L_d(\xi). \tag{E86.1}$$

Let S mean all primes above $3,17$ in the indicated field. All class groups below are their three-primary parts. Put

$$A_d=\mathrm{Cl}_S(B_d)\{3\},\qquad C_d^S=\mathrm{Cl}_S(L_d)\{3\},\qquad d\ge 1. \tag{E86.2}$$

These entire towers, including their actual ideal maps, are

$$\begin{gathered} A_d=\mathbb{Z}/3,\qquad C_d^S=(\mathbb{Z}/3)^2,\\ N_{B_{d+1}/B_d}=1,\quad j_{B_{d+1}/B_d}=0,\qquad N_{L_{d+1}/L_d}=I_2,\quad j_{L_{d+1}/L_d}=0. \end{gathered} \tag{E86.3}$$

Here are explicit compatible bases for these formulas. Let $\mathfrak{p}_d$ be the degree-one prime of B_d above 73 with $\xi=55$ and $\beta_d=22$ for odd d, 63 for even d, as in E79.10. Its three-primary S-class is the displayed generator of A_d. In L_d, let

$$\begin{gathered} J_d=(73,\beta_d-r_d),\qquad P_d=(7,\beta_d+1),\\ a_d=[J_d/P_d]_3,\quad b_d=[J_d]_3,\qquad r_d=\begin{cases}22 & d \text{ odd},\\ 63 & d \text{ even}.\end{cases} \end{gathered} \tag{E86.4}$$

The ideals mean the kernels of the indicated residue maps on the maximal orders. Their degree is one and their adjacent norms are their predecessors. The pair (a_d,b_d) is the specified S-class basis. The remaining maps at each level are

$$\begin{gathered} N_{B_d/L_d}:\mathbb{Z}/3\longrightarrow(\mathbb{Z}/3)^2,\qquad z\longmapsto(0,z),\\ j_{B_d/L_d}=0,\qquad \mathrm{Art}_{B_d/L_d}(xa_d+yb_d)=x\pmod 3. \end{gathered} \tag{E86.5}$$

Arithmetic Artin is used, and $1\in\mathbb{Z}/3$ sends ξ to $\zeta\xi$. Thus the image of norm in (E86.5) is exactly the kernel of this Artin map.

The first ordinary class groups are completely evaluated:

$$\mathrm{Cl}(B_2)\{3\}=C_9\oplus C_3^2,\qquad \mathrm{Cl}(L_2)\{3\}=C_9\oplus C_3. \tag{E86.6}$$

Here and below $C_m=\mathbb{Z}/m$. These are actual Artin coordinates in the computed genus fields. In the coordinates of E.76, the restriction and kernel maps for B_2 are

$$\begin{gathered} \Gamma_B=C_9\oplus C_3^3,\qquad \chi_B(x,y,z,t)=x+2y+z+t\pmod 3,\\ \mathrm{Cl}(B_2)\{3\}\xrightarrow{\sim}\ker\chi_B,\quad (s,u,v)\longmapsto(-2s-3u,u,v,-s-2u-v),\\ (x,y,z,t)\longmapsto(x-3y-3z-3t,y,z). \end{gathered} \tag{E86.7}$$

The two displayed arrows are inverse modulo the indicated cyclic orders. For the lower genus field computed in the certificate, the corresponding maps are

$$\begin{gathered} \Gamma_L=C_9\oplus C_3^2,\qquad \chi_L(x,y,z)=2(x+y+z)\pmod 3,\\ \mathrm{Cl}(L_2)\{3\}\xrightarrow{\sim}\ker\chi_L,\quad (a,b)\longmapsto(-2a,b,-a-b),\\ (x,y,z)\longmapsto(x-3y-3z,y). \end{gathered} \tag{E86.8}$$

In (E86.8), the ordinary classes of the ideals in (E86.4) are exactly $a_2 = (1, 0)$ and $b_2 = (0, 1)$, and $[\mathfrak{q}_{3,2}] = (3, 0) = 3a_2$. Both computed genus fields are the full Hilbert three-class fields of their respective upper fields.

For clarity, the ordinary norm and extension matrices are also explicit. Columns are source coordinates, the upper coordinates are (s, u, v) from (E86.7), the L_2 coordinates are (a_2, b_2), and the B_1 coordinates are the certified ones of E.76:

extension	N	j
B_2/L_2	$\begin{pmatrix} 3 & 3 & 6 \\ 2 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 0 \end{pmatrix}$
B_2/B_1	$\begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}$	$\begin{pmatrix} 0 & 6 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$.

(E86.9)

The subgroup generated by the six S-primes in B_2 is exactly $(\mathrm{Cl}(B_2)\{3\})[3]$, of order 27. In the ordering of E.81 their classes are

$$[Q_1] = [Q_2] = [Q_3] = (6, 1, 0), \qquad [T_1] = (6, 2, 2), \quad [T_2] = (0, 2, 2), \quad [T_3] = (3, 2, 2). \tag{E86.10}$$

Consequently the full principal S-divisor lattice is

$$L_S = \{w \in \mathbb{Z}^6 :; w_1 + w_2 + w_3 \equiv 0 \pmod 3, \quad w_4 \equiv w_5 \equiv w_6 \pmod 3\},$$
$$[\mathbb{Z}^6 : L_S] = 27, \qquad H_S = \begin{pmatrix} 3 & 2 & 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E86.11}$$

This evaluates the exact residual classes of E.84: $\delta = w = 0$, its group $W = 0$, and all twenty-seven literal ideals $J_{a,b,c}$ of E.81 are principal. The complete Smith coordinate maps for (E86.7)–(E86.11) and actual prime-ideal generators are retained.

There are exact evaluated unit-norm defects at every level. Write $\mathscr{U}_d = \mathcal{O}_{B_d}[1/3, 1/17]^\times$ and $\mathscr{V}_d = \mathcal{O}_{L_d}[1/3, 1/17]^\times$. For $r = 1, 2$, $n = 3^r$, and $K_{d,r} = \mathscr{U}_d \cap N_{B_{d+r}/B_d} B_{d+r}^\times$,

$$K_{d,r}/N\mathscr{U}_{d+r} = C_3, \qquad \mathscr{U}_d/N\mathscr{U}_{d+r} \simeq C_n^5 \oplus C_3, \qquad [\mathscr{U}_d : N\mathscr{U}_{d+r}] = 3^{5r+1},$$
$$\mathrm{Ann}_{\mathbb{Z}_3[\mathrm{Gal}(B_{d+r}/B_d)]}[\mathfrak{p}_{d+r}]_3 = (3, \sigma - 1). \tag{E86.12}$$

For the actual adjacent cyclic extension L_{d+1}/L_d, put $K'_d = \mathscr{V}_d \cap NL_{d+1}^\times$. Then

$$[\mathscr{V}_d : K'_d] = 3, \qquad K'_d/N\mathscr{V}_{d+1} = C_3^2, \qquad \mathscr{V}_d/N\mathscr{V}_{d+1} = C_3^3. \tag{E86.13}$$

In (E86.12) the first quotient is identified with the prime-73 class, and in (E86.13) with the actual pair (a_{d+1}, b_{d+1}). The identifying map is explicit: given a field norm $u = Nx$, solve

$$\mathrm{div}_{\notin S}(x) = (\sigma - 1)D; \qquad \omega(u) = [D]_3 \quad \text{in the indicated S-class coordinates.} \tag{E86.14}$$

Its kernel is exactly the S-unit norms. It is independent of both choices, and is onto. Thus the groups in (E86.12)–(E86.13) come with a membership map, not just their orders. The elementary-divisor identifications of the larger quotients allow choices of character lifts.

For the horizontal unramified extension B_d/L_d, the complete S-unit norm condition is

$$N_{B_d/L_d}\mathscr{U}_d = \begin{cases} \{u \in \mathscr{V}_1 : v_3(u) + 2v_{17}(u) \equiv 0 \pmod 3\}, & d = 1, \\ \{u \in \mathscr{V}_d : v_{17}(u) \equiv 0 \pmod 3\}, & d \geq 2. \end{cases} \tag{E86.15}$$

The valuations are normalized at the unique primes of L_d above the indicated rational primes.

All these finite quotients, maps and zero criteria remain exact after completion for multiplication by 3^m. Every nonempty completed unit-norm fiber is a translate of the actual norm kernel. Its abstract compact group and rank are

norm on S-units	kernel
$B_{d+r}/B_d, r = 1, 2$	$C_{3^{\min(r,2)}} \oplus \mathbb{Z}_3^{3^{d+1}(3^r-1)}$
L_{d+1}/L_d	$C_3 \oplus \mathbb{Z}_3^{2\cdot 3^d}$
B_d/L_d	$C_3 \oplus \mathbb{Z}_3^{2\cdot 3^d+4}$.

(E86.16)

The finite torsion is the literal kernel on roots of unity; the free coordinates are choices of an integral basis of the displayed norm kernel. Norms and extensions on the S-class groups at arbitrary finite distances are the compositions of (E86.3), with these same specified prime bases. The assertions about cyclic unit defects in (E86.12) use exactly $r = 1, 2$.

This evaluates all previously retained S-class annihilators and capitulation maps in E.79. The actual values of the remaining ordinary prime classes $[\mathfrak{q}_{3,d}]$ for $d \geq 3$, and their eventual exponent h in E.69, are not computed by (E86.3). Their comparison is the actual norm/extension square E76.11 and the exact S-class quotient $\mathrm{Cl}(L_d)\{3\} \twoheadrightarrow C_d^S$, whose kernel is generated by $[\mathfrak{q}_{3,d}]$. The manuscript anchor remains LM §51.16, fols. 733v–736v, scans 172–178, through the actual positional norm maps of E.46, E.69 and E.72.

Proof. The all-level automorphisms and their class maps. We first recall the exact field comparison behind the horizontal arrows. The element 17 has a cube root in $\mathbb{Q}_3$, since $-17 \in 1 + 9\mathbb{Z}_3$ and the cube map $1 + 3\mathbb{Z}_3 \to 1 + 9\mathbb{Z}_3$ is bijective. Hence $\xi = \beta_1/a$ already belongs to the completion of L_1 at three, and therefore to every later such completion. At seventeen the completion of K has residue field $\mathbb{F}_{289}$, which contains μ_9; its Teichmuller lifts give the corresponding local inclusion of ξ. Thus the base change of $K(\xi)/K$ to L_d is split at these two primes. It is unramified elsewhere because $K(\xi)/K$ is unramified away from three. It has degree three: the unique prime over seventeen is totally ramified in L_d/K, while it splits into three primes in $K(\xi)/K$, excluding a global inclusion. This proves the actual unramified, S-split cubic comparison at every level, with the embeddings fixed by (E86.1).

The adjacent B extensions are cyclic cubic, and the two-step extensions cyclic ninth-degree, because $\mu_9 \subset B_d$. They are totally ramified at their six S-primes. At seventeen use $Y^{3^r} - \beta_d$; at three use $(Y-1)^{3^r} - \beta_d$ and the uniformizer $\beta_d + 1$ established in E.76. Their discriminants have no other prime support. The adjacent L extension is cyclic cubic, totally ramified at its two S-primes. These observations justify every cyclic and ramification hypothesis used below.

On B_{d+1} define the literal substitutions

	ξ	β_{d+1}
g_d	ξ	$\xi\beta_{d+1}$
h_d	$\zeta\xi$	β_{d+1}

$$g_d^9 = h_d^3 = 1, \quad g_d^3 = \sigma_d, \quad h_d g_d h_d^{-1} = g_d^4. \tag{E86.17}$$

Here σ_d is the specified generator of $\mathrm{Gal}(B_{d+1}/B_d)$. Both substitutions preserve $\xi^6 + \xi^3 + 1 = 0$ and $\beta_{d+1}^{3^{d+1}} = 17\xi^3$; their inverse substitutions are powers of themselves. Thus they are actual automorphisms. They preserve B_d: $g_d(\beta_d) = \zeta\beta_d$ and $h_d(\beta_d) = \beta_d$. Both commute with σ_d. Their restrictions satisfy $h_d|_{B_d} = h_{d-1}$ and, for $d \geq 2$, $g_d|_{B_d} = g_{d-1}^3 = \sigma_{d-1}$. At $d = 1$ the

restriction is exactly the base automorphism g_F in E84.13–15. These are the restriction maps, without an assertion that the sequence g_d itself is compatible.

E.79 proves that actual ideal norm induces $(A_{d+1})_{\sigma_d} \simeq A_d$, and proves $A_1 = C_3$. Suppose the latter group at level d has the specified generator. The coherent prime $\mathfrak{p}_{d+1}$ maps to it, so its class generates A_{d+1} as a module over $R = \mathbb{Z}_3[C_3]$. Here is the finite Nakayama argument: quotient by the generated module; the quotient Q satisfies $Q = (3, \sigma_d - 1)Q$. Modulo three the augmentation ideal is nilpotent, so iteration gives $Q = 3Q$, and finiteness gives $Q = 0$.

Write this cyclic module as R/I. Since R is commutative, every endomorphism of R/I is multiplication by the image of one. Its endomorphism ring is therefore commutative. The automorphisms g_d, h_d are R-linear, since they commute with σ_d and preserve S. Their induced endomorphisms commute. Their actual commutator in (E86.17) is σ_d, which consequently acts identically. Thus $A_{d+1} = (A_{d+1})_{\sigma_d} = C_3$, and the actual norm is an isomorphism. Extension and norm satisfy $jN = 1 + \sigma_d + \sigma_d^2 = 3$ on this group; norm surjectivity therefore gives $j = 0$. This proves the B part of (E86.3) by induction. The same group of order three has trivial action under every three-power order automorphism. For the actual ninth-degree cyclic steps this gives the stated annihilator $(3, \sigma - 1)$; norm and extension at longer distances are their literal compositions.

The residue equalities $22^3 = 63$ and $63^3 = 22$ in $\mathbb{F}_{73}$ prove the compatibility of the prime-73 chain at every level. These roots have order eight, so the indicated residue maps are defined at all levels. The defining polynomials are etale at 73, and their degree-one residue maps extend to the maximal orders. Norm of each prime is its predecessor. This also proves that the induction uses the specified prime class, not an unspecified cyclic generator.

The entire ordinary class group of B_2. Put $C = \mathrm{Cl}(B_2)\{3\}$ and let B be its subgroup generated by the six S-primes. Then $C/B = A_2 = C_3$. The action of $g = g_1$ on this quotient is trivial, because its order divides nine. Hence $(g-1)C \subset B$. E.84 proves the literal permutations $g : Q_1 \mapsto Q_3 \mapsto Q_2 \mapsto Q_1$, $g(T_i) = T_i$, each S-class has order three, and $[Q_1Q_2Q_3] = 0$. Therefore $1 + g + g^2$ is zero on B. It follows that

$$(\sigma - 1)C = (g^3 - 1)C = (1 + g + g^2)(g - 1)C = 0. \tag{E86.18}$$

The computed ambiguous order in E.76 is 81, so $|C| = 81$ and $|B| = 27$.

The genus map of E.81 has image of order 27 on B, hence is injective there. Its zero column difference gives $\delta = [Q_2] - [Q_1] = 0$. The already proved S-capitulation means $[I] \in B$, for $I = P_{73}\mathcal{O}_{B_2}$. Thus $w = [I] + [T_2] - [T_1] \in B$, and its zero genus value gives $w = 0$. E84.8 now proves principality of all twenty-seven literal ideals.

To identify the full group and its actual map, let $\mathcal{H}$ be the Hilbert three-class field of B_2. The extension $\mathcal{H}/B_1$ is Galois; its commutator subgroup is $(\sigma - 1)C$, because its quotient by C is cyclic. By (E86.18) it is abelian. Hence $\mathcal{H}$ equals the maximal abelian-over-B_1 unramified-over-B_2 field computed in E.76. Its group is $\Gamma_B = C_9 \oplus C_3^3$. The actual restriction character is computed on prime Frobenii by

$$\overline{\beta_1}^{(N\mathfrak{l}-1)/3} = \overline{\zeta}^{\chi_B(\mathfrak{l})}. \tag{E86.19}$$

The four basis primes in the certificate have norms $2^6, 7^3, 19^3, 73$, respectively. Their genus-coordinate matrix and character values are

$$V_B = \begin{pmatrix} 3 & 3 & 6 & 5 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 2 \\ 1 & 0 & 2 & 2 \end{pmatrix}, \qquad (1, 1, 1, 0). \tag{E86.20}$$

The matrix is invertible modulo three, so these primes generate Γ_B. Formula (E86.19) therefore determines the character uniquely as $(1, 2, 1, 1)$; the program checks another 188 actual prime

Frobenii. Its full lattice kernel, with integer Smith transformations and inverses, is precisely (E86.7). Substituting the six Artin columns of E.81 gives (E86.10), and gives $[I] = (6, 0, 0)$. These classes generate the full three-torsion subgroup of C, proving (E86.11) in both directions. The full genus map is now injective on C, so the earlier group W is zero.

The horizontal S-class groups and all lower bases. For any d, let $\mathcal{H}_B^S$ be the S-split Hilbert three-class field of B_d. It is Galois over L_d. Its kernel over B_d is $A_d = C_3$, and conjugation has order dividing three, so acts trivially on this kernel. An extension of a cyclic group by a central cyclic group is abelian. Thus $\mathcal{H}_B^S/L_d$ is abelian of degree nine. It is everywhere unramified and S-split.

The two field inclusions are useful here. The extension B_d/L_d is itself unramified, abelian and S-split, so $B_d \subset \mathcal{H}_L^S$. The extension $\mathcal{H}_L^S/B_d$ then has the same properties, giving $\mathcal{H}_L^S \subset \mathcal{H}_B^S$. The abelian property just proved gives the reverse inclusion. Hence

$$\mathcal{H}_L^S = \mathcal{H}_B^S, \qquad |C_d^S| = 9. \tag{E86.21}$$

Restriction identifies actual norm $A_d \to C_d^S$ with the inclusion of the subgroup over B_d: for a prime of residue degree f it sends its Frobenius to the fth power of the lower Frobenius, exactly its ideal norm. Extension in this common abelian class field is cubing. Indeed a split lower prime extends to three primes with the same Frobenius; an inert prime has a single upper Frobenius equal to the cube. Multiplication proves the assertion on all ideal classes.

We determine its exponent without assuming an upper unit basis. For an unramified S-split cyclic cubic E/F, the invariant divisors outside S are exactly extended divisors. Hilbert 90 therefore gives the actual isomorphism

$$\begin{aligned} \ker(j : \mathrm{Cl}_S(F) \to \mathrm{Cl}_S(E)) &\simeq H^1(\mathrm{Gal}(E/F), \mathcal{O}_{E,S}^\times), \\ [D] &\longmapsto (g \mapsto g(x)/x), \quad \mathrm{div}_{\notin S}(x) = jD. \end{aligned} \tag{E86.22}$$

Changing x by an S-unit changes a coboundary; Hilbert 90 gives the inverse, and invariant divisors give precisely the displayed kernel. The S-unit Herbrand quotient in this extension is $1/3$. All archimedean and S-place modules are regular permutation modules, and the product formula removes one trivial module. The lattice proof of Milne VII Proposition 3.1 thus gives

$$|\ker j| = 3[\mathcal{O}_{F,S}^\times : N\mathcal{O}_{E,S}^\times]. \tag{E86.23}$$

Finite-index changes of logarithmic lattices and finite torsion have Herbrand quotient one [Mil20b].

For $d \geq 2$, norm from B_d to B_2 preserves each of its six S-valuation coefficients: there is one prime above each and residue degree one. Consequently every principal S-divisor in B_d satisfies the congruences (E86.11), after applying norm. In particular the total exponent of its three primes above seventeen is divisible by three. In B_d/L_d this total is the normalized valuation of the norm at $\mathfrak{q}_{17,d}$. But β_d is an S-unit of L_d with that valuation one. It is therefore excluded from the S-unit norm image. Thus its index is at least three. By (E86.21)–(E86.23) the index is at most three. It is exactly three, the whole group C_d^S capitulates, and the necessary valuation congruence is sufficient. Since extension was cubing in (E86.21), C_d^S has exponent three. This proves $C_d^S = C_3^2$ and (E86.15) for $d \geq 2$.

At $d = 1$, the certified class group of L_1 is C_3^2 and its primes at three and seventeen are principal (E.57). Thus $C_1^S = C_3^2$; cubing again gives zero extension, and (E86.23) gives unit-norm index three. The certified principal S-divisor lattice in B_1 is E79.17, whose necessary norm congruence is exactly $v_3 + 2v_{17} = 0$ modulo three. It has index three, proving (E86.15) at this remaining level as well.

Norm on S-classes in the adjacent L extension is onto. One direct class-field proof is to base change its lower S-split Hilbert field; its intersection with the cyclic ramified extension is the

lower field. The resulting unramified S-split extension upstairs has restriction map on prime Frobenii equal to ideal norm. Both S-class groups have order nine, so norm is an isomorphism. It follows that the cyclic action on the upper group is trivial, and $jN = 3$ gives zero extension.

At seven the residue map $\beta_d = -1$ is defined at every level, with $\zeta = 2$. It is etale and degree one, and cubing sends it to its predecessor. At seventy-three the same assertion was already proved. The lower calculation below gives the indicated basis at $d = 2$; norm isomorphisms propagate it upward and downward. At L_1, the certificate independently gives the columns $(2, 0)$ and $(2, 1)$ in its certified class basis, an invertible matrix modulo three. For the extension adjoining ξ, the Frobenius at P_d sends ξ to $\xi^7 = \zeta^2\xi$, while at J_d it sends ξ to $\xi^{73} = \xi$. Thus its Artin coordinates on a_d, b_d are exactly $1, 0$. The prime $\mathfrak{p}_d$ has norm J_d, proving every map in (E86.3)–(E86.5).

The ordinary lower group and the actual matrix comparison. The kernel of the ordinary-to-S class quotient for L_2 is generated by $\mathfrak{q}_{3,2}$, since $\mathfrak{q}_{17,2} = (\beta_2)$. E.72 proves that the former has exact order three. Therefore $|\operatorname{Cl}(L_2)\{3\}| = 27$. For its lower genus field the retained program uses only the certified degree-six base L_1, with polynomial $X^6 + 17X^3 + 289$. It computes the local norm units for the actual cubic $Y^3 - \beta_1$, using the forty literal norms $N(1 + (Y + 1)^k)$. The seven-coordinate local unit quotient modulo cubes has a norm subspace of rank six; local reciprocity proves that this is its full norm kernel. At seventeen the kernel is the residue cubes. Quotienting the ray class group by these local norm unit groups gives exactly the maximal abelian-over-L_1 field unramified over L_2, by the idelic map and converse proved in E76.17. Its order over L_1 is 81 and its Smith group is $C_9 \oplus C_3^2$. Its degree 27 over L_2 now equals the three-primary class number, so it is the Hilbert three-class field. This also proves that the cyclic action of $\operatorname{Gal}(L_2/L_1)$ on the ordinary class group is trivial.

The three prime Frobenii with genus-coordinate columns

$$V_L = \begin{pmatrix} 0 & 8 & 2 \\ 0 & 0 & 1 \\ 2 & 0 & 0 \end{pmatrix}, \qquad (1, 1, 0) \tag{E86.24}$$

have norms $2^6, 7, 7$. Their matrix is invertible modulo three. The literal cubic-residue formula (E86.19), now in L_1, gives the character $(2, 2, 2)$. The program checks all 82 sampled prime Frobenii and retains the full integer Smith maps, yielding (E86.8).

The degree-one primes $(7, \beta_2 - 3)$ and $(73, \beta_2 - 63)$ have ordinary coordinates $(8, 1)$ and $(0, 1)$, respectively. The first differs in its residue label from the coherent $P_2 = (7, \beta_2 + 1)$. There is an explicit comparison: with $\sigma(\beta_2) = \zeta\beta_2$ and $\zeta = 2$ modulo this prime, application of σ^2 changes the root 3 to $3\zeta^{-2} = 6 = -1$. Thus

$$\sigma^2(7, \beta_2 - 3) = (7, \beta_2 + 1). \tag{E86.25}$$

Their classes agree by the just proved trivial cyclic action, equivalently by their identical lower genus Artin value $(2, 1, 0)$. Taking their actual ideal ratio with J_2 gives $(1, 0)$, as asserted. The wild prime has lower genus value $(3, 0, 0)$, giving $[\mathfrak{q}_{3,2}] = 3a_2$.

We give enough arithmetic data to reproduce all four ordinary maps in (E86.9). Three degree-one primes of B_2 above 73 are specified by (t, β_2) equal to $(33, 17), (31, 17), (29, 33)$, with $t = a - \xi$ in B_1. Their upper class coordinates, and their norms to L_2 and B_1, are the respective columns

$$M = \begin{pmatrix} 2 & 5 & 8 \\ 0 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}, \quad N_L M = \begin{pmatrix} 0 & 0 & 3 \\ 1 & 1 & 1 \end{pmatrix}, \quad N_B M = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \tag{E86.26}$$

Here M is invertible modulo nine, with inverse $\left(\begin{smallmatrix} 6 & 8 & 8 \\ 1 & 3 & 8 \\ 7 & 4 & 2 \end{smallmatrix}\right)$. Hence these are actual ideal generators and determine both norm matrices. The lower coordinates of their contractions are obtained by the residue maps $\beta_1 = 22, 22, 21$, respectively, followed by (E86.8); all ideal HNFs are retained.

The actual h and g actions on the upper class group, obtained by their action on the retained ray ideals, are

$$h = \begin{pmatrix} 1 & 0 & 6 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \qquad g = \begin{pmatrix} 4 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{on } C_9 \oplus C_3^2. \tag{E86.27}$$

They satisfy $1 + h + h^2 = 0$. Thus extension of $b_2 = N[\mathfrak{p}_2]_3$ is zero. The prime P_2 at seven is inert in B_2/L_2, since μ_9 is not in $\mathbb{F}_7$. Its unique upper prime contracts in B_1 to the degree-three prime with $\beta_1 = 6$; its upper genus value is $(3, 2, 0, 2)$, hence its class is $(0, 2, 0)$. Therefore extension of $a_2 = [J_2/P_2]$ is $(0, 1, 0)$. This proves the first extension matrix. The second is E84.2 transported through (E86.7), using $[I] = (6, 0, 0)$. The certificate verifies both identities $Nj = 3$ and $jN = \sum \sigma$ with their actual source and target moduli.

The cyclic divisor sequences and completed fibers. For each cyclic step used in (E86.12) or (E86.13), every ramified prime belongs to S. On the divisor group outside S, norm is residue degree times the sum on each orbit. Consequently

$$\ker N = (\sigma - 1)\operatorname{Div}_{\notin S}(E), \qquad \operatorname{Div}_{\notin S}(E)^{\sigma} = j \operatorname{Div}_{\notin S}(F). \tag{E86.28}$$

For $u = Nx$ an S-unit, these identities produce the divisor D in (E86.14). Changing D changes its class by an extended base class, now zero by (E86.3). Changing x by a norm-one element changes D by a principal divisor, by Hilbert 90. This proves well-definedness. Every invariant S-class has a representative D for which $(\sigma - 1)D$ is principal outside S; its generator has S-unit norm. Thus the map is onto. The quotient of invariant full S-classes by extended classes is killed by the cyclic degree: the ideal norm identity multiplies any invariant class by that degree. Hence its prime-to-three part is zero, justifying the three-primary projection. Finally a zero class in this quotient means $D = \operatorname{div}_{\notin S}(z) + jD_0$; dividing x by $\sigma(z)/z$ gives an S-unit with the same norm. The reverse inclusion uses $D = 0$. This proves the exact kernel and image of (E86.14). It is the exact Hilbert 90 map E79.22, with both class groups and the extension map now evaluated.

For the B steps E.79 proves the full five-coordinate local Hilbert-symbol image C_n^5 and the S-unit Herbrand quotient n^5. The kernel of class extension is now C_3, so (E86.22) and that Herbrand quotient give $[\mathscr{U}_d : N\mathscr{U}_{d+r}] = 3n^5$. Together with (E86.14) this proves the exact sequence $0 \to C_3 \to \mathscr{U}_d/N\mathscr{U}_{d+r} \to C_n^5 \to 0$. The middle group is killed by n, since inclusion of a base S-unit has norm its nth power. Choose preimages of the five standard generators; their orders divide n, so they split this sequence as modules over $\mathbb{Z}/n$. This proves (E86.12) with its stated choice of elementary-divisor coordinates.

For the L step the S-unit Herbrand quotient is 3: its two totally ramified S-places contribute two trivial modules, and the product formula removes one. Class extension has kernel C_3^2, so the unit-norm index is 27. Formula (E86.14) has kernel quotient C_3^2, hence the field-norm index is three. Equivalently, the two actual local Hilbert symbols have the full product-one image C_3. The cyclic Hasse norm theorem identifies their simultaneous kernel with K_d' [Mil20b, VIII, Theorem 3.1]. The full S-unit quotient is killed by three, so is C_3^3, proving (E86.13).

There are literal generator procedures for the defect coordinates. For the B step first represent $[\mathfrak{p}_{d+r}]_3$ by an actual ideal. Write the exponent of the full S-class group as $3^s m$ with $(3, m) = 1$, choose $e \equiv 1 \pmod{3^s}$ and $e \equiv 0 \pmod{m}$, and take $D = \mathfrak{p}_{d+r}^e$. This ideal represents the desired primary projection. Choose a generator outside S for $(\sigma - 1)D$; its field norm represents 1 under ω. For the L step do the same with the actual ideals J_{d+1}/P_{d+1} and

J_{d+1}, applying the same primary projection. Different choices change only the S-unit-norm coset, by the proved kernel statement. These procedures and (E86.14) specify all quotient maps with no ambiguity from the prime-to-three class components. Alternatively the ideal representative can be chosen directly by its specified three-primary S-class.

Every S-unit group here is finitely generated. Norm image has finite index, since it contains the appropriate powers of the base group. Completion for multiplication by 3^m therefore preserves all these kernels, images and finite quotients. The ranks of the completed S-unit groups are $3^{d+1}+5$ for B_d and 3^d+1 for L_d. Surjectivity after tensoring with $\mathbb{Q}_3$ gives the free-rank differences in (E86.16).

The three-power roots of unity of B_d are exactly μ_9: at seventeen any root of unity of order prime to seventeen injects into $\mathbb{F}_{289}^{\times}$, whose three-part has order nine. For L_d they are exactly μ_3, since adjoining ξ has degree three. In the vertical B steps the root-of-unity norm is raising to 3^r; in the vertical L step it is cubing. Horizontally $N_{B_d/L_d}\xi = \xi^{1+4+7} = \zeta$. These give exactly the torsion subgroups shown in (E86.16). A finitely generated module over $\mathbb{Z}_3$ is its finite torsion plus a free module, so choosing an integral basis of its torsion-free quotient and lifts gives the asserted compact coordinates for every actual kernel. Translation by one norm preimage gives, with its inverse, every nonempty fiber. □

All abelian preimages, their horizontal obstruction, and the filled elliptic target

E.87. The rank-seven relative lattice and an evaluated free wild obstruction [Proved]

Retain E.85's actual source marking, its section (u_0, d), and the intrinsic abelian class α_0. The actual map from $V = N^{\mathrm{ab}}$ to the joint abelian lattice $\mathcal{J}_1$, in its coordinates (L, p, m), is

$$R = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 \\ 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -3 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}. \tag{E87.1}$$

Its actual local cohomology map is onto, with the following complete kernel and fibre type:

$$\begin{gathered} H^1(\mathbb{Q}_3, V) = \mathbb{Z}_3^{15} \oplus C_3^3 \twoheadrightarrow H^1(\mathbb{Q}_3, \mathcal{J}_1) = \mathbb{Z}_3^5 \oplus C_3, \\ \ker H^1(R) \simeq \mathbb{Z}_3^{10} \oplus C_3^3. \end{gathered} \tag{E87.2}$$

All three source torsion coordinates map to zero; the target torsion coordinate is reached from a free coordinate. In particular this exact sequence has no $\mathbb{Z}_3$-linear section. The following actual inverse coordinates include its finite carry.

Use the retained Smith matrix L_1 for $(I - T_1)(I + J_1)$, with diagonal $(1, 3, 3, 3, 0^6)$. Write a horizontal class as

$$L_1 U = (0, \epsilon_1, \epsilon_2, \epsilon_3, q_1, \ldots, q_6), \qquad \epsilon_i \in C_3. \tag{E87.3}$$

Its target horizontal coordinates are

$$L_u = -q_2 + q_3 - q_4 - q_5, \quad m_u = q_3 - q_5 - q_6, \quad \epsilon = 2q_3 - q_5 \pmod 3. \tag{E87.4}$$

For a prescribed (L_u, m_u, ϵ), take $\epsilon \in \{0, 1, 2\}$, retain $q_1, q_3, q_4, k \in \mathbb{Z}_3$ and all three source torsion coordinates, and set

$$q_5 = 2q_3 - \epsilon - 3k, \qquad q_6 = -q_3 + \epsilon + 3k - m_u, \qquad q_2 = -q_3 - q_4 + \epsilon + 3k - L_u. \tag{E87.5}$$

The inverse recovers $k = (2q_3 - q_5 - \epsilon)/3$. For the relative coordinate use E.75's basis

$$A = \theta(3f_1 + f_2) + a_0 A_0 + a_1 A_1 + a_2 A_2 + b_0 B_0 + b_1 B_1 + b_2 B_2 + c_1 C_1 + c_2 C_2. \tag{E87.6}$$

Its image is

$$(L_a, p_a, m_a) = (2\theta - b_0 - b_1 - b_2, 5\theta - 3c_2, 2\theta - c_1 - c_2). \tag{E87.7}$$

Retain freely $a_0, a_1, a_2, b_1, b_2, c_2$ and solve

$$\theta = (p_a + 3c_2)/5, \quad c_1 = 2\theta - c_2 - m_a, \quad b_0 = 2\theta - b_1 - b_2 - L_a. \tag{E87.8}$$

Equations (E87.3)–(E87.8), with the literal inverse Smith matrix, are mutually inverse coordinates on every fibre in (E87.2).

There is also an exact coefficient-kernel description. Put $K_1 = \ker R$ and use its following seven basis vectors:

$$\begin{aligned} &k_1 = f_1 + 2f_2 - B_0 - C_1, \quad k_2 = -3f_2 + 3B_0 + 2C_1 + C_2, \\ &k_3 = A_0, \qquad\qquad k_4 = A_1, \qquad\qquad k_5 = A_2, \\ &k_6 = B_1 - B_0, \qquad\qquad k_7 = B_2 - B_0. \end{aligned} \tag{E87.9}$$

If K is this ten-by-seven matrix, its left inverse selects rows $f_1, C_2, A_0, A_1, A_2, B_1, B_2$. The exact operators are $T_K = K^\ell T_1 K$ and $J_K = K^\ell J_1 K$, saved as integral matrices, and

$$\begin{gathered} H^1(\mathbb{Q}_3, K_1) \xrightarrow{\sim} \ker H^1(R), \\ \mathcal{P}_K = K_1/(I - T_K)K_1^{J_K} \simeq \mathbb{Z}_3^4 \oplus C_3^3, \\ \mathcal{K}_K = \ker((I - J_K)(T_K - 7I)) = \mathbb{Z}_3(9k_1 + 5k_2) \oplus \bigoplus_{i=3}^{7} \mathbb{Z}_3 k_i, \\ H^1(\mathbb{Q}_3, K_1) = \mathcal{P}_K \oplus \mathcal{K}_K. \end{gathered} \tag{E87.10}$$

These are the coefficient and cohomology maps of the actual exact sequence $0 \to K_1 \to V \to \mathcal{J}_1 \to 0$.

A complete larger intrinsic locus. Consider all intrinsic abelian classes whose relative value is still $\bar{d} = e_{B_0}$ and whose joint abelian image is β_0. They are exactly $\alpha_0 + \mathcal{P}_K$. For $v = \sum_{i=1}^{7} x_i k_i$, their full class-two obstruction is the evaluated homomorphism

$$\begin{gathered} \delta_{B_0} : \mathcal{P}_K \longrightarrow \mathcal{B}_2 = \mathbb{Z}_3 \oplus C_3^{11} \oplus C_9^4, \qquad [v] \longmapsto [7[v, e_{B_0}]], \\ \operatorname{im} \delta_{B_0} \simeq C_3^3, \qquad \ker \delta_{B_0} \simeq \mathbb{Z}_3^4. \end{gathered} \tag{E87.11}$$

There are exactly three scalar conditions, all modulo three:

$$x_1 - 2x_6 - x_7 = 0, \qquad x_2 = 0, \qquad x_4 - x_5 - x_6 - 2x_7 = 0. \tag{E87.12}$$

Their full integral solution lattice is the column lattice of

$$Q_K = \begin{pmatrix} 3 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \det Q_K = 27. \tag{E87.13}$$

The exact Smith matrix of $Q_K^{-1}(I - T_K)(I + J_K)$ has three unit factors and four zero factors. It supplies both inverse coordinates on the free four-dimensional zero locus.

All these abelian classes have actual intrinsic class-two lifts. Their entire class-two section locus maps to the same index-nine part

$$\mathbb{Z}_3 \oplus 3\mathbb{Z}_3 \oplus 0 \subset \mathfrak{G}_{\beta_0} \tag{E87.14}$$

computed in E.85. Every point of this image has, within this whole inertia-fixed locus, a complete geometric-conjugacy fibre

$$\mathbb{Z}_3^{48} \oplus C_3^9 \oplus C_9^5. \tag{E87.15}$$

The four additional coordinates are the actual kernel in (E87.11); the other forty-four and the fourteen cyclic coordinates are E.85's full central fibre coordinates. The proof specifies the particular central corrections, the joint normalization and the remaining actual geometric conjugators.

An evaluated quotient for all intrinsic abelian inputs. Fill the eight cusps other than $O = C_0$ and put

$$F_E = \pi_1^{(3)}(E \setminus \{O\}), \qquad C_E = F_E/\gamma_3 F_E. \tag{E87.16}$$

There is the actual equivariant quotient

$$N_2 \twoheadrightarrow C_E, \qquad \ker = \frac{\langle\langle d_i : i \neq O \rangle\rangle \gamma_3 N}{\gamma_3 N}. \tag{E87.17}$$

Its abelian map sends f_1, f_2 to the two elliptic handle generators and sends the eight other basis vectors to zero. Its central map is projection to the coefficient of $[f_2, f_1]$, precisely E.74's free $\mathcal{B}_2$ coordinate.

The actual arithmetic action on C_E is completely evaluated after the following explicit central coordinate morphism. Choose handle generators g_1, g_2 with the stated abelian images, and put $X_i = \log g_i$, $Z = [X_2, X_1]$. The rational involution $\iota[x : y : z] = [y : x : z]$, with its coherent tangential comparison below, has

$$\iota X_i = -X_i + b_i Z, \qquad \iota Z = Z. \tag{E87.18}$$

The coefficients b_i here are the actual central coordinates of $\iota(g_i)g_i$. Retain them in the literal invertible change of basis

$$X_i^\sharp = X_i - \tfrac{1}{2} b_i Z, \qquad g_i^\sharp = g_i [g_2, g_1]^{-b_i/2}. \tag{E87.19}$$

For old group Hall coordinates $g_1^x g_2^y [g_2, g_1]^n$, the new Lie central coordinate and its inverse are

$$r^\sharp = n - xy/2 + (b_1 x + b_2 y)/2, \qquad n = r^\sharp + xy/2 - (b_1 x + b_2 y)/2. \tag{E87.20}$$

Thus the comparison preserves the E.37 abelian marking and retains every central correction.

In coordinates $(X_1^\sharp, X_2^\sharp, Z)$ the genuine arithmetic operators are exactly

$$T = (A_E, 7), \quad J = (C_E^{\mathrm{mat}}, -1), \qquad A_E = \begin{pmatrix} 1 & -3 \\ 3 & -2 \end{pmatrix}, \quad C_E^{\mathrm{mat}} = \begin{pmatrix} 1 & -1 \\ 0 & -1 \end{pmatrix}. \tag{E87.21}$$

The first-degree section coordinates of this quotient are $(U_1, t, \epsilon) \in \mathbb{Z}_3^2 \times C_3$, where its relative value is $(3t, t)$ and its horizontal value is (U_1, U_2) with $\epsilon = U_2 \bmod 3$. Its complete class-two lifting image is

$$\begin{gathered} -7t(U_1 + 21t/2) = 0, \\ t = 0 \quad \text{or} \quad U_1 = -21t/2. \end{gathered} \tag{E87.22}$$

Each point of this exact closed locus has fibre $\mathbb{Z}_3^2$. Its inverse takes arbitrary central Lie coordinates (r_u, r_a) and uses

$$u = (U_1, \epsilon, r_u), \qquad a = (3t, t, r_a) \tag{E87.23}$$

as E.77's two values in C_E. For an unnormalized representative the actual fixed conjugator is $\exp(bX_1^\sharp)$, where $b=(U_2-\epsilon)/3$. Its full central corrections are

$$r_u \longmapsto r_u+\frac{3b^2+3bU_1-2bU_2}{2}, \qquad r_a \longmapsto r_a-bt. \tag{E87.24}$$

There is no remaining lower stabilizer. These coordinates therefore include the complete geometric-conjugacy fibres.

For any intrinsic abelian class, projection through (E87.17) makes (E87.22) its exact free class-two obstruction. Consequently every intrinsic class-two section lies over these two evaluated elliptic branches. The fifteen finite coordinates of $\mathcal{B}_2$ for general relative inputs remain separate from this evaluated free coordinate; on the horizontal locus they have already been evaluated by (E87.11)–(E87.13).

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). All intrinsic, joint and filled maps in this result are the actual displayed quotient or coefficient maps.

Proof. The complete first-degree comparison. The rows in (E87.1) are obtained from the four actual unit characters in E.83 as X_h-Y_h, X_h and $(X_h+X_{\bar h})/3$. They are integral, and their columns f_2,B_0,C_1 have determinant a unit. Hence R is onto. Direct multiplication gives

$$RT_1=7R, \qquad RJ_1=J_{\mathcal{J}_1}R. \tag{E87.25}$$

The actual E.77 source, reduced to any abelian lattice M, has coordinates

$$\frac{M}{(I-T)M^J}\oplus\ker((I-J)(T-7):M\to M), \tag{E87.26}$$

with the source values (U,A) and the complete fixed geometric gauge. This follows either directly from its relator or from E.75's exact cochain complex, using the specified source marking throughout.

The first-degree Smith matrix has the stated diagonal and the following literal coefficient product:

$$RL_1^{-1}=\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -1 & 1 & -1 & -1 & 0\\ -6 & -6 & 6 & 6 & 0 & 0 & 2 & 0 & -1 & -3\\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & -1 & -1 \end{pmatrix}. \tag{E87.27}$$

The target fixed lattice is generated by its p coordinate, and its gauge is multiplication by six on that coordinate. Thus (E87.27) gives exactly (E87.4). It proves that all source torsion maps to zero. Equations (E87.5) give all horizontal preimages and their inverse, including the residue carry. The actual relative kernel on V is generated by $3f_1+f_2$ and the eight cusp vectors. Multiplication by R gives (E87.7), and (E87.8) proves the full relative surjectivity with its inverse. This proves (E87.2) and all the claimed fibre coordinates. A linear section would send the target element of order three to a source element of order three, contradicting the computed zero map on source torsion. This proves the stated non-splitting assertion.

Solving the three rows of R gives exactly the seven vectors (E87.9) and the displayed left inverse. They provide a saturated coefficient kernel. In these coordinates its operators are

$$T_K=\begin{pmatrix} -5 & 9 & 0 & 0 & 0 & 0 & 0\\ -3 & 4 & 0 & 0 & 0 & 0 & 0\\ 5 & -9 & 7 & 0 & 0 & 0 & 0\\ 2 & -12 & 0 & 7 & 0 & 0 & 0\\ -4 & 3 & 0 & 0 & 7 & 0 & 0\\ 1 & -6 & 0 & 0 & 0 & 7 & 0\\ -8 & 6 & 0 & 0 & 0 & 0 & 7 \end{pmatrix}, \qquad J_K=\begin{pmatrix} -1 & 3 & 0 & 0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & -1 & 0 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & -1 & 0 & 0\\ -1 & 3 & 0 & -1 & 0 & 0 & 0\\ 1 & 0 & 0 & 0 & 0 & 0 & -1\\ -1 & 3 & 0 & 0 & 0 & -1 & 0 \end{pmatrix}. \tag{E87.28}$$

The horizontal Smith factors are $3,3,3,0^4$. The relative equation is the single saturated condition $5x_1 = 9x_2$, giving the relative basis in (E87.10). Its inverse divides x_2 by the unit five. Since $H^0(\mathbb{Q}_3, \mathcal{J}_1) = 0$ and $H^1(R)$ is onto, the exact cohomology sequence identifies $H^1(K_1)$ with the full kernel. This also proves that the displayed coefficient description includes every abelian preimage.

The complete horizontal obstruction. Fix the actual section (u_0, d) of E.85 and choose a continuous geometric Hall lift g_v of $v \in K_1$ into N_2. In class two the commutator depends only on abelian coordinates. The full cusp norm identity therefore gives

$$(u_0 g_v)T(d)(u_0 g_v)^{-1} d^{-7} = 7[v, e_{B_0}] \quad \text{in } A_2. \tag{E87.29}$$

Replacing the relative value by dw_a adds $(T_2 - 7I)a$. Hence the exact source condition is

$$(I - J_2)(7[v, e_{B_0}] + (T_2 - 7I)a) = 0. \tag{E87.30}$$

E.77's factor of one half identifies its obstruction with the class in (E87.11). For a horizontal gauge vector $b \in K_1^{J_K}$, the bracket identity

$$7[(I - T_1)b, e_{B_0}] = -(T_2 - 7I)[b, e_{B_0}] \tag{E87.31}$$

proves that this map descends through the full abelian gauge. Its free coefficient is zero, because every bracket in (E87.29) contains a cusp vector.

The certificate computes its fifteen retained E.74 coordinates and their moduli. All its values have order dividing three, including the components landing in $3C_9$. Dividing those four coordinate subgroups by three, the three selected target rows numbered $4, 8, 15$ have an invertible three-by-three matrix over $\mathbb{F}_3$. The entire matrix factors through the three characters in (E87.12), with that inverse. This proves its exact image C_3^3 and exact kernel lattice (E87.13). The actual transformed gauge matrix has three unit factors, proving that the cohomological zero locus is free of rank four. Both inverse Smith maps are retained, rather than just these group orders. Explicitly, if $L_Q(Q_K^{-1}(I - T_K)(I + J_K))R_Q$ has diagonal $(1,1,1,0,0,0,0)$, the chosen representative of $(q_1, q_2, q_3, q_4) \in \mathbb{Z}_3^4$ is $v = KQ_K L_Q^{-1}(0,0,0,q_1,q_2,q_3,q_4)^{\mathsf{t}}$.

Here is the actual particular lift for every point of that lattice. Use the exact E.75 Smith system for

$$D^- = (I - J_2)(T_2 - 7I) : A_2 \longrightarrow A_2^-. \tag{E87.32}$$

Let B^- be its minus-eigenspace basis, $(B^-)^\ell$ its left inverse, and $L^- D^-_{\mathrm{coord}} R^- = \Delta^-$ its exact Smith system. The first twenty-three diagonal entries d_i are nonzero; the remaining row is zero. For $v = KQ_K x$ put

$$\begin{aligned} \omega = (I - J_2)7[v, e_{B_0}], \qquad y = L^-(B^-)^\ell \omega, \\ a(v) = -R^-_{*,1:23}\,\mathrm{diag}(d_i^{-1}) y_{1:23}. \end{aligned} \tag{E87.33}$$

Every entry of this forty-five-by-seven matrix is in $\mathbb{Z}_{(3)}$; its exact product with D^- is $-\omega$. These are verified rational matrix identities. Thus (E87.33) is an actual integral lift, not an inverse-limit inference from finite vanishings.

The image $F_2 a(v)$ is diagonal, because $F_2[v, e_{B_0}] = 0$ and (E87.30) holds. It lies in L_2, so write it as $(3s, 3s)$. Replace $a(v)$ by $a(v) - s a_*$, with the literal vector of E.85. This preserves (E87.30) and makes its joint central image zero. The resulting exact correction matrix and all its columns are saved.

It remains to normalize the horizontal joint value without changing its intrinsic abelian coordinate. The pair g_v has zero joint abelian image, hence has central image (c^n, c^m). The section just constructed has relative joint value (y, y). Its joint horizontal class is E.80's family

with parameters $(2+n, 3+m)$. Applying E.83 to this actual intrinsic section proves $n+m \in 3\mathbb{Z}_3$. The literal Hall right inverse (E85.22) therefore gives z_v with $F_2 z_v = (n, m)$. Replacing g_v by $g_v w_{-z_v}$ makes its whole class-two joint image one; it leaves its abelian coordinate and (E87.29) unchanged. This is an actual continuous normalization: the joint coordinates of the chosen Hall lift are continuous, and division of $n + m$ by three is taken on its proved image. No value of a geometric path is replaced by an assumed zero.

The two resulting values now give an intrinsic section mapping exactly to the E.85 origin. E.85's central splitting and kernel coordinates supply every target point in (E87.14) and every central fibre. Conversely any section in this inertia-fixed abelian locus has zero obstruction (E87.11), so its horizontal abelian class belongs to the computed free four-dimensional kernel. The abelian stabilizer is zero, as in E.85. Its unique aligning abelian conjugator is lifted to the actual fixed subgroup; the gauge uses its literal $guT(g)^{-1}$ and gag^{-1}, retaining any nonlinear central correction. The remaining gauge is precisely the central one of E.85. These observations prove both directions of the full fibre description (E87.15). For a completely specified fixed lift of an aligning vector b, choose its ordered Hall lift g_b and put $\delta = J(g_b)g_b^{-1} \in A_2$. Then $J\delta = -\delta$ and $g = \delta^{1/2} g_b$ is fixed. Thus this alignment too is an actual continuous coordinate map, with the indicated central correction retained.

The rational involution and its exact tangent. The map $\iota[x : y : z] = [y : x : z]$ is a rational order-two automorphism of the cubic fixing $O = [1 : -1 : 0]$. It interchanges A_i with B_i and sends C_i to C_{-i}, so it preserves the eight-cusp filling kernel. Here is an exact verification that it is $[-1]$ on the elliptic curve with origin O. Put $r = (x + y)/z$ and $v = (x - y)/z$. The cubic equation gives $v^2 = -(r^3 + 4)/(3r)$, so the quotient by this involution is the rational r-line and has degree two. On $\mathrm{Pic}^0(E)$, pullback after the divisor norm is $1 + \iota$. It factors through $\mathrm{Pic}^0(\mathbb{P}^1) = 0$, hence $1 + \iota = 0$. The Abel map $P \mapsto [P - O]$ intertwines the involution because O is fixed. This proves the asserted multiplication map and its action $-I$ on the actual elliptic Tate module.

The based comparison is fully specified. Near O write $s = z/x$ and $y/x = -(1 + s^3)^{1/3}$ with constant term -1. Then

$$\iota(s) = -s(1 + s^3)^{-1/3}, \qquad s^{1/3^n} \longmapsto -s^{1/3^n}(1 + s^3)^{-1/3^{n+1}}. \tag{E87.34}$$

All unit roots in this formula have constant term one and rational coefficients. The constant roots of -1 are all -1, and the displayed maps are compatible under cubing. Moreover $\iota(s)^3 = -s^3/(1 + s^3)$, so $1 + \iota(s)^3 = (1 + s^3)^{-1}$. Applying the displayed root map twice therefore gives exactly $s^{1/3^n}$. Thus the comparison is an actual based involution and commutes with the entire arithmetic action. It descends through the filling map, which is defined over $\mathbb{Q}$ with the same tangent.

For the filling map the complex fundamental-polygon presentation has two handle generators and nine peripheral generators, with one relation equating their product to the handle commutator. Killing the eight specified peripherals leaves the free group on the two handles. Riemann existence and its pointed comparison [GR71, Exposé XII, Théorème 5.1 and Corollaire 5.2], followed by the universal property of pro-three completion, therefore give exactly the stated closed normal kernel and the surjection on geometric groups. The images of the two chosen intrinsic handle generators form its abelian basis; they form an actual free basis by the pro-three Frattini criterion and the Hopfian property. This proves (E87.17), including its abelian and central maps. The quotient kills all the other second-degree Hall vectors.

The actual central coordinate morphism. For a class-two pro-three group, the integral Baker–Campbell–Hausdorff law uses only division by two. In the stated rank-two basis it is the explicit law

$$(X, Y, r)(X', Y', r') = (X + X', Y + Y', r + r' - (XY' - YX')/2). \tag{E87.35}$$

It follows directly from the class-two Hall collection law, so logarithm and exponential here are mutually inverse continuous integral coordinate maps. Since ι is $-I$ on the elliptic Tate module and fixes its central wedge, its general action has the form (E87.18). Applying (E87.19) makes it exactly $(-I,+I)$. Direct substitution gives the inverse formula (E87.20).

Every genuine arithmetic automorphism commutes with this involution. On the new handle basis it has the form

$$X_i^\sharp \longmapsto \sum_j A_{ji}(g) X_j^\sharp + \eta_i(g) Z. \tag{E87.36}$$

Commuting with $(-I,+I)$ gives $2\eta_i(g)=0$, hence $\eta_i(g)=0$ in the torsion-free central lattice. Its action on Z is $\det A(g)=\chi(g)$. This proves the exact graded action through the displayed coordinate morphism, with no replacement of the original based group. In the old basis the same computation gives the actual cross row

$$\eta(g) = \big(\chi(g)b - bA(g)\big)/2, \qquad b=(b_1,b_2). \tag{E87.37}$$

Thus the old and new arithmetic actions are related by the full central formula, not just by their abelian matrices. The actual E.37 matrices now give (E87.21).

The complete filled local section set. The abelian E.77 relation is $(I-C_E^{\mathrm{mat}})(A_E-7I)A=0$. Its exact kernel is $A=(3t,t)$. The fixed abelian gauge is generated by $(1,0)$, and $(I-A_E)(b,0)=(0,-3b)$. This gives the three first-degree coordinates used in the statement.

Let $u=(U_1,U_2,r_u)$ and $a=(3t,t,r_a)$ in (E87.35). Direct multiplication of the actual source norm word gives

$$uT(a)u^{-1}a^{-7} = (-21t,0,-7t(U_1+21t/2)). \tag{E87.38}$$

Its abelian coordinate is fixed by C_E^{mat}, and its central coordinate is negated by J. The complete source relation is therefore exactly (E87.22). The central obstruction normalization of E.77 divides the doubled central defect by two, giving the displayed free obstruction with its sign. Since $\mathbb{Z}_3$ is an integral domain, its vanishing locus is exactly the union of the two stated branches. Both central coordinates are unrestricted.

The fixed geometric conjugator has coordinates $(b,0,0)$: its central coordinate must vanish because J negates it. Applying the full group law to $guT(g)^{-1}$ and gag^{-1} gives respectively

$$\begin{aligned} &(U_1, U_2-3b, r_u+(3b^2+3bU_1-2bU_2)/2),\\ &(3t,t,r_a-bt). \end{aligned} \tag{E87.39}$$

Taking $b=(U_2-\epsilon)/3$ proves (E87.24) and its inverse. There is no nonzero abelian fixed vector because $\det(I-A_E)=9$; there is no central fixed vector because $7-1$ is injective. Thus no further stabilizer acts on the two central coordinates. This proves the full fibre and homeomorphism assertions.

Finally the central projection in (E87.17) is the exact free coordinate used in E.74. The natural obstruction square for these actual central extensions, proved by projecting a continuous Hall lift and its defect as in E.83, identifies (E87.22) with the free component of every intrinsic obstruction. This justifies each intrinsic exclusion through the actual filled map.

All matrix divisions above take place on proved integral images. The retained finite Smith systems contain both inverse maps, and the displayed particular lift is an exact rational identity over $\mathbb{Z}_{(3)}$. All coordinate spaces are compact; their explicit continuous inverses give the asserted compact topologies. The actual code checks the full first-degree maps, all $2,187$ residue inputs for the horizontal obstruction, every particular correction column, the coherent formal-disc square, the central averaging morphism, the symbolic filled norm word and 729 complete gauge/inverse identities. The source and input hashes and all actual outputs are retained. □

Two finite class congruences for the entire complex norm system

E.88. All outside-S divisor lifts, convergent product coordinates, and the complex positional cube condition [Proved]

Use the actual complex fields L_d and roots of E.72, including $L_0 = K = \mathbb{Q}(\zeta)$. Put

$$\mathcal{X} = \varprojlim_{d,N} \widehat{L_d^{\times}}_{\,3}, \qquad \mathcal{X}_S = \varprojlim_{d,N} (\mathcal{O}_{L_d}[1/3, 1/17]^{\times} \otimes \mathbb{Z}_3). \tag{E88.1}$$

The inverse limits use the actual field norms. Let $\mathcal{P}$ be the rational primes $p \neq 3, 17$ with $v_3(\mathrm{ord}_p\, 17) = 1$. For each such p, choose the unique residue value of ζ for which 17ζ has order prime to three, and the unique prime-to-three roots $r_{d,p}^{3^d} = 17\zeta$ in $\mathbb{F}_p^{\times}$. Write $\mathfrak{P}_{d,p} = (p, \beta_d - r_{d,p})$. These are precisely the coherent prime chains of E72.3. The notation $\mathcal{P}$ is the rational-prime indexing of the set $\mathcal{P}_1$ there.

Here is a finite-field and degree-six algorithm for the entire class map. Compute the class of $\mathfrak{P}_{1,p}$ in the certified class group $\mathrm{Cl}(L_1) = C_3^2$ of E.57, obtaining (u_p, v_p). In the coherent basis of E.86 put

$$\rho(p) = (c_p, d_p) = (2u_p + 2v_p, v_p) \pmod 3, \qquad c_p, d_p \in \{0, 1, 2\}. \tag{E88.2}$$

This is an evaluation in one fixed number field, with its retained integral basis, ideal class generators and unconditionally certified class map. In particular $\rho(7) = (2, 1)$ and $\rho(73) = (0, 1)$. For

$$\mathscr{R}(\mathcal{P}) = \{(t_p) \in \prod_{p \in \mathcal{P}} \mathbb{Z}_3 : t_p \longrightarrow 0 \pmod{3^m} \text{ for every } m\},$$

define the continuous surjection

$$\rho : \mathscr{R}(\mathcal{P}) \longrightarrow C_3^2, \qquad (t_p) \longmapsto \Big(\sum c_p t_p, \sum d_p t_p\Big) \pmod 3, \qquad \mathcal{K} = \ker \rho. \tag{E88.3}$$

The sums modulo three have finite support. The exact global divisor sequence is

$$0 \longrightarrow \mathcal{X}_S \longrightarrow \mathcal{X} \xrightarrow{\mathrm{div}_{\notin S}} \mathscr{R}(\mathcal{P}) \xrightarrow{\rho} C_3^2 \longrightarrow 0. \tag{E88.4}$$

Thus the two congruences in (E88.3) are necessary and sufficient for a coherent multiplicative lift at all levels, allowing valuations at $3, 17$.

There are explicit topological coordinates on their kernel. If e_p denotes its divisor coefficient vector, put

$$E_7 = 3e_7, \qquad E_{73} = 3e_{73}, \qquad E_p = e_p + c_p e_7 - (c_p + d_p) e_{73} \quad (p \neq 7, 73). \tag{E88.5}$$

The following map and inverse are topological isomorphisms:

$$\begin{aligned} \mathbb{Z}_3^2 \times \mathscr{R}(\mathcal{P} \setminus \{7, 73\}) &\longrightarrow \mathcal{K}, \\ (z_7, z_{73}, (r_p)) &\longmapsto z_7 E_7 + z_{73} E_{73} + \sum r_p E_p, \\ r_p &= t_p \quad (p \neq 7, 73), \\ z_7 &= \frac{t_7 - \sum c_p t_p}{3}, \qquad z_{73} = \frac{t_{73} + \sum (c_p + d_p) t_p}{3}. \end{aligned} \tag{E88.6}$$

The sums in the inverse run over $p \neq 7, 73$ and converge in $\mathbb{Z}_3$. Their numerators are divisible by three by exactly (E88.3).

Choose, for each vector E_i in (E88.5), one actual coherent lift $x^{(i)} \in \mathcal{X}$ supplied by (E88.4). Remove its common valuation at seventeen by dividing by the corresponding power of the coherent tuple $\beta = (\beta_d)_{d\ge 0}$; all chosen lifts will have valuation zero there. Then

$$\begin{aligned} s : \mathcal{K} &\longrightarrow \mathcal{X}, \qquad s(t) = \bigl(x^{(7)}\bigr)^{z_7} \bigl(x^{(73)}\bigr)^{z_{73}} \prod_{p\neq 7,73} \bigl(x^{(p)}\bigr)^{r_p}, \\ \mathcal{X}_S \times \mathcal{K} &\xrightarrow{\sim} \mathcal{X}, \qquad (u,t) \longmapsto u\, s(t), \\ x &\longmapsto \bigl(x\, s(\mathrm{div}_{\notin S}\, x)^{-1}, \mathrm{div}_{\notin S}\, x\bigr). \end{aligned} \tag{E88.7}$$

The product is convergent in every finite field and every power quotient, hence in the displayed inverse limit. Both maps are continuous. Changing the choices multiplies s by a continuous homomorphism $\mathcal{K} \to \mathcal{U}_\infty$, where $\mathcal{U}_\infty$ is the coherent 3-unit group of E.69. Conversely every such homomorphism gives another normalized splitting.

In particular E.69 and E72.10 give the full compact kernel and its literal coordinates:

$$\mathcal{X}_S = \beta^{\mathbb{Z}_3}\mathcal{U}_\infty \simeq \mathbb{Z}_3^2 \times \prod_{d\ge 0} \mathbb{Z}_3^{2\cdot 3^d}. \tag{E88.8}$$

The two factors in (E88.7), together with (E88.6) and E69.6, compute the entire complex multiplicative norm system as a topological group with specified forward and inverse maps. The restricted sequence factor is given its multiplication-by-3^m prodiscrete topology; it is not declared compact.

The rational base norm has a precise additional coordinate. Define

$$\Theta(t) = v_{1-\zeta}\bigl(s(t)_0\bigr) \in \mathbb{Z}_3. \tag{E88.9}$$

This is a continuous linear map. A normalized splitting change modifies it by 3^h times a continuous linear map to $\mathbb{Z}_3$, with the actual finite integer h of E.69. Thus $\Theta \bmod 3^h$ is independent of the choices. There is a fully evaluated reduction modulo three. Let $(\ell_p, m_p) \in C_9 \oplus C_3$ be the ordinary class of $\mathfrak{P}_{2,p}$ in E86.8, with $0 \le \ell_p < 9$. It is computed from the same lower prime $\mathfrak{P}_{1,p}$ by the retained genus Artin map; its first coordinate modulo three is c_p. Then

$$\Theta(t) \equiv -\frac{\sum_{p\in\mathcal{P}} \ell_p t_p}{3} \pmod 3 \qquad (t \in \mathcal{K}). \tag{E88.10}$$

The numerator belongs to $3\mathbb{Z}_3$, and changing an integer representative of ℓ_p does not change the residue.

Here is the exact transfer to global H-section classes. In E.46 fix $A = 3\kappa_3(17)$ and take pure positional component 1. The entire complex component locus satisfying the total-norm cube condition consists of

$$\begin{aligned} x = \beta^{\,3b}\, u(z,\mathbf{a})\, s(3r), \qquad & b, z \in \mathbb{Z}_3, \quad \mathbf{a} \in \prod_{d\ge 0} \mathbb{Z}_3^{2\cdot 3^d}, \\ r \in \mathscr{R}(\mathcal{P}), \qquad & \sum c_p r_p = 0 \pmod 3. \end{aligned} \tag{E88.11}$$

Here $u(z,\mathbf{a})$ is the actual recursive coherent 3-unit map E69.6. Formula (E88.7) gives the inverse and proves uniqueness of these coordinates. In these coordinates the scaled toric map E46.7 is

$$\left(\kappa_3(17),\ 2b\,\kappa_3(17) + \left(3^{h-1}z + \frac{\Theta(3r)}{3}\right)\kappa_3(3) + \sum_{p\in\mathcal{P}} r_p \kappa_3(p)\right). \tag{E88.12}$$

The quotient $\Theta(3r)/3$ is integral by (E88.10)–(E88.11). The subset of r in (E88.11) itself has the explicit restricted basis $3e_7$ and $e_p - \widetilde{2c_p}e_7$ for $p \neq 7$, where $\widetilde{2c_p} \in \{0,1,2\}$ is the indicated

residue. Thus this entire locus has coordinates $\mathbb{Z}_3^3 \times \mathscr{R}(\mathcal{P} \setminus \{7\}) \times \prod_{d\geq 0} \mathbb{Z}_3^{2\cdot 3^d}$. These are the section classes in the actual target H of the quotient maps E.38 and the Kummer–Shapiro comparison E.46.

Two literal norm-coherent examples fix useful arithmetic origins:

$$x_d^+ = \beta_d + 1, \qquad x_d^- = \beta_d - 1, \qquad N_{K/\mathbb{Q}} x_0^+ = 273 = 3 \cdot 7 \cdot 13, \quad N_{K/\mathbb{Q}} x_0^- = 307. \tag{E88.13}$$

Their outside-S divisors are $e_7 + e_{13}$ and e_{307}, and their valuations at three are 1 and 0. Their cubes therefore give actual global H sections with second toric coordinates $\kappa_3(273)$ and $\kappa_3(307)$, respectively. The certificate evaluates all 79 eligible primes at most 2000, the complete finite pivot matrix and its inverse, and the phase (E88.10).

The remaining integer h is retained exactly as in E.69. The values of the choice-independent map $\Theta \bmod 3^h$ beyond its displayed modulo-three evaluation are not claimed numerically evaluated here. The previous all-level principal-class equations away S are replaced by the two explicitly evaluated conditions (E88.3); no unknown higher S-class matrix remains. The manuscript anchor is LM §51.16, fols. 733v–736v, scans 172–178, with the actual maps E.46, E.69, E.72 and E.86.

Proof. The actual class maps at every level. For $p \in \mathcal{P}$, its indicated residue root and its cube define the exact prime norm $N_{L_{d+1}/L_d} \mathfrak{P}_{d+1,p} = \mathfrak{P}_{d,p}$, with residue degree one. E.86 identifies every S-class norm with the identity in the coherent (a_d, b_d) basis. Therefore the S-class of this prime has the same two coordinates at every level. At level one the columns of (a_1, b_1) in the certified class basis are $(2,0), (2,1)$. The inverse matrix modulo three is $\left(\begin{smallmatrix} 2 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, which proves (E88.2). It proves the two necessary class congruences at every level, not just at the bottom.

There is a complete finite algorithm for these coordinates for any eligible prime. Set $q = (p-1)/3^{v_3(p-1)}$. For the two primitive cubic roots in $\mathbb{F}_p$, exactly one makes 17ζ have order prime to three. Exponentiation by the inverse of 3^d modulo q gives $r_{d,p}$. The ideal $(p, \beta_1 - r_{1,p})$ is then evaluated by the certified degree-six class map. The code retains the maximal-order ideal interpretation and checks its norm p. The official algorithm returns the coordinates in the retained class-generator ideals [The26, entries `bnfcertify`, `bnfisprincipal`, `bnrisprincipal`]. At the next level the Frobenius in the lower genus field of E.86 is computed on this same prime. It lies in the kernel of the actual cubic restriction character, because the specified residue is a cube. Since that genus field is the full Hilbert three-class field of L_2, E86.8 gives the entire ordinary class. Norm compatibility gives $\ell_p \bmod 3 = c_p$, $m_p = d_p$.

Completion and sufficiency. For one number field F, the ordinary outside-S divisor sequence is

$$0 \longrightarrow \mathcal{O}_{F,S}^\times \longrightarrow F^\times \longrightarrow \mathrm{Div}_{\notin S}(F) \longrightarrow \mathrm{Cl}_S(F) \longrightarrow 0. \tag{E88.14}$$

Its image of principal divisors is a subgroup of a free abelian group, hence is free. The first short exact sequence therefore splits as abelian groups. For the second short exact sequence the class quotient is finite. More explicitly, let D' be the preimage of the prime-to-three subgroup of the class group in the divisor group. Then D'/Prin has prime-to-three order, so the two subgroups have the same three-adic completion. Choose 3^e annihilating the three-primary class group. It gives the cofinal inclusions $3^{m+e}\operatorname{Div} \subset 3^m D' \subset 3^m \operatorname{Div}$. Thus the completed sequence remains exact:

$$0 \longrightarrow \widehat{\mathcal{O}_{F,S}^\times}_3 \longrightarrow \widehat{F^\times}_3 \longrightarrow \widehat{\mathrm{Div}_{\notin S}(F)}_3 \longrightarrow \mathrm{Cl}_S(F)\{3\} \longrightarrow 0. \tag{E88.15}$$

The first term is compact, because the S-unit group is finitely generated. Every nonempty fiber of the completed divisor map is a translate of that compact group. In particular it is compact even though the completed divisor group is prodiscrete with infinite quotients.

E72.4, with zero coefficients on S, identifies the outside-S divisor norm limit with $\mathscr{R}(\mathcal{P})$. For $t \in \mathcal{K}$, its divisor at every level has zero S-class, by the norm-identical class maps already proved. Each completed principal-generator fiber is therefore nonempty and compact. Norm maps carry a fiber into its predecessor. Any finite collection of compatibility equations has a solution: choose an element in the highest involved fiber and take its successive actual norms. The product of all fibers is compact, and these are closed conditions with the finite intersection property. Hence their entire intersection is nonempty. This supplies one coherent lift with all the prescribed outside-S divisors. Necessity follows from (E88.15). The kernel is exactly $\mathcal{X}_S$, component by component. Surjectivity of ρ follows from the two actual prime columns $(2,1)$, $(0,1)$ at $7, 73$. This proves every kernel and image in (E88.4).

Restricted coordinates and convergent splittings. Substitute the two pivot columns into (E88.5). Every displayed vector has zero S-class. For a restricted sequence t, the two sums in (E88.6) converge: their coefficients are bounded integers and t_p tends to zero modulo every power of three. The two class equations say exactly that both resulting numerators are divisible by three. Reading all nonpivot coordinates proves the formula and inverse. Reduction modulo 3^m and, for the inverse, modulo 3^{m+1} proves their continuity. This proves (E88.6) in its actual prodiscrete topology.

Choose one lift of each basis vector using the preceding compact-fiber argument. The seventeen-valuation is the same at every level because that prime has residue degree one; dividing by the appropriate power of β removes it compatibly. For each fixed level d and quotient by 3^mth powers, all but finitely many factors in (E88.7) equal one: their exponents r_p belong to $3^m\mathbb{Z}_3$. Thus the products converge, independently of their ordering. Norms commute with the finite products and with these limits. The product therefore belongs to $\mathcal{X}$ and has outside-S divisor t. This proves the homomorphism and section assertions.

Continuity is equally explicit. A basic open condition in $\mathcal{X}$ imposes a power quotient at finitely many field levels. Exponents divisible by a sufficiently large power of three send every factor to one in all these quotients. Thus the linear product map is continuous in the uniform power topology on the restricted sequence module, and in the usual topology on the two finite coordinates. The inverse in (E88.7) is a composition of continuous divisor, section and multiplication maps. This proves the topological isomorphism without a compactness claim for its restricted factor.

Two choices of a normalized lift of a basis vector differ by an actual coherent 3-unit. Their products define a continuous homomorphism $\mathcal{K} \to \mathcal{U}_\infty$ by the same convergence argument. Conversely any continuous homomorphism of these modules is determined by its values on the displayed basis: finitely supported coordinate vectors are dense. Multiplying the chosen lifts by those values gives exactly the corresponding changed splitting. This proves the full choice ambiguity. Finally remove the common seventeen valuation from an arbitrary element of $\mathcal{X}_S$; the remainder is precisely the coherent 3-unit group. E69.6 supplies its actual product coordinates and their inverse, proving (E88.8).

The base phase and the positional condition. Valuation is continuous and commutes with the degree-one norms at three, so (E88.9) is continuous and linear. E69.3 says that the base projection of every coherent 3-unit is a power $\lambda^{3^h z}$, $\lambda = 1 - \zeta$. This proves the stated change of Θ. Conversely the chosen E69.6 section of this base projection lifts every continuous linear $\mathcal{K} \to \mathbb{Z}_3$ to a possible splitting change. Thus the ambiguity is exactly the one displayed.

At level two the full divisor of $s(t)_2$ is $\Theta(t)\mathfrak{q}_{3,2} + \sum t_p \mathfrak{P}_{2,p}$, with zero seventeen coefficient. Its ordinary class must vanish. E.86 gives $[\mathfrak{q}_{3,2}] = (3,0)$ and the coordinates (ℓ_p, m_p). The resulting first coordinate equation is $3\Theta(t) + \sum \ell_p t_p = 0$ in C_9. The second is already the S-class condition. This proves (E88.10), including the integrality of its division and its independence of coordinate representatives.

For $x = \beta^{\,b} u(z, \mathbf{a}) s(t)$, the actual rational base norm is

$$N_{K/\mathbb{Q}} x_0 = 17^{2b}\, 3^{\,3^h z + \Theta(t)} \prod_{p \in \mathcal{P}} p^{t_p} \quad \text{in } \widehat{\mathbb{Q}^\times}_3. \tag{E88.16}$$

Its divisor follows from the displayed valuations and residue degrees; the possible μ_3 factor in K has rational norm one. The pro-three completion of the rational unit group is zero, so these prime coordinates determine the entire rational element. The product converges in that completion by the restricted-support condition.

This element is a cube exactly when $b \in 3\mathbb{Z}_3$, each $t_p \in 3\mathbb{Z}_3$, and $\Theta(t) \in 3\mathbb{Z}_3$, since $h \geq 1$. Write $t = 3r$. It automatically belongs to $\mathcal{K}$. Equation (E88.10) gives $\Theta(3r) = -\sum c_p r_p$ modulo three. This proves both directions of (E88.11). The first-coordinate cocycle is still the specified $A = 3\kappa_3(17)$ in E.46; its actual scaled toric map divides the rational norm coordinate by three. Applying that map to (E88.16) proves (E88.12), with all factors and divisibilities typed. The one-row pivot basis for the condition on r follows from $c_7 = 2$, whose inverse in $\mathbb{F}_3$ is 2. The same restricted-product argument proves its forward and inverse topology.

For the two examples, $N_{L_{d+1}/L_d}(\beta_{d+1} \pm 1) = \beta_d \pm 1$ follows directly from the odd cubic polynomial. Their rational base norms are $17^2 - 17 + 1 = 273$ and $17^2 + 17 + 1 = 307$. The unique degree-one coherent branches give the stated outside-S divisors, and $\beta_d + 1$ is the uniformizer at three while $\beta_d - 1$ is a unit there. These literal maps prove (E88.13) and the two asserted section classes. The finite certificate checks the same phase and class equations, all inverse pivot coordinates, and the norm identities independently. □

The complete horizontal intrinsic family and its varying joint image

E.89. All horizontal inputs over the actual rational-cusp relative value [Proved]

Retain E.85's actual local source and full intrinsic section (u_0, d), with $\bar{d} = e_{B_0}$, and all coefficient conventions of E.87. Write $V = \langle f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2 \rangle$ and $A_2 = \Lambda^2 V$. The following result includes every intrinsic abelian class whose relative value is e_{B_0}, with arbitrary horizontal value.

Put

$$\mathcal{P}_1 = V/(I - T_1)V^{J_1} \simeq \mathbb{Z}_3^6 \oplus C_3^3. \tag{E89.1}$$

The actual full class-two obstruction on these abelian classes is

$$\begin{aligned} \delta : \mathcal{P}_1 \longrightarrow \mathcal{B}_2, \qquad [v] \longmapsto [7[v, e_{B_0}]], \\ \operatorname{im}\delta \simeq C_3^2 \oplus C_9, \qquad \ker\delta \simeq \mathbb{Z}_3^6. \end{aligned} \tag{E89.2}$$

For a column v in the displayed ten-generator order, its three exact conditions are

$$\begin{pmatrix} 2 & 0 & 0 & 1 & 2 & 0 & 1 & 2 & 0 & 0 \\ 2 & 0 & 0 & 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 2 & 8 & 0 & 3 & 6 & 0 & 6 & 3 & 0 & 0 \end{pmatrix} v \equiv 0 \pmod{(3,3,9)}. \tag{E89.3}$$

These conditions include every finite obstruction; the free obstruction coordinate is zero on this whole family.

Here are actual integral representatives of the six free coordinates on the zero locus. Set

$$\begin{aligned} v_L &= -B_0, \\ v_m &= 2f_1 - 2f_2 + 3B_0 + B_2 + C_1 + 2C_2, \\ v_1 &= A_0, & v_2 &= A_1 + A_2, \\ v_3 &= 2f_1 + 4f_2 + A_1 - 3B_0 + B_1 - 2C_1, \\ v_4 &= f_1 + 2f_2 + 2A_1 - 2B_0 + B_2 - C_1, \end{aligned} \tag{E89.4}$$

and write

$$v = Lv_L + mv_m + q_1v_1 + q_2v_2 + q_3v_3 + q_4v_4. \tag{E89.5}$$

For R of E.87 these vectors have the exact images

$$Rv = (L, 0, m), \qquad Rv_j = 0 \quad (1 \le j \le 4). \tag{E89.6}$$

The inverse on the full integral solution lattice of (E89.3), modulo its complete fixed abelian gauge, is the following literal matrix:

$$\begin{pmatrix} L \\ m \\ q_1 \\ q_2 \\ q_3 \\ q_4 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 & 0 & 0 & -1 & -1 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 4/3 & -5/3 & 1 & 0 & 0 & 0 & 0 & 0 & -2 & -2 \\ 1/3 & 1/3 & 0 & 1/2 & 1/2 & 0 & -3/2 & -3/2 & 1/2 & 1/2 \\ -4/3 & 8/3 & 0 & 0 & 0 & 0 & -1 & -1 & 3 & 3 \\ 5/3 & -10/3 & 0 & 0 & 0 & 0 & 2 & 2 & -4 & -4 \end{pmatrix} v. \tag{E89.7}$$

All divided entries are integral on that proved solution lattice. The matrix kills the entire gauge and is inverse to (E89.4).

For the particular relative central correction, use the actual Hall brackets and put

$$\begin{aligned} a_m &= -\tfrac{3}{4}[B_0, f_1] - \tfrac{5}{4}[B_0, f_2] + \tfrac{1}{4}[B_0, A_1] - \tfrac{1}{2}[B_1, B_0] - \tfrac{1}{2}[C_1, B_0], \\ a_3 &= \tfrac{1}{2}[B_0, A_1] - \tfrac{1}{2}[B_1, B_0], \qquad a_4 = \tfrac{1}{2}[B_0, A_1], \\ a(v) &= ma_m + q_3a_3 + q_4a_4. \end{aligned} \tag{E89.8}$$

These are literal integral coordinates in A_2. They satisfy the exact identities

$$\begin{gathered} (I - J_2)(T_2 - 7I)a(v) = -(I - J_2)7[v, e_{B_0}], \\ F_2a(v) = (m/4, -m/4). \end{gathered} \tag{E89.9}$$

The complete joint image can also be evaluated, including its changing finite central residue. Use $\mathcal{J}_2$ and its two Hall factors from E.80. For $L, m, \rho, \sigma \in \mathbb{Z}_3$, let $\epsilon(m) \in \{0, 1, 2\}$ be the residue of m. The image is exactly the classes with the following representatives:

$$\begin{aligned} u_{L,m,\rho} &= (x^{5/2}y^{-L}c^{2+\epsilon(m)+\rho/2}, x^{1/2+3m}y^{3m-L-2}c^{3+\epsilon(m)-\rho/2}), \\ a_{m,\sigma} &= (yc^{3\sigma+m/4}, yc^{3\sigma-m/4}). \end{aligned} \tag{E89.10}$$

They are values on the actual two generators in E.77. Every pair in (E89.10) satisfies its full local source relation. The four displayed parameters are unique after complete geometric conjugacy. Thus the joint image is homeomorphic to $\mathbb{Z}_3^4$.

More explicitly, over a fixed first-stage pair (L, m), write s for the relative central mean, ρ for the difference of the horizontal central changes from $(2, 3)$, and ϵ for their mean modulo three. In the full target central fibre $\mathbb{Z}_3^2 \times C_3$ the exact intrinsic image is

$$s \in 3\mathbb{Z}_3, \qquad \epsilon = m \pmod 3, \qquad \rho \in \mathbb{Z}_3. \tag{E89.11}$$

The full intrinsic class-two locus under consideration has complete compact coordinates

$$\mathbb{Z}_3^{52} \times C_3^9 \times C_9^5. \tag{E89.12}$$

Every point of its joint image has the complete geometric-conjugacy fibre

$$\mathbb{Z}_3^{48} \times C_3^9 \times C_9^5. \tag{E89.13}$$

The proof gives both coordinate directions, including the actual Hall-image correction, the unique lower fixed conjugator, and the full central gauge. At $L = m = 0$ this is the entire horizontal locus of E.87.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). The relative abelian value is specified throughout; arbitrary other relative inputs are not identified with this family.

Proof. The actual obstruction on every horizontal input. Choose an ordered geometric Hall lift g_v of any $v \in V$ into N_2. The actual cusp norm identity of E.85 gives, in the central lattice,

$$(u_0 g_v)T(d)(u_0 g_v)^{-1} d^{-7} = 7[v, e_{B_0}]. \tag{E89.14}$$

This identity uses only the abelian coordinates of g_v, so its proof applies to all of V. Adding a relative central factor w_a adds $(T_2 - 7I)a$. The actual E.77 relation is therefore precisely

$$(I - J_2)(7[v, e_{B_0}] + (T_2 - 7I)a) = 0. \tag{E89.15}$$

Its factor-one-half cohomology normalization gives (E89.2). The bracket identity

$$7[(I - T_1)b, e_{B_0}] = -(T_2 - 7I)[b, e_{B_0}] \quad (b \in V^{J_1}) \tag{E89.16}$$

proves descent through the entire geometric abelian gauge. Every bracket contains a cusp vector, so its free elliptic coordinate is zero.

The exact retained E.74 Smith matrix sends this ten-column defect to a fifteen-row matrix with moduli $(3^{11}, 9^4)$. Rows $4, 8, 15$ are exactly (E89.3), with moduli $(3, 3, 9)$. Their columns f_1, B_1, f_2 form a matrix of determinant -16. Its inverse gives a literal section onto these three residue coordinates. The complete fifteen-row map factors through them, with the entrywise moduli retained in the certificate. Thus they give both the full image $C_3^2 \oplus C_9$ and its exact integral kernel $Q\mathbb{Z}_3^{10}$. Here Q is the saved Hermite matrix, with $\det Q = 81$ and local Smith factors $1^7, 3, 3, 9$.

The actual gauge matrix is $G = (I - T_1)(I + J_1)$; multiplication by $I + J_1$ has image V^{J_1} because two is a unit. The exact matrix $Q^{-1}G$ has four unit Smith factors and six zero factors. Both inverse matrices are saved. This proves the free rank-six zero locus and the absence of a remaining torsion coordinate there. Direct multiplication proves that (E89.7) is integral on $Q\mathbb{Z}_3^{10}$, kills G, and is inverse to the six columns (E89.4). It therefore gives every cohomological coordinate, including every preimage outside the coefficient kernel of E.87.

The relative correction (E89.8) is obtained from the exact E.75 Smith inverse for $(I - J_2)(T_2 - 7I)$, followed by subtraction of its common joint mean using a_* of E.85. More directly, both equations in (E89.9) are exact rational matrix identities over $\mathbb{Z}_{(3)}$, verified for all six columns. Thus they construct actual integral lifts for all six parameters.

The actual finite character fixing the target residue. For $((x^A y^B c^n), (x^{A'} y^{B'} c^{n'})) \in \mathcal{J}_2$ put $p = A$ and define

$$\vartheta = n + n' + p(p - 1) \pmod 3. \tag{E89.17}$$

This is a genuine continuous homomorphism $\mathcal{J}_2 \to C_3$. To verify it, use the exact Hall product

$$(A, B, n)(D, E, r) = (A + D, B + E, n + r - DB). \tag{E89.18}$$

In the joint abelian lattice the two pairs are $(p, p - L)$ and $(3m - p, 3m - p - L)$. The two central cross terms reduce to pq modulo three. The difference of the two polynomial terms $p(p - 1)$ under addition is $2pq = -pq$ modulo three. They cancel, proving the homomorphism. The genuine operators of E.80 give

$$\vartheta\mathcal{T} = \vartheta + p \pmod 3, \qquad \vartheta_\mathcal{J} = \vartheta. \tag{E89.19}$$

These follow by substituting $T_\mu(A,B,n) = (7A, 7B, 49n + 7\mu B)$ and $I(A,B,n) = (-A,-B,n)$ in (E89.17).

The pullback of this character to N_2 factors through V, since it is a homomorphism to an abelian group. Its values on A_i are zero: the two maps fill these cusps. Its values on C_1, C_2 are zero as well: one factor is a conjugate of a third power of the infinity peripheral generator, and the other is trivial. Formula (E89.17) vanishes on every such conjugate. Its value on B_0 is zero by the actual normalization $d \mapsto (y, y)$.

Write the remaining possible character row as $(a, b, 0, 0, 0, 0, c, e, 0, 0)$. The exact matrices T_1, J_1 and (E89.19) impose

$$wT_1 = w + R_{p,*}, \qquad wJ_1 = w \pmod 3. \tag{E89.20}$$

Solving these four scalar unknowns over $\mathbb{F}_3$ gives precisely

$$w = (a, a+1, 0, 0, 0, 0, 2, 1, 0, 0), \qquad a \in \mathbb{F}_3. \tag{E89.21}$$

This retains every possibility for the actual handle marking. All three rows have the same value on (E89.4), namely

$$\vartheta(g_v) = 2m \pmod 3. \tag{E89.22}$$

Thus no unevaluated handle coefficient enters the asserted image. This conclusion is a consequence of the actual homomorphism and its arithmetic intertwiners.

An actual origin over each varying abelian target. For the representative (E89.5), its Hall lift has actual joint image

$$g_v \longmapsto (y^{-L}c^n,\ x^{3m}y^{3m-L}c^{n'}). \tag{E89.23}$$

The two central coordinates here mean the actual evaluation of the chosen Hall lift. They are continuous functions of its six coordinates. Equation (E89.22) proves $n + n' \equiv 2m \pmod 3$.

Multiplying the actual E.85 value u_0 by (E89.23) gives central coordinates $(2+n, 3+n'+6m)$. Put $\epsilon = \epsilon(m)$. The difference from the desired central pair $(2+\epsilon, 3+\epsilon)$ is

$$(\epsilon - n, \epsilon - n' - 6m), \tag{E89.24}$$

whose sum is divisible by three. The exact image of $F_2 : A_2 \to \mathbb{Z}_3^2$ is $L_2 = \{(r,s) : r+s \in 3\mathbb{Z}_3\}$. For any such pair its literal right inverse from E.85 is

$$(r,s) \longmapsto (2\ell - r)[f_2, f_1] - \ell[B_0, f_1], \qquad \ell = (r+s)/3. \tag{E89.25}$$

Apply this actual map to (E89.24) and multiply g_v by the resulting central factor. It does not change v or (E89.14), and makes the entire joint horizontal value exactly the one in (E89.10) with $\rho = 0$. The relative value $dw_{a(v)}$ already has the required joint image with $\sigma = 0$ by (E89.9). This constructs an actual intrinsic section mapping to the indicated origin for every $L, m, q_1, \ldots, q_4$. The construction evaluates the Hall image before correcting it; no path coordinate is replaced by zero without that map.

Adding ρz_* to the horizontal central value and σa_* to the relative central value, with the actual E.85 vectors

$$F_2 z_* = (1/2, -1/2), \qquad F_2 a_* = (3, 3), \tag{E89.26}$$

gives all four parameters in (E89.10). Every remaining central variation is exactly E.85's full central coefficient map. It supplies its complete kernel coordinates and no additional image.

Full target conjugacy and the inverse maps. For reference, an arbitrary representative with the same relative abelian value has

$$\begin{aligned} u &= (x^{5/2+p}y^{p-L}c^N, x^{1/2+3m-p}y^{3m-p-L-2}c^M), \\ a &= (yc^{a_1}, yc^{a_2}). \end{aligned} \tag{E89.27}$$

Its source relation is exactly

$$2p - 3m + 6(a_1 - a_2) = 0. \tag{E89.28}$$

It follows that $p \in 3\mathbb{Z}_3$. The complete fixed geometric group consists of (w, Iw) with $\ell(w) = 0$, so write $w = x^b y^b c^r$. Its actual gauge sends $p \mapsto p - 6b$, $L \mapsto L$, $m \mapsto m$, and its full central formulas are

$$\begin{aligned} N &\longmapsto N - 48r - 7Lb - 42b^2 + 6bp + 15b, \\ M &\longmapsto M - 48r + 7Lb - 42b^2 - 18bm + 6bp + 11b, \\ a_1 &\longmapsto a_1 + b, \qquad a_2 \longmapsto a_2 - b. \end{aligned} \tag{E89.29}$$

They follow by (E89.18), using the actual $T_{-5/2}, T_{-1/2}$ on the two factors. Taking $b = p/6$ makes $p = 0$, and (E89.28) then gives $a_1 - a_2 = m/2$. The remaining central gauge changes both N, M by $-48r$. Thus the complete invariants are

$$L,\ m,\ \rho = N - M + 1,\ s = (a_1 + a_2)/2,\ \epsilon = (N + M - 5)/2 \pmod 3. \tag{E89.30}$$

The corresponding inverse is (E89.10) with 3σ replaced by s and $\epsilon(m)$ replaced by the indicated residue representative. There is no nonzero lower stabilizer: p changes by $-6b$ in the torsion-free lattice. The remaining central gauge is exactly the one just divided out. This proves the full target fibre coordinates.

For the intrinsic family, (E89.23)–(E89.25) give $\epsilon = m \pmod 3$. The particular relative image has mean zero, while the full relative central coefficient image has mean in $3\mathbb{Z}_3$ by E.85. The horizontal central map changes ρ arbitrarily and changes the mean residue by zero. This proves both necessity and sufficiency in (E89.11).

Here is the full source inverse. Start with an arbitrary actual intrinsic section in the stated locus and apply E.77's order-two normalization. Subtract the actual horizontal abelian value of u_0 to obtain v. Its vanishing obstruction gives (E89.3), and (E89.7) recovers all six free parameters. The saved exact Smith inverse for G gives the unique fixed abelian aligning vector b taking v to (E89.5). Concretely, if $L_G G R_G = D_G$ is the exact Smith system and Δ is that difference, one chosen formula is

$$b = (I + J_1)(R_G)_{*,1:4}(D_G)^{-1}_{1:4,1:4}(L_G\Delta)_{1:4}. \tag{E89.31}$$

Every division is integral on the proved image. The vector is unique because $(I - T_1)b = 0$ forces $b = 0$. Choose its ordered Hall lift g_b. Then $\delta_b = J(g_b)g_b^{-1}$ is central and anti-invariant, and the actual lift $g = \delta_b^{1/2} g_b$ is fixed. Apply its literal conjugation $(u, a) \mapsto (guT(g)^{-1}, gag^{-1})$. All resulting central terms are retained.

Subtract the particular section obtained from (E89.8), (E89.23)–(E89.25). The remaining coordinates are precisely E.85's central lattice quotient and saturated relative kernel. Its explicit forward and inverse Smith maps recover ρ, σ and the forty-four free and fourteen finite kernel coordinates. The four parameters q_i are retained in the full fibre. Conversely, these instructions construct a section from every such collection of coordinates. There is no residual lower stabilizer, so no further quotient of that fibre is present. This proves (E89.12)–(E89.13) with the complete inverse maps.

All finite carries use the specified residue representatives. Every other map is an actual continuous group map, an integral linear map on its stated image, or the indicated continuous Hall lift. The parameter spaces are compact and the displayed inverses are continuous. These give exactly the stated compact topologies.

The certificate retains the full fifteen-coordinate map, its actual three-coordinate section, all Smith maps and inverses, the six-column particular correction and its joint image, all three possible handle character rows, and the full symbolic norm and conjugation formulas. It checks all 81 obstruction values and $2,187$ complete inverse identities. The exact input hashes and actual outputs are retained. □

All rational-cusp relative multiples and their actual global restrictions

E.90. The nonunit relative strata, complete joint fibres, and global cusp and unramified comparisons [Proved]

Use the actual source of E.77, the positive rational-cusp origin (u_0, d) of E.85, and the unchanged ten-generator cubic lattice V of E.89. For each $r \in \mathbb{Z}_3$, including zero, (u_0, d^r) is an actual full intrinsic local section. In this statement the relative value means the value on E.77's marked relative generator; its abelian coordinate is re_{B_0}.

Allow every horizontal intrinsic abelian input $[v] \in \mathcal{P}_1 = V/(I - T_1)V^{J_1}$ over this relative value. Its entire class-two obstruction is

$$\delta_r([v]) = r\,[7[v, e_{B_0}]] \quad \text{in } \mathcal{B}_2. \tag{E90.1}$$

Write $p = (Rv)_p = 2v_{f_1} - v_{f_2} - 3v_{C_2}$, using E.87's actual map $R : V \to \mathcal{J}_1$. The exact zero loci are as follows:

r	condition on $[v]$	$\ker \delta_r$
$r \in \mathbb{Z}_3^\times$	all three conditions (E89.3)	$\mathbb{Z}_3^6$
$r \in 3\mathbb{Z}_3^\times$	$p \equiv 0 \pmod 3$	$\mathbb{Z}_3^6 \oplus C_3^3$
$r \in 9\mathbb{Z}_3$	every $[v] \in \mathcal{P}_1$	$\mathbb{Z}_3^6 \oplus C_3^3$.

(E90.2)

The corresponding obstruction images have orders $81, 3, 1$. The free obstruction coordinate is zero throughout these strata, and all fifteen finite coordinates are evaluated by the scalar multiple of E.89's retained matrix.

Here is the complete image in the actual joint class-two quotient $\mathcal{J}_2$ of E.80. Take $L, m, \rho, \sigma \in \mathbb{Z}_3$ and the indicated residue representatives $p, \epsilon \in \{0, 1, 2\}$. Its representatives on E.77's two marked generators are

$$\begin{aligned} u &= (x^{5/2+p} y^{p-L} c^{2+\epsilon+\rho/2}, x^{1/2+3m-p} y^{3m-p-L-2} c^{3+\epsilon-\rho/2}), \\ a &= (y^r c^{3\sigma+rm/4-rp/6}, y^r c^{3\sigma-rm/4+rp/6}), \end{aligned} \tag{E90.3}$$

where the complete ranges are

r	p	ϵ	joint image
$r \in \mathbb{Z}_3^\times$	0	$m \bmod 3$	$\mathbb{Z}_3^4$
$r \in 3\mathbb{Z}_3^\times$	0	arbitrary	$\mathbb{Z}_3^4 \times C_3$
$r \in 9\mathbb{Z}_3$	arbitrary	arbitrary	$\mathbb{Z}_3^4 \times C_3^2$.

(E90.4)

Every divided exponent in (E90.3) is integral on its stated domain. These are unique coordinates after the full geometric conjugacy. The entire intrinsic class-two source locus and every fibre of its map onto this joint image have the types

r	source locus	each joint fibre
$r \in \mathbb{Z}_3^\times$	$\mathbb{Z}_3^{52} \times C_3^9 \times C_9^5$	$\mathbb{Z}_3^{48} \times C_3^9 \times C_9^5$
$r \in 3\mathbb{Z}_3$	$\mathbb{Z}_3^{52} \times C_3^{12} \times C_9^5$	$\mathbb{Z}_3^{48} \times C_3^{11} \times C_9^5$.

(E90.5)

The proof gives actual inverse coordinates, retaining the full lower conjugator, the nonlinear Hall terms, and the uncomputed handle character only until an explicit integral correction removes it.

There is also an exact global comparison. Choose the allowed rational tangent y/x of value one at B_0, while the tangent at O is z/x of value one. The rational automorphism $[x : y : z] \mapsto [x : z : y]$ carries these tangents exactly. Both based actions factor through the

same Γ of E.34, and the entire pro-three path torsor between them also factors through Γ. Thus the normalized path of E.85 gives a global cusp cocycle u_g satisfying

$$u_g\, g(d)\, u_g^{-1} = d^{\chi(g)}, \qquad z_\beta(g) = d^{\beta(g)} u_g \quad (\beta \in Z^1(\mathbb{Q}, \mathbb{Z}_3(1))). \tag{E90.6}$$

Its map on Kummer classes is injective into full intrinsic geometric-conjugacy classes. The same assertion holds after restriction to $\mathbb{Q}_3$.

All global-to-local fibres in this cusp family are explicit. Define the complete restricted exponent module

$$\mathscr{R}_{\mathcal{P}} = \left\{ (a_\ell)_{\ell \text{ prime}} \in \prod_\ell \mathbb{Z}_3 : \text{for every } n,\ a_\ell \in 3^n\mathbb{Z}_3 \text{ for all but finitely many } \ell \right\}. \tag{E90.7}$$

Its topology is the inverse limit of the discrete groups $\bigoplus_\ell \mathbb{Z}/3^n$, with the literal reduction maps. Kummer theory identifies it with $H^1(\mathbb{Q}, \mathbb{Z}_3(1))$. For $\ell \neq 3$ put

$$\langle \ell \rangle = \ell/\omega(\ell) \in 1 + 3\mathbb{Z}_3, \quad \omega(\ell) \in \{1, -1\}, \qquad \ell_\ell = \frac{\log\langle \ell \rangle}{\log 4}. \tag{E90.8}$$

Here $\omega(\ell) \equiv \ell \pmod 3$. In the actual local Kummer basis $(3, 4)$, restriction and all its inverse fibres are

$$\begin{aligned} H^1(\mathbb{Q}, \mathbb{Z}_3(1)) &\longrightarrow H^1(\mathbb{Q}_3, \mathbb{Z}_3(1)) \simeq \mathbb{Z}_3^2, \\ (a_\ell) &\longmapsto \left(a_3, \sum_{\ell \neq 3} a_\ell \ell_\ell \right), \\ a_3 = u, \qquad a_2 &= 2v - 2 \sum_{\ell \neq 2,3} a_\ell \ell_\ell, \qquad (a_\ell)_{\ell \neq 2,3} \in \mathscr{R}_{\mathcal{P}\setminus\{2,3\}}. \end{aligned} \tag{E90.9}$$

The last line is the full inverse fibre over (u, v). Both sums converge in the stated topology. In particular this is a split topological map, with section the actual completed class $3^u 4^v$. There is no surviving finite Kummer torsion in either the global or the local group in (E90.9).

Retain the actual E.77 marking: normalize the local Kummer representatives at its order-two element, and write the two values on (t, r_s) as columns. Here r_s denotes its marked relative generator. Then

$$\kappa_3 : (1, 0), \qquad \kappa_4 : (b, \alpha), \quad \alpha \in \mathbb{Z}_3^\times, \qquad \begin{pmatrix} U \\ A \end{pmatrix} = \begin{pmatrix} 1 & b \\ 0 & \alpha \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}, \quad \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} U - bA/\alpha \\ A/\alpha \end{pmatrix}. \tag{E90.10}$$

The constants b, α are the actual evaluations in this fixed source. Formula (E90.10) is the exact comparison with the logarithmic basis, without assigning substitute values to them. If $u = pT(p)^{-1}$ has the actual intermediate coordinate V in (E85.20), the global class

$$\beta_r = (-V - br/\alpha)\kappa_3 + (r/\alpha)\kappa_4 \tag{E90.11}$$

gives a full global cusp section restricting to (u_0, d^r). All its global cusp preimages over this local class are exactly the fibre (E90.9), transported by (E90.10).

Finally the whole family unramified outside three has a complete global comparison through these coordinates. Its intrinsic class-two set, its joint image, and every joint fibre are

$$\begin{aligned} H^1(\Gamma, N_2) &\simeq \mathbb{Z}_3^{30} \times C_3^{12} \times C_9^5, \\ H^1(\Gamma, \mathcal{J}_2) &\simeq \mathbb{Z}_3^3 \times C_3^2, \\ \mathrm{Fib}\big(H^1(\Gamma, N_2) \longrightarrow H^1(\Gamma, \mathcal{J}_2)\big) &\simeq \mathbb{Z}_3^{27} \times C_3^{11} \times C_9^5. \end{aligned} \tag{E90.12}$$

The last line holds for every target point. In (E90.3) the target is exactly $r = 0$, $\sigma = 0$, with all L, m, ρ, p, ϵ. The source map is inflation along the actual global and local surjections to Γ; it is injective on geometric-conjugacy classes. Every source in (E90.12) lifts to the full N. With E.41's fixed recursive Hall–Smith coordinates, every full unramified intrinsic-to-joint fibre is

$$\left(\mathbb{Z}_3^{27} \times C_3^{11} \times C_9^5\right) \times \prod_{j \geq 3} \mathcal{C}_j(N). \tag{E90.13}$$

The coordinate maps are the actual truncation, the inverse maps proved below, and E.41's successive corrected lifts. General global ramified sections still use the actual relations of E.73; (E90.9)–(E90.13) give the full global fibres for the explicitly specified families.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up(4). All local relative multiples, their full intrinsic class-two fibres, and the stated global restriction maps are evaluated.

Proof. Every horizontal input and every multiplier. The full identity $u_0 T(d) u_0^{-1} = d^7$ implies $u_0 T(d^r) u_0^{-1} = d^{7r}$ for each $r \in \mathbb{Z}_3$. Together with $J(d) = d^{-1}$ this verifies E.77's actual source relation before taking a geometric quotient. For an ordered Hall lift g_v of v and a relative central factor w_a, the defect in N_2 is exactly

$$(u_0 g_v) T(d^r w_a)(u_0 g_v)^{-1} (d^r w_a)^{-7} = 7r[v, e_{B_0}] + (T_2 - 7I)a. \tag{E90.14}$$

Thus the source relation is the vanishing of $(I - J_2)$ applied to the right side. E.75's actual coefficient quotient proves (E90.1). E.89 evaluates its image as $C_3^2 \oplus C_9$. Multiplication by a unit preserves its kernel; multiplication by three leaves just its last character modulo three. That character is $(Rv)_p$ modulo three, by the literal rows of (E89.3). Multiplication by nine is zero.

For clarity the integral quotient calculation is also retained. Let Q_1 be E.89's ten-by-ten kernel lattice, let Q_3 be the Hermite basis of $\{v : (Rv)_p \in 3\mathbb{Z}_3\}$, and put $Q_9 = I$. For $G = (I - T_1)(I + J_1)$, the exact Smith valuations of $Q_j^{-1} G$ are respectively

$$(0,0,0,0), \qquad (0,1,1,1), \qquad (0,1,1,1), \quad \text{with free rank six in each case.} \tag{E90.15}$$

The certificate retains all left and right matrices and their inverses over $\mathbb{Z}_{(3)}$. They give both directions of every horizontal coordinate map in (E90.2). Their four nonzero pivots also give the unique fixed abelian aligning vector exactly as in (E89.31).

There is a single rational correction matrix which is integral on all the required domains. For $v = (x, y, a_0, a_1, a_2, b_0, b_1, b_2, c_1, c_2)$, define $B(v) \in A_2 \otimes \mathbb{Q}_3$ by the following five Hall coefficients, with all other coefficients zero:

$$\begin{aligned}
[B_0, f_1] &: \quad (-7x + 5y + 3c_1 - 3c_2)/36, \\
[B_0, f_2] &: \quad (-17x + 7y - 3c_1 + 3c_2)/36, \\
[B_0, A_1] &: \quad (7x + y + 6a_1 - 6a_2 + 3c_1 - 3c_2)/36, \\
[B_1, B_0] &: \quad (-5x + y - 3b_1 + 3b_2)/18, \\
[C_1, B_0] &: \quad (-5x + y - 3c_1 + 3c_2)/18.
\end{aligned} \tag{E90.16}$$

Direct substitution in the retained genuine matrices gives the exact identities

$$\begin{gathered}
(I - J_2)(T_2 - 7I)B(v) = -(I - J_2)7[v, e_{B_0}], \\
F_2 B(v) = (m/4 - p/6, -m/4 + p/6), \qquad Rv = (L, p, m), \\
BQ_1 \in M_{45,10}(\mathbb{Z}_3), \quad 3BQ_3 \in M_{45,10}(\mathbb{Z}_3), \quad 9B \in M_{45,10}(\mathbb{Z}_3).
\end{gathered} \tag{E90.17}$$

The last assertions can be checked coefficient by coefficient in (E90.16), or from the saved exact matrix products. Hence $rB(v)$ is an actual relative central correction on precisely the proved zero loci. On E.89's six vectors it is r times (E89.8). No division outside an established integral image is used.

After choosing horizontal Smith representatives and this particular correction, the remaining source coordinates are exactly

$$\mathcal{P}_2 \oplus \mathcal{K}_2 = (\mathbb{Z}_3^{24} \oplus C_3^9 \oplus C_9^5) \oplus \mathbb{Z}_3^{22} \tag{E90.18}$$

from E.85. A geometric conjugator first aligns the abelian input; its fixed Hall lift is $\delta^{1/2} g_b$, where $\delta = J(g_b) g_b^{-1}$. Apply its literal action on both section values, retaining the central terms. The only remaining gauge is the central one in E.85. These steps prove the source types in (E90.5) and give their forward and inverse coordinates.

The exact joint relation and its full conjugacy. Before choosing residues, use the two horizontal words in (E90.3) with arbitrary central exponents N, M, and write the relative value as $(y^r c^{a_1}, y^r c^{a_2})$. E.80's actual Hall multiplication and the two operators $T_{-5/2}, T_{-1/2}$ give the source relation

$$2rp - 3rm + 6(a_1 - a_2) = 0. \tag{E90.19}$$

Its complete fixed geometric group is (w, Iw), where $w = x^b y^b c^z$. The exact action is

$$\begin{aligned}
&p \longmapsto p - 6b, && L \longmapsto L, && m \longmapsto m,\\
&N \longmapsto N - 48z - 7Lb - 42b^2 + 6bp + 15b,\\
&M \longmapsto M - 48z + 7Lb - 42b^2 - 18bm + 6bp + 11b,\\
&a_1 \longmapsto a_1 + rb, && a_2 \longmapsto a_2 - rb.
\end{aligned} \tag{E90.20}$$

First take $b = (p - p_0)/6$, where p_0 is the chosen residue in $\{0, 1, 2\}$. Then normalize the horizontal central mean by its actual $-48z$ change. This gives (E90.3) with the invariants

$$L,\ m,\ p_0, \quad \rho = N - M + 1, \quad s = (a_1 + a_2)/2, \quad \epsilon = (N + M - 5)/2 \pmod 3. \tag{E90.21}$$

All central expressions here are evaluated after the first, nonlinear, change in (E90.20). There is no nonzero lower stabilizer, since $6b = 0$ implies $b = 0$ in $\mathbb{Z}_3$. The remaining central normalization is exactly the stated central gauge. Substitution of the representatives (E90.3) gives the two equal norm defects

$$u_i T_{\mu_i}(a_i) u_i^{-1} a_i^{-7} = c^{126\sigma + 21rm/2}, \qquad (\mu_1, \mu_2) = (-5/2, -1/2). \tag{E90.22}$$

This is a direct verification of the full local source relation for every asserted target point.

For a unit r, the horizontal zero locus is exactly that of E.89. Its genuine character ϑ has value $2m$, so the actual mean residue is m. The particular correction (E90.17) has relative mean zero. E.85's complete coefficient map makes that mean run through $3\mathbb{Z}_3$ and makes ρ arbitrary. This proves the first row of (E90.4), with all E.89 inverse maps unchanged except for the factor r in the particular correction.

For the two nonunit cases we must retain the additional finite horizontal freedom. The kernel of the horizontal map to $H^1(\Gamma, \mathcal{J}_1)$ is E.87's actual

$$\mathcal{P}_K \simeq \mathbb{Z}_3^4 \oplus C_3^3. \tag{E90.23}$$

Its obstruction under δ has exponent three, and therefore every class of this kernel is permitted for nonunit r. In E.87's retained Smith coordinates $(t_1, t_2, t_3, z_1, z_2, z_3, z_4)$, the genuine horizontal mean phase is the following evaluated character:

$$\psi = \vartheta/2 = t_1/2 - t_2 + t_3 + z_1 + z_4/2 \pmod 3. \tag{E90.24}$$

This follows by applying any of the three actual possible rows (E89.21) to the exact kernel Smith inverse. Their restrictions are identical. The third torsion vector is literally $2(B_1 - B_2)$, has phase one, and its triple is in the displayed fixed gauge. Thus ψ is onto, with complete inverse fibres

$$t_3 = \epsilon - t_1/2 + t_2 - z_1 - z_4/2 \pmod 3, \qquad (t_1, t_2) \in C_3^2, \quad (z_1, z_2, z_3, z_4) \in \mathbb{Z}_3^4. \tag{E90.25}$$

Take the right side's residue in $\{0, 1, 2\}$. This gives the actual type $\ker \psi \simeq \mathbb{Z}_3^4 \oplus C_3^2$.

Here is an inverse construction over every target first-stage coordinate. A literal lift is

$$\begin{gathered} v_0 = -pf_2 + (p - L)B_0 + (p - m)C_1, \qquad Rv_0 = (L, p, m), \\ \eta = \vartheta(g_{f_1}) \in \mathbb{F}_3, \qquad \epsilon_0 = -p(p + \eta + 2)/2 \pmod 3. \end{gathered} \tag{E90.26}$$

For valuation-one r, take $p = 0$; for $r \in 9\mathbb{Z}_3$ take any of its three residues. These are valid horizontal inputs by (E90.2). Evaluate the actual joint Hall image of g_{v_0} and of the actual horizontal value $u_0 g_{v_0}$. Its finite mean phase is exactly ϵ_0: (E89.21) gives $\vartheta(g_{v_0}) = -(\eta + 1)p$, and multiplication by u_0 adds $6m - 2p$ to the sum of the two central changes. The term $p(p - 1)$ in ϑ then gives the displayed formula. This retains the actual marking coefficient η; it is not assigned a value. Add the required residue multiple of $2(B_1 - B_2)$ to make that phase ϵ. All other horizontal preimages are obtained by (E90.25). This step uses the actual Hall image and does not assign a value to the remaining handle coefficient in (E89.21).

The difference between the resulting actual central pair and the desired horizontal pair in (E90.3) has sum divisible by three. Apply the literal right inverse (E89.25) of $F_2 : A_2 \to L_2$ to that difference. It changes no abelian coordinate or norm defect. The relative correction $rB(v)$ already has the required difference and mean zero. Finally add ρz_* and σa_* of (E85.13). This constructs every claimed target point and proves the second and third rows of (E90.4).

Conversely apply (E90.20) to an actual target image and the unique fixed abelian lift of that gauge to its intrinsic source. Formula (E90.17), the actual phase (E90.24), and E.85's full coefficient image force exactly the displayed restrictions. After subtracting the constructed origin, E.85's explicit Smith and saturated-kernel inverses recover the remaining forty-four free, nine order-three, and five order-nine coordinates. In the nonunit cases append the four free and two order-three coordinates of (E90.25); in the unit case append E.89's four free coordinates. This proves the fibre types (E90.5), including their complete inverses and absence of a further stabilizer quotient. Every map is continuous: the residue choices are on finite sets and the other maps are continuous Hall lifts or integral maps on their proved images. The compact parameter spaces give exactly the stated orbit topologies.

The actual global path comparison. On the cubic let $a[x : y : z] = [x : z : y]$. It maps $O = [1 : -1 : 0]$ to $B_0 = [1 : 0 : -1]$ and satisfies

$$(y/x) \circ a = z/x, \qquad a^2 = 1. \tag{E90.27}$$

At every three-power level this sends the chosen formal parameter root to the same root. It is therefore an exact rational tangential basepoint isomorphism, whose composition square is the identity. E.34 and the actual tangent transport in E.78 prove that the action at O factors through the fixed Γ. Conjugating by this rational isomorphism proves the same factorization for the action at B_0.

Let $g \in \ker(G_{\mathbb{Q}} \to \Gamma)$ and let p be any path between these tangents. Both based actions of g are the identity. The difference between $g(p)$ and p therefore centralizes the full geometric group at either endpoint. That group is free pro-three of rank ten and has trivial centre. Indeed use the injective Magnus map of E.41. The first nonzero homogeneous term of a nonidentity central element would commute with all the independent variables. The centre of

the free associative algebra on two or more variables consists of scalars: commuting with one variable forces each word to be a power of it, and commuting with a second then kills every positive-degree term. This contradicts that first homogeneous term. Hence $g(p) = p$. This proves factorization of the entire path torsor, with its actual endpoint actions. Choose the order-two-fixed path and its fixed geometric normalization as in E.85. The latter still factors through Γ, since its kernel fixes every geometric element as well. This strengthens E.85's path construction by the displayed basepoint morphism and centre calculation.

Put $u_g = pg(p)^{-1}$. The literal path composition law proves its cocycle identity and the first equality in (E90.6). For any Tate cocycle β one then has

$$\begin{aligned} d^{\beta(g)} u_g \, g(d^{\beta(h)} u_h) &= d^{\beta(g)+\chi(g)\beta(h)} u_{gh} \\ &= d^{\beta(gh)} u_{gh}. \end{aligned} \tag{E90.28}$$

Changing β by the coboundary $(\chi-1)k$ changes this section by the actual geometric conjugation by d^{-k}. Thus (E90.6) is a well-defined map on cohomology classes.

It is injective by a concrete morphism. The rational unit $f = -y/x : U_3 \to \mathbb{G}_m$ has value one at O, has order one at B_0, and has leading coefficient -1 in the chosen parameter y/x. Its induced map on pro-three fundamental groups sends d to $1 \in \mathbb{Z}_3(1)$. The coherent 3^n-th roots of this leading coefficient can all be chosen as -1. Consequently the image path cocycle is a coboundary: at every Kummer cover the source value one and this tangent value have rational chosen roots. The induced cohomology map sends $[z_\beta]$ exactly to $[\beta]$. This is a left inverse to the global map and, after restriction, to its local map.

The complete Kummer restriction and its topology. The Kummer exact sequence and Hilbert 90 give $H^1(F, \mu_{3^n}) = F^\times / F^{\times 3^n}$ for $F = \mathbb{Q}, \mathbb{Q}_3$; the original proofs used here are [Mil20b, II, Proposition1.22; III, §4, Step2]. The inverse-limit comparison is exact here. Indeed $H^0(F, \mu_{3^n}) = 1$ for both fields. Choose representatives of compatible finite cocycle classes. At the next level their difference is a coboundary; lift its coefficient and correct it. This produces compatible continuous cocycles. For a cocycle whose classes are all zero, the boundary coefficients are unique and therefore compatible, giving one continuous boundary. Matching the same finite-level kernels proves the topological comparison, as well as its algebraic bijection.

Unique rational factorization, with the sign removed by the odd power maps, now gives (E90.7). A residue modulo 3^n has finite support, and compatibility is exactly the restricted condition stated there. These finite-exponent quotients are discrete; in particular reduction modulo three has the full infinite direct sum as its quotient. This specifies the topology used in every global fibre assertion.

Locally the exact multiplicative decomposition is

$$\mathbb{Q}_3^\times = 3^{\mathbb{Z}} \times \{1, -1\} \times (1 + 3\mathbb{Z}_3). \tag{E90.29}$$

The logarithm converges on the last factor and gives an isomorphism with $3\mathbb{Z}_3$. This follows directly from the logarithm and exponential series: their terms tend to zero there, their composition identities hold by convergent formal substitution, and their first terms are units times the argument. Since $v_3(\log 4) = 1$, 4 gives its $\mathbb{Z}_3$ basis. The sign factor has bijective three-power maps and contributes zero to completion. This proves the stated local basis, including the torsion assertion.

Every ℓ_ℓ in (E90.8) is integral. For a restricted exponent family, the terms $a_\ell \ell_\ell$ are uniformly zero modulo any fixed 3^n except at finitely many primes. Their sum therefore converges, independently of ordering, and is continuous for (E90.7). The equality $(-2)^2 = 4$ proves $\ell_2 = 1/2$ exactly. Substitution gives both directions of (E90.9). The inverse preserves

restrictedness and each finite-level kernel, so this is a split topological isomorphism

$$\mathscr{R}_{\mathcal{P}} \simeq \mathbb{Z}_3^2 \times \mathscr{R}_{\mathcal{P}\setminus\{2,3\}}. \tag{E90.30}$$

The scalar instance of E.77, with coefficient $\mathbb{Z}_3(1)$, makes its normalized two values arbitrary in $\mathbb{Z}_3^2$: the relation is zero and the fixed geometric gauge is zero. Thus evaluation on (t, r_s) is an isomorphism. The actual κ_3 already has values $(1, 0)$, because its marked quotient value is one and it factors through Γ. The local basis $(3, 4)$ then implies that the second value α of κ_4 is a unit. This proves the actual change of basis (E90.10).

For completeness, a chosen global root cocycle can always be normalized at the local order-two element by the actual root change $k = \kappa(c)/2$: adding $(\chi - 1)k$ makes its value there zero. This is also a global coboundary, and its geometric transport is d^{-k} as in (E90.28). It does not require identifying that local involution with a torsion element of an absolute local Galois group. All evaluations are in the actual relative source of E.74–E.77.

The path factorization gives $u_{r_s} = 1$, and $u_t = u$ of (E85.20). The normalized local values of (E90.11) are $(-V, r)$ by (E90.10). Consequently (E90.6) restricts literally to $(d^{-V}u, d^r) = (u_0, d^r)$. The injective Kummer comparison and (E90.9) now give all global cusp preimages, including their exact completion topology.

The full global unramified map and its fibres. The actual global and local maps onto Γ are surjective by E.34 and E.74. Inflation on nonabelian cohomology is injective: an equality up to one geometric conjugator after inflation gives the same equality on every element of Γ by choosing a preimage. A normalized section factors through Γ precisely when its E.77 relative value is the identity. This uses the actual presentation and the closed normal generation of its kernel.

For N_2, E.41's recursive coordinates are $\mathcal{P}_1 \times \mathcal{P}_2$, of the type displayed in (E90.12). The full joint map is onto: lift any normalized horizontal joint value through the actual surjection $N_2 \to \mathcal{J}_2$, and assign its conjugate value to τ'. The free group P and its actual order-two swap give a continuous Γ section. In the local coordinates this is exactly $r = 0$, $a = 1$, hence $\sigma = 0$, with the remaining five coordinates of (E90.3).

The complete inverse fibre uses the four free and two order-three horizontal phase kernel coordinates (E90.25). Once its actual Hall image is aligned, the horizontal central coefficient map has kernel $\mathbb{Z}_3^{23} \times C_3^9 \times C_9^5$ by E.85. Its explicit inverse (E85.17), with the relative coordinates omitted, recovers all these coordinates. Their product is exactly the last type in (E90.12). This calculation retains the same actual Hall correction and full lower gauge as the local map; it is not inferred by subtracting dimensions.

Every such N_2 section lifts to N by the free-source construction of E.41. Its recursive normal form uses actual corrected lifts in each degree and has unique aligning conjugators. The map to $\mathcal{J}_2$ factors through N_2, so the remaining independent coordinates are exactly $\mathcal{C}_j(N)$ for $j \geq 3$. This proves the literal comparison map, its inverse, and (E90.13). For arbitrary global sections the restriction of the actual Hall defect is its local defect; thus (E90.1)–(E90.5) remain necessary at this place whenever their relative abelian value is specified. Sufficiency and global fibres have been proved above for the cusp and unramified families.

The certificate checks all 81 obstruction values in eight multiplier cases, the three integral Smith systems and all inverses, 567 actual particular corrections, 2,187 phase inverse coordinates, and the entire symbolic target norm and gauge. It also checks the exact tangent swap, 168 prime–precision logarithmic identities, 1,134 localization inverses, and the literal Kummer source change of basis. All inputs, exact outputs and hashes are retained. □

Ordinary class actions and the remaining prime-class transvection

E.91. All vertical ordinary actions, the original class-group types, and an exact ninth-genus refinement [Proved]

Use the literal fields, roots, prime labels and norm maps of E.86:

$$L_d = \mathbb{Q}(\zeta, \beta_d), \qquad B_d = L_d(\xi), \qquad \xi^3 = \zeta, \quad \beta_{d+1}^3 = \beta_d, \quad \beta_d^{3^d} = 17\zeta. \tag{E91.1}$$

Write $\mathcal{C}_d = \mathrm{Cl}(B_d)\{3\}$, let $R_d \subset \mathcal{C}_d$ be the subgroup generated by its six primes above $3, 17$, and put $D_d = \mathrm{Cl}(L_d)\{3\}$. The three-primary ideal classes are written additively. The actual vertical generator $\sigma_d(\beta_{d+1}) = \zeta\beta_{d+1}$ acts trivially on the *entire ordinary* group $\mathcal{C}_{d+1}$, for every $d \geq 1$. Consequently the adjacent ordinary norm and extension maps satisfy

$$\begin{gathered} N_d : \mathcal{C}_{d+1} \twoheadrightarrow \mathcal{C}_d, \qquad j_d N_d = 3, \qquad \ker N_d \subset \mathcal{C}_{d+1}[3], \\ \ker j_d = N_d(\mathcal{C}_{d+1}[3]). \end{gathered} \tag{E91.2}$$

These are maps of ordinary ideal classes; their further quotients by R_d are the specified S-class maps of E.86.

Let $q_d = [\mathfrak{q}_{3,d}]$, and write α_d for the exponent called a_d in E69.5, so that $\mathrm{ord}(q_d) = 3^{\alpha_d}$. For every $d \geq 2$,

$$D_d \simeq C_{3^{\alpha_d+1}} \oplus C_3, \qquad \langle q_d \rangle = 3D_d, \qquad D_d/3D_d \xrightarrow{\sim} \mathrm{Cl}_S(L_d)\{3\}. \tag{E91.3}$$

The last arrow is the actual removal-of-S-primes map; its basis is the pair of specified ideal classes in E86.4. In particular those two ordinary ideal classes generate D_d. Adjacent norms on the D_d are surjective and send q_{d+1} to q_d. Thus the whole ordinary tower, as well as its specified prime, stabilizes under norm once the proved finite exponent h of E.69 stabilizes:

$$\varprojlim_{N} D_d \simeq C_{3^{h+1}} \oplus C_3, \qquad (q_d)_d \text{ generates its subgroup of triples.} \tag{E91.4}$$

The isomorphism is a choice of a cyclic basis in the finite stable group. Its map to every level is the actual iterated ideal norm.

The next unevaluated ordinary class group has a small, explicit residual comparison. On B_3 put

$$g(\xi) = \xi, \quad g(\beta_3) = \xi\beta_3, \qquad h(\xi) = \zeta\xi, \quad h(\beta_3) = \beta_3. \tag{E91.5}$$

Let $\mathfrak{p}_3$ be the E.86 prime above 73 with $\xi = 55$ and $\beta_3 = 22$. Then

$$\begin{gathered} W = (g-1)\mathcal{C}_3 = \langle (g-1)[\mathfrak{p}_3]_3 \rangle, \qquad 3W = 0, \quad |W| \leq 3, \quad h|_W = 1, \\ 0 \longrightarrow W \longrightarrow \mathcal{C}_3 \xrightarrow{\mathrm{Art}} A_9 \longrightarrow 0, \qquad A_9 = C_9^2 \oplus C_3. \end{gathered} \tag{E91.6}$$

Here Art is restriction to the actual ninth-genus field $\mathcal{G}_9$ of E.76, not an abstract quotient of the same order. In its coordinates $\Gamma_9 = C_9^3 \oplus C_3$, the actual Kummer character and mutually inverse kernel coordinates are

$$\begin{gathered} \chi_9(x,y,z,t) = 8x + 7y + 2z + 3t \pmod 9, \\ A_9 \longrightarrow \ker \chi_9 : \quad (y,z,t) \longmapsto (-2y + 2z + 3t, y, z, t), \\ (x,y,z,t) \longmapsto (y,z,t). \end{gathered} \tag{E91.7}$$

All cyclic moduli are those in the preceding display. In these A_9 coordinates the action, primes and S-quotient are

$$H=\begin{pmatrix}4&3&0\\8&7&6\\1&0&1\end{pmatrix},\qquad \mathrm{Art}[\mathfrak{p}_3]=(1,8,2),$$

$$\begin{array}{c|ccc|ccc}&Q_1&Q_2&Q_3&T_1&T_2&T_3\\\hline y&6&6&6&6&3&0\\z&0&6&3&5&8&5\\t&2&2&2&1&1&1\end{array},\qquad \mathrm{Art}(R_3)=3C_9\oplus C_9\oplus C_3, \tag{E91.8}$$

$$A_9/\mathrm{Art}(R_3)\longrightarrow C_3:(y,z,t)\longmapsto y\pmod 3.$$

The prime labels are transported from E.81 by their unique totally ramified extensions; h cycles both displayed triples.

These actual action maps impose further unconditional bounds:

$$\begin{gathered}\exp(\mathcal{C}_3)=9,\qquad \mathcal{C}_3\in\{C_9^2\oplus C_3,C_9^3,C_9^2\oplus C_3^2\},\\ [Q_1]+[Q_2]+[Q_3]=3[Q_i]\in W\quad(i=1,2,3),\\ \mathrm{ord}([Q_2]-[Q_1])=3,\qquad [Q_3]-[Q_1]=-([Q_2]-[Q_1]),\\ \langle[T_1],[T_2],[T_3]\rangle\simeq C_9\oplus C_3,\quad \text{Art is injective on this subgroup.}\end{gathered} \tag{E91.9}$$

The membership in the displayed finite set is a proved group-type bound. It does not select its remaining alternative.

The exact comparison with the original prime is

$$\begin{gathered}N_{B_3/L_3}[Q_i]=q_3,\qquad j_{B_3/L_3}(q_3)=3[Q_i],\\ N_{B_3/L_3}(\mathcal{C}_3)=\ker\bigl(\mathrm{Art}_{B_3/L_3}:D_3\to C_3\bigr),\qquad [D_3:N(\mathcal{C}_3)]=3.\end{gathered} \tag{E91.10}$$

In the S-coordinates of E86.4 the last Artin character reads the first coordinate. In particular the exponent-nine bound for its norm image remains compatible with both original possibilities $D_3=C_9\oplus C_3$ and $D_3=C_{27}\oplus C_3$. The actual value of $3[Q_i]\in W$, and therefore the proposed all-level principality of the three-prime product, remain a specified arithmetic step.

One additional computation records the precise result of an attempted unit construction. Take the full B1 S-unit basis of E.79, its six valuation rows V, and its six cubic local norm characters M. The class matrix of the six B2 S-primes in their exponent-three subgroup is

$$C_S=\begin{pmatrix}2&2&2&2&0&1\\1&1&1&2&2&2\\0&0&0&2&2&2\end{pmatrix}.\quad \dim\ker(C_SV \bmod 3)=13,\quad \mathrm{rank}_{\mathbb{F}_3}M\ker(C_SV)=3. \tag{E91.11}$$

The proof constructs the corresponding actual ordinary B2 units and compares their B3/B2 symbols. This image equals the already computed rank-three image of the included ordinary B1 units. Thus this particular construction has an evaluated image; the image of all ordinary B2 units is not assigned this rank.

The manuscript anchor remains LM §51.16, folios733v–736v, scans172–178, via the literal positional field norms of E.46, E.69, E.72 and the all-level divisor comparison of E.88.

Proof. The actual automorphisms and prime restrictions. The substitutions on B_{d+1} are $g_d(\xi)=\xi$, $g_d(\beta_{d+1})=\xi\beta_{d+1}$ and $h_d(\xi)=\zeta\xi$, $h_d(\beta_{d+1})=\beta_{d+1}$. They preserve the literal defining relations, have orders nine and three, and satisfy

$$g_d^3=\sigma_d,\qquad h_dg_dh_d^{-1}=g_d^4,\qquad g_d|_{B_d}=\sigma_{d-1}\quad(d\geq 2),\qquad h_d|_{B_d}=h_{d-1}. \tag{E91.12}$$

For example the first restriction sends β_d to $\xi^3\beta_d = \zeta\beta_d$ and fixes ξ. These are maps of the indicated fields, with no claim that the g_d themselves form a compatible sequence.

Every S-prime is totally ramified at every adjacent vertical step. Hence σ_{d-1} fixes each of the six S-primes in B_d, and g_d fixes its unique upper prime in B_{d+1}. The h_d action permutes the three primes over each original prime in L_{d+1}, since B_{d+1}/L_{d+1} is the actual S-split unramified cubic extension. It is the same permutation as at B1, transported by the unique vertical prime extension.

For $d \geq 2$ put $C = \mathcal{C}_{d+1}$, $R = R_{d+1}$ and $n = g_d - 1$. E.86 gives $C/R = C_3$. An automorphism of three-power order acts identically on C_3, so $nC \subset R$. The preceding prime calculation gives $nR = 0$. Since $3C \subset R$, it also gives $3nC = 0$. Thus $n^2 = 0$ and

$$(\sigma_d - 1)C = ((1+n)^3 - 1)C = 0. \tag{E91.13}$$

For $d = 1$ the exact action on $\mathcal{C}_2$ was proved in E86.18.

For either adjacent B or L extension, ordinary ideal norm is surjective. Here is the class-field map underlying this fact. If E/F is the indicated cyclic cubic extension and H/F the Hilbert three-class field, then $E \cap H = F$: a nontrivial intersection would be all of E, contradicting its total ramification at an S-prime. Restriction of Frobenius identifies the class norm image with $\mathrm{Gal}(H/(E \cap H))$; equivalently, in the compositum EH every Frobenius $(a, 1)$ supplies a degree-one prime of E whose ideal norm represents a. This is the actual norm/Artin comparison of class field theory [Mil20b, V, Proposition3.3, Theorem3.6 and Example3.9; VIII, Theorem7.4]. The identity $jN = \sum \sigma^i$ follows on each prime ideal from its ramification and residue degrees, and hence on all classes. Equation (E91.13) gives $jN = 3$. It kills the norm kernel by three. Finally $j(x) = 0$ and $N(y) = x$ are equivalent to $3y = 0$, proving the last equality of (E91.2).

The full abelian-group alternatives for the L tower. The ideal (β_d) is the unique prime over seventeen. Thus removing S quotients D_d by precisely $\langle q_d \rangle$. E.86 identifies that quotient with C_3^2, and E.72 gives the exact order 3^{α_d} of its kernel.

For completeness, let a finite abelian three-group D have a cyclic subgroup $Q = \langle q \rangle$ of order 3^a, with $D/Q = C_3^2$ and $a \geq 1$. Then $3D \subset Q$, so $3D$ is cyclic. In the elementary-divisor decomposition at most one factor can have order greater than three. Counting the order $|D| = 3^{a+2}$, and checking whether q is primitive modulo $3D$, leaves exactly the two possibilities

D	Q
$C_{3^{a+1}} \oplus C_3$	$3D$,
$C_{3^a} \oplus C_3^2$	a primitive cyclic direct summand.

(E91.14)

To verify the second row directly, a generator q of maximal order with nonzero reduction can be made the generator of the long cyclic factor by an automorphism of the displayed group; its remaining two generators have order three. In the first row both Q and $3D$ have order 3^a, giving their equality. This also treats $a = 1$, where the second group is C_3^3.

The first row holds at $d = 2$ by E.86. Suppose it holds at the preceding level. In the second alternative at the next level, the group would be generated by q_d and elements of order three. Its actual norm, which sends q_d to q_{d-1}, could therefore only have image in

$$\langle q_{d-1} \rangle + D_{d-1}[3] = 3C_{3^{\alpha_{d-1}+1}} \oplus C_3. \tag{E91.15}$$

This has index three in D_{d-1}, contradicting the proved norm surjectivity. Induction proves (E91.3). Its two specified ideal classes generate because their images form a basis of $D_d/3D_d$; the elementary finite Nakayama argument lifts that basis to generators of D_d. E.69 proves $\alpha_d = h$ for all sufficiently large d. The surjective norms then connect finite groups of the same

order and are isomorphisms. Projection onto any such stable level proves (E91.4), including the actual compatible prime.

The exact genus quotient and its one possible kernel. On B_3 the preceding calculation gives $n = g - 1$ with $nR_3 = 0$, $3n\mathcal{C}_3 = 0$ and $n^2 = 0$. The prime $\mathfrak{p}_3$ generates $\mathcal{C}_3/R_3$ by E.86, so it generates the image of n. The actions of g and h commute on $\mathcal{C}_3$, since their field commutator is $g^3 = \sigma_2$, whose class action is trivial. Also $(h-1)\mathcal{C}_3 \subset R_3$. Therefore $h(n[\mathfrak{p}_3]) = n[\mathfrak{p}_3]$, giving every assertion about W in (E91.6).

Let $\mathcal{H}$ be the full Hilbert three-class field of B_3. It is Galois over B_1, since B_3/B_1 is cyclic of degree nine. The commutator subgroup of $\mathrm{Gal}(\mathcal{H}/B_1)$ is exactly $(g-1)\mathcal{C}_3$: the subgroup $\mathcal{C}_3$ is abelian and the quotient is cyclic, so commutators are precisely those with a lift of its generator. Consequently its maximal abelian-over-B_1 subfield is the genus field $\mathcal{G}_9$ computed in E.76. Restriction proves the exact sequence (E91.6), including its kernel.

The retained computation re-evaluates the Kummer character on the actual ray ideal generators. For a prime coprime to S it is determined by $\overline{\beta}_1^{(N\mathfrak{p}-1)/9} = \overline{\xi}^{\chi_9(\mathfrak{p})}$. Transport through the full integral Smith maps gives $(8,7,2,3)$ in the four nontrivial genus coordinates. Substitution proves both inverse arrows of (E91.7). The actual cyclotomic automorphism h on the ray quotient gives the matrix H. The six ramified Frobenii are represented by the explicit Chinese-remainder local uniformizers and coprime ideals of E.76; their columns are (E91.8). Their lattice image is exactly $3\mathbb{Z} \oplus \mathbb{Z} \oplus \mathbb{Z}$ modulo $(9,9,3)$. The prime with $\xi = 55$, $\beta_1 = 22$ has ray residue $a - \xi = 33$ modulo 73 and gives $(1,8,2)$. The literal equalities $22^3 = 63$, $63^3 = 22$ modulo 73 identify its norm with the specified $\mathfrak{p}_3$. This proves the final S-quotient and every displayed prime map.

Retaining every central extension and action parameter. If $W = 0$, the remaining assertions about exponent and primes follow directly from (E91.8). For the other case let w denote its actual generator $(g-1)[\mathfrak{p}_3]_3$, and choose any lifts Y, Z, T of the three coordinates of A_9. The entire abelian extension has the presentation

$$3w = 0, \qquad 9Y = aw, \qquad 9Z = bw, \qquad 3T = cw, \quad a,b,c \in \mathbb{F}_3. \tag{E91.16}$$

Indeed these relations give a group of order $3|A_9|$, and the surjection to $\mathcal{C}_3$ is therefore an isomorphism. Since $h(w) = w$, its most general lift of the actual matrix is

$$\begin{aligned} hY &= 4Y + 8Z + T + ew, \\ hZ &= 3Y + 7Z + fw, \\ hT &= 6Z + T + kw, \qquad e,f,k \in \mathbb{F}_3. \end{aligned} \tag{E91.17}$$

Applying h to $9Y = aw$ forces $b = 0$. Put $A = \mathrm{diag}(9,9,3)$. The exact matrix identity

$$A^{-1}(H^3 - I) = \begin{pmatrix} 49 & 39 & 24 \\ 112 & 88 & 54 \\ 15 & 12 & 6 \end{pmatrix} \tag{E91.18}$$

then shows that $h^3 = 1$ forces $a = 0$. The correction from (e,f,k) is zero, because $I + H + H^2$ is zero modulo three. No value is assigned to c. The resulting presentations are $C_9^2 \oplus C_3^2$ for $c = 0$ and C_9^3 for $c \neq 0$. Together with $W = 0$ these prove the exponent and type bounds.

Write $v = (6,0,2)^t$, the actual image of $[Q_1]$. A lift differs from $6Y + 2T$ only by a multiple of w. That difference contributes zero to its h-norm. The possible correction from (E91.17) also vanishes, since $(2I + H)v = 0 \pmod 3$. The exact relation vectors are

$$\begin{aligned} A^{-1}(I + H + H^2)v &= (34,80,14)^t, \\ A^{-1}(3v) &= (2,0,2)^t. \end{aligned} \tag{E91.19}$$

With $a = b = 0$ both expressions represent $2cw$. Since h cycles the Q-primes, this proves their sum equals three times each prime class. Their differences are therefore killed by three; the nonzero genus difference proves the asserted exact order and the relation between the two differences.

For the T-primes the literal divisor of β_3 gives $[T_1] + [T_2] + [T_3] = 0$. The matrix image of $[T_1]$ is $(6, 5, 1)$, of order nine, while that of $[T_2] - [T_1]$ is $(6, 3, 0)$, of order three. In (E91.16), their triples differ by $-9Y + 9Z$ and a multiple of $3w$, which vanish because $a = b = 0$. They consequently generate $C_9 \oplus C_3$; their two genus images already generate a group of order27. This proves injectivity on the T-subgroup.

The original prime and the precise limit of the unit attempt. The horizontal extension is unramified and each original S-prime splits into three degree-one primes. Ideal norm and extension give exactly (E91.10), with the sum now evaluated by (E91.19). The class norm image has index three, because B_3 itself is the indicated subfield of the Hilbert class field of L_3. Restriction of Frobenius identifies it with the stated Artin kernel. Thus an exponent-nine norm subgroup does not give an exponent-nine bound on all of D_3; the actual index-three kernel in $C_{27} \oplus C_3$ can have exponent nine. The remaining value is stated explicitly.

Finally let u be a B1 S-unit whose valuation vector v has zero class in the B2 S-prime subgroup. The full principal S-lattice in E86.11 supplies an actual S-unit w_0 in B2 with divisor v in the transported prime basis. The included element u has divisor $3v$, so $\varepsilon = u/w_0^3$ is an ordinary B2 unit. Its cubic norm characters for B3/B2 are exactly those of u for B2/B1. At each of the six completions this is the restriction/norm identity for the Hilbert pairing,

$$(\operatorname{res} u, \beta_2)_{B_2,3} = (u, N_{B_2/B_1}\beta_2)_{B_1,3} = (u, \beta_1)_{B_1,3}. \tag{E91.20}$$

The cube w_0^3 contributes zero. This is the exact pairing square proved from the restriction/corestriction identity in E.79 and [Mil20b, III,Theorem3.1]. Thus the full image of this construction is $M \ker(C_S V \bmod 3)$. The complete fifteen-column matrices, the thirteen-dimensional kernel, and its image are retained. Their ranks are respectively the values in (E91.11), with the same rank-three image as the first nine ordinary-unit columns. This records what the attempt computes without assigning its image to the full ordinary-unit group.

The portable program reconstructs every genus input from the certified degree-eighteen B1 field and retains full Smith maps and inverses. It verifies all 243 simple-coordinate round trips, all 27 abelian extension parameters and all 81 remaining lifts of h, as well as the exact class-type alternatives and the unit-lifting matrix. The initial row-shape comparison error and its actual output are retained separately; no assertion uses that failed execution. □

The entire global intrinsic abelian localization image

E.92. Exact global abelian localization, its finite carry and every fibre [Proved]

Retain the actual cubic group, source marking and integral lattices of E.37, E.74 and E.77. Put

$$W = T_3 T_D, \qquad V = T_3 J_{E,D}, \qquad E_3 = T_3 E, \qquad 0 \longrightarrow W \longrightarrow V \xrightarrow{\pi_E} E_3 \longrightarrow 0. \tag{E92.1}$$

Here W is the span of the eight positive peripheral vectors, and π_E is the literal first-two-coordinate map in the basis of E.37. Its rational torus is $T_D = \mathbb{G}_m^2 \times (\operatorname{Res}_{K/\mathbb{Q}} \mathbb{G}_m)^3$, where $K = \mathbb{Q}(\zeta_3)$. The two rational factors are A_0, B_0, and the three ordered pairs are $(A_1, A_2), (B_1, B_2), (C_1, C_2)$.

Localization of the full global intrinsic abelian cohomology has the exact sequence

$$\begin{gathered}0 \longrightarrow \mathscr{L} \longrightarrow H^1(\mathbb{Q}, V) \xrightarrow{\mathrm{loc}_3} \mathcal{I} \longrightarrow 0,\\ \mathcal{I} = \{[z] \in H^1(\mathbb{Q}_3, V) : \text{the relative value in E.77 belongs to } W\},\\ \mathcal{I} \simeq \mathbb{Z}_3^{14} \oplus C_3^3,\\ \mathscr{L} \simeq \mathscr{R}_{\mathbb{Q}}(\mathcal{P}_{\mathbb{Q}} \setminus \{2,3\})^2 \times \mathscr{R}_K(\mathcal{P}_K \setminus \{\mathfrak{l},\mathfrak{u},\mathfrak{t}\})^3.\end{gathered} \tag{E92.2}$$

The notation $\mathscr{R}_F$ means restricted $\mathbb{Z}_3$ exponent sequences on the displayed prime ideals, with the inverse-limit topology of discrete finite-exponent direct sums. The three omitted prime ideals of K have the actual generators

$$\lambda = 1 - \zeta_3, \qquad u_2 = 1 + \lambda^2, \qquad 2, \qquad \mathfrak{l} = (\lambda), \quad \mathfrak{u} = (u_2), \quad \mathfrak{t} = (2). \tag{E92.3}$$

Their norms are $3, 13, 4$. The sequence (E92.2) has an explicit continuous $\mathbb{Z}_3$-linear section described below. Every localization fibre is exactly the displayed $\mathscr{L}$, with its full forward and inverse coordinates.

In E.87's actual local coordinates the relative value is

$$A = \theta(3f_1 + f_2) + \sum_{d \neq O} a_d e_d. \tag{E92.4}$$

Thus the complete global image is the single exact condition $\theta = 0$. This is an abelian localization criterion, including both image inclusions. In particular every global intrinsic section at any geometric level projects to a local abelian input with $\theta = 0$. The local branch with nonzero relative compact handle in E.87 has no global abelian preimage. Its exclusion uses the actual compact filling and localization maps in (E92.1).

All toric localization maps and their inverses are concrete. The two rational factors use (E90.7)–(E90.9). For a quadratic factor put $k = \mathbb{Q}_3(\zeta_3)$. Choose one generator $\pi_{\mathfrak{p}}$ of every prime ideal of K, using the three literal choices (E92.3). There are exact topological identifications

$$\widehat{K^\times}_3 = C_3 \oplus \mathscr{R}_K(\mathcal{P}_K), \qquad \widehat{k^\times}_3 = \lambda^{\mathbb{Z}_3} \times \mu_3 \times u_2^{\mathbb{Z}_3} \times 2^{\mathbb{Z}_3}. \tag{E92.5}$$

The C_3 coordinate is the exponent of the actual root ζ_3. For $\mathfrak{p} \neq \mathfrak{l}$, write the unique literal local decomposition

$$\begin{gathered}\pi_{\mathfrak{p}} = \omega_{\mathfrak{p}} \zeta_3^{e_{\mathfrak{p}}} v_{\mathfrak{p}},\\ \omega_{\mathfrak{p}} \in \{1, -1\}, \quad e_{\mathfrak{p}} \in \{0, 1, 2\}, \quad v_{\mathfrak{p}} \in 1 + 3\mathcal{O}_k,\\ \log v_{\mathfrak{p}} = a_{\mathfrak{p}} \log u_2 + b_{\mathfrak{p}} \log(-2).\end{gathered} \tag{E92.6}$$

The coefficients $a_{\mathfrak{p}}, b_{\mathfrak{p}}$ belong to $\mathbb{Z}_3$. The actual completed localization is

$$(\epsilon, (t_{\mathfrak{p}})) \longmapsto \left(t_{\mathfrak{l}},\ \epsilon + \sum_{\mathfrak{p} \neq \mathfrak{l}} e_{\mathfrak{p}} t_{\mathfrak{p}} \bmod 3,\ \sum_{\mathfrak{p} \neq \mathfrak{l}} a_{\mathfrak{p}} t_{\mathfrak{p}},\ \sum_{\mathfrak{p} \neq \mathfrak{l}} b_{\mathfrak{p}} t_{\mathfrak{p}} \right). \tag{E92.7}$$

The finite torsion sum and both convergent logarithmic sums are retained. Over a local point (v, e, a, b), choose arbitrary restricted exponents on all primes except the three pivots. The complete inverse fibre is

$$\begin{aligned} t_{\mathfrak{l}} &= v,\\ t_{\mathfrak{u}} &= a - {\sum}' a_{\mathfrak{p}} t_{\mathfrak{p}}, \qquad t_{\mathfrak{t}} = b - {\sum}' b_{\mathfrak{p}} t_{\mathfrak{p}},\\ \epsilon &= e - {\sum}' e_{\mathfrak{p}} t_{\mathfrak{p}} \pmod 3. \end{aligned} \tag{E92.8}$$

The primes on the sums exclude all three pivots. The section takes the literal global completed element $\lambda^{v}\zeta_3^{e}u_2^{a}2^{b}$. In particular the finite μ_3 coordinate is mapped identically, and no torsion remains in the kernel of (E92.7).

The finite elliptic carry is also explicit. Use E.37's representatives of a global class

$$(b,\epsilon,\beta),\qquad b\in\mathbb{Z}_3,\quad \epsilon\in\{0,1,2\},\quad \beta\in H^1(\mathbb{Q},W),\tag{E92.9}$$

where the elliptic lift has value $bf_1+\epsilon f_2$ on τ, its conjugate on τ', and zero on c. Define the following actual toric Kummer tuple:

$$c_T=(3,9,\lambda^2,\lambda,\zeta_3^{-1}\lambda^3)\in\widehat{\mathbb{Q}^{\times}}_3^{\,2}\times\widehat{K^{\times}}_3^{\,3}.\tag{E92.10}$$

If $\epsilon+\epsilon'=\epsilon''+3k$ with $\epsilon''\in\{0,1,2\}$, the exact addition law is

$$(b,\epsilon,\beta)+(b',\epsilon',\beta')=(b+b',\epsilon'',\beta\beta' c_T^{k}).\tag{E92.11}$$

This formula holds for the global set and its local image, with the actual localization of c_T. To give additive coordinates, let $a=v_3(\beta_{A_0})$ and set

$$w=3a+\epsilon,\qquad \gamma=\beta c_T^{-a}.\tag{E92.12}$$

Then $v_3(\gamma_{A_0})=0$. The map $(b,\epsilon,\beta)\mapsto(b,w,\gamma)$ and the inverse

$$\epsilon=w \bmod 3 \text{ in } \{0,1,2\},\qquad a=(w-\epsilon)/3,\qquad \beta=\gamma c_T^{a}\tag{E92.13}$$

are mutually inverse topological group coordinates. Localization is the identity on (b,w) and is the literal toric localization on γ. For the A_0 factor its local section is 4^{z}; for the B_0 factor it is $3^{z}4^{z'}$; for each quadratic factor it is the section following (E92.8). These give the promised actual linear section in (E92.2), including every finite carry.

All maps in the statement are maps on the complete global abelian group. They do not assert a class-two lift for all of its points. The remaining intrinsic wild class-two calculation is thereby reduced to the eight toric relative directions in (E92.4), with the actual global ramified relations of E.73 still retained. The manuscript anchor is §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d), follow-up(4).

Proof. The exact global and local compact comparisons. The sequence (E92.1) is the actual Tate-module sequence of the generalized Jacobian in E.37. Its inclusion uses the eight peripheral columns; its projection selects f_1,f_2. The retained matrices intertwine both maps. In particular T acts on W by $7I$, and J by minus the permutation of the three indicated pairs, with -1 on A_0,B_0.

We first check the full scope of the global proper elliptic assertion. At every finite prime $p\neq 3$, E has good reduction and E_3 is unramified. The restriction of a continuous E_3-valued cocycle to inertia is a homomorphism. Its wild prime-to-three part has zero image, and its tame pro-three part is determined by a vector satisfying $Fv=pv$. The good-reduction eigenvalues have complex magnitude $\sqrt{p}$, so this vector is zero. This argument applies integrally: $F-p$ is invertible over $\mathbb{Q}_3$ and the lattice is torsion free. The cocycle itself is therefore unramified away from three.

Over K, both its arithmetic action and its cocycle form a homomorphism into a pro-three semidirect group: $E[3]$ is K-rational. Consequently this homomorphism factors through the actual maximal pro-three group P used in E.34. The order-two descent gives precisely Γ. Thus E37.22 computes the entire global group, in its inverse-limit cohomology topology:

$$H^1(\mathbb{Q},E_3)=H^1(\Gamma,E_3)=\mathbb{Z}_3t_1\oplus C_3t_2.\tag{E92.14}$$

The local map onto Γ is surjective by E.74. Hence inflation of (E92.14) to the local group is injective, and in the actual E.77 coordinates its image is exactly zero relative value. Indeed the two relative generators normally generate the kernel of the marked quotient. The proper elliptic local coordinates are explicitly

$$U_E = (b, s), \qquad A_E = \theta(3, 1), \qquad (b, s, \theta) \sim (b, s + 3z, \theta). \tag{E92.15}$$

Thus the image of (E92.14) is precisely $\theta = 0$, with coordinates $(b, s \bmod 3)$.

There are actual short exact cohomology sequences

$$\begin{aligned} 0 &\longrightarrow H^1(\mathbb{Q}, W) \longrightarrow H^1(\mathbb{Q}, V) \longrightarrow H^1(\mathbb{Q}, E_3) \longrightarrow 0, \\ 0 &\longrightarrow H^1(\mathbb{Q}_3, W) \longrightarrow H^1(\mathbb{Q}_3, V) \longrightarrow H^1(\mathbb{Q}_3, E_3) \longrightarrow 0. \end{aligned} \tag{E92.16}$$

Both left injections follow from $H^0(E_3) = 0$: the actual matrix $A_E(\tau) - I$ has determinant nine. Globally, every class (E92.14) has the specified V lift $bf_1 + \epsilon f_2$ on the free source P, proving the last surjection. Locally, E.37 gives an actual map $j : E_3 \to V$ with $\pi_E j = 3$. Hence the connecting homomorphism is killed by three. Its target is torsion free: Shapiro and local duality give

$$H^2(\mathbb{Q}_3, W) = H^2(\mathbb{Q}_3, \mathbb{Z}_3(1))^2 \oplus H^2(k, \mathbb{Z}_3(1))^3 = \mathbb{Z}_3^5. \tag{E92.17}$$

This is the finite-to-integral duality proved in E.74; alternatively the local Brauer invariant and the finite Kummer sequence give each copy of $\mathbb{Z}_3$ under its actual inverse transition. Thus the local connecting map is zero. Restriction commutes with every arrow in (E92.16).

The toric restriction map is onto by the explicit field constructions below. Therefore any local V-class whose compact image has $\theta = 0$ is global: lift that compact class by (E92.14), subtract its actual global V lift locally, and lift the unique remaining toric class. Conversely every global class has compact image in (E92.14). This proves both image inclusions in (E92.2). If a global V-class restricts to zero, its compact image is zero by injectivity of (E92.14). Both injections in (E92.16) then identify the complete localization kernel with the toric kernel. This proves the required kernel comparison, before any coordinates or ranks are imposed.

Actual toric Shapiro maps. On a quadratic pair write its two cocharacters in the order stated after (E92.1). For the global group use the actual complex conjugation c, and for the local calculation use E.74's actual relative group and its chosen c. On the index-two subgroup the pair has scalar action χ, while c acts by minus swap. The normalized Shapiro inverse of a Tate cocycle b on that subgroup is

$$z(h) = \big(b(h), -b(chc)\big), \qquad z(c) = 0. \tag{E92.18}$$

Its first coordinate on the subgroup is the inverse map. The cocycle and conjugation identities follow by substitution; a boundary with coefficient a is carried to the boundary with coefficient $(a, -a)$. Averaging over the order-two element proves that these formulas give the full cohomology isomorphism. E.74's actual maximal-pro-three comparison identifies the local computation with absolute local cohomology, so no order-two element of an absolute nonreal local Galois group is assumed. These are the exact maps used in (E92.16).

The quadratic field, units and root carries. The Eisenstein integer ring $\mathbb{Z}[\zeta_3]$ is Euclidean for its norm. For completeness, write any complex quotient as $a + b\zeta_3$ and round both real coefficients to integers. The remainder has $|a|, |b| \le 1/2$, whence $a^2 - ab + b^2 \le 3/4 < 1$. Division with strictly smaller norm follows. Thus every prime ideal has a generator and the units are the six solutions of norm one, μ_6. This proves the global multiplicative decomposition used in (E92.5). The norm of $u_2 = 1 - 3\zeta_3$ is thirteen, and $X^2 + X + 1$ is irreducible modulo two; these prove all three prime assertions in (E92.3).

In the local ring use the actual basis $(1, \lambda)$. The identities

$$\lambda^2 = 3\lambda - 3 = -3\zeta_3, \quad \overline{\lambda} = 3 - \lambda, \quad N(a + b\lambda) = a^2 + 3ab + 3b^2 \tag{E92.19}$$

fix all signs and products. For a unit $a + b\lambda$, first choose $\omega \in \{1, -1\}$ so that $\omega a \equiv 1 \pmod 3$. Then put $e = -\omega b \bmod 3$ in $\{0, 1, 2\}$. Multiplication by ζ_3^{-e} gives a unit congruent to one modulo $3\mathcal{O}_k$. The choices are unique, since $\mathcal{O}_k/(\lambda^2) = \mathbb{F}_3[\lambda]/(\lambda^2)$ and $\zeta_3^e \equiv 1 - e\lambda$ there. This proves the literal decomposition (E92.6).

Logarithm and exponential converge and are inverse on $1 + 3\mathcal{O}_k$ and $3\mathcal{O}_k$. The termwise valuations prove convergence; formal composition and multiplication identities then hold by their convergent substitutions. In the basis $(1, \lambda)$, the two columns are

$$\begin{bmatrix} \dfrac{\log u_2}{3} & \dfrac{\log(-2)}{3} \end{bmatrix} \equiv \begin{pmatrix} -1 & -1 \\ 1 & 0 \end{pmatrix} \pmod 3. \tag{E92.20}$$

Its determinant is a unit. The matrix geometric series, or the adjugate divided by its unit determinant, gives its actual inverse over $\mathbb{Z}_3$. Thus $u_2, -2$ are a basis of the whole principal unit group. The factor -1 is removed by odd-power completion, whereas μ_3 survives. This proves the local identification (E92.5), with all torsion and all logarithmic coefficients.

Kummer theory gives the corresponding cohomology maps at each finite power by the exact sequence and Hilbert 90, as in [Mil20b, II, Proposition1.22; III, §4, Step2]. Here the groups of invariant roots can contain μ_3. The continuous comparison nevertheless has a direct proof: recursively align compatible finite cocycles by lifting their boundary coefficients. For injectivity, the sets of boundary coefficients of a zero class are nonempty finite sets; their compatible inverse limit is nonempty by the finite intersection property. This proves the inverse-limit isomorphism, including the finite μ_3. The same finite-level kernels prove its topological assertion.

Every coefficient in (E92.6) is integral. For a restricted exponent sequence the torsion sum has finite support modulo three, and both logarithmic sums converge uniformly modulo every power of three. The three pivot images are literally

$$\lambda \mapsto (1, 0, 0, 0), \quad u_2 \mapsto (0, 0, 1, 0), \quad 2 \mapsto (0, 0, 0, 1), \quad \zeta_3 \mapsto (0, 1, 0, 0). \tag{E92.21}$$

Substitution now proves (E92.7)–(E92.8) in both directions, with every finite torsion carry. Their continuity is reduction modulo the same fixed power, using uniform tails. The rational maps of E.90 have the same property. This computes the whole toric kernel in (E92.2).

The literal elliptic carry and additive coordinates. In the retained integral basis put

$$h_+ = \tfrac{1}{2}(I + J_1)f_1 = f_1 + \tfrac{1}{2}(A_1 + A_2 + B_1 + B_2 + C_1 + C_2). \tag{E92.22}$$

It is fixed by J_1. The exact matrices give

$$(T_1 - I)h_+ = 3f_2 + d_T, \qquad d_T = (-1, -3, 1, -2, 0, -1, -1, -2) \in W. \tag{E92.23}$$

This is an actual global and local coboundary in the normalized coordinates. Thus $3[f_2] = -[d_T]$ in both cohomology groups.

We verify the precise Kummer tuple for this carry. On the marked unramified generators the actual classes have values

$$\kappa_3 : (1, -1), \qquad \kappa_{\zeta_3} : (2, 2), \qquad \kappa_\lambda : (3/2, 1/2). \tag{E92.24}$$

The first is the fixed E.49/E.78 normalization; the second follows from the coherent roots of ζ_3 and $\chi(\tau) = \chi(\tau') = 7$; the third follows from $\lambda^2 = -3\zeta_3$ with the coherent rational roots of -1.

These are cohomology identities with their indicated compatible representatives. Applying (E92.18) to (E92.10) gives the eight-coordinate value

$$v(c_T) = (1, 3, -1, 2, 3/2, -1/2, 5/2, 1/2). \tag{E92.25}$$

The difference $-d_T - v(c_T)$ is exactly $(I - 7I)q$, where

$$q = (0, 0, 0, 0, 1/4, -1/4, 1/4, -1/4), \qquad J_W q = q. \tag{E92.26}$$

Thus the two normalized cocycles differ by the displayed actual fixed coboundary. This proves $3[f_2] = [c_T]$, with the toric/elliptic basis map and all signs fixed. It proves (E92.11) and hence both additive-coordinate directions (E92.12)–(E92.13).

For precision, the actual extraction of the toric remainder is also explicit. Normalize a cocycle to vanish on c. Its compact projection factors through Γ globally; for a local point in the stated image its relative compact value is zero and the same factorization holds. Write its horizontal value as U. Put

$$b = U_{f_1}, \quad \epsilon = U_{f_2} \bmod 3, \quad k = (U_{f_2} - \epsilon)/3, \qquad U_T = U - k(T_1 - I)h_+ - bf_1 - \epsilon f_2. \tag{E92.27}$$

The first two entries of U_T are zero. Apply the actual boundary kh_+ to the whole cocycle, and then subtract the specified global elliptic lift. The result is the unique toric cohomology class in (E92.16); (E92.18) and the Kummer inverse give its five field classes. Conversely these steps are reversible. They include the boundary on all generators, not just its value at τ.

The topology is equally explicit. Evaluation on c, τ, the finite residue extraction, and division by three on its stated image are continuous. The toric Kummer maps have the topology already proved. Therefore (E92.9) and (E92.12) are continuous bijections with the displayed continuous inverses, even though their global restricted factors are not compact. In the additive coordinates, the local image is

$$\mathbb{Z}_3^2 \times \mathbb{Z}_3 \times \mathbb{Z}_3^2 \times (C_3 \times \mathbb{Z}_3^3)^3 = \mathbb{Z}_3^{14} \times C_3^3. \tag{E92.28}$$

The first two factors are (b, w); the next two are the rational torus factors, the first with valuation at three zero. The actual field sections described after (E92.13) preserve this condition. They give a continuous linear section of localization. The two rational and three quadratic kernel formulas give every fibre, including its inverse, in (E92.2).

One may also read this image directly in E.74's Smith coordinates. The horizontal group is $\mathbb{Z}_3^6 \oplus C_3^3$, and the actual relative kernel has the basis $(3f_1 + f_2, e_d : d \neq O)$. Setting its first coordinate to zero leaves eight free coordinates. The three global μ_3 generators map to the three intrinsic torsion coordinates by the invertible matrix and inverse

$$\begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 1 & 1 \end{pmatrix} \pmod 3. \tag{E92.29}$$

These are obtained by applying the retained Smith map to $2(A_1 - A_2)$, $2(B_1 - B_2)$, $2(C_1 - C_2)$. Their triples are literal fixed coboundaries, so the calculation identifies the actual torsion classes. It agrees with (E92.28) and proves the same image using the original local source marking.

The certificate retains both actual coefficient maps, the marked relative basis, all carry vectors and coboundaries, and the torsion matrix/inverse. It verifies 243 complete elliptic/toric inverse calculations and 729 finite-carry identities. In the actual ring (E92.19) it checks 67 prime-ideal generators, 462 prime–precision logarithmic identities, and 567 full localization inverses with their 567 finite μ_3 carries. All computation uses the stated integral ring, actual input matrices and recorded hashes. □

The complete global central localization map and its joint quotient

E.93. Every central localization fibre and its exact action on global class-two sections [Proved]

Keep the cubic curve, rational tangent, genuine arithmetic action, ten-generator lattice V, and class-two group N_2 of E.37 and E.74. Put $M = \Gamma_2 N/\Gamma_3 N = \bigwedge^2 V$, with the actual Hall convention of E.74, and write

$$\begin{aligned} \mathcal{P}_2 &= M/(I - T_2)M^{J_2} \simeq \mathbb{Z}_3^{24} \oplus C_3^9 \oplus C_9^5, \\ \mathcal{K}_2 &= \ker\bigl((I - J_2)(T_2 - 7I) : M \longrightarrow M\bigr) \simeq \mathbb{Z}_3^{22}. \end{aligned} \tag{E93.1}$$

All cohomology here is continuous integral cohomology. For a set of rational primes $\mathcal{S}$, let

$$\mathcal{R}_{\mathcal{S}}(\mathbb{Z}_3) = \{(a_p)_{p\in\mathcal{S}} : \text{for every } n,\ a_p \in 3^n\mathbb{Z}_3 \text{ for all but finitely many } p\} \tag{E93.2}$$

carry the inverse-limit topology of the discrete groups $\bigoplus_{p\in\mathcal{S}} \mathbb{Z}/3^n$.

Let w be the primitive integral Tate vector of E.37: $w = (3f_1 + U_1) \wedge (3f_2 + U_2)$. Its actual based actions are $T_2 w = 7w$ and $J_2 w = -w$. Write $[w]$ for its class in $\mathcal{P}_2$; no choice of a free basis vector for that class is made. In the actual global splitting of E.37 and the actual local coordinates of E.75 the restriction map is

$$\begin{aligned} H^1(\mathbb{Q}, M) &= \mathcal{P}_2 \oplus \mathcal{R}_{\{p\neq 3\}}(\mathbb{Z}_3), \\ H^1(\mathbb{Q}_3, M) &= \mathcal{P}_2 \oplus \mathcal{K}_2, \\ \operatorname{res}\bigl(z, (a_p)\bigr) &= \bigl(z + b\mathcal{L}(a)[w],\ \alpha\mathcal{L}(a)w\bigr), \\ \mathcal{L}(a) = \sum_{p\neq 3} a_p \ell_p, &\qquad \ell_p = \frac{\log\langle p\rangle}{\log 4} \in \mathbb{Z}_3, \qquad \ell_2 = \tfrac{1}{2}. \end{aligned} \tag{E93.3}$$

The brackets $\langle p\rangle$ remove the sign of the three-adic unit. The constants $b \in \mathbb{Z}_3$ and $\alpha \in \mathbb{Z}_3^\times$ are the actual unchanged marked Kummer evaluations of E.90: $\kappa(3) = (1, 0)$ and $\kappa(4) = (b, \alpha)$. Their numerical values are not assigned.

The entire image, kernel and cokernel of this specific continuous homomorphism are

$$\begin{aligned} \operatorname{im}(\operatorname{res}) &= \mathcal{P}_2 \oplus \mathbb{Z}_3 w \simeq \mathbb{Z}_3^{25} \oplus C_3^9 \oplus C_9^5, \\ \ker(\operatorname{res}) &\simeq \mathcal{R}_{\{p\neq 2,3\}}(\mathbb{Z}_3), \\ \operatorname{coker}(\operatorname{res}) &= \mathcal{K}_2/\mathbb{Z}_3 w \simeq \mathbb{Z}_3^{21}. \end{aligned} \tag{E93.4}$$

The isomorphism for the kernel sends an arbitrary restricted family $(a_p)_{p\neq 2,3}$ to

$$z = 0, \qquad a_2 = -2 \sum_{p\neq 2,3} a_p \ell_p, \tag{E93.5}$$

with the other coordinates unchanged. A continuous additive section onto the image sends (u, tw) to

$$z = u - \frac{bt}{\alpha}[w], \qquad a_2 = \frac{2t}{\alpha}, \qquad a_p = 0 \quad (p \neq 2, 3). \tag{E93.6}$$

Adding (E93.5) to (E93.6) gives every localization fibre.

Here are exact coordinates for the cokernel. Let K and K^ℓ be E.75's retained 45×22 saturated kernel basis and its left inverse. The actual column $k = K^\ell w$ is

$$\begin{aligned} k = (&-2, -4, 0, -4, -2, 0, 0, 1, -2, 3, 0, \\ &2, 4, 6, 6, -2, -2, 1, -4, -1, 2, -3)^t. \end{aligned} \tag{E93.7}$$

For $a = K\eta$, define

$$t = -\eta_1/2, \qquad \xi_j = \eta_j + (k_j/2)\eta_1 \quad (2 \le j \le 22). \tag{E93.8}$$

Then $a = tw + \sum_{j=2}^{22} \xi_j K e_j$. Thus $(\xi_2, \dots, \xi_{22})$ is the full quotient map and $\xi \mapsto \sum_{j=2}^{22} \xi_j K e_j$ is its continuous integral section. These maps, including their full 45-coordinate matrices, are retained in the certificate.

For the two actual quotient sections used in E.85, let $F_2 : M \to A = \mathbb{Z}_3^2$ be its exact alternating coefficient map. Give $H^1(\mathbb{Q}_3, A)$ the complete coordinates (ρ, σ, ϵ) of E.85, including geometric conjugacy. The composite map is exactly

$$\begin{gathered} H^1(\mathbb{Q}, M) \xrightarrow{\text{res}} H^1(\mathbb{Q}_3, M) \xrightarrow{F_{2,*}} H^1(\mathbb{Q}_3, A), \\ (z, (a_p)) \longmapsto (\rho(z), 0, 0), \qquad \rho(z) = (F_{2,1} - F_{2,2})z, \\ \text{im} = \mathbb{Z}_3 \oplus 0 \oplus 0, \\ \ker = \mathcal{R}_{\{p \neq 3\}}(\mathbb{Z}_3) \oplus \mathbb{Z}_3^{23} \oplus C_3^9 \oplus C_9^5. \end{gathered} \tag{E93.9}$$

The actual horizontal splitting vector is $z_* = -\frac{1}{2}[f_2, f_1]$, with $F_2 z_* = (1/2, -1/2)$. In the Smith coordinates $p = L_G z$ of E.85, retain all fourteen finite coordinates and the final twenty-three free coordinates. The missing free coordinate is obtained from

$$\begin{gathered} p_{22} = \sum_{j=2}^{24} r_j p_{21+j} - \rho, \\ r = (-1, -1, -1, 0, 0, 0, 0, -3, 3, 0, -1, 0, -1, 0, 0, 3, -3, 0, 0, -2, -3, -3, -1, 0). \end{gathered} \tag{E93.10}$$

The exact full inverse matrices occur in the certificate.

These maps apply to the actual nonabelian fibres as follows. Let $\mathcal{I} \subset H^1(\mathbb{Q}, V)$ be the full image of $H^1(\mathbb{Q}, N_2)$ computed by the explicit obstruction criterion in E.37. For each $\gamma \in \mathcal{I}$, let $\mathfrak{F}_\gamma$ be its nonempty global section fibre. Its restriction to the local intrinsic fibre is equivariant for (E93.3). After choosing any actual class in $\mathfrak{F}_\gamma$ as origin, the image is its translate of the group in (E93.4), and every fibre is the group in (E93.5). The image of $\mathfrak{F}_\gamma$ in the pair of local quotient-section classes is its origin's translate of $\mathbb{Z}_3 \oplus 0 \oplus 0$, and every fibre has exactly the kernel in (E93.9). A change of global origin translates these coordinates by its unique class in $H^1(\mathbb{Q}, M)$. No arbitrary origin is assigned a numerical joint value.

This computes central variations on every actual global class-two fibre. Evaluating the particular joint origin for every global abelian input, and all higher nonabelian lifts, remain the stated steps of E.23. The manuscript passages are section51.12, folios717–718, scans136–138; section51.15, folios725–727, scans156–160; and section51.16, folio734v, scan174, with the exact comparisons of E.37 and E.85.

Proof. The genuine coefficient line and its basis transport. The module M is the central kernel in $1 \to M \to N_2 \to V \to 1$, with conjugation action induced by the retained rational tangential section. E.37 constructs the integral vector w in the lexicographic exterior basis $(q_i \wedge q_j)_{i<j}$. E.74 uses Hall generators $[q_a, q_b]$ with $a > b$. We use the explicit signed permutation matrix S whose row for $[q_a, q_b]$ has entry -1 in column (b, a) and zero elsewhere. Thus

$$SS^t = I, \qquad T_2 S = S \bigwedge^2 T_1, \qquad J_2 S = S \bigwedge^2 J_1. \tag{E93.11}$$

These are equalities of maps between the two specified free $\mathbb{Z}_3$-modules. The certificate checks every entry and transports w by S before applying any Hall matrix. Its Hall coordinate numbered nine is -1, proving primitivity in M directly. The actual arithmetic action then

gives $T_2 w = 7w$ and $J_2 w = -w$. In particular the map $\mathbb{Z}_3(1) \to M$, $1 \mapsto w$, is equivariant for the full global action, as proved in E.37.

The global and local decompositions and restriction. E.37.20 gives an exact continuous localization sequence whose residue module is (E93.2) for $p \neq 3$. Its proof uses the actual inertia maps: the p-eigenlattice of Frobenius in M is exactly $\mathbb{Z}_3 w$. The explicit splitting is

$$(a_p) \longmapsto \sum_{p \neq 3} a_p \kappa(p) w. \tag{E93.12}$$

Every reduction modulo 3^n in this sum has only finitely many nonzero terms. These reductions are compatible, so the sum is a uniformly convergent continuous cocycle. It has the specified inertia residues at all primes. Its complement is the unramified term evaluated at the actual free generator τ, namely $\mathcal{P}_2$ in (E93.1). The image of this unramified cocycle under the actual local quotient map of E.74 is $(z, 0)$: the marked relative generator dies and the marked τ is unchanged. This identifies the first summand through an actual cocycle map.

E.75 evaluates the local normalized cocycles as $\mathcal{P}_2 \oplus \mathcal{K}_2$, with horizontal value modulo $(I - T_2)M^{J_2}$ and literal relative value in $\mathcal{K}_2$. The evaluated Kummer comparison of E.90 identifies $H^1(\mathbb{Q}_3, \mathbb{Z}_3(1))$ with $\mathbb{Z}_3\kappa(3) \oplus \mathbb{Z}_3\kappa(4)$. For the rational prime $p \neq 3$ its valuation coordinate is zero and its unit coordinate is ℓ_p. Hence

$$\operatorname{res} \kappa(p) = \ell_p \kappa(4) = (b\ell_p, \alpha\ell_p) \tag{E93.13}$$

in the actual marked evaluations. This is an equality of cohomology classes with the same chosen generators. Multiplying by the equivariant vector w, then adding the unramified class, proves (E93.3).

For completeness, $\log 4$ has three-adic valuation one, and $\log\langle p \rangle \in 3\mathbb{Z}_3$; therefore every ℓ_p is integral. The identity $(-2)^2 = 4$ in $1 + 3\mathbb{Z}_3$ gives $\ell_2 = 1/2$. The uniform residue condition (E93.2) makes $\mathcal{L}$ converge and makes it continuous. Thus $a_2 = 2t$ with all other entries zero is a continuous additive section of $\mathcal{L}$ onto $\mathbb{Z}_3$.

The second coordinate of (E93.3) now has image exactly $\mathbb{Z}_3 w$, since α is a unit. Its first coordinate can be assigned freely by changing z. This proves the image assertion and verifies (E93.6) by literal substitution. In the kernel, the torsion-free line $\mathbb{Z}_3 w$ forces $\mathcal{L}(a) = 0$ and then $z = 0$. Solving its single unit equation using $\ell_2 = 1/2$ gives precisely (E93.5), including its inverse projection and topology.

The full integral quotient map. The matrices K, K^ℓ satisfy $K^\ell K = I$; E.75 proves that K maps $\mathbb{Z}_3^{22}$ onto the entire kernel $\mathcal{K}_2$. Their exact product with the transported w is the column (E93.7), and $Kk = w$. Replace the first column of the identity matrix of order22 by k. Its determinant is -2, a three-adic unit, so its matrix and inverse are integral over $\mathbb{Z}_{(3)}$. The inverse equations are precisely (E93.8). They prove directly that the relative Tate line is a direct summand and give the entire rank21 cokernel in (E93.4).

This relative primitivity does not enter as an unproved primitivity assertion for the horizontal class $[w]$. Indeed the program evaluates $L_G w$; its free entries are all divisible by three and at least one is not divisible by nine. Formula (E93.6) uses the actual class $[w]$ with its entire Smith vector, so its correctness retains this divisibility and all fourteen finite coordinates.

The exact joint coefficient map. E.85 constructs F_2 from the four actual unit characters, with image $L_2 = \{(x, y) : x + y \in 3\mathbb{Z}_3\} \subset A$. It proves the genuine arithmetic intertwiners $F_2 T_2 = 49 F_2$ and $F_2 J_2 = \operatorname{swap} F_2$. Applying the first equality to w gives

$$7F_2 w = 49 F_2 w, \qquad F_2 w = 0, \tag{E93.14}$$

the last implication following from the torsion-free target and the nonzero integer42. The retained matrix multiplication verifies the same zero column after the exact transport (E93.11).

The full map on local cohomology in E85.12 is

$$([z], a) \longmapsto \big((F_{2,1} - F_{2,2})z,\ F_{2,1}a,\ 0\big). \tag{E93.15}$$

Its last entry is zero because $F_2 z \in L_2$; the actual central conjugator removes its mean modulo three as in E85.25. Substituting (E93.3) and (E93.14) proves (E93.9). At the cocycle level every summand $a_p \kappa(p) w$ already has zero image under F_2; no limiting ambiguity is used.

For every Smith coordinate the exact identity $L_G(I - T_2)(I + J_2)R_G = \mathrm{diag}(1^7, 3^9, 9^5, 0^{24})$ and both retained inverse matrices are checked. Multiplying ρ by L_G^{-1} gives zero in the first twenty-one columns and the row r in (E93.10) in the last twenty-four. Its first coefficient is -1. Solving that equation proves the full 23-dimensional free kernel while leaving all $C_3^9 \oplus C_9^5$ coordinates arbitrary. The vector z_* gives the displayed continuous additive section of ρ. Combining these coordinates with every restricted prime family proves the entire kernel and every inverse fibre in (E93.9).

Transport to actual section classes. The class-two central extension identifies the difference between any two lifts of the same abelian cocycle with a unique M-cocycle modulo central coboundaries. Changing the representative of the fixed abelian class introduces its stabilizer $H^0(\mathbb{Q}, V)$. This group is zero by E.36. The corresponding local stabilizer is zero by E.74, and the target abelian stabilizer is zero because its actual τ-operator is $7I$, as verified in E.85. Thus no extra stabilizer quotient changes any of the displayed coefficient fibres. The same cocycle multiplication commutes with restriction and with the two quotient homomorphisms, proving equivariance and all the stated torsor assertions.

The choice of an origin is a translation within the already nonempty fibre over $\gamma \in \mathcal{I}$. Its change is the unique central cohomology class just constructed, whose restriction and joint image are (E93.3) and (E93.9). This gives the exact ambiguity as well as both transfer maps.

All finite matrices in this proof are over $\mathbb{Z}_{(3)}$ with the displayed integral inverses. The restricted sums are continuous in the topology (E93.2). The certificate checks the two basis transports and all matrix identities, the symbolic inverse retaining b, α, 1134 finite-precision localization identities, and 1458 full relative and horizontal coordinate round trips. The algebraic identities and uniform convergence above prove the full integral statements. □

The next cyclotomic base change and its actual class quotients

E.94. The μ_{27} comparison, a noncyclic S-class quotient, and its exact norm and extension maps [Proved]

Keep the literal roots and primes of E.86 and E.91. Put

$$\eta^3 = \xi, \qquad E = B_3(\eta), \qquad F = B_1, \qquad \beta_3^9 = \beta_1, \qquad \xi^3 = \zeta. \tag{E94.1}$$

Then $[E : F] = 27$, $[E : B_3] = 3$, and $\mathrm{Gal}(E/F) = C_9 \oplus C_3$. At each of the three primes Q_i over three, E/B_3 is totally ramified. At each of the three primes T_i over seventeen it is unramified of residue degree three. All six primes have a unique prolongation, denoted Q_i^E, T_i^E. The ideal norm and extension coefficients are therefore

$$\begin{aligned} N_{E/B_3} Q_i^E &= Q_i, \quad j_{E/B_3} Q_i = 3Q_i^E, \\ N_{E/B_3} T_i^E &= 3T_i, \quad j_{E/B_3} T_i = T_i^E. \end{aligned} \tag{E94.2}$$

Both ordinary and S-class norms onto B_3 are surjective, where $S = \{3, 17\}$.

There are literal K-automorphisms

$$\begin{aligned} G(\beta_3) &= \eta\beta_3, & G(\eta) &= \eta, & |G| &= 27, \\ H(\beta_3) &= \beta_3, & H(\eta) &= \eta^4, & |H| &= 9, \\ HGH^{-1} &= G^4, & \mathrm{Gal}(E/B_3) &= \langle H^3 \rangle. \end{aligned} \tag{E94.3}$$

They generate $\mathrm{Gal}(E/K)$, of order 243. The restrictions of G^3, H to B_3 are exactly g_2, h_2 of E.91. The element G does not preserve B_3; indeed $GH^3G^{-1} = G^{-9}H^3$.

Let $\mathcal{G}_E/F$ be the maximal abelian three-extension containing E that is unramified over E. Let $\mathcal{G}_E^S$ be its maximal subextension in which all six S-primes of E split. These fields and their actual restriction characters are computed by the ray presentations below. Their groups are

$$\begin{array}{c|c|c} & \mathrm{Gal}(\mathcal{G}/F) & \mathrm{Gal}(\mathcal{G}/E) \\ \hline \mathcal{G}_E & C_9^4 \oplus C_3^2 & C_9^3 \oplus C_3 \\ \mathcal{G}_E^S & C_9^2 \oplus C_3^2 & C_9 \oplus C_3. \end{array} \tag{E94.4}$$

Write the right-hand groups as $\mathcal{A} = C_9^3 \oplus C_3$ and $\mathcal{A}_S = C_9 \oplus C_3$. Restriction of the Hilbert class fields gives actual surjections

$$\mathrm{Cl}(E)\{3\} \longrightarrow \mathcal{A}, \qquad \mathrm{Cl}_S(E)\{3\} \longrightarrow \mathcal{A}_S. \tag{E94.5}$$

These are evaluated class quotients. The statement does not assign either full class group to its quotient.

The bases in (E94.4) are specified by the integer Smith maps in the certificate. In these bases, removal of the S-primes is the exact matrix

$$Q : \mathcal{A} \longrightarrow \mathcal{A}_S, \qquad Q = \begin{pmatrix} 0 & 8 & 3 & 6 \\ 2 & 2 & 1 & 0 \end{pmatrix}. \tag{E94.6}$$

Its kernel, of order 81, is generated by the following six actual prime classes:

$$C_E = \begin{array}{c|cccccc} & Q_1^E & Q_2^E & Q_3^E & T_1^E & T_2^E & T_3^E \\ \hline C_9 & 1 & 7 & 1 & 3 & 3 & 3 \\ C_9 & 3 & 6 & 0 & 0 & 0 & 0 \\ C_9 & 1 & 1 & 7 & 6 & 6 & 6 \\ C_3 & 0 & 2 & 1 & 0 & 0 & 0 \end{array}. \tag{E94.7}$$

The first three columns have order nine and the last three have order three in this genus quotient. Their images are computed from local norm uniformizers, with the residue-degree coefficients in (E94.2) retained.

Let $A_9 = C_9^2 \oplus C_3$ have the simple (y, z, t) coordinates of E.91. The original ninth-genus field embeds in $\mathcal{G}_E$. The resulting norm and extension comparisons are

$$\begin{gathered} N : \mathcal{A} \twoheadrightarrow A_9, \qquad N = \begin{pmatrix} 3 & 1 & 0 & 3 \\ 7 & 0 & 2 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}, \\ J : A_9 \longrightarrow \mathcal{A}, \qquad J = \begin{pmatrix} 0 & 3 & 6 \\ 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 0 & 0 \end{pmatrix}, \qquad JN = 3, \quad NJ = 3, \quad \ker N = C_3^2. \end{gathered} \tag{E94.8}$$

The map N is induced by actual ideal norm. The map J is induced by actual ideal extension: the E.91 residual

$$W = (g_2 - 1)\,\mathrm{Cl}(B_3)\{3\} \quad \text{maps to zero in } \mathcal{A}. \tag{E94.9}$$

Thus the extension comparison factors through the already evaluated quotient A_9; the proof identifies that factor rather than choosing an unrelated homomorphism between groups of the same order.

The normalizer action on the relative S-genus quotient, and its actual norm to the original S-class group, are

$$\begin{gathered} G|_{\mathcal{A}_S} = I, \qquad H|_{\mathcal{A}_S} = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}, \qquad \nu : \mathcal{A}_S \twoheadrightarrow \mathrm{Cl}_S(B_3)\{3\} = C_3, \\ \nu(x,y) = -x \pmod 3, \qquad \ker \nu = C_3^2. \end{gathered} \tag{E94.10}$$

The norm is identified using the coherent prime-73 basis of E.86. In particular $\mathrm{Cl}_S(E)/(H^3 - 1)\,\mathrm{Cl}_S(E)$ has the noncyclic quotient $C_9 \oplus C_3$. Consequently $\mathrm{Cl}_S(E)\{3\}$ is not a cyclic module over $\mathbb{Z}_3[\mathrm{Gal}(E/B_3)]$. This follows through (E94.5), on which H^3 acts identically.

The computation uses the same certified degree-eighteen field F as E.76. It introduces no BNF calculation in B_3, E, or the original degree-eighteen field L_2. Every local kernel, ray relation, Kummer character, Smith transformation and inverse is retained. It also records the unsuccessful initial local-unit spanning family and the corrected literal norm family. The full class groups of E, the value of the E.91 transvection, and the eventual exponent h of E.69 remain outside the evaluations claimed here; equation (E94.9) states the exact map by which that transvection disappears in this particular comparison.

The manuscript anchor is section51.16, folios733v–736v, scans172–178, via the actual positional norm systems of E.46, E.69, E.72 and E.88.

Proof. The fields, local degrees, and actual normalizer. The residue field at each prime above seventeen in B_3 is $\mathbb{F}_{289}$. It contains μ_9 and does not contain μ_{27}, since $v_3(289 - 1) = 2$. Adjoining η is therefore unramified of degree three there. This proves the global degree $[E : B_3] = 3$, and proves the assertion at seventeen in (E94.2).

At a prime of F above three the completion is $F_v = \mathbb{Q}_3(\mu_9)$. Indeed 17 has a cube root $a_0 \in \mathbb{Q}_3$, because $-17 \in 1 + 9\mathbb{Z}_3$, and the three embeddings send a to $a_0\zeta^j$. The local radical extension B_3/F is cyclic of degree nine, generated by a ninth root of β_1. Its local norm character will be denoted $\chi_v : F_v^\times \to \mathbb{Z}/9$. The extension $F_v(\eta)/F_v$ has its cyclotomic character $\psi_v : F_v^\times \to \mathbb{Z}/3$, where the latter coordinate means $\eta \mapsto \zeta\eta$.

Their normalizations can be checked directly. The actual arithmetic local Artin map on $\mathbb{Q}_3$-units acts on μ_{27} by the inverse unit. This follows also by applying global reciprocity to a rational element and the prime Frobenii in the cyclotomic field [Mil20b, V,Example3.2,Proposition5.2,Theorem5.3 and Summary5.10; III,Theorem3.4]. Local norm compatibility then gives

$$\begin{gathered} N_{F_v/\mathbb{Q}_3}(\xi) = 1, \qquad N_{F_v/\mathbb{Q}_3}(\beta_1) = 17^2 \equiv 19 \pmod{27}, \\ N_{F_v/\mathbb{Q}_3}(1 + \beta_1) = 1 - 17 + 17^2 = 273, \\ \psi_v(\xi) = 0, \qquad \psi_v(\beta_1) = 1, \qquad \psi_v(1 + \beta_1) = 2. \end{gathered} \tag{E94.11}$$

For the last value one removes the factor three from 273; its remaining unit is $91 \equiv 10$, whose inverse is 19 modulo27. The norm identity is the resultant $\mathrm{Res}_Z(Z^6 + Z^3 + 1, 1 + a_0 Z) = a_0^6 - a_0^3 + 1$; multiplication of the exponent of ξ by any unit modulo9 does not change it.

The actual ninth-radical character has $\chi_v(\xi) = 5$, a unit modulo9, and $\chi_v(\beta_1) = 0$. The former follows from the normalized E.76 local character and the reciprocity calculation below. The latter follows from $N(\beta_3) = \beta_1$. Thus the two characters are independent on local units. Their joint inertia image is $C_9 \oplus C_3$, of order27. The compositum E/F is totally ramified at these three primes, so E/B_3 is totally ramified there. All ramification is in S. This proves (E94.2).

The displayed substitutions in (E94.3) respect $\eta^3 = \xi$ and $\beta_3^9 = \beta_1$. Their $27 \cdot 9$ distinct actions are $G^i H^j$, with multiplication

$$(i,j)(k,l) = (i + 4^j k,\ j + l) \quad \text{in } \mathbb{Z}/27 \rtimes \mathbb{Z}/9. \tag{E94.12}$$

Their number equals $[E:K]$. This proves the asserted full Galois group and every restriction in (E94.3). In particular G^3, H preserve B_3, while the explicitly different conjugate of $\langle H^3\rangle$ proves that G does not.

For either ordinary or S-classes, let $\mathcal{H}$ be the corresponding Hilbert three-class field of B_3. Its intersection with the cyclic cubic extension E/B_3 is B_3, because that extension ramifies at Q_i. For any class in $\mathrm{Gal}(\mathcal{H}/B_3)$, Chebotarev applied to $E\mathcal{H}/B_3$ gives a prime whose Frobenius is that class on $\mathcal{H}$ and identity on E. A degree-one prime above it has ideal norm representing the given class. This proves actual norm surjectivity. The precise source anchors are the norm/Frobenius square, the Hilbert-class-field correspondence at modulus1, and Chebotarev [Mil20b, V,Proposition3.3,Theorem3.6 and Example3.9; VIII,Theorem7.4].

The exact joint local norm kernels. Use the E.76 bases for the three local unit groups modulo ninth powers. Each is C_9^7. The three unscaled radical rows and the common cyclotomic row are

$$\begin{pmatrix} 1 & 1 & 5 & 2 & 8 & 3 & 6 \\ 1 & 1 & 2 & 8 & 5 & 3 & 6 \\ 1 & 1 & 8 & 5 & 2 & 3 & 6 \end{pmatrix} \pmod 9, \qquad (0,0,1,1,1,0,0) \pmod 3. \tag{E94.13}$$

The actual radical rows at these primes are five times the rows displayed modulo9. For every one of the three primes the common integer kernel has the same basis

$$K_{\mathrm{loc}} = \begin{pmatrix} 9 & 8 & 3 & 6 & 0 & 6 & 3 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}. \tag{E94.14}$$

The determinant is27, and the exact Smith quotient is $C_9 \oplus C_3$.

Here is a literal construction of the new character, including the exactness test. In $F_v[Y]/(Y^3-\xi)$ use the actual norms of Y and $1+(1-Y)^k$. A norm of $r+qY+pY^2$ is

$$r^3+\xi q^3+\xi^2 p^3-3\xi pqr. \tag{E94.15}$$

The first eleven recorded norms, including $N(Y)=\xi$, span a six-dimensional subspace modulo cubes in each of the seven-dimensional local unit groups. Every column is an actual norm. Local class field theory gives norm index three for this totally ramified cubic extension. Thus these columns give its entire unit norm kernel. The unique annihilating row is normalized by its nonzero value on β_1, giving (E94.13) and (E94.11). No unsuccessful bounded search is used to assert a norm obstruction.

At the primes over seventeen the radical character on a unit u is $-\log_\xi(\bar{u}^{32})$ modulo9; the cyclotomic character is twice the valuation modulo3. On the fifteen specified S-unit generators of F, the program evaluates all twelve characters. Their exact reciprocity rows determine the scale five in (E94.13). With the actual local generator choices in the run, the six scaling coefficients for the radical rows are $(5,5,5,8,5,4)$. Both sums of actual local characters vanish on every global S-unit.

The combined character image on all S-units has order 3^{13}; on ordinary units it has order 3^7. On the kernel of the six radical characters modulo9, the six cyclotomic characters have rank three. These are integer kernel/image calculations, with the full fifteen-column matrices and all Smith maps retained. The local ideal modulus is $Q_1^{16}Q_2^{16}Q_3^{16}T_1T_2T_3$. It suffices: at a

wild prime, logarithm sends the sixteenth unit filtration inside ninth powers, since division by nine leaves filtration greater than $6/(3-1)=3$.

The ray-class construction and its maximality. Let $\mathcal{R}$ be the three-primary ray class group for that modulus. At each Q_i, impose the local unit relations (E94.14); at each T_i, impose ninth powers of units. Chinese remainder representatives give actual principal-ideal columns in the ray class group. Their quotient is $\Gamma_E=\mathrm{Gal}(\mathcal{G}_E/F)$. Indeed an abelian extension over F becomes unramified over E exactly when its local Artin maps kill $N_{E_w/F_v}\mathcal{O}_{E_w}^{\times}$. The relations just imposed are those subgroups: both characters occur at Q_i, and the unramified cubic cyclotomic factor contributes no unit condition at T_i. The local norm/Artin compatibility follows from the proved restriction/corestriction pairing [Mil20b, III,Theorem3.1]; the global ray-class existence and reciprocity identify the corresponding field [Mil20b, V,Theorems3.5–3.6].

This is the maximal such three-extension. Away from S no ramification can disappear under the unramified extension E/F, so every such abelian extension is included in the chosen ray field. At S the imposed subgroups are the full local unit norm groups, not merely subgroups generated by some found elements. The exponent bound used in the computation is also exact: the remaining local unit quotients have exponent nine and $\mathrm{Cl}(F)\{3\}$ has exponent three. Hence the entire ray quotient has exponent dividing27. Taking the exponent-27 ray quotient loses no part of Γ_E.

On a prime $\mathfrak{l}\nmid 3\cdot 17$ its two restriction coordinates to E/F are computed from

$$\bar{\beta}_1^{(N\mathfrak{l}-1)/9}=\bar{\xi}^{\chi(\mathfrak{l})},\qquad \bar{\xi}^{(N\mathfrak{l}-1)/3}=\bar{\zeta}^{\psi(\mathfrak{l})}. \tag{E94.16}$$

The certificate factors actual ray basis ideals into primes and evaluates these two equalities, including their integer exponents. The kernel has index27. If L_{ord} is the full ray relation lattice, and K is this integer character-kernel lattice, the two presentations are

$$\Gamma_E=\mathbb{Z}^{16}/L_{\mathrm{ord}},\qquad \mathcal{A}=K/L_{\mathrm{ord}}. \tag{E94.17}$$

The exported bases satisfy $KP=L_{\mathrm{ord}}$; their Smith factors are respectively $(9,9,9,9,3,3)$ and $(9,9,9,3)$. Every Smith left and right matrix has determinant ±1, and its inverse is retained.

For complete splitting over E, one also imposes the full local norm uniformizers. They can be chosen in $F_v^{\times}$ as

$$u_{Q_i}=(1+\beta_1)\beta_1^{-2},\qquad u_{T_i}=\beta_1^3. \tag{E94.18}$$

The first has valuation one and both character values zero by (E94.11) and $N(1+\beta_3)=1+\beta_1$. The second has valuation three and both character values zero. Thus these elements, together with the local unit norm groups, generate the full local norm groups. A Chinese remainder element z agreeing with u_v at v, and with one at the other S-primes, gives the prime-to-S ideal $I=(z)/v^{v(u_v)}$. Its negative ray class is the local Artin class of u_v. The program constructs all six such ideals and checks their S-valuations exactly.

Adjoining these six columns to L_{ord} gives a lattice $L_S\subset K$. Its quotient over F has Smith factors $(9,9,3,3)$; K/L_S has factors $(9,3)$. This proves the second row of (E94.4). The inclusion $L_{\mathrm{ord}}\subset L_S$ gives the actual quotient matrix (E94.6), and its six columns, written in the first relative Smith basis, are exactly (E94.7). They are also the Frobenii of Q_i^E,T_i^E: local norm compatibility takes a uniformizer upstairs to a local norm uniformizer downstairs. Their generated subgroup has order81, equal to the kernel order of (E94.6).

The original-genus norm and the genuine extension map. The local unit norms from E/F lie in those from B_3/F. Hence the original ninth-genus field is a subfield of $\mathcal{G}_E$. The program constructs its restriction matrix by evaluating each actual new ray basis ideal in the original ray class group. Equality of cyclic-factor lists is not used to identify the two ray bases. Intersecting with the two Kummer kernels gives the onto matrix N in (E94.8), in exactly the simple

coordinates of E.91. For an upper prime of residue degree f, restriction of its Frobenius is the fth power of the lower Frobenius. Thus this restriction is the actual ideal norm map.

Let j be ordinary ideal extension from B_3 to E, followed by (E94.5). The lift G^3 of g_2 lies in $\mathrm{Gal}(E/F)$, so it acts identically on the abelian-over-F genus group. Equivariance therefore proves $j((g_2-1)c)=0$ for every actual class c. E.91 identifies that subgroup with the complete kernel W of its original genus map. This proves (E94.9), and gives a well-defined map $J : A_9 \to \mathcal{A}$.

The norm from $\mathrm{Cl}(E)\{3\}$ onto $\mathrm{Cl}(B_3)\{3\}$ is onto, as proved above. On the new genus quotient the deck group $\langle H^3\rangle$ acts identically. The actual ideal identities give $JN = 1 + H^3 + H^6 = 3$ and $NJ = 3$. The full Smith calculation of $\ker N$ gives C_3^2; hence three times any lift under N is independent of the chosen lift. Applying this to the three specified old basis classes gives exactly the displayed matrix J. This proves that the finite matrix is the actual extension comparison.

The six prime columns provide independent checks of the coefficients:

$$NC_E = C_{\mathrm{old}}\,\mathrm{diag}(1,1,1,3,3,3), \qquad JC_{\mathrm{old}} = C_E\,\mathrm{diag}(3,3,3,1,1,1). \tag{E94.19}$$

Here C_{old} is the E.91 matrix in its (y,z,t) coordinates. Thus the new prime norms do not silently identify the inert and ramified prolongations.

Normalizer action and the exact noncyclic comparison. Both G, H preserve F and normalize the defining norm subgroups. Their actions on F are the literal substitutions $a \mapsto \zeta a, \xi \mapsto \xi$ and $a \mapsto \zeta^{-1}a, \xi \mapsto \xi^4$. The actual ray Galois matrices preserve L_{ord}, L_S, K, and induce on $\mathcal{A}$ the matrices

$$G_{\mathcal{A}} = \begin{pmatrix} 7 & 1 & 0 & 0 \\ 0 & 7 & 6 & 0 \\ 6 & 4 & 7 & 0 \\ 0 & 2 & 1 & 1 \end{pmatrix}, \qquad H_{\mathcal{A}} = \begin{pmatrix} 1 & 1 & 3 & 6 \\ 0 & 7 & 3 & 0 \\ 6 & 5 & 7 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}. \tag{E94.20}$$

They commute and their cubes are identity, with the indicated row moduli. Taking the quotient (E94.6) gives precisely (E94.10).

The S-class norm to F, evaluated on the same actual ray basis ideals, is its certified second class coordinate modulo three. E.86 identifies the norm $\mathrm{Cl}_S(B_3) \to \mathrm{Cl}_S(F)$ with the identity in the coherent prime-73 bases. Therefore this character is exactly the norm to the original C_3, not just some nonzero character of that order. Its value in the two relative Smith coordinates is $(-1, 0)$, proving (E94.10).

The source of (E94.5) modulo $H^3 - 1$ still surjects onto $\mathcal{A}_S$, since H^3 belongs to the abelian genus base group. A cyclic $\mathbb{Z}_3[\langle H^3\rangle]$-module has cyclic $\mathbb{Z}_3$-coinvariants: the image of a single module generator generates the quotient. Its quotients are cyclic. The explicitly proved quotient $C_9 \oplus C_3$ has two independent modulo-three generators. This proves the noncyclic assertion through the actual surjective map, and identifies the exact premise that fails in transporting the earlier cyclic-module computation to E.

Finally the Python verifier checks ten full Smith systems, both ideal norm/extension identities, all 2187 relative ordinary coordinates, all 27 relative S coordinates, and the 243 normalizer group elements. The six-stage portable rebuild begins with the certified F-BNF, reconstructs every new local and ray object, and retains its actual stdout and binary mathematical objects. The remaining values excluded from the statement are not inferred from the genus quotients or from the two retained failed local spanning attempts. □

All eight toric relative inputs in the wild class-two relation

E.95. Evaluated cusp cochains, the complete toric wild obstruction and every fibre [Proved]

Use the actual source of E.77, the compact filling and localization of E.92, and the following ordered lattices:

$$\begin{aligned} V &= \langle f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2\rangle, \\ W &= \langle A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2\rangle, \qquad M = \Lambda^2 V. \end{aligned} \tag{E95.1}$$

Write T_r, J_r for the retained actual graded actions of τ, c. Thus $T_1|_W = 7I$, and $J_1|_W$ is minus the permutation fixing A_0, B_0 and interchanging the three other pairs. Let $N_2 = N/\gamma_3 N$ be the intrinsic class-two quotient. All changes of its marking below are actual continuous automorphisms; the full action and every quotient map are transported along them.

There are explicit c-equivariant framed-cusp cocycles k_d, representing the actual generalized-Jacobian points z_d of E.37. On the actual G_K, for $K = \mathbb{Q}(\zeta_3)$, they are

$$k_d(g) = (A_g - I)b_d + r_d\kappa_3(g) + s_d\kappa_\zeta(g), \qquad \kappa_\zeta(g) = \frac{\chi(g) - 1}{3}. \tag{E95.2}$$

The three rational matrices $(b_d), (r_d), (s_d)$ are displayed in the proof and retained entrywise in the certificate. Their sums in (E95.2) are integral actual cocycles. The ordinary Kummer cocycle κ_3 uses the E.49 coherent positive roots, so its values on τ, τ' are $1, -1$. Put $K = (k_d(\tau))_d$. Its exact value is

$$4K = \begin{pmatrix} 0 & -2 & 2 & 0 & 2 & -2 & 2 & -2 \\ 4 & 0 & 2 & 8 & 0 & -2 & 4 & 2 \\ -4 & -12 & 10 & -4 & 6 & -8 & 8 & -14 \\ 2 & -28 & 8 & 4 & -4 & -22 & 2 & -20 \\ -6 & -10 & 22 & -12 & 20 & 2 & 14 & -8 \\ -4 & -15 & 11 & -8 & 9 & -9 & 5 & -13 \\ -6 & -6 & 5 & -6 & -2 & 5 & -10 & -5 \\ 2 & -9 & 4 & 2 & -5 & -4 & 1 & 4 \\ -4 & -11 & 9 & -10 & 8 & 4 & 0 & 6 \\ -4 & -15 & 7 & -2 & -6 & -12 & -12 & -12 \end{pmatrix}. \tag{E95.3}$$

The columns have the eight-cusp order in (E95.1).

Here is the complete wild class-two image over the entire global abelian localization image of E.92. Its local abelian coordinates are

$$([U], A) \in \mathcal{P}_1 \times W, \qquad \mathcal{P}_1 = V/(I - T_1)V^{J_1} \simeq C_3^3 \oplus \mathbb{Z}_3^6. \tag{E95.4}$$

For $A = \sum_d a_d e_d$, the actual obstruction is

$$\begin{aligned} \omega(U, A) &= 7\sum_d a_d[U - k_d(\tau), e_d] \in M, \\ \delta(U, A) &= [\omega(U, A)] \in \mathcal{B}_2 = \frac{M}{(T_2 - 7I)M + (J_2 + I)M}. \end{aligned} \tag{E95.5}$$

Its free coordinate is zero. Its fifteen finite coordinates have moduli

$$(m_1, \ldots, m_{15}) = (3^{[11]}, 9^{[4]}). \tag{E95.6}$$

The following formulas specify every coordinate and its inverse using the retained exact integral Smith maps. Let L_2 be the E.74 left Smith matrix of $[T_2 - 7I \mid J_2 + I]$, and let S select its fifteen nonunit nonzero rows. Let R_1 be the nine representative columns of the inverse first-degree left Smith matrix in E.87, after its unit coordinate has been set to zero. For $p = (p_1, \dots, p_9)$, with $p_1, p_2, p_3 \in C_3$, write $U = R_1 p$. Define

$$\begin{aligned} B_d(v) &= 7[v, e_d], \\ F_d &= SL_2B_dR_1, \\ c_d &= -SL_2B_dk_d(\tau), \\ f(p,a) &= \sum_d a_d(c_d + F_d p) \pmod{(m_i)}. \end{aligned} \tag{E95.7}$$

All eight 15×10 matrices $[c_d \mid F_d]$ are evaluated in `a_toric_wild.json`; the formula uses only the actual matrices and the numerical columns (E95.3). It contains no unevaluated arithmetic cochain. The exact image condition is the fifteen congruences

$$f(p,a) = 0. \tag{E95.8}$$

Their constant 15×8 matrix is

$$(c_d) = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 & 2 & 1 & 0 & 0 \\ 0 & 2 & 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 2 & 2 & 1 & 1 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 0 & 1 & 1 & 1 \\ 2 & 0 & 0 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 2 & 1 \\ 1 & 0 & 2 & 1 & 1 & 2 & 1 & 1 \\ 8 & 8 & 2 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 3 & 3 \\ 3 & 4 & 7 & 3 & 5 & 8 & 3 & 3 \\ 0 & 7 & 7 & 2 & 1 & 7 & 6 & 6 \end{pmatrix}. \tag{E95.9}$$

Every nonempty fibre over a fixed abelian class (p, A) has the complete compact coordinates

$$\mathcal{P}_2 \oplus \mathcal{K}_2 \simeq \mathbb{Z}_3^{46} \oplus C_3^9 \oplus C_9^5, \quad \mathcal{P}_2 = M/(I - T_2)M^{J_2}, \quad \mathcal{K}_2 = \ker((I - J_2)(T_2 - 7I)). \tag{E95.10}$$

A single saved rational matrix Q gives an integral particular relative correction precisely on (E95.8):

$$y_0 = Q\omega(U, A), \qquad (I - J_2)(T_2 - 7I)y_0 = -(I - J_2)\omega(U, A). \tag{E95.11}$$

With the saved saturated basis K_2 and inverse K_2^{-1} of E.75, the forward map takes

$$u = (U, z), \qquad a = (A, y_0 + K_2\eta), \qquad [z] \in \mathcal{P}_2, \quad \eta \in \mathbb{Z}_3^{22}, \tag{E95.12}$$

in the actual J-linear class-two coordinates constructed below. The inverse is the complete geometric gauge normalization followed by $[z]$ and $K_2^{-1}(y - y_0)$. All finite carries are retained by the saved Smith maps and their inverses.

The projection of (E95.8) to W is a completely evaluated finite congruence locus: exactly 16713 of the 9^8 residues modulo $9W$ occur. Its full two-digit list, the row-reduction maps, and

reconstructed horizontal solutions are retained in `a_toric_residues.json`. The following table also gives every horizontal kernel:

first digit	number	second digits per first digit	horizontal zero-locus type
0	1	675	$\mathbb{Z}_3^6 \oplus C_3^3$
nonzero axes of W^{J_1} mod 3	6	486	$\mathbb{Z}_3^6 \oplus C_3^3$
other admissible digits	54	243	$\mathbb{Z}_3^6$

(E95.13)

An axis here means one of the three lines generated by $A_1 - A_2, B_1 - B_2, C_1 - C_2$. The table concerns admissible second digits in each row; all other residues are excluded. Consequently the complete section fibre over an admissible fixed A is, noncanonically and with explicit Smith coordinate maps,

$$\mathbb{Z}_3^{52} \times C_3^{9+t(A)} \times C_9^5, \qquad t(A) = 3 \text{ in the first two rows of (E95.13), and 0 in the third.} \tag{E95.14}$$

For all nonunit relative values there is a short closed form. Write $A = 3a$ and put $\ell = p_6 + p_8 \pmod 3$. Then (E95.8) is exactly

$$(C + \ell D)\bar{a} = 0 \quad \text{in } \mathbb{F}_3^4, \tag{E95.15}$$

where

$$C = \begin{pmatrix} 2 & 2 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 2 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 & 1 & 1 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 2 & 2 & 0 & 2 & 2 & 0 & 0 \\ 0 & 2 & 2 & 2 & 1 & 1 & 0 & 0 \end{pmatrix}. \tag{E95.16}$$

Each of the three kernels has dimension five, and their pairwise intersections are the same three-dimensional $W^{J_1}/3W^{J_1}$. Their union has 675 elements. In particular all $9W$ admits every horizontal input. More precisely, the largest submodule of W with this last property is exactly

$$9W + 3W^{J_1}. \tag{E95.17}$$

There are also explicit exclusions with a global abelian preimage. For $A = A_0 + B_0$, the sum of rows three and four of $f(p, a)$ is 1 (mod 3) for every p. Thus every global abelian class localizing to that relative value fails to lift to a global intrinsic class-two section. Existence and all fibres of its global abelian preimages are the actual localization statement E.92. The other twenty-seven two-distinct-cusp sums are excluded as well by the retained exact affine systems.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d), follow-up(4). This evaluates the wild class-two obstruction on the entire global abelian localization image. Global good-place relations from E.73 and higher intrinsic obstructions remain part of E.23.

Proof. The actual finite marking and its coordinate morphism. Let $P = A_0$, $Q = A_2$, and use the actual framed point z_P of E.24 and E.37. Its compact Kummer class has the representative f_2 on τ: if the c-fixed $\mathcal{O}_3$ Tate basis t_E lifts P, the compatible division points give $(g-1)t_E/3$, and $(\tau - 1)t_E/3 = \zeta_3 t_E$. The actual triple relation is

$$3z_P = (3^{-2}, -3^{-1}, \lambda^{-1}, \lambda^{-2}, \zeta_3^2\lambda^{-3}), \quad \lambda = 1 - \zeta_3. \tag{E95.18}$$

Divide this relation in cohomology by the literal elliptic carry c_T of E.92. The free toric part of $\kappa(z_P) - [f_2]$ is then $(3^{-1}, 3^{-1}, \lambda^{-1}, \lambda^{-1}, \lambda^{-2})$. Its only remaining ambiguity is the three actual global μ_3 coordinates, with exponents $(\epsilon_1, \epsilon_2, \epsilon_3)$. The sign in the second rational factor has the coherent odd roots all equal to -1, so its pro-three Kummer cocycle is zero.

The two actual functions

$$f_{10} = \frac{x + \zeta_3^2 y + z}{x + y}, \qquad f_{01} = \frac{\zeta_3^2 x + y + z}{x + y} \tag{E95.19}$$

have divisors $A_1 + B_0 + C_1 - 3O$ and $A_0 + B_1 + C_2 - 3O$. Their vector-frame evaluations at z_P are $1/3$ and $1/(3\lambda)$, up to the literal sign $(-1)^{\mathrm{ord}_P f + \mathrm{ord}_O f}$. The coherent odd-root comparison for that sign is fixed. The toric character row is the divisor row; its two compact columns are $-$(divisor row)$U/3$, because the character kills the actual map $j_* = [3I; U]$. These are integral rows. The same possible Tate boundary $6a$ must occur on both free generators τ, τ'. Substitution of the two actual Kummer values therefore gives exactly

$$(\epsilon_1, \epsilon_2, \epsilon_3) = (1 - t, t, 1 + t), \qquad t \in \mathbb{F}_3. \tag{E95.20}$$

The remaining vector $(-1, 1, 1)$ is $j(P)$ in the three toric torsion factors, by the actual divisor Weil pairing of E.37. The certificate also verifies it through the same coefficient and cohomology maps.

Choose the representative $t \in \{0, 1, 2\}$ of this actual phase and set $u = 1 + 3t$. Changing the actual Tate basis to ut_E gives the following map on V:

$$P_u = I + t j_* \pi_E, \qquad P_u P_v = P_{uv}, \qquad P_u^{-1} = I - \frac{t}{1 + 3t} j_* \pi_E. \tag{E95.21}$$

It fixes W, commutes with T_1, J_1, and fixes both unit-character rows in (E95.19). The changed cohomology coordinates are P_u^{-1}. The certificate gives the actual integral J_1-fixed column h satisfying

$$P_u^{-1} k_P^{\mathrm{old}}(\tau) - k_P(\tau) = (T_1 - I)h, \qquad k_P(\tau) = (0, 1, -1, 1/2, -3/2, -1, -3/2, 1/2, -1, -1). \tag{E95.22}$$

Thus the new exponents are $(1, 0, 1)$.

This is also an actual full-group change of marking. For each of the two handle generators x_i, send x_i to x_i^u followed by the ordered product of peripheral powers tU_{di}, and fix each peripheral generator. Its Frattini map is invertible, so the resulting continuous endomorphism of the finite-rank free pro-three group is an automorphism Ψ_u. Transport the full action and all geometric maps by

$$\rho_{\mathrm{new}}(g) = \Psi_u^{-1} \rho_{\mathrm{old}}(g) \Psi_u, \qquad F_{\mathrm{new}} = F_{\mathrm{old}} \Psi_u. \tag{E95.23}$$

The inverse automorphism follows from the universal property and the inverse Frattini map; the displayed matrix is its abelianization. Formula (E95.23), rather than an equality of higher actions deduced from matrices, specifies the higher comparison. The previously retained finite joint character changes by $\eta_{\mathrm{new}} = \eta_{\mathrm{old}} + t \pmod 3$, as direct multiplication of its E.89 character row by P_u shows. The old phase and old character are retained through this map, without a numerical assignment to either.

All eight actual framed points. Use coherent roots with $\kappa_\lambda = (\kappa_3 + \kappa_\zeta)/2$, from the literal identity $\lambda^2 = -3\zeta_3$. E.92 gives the full cochain identity

$$3u_{f_2}(g) = c_{T,3}\kappa_3(g) + c_{T,\zeta}\kappa_\zeta(g) + (A_g - I)h_*. \tag{E95.24}$$

It holds on both τ, τ' and at c; both sides are cocycles on the actual free split source, so it holds everywhere. The exact columns $h_*, c_{T,3}, c_{T,\zeta}$ are retained in the certificate. Adding the phase-zero toric tuple gives the full k_P, not only its value on one Frobenius.

The actual CM automorphism $\sigma : [x : y : z] \mapsto [x : y : \zeta_3 z]$ sends P to Q and fixes O. If S is its literal cusp permutation and $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$, its map on V is

$$\Sigma = \begin{pmatrix} R & 0 \\ (UR - SU)/3 & S \end{pmatrix}. \tag{E95.25}$$

This follows from the torus permutation, the compact CM map, and equivariance of the specified j; these three maps uniquely determine the lower block. The matrix is integral, has cube I, and commutes with the actual G_K action. The vector-frame factors at Q and O are both ζ_3^{-1} and are 1 elsewhere. Hence

$$\sigma_* z_P = z_Q + \Delta, \quad \Delta_Q = 1, \quad \Delta_d = \zeta_3 \ (d \neq O, Q), \qquad k_Q = \Sigma k_P - \kappa(\Delta). \tag{E95.26}$$

For $a + b = d$ in $E[3]$, divide the chord or tangent through a, b by the line through $d, -d, O$; when $d = O$, use the tangent at O as denominator. The resulting function f has divisor $a + b - d - O$. Multiplication by this actual function in the framed line bundles proves

$$z_a + z_b - z_d = (\operatorname{lead}_r(f) / \operatorname{lead}_O(f))_{r \neq O}. \tag{E95.27}$$

All leading coefficients are computed in the literal parameters $x/z, y/z, z/x$ at A_i, B_i, C_i. For a line $L = ax + by + cz$, the first nonzero entries of the following lists give its order and leading coefficient:

$$\begin{array}{c|ccc} A_i & (0, c - b\zeta_3^i) & (1, a) & (3, -b\zeta_3^i/3) \\ B_i & (0, c - a\zeta_3^i) & (1, b) & (3, -a\zeta_3^i/3) \\ C_i & (0, a - b\zeta_3^i) & (1, c) & (3, -b\zeta_3^i/3). \end{array} \tag{E95.28}$$

They follow directly by substituting the cubic equation in the indicated local parameter. Every value is a literal $\pm\zeta_3^e \lambda^n$. The operations $P+P, Q+Q, P+Q, P+B_1, P+B_2, B_0+B_2$ generate all eight points from P, Q. The certificate verifies all sixty-four addition identities. In each comparison the same integral boundary column works on τ and τ'. It also verifies every triple relation, proper Kummer projection, and conjugation identity. Averaging the cocycle at d with the c-transform of that at cd preserves its actual class and gives the chosen c-equivariant representative.

For completeness the entire matrices in (E95.2) are as follows, with the same row and column orders as K:

$$24(b_d) = \begin{pmatrix} 8 & 4 & 0 & 16 & -4 & 0 & 4 & 8 \\ 0 & 4 & -4 & 0 & -4 & 4 & -4 & 4 \\ 0 & -2 & 2 & 0 & 2 & -2 & 2 & -2 \\ 4 & 0 & 2 & 8 & 0 & -2 & 4 & 2 \\ 4 & 2 & 0 & 8 & -2 & 0 & 2 & 4 \\ 0 & -3 & 3 & 0 & 3 & -3 & 3 & -3 \\ 2 & 0 & 1 & 4 & 0 & -1 & 2 & 1 \\ 6 & 3 & 0 & 12 & -3 & 0 & 3 & 6 \\ 2 & -3 & 1 & 4 & 0 & -4 & 2 & -2 \\ 6 & 3 & 3 & 12 & 0 & 0 & 6 & 6 \end{pmatrix}, \tag{E95.29}$$

$$6(r_d) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -4 & -1 & -1 & -2 & -2 & -2 & -3 & -3 \\ -1 & -4 & -1 & -2 & -2 & -2 & -3 & -3 \\ -1 & -1 & -4 & -2 & -2 & -2 & -3 & -3 \\ -2 & -2 & -2 & -4 & -1 & -1 & -3 & -3 \\ -2 & -2 & -2 & -1 & -4 & -1 & -3 & -3 \\ -2 & -2 & -2 & -1 & -1 & -4 & -3 & -3 \\ -3 & -3 & -3 & -3 & -3 & -3 & -6 & -3 \\ -3 & -3 & -3 & -3 & -3 & -3 & -3 & -6 \end{pmatrix}, \tag{E95.30}$$

$$6(s_d) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -8 & 8 & 0 & 5 & -5 & 8 & -8 \\ 5 & -18 & 7 & 10 & -3 & -16 & 5 & -11 \\ -5 & -7 & 18 & -10 & 16 & 3 & 11 & -5 \\ 0 & -10 & 10 & 0 & 7 & -7 & 7 & -7 \\ -2 & -3 & 5 & -1 & 0 & 4 & -5 & -1 \\ 2 & -5 & 3 & 1 & -4 & 0 & 1 & 5 \\ 1 & -5 & 10 & -1 & 8 & 5 & 6 & 9 \\ -1 & -10 & 5 & 1 & -5 & -8 & -9 & -6 \end{pmatrix}. \tag{E95.31}$$

These rational summands have the exact symmetries $b_{cd} = J_1 b_d$, $r_{cd} = -J_1 r_d$, $s_{cd} = J_1 s_d$. Their actual cocycle sum is integral, as established by the point and frame construction. Substitution of $\kappa_3(\tau) = 1$, $\kappa_\zeta(\tau) = 2$ proves (E95.3). In particular (E95.2) replaces the framed division input at every actual arithmetic element on this toric cusp subspace.

Literal geometric paths and class-two coordinates. Start with actual c-equivariant peripheral paths. For a rational cusp, coprime complement conjugacy in its pro-three path torsor gives a fixed path; for a conjugate pair, choose one path and its conjugate. The framed Abel–Jacobi comparison of E.37, including its coherent odd-root sign transport, identifies the abelian path class with $\kappa(z_d)$. If its initial representative is k_d^{old}, the unique boundary column carrying it to (E95.2) is obtained from

$$b_d^{\mathrm{path}} = (T_1 - I)^{-1}(k_d^{\mathrm{old}}(\tau) - k_d(\tau)). \tag{E95.32}$$

It is integral because the two actual cocycles have the same class. Uniqueness uses $H^0(P, V) = 0$, or equivalently the invertibility of $T_1 - I$ over $\mathbb{Q}_3$. The corresponding equation on τ' and the c-equivariance are retained. Choose group lifts of these columns equivariantly; fixed columns lift to fixed elements by successive coprime lifting through the lower central quotients. Conjugating the peripheral generators by these actual lifts performs the required change. Its Frattini map is the identity, so it extends to an automorphism of the full free group. All higher actions and maps are again transported as in (E95.23).

At class two, use the integral BCH lattice, valid because 2 is a unit in $\mathbb{Z}_3$. Choose a linear lift of V, and average it with its J-transform:

$$l(v) = \tfrac{1}{2}(l_0(v) + J l_0(J_1 v)), \qquad (v, m)(v', m') = (v + v', m + m' + \tfrac{1}{2}[v, v']). \tag{E95.33}$$

The lift remains the specified peripheral lift on W. It gives an actual Lie and group coordinate isomorphism, with $J(v, m) = (J_1 v, J_2 m)$. It can also be lifted to a full free-group automorphism by choosing lifts of its ten generator images. The peripheral action in these coordinates is the literal formula

$$T(e_d, 0) = (7e_d, -7[k_d(\tau), e_d]). \tag{E95.34}$$

It follows by expanding the actual path conjugation of the positive peripheral generator. Linearity of the Lie action gives its value on every $A \in W$. The action on the other handle lifts is retained as its actual transported action; it is not replaced by the graded matrix when a gauge operation uses it.

The obstruction and its integral inverse. Normalize a local section to have value 1 at c. Write its two independent values as $u = (U, z)$ and $a = (A, y)$. The actual source relation of E.77 says that $uT(a)u^{-1}a^{-7}$ is fixed by J. For $A \in W$, its abelian coordinate is zero and its central coordinate, by (E95.33)–(E95.34), is

$$(T_2 - 7I)y + \omega(U, A). \tag{E95.35}$$

The powers contribute no additional BCH term because $T_1 A = 7A$. Applying $I - J_2$ gives the claimed exact relation. The factor-one-half map (E75.7) sends its cohomology class to $[\omega]$ in $\mathcal{B}_2$. Every bracket in ω contains a peripheral vector, so the free compact elliptic coordinate vanishes.

For $v \in V^{J_1}$ the literal identity

$$7[(I - T_1)v, A] = -(T_2 - 7I)[v, A] \tag{E95.36}$$

proves descent through the complete first-degree gauge. A change $k_d \mapsto k_d + (T_1 - I)v_d$ also changes its contribution by the displayed image of $T_2 - 7I$. Thus both the path morphisms and the finite carries in $\mathcal{P}_1$ are accounted for. The certificate checks them as literal coefficient identities.

Let H^- be the last block of the inverse splitting matrix of J_2 in E.75, and write its exact Smith identity as

$$L^- H^- (I - J_2)(T_2 - 7I)R^- = D^-. \tag{E95.37}$$

There are twenty-three nonzero diagonal entries, with valuations $0^{[8]}, 1^{[11]}, 2^{[4]}$. Taking the inverse of these entries and setting the remaining coordinates to zero gives the actual rational matrix

$$Q = -R^-_{*,1:23}(D^-_{1:23,1:23})^{-1} L^-_{1:23,*} H^-(I - J_2). \tag{E95.38}$$

Its product with (E95.37) proves (E95.11) on the forty-four-dimensional cusp-containing summand. The exact diagonal divisibilities say that $Q\omega$ is integral precisely when (E95.8) holds. This proves both directions of the image condition and supplies an actual section whenever it holds. The saturated kernel and its inverse give all relative solutions.

To give the complete geometric quotient, first use the unique fixed abelian conjugator to carry U to its Smith representative $R_1 p$. Lift that conjugator to the actual J-fixed subgroup of N_2 and apply the literal operations $u \mapsto wuT(w)^{-1}$, $a \mapsto waw^{-1}$. These operations use the full transported action on w and retain its central terms. The remaining fixed conjugators are central and change only z by $(I - T_2)M^{J_2}$. Their quotient has Smith valuations $0^{[7]}, 1^{[9]}, 2^{[5]}$, with twenty-four free coordinates. The actual inverse Smith matrices and the saturated relative kernel inverse prove (E95.10)–(E95.12). No stabilizer is divided out after this step: its first nonzero graded coordinate would be a fixed vector for T_r, whose eigenvalues are the positive geometric weights of E.37, and none is 1. This also proves uniqueness of the aligning conjugators.

The complete finite image and its inverse algorithm. Equations (E95.7) are affine linear in p and linear in a. Their coefficients are reduced row by row with the actual moduli (E95.6). The three finite p coordinates can be changed by 3 without altering any residue. Thus the system factors through A mod $9W$, with the other six horizontal residues taken modulo 9. For a fixed A, exact Smith elimination on

$$[\,\textstyle\sum a_d S L_2 B_d \mid \operatorname{diag}(m_i)\,] \tag{E95.39}$$

decides the affine consistency, gives a particular U, and gives its complete integral kernel modulo the fixed gauge. The certificate retains both inverse matrices for forty-eight actual systems, including every single cusp and every two-distinct-cusp sum. Formula (E95.39) is the same finite algorithm for arbitrary input coefficients.

The exhaustive residue calculation has a shorter two-stage implementation whose intermediate maps are retained. For each $a_0 \in \mathbb{F}_3^8$, solve the fifteen equations modulo 3 as $p_0 = x_0 + Nx$. There are exactly sixty-one consistent a_0. Write $a = a_0 + 3a_1$, use the integer lifts of these matrices, and let F_0, c_0 be the four modulo-nine rows at a_0. Set

$$t_0 = (F_0x_0 + c_0)/3, \qquad Q_0 = F_0N/3, \qquad Z \text{ a basis of the left kernel of } F_0 \bmod 3. \tag{E95.40}$$

The divisions are exact integer divisions. If F_1, c_1 are the same rows evaluated on a_1, the remaining condition is precisely

$$Z(Q_0 + F_1N)x = -Z(t_0 + F_1x_0 + c_1) \pmod 3. \tag{E95.41}$$

A solution gives $p_0 = x_0 + Nx$, and the literal equation $F_0p_1 = -(t_0 + Q_0x + F_1p_0 + c_1)$ (mod 3) reconstructs $p = p_0 + 3p_1$. Thus the finite computation has a proved inverse at every step. Gaussian elimination over $\mathbb{F}_3$ on all 6561 first digits, and on all 6561 second digits for each consistent first digit, gives (E95.13). The saved lists are complete, and reconstructed witnesses are checked in the original fifteen congruences.

The kernel on the three finite horizontal coordinates depends only on a_0, because their modulo-nine coefficients are divisible by 3. Its dimension is exactly $t(A)$ in (E95.14). A closed subgroup of $C_3^3 \oplus \mathbb{Z}_3^6$ defined by finite congruences has free rank six; its torsion is exactly that computed kernel. Its Smith maps give continuous forward and inverse coordinates. Combining them with the linear particular correction (E95.38), and then the central fibre maps, proves the full fibre (E95.14), including its topology.

For $a_0 = 0$, divide the last four rows by 3. Their complete bilinear tensor factors as the single scalar $p_6 + p_8$, giving (E95.15)–(E95.16). The three matrices have rank three and any two stacked have rank five. Their common kernel is the span of the three cusp differences fixed by J_1. This proves the count $3 \cdot 3^5 - 2 \cdot 3^3 = 675$ and every nonunit assertion. Finally intersect the kernels of all constant and all horizontal coefficient rows, with their actual moduli. Its exact Hermite basis is

$$\begin{pmatrix} 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 9 & 6 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 9 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 9 & 6 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 \end{pmatrix}. \tag{E95.42}$$

It is exactly the Hermite basis of $9W + 3W^{J_1}$. Both inclusions and the inverse lattice coordinates are verified by integer multiplication. For the explicit exclusion $A_0 + B_0$, adding rows three and four of the original system gives the constant 1 (mod 3), with every horizontal coefficient zero. E.92 supplies actual global abelian preimages; functoriality of localization transfers the failure of the local intrinsic class-two lift to each of them. This concludes the exact wild calculation while retaining the other global relations of E.73. □

Global congruences on the zero free elliptic slice

E.96. An explicit global class-two locus and its exact modulo-three inertia image [Proved]

Keep the cubic curve, rational base tangent, genuine arithmetic action, and pro-three groups of E.37. Use the actual change of marking of E.95: the full action is transported by $\Psi^{-1}\rho\Psi$, every quotient map by $F\Psi$, and every path by the same isomorphism. Its Tate matrix fixes the eight peripheral vectors and commutes with the displayed first-degree arithmetic matrices. The following formulas use the resulting phase-zero path constants.

Let $W \subset V$ be the rank-eight toric lattice and put $M = \bigwedge^2 V$, $N_2 = N/\gamma_3 N$. Define the following actual subset of global continuous cohomology:

$$\mathcal{H}_0 = \ker\left(H^1(\mathbb{Q}, V) \longrightarrow H^1(\mathbb{Q}, T_3E) = \mathbb{Z}_3 t_1 \oplus C_3 t_2 \longrightarrow \mathbb{Z}_3 t_1\right). \tag{E96.1}$$

The arrows are induced by the compact projection and by E.37's specified coordinates. Every element of $\mathcal{H}_0$ has unique coordinates (ϵ, β), with $\epsilon \in \{0,1,2\}$ and $\beta \in H^1(\mathbb{Q}, W)$, represented by

$$\alpha = \epsilon u_{f_2} + \kappa(\beta). \tag{E96.2}$$

Here $u_{f_2}(\tau) = f_2$, its value at τ' is Jf_2, and its value at c is zero. The toric group is the actual Kummer group $(\widehat{\mathbb{Q}^\times}^3)^2 \times (\widehat{K^\times}^3)^3$ of E.92; its addition with the representative ϵ retains the finite carry c_T in that statement.

Use the order $(f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2)$. The following fixed columns occur in the formula:

$$\begin{aligned} h_* &= \tfrac{1}{2}(I+J)f_1 - q_0, \\ q_0 &= (0,0,0,0,0,0,\tfrac{1}{4},-\tfrac{1}{4},\tfrac{1}{4},-\tfrac{1}{4})^t, \\ c_3 &= (0,0,1,1,1,2,\tfrac{1}{2},\tfrac{1}{2},\tfrac{3}{2},\tfrac{3}{2})^t, \\ c_\zeta &= (0,0,0,1,-1,0,\tfrac{1}{2},-\tfrac{1}{2},\tfrac{1}{2},-\tfrac{1}{2})^t. \end{aligned} \tag{E96.3}$$

On G_K, the literal toric carry satisfies $\kappa(c_T) = c_3\kappa_3 + c_\zeta\kappa_\zeta$, where $\kappa_\zeta(g) = (\chi(g)-1)/3$ and κ_3 is the unchanged ordinary Kummer cocycle of three. E.95's eight actual framed-point cocycles have the specified rational columns

$$k_d(g) = (A_g - I)b_d + r_d\kappa_3(g) + s_d(\chi(g)-1)/3. \tag{E96.4}$$

All $3b_d, 3r_d, 3s_d$ are integral over $\mathbb{Z}_3$, and $r_d, s_d \in W \otimes \mathbb{Q}_3$. These are the explicit columns retained with the E.95 path calculation.

The quotient lattice used here is exactly $M/\mathbb{Z}_3 w$, with w from E.37. Let p denote the coordinate $e_{A_0} \wedge e_{A_2}$, whose coefficient in w is one. Define the integral quotient map and its integral section by

$$Qm = (m - m_p w)_{i \neq p} \in \mathbb{Z}_3^{44}, \qquad Sx = \text{the vector obtained by inserting zero at } p. \tag{E96.5}$$

Thus $QS = I$, $SQ = I - we_p^t$, and $Qw = 0$.

For each prime $\mathfrak{q} \nmid 3$ of K, use the compatible Frobenius and tame generator of E.37 and put

$$\begin{gathered} q = N\mathfrak{q}, \qquad k = q-1, \qquad n = k/3, \qquad t = \kappa_3(F_\mathfrak{q}), \\ y = \kappa(\beta)(F_\mathfrak{q}) \in W, \qquad v = \alpha(\tau_\mathfrak{q}) = \sum_d v_d e_d \in W, \\ L_b(v) = \sum_d v_d b_d \wedge e_d, \qquad L_r(v) = \sum_d v_d r_d \wedge e_d, \qquad L_s(v) = \sum_d v_d s_d \wedge e_d. \end{gathered} \tag{E96.6}$$

The valuations defining v and the Kummer values defining y, t use the actual field classes and source marking. Define the following integral column of forty-four coordinates:

$$\begin{aligned}\mathcal{N}_{\mathfrak{q}} = Q\big(&\epsilon k\, h_* \wedge v - 3kL_b(v) + 3y \wedge v + \epsilon t\, c_3 \wedge v \\ &+ \epsilon n\, c_\zeta \wedge v - 3tL_r(v) - kL_s(v)\big).\end{aligned} \tag{E96.7}$$

In particular the elliptic Frobenius matrix has disappeared from this exact formula. The original E.37 correction is

$$c_{\mathfrak{q}} = -\frac{\mathcal{N}_{\mathfrak{q}}}{3(q-1)}. \tag{E96.8}$$

Consequently the entire image of $H^1(\mathbb{Q}, N_2) \to H^1(\mathbb{Q}, V)$ inside $\mathcal{H}_0$ is the following explicitly specified set:

$$\begin{gathered}\mathcal{N}_{\mathfrak{q}} \in 3^{v_3(q-1)+1}\mathbb{Z}_3^{44} \quad \text{for every } \mathfrak{q} \nmid 3, \\ \text{for every } m \geq 0, \quad \mathcal{N}_{\mathfrak{q}} \in 3^{v_3(q-1)+1+m}\mathbb{Z}_3^{44} \quad \text{outside a finite set of primes.}\end{gathered} \tag{E96.9}$$

On the subgroup whose toric valuation vectors have a fixed finite support, the second line is automatic and the first line has only that finite support to check. Formula (E96.7), or its forty-four expanded polynomial rows in the certificate, evaluates every congruence.

There is also a small exact quotient of this image. Put

$$D_0 = O = C_0, \qquad D_1 = A_0, \qquad D_2 = B_0, \qquad e_O = -\sum_{d \neq O} e_d. \tag{E96.10}$$

On the stratum with compact coordinate ϵt_2, the image on all prime inertia vectors modulo three is exactly

$$\bigoplus_{p \neq 3} \mathbb{F}_3 e_{D_\epsilon}. \tag{E96.11}$$

The sum is indexed by rational primes. At every prime of K above p use its corresponding valuation vector, with the conjugate-prime equivariance induced by the actual torus. In the displayed eight coordinates this says respectively

$$\overline{v}_{\mathfrak{q}} \in \begin{cases} \mathbb{F}_3(1,1,1,1,1,1,1,1), & \epsilon = 0, \\ \mathbb{F}_3(1,0,0,0,0,0,0,0), & \epsilon = 1, \\ \mathbb{F}_3(0,0,0,1,0,0,0,0), & \epsilon = 2. \end{cases} \tag{E96.12}$$

Actual rational tangential sections at the three indicated cusps, twisted by rational Kummer classes, give all elements of (E96.11). The pure toric class $\kappa(7)e_{A_0}$ is excluded on the $\epsilon = 0$ stratum; the two classes obtained by twisting the actual A_0 and B_0 tangential sections by $\kappa(7)$ have zero correction at the marked prime of norm seven.

Every nonempty full class-two fibre over a point of (E96.9) is the torsor of E.37 under

$$\mathcal{R}_{\{p \neq 3\}}(\mathbb{Z}_3) \oplus \mathbb{Z}_3^{24} \oplus C_3^9 \oplus C_9^5. \tag{E96.13}$$

E.93 gives its exact central localization map. The slice with nonzero free elliptic coordinate, further geometric quotients, and the quartic curve remain in the scope of the active E.23 task.

Manuscript: section51.16, folios733v–736v, scans172–178, formulas(371)–(379); Part E(c),(d) and follow-up(4).

Proof. Coordinates and the actual marking. The compact projection and the vanishing of its global degree-zero group give the injective toric Kummer map used in E.92. Its description of the compact cohomology then gives the unique representatives (E96.2). The normalization

$u_{f_2}(c) = 0$ and the chosen two free generators specify an actual cocycle. In particular this choice uses the finite carry rather than treating the three possible representatives as an additive direct summand.

The E.95 change of marking has an actual lift Ψ on the free pro-three group. Sending a cocycle $z(g)$ to $\Psi^{-1}z(g)$ is a bijection of continuous cocycles for the actions ρ and $\Psi^{-1}\rho\Psi$, and sends geometric conjugations to geometric conjugations. Its inverse is induced by Ψ. The lower central series is characteristic, so this isomorphism descends to N_2 and to its abelian quotient. On the latter it fixes every e_d; on the compact Tate module it is the scalar u^{-1} with $u \equiv 1 \pmod 3$. Thus it preserves the zero free-coordinate slice and its compact residue ϵ. All obstruction and inertia assertions therefore transfer through this specified isomorphism. The actual transported peripheral loop at O still has abelian value e_O in (E96.10).

E.92 and the full cochain comparison in E.95 give

$$3u_{f_2}(g) = c_3\kappa_3(g) + c_\zeta(\chi(g) - 1)/3 + (A_g - I)h_*. \tag{E96.14}$$

This is an identity on the actual arithmetic source. The two sides are cocycles, agree on both free generators, and respect complex conjugation. The columns in (E96.3) are those obtained by the literal Shapiro map on c_T, including its third roots of unity. Equation (E96.4) has the same full-source meaning by the framed-point construction of E.95.

The exact inverse of the local differential. Fix $\mathfrak{q} \nmid 3$. On W the Frobenius matrix is qI. For $d = q^{-1}\bigwedge^2 A - I$, the following identities hold over $\mathbb{Q}_3$:

$$d(h \wedge e_a) = (A - I)h \wedge e_a, \qquad d|_{\wedge^2 W} = (q - 1)I. \tag{E96.15}$$

They follow by applying the exterior square to the two factors; no splitting of the integral extension $0 \to W \to V \to T_3E \to 0$ is used. Write

$$h = \epsilon h_*/3, \qquad u = y + \epsilon(c_3 t + c_\zeta k/3)/3. \tag{E96.16}$$

Equation (E96.14) says exactly that $\alpha(F) = (A - I)h + u$. Define the rational exterior column

$$X = h \wedge v - L_b(v) + \frac{u \wedge v - tL_r(v) - (k/3)L_s(v)}{k}. \tag{E96.17}$$

Apply (E96.15) to every term. Substitution of (E96.4) then gives the literal equality

$$dX = \alpha(F) \wedge v - \sum_a v_a k_a(F) \wedge e_a = D_{\mathfrak{q}}. \tag{E96.18}$$

The last column is the actual local extension defect in E37.10, including its peripheral sign.

The coefficient $w_p = 1$ proves all three identities following (E96.5). It also proves that $\ker Q$ on the integral lattice is exactly $\mathbb{Z}_3 w$: $Qm = 0$ gives $m = m_p w$ with integral m_p. Thus Q is the full quotient map to the actual domain of $d_{\mathfrak{q}}$. Its rational kernel is $\mathbb{Q}_3 w$. By E.37, the induced differential from this quotient to M_0 is invertible over $\mathbb{Q}_3$. Equation (E96.18) therefore proves that its unique negative inverse is $-QX$. Expanding $3kQX$ using (E96.16) is exactly (E96.7), which proves (E96.8).

For integrality, the residue field of every such K-prime contains μ_3, hence $k \in 3\mathbb{Z}_3$. Every coefficient in (E96.7) is integral: h_*, c_3, c_ζ are integral, the three path columns become integral after multiplication by three, and $n = k/3$ is integral. The quotient Q is integral as just proved. This proves the asserted numerator lattice without replacing $M/\mathbb{Z}_3 w$ by M_0. In particular the first row of Q retains the coefficient -9 at the pivot column.

Global completeness and convergence. All the Kummer cochains in (E96.6) are continuous with values in the indicated integral lattices. The inertia vectors tend to zero modulo every

power of three, and the other coefficients are bounded in $\mathbb{Z}_3$. Formula (E96.7) is therefore an integral restricted sequence. Equation (E96.8) identifies it with the very corrections appearing in E37.11. Division by $3(q-1)$ gives precisely the two exponents in (E96.9), including the variation of $v_3(q-1)$ with the prime.

E.37 proves both directions of the global extension criterion through its continuous localization complex; restriction from $\mathbb{Q}$ to K is injective on the obstruction because restriction followed by corestriction is multiplication by two. Hence the set in (E96.9) is the entire global image on the specified slice. Changing an integral path representative adds the differential of an integral restricted sequence, as in E.37. The same applies to the representative of the abelian cocycle. Thus the description is on cohomology classes with the displayed coordinates, rather than on an unrecorded choice of local representatives. For a valuation vector with finite support, $v = 0$ makes the entire column vanish at every other prime, proving the finite-support assertion. The degree-zero stabilizer vanishes by E.36, so the already evaluated central cohomology gives exactly (E96.13).

The complete first inertia obstruction. Let $x = X_p$ be the pivot coefficient of (E96.17). The $f_1 \wedge f_2$ coefficient of X is zero, because every term has a toric factor. The same coefficient of w is nine. Thus integrality of QX gives $9x \in \mathbb{Z}_3$. Put $\theta = 9x \bmod 3$.

Write (u_a, v_a^{pt}) for the actual coordinates of the cusp a in E37.2. The compact part of h_* is $(1,0)$, and the retained path columns satisfy

$$3\pi_E(b_a) \equiv (u_a, v_a^{\mathrm{pt}}) \pmod 3. \tag{E96.19}$$

This follows from the proper Kummer class of the actual point a; it is also checked for all eight literal columns. The coefficients of w at $f_1 \wedge e_a$ and $f_2 \wedge e_a$ are $-3u_a$ and $-3v_a^{\mathrm{pt}}$, respectively. Multiplying the two corresponding coordinates of QX by three and reducing modulo three gives

$$(\epsilon - u_a)\overline{v}_a + \theta u_a = 0, \qquad v_a^{\mathrm{pt}}(\overline{v}_a - \theta) = 0. \tag{E96.20}$$

These equations retain the pivot contribution.

For $\epsilon = 0$, every cusp has a nonzero point coordinate, so (E96.20) gives $\overline{v}_a = \theta$ for all eight cusps. For $\epsilon = 1, 2$, use the cusp B_1 with point coordinates $(1,1)$: its two equations give $\epsilon\theta = 0$, hence $\theta = 0$. All six cusps with $v_a^{\mathrm{pt}} \neq 0$ then have $\overline{v}_a = 0$. The remaining two coordinates are $A_0 = (1,0)$ and $B_0 = (2,0)$; their first equations leave exactly the coordinate asserted in (E96.12). This proves the necessary image inclusion.

For the reverse image inclusion, take the actual rational tangent at D_ϵ. Its path Kummer class has compact coordinate ϵt_2 and is unramified away from three. The local geometric inertia at that tangent is $\mathbb{Z}_3(1)$, with the actual cyclotomic Galois action. Twisting its section by the continuous cocycle $\kappa(a)$ for $a \in \mathbb{Q}^\times$ gives a genuine section of the full geometric group, before taking N_2. After transport to the base tangent its abelian class is $k_{D_\epsilon} + \kappa(a)e_{D_\epsilon}$. Its good-prime inertia vector is $v_p(a)e_{D_\epsilon}$.

Every finitely supported family in $\bigoplus_{p\neq 3} \mathbb{F}_3$ is obtained from the finite rational product $a = \prod p^{a_p}$, with $a_p \in \{0,1,2\}$. Since $K/\mathbb{Q}$ is unramified away from three, the valuation at each prime above p is the same. The cusp directions in (E96.10) are fixed by the actual cusp permutation, so this construction has precisely the required conjugate-prime equivariance. It realizes every family in (E96.11). For $\epsilon = 0$, the sign in $e_O = -\sum e_d$ is absorbed by replacing a with a^{-1}. This proves equality of the images.

The class $\kappa(7)e_{A_0}$ has compact residue zero and has inertia vector e_{A_0} at a prime over seven, which fails the first line of (E96.12). The certificate also evaluates its full correction at the actual marked norm-seven Frobenius, and finds thirty-six nonintegral coordinates. For the two rational cusp twists, the same local defect is $k_d(F) \wedge e_d - k_d(F) \wedge e_d = 0$; their forty-four corrections are all zero.

The program retains the exact quotient and its section, all fixed columns, and all forty-four expanded numerator polynomials. Their 569 terms have denominators prime to three. Seventy-two complete matrix comparisons agree with the original 44-dimensional inverse. These additional matrix inputs are explicitly labelled algebraic samples; the separate norm-seven witnesses use the actual marked arithmetic values. The finite calculation checks all 59,049 pairs of an inertia vector and pivot residue across the three strata, giving exactly three solutions on each stratum. The universal identities and the global proof above establish the result for the full continuous groups. □

An evaluated ordinary-unit obstruction and the corrected three-unit norm coset

E.97. The exact nonzero norm defect of ξc_1 [Proved]

Use the literal roots and fields of E.86:

$$\zeta^2+\zeta+1=0,\quad \xi^3=\zeta,\quad \beta_1^3=17\zeta,\quad \beta_2^3=\beta_1,\qquad L_i=K(\beta_i),\quad B_i=L_i(\xi),\quad K=\mathbb{Q}(\zeta). \tag{E97.1}$$

Put $c_1=\beta_1+1+3\zeta$, let S consist of the primes above $3,17$, and write $E_i=\mathcal{O}_{L_i}^\times$, $U_i=\mathcal{O}_{B_i,S}^\times$. The cyclic generator is always $\sigma(\beta_2)=\zeta\beta_2$, fixing B_1. Its restriction generates L_2/L_1.

In the actual ideal-class coordinates (a_2,b_2) of E86.4 and E86.8, the ordinary-unit norm defect is

$$\begin{gathered} E_1/N_{L_2/L_1}E_2 \xrightarrow{\omega_L} (C_9a_2\oplus C_3b_2)/\langle 3a_2\rangle \xrightarrow{\sim} C_3^2,\\ \omega_L(\zeta)=(0,1). \end{gathered} \tag{E97.2}$$

The first arrow is an isomorphism. Its literal definition is: choose $w\in L_2^\times$ of norm u, solve $\operatorname{div}(w)=(\sigma-1)D$, and take $[D]_3 \bmod \langle 3a_2\rangle$. In particular ζ is a field norm and has no ordinary-unit norm preimage.

The corrected field-norm target from E.84 has the exact S-unit defect

$$\begin{gathered} \xi c_1\in U_1\cap N_{B_2/B_1}B_2^\times,\qquad \omega_B(\xi c_1)=1 \quad \text{in } \mathrm{Cl}_S(B_2)\{3\}=C_3,\\ \xi c_1\notin N_{B_2/B_1}U_2. \end{gathered} \tag{E97.3}$$

Here ω_B is exactly the divisor map of E86.14, and 1 is the specified class of the prime $(73,\xi-55,\beta_2-63)$. Thus (E97.3) evaluates the remaining coset in E84.14. For every generator γ of the proved principal ideal $Q_1Q_2Q_3\subset\mathcal{O}_{B_2}$, the ordinary B_1-unit $\xi c_1/N\gamma$ has this same nonzero S-unit norm defect.

All these nonmembership assertions persist on completion for multiplication by 3^m. There is a full calculation on the resulting one-parameter line:

$$\begin{gathered} \left\{x\in\widehat{U}_2 : Nx=(\xi c_1)^r\right\}=\begin{cases}\varnothing, & r\notin 3\mathbb{Z}_3,\\ (\xi c_1)^z\ \ker(N:\widehat{U}_2\to\widehat{U}_1), & r=3z,\end{cases}\\ \ker N\simeq C_3\oplus\mathbb{Z}_3^{18}. \end{gathered} \tag{E97.4}$$

In the second row ξc_1 is included from B_1 into B_2, so its norm is its cube. The kernel in (E97.4) consists of three-units: its seventeen-valuations are zero. The abstract direct sum records its compact group type; the displayed coset uses the actual field norm.

The proof supplies a literal field-norm witness for ζ, its complete four-orbit divisor, and its arithmetic Artin coordinates. It evaluates this finite norm coset without assigning a value to the eventual exponent h of E.69 or to the transvection W of E.91. The comparison to the positional section coordinates remains the actual norm map of E.46 and E.88. The manuscript anchor is LM §51.16, folios733v–736v, scans172–178.

Proof. The ordinary divisor map and its exact ambiguity. E.86 proves $\mathrm{Cl}(L_2)\{3\} = C_9 a_2 \oplus C_3 b_2$, that σ acts trivially on this group, and that $[\mathfrak{q}_{3,2}] = 3a_2$. The other ramified prime is (β_2) and is principal. Norm on the ordinary class groups is onto by E.91. The identity $jN = 1+\sigma+\sigma^2 = 3$ therefore gives

$$j\,\mathrm{Cl}(L_1)\{3\} = 3\,\mathrm{Cl}(L_2)\{3\} = \langle 3a_2 \rangle. \tag{E97.5}$$

An invariant fractional ideal is generated by extended base ideals and the ramified primes. To see this on each prime orbit, its coefficients must all be equal; the orbit sum is an extended ideal at an unramified prime, while at a totally ramified prime the fixed upper prime generates the quotient by the extended ideal. Consequently the three-primary strongly ambiguous subgroup is exactly $\langle 3a_2 \rangle$.

For completeness, the map in (E97.2) is the inverse of the exact map in Lemmermeyer, Proposition1, whose full proof is read here [Lem13, Proposition1]. Its orientation can be checked directly. A norm-zero divisor is in the image of $\sigma - 1$, separately on each finite prime orbit. Thus $Nw = u \in E_1$ gives the indicated D. Replacing D by another solution adds an invariant ideal. Replacing w by a second norm preimage multiplies it by $\sigma(v)/v$, by Hilbert90 [Mil20b, II, Proposition1.22 and Corollary1.23], and changes D by a principal ideal, followed by that invariant ambiguity. Conversely, a zero class modulo invariant ideals permits D to be adjusted to a principal divisor; dividing w by the corresponding Hilbert90 element gives an ordinary unit of the same norm. Every ambiguous ideal class gives a norm of a generator of its $\sigma - 1$ divisor. This proves both directions of the isomorphism and its exact kernel.

In this extension every ordinary L_1-unit is a field norm. The certified basis in E.64 consists of two free units and ζ; their residues at the unique prime above seventeen are respectively ζ^2, ζ^2, ζ. All are cubes in $\mathbb{F}_{289}^{\times}$. Thus they are local norms there. At unramified places ordinary units are local norms, and all infinite places are complex. The global reciprocity product then kills the only remaining local norm character, at three. The cyclic Hasse norm theorem [Mil20b, VIII, Theorem3.1] gives the field-norm assertion. This verifies the domain E_1 in (E97.2). The quotient has order nine and is killed by three.

An explicit cyclic-algebra switch. Set $b = \beta_1$, so $b^6 + 17b^3 + 289 = 0$ and $\zeta = b^3/17$. Define $u_j \in L_1$ by the rows of this integer matrix:

$$\begin{pmatrix} u_0 \\ u_1 \\ u_2 \end{pmatrix} = \frac{1}{11169} \begin{pmatrix} -717026 & -1721029 & -474164 & 29437 & -20479 & -31853 \\ -2831945 & 130475 & 550222 & 53428 & -19498 & -6494 \\ -2266763 & 678062 & 13447 & -275174 & -49207 & 9805 \end{pmatrix} \begin{pmatrix} 1 \\ b \\ b^2 \\ b^3 \\ b^4 \\ b^5 \end{pmatrix}. \tag{E97.6}$$

Direct reduction modulo the displayed degree-six polynomial proves the exact identity

$$u_0^3 + \zeta u_1^3 + \zeta^2 u_2^3 - 3\zeta u_0 u_1 u_2 = b. \tag{E97.7}$$

Thus $u_0 + u_1\xi + u_2\xi^2$ is a literal norm preimage of b in B_1/L_1. The exploratory computation obtained it from the already certified B_1 S-units; (E97.6)–(E97.7) verify it independently of that discovery.

The switch to the required extension is given by actual matrices over L_1:

$$A = \begin{pmatrix} 0 & 0 & \zeta \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B = (u_0 I + u_1 A + u_2 A^2)\,\mathrm{diag}(1, \zeta, \zeta^2). \quad A^3 = \zeta I,\ B^3 = bI,\ BA = \zeta AB. \tag{E97.8}$$

In L_2 take the unique vector v with $(B-\beta_2 I)v=0$ and $v_3=1$. The certificate checks that this eigenspace is one-dimensional and that this normalization is defined. Both Av and $\sigma(v)$ lie in the $\zeta\beta_2$-eigenline, hence

$$Av = w\,\sigma(v), \qquad w = \frac{(Av)_1}{\sigma(v)_1} \in L_2^{\times}. \tag{E97.9}$$

This is a literal definition of the retained witness. Applying A three times and using that it commutes with coefficient-wise σ gives $\zeta v = w\,\sigma(w)\sigma^2(w)v$. Consequently $N_{L_2/L_1}w=\zeta$. The program checks this both in the relative field $L_1[T]/(T^3-b)$ and in the absolute field $\mathbb{Q}[T]/(T^{18}+17T^9+289)$.

The full divisor and its actual class. For each row of the following table let $P=(p,\beta_2-r)$, and order its three primes as $P,\sigma P,\sigma^2 P$. The vector v_P gives their valuations of w; d_P gives their coefficients in D. The last column is the arithmetic Artin value of the contracted prime (p,β_1-r^3) in the lower genus coordinates $\Gamma_L=C_9\oplus C_3^2$ of E86.8:

i	p_i	r_i
1	7	3
2	13	4
3	988728187	449890749
4	212464545716738508121	96597925880218313619

i	v_{P_i}	d_{P_i}	$\mathrm{Art}(P_i\cap L_1)$
1	$(-1,0,1)$	$(0,0,-1)$	$(2,1,0)$
2	$(1,0,-1)$	$(0,0,1)$	$(4,2,0)$
3	$(-1,1,0)$	$(0,-1,-1)$	$(1,0,2)$
4	$(-1,1,0)$	$(0,-1,-1)$	$(0,0,0)$

(E97.10)

All four rational integers p are proved prime in the certificate. The prime ideals are kernels of the actual residue maps on the maximal orders. Their residue degrees are one, their relative orbits have length three, and these rows are the complete factorization of (w). For each row the coefficient identity is

$$(d_2-d_0,\,d_0-d_1,\,d_1-d_2)=v_P. \tag{E97.11}$$

Multiplication of the full HNF ideals independently checks $\sigma(D)/D=(w)$.

The genus field is abelian over L_1; therefore the three conjugate primes in a row have the same Artin value in its subgroup over L_2. Their relative residue degrees are one, so this is exactly the Artin value of the contracted lower prime. Summing the last column with coefficients $\sum_i d_{P,i}=-1,1,-2,-2$ gives

$$\mathrm{Art}(D)=(0,1,2)\in\Gamma_L, \qquad (x,y,z)\longmapsto(x-3y-3z,\,y) \quad\Longrightarrow\quad [D]_3=(0,1). \tag{E97.12}$$

This uses the actual E86.8 class map. Its integral kernel coordinates, including the redundant order-one coordinate, are checked by the inverse matrices

$$C=\begin{pmatrix} 1 & -3 & -3\\ 0 & 1 & 0\\ 1/3 & -2/3 & -2/3\end{pmatrix}, \qquad C^{-1}=\begin{pmatrix} -2 & 0 & 9\\ 0 & 1 & 0\\ -1 & -1 & 3\end{pmatrix}. \tag{E97.13}$$

The source of C is $\ker(2(x+y+z) \bmod 3)$, so its last coordinate is integral. Its target orders are $(9,3,1)$. Quotienting by the ramified class $(3,0,0)$ gives relation lattice $\mathrm{diag}(3,3,1)$; the class of D has lattice coordinate $(0,1/3,0)$. This proves the nonzero value in (E97.2).

The horizontal transfer of the obstruction. On B_2 let

$$h(\xi)=\zeta\xi, \qquad h(\beta_2)=\beta_2. \tag{E97.14}$$

It generates B_2/L_2, and commutes with σ. Thus the actual norms form the commutative square

$$\begin{array}{ccc} B_2^\times & \xrightarrow{N_\sigma} & B_1^\times \\ \downarrow N_h & & \downarrow N_h \\ L_2^\times & \xrightarrow{N_\sigma} & L_1^\times . \end{array} \tag{E97.15}$$

E.84 proves the field-norm membership of ξc_1. Choose a field preimage x. The unique vertical prime extensions over each of the three Q_i and three T_i have residue degree one. Since $(c_1) = \mathfrak{q}_{3,1}$ in L_1, the six valuations of x are $(1,1,1,0,0,0)$. This statement concerns every field preimage, even when it has additional support outside S. The horizontal extension is S-split. Consequently

$$y = \frac{N_h x}{c_1} \in L_2^\times, \qquad v_{\mathfrak{q}_{3,2}}(y) = v_{\mathfrak{q}_{17,2}}(y) = 0, \qquad N_\sigma y = \frac{N_h(\xi c_1)}{c_1^3} = \frac{\zeta c_1^3}{c_1^3} = \zeta. \tag{E97.16}$$

An S-unit x would make y an ordinary unit, contrary to (E97.2). This already proves the nonmembership in (E97.3).

The same square fixes its exact sign and S-class coordinate. In the free group of divisors outside S, write $\mathrm{div}_S(x) = (\sigma - 1)D_B$. Then

$$\mathrm{div}(y) = (\sigma - 1)N_h D_B, \qquad N_{B_2/L_2} : \mathrm{Cl}_S(B_2)\{3\} \longrightarrow \mathrm{Cl}_S(L_2)\{3\}, \quad z \longmapsto (0, z). \tag{E97.17}$$

The first equality is an equality of full divisors, because y has zero S-valuations. The second is the literal E86.5 map in the coherent prime bases. The calculation (E97.12) says that its image on $[D_B]_3 = \omega_B(\xi c_1)$ is $(0,1)$. Thus $\omega_B(\xi c_1) = 1$, as claimed.

E.86 proves that $Q_1Q_2Q_3$ is principal. A generator γ has the same six valuations as x. Accordingly $\xi c_1/N_\sigma\gamma$ is an ordinary B_1-unit. Multiplying by this actual S-unit norm does not change the class in E97.3, proving the assertion for every such generator.

Completion and all powers in this line. The unit and S-unit groups here are finitely generated. Every norm image under consideration contains the cubes of its target group, through inclusion of a base element. Their finite cokernels are therefore killed by three; completion for 3^m-multiplication preserves those cokernels and the displayed obstruction homomorphisms. Their completed norm images are also closed, being images of compact finitely generated $\mathbb{Z}_3$-modules. Thus the value $1 \in C_3$ in E97.3 is still nonzero after completion. Its restriction to the powers of ξc_1 is $r \mapsto r \bmod 3$, whose kernel is exactly $3\mathbb{Z}_3$.

For $r = 3z$, inclusion of the element $(\xi c_1)^z$ gives the stated preimage; multiplication by the actual norm kernel gives every other preimage. The norm on the six S-valuations has coefficient one at each prime. Hence a norm-one S-unit has zero seventeen-valuations. The kernel has free rank $(27 - 1 + 6) - (9 - 1 + 6) = 18$, and its torsion is exactly the kernel of cubing on μ_9, namely μ_3. This also follows from the actual kernel calculation in E.86 and proves (E97.4).

The portable certificate rebuilds only the degree-six lower genus computation and then verifies (E97.6)–(E97.17). It retains every field element, eigenvector, ideal HNF, prime orbit, Artin coordinate, and inverse class map. The two earlier degree-eighteen BNF attempts remain separately recorded as running attempts; no upper BNF or unproved unit basis is used in this proof. □

All sections over the split-prime source at seven

E.98. The complete seven-and-three class-two locus and the full-to-metabelian ramified fibres [Proved]

We retain the actual based geometric pro-three group N of the cubic with its nine cusps

removed, the marking of E.95, and its genuine arithmetic action. Put

$$N_2 = N/\gamma_3 N, \qquad N_{\mathrm{m}} = N/N'', \qquad V = N^{\mathrm{ab}}, \quad M = \bigwedge^2 V, \quad W = \langle e_d : d \neq O \rangle \subset V. \tag{E98.1}$$

Here V has rank ten and W has rank eight over $\mathbb{Z}_3$. For each of these three target groups H, let $\mathrm{Sect}_{37}(H)$ be the geometric conjugacy classes of continuous global sections whose cocycles are unramified outside $\{3, 7\}$. The following coordinates compute these stated global section spaces.

Write $T = \Phi_\tau$ and $J = \Phi_c$ for the actual Frobenius and complex-conjugation actions specified below. With $T_1 = T|V$, the horizontal abelian space is

$$P_1 = V/(I - T_1)V^{J_1} \simeq C_3^3 \oplus \mathbb{Z}_3^6. \tag{E98.2}$$

Use the retained E.37 Smith representatives $U \in V$ for P_1. The entire abelian section space is $P_1 \times W$. Set $e_O = -\sum_{d \neq O} e_d$, and let $k_d \in V$ be the actual E.95 Frobenius cusp columns; set $k_O = 0$. The image of $\mathrm{Sect}_{37}(N_2)$ in this abelian space consists exactly of the following pairs:

$$\begin{aligned} &v \in 9W : && U \text{ arbitrary;} \\ &v \equiv r e_D \pmod{9W}, \quad 1 \leq r \leq 8 : && r(U - k_D) \in (T_1 - I)V + \mathbb{Z}_3 e_D. \end{aligned} \tag{E98.3}$$

The nine lines have disjoint nonzero residues modulo nine, so D, r in the second row are unique. Every local quotient in this formula is given by an explicit Smith map with both inverse transformations:

$$V/((T_1 - I)V + \mathbb{Z}_3 e_D) \simeq C_3^5 \oplus C_9^2 \qquad (D \in E[3]). \tag{E98.4}$$

In particular the complete relative image and the lattice admitting every horizontal input are, respectively,

$$\bigcup_{D \in E[3]} (\mathbb{Z}_3 e_D + 9W), \qquad 9W. \tag{E98.5}$$

The first set has exactly 73 residues modulo nine.

Over every pair in (E98.3), the complete geometric fibre is

$$\mathbb{Z}_3^{25} \times C_3^9 \times C_9^5. \tag{E98.6}$$

Over a fixed v in (E98.5), the complete fibre of the relative-inertia map is

$$\mathbb{Z}_3^{31} \times C_3^{9+t(v)} \times C_9^5, \qquad t(v) = \begin{cases} 3 & v \in 3W, \\ 0 & v \notin 3W. \end{cases} \tag{E98.7}$$

These are chosen continuous coordinates with explicit inverses. The proof gives the genuine conjugation changes between representatives and all origins; it does not assign an abelian group operation to the nonabelian section space.

For the full and intrinsic metabelian targets, the zero-inertia subspace is the complete unramified space computed in E.41. Every ramified class has exactly the coordinates

$$\coprod_{D \in E[3]} (\mathbb{Z}_3 \setminus \{0\}) \times \{0, 1, 2\} \times H^J \backslash H, \qquad H = N, \ N_{\mathrm{m}}. \tag{E98.8}$$

The actual forward map uses the chosen positive peripheral element δ_D and its tangential Frobenius $z_D \tau$:

$$\begin{aligned} a(t) &= g \delta_D^r g^{-1}, \\ a(\tau) &= g \delta_D^b z_D T(g)^{-1}, \qquad a(c) = 1, \\ a(t') &= J(a(t))^{-1}, \qquad a(\tau') = J(a(\tau)). \end{aligned} \tag{E98.9}$$

The path and peripheral elements for N_{m} are the actual projections of those for N.

There are explicit coordinates for the last factor:

$$\begin{aligned} H^J\backslash H &\longrightarrow \mathrm{Anti}_J(H) := \{x \in H : J(x) = x^{-1}\}, \\ [g] &\longmapsto g^{-1}J(g), \qquad x \longmapsto [x^{-1/2}]. \end{aligned} \tag{E98.10}$$

Moreover these spaces have recursive Hall and Chen coordinates

$$\mathrm{Anti}_J(N) \simeq \prod_{m\geq 1} \mathbb{Z}_3^{\nu_m}, \qquad \mathrm{Anti}_J(N_{\mathrm{m}}) \simeq \prod_{m\geq 1} \mathbb{Z}_3^{\bar{\nu}_m}. \tag{E98.11}$$

All ranks are evaluated by the following formulas. Put $c_m = [z^m](1-z)^{-4}(1+z)^{-6}$ and $a_d = -2$ for odd d, $a_d = 10$ for even d. Then

$$\begin{aligned} \nu_m &= \frac{1}{2m}\sum_{d\mid m} \mu(d)(10^{m/d} - a_d^{m/d}), \\ \bar{\nu}_1 &= 6, \qquad \bar{\nu}_m = \frac{1}{2}\left((m-1)\binom{m+8}{m} + 2c_{m-1} + c_m\right) \quad (m \geq 2). \end{aligned} \tag{E98.12}$$

These coordinates retain the action on each actual nilpotent quotient, including its higher commutator terms.

The map from full to metabelian ramified sections is

$$(D, r, b, [g]) \longmapsto (D, r, b, [\bar{g}]). \tag{E98.13}$$

It is onto, and every fibre, with its explicit origin, is

$$(N'')^J\backslash N'' \simeq \prod_{m\geq 4} \mathbb{Z}_3^{\nu_m - \bar{\nu}_m}. \tag{E98.14}$$

The degree-four factor has rank 504. Thus (E98.14) evaluates the full global fibre of the displayed map.

There are also actual global class-two sections in this source with inertia $9(e_{A_0} + e_{B_0})$ and horizontal coordinate zero. Every such section has no intrinsic metabelian lift and no full lift. The explicit central inertia and the complete tame relation check appear below.

Manuscript: §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and the fourth exactness follow-up. The unrestricted global and higher-stage requirements remain recorded in E.23.

Proof. The actual arithmetic source and both comparison maps. Let $K = \mathbb{Q}(\zeta_3)$, let $\mathfrak{p}, \bar{\mathfrak{p}}$ be the primes over seven, and put $\pi = 1 + 3\zeta_3$. We use the actual normalized Frobenii of E.56 at these two prolongations. Its source isomorphism, specialized to seven, is

$$G_{37} = \left(D_7 \underset{3}{*} D_7'\right) \rtimes \langle c\rangle, \quad \begin{cases} D_7 = \langle t, \tau \mid \tau t \tau^{-1} = t^7\rangle_3, \\ D_7' = \langle t', \tau' \mid \tau' t' (\tau')^{-1} = (t')^7\rangle_3, \\ c\tau c = \tau', \qquad ctc = (t')^{-1}. \end{cases} \tag{E98.15}$$

Its characters are $\chi(\tau) = \chi(\tau') = 7$, $\kappa_3(3)(\tau) = 1$, $\kappa_3(3)(\tau') = -1$, and $\kappa_3(7)(t) = \kappa_3(7)(t') = 1$. The latter cocycle vanishes at both chosen Frobenii. The two former Frattini characters have determinant $-2(7-1)/3$, a three-adic unit. Their actual images therefore form a free basis of the group unramified outside three computed in E.34.

We recall the exact kernel comparison used in E.56. [Win07, Theorem2.4, its proof and the remark after Proposition2.1] identifies the kernel with the continuous free product of the actual inertia groups over $P/\langle\tau\rangle$ and $P/\langle\tau'\rangle$. The two bundle relations are $(p\tau^z, t^u) = (p, t^{7^z u})$ and their primed versions. The coset tree for $D_7 *_3 D_7'$ has these same stabilizers and these same conjugation maps. Maps induced by the four actual generator inclusions give inverse maps on these bundles and the identity on the quotient P. This proves (E98.15) with its specified local subgroups. All primes above three are in the ramification set, and the cyclotomic tower has nontrivial unramified completion at both added primes, as required in that theorem. The actual Frobenii and inertias also belong to the full relative arithmetic group; the same relations give a reverse map whose composite with the arithmetic projection is identity. It commutes with c.

The actual geometric action is unramified outside three by E.34 and the based tangent comparison of E.78. Its image over K is pro-three. Pairing a cocycle with this action gives a homomorphism into a pro-three semidirect product. Consequently a cocycle in the stated ramification locus factors through (E98.15); inflation gives the inverse construction on the original arithmetic source. Both maps preserve geometric conjugacy. Thus relations in (E98.15) construct the requested global sections in this locus.

An odd profinite group has unique continuous square roots: on each finite quotient of odd exponent e, the inverse of squaring sends x to x^k for $2k \equiv 1 \pmod{e}$. These inverses commute with quotient maps. A cocycle value $z = a(c)$ satisfies $J(z) = z^{-1}$. Geometric conjugation by $z^{-1/2}$ makes this value one. The residual conjugators are exactly H^J. Set $u = a(\tau)$ and $a = a(t)$. The other two values are forced as in (E98.9), and the first tame relation becomes

$$uT(a)u^{-1} = a^7. \tag{E98.16}$$

The second is its J-image. Hence normalized sections are exactly the solutions of (E98.16), modulo $(u, a) \mapsto (huT(h)^{-1}, hah^{-1})$, $h \in H^J$. The free-product universal property proves sufficiency and continuity, in addition to necessity.

The exact class-two differential. In the E.95 basis $(f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2)$, put $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and retain the gluing matrix U_{gl} of E.37. The actual abelian Frobenius is

$$T_1 = \begin{pmatrix} I_2 + 3R & 0 \\ U_{\mathrm{gl}}R - 2U_{\mathrm{gl}} & 7I_8 \end{pmatrix}. \tag{E98.17}$$

Use the actual J-linear class-two coordinates of E.95, with product $(v, m)(v', m') = (v + v', m + m' + [v, v']/2)$. Write $B = \bigwedge^2 T_1$ and $J_2 = \bigwedge^2 J_1$. For the actual cusp path columns, $T(e_d, 0) = (7e_d, -7[k_d, e_d])$. The abelian part of (E98.16) forces $v \in W$; the proper block $T_1 - 7I$ is rationally invertible. Its central part is exactly

$$\lceil m + \Delta(U, v) = 0, \qquad \lceil = B/7 - I, \quad \Delta(U, v) = \sum_{d \neq O} v_d[U - k_d, e_d]. \tag{E98.18}$$

Here $u = (U, z)$ and $a = (v, m)$; the central z cancels. Only the actual action on peripheral inertia occurs in (E98.18), and those columns are fully evaluated in E.95.

Let $j = (3I_2; U_{\mathrm{gl}})$ and $w = j(f_1) \wedge j(f_2)$. It is primitive, $Bw = 7w$, and $J_2 w = -w$. Order the forty-five wedge coordinates by increasing pairs of the displayed ten indices. The coefficient of w at $A_0 \wedge A_2$ is one; call this position p. Let Q subtract the p-th coordinate times w and then delete that coordinate, and let S insert zero at that position. Let P delete the $f_1 \wedge f_2$ coordinate. The actual maps satisfy

$$QS = I_{44}, \quad SQ = I_{45} - we_p^{\mathsf{t}}, \quad Qw = 0, \qquad D = P\lceil S. \tag{E98.19}$$

The image of $\lceil$ is in $M_0 = \ker(e^{\mathsf{t}}_{f_1 \wedge f_2})$. The retained exact Smith identity $LDR = D_0$ has

$$D_0 = \operatorname{diag}(1^{[14]}, 3^{[18]}, 9^{[12]}), \qquad D^{-1} = RD_0^{-1}L. \tag{E98.20}$$

The certificate checks both inverse products and all four change-of-basis inverse products. Thus $\lceil$ has precisely kernel $\mathbb{Z}_3 w$, and a particular solution is

$$m_0(U, v) = -SD^{-1}P\Delta(U, v). \tag{E98.21}$$

It is integral exactly when the eighteen modulo-three and twelve modulo-nine Smith congruences vanish. Every integral solution is uniquely $m_0(U, v) + \lambda w$, $\lambda \in \mathbb{Z}_3$; its inverse scalar is its p-th coordinate. The full cokernel obstruction in this equation is $C_3^{18} \oplus C_9^{12}$.

Every residue and the nine exact cusp lattices. The E.37 first-degree Smith map for $(I - T_1)(I + J_1)$ has nonzero diagonal $1, 3, 3, 3$. Its inverse target matrix, with the unit column removed, gives the ten-by-nine representative matrix R_1 used here. Its first three coordinates are reduced modulo three and its last six are arbitrary in $\mathbb{Z}_3$. For each of the eight d, the certificate records the thirty-by-ten matrix of the nonunit rows of

$$LP[\,\cdot\,, e_d], [-k_d \mid R_1]. \tag{E98.22}$$

All gauge columns vanish in their exact cokernel, and three times each finite horizontal column vanishes. This proves the descent to $P_1 \times W$, with its finite carries included.

All coefficients of (E98.21) have three-adic denominator dividing nine. Therefore its integrality depends on v mod 9 and on the indicated horizontal residues. The retained two-digit elimination tests all 6561 first digits. Exactly nineteen admit a horizontal value. For each of those it tests all 6561 second digits, retaining the divided constant and divided-kernel carry, and solves the remaining linear system over $\mathbb{F}_3$. The zero first digit has nineteen admissible second digits; each of the eighteen nonzero first digits has three. All seventy-three resulting horizontal witnesses are substituted into every original congruence and into the full rational inverse (E98.21). Their inertia vectors are exactly

$$\{re_D \bmod 9W : D \in E[3],\ 0 \le r < 9\}. \tag{E98.23}$$

The set equality is checked directly, with all actual vectors retained. Thus the finite enumeration proves exclusion as well as construction of the relative image.

Here is the exact horizontal criterion on these residues. For every D, the projection of w in $\bigwedge^2(V/\mathbb{Z}_3 e_D)$ is primitive. The certificate supplies an integral unimodular basis containing e_D, its inverse, and a unit wedge coefficient after deleting that direction. Its ten-by-eleven matrix $[T_1 - I \mid e_D]$ has Smith diagonal $1^{[3]}, 3^{[5]}, 9^{[2]}$, with an explicit rational right inverse. This proves (E98.4) as an isomorphism with specified maps.

The actual cusp origin for $D \neq O$ is $U = k_D$, and for O it is $U = 0$ with its retained central peripheral coordinate. In either case (E98.21) at the origin is integral. Put $h = (T_1 - I)^{-1}(U - k_D)$ over $\mathbb{Q}_3$; this inverse exists because none of the eigenvalues of T_1 is one. The identity

$$\lceil [h, e_D] = [(T_1 - I)h, e_D] \tag{E98.24}$$

gives the change of particular solution as $-SQ[rh, e_D]$. Integrality of this expression is equivalent to $rh \in V + \mathbb{Q}_3 e_D$. Indeed a rational multiple of w making the wedge integral projects to a multiple of the primitive projected w; that multiple is integral. Removing it shows $[rh, e_D] \in \bigwedge^2 V$, and the integral basis containing e_D identifies this with the stated condition on rh. Applying $T_1 - I$, and using $(T_1 - I)e_D = 6e_D$, identifies it exactly with $r(U - k_D) \in (T_1 - I)V + \mathbb{Z}_3 e_D$. Here $V \cap \mathbb{Q}_3 e_D = \mathbb{Z}_3 e_D$ by primitivity. Adding $9W$ to the inertia changes (E98.21) integrally for every horizontal value. This proves both rows of (E98.3).

The map $P_1 \to V/((T_1 - I)V + \mathbb{Z}_3 e_D)$ is onto, since it is induced by the identity on V. For a unit r its kernel has index 3^9, and for $r \in 3\mathbb{Z}_3 \setminus 9\mathbb{Z}_3$ the kernel of multiplication by r after that map has index nine. The certificate evaluates all twenty-seven torsion inputs of P_1: only zero survives in the first kernel, and all survive in the second. Thus their additive types are respectively $\mathbb{Z}_3^6$ and $\mathbb{Z}_3^6 \oplus C_3^3$; for inertia in $9W$ the whole P_1 occurs. Exact row-kernel intersection of every constant and horizontal coefficient in (E98.22) has Hermite basis $9I_8$. This independently proves the second claim of (E98.5). The quotient (E98.4) also shows why every nonzero inertia residue imposes a nonzero horizontal condition.

All class-two fibres and actual inverse coordinates. For every allowed (U, v), choose m from (E98.21) and an arbitrary central z. After the horizontal representative is fixed, residual conjugators are exactly M^{J_2}, and they act by $z \mapsto z + (I - B)k$. Use the integral J_1 eigenspace basis of E.95, of ranks four and six, and its wedge splitting. The twenty-one same-sign wedges give the exact basis matrix M_+ of M^{J_2}. For $C = (I - B)M_+$, the full Smith data are

$$L_C C R_C = D_C, \qquad D_C = \operatorname{diag}(1^{[7]}, 3^{[9]}, 9^{[5]}), \quad C \text{ has size } 45 \times 21. \tag{E98.25}$$

All four inverse products and forty complete coordinate-and-conjugator round trips are retained. Use zero at the seven unit entries, standard residues at the fourteen finite entries, and arbitrary three-adic values at the last twenty-four entries, then multiply by L_C^{-1}. The inverse multiplies by L_C and reduces the first twenty-one entries. Its correcting conjugator is $R_C D_C^{-1}$ times that difference, in those twenty-one coordinates. Together with the scalar λ of (E98.21), this proves (E98.6).

For completeness the first-degree change also has a literal class-two inverse. The unique $h \in V^{J_1}$ carrying an arbitrary U to its Smith representative is obtained from the first-degree Smith inverse. Conjugate by $(h, 0)$, using the genuine action $T(h, 0) = (T_1 h, \eta_T(h))$. The central changes are exactly

$$\begin{aligned} z &\longmapsto z - \eta_T(h) + \tfrac{1}{2}[h, U] - \tfrac{1}{2}[h + U, T_1 h], \\ m &\longmapsto m + [h, v]. \end{aligned} \tag{E98.26}$$

The handle-action term is the actual arithmetic conjugation of the chosen lift. Formula (E98.26) specifies its use and preserves it. The inverse does not replace that action by its graded matrix. After this change apply (E98.21) and (E98.25). Conversely these formulas construct every pair satisfying the actual tame relation and hence every global section in this locus. Uniqueness of h follows from rational injectivity of $(I - T_1)|V^{J_1}$, and uniqueness of the residual central conjugator follows from the full column rank in (E98.25). Thus no stabilizer quotient is omitted. Combining the horizontal kernels already computed with this central fibre gives (E98.7). All maps are continuous on the indicated integral lattices and their finitely many residue strata.

For a concrete nonlifting section, put

$$U = 0, \quad v = 9(e_{A_0} + e_{B_0}), \quad u = 1, \quad a(t) = (v, m_0(0, v)), \quad a(c) = 1, \tag{E98.27}$$

and use (E98.9) for the primed values. The full forty-five-coordinate vector (E98.21) is integral and is retained in the certificate; direct substitution gives $\ulcorner m_0 = -\Delta(0, v)$. This proves the tame word exactly, including all central coordinates. Every one of the nine matrices $[v \mid e_D]$ has rank two. E.61 and E.66 classify the full and intrinsic metabelian local tame sections and put every nonzero abelian inertia on exactly one of these cusp lines. The actual projections $N \to N_{\mathrm{m}} \to N_2$ preserve these inertia vectors. Restricting a global lift to the chosen place over seven would give such a local lift, which the computed vector excludes. The same argument applies to every central class over this abelian input.

The full and metabelian ramified packets. For either $H = N$ or $H = N_{\mathrm{m}}$, E.61 and E.66 apply to every nonidentity inertia in (E98.16). In the metabelian case their proof first excludes nonidentity inertia with zero abelianization and then treats all nonunit exponents. They give $a = g\delta_D^r g^{-1}$, $r \neq 0$, and $C_H(\delta_D^r) = \langle \delta_D \rangle$. The actual cusp path satisfies $z_D T(\delta_D) z_D^{-1} = \delta_D^7$. Substitution in (E98.16) and this centralizer equality give exactly its second value in (E98.9), with $b \in \mathbb{Z}_3$. The abelian inertia determines D and r.

Two representatives for fixed D, r have the same inertia precisely when $g' = g\delta_D^k$. Using the displayed actual Frobenius relation in the second value shows that their values of u coincide precisely when $b' = b + 6k$. Choose the unique $b \in \{0, 1, 2\}$ representing its class modulo $6\mathbb{Z}_3$. This determines k uniquely, so removes the whole right-cusp ambiguity. The residual global conjugations are exactly left multiplication by H^J. This proves (E98.8), with forward map (E98.9) and inverse obtained from these unique data. The $a = 1$ part of (E98.16) is exactly E.41's unramified section quotient. These two parts exhaust the stated full and metabelian spaces.

The map (E98.10) is left-H^J invariant. Equality of the images of g, g' gives $J(g'g^{-1}) = g'g^{-1}$, proving injectivity. For an anti element x, its inverse image $x^{-1/2}$ satisfies $g^{-1}J(g) = x$. This proves surjectivity and both inverse formulas; continuity follows from that of the unique roots. On every shell $v_3(r) = n$, the parameter space (E98.8) is compact and its target geometric orbit space is Hausdorff, so the continuous bijection is a homeomorphism. The shells are clopen in the ramified subspace, as detected by abelian inertia. Their union therefore has the stated topology.

The exact full-to-metabelian global fibre. Put $R_0 = N''$. Formula (E98.13) follows by applying the actual quotient to every word in (E98.9). It is onto. We also have a literal lift of every fixed element: for $\bar{h} \in N_{\mathrm{m}}^J$, choose $h_0 \in N$, put $k = h_0^{-1} J(h_0) \in R_0$, and set $h = h_0 k^{1/2}$. Since $J(k) = k^{-1}$, $J(h) = h$ and h projects to $\bar{h}$. Choose a lift g_0 of a target representative $\bar{g}$. Every source coset over it has a representative kg_0, $k \in R_0$: lift and remove its left fixed factor by the preceding formula. Two such representatives are equivalent exactly when $k' = hk$ with $h \in R_0^J$. The actual fibre map is thus

$$R_0^J \backslash R_0 \longrightarrow \mathrm{fib}_{[\bar{g}]}, \qquad [k] \longmapsto [kg_0]. \tag{E98.28}$$

Its inverse is the described removal of the fixed factor. It is a continuous bijection between compact Hausdorff spaces, hence a homeomorphism.

All-degree coordinates for the remaining explicit quotients. We give the construction underlying (E98.11)–(E98.14). The integral lower-central comparison for N is the free Lie algebra on V, as proved in E.40–E.41. For N_{m} we use the exact completed Fox map of (E41.13),

$$N'/N'' \longrightarrow \ker\bigl(\mathbb{Z}_3[[X_1, \dots, X_{10}]]^{10} \xrightarrow{(X_i)} \mathbb{Z}_3[[X_1, \dots, X_{10}]]\bigr). \tag{E98.29}$$

The homogeneous filtration equality proved in (E41.15) identifies its degree $m \geq 2$ quotient with $\ker(V \otimes \mathrm{Sym}^{m-1} V \to \mathrm{Sym}^m V)$. That proof gives the actual commutator-to-syzygy map, its integral Chen basis and its induced pro-three topology. Thus all these lattices are free, with no extra completion torsion. For R_0 use the induced filtration $R_0 \cap \gamma_m N$. Its graded quotient is exactly the kernel of the two preceding graded maps: an element with zero graded image has image in $\gamma_{m+1} N_{\mathrm{m}}$; lift that image in $\gamma_{m+1} N$ and divide to obtain an element of R_0 with the same graded coordinate. This also proves the map direction needed for surjectivity onto that kernel. It is saturated because the target graded lattice is free.

Here is an explicit coordinate procedure for each of the three resulting groups. At one central filtration step, start with an anti element in the lower quotient. Use the ordered Hall or Chen lifts, with zero new coordinates, to obtain a lift u. For R_0 use the saturated kernel

basis of the actual Hall-to-Chen matrix and its lifted commutators. The element $d = uJ(u)$ belongs to the central new lattice. Centrality gives $J(d) = d$. Consequently

$$ud^{-1/2} \tag{E98.30}$$

is an anti lift. All other anti lifts are uniquely its products by the negative eigenspace of that central lattice. The projectors $(I + J_m)/2$ and $(I - J_m)/2$ are integral, so their exact lattice bases and inverses give the new coordinates. For the inverse procedure, subtract the previous canonical lift and read the negative central coordinates. This proves the two mutually inverse recursive maps, including every nonlinear carry.

Ordered commutator power maps are continuous for three-adic exponents; the integral collection identities give continuous Hall and Chen coordinate sections at each finite nilpotent stage. The corresponding kernel bases supply the same sections for R_0. Every finite three-group is nilpotent. Hence the intersection of the closed lower-central terms of these groups is trivial, and compactness makes the map onto the inverse limit of these quotients surjective. For R_0 the induced filtration has the same properties by its inclusion in N. Passing to that inverse limit in (E98.30) proves the product homeomorphisms.

It remains to evaluate their ranks. On V, J_1 has positive rank four and negative rank six. Integral Hall independence and the tensor-algebra PBW identity give

$$\operatorname{rank} \operatorname{Lie}_m(V) = \frac{1}{m} \sum_{d|m} \mu(d) 10^{m/d}, \qquad \operatorname{tr}(J_m) = \frac{1}{m} \sum_{d|m} \mu(d) a_d^{m/d}. \tag{E98.31}$$

For example the second identity follows by taking the trace generating series in PBW, expanding the formal logarithm, and applying Möbius inversion; $\operatorname{tr}(J_1^d) = a_d$. In the metabelian syzygy sequence the multiplication map onto $\operatorname{Sym}^m V$ is onto over $\mathbb{Z}_3$, since each monomial is the product of a variable and a monomial of degree $m-1$. Its kernel therefore has rank $(m-1)\binom{m+8}{m}$ and trace $-2c_{m-1} - c_m$. Taking one half of rank minus trace gives (E98.12). Exactness of the saturated kernel sequence gives negative rank $\nu_m - \bar{\nu}_m$ for R_0. The derived ideal has its first nonzero second-derived component in degree four; the same formulas give zero in degrees one through three and $1248 - 744 = 504$ in degree four. The program evaluates the fourteen first exact character rows and their integrality. Applying (E98.10) to R_0, followed by (E98.30), proves the final product formula (E98.14). □

The complete intrinsic class-two locus unramified outside two and three

E.99. All global class-two sections over the actual two-prime arithmetic source [Proved]

Let N be the based geometric pro-three group of the cubic with its nine cusps removed, in the actual marking of E.95. Put $N_2 = N/\gamma_3 N$, $V = N^{\mathrm{ab}}$ and $M = \gamma_2 N/\gamma_3 N = \bigwedge^2 V$. Write $\operatorname{Sect}_{23}(N_2)$ for the geometric conjugacy classes of continuous sections whose cocycles factor through the relative arithmetic quotient unramified outside $\{2, 3\}$. There are explicit continuous mutually inverse coordinates

$$\boxed{\operatorname{Sect}_{23}(N_2) \simeq \coprod_{(a,b,c) \in \mathbb{F}_3^2 \times \mathbb{Z}/9} \left(\mathbb{Z}_3^{36} \times (\mathbb{Z}/3)^9 \times (\mathbb{Z}/9)^5 \right).} \tag{E99.1}$$

The finite representatives, all coefficient maps, the exact abelian image, and the nonlinear changes of representative are specified below. In particular (E99.1) is a statement about actual global sections and their complete geometric quotient.

Use the basis $(f_1, f_2, A_0, A_1, A_2, B_0, B_1, B_2, C_1, C_2)$ of E.95; a cusp label in a linear formula means its positive peripheral vector. Set

$$\begin{aligned} h &= f_1 + \tfrac{1}{2}(A_1 + A_2 + B_1 + B_2 + C_1 + C_2), \\ V^+ &= \langle h, A_1 - A_2, B_1 - B_2, C_1 - C_2 \rangle, \\ W^- &= \langle A_0, B_0, A_1 + A_2, B_1 + B_2, C_1 + C_2 \rangle, \\ h_0 &= (1,1,1,1,1), \qquad h_1 = (1,0,0,0,0), \qquad h_2 = (0,1,0,0,0). \end{aligned} \tag{E99.2}$$

Here $V^+ = V^{J_1}$ and $W^- = W \cap V^{-J_1}$. Thus $h_0 = -e_O$, $h_1 = e_{A_0}$ and $h_2 = e_{B_0}$ after the displayed inclusion of $\mathbb{Z}_3^5$ in W. For $p = (a, b, c)$ in (E99.1), use the representatives $0 \le a, b < 3$, $0 \le c < 9$ and put

$$U_p = (3a - 3b - c)h - a(B_1 - B_2) + (-a + b)(C_1 - C_2). \tag{E99.3}$$

Define $\epsilon_p = 0$ at

$$p_O = (0,0,0), \qquad p_{A_0} = (2,2,7), \qquad p_{B_0} = (2,0,2), \tag{E99.4}$$

and $\epsilon_p = 1$ at the other seventy-eight points. The complete inertia lattice in the p-th stratum is

$$\boxed{\mathscr{K}_p = 9\mathbb{Z}_3^5 + 3^{\epsilon_p}\mathbb{Z}_3 h_{\bar{c}}.} \tag{E99.5}$$

For every abelian section in the coordinates below, its class-two preimage is nonempty exactly when its inertia coordinate belongs to (E99.5). Each nonempty preimage is

$$\mathbb{Z}_3^{25} \times (\mathbb{Z}/3)^9 \times (\mathbb{Z}/9)^5. \tag{E99.6}$$

Its coordinate origin and inverse are part of the construction. The other six abelian coordinates are unrestricted. The full abelian source has additive type $\mathbb{Z}_3^{11} \oplus (\mathbb{Z}/3)^3$; its finite carry in the representatives (E99.3) is retained by the exact conjugation map in the proof.

The projection onto the relative inertia coordinate is therefore the union of the three rational-cusp lines modulo nine. More precisely, among all $81 \cdot 9^5 = 4\,782\,969$ finite pairs $(p, v \bmod 9)$, exactly 261 occur. Three strata have nine such residues, and seventy-eight have three. This is the exact finite classification, including all nonunit inertia parameters.

There are global class-two sections with abelian class

$$9\kappa_3(2)(e_{A_0} + e_{B_0}) \tag{E99.7}$$

which have no global lift to N/N'', and consequently no global lift to N. An actual three-generator section and its central inertia word are given in (E99.32).

The certificate `a_s23_intrinsic.py` retains every Smith matrix and inverse, all eighty-one inertia lattices and their particular corrections, all finite residue checks, the actual tame word checks, and the full central coordinate inverse. Its input hashes are recorded in the output.

Manuscript: §51.16, folios733v–736v, scans172–178, (371)–(379), Part E(c),(d) and the fourth exactness follow-up.

Proof. The actual arithmetic source and the original marking. Let $K = \mathbb{Q}(\zeta_3)$ and let G_{23} denote the relative maximal pro-three quotient of the arithmetic group unramified outside two and three, over $\mathrm{Gal}(K/\mathbb{Q})$. The actual source comparison of E.55 gives

$$G_{23} = \Big(\langle t, \sigma \mid \sigma t \sigma^{-1} = t^4 \rangle_3 \underset{3}{*} \langle \rho \rangle \Big) \rtimes \langle c \rangle, \quad ctc = t^{-1}, \quad c\sigma c = \sigma, \quad c\rho c = \rho^{-1}. \tag{E99.8}$$

We use the original complex-conjugation marking c and the original rational Kummer roots, rather than leaving a change of these choices implicit.

Here is the exact transport from the construction in E.55. Suppose its local order-two element is $d = gcg^{-1}$. For either rational Kummer cocycle κ, with $\kappa(c) = 0$, one has $\kappa(d) = 2\kappa(g)$. The root change making the new roots d-fixed has exponent $e = \kappa(g)$ and changes the cocycle to $\kappa + (\chi - 1)e$. Its Frobenius $\widehat{\sigma}$ has character four and fixes these roots. Consequently

$$\kappa(\widehat{\sigma}) = -3\kappa(g), \qquad \kappa(g^{-1}\widehat{\sigma}g) = 0. \tag{E99.9}$$

Conjugate the entire arithmetic presentation by g^{-1}. The old c is now its order-two element. Conjugated tame inertia has Kummer value $\chi(g)^{-1}$; raising it to the unit power $\chi(g)$ gives $\kappa_3(2)(t) = 1$. This preserves the tame relation. The same calculation applies simultaneously to the roots of two and three. It is a specified isomorphism of the arithmetic sources, with inverse conjugation and inverse unit power.

For a useful exact choice of the other free generator, lift the old marked τ of E.37 and E.49 to this group. Multiply its lift on the left by the inverse of its $\kappa_3(2)$-power of t, and call the result γ. It has the same geometric action as τ, because the actual based action is unramified at two. Set

$$\rho = \gamma c \gamma^{-1} c. \tag{E99.10}$$

It satisfies the last identity of (E99.8) exactly. The characters on the three generators are

	t	σ	ρ
χ_3	1	4	1
$\kappa_3(2)$	1	0	0
$\kappa_3(3)$	0	0	2.

(E99.11)

The images of σ, ρ freely generate the unramified group P of E.34: their two Frattini characters are independent. Define $\rho_1 := \rho^{1/2}$ in this actual free factor. It has the last Kummer value one. The literal free-factor automorphism $\rho_1 \mapsto \rho_1^2$, with inverse $\rho \mapsto \rho^{1/2}$.

For completeness, (E99.8) retains all relations of the actual arithmetic source. The opened [Win07, Theorem2.4 and its proof, pp13–14] identifies the kernel over P with the continuous free product of actual inertia groups, indexed by $P/\langle\sigma\rangle$. The coset tree of the displayed free product has exactly the same vertex stabilizers and the same map on each inertia factor. Its universal property gives inverse maps on the kernels; the quotient map is the identity on P. This is the E.55 source isomorphism, now with the explicit transport (E99.9)–(E99.10). It also gives a reciprocal section into the full relative arithmetic group. Thus solving its relations constructs global sections, not only compatible local restrictions. The actual action on N factors through the same based quotient by E.34 and the tangent transport of E.78.

The evaluated coefficient operators. Write U_{gl} for the eight-by-two gluing matrix of E.37, in the sign convention retained in E.95:

$$U_{\mathrm{gl}} = \begin{pmatrix} 0 & -1 \\ 2 & -2 \\ 1 & 0 \\ 0 & -2 \\ 1 & -1 \\ 2 & 0 \\ 1 & -2 \\ 2 & -1 \end{pmatrix}. \tag{E99.12}$$

The actual Frobenius at the K-prime over two acts on the proper Tate module by -2. The action on the torus is four. The conjugation in (E99.9) does not change its matrix on V: the

G_K coefficient representation in E.37 is abelian. Hence, with $F = \Phi_\sigma|V$,

$$F = \begin{pmatrix} -2I_2 & 0 \\ -2U_{\mathrm{gl}} & 4I_8 \end{pmatrix}, \qquad FJ_1 = J_1F. \tag{E99.13}$$

Let $Z = \left(\begin{smallmatrix} 0 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ be multiplication by ζ_3 in the retained Tate basis. Since γ has the actual proper action $1 + 3\zeta_3$, the operator $A = \Phi_\rho|V$ is exactly

$$E = (I_2 + 3Z)(I_2 + 3Z^2)^{-1} = \frac{1}{7}\begin{pmatrix} -8 & 3 \\ -3 & -5 \end{pmatrix}, \qquad A = \begin{pmatrix} E & 0 \\ (U_{\mathrm{gl}}E - U_{\mathrm{gl}})/3 & I_8 \end{pmatrix}. \tag{E99.14}$$

All entries belong to $\mathbb{Z}_{(3)}$. The actual automorphism $\mathcal{L} = \Phi_\rho J$ of N_2 has square one. Its graded operators are $L = AJ_1$ and $L_2 = (\bigwedge^2 A)J_2$.

Write the E.95 cusp columns as b_d, r_d, s_d, so that their actual cocycles are $(A_g - I)b_d + r_d\kappa_3(3)(g) + s_d\kappa_3(\zeta_3)(g)$. Equation (E99.11) and $\kappa_3(\zeta_3) = (\chi_3 - 1)/3$ give the evaluated columns

$$k_d = (F - I)b_d + s_d. \tag{E99.15}$$

They are integral, and the certificate records their entire ten-by-eight matrix. These are the actual cusp path values for this Frobenius; no modular-root coefficient is substituted for them.

Use the actual J-linear BCH coordinates of E.95:

$$(v, m)(v', m') = (v + v', m + m' + \tfrac{1}{2}[v, v']), \qquad J(v, m) = (J_1v, J_2m). \tag{E99.16}$$

The peripheral action is $\Phi_\sigma(e_d, 0) = (4e_d, -4[k_d, e_d])$. Let $\ell_0(v) = (v, 0)$ be the chosen linear Lie lift. For $R \in V^{-L}$ the actual Lie element

$$\ell_-(R) = \tfrac{1}{2}(\ell_0(R) - \mathcal{L}\ell_0(R)) \tag{E99.17}$$

has abelian image R and is negated by $\mathcal{L}$. This uses the genuine automorphism on the whole Lie algebra. It does not identify its action on the two handle lifts with its graded matrix. If $\mathcal{L}(v, m) = (Lv, L_2m + \eta_{\mathcal{L}}(v))$, then (E99.17) is the explicit element $(R, -\eta_{\mathcal{L}}(R)/2)$. Equivalently, its exponential is the literal group word $x\mathcal{L}(x)^{-1}$ with $x = \exp(\ell_0(R)/2)$. Thus the construction and inverse can use the actual arithmetic conjugation word directly, without assigning uncomputed handle coefficients.

An integral basis of V^{-L} consists of the five columns of W^- and the column

$$\ell = \frac{1}{14}(-5, -1, 5, 2, 3, 10, 1, -1, 6, 4)^{\mathsf{t}}. \tag{E99.18}$$

Together with $(I + L)f_1/2$ and the three cusp differences in (E99.2), these form an invertible integral splitting matrix. Its inverse, and both exterior-square eigenspace matrices, are in the certificate.

Every abelian representative and its exact carry. Normalize a section at c to have value one. This has a continuous explicit inverse operation: if its old value is $\exp(X)$, the cocycle relation and (E99.16) give $JX = -X$; conjugation by $\exp(-X/2)$ makes its new value one. Its residual geometric group is N_2^J. At the abelian level, the source relations say that its three values have the form

$$a(t) = v \in W^-, \qquad a(\sigma) = U \in V^+, \qquad a(\rho) = R \in V^{-L}. \tag{E99.19}$$

Indeed $Fv = 4v$ forces the two proper coordinates of v to vanish; the remaining order-two equations give the indicated signs. A fixed abelian conjugator h' acts by

$$(v, U, R) \longmapsto (v, U + (I - F)h', R + (I - A)h'), \qquad h' \in V^+. \tag{E99.20}$$

In the basis of (E99.2), the first matrix is

$$G = (I - F)|V^+ = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 1 & -3 & 0 & 0 \\ -1 & 0 & -3 & 0 \\ -1 & 0 & 0 & -3 \end{pmatrix}. \tag{E99.21}$$

Its retained exact Smith identity is $L_G G R_G = D_G$, where $D_G = \operatorname{diag}(1, 3, 3, 9)$ and

$$L_G^{-1} = \begin{pmatrix} 3 & 3 & -3 & -1 \\ 1 & 0 & 0 & 0 \\ -1 & -1 & 0 & 0 \\ -1 & -1 & 1 & 0 \end{pmatrix}. \tag{E99.22}$$

Write $u_+ = [U]_{V^+}$ for the four coordinates of U in (E99.2). Reduce the four entries of $L_G u_+$ modulo these diagonal entries, choosing zero at the unit entry. This gives $(0, a, b, c)$ and exactly (E99.3). The unique conjugator carrying U to it is $[h']_{V^+} = R_G D_G^{-1}((0, a, b, c)^{\mathsf{t}} - L_G u_+)$. It is integral and is unique since $I - F$ is rationally invertible. Apply its second component in (E99.20) to retain the actual R coordinate. This supplies both the abelian inverse and its finite carry. The complete ten-by-four gauge matrix in (E99.20) has Smith entries $1, 3, 3, 3$, so adding the five free inertia coordinates gives the asserted additive abelian type.

The sole class-two relation and its complete finite image. With $U = U_p$, every normalized class-two candidate is specified by

$$a(t) = (v, m), \qquad a(\sigma) = (U, z), \qquad a(\rho) = \exp(\ell_-(R) + s), \qquad \begin{matrix} m \in M^{-J_2},\ z \in M^{J_2}, \\ s \in M^{-L_2},\ R \in V^{-L}. \end{matrix} \tag{E99.23}$$

The order-two relations hold exactly by (E99.16)–(E99.17). Expanding the tame relation gives the single equation

$$(\bigwedge^2 F - 4I)m + \omega(U, v) = 0, \qquad \omega(U, v) = 4 \sum_d v_d [U - k_d, e_d]. \tag{E99.24}$$

The central z cancels. No R or s occurs because the ρ factor of the arithmetic source is free. In particular solving (E99.24) is sufficient as well as necessary for an actual global section.

Here are complete finite matrices for this test, including its integral inverse. Form S_V by concatenating the V^+ basis with $(I - J_1)f_2/2$ and the W^- basis. Form S_M from brackets of its columns, placing first the twenty-one pairs with equal signs and then the twenty-four pairs with opposite signs. Write M_- for the last twenty-four columns and H_- for the last twenty-four rows of S_M^{-1}. All these matrices and inverses are integral. For

$$D = H_-(\bigwedge^2 F - 4I)M_-, \qquad L_D D R_D = D_D, \tag{E99.25}$$

the exact diagonal is $1^{[8]}, 3^{[9]}, 9^{[6]}, 0$. The final row of $L_D H_- \omega(U, v)$ is identically zero; this is also checked as an exact tensor identity. The other fifteen nonunit rows give nine congruences modulo three and six modulo nine. Define

$$Q = -M_-(R_D)_{*,1:23}(D_D)^{-1}_{1:23,1:23}(L_D)_{1:23,*}H_-. \tag{E99.26}$$

Then $(\bigwedge^2 F - 4I)Q\omega = -\omega$ exactly. Its value is integral exactly on those fifteen congruences.

To evaluate them completely, for each of the eighty-one values (E99.3), intersect their row kernels in $\mathbb{Z}_3^5$. The certificate does this over $\mathbb{Z}_{(3)}$, retaining a Hermite basis X_p, its inverse, and $Q\omega(U_p, W^- X_p)$. It proves the literal equality

$$X_p = \operatorname{HNF}\left[9I_5 \mid 3^{\epsilon_p} h_{\bar{c}}\right]. \tag{E99.27}$$

Each kernel contains $9\mathbb{Z}_3^5$, so the exact enumeration modulo nine exhausts it. The code independently tests all 4 782 969 finite pairs by the retained rational inverses; it finds precisely the cyclic groups in (E99.27). This proves (E99.5), not only their orders. It also reduces the actual k_{A_0}, k_{B_0} by (E99.22), obtaining exactly (E99.4); the base cusp has column zero. Thus the three exceptional origins have the stated actual cusp interpretation.

The kernel of (E99.25) is generated by the primitive compact class

$$w = (3f_1 + (U_{\mathrm{gl}})_1) \wedge (3f_2 + (U_{\mathrm{gl}})_2), \qquad w = -\tfrac{1}{4} M_-(R_D)_{*,24}. \tag{E99.28}$$

Consequently every solution has the unique form $m = Q\omega(U_p, v) + \lambda w$. Given m, its inverse is minus four times the last coordinate of $R_D^{-1} H_-(m - Q\omega)$. This gives the first free central coordinate without an unspecified division.

The complete central quotient and both nonlinear inverses. Once U has its unique representative, the first-degree part of every remaining conjugator is zero. The residual group is exactly M^{J_2}. Its action on (E99.23) is

$$(m, z, s) \longmapsto (m, z + (I - \bigwedge^2 F)k, s + (I - \bigwedge^2 A)k), \qquad k \in M^{J_2}. \tag{E99.29}$$

The last expression belongs to M^{-L_2}, by $J_2(\bigwedge^2 A)J_2 = (\bigwedge^2 A)^{-1}$. Using the two explicit integral eigenspace splittings, its matrix H has size forty-five by twenty-one. The certificate retains $L_H H R_H = D_H$, with all inverse matrices, and proves

$$D_H = \mathrm{diag}(1^{[7]}, 3^{[9]}, 9^{[5]}), \qquad \operatorname{rank}\operatorname{coker} H = 24. \tag{E99.30}$$

To construct a representative pair (z, s), use zero in the seven unit coordinates, the standard residues in the nine three-torsion and five nine-torsion coordinates, and arbitrary entries in the final twenty-four coordinates. Multiply by L_H^{-1} and then by the two target eigenspace bases. Conversely, multiply a given pair by L_H and reduce those entries. The correcting conjugator is exactly $R_H D_H^{-1}$ times the difference in the first twenty-one entries. The certificate checks its integrality and both inverse identities. Together with (E99.28), this proves (E99.6).

The first-degree alignment in (E99.22) also has an exact class-two lift, so the above coordinate inverse applies to every representative. Namely use $w' = (h', 0) \in N_2^J$ and perform the literal operations $a(g) \mapsto w' a(g) \Phi_g(w')^{-1}$ on the three actual generator values. If the value at σ is (U, z), its new central coordinate is

$$z - \eta_\sigma(h') + \tfrac{1}{2}[h', U] - \tfrac{1}{2}[h' + U, Fh'], \quad \Phi_\sigma(h', 0) = (Fh', \eta_\sigma(h')), \tag{E99.31}$$

and the inertia central coordinate changes by $[h', v]$. For the ρ value the identical formula uses A and the actual η_ρ; after it, recover s by subtracting the actual lift (E99.17) at the new R. These are fully specified genuine action maps. They retain the transported handle corrections of E.95 throughout; only the evaluations on peripheral inertia were needed for the numerical obstruction (E99.24). The identity $4[(I - F)h', v] = -(\bigwedge^2 F - 4I)[h', v]$ also verifies the obstruction's exact descent. It is checked on all twenty basis pairs in the certificate.

For the forward map in (E99.1), choose p, write $v = W^- X_p x$ with $x \in \mathbb{Z}_3^5$, choose the six coordinates of R in (E99.18), the scalar λ of (E99.28), and the central coordinates of (E99.30). Equations (E99.23)–(E99.28) then give a cocycle on the actual presentation (E99.8), and hence a global section. For the inverse, first normalize at c, next use (E99.22) and the literal action (E99.31), then apply X_p^{-1}, (E99.28), and (E99.30). Uniqueness of the first conjugator follows from $I - F$; uniqueness of the residual central conjugator follows from $I - \bigwedge^2 F$, which is rationally invertible. Thus no stabilizer quotient is omitted. All displayed maps are continuous on compact integral lattices, and the finitely many residue representatives are clopen. The resulting bijections and their explicit inverses are homeomorphisms. This proves (E99.1).

An explicit global obstruction to further lifting. Take $p = (0,0,0)$, $R = 0$, all central Frobenius coordinates zero, and $v = 9(e_{A_0} + e_{B_0})$. Set

$$\begin{aligned} m = {} & -3[A_0, f_1] - 6[B_0, f_1] + 4[A_1, A_0] - [A_2, A_0] - 8[B_0, A_1] + 2[B_0, A_2] \\ & - \tfrac{1}{4}[B_1, A_0] + \tfrac{13}{4}[B_2, A_0] + \tfrac{5}{4}[C_1, A_0] + \tfrac{7}{4}[C_2, A_0] \\ & + [B_1, B_0] + 5[B_2, B_0] + [C_1, B_0] + 5[C_2, B_0], \\ a(t) = \exp(v + m), \qquad & a(\sigma) = a(\rho) = a(c) = 1. \end{aligned} \tag{E99.32}$$

This is the exact particular solution (E99.26). The program evaluates its entire BCH tame word and obtains the identity; it also checks its order-two equation. Therefore it defines an actual global section with (E99.7). At the place two its abelian inertia is on none of the nine cusp lines: two independent positive peripheral coordinates are nonzero, and the other six are zero. E.61 and E.66 prove that every local intrinsic metabelian section has abelian inertia on one of those nine lines. The actual maps $N \to N/N'' \to N/\gamma_3 N$ preserve this inertia vector. A global lift would restrict to such a local lift, which is impossible. This proves the asserted global nonlifting statement for every central class above this abelian input. It leaves the unrestricted global and higher-stage targets of E.23 active while evaluating the complete stated two-prime class-two locus. □

Every full and intrinsic-metabelian section over the actual two-prime source

E.100. Complete cubic sections unramified outside two and three, with every full-to-metabelian fibre [Proved]

Let N be the actual based cubic pro-three group of E.34 and E.78, let $M = N/N''$, and retain E.99's original arithmetic marking and rational Kummer roots. Write

$$\mathscr{S}_{23}(H) = H^1_{\mathrm{cts}}(G_{23}, H), \qquad \mathscr{S}_3(H) = H^1_{\mathrm{cts}}(\Gamma, H) \quad (H = N, M), \tag{E100.1}$$

where both cohomology sets mean geometric-conjugacy classes for the genuine based action. The maps from the corresponding absolute arithmetic groups are the actual quotient maps proved in E.34, E.55 and E.99. In particular

$$\begin{aligned} G_{23} = (D_4 \underset{3}{*} \langle\rho\rangle) \rtimes \langle c\rangle, \qquad & D_4 = \langle t, \sigma \mid \sigma t \sigma^{-1} = t^4 \rangle_3, \\ ctc = t^{-1}, \quad c\sigma c = \sigma, \quad c\rho c = \rho^{-1}, \qquad & G_{23}/\overline{\langle\langle t \rangle\rangle} = \Gamma = P \rtimes \langle c\rangle. \end{aligned} \tag{E100.2}$$

Put $J = \Phi_c$, $F = \Phi_\sigma$, $T = \Phi_\tau$ and $A = \Phi_\rho$. These denote actual group automorphisms, and satisfy

$$\Phi_t = 1, \qquad FJ = JF, \qquad A = TJT^{-1}J, \qquad \mathcal{L} := AJ = TJT^{-1}. \tag{E100.3}$$

The section set (E100.1) has the following complete partition: its zero-inertia part is precisely the inflated $\mathscr{S}_3(H)$, and its nonzero-inertia part consists of three strata, indexed by the rational cusps

$$O = [1 : -1 : 0], \qquad A_0 = [0 : -1 : 1], \qquad B_0 = [1 : 0 : -1]. \tag{E100.4}$$

Choose the actual rational tangents and c-fixed paths constructed below. Let d_i be the resulting positive peripheral element and $u_i = p_i F(p_i)^{-1}$. The entire stratum at i has unique representatives

	t	σ	ρ	c
$a_{i,a,y}$	d_i^a	u_i	$T(y)$	1

$$a \in \mathbb{Z}_3 \setminus \{0\}, \qquad J(y) = y^{-1}. \tag{E100.5}$$

Here $J(d_i)=d_i^{-1}$, $J(u_i)=u_i$ and $u_iF(d_i)u_i^{-1}=d_i^4$. Every row of (E100.5) is an actual global section. Its geometric stabilizer is trivial, and every section in its stratum has one and only one J-fixed geometric conjugator carrying it to that row.

Define $\mathrm{Anti}_J(H)=\{y\in H : J(y)=y^{-1}\}$. There are explicit compatible Hall coordinates

$$\mathrm{Anti}_J(H)\simeq\prod_{r\geq1}\mathbb{Z}_3^{\nu_r(H)},\qquad \mathscr{S}_{23}(H)_i^{\mathrm{ram}}\simeq(\mathbb{Z}_3\setminus\{0\})\times\prod_{r\geq1}\mathbb{Z}_3^{\nu_r(H)}. \tag{E100.6}$$

These are homeomorphisms of spaces. The anti set is used with its displayed group equation; no group operation on that set is imposed. The ranks are completely evaluated in every degree. For $r\geq1$ set

$$\begin{aligned}
&n_r(N)=\frac{1}{r}\sum_{d|r}\mu(d)10^{r/d},\qquad && j_r(N)=\frac{1}{r}\sum_{d|r}\mu(d)(4+(-1)^d6)^{r/d},\\
&n_1(M)=10,\quad j_1(M)=-2,\qquad && n_r(M)=(r-1)\binom{r+8}{r}\quad(r\geq2),\\
&j_r(M)=-2c_{r-1}-c_r\quad(r\geq2),\qquad && c_s=[z^s](1-z)^{-4}(1+z)^{-6},\\
&\nu_r(H)=(n_r(H)-j_r(H))/2.
\end{aligned} \tag{E100.7}$$

For example,

r	1	2	3	4	5	6
$\nu_r(N)$	6	24	166	1248	10002	83320
$\nu_r(M)$	6	24	166	744	2580	7504
$\nu_r(N)-\nu_r(M)$	0	0	0	504	7422	75816.

(E100.8)

The actual map $\mathscr{S}_{23}(N)\to\mathscr{S}_{23}(M)$ is onto. On each ramified stratum it preserves i,a. For the matched Hall and Chen coordinates constructed below, its remaining map literally deletes the negative Hall coordinates not belonging to the Chen subset. Consequently every ramified target has full fibre

$$\prod_{r\geq4}\mathbb{Z}_3^{\nu_r(N)-\nu_r(M)}. \tag{E100.9}$$

There is also a coordinate-independent description with its actual map. Put $R=N''$, fix the target's ρ-value x_M, and choose an actual lift $x_0\in\mathrm{Anti}_{\mathcal{L}}(N)$. Then

$$\begin{aligned}
\mathrm{Fib}(x_M)=\mathrm{Anti}_{\mathcal{L}_0}(R),\qquad \mathcal{L}_0=\mathrm{Ad}(x_0)\mathcal{L},\\
r\longmapsto rx_0,\qquad R/R^{\mathcal{L}_0}\xrightarrow{\sim}\mathrm{Anti}_{\mathcal{L}_0}(R),\quad gR^{\mathcal{L}_0}\longmapsto g\mathcal{L}_0(g)^{-1}.
\end{aligned} \tag{E100.10}$$

The quotient in the last line is the space of right cosets. The map has the continuous section $r\mapsto r^{1/2}$ before passing to those cosets.

Here are all zero-inertia fibres as well. Give R its induced filtration and put

$$\begin{aligned}
K_r=(R\cap\gamma_rN)/(R\cap\gamma_{r+1}N)=\ker(L_r(N)\to L_r(M)),\\
\mathcal{C}_r(R)=K_r/(T_r-I)K_r^J,\qquad L_r(H)=\gamma_rH/\gamma_{r+1}H.
\end{aligned} \tag{E100.11}$$

These are finite-rank, explicitly presented compact $\mathbb{Z}_3$-modules, with free rank $\nu_r(N)-\nu_r(M)$. Their matrices are restrictions of the actual integral Hall matrices of E.37 and E.41 to the non-Chen Hall basis proved below. Their torsion is killed by 3^{r+b_r}, where $b_r=\max_{r\leq j\leq2r}(1+v_3(j))$. Thus Smith reduction modulo 3^{r+b_r+1}, with the retained integral basis maps, determines every finite factor and its inverse coordinates. There are exact sequences

$$0\longrightarrow\mathcal{C}_r(R)\xrightarrow{\iota_r}\mathcal{C}_r(N)\xrightarrow{\bar{\pi}_r}\mathcal{C}_r(M)\longrightarrow0. \tag{E100.12}$$

The notation $\mathcal{C}_r(N), \mathcal{C}_r(M)$ is that of E.41. Its full unramified coordinates give

$$\begin{aligned} \mathscr{S}_3(H) &\simeq \prod_{r\geq 1} \mathcal{C}_r(H), \\ \operatorname{Fib}\big(\mathscr{S}_3(N) \to \mathscr{S}_3(M)\big) &\simeq \prod_{r\geq 4} \mathcal{C}_r(R). \end{aligned} \tag{E100.13}$$

Every fibre in the second line is nonempty. Equations (E100.24)–(E100.26) below give the actual triangular map and every inverse fibre, including the nonlinear arithmetic corrections. A full ramified source cannot lie over a zero-inertia metabelian target, so (E100.13) is also that target's entire fibre in $\mathscr{S}_{23}(N)$.

The topology on the whole partition is the actual quotient topology specified in (E100.14) below. In particular its zero-inertia boundary is retained: in (E100.5), $a \to 0$ gives the unramified section with values $(1, u_i, T(y), 1)$. The statement evaluates the entire cubic source unramified outside $\{2, 3\}$, within the global maps of E.23. The manuscript anchor is §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d).

Proof. The actual arithmetic presentation and its normalized values. E.99 proves (E100.2), including its actual maps to the relative global arithmetic group and to the unramified quotient. Its proof retains the maps on each inertia factor in the continuous free-product theorem [Win07, Lemma1.1, Theorem2.4 and their proofs, pp4–5,13–15]. We use precisely those maps. The element $\rho = \gamma c \gamma^{-1} c$ there has $\Phi_\gamma = T$, because multiplying the chosen lift of τ by a tame inertia power changes neither its actual geometric action nor its image in P. Hence (E100.3) is an equality of full automorphisms. Both $R = N''$ and its quotient are stable, so the same equality holds on M, on R, and on all the induced filtrations.

Every continuous cocycle under consideration factors through this source. Over K its cocycle together with the genuine action has image in a pro-three semidirect product; the maximal pro-three arithmetic quotient therefore supplies the factorization. The relative order-two quotient is then exactly the one retained in E.99. Conversely its actual source map inflates every cocycle constructed here to the original global arithmetic group.

If a cocycle has value v at c, its order-two relation is $vJ(v) = 1$. Conjugation by $v^{-1/2}$ makes that value one. Every element of a pro-three group has a unique square root, continuously: this is the inverse of squaring on its procyclic subgroup, and the inverses agree in every finite three-group quotient. The remaining geometric conjugators are exactly H^J. Thus the whole section space is the compact Hausdorff quotient

$$\begin{aligned} \mathscr{S}_{23}(H) = \{(\nu, u, x) \in H^3 : J(\nu) = \nu^{-1},\ J(u) = u,\ \mathcal{L}(x) = x^{-1}, \\ uF(\nu)u^{-1} = \nu^4\}/H^J, \\ g \cdot (\nu, u, x) = (g\nu g^{-1},\ guF(g)^{-1},\ gxA(g)^{-1}). \end{aligned} \tag{E100.14}$$

The last norm equation follows from $c\rho c = \rho^{-1}$ and the cocycle identity: $J(x) = A^{-1}(x^{-1})$. The free-product presentation proves sufficiency of these four equations, as well as necessity.

Rational cusp maps, their whole path torsors, and descent. Use $s = z/x$ at O, $s_A = x/z$ at A_0 and $s_B = y/x$ at B_0, with tangent value one. The rational coordinate maps

$$\begin{aligned} a_A[x : y : z] = [z : y : x], \qquad a_B[x : y : z] = [x : z : y], \\ s_A a_A = s_B a_B = s, \qquad a_A^2 = a_B^2 = 1 \end{aligned} \tag{E100.15}$$

preserve the actual cubic and its cusp divisor. They identify the corresponding tangential fibre functors at every three-power level. For the comparison with E.34's parameter $-x/z$ at A_0,

the compatible roots of -1 are all -1. They are rational, and the two transports compose to the identity. This is the same precise basepoint transport used in E.78 and E90.27.

The based action at each of these tangents therefore factors through the same Γ. Its entire pro-three path torsor does also. Indeed an element of the kernel acts trivially on both endpoint groups; the difference between its translate of a path and that path centralizes the full endpoint group. The latter is free pro-three of rank ten and has trivial centre, as proved by the Magnus first-term argument in E.90. Thus the difference is one. This proves the asserted factorization of the path itself, not just its abelianization.

Choose any such path p. For $\eta = pJ(p)^{-1}$ one has $\eta J(\eta) = 1$, so $p_i = \eta^{-1/2}p$ is literally J-fixed. Transport the positive local inertia along p_i to obtain d_i. Its full path cocycle is $p_i g(p_i)^{-1}$, and therefore

$$J(d_i) = d_i^{-1}, \quad J(u_i) = u_i, \quad u_i F(d_i) u_i^{-1} = d_i^4, \quad p_i t(p_i)^{-1} = 1. \tag{E100.16}$$

The last equality uses the whole path factorization just proved and $t \in \ker(G_{23} \to \Gamma)$. This supplies the actual source and target maps for each row of (E100.5).

Restrict a normalized section to D_4. For $H = N$, E.61 gives the nine full local cusp packets over $\mathbb{Q}_2(\zeta_3)$; for $H = M$, E.66 gives the identical complete ramified packets. Their parameters are $(i, a, \bar{b})$ with $a \neq 0$ and $\bar{b} \in \mathbb{Z}_3/3$. Choose conjugate paths for each nonrational pair. On those packets the actual involution acts as

$$(i, a, \bar{b}) \longmapsto (\bar{i}, a, -\bar{b}). \tag{E100.17}$$

The source inversion of t and the positive peripheral inversion cancel on a. The source σ is fixed, while the peripheral factor d_i^b is inverted. These are exactly the three sign and label changes in (E100.17). The only fixed cusp labels are (E100.4), and multiplication by two is invertible on $\mathbb{Z}_3/3$. Thus a descended ramified class has exactly $i \in \{O, A_0, B_0\}$ and $\bar{b} = 0$.

There is an actual unique conjugator aligning its D_4-restriction with (d_i^a, u_i). Existence follows from the local classification and the change $b \mapsto b - 3k$ by d_i^k. For uniqueness, the centralizer of d_i^a, including nonunit a, is $\langle d_i \rangle$ in both N and M: E.61 proves the full case, and E66.18–21 proves the metabelian case through its injective continuous Magnus map. A stabilizer of the full local section would then have exponent $k = 4k$, hence $k = 0$. Since both local sections are J-normalized, $J(g)$ is another aligning conjugator whenever g is one. Uniqueness forces $J(g) = g$. This proves actual descent and its full conjugacy assertion, without identifying an invariant class with a descent datum by assumption.

After that alignment the free ρ-value is any $x \in \mathrm{Anti}_{\mathcal{L}}(H)$. The exact equality (E100.3) identifies it with $x = T(y)$ for a unique $y \in \mathrm{Anti}_J(H)$. All rows (E100.5) follow, with no remaining geometric stabilizer. A different fixed path $p_i' = hp_i$, $h \in N^J$, changes the row by the actual conjugation h. In the y-coordinate its change is

$$y \longmapsto g\, y\, J(g)^{-1}, \qquad g = T^{-1}(h). \tag{E100.18}$$

This gives the exact coordinate morphism for every allowed path change. The map to E.99 is the actual truncation of the four entries in (E100.5), followed by E99.20–22 and E99.29–31. Its three Frobenius abelian labels are $p_O = (0, 0, 0)$, $p_{A_0} = (2, 2, 7)$ and $p_{B_0} = (2, 0, 2)$, and its inertia is the literal vector $a[d_i]$. The higher coordinates use the truncated actual u_i and $T(y)$ in that specified normalization algorithm. Thus the comparison retains the actual path origins and all its central carries.

An actual J-adapted free basis. Use the integral J-eigenbasis of V in E99.2 and E99.19: it has four plus columns and six minus columns. Its matrix and inverse over $\mathbb{Z}_{(3)}$ are retained in the certificate. A fixed abelian column has a full fixed lift. Inductively, if w lifts a fixed

element through a central lower-central layer, put $\delta = J(w)w^{-1}$ in that layer. Then $J(\delta) = -\delta$, and $\delta^{1/2}w$ is fixed. Ordered Hall lifts and this correction give a compatible inverse sequence, hence a fixed element of N. For a minus column, choose any abelian lift w and use instead $w^{1/2}J(w)^{-1/2}$. It has exactly that column and is inverted by J.

The resulting ten elements $y_1, \dots, y_{10}$ form a free pro-three basis, with $J(y_i) = y_i$ for $i \le 4$ and $J(y_i) = y_i^{-1}$ for $i > 4$. Here is the group-level justification. Their reductions form a basis of $N/\Phi(N)$, so their closed subgroup is all of N: a proper closed subgroup lies in a maximal open subgroup of index three. The resulting endomorphism from the old free basis is surjective and injective. To see the latter directly, a finitely generated profinite group has finitely many open normal subgroups of each bounded index. Pullback by a surjective endomorphism permutes each such finite set. Some iterate therefore fixes each chosen open normal subgroup, and the endomorphism kernel belongs to it. The intersection of those subgroups is one. This also describes the inverse as the compatible inverses on characteristic finite three-group quotients. Thus the construction is an actual continuous change of free basis; the original arithmetic operators are transported by that isomorphism throughout.

Integral Hall deletion and the strict kernel. Take the standard degree-ordered Hall basis in these generators. Its left-comb words

$$[y_i, y_j, y_{k_3}, \dots, y_{k_r}], \qquad i > j \le k_3 \le \cdots \le k_r \tag{E100.19}$$

are exactly its Chen subset. Every other Hall word contains an internal bracket whose two children have degree at least two. That internal group word is in N''; normality then puts the whole Hall word in N'', as an actual group element.

The continuous Fox isomorphism E41.13 identifies N'/N'' with $\ker(\Lambda^{10} \xrightarrow{(X_i)} \Lambda)$. E41.15 proves its exact homogeneous filtration and saturation. Under this actual map, (E100.19) has leading syzygy

$$(-1)^{r-2} X_{k_3} \cdots X_{k_r}(X_i e_j - X_j e_i).$$

Those are the integral Chen basis from the homogeneous Koszul presentation; the same basis and the literal Hall subset are also displayed in [Sat10, §2.4, equation(2) and the following paragraph, p5]. Thus no further integral relation occurs. At every lower-central level, the actual quotient of an ordered Hall product is the ordered product obtained by deleting all non-Chen factors. The remaining Chen coordinates are unique, degree by degree, because their graded modules are free with the displayed basis. Continuity extends the equality to all $\mathbb{Z}_3$ powers and all limits.

In particular the non-Chen Hall words give an integral basis of K_r in (E100.11). This is the induced group filtration: if a leading Hall class maps to zero, lift its higher-degree image to $\gamma_{r+1}N$ and subtract that lift. Conversely every element of $R \cap \gamma_r N$ maps to zero. The quotient $L_r(M)$ is free, so this kernel is saturated. The preceding literal Hall-word argument supplies actual lifts in R for all its basis vectors. Both groups and R are their inverse limits for these filtrations: intersection is one since finite three-group quotients are nilpotent, and compactness gives surjectivity onto the nested inverse systems.

The anti coordinates and their exact quotient map. Write $H_r = H/\gamma_{r+1}H$ and let $h_r : H_{r-1} \to H_r$ be its ordered Hall or Chen lift with last-degree coordinates zero. These lifts commute with $N_r \to M_r$ by the just-proved literal deletion map. For an anti point $z \in H_{r-1}$, put

$$\begin{aligned} u &= h_r(z), \qquad d = uJ(u) \in L_r(H), \\ \widehat{h}_r(z) &= u\, d^{-1/2}, \\ z_r &= \widehat{h}_r(z_{r-1})\, \beta_r(a_r), \qquad a_r \in L_r(H)^-, \quad z_0 = 1. \end{aligned} \tag{E100.20}$$

Here β_r is the actual ordered product of the negative degree-r Hall or Chen commutators, in the central top layer. One has $J(d) = d$: indeed $J(u) = u^{-1}d$ and d is central. Hence

$\widehat{h}_r(z)J(\widehat{h}_r(z)) = 1$. Every other anti lift differs by one and only one element of $L_r(H)^-$. The inverse of (E100.20) takes the lower image and then the Hall coordinates of $z_r\widehat{h}_r(z_{r-1})^{-1}$. This proves both directions at every level, and the compatible limit gives the homeomorphism (E100.6).

The sign of a Hall word is the product of the signs of its letters. In the chosen basis J is diagonal on each graded Hall or Chen basis, even though its action on a full noncentral Hall word also has higher terms. Formula (E100.20) uses that full action. Every operation in (E100.20) commutes with the quotient: the zero-top lifts, J, the central product, its unique square root, and the retained negative Hall coordinates. Consequently the map on the anti coordinates is exactly deletion of the non-Chen negative coordinates. Applying the actual T to both source and target gives the same assertion for (E100.5). This proves (E100.9) and constructs an actual full lift of every ramified metabelian target.

The ranks of the full graded basis and its involution trace are the Witt formulas in (E100.7). The exact syzygy sequence

$$0 \longrightarrow L_r(M) \longrightarrow V \otimes \operatorname{Sym}^{r-1} V \longrightarrow \operatorname{Sym}^r V \longrightarrow 0 \tag{E100.21}$$

is an integral sequence, by the preceding Fox and Koszul map. Its rank and trace give the metabelian formulas in (E100.7). The non-Chen negative count is therefore their exact difference, not a rational dimension substituted for an integral kernel. This proves the table.

For (E100.10), write a second lift of x_M as rx_0. Its anti equation is exactly $(\operatorname{Ad}(x_0)\mathcal{L})(r) = r^{-1}$. The operator $\mathcal{L}_0$ is an involution because $x_0\mathcal{L}(x_0) = 1$. The map $g \mapsto g\mathcal{L}_0(g)^{-1}$ is onto: the preimage of r can be chosen to be $r^{1/2}$. Two such preimages differ on the right by $R^{\mathcal{L}_0}$. These are literal equalities in R, and prove the claimed fibre and quotient maps.

Zero inertia and every unramified lifting fibre. If $\nu = 1$, the section kills the geometric cocycle on the full closed normal inertia subgroup generated by t. Indeed $\Phi_t = 1$, and the cocycle identity then kills every conjugate of t and its closed normal closure. It factors through the actual quotient in (E100.2), so the zero-inertia set is exactly the inflated E.41 set. Inflation is injective on geometric conjugacy since the source quotient is onto. The transition to E.41's (τ, τ') coordinates is evaluation on those actual elements of P, using the inverse of its free-basis map $(\sigma, \rho) \to P$. That inverse exists by the E.99 Frattini-basis proof and is the compatible inverse on finite characteristic quotients. This retains the same source marking and all Frobenius words.

Fix an unramified target with E.41 representative x_M at τ. Choose any actual lift $x_0 \in N$. The free arithmetic group P gives a full normalized section with that value, as in E41.9. Put $\Theta = \operatorname{Ad}(x_0)T$ on R. The entire full fibre is the following actual orbit space:

$$R/\!\sim_\Theta, \qquad k \cdot r = kr\Theta(k)^{-1} \quad (k \in R^J), \qquad r \longmapsto rx_0. \tag{E100.22}$$

Here $\sim_\Theta$ is the displayed orbit relation. To check that it is the whole fibre, first align the projected target by its unique M^J-conjugator from E.41. It has a fixed lift in N, by the same central half-correction used above. All remaining conjugators project to a target stabilizer, which is one by E41.17. They therefore belong to R^J.

The action of Θ on K_r is exactly T_r, since its inner factor is trivial on this induced grade. Every twisted stabilizer in (E100.22) is trivial: its first nonzero graded coordinate would be fixed by the actual arithmetic action, whereas the Frobenius weight argument of E41.17 excludes such a vector. Fixed lifts through the central layers exist by division by two. E.41's normal-form induction therefore applies to the induced filtration of R with precisely the coefficient quotients (E100.11), proving the second line of (E100.13).

For clarity its linear maps and its nonlinear comparison are both explicit. Taking fixed parts of the strict sequence $0 \to K_r \to L_r(N) \to L_r(M) \to 0$ is exact because two is a unit.

Apply $T_r - I$ to those fixed parts. It is rationally injective on every term, since the actual Frobenius at seven has no eigenvalue one in positive degree. The resulting kernel-cokernel sequence is (E100.12). For its inverse on the kernel, represent a class by $v \in L_r(N)$ and solve

$$\pi_r(v) = (T_r(M) - I)b, \quad b \in L_r(M)^J; \quad v - (T_r(N) - I)\widetilde{b} \in K_r, \tag{E100.23}$$

where $\widetilde{b}$ is any integral fixed lift of b. Uniqueness of b and exactness prove independence of the lift modulo $(T_r - I)K_r^J$. This specifies every arrow in (E100.12).

Choose the pointed Hall–Smith representatives of E.41 and a continuous set section λ_r of $\bar{\pi}_r$. The latter is obtained from the actual finite-rank Smith maps: lift free basis vectors and use the chosen residue digits for finite factors. Suppose the source's lower coordinates are fixed, with canonical word $x_{N,r-1}$, and let $x_{M,r-1}$ be its target normal form. Let $g_{r-1} \in M_{r-1}^J$ be the unique lower aligning conjugator. Lift it by the actual fixed half-correction to $\widehat{g} \in M_r^J$, and define the central element

$$d_r = \widehat{g}\,\pi(h_r^N(x_{N,r-1}))T(\widehat{g})^{-1}h_r^M(x_{M,r-1})^{-1} \in L_r(M). \tag{E100.24}$$

For the genuine coordinate map one has

$$b_r = \bar{\pi}_r(a_r) + [d_r], \qquad (T_r(M) - I)k = d_r + \pi_r(s_r^N(a_r)) - s_r^M(b_r). \tag{E100.25}$$

The last equation has a unique integral solution $k \in L_r(M)^J$, and $k\widehat{g}$ is the next aligning conjugator. It retains the full arithmetic action on the lifted word in (E100.24). To obtain every full preimage of a specified target (b_r), choose arbitrary $v_r \in \mathcal{C}_r(R)$ and set, successively,

$$a_r = \lambda_r(b_r - [d_r]) + \iota_r(v_r). \tag{E100.26}$$

Equations (E100.24)–(E100.26), including (E100.23), are the forward and inverse maps underlying the second product in (E100.13). No lower correction is removed by replacing its arithmetic operator with a graded matrix.

The free rank in (E100.11) is the negative rank of K_r. For its stated finite bound, use E41.20 on $L_r(N)$: the split cubic lattice contains $3V$, and its degree-r Frobenius eigenvalues are $(-2)^j$, $r \le j \le 2r$. The inverse of $F_r - I$ preserves $K_r \otimes \mathbb{Q}_3$. Saturation therefore restricts that same integral denominator bound to K_r. The torsion in its normalized $H^1(\Gamma, K_r)$ is $H^0(\Gamma, K_r \otimes \mathbb{Q}_3/K_r)$, by the exact rational coefficient sequence and the absence of invariants. The Frobenius bound kills this group by 3^{r+b_r}, proving the finite Smith precision claimed in the statement.

Topology, boundary, and the retained certificate. All Hall products, square roots, fixed lifts, and finite-rank Smith operations used above are continuous on their proved integral domains. The inverse limits are compact. This proves the anti and unramified homeomorphisms. On each annulus $v_3(a) = e$, the ramified parametrization is a continuous bijection from a compact space into the Hausdorff quotient (E100.14), hence a homeomorphism. The annuli give the topology on $\mathbb{Z}_3 \setminus \{0\}$. Their zero-inertia limits are obtained by the actual rows (E100.5), proving the boundary assertion. Every stabilizer is trivial: for ramified rows this was proved locally, and for zero inertia it is E41.17. Finally a full ramified row has nonzero abelian inertia $a[d_i]$; its metabelian image remains ramified. This proves the full scope of the fibre statements.

The retained program `a_s23_full.py` constructs the literal Hall-to-Chen deletion map through degree five, its integral right inverse, and every signed kernel basis. It checks all 44 385 bracket comparisons, and all 2 475 degree-four columns for each of the actual T, J, F actions. It saves their restrictions to the actual rank-990 kernel. The first negative kernel rank is 504. A separate exact rational Magnus calculation through degree four verifies 128 nonlinear correction identities, 64 complete coordinate round trips, 120 literal conjugate- involution

identities and 25 complete kernel-fibre comparisons. Every denominator in that calculation is prime to three. Its auxiliary Nielsen automorphism checks the group identity used in (E100.3); the actual arithmetic identity is the source word (E99.10). The certificate also checks all twenty-seven local descent labels and both exact rational tangent maps. □

Complete first ordinary-unit and S-unit norm images

E.101. The remaining first-layer scalar and all four unit norm images [Proved]

Keep the literal fields and generators of E.86 and E.97:

$$\begin{aligned} &\zeta^2+\zeta+1=0, \quad \xi^3=\zeta, \quad \beta_1^3=17\zeta, \quad \beta_2^3=\beta_1,\\ &L_i=\mathbb{Q}(\zeta,\beta_i), \quad B_i=L_i(\xi), \quad c_1=\beta_1+1+3\zeta. \end{aligned} \tag{E101.1}$$

Let S be the primes above $3,17$, and let a hat denote completion for multiplication by 3^m. The two certified free ordinary-unit generators of L_1 used in E.64 and E.97 are, with $b_0=\beta_1$,

$$\begin{aligned} e_1&=-18+3b_0+3b_0^2-\frac{18b_0^3+5b_0^4-2b_0^5}{17},\\ e_2&=-18-5b_0+2b_0^2-\frac{b_0^3+8b_0^4+b_0^5}{17}, \qquad \epsilon=e_1e_2. \end{aligned} \tag{E101.2}$$

For the exact ordinary divisor defect of E97.2, with $\sigma(\beta_2)=\zeta\beta_2$, one has

$$\omega_L(\epsilon)=(0,2), \qquad \omega_L(\zeta)=(0,1). \tag{E101.3}$$

These coordinates are the specified $(a_2 \bmod 3, b_2)$ coordinates of $\mathrm{Cl}(L_2)\{3\}/\langle\mathfrak{q}_{3,2}\rangle$.

Here are complete norm-image tests, including both directions. Write an ordinary completed L_1-unit as $u=\zeta^c e_1^a e_2^b$, with $c\in\mathbb{Z}/3$, $a,b\in\mathbb{Z}_3$. For an S-unit append the factors $c_1^e\beta_1^f$, with $e,f\in\mathbb{Z}_3$. Then

$$\begin{aligned} u\in N_{L_2/L_1}\widehat{\mathcal{O}_{L_2}^{\times}} &\iff a\equiv b\equiv c \pmod 3,\\ uc_1^e\beta_1^f\in N_{L_2/L_1}\widehat{\mathcal{O}_{L_2,S}^{\times}} &\iff a\equiv b\equiv c \pmod 3 \quad\text{and}\quad e\equiv 0 \pmod 3. \end{aligned} \tag{E101.4}$$

Thus the ordinary image has free basis $e_1^3,\zeta e_1e_2$; the S-unit image has free basis $e_1^3,\zeta e_1e_2,c_1^3,\beta_1$. The actual quotient coordinate maps are respectively

$$\begin{aligned} \zeta^c e_1^a e_2^b &\longmapsto (a-b,c-b) \pmod 3,\\ \zeta^c e_1^a e_2^b c_1^e\beta_1^f &\longmapsto (a-b,c-b,e) \pmod 3. \end{aligned} \tag{E101.5}$$

Their inverses on quotient classes send (r,s) to $[e_1^r\zeta^s]$, and (r,s,t) to $[e_1^r\zeta^s c_1^t]$. In particular these quotients are C_3^2 and C_3^3.

There is also a complete calculation over B_1. Use the exact E.79 basis

$$\mathcal{B}=(u_1,\dots,u_{15})=(\xi,f_1,\dots,f_8,s_1,\dots,s_6), \qquad \widehat{\mathcal{O}_{B_1,S}^{\times}}=\mathbb{Z}_3^{15}/9\mathbb{Z}_3\mathbf{e}_1. \tag{E101.6}$$

The f_i are its ordered certified fundamental units, and s_i its ordered principal S-divisor generators. The certificate retains their literal eighteen-term coefficient vectors in the primitive element $X=\sqrt[3]{17}-\xi$; its polynomial is $X^{18}-105X^{15}+4137X^{12}-81182X^9+1128687X^6-6822222X^3+28934443$. Thus (E101.6) fixes actual field elements and the coordinate map $x\mapsto\prod_i u_i^{x_i}$.

The norm image from B_2, in these coordinates, is exactly the six-congruence lattice

$$\begin{aligned} x_1-x_2-x_4-x_{10}-x_{12}\equiv 0,\quad & x_5-2x_6-x_{10}-2x_{11}\equiv 0,\\ x_8-x_{10}-x_{11}\equiv 0,\quad & x_9-x_{10}-x_{12}-x_{15}\equiv 0,\\ x_{13}-x_{15}\equiv 0,\quad & x_{14}-x_{15}\equiv 0 \pmod 3. \end{aligned} \tag{E101.7}$$

Restricting to ordinary units, by setting $x_{10}=\cdots=x_{15}=0$, gives the complete test

$$x_1-x_2-x_4\equiv 0,\qquad x_5-2x_6\equiv 0,\qquad x_8\equiv x_9\equiv 0 \pmod 3. \tag{E101.8}$$

The two actual norm quotients are therefore C_3^6 and C_3^4, respectively. The quotient map is the ordered list of the six expressions in (E101.7), or the four in (E101.8); a section on quotient classes uses the coordinate vectors $(\mathbf{e}_1,\mathbf{e}_5,\mathbf{e}_8,\mathbf{e}_9,\mathbf{e}_{13},\mathbf{e}_{14})$, or its first four vectors. In these sentences a section means a map into the quotient of units by actual norms, so reducing an exponent modulo three is justified by the included-base cube norm.

Here are explicit inverse coordinates on the S-unit image itself. Define $y_i=x_i$ for $i=2,3,4,6,7,10,11,12,15$, and set

$$\begin{aligned} y_1&=(x_1-x_2-x_4-x_{10}-x_{12})/3,\\ y_5&=(x_5-2x_6-x_{10}-2x_{11})/3,\\ y_8&=(x_8-x_{10}-x_{11})/3,\\ y_9&=(x_9-x_{10}-x_{12}-x_{15})/3,\\ y_{13}&=(x_{13}-x_{15})/3,\qquad y_{14}=(x_{14}-x_{15})/3. \end{aligned} \tag{E101.9}$$

Recover x by these six equations and the unchanged nine coordinates. The sole relation $9\mathbf{e}_1$ becomes $3\mathbf{e}_1$ in the y coordinates. Consequently the S-unit norm image is explicitly $C_3\oplus\mathbb{Z}_3^{14}$; the same formulas with the last six coordinates zero identify the ordinary image with $C_3\oplus\mathbb{Z}_3^{8}$.

Every nonempty norm fibre is a translate of its actual norm-one subgroup. For either ordinary units or S-units the compact group types of these subgroups are

$$\begin{array}{c|c} L_2/L_1 & C_3\oplus\mathbb{Z}_3^{6}\\ B_2/B_1 & C_3\oplus\mathbb{Z}_3^{18}. \end{array} \tag{E101.10}$$

The proof gives the actual ideal equation defining an ordinary L_2-unit η of norm ζe_1e_2. Its residue at seventeen has $\bar\eta^3=\zeta^2$ and $\bar\eta^{96}=\zeta$, so the ordinary upper unit group has a noncube residue. The comparison to the positional section norm system is the actual field norm of E.46 and E.88. The manuscript anchor is LM §51.16, folios733v–736v, scans172–178.

Proof. A literal reversed norm and the actual eigenline. The lower polynomial is $b^6+17b^3+289=0$. Let $U_0,U_1,U_2\in L_1$ be given by the following matrix, whose rows multiply $(1,b,\dots,b^5)^t$:

$$(U_0,U_1,U_2)^t=\frac{1}{16578519}\mathcal{U}(1,b,b^2,b^3,b^4,b^5)^t,$$
$$\mathcal{U}^t=\begin{pmatrix} -37103278413 & 161479643843 & -551544947582\\ 25225148828 & -70558353324 & -108462373370\\ -5743099426 & -3192102308 & 26395251042\\ -1117338456 & 5646166658 & -53737480949\\ 415980086 & -6279927666 & 14839765429\\ -548768638 & 1260332509 & -2202740637 \end{pmatrix}. \tag{E101.11}$$

Exact polynomial reduction gives

$$U_0^3+\epsilon U_1^3+\epsilon^2U_2^3-3\epsilon U_0U_1U_2=b. \tag{E101.12}$$

Thus $U_0 + U_1\delta + U_2\delta^2$, for $\delta^3 = \epsilon$, is a literal reversed norm preimage of b. Both GP and the independent rational Python implementation verify (E101.12) from its coefficients. The discovery computation used generators of an uncertified BNF in this auxiliary field. Its class-group output is not an input to (E101.11)–(E101.12) or to any subsequent assertion.

Define the following matrices over L_1:

$$\begin{gathered} A = \begin{pmatrix} 0 & 0 & \epsilon \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad S = \operatorname{diag}(1, \zeta, \zeta^2), \\ B = (U_0 I + U_1 A + U_2 A^2) S, \\ A^3 = \epsilon I, \quad B^3 = bI, \quad BA = \zeta AB. \end{gathered} \tag{E101.13}$$

The certificate verifies each identity. In L_2, the β_2-eigenline of B is one-dimensional and has a unique vector v with $v_3 = 1$. The same eigenline calculation as in E97.9 gives

$$Av = w\sigma(v), \qquad w = v_2, \qquad Nw = \epsilon. \tag{E101.14}$$

Here the third coordinate gives the displayed formula for w. The certificate retains every coefficient of v and w in the absolute degree-eighteen field and independently verifies the relative and absolute norm identities.

There is a short exact divisor construction which avoids factoring the large numerator of w. Let I be the fractional ideal generated by the three entries of v, and put $D = I^{-1}$. Since ϵ is an ordinary unit, both A and A^{-1} have integral entries. Therefore the ideal generated by Av is exactly I. Equation (E101.14) implies

$$I = (w)\sigma(I), \qquad (w) = \frac{\sigma(D)}{D}. \tag{E101.15}$$

This is also checked by equality of the full ideal HNF matrices. Thus $[D]$ is the actual class defining $\omega_L(\epsilon)$, with its sign fixed.

The complete class evaluation. An exact ideal reduction yields $D = (m)J$, with $m \in L_2^{\times}$, and $NJ = 70782607 = 7^2 \cdot 1444543$. The retained full ideal factorization has these three rows:

p	$f(P/p)$	$v_P(J)$	$f(P/(P \cap L_1))$	$\mathrm{Art}_{\Gamma_L}(P \cap L_1)$
7	1	1	1	$(2,1,0)$
7	1	1	1	$(2,1,0)$
1444543	1	1	1	$(8,0,1)$.

(E101.16)

Both rational primes are certified prime. The two rows over seven denote the two actual prime ideals stored as HNF matrices. Contraction is tested by extension of each lower prime's integral-basis generators and its valuation at the upper prime; no root or prime identification is inferred from the rational prime alone. The Hilbert three-class field in E.86 is abelian over L_1, so the Artin values are the indicated lower values multiplied by the relative residue degrees. This is precisely the norm/Frobenius compatibility [Mil20b, V, Proposition3.3, Theorem3.6 and Example3.9].

The sum of the last column is $(3,2,1)$ in $\Gamma_L = C_9 \oplus C_3^2$. Apply the actual E97.13 map and its retained inverse:

$$C = \begin{pmatrix} 1 & -3 & -3 \\ 0 & 1 & 0 \\ 1/3 & -2/3 & -2/3 \end{pmatrix}, \qquad C^{-1} = \begin{pmatrix} -2 & 0 & 9 \\ 0 & 1 & 0 \\ -1 & -1 & 3 \end{pmatrix}. \quad [D]_3 = (3,2) \in C_9 \oplus C_3. \tag{E101.17}$$

The source is the subgroup where the third row of C is integral. The target orders, including its redundant coordinate, are $(9,3,1)$. Quotienting by the strongly ambiguous generator $(3,0)$ gives $(0,2)$, proving (E101.3).

E.97 proves that every ordinary lower unit is a field norm and that the ordinary norm quotient has order nine, with the exact Hilbert90 divisor map used here. It also evaluates $\omega_L(\zeta) = (0, 1)$. Thus $\zeta\epsilon$ has zero defect. The group of lower units modulo cubes has dimension three over $\mathbb{F}_3$; its norm kernel consequently has dimension one. The nonzero vector $(c, a, b) = (1, 1, 1)$ spans it. This proves the first equivalence in (E101.4), and the quotient map and inverse in (E101.5) follow by dividing by $(\zeta e_1 e_2)^b$.

The S-unit norm image contains $\beta_1 = N\beta_2$ and $c_1^3 = N(c_1)$, in addition to the ordinary image. The subgroup they generate has index 27 in the full lower S-unit group. E86.13 proves that the actual S-unit norm quotient has order 27. This inclusion and equality of indices prove the second equivalence of (E101.4), its displayed free basis, and the second quotient inverse.

An actual ordinary upper-unit preimage. Let D_ζ, w_ζ be the exact ideal and element of E.97, and let D, w be those of (E101.14)–(E101.15). The class of $DD_\zeta/\mathfrak{q}_{3,2}$ is zero in the three-primary coordinates. This ideal class is fixed by σ. It is also zero in the entire class group: the ambiguous-class formula gives

$$\left|\mathrm{Cl}(L_2)^{\langle\sigma\rangle}\right| = \frac{h(L_1)\, e_3 e_{17}}{3\,[E_1 : E_1 \cap N L_2^\times]} = \frac{9 \cdot 3 \cdot 3}{3} = 27. \tag{E101.18}$$

Every lower unit is a field norm, as established in E.97, so the denominator's unit index is one. There are no real ramified places. This applies the full ordinary ambiguous-class formula, including its proof [Lem13, Theorem1]. Thus no prime-to-three component is being discarded when concluding principality.

Choose a generator d of $DD_\zeta/\mathfrak{q}_{3,2}$. Then the literal map

$$\eta = w w_\zeta \frac{d}{\sigma(d)} \in \mathcal{O}_{L_2}^\times, \qquad N\eta = \zeta e_1 e_2 \tag{E101.19}$$

follows by cancellation in the full divisor group. This defines a preimage by an exact principal-ideal equation whose solvability has just been proved. Changing d by an ordinary unit changes η by the corresponding norm-one Hilbert90 element. For the ordinary image in (E101.4), write $a = b + 3s$, $c = b \bmod 3$; its preimage is then $\eta^b e_1^s$. Appending $c_1^{3t}\beta_1^f$ gives the S-unit preimage $\eta^b e_1^s c_1^t \beta_2^f$. These formulas are maps on the completed groups as well. The residue computations of E.97 give $\bar{e}_1 = \bar{e}_2 = \zeta^2$ at seventeen. The vertical extension is totally ramified with unchanged residue field $\mathbb{F}_{289}$, so its unit norm reduces to cubing. Hence $\bar{\eta}^3 = \zeta^2$ and $\bar{\eta}^{96} = (\zeta^2)^{32} = \zeta$.

The full B-unit defect through the actual horizontal norm. Let R be the coefficient matrix of $N_h : B_1^\times \to L_1^\times$, in the basis (E101.6) and target basis $(\zeta, e_1, e_2, c_1, \beta_1)$:

$$R = \begin{pmatrix} 1 & 2 & 0 & 2 & 1 & 1 & 0 & 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 0 & 0 & 0 & 0 & -1 & -1 & 0 & 0 & -1 & 1 & 2 & 2 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 2 & -1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 3 & 3 & 3 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}. \tag{E101.20}$$

The first row is interpreted modulo three. The field morphism is given by $b \mapsto \sqrt[3]{17}\xi$; the certificate stores its integral-basis matrix E, together with $(E^tE)^{-1}E^t$, and checks that this projection recovers every actual horizontal norm. For every column, multiplying its three literal h-conjugates and reconstructing the target from R agrees up to the prime-to-three sign. This proves that (E101.20) is the actual norm map.

For reference the six local norm characters of E.79 are

$$M = \begin{pmatrix} 2 & 1 & 0 & 1 & 0 & 0 & 0 & 2 & 1 & 1 & 1 & 0 & 0 & 0 & 2 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 2 & 1 & 2 & 2 & 0 & 0 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 1 & 0 & 1 \\ 2 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 2 & 2 & 0 & 0 & 1 & 1 \\ 1 & 2 & 0 & 2 & 0 & 0 & 0 & 2 & 2 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix} \quad (\mathrm{mod}3). \tag{E101.21}$$

Its rank is five. For $H = \ker M \subset \mathbb{F}_3^{15}$, the exact matrix calculation gives

$$R(H) = \{(c, a, a, 0, 0) : c, a \in \mathbb{F}_3\}. \tag{E101.22}$$

Both inclusions and rank two are checked on a retained ten-column basis of H. Thus for a field-norm S-unit u, the horizontal norm can be divided by included-base cubes and the known β_1-norm to leave the ordinary target $\zeta^c \epsilon^a$. The commuting norm square of E97.15 and the exact class map of E97.17 then give

$$\begin{aligned} \mathrm{Cl}_S(B_2)\{3\} \longrightarrow \mathrm{Cl}_S(L_2)\{3\}, \qquad & z \longmapsto (0, z), \\ \omega_L(N_h u) = (0, c + 2a), \qquad & \omega_B(u) = c + 2a. \end{aligned} \tag{E101.23}$$

This follows directly on divisors: $\mathrm{div}_S\, x = (\sigma - 1)D_B$ is sent by N_h to $(\sigma - 1)N_h D_B$; stripping the known norm factors changes no defect class. The injection $z \mapsto (0, z)$ fixes the sign and identifies the final scalar exactly.

Consequently the full B-S-unit norm image is the kernel modulo three of the seven-row matrix

$$\Phi = \begin{pmatrix} M \\ R_1 + 2R_2 \end{pmatrix}, \qquad \mathrm{rank}_{\mathbb{F}_3} \Phi = 6. \tag{E101.24}$$

The displayed R_i denote rows, indexed from one. The first six rows impose field-norm membership by the cyclic Hasse norm theorem; the last imposes zero in the exact E86.14 S-unit norm defect. This proves both directions before any lattice calculation.

The integral kernel has the HNF basis recovered by reversing (E101.9). Its determinant is 3^6, and Φ kills each column modulo three. The index therefore proves equality with the full kernel. The retained Smith matrices have diagonal $3^6, 1^9$, with both unimodular inverses checked. Equations (E101.7) and (E101.9) follow. The analogous ordinary submatrix has rank four; its kernel HNF has diagonal $3^4, 1^5$. Every S-unit norm preimage of an ordinary target is already ordinary: each vertical S-prime is unique above its predecessor with residue degree one, so its norm valuation is exactly its own valuation. This proves (E101.8) for ordinary norms.

Completion and fibre maps. All four unit groups are finitely generated. Inclusion of a lower unit has norm its cube; hence all norm cokernels here are finite and killed by three. Completion for multiplication by 3^m preserves the calculated cokernels. The completed norm images are closed images of compact $\mathbb{Z}_3$-modules, so the congruence tests and their inverse maps hold with exactly the topology in (E101.4)–(E101.9). Translation by any norm preimage gives an actual homeomorphism from the norm-one subgroup onto its fibre. In the L tower the unit ranks are eight and two, and in the B tower they are twenty-six and eight. The norm has full target rank because its image contains cubes. The torsion kernel is μ_3 in both cases: cubing on μ_3 or μ_9, respectively. Thus (E101.10) follows by the structure theorem for finitely generated $\mathbb{Z}_3$-modules. The same S-valuation argument shows that the ordinary and S-unit norm-one subgroups are equal, as actual subgroups of the upper field, not merely abstractly isomorphic.

The portable certificate independently checks the literal polynomial norms, both eigenvector equations, the class-coordinate inverses, every one of the 3^{10} field-norm residue vectors, and the

full Smith and projection inverses. It rebuilds only the previously certified lower BNF/genus inputs. The separate bounded search for a numerical generator in (E101.19) is retained with its actual no-hit output; its absence supplies no additional mathematical assertion. No value of the eventual exponent h or the E.91 transvection is deduced from these first-layer norm calculations. □

The exact first integral kernel of the unramified comparison

E.102. All finite factors and inverse maps in the degree-four full-to-metabelian fibre [Proved]

Retain the actual cubic geometric pro-three group N, its intrinsic metabelian quotient $M = N/N''$, and the original arithmetic automorphisms T, J of E.100. For the induced degree-four lattice

$$K_4 = (N'' \cap \gamma_4 N)/(N'' \cap \gamma_5 N) = \ker(L_4(N) \longrightarrow L_4(M)), \tag{E102.1}$$

the complete integral coefficient quotient is

$$\mathcal{C}_4(N'') = K_4/(T_4 - I)K_4^J \simeq \mathbb{Z}_3^{504} \oplus C_3^{144} \oplus C_9^{56} \oplus C_{27}^{30} \oplus C_{81}^{9}. \tag{E102.2}$$

Here $C_m = \mathbb{Z}/m\mathbb{Z}$ with its finite topology. The finite subgroup has order 3^{382} and exponent 81. All isomorphisms below are continuous homomorphisms of compact $\mathbb{Z}_3$-modules, and their complete integral basis maps and inverses are retained.

This computes the entire first nonzero coefficient kernel in E100.12. Through the actual triangular comparison E100.24–E100.26, it gives every degree-four fibre above the full unramified metabelian section space, including the zero-inertia loci in E.98 and E.100. The higher-degree factors keep their exact presentations and inverse maps of E.100. The manuscript anchor is §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up4.

Proof. The actual integer map. We use the ordered non-Chen Hall basis of E.100. It identifies K_4 with $\mathbb{Z}_3^{990}$; the integral arithmetic matrices T_4, J_4 in this very basis are the saved restrictions of the actual Hall actions. In particular this basis retains the original marking, without replacing the arithmetic operators by chosen operators with the same ranks. The complete matrices satisfy

$$J_4^2 = I, \qquad \operatorname{tr}(J_4) = -18, \qquad \operatorname{rank} K_4^J = 486. \tag{E102.3}$$

The identity $J_4^2 = I$ gives the direct decomposition by the two integral projectors $(I \pm J_4)/2$. Therefore $\operatorname{im}(I + J_4) = K_4^J$ and

$$C = (T_4 - I)(I + J_4) : \mathbb{Z}_3^{990} \longrightarrow K_4 \quad \text{has image } (T_4 - I)K_4^J. \tag{E102.4}$$

The fixed rank in (E102.3) gives $\operatorname{rank}_{\mathbb{Q}_3} C \le 486$.

Finite exact computation and its full integral lift. The retained program performs integer row and column operations modulo $q = 3^7 = 2187$. At each step it takes an entry of minimal three-adic valuation, moves it to the next pivot, multiplies its row by the inverse of its unit part, and clears that column and row. All coefficients and both inverse transformations are recorded. The result is

$$UCV \equiv \operatorname{diag}(D, 0_{504}) \pmod q, \qquad D = \operatorname{diag}(1^{247}, 3^{144}, 9^{56}, 27^{30}, 81^{9}). \tag{E102.5}$$

The certificate verifies both products of U and its retained modular inverse, both products of V and its retained modular inverse, and every entry of (E102.5). These give 486 nonzero pivots

modulo q, so the upper rank bound is attained: $\operatorname{rank}_{\mathbb{Q}_3} C = 486$. In particular no remaining nonzero invariant factor is hidden by the finite precision.

Here is an explicit lift of these maps over $\mathbb{Z}_3$. Choose the integer representatives between zero and $q-1$ of the saved matrices U, V. They are invertible over $\mathbb{Z}_3$: their displayed modular inverse identities already give invertibility modulo three. Write their exact integer product as

$$B = UCV = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}, \qquad B_{11} = DS, \quad S = I + E, \quad E = D^{-1}(B_{11} - D). \tag{E102.6}$$

Every entry of $B - \operatorname{diag}(D, 0)$ is divisible by 3^7. Every diagonal entry of D divides 3^4. Thus E is integral and divisible by 3^3, and

$$S^{-1} = \sum_{j \geq 0} (-E)^j \tag{E102.7}$$

converges in the integral matrix ring. Both $S^{-1}D^{-1}B_{12}$ and $B_{21}S^{-1}D^{-1}$ are integral. Consequently the explicit triangular matrices

$$\begin{aligned} V_1 &= \begin{pmatrix} S^{-1} & -S^{-1}D^{-1}B_{12} \\ 0 & I \end{pmatrix}, \\ U_1 &= \begin{pmatrix} I & 0 \\ -B_{21}S^{-1}D^{-1} & I \end{pmatrix} \end{aligned} \tag{E102.8}$$

and their block-triangular inverses belong to $\mathrm{GL}_{990}(\mathbb{Z}_3)$. Their exact product is

$$U_1 B V_1 = \begin{pmatrix} D & 0 \\ 0 & B_{22} - B_{21}S^{-1}D^{-1}B_{12} \end{pmatrix} = \operatorname{diag}(D, 0_{504}). \tag{E102.9}$$

The final lower block vanishes over $\mathbb{Q}_3$ because the full matrix has rank 486 and its upper block already has that rank. All its entries are integral, so it vanishes over $\mathbb{Z}_3$ as well. This proves the complete integral Smith equality, with the actual matrices $U_1 U$ and $V V_1$.

For completeness, inverses of the integer lifts U, V can also be recovered directly from the retained modular inverse matrices. Let $\widetilde{U}^{-1}$ denote the stored integer representative and put $E_U = I - U\widetilde{U}^{-1}$. Then

$$U^{-1} = \widetilde{U}^{-1} \sum_{j \geq 0} E_U^j, \qquad E_U \in 3^7 M_{990}(\mathbb{Z}_3), \tag{E102.10}$$

and the same formula gives V^{-1}. Every series in these formulas has a stated integral convergence rate and determines all digits of its map.

Quotient coordinates and the section-space comparison. For $x \in K_4$, apply $U_1 U$ to its coordinate column. The first 247 coordinates have quotient zero; reduce the next $144, 56, 30, 9$ coordinates modulo $3, 9, 27, 81$, respectively, and retain the last 504 coordinates in $\mathbb{Z}_3$. Equation (E102.9) proves that the exact kernel of this continuous homomorphism is $(T_4 - I)K_4^J$. To recover a preimage, choose the standard integer representatives in the finite coordinates, insert zeros in the first 247 positions, and apply $U^{-1}U_1^{-1}$. This is a continuous set section; on the quotient it gives the inverse module isomorphism, independent of the chosen finite lifts. These are the two maps in (E102.2). Their torsion order is $3^{144+2\cdot 56+3\cdot 30+4\cdot 9} = 3^{382}$, and the nine C_{81} factors prove the exponent claim.

The strict kernel and fixed-lift maps in E100.11–12 identify this quotient with the entire kernel of $\mathcal{C}_4(N) \to \mathcal{C}_4(M)$. All earlier induced kernel layers vanish. The nonlinear aligning conjugator and arithmetic correction at the next step are exactly E100.24–25; inserting the

representatives just constructed into E100.26 gives every degree-four preimage. Thus the calculation transfers to the stated section fibres by those actual maps. The entire inverse limit continues with E.100's subsequent integral coordinates.

The files `e_kernel_degree4_smith.py`, its actual stdout, its JSON operation log, and its NumPy archive retain every matrix in (E102.3)–(E102.6), all modular inverses and the original Hall indices. All arithmetic is integer arithmetic. The largest absolute entry of C is 4800, and the exact integer-lift product B is checked with explicit overflow bounds before each machine-integer product. The complete identities, rather than only the Smith list, are the finite certificate for (E102.5). □

Every odd-primary abelian section of the plus quartic

E.103. The complete quartic odd-abelian section space, its toric map, and every fibre [Proved]

Put

$$C : x^4 + y^4 + z^4 = 0, \qquad U = C \setminus \{xyz = 0\}, \qquad F = \mathbb{Q}(\zeta_8), \qquad E_+ : Y^2Z = X^3 + XZ^2. \tag{E103.1}$$

Let $R = \mathbb{Z}_\ell$ for an odd prime ℓ, or let $R = \prod_{\ell \neq 2} \mathbb{Z}_\ell$. Write $A_R = H_1(U_{\overline{\mathbb{Q}}}, R) = T_R J_{C,D}$, where $D = C \setminus U$, and let $\mathcal{E}_R \to G_{\mathbb{Q}}$ be the actual arithmetic fundamental-group extension with geometric kernel A_R. Its source map is the quotient of $\pi_1(U)$ by the kernel of the geometric abelian map to A_R. There is a continuous section s_R constructed by the degree-four coset formula (E103.9) below. It has zero image under the actual coordinate-unit map $(x/z, y/z)$ to the two-dimensional toric extension.

For a field L write $\widehat{L^\times}_R$ for its power completion at the integers invertible away from the primes occurring in R; thus the indices are ℓ^m in the first case and all odd positive integers in the second. Set

$$K_R = \ker\big(N_{F/\mathbb{Q}} : \widehat{F^\times}_R \longrightarrow \widehat{\mathbb{Q}^\times}_R\big). \tag{E103.2}$$

Relative to s_R, the entire geometric-conjugacy section space is the following topological R-module:

$$\mathrm{Sect}(\mathcal{E}_R)/A_R \simeq (\widehat{F^\times}_R)^2 \times K_R \times R^3. \tag{E103.3}$$

The three last coordinates are the three evaluated elliptic cohomology factors, each isomorphic to R by (E103.19). The map to the toric minimum and every one of its fibres are

$$(a, b, k, e) \longmapsto (Na, Nb), \qquad \mathrm{Fib}_{(u,v)} \simeq K_R^3 \times R^3 \quad ((u, v) \in (\widehat{\mathbb{Q}^\times}_R)^2). \tag{E103.4}$$

In particular this map is onto. Equations (E103.23)–(E103.25) give the actual forward and inverse fibre coordinates.

Here is a prime-indexed evaluation of all the factors. For a countable set I let

$$\mathcal{R}_I(R) = \{(a_i) \in R^I : \text{ for every open ideal } nR,\ a_i \in nR \text{ for all but finitely many } i\}, \tag{E103.5}$$

with the uniform congruence topology, namely the inverse limit of the discrete modules $\bigoplus_I R/nR$. Choose a generator of each prime ideal of F, and use the unit $\epsilon = 1 + \sqrt{2}$. Then (E103.3) is $R^6 \times \mathcal{R}_I(R)$, and every fibre in (E103.4) is $R^6 \times \mathcal{R}_{I'}(R)$, with the following literal multiplicities over each rational prime:

p	$g_p = \#\{\mathfrak{p} \mid p\}$	f_p	$\#I_p = 3g_p - 1$	$\#I'_p = 3(g_p - 1)$
2	1	1	2	0
$p \equiv 1 \pmod 8$	4	1	11	9
$p \equiv 3, 5, 7 \pmod 8$	2	2	5	3

(E103.6)

The completions and their topology in this statement retain infinite supports tending to zero. They do not impose a fixed finite ramification set.

For every prime ℓ, including two, the pushout of the actual abelian extension along $T_\ell J_{C,D} \to T_\ell J_{C,D} \otimes \mathbb{Q}_\ell$ also splits. Its section space and its toric fibres are, respectively,

$$\mathbb{Q}_\ell^6 \times c_0(I, \mathbb{Q}_\ell), \qquad \mathbb{Q}_\ell^6 \times c_0(I', \mathbb{Q}_\ell), \tag{E103.7}$$

with the same indices and explicit maps, now over $\mathbb{Q}_\ell$. These are statements about the specified abelian pushout. The source maps from the full and intrinsic-metabelian quartic extensions to their integral two-primary abelian quotient retain the obstruction proved in E.18. Consequently their global section sets are empty. Formulas (E103.3) and (E103.7) classify the sections of the target extensions after the displayed coefficient maps; the induced map from the empty full profinite section set has empty image. The Fermat covering coordinate $t = -(x/z)^4$ induces 4 times (E103.4), because $1 - t = -(y/z)^4$ and the class of -1 is zero in these odd or rational Kummer groups. The manuscript anchor is §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and the quartic case retained in E.23.

Proof. An actual arithmetic section by a finite coset formula. The point $P = [\zeta_8 : 0 : 1]$ is an F-rational cusp. With parameter $s = y/z$ the formal branch at P is $x/z = \zeta_8(1 + s^4)^{1/4}$, with the fourth root having constant coefficient one. Its tangent of value one gives an actual section s_H over $H = G_F$ of every extension under consideration. Fix the same geometric fibre functor throughout. Let g_i, $0 \le i < 4$, represent the left cosets $G_\mathbb{Q}/H$, with $g_0 = 1$, and choose lifts e_i in $\mathcal{E}_R$. For $g \in G_\mathbb{Q}$ set $j = g \cdot i$ and $h_i(g) = g_j^{-1} g g_i \in H$. For a lift e of g define, additively in the geometric kernel,

$$b_i(e) = e e_i s_H(h_i(g))^{-1} e_j^{-1} \in A_R, \qquad B(e) = \sum_{i=0}^{3} b_i(e). \tag{E103.8}$$

If f projects to h, multiplication of these literal words gives $b_i(ef) = g b_i(f) + b_{h\cdot i}(e)$. Thus

$$B(ef) = B(e) + gB(f), \qquad B(a) = 4a \ (a \in A_R), \qquad s_0(g) = (-\tfrac{1}{4}B(e))e. \tag{E103.9}$$

The last expression is independent of the chosen lift e and is a homomorphism. It is continuous: on each of the four clopen cosets use the lift $e_i s_H(g_i^{-1} g)$, and apply the displayed finite continuous formula. All divisions by four are integral in R. The identical construction works in the rational pushout, where division by four is again defined. This proves splitting of the actual extension, with no compatibility assumption on separately chosen finite sections.

Choose compatible roots of the coordinate units in the fixed tangential fibre. They give the actual homomorphism $\mathcal{E}_R \to R(1)^2 \rtimes G_\mathbb{Q}$. Its restriction to A_R is denoted r. Below we construct a $G_\mathbb{Q}$-equivariant module section $\iota : R(1)^2 \to A_R$ of r. If $r_0(g)$ is the first component of the image of $s_0(g)$, then

$$s_R(g) = (-\iota(r_0(g)))s_0(g) \tag{E103.10}$$

is a section with toric value zero. This retains the actual root and basepoint choices: a different choice changes r_0 by its specified coboundary and (E103.10) transports the origin with it. For odd coefficients the chosen compatible roots of ζ_8 already lie in $\mu_8 \subset F$, but this observation is not needed for the averaging argument.

The compact and toric coefficient maps. Let $J = \mathrm{Jac}(C)$ and $G = J_{C,D}$. C.5 gives the unique homomorphism $j : J \to G$ with $pj = [4]$, where $p : G \to J$ is the projection. Here the pullback of the extension by $[4]$ is split because all cusp differences are killed by four. Its splitting is unique since $\mathrm{Hom}(J, T_D) = 0$, so it descends to $\mathbb{Q}$. This is precisely the map used below.

There are three everywhere-defined maps $C \to E_+$:

$$\begin{aligned} f_0[x:y:z] &= [-x^2y : z^2x : y^3], \\ f_1[x:y:z] &= [-y^2z : x^2y : z^3], \\ f_2[x:y:z] &= [-z^2x : y^2z : x^3]. \end{aligned} \tag{E103.11}$$

Their coordinate triples have no common zero on C and satisfy the elliptic equation. For example, on $y \neq 0$, writing $u = x/y$, $v = z/y$, the first map is $X = -u^2$, $Y = uv^2$; a generic target point has two choices of u and two of v. Thus its degree is four, as are those of its cyclic conjugates. In the affine chart $z = 1$, a basis of holomorphic differentials on C is dx/y^3, $x\,dx/y^3$, $y\,dx/y^3$. The three pullbacks of dX/Y are twice these differentials, in that order. In particular the maps are independent. The first map also agrees on the common domain with the homogeneous map in [FLGS18, Proposition3.1 and its proof]; (E103.11) removes its common factor and records the extension at the cusps.

The sign changes of x, y, z act on these maps by the three distinct nontrivial characters of their group C_2^2. The character table, with rows the three coordinate sign changes, is

$$\begin{pmatrix} -1 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & -1 & -1 \end{pmatrix}. \tag{E103.12}$$

For $i \neq j$ a sign change with different characters makes $f_{i*}f_j^*$ equal its negative. The group of homomorphisms of abelian varieties is torsion-free, so this composite is zero. For $i = j$, the push–pull composite is multiplication by the degree, namely four. Therefore, with $\Phi = (f_{0*}, f_{1*}, f_{2*})$ and $\Psi = \sum_i f_i^*$,

$$\Phi : J \longrightarrow E_+^3, \quad \Psi : E_+^3 \longrightarrow J, \quad \Phi\Psi = [4], \qquad \Psi\Phi = [4]. \tag{E103.13}$$

For the second equality, the first makes Φ an isogeny; the image of $\Psi\Phi - [4]$ is connected and contained in its finite kernel. Since Ψ is dual to Φ for the canonical polarizations, both have degree64.

Let $\Lambda_R = R[G_{\mathbb{Q}}/G_F](1)$, and write $\delta : R(1) \to \Lambda_R$ for the constant vector and aug for the sum of its four coordinates. The three cusp orbits are the divisors $x = 0$, $y = 0$, $z = 0$, each isomorphic to $\operatorname{Spec} F$. Positive peripheral cocharacters identify

$$T_R T_D = \Lambda_R^3/\delta_{\mathrm{diag}} R(1), \qquad I_R = \ker(\mathrm{aug} : \Lambda_R \to R(1)). \tag{E103.14}$$

If $t = \mathrm{aug}(v_3)/4$, the map

$$[v_1, v_2, v_3] \longmapsto (v_1 - \delta t,\ v_2 - \delta t,\ v_3 - \delta t) \in \Lambda_R^2 \oplus I_R \tag{E103.15}$$

is well-defined. Its inverse takes (a, b, k) to its class in the quotient. These are equivariant inverse maps, since the augmentation of the third component is zero. Combining them with (E103.13) and the actual j gives

$$\begin{aligned} A_R &\simeq T_R T_D \oplus (T_R E_+)^3, \\ v &\longmapsto (v - \tfrac{1}{4} j_* p_* v,\ \Phi_* p_* v), \\ (w, e) &\longmapsto w + \tfrac{1}{16} j_* \Psi_* e. \end{aligned} \tag{E103.16}$$

The inverse coefficients follow from $p_* j_* = 4$ and $\Phi_* \Psi_* = 4$, which specify the integral odd-primary comparison and its inverse.

The two divisors of $x/z, y/z$ give on (E103.14) the map $(v_1, v_2, v_3) \mapsto (\mathrm{aug}\, v_1 - \mathrm{aug}\, v_3, \mathrm{aug}\, v_2 - \mathrm{aug}\, v_3)$. The composite with j is zero because it is a homomorphism

from the proper abelian variety J to $\mathbb{G}_m^2$. Thus under (E103.15)–(E103.16) it is $(a,b,k,e)\mapsto(\operatorname{aug} a,\operatorname{aug} b)$. The equivariant section used in (E103.10) is $(u,v)\mapsto[\delta u/4,\delta v/4,0]$ in the toric summand.

Every elliptic cohomology factor is evaluated. First we give a proof of both required rank-zero inputs. The degree-two isogeny and its dual are the actual rational maps

$$\begin{aligned}\phi : E_+ \longrightarrow E_- :&\quad (x,y)\longmapsto (x+x^{-1},\ y(1-x^{-2})),\\ E_- :&\quad Y^2=X^3-4X,\\ \psi : E_- \longrightarrow E_+ :&\quad (X,Y)\longmapsto \bigl((X-4X^{-1})/4, Y(1+4X^{-2})/8\bigr),\\ &\quad \psi\phi=[2],\qquad \phi\psi=[2].\end{aligned}\tag{E103.17}$$

The formulas extend at their finite kernels and at infinity. The induced rational Tate-module maps are invertible, so every local Euler factor agrees for the two curves. The original Martin–Ono Theorems1–2 identify the L-function of E_- with the Mellin transform of

$$f(\tau)=\frac{\eta(8\tau)^8}{\eta(4\tau)^2\eta(16\tau)^2},\qquad L(E_+,1)=L(E_-,1)=2\pi\int_0^\infty f(it)\,dt>0.\tag{E103.18}$$

Every eta factor is positive on the positive imaginary axis. The cusp-form decay at zero and infinity makes the integral convergent; the Mellin identity first holds in an absolute convergence half-plane and then continues to $s=1$. The integral is strictly positive, proving analytic rank zero without a numerical nonvanishing test [MO97, Theorems1–2, tables and proof, pp.2–5 of the opened author PDF].

For completeness, the algebraic rank and torsion are elementary here. Write a rational point of E_+ as $x=m/n^2$, $y=k/n^3$, with $(m,n)=1$. The equation gives $k^2=m(m^2+n^4)$, whose two factors are coprime; real positivity gives $m\geq 0$. Thus x is a square. The image of ψ is exactly the points with square x: if $x=s^2\neq 0$, an inverse point is $X=2s^2+2y/s$, $Y=2sX$; the exceptional rational two-torsion point also has rational inverse. Similarly the image of ϕ consists of the points of E_- with square X. The squareclass map on E_-, with values 1 at infinity and -4 at $(0,0)$, is a homomorphism: the product of the three intersection abscissas of a line is its intercept squared. Its possible values are $\{1,-1,2,-2\}$ by the same coprime-factor argument, and all four occur at the rational two-torsion points. Hence $E_-(\mathbb{Q})/\phi E_+(\mathbb{Q})$ has order four, whereas $E_+(\mathbb{Q})/\psi E_-(\mathbb{Q})$ is trivial. The nontrivial element of $\ker\psi$ has squareclass -1, so (E103.17) gives $|E_+(\mathbb{Q})/2E_+(\mathbb{Q})|=2$. Mordell–Weil then gives rank zero. Reduction at three and five, both with four points, excludes odd rational torsion. There is no rational point of order four, since its double would be $(0,0)$ and the duplication formula would force $x=\pm1$, with $y^2=\pm2$. Thus $E_+(\mathbb{Q})=C_2$.

By Çiperiani–Stix, Theorem18(2), analytic rank zero gives $\operatorname{Div} H^1(\mathbb{Q},E_+)\simeq\mathbb{Q}/\mathbb{Z}$ [ÇS13]. The additional prime restrictions in their theorem concern the separate assertion about its intersection with the Tate–Shafarevich group; none is used here. Fix one isomorphism $\vartheta:\mathbb{Q}/\mathbb{Z}\to\operatorname{Div} H^1(\mathbb{Q},E_+)$. The Kummer sequences and inverse limit give, for odd ℓ,

$$\begin{aligned}H^1(\mathbb{Q},T_\ell E_+)&\simeq \operatorname{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, H^1(\mathbb{Q},E_+))\\ &=\operatorname{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell,\operatorname{Div} H^1(\mathbb{Q},E_+))\simeq\mathbb{Z}_\ell.\end{aligned}\tag{E103.19}$$

Indeed $E_+(\mathbb{Q})/\ell^m E_+(\mathbb{Q})=0$, the finite H^0 groups satisfy the Mittag–Leffler condition, and every image of a divisible group is divisible. The inverse of the last coordinate map sends $a\in\mathbb{Z}_\ell$ to the compatible classes lifting $\vartheta(a/\ell^m)$ in the finite Kummer sequences. These lifts are unique as cohomology classes. This specifies the elliptic coordinate choice and its inverse; the theorem does not require a named but unevaluated elliptic H^1 factor. For the prime-to-two coefficient product take the product of these maps using the same ϑ.

For its topology, an integral elliptic cocycle is unramified outside $\{2, \ell\}$. At a good prime $p \neq \ell$, its inertia restriction is a homomorphism from $\mathbb{Z}_\ell(1)$ to $T_\ell E_+$ with Frobenius eigenvalue p. The two Frobenius eigenvalues have absolute value $\sqrt{p}$, so this homomorphism is zero. Finite-coefficient cohomology over that fixed finite ramification set is finite, by the finiteness of extensions of bounded degree and ramification. It follows that the left side of (E103.19) is compact in its inverse-limit topology. Its reduction kernels, transported by (E103.19), are $\ell^m\mathbb{Z}_\ell$, since $H^0(\mathbb{Q}, E_+[\ell^m]) = 0$. Thus the displayed isomorphism and its inverse are continuous.

Shapiro with the actual norm and its cochain inverse. Use the same coset representatives as in (E103.8). The action on $\Lambda_R = R(1)^4$ is $(ga)_{g\cdot i} = \chi(g)a_i$. For a continuous cocycle $u : H \to R(1)$ put

$$z_u(g)_{g\cdot i} = \chi(g_{g\cdot i})\, u(g_{g\cdot i}^{-1} g g_i). \tag{E103.20}$$

The identity $h_i(gh) = h_{h\cdot i}(g)h_i(h)$ proves the cocycle identity directly. Restriction to H followed by the zeroth coordinate recovers u. Conversely, for a Λ_R-cocycle z, let $u(h) = z(h)_0$ and $m_i = z(g_i)_i$. The cocycle identity at $gg_i = g_{g\cdot i}h_i(g)$ gives $z = z_u - \delta m$, where $\delta m(g) = gm - m$. This proves the cohomology isomorphism and its inverse with their exact signs. It is the same finite-coset Shapiro map used in E.46.

The sum of the coordinates in (E103.20) is the Kummer cocycle of the norm. To check this without a convention choice, choose compatible roots $a^{1/n}$ for $a \in F^\times$. Then $\prod_i g_i(a^{1/n})$ is a compatible root of Na; applying g gives exactly the sum in (E103.20). The augmentation is split by $\delta/4$, so

$$\begin{aligned} H^1(\mathbb{Q}, \Lambda_R) &\simeq \widehat{F^\times}_R, \\ H^1(\mathbb{Q}, I_R) &\simeq K_R, \\ H^1(\mathbb{Q}, A_R) &\simeq (\widehat{F^\times}_R)^2 \times K_R \times R^3. \end{aligned} \tag{E103.21}$$

The Kummer limit and all maps here have the uniform topology. One can also construct a cocycle from the finite-level classes: choose representatives successively, and remove the coboundary difference from their reductions by a lifted coefficient vector. The aligning vector is unique modulo the current level because all the relevant H^0 groups vanish. For Λ_R this is the absence of odd roots of unity in F; for I_R it follows by inclusion; for the elliptic factors it was proved above. The compatible representatives converge uniformly to the required continuous cocycle. The same construction proves continuity of the inverse maps. The finite H^0 argument also gives the inverse-limit statement directly, without assuming finiteness of global Kummer groups.

The section corresponding to a cocycle z is literally $g \mapsto z(g)s_R(g)$. Geometric conjugation replaces z by a coboundary. Conversely every section has this form, so (E103.21) proves (E103.3). The actual divisor computation after (E103.16), followed by the norm calculation just made, proves (E103.4), including its zero origin.

Complete prime coordinates and all inverses. We include the field calculation to specify every factor in (E103.21). The ring of integers of F is $\mathbb{Z}[\zeta_8]$ and its discriminant is256. For example, the polynomial $\Phi_8(X + 1) = X^4 + 4X^3 + 6X^2 + 4X + 2$ gives a uniformizer at two; away from two its discriminant is a unit. These two local statements give the asserted full ring of integers. The Minkowski bound is $24/\pi^2 < 3$ [Mil20a, Theorem4.3, printed p.70]. Every ideal class therefore has an integral representative of norm one or two. The unique prime above two is $(1 - \zeta_8)$, whose norm is two. Thus the class number is one.

The unit group is $\mu_8 \times \langle\epsilon\rangle$, $\epsilon = 1 + \sqrt{2}$. Here is an elementary check of the unit index. The positive generator of $\mathbb{Z}[\sqrt{2}]^\times/\{\pm 1\}$ is ϵ: a smaller unit greater than one would, by its norm ± 1, have a nonzero integral $\sqrt{2}$ coefficient of absolute value less than one. For a unit $u \in F$, the element $u\bar{u}$ is a totally positive real unit, hence ϵ^{2k}. Then all conjugates of $u\epsilon^{-k}$ have

absolute value one. Powers of an algebraic integer with this property have bounded integral characteristic coefficients, so two powers coincide; the element is a root of unity. The roots of unity in F are exactly μ_8: the field is unramified at odd primes and has degree four, excluding both an odd-order root and a primitive sixteenth root.

Choose an integral generator $\pi_{\mathfrak{p}}$ of each prime ideal. Its norm is p^{f_p}, with positive sign because F is totally imaginary. Unique ideal factorization now gives the actual multiplicative decomposition and its completion

$$F^{\times} = \mu_8 \times \langle \epsilon \rangle \times \bigoplus_{\mathfrak{p}} \langle \pi_{\mathfrak{p}} \rangle, \qquad \widehat{F^{\times}}_R = R \times \mathcal{R}_{\{\mathfrak{p}\}}(R). \tag{E103.22}$$

The first map recovers the prime exponents by valuations and then the unit exponent. Its inverse multiplies the chosen factors. In the completion this means the compatible finite products modulo each n; they converge in the uniform completion topology. The torsion μ_8 disappears. Likewise $\widehat{\mathbb{Q}^{\times}}_R = \mathcal{R}_{\{p\}}(R)$.

The Galois group $(\mathbb{Z}/8)^{\times}$ has exponent two. At an odd prime its Frobenius is p mod 8, so the residue degrees and numbers of factors are precisely those in (E103.6). At two the unique prime has residue degree one and ramification degree four. The norm on (E103.22) kills the unit coordinate and, over each p, is

$$(a_{p,0}, \dots, a_{p,g_p-1}) \longmapsto d_p = f_p \sum_{j=0}^{g_p-1} a_{p,j}. \tag{E103.23}$$

Choose one distinguished prime $\mathfrak{p}_{p,0}$ for each p. Since f_p is one or two, the inverse coordinates for this row are

$$(d_p, k_{p,1}, \dots, k_{p,g_p-1}) \longmapsto (d_p/f_p - \sum_{j>0} k_{p,j},\ k_{p,1}, \dots, k_{p,g_p-1}). \tag{E103.24}$$

Both directions preserve the restricted condition uniformly: there are at most four entries in each block, and every denominator is a unit of R. Thus $K_R = R \times \mathcal{R}_{\{(p,j):1\le j<g_p\}}(R)$. Let ν be the norm section setting the unit coordinate and all $k_{p,j}$ to zero in (E103.24). It is a genuine continuous map on the completed multiplicative groups, with $N \circ \nu = \mathrm{id}$. For any (u, v) the forward and inverse fibre maps are

$$\begin{aligned} (k_1, k_2, k_3, e) &\longmapsto (\nu(u)k_1,\ \nu(v)k_2,\ k_3,\ e), \\ (a, b, k, e) &\longmapsto (a/\nu(u),\ b/\nu(v),\ k,\ e). \end{aligned} \tag{E103.25}$$

Products here denote the completed multiplicative groups; the prime and unit coordinates themselves are additive. These formulas prove both image inclusions, every fibre, and all the multiplicities in (E103.6).

Rational coefficient pushouts and exact scope. For a finite-rank $\mathbb{Z}_\ell$ lattice M, every continuous cocycle from the compact group $G_{\mathbb{Q}}$ to $M \otimes \mathbb{Q}_\ell$ has bounded image and therefore lies in some $\ell^{-a}M$. The same boundedness applies to coboundaries. Consequently

$$H^1(\mathbb{Q}, M \otimes \mathbb{Q}_\ell) = H^1(\mathbb{Q}, M) \otimes \mathbb{Q}_\ell. \tag{E103.26}$$

All the actual maps (E103.11)–(E103.16) remain valid after this scalar extension, also for $\ell = 2$. For the elliptic factor the Kummer kernel is $E_+(\mathbb{Q})^{\wedge}_{\ell}$, which is zero for odd ℓ and finite for $\ell = 2$; it disappears in (E103.26). The same maximal-divisible subgroup therefore gives $H^1(\mathbb{Q}, V_\ell E_+) = \mathbb{Q}_\ell$ for every ℓ. For the torus, the finite roots of unity likewise disappear. The formulas (E103.23)–(E103.25) still apply, with denominators bounded by two. A family in the rationalized restricted module has uniformly bounded denominators and tends to zero;

conversely a $\mathbb{Q}_\ell$-valued family tending to zero has a uniform lower valuation bound. This identifies the module and its uniform topology with c_0 and proves (E103.7). The cochain and coordinate maps above and their inverses are continuous in that topology.

Finally, the actual quotient maps from the full and intrinsic-metabelian geometric groups to the integral two-primary abelian lattice send any section to a section. E.18 proves that target empty, so those sources are empty. The maps to (E103.3) discard that two-primary coefficient, and the rational pushout kills the real order-two factor by the literal correction $-a_0/2$ in E.18's fixed first coordinate: its new square coordinate is $1 + 2(-1/2) = 0$. The coset construction proves existence in these new targets directly. No assertion about lifting the computed odd-abelian sections to a full odd nonabelian target is made by this coefficient computation; that part of the global higher-section question in E.23 retains its stated target.

The retained program `a_quartic_odd.py` checks the nine affine basepoint charts of (E103.11), all three differential pullbacks, the character table, the actual two-isogeny and dual identities, and both square-coordinate inverses used in the elementary descent. It saves the full eleven-dimensional torus matrices and inverse, checks all four Galois permutations, the norm rows and their section. It computes every prime factorization of Φ_8 through250 and compares128 exact eta coefficients with point counts and with the owned PARI calculation. The latter unconditionally certifies the field's class number, the unit generator, and the elliptic rank bounds. The program also checks10000 literal finite coset-average identities,400 section products,2000 Shapiro cocycle identities, their cochain inverses and traces, and the actual finite congruence transitions of (E103.24). These certificates check the displayed maps; the infinite cohomology and completion claims follow from the proofs above. □

Complete full and metabelian sections over two families of arithmetic sources

E.104. All cubic sections over the two-prime source families of E.55 and E.56 [Proved]

Keep the actual based cubic geometric pro-three group N, its intrinsic metabelian quotient $M = N/N''$, the original complex conjugation c, and its genuine action J. Define the following explicit sets of rational primes:

$$\begin{aligned} \mathscr{I} &= \{\ell \text{ prime} : \ell \equiv 2, 5 \pmod 9\}, \\ \mathscr{P} &= \{\ell \text{ prime} : v_3(\ell - 1) = 1,\ 3^{(\ell-1)/3} \not\equiv 1 \pmod \ell\}. \end{aligned} \tag{E104.1}$$

For every $\ell \in \mathscr{I} \cup \mathscr{P}$, let $G_{3\ell}$ be the actual relative pro-three arithmetic group over $K = \mathbb{Q}(\zeta_3)$, unramified outside the primes over three and ℓ, with its retained order-two quotient. Write

$$\mathscr{S}_{3\ell}(H) = H^1_{\mathrm{cts}}(G_{3\ell}, H) \quad (H = N, M) \tag{E104.2}$$

for geometric-conjugacy classes under the genuine based action. Inflation along the specified arithmetic quotient identifies this with the corresponding locus of global sections. The following coordinates compute the entire space (E104.2), and every full-to-metabelian fibre.

For $\ell \in \mathscr{I}$, put $q = \ell^2$. There are actual marked generators

$$\begin{gathered} G_{3\ell} = (D_q \underset{3}{*} \langle\rho\rangle) \rtimes \langle c\rangle, \quad D_q = \langle t, \sigma \mid \sigma t \sigma^{-1} = t^q\rangle_3, \\ ctc = t^{-1}, \quad c\sigma c = \sigma, \quad c\rho c = \rho^{-1}. \end{gathered} \tag{E104.3}$$

In the original Kummer marking their columns are

	t	σ	ρ
χ_3	1	q	1
$\kappa_3(\ell)$	1	0	0
$\kappa_3(3)$	0	0	2

. (E104.4)

Let $T=\Phi_\tau$ be the original arithmetic operator of E.37, and set $F_\ell=\Phi_\sigma$. The free generator is chosen so that

$$\Phi_t=1,\quad \Phi_\rho=TJT^{-1}J,\quad \mathcal{L}:=\Phi_\rho J=TJT^{-1}. \tag{E104.5}$$

Use the same three rational cusps, rational tangents and J-fixed paths p_i as in E.100, and put $u_{i,\ell}=p_iF_\ell(p_i)^{-1}$. Every ramified section has a unique representative

$$\begin{array}{c|cccc} & t & \sigma & \rho & c \\ \hline a_{i,r,y} & d_i^r & u_{i,\ell} & T(y) & 1 \end{array},\qquad i\in\{O,A_0,B_0\},\quad r\in\mathbb{Z}_3\setminus\{0\},\quad J(y)=y^{-1}. \tag{E104.6}$$

Uniqueness here means uniqueness after the unique J-fixed geometric conjugator aligning its tame restriction with $(d_i^r,u_{i,\ell})$.

For $\ell\in\mathscr{P}$, the marked source is instead

$$\begin{aligned} G_{3\ell}&=(D_\ell\underset{3}{*}D'_\ell)\rtimes\langle c\rangle,\\ D_\ell&=\langle t,\sigma\mid \sigma t\sigma^{-1}=t^\ell\rangle_3,\quad c\sigma c=\sigma',\quad ctc=(t')^{-1}. \end{aligned} \tag{E104.7}$$

Here the exact root-normalized generators are those of E56.11–13. Put $T_\ell=\Phi_\sigma$. For each of the nine K-rational cusps D, use its actual positive peripheral element δ_D and path value $z_{D,\ell}$, so that

$$z_{D,\ell}T_\ell(\delta_D)z_{D,\ell}^{-1}=\delta_D^\ell. \tag{E104.8}$$

The complete ramified coordinates are

$$(D,r,b,[g])\in\{\text{nine cusps}\}\times(\mathbb{Z}_3\setminus\{0\})\times\{0,1,2\}\times H^J\backslash H, \tag{E104.9}$$

with actual values

$$\begin{aligned} a(t)=g\delta_D^rg^{-1},\qquad a(\sigma)=g\delta_D^bz_{D,\ell}T_\ell(g)^{-1},\qquad a(c)=1,\\ a(t')=J(a(t))^{-1},\qquad a(\sigma')=J(a(\sigma)). \end{aligned} \tag{E104.10}$$

Every row in (E104.6) and (E104.10) is a global section in its stated source; every geometric stabilizer is trivial.

In both families the zero-inertia part is exactly $\mathscr{S}_3(H)=H^1_{\mathrm{cts}}(\Gamma,H)$ of E.41. The actual full-to-metabelian map is onto on the whole space. It preserves the displayed cusp, exponent and finite residue coordinates. Its ramified fibres are

$$\prod_{m\ge4}\mathbb{Z}_3^{\nu_m(N)-\nu_m(M)}, \tag{E104.11}$$

with the exact all-degree ranks and mutually inverse Hall maps of E100.7 and E98.30–31. Every zero-inertia fibre is

$$\prod_{m\ge4}\mathcal{C}_m(N''),\qquad \mathcal{C}_m(N'')=K_m/(T_m-I)K_m^J, \tag{E104.12}$$

with the actual nonlinear comparison maps E100.23–26. In particular its first factor is exactly $\mathbb{Z}_3^{504}\oplus C_3^{144}\oplus C_9^{56}\oplus C_{27}^{30}\oplus C_{81}^{9}$ by E.102. No full ramified class maps to a zero-inertia target. The whole space retains the quotient topology and the $r\to0$ boundaries described in the proof.

The arithmetic source conditions in (E104.1) specify the scope of this classification. E.23 continues to retain the unrestricted global source and the other requested geometric prime components. The manuscript anchor is §51.16, folios733v–736v, scans172–178, formulas(371)–(379), Part E(c),(d) and follow-up4.

Proof. Original-marking transport for every inert prime. E.55 constructs the actual tame factor at the unique K-prime over ℓ, with residue cardinality q. We specify its transport to the original c and roots. Let its local order-two element be $d = gcg^{-1}$. For either original rational Kummer cocycle κ we have $\kappa(c) = 0$ and $\kappa(d) = 2\kappa(g)$. Multiplying the root system by the compatible roots of unity with exponent $e = \kappa(g)$ changes the cocycle to $\kappa + (\chi_3 - 1)e$ and makes those roots d-fixed. The local Frobenius $\widehat{\sigma}$, chosen as in E.55, fixes these roots. Therefore

$$\begin{aligned} \kappa(\widehat{\sigma}) &= -(q-1)\kappa(g), \\ \kappa(g^{-1}\widehat{\sigma}g) &= \chi_3(g)^{-1}\big(\kappa(\widehat{\sigma}) + (q-1)\kappa(g)\big) = 0. \end{aligned} \tag{E104.13}$$

Conjugate the entire source by g^{-1} and raise its tame inertia to the unit power $\chi_3(g)$. These operations have the stated inverse conjugation and inverse unit power, preserve the tame relation, and give the first two columns of (E104.4) for the original roots and the original c.

Lift the original τ to this arithmetic source. Multiply its lift on the left by the negative of its $\kappa_3(\ell)$-power of t, obtaining γ. Its image in the unramified quotient is still τ. The actual based geometric action is unramified at ℓ by E.34 and E.78, so $\Phi_\gamma = T$. Set $\rho = \gamma c \gamma^{-1} c$. The cocycle identities give its last column in (E104.4), the exact relation $c\rho c = \rho^{-1}$ and (E104.5). The two Frattini characters of σ, ρ in the free rank-two unramified group P have matrix

$$\begin{pmatrix} (q-1)/3 & 0 \\ 0 & 2 \end{pmatrix} \pmod 3. \tag{E104.14}$$

Its determinant is a unit because $v_3(q-1) = 1$. Thus these are an actual free basis of P. The inverse basis map is the compatible inverse on finite characteristic quotients, by the finite-index Hopfian argument in E.100.

The decomposition factor and this particular ρ give (E104.3), including both inverse maps. Indeed the kernel of its map to P is the continuous free product of the actual tame inertia groups over $P/\langle\sigma\rangle$: on the arithmetic side this is the opened Wingberg theorem [Win07, Lemma1.1, Theorem2.4 and their proofs]; on the displayed source it is its pro-three coset tree. The generator inclusions give the identity on P and the same isomorphism on every inertia factor. The universal properties give inverse maps on the kernel, hence on the whole source. Choosing the same generators in the original arithmetic group gives the reverse continuous section of its projection. Their conjugation relations make this map c-equivariant. This proves the required source maps for every $\ell \in \mathscr{I}$, with all original markings retained.

The actual split-prime sources. For $\ell \in \mathscr{P}$, E.56 constructs actual Frobenii with cyclotomic character ℓ and Kummer values of three equal to $1, -1$. Its explicit conjugator E56.12 changes the original cubic value $t_0 \equiv 1 \pmod 3$ to exactly one; it preserves the original complex conjugation and root systems. The two unramified Frattini columns have determinant $-2(\ell-1)/3$ modulo three, hence give a free basis of the same P. The two actual decomposition groups then have the free-product presentation (E104.7). For clarity, its kernel comparison uses the two induced inertia bundles with relations $(p\sigma^z, t^u) \sim (p, t^{\ell^z u})$ and the conjugate relations. The coset-tree inclusions and the actual arithmetic inertia inclusions are inverse on those bundles and on P. This is the precise E.56 source isomorphism and its reciprocal arithmetic section.

For both families, the genuine action on N factors through the original unramified $\Gamma = P \rtimes \langle c \rangle$. Over K its image is pro-three. A cocycle paired with this action is therefore a homomorphism to a pro-three semidirect product; it factors through the displayed relative pro-three source. Inflation along the actual arithmetic quotient is the inverse construction. A geometric conjugacy after inflation already conjugates the values on its onto quotient, so inflation is injective on the stated classes.

The exact section equations and their topology. The order-two equation for a cocycle value v at c is $vJ(v) = 1$. Conjugation by $v^{-1/2}$ makes that value one; all remaining conjugators are H^J. For the inert family, write its other values as (ν, u, x) at (t, σ, ρ). The complete normalized space is the actual quotient

$$\left\{(\nu, u, x) : J(\nu) = \nu^{-1},\ J(u) = u,\ \mathcal{L}(x) = x^{-1},\ uF_\ell(\nu)u^{-1} = \nu^q\right\}/H^J. \tag{E104.15}$$

The action is $(\nu, u, x) \mapsto (h\nu h^{-1}, huF_\ell(h)^{-1}, hx\Phi_\rho(h)^{-1})$. For the split family the complete normalized space is

$$\{(\nu, u) : uT_\ell(\nu)u^{-1} = \nu^\ell\}/H^J, \quad h(\nu, u) = (h\nu h^{-1}, huT_\ell(h)^{-1}). \tag{E104.16}$$

The primed values are forced by c, and the second tame equation is exactly the J-image of the first. The free-product universal property proves sufficiency as well as necessity of these equations. Their solution sets are closed in compact powers of H; the compact geometric actions give Hausdorff quotient spaces. These are the topologies used throughout the statement.

Every inert ramified class. E.61 and E.66 classify the complete ramified full and metabelian local sections at every good prime away from three. Their maps apply to the actual local factor D_q, with all nonunit exponents included. The nine local labels are $(D, r, \bar{b})$, with $r \neq 0$ and $\bar{b} \in \mathbb{Z}_3/(q-1)\mathbb{Z}_3 = \mathbb{F}_3$. Choose conjugate paths for each nonrational cusp pair, and the J-fixed paths of E.100 at the three rational cusps. The actual involution on these packets is $(D, r, \bar{b}) \mapsto (\bar{D}, r, -\bar{b})$: its source inertia inversion cancels its peripheral inversion, whereas σ is fixed and the peripheral factor in its value is inverted. The fixed labels are exactly O, A_0, B_0 and $\bar{b} = 0$. This derives the descent labels from the actual source equations.

The rational path torsors factor through Γ by the centre-free endpoint argument of E.100. Thus their σ-values are precisely $u_{i,\ell}$, with $J(u_{i,\ell}) = u_{i,\ell}$ and $u_{i,\ell}F_\ell(d_i)u_{i,\ell}^{-1} = d_i^q$. The local class has an aligning conjugator to $(d_i^r, u_{i,\ell})$. Its ambiguity centralizes d_i^r, hence equals d_i^k, by the exact full and metabelian centralizer maps of E.61 and E.66. Preserving the Frobenius value forces $k = qk$, so $k = 0$. Uniqueness also makes the conjugator J-fixed, since its J-image aligns the same two normalized sections. The remaining free value is exactly $x \in \mathrm{Anti}_{\mathcal{L}}(H)$. The literal conjugacy $\mathcal{L} = TJT^{-1}$ gives $x = T(y)$ with $y \in \mathrm{Anti}_J(H)$. This proves (E104.6) and its inverse, including triviality of the full geometric stabilizer.

Every split ramified class. Apply the same complete local classification to D_ℓ. For each cusp its whole path torsor is unramified outside three: its endpoint is obtained from the base tangent by a coordinate permutation and multiplication by third roots of unity over K. The centre-free endpoint argument therefore gives the actual value $z_{D,\ell}$ in (E104.8). The local inertia and Frobenius values are exactly the first line of (E104.10), initially with $b \in \mathbb{Z}_3$. For fixed D, r, their entire right ambiguity is

$$(g, b) \longmapsto (g\delta_D^k, b + (\ell - 1)k). \tag{E104.17}$$

Substitution using (E104.8) proves this equality of both actual cocycle values. As $(\ell - 1)\mathbb{Z}_3 = 3\mathbb{Z}_3$, there is a unique such k making b its representative in $\{0, 1, 2\}$. The residual global conjugations are exactly left multiplication by H^J. The inertia abelianization determines D, r; the centralizer proof and (E104.17) determine the rest uniquely. This proves the maps (E104.9)–(E104.10). A stabilizer would have the same exponent equation $k = \ell k$, so is trivial.

The quotient $H^J\backslash H$ has the actual inverse coordinates $[g] \mapsto g^{-1}J(g)$ and $x \mapsto [x^{-1/2}]$ from E98.10. Its Hall coordinates and those of the inert anti set are the integral ones of E.100. On each shell $v_3(r) = e$ the parameter maps are continuous bijections from compact spaces to Hausdorff spaces and hence homeomorphisms. For fixed remaining parameters, $r \to 0$ in (E104.6) or (E104.10) gives their actual zero-inertia row. Thus these coordinates keep the boundary topology of (E104.15)–(E104.16).

All comparison fibres, including zero inertia. Apply $N \to M$ to each actual cocycle value. For the inert family the compatible J-adapted Hall and Chen coordinates of E.100 delete exactly the non-Chen negative coordinates of y. This gives (E104.11) with both inverse maps. For the split family the map is $(D, r, b, [g]) \mapsto (D, r, b, [\bar{g}])$. Every fixed element has an actual fixed lift: from h_0 lifting $\bar{h} \in M^J$, put $k = h_0^{-1} J(h_0) \in N''$ and use $h_0 k^{1/2}$. Above a chosen lift g_0 of $\bar{g}$, the full fibre map is

$$(N'')^J \backslash N'' \longrightarrow \mathrm{Fib}([\bar{g}]), \quad [k] \longmapsto [k g_0]. \tag{E104.18}$$

Removing the lifted fixed factor gives its inverse. The integral kernel anti coordinates of E98.28–31 then give exactly (E104.11), with all nonlinear central corrections retained.

With inertia value one, the cocycle kills the entire closed normal inertia subgroup, because the based action of that subgroup is trivial. It therefore factors through the actual quotient Γ, and every cocycle on Γ inflates. The new unramified free basis in either family is transported to the original basis of E.41 by the actual inverse free-basis maps above. Thus its section quotient and every full-to-metabelian fibre are exactly those of E100.22–26, with the original T in (E104.12). This comparison does not replace T by the prime-dependent F_ℓ or T_ℓ. E.102 evaluates its first kernel factor by actual integral Smith maps. Finally every ramified full class has nonzero peripheral abelianization $r[d_D]$. The actual quotient to M preserves it, so every fibre over zero inertia has already been included.

The retained program checks every prime through 2000 in the two sets (E104.1), the complete two-by-two Frattini inverses, original-root conjugation identities, all twenty-seven inert descent labels, and the finite-precision forward and inverse exponent maps in (E104.17). It independently evaluates the Hall character ranks through degree twenty and compares the first twelve rows with the frozen actual E.100 certificate. These checks support the displayed generator and coordinate formulas; the source isomorphisms, complete local classification and all-degree inverse maps above prove the statement for the full prime sets (E104.1). □

The actual quartic origin at class two and its complete toric image

E.105. A class-two lift of the quartic coset origin, with an explicit continuous toric criterion [Proved]

Retain the plus quartic, its twelve cusps, the field $F = \mathbb{Q}(\zeta_8)$, and every actual coefficient map of E.103. Let R be $\mathbb{Z}_\ell$ for an odd prime, the product of these rings over all odd primes, or $\mathbb{Q}_\ell$ for any prime. Write $A = A_R$ and $W = T_R T_D \subset A$. The geometric class-two group and its central kernel are

$$N_{R,2}, \qquad B_R = \gamma_2 N_R / \gamma_3 N_R = \Lambda_R^2 A_R, \qquad \operatorname{rank} A_R = 17, \quad \operatorname{rank} B_R = 136. \tag{E105.1}$$

For integral coefficients this is the indicated geometric pro-ℓ, or product of pro-ℓ, quotient of $\pi_1(U)$. For rational coefficients it is the geometric two-step Malcev pushout specified in the proof. Let $\mathcal{E}_{R,2}$ be the resulting actual arithmetic extension.

The section s_R constructed in E.103 lifts to a continuous section $S_R : G_{\mathbb{Q}} \to \mathcal{E}_{R,2}$. This statement applies to the actual choices of coset lifts and unit roots in E.103: (E105.18) transports the construction to each such choice. No new abelian origin is substituted.

Let $\iota : R(1)^2 \to A_R$ be the actual coefficient section in (E103.16), and write α, β for continuous Kummer classes. The complete image of the class-two section space under the coordinate units $(x/z, y/z)$ is

$$\left\{ (\alpha, \beta) \in H^1_{\mathrm{cts}}(\mathbb{Q}, R(1))^2 : \alpha \cup \beta = 0 \text{ in } H^2_{\mathrm{cts}}(\mathbb{Q}, R(2)) \right\}. \tag{E105.2}$$

Every pair in this set is attained above the specific E.103 abelian section $\iota(\alpha,\beta)s_R$. The following formulas evaluate the condition in (E105.2) on every completed input, including unrestricted ramification.

First let $R=\mathbb{Z}_\ell$, with ℓ odd. Write

$$\alpha=\sum_p a_p\kappa_p,\qquad \beta=\sum_p b_p\kappa_p,\quad a_p,b_p\in\mathbb{Z}_\ell,\quad a_p,b_p\longrightarrow 0, \tag{E105.3}$$

where p runs over the rational primes and the sums converge uniformly as cocycles. For $q\neq\ell$ put $e_q=v_\ell(q-1)$. Choose a primitive ℓ^{e_q}th root $\eta_q\in\mathbb{F}_q^*$, and, for $p\neq q$, define $r_{q,p}\in\{0,\dots,\ell^{e_q}-1\}$ by

$$\overline{p}^{(q-1)/\ell^{e_q}}=\eta_q^{r_{q,p}}.\quad A_q=\sum_{p\neq q}a_pr_{q,p},\quad B_q=\sum_{p\neq q}b_pr_{q,p},\quad c_q=-\frac{A_qb_q-B_qa_q}{q-1}. \tag{E105.4}$$

When $e_q=0$, all $r_{q,p}$ are zero. Then (E105.2) is exactly the condition

$$c_q\in\mathbb{Z}_\ell\quad(q\neq\ell),\qquad c_q\longrightarrow 0. \tag{E105.5}$$

For the product of all odd coefficient rings apply (E105.5) in every component, with the uniform congruence topology of (E103.5). For $R=\mathbb{Q}_\ell$, including $\ell=2$, use $a_p,b_p\in\mathbb{Q}_\ell$ tending to zero and the same formula (E105.4); the exact condition is

$$c_q\longrightarrow 0\text{ in }\mathbb{Q}_\ell. \tag{E105.6}$$

This includes a common bound on the denominators of the c_q. The conditions are independent of the specified roots and residue generators through the explicit changes in the proof.

For rational numbers $a,b\in\mathbb{Q}^*$ and odd integral coefficients, (E105.5) is the finite test that, at every prime q, the ℓ-primary component of

$$(-1)^{v_q(a)v_q(b)}\frac{a^{v_q(b)}}{b^{v_q(a)}}\pmod q\quad\text{in }\mathbb{F}_q^* \tag{E105.7}$$

is trivial. Only the union of the numerator and denominator prime supports of a and b needs to be checked. For the product of all odd coefficients the order of every residue in (E105.7) must be a power of two. Every rational-number pair passes the rational-coefficient test (E105.6). In contrast, (E105.29) below is an actual completed pair that passes every finite-coefficient and every local test and fails (E105.5) and (E105.6). The pair $(a,b)=(2,7)$ fails the integral test at $\ell=3,q=7$.

Over each specific liftable abelian section, the entire geometric-conjugacy fibre is the torsor under $H^1_{\mathrm{cts}}(\mathbb{Q},B_R)$ with the literal action $(z,S)\mapsto(g\mapsto z(g)S(g))$. Its coefficient maps include the following actual direct sum:

$$\Lambda^2A_R=\Lambda^2W\ \oplus\ (W\otimes(T_RE_+)^3)\ \oplus\ R(1)^6\ \oplus\left(\operatorname{Sym}^2T_RE_+\right)^3. \tag{E105.8}$$

The ranks are $55,66,6,9$. The theorem evaluates the origin and the complete toric image; it retains the further obstruction for the other abelian preimages in (E103.3), and the unevaluated cohomology factors in (E105.8). The full arbitrary-input quartic class-two classification in E.23 therefore remains active. The source map to the integral two-primary abelian quotient still carries E.18's obstruction; no section of the full profinite quartic extension is inferred.

Proof. Literal cusps, frames, and fourth-division paths. Put $\zeta=\zeta_8$, $\alpha_i=\zeta^{1+2i}$, and use

d	point	s_d	L_d
A_i	$[0:\alpha_i:1]$	x/z	$y-\alpha_iz$
B_i	$[\alpha_i:0:1]$	y/z	$x-\alpha_iz$
C_i	$[\alpha_i:1:0]$	z/y	$x-\alpha_iy$

$$O=B_0. \tag{E105.9}$$

All tangents have value one. In each displayed affine chart the nonzero coordinate is $\alpha_i(1+s_d^4)^{1/4}$, where the fourth root has constant coefficient one. Consequently the leading value of L_d at d is $\alpha_i/4$, in order four. Substituting $L_d = 0$ in the quartic equation leaves the fourth power of the omitted coordinate, proving $\operatorname{div}(L_d/L_O) = 4d - 4O$ including multiplicities.

Let z_d be the framed point $\mathcal{O}(d-O)$ of $J_{C,D}$, with vector frames s_d^{-1} at d, s_O at O, and the canonical section at the other cusps. Write $f_d = L_d/L_O$. Multiplication by f_d trivializes $4z_d$ and gives the actual torus coordinates

$$t_{q,d} = \frac{\operatorname{lead}_q(f_d)}{\operatorname{lead}_O(f_d)}, \qquad q, d \neq O. \tag{E105.10}$$

Here leading values use the exact parameters in (E105.9), with the appropriate zero or pole order removed. The program evaluates all 121 entries in $\mathbb{Q}[\zeta]/(\zeta^4+1)$ and retains their four rational coefficients and their exact root phases: $t_{q,d} = \zeta^{K_{qd}}(\sqrt{2})^{N_{qd}}$. In the order

$$A_0, A_1, A_2, A_3, B_1, B_2, B_3, C_0, C_1, C_2, C_3 \tag{E105.11}$$

the integer matrix N is

$$\begin{pmatrix}
-8 & -3 & -2 & -3 & -5 & -6 & -5 & -4 & -4 & -4 & -4 \\
-3 & -8 & -3 & -2 & -5 & -6 & -5 & -4 & -4 & -4 & -4 \\
-2 & -3 & -8 & -3 & -5 & -6 & -5 & -4 & -4 & -4 & -4 \\
-3 & -2 & -3 & -8 & -5 & -6 & -5 & -4 & -4 & -4 & -4 \\
-5 & -5 & -5 & -5 & -10 & -6 & -4 & -5 & -5 & -5 & -5 \\
-6 & -6 & -6 & -6 & -6 & -12 & -6 & -6 & -6 & -6 & -6 \\
-5 & -5 & -5 & -5 & -4 & -6 & -10 & -5 & -5 & -5 & -5 \\
-4 & -4 & -4 & -4 & -5 & -6 & -5 & -8 & -3 & -2 & -3 \\
-4 & -4 & -4 & -4 & -5 & -6 & -5 & -3 & -8 & -3 & -2 \\
-4 & -4 & -4 & -4 & -5 & -6 & -5 & -2 & -3 & -8 & -3 \\
-4 & -4 & -4 & -4 & -5 & -6 & -5 & -3 & -2 & -3 & -8
\end{pmatrix}. \tag{E105.12}$$

The retained $K_{qd} \in \{0, \dots, 7\}$, rather than a discarded root-of-unity factor, is used below for $\mathbb{Q}_2$.

First work with odd coefficients and put $H = G_F$. Choose coherent roots of two. Coherent odd roots of $\sqrt{2}$ can and will be chosen to square to them: the equation $2a + mb = 1$, for odd m, gives the unique compatible choice at level m. Thus $\kappa_{\sqrt{2}} = \kappa_2/2$ literally. All coherent odd roots of μ_8 lie in $\mu_8 \subset F$, and their Kummer cocycles on H vanish.

Here is the exact fourth-division construction of the cusp path class. Choose an m-division v_m of the torus point $4z_d = t_d$ using the roots just specified. For integers a, b with $4a + mb = 1$ set $u_m = av_m + bz_d$ in the generalized Jacobian. Then $mu_m = z_d$ and $4u_m = v_m$. These points are compatible: the two required equations determine u_m uniquely, since multiplication by four is an automorphism of the m-torsion. Their proper images are $p(u_m) = b[d-O]$, which are H-fixed four-torsion points; thus their proper Kummer cocycle is literally zero. They prove the exact cocycle identity, in the fixed splitting (E103.16),

$$\kappa(z_d)(g) = v_d\kappa_2(g), \qquad v_d = \tfrac{1}{8}N_{\bullet d} \in W, \qquad p_*\kappa(z_d) = 0. \tag{E105.13}$$

In particular the vanishing compact coordinate follows from the actual division construction. The fact that $d - O$ is killed by four alone would specify only its cohomology class.

The framed Abel–Jacobi comparison has the same explicit vector-frame convention as (E24.3): for a function f whose divisor is supported at the cusps, its character is its value on a divisor times the product of the frame ratios with the specified divisor exponents. Applied to f_q

this identifies four times the qth torus coordinate; it is zero on $j(J)$ because $\mathrm{Hom}(J,\mathbb{G}_m)=0$. Together with the proper Abel–Jacobi map these characters detect A_R, by the actual inverse (E103.16). Regularizing at both tangents yields (E105.10), with the following exact vector-frame correction. Put ϵ_d equal to $+1$ at d and -1 at each other non-O cusp. For a unit with divisor $\sum m_q q$ its character on this tuple is $(-1)^{m_d+m_O}$, since $\sum m_q=0$. This is precisely the two endpoint sign from replacing the local difference $s-t$ by $t-s$ in regularizing the divisor frames, as in the framed comparison of E.37. Thus the tangential class is $\kappa(z_d\epsilon_d)$; the coherent odd divisions of ϵ_d are H-fixed. This proves that (E105.13) is the actual abelian path cocycle with those tangent and root choices. It also proves its sign without identifying an unframed divisor with a tangential path.

Choose a full return path p_d from the tangent at d to the tangent at O lifting this abelian path. The map from the full path torsor to the abelian path torsor is onto; starting with any path, multiply it by a lift of the required abelian difference. Let $d_d=p_d t_d^{\mathrm{per}} p_d^{-1}$, where t_d^{per} is the positive local peripheral generator. The actual conjugator is $u_d(g)=p_d g(p_d)^{-1}$, so $u_d(g)g(d_d)u_d(g)^{-1}=d_d^{\chi(g)}$ and its abelian coordinate is the positive cocycle in (E105.13). These eleven peripherals extend to a free pro-ℓ basis. Choose the six remaining abelian basis vectors in the specific compact summand (E103.16) and lift them. The Frattini criterion gives a free basis, since the resulting abelian basis change is invertible. This is an actual free-group basis morphism, retaining the peripheral paths.

The class-two action needed for the origin. For odd coefficients use BCH coordinates $N_{R,2}=A\oplus\Lambda^2A$ with multiplication

$$(x,z)(y,w)=(x+y,\,z+w+\tfrac12 x\wedge y). \tag{E105.14}$$

The conversion from ordered Hall coordinates is $(a_i;b_{ij})\mapsto(\sum a_ie_i,\sum b_{ij}e_i\wedge e_j+\frac12\sum_{i<j}a_ia_je_i\wedge e_j)$. This gives its integral inverse and proves the claimed class-two group identification. Let $a(g)=\kappa_2(g)$ and define $L:W\to\Lambda^2W$ by $L(e_d)=e_d\wedge v_d$. Expanding the actual peripheral conjugation gives, on $W\oplus\Lambda^2W$,

$$g(x,z)=(\chi_g x,\chi_g^2 z+\chi_g a(g)L(x)). \tag{E105.15}$$

Indeed $u_d^{-1}d_d^{\chi_g}u_d$ has central term $\chi_g a(g)e_d\wedge v_d$. The central coordinate of u_d cancels. Unknown action on the six other free generators does not enter (E105.15); the formula proves that this displayed subgroup is H-stable.

Choose the E.103 coset lifts to be the return paths from the four conjugates B_i of O, with the abelian normalization just proved. The averaged abelian origin restricts to H as $a(g)v$ relative to s_H, where

$$v=\tfrac14(v_{B_1}+v_{B_2}+v_{B_3})=\left(-\tfrac12,-\tfrac12,-\tfrac12,-\tfrac12,-\tfrac58,-\tfrac34,-\tfrac58,-\tfrac12,-\tfrac12,-\tfrac12,-\tfrac12\right). \tag{E105.16}$$

To check the sign, its ith conjugate cusp section is $u_i(h)s_H(h)$. Thus the word $b_i(s_H(h))$ in (E103.8) is $-u_i(h)$ in A, and (E103.9) adds their average. Both coordinate-unit rows kill v. The tangential values at O are ζ_8 and the unit leading coefficient one; their odd Kummer cocycles are zero on H. Hence $r_0|_H=0$. A global $R(1)^2$ cocycle vanishing on H is zero: normality gives $hc(g)=c(g)$ for every $h\in H$, while $R(1)^H=0$. Consequently the toric correction in (E103.10) vanishes for this normalization.

There is now a literal class-two H-section above this abelian restriction. Its geometric cocycle relative to the original tangent section is

$$c_v(g)=(a(g)v,\ \tfrac12 a(g)^2L(v)). \tag{E105.17}$$

Substitution in (E105.14)–(E105.15) and $a(gh)=a(g)+\chi_g a(h)$ proves $c_v(gh)=c_v(g)g(c_v(h))$: the wedge of v with itself is zero and the cross term is $\chi_g a(g)a(h)L(v)$.

Pull back $\mathcal{E}_{R,2} \to \mathcal{E}_R$ along the *fixed* abelian section s_R. Its kernel is the abelian module B_R; the geometric abelian part of an arithmetic lift acts trivially on this central kernel. Its H-section is (E105.17). Apply the literal formula (E103.8)–(E103.9), now to this abelian-kernel extension, using its four actual cosets. It produces a continuous global section of the pullback and therefore the required S_R. This is where division by four is used a second time. It averages an extension with abelian kernel B_R, not four sections of a nonabelian geometric extension.

Here is the promised comparison to every originally chosen E.103 origin. If $e'_i = a_i e_i$ are the aligned coset lifts and $q = \frac{1}{4}\sum_i a_i$, direct use of (E103.8) gives

$$B'(e) = B(e) + g(4q) - 4q, \qquad s'_0 = \mathrm{Int}(q)s_0, \qquad s'_R = \mathrm{Int}(q - \iota r(q))s_R. \tag{E105.18}$$

Lift $q - \iota r(q)$ to $N_{R,2}$ and conjugate the constructed class-two section by its inverse. Its abelian image is exactly the old s_R. Changing unit roots by the cocycle coboundary dk changes the toric-zero origin by $\mathrm{Int}(\iota k)$, with the same lift rule. These formulas retain all old choices, even when their numerical path coordinates were not separately assigned.

The rational two-primary construction. For any ℓ, the class-two Malcev group is the BCH group of $A_{\mathbb{Q}_\ell} \oplus \Lambda^2 A_{\mathbb{Q}_\ell}$. The original $N_{\ell,2}$ maps into it by the displayed Hall/BCH conversion. Its arithmetic automorphisms extend uniquely to this group. The actual arithmetic pushout is the quotient of $N_{\mathbb{Q}_\ell,2} \rtimes \mathcal{E}_{\ell,2}$ by the diagonal normal subgroup $\{(n^{-1}, n) : n \in N_{\ell,2}\}$. The diagonal subgroup is closed, and its intersection with the first factor is trivial; this gives the specified extension and the literal map from the original one.

Only the root normalization needs a new check at two. On H, choose the coherent roots $\zeta_{8\cdot 2^n}$ of ζ_8. Then $\kappa(\zeta_8^k) = k(\chi - 1)/8$. Choose compatible roots of $\sqrt{2}$ squaring to the chosen roots of two. If u_n are any compatible 2^n-division points of z_d and v_n are the explicit torus roots of $4z_d$, then $b_d = (4u_n - v_n)_n \in T_2 J_{C,D}$ is an actual Tate vector, and

$$\kappa(z_d) = v_d\kappa_2 + d\left(\tfrac{1}{4}b_d + \tfrac{1}{32}K_{\bullet d}\right) \quad \text{in } A_{\mathbb{Q}_2}. \tag{E105.19}$$

The compact coordinate in the first term is zero in the fixed $j/4$ splitting. Multiplying the return path by the displayed vector subtracts precisely this coboundary, since $p \mapsto mp$ changes $pg(p)^{-1}$ by $m - gm$ in abelian coordinates. For the actual tangential class add the endpoint correction just specified: let $\sigma_d \in W$ have entry zero at d and entry one at every other non-O cusp. Coherent roots of -1 give $\kappa(\epsilon_d) = d(\sigma_d/2)$. Hence the literal path shift is $b_d/4 + K_{\bullet d}/32 + \sigma_d/2$. The last term is retained separately because $\epsilon_d^4 = 1$ and it is not detected by the matrix of $4z_d$. This is an actual rational class-two inner transport; it need not be an integral pro-two path.

At O the first coordinate unit has Kummer cocycle $(\chi - 1)/8$ and the second has zero. Thus the aligned E.103 r_0 is exactly $d(1/8, 0)$ globally: its difference has zero restriction to H and $\mathbb{Q}_2(1)^H = 0$. Put $h_0 = \iota(1/8, 0)$ and transport both the tangent section and the peripheral coordinates by $\mathrm{Int}(h_0)$. In these transported coordinates (E105.15)–(E105.17) hold literally for the actual toric-zero origin. All divisions by two and four exist, so the same pulled-back central average proves the rational assertion. Equation (E105.18) gives the inverse transport to every specified old marking. For odd rational coefficients the earlier odd integral comparison followed by this pushout gives the same construction.

Sufficiency and necessity for the complete toric image. The action relative to the new origin, restricted to H, replaces L by

$$L_v(w) = L(w) + v \wedge w. \tag{E105.20}$$

The further central averaging does not change this action: two lifts of the same abelian section differ centrally. Let $w_A = \sum_i e_{A_i}$ and $w_C = \sum_i e_{C_i}$. In W the remaining orbit sum is $w_B = -w_A - w_C$. The exact columns of (E105.12) give

$$L_v(w_A) = L_v(w_C) = 0, \qquad \iota(u, v) = \tfrac{1}{4}(uw_A + vw_B). \tag{E105.21}$$

The program retains all 55×11 entries of L, L_v, the two columns of ι, their actual retraction, and the literal zero matrix in (E105.21).

Suppose $\alpha \cup \beta = 0$ and choose a continuous cochain k with $dk = \alpha \cup \beta$, using $dk(g,h) = gk(h) - k(gh) + k(g)$. On H, put $b = \iota(\alpha, \beta)$ and $\omega = \iota(1,0) \wedge \iota(0,1)$. The explicit central cochain

$$z(g) = -\big(k(g) + \alpha(g)\beta(g)/2\big)\omega \tag{E105.22}$$

has differential equal to minus the BCH defect $\frac{1}{2}b(g) \wedge gb(h)$. Indeed $d(\alpha\beta) = -(\alpha \cup \beta + \beta \cup \alpha)$, and (E105.21) removes the additional action term. Thus $(b, z)S_R|_H$ is a genuine H-section above the desired abelian input. Pull back over that input and apply the abelian-kernel four-coset formula once more. This proves global sufficiency and gives a literal construction of its lift.

For necessity use the actual morphism $U \to \mathbb{P}^1 - \{0, 1, \infty\}$, $t = -(x/z)^4$, $1 - t = -(y/z)^4$. It induces the class-two map and multiplication by four on both Kummer coordinates. The coherent roots of -1 give zero odd Kummer cocycle; over $\mathbb{Q}_2$ their cocycle is the coboundary $d(1/2)$, transported by the corresponding actual inner change in the target BCH group. The rational tangential paths at zero and one have abelian Kummer class zero. The standard based Galois formula $x \mapsto x^\chi$, $y \mapsto f_g^{-1} y^\chi f_g$ has f_g in the commutator subgroup, so its class-two action is pure cyclotomic. The explicit target obstruction is therefore $16\alpha \cup \beta$; equivalently this follows by multiplying its rank-two BCH law. See also the actual formula and proof in [Wic12b, Proposition7, §12.4.5]. Since sixteen is invertible in every coefficient ring in this theorem, a class-two section forces $\alpha \cup \beta = 0$. This proves both image inclusions.

More generally the actual defect of a W-valued cocycle b on H, relative to S_R, is

$$\Omega_b(g,h) = \chi_g \kappa_2(g) L_v(b(h)) + \tfrac{1}{2} b(g) \wedge \chi_g b(h). \tag{E105.23}$$

This specifies the action and cup terms for the other toric directions, without replacing the handle action by the associated-graded representation.

The continuous cup group, for every odd prime. We give the source calculation and its support maps, since taking only finite or local cup tests would lose part of (E105.5). Put $G_S = G_{\mathbb{Q},\{\ell,\infty\}}$ and $T_n = \mathbb{Z}/\ell^n(2)$. For $\ell = 3$, the exact proof (E.32g) gives $H^2(G_S, T_n) = H^3(G_S, T_n) = 0$. For $\ell \geq 5$ let $K = \mathbb{Q}(\mu_\ell)$ and $\Delta = \mathrm{Gal}(K/\mathbb{Q})$, with character ω on μ_ℓ. Restriction and the Δ-invariants functor are exact here because $|\Delta| = \ell - 1$ is invertible. The literal coefficient identity $\mathbb{F}_\ell(2) = \mu_\ell \otimes \mathbb{F}_\ell(1)$ and the Kummer sequence show that $H^1(G_S, \mathbb{F}_\ell(2))$ is the ω^{-1} part of the S-unit and ℓ-torsion class groups of K. The unique prime over ℓ is principal, so inverting it does not change the class group.

The odd-character part of the S-unit group modulo ℓ comes only from μ_ℓ, of character ω. For completeness, an S-unit u has $u/\bar{u}$ integral at every finite prime (the unique removed prime is fixed by conjugation), and all its conjugates have absolute value one. It is a root of unity. Hence conjugation is trivial on the torsion-free unit quotient, and the stated odd part follows. Since $\omega^{-1} \neq \omega$ for $\ell \geq 5$, that S-unit component vanishes. Ribet's original Main Theorem(1.1), with $k = 2$, gives the vanishing of the class component because $B_2 = 1/6$; the used direction is Herbrand's theorem [Rib76, p.151]. The theorem is stated for the class group modulo ℓ; its vanishing is equivalent to the vanishing of the ℓ-torsion component, since the Δ idempotents act on the finite ℓ-primary class group and Nakayama applies to each summand. We have proved $H^1(G_S, \mathbb{F}_\ell(2)) = 0$; $H^0 = 0$ as well. The real factor in the finite global Euler characteristic is one, because this is an even character. Milne I.5.1 gives $H^2 = 0$. Devissage and the odd-coefficient real-place formula I.4.10(c), with finiteness I.4.15, now give

$$H^2(G_S, T_n) = H^3(G_S, T_n) = 0, \qquad H^1(G_S, T_n) \text{ finite.} \tag{E105.24}$$

Apply the finite localization triangle on $\operatorname{Spec} \mathbb{Z}[1/\ell]$ and remove a finite set of primes $q \neq \ell$. The support complex at q is $R\Gamma(\mathbb{F}_q, \mathbb{Z}/\ell^n(1))[-1]$: tame inertia is $\mathbb{Z}_\ell(1)$, its wild subgroup

has prime-to-ℓ cohomology zero, and purity gives this specified twist and shift. First take the filtered direct limit over the removed sets. Finite continuous cochains factor through finite extensions, so the generic-field term is exactly $R\Gamma(G_{\mathbb{Q}}, T_n)$ and the support term is a direct sum. Then take the derived inverse limit in n, using [Jan88, Proposition1.6, Theorem2.2]. The two terms in each residue complex are

$$[\,\mathbb{Z}/\ell^n \xrightarrow{q-1} \mathbb{Z}/\ell^n\,] \quad \text{in degrees } 0, 1, \tag{E105.25}$$

with arithmetic Frobenius. The direct sums have surjective coefficient transitions in both terms. Their derived inverse limit is the literal complex $[\mathcal{R} \xrightarrow{d} \mathcal{R}]$, where $\mathcal{R}$ consists of the $\mathbb{Z}_\ell$ sequences tending to zero, indexed by $q \neq \ell$, and $d(z)_q = (q-1)z_q$. Equation (E105.24) and the finite degree-one groups kill the adjacent base terms. We obtain the actual module isomorphism

$$H^2_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_\ell(2)) \simeq \mathcal{R}/d\mathcal{R}. \tag{E105.26}$$

There is no closure in this denominator. The localization, purity and arithmetic-étale maps used here are Milne II.0.1(f), II.1.7(b), II.2.9; these are the exact same maps retained in E.32.

To identify the cup at chain level, choose at q tame inertia τ and arithmetic Frobenius ϕ with $\phi\tau\phi^{-1} = \tau^q$, $\kappa_q(\tau) = 1$, $\kappa_q(\phi) = 0$. The two Kummer cocycles on inertia are proportional. Their alternating BCH cup therefore has a canonical central-zero inertia splitting. Write a_q, b_q for the two inertia values and A'_q, B'_q for the Frobenius values. In the rank-two BCH pullback, with central inertia value w, its tame relation has central defect

$$q((q-1)w + A'_q b_q - B'_q a_q). \tag{E105.27}$$

The central Frobenius value cancels. The support map divides this defect by q, since $\mathrm{Hom}(\mathbb{Z}_\ell(1), T) = T(-1)$ compares its value at τ^q with its value at τ. Thus the cup maps to $[A'_q b_q - B'_q a_q]_q$ in (E105.26). These inertia splittings are unramified outside a finite set at each coefficient level. They give a representative in the direct sums before taking their derived limit.

Choose the place at q to send the fixed ℓ^{e_q}th root of unity to η_q. Finite-field Kummer theory then gives $\kappa_p(\phi) \equiv r_{q,p} \pmod{q-1}$. Write their difference as $(q-1)t_{q,p}$, with $t_{q,p} \in \mathbb{Z}_\ell$. The sums $\sum a_p t_{q,p}$ and $\sum b_p t_{q,p}$ converge and are uniformly bounded. Replacing A'_q, B'_q by A_q, B_q changes the representative by d of a restricted family: every term contains a_q or b_q. Changing global cocycles by $\alpha \mapsto \alpha + du$, $\beta \mapsto \beta + dv$ adds $(q-1)(ub_q - va_q)$, again the image of a restricted family. Replacing ϕ by $\tau^r\phi$ leaves $A'_q b_q - B'_q a_q$ unchanged. Changing η_q multiplies the coordinate by a unit and adds the preceding bounded-root correction. These formulas prove the claimed independence with the actual coboundary sign. The unique possible inertia correction is (E105.4). It belongs to $\mathcal{R}$ exactly under (E105.5), proving the odd integral criterion.

Rational coefficients, with the same support topology. For every prime ℓ, including two, Hain–Matsumoto Theorem B.1 gives $H^2_{\mathrm{cts}}(G_S, \mathbb{Q}_\ell(2)) = 0$; their Proposition B.2 and Lemma B.3 prove the finite coefficient comparison, and the proof on pp.38–39 includes the case $\ell = 2$ [HM03]. The degree-three base term also vanishes rationally: Milne I.4.10(c) identifies finite-coefficient degrees at least three with the real-place groups, which are zero for odd ℓ and killed by two for $\ell = 2$. The finite-S inverse systems are finite, so no additional first derived limit occurs there.

For the unrestricted source one must retain a separate cochain comparison. For every degree i, compactness of $G^i_{\mathbb{Q}}$ gives the isomorphism

$$C^i_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Z}_\ell(2)) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell \xrightarrow{\sim} C^i_{\mathrm{cts}}(G_{\mathbb{Q}}, \mathbb{Q}_\ell(2)). \tag{E105.28}$$

Every continuous cochain on the right has compact, hence bounded, image, so multiplication by one power of ℓ puts all its values in the integral lattice. The same maps commute with

differentials. Apply this scalar extension to the finite-level localization construction, after the derived inverse limit. The support complex becomes $[c_0(\mathbb{Q}_\ell) \xrightarrow{d} c_0(\mathbb{Q}_\ell)]$, and the rational base terms just computed vanish. This proves the rational version of (E105.26). The chain calculation (E105.27) remains literal; it proves (E105.6). A family tending to zero in $\mathbb{Q}_\ell$ is bounded, so the condition retains both a common denominator and convergence. For the product of odd rings, continuous cochains, their differentials and products are componentwise; the open-ideal topology in E.103 is exactly the resulting product of the uniform primewise topologies.

For rational-number inputs the expression $A_q b_q - B_q a_q$ is the discrete logarithm of (E105.7) in its ℓ-primary group. This is a direct finite-field calculation with the units obtained by removing the q-valuation. The sign has order two and therefore vanishes for odd ℓ. There is finite support, so integrality is the entire odd integral test and the rational test always holds. This also agrees with the actual Milnor-symbol map and tame-symbol computation in [Wic12b, Proposition26 and Remark28]; the present proof does not infer an inverse-limit injection from that finite-symbol statement.

To exhibit why convergence is necessary, use the actual increasing prime sequence constructed in E.32: $q_j \equiv 1 \pmod{3^j}$ and three is not a cube modulo q_j. Put $r_j = r_{q_j,3}$, which is a 3-adic unit, and set

$$a_3 = 1, \quad a_p = 0 \ (p \neq 3), \qquad b_{q_j} = (q_j - 1)/r_j, \quad b_p = 0 \text{ otherwise.} \tag{E105.29}$$

This is a genuine restricted pair because $v_3(b_{q_j}) \geq j$. Its support representative is $q_j - 1$ at every q_j, and its unique correction is -1 there. At each finite coefficient level a finitely supported primitive is obtained by putting -1 only where $v_3(q_j - 1)$ is smaller than that level. Every local reduction is also zero. The correction fails to tend to zero even after multiplication by any nonzero rational 3-adic scalar. Thus this pair has no integral or rational class-two quartic lift. Its abelian preimage exists by E.103, so the obstruction has the asserted actual source.

Central fibres and the retained further target. The E.103 splitting identifies $A = W \oplus V^3$, where $V = T_R E_+$ with its actual Weil pairing. Exterior square gives $\Lambda^2 W \oplus (W \otimes V^3) \oplus \Lambda^2(V^3)$. Each individual $\Lambda^2 V_i$ is $R(1)$. For each pair $i < j$, the two identical copies of V identify $V_i \otimes V_j$ with $\mathrm{Sym}^2 V \oplus \Lambda^2 V$ using the projectors $(1 \pm \mathrm{swap})/2$. These are integral for odd R and valid in every rational target. In a symplectic basis (e, f), the inverse column for the cross determinant is $(e_i \wedge f_j - f_i \wedge e_j)/2$; contraction sends it to one. These six contraction/section maps and their inverses are retained in the certificate. This proves (E105.8) with actual coefficient maps.

There are no invariant vectors in A. The toric part has none after restriction to H, where it is $R(1)^{11}$. For the proper part, the isogeny in E.103 and the finite group $E_+(\mathbb{Q})$ show that its integral Tate invariants vanish; rational invariants vanish after multiplying by a common denominator. Consequently the stabilizer of an abelian section under geometric conjugation is zero. The exact central-extension sequence then makes every nonempty fibre a torsor under the displayed $H^1(\mathbb{Q}, B_R)$, with the stated continuous cocycle action and its inverse given by the ratio of two lifts. To compare classes over an abelian conjugacy class, first use the unique abelian conjugator to align their actual abelian sections, and lift that vector by the BCH coordinates (E105.14). Its ambiguity is central and is exactly the usual coboundary in B_R. The alignment is continuous: for integral coefficients the injective coboundary map has compact domain, and for rational coefficients a finite collection of Galois elements gives an injective linear matrix with a continuous left inverse. Thus the fibre maps also retain the stated cochain-quotient topology. The actual maps in (E105.8), and the explicit toric defect (E105.23), specify what is evaluated here. The remaining handle-action and cohomology calculations for all abelian preimages are retained in E.23. □

The complete next ordinary class groups from the actual norm lattice

E.106. The ninth-symbol transfer, vanishing transvection, and all third-level class maps [Proved]

Retain the literal roots, prime labels and unit basis of E.76, E.86, E.91 and E.101:

$$K=\mathbb{Q}(\zeta),\quad \xi^3=\zeta,\quad L_d=K(\beta_d),\quad B_d=L_d(\xi),\qquad \beta_{d+1}^3=\beta_d,\quad \beta_1^3=17\zeta. \tag{E106.1}$$

Write $E_F=\mathcal{O}_F^\times$ and put a hat on a unit group for its completion for multiplication by 3^m. The first nine E.101 generators of E_{B_1} are $u_1=\xi,u_2,\dots,u_9$; their only three-primary relation is $u_1^9=1$. The complete ordinary norm image from B_2 has the specified coordinates

$$\begin{aligned} N_{B_2/B_1}\widehat{E}_{B_2}&=\left\{\prod_{i=1}^{9}u_i^{x_i}:x=Py\right\},\\ Py&=(3y_1+y_2+y_4,\ y_2,\ y_3,\ y_4,3y_5+2y_6,\ y_6,\ y_7,\ 3y_8,\ 3y_9)^t,\\ y&\in\mathbb{Z}_3^9/3\mathbb{Z}_3e_1. \end{aligned} \tag{E106.2}$$

For $v\in\widehat{E}_{B_2}$, take the actual y-coordinates of $N_{B_2/B_1}v$ in (E106.2). Its six arithmetic cubic local norm characters for B_3/B_2, ordered by the three primes over three and then the three over seventeen, are

$$\chi(v)=Dy\pmod 3,\qquad D=\begin{pmatrix}2&1&2&2&0&2&2&2&1\\2&1&2&0&0&2&0&0&0\\2&1&2&1&0&2&1&1&2\\1&2&1&0&0&0&2&2&2\\1&2&1&1&0&0&0&2&2\\1&2&1&2&0&0&1&2&2\end{pmatrix}. \tag{E106.3}$$

The arithmetic generator sends β_3 to $\zeta\beta_3$. The image of this actual character map has dimension four. Its kernel is the completed ordinary-unit subgroup of field norms from B_3. The certificate gives the full finite row and column transformations, their inverses, a four-column image basis, and its coordinate inverse.

The entire ordinary three-class groups are

$$\begin{aligned} \mathrm{Cl}(B_3)\{3\}\xrightarrow[\mathrm{Art}]{\sim}A_9=C_9^2\oplus C_3,&\qquad W=(g_2-1)\,\mathrm{Cl}(B_3)\{3\}=0,\\ \mathrm{Cl}(L_3)\{3\}=C_9a_3\oplus C_3b_3,&\qquad [\mathfrak{q}_{3,3}]=3a_3,\qquad \mathrm{ord}[\mathfrak{q}_{3,3}]=3,\\ N_{L_3/K}\widehat{\mathcal{O}_{L_3}[1/3]^\times}_3=\lambda^{3\mathbb{Z}_3},&\qquad a_3^{\mathrm{E69}}=1. \end{aligned} \tag{E106.4}$$

Here a_3,b_3 are the literal coherent ideal classes of E86.4; a_3^{E69} distinguishes the projection exponent from that ideal class. The first arrow is the existing restriction to the actual ninth-genus field. That field is therefore the full Hilbert three-class field of B_3.

Use (y,z,t) for the A_9 coordinates, (s,u,v) for the $C_9\oplus C_3^2$ coordinates of $\mathrm{Cl}(B_2)\{3\}$, and (a_d,b_d) for the specified L_d coordinates. All adjacent and horizontal ordinary class maps at this level are

extension	ideal norm	ideal extension
B_3/B_2	$\begin{pmatrix}2&3&6\\2&0&2\\0&1&0\end{pmatrix}$	$\begin{pmatrix}6&0&0\\0&0&3\\0&0&0\end{pmatrix}$
L_3/L_2	I_2	$3I_2$
B_3/L_3	$\begin{pmatrix}3&6&6\\1&0&0\end{pmatrix}$	$\begin{pmatrix}3&0\\3&3\\2&0\end{pmatrix}$.

(E106.5)

Every row is reduced modulo its target cyclic order. The vertical B norm kernel is $\langle(0,3,0)\rangle = C_3$. The horizontal extension kernel is precisely $\langle 3a_3\rangle = \langle[\mathfrak{q}_{3,3}]\rangle$. The actual ordinary unit norm is surjective:

$$N_{B_3/L_3} E_{B_3} = E_{L_3}. \tag{E106.6}$$

The six actual S-prime classes are now completely determined:

$$C_S = \begin{array}{c|cccccc} & Q_1 & Q_2 & Q_3 & T_1 & T_2 & T_3 \\ \hline C_9 & 6 & 6 & 6 & 6 & 3 & 0 \\ C_9 & 0 & 6 & 3 & 5 & 8 & 5 \\ C_3 & 2 & 2 & 2 & 1 & 1 & 1 \end{array}. \tag{E106.7}$$

They generate the order-81 subgroup $R_3 = 3C_9 \oplus C_9 \oplus C_3$. Its actual quotient map is $(y,z,t) \mapsto y \bmod 3$, in the coherent prime-73 S-basis. All three Q_i have order three, and their product is principal. The full principal S-divisor lattice has basis and index

$$\mathcal{L}_3 = H_S\mathbb{Z}^6, \qquad H_S = \begin{pmatrix} 3 & 1 & -3 & 2 & 2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -2 & 3 & -2 & -2 & 0 \\ 0 & 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad [\mathbb{Z}^6 : \mathcal{L}_3] = 81. \tag{E106.8}$$

For a divisor vector $(q_1,q_2,q_3,t_1,t_2,t_3)$ the inverse coordinates are

$$\left(\frac{q_1+q_2+q_3}{3},\ q_2,\ \frac{q_3+2q_2+2B+2C}{3},\ B,\ C,\ t_3\right), \quad B = \frac{t_1-t_3}{3}, \quad C = \frac{t_2-t_3}{3}. \tag{E106.9}$$

Thus an integral S-divisor is principal exactly when all these coordinates are integral. The same criterion holds after completion, with integrality in $\mathbb{Z}_3$.

There is an exact complete ordinary-unit norm test from B_3 to B_2. First apply (E106.3). On its zero kernel, for an ordinary unit v choose a field norm preimage $w \in B_3^\times$ and solve $\operatorname{div}(w) = (\sigma_2 - 1)\mathfrak{D}$. Write $\operatorname{Art}[\mathfrak{D}]_3 = (y,z,t)$ in (E106.4). Then

$$\begin{aligned} v \in N_{B_3/B_2} E_{B_3} \quad &\Longleftrightarrow \quad \chi(v) = 0 \text{ and } y(\operatorname{Art}[\mathfrak{D}]_3) = 0 \pmod 3, \\ E_{B_2}/N_{B_3/B_2}E_{B_3} \simeq C_3^5, \qquad &\ker(N : \widehat{E}_{B_3} \to \widehat{E}_{B_2}) \simeq C_3 \oplus \mathbb{Z}_3^{54}. \end{aligned} \tag{E106.10}$$

The second test is independent of both choices, and both directions are proved below. On completion it is the continuous extension of this finite quotient test. Each nonempty completed norm fiber is the coset of the actual norm kernel through any of its preimages. The corresponding L_3/L_2 ordinary norm quotient is C_3^3, with norm kernel $C_3 \oplus \mathbb{Z}_3^{18}$.

These conclusions evaluate the third finite projection and the E.91 transvection. The eventual exponent h of E.69 and the later ordinary prime orders are not assigned a value by this finite computation. The manuscript anchor is LM §51.16, folios733v–736v, scans172–178, through the actual positional norm maps of E.46, E.69, E.72 and E.88.

Proof. The local norm square and its precise exponent. At each of the six S-primes of B_1 there is one prime in each vertical upper field, with residue degree one. The local extension B_3/B_1 is cyclic of degree nine: $\mu_9 \subset B_1$, $\beta_3^9 = \beta_1$, and its generator g sends β_3 to $\xi\beta_3$. Its subgroup over B_2 is generated by $g^3 = \sigma_2$. The specified local Artin maps give the commutative square

$$\begin{array}{ccc} B_{2,w}^\times & \xrightarrow{\text{Art}} & \operatorname{Gal}(B_{3,W}/B_{2,w}) \\ \downarrow N & & \downarrow \text{inclusion} \\ B_{1,v}^\times & \xrightarrow{\text{Art}} & \operatorname{Gal}(B_{3,W}/B_{1,v}). \end{array} \tag{E106.11}$$

The right arrow takes σ_2^j to g^{3j}. Hence

$$\chi_{9,v}(Nu) = 3\chi_{3,w}(u) \quad \text{in } \mathbb{Z}/9. \tag{E106.12}$$

This factor three is the literal inclusion $\mu_3 \hookrightarrow \mu_9$, $\zeta^j = \xi^{3j}$. With the norm variable first, the equivalent Hilbert-symbol identity is $(u, \beta_2)_{B_{2,w},3} = (u, \beta_1)_{B_{2,w},9} = (Nu, \beta_1)_{B_{1,v},9}$. The Artin square fixes the orientation regardless of which Hilbert-symbol argument convention is used. It follows from the restriction/corestriction compatibility of the proved Tate cup-product isomorphism [Mil20b, III,Theorem3.1 and its following proof; III,§4]. This uses that proved compatibility; III,Lemma3.2 itself prints that its proof is to be added.

The E.76 ninth-character rows on the literal nine units are

$$M_9 = \begin{pmatrix} 1 & 5 & 3 & 2 & 6 & 0 & 3 & 4 & 2 \\ 1 & 5 & 3 & 8 & 3 & 6 & 0 & 3 & 6 \\ 1 & 5 & 3 & 5 & 0 & 3 & 6 & 8 & 7 \\ 5 & 7 & 6 & 4 & 0 & 0 & 3 & 1 & 4 \\ 8 & 4 & 6 & 7 & 3 & 3 & 0 & 1 & 1 \\ 1 & 5 & 3 & 5 & 3 & 3 & 3 & 5 & 8 \end{pmatrix} \pmod 9. \tag{E106.13}$$

The exact arithmetic normalization of E94.13 multiplies these rows by $(5, 5, 5, 8, 5, 4)$ modulo nine. Their normalized sum is zero on every global unit. The certificate checks the equality of all nine literal unit polynomials in E.76 and E.101 before composing these maps.

E.101 proves that P is the complete ordinary norm-image lattice. Each of its nine generators is an actual ordinary norm. Every ordinary norm has these coordinates, with the sole relation $3e_1$. Equation (E106.12) consequently proves both containments in the character image, and gives

$$D = \left(\frac{1}{3} \operatorname{diag}(5, 5, 5, 8, 5, 4) M_9 P \right) \bmod 3. \tag{E106.14}$$

Every entry before division is divisible by three. Changing a ninth-character representative changes the divided entry by a multiple of three. The torsion relation is killed. The complete finite row and column transformations reduce D to a rank-four identity block; their inverses are retained. Enumeration of all 3^9 source residues gives 81 image vectors, each with 243 preimages.

Thus the local field-norm index on ordinary units is exactly 81. Units are norms at unramified finite primes, and all infinite primes are complex. The cyclic Hasse norm theorem gives the asserted global kernel [Mil20b, VIII,Theorem3.1 and proof]. The same finite test holds on completion: the unit group is dense and the finite character map is continuous. Equivalently, the completed norm image in the full multiplicative group is the kernel of the completed cyclic norm-obstruction map of E.64.

The entire class group and the prime product. There are precisely six ramified finite primes in B_3/B_2, all of index three. E.91 proves that the vertical generator acts identically on the entire ordinary three-class group. The ambiguous class number formula therefore computes the full order:

$$|\operatorname{Cl}(B_3)\{3\}| = |\operatorname{Cl}(B_2)\{3\}| \, \frac{3^{6-1}}{3^4} = 81 \cdot 3 = 243. \tag{E106.15}$$

The denominator is the ordinary-unit *field*-norm index, not the ordinary-unit norm index [Lem13, Theorem1(1), with its ideal/Herbrand derivation]. The actual Artin map of E.91 is onto A_9, also of order 243, with kernel W. Thus it is an isomorphism and $W = 0$. Its existing prime columns give the full classes (E106.7).

Their subgroup has order 81. The determinant of H_S is 81, and its columns map to zero under C_S with target moduli $(9, 9, 3)$. Equality of the indices proves that it is the full kernel.

Solving its six equations gives (E106.9). Each S-prime class is three-primary: a suitable power extends the corresponding B1 S-prime, whose class has order dividing three. Hence this kernel is actual principality for S-divisors. In particular $Q_1Q_2Q_3$ and every Q_i^3 are principal.

The horizontal cubic B_3/L_3 is the actual S-split unramified extension of E.86. Its prime norms send each Q_i to $\mathfrak{q}_{3,3}$. Thus $\mathfrak{q}_{3,3}^3$ is principal. Its norm to L_2 has order three, proving that its order is exactly three. E.91's group-type theorem now gives $\mathrm{Cl}(L_3)\{3\} = C_9 \oplus C_3$. Norm to L_2 is onto between groups of order 27, so is an isomorphism; it sends the literal coherent a_3, b_3 to a_2, b_2. Therefore $[\mathfrak{q}_{3,3}] = 3a_3$. The exact valuation/norm comparison in E.72 identifies this order with the E.69 finite projection exponent.

All ordinary norm and extension matrices. The vertical B norm uses the actual ray restriction of E.76. Write U_3, U_9 for its integral Smith left maps, R for its ray restriction matrix, ι for E91.7's simple inclusion, and p for E86.7's simple projection. The literal composition is

$$N = pU_3RU_9^{-1}\iota. \tag{E106.16}$$

Its cyclic reduction is the first matrix of (E106.5). It is onto and its kernel is exactly $\{(0,0,0),(0,3,0),(0,6,0)\}$. Since the vertical class action is identity, ideal extension obeys $jN = 3$. Three times any norm preimage is independent of its choice, yielding the displayed matrix j. The certificate checks both identities $jN = 3$, $Nj = 3$ on every class and checks all six specified prime norms.

The L norm is the isomorphism already proved. Its equivariance, $N\sigma = N$, makes the vertical class action identity, hence its extension is $3I_2$. Commuting the actual field norms in (B_3, B_2, L_3, L_2), with E86.9, gives the horizontal norm matrix.

For the horizontal extension the actual primes remove the possible lift ambiguity. The coherent prime above 73 has class $(1,8,2)$; its norm is J_3. The extension of $b_3 = [J_3]_3$ is consequently its h-trace:

$$H = \begin{pmatrix} 4 & 3 & 0 \\ 8 & 7 & 6 \\ 1 & 0 & 1 \end{pmatrix}, \qquad (1 + H + H^2)(1,8,2)^t = (0,3,0)^t \quad \text{in } A_9. \tag{E106.17}$$

The prime $P_3 = (7, \beta_3 + 1)$ is inert in B_3/L_3: its residue field does not contain μ_9. Its extended prime has residue field $\mathbb{F}_{343}$. Its norm to B1 is $(7, \beta_1 + 1)$, with residue degree one in that vertical comparison. This B1 prime splits in B_3/B_1 because $\beta_1 = -1$ is a ninth power in its residue field. The fresh certified B1 ray calculation gives

$$\mathrm{Art}[(7, \beta_1 + 1)] = (6,0,1), \qquad \mathrm{Art}[(73, \xi - 55, \beta_1 - 22)] = (1,8,2). \tag{E106.18}$$

Both complete ideal HNFs and ray/Smith coordinates are retained. Arithmetic Frobenius restriction is the actual norm comparison. Since $a_3 = [J_3/P_3]_3$, subtracting the first column in (E106.18) from (E106.17) yields $j(a_3) = (3,3,2)$, $j(b_3) = (0,3,0)$. The identities $jN = 1 + H + H^2$, $Nj = 3$ hold on every class, and $\ker j = \langle 3a_3 \rangle$.

For this unramified cubic extension the invariant fractional ideals are exactly extended base ideals. Hilbert90 and the principal-ideal sequence identify the capitulation kernel with $H^1(C_3, E_{B_3})$. The unit Herbrand quotient is $1/3$, since all infinite primes are complex. Therefore

$$|\ker j| = 3[E_{L_3} : N_{B_3/L_3} E_{B_3}]. \tag{E106.19}$$

Its left side is three. This proves (E106.6) for the actual unit norm. These are the ideal and Herbrand maps used in E.86 and in the cited ambiguous-class derivation.

The remaining ordinary-unit defect and completion. For B_3/B_2, every invariant ideal is a product of an extended base ideal and powers of the six ramified primes. Thus its strongly

ambiguous three-class subgroup is $j\,\mathrm{Cl}(B_2)\{3\} + R_3 = R_3$, while the full ambiguous group is A_9. For a unit that is a field norm, the ideal Hilbert90 equation in the statement exists, orbit by orbit on primes. Changing its solution adds an invariant ideal. Changing the norm preimage changes it by a principal divisor followed by that invariant ambiguity. Its exact choice-independent obstruction is

$$A_9/R_3 \xrightarrow{\sim} C_3, \qquad (y,z,t) \longmapsto y \bmod 3. \tag{E106.20}$$

Vanishing lets the divisor be adjusted to a principal one; the resulting Hilbert90 correction makes the norm preimage an ordinary unit. An ordinary unit preimage conversely has zero divisor. Every ambiguous ideal class yields a field norm by this construction, proving surjectivity of the extra obstruction. This proves both directions in (E106.10), with the actual ideal-class map [Lem13, Proposition1].

The extra index three times the local index 81 gives ordinary norm index 243. Inclusion of a base unit has norm its cube, so this quotient is killed by three and is C_3^5. For L_3/L_2, the now trivial class action and its two ramified primes give field-unit norm index three by the ambiguous formula. Its strongly ambiguous subgroup is $3D_3 = \langle \mathfrak{q}_{3,3} \rangle$, of order three; the other ramified prime is principal. Its additional defect has order nine. The ordinary norm quotient is therefore C_3^3.

Each norm image contains the cubes and has finite cokernel. Completion of these finitely generated unit groups preserves every displayed image and finite test. The kernel free ranks are $80 - 26 = 54$ for B and $26 - 8 = 18$ for L. Their torsion kernels are the kernels of cubing on μ_9 or μ_3, respectively, namely μ_3. This proves the fiber groups. Unique-prime, residue-degree-one valuation maps also show that an S-unit of norm one has zero S-valuations.

The computation uses the complete E.101 norm image. All finite transformations, the 19 683 character inputs, all 243 upper and 81 lower class coordinates, and the two actual prime Frobenii are retained. It uses no BNF computation in B_2, B_3, L_2 or L_3. □

Stable ordinary classes and the complete complex positional phase

E.107. The fifth character, the exact exponent $h = 1$, and every complex toric fiber [Proved]

Use the literal roots, prime labels and ordinary class coordinates of E.106. For the cubic B_{d+1}/B_d, write $\chi_d : E_{B_d} \to \mathbb{F}_3^6$ for the six arithmetic local norm characters, in the order $Q_1, Q_2, Q_3, T_1, T_2, T_3$. Every character uses the actual generator $\beta_{d+1} \mapsto \zeta\beta_{d+1}$. Put

$$S_Q(x) = x_1 + x_2 + x_3, \qquad S_T(x) = x_4 + x_5 + x_6, \qquad \mathcal{H} = \ker(S_Q + S_T). \tag{E107.1}$$

The complete images are

$$\chi_2(E_{B_2}) = \ker S_Q \cap \ker S_T, \qquad \chi_d(E_{B_d}) = \mathcal{H} \quad (d \geq 3). \tag{E107.2}$$

Their dimensions are four and five, respectively. Five actual ordinary units in B_3, included in each later field, generate every latter image. The proof constructs their classes from the complete E.106 image and an ordinary preimage of E.101's η under N_{B_3/L_3}. The latter unit has the exact triple sums $(S_Q, S_T) = (1, 2)$.

All later ordinary class groups and their actual norm maps stabilize:

$$\begin{gathered} \mathcal{C}_d = \mathrm{Cl}(B_d)\{3\} = C_9^2 \oplus C_3 \quad (d \geq 3), \qquad D_d = \mathrm{Cl}(L_d)\{3\} = C_9 a_d \oplus C_3 b_d \quad (d \geq 2), \\ N_{B_{d+1}/B_d} = I_3, \quad j_{B_{d+1}/B_d} = 3I_3 \quad (d \geq 3), \\ N_{L_{d+1}/L_d} = I_2, \quad j_{L_{d+1}/L_d} = 3I_2 \quad (d \geq 2). \end{gathered} \tag{E107.3}$$

The coordinates are transported by the actual iterated ideal norms, starting with E.106; these equalities specify the maps. The horizontal ordinary class maps for every $d \geq 3$ are

$$N_{B_d/L_d} = \begin{pmatrix} 3 & 6 & 6 \\ 1 & 0 & 0 \end{pmatrix}, \qquad j_{B_d/L_d} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \\ 2 & 0 \end{pmatrix}. \tag{E107.4}$$

The h-action is the E.106 matrix $H = \left(\begin{smallmatrix} 4 & 3 & 0 \\ 8 & 7 & 6 \\ 1 & 0 & 1 \end{smallmatrix}\right)$; the actual g_{d-1}-action is identity on $\mathcal{C}_d$. The six S-prime columns and the principal S-divisor lattice are exactly C_S, H_S of E106.7–9 at every $d \geq 3$. In particular

$$\operatorname{ord}[Q_i] = 3, \qquad Q_1Q_2Q_3 \text{ is principal}, \qquad q_d = [\mathfrak{q}_{3,d}] = 3a_d, \quad \operatorname{ord}(q_d) = 3 \quad (d \geq 2). \tag{E107.5}$$

The actual horizontal ordinary unit norm is onto for every $d \geq 3$. The vertical ordinary norm quotients are

$$E_{B_d}/NE_{B_{d+1}} \simeq C_3^6 \quad (d \geq 3), \qquad E_{L_d}/NE_{L_{d+1}} \simeq C_3^3 \quad (d \geq 2). \tag{E107.6}$$

Their complete norm tests are the local field-norm characters followed, on the zero character kernel, by the actual Hilbert90 divisor quotient $\mathcal{C}_{d+1}/R_{d+1} = C_3$, or $D_{d+1}/\langle q_{d+1}\rangle = C_3^2$, respectively. Every class coordinate is computed by the iterated norm to the fixed E.106 class group. The completed norm-one kernels have types $C_3 \oplus \mathbb{Z}_3^{2\cdot 3^{d+1}}$ and $C_3 \oplus \mathbb{Z}_3^{2\cdot 3^d}$.

For the actual coherent three-unit group of E.69 one has

$$\begin{aligned} N_{L_d/K}\widehat{\mathcal{O}_{L_d}[1/3]^\times}_3 &= \lambda^{3\mathbb{Z}_3} \quad (d \geq 2), \\ I_0 = \operatorname{pr}_0 \mathcal{U}_\infty &= \lambda^{3\mathbb{Z}_3}, \qquad h = 1. \end{aligned} \tag{E107.7}$$

Choose one actual coherent tuple $e \in \mathcal{U}_\infty$ with $e_0 = \lambda^3$, and put

$$\mathcal{U}_\infty^0 = \ker(\operatorname{pr}_0 : \mathcal{U}_\infty \to I_0). \qquad \mathbb{Z}_3 \times \mathcal{U}_\infty^0 \xrightarrow{\sim} \mathcal{U}_\infty, \quad (z,k) \longmapsto e^z k. \tag{E107.8}$$

The inverse extracts $z = v_\lambda(u_0)/3$ and then $k = ue^{-z}$. The compact kernel has the E.69 recursive coordinates

$$\mathcal{U}_\infty^0 \simeq \prod_{d \geq 0} \mathbb{Z}_3^{2\cdot 3^d}. \tag{E107.9}$$

Changing e multiplies it by one element of this exact kernel.

The remaining complex positional phase is now fully evaluated. Use $\mathcal{P}, \mathscr{R}(\mathcal{P}), \mathcal{K}$ and the actual classes $(\ell_p, m_p) \in C_9 \oplus C_3$ of E.88, with $0 \leq \ell_p < 9$. Define on its specified domain

$$\mathcal{K} = \left\{ t \in \mathscr{R}(\mathcal{P}) : \sum c_p t_p = \sum d_p t_p = 0 \pmod 3 \right\}, \qquad \Theta_*(t) = -\frac{\sum_{p \in \mathcal{P}} \ell_p t_p}{3}. \tag{E107.10}$$

Here $c_p = \ell_p \bmod 3$. The numerator belongs to $3\mathbb{Z}_3$; the sum converges in the restricted multiplication-by-3^m topology. There is a continuous coherent divisor splitting s_* with valuation zero at seventeen and phase exactly Θ_*. The following are full forward and inverse coordinates:

$$\begin{aligned} &\mathbb{Z}_3^2 \times \mathcal{U}_\infty^0 \times \mathcal{K} \xrightarrow{\sim} \mathcal{X}, && (b,z,k,t) \longmapsto \beta^b e^z k\, s_*(t), \\ &\qquad t = \operatorname{div}_{\notin S} x, && b = v_{17}(x_0), \qquad z = \frac{v_\lambda(x_0) - \Theta_*(t)}{3}, \\ &\qquad k = x\beta^{-b} e^{-z} s_*(t)^{-1}. \end{aligned} \tag{E107.11}$$

In this formula v_{17} is the normalized valuation at the unique prime of K above seventeen; $\beta_0 = 17\zeta$. The resulting actual rational base norm, in $\widehat{\mathbb{Q}^\times}_3$, is

$$N_{K/\mathbb{Q}}(x_0) = 17^{2b}\, 3^{\,3z-\frac{1}{3}\sum \ell_p t_p} \prod_{p\in\mathcal{P}} p^{t_p}. \tag{E107.12}$$

Changing the normalized splitting while retaining the exact phase Θ_* is precisely multiplication by a continuous homomorphism $\mathcal{K} \to \mathcal{U}_\infty^0$. Both directions of this description hold.

Here is the exact complete fiber statement in the section target H. Fix $A = 3\kappa_3(17)$ in E.46 and the pure positional component 1. The image on the scaled toric coordinates consists exactly of

$$\begin{gathered}\left(\kappa_3(17),\ B_{17}\kappa_3(17) + B_3\kappa_3(3) + \sum_{p\in\mathcal{P}} r_p\kappa_3(p)\right), \\ B_{17}, B_3 \in \mathbb{Z}_3, \qquad r \in \mathscr{R}(\mathcal{P}), \qquad \sum c_p r_p = 0 \pmod 3.\end{gathered} \tag{E107.13}$$

For every such pair its entire fiber is $\mathcal{U}_\infty^0$. The explicit map of that fiber and its inverse are

$$\begin{aligned} k \longmapsto x &= \beta^{\,3B_{17}/2}\, e^{\,B_3+\frac{1}{3}\sum \ell_p r_p}\, k\, s_*(3r), \\ k &= x\beta^{-3B_{17}/2} e^{-B_3-\frac{1}{3}\sum \ell_p r_p} s_*(3r)^{-1}. \end{aligned} \tag{E107.14}$$

Every exponent in this display is integral in $\mathbb{Z}_3$. The comparison to section classes is the actual Kummer–Shapiro map E.46 and its total-norm cube condition; the pure component has been specified. This evaluates the complete complex component and its toric fibers, without claiming an intrinsic lift or an evaluation of the separate pure-field norm system. The manuscript anchor is LM §51.16, folios733v–736v, scans172–178 through E.46, E.69, E.72 and E.88.

Proof. Inclusion of consecutive actual character maps. For every $d \geq 2$, the extension B_{d+1}/B_{d-1} is cyclic of degree nine. Its generator g sends β_{d+1} to $\xi\beta_{d+1}$; its subgroup over B_d is generated by g^3. All six primes are totally ramified, with residue degree one and the previously fixed unique vertical prolongations. For $u \in B_{d-1}^\times$ included in B_d, the E.106 local Artin square consequently gives

$$\begin{aligned} 3\chi_{d,w}(u) = \chi_{9,v}(N_{B_{d,w}/B_{d-1,v}}u) = \chi_{9,v}(u^3) = 3\chi_{9,v}(u) \pmod 9, \\ \chi_{9,v}(u) \pmod 3 = \chi_{d-1,v}(u). \end{aligned} \tag{E107.15}$$

The second equality is literal restriction: g sends $\beta_d = \beta_{d+1}^3$ to $\zeta\beta_d$. Division by the injection $\mathbb{Z}/3 \xrightarrow{3} \mathbb{Z}/9$ therefore proves

$$\chi_d(u) = \chi_{d-1}(u) \quad \text{for every } u \in E_{B_{d-1}}. \tag{E107.16}$$

Thus the images are nested, with precisely the same arithmetic generators and labels. This is the actual norm/inclusion square, obtained from the proved restriction/corestriction compatibility of the Tate cup product [Mil20b, III,Theorem3.1 and its proof; III,§4]. It does not require an identification of different abstract norm characters.

The actual fifth character and both split-prime sums. The complete matrix D in E106.3 has rank four and each of its two triple sums is zero. Hence its image is exactly $\ker S_Q \cap \ker S_T$, an explicitly specified subspace of dimension four.

Let $\eta \in E_{L_2}$ be the actual ordinary unit constructed in E101.19 and its following residue paragraph, and include it in L_3. At seventeen its residue lies in $\mathbb{F}_{289}$ and satisfies

$$\bar\eta^3 = \zeta^2, \qquad \bar\eta^{96} = \zeta. \tag{E107.17}$$

The residue field and residue value remain unchanged under every later totally ramified vertical inclusion. The tame unit character at seventeen is calibrated in the established arithmetic convention by ξ: E106.3 gives $\chi_{2,T_i}(\xi) = 1$, while $\bar{\xi}^{96} = \zeta^2$. Since the norm on residues of a totally ramified tame cubic is cubing, its unit character is therefore

$$\chi_{17}(a) = -\log_\zeta(\bar{a}^{96}) \pmod 3. \tag{E107.18}$$

This determines the character on the entire residue field, whose cube quotient has order three. Principal units are cubes because their pro-seventeen group admits unique division by three. The uniformizer β_d has valuation one, so these are the characters for the actual cubic radical generators at every level. The residue calculation gives $\chi_{17}(\eta) = 2$.

For L_4/L_3, all other finite primes are unramified and η is an ordinary unit, so their Artin characters vanish. All infinite places are complex. Global reciprocity with the actual generator $\beta_4 \mapsto \zeta\beta_4$ thus gives the character at the unique prime above three as 1.

E.106 proves the actual surjection $N_{B_3/L_3}E_{B_3} = E_{L_3}$. Choose $v \in E_{B_3}$ with $N_{B_3/L_3}v = \eta$. Each S-prime of L_3 splits into three degree-one primes in the unramified extension B_3/L_3. The equality $B_4 = B_3L_4$ identifies the local cubic groups by the same generator on β_4. Writing v_i for the three actual images in the common completion, the local norm is $\prod_i v_i$, and the Artin maps are homomorphisms. Hence the two exact squares give

$$\begin{aligned} S_Q(\chi_3(v)) &= \chi_{L_4/L_3,3}\left(\prod_{i=1}^{3} v_{Q_i}\right) = \chi_{L_4/L_3,3}(\eta) = 1, \\ S_T(\chi_3(v)) &= \chi_{L_4/L_3,17}\left(\prod_{i=1}^{3} v_{T_i}\right) = \chi_{L_4/L_3,17}(\eta) = 2. \end{aligned} \tag{E107.19}$$

There is no residue-degree multiplier in either square: all three horizontal completions have degree one. The horizontal automorphism fixes ζ and β_4, so it does not change the target generator between the three maps.

This fifth vector is independent of the rank-four included image in (E107.16). For every ordinary unit of B_d, all local characters away the six S-primes vanish. Global reciprocity therefore gives $S_Q + S_T = 0$, proving the rank-five upper bound at every stage. This proves (E107.2), including its persistence under all later inclusions. More explicitly, put $w_0 = (1, 0, 0, 2, 0, 0)^t$. The difference $\chi_3(v) - w_0$ belongs to the actual rank-four image. E.106's image inverse and ordinary unit preimages therefore supply $u \in E_{B_2}$ with that character. Replacing v by v/u gives the literal fifth generator of character w_0. Together with four units realizing the retained E.106 image columns, it proves the five-unit assertion. The certificate records a coordinate inverse for these five columns on the entire reciprocity hyperplane.

Stable ordinary classes and exact prime maps. The cyclic Hasse norm theorem converts the complete image $\mathcal{H}$ into ordinary-unit field-norm index 3^5 for every B_{d+1}/B_d, $d \geq 3$ [Mil20b, VIII,Theorem3.1 and proof]. There are six ramified primes, each of ramification index three, and no real places. E.91 proves that the actual vertical generator acts identically on the entire ordinary three-class group. The ambiguous class number formula therefore gives

$$|\mathcal{C}_{d+1}| = |\mathcal{C}_d|\frac{3^{6-1}}{3^5} = |\mathcal{C}_d| \qquad (d \geq 3). \tag{E107.20}$$

The formula uses the field-norm index just evaluated [Lem13, Theorem1(1) and its ideal/Herbrand derivation].

Every adjacent ordinary class norm is onto. Indeed, for either ramified cyclic cubic E/F, its intersection with the Hilbert three-class field H_F is F: otherwise the cyclic extension would itself be unramified. In EH_F/F, the Frobenius elements $(1, a)$ give degree-one primes of E

whose ideal norms represent every $a \in \mathrm{Gal}(H_F/F)$. This is the actual ideal-norm/Artin square, not an order comparison without a map [Mil20b, V,Proposition3.3,Theorem3.6,Example3.9; VIII,Theorem7.4]. Consequently (E107.20) makes all B norms from level three isomorphisms.

The unique vertical S-prime prolongations have residue degree one, so these isomorphisms send each named prime to the same named prime at the preceding level. Every prime column in E106.7 is therefore retained, and its integral relation lattice is exactly E106.8–9 at every level. All these prime classes are three-primary, as proved there from their powers extending B1 primes. Thus these relations are actual principality. In particular every Q_i has order three and their product is principal. The horizontal norm sends Q_i to $\mathfrak{q}_{3,d}$; its norm down to L_2 has order three. The original prime therefore has order exactly three at every $d \geq 2$. E91.3 now gives $D_d = C_9 \oplus C_3$. Its surjective norms are isomorphisms too.

Transporting the actual coordinate bases by these norm isomorphisms proves every identity in (E107.3). In particular the vertical actions on D_d are identity, and the ideal identity $jN = \sum \sigma^i$ gives $j = 3I$. On the B groups this action identity was already E.91. The restriction of g_{d-1} to B_{d-1} is the vertical generator there, whose class action is identity. Norm equivariance and its injectivity prove that g_{d-1} also acts identically for $d \geq 4$; the case $d = 3$ is the vanishing $W = 0$ in E.106. The h-action transports to the same matrix H.

The norm squares commute by transitivity. Extension also commutes with the other vertical norm in the horizontal square: in this linearly disjoint square, h and the vertical generator commute, and the prime ramification/residue coefficients agree; equivalently it is the ideal restriction/transfer identity for the two subgroups. Thus both horizontal matrices remain E106.5. Their extension kernel has order three at every level. Hilbert90 for principal ideals in this unramified cubic, and the unit Herbrand quotient 1/3, give

$$3 = |\ker j_{B_d/L_d}| = 3[E_{L_d} : N_{B_d/L_d} E_{B_d}] \quad (d \geq 3). \tag{E107.21}$$

This proves all the horizontal ordinary norm surjections. The Herbrand quotient here is the ordinary-unit case of the proved lattice comparison [Mil20b, VII,Proposition3.1 and its complete proof].

The six S-primes generate the same order-81 subgroup $R_d \subset \mathcal{C}_d$. The strongly ambiguous subgroup for an adjacent B step is $j\mathcal{C}_d + R_{d+1} = R_{d+1}$; the inclusion $3\mathcal{C}_{d+1} \subset R_{d+1}$ follows from the displayed prime columns. Its quotient in the full ambiguous group has order three and is the coordinate y mod 3. The exact divisor Hilbert90 map of E106.20 therefore gives the additional ordinary-unit norm index three. Multiplied by the field index 3^5, this is 3^6.

For an L step, the included η has noncube residue at seventeen. Global reciprocity at the two ramified primes bounds its ordinary-unit field-norm image by dimension one and this residue makes that dimension exactly one. The strongly ambiguous subgroup is $3D_{d+1} = \langle q_{d+1} \rangle$, of order three; the other ramified prime is the principal divisor of β_{d+1}. The remaining quotient has order nine. The full ordinary norm index is therefore $3 \cdot 9 = 27$. These quotient groups are killed by three because an included base unit has norm its cube, proving (E107.6). The same explicit Hilbert90 divisor argument as E.106 proves necessity and sufficiency of each stated two-step test. Completion preserves these finite-index norm images. Unit ranks and the cubing kernels on μ_9 or μ_3 give the stated completed kernel types.

The exact coherent three-unit exponent. The distinguished prime now has order three at every $d \geq 2$. The principal-valuation/norm identity of E.72 consequently gives the first equality of (E107.7). There is no missing root-of-unity coordinate: all these images factor through the literal $N_{L_1/K}\mathcal{U}_1 = \lambda^{\mathbb{Z}_3}$ of E69.16.

For clarity, this finite-level assertion supplies a compatible tuple of base norm λ^3. Each completed three-unit group $\mathcal{U}_d$ is compact. In their product impose the closed adjacent norm equations and the equation $u_0 = \lambda^3$. Any finite set of these conditions is satisfied by an element

at the largest specified level, of norm λ^3, and all its lower norms. The finite-intersection property gives a tuple in the full product. Conversely every coherent tuple projects to each finite norm image, hence to $\lambda^{3\mathbb{Z}_3}$. This proves $I_0 = \lambda^{3\mathbb{Z}_3}$ and $h = 1$ with the actual transition maps. It proves the existence of e in (E107.8). The displayed maps there are continuous inverses; changing e uses exactly $\ker \mathrm{pr}_0$. The already proved recursive norm-kernel construction E69.6 gives (E107.9), with its specified continuous forward and inverse maps. No compactness is asserted for a restricted divisor module.

Exact phase normalization and every toric fiber. Start with an actual continuous splitting s supplied by E88.7, normalized to have valuation zero at seventeen. Its phase $\Theta(t)$ satisfies $\Theta(t) \equiv -\sum \ell_p t_p/3 \pmod 3$ by E88.10. The right side is an integral continuous linear map on $\mathcal{K}$: its numerator is divisible by three because $\ell_p \equiv c_p \pmod 3$, and the restricted sequence condition makes its sum convergent. Define

$$q(t) = \frac{\Theta_*(t) - \Theta(t)}{3} \in \mathbb{Z}_3, \qquad s_*(t) = s(t)e^{q(t)}. \tag{E107.22}$$

This is an actual continuous coherent map. It preserves the outside-S divisor and the seventeen valuation, and adds $3q(t)$ to the base three valuation. Hence its phase is exactly Θ_*. Two splittings with these same data differ by a continuous homomorphism to $\mathcal{U}_\infty$ with base valuation zero. Since its base image is exactly $\lambda^{3\mathbb{Z}_3}$, that condition is precisely membership in $\mathcal{U}_\infty^0$. Conversely multiplying by any such homomorphism preserves all the data. This proves the full normalization ambiguity.

Now E88.7 and (E107.8) give (E107.11), including its inverse: after removing the outside divisor and the seventeen coordinate, the remainder is a coherent three-unit, whose base valuation is divisible by three. The compact factors are continuous under $\mathbb{Z}_3$-powers. The restricted factor is continuous by the convergent products proved in E.88, and (E107.22) is a continuous correction. These facts prove the asserted topological isomorphism and all inverse-coordinate claims. Taking the actual norm of the zeroth coordinate uses $N\beta_0 = 17^2$, $N\lambda = 3$, and degree-one norm p for each chosen outside-S prime, giving (E107.12). Prime-to-three rational torsion has already disappeared in $\widehat{\mathbb{Q}^\times}_3$.

The total-norm cube condition in E.46 can now be tested coefficient by coefficient in that same completed multiplicative group. Equation (E107.12) is a cube exactly when

$$b \in 3\mathbb{Z}_3, \qquad t = 3r, \qquad \sum \ell_p r_p = 0 \pmod 3 \iff \sum c_p r_p = 0 \pmod 3. \tag{E107.23}$$

When $t = 3r$, both defining class congruences for $t \in \mathcal{K}$ already hold. Dividing the norm exponents by three, as prescribed by the actual toric map E46.7, gives

$$B_{17} = 2b/3, \qquad B_3 = z - \frac{1}{3}\sum \ell_p r_p. \tag{E107.24}$$

These two equations have the unique inverse used in (E107.14), because two is a unit in $\mathbb{Z}_3$. All outside coefficients are the specified r_p. The remaining coordinate is exactly $k \in \mathcal{U}_\infty^0$, with no additional stabilizer in the E.46 section comparison. This proves both directions of (E107.13) and the complete fiber statement with its actual maps.

The certificate exhausts the rank-four kernel, all eighty-one possible fifth vectors with triple sums $(1, 2)$, the entire reciprocity hyperplane and its coordinate inverse. It checks all residue units and their character products in $\mathbb{F}_{289}$, every stable B class, the principal S-lattice, the full retained seventy-nine-prime phase basis, and the complete modulo-nine cube condition on five actual prime coordinates. The all-level conclusions follow from the norm squares and induction just proved; no upper BNF is used. □

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[Mil08] James S. Milne. Abelian varieties. Version 2.00, March 2008, 2008. Verified: author PDF opened; II Theorem1.1(b) and complete proofs through Lemmas1.2–1.3, printed pages75–78; III Proposition9.1 and its prime-to-characteristic Kummer proof through Lemmas9.2–9.3 and Corollary9.6, printed pages113–115, read.

[Mil20a] James S. Milne. Algebraic number theory, 2020. Verified: original author notes, version3.08 dated19July2020, opened; Theorem4.3, printed p.70 (PDF page72), is the ideal-class Minkowski bound used for Q(zeta8).

[Mil20b] James S. Milne. Class field theory. Author's course notes, version 4.03, 2020. verified: Opened primary text: Theorem I.4.16 and its proof, Theorem V.3.23.

[Mil21] James S. Milne. *Elliptic Curves*. World Scientific, 2 edition, 2021. verified: Author-hosted second edition opened. Chapter II Proposition 6.4 identifies the Tate characteristic polynomial and determinant with the endomorphism polynomial and degree; Chapter IV Proposition 9.1 and Theorem 9.4 proof give Frobenius degree and the point-count trace; Chapter V Proposition 8.1 gives unramified prime-to-residue torsion. D.35 performs its own complete point counts and exact unit-group generation.

[MO97] Yves Martin and Ken Ono. Eta-quotients and elliptic curves. *Proceedings of the American Mathematical Society*, 125(11):3169–3176, 1997. Verified: full original author PDF opened; Theorems1–2 and their proofs, author-PDF pp.2–5, give the conductor64 eta quotient and the model y squared equals x cubed minus4x. The author's publication page independently verifies the journal, year, volume and pages.

[Nak95] Hiroaki Nakamura. On exterior Galois representations associated with open elliptic curves. *Journal of Mathematical Sciences, the University of Tokyo*, 2:197–231, 1995.

[Nak13] Hiroaki Nakamura. On arithmetic monodromy representations of eisenstein type in fundamental groups of once punctured elliptic curves. *Publications of the Research Institute for Mathematical Sciences*, 49(2):413–496, 2013. verified: official complete original opened and read, including equations(3.6.1)–(3.6.3), Theorem5.2.4 and equation(5.2.5), Proposition5.7.3 with proof, Lemma5.9.1, Section5.10 and Theorem6.10.3. E58 retains the actual modular-path, discriminant-root and tangent comparison.

[NS07] Nikolay Nikolov and Dan Segal. On finitely generated profinite groups, I: Strong completeness and uniform bounds. *Annals of Mathematics*, 165(1):171–238, 2007. verified: Theorem 1.4 opened in the original article; abstract commutator subgroups are closed.

[NTY10] Hiroaki Nakamura, Hiroshi Tsunogai, and Seidai Yasuda. Harmonic and equianharmonic equations in the Grothendieck–Teichmüller group. III. *Journal of the Institute of Mathematics of Jussieu*, 9(2):431–448, 2010. verified: Complete original opened from the author's public ResearchGate deposit. Proposition4.4 is a full-GT cover-Tate formula; Proposition4.13 and Theorem4.16 are explicitly Galois statements. Introduction p432 states the unresolved broader power-conjugacy question. No extension to GT0 is inferred.

[Ots16] Noriyuki Otsubo. Homology of the Fermat tower and universal measures for Jacobi sums. *Canadian Mathematical Bulletin*, 59(3):624–640, 2016. Verified: original PDF opened; sections 4–6, especially Proposition 4.4 and Theorem 6.1, checked.

[PS03] Stefan Papadima and Alexander I. Suciu. Chen lie algebras, 2003. verified: Corollary 5.6 and Theorem 6.2, including the integral Koszul maps (6.3)–(6.4), opened in version 1, pp. 11–12.

[Rib76] Kenneth A. Ribet. A modular construction of unramified p-extensions of q(mu-p). *Inventiones Mathematicae*, 34:151–162, 1976. verified: original author-hosted full PDF opened; printed p.151, Main Theorem (1.1) and its stated Herbrand direction visually checked. The k=2 specialization is used only for primes at least five.

[Roh77] David E. Rohrlich. Points at infinity on the Fermat curves. *Inventiones Mathematicae*, 39:95–127, 1977. Verified: original digitized article opened; definitions on p.95 and Theorem 1 on p.101 checked, including the even-level factor.

[Sat10] Takao Satoh. On the lower central series of a free abelian by polynilpotent group. Kyoto University preprint 2010-05, 2010. verified: Section 2.4, equation (2) and the following integral Chen basis opened on p. 5; the pro-three comparison is proved separately in E.41.

[Shaa] Romyar Sharifi. Iwasawa theory. Author lecture notes, Chapter 1: Class groups and units. Verified author text and proofs opened 13 November 2026: Theorems 1.5.4, 1.5.7, 1.5.17 and 1.5.21; the last proves the abelian-field reduction from Brumer's logarithm theorem.

[Shab] Romyar Sharifi. Iwasawa theory: Cyclotomic fields. Author lecture notes, Chapter 4. verified: full author PDF and HTML opened, Theorem 4.3.3 and its regulator proof on printed pp.108–109 state the exact real circular-unit index and the independent normalized generators; Lemma 4.3.6 on p.110 proves the exact relative norm of 1 minus a cyclotomic root.

[Sti12] Jakob Stix. On cuspidal sections of algebraic fundamental groups. In *Galois–Teichmüller Theory and Arithmetic Geometry*, volume 63 of *Advanced Studies in Pure Mathematics*, pages 519–563. 2012. Verified: author PDF opened; Definition26, Lemma28, Theorem30 and its proof, Corollaries31–32 read, with the characteristic-zero/full-profinite scope on page1 retained. Publication metadata checked on the author's publication page. E.61 proves the required pro-three finite-field variant with its actual cover maps.

[Stö08] Ralph Stöhr. Bases, filtrations and module decompositions of free lie algebras. *Journal of Pure and Applied Algebra*, 212(5):1187–1206, 2008. verified: Lazard Elimination Theorem over an arbitrary commutative ring, Section 2, p. 3 of the opened author preprint.

[Sut19] Andrew V. Sutherland. 18.783 elliptic curves: Lecture 24, divisors and the weil pairing, 2019. verified: Corollary 24.27, p. 12: the normalized tangent-function formula, including its sign.

[The26] The PARI Group. *PARI/GP 2.19.0: General Number Fields,* `bnfinit` *and* `bnfcertify`. Institut de Mathématiques de Bordeaux, 2026. verified: opened the official descriptions of bnfinit and bnfcertify, including the exact certification meaning of return value one; downloaded PARI 2.19.0 from the official source archive, checked the published SHA-256, compiled it, and executed all eight unconditional certifications and five unit norm calculations.

[Vél79] Jacques Vélu. Le groupe cuspidal des courbes de Fermat. *Séminaire Delange–Pisot–Poitou. Théorie des nombres*, 20(2):1–11, 1978–1979. Exposé 28. Verified: original PDF opened; Theorem 1 on p.28-02 and the relation calculation on pp.28-08–28-10 checked.

[Wei13] Charles A. Weibel. *The K-book: An Introduction to Algebraic K-theory*, volume 145 of *Graduate Studies in Mathematics*. American Mathematical Society, 2013. verified: opened author PDF; III.5.2.2, III.6.1, III.6.3, III.6.5–III.6.5.1 checked.

[Wic12a] Kirsten Wickelgren. On 3-nilpotent obstructions to fundamental group sections for the projective line minus three points. In Jakob Stix, editor, *The Arithmetic of Fundamental Groups—PIA 2010*, volume 2 of *Contributions in Mathematical and Computational Sciences*. Springer, 2012. verified: Section 12.2.1, equations (12.4)–(12.10), opened for tangential paths and the exact Kummer cocycle convention.

[Wic12b] Kirsten Wickelgren. On 3-nilpotent obstructions to pi1 sections for P1 over the rationals minus three points. In *The Arithmetic of Fundamental Groups—PIA 2010*, volume 2 of *Contributions in Mathematical and Computational Sciences*. Springer, 2012. verified: author's full text opened; Lemma 4 and equations (12.13)–(12.15). It computes the thrice-punctured-line obstructions; no cubic section classification is inferred from it.

[Wil20] Gareth Wilkes. Classification of pro-p PD^2 pairs and the pro-p curve complex. *Groups, Geometry, and Dynamics*, 14(3):917–948, 2020. verified: Original author preprint arXiv:1711.02953v2 opened, especially section 1.3, Theorem 3.3 and its proof on preprint pages 15–19, and Theorem 4.6. Publisher metadata checked at EMS Press. Used for the relative surface pair and orientation; its standard-form classification is not asserted to preserve an additional finite deck action.

[Win] Kay Wingberg. Free quotients of demuškin groups with operators. Author-hosted original. verified: Full author original opened, including the introduction, Proposition3.1, Lemma3.2, Proposition3.3 and Theorem3.4 with their proofs. The orientation factorization is Proposition3.3 in this version.

[Win07] Kay Wingberg. Riemann's existence theorem and the $K(\pi, 1)$-property of rings of integers, 2007. verified: Author's preliminary version, December 6, 2007. Preceding continuous-bundle definition and Lemma 1.1 with proof, Proposition 2.1, and Theorem 2.4 including the infinite-set clause and proof opened and checked.